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Preface

Linear Algebra is often the first place where a student encounters the true power of mathematical abstraction. In an introductory course, one learns to manipulate matrices, solve systems of linear equations, and perhaps calculate eigenvalues. But *Advanced* Linear Algebra is about something deeper: it is the study of structures.

The Philosophy of this Book

The central theme of these notes is the transition from concrete calculation to abstract reasoning. We begin not with numbers, but with axioms. By choosing to define mathematical objects—be they vectors, linear transformations, or inner products—entirely by how they behave rather than what they are, we unlock a universal language.

A theorem proved for an abstract vector space is a theorem proved simultaneously for geometric arrows, for polynomials, for continuous functions, and for quantum states. This “solve once, apply everywhere” philosophy is the heartbeat of modern mathematics.

How to Use These Notes

This book is designed to be a rigorous companion for a second course in linear algebra. Each chapter is structured to build a solid theoretical foundation:

- **Rigorous Definitions:** We start with precise mathematical statements to ensure no ambiguity in our logic.
- **Complete Proofs:** Every major result is accompanied by a full proof. Mathematics is not a spectator sport; I encourage you to attempt the proofs yourself before reading the solutions.
- **Illustrative Examples:** Abstract theory is balanced with concrete examples to build intuition.
- **Exercises:** Practice problems are provided at the end of sections to test your understanding.

Prerequisites

To get the most out of these notes, you should have a solid grasp of:

- **Introductory Linear Algebra:** You should be comfortable with matrix multiplication, determinants, and Gaussian elimination.

- **Calculus:** Familiarity with derivatives and integrals is assumed for some of our more advanced examples.
 - **Mathematical Maturity:** A basic understanding of logic and proof techniques (induction, contradiction, contrapositive) is essential. These are reviewed briefly in Chapter 0.
-

Welcome to the world of abstract linear algebra. I hope you find as much beauty in these structures as I do.

Chapter 0

Preliminaries

Most people meet linear algebra as a way of computing: row reduction, determinants, maybe eigenvalues. This book treats it as a subject you **prove**. Every statement comes with a reason, and those reasons are built from a small set of ideas: sets, functions, equivalence relations, fields, polynomials and matrices. This chapter collects that toolkit in one place.

You have probably seen some of this before. Some of it, such as negating a statement with several quantifiers, or checking that a rule on equivalence classes is well defined, is exactly where careful readers later stumble. So each section spends its time on the delicate points and moves quickly past the rest.

How to use this chapter. Skim the sections you know, but try the *A. Check your understanding* exercises at the end of each one. If you can do them without looking back, move on. If not, that section is worth a slower read.

What you need. Nothing from this book. School algebra is enough, plus some familiarity with numbers such as $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$. One convention to note now: in this book the natural numbers $\mathbb{N}$ start at 0.

0.0.1 Roadmap

- **Statements, implications and quantifiers.** How mathematical statements are built, and how to negate them correctly.
- **How proofs are built.** Direct proof, contrapositive, contradiction, cases, uniqueness, counterexamples and induction, and the conventions this book's proofs follow.
- **Sets.** Subsets, operations, indexed families, and the habit of proving two sets equal by double inclusion.
- **Functions.** Injective, surjective and bijective maps; composition; left, right and two-sided inverses.
- **Relations, equivalence classes and quotients.** Sorting a set into classes, and defining functions on classes without ambiguity.
- **Complex numbers.** Arithmetic, conjugates, the modulus, polar form and roots of unity.
- **Fields.** The number systems in which linear algebra works, from $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ to the finite fields $\mathbb{F}_p$.
- **Polynomials.** Degree, evaluation, and why a polynomial is not the same thing as a function.

- **Matrices.** Addition, the three ways to read a matrix product, transpose, trace and inverses.
- **Groups and permutations.** One structure shared by invertible matrices, the non-zero elements of a field, and rearrangements of a set.

0.0.2 Named moves

These proof techniques appear throughout the book. Each is introduced in this chapter.

- **Contrapositive.** To prove “if P then Q ”, prove “if not Q then not P ”.
- **Assume two.** To prove something is unique, suppose two objects have the property and show they are equal.
- **Double inclusion.** To prove $A = B$, prove $A \subseteq B$ and $B \subseteq A$.
- **Suppose $f(a) = f(b)$.** The standard first line when proving that a function is injective.
- **Check well-definedness.** A rule given through a representative must not depend on which representative is chosen.
- **Induction on n .** Find a smaller instance of the problem inside the larger one.

0.1 Statements, Implications and Quantifiers

Linear algebra is a subject of proofs. Every theorem in this book has the shape “if these hypotheses hold, then this conclusion holds”, and most definitions hide a “for every” or a “there exists”. Before we can prove anything, we need to read such sentences exactly: when they are true, what they claim, and what their opposite says. This section builds that reading skill. It is the grammar that the rest of the book is written in.

0.1.1 Statements and connectives

A **statement** is a sentence that is either true or false, but not both. “7 is prime” is a statement, and it is true. “ $2 + 2 = 5$ ” is a statement, and it is false. “Is 7 prime?” is not a statement, and neither is “ $x > 3$ ” on its own: its truth depends on x , which nobody has chosen yet. Sentences with a free variable are handled by quantifiers later in this section.

We build longer statements from shorter ones with a few fixed words. Everyday language uses these words loosely, so mathematics pins each one down by saying exactly when the combined statement is true.

A connective builds a new statement whose truth depends only on the truth of its parts.

Definition 0.1.1 — Negation, conjunction, disjunction

Let P and Q be statements.

- The **negation** $\neg P$ (“not P ”) is true when P is false, and false when P is true.
- The **conjunction** $P \wedge Q$ (“ P and Q ”) is true when **both** P and Q are true, and false otherwise.
- The **disjunction** $P \vee Q$ (“ P or Q ”) is true when **at least one** of P , Q is true,

and false only when both are false.

A **truth table** lists every combination of truth values of the parts, one per row, and records the value of the compound statement. With T for true and F for false:

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$
T	T	F	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	F

Two statement forms are **logically equivalent**, written $\equiv$, if they have the same truth value in every row of the truth table. For example, $\neg(\neg P) \equiv P$: negating twice returns the original value in both rows.

A few examples, checked against the definition:

- “7 is prime **and** 7 is even” is false, because its second part is false.
- “7 is prime **or** 7 is even” is true, because its first part is true.
- “ $2 < 3$ **or** $3 < 4$ ” is true. Both parts are true, and that is allowed.

Warning. In mathematics “or” is **inclusive**: $P \vee Q$ is true when both parts are true. The menu phrase “soup or salad”, which forbids having both, is a different connective. So “6 is even or 6 is divisible by 3” is true, even though both halves hold.

0.1.2 Implication

Almost every theorem is an implication: “if P , then Q ”. It is worth being completely precise about when such a sentence counts as true, because the answer surprises many readers in one row of the table.

Think of an implication as a **promise**. A parent says: “If you finish your homework, then you may watch a film.” When has the parent broken the promise? Only when the homework is finished and the film is still refused. If the homework is not finished, the promise says nothing about what happens, so whatever happens, the promise was not broken.

An implication is false only when the hypothesis holds and the conclusion fails.

Definition 0.1.2 — Implication, converse, contrapositive

Let P and Q be statements.

- The **implication** $P \Rightarrow Q$ (“if P then Q ”) is false when P is true and Q is false, and it is true in **every** other case. We call P the **hypothesis** and Q the **conclusion**.
- The **converse** of $P \Rightarrow Q$ is the implication $Q \Rightarrow P$.
- The **contrapositive** of $P \Rightarrow Q$ is the implication $\neg Q \Rightarrow \neg P$.

In words: the first clause says an implication can only fail in one way, by a true hypothesis leading to a false conclusion. The converse swaps the roles of hypothesis and conclusion. The

contrapositive swaps them **and** negates both.

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

The implication $P \Rightarrow Q$ is read in many ways, and all of them mean the same thing:

- “ P implies Q ”, “ Q whenever P ”, “ Q if P ”;
- “ P only if Q ”;
- “ P is **sufficient** for Q ”, “ Q is **necessary** for P ”.

The phrase “only if” trips people up. “ P only if Q ” says that P cannot happen without Q , which is exactly $P \Rightarrow Q$.

The last two rows of the table, where the hypothesis is false, are called **vacuous** cases. An implication with a false hypothesis is **vacuously true**.

Example. (A vacuously true implication)

Let n be an integer. Is the statement “if n is odd and n is even, then $n = 7$ ” true?

Solution. Yes. No integer is both odd and even, so the hypothesis “ n is odd and n is even” is false for every integer n . By [Definition 0.1.2](#), an implication with a false hypothesis is true, whatever the conclusion says. So the statement is true for every integer n , including $n = 7$ and $n = 0$. It is true, but it tells us nothing.

Why this definition. The two vacuous rows may look like a convention picked for convenience. They are forced once we want implications to work inside “for every” statements. Consider the true fact

for every integer n , if $n > 2$ then $n^2 > 4$.

For this to be true, the implication must be true for **every** integer n . Try $n = -3$: the hypothesis $-3 > 2$ is false, and the conclusion $9 > 4$ is true. Try $n = 1$: the hypothesis $1 > 2$ is false, and the conclusion $1 > 4$ is false. If either row were declared false, this perfectly good theorem would become false. So both vacuous rows must be T. In the same way, a theorem with hypotheses says nothing about objects that fail them, and that is exactly how we want to read it.

Remark. The implication $P \Rightarrow Q$ does **not** claim that P causes Q , or that the two are related at all. “If $2 + 2 = 4$, then 7 is prime” is a true implication, because both parts are true. In practice we only ever prove implications where the hypothesis is used, but the truth value does not depend on that.

The contrapositive is the most useful rewriting of an implication, because it is always interchangeable with the original.

Theorem 0.1.1 — Contrapositive equivalence

For all statements P and Q , the implication $P \Rightarrow Q$ is logically equivalent to its contrapositive $\neg Q \Rightarrow \neg P$.

Proof. Let P and Q be statements. We compute both sides in each of the four rows, using Definition 0.1.1 for the negations and Definition 0.1.2 for the implications.

P	Q	$\neg Q$	$\neg P$	$P \Rightarrow Q$	$\neg Q \Rightarrow \neg P$
T	T	F	F	T	T
T	F	T	F	F	F
F	T	F	T	T	T
F	F	T	T	T	T

In the second row, $\neg Q$ is true and $\neg P$ is false, so $\neg Q \Rightarrow \neg P$ is false. In every other row the hypothesis $\neg Q$ is false or the conclusion $\neg P$ is true, so the contrapositive is true. The last two columns agree in every row. This proves that the two statements are logically equivalent. ■

So to prove “if P then Q ” we may instead prove “if Q fails then P fails”. The next section turns this into a proof technique.

The converse is a different matter. It is a separate statement, and it can be false while the original is true.

Warning. The converse is not the contrapositive. “If $x = 2$, then $x^2 = 4$ ” is true for every real number x . Its converse “if $x^2 = 4$, then $x = 2$ ” is false: $x = -2$ satisfies the hypothesis and fails the conclusion. Proving $P \Rightarrow Q$ never proves $Q \Rightarrow P$.

Later in the book the converse question comes up constantly. In Chapter 2 we will see that if a square matrix A is invertible, then $A\mathbf{x} = \mathbf{0}$ has only the solution $\mathbf{x} = \mathbf{0}$. The converse, “if $A\mathbf{x} = \mathbf{0}$ has only the solution $\mathbf{x} = \mathbf{0}$, then A is invertible”, also turns out to be true for square matrices. But it is a separate theorem, it needs its own proof, and that proof takes real work.

Quick check. Consider “if an integer n is divisible by 6, then n is even”. Write its converse and its contrapositive, and decide which of the three statements are true for every integer n .

Solution. The converse is “if n is even, then n is divisible by 6”. The contrapositive is “if n is not even, then n is not divisible by 6”, that is, “if n is odd, then $6 \nmid n$ ”.

The original is true: if $n = 6k$ then $n = 2(3k)$ is even. The contrapositive is true, by Theorem 0.1.1. The converse is false: $n = 2$ is even but not divisible by 6.

When an implication and its converse are both true, we say the two statements are equivalent.

The **biconditional** $P \Leftrightarrow Q$ (“ P if and only if Q ”, often written “ P iff Q ”) is the statement $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$. It is true exactly when P and Q have the same truth value:

P	Q	$P \Rightarrow Q$	$Q \Rightarrow P$	$P \Leftrightarrow Q$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

The phrase splits in two: “ P if Q ” is $Q \Rightarrow P$, and “ P only if Q ” is $P \Rightarrow Q$. Other phrasings are “ P is necessary and sufficient for Q ” and “ P and Q are equivalent”. Proving an “iff” therefore always means proving two implications.

0.1.3 Quantifiers

The sentence “ $x^2 \geq 0$ ” is not yet a statement, because x is not specified. A sentence $P(x)$ that becomes a statement once a value of x is substituted is called a **predicate** (or open sentence) in the variable x . There are two standard ways to turn a predicate into a statement: claim it for every value, or claim it for at least one.

“For every” makes a claim about each element; “there exists” asks for a single witness.

Definition 0.1.3 — Quantifiers

Let $P(x)$ be a predicate whose variable x ranges over a fixed collection S .

- The **universal** statement $\forall x \in S : P(x)$ (“for every x in S , $P(x)$ ”) is true when $P(x)$ is true for **every** element x of S , and false otherwise.
- The **existential** statement $\exists x \in S : P(x)$ (“there exists x in S such that $P(x)$ ”) is true when $P(x)$ is true for **at least one** element x of S , and false otherwise.

The symbols $\forall$ and $\exists$ are the **universal** and **existential quantifiers**.

In words: a universal statement is a promise about every element, so a single element where P fails makes it false. An existential statement needs one element that works. It does not say which one, and it does not say there is only one.

The collection S is part of the statement. Changing S can change the truth value, so always ask: *ranging over what?*

Example. (Quantified statements)

Decide whether each statement is true.

- $\forall x \in \mathbb{R} : x^2 \geq 0$.
- $\exists x \in \mathbb{R} : x^2 = -1$.
- $\exists n \in \mathbb{Z} : n^2 = 2$.
- $\forall n \in \mathbb{Z} : n^2 \neq 2$.

Solution.

- (a) True. Every real number x is either ≥ 0 or < 0 , and in both cases $x^2 = x \cdot x \geq 0$.
- (b) False. By (a), $x^2 \geq 0 > -1$ for every real x , so no real x is a witness. Over the complex numbers (introduced later in this chapter), $x = i$ is a witness, so the same predicate becomes true when the range changes from $\mathbb{R}$ to $\mathbb{C}$.
- (c) False. If $|n| \leq 1$ then $n^2 \leq 1 < 2$, and if $|n| \geq 2$ then $n^2 \geq 4 > 2$. So no integer squares to 2.
- (d) True. This is the statement shown in (c), said the other way round: every integer fails to square to 2.

Parts (c) and (d) already show the pattern of the next subsection: “there is no witness” is the same as “everything fails”.

A degenerate case. What if the range contains no elements at all? Take “every integer n with $n^2 < 0$ satisfies $n = 7$ ”. There are no such integers, so there is nothing that could fail, and the statement is true. It is the quantifier version of vacuous truth: it reads $\forall n \in \mathbb{Z} : (n^2 < 0 \Rightarrow n = 7)$, and every instance has a false hypothesis. By contrast, “there exists an integer n with $n^2 < 0$ ” is false, because there is no witness. This case matters later: facts about “every element of an empty list” will be vacuously true, and that is what makes certain edge cases of definitions work.

Remark. We write $\exists! x \in S : P(x)$ for “there exists a **unique** x in S with $P(x)$ ”. It means two things: some x satisfies P , and any two elements satisfying P are equal. The next section shows how each half is proved.

0.1.4 The order of quantifiers

Most definitions in this book use several quantifiers in a row. Their order is part of the meaning.

Example. (Swapping two quantifiers)

Compare the two statements

$$(i) \quad \forall x \in \mathbb{R} \exists y \in \mathbb{R} : y > x, \quad (ii) \quad \exists y \in \mathbb{R} \forall x \in \mathbb{R} : y > x.$$

Decide which is true.

Solution. Statement (i) says: whichever real x you are given, you can find a real y larger than it. This is true. Given x , take $y = x + 1$, and then $y > x$.

Statement (ii) says: there is one real number y that is larger than every real x . This is false. Whatever y is proposed, the choice $x = y + 1$ gives $y > x$ false. So no y is a witness.

The difference is **who chooses first**. In (i), x is chosen first and y may depend on it: $y = x + 1$ changes with x . In (ii), y must be fixed before x is known, so it has to work for all x at once. Reading quantifiers as a game helps: $\forall$ is a move by an opponent, $\exists$ is your reply, and you may only use information already on the table.

Warning. Never swap a $\forall$ and an $\exists$ without a proof. The implication “ $\exists y \forall x \Rightarrow \forall x \exists y$ ” is always valid, since the single y works for each x . The reverse direction fails, as [Example \(Swapping two quantifiers\)](#) shows. Two quantifiers of the **same** kind, such as $\forall x \forall y$, may be swapped freely.

Quick check. Over the positive real numbers, statement (A) is “ $\forall \varepsilon > 0 \exists \delta > 0 : \delta < \varepsilon$ ”. Statement (B) is obtained by swapping the quantifiers: “ $\exists \delta > 0 \forall \varepsilon > 0 : \delta < \varepsilon$ ”. Are (A) and (B) both true?

Solution. No. (A) is true: given $\varepsilon > 0$, take $\delta = \varepsilon/2$, which is positive and smaller than ε . (B) is false: whatever $\delta > 0$ is proposed, the choice $\varepsilon = \delta$ makes $\delta < \varepsilon$ false. In (A) the number δ may depend on ε ; in (B) it may not.

0.1.5 Negation

Much of mathematics consists of showing that something **fails**: a function is not injective, a list is not independent, a claimed theorem is false. To do this we need the negation of the statement written in a usable form. Putting “it is not the case that” in front is correct but useless; we want the negation with the “not” pushed all the way inside, next to a simple condition we can check.

First, the connectives.

Theorem 0.1.2 — De Morgan’s laws for statements

For all statements P and Q :

- (a) $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$;
- (b) $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$;
- (c) $\neg(P \Rightarrow Q) \equiv P \wedge \neg Q$.

Proof. Let P and Q be statements. We compare truth tables, using [Definition 0.1.1](#) and [Definition 0.1.2](#).

P	Q	$\neg(P \wedge Q)$	$\neg P \vee \neg Q$	$\neg(P \vee Q)$	$\neg P \wedge \neg Q$	$\neg(P \Rightarrow Q)$	$P \wedge \neg Q$
T	T	F	F	F	F	F	F
T	F	T	T	F	F	T	T
F	T	T	T	F	F	F	F
F	F	T	T	T	T	F	F

For (a), $P \wedge Q$ is true only in the first row, so its negation is false only there. The statement $\neg P \vee \neg Q$ is false only when $\neg P$ and $\neg Q$ are both false, which is again only the first row. For (b), $\neg(P \vee Q)$ is true only in the last row, where both parts are false, and so is $\neg P \wedge \neg Q$. For (c), $P \Rightarrow Q$ is false only in the second row, so $\neg(P \Rightarrow Q)$ is true only there, and $P \wedge \neg Q$ is also true only there. In each case the two columns agree in all four rows, as claimed. ■

Law (c) deserves a sentence. The negation of an implication is **not** another implication. To refute “if P then Q ”, you must exhibit a situation where P holds **and** Q fails.

Now the quantifiers. A universal statement fails exactly when some element fails it, and an existential statement fails exactly when every element fails it. We record this as the rule we will use every time.

To negate, flip each quantifier and negate what is inside.

Definition 0.1.4 — Negation of quantifiers

Let $P(x)$ be a predicate with x ranging over S . Then

$$\neg(\forall x \in S : P(x)) \equiv \exists x \in S : \neg P(x), \quad \neg(\exists x \in S : P(x)) \equiv \forall x \in S : \neg P(x).$$

For a statement with several quantifiers, apply these rules from the outside in: each $\forall$ becomes $\exists$, each $\exists$ becomes $\forall$, the ranges stay the same, and the innermost condition is negated using [Theorem 0.1.2](#).

In words: “not every x works” means “some x fails”, and “no x works” means “every x fails”. Both rules are just [Definition 0.1.3](#) read carefully. The universal statement is false precisely when it is not the case that every element satisfies P , which is precisely when at least one element satisfies $\neg P$. The **ranges do not change**: the negation of “every real number squared is non-negative” talks about real numbers, not about some other collection.

Example. (Negating simple quantified statements)

Negate each statement, and say which of the statement and its negation is true.

- (a) $\forall x \in \mathbb{R} : x^2 \geq 0$.
- (b) $\exists n \in \mathbb{Z} : n$ is even and $n > 10$.
- (c) $\forall x \in \mathbb{R} \exists y \in \mathbb{R} : y > x$.

Solution.

- (a) The negation is $\exists x \in \mathbb{R} : x^2 < 0$. The original is true, by [Example \(Quantified statements\)](#), so the negation is false.
- (b) By [Definition 0.1.4](#) and law (a) of [Theorem 0.1.2](#), the negation is $\forall n \in \mathbb{Z} : n$ is odd or $n \leq 10$. The original is true, with witness $n = 12$, so the negation

is false: $n = 12$ is neither odd nor ≤ 10 .

- (c) Flip both quantifiers and negate the inside: $\exists x \in \mathbb{R} \forall y \in \mathbb{R} : y \leq x$. This says that some real number is at least as large as every real number. The original is true by [Example \(Swapping two quantifiers\)](#), so the negation is false.

The next two examples negate definitions that will appear over and over. We use the word **function** informally here, as a rule f that assigns to each a in a collection A one value $f(a)$ in a collection B ; functions are treated properly later in this chapter.

Example. (What “not injective” means)

A function f from A to B is called **injective** if

$$\forall a_1 \in A \forall a_2 \in A : (f(a_1) = f(a_2) \Rightarrow a_1 = a_2).$$

Write down, with no “not” in front, what it means for f to be **not** injective. Then show that $f(x) = x^2$, from $\mathbb{R}$ to $\mathbb{R}$, is not injective.

Solution. By [Definition 0.1.4](#), both $\forall$ become $\exists$, and the inside implication is negated. By law (c) of [Theorem 0.1.2](#), $\neg(f(a_1) = f(a_2) \Rightarrow a_1 = a_2) \equiv f(a_1) = f(a_2) \wedge a_1 \neq a_2$. So f is not injective if and only if

$$\exists a_1 \in A \exists a_2 \in A : f(a_1) = f(a_2) \text{ and } a_1 \neq a_2.$$

In words, two **different** inputs have the same output. For $f(x) = x^2$, take $a_1 = 1$ and $a_2 = -1$. Then $f(1) = 1 = f(-1)$ and $1 \neq -1$, so f is not injective.

Example. (What “not surjective” means)

A function f from A to B is called **surjective** if

$$\forall b \in B \exists a \in A : f(a) = b.$$

Negate this, and show that $f(x) = x^2$, from $\mathbb{R}$ to $\mathbb{R}$, is not surjective.

Solution. Flipping the two quantifiers and negating the equation gives: f is not surjective if and only if

$$\exists b \in B \forall a \in A : f(a) \neq b.$$

In words, **one** value in B is never reached. For $f(x) = x^2$, take $b = -1$. For every real a we have $f(a) = a^2 \geq 0 > -1$ by [Example \(Quantified statements\)](#), so $f(a) \neq -1$. Hence f is not surjective.

The most important negation in the early chapters is that of a statement ending in “all zero”. We meet it here without any vector spaces, using pairs of real numbers added and scaled entry

by entry.

Example. (Negating “only the zero combination”)

Consider the pairs $(1, 2)$, $(2, 4)$, $(0, 1)$, and the statement

$$\forall a, b, c \in \mathbb{R} : (a(1, 2) + b(2, 4) + c(0, 1) = (0, 0) \Rightarrow a = 0 \wedge b = 0 \wedge c = 0).$$

Negate it, and decide which of the statement and its negation is true.

Solution. Flip the quantifier, and use law (c) of [Theorem 0.1.2](#) for the implication and law (a) for the conjunction. The negation is

$$\exists a, b, c \in \mathbb{R} : a(1, 2) + b(2, 4) + c(0, 1) = (0, 0) \text{ and } (a \neq 0 \vee b \neq 0 \vee c \neq 0).$$

The last condition says that a, b, c are **not all zero**: at least one of them is non-zero.

The negation is true. Take $a = 2$, $b = -1$, $c = 0$. Then

$$2(1, 2) - (2, 4) + 0(0, 1) = (2 - 2 + 0, 4 - 4 + 0) = (0, 0),$$

and $a = 2 \neq 0$. So the original statement is false.

In Chapter 1 the original statement is what “the list is linearly independent” means, and its negation is “linearly dependent”. The small words in the negation carry the weight.

Warning. **“Not all zero” is not “all non-zero”.** The negation of “ $a = b = c = 0$ ” is “at least one of a, b, c is non-zero”. In [Example \(Negating “only the zero combination”\)](#) the witness is $(2, -1, 0)$, which has a zero entry. In fact **no** witness has all three entries non-zero: comparing entries, $a + 2b = 0$ and $2a + 4b + c = 0$, so $c = -2(a + 2b) = 0$. Negating with “all non-zero” would wrongly conclude that the original statement is true.

Quick check. Negate “every student passed some exam”, with no “not” in front of the whole sentence.

Solution. Write it as “ $\forall$ students $s \exists$ exam $e : s$ passed e ”. Flipping both quantifiers and negating the inside gives “ $\exists$ student $s \forall$ exams $e : s$ did not pass e ”. In words: **some student passed no exam**. The tempting answer “no student passed any exam” is the negation of “some student passed some exam”, a different statement.

Remark. Each form of statement suggests how to prove or refute it. To prove $\forall x \in S : P(x)$, take an arbitrary $x \in S$ and show $P(x)$. To prove $\exists x \in S : P(x)$, produce a witness. To refute a $\forall$, produce one counterexample. To refute an $\exists$, show every candidate fails. The next section develops these moves into proof techniques.

0.1.6 Exercises

0.1.6.1 A. Check your understanding

Exercise A1.

- (a) Decide whether each statement is true: (i) “ $3 < 2$ or 3 is odd”; (ii) “if $3 < 2$, then $3 = 5$ ”; (iii) “if 3 is odd, then $3 < 2$ ”.
- (b) Write the converse and the contrapositive of “if an integer n is divisible by 4, then n is even”. Which of the three statements are true for every integer n ?
- (c) Negate $\forall x \in \mathbb{R} \exists n \in \mathbb{Z} : n > x$, with no $\neg$ in front. Which of the statement and its negation is true?
- (d) True or false: $\neg(P \wedge Q) \equiv \neg P \wedge \neg Q$ for all statements P, Q . Give a reason.

Solution.

- (a) (i) True: the second part is true, and a disjunction needs only one true part. (ii) True: the hypothesis $3 < 2$ is false, so the implication is vacuously true. (iii) False: the hypothesis is true and the conclusion is false, which is the one case where an implication fails.
- (b) The converse is “if n is even, then n is divisible by 4”. The contrapositive is “if n is odd, then n is not divisible by 4”. The original is true: $n = 4k$ gives $n = 2(2k)$. The contrapositive is true by [Theorem 0.1.1](#). The converse is false: $n = 2$ is even and not divisible by 4.
- (c) By [Definition 0.1.4](#), the negation is $\exists x \in \mathbb{R} \forall n \in \mathbb{Z} : n \leq x$. The original is true: given x , some integer exceeds x (for instance the integer part of $|x|$ plus 1). So the negation is false.
- (d) False. Take P true and Q false. Then $P \wedge Q$ is false, so $\neg(P \wedge Q)$ is true, while $\neg P \wedge \neg Q$ is false because $\neg P$ is false. The correct law is $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$, by [Theorem 0.1.2](#).

0.1.6.2 B. Practice

Exercise B1: Negating definitions.

Write the negation of each definition as a quantified statement with no $\neg$ in front, and then say it in words.

- (a) A function f from $\mathbb{R}$ to $\mathbb{R}$ is **bounded** if $\exists M \in \mathbb{R} \forall x \in \mathbb{R} : |f(x)| \leq M$.
- (b) A function f from A to B is injective if $\forall a_1, a_2 \in A : (a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2))$. Compare your answer with [Example \(What “not injective” means\)](#).
- (c) “Every pair of reals is a combination of $(1, 0)$ and $(1, 1)$ ”: $\forall p, q \in \mathbb{R} \exists a, b \in \mathbb{R} : (p, q) = a(1, 0) + b(1, 1)$, where pairs are added and scaled entry by entry.

Solution.

- (a) By [Definition 0.1.4](#), f is not bounded if and only if $\forall M \in \mathbb{R} \exists x \in \mathbb{R} : |f(x)| > M$. In words: whatever bound M is proposed, some value of f exceeds it in absolute value.

- (b) Flip both quantifiers and negate the implication using law (c) of [Theorem 0.1.2](#): $\exists a_1, a_2 \in A : a_1 \neq a_2 \wedge f(a_1) = f(a_2)$. In words: two different inputs share an output. This is the same condition as in [Example \(What “not injective” means\)](#). That is expected: the given implication is the contrapositive of $f(a_1) = f(a_2) \Rightarrow a_1 = a_2$, so by [Theorem 0.1.1](#) the two definitions agree, and so do their negations.
- (c) Flip all the quantifiers and negate the equation: $\exists p, q \in \mathbb{R} \forall a, b \in \mathbb{R} : (p, q) \neq a(1, 0) + b(1, 1)$. In words: some pair is not a combination of $(1, 0)$ and $(1, 1)$, whatever coefficients are used. (The original is in fact true, since $(p - q)(1, 0) + q(1, 1) = (p, q)$, so this negation is false.)

Exercise B2: Integers versus reals.

For each statement, decide whether it is true when x and y range over $\mathbb{Z}$, and whether it is true when they range over $\mathbb{R}$. Justify your answer.

- (a) $\forall x \exists y : x < y < x + 1$.
- (b) $\forall x (x \neq 0 \Rightarrow \exists y : xy = 1)$.
- (c) $\exists x \forall y : xy = y$.
- (d) $\forall x \exists y : y^2 = x$.

Solution.

- (a) Over $\mathbb{Z}$: false. Take $x = 0$. No integer y satisfies $0 < y < 1$. Over $\mathbb{R}$: true. Given x , take $y = x + \frac{1}{2}$; then $x < y < x + 1$.
- (b) Over $\mathbb{Z}$: false. Take $x = 2$, which is non-zero. If $2y = 1$ for an integer y , then 1 would be even, which is false. Over $\mathbb{R}$: true. Given $x \neq 0$, take $y = 1/x$, which exists because $x \neq 0$, and then $xy = 1$.
- (c) True over both. Take $x = 1$, which lies in $\mathbb{Z}$ and in $\mathbb{R}$. Then $xy = y$ for every y . Here one x works for all y , so the order $\exists x \forall y$ causes no difficulty.
- (d) False over both. Take $x = -1$. For every y , integer or real, $y^2 \geq 0 > -1$ by [Example \(Quantified statements\)](#), so $y^2 \neq -1$.

0.1.6.3 C. Going deeper

Exercise C1: Exporting a hypothesis.

Prove that $P \Rightarrow (Q \Rightarrow R)$ is logically equivalent to $(P \wedge Q) \Rightarrow R$ for all statements P, Q, R . Explain in one sentence why this lets a theorem with two hypotheses be stated either way.

Solution. Let P, Q, R be statements. By [Definition 0.1.2](#), an implication is false exactly when its hypothesis is true and its conclusion is false.

The statement $(P \wedge Q) \Rightarrow R$ is false exactly when $P \wedge Q$ is true and R is false, that is, when P is true, Q is true and R is false.

The statement $P \Rightarrow (Q \Rightarrow R)$ is false exactly when P is true and $Q \Rightarrow R$ is false. By the same definition, $Q \Rightarrow R$ is false exactly when Q is true and R is false. So $P \Rightarrow (Q \Rightarrow R)$ is false exactly when P is true, Q is true and R is false.

Both statements are false in exactly the same row of the eight-row truth table, and true in the other seven. Hence they are logically equivalent.

Consequently “if P , then (if Q , then R)” and “if P and Q , then R ” say the same thing, so a theorem may list its hypotheses one after another or all together.

Exercise C2: When can a converse fail too?.

- (a) Determine whether there exist statements P and Q such that both $P \Rightarrow Q$ and its converse $Q \Rightarrow P$ are false. Justify your answer.
- (b) Give predicates $P(n)$ and $Q(n)$ about integers such that both $\forall n \in \mathbb{Z} : (P(n) \Rightarrow Q(n))$ and $\forall n \in \mathbb{Z} : (Q(n) \Rightarrow P(n))$ are false.
- (c) Explain why (a) and (b) do not contradict each other.

Solution.

- (a) No such statements exist. Suppose $P \Rightarrow Q$ is false. By [Definition 0.1.2](#), P is true and Q is false. Then $Q \Rightarrow P$ has a false hypothesis, so it is true. Hence an implication and its converse can never both be false.
- (b) Let $P(n)$ be “ n is even” and $Q(n)$ be “ n is divisible by 3”. The first universal statement is false: $n = 2$ is even and not divisible by 3. The second is false: $n = 3$ is divisible by 3 and not even.
- (c) In (a), P and Q are single statements, each with one fixed truth value. In (b), each universal statement is refuted by its own counterexample, and the two counterexamples are **different** integers. At $n = 2$ the converse $Q(2) \Rightarrow P(2)$ is vacuously true, and at $n = 3$ the original $P(3) \Rightarrow Q(3)$ is vacuously true, in line with (a). A “for every” statement can fail at one place while its converse fails somewhere else.

0.2 How Proofs Are Built

The previous section taught us to read statements: what an implication claims, what the quantifiers demand, and how to negate. Now we write proofs of them. A proof is a chain of statements, each one justified by a definition, a hypothesis or an earlier result, that ends in the statement we want. This section collects the handful of shapes that almost every proof in this book takes, shows each on a small fact about integers, and ends with the conventions that the book’s proofs follow.

Throughout, a theorem will usually come with two blocks. The **Idea** says how the argument was found, often by working backwards from the goal. The **Proof** then presents the argument forwards, one justified step at a time. Read the Idea first, then check the Proof line by line.

0.2.1 Direct proof

Most statements have the form “for every x , if $P(x)$ then $Q(x)$ ”. The direct way to prove this follows the shape of the statement:

1. **Let** x be arbitrary. By [Definition 0.1.3](#), a universal statement is a claim about each element, so we must not pick a convenient one.
2. **Suppose** $P(x)$. By [Definition 0.1.2](#), the implication can only fail when $P(x)$ is true, so that is the only case we need to handle.
3. **Unwind** $P(x)$ and $Q(x)$ into their definitions. Then push forward from what we have towards what we want, often working backwards from the goal on scrap paper first.

Before we can prove anything about integers, we need definitions to unwind.

Definition 0.2.1 — Even and odd

An integer n is **even** if there **exists** an integer k with $n = 2k$. An integer n is **odd** if there **exists** an integer k with $n = 2k + 1$.

So $0 = 2 \cdot 0$ is even, $-6 = 2 \cdot (-3)$ is even, and $7 = 2 \cdot 3 + 1$ is odd. Each definition is an existential statement: to show a number is even, we must produce the integer k . We use the school fact that **every integer is even or odd, and not both**; an exercise at the end of this section proves a more general fact about remainders.

Proposition 0.2.1 — Sum of even integers

For all integers m and n , if m and n are even, then $m + n$ is even.

Idea. Givens: $m = 2a$ and $n = 2b$ for some integers a, b . Goal: $m + n = 2k$ for some integer k . So we must *find* k . Adding the givens gives $m + n = 2a + 2b = 2(a + b)$, and $k = a + b$ is the witness.

Proof. Let m and n be even integers. By [Definition 0.2.1](#), there exist integers a and b with $m = 2a$ and $n = 2b$. Then

$$m + n = 2a + 2b = 2(a + b).$$

Since $a + b$ is an integer, [Definition 0.2.1](#) shows that $m + n$ is even. This proves the claim. ■

The pattern is worth naming. The hypothesis “ m is even” is an **existence** statement, so using it hands us an integer a to work with. The goal “ $m + n$ is even” is also an existence statement, so proving it means **producing** an integer, here $a + b$. The two letters a and b must be different: m and n are arbitrary, so their halves have nothing to do with each other.

Warning. A **proof of a “for every” statement cannot start with a specific example**. Writing “ $4 + 6 = 10$, which is even” checks one case and proves nothing about all even integers. Equally, writing $m = 2k$ and $n = 2k$ with the **same** k silently assumes $m = n$, so it only proves that $2m$ is even.

0.2.2 Proof by contrapositive

Sometimes the direct route stalls. Consider the statement “if n^2 is even, then n is even”. Going directly, we would suppose $n^2 = 2k$ and try to say something about n . But $n = \pm\sqrt{2k}$ is not visibly of the form $2 \cdot (\text{integer})$, and we are stuck.

By [Theorem 0.1.1](#), “if P then Q ” is equivalent to “if not Q then not P ”. Here the contrapositive is “if n is not even, then n^2 is not even”, that is, “if n is odd, then n^2 is odd”. Now the hypothesis hands us a formula for n , and squaring a formula is easy.

Proposition 0.2.2 — Even squares

For every integer n , if n^2 is even, then n is even.

Idea. The hypothesis n^2 even says little about n , but “ n odd” gives $n = 2k + 1$ explicitly. So prove the contrapositive: from $n = 2k + 1$, expand n^2 and look for the form $2(\dots) + 1$.

Proof. Let n be an integer. By [Theorem 0.1.1](#), it suffices to prove that if n is not even, then n^2 is not even. Suppose n is not even. Since every integer is even or odd, n is odd, so by [Definition 0.2.1](#) there is an integer k with $n = 2k + 1$. Then

$$n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Since $2k^2 + 2k$ is an integer, n^2 is odd by [Definition 0.2.1](#). Since no integer is both even and odd, n^2 is not even. This proves the contrapositive, and hence the statement. ■

A contrapositive proof is a direct proof of a different, equivalent implication. Use it when the negated conclusion gives more to work with than the hypothesis does.

0.2.3 Proof by contradiction

To prove a statement S by **contradiction**, suppose that S is false and deduce something impossible: a statement together with its negation. Why does this work? We have shown that the implication $\neg S \Rightarrow (R \wedge \neg R)$ is true, for some statement R . Its conclusion $R \wedge \neg R$ is false in every row, so by [Definition 0.1.2](#) the implication can only be true if its hypothesis $\neg S$ is false. Hence S is true.

Contradiction is the natural tool for statements that say something **cannot** happen, or that there is **no** object of some kind. Such statements give nothing to start from directly, while their negation hands us an object to study.

Statements about fractions and primes need the language of divisibility.

Definition 0.2.2 — Divisibility and primes

Let a and b be integers. We say a **divides** b , written $a \mid b$, if there **exists** an integer k with $b = ak$. An integer p is **prime** if $p \geq 2$ and the only **positive** divisors of p are 1

and p .

For example, $3 \mid 12$ since $12 = 3 \cdot 4$, and $3 \nmid 7$. The first primes are 2, 3, 5, 7, 11. The integer 1 is **not** prime, because the definition requires $p \geq 2$; this convention is what makes factorizations into primes unique. In this language, n is even exactly when $2 \mid n$.

A real number is **rational** if it equals a/b for some integers a and b with $b \neq 0$, and **irrational** otherwise. We use the school fact that every rational number can be written **in lowest terms**: as a/b with $b \geq 1$, where no integer $d \geq 2$ divides both a and b . (We justify this fact at the end of this section.)

Theorem 0.2.3 — Irrationality of $\sqrt{2}$

There is no rational number whose square is 2. That is, $\sqrt{2}$ is irrational.

Idea. “There is no rational x with $x^2 = 2$ ” gives nothing to hold, so suppose there is one, and write it in lowest terms a/b . Clearing denominators gives $a^2 = 2b^2$, an integer equation. Then Proposition 0.2.2 applies twice: first a is even, then (after substituting $a = 2c$) b is even. That contradicts lowest terms.

Proof. Suppose, for a contradiction, that some rational number x satisfies $x^2 = 2$. Write $x = a/b$ in lowest terms, so $b \geq 1$ and no integer $d \geq 2$ divides both a and b . Then $a^2/b^2 = 2$, and multiplying by b^2 , which is non-zero, gives

$$a^2 = 2b^2.$$

Therefore a^2 is even by Definition 0.2.1, and hence a is even by Proposition 0.2.2. Write $a = 2c$ with c an integer. Substituting gives $4c^2 = 2b^2$, so $b^2 = 2c^2$ is even. By Proposition 0.2.2 again, b is even. Thus 2 divides both a and b , which contradicts the choice of a/b in lowest terms. Hence no rational number has square 2. ■

The contradiction came from the lowest-terms condition, which is the hypothesis we chose to impose. That is typical: in a contradiction proof, look for the assumption you have not yet used, because the contradiction usually comes from there.

Remark. If a contradiction proof of “ $P \Rightarrow Q$ ” assumes P and $\neg Q$ and then only ever derives $\neg P$, it is really a contrapositive proof in disguise. Write it as a contrapositive; it is shorter and clearer.

A second classic contradiction proof, that there are infinitely many primes, needs one more tool and appears at the end of this section.

0.2.4 Proof by cases

When a hypothesis splits naturally into possibilities, handle each one separately. If $P_1 \vee P_2$ is true, and both $P_1 \Rightarrow Q$ and $P_2 \Rightarrow Q$ are true, then Q is true: whichever of P_1, P_2 holds, its implication delivers Q . The cases must **cover every possibility**. They are allowed to overlap.

Proposition 0.2.4 — $n^2 + n$ is even

For every integer n , the integer $n^2 + n$ is even.

Idea. Nothing about n is given, but every integer is even or odd, and each case gives a formula for n . Factoring first, $n^2 + n = n(n+1)$, makes each case a one-line computation.

Proof. Let n be an integer. Then $n^2 + n = n(n+1)$. Since every integer is even or odd, there are two cases.

Case 1: n is even. By [Definition 0.2.1](#), $n = 2k$ for some integer k . Then $n(n+1) = 2(k(n+1))$, and $k(n+1)$ is an integer.

Case 2: n is odd. By [Definition 0.2.1](#), $n = 2k+1$ for some integer k . Then $n+1 = 2(k+1)$, so $n(n+1) = 2(n(k+1))$, and $n(k+1)$ is an integer.

In both cases $n^2 + n$ is even by [Definition 0.2.1](#). This shows the claim for every integer n . ■

0.2.5 Proving “if and only if”

By the definition of the biconditional in the previous section, $P \Leftrightarrow Q$ means $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$. So a proof of an “iff” has **two directions**, labeled $(\Rightarrow)$ and $(\Leftarrow)$, and each is proved by any method we like. Often one direction is already known.

Proposition 0.2.5 — Parity of a square

For every integer n , n is even if and only if n^2 is even.

Proof. Let n be an integer.

$(\Rightarrow)$ Suppose n is even, so $n = 2k$ for some integer k by [Definition 0.2.1](#). Then $n^2 = 4k^2 = 2(2k^2)$, and $2k^2$ is an integer, so n^2 is even.

$(\Leftarrow)$ This is [Proposition 0.2.2](#).

Both directions hold, so n is even if and only if n^2 is even. ■

Warning. Reversing the steps is not automatic. A common slip is to “prove the converse” by rewriting the same argument backwards without checking that every step reverses. For instance, from $x = 2$ we get $x^2 = 4$; but from $x^2 = 4$ we only get $x = 2$ **or** $x = -2$. Each direction needs its own justification.

0.2.6 Existence and uniqueness

To prove “there exists x with $P(x)$ ”, the most direct method is to **produce a witness** and check it. For example, “there exists an integer n with $n^2 = n + 2$ ” is proved by $n = 2$: indeed $2^2 = 4 = 2 + 2$.

To prove “there is **at most one** x with $P(x)$ ”, the standard move is: *suppose x and x' both satisfy P , and show $x = x'$* . Notice that we do not assume $x \neq x'$; we simply take two objects with the property and prove they are the same. “There exists a **unique** x ” means both existence and at most one, so it takes both arguments.

Here is an example that we will meet again when we study fields and groups. A **binary operation** $*$ on a collection S is a rule that assigns to each ordered pair a, b of elements of S an element $a * b$ of S . Addition and multiplication of integers are binary operations on $\mathbb{Z}$. An **identity element** for $*$ is an element e of S such that $e * a = a$ and $a * e = a$ for **every** a in S .

Proposition 0.2.6 — Uniqueness of the identity

Let $*$ be a binary operation on S . Then $*$ has at most one identity element.

Idea. Suppose e and e' are both identities. The only thing we know about them is how they behave in a product, so look at the one product that involves both, $e * e'$, and evaluate it in two ways.

Proof. Suppose e and e' are both identity elements for $*$. Since e' is an identity, $e * e' = e$, taking $a = e$ in its defining property. Since e is an identity, $e * e' = e'$, taking $a = e'$ in its defining property. Therefore $e = e * e' = e'$. This shows that any two identity elements are equal. ■

So 0 is **the** identity for addition on $\mathbb{Z}$, and 1 is **the** identity for multiplication. The result says nothing about existence. Subtraction on $\mathbb{Z}$ has no identity at all: $a - e = a$ for every a forces $e = 0$, but then $e - a = -a$, which is not a when $a = 1$. Uniqueness proofs of exactly this shape will recur for zero vectors, negatives, inverse matrices and inverse functions.

0.2.7 Counterexamples

By [Definition 0.1.4](#), the negation of “ $\forall x : P(x)$ ” is “ $\exists x : \neg P(x)$ ”. So to **refute** a universal statement, one witness suffices: a single x where $P(x)$ fails. Such an x is a **counterexample**. Conversely, no number of examples can prove a universal statement over an infinite range.

Example. (Refuting “every prime is odd”)

Decide whether every prime number is odd. Justify your answer.

Solution. False. The statement is $\forall p \text{ prime} : p \text{ is odd}$, so by [Definition 0.1.4](#) one prime that is not odd refutes it. Take $p = 2$. It is prime, since $2 \geq 2$ and a positive divisor of 2 is at most 2, so it is 1 or 2. It is even, since $2 = 2 \cdot 1$, and hence not odd. So 2 is a

counterexample.

Warning. Many examples prove nothing. The number $n^2 + n + 41$ is prime for every integer n from 0 to 39. Yet the statement “ $n^2 + n + 41$ is prime for every $n \geq 0$ ” is false: at $n = 40$ we get $40^2 + 40 + 41 = 1681 = 41^2$, which is not prime.

0.2.8 Mathematical induction

Some statements are really infinitely many statements, one for each natural number: “for every $n \geq 1$, $1 + 2 + \cdots + n = n(n+1)/2$ ”. Checking cases one at a time never finishes. Induction replaces the infinite list of checks by two finite tasks, like knocking over a row of dominoes: push the first one, and make sure each domino knocks over the next.

We take this as a basic property of the integers, not something to prove.

Theorem 0.2.7 — Principle of Mathematical Induction

Let n_0 be an integer, and for each integer $n \geq n_0$ let $P(n)$ be a statement. Suppose

- (i) (**base case**) $P(n_0)$ is true, and
- (ii) (**inductive step**) for **every** integer $n \geq n_0$, if $P(n)$ is true then $P(n+1)$ is true.

Then $P(n)$ is true for every integer $n \geq n_0$.

In the inductive step, the assumption “ $P(n)$ is true” is called the **induction hypothesis**. It is not circular: we do not assume the whole conclusion, only that the statement holds at one arbitrary n , and we prove that it then holds at $n+1$.

Proposition 0.2.8 — Sum of the first n integers

For every integer $n \geq 1$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Idea. Passing from n to $n+1$ adds exactly one term, $n+1$, to the left side. So the step reduces to the algebra identity $\frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}$, which holds after taking out the factor $n+1$.

Proof. We prove this by induction on n , using Theorem 0.2.7 with $n_0 = 1$. Let $P(n)$ be the statement $1 + 2 + \cdots + n = n(n+1)/2$.

Base case. For $n = 1$ the left side is 1 and the right side is $1 \cdot 2/2 = 1$. So $P(1)$ is true.

Inductive step. Let $n \geq 1$ and suppose $P(n)$ is true. Then

$$1 + 2 + \cdots + n + (n+1) = \frac{n(n+1)}{2} + (n+1) = (n+1)\left(\frac{n}{2} + 1\right) = \frac{(n+1)(n+2)}{2},$$

where the first equality uses the induction hypothesis $P(n)$. This is $P(n+1)$.

By [Theorem 0.2.7](#), $P(n)$ is true for every integer $n \geq 1$, as claimed. ■

The base case does not have to be 1, and the step does not have to be an equation. In this book the **natural numbers** are $\mathbb{N} = \{0, 1, 2, \dots\}$: they **include** 0. (Some books start at 1; the next section fixes our convention formally.)

Proposition 0.2.9 — $2^n > n$

For every natural number n , we have $2^n > n$.

Proof. We prove this by induction on n , with $n_0 = 0$. For $n = 0$, $2^0 = 1 > 0$. Now let $n \geq 0$ and suppose $2^n > n$. Then

$$2^{n+1} = 2^n + 2^n > n + 2^n \geq n + 1,$$

where the strict inequality uses the induction hypothesis and the last step uses $2^n \geq 1$. Hence $2^{n+1} > n + 1$. By [Theorem 0.2.7](#), $2^n > n$ for every $n \in \mathbb{N}$. ■

The inductive step must work for **every** $n \geq n_0$, including the smallest ones. The following fake proof fails exactly there.

Warning. The “**all horses are the same color**” fallacy. *Claim:* in every group of $n \geq 1$ horses, all horses have the same color. *“Proof”:* for $n = 1$ there is nothing to show. Suppose the claim holds for groups of n horses, and take a group of $n + 1$ horses $h_1, \dots, h_{n+1}$. The group $h_1, \dots, h_n$ is single-colored by the induction hypothesis, and so is $h_2, \dots, h_{n+1}$. The two groups share a horse, so all $n + 1$ horses have its color. The claim is plainly false, so this “proof” has a gap. Before opening the check below, try to find the exact value of n where the step breaks.

Quick check. Where does the inductive step in the horse argument fail?

Solution. It fails when passing from $n = 1$ to $n + 1 = 2$. The two groups are then $\{h_1\}$ and $\{h_2\}$, which share **no** horse, so nothing links the color of h_1 to that of h_2 . The sentence “the two groups share a horse” is true only when $n \geq 2$. Since the step $P(1) \Rightarrow P(2)$ is not proved, the chain of dominoes breaks at the very first push, and the base case alone gives nothing.

0.2.9 Strong induction and well-ordering

In some arguments, knowing $P(n)$ alone is not enough to reach $P(n + 1)$: we need the statement at some **earlier** value, and we do not know in advance which one. Strong induction allows the hypothesis at all earlier values at once.

Theorem 0.2.10 — Strong induction

Let n_0 be an integer, and for each integer $n \geq n_0$ let $P(n)$ be a statement. Suppose that for **every** integer $n \geq n_0$,

if $P(k)$ is true for all integers k with $n_0 \leq k < n$, then $P(n)$ is true.

Then $P(n)$ is true for every integer $n \geq n_0$.

Idea. Ordinary induction applies to the stronger statement $Q(n)$: “ $P(k)$ holds for all k from n_0 to n ”. The hypothesis is exactly what pushes $Q(n)$ to $Q(n+1)$. At $n = n_0$ the condition “for all k with $n_0 \leq k < n_0$ ” is vacuous, so the hypothesis already contains the base case.

Proof. For each integer $n \geq n_0$, let $Q(n)$ be the statement “ $P(k)$ is true for all integers k with $n_0 \leq k \leq n$ ”. We prove $Q(n)$ for all $n \geq n_0$ by [Theorem 0.2.7](#).

Base case. There is no integer k with $n_0 \leq k < n_0$, so the condition “ $P(k)$ is true for all such k ” is vacuously true. By the hypothesis with $n = n_0$, $P(n_0)$ is true. Hence $Q(n_0)$ is true.

Inductive step. Let $n \geq n_0$ and suppose $Q(n)$ is true, so $P(k)$ holds for all k with $n_0 \leq k < n+1$. By the hypothesis applied to $n+1$, $P(n+1)$ is true. Together with $Q(n)$, this gives $Q(n+1)$.

By [Theorem 0.2.7](#), $Q(n)$ holds for all $n \geq n_0$. In particular $P(n)$ holds for all $n \geq n_0$, since $Q(n)$ includes $k = n$. This proves the theorem. ■

The same property of the natural numbers can be phrased without any statements $P(n)$ at all: there is no endless strictly decreasing sequence of natural numbers, so every search for a smallest example ends.

Theorem 0.2.11 — Well-Ordering Principle

Every non-empty collection of natural numbers has a least element: an element m of the collection with $m \leq s$ for every s in the collection.

We take this as a basic property too. It is equivalent to the Principle of Mathematical Induction, in the sense that each can be proved from the other, but we will not need that. Well-ordering is the natural language for “take a smallest counterexample” arguments, which are contradiction proofs in disguise.

Here is a result that needs strong induction, because the smaller number we reach is not $n-1$.

Proposition 0.2.12 — Existence of a prime factor

Every integer $n \geq 2$ has a prime divisor.

Idea. If n is prime, n itself works. If not, $n = ab$ with $1 < a < n$, and a prime dividing a also divides n . The factor a can be any size below n , so we need the statement for all smaller values, which is strong induction.

Proof. We use [Theorem 0.2.10](#) with $n_0 = 2$. Let $n \geq 2$, and suppose every integer k with $2 \leq k < n$ has a prime divisor.

Case 1: n is prime. Then $n \mid n$, so n is a prime divisor of n .

Case 2: n is not prime. Since $n \geq 2$, [Definition 0.2.2](#) gives a positive divisor a of n with $a \neq 1$ and $a \neq n$. Write $n = ab$ with b an integer. Since a and n are positive, $b \geq 1$, and $b \neq 1$ because $a \neq n$; so $b \geq 2$ and $a = n/b < n$. Thus $2 \leq a < n$, and by the hypothesis a has a prime divisor p , say $a = pc$. Then $n = p(cb)$, so $p \mid n$.

In both cases n has a prime divisor. By [Theorem 0.2.10](#), this holds for every $n \geq 2$. ■

Now the promised contradiction proof.

Theorem 0.2.13 — Infinitely many primes

There are infinitely many prime numbers.

Idea. Suppose the primes form a finite list. Build a number that no prime on the list can divide: multiply them all and add 1. Dividing by any p_i leaves remainder 1. But [Proposition 0.2.12](#) says this number has some prime divisor, and that prime must be on the list.

Proof. Suppose, for a contradiction, that there are only finitely many primes, and list them all as $p_1, p_2, \dots, p_k$. Let

$$N = p_1 p_2 \cdots p_k + 1.$$

The list is non-empty, since 2 is prime, and each $p_i \geq 2$; hence $N \geq 3$. By [Proposition 0.2.12](#), N has a prime divisor p . Since the list contains every prime, $p = p_i$ for some i . Then $p_i \mid N$ and $p_i \mid p_1 p_2 \cdots p_k$, so there are integers s, t with $N = p_i s$ and $p_1 \cdots p_k = p_i t$. Subtracting gives $1 = p_i(s - t)$. Since $s - t$ is an integer and $p_i \geq 2$, this is impossible: $p_i(s - t)$ is 0 when $s = t$, and has absolute value at least 2 otherwise. This contradiction shows that there are infinitely many primes. ■

Remark. Well-ordering also justifies the “lowest terms” fact used in [Theorem 0.2.3](#). Let x be rational. There is at least one way of writing $x = a/b$ with $b \geq 1$: if $x = a/b$ with $b < 0$, then also $x = (-a)/(-b)$ with $-b \geq 1$. Among all such ways, the collection of possible denominators b is a non-empty collection of natural numbers, so by [Theorem 0.2.11](#) it has a least element. Take a representation a/b with that least b . If some $d \geq 2$ divided both, say $a = da'$ and $b = db'$, then $x = a'/b'$ with $1 \leq b' < b$, contradicting minimality. So a/b is in lowest terms.

0.2.10 Writing proofs in this book

The proofs in this book follow a few conventions. Knowing them makes proofs faster to read, and copying them makes your own proofs easier to check.

- **The first sentence does mathematics.** Proofs open with “Let ...”, “Suppose ...”, “Since ...” or “By ...”. The motivation has already been given in the Idea block.
- **Every step names its reason.** Each line cites a hypothesis, a definition, or an earlier result by name, as in “by [Proposition 0.2.2](#)”. When a display hides a reason, a clause such as “where the first equality uses the induction hypothesis” points to it.
- **Structure is marked.** Two directions of an iff are labeled $(\Rightarrow)$ and $(\Leftarrow)$, cases are labeled *Case 1* and *Case 2*, and inductions name their base case and inductive step.
- **No hidden gaps.** Words like “obviously” and “clearly” never stand in for an argument. If a step is short, it is written out; if it is routine and has been done before, the proof says so.
- **The ending restates the result:** “This shows ...”, “This proves ...”, “as claimed”.

How to read a proof here. First read the statement and identify the givens and the goal. Then read the Idea, which says why the argument has the shape it has. Then read the Proof with a pencil. At each line, ask which earlier line or result justifies it; if you cannot answer, that is the line to reread. Finally, ask which hypothesis each step used. If a hypothesis was never used, either it was not needed or you have missed a step.

0.2.11 Exercises

0.2.11.1 A. Check your understanding

Exercise A1.

For each statement, name the proof technique you would use first (direct, contrapositive, contradiction, cases, induction, or counterexample), and give a one-sentence reason. Then carry out the proof or refutation.

- The sum of two odd integers is even.
- For every integer n , if $3n + 2$ is odd, then n is odd.
- For every integer $n \geq 1$, $3 \mid 4^n - 1$.
- For every integer $n \geq 1$, $n^2 \geq 2n$.
- There is no largest integer.

Solution.

- Direct.* The hypothesis gives formulas to add. Let $m = 2a + 1$ and $n = 2b + 1$ with a, b integers, by [Definition 0.2.1](#). Then $m + n = 2(a + b + 1)$, which is even.
- Contrapositive.* The conclusion’s negation “ n even” gives a formula, while “ $3n + 2$ odd” does not. By [Theorem 0.1.1](#) it suffices to show: if n is even then $3n + 2$ is even. If $n = 2k$, then $3n + 2 = 2(3k + 1)$, which is even, hence not odd.
- Induction.* It is a statement for every $n \geq 1$, and $4^{n+1} - 1$ is built from $4^n - 1$. For $n = 1$, $4 - 1 = 3 = 3 \cdot 1$. If $4^n - 1 = 3k$, then $4^{n+1} - 1 = 4(4^n - 1) + 3 = 3(4k + 1)$.

By [Theorem 0.2.7](#) the claim holds for all $n \geq 1$.

- (d) *Counterexample.* The statement is false, so a single witness refutes it. At $n = 1$, $n^2 = 1 < 2 = 2n$.
- (e) *Contradiction.* It says an object does not exist, so assume it does. Suppose m is a largest integer. Then $m + 1$ is an integer with $m + 1 > m$, contradicting the choice of m . Hence there is no largest integer.

0.2.11.2 B. Practice

Exercise B1: Contrapositive.

Prove that for every integer n , if n^2 is odd, then n is odd.

Solution. Let n be an integer. By [Theorem 0.1.1](#), it suffices to prove that if n is not odd, then n^2 is not odd. Suppose n is not odd. Since every integer is even or odd, n is even, so $n = 2k$ for some integer k by [Definition 0.2.1](#). Then $n^2 = 4k^2 = 2(2k^2)$, so n^2 is even. Since no integer is both even and odd, n^2 is not odd. This proves the statement.

Exercise B2: Induction.

Prove that for every integer $n \geq 1$, the sum of the first n odd numbers is n^2 :

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2.$$

Solution. We prove this by induction on n , using [Theorem 0.2.7](#) with $n_0 = 1$. Let $P(n)$ be the displayed statement.

Base case. For $n = 1$ the left side is 1 and the right side is $1^2 = 1$, so $P(1)$ is true.

Inductive step. Let $n \geq 1$ and suppose $P(n)$ is true. The $(n + 1)$ -st odd number is $2(n + 1) - 1 = 2n + 1$. Then

$$1 + 3 + \cdots + (2n - 1) + (2n + 1) = n^2 + 2n + 1 = (n + 1)^2,$$

where the first equality uses the induction hypothesis. This is $P(n + 1)$.

By [Theorem 0.2.7](#), $P(n)$ holds for every $n \geq 1$.

Exercise B3: $\sqrt{3}$ is irrational.

You may use the school fact that every integer has exactly one of the forms $3k$, $3k + 1$, $3k + 2$ with k an integer.

- (a) Prove that for every integer n , if $3 \mid n^2$ then $3 \mid n$.
- (b) Hence prove that there is no rational number whose square is 3.

Solution.

- (a) Let n be an integer. By [Theorem 0.1.1](#), it suffices to show that if $3 \nmid n$ then $3 \nmid n^2$. Suppose $3 \nmid n$. Then $n = 3k + 1$ or $n = 3k + 2$ for some integer k .

Case 1: $n = 3k + 1$. Then $n^2 = 9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1$.

Case 2: $n = 3k + 2$. Then $n^2 = 9k^2 + 12k + 4 = 3(3k^2 + 4k + 1) + 1$.

In both cases n^2 has the form $3j + 1$ with j an integer. Since an integer has only one of the three forms, n^2 is not of the form $3j$, so $3 \nmid n^2$. This proves (a).

- (b) Suppose, for a contradiction, that a rational number x satisfies $x^2 = 3$. Write $x = a/b$ in lowest terms, so $b \geq 1$ and no integer $d \geq 2$ divides both a and b . Multiplying $a^2/b^2 = 3$ by $b^2 \neq 0$ gives $a^2 = 3b^2$. Therefore $3 \mid a^2$, and by (a), $3 \mid a$. Write $a = 3c$. Then $9c^2 = 3b^2$, so $b^2 = 3c^2$ and $3 \mid b^2$. By (a) again, $3 \mid b$. Thus 3 divides both a and b , contradicting lowest terms. Hence no rational number has square 3.

0.2.11.3 C. Going deeper

Exercise C1: Stamps of 4 and 5.

- (a) Show that each of 12, 13, 14, 15 can be written as $4a + 5b$ with $a, b \in \mathbb{N}$.
- (b) Prove by strong induction that every integer $n \geq 12$ can be written as $4a + 5b$ with $a, b \in \mathbb{N}$.
- (c) Show that 11 cannot be written in this form, so the bound 12 in (b) cannot be lowered to 11.

Hint: for $n \geq 16$, look at $n - 4$.

Solution.

- (a) $12 = 4 \cdot 3 + 5 \cdot 0$, $13 = 4 \cdot 2 + 5 \cdot 1$, $14 = 4 \cdot 1 + 5 \cdot 2$, and $15 = 4 \cdot 0 + 5 \cdot 3$.
- (b) We use [Theorem 0.2.10](#) with $n_0 = 12$. Let $n \geq 12$, and suppose every integer k with $12 \leq k < n$ can be written as $4a + 5b$ with $a, b \in \mathbb{N}$.
Case 1: $12 \leq n \leq 15$. Then n has the required form by (a).
Case 2: $n \geq 16$. Then $k = n - 4$ satisfies $12 \leq k < n$, so by the hypothesis $n - 4 = 4a + 5b$ with $a, b \in \mathbb{N}$. Hence $n = 4(a + 1) + 5b$, and $a + 1 \in \mathbb{N}$.
 In both cases n has the required form. By [Theorem 0.2.10](#), every integer $n \geq 12$ does.
- (c) Suppose $11 = 4a + 5b$ with $a, b \in \mathbb{N}$. Since $5b \leq 11$, $b \in \{0, 1, 2\}$. If $b = 0$ then $4a = 11$; if $b = 1$ then $4a = 6$; if $b = 2$ then $4a = 1$. None of 11, 6, 1 is a multiple of 4, so no such a exists. Hence 11 is not of the form $4a + 5b$.

Exercise C2: Division with remainder.

Let a and b be integers with $b \geq 1$. Let R be the collection of all **non-negative** integers of the form $a - qb$ with q an integer.

- (a) Show that R is non-empty.
- (b) Let r be the least element of R , which exists by [Theorem 0.2.11](#), and write $r = a - qb$. Prove that $0 \leq r < b$.
- (c) Prove that the pair (q, r) is unique: if $a = qb + r = q'b + r'$ with $0 \leq r < b$ and $0 \leq r' < b$, then $q = q'$ and $r = r'$.

Hint for (a): try $q = -|a|$.

Solution.

- (a) Take $q = -|a|$. Then $a - qb = a + |a|b$. Since $b \geq 1$ and $|a| \geq 0$, we have $|a|b \geq |a|$, so $a + |a|b \geq a + |a| \geq 0$, where the last step uses $|a| \geq -a$. Hence $a + |a|b$ belongs to R , and R is non-empty.
- (b) By (a) and [Theorem 0.2.11](#), R has a least element $r = a - qb$ with $q \in \mathbb{Z}$. Since r belongs to R , $r \geq 0$. Suppose, for a contradiction, that $r \geq b$. Then $r - b = a - (q + 1)b \geq 0$, and $q + 1 \in \mathbb{Z}$, so $r - b$ belongs to R . But $b \geq 1$ gives $r - b < r$, contradicting the choice of r as the least element of R . Hence $0 \leq r < b$.
- (c) Suppose $qb + r = q'b + r'$ with $0 \leq r, r' < b$. Then $r' - r = (q - q')b$, so $b \mid r' - r$. From the bounds, $-b < r' - r < b$. Suppose $q \neq q'$. Then $|q - q'| \geq 1$, so $|r' - r| = |q - q'|b \geq b$, contradicting $-b < r' - r < b$. Hence $q = q'$, and then $r' - r = 0$, so $r = r'$. This proves uniqueness.

Together, (a)–(c) show that for integers a and $b \geq 1$ there are unique integers q, r with $a = qb + r$ and $0 \leq r < b$. With $b = 2$, this is the fact that every integer is even or odd, and not both.

0.3 Sets

Every object in linear algebra is a set with some extra structure. A vector space is a set on which we can add and scale, a subspace is a subset of it, and a linear map sends elements of one set to elements of another. Before the structure arrives, we need the plain language of sets: how to describe them, when two of them are equal, and how to combine them. The most important move in this section is proving that two sets are equal by proving two inclusions, a move we will use for subspaces, spans, kernels and images throughout the book.

0.3.1 Sets, elements and set-builder notation

A **set** is a collection of objects, called its **elements**. We write $x \in A$ for “ x is an element of A ” (also “ x belongs to A ”, “ x lies in A ”), and $x \notin A$ for its negation. A set is determined by its elements and nothing else: it has no order and no repetition. So $\{1, 2\}$, $\{2, 1\}$ and $\{1, 1, 2\}$ all describe the same set, whose elements are 1 and 2. We make this precise below.

Small sets can be listed between braces. Infinite sets we meet constantly have fixed names.

Definition 0.3.1 — Common number sets

We write

- $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ for the **natural numbers**; in this book $\mathbb{N}$ **includes** 0;
- $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ for the **integers**;
- $\mathbb{Q}$ for the **rational numbers**, the numbers a/b with $a, b \in \mathbb{Z}$ and $b \neq 0$;
- $\mathbb{R}$ for the **real numbers**;

- $\mathbb{C}$ for the **complex numbers**, studied later in this chapter.

Books disagree about whether $0 \in \mathbb{N}$. With our convention, the Well-Ordering Principle ([Theorem 0.2.11](#)) applies to collections that contain 0, and “ $n \in \mathbb{N}$ ” means $n \geq 0$. When we need positive integers only, we say “ $n \geq 1$ ”. For real numbers $a \leq b$ we also use **intervals**: $[a, b]$ is the set of real x with $a \leq x \leq b$, (a, b) the set with $a < x < b$, and $[a, b)$, $(a, b]$ are defined in the same way.

Listing elements fails for most sets we care about, such as “the real numbers whose square is less than 4”. Instead we describe a set by a property, and select from a set we already have.

Definition 0.3.2 — Set-builder notation

Let S be a set and let $P(x)$ be a predicate for $x \in S$. The set

$$\{x \in S : P(x)\}$$

is the set whose elements are **exactly** those $x \in S$ for which $P(x)$ is true.

In words: an object y belongs to $\{x \in S : P(x)\}$ if and only if $y \in S$ **and** $P(y)$ is true. Both parts matter. The ambient set S says where we are looking, and the predicate says what we keep. The colon is read “such that”; some books use a vertical bar instead. When it is plain which sets the elements come from, as in the unions and intersections below, we sometimes leave the ambient set out and write $\{x : P(x)\}$.

A second form describes the elements by a formula: $\{2k : k \in \mathbb{Z}\}$ is the set of all numbers of the form $2k$ with k an integer. An object y belongs to it exactly when $y = 2k$ for **some** $k \in \mathbb{Z}$. This form will describe sets such as “all combinations of given vectors” in Chapter 1.

Example. (Reading set-builder notation)

Describe each set more simply.

- $\{x \in \mathbb{R} : x^2 < 4\}$.
- $\{x \in \mathbb{Z} : x^2 < 4\}$.
- $\{n \in \mathbb{Z} : n \text{ is even}\}$ and $\{2k : k \in \mathbb{Z}\}$.
- $\{(x, y) : x, y \in \mathbb{R}, x + y = 1\}$, a set of ordered pairs (made precise under Cartesian products below).

Solution.

- A real number satisfies $x^2 < 4$ exactly when $-2 < x < 2$, so the set is the interval $(-2, 2)$.
- The same predicate, selected from $\mathbb{Z}$ instead, keeps only integers: the set is $\{-1, 0, 1\}$. Changing the ambient set changes the set.
- Both are the set $\{\dots, -4, -2, 0, 2, 4, \dots\}$ of even integers. The first selects the integers with a property; the second produces each element by the formula $2k$. By the definition of “even” ([Definition 0.2.1](#)), an integer is even exactly when it equals $2k$ for some $k \in \mathbb{Z}$.
- The pairs of reals whose coordinates add to 1: the line in the plane through $(1, 0)$

and $(0, 1)$.

What if no element of S satisfies the predicate? We still want a set, and it has no elements.

Definition 0.3.3 — Empty set

The **empty set** $\emptyset$ is the set with **no** elements: $x \notin \emptyset$ for every object x .

For example, $\{x \in \mathbb{R} : x^2 = -1\} = \emptyset$, since $x^2 \geq 0$ for every real x (Example (Quantified statements)). Also $\{n \in \mathbb{Z} : 0 < n < 1\} = \emptyset$. We will see shortly that there is only **one** empty set, so these two descriptions give the same set.

Statements about all elements of $\emptyset$ are vacuously true: “every element of $\emptyset$ is a prime number” has no element that could fail, just as in the degenerate case discussed after Example (Quantified statements).

Warning. $\emptyset$, $\{\emptyset\}$ and $\{0\}$ are three different sets. The set $\emptyset$ has no elements. The set $\{\emptyset\}$ has exactly one element, namely the empty set. The set $\{0\}$ has exactly one element, namely the number 0. A box containing an empty box is not an empty box. In Chapter 1 this matters: $\{0\}$ will be a subspace, and $\emptyset$ never is.

0.3.2 Subsets

Most sets in this book live inside a bigger set: the even integers inside $\mathbb{Z}$, a line inside the plane, and later a subspace inside a vector space. The first thing to check about a candidate subspace will be that it sits inside the space at all. We need a word for “sits inside”.

A is a subset of B when everything in A is also in B.

Definition 0.3.4 — Subset

Let A and B be sets. We say A is a **subset** of B , and write $A \subseteq B$, if

$$\forall x : (x \in A \Rightarrow x \in B).$$

If $A \subseteq B$ and $A \neq B$, then A is a **proper** subset of B , written $A \subsetneq B$.

In words: $A \subseteq B$ says that **every** element of A is an element of B . It says nothing about elements of B ; B may contain many more. The definition is an implication inside a “for every”, so the proof template of the previous section applies: to show $A \subseteq B$, **let** $x \in A$ and show $x \in B$.

By Definition 0.1.4 and Theorem 0.1.2, the negation reads

$$A \not\subseteq B \iff \exists x : x \in A \text{ and } x \notin B.$$

So to show A is **not** a subset of B , exhibit one element of A outside B .

Example. (Subsets)

Decide whether each inclusion holds.

- (a) $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$.
- (b) $\{1, 2\} \subseteq \{1, 2, 3\}$ and $\{1, 4\} \subseteq \{1, 2, 3\}$.
- (c) $A \subseteq A$ and $\emptyset \subseteq A$, for any set A .
- (d) $\{x \in \mathbb{R} : x^2 = 1\} \subseteq \mathbb{Z}$.

Solution.

- (a) All hold. Every natural number is an integer; every integer n equals $n/1$, so it is rational; every rational number is real.
- (b) The first holds: each of 1, 2 is an element of $\{1, 2, 3\}$. The second fails, although it differs from the first in only one element: 1 passes, but $4 \in \{1, 4\}$ and $4 \notin \{1, 2, 3\}$, so 4 is a witness for $\{1, 4\} \not\subseteq \{1, 2, 3\}$.
- (c) Both hold. For $A \subseteq A$: let $x \in A$; then $x \in A$. For $\emptyset \subseteq A$: the implication $x \in \emptyset \Rightarrow x \in A$ has a false hypothesis for every x by [Definition 0.3.3](#), so it is vacuously true ([Definition 0.1.2](#)). The empty set is a subset of every set, including itself.
- (d) Holds. Let $x \in \mathbb{R}$ with $x^2 = 1$. Then $(x - 1)(x + 1) = 0$, so $x = 1$ or $x = -1$, and both are integers.

Warning. $\in$ and $\subseteq$ are different relations. $x \in A$ says the object x is one of the elements of A . $A \subseteq B$ compares two sets, element by element. So $1 \in \{1, 2\}$ is true while $1 \subseteq \{1, 2\}$ makes no sense (1 is not a set here); and $\{1\} \subseteq \{1, 2\}$ is true while $\{1\} \in \{1, 2\}$ is false.

Quick check. Let $X = \{\{1, 2\}, 3\}$. Is $\{1, 2\} \subseteq X$? Is $\{1, 2\} \in X$?

Solution. The set X has exactly two elements: the set $\{1, 2\}$ and the number 3. So $\{1, 2\} \in X$ is **true**. But $\{1, 2\} \subseteq X$ is **false**: $1 \in \{1, 2\}$, and 1 is neither the set $\{1, 2\}$ nor the number 3, so $1 \notin X$.

0.3.3 Equality of sets and double inclusion

We said a set is determined by its elements. Here is the precise version.

Definition 0.3.5 — Set equality

Two sets A and B are **equal**, written $A = B$, if they have exactly the same elements:

$$\forall x : (x \in A \Leftrightarrow x \in B).$$

So $\{1, 2\} = \{2, 1\} = \{1, 1, 2\}$: each object belongs to one of these sets exactly when it is 1 or 2. Order and repetition in a list are features of how we **write** a set, not of the set.

Proving a biconditional for every x at once is awkward. It is much easier to prove two implications separately, and that is exactly what the next result allows.

Theorem 0.3.1 — Double inclusion

Let A and B be sets. Then $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

Proof. By [Definition 0.3.5](#), $A = B$ means that for every x , both $x \in A \Rightarrow x \in B$ and $x \in B \Rightarrow x \in A$ hold, since a biconditional is the conjunction of an implication and its converse. A statement of the form “for every x , $P(x)$ and $Q(x)$ ” holds exactly when “for every x , $P(x)$ ” and “for every x , $Q(x)$ ” both hold, by [Definition 0.1.3](#). Hence $A = B$ holds exactly when $\forall x (x \in A \Rightarrow x \in B)$ and $\forall x (x \in B \Rightarrow x \in A)$ both hold. By [Definition 0.3.4](#), these are $A \subseteq B$ and $B \subseteq A$. This proves the theorem. ■

This is the move “**prove set equality by double inclusion**”. The proof is written in two labeled halves, $(\subseteq)$ and $(\supseteq)$, each starting “Let $x \in \dots$ ”. Here is a first consequence.

Remark. There is only one empty set. If E and E' are sets with no elements, then $E \subseteq E'$ and $E' \subseteq E$ hold vacuously, as in [Example \(Subsets\)](#) (c), so $E = E'$ by [Theorem 0.3.1](#). This is why we may say **the** empty set.

Example. (A solution set by double inclusion)

Prove that $\{x \in \mathbb{R} : x^2 - 3x + 2 = 0\} = \{1, 2\}$.

Solution. Write $S = \{x \in \mathbb{R} : x^2 - 3x + 2 = 0\}$. By [Theorem 0.3.1](#) it suffices to prove $S \subseteq \{1, 2\}$ and $\{1, 2\} \subseteq S$.

$(\subseteq)$ Let $x \in S$. Then x is real and $x^2 - 3x + 2 = 0$. Factoring, $(x - 1)(x - 2) = 0$. A product of two real numbers is zero only if one factor is zero, so $x = 1$ or $x = 2$. Hence $x \in \{1, 2\}$.

$(\supseteq)$ Let $x \in \{1, 2\}$, so $x = 1$ or $x = 2$. Both are real. If $x = 1$, then $1 - 3 + 2 = 0$; if $x = 2$, then $4 - 6 + 2 = 0$. In either case $x \in S$.

This shows $S = \{1, 2\}$.

Solving an equation “at school” usually produces only the $(\subseteq)$ half: “if x is a solution, then x is 1 or 2”. The $(\supseteq)$ half, checking that the candidates really are solutions, is the part that gets forgotten. It matters whenever a step in the solving is not reversible, such as squaring both sides.

0.3.4 Union, intersection and difference

Given two sets, there are three basic ways to form a new one: keep what is in either, keep what is in both, or keep what is in the first but not the second. Each is a set-builder description whose predicate uses a connective from the section on logic.

Definition 0.3.6 — Union

The **union** of sets A and B is

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$

Definition 0.3.7 — Intersection

The **intersection** of sets A and B is

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

We call A and B **disjoint** if $A \cap B = \emptyset$.

Definition 0.3.8 — Set difference and complement

The **difference** of sets A and B is

$$A \setminus B = \{x \in A : x \notin B\}.$$

When all sets under discussion are subsets of one fixed set U , the **complement** of $B \subseteq U$ is $B^c = U \setminus B$.

In words: the union keeps an element that lies in **at least one** of the sets (the “or” is inclusive, as always); the intersection keeps an element that lies in **both**; the difference $A \setminus B$ keeps the elements of A that are **not** in B . The complement depends on the choice of U : the complement of $\mathbb{N}$ is the set of negative integers inside $\mathbb{Z}$, but a much bigger set inside $\mathbb{R}$.

Example. (Operations on finite sets)

Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$. Compute $A \cup B$, $A \cap B$, $A \setminus B$ and $B \setminus A$.

Solution. Checking each element against the definitions: $A \cup B = \{1, 2, 3, 4, 5, 6\}$; $A \cap B = \{3, 4\}$; $A \setminus B = \{1, 2\}$; and $B \setminus A = \{5, 6\}$. Since $A \setminus B \neq B \setminus A$, the difference is not symmetric in its two arguments, unlike union and intersection.

Example. (Operations on intervals)

Let $A = [0, 2]$ and $B = [1, 3]$. Compute $A \cup B$, $A \cap B$ and $A \setminus B$.

Solution. A real x lies in $A \cup B$ when $0 \leq x \leq 2$ or $1 \leq x \leq 3$, which happens exactly when $0 \leq x \leq 3$; so $A \cup B = [0, 3]$. It lies in $A \cap B$ when both hold, that is, when $1 \leq x \leq 2$; so $A \cap B = [1, 2]$. It lies in $A \setminus B$ when $0 \leq x \leq 2$ and not $1 \leq x \leq 3$. Given $x \leq 2$, “not $1 \leq x \leq 3$ ” means $x < 1$; so $A \setminus B = [0, 1)$. The endpoint 1 is excluded because $1 \in B$.

Unions, intersections and differences interact through laws that mirror those for “or”, “and”

and “not”. The first pair is De Morgan’s laws, the set version of [Theorem 0.1.2](#).

Theorem 0.3.2 — De Morgan’s laws for sets

Let A , B and C be sets. Then

- (a) $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$;
- (b) $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$.

In particular, if $B, C \subseteq U$, then $(B \cup C)^c = B^c \cap C^c$ and $(B \cap C)^c = B^c \cup C^c$.

Idea. Unwind membership into connectives. “ $x \notin B \cup C$ ” is “not ($x \in B$ or $x \in C$)”, and De Morgan for statements turns that into “ $x \notin B$ and $x \notin C$ ”. So each law is its logical counterpart, applied one element at a time. We prove (a) by double inclusion; (b) is an exercise.

Proof. We prove (a) using [Theorem 0.3.1](#).

($\subseteq$) Let $x \in A \setminus (B \cup C)$. By [Definition 0.3.8](#), $x \in A$ and $x \notin B \cup C$. By [Definition 0.3.6](#), the second says that “ $x \in B$ or $x \in C$ ” is false, so by [Theorem 0.1.2](#), $x \notin B$ and $x \notin C$. Since $x \in A$ and $x \notin B$, we have $x \in A \setminus B$; since $x \in A$ and $x \notin C$, we have $x \in A \setminus C$. Hence $x \in (A \setminus B) \cap (A \setminus C)$ by [Definition 0.3.7](#).

($\supseteq$) Let $x \in (A \setminus B) \cap (A \setminus C)$. By [Definition 0.3.7](#) and [Definition 0.3.8](#), $x \in A$, $x \notin B$ and $x \notin C$. By [Theorem 0.1.2](#), “ $x \in B$ or $x \in C$ ” is false, so $x \notin B \cup C$. Together with $x \in A$, this gives $x \in A \setminus (B \cup C)$.

This proves (a). For the complement form, take $A = U$ and use [Definition 0.3.8](#): $(B \cup C)^c = U \setminus (B \cup C) = (U \setminus B) \cap (U \setminus C) = B^c \cap C^c$, and similarly for (b). ■

The second pair of laws says each of $\cap$ and $\cup$ distributes over the other, like multiplication over addition, except that here **both** directions hold.

Theorem 0.3.3 — Distributive laws for sets

Let A , B and C be sets. Then

- (a) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$;
- (b) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Idea. Double inclusion again, and in each half the “or” in a union forces a split into cases. For (b) ($\supseteq$), an element of both $A \cup B$ and $A \cup C$ is either in A , and then done, or not in A , and then it must be in B and in C . We prove (b); (a) is an exercise.

Proof. We prove (b) using [Theorem 0.3.1](#).

($\subseteq$) Let $x \in A \cup (B \cap C)$. By [Definition 0.3.6](#), $x \in A$ or $x \in B \cap C$.

Case 1: $x \in A$. Then $x \in A \cup B$ and $x \in A \cup C$ by [Definition 0.3.6](#).

Case 2: $x \in B \cap C$. Then $x \in B$ and $x \in C$ by Definition 0.3.7, so again $x \in A \cup B$ and $x \in A \cup C$.

In both cases $x \in (A \cup B) \cap (A \cup C)$ by Definition 0.3.7.

($\supseteq$) Let $x \in (A \cup B) \cap (A \cup C)$, so $x \in A \cup B$ and $x \in A \cup C$.

Case 1: $x \in A$. Then $x \in A \cup (B \cap C)$.

Case 2: $x \notin A$. Since $x \in A \cup B$ and $x \notin A$, we get $x \in B$. Since $x \in A \cup C$ and $x \notin A$, we get $x \in C$. Hence $x \in B \cap C$, so $x \in A \cup (B \cap C)$.

In both cases $x \in A \cup (B \cap C)$. This proves (b). ■

Remark. In Chapter 1 we will see that the intersection of two subspaces is always a subspace, while their union usually is not: the two axes in the plane are subspaces, but $(1, 0) + (0, 1) = (1, 1)$ lies on neither. The union is replaced there by a different construction, the sum of subspaces.

0.3.5 Indexed families of sets

Two sets are not always enough. In Chapter 1 we will need “the intersection of **all** subspaces that contain a given set”, and there are usually infinitely many of them. So we need unions and intersections of any number of sets at once.

An indexed family is a list of sets with one set for each label; its union keeps what lies in some member, its intersection what lies in every member.

Definition 0.3.9 — Unions and intersections of indexed families

Let I be a set, and for each $i \in I$ let A_i be a set. The collection $(A_i)_{i \in I}$ is an **indexed family** of sets, with **index set** I . Its **union** is

$$\bigcup_{i \in I} A_i = \{x : \exists i \in I : x \in A_i\},$$

and, if I is **non-empty**, its **intersection** is

$$\bigcap_{i \in I} A_i = \{x : \forall i \in I : x \in A_i\}.$$

In words: x lies in the union when it lies in **at least one** A_i , and in the intersection when it lies in **every** A_i . When $I = \{1, 2\}$, these are $A_1 \cup A_2$ and $A_1 \cap A_2$. When $I = \{1, 2, 3, \dots\}$, we also write $\bigcup_{n=1}^{\infty} A_n$ and $\bigcap_{n=1}^{\infty} A_n$.

Why I must be non-empty for the intersection. If $I = \emptyset$, the condition “ $\forall i \in I : x \in A_i$ ” is vacuously true for **every** object x , so the “intersection” would contain everything, which is not a set we want. (When all sets live inside a fixed U , one sometimes defines the empty intersection to be U .) The empty union causes no such trouble: “ $\exists i \in \emptyset$ ” is always false, so $\bigcup_{i \in \emptyset} A_i = \emptyset$.

Example. (Indexed unions and intersections of intervals)

For each integer $n \geq 1$, let $A_n = [1/n, 1]$ and $B_n = (0, 1/n)$. Show that

$$\bigcup_{n=1}^{\infty} A_n = (0, 1] \quad \text{and} \quad \bigcap_{n=1}^{\infty} B_n = \emptyset.$$

You may use the fact that for every real $x > 0$ there is an integer $n \geq 1$ with $1/n < x$.

Solution. The union. ($\subseteq$) Let $x \in \bigcup_n A_n$. By [Definition 0.3.9](#), $x \in A_n$ for some $n \geq 1$, so $0 < 1/n \leq x \leq 1$. Hence $x \in (0, 1]$. ($\supseteq$) Let $x \in (0, 1]$. Since $x > 0$, there is an integer $n \geq 1$ with $1/n < x$. Then $1/n \leq x \leq 1$, so $x \in A_n$, and hence x lies in the union. By [Theorem 0.3.1](#), the union is $(0, 1]$.

The intersection. Suppose, for a contradiction, that some x lies in $\bigcap_n B_n$. Then $x \in B_1 = (0, 1)$, so $x > 0$. Choose an integer $n \geq 1$ with $1/n < x$. Then $x \notin (0, 1/n) = B_n$, contradicting $x \in B_n$ for every n . Hence the intersection has no elements, so it equals $\emptyset$.

Each B_n is non-empty (it contains $1/(2n)$), and each is contained in the one before. Yet their intersection is empty. Infinite intersections can shrink to nothing even when no single step does.

De Morgan's laws hold for indexed families with the same proofs: for a non-empty index set I , $A \setminus \bigcup_i B_i = \bigcap_i (A \setminus B_i)$ and $A \setminus \bigcap_i B_i = \bigcup_i (A \setminus B_i)$. The “or” of the finite version becomes “there exists”, and the “and” becomes “for every”, exactly as in [Definition 0.1.4](#).

0.3.6 Cartesian products and n -tuples

A set forgets order: $\{1, 2\} = \{2, 1\}$. But a point in the plane is not like that. The point $(1, 2)$ is different from $(2, 1)$. We need objects that remember order.

An **ordered pair** (a, b) has a first entry a and a second entry b , and two ordered pairs are equal exactly when their entries agree in order:

$$(a, b) = (a', b') \iff a = a' \text{ and } b = b'.$$

Definition 0.3.10 — Cartesian product

Let A and B be sets. The **Cartesian product** of A and B is the set of ordered pairs

$$A \times B = \{(a, b) : a \in A, b \in B\}.$$

More generally, for sets $A_1, \dots, A_n$, the product $A_1 \times \dots \times A_n$ is the set of all ordered lists $(a_1, \dots, a_n)$ with $a_i \in A_i$ for each i .

For example, $\{1, 2\} \times \{x, y\} = \{(1, x), (1, y), (2, x), (2, y)\}$. The order of the factors matters: $(1, x) \in \{1, 2\} \times \{x, y\}$, but $(1, x) \notin \{x, y\} \times \{1, 2\}$, whose pairs have a letter first. A degenerate case: $A \times \emptyset = \emptyset$ for every set A , since no pair can have a second entry from $\emptyset$.

Definition 0.3.11 — n -tuples

Let A be a set and $n \geq 1$ an integer. An n -**tuple** of elements of A is an ordered list $(a_1, \dots, a_n)$ with each $a_i \in A$. The set of all of them is

$$A^n = \underbrace{A \times A \times \cdots \times A}_{n \text{ factors}}.$$

Two n -tuples are equal exactly when $a_i = b_i$ for **every** $i = 1, \dots, n$.

Example. (Sets of tuples)

Describe some elements of $\mathbb{R}^2$, $\mathbb{Z}^3$ and $\{0, 1\}^3$, and count the elements of $\{0, 1\}^3$.

Solution. The set $\mathbb{R}^2$ is the plane: it contains $(0, 0)$, $(1, -2)$ and $(\sqrt{2}, \pi)$. The set $\mathbb{Z}^3$ contains $(1, 0, -4)$ but not $(1, \frac{1}{2}, 0)$, whose second entry is not an integer. The set $\{0, 1\}^3$ consists of the lists of three bits:

$$\{0, 1\}^3 = \{(0, 0, 0), (0, 0, 1), (0, 1, 0), (0, 1, 1), (1, 0, 0), (1, 0, 1), (1, 1, 0), (1, 1, 1)\}.$$

There are two choices for each of the three entries, giving $2 \cdot 2 \cdot 2 = 8$ elements. In the same way, if A has k elements then A^n has k^n elements.

In Chapter 1, $\mathbb{R}^n$ and more generally F^n become the first examples of vector spaces, with n -tuples added and scaled entry by entry.

0.3.7 Power sets

Sets can themselves be elements of sets, as the warning about $\{\emptyset\}$ showed. One such set collects every subset of a given set.

Definition 0.3.12 — Power set

The **power set** of a set S is the set of all subsets of S :

$$\mathcal{P}(S) = \{X : X \subseteq S\}.$$

So $X \in \mathcal{P}(S)$ means exactly $X \subseteq S$. For example,

$$\mathcal{P}(\{1, 2\}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\},$$

which has 4 elements. The degenerate case is instructive: $\mathcal{P}(\emptyset) = \{\emptyset\}$ has **one** element, because the empty set has exactly one subset, itself. For a set with n elements the power set has 2^n elements; the last exercise proves this by induction.

0.3.8 Exercises

0.3.8.1 A. Check your understanding

Exercise A1.

Decide whether each statement is true or false. Give a one-line reason.

- (a) $0 \in \mathbb{N}$.
- (b) $\emptyset \in \emptyset$.
- (c) $\emptyset \subseteq \emptyset$.
- (d) $\{0\} = \emptyset$.
- (e) $\{1, 2\} = \{2, 1, 1\}$ and $(1, 2) = (2, 1)$.
- (f) $\{1\} \in \mathcal{P}(\{1, 2\})$ and $\{1\} \subseteq \mathcal{P}(\{1, 2\})$.

Solution.

- (a) True. By [Definition 0.3.1](#), $\mathbb{N}$ includes 0 in this book.
- (b) False. By [Definition 0.3.3](#), nothing is an element of $\emptyset$, not even $\emptyset$.
- (c) True. The empty set is a subset of every set, vacuously, as in [Example \(Subsets\)](#).
- (d) False. $0 \in \{0\}$ but $0 \notin \emptyset$, so the two sets do not have the same elements ([Definition 0.3.5](#)).
- (e) The first is true: both sets have exactly the elements 1 and 2. The second is false: ordered pairs are equal only when their first entries agree, and $1 \neq 2$.
- (f) The first is true: $\{1\} \subseteq \{1, 2\}$, so $\{1\} \in \mathcal{P}(\{1, 2\})$ by [Definition 0.3.12](#). The second is false: $1 \in \{1\}$, but 1 is a number, not a subset of $\{1, 2\}$, so $1 \notin \mathcal{P}(\{1, 2\})$.

0.3.8.2 B. Practice

Exercise B1: The second De Morgan law.

Let A, B, C be sets. Prove that $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$, which is part (b) of [Theorem 0.3.2](#).

Solution. We use [Theorem 0.3.1](#).

($\subseteq$) Let $x \in A \setminus (B \cap C)$. By [Definition 0.3.8](#), $x \in A$ and $x \notin B \cap C$. By [Definition 0.3.7](#), “ $x \in B$ and $x \in C$ ” is false, so by [Theorem 0.1.2](#), $x \notin B$ or $x \notin C$. If $x \notin B$, then $x \in A \setminus B$; if $x \notin C$, then $x \in A \setminus C$. In either case $x \in (A \setminus B) \cup (A \setminus C)$ by [Definition 0.3.6](#).

($\supseteq$) Let $x \in (A \setminus B) \cup (A \setminus C)$. By [Definition 0.3.6](#), $x \in A \setminus B$ or $x \in A \setminus C$. In the first case $x \in A$ and $x \notin B$; in the second, $x \in A$ and $x \notin C$. In both cases $x \in A$, and “ $x \notin B$ or $x \notin C$ ” is true, so by [Theorem 0.1.2](#) “ $x \in B$ and $x \in C$ ” is false, that is, $x \notin B \cap C$. Hence $x \in A \setminus (B \cap C)$.

This proves the equality.

Exercise B2: A distributive law.

Let A, B, C be sets. Prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, which is part (a) of [Theorem 0.3.3](#).

Solution. We use [Theorem 0.3.1](#).

($\subseteq$) Let $x \in A \cap (B \cup C)$. Then $x \in A$ and $x \in B \cup C$, so $x \in B$ or $x \in C$. If $x \in B$, then $x \in A \cap B$; if $x \in C$, then $x \in A \cap C$. In either case $x \in (A \cap B) \cup (A \cap C)$.

($\supseteq$) Let $x \in (A \cap B) \cup (A \cap C)$. If $x \in A \cap B$, then $x \in A$ and $x \in B \subseteq B \cup C$. If $x \in A \cap C$, then $x \in A$ and $x \in C \subseteq B \cup C$. In either case $x \in A$ and $x \in B \cup C$, so $x \in A \cap (B \cup C)$.

This proves the equality.

Exercise B3: An indexed intersection.

Prove that

$$\bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}.$$

You may use the fact that for every real $x > 0$ there is an integer $n \geq 1$ with $1/n < x$.

Solution. Write $B_n = (-1/n, 1/n)$ and $B = \bigcap_{n \geq 1} B_n$. We use [Theorem 0.3.1](#).

($\supseteq$) For every $n \geq 1$, $-1/n < 0 < 1/n$, so $0 \in B_n$. By [Definition 0.3.9](#), $0 \in B$.

($\subseteq$) Let $x \in B$. Suppose, for a contradiction, that $x \neq 0$. Then $|x| > 0$, so there is an integer $n \geq 1$ with $1/n < |x|$. Then $x \geq 1/n$ or $x \leq -1/n$, so $x \notin B_n$. This contradicts $x \in B$, which requires $x \in B_n$ for **every** n . Hence $x = 0$, that is, $x \in \{0\}$.

This shows $B = \{0\}$.

0.3.8.3 C. Going deeper**Exercise C1: When union equals intersection.**

Let A and B be sets. Prove that $A \cup B = A \cap B$ if and only if $A = B$.

Solution. ($\Leftarrow$) Suppose $A = B$. Then $A \cup B = A \cup A$ and $A \cap B = A \cap A$. By [Definition 0.3.6](#) and [Definition 0.3.7](#), “ $x \in A$ or $x \in A$ ” and “ $x \in A$ and $x \in A$ ” are both equivalent to $x \in A$, so $A \cup A = A = A \cap A$. Hence $A \cup B = A \cap B$.

($\Rightarrow$) Suppose $A \cup B = A \cap B$. We show $A = B$ using [Theorem 0.3.1](#).

($\subseteq$) Let $x \in A$. Then $x \in A \cup B$ by [Definition 0.3.6](#). Since $A \cup B = A \cap B$, we get $x \in A \cap B$, and in particular $x \in B$.

($\supseteq$) Swapping the roles of A and B in the previous paragraph, which is allowed since $A \cup B = B \cup A$ and $A \cap B = B \cap A$, gives $B \subseteq A$.

Hence $A = B$. This proves the equivalence.

Exercise C2: Counting subsets.

For a finite set S , write $|S|$ for its number of elements. The aim is to prove that $|\mathcal{P}(S)| = 2^{|S|}$ for every finite set S .

- (a) Check the claim when $S = \emptyset$.
- (b) Let S have $n+1$ elements, fix $s \in S$, and let $S' = S \setminus \{s\}$. Show that every subset X of S belongs to exactly one of the two collections: the subsets of S with $s \notin X$, and the subsets of S with $s \in X$.
- (c) Show that the subsets of S with $s \notin X$ are exactly the subsets of S' , and that $X \mapsto X \setminus \{s\}$ and $T \mapsto T \cup \{s\}$ pair off the subsets of S containing s with the subsets of S' , each undoing the other.
- (d) Hence prove the claim by induction on $n = |S|$.

Hint: in (c), use double inclusion to check that $(X \setminus \{s\}) \cup \{s\} = X$ when $s \in X$.

Solution.

- (a) The only subset of $\emptyset$ is $\emptyset$ itself, since any $X \subseteq \emptyset$ has no elements and so equals $\emptyset$ by [Theorem 0.3.1](#). So $\mathcal{P}(\emptyset) = \{\emptyset\}$ has $1 = 2^0$ element.
- (b) Let $X \subseteq S$. The statement $s \in X$ is either true or false, and not both. So X lies in exactly one of the two collections.
- (c) *First collection.* If $X \subseteq S$ and $s \notin X$, then every $x \in X$ lies in S and is not s , so $x \in S'$; hence $X \subseteq S'$. Conversely, if $T \subseteq S'$, then $T \subseteq S$ and $s \notin T$, since no element of S' equals s . So the first collection is exactly $\mathcal{P}(S')$.
Second collection. If $X \subseteq S$ with $s \in X$, then $X \setminus \{s\} \subseteq S'$. If $T \subseteq S'$, then $T \cup \{s\} \subseteq S$ contains s . Now we check that the two operations undo each other. For X containing s : an element lies in $(X \setminus \{s\}) \cup \{s\}$ exactly when it is in X and not s , or it is s ; since $s \in X$, this is exactly when it is in X . So $(X \setminus \{s\}) \cup \{s\} = X$. For $T \subseteq S'$: an element lies in $(T \cup \{s\}) \setminus \{s\}$ exactly when it is in T or equal to s , and not equal to s ; since $s \notin T$, this is exactly when it is in T . So $(T \cup \{s\}) \setminus \{s\} = T$. Hence the two operations pair the subsets of S containing s one-to-one with the subsets of S' , and the two collections have the same number of elements.
- (d) Let $P(n)$ be the statement “every set with n elements has exactly 2^n subsets”. We use [Theorem 0.2.7](#) with $n_0 = 0$. The base case $P(0)$ is (a), since the only set with no elements is $\emptyset$. Let $n \geq 0$, suppose $P(n)$, and let S have $n+1$ elements. Fix $s \in S$, which exists since $n+1 \geq 1$, and let $S' = S \setminus \{s\}$, a set with n elements. By (b), the number of subsets of S is the size of the first collection plus the size of the second. By (c) and the induction hypothesis, each of these sizes equals $|\mathcal{P}(S')| = 2^n$. Hence S has $2^n + 2^n = 2^{n+1}$ subsets, which is $P(n+1)$. By [Theorem 0.2.7](#), $P(n)$ holds for every $n \in \mathbb{N}$, as claimed.

0.4 Functions

Sets are the objects of this chapter. The next question is how to move between them, and the tool is a function. Almost everything in linear algebra is a question about a function: a linear map is a function, a matrix will turn out to encode one, and “does $A\mathbf{x} = \mathbf{b}$ have a solution, and

is it unique?” is a question about whether a function is surjective and injective. This section fixes the language of functions, composition and inverses, and proves the handful of facts about them that the rest of the book uses without comment.

0.4.1 What a function is

At school a function is usually a formula, such as $x^2 + 1$. A formula alone is not enough. The same formula x^2 behaves differently depending on which numbers we feed it and where we allow the answers to land, and whether an equation $x^2 = c$ can be solved depends on exactly that information. So a function carries three pieces of data, not one.

A function is a machine with a fixed input set and a fixed target set that turns each input into exactly one output.

Definition 0.4.1 — Function

Let X and Y be sets. A **function** (or **map**) $f: X \rightarrow Y$ is a rule that assigns to **every** element $x \in X$ **exactly one** element $f(x) \in Y$. The set X is the **domain** of f , the set Y is its **codomain**, and $f(x)$ is the **value** of f at x . We also write $x \mapsto f(x)$.

Two functions $f: X \rightarrow Y$ and $g: X' \rightarrow Y'$ are **equal** if $X = X'$, $Y = Y'$, and $f(x) = g(x)$ for every $x \in X$.

In words: the first clause says every input gets an output, so nothing in the domain is left out. The second says the output is unique, so the rule never hesitates between two values. The codomain is where outputs are *allowed* to land; nothing says every element of Y must be hit.

If you want a definition that avoids the vague word “rule”, use the Cartesian product ([Definition 0.3.10](#)): a function $X \rightarrow Y$ is a subset $G \subseteq X \times Y$ such that for every $x \in X$ there is exactly one $y \in Y$ with $(x, y) \in G$. The set G is the **graph** of f , and $f(x)$ is that unique y . Nothing in this book depends on which version you prefer.

Example. (First examples)

Check each of the following against the two clauses of [Definition 0.4.1](#).

- (a) $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 3x - 5$. Every real x gives one real number $3x - 5$.
- (b) $s: \mathbb{R}^2 \rightarrow \mathbb{R}$, $s(x, y) = x + y$. The domain is a Cartesian product; each pair gives one sum.
- (c) $\ell: \{\text{finite strings of the letters } a, b\} \rightarrow \mathbb{N}$, sending a string to its length. The empty string has length 0, which is why it matters that $\mathbb{N}$ contains 0.
- (d) For any set X , the rule $x \mapsto x$ is a function $X \rightarrow X$. It looks too simple to matter, but it will play the role that the number 1 plays for multiplication.
- (e) For any set Y , there is exactly one function $\emptyset \rightarrow Y$, the **empty function**. Both clauses are about elements of the domain, and there are none, so they hold vacuously. It is a degenerate case that makes statements like “for every set X ” true without exceptions.

The failures are just as instructive. The rule “ $x \mapsto$ the real number y with $y^2 = x$ ” is **not** a

function $\mathbb{R} \rightarrow \mathbb{R}$. At $x = -1$ there is no such y , so the first clause fails. At $x = 4$ there are two, $y = 2$ and $y = -2$, so the second clause fails. Repairing both problems gives a genuine function $[0, \infty) \rightarrow \mathbb{R}$, $x \mapsto \sqrt{x}$, where $\sqrt{x}$ means the non-negative root.

The equality clause says the codomain is part of the function. The functions $\mathbb{R} \rightarrow \mathbb{R}$, $x \mapsto x^2$ and $\mathbb{R} \rightarrow [0, \infty)$, $x \mapsto x^2$ have the same values but are **different** functions. This is not pedantry: we will see in a moment that one of them is surjective and the other is not.

Quick check. Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be $f(x) = |x|$ and $g(x) = \sqrt{x^2}$. Are f and g equal?

Solution. Yes. They have the same domain and the same codomain, and for every real x we have $\sqrt{x^2} = |x|$, because $|x|$ is the non-negative number whose square is x^2 . Different formulas can describe the same function; what counts is the values.

0.4.2 Images and preimages

Often we want to push a whole set through a function, or ask which inputs land in a given region. Both operations get a name.

Definition 0.4.2 — Image and preimage

Let $f: X \rightarrow Y$ be a function, $A \subseteq X$ and $B \subseteq Y$.

- The **image** of A under f is $f(A) := \{f(a) : a \in A\} \subseteq Y$. The set $f(X)$ is called the **image of f** .
- The **preimage** of B under f is $f^{-1}(B) := \{x \in X : f(x) \in B\} \subseteq X$.

In words: $f(A)$ collects the outputs of the inputs in A . The preimage $f^{-1}(B)$ collects every input whose output lies in B . So $y \in f(A)$ means “ $y = f(a)$ for **some** $a \in A$ ”, while $x \in f^{-1}(B)$ means “ $f(x) \in B$ ”, with no existential quantifier at all.

Example. (Images and preimages of the squaring map)

Let $q: \mathbb{R} \rightarrow \mathbb{R}$, $q(x) = x^2$. Compute $q([-1, 2])$, $q^{-1}([1, 4])$ and $q^{-1}([-5, -1])$.

Solution. As x runs over $[-1, 2]$, the square x^2 takes every value in $[0, 4]$: values in $[0, 1]$ come from $[0, 1]$, values in $[1, 4]$ from $[1, 2]$, and no square is negative or exceeds $2^2 = 4$ on this interval. Hence $q([-1, 2]) = [0, 4]$.

Next, $x^2 \in [1, 4]$ means $1 \leq |x| \leq 2$. Hence $q^{-1}([1, 4]) = [-2, -1] \cup [1, 2]$.

Finally, no real square lies in $[-5, -1]$, so $q^{-1}([-5, -1]) = \emptyset$. A preimage may be empty; that is not an error.

Warning. The symbol $f^{-1}(B)$ is defined for **every** function, invertible or not; it does not mean that an inverse function exists. For the same reason, pushing forward and pulling back need not undo each other. With $q(x) = x^2$ on $\mathbb{R}$: for $B = [-1, 4]$ we get $q^{-1}(B) = [-2, 2]$ and $q(q^{-1}(B)) = [0, 4] \neq B$; for $A = [0, 1]$ we get $q(A) = [0, 1]$ and $q^{-1}(q(A)) = [-1, 1] \neq A$.

What does survive is one inclusion in each direction.

Proposition 0.4.1 — Image and preimage inclusions

Let $f: X \rightarrow Y$, $A \subseteq X$ and $B \subseteq Y$. Then

$$f(f^{-1}(B)) \subseteq B \quad \text{and} \quad A \subseteq f^{-1}(f(A)).$$

Proof. Let $y \in f(f^{-1}(B))$. By Definition 0.4.2, $y = f(x)$ for some $x \in f^{-1}(B)$, and $x \in f^{-1}(B)$ means $f(x) \in B$. Hence $y \in B$.

Let $a \in A$. Then $f(a) \in f(A)$ by definition of the image, so $a \in f^{-1}(f(A))$ by definition of the preimage. This proves both inclusions. ■

The warning shows both inclusions can be strict. In [Exercise \(C3: When images and preimages undo each other\)](#) you will find exactly which property of f turns each one into an equality.

0.4.3 Injective, surjective, bijective

Given $f: X \rightarrow Y$ and a target $y \in Y$, the most basic question is how many solutions the equation $f(x) = y$ has. There are two separate issues: whether there is **at least one** solution, and whether there is **at most one**. In linear algebra these become “does $A\mathbf{x} = \mathbf{b}$ have a solution?” and “is the solution unique?”, and they are answered by different tools. So the two halves deserve separate names.

Injective means no two inputs collide; surjective means every target is hit; bijective means both, so every equation $f(x) = y$ has exactly one solution.

Definition 0.4.3 — Injective, surjective, bijective

Let $f: X \rightarrow Y$ be a function.

- f is **injective** (one-to-one) if **for all** $x_1, x_2 \in X$, $f(x_1) = f(x_2)$ implies $x_1 = x_2$.
- f is **surjective** (onto) if **for every** $y \in Y$ **there exists** $x \in X$ with $f(x) = y$.
Equivalently, $f(X) = Y$.
- f is **bijective** if it is both injective and surjective.

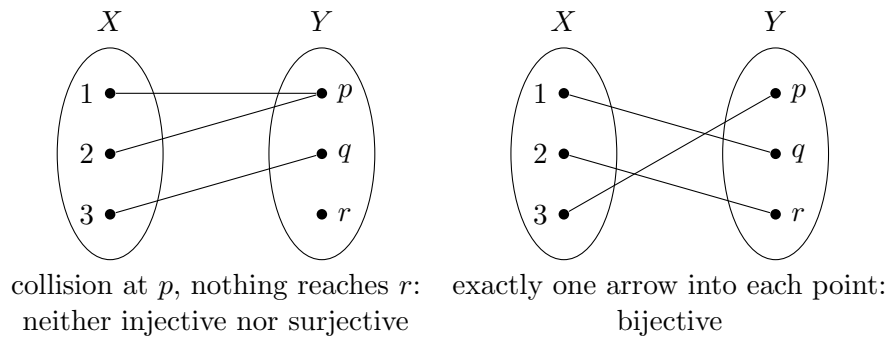
In words: injectivity says that if two inputs give the same output, they were the same input. Equivalently, by the contrapositive ([Theorem 0.1.1](#)), distinct inputs give distinct outputs. Surjectivity says the image of f fills the whole codomain. The quantifiers matter: in surjectivity the x is allowed to depend on y .

Negating these, as in [Example \(What “not injective” means\)](#) and [Example \(What “not surjective” means\)](#), tells us what a counterexample looks like. The map f is **not injective** when there exist $x_1 \neq x_2$ with $f(x_1) = f(x_2)$: one collision is enough. It is **not surjective** when there exists $y \in Y$ that equals $f(x)$ for no $x \in X$: one missed target is enough.

The standard moves. To prove f injective, write “Suppose $f(x_1) = f(x_2)$ ” and work towards

$x_1 = x_2$. To prove f surjective, write “Let $y \in Y$ ” and then produce an x , usually found backwards by solving $f(x) = y$ on scrap paper, and check forwards that $f(x) = y$.

The picture below shows maps between three-element sets, drawn as arrows. A collision is two arrows landing on one point; a missed target is a point with no arrow.



Example. (Deciding injectivity and surjectivity)

Determine which of the following functions are injective, surjective or bijective.

- (a) $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 3x - 5$.
- (b) $g: \mathbb{Z} \rightarrow \mathbb{Z}, g(n) = 2n$.
- (c) $s: \mathbb{R}^2 \rightarrow \mathbb{R}, s(x, y) = x + y$.
- (d) $\text{id}_X: X \rightarrow X, x \mapsto x$, for any set X .

Solution.

- (a) Bijective. Suppose $f(x_1) = f(x_2)$. Then $3x_1 - 5 = 3x_2 - 5$, so $3x_1 = 3x_2$, and dividing by $3 \neq 0$ gives $x_1 = x_2$. Hence f is injective. Let $y \in \mathbb{R}$. Solving $3x - 5 = y$ suggests $x = (y + 5)/3$, and indeed $f((y + 5)/3) = (y + 5) - 5 = y$. Hence f is surjective.
- (b) Injective, not surjective. Suppose $2n_1 = 2n_2$; then $n_1 = n_2$. But 1 is not hit: $2n = 1$ has no integer solution, since $2n$ is even. Note that the same formula $x \mapsto 2x$ on $\mathbb{R} \rightarrow \mathbb{R}$ is surjective; the failure comes from the domain and codomain, not the formula.
- (c) Surjective, not injective. Let $t \in \mathbb{R}$; then $s(t, 0) = t$, so s is surjective. But $s(1, 0) = 1 = s(0, 1)$ while $(1, 0) \neq (0, 1)$, a collision, so s is not injective.
- (d) Bijective. If $\text{id}_X(x_1) = \text{id}_X(x_2)$ then $x_1 = x_2$ directly, and every $x \in X$ is hit by itself. This degenerate case is the model of a bijection and reappears as the identity function below.

The cleanest way to see that domain and codomain are part of the question is to keep the formula $x \mapsto x^2$ fixed and change only the sets. Write $[0, \infty)$ for the non-negative reals.

Function	Injective?	Surjective?
$\mathbb{R} \rightarrow \mathbb{R}$	no: $(-1)^2 = 1^2$	no: -1 is not a square
$\mathbb{R} \rightarrow [0, \infty)$	no: $(-1)^2 = 1^2$	yes: $y = (\sqrt{y})^2$

Function	Injective?	Surjective?
$[0, \infty) \rightarrow \mathbb{R}$	yes	no: -1 is not a square
$[0, \infty) \rightarrow [0, \infty)$	yes	yes

For the injective rows: if $x_1, x_2 \geq 0$ and $x_1^2 = x_2^2$, then $(x_1 - x_2)(x_1 + x_2) = 0$. Either $x_1 = x_2$, or $x_1 + x_2 = 0$, which for non-negative numbers forces $x_1 = x_2 = 0$. In both cases $x_1 = x_2$.

Read the table as a set of minimal changes. Shrinking the domain from $\mathbb{R}$ to $[0, \infty)$ removes the collision between -1 and 1 , so the injectivity clause now holds. Shrinking the codomain removes the missed targets, so the surjectivity clause now holds. The formula never changed.

Warning. Do not confuse injectivity with the statement “ $x_1 = x_2 \Rightarrow f(x_1) = f(x_2)$ ”. That implication is true for **every** function; it is the second clause of [Definition 0.4.1](#). Injectivity is its converse, “ $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ ”, and $x \mapsto x^2$ on $\mathbb{R}$ satisfies the first but not the second.

Why these definitions. Surjectivity depends on the codomain, and injectivity on the domain, so neither is a property of a formula alone; this is why [Definition 0.4.1](#) insists on both sets. The names are descriptive: “in-jective” throws the domain *into* the codomain without collisions, and “sur-jective” (French *sur*, onto) lands *onto* all of it. In Chapter 3 we will meet a fast test for linear maps: a linear map is injective exactly when the only input sent to $\mathbf{0}$ is $\mathbf{0}$. That test is special to linear maps and fails for general functions, which is one reason the general definitions come first.

0.4.4 Composition

If the outputs of one function can be fed into another, we can run them in sequence. This is the operation that matrix multiplication will describe in Chapter 3.

Definition 0.4.4 — Composition

Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be functions. The **composition** $g \circ f: X \rightarrow Z$ is defined by

$$(g \circ f)(x) := g(f(x)) \quad \text{for every } x \in X.$$

Read $g \circ f$ as “ g after f ”: the function written on the **right** acts **first**. The definition needs the codomain of f to equal the domain of g , so that $g(f(x))$ makes sense.

Example. (Order matters)

- (a) Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be $f(x) = x + 3$ and $g(x) = 2x$. Compute $g \circ f$ and $f \circ g$.
- (b) Let $q: \mathbb{R} \rightarrow [0, \infty)$, $q(x) = x^2$, and $r: [0, \infty) \rightarrow \mathbb{R}$, $r(y) = \sqrt{y}$. Compute $r \circ q$ and $q \circ r$.

Solution.

- (a) $(g \circ f)(x) = g(x + 3) = 2x + 6$, while $(f \circ g)(x) = f(2x) = 2x + 3$. At $x = 0$ these

are $6 \neq 3$, so $g \circ f \neq f \circ g$. Composition is not commutative, even when both orders make sense.

- (b) $r \circ q: \mathbb{R} \rightarrow \mathbb{R}$ is $x \mapsto \sqrt{x^2} = |x|$. In the other order, $q \circ r: [0, \infty) \rightarrow [0, \infty)$ is $y \mapsto (\sqrt{y})^2 = y$. So $q \circ r$ does nothing, while $r \circ q$ forgets the sign of x . The two orders do not even have the same domain. We will return to this pair when we discuss one-sided inverses.

Although the order matters, the grouping does not. This is what lets us write $h \circ g \circ f$ without brackets.

Theorem 0.4.2 — Associativity of composition

Let $f: X \rightarrow Y$, $g: Y \rightarrow Z$ and $h: Z \rightarrow W$ be functions. Then $(h \circ g) \circ f = h \circ (g \circ f)$.

Proof. Both sides are functions $X \rightarrow W$. Let $x \in X$. By Definition 0.4.4 applied twice on each side,

$$((h \circ g) \circ f)(x) = (h \circ g)(f(x)) = h(g(f(x))) = h((g \circ f)(x)) = (h \circ (g \circ f))(x).$$

Since the values agree at every x , the two functions are equal by Definition 0.4.1. ■

Next, the properties of the previous subsection pass through composition.

Theorem 0.4.3 — Composition preserves injectivity and surjectivity

Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be functions.

- (a) If f and g are injective, then $g \circ f$ is injective.
- (b) If f and g are surjective, then $g \circ f$ is surjective.
- (c) If f and g are bijective, then $g \circ f$ is bijective.

Idea. For (a), start from a collision of $g \circ f$ and peel off one function at a time, outermost first: injectivity of g removes g , then injectivity of f removes f . For (b), run backwards: reach z from some y using g , then reach that y from some x using f .

Proof.

- (a) Suppose $(g \circ f)(x_1) = (g \circ f)(x_2)$, that is, $g(f(x_1)) = g(f(x_2))$. Since g is injective, $f(x_1) = f(x_2)$. Since f is injective, $x_1 = x_2$. Hence $g \circ f$ is injective.
- (b) Let $z \in Z$. Since g is surjective, there is $y \in Y$ with $g(y) = z$. Since f is surjective, there is $x \in X$ with $f(x) = y$. Then $(g \circ f)(x) = g(y) = z$. Hence $g \circ f$ is surjective.
- (c) If f and g are bijective, then $g \circ f$ is injective by (a) and surjective by (b), so it is bijective. ■

The converse of (a) is false, but half of it survives, and that half is surprisingly useful. Suppose $g \circ f$ is injective. A collision of f would be a collision of $g \circ f$, so f must be injective; but g is only used on the image $f(X)$, so g may still collide outside it. The surjective case is dual.

Theorem 0.4.4 — Partial converse for compositions

Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be functions.

- (a) If $g \circ f$ is injective, then f is injective.
- (b) If $g \circ f$ is surjective, then g is surjective.

Proof.

- (a) Suppose $f(x_1) = f(x_2)$. Applying g to both sides gives $(g \circ f)(x_1) = (g \circ f)(x_2)$. Since $g \circ f$ is injective, $x_1 = x_2$. Hence f is injective.
- (b) Let $z \in Z$. Since $g \circ f$ is surjective, there is $x \in X$ with $g(f(x)) = z$. Put $y = f(x) \in Y$; then $g(y) = z$. Hence g is surjective.

■

Remark. The other halves really fail. Take $X = Z = \{1\}$, $Y = \{1, 2\}$, $f(1) = 1$ and $g(1) = g(2) = 1$. Then $g \circ f = \text{id}_{\{1\}}$ is bijective, yet g is not injective ($g(1) = g(2)$) and f is not surjective (nothing maps to 2).

0.4.5 Identity and one-sided inverses

The function that does nothing deserves a name, because “undoing f ” means “composing with f gives the function that does nothing”.

Definition 0.4.5 — Identity function

For a set X , the **identity function** on X is $\text{id}_X: X \rightarrow X$, $\text{id}_X(x) = x$.

For every $f: X \rightarrow Y$ we have $f \circ \text{id}_X = f$ and $\text{id}_Y \circ f = f$, since both sides send x to $f(x)$. Notice the subscripts: on the right of f the identity lives on the domain, on the left on the codomain.

Now look again at [Example \(Order matters\)](#) (b). There $q \circ r = \text{id}_{[0, \infty)}$, but $r \circ q = |\cdot| \neq \text{id}_{\mathbb{R}}$. So r undoes q from one side only. This one-sided behavior is common enough to need a word.

Definition 0.4.6 — Left and right inverses

Let $f: X \rightarrow Y$. A function $g: Y \rightarrow X$ is a **left inverse** of f if $g \circ f = \text{id}_X$, and a **right inverse** of f if $f \circ g = \text{id}_Y$.

“Left” and “right” refer to where g is written in the composition. A left inverse recovers the

input from the output: $g(f(x)) = x$. A right inverse picks, for each target y , an input that reaches it: $f(g(y)) = y$. In [Example \(Order matters\)](#) (b), r is a right inverse of q , and q is a left inverse of r . But r is **not** a left inverse of q , because $r(q(-1)) = 1 \neq -1$.

The descriptions “recovers the input” and “reaches every target” should remind you of injective and surjective. They are exactly that.

Theorem 0.4.5 — Left inverses and injectivity

Let $f: X \rightarrow Y$ with X **non-empty**. Then f has a left inverse if and only if f is injective.

Idea. $(\Rightarrow)$ is [Theorem 0.4.4](#) with $g \circ f = \text{id}_X$, which is injective. For $(\Leftarrow)$ we must define $g(y)$ for every $y \in Y$. If y is in the image, injectivity gives a **unique** x with $f(x) = y$, and $g(y)$ must be that x . If y is not in the image, the condition $g \circ f = \text{id}_X$ says nothing about $g(y)$, so we send y anywhere in X . That is where non-emptiness is used.

Proof. $(\Rightarrow)$ Suppose $g \circ f = \text{id}_X$. The identity id_X is injective, so f is injective by [Theorem 0.4.4](#) (a).

$(\Leftarrow)$ Suppose f is injective. Since $X \neq \emptyset$, we may fix some $x_0 \in X$. Define $g: Y \rightarrow X$ as follows. If $y \in f(X)$, there is some $x \in X$ with $f(x) = y$, and this x is unique because f is injective; set $g(y) = x$. If $y \notin f(X)$, set $g(y) = x_0$. Every y receives exactly one value, so g is a function. For $x \in X$, the element $y = f(x)$ lies in $f(X)$, and the unique input sent to it is x , so $g(f(x)) = x$. Hence $g \circ f = \text{id}_X$, as claimed. ■

The hypothesis $X \neq \emptyset$ is not decoration. The empty function $\emptyset \rightarrow \{1\}$ is injective (there are no two inputs to collide), but there is no function $\{1\} \rightarrow \emptyset$ at all, since 1 would have nowhere to go.

Theorem 0.4.6 — Right inverses and surjectivity

Let $f: X \rightarrow Y$. Then f has a right inverse if and only if f is surjective.

Proof. $(\Rightarrow)$ Suppose $f \circ g = \text{id}_Y$. The identity id_Y is surjective, so f is surjective by [Theorem 0.4.4](#) (b).

$(\Leftarrow)$ Suppose f is surjective. For each $y \in Y$ the set $f^{-1}(\{y\})$ is non-empty, so we may choose one element of it and call it $g(y)$. This defines $g: Y \rightarrow X$ with $f(g(y)) = y$ for every $y \in Y$, that is, $f \circ g = \text{id}_Y$. ■

When Y is infinite, the step “choose one element from each of infinitely many non-empty sets at once” is an axiom of set theory, the **axiom of choice**; we use it freely and never mention it again in this chapter.

Unlike an inverse in the usual sense, one-sided inverses are rarely unique. For $q: \mathbb{R} \rightarrow [0, \infty)$, $x \mapsto x^2$, both $y \mapsto \sqrt{y}$ and $y \mapsto -\sqrt{y}$ are right inverses: each picks one of the two square roots.

Quick check. Give a function $f: \mathbb{N} \rightarrow \mathbb{N}$ that has a left inverse but no right inverse.

Solution. Take $f(n) = n + 1$. It is injective, since $n_1 + 1 = n_2 + 1$ gives $n_1 = n_2$, so by [Theorem 0.4.5](#) it has a left inverse; explicitly, $g(m) = m - 1$ for $m \geq 1$ and $g(0) = 0$ satisfies $g(f(n)) = n$. It is not surjective, since $n + 1 = 0$ has no solution in $\mathbb{N}$. So by [Theorem 0.4.6](#) it has no right inverse.

0.4.6 Inverses

When a function has an inverse on both sides, the two one-sided inverses turn out to coincide, and we get the notion of inverse we know from e^x and $\ln x$.

Definition 0.4.7 — Inverse function

Let $f: X \rightarrow Y$. A function $g: Y \rightarrow X$ is an **inverse** of f if

$$g \circ f = \text{id}_X \quad \text{and} \quad f \circ g = \text{id}_Y.$$

If f has an inverse, f is **invertible**.

Both conditions are needed: [Example \(Order matters\)](#) (b) shows a pair with one but not the other. The next theorem says which functions are invertible and that “the” inverse is justified.

Theorem 0.4.7 — Bijective if and only if invertible

A function $f: X \rightarrow Y$ is invertible if and only if it is bijective. Moreover, an inverse of f , when it exists, is **unique**; we denote it $f^{-1}: Y \rightarrow X$.

Idea. For $(\Leftarrow)$ one could combine the two one-sided theorems, but that route needs $X \neq \emptyset$ and a choice. A bijection makes both unnecessary: every y has **exactly one** preimage, so $f^{-1}(y)$ is forced. For uniqueness, the standard move is to assume two inverses g and h , and sandwich: $g = g \circ (f \circ h) = (g \circ f) \circ h = h$.

Proof. $(\Rightarrow)$ Suppose g is an inverse of f . Since $g \circ f = \text{id}_X$ is injective, f is injective by [Theorem 0.4.4](#) (a). Since $f \circ g = \text{id}_Y$ is surjective, f is surjective by [Theorem 0.4.4](#) (b). Hence f is bijective.

$(\Leftarrow)$ Suppose f is bijective, and let $y \in Y$. Since f is surjective, some $x \in X$ has $f(x) = y$, and since f is injective, this x is unique. Define $g(y)$ to be this x . Then $f(g(y)) = y$ for every $y \in Y$, so $f \circ g = \text{id}_Y$. For $x \in X$, the unique input sent to $f(x)$ is x itself, so $g(f(x)) = x$, and $g \circ f = \text{id}_X$. Hence g is an inverse of f .

(Uniqueness) Suppose g and h are both inverses of f . Then

$$g = g \circ \text{id}_Y = g \circ (f \circ h) = (g \circ f) \circ h = \text{id}_X \circ h = h,$$

where the third equality is [Theorem 0.4.2](#). This proves the theorem. ■

The uniqueness computation used only $g \circ f = \text{id}_X$ and $f \circ h = \text{id}_Y$. So it proves more: **if f has a left inverse g and a right inverse h , then $g = h$** , and this common function is the inverse. The same sandwich will prove that a matrix has at most one inverse.

Warning. The notation f^{-1} now has two meanings. The preimage $f^{-1}(B)$ of a **set** $B \subseteq Y$ exists for every f . The inverse function f^{-1} , applied to an **element** $y \in Y$, exists only for bijective f . Writing $q^{-1}(4)$ for $q(x) = x^2$ on $\mathbb{R}$ is meaningless, while $q^{-1}(\{4\}) = \{-2, 2\}$ is fine. When f is bijective the two agree: $f^{-1}(\{y\}) = \{f^{-1}(y)\}$.

Example. (Finding inverses)

For each function, decide whether it is invertible, and if so find its inverse.

- (a) $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 5 - 2x$.
- (b) $t: \mathbb{R}^2 \rightarrow \mathbb{R}^2, t(x, y) = (y, x)$.
- (c) $g: \mathbb{R} \rightarrow \mathbb{R}, g(x) = x^2$.
- (d) $h: [0, \infty) \rightarrow [0, \infty), h(x) = x^2$.

Solution.

- (a) Solving $5 - 2x = y$ gives $x = (5 - y)/2$. Put $k(y) = (5 - y)/2$. Check: $k(f(x)) = (5 - (5 - 2x))/2 = x$ and $f(k(y)) = 5 - (5 - y) = y$. So k is an inverse of f , and $f^{-1}(y) = (5 - y)/2$.
- (b) $t(t(x, y)) = t(y, x) = (x, y)$, so $t \circ t = \text{id}_{\mathbb{R}^2}$. Hence t is its own inverse: $t^{-1} = t$.
- (c) Not invertible. $g(-1) = g(1)$, so g is not injective, hence not bijective, and [Theorem 0.4.7](#) says it has no inverse.
- (d) Invertible. By the table in the previous subsection h is bijective, and the inverse is $h^{-1}(y) = \sqrt{y}$: indeed $\sqrt{x^2} = x$ for $x \geq 0$ and $(\sqrt{y})^2 = y$ for $y \geq 0$.

To undo two steps, undo the last one first. When you dress, the shirt goes on before the jacket; to undress, the jacket comes off first.

Theorem 0.4.8 — Inverse of a composition

Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be bijective. Then $g \circ f$ is bijective and

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

Proof. Using [Theorem 0.4.2](#) to regroup,

$$(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ \text{id}_Y \circ f = f^{-1} \circ f = \text{id}_X,$$

and in the same way $(g \circ f) \circ (f^{-1} \circ g^{-1}) = g \circ (f \circ f^{-1}) \circ g^{-1} = g \circ \text{id}_X \circ g^{-1} = g \circ g^{-1} = \text{id}_Z$. Hence $f^{-1} \circ g^{-1}$ is an inverse of $g \circ f$. By [Theorem 0.4.7](#), $g \circ f$ is bijective and its unique inverse is $f^{-1} \circ g^{-1}$, as claimed. ■

The same reversal will appear for matrices as $(AB)^{-1} = B^{-1}A^{-1}$, and for the same reason.

0.4.7 Maps between finite sets

For functions between two finite sets **of the same size**, one of the two conditions comes for free. This is the pigeonhole principle, and it is the ancestor of a move used constantly from Chapter 3 on: instead of checking both injectivity and surjectivity, count, and check only one.

We write $|X|$ for the number of elements of a finite set X . We use two counting facts about lists without proof: a list of n entries with a repeated entry contains at most $n - 1$ distinct elements, and a subset of a finite set Y with $|Y|$ elements is all of Y .

Theorem 0.4.9 — Injective iff surjective for finite sets of equal size

Let X and Y be finite sets with $|X| = |Y|$, and let $f: X \rightarrow Y$. Then f is injective if and only if f is surjective. In that case f is bijective.

Idea. List the elements of X as $x_1, \dots, x_n$; the image $f(X)$ is then listed by $f(x_1), \dots, f(x_n)$. Injective means this list has no repeats, so it has n distinct entries and must fill Y . Surjective means the list fills Y , which has n elements, so there is no room for a repeat.

Proof. Let $n = |X| = |Y|$, and write $X = \{x_1, \dots, x_n\}$ with the x_i distinct. By Definition 0.4.2, $f(X)$ consists of the entries of the list $f(x_1), \dots, f(x_n)$.

($\Rightarrow$) Suppose f is injective. Since the x_i are distinct, the entries $f(x_1), \dots, f(x_n)$ are distinct. Hence $f(X)$ is a subset of Y with $n = |Y|$ elements, so $f(X) = Y$, and f is surjective.

($\Leftarrow$) We prove the contrapositive (Theorem 0.1.1). Suppose f is not injective. Then $f(x_i) = f(x_j)$ for some $i \neq j$, so the list $f(x_1), \dots, f(x_n)$ has a repeated entry and $f(X)$ has at most $n - 1$ elements. Hence $f(X) \neq Y$, and f is not surjective.

So if f is injective or surjective, the two directions show that it is both, that is, bijective. ■

For a finite set X , this applies to every $f: X \rightarrow X$: such a map is injective if and only if it is surjective. In Chapter 3 the same shape of statement returns for linear maps $V \rightarrow V$ on a finite-dimensional space, with dimension playing the role of the number of elements.

Warning. The theorem needs finiteness. The map $g: \mathbb{N} \rightarrow \mathbb{N}$, $g(n) = 2n$, is injective but misses every odd number, so it is not surjective, even though domain and codomain are the same set. Infinite sets have room for an injection into a proper part of themselves.

0.4.8 Exercises

0.4.8.1 A. Check your understanding

Exercise A1.

- (a) State what it means for $f: X \rightarrow Y$ to be injective, and what it means to be surjective.
- (b) True or false: the functions $\mathbb{R} \rightarrow \mathbb{R}$, $x \mapsto x^3$ and $\mathbb{R} \rightarrow \mathbb{R}$, $x \mapsto x \cdot x \cdot x$ are equal. Give a reason.
- (c) True or false: if $g \circ f$ is injective, then g is injective. Give a reason.
- (d) Write down the first line of a proof that a function f is injective, and the first line of a proof that it is surjective.
- (e) True or false: the preimage $f^{-1}(B)$ is defined only when f is bijective.
- (f) True or false: every injective function $\{1, 2, 3\} \rightarrow \{1, 2, 3\}$ is bijective. Name the result you use.

Solution.

- (a) f is injective if for all $x_1, x_2 \in X$, $f(x_1) = f(x_2)$ implies $x_1 = x_2$. It is surjective if for every $y \in Y$ there exists $x \in X$ with $f(x) = y$.
- (b) True. They have the same domain and codomain, and $x^3 = x \cdot x \cdot x$ for every real x . Equality of functions is about values, not formulas.
- (c) False. In the remark after [Theorem 0.4.4](#), $g \circ f = \text{id}_{\{1\}}$ is injective but $g(1) = g(2)$. Only f is forced to be injective.
- (d) Injective: “Suppose $x_1, x_2 \in X$ satisfy $f(x_1) = f(x_2)$.” Surjective: “Let $y \in Y$.”
- (e) False. $f^{-1}(B) = \{x \in X : f(x) \in B\}$ makes sense for every function; see [Definition 0.4.2](#) and the warning after it.
- (f) True, by [Theorem 0.4.9](#): domain and codomain are finite with the same number of elements, so an injective map is surjective, hence bijective.

0.4.8.2 B. Practice

Exercise B1: Injective or surjective?.

Determine whether each function is injective, and whether it is surjective. Justify your answers.

- (a) $f: \mathbb{Z} \rightarrow \mathbb{Z}$, $f(n) = 3n + 1$.
- (b) $g: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $g(x, y) = (x + y, x - y)$.
- (c) $h: \mathbb{N} \rightarrow \mathbb{N}$, $h(n) = \lfloor n/2 \rfloor$, the largest integer not exceeding $n/2$.
- (d) $p: \mathbb{R} \rightarrow \mathbb{R}^2$, $p(t) = (t, t^2)$.

Solution.

- (a) Injective, not surjective. Suppose $3n_1 + 1 = 3n_2 + 1$. Then $3n_1 = 3n_2$, so $n_1 = n_2$. But 0 is not hit: $3n + 1 = 0$ would give $n = -1/3 \notin \mathbb{Z}$.
- (b) Bijective. Suppose $g(x_1, y_1) = g(x_2, y_2)$. Then $x_1 + y_1 = x_2 + y_2$ and $x_1 - y_1 = x_2 - y_2$. Adding the two equations gives $2x_1 = 2x_2$, and subtracting gives $2y_1 = 2y_2$, so $(x_1, y_1) = (x_2, y_2)$. Hence g is injective. Let $(u, v) \in \mathbb{R}^2$. Then

$$g\left(\frac{u+v}{2}, \frac{u-v}{2}\right) = \left(\frac{u+v}{2} + \frac{u-v}{2}, \frac{u+v}{2} - \frac{u-v}{2}\right) = (u, v).$$

Hence g is surjective.

- (c) Surjective, not injective. Let $m \in \mathbb{N}$. Then $h(2m) = \lfloor m \rfloor = m$, so h is surjective. But $h(0) = 0 = h(1)$, a collision, so h is not injective.
- (d) Injective, not surjective. Suppose $(t_1, t_1^2) = (t_2, t_2^2)$. Comparing first coordinates gives $t_1 = t_2$. But $(0, 1)$ is not hit: $p(t) = (0, 1)$ forces $t = 0$ and then $t^2 = 0 \neq 1$.

Exercise B2: Inverses.

- (a) Show that $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 7 - 4x$, is invertible and find f^{-1} .
- (b) Let $D = \mathbb{R} \setminus \{1\}$ and define $k: D \rightarrow D$ by $k(x) = \frac{x+1}{x-1}$. Show that k really takes values in D , and that $k \circ k = \text{id}_D$. Hence deduce that k is bijective and find k^{-1} .
- (c) With g as in B1(b), find g^{-1} . Hence find the inverse of $g \circ g$ without computing $g \circ g$ first.

Solution.

- (a) Put $e(y) = (7 - y)/4$. Then $e(f(x)) = (7 - (7 - 4x))/4 = x$ and $f(e(y)) = 7 - (7 - y) = y$. So e is an inverse of f , and by [Theorem 0.4.7](#) f is invertible with $f^{-1}(y) = (7 - y)/4$.
- (b) For $x \in D$ the denominator $x - 1$ is non-zero, so $k(x)$ is a real number. If $k(x) = 1$, then $x + 1 = x - 1$, that is, $1 = -1$, which is false. Hence $k(x) \in D$. Next, for $x \in D$,

$$k(k(x)) = \frac{\frac{x+1}{x-1} + 1}{\frac{x+1}{x-1} - 1} = \frac{(x+1) + (x-1)}{(x+1) - (x-1)} = \frac{2x}{2} = x,$$

where the second equality multiplies numerator and denominator by $x - 1 \neq 0$. So $k \circ k = \text{id}_D$. This says k is an inverse of itself, so k is invertible, hence bijective by [Theorem 0.4.7](#), and $k^{-1} = k$.

- (c) The solution of B1(b) shows that $e(u, v) = (\frac{u+v}{2}, \frac{u-v}{2})$ satisfies $g \circ e = \text{id}_{\mathbb{R}^2}$. Also $e(g(x, y)) = e(x + y, x - y) = (\frac{2x}{2}, \frac{2y}{2}) = (x, y)$, so $e \circ g = \text{id}_{\mathbb{R}^2}$. Hence $g^{-1} = e$. By [Theorem 0.4.8](#),

$$(g \circ g)^{-1} = g^{-1} \circ g^{-1}, \quad g^{-1}(g^{-1}(u, v)) = g^{-1}\left(\frac{u+v}{2}, \frac{u-v}{2}\right) = \left(\frac{u}{2}, \frac{v}{2}\right).$$

(As a check, $g(g(x, y)) = g(x + y, x - y) = (2x, 2y)$, which $(u, v) \mapsto (u/2, v/2)$ indeed undoes.)

Exercise B3: Images of unions and intersections.

Let $f: X \rightarrow Y$ and $A_1, A_2 \subseteq X$.

- (a) Prove that $f(A_1 \cup A_2) = f(A_1) \cup f(A_2)$.
- (b) Prove that $f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2)$.
- (c) Using $q(x) = x^2$ on $\mathbb{R}$, give an example where the inclusion in (b) is strict.

Solution.

- (a) We use double inclusion ([Theorem 0.3.1](#)).
- ($\subseteq$) Let $y \in f(A_1 \cup A_2)$. Then $y = f(a)$ for some $a \in A_1 \cup A_2$. If $a \in A_1$, then $y \in f(A_1)$; if $a \in A_2$, then $y \in f(A_2)$. In either case $y \in f(A_1) \cup f(A_2)$.
- ($\supseteq$) Let $y \in f(A_1) \cup f(A_2)$. If $y \in f(A_1)$, then $y = f(a)$ for some $a \in A_1 \subseteq A_1 \cup A_2$, so $y \in f(A_1 \cup A_2)$. The case $y \in f(A_2)$ is the same with the roles of A_1 and A_2 swapped. This proves the equality.
- (b) Let $y \in f(A_1 \cap A_2)$. Then $y = f(a)$ for some $a \in A_1 \cap A_2$. Since $a \in A_1$, $y \in f(A_1)$, and since $a \in A_2$, $y \in f(A_2)$. Hence $y \in f(A_1) \cap f(A_2)$.
- (c) Take $A_1 = \{-1\}$ and $A_2 = \{1\}$. Then $A_1 \cap A_2 = \emptyset$, so $q(A_1 \cap A_2) = \emptyset$, while $q(A_1) \cap q(A_2) = \{1\} \cap \{1\} = \{1\}$. The inclusion is strict. The collision $q(-1) = q(1)$ is what goes wrong; for an injective f , equality holds in (b).

0.4.8.3 C. Going deeper

Exercise C1: Left inverses are not unique.

- (a) Let $f: X \rightarrow Y$ be injective but **not** surjective, with X having at least two elements. Prove that f has at least two different left inverses.
- (b) Find two different left inverses of $f: \mathbb{N} \rightarrow \mathbb{N}$, $f(n) = n + 1$.
- (c) Explain why a bijective function has exactly one left inverse.

Hint for (a): look at the proof of [Theorem 0.4.5](#) and ask where it made a free choice.

Solution.

- (a) Since f is not surjective, there is $y_0 \in Y \setminus f(X)$. Since X has at least two elements, there are $x_0 \neq x_1$ in X . As in the proof of [Theorem 0.4.5](#), for $y \in f(X)$ let $\bar{y}$ be the unique $x \in X$ with $f(x) = y$. Define $g_0, g_1: Y \rightarrow X$ by $g_i(y) = \bar{y}$ for $y \in f(X)$, and $g_i(y) = x_i$ for $y \notin f(X)$. The proof of [Theorem 0.4.5](#) shows that each g_i is a left inverse of f , since $g_i(f(x)) = x$ uses only the values on $f(X)$. But $g_0(y_0) = x_0 \neq x_1 = g_1(y_0)$, so $g_0 \neq g_1$.
- (b) The only element missed by f is 0. Define $g_0(m) = m - 1$ for $m \geq 1$ and $g_0(0) = 0$; define $g_1(m) = m - 1$ for $m \geq 1$ and $g_1(0) = 5$. Both satisfy $g_i(n + 1) = n$ for every $n \in \mathbb{N}$, and $g_0(0) \neq g_1(0)$.
- (c) Let f be bijective, and let g be any left inverse. By [Theorem 0.4.7](#), f has an inverse f^{-1} , which in particular is a right inverse. By the observation following that theorem, a left inverse and a right inverse of the same function are equal, so $g = f^{-1}$. Hence every left inverse equals f^{-1} . Conversely, f^{-1} is itself a left inverse, since $f^{-1} \circ f = \text{id}_X$. So f has exactly one left inverse.

Exercise C2: Self-maps of finite and infinite sets.

- (a) Let X be a finite set and $f: X \rightarrow X$ injective. Prove that f is bijective.
- (b) Show that (a) fails for $X = \mathbb{N}$: give an injective $f: \mathbb{N} \rightarrow \mathbb{N}$ that is not bijective.
- (c) Give a surjective $h: \mathbb{N} \rightarrow \mathbb{N}$ that is not bijective.
- (d) Let X be finite and $f, g: X \rightarrow X$ with $g \circ f = \text{id}_X$. Prove that $f \circ g = \text{id}_X$ as well.

Hint for (d): first show that f is bijective.

Solution.

- (a) The domain and codomain are the same finite set, so $|X| = |X|$. By [Theorem 0.4.9](#), the injective map f is bijective.
- (b) $f(n) = n + 1$ is injective, since $n_1 + 1 = n_2 + 1$ gives $n_1 = n_2$, but 0 is not of the form $n + 1$ with $n \in \mathbb{N}$. Hence f is not surjective, so not bijective.
- (c) Take $h(n) = \lfloor n/2 \rfloor$. By B1(c) it is surjective but not injective.
- (d) Since $g \circ f = \text{id}_X$ is injective, f is injective by [Theorem 0.4.4](#) (a). By (a), f is bijective, so it has an inverse f^{-1} by [Theorem 0.4.7](#). Then

$$g = g \circ \text{id}_X = g \circ (f \circ f^{-1}) = (g \circ f) \circ f^{-1} = \text{id}_X \circ f^{-1} = f^{-1},$$

using [Theorem 0.4.2](#). Hence $f \circ g = f \circ f^{-1} = \text{id}_X$, as claimed. (For $X = \mathbb{N}$ this fails: with $f(n) = n + 1$ and g_0 from C1(b), $g_0 \circ f = \text{id}_{\mathbb{N}}$ but $f(g_0(0)) = 1 \neq 0$. The matrix version of (d), “ $AB = I$ implies $BA = I$ for square matrices”, is proved in Chapter 2.)

Exercise C3: When images and preimages undo each other.

Let $f: X \rightarrow Y$.

- (a) Prove that $f(f^{-1}(B)) = B$ for **every** $B \subseteq Y$ if and only if f is surjective.
- (b) Prove that $f^{-1}(f(A)) = A$ for **every** $A \subseteq X$ if and only if f is injective.

Hint: for each direction you need only one well-chosen set; try a one-element set.

Solution.

- (a) ($\Rightarrow$) Suppose the equality holds for every B , and let $y \in Y$. Taking $B = \{y\}$ gives $f(f^{-1}(\{y\})) = \{y\}$, so $y \in f(f^{-1}(\{y\}))$. Hence $y = f(x)$ for some x , and f is surjective.
- ($\Leftarrow$) Suppose f is surjective, and let $B \subseteq Y$. The inclusion $f(f^{-1}(B)) \subseteq B$ is [Proposition 0.4.1](#). For the reverse, let $y \in B$. Since f is surjective, $y = f(x)$ for some $x \in X$. Then $f(x) \in B$, so $x \in f^{-1}(B)$, and hence $y = f(x) \in f(f^{-1}(B))$. By [Theorem 0.3.1](#), $f(f^{-1}(B)) = B$.
- (b) ($\Rightarrow$) Suppose the equality holds for every A , and suppose $f(x_1) = f(x_2)$. Taking $A = \{x_1\}$ gives $f^{-1}(\{f(x_1)\}) = \{x_1\}$. Since $f(x_2) = f(x_1)$, we have $x_2 \in f^{-1}(\{f(x_1)\}) = \{x_1\}$, so $x_2 = x_1$. Hence f is injective.
- ($\Leftarrow$) Suppose f is injective, and let $A \subseteq X$. The inclusion $A \subseteq f^{-1}(f(A))$ is [Proposition 0.4.1](#). For the reverse, let $x \in f^{-1}(f(A))$. Then $f(x) \in f(A)$, so $f(x) = f(a)$ for some $a \in A$. Since f is injective, $x = a \in A$. By [Theorem 0.3.1](#), $f^{-1}(f(A)) = A$.

0.5 Relations, Equivalence Classes and Quotients

Mathematics constantly declares different-looking things to be “the same”: $\frac{1}{2}$ and $\frac{2}{4}$ are the same number, 14:00 and 2 p.m. are the same time, and in Chapter 3 two vectors will count as the same when their difference lies in a fixed subspace. This section makes “the same, for present purposes” precise. The notion is an equivalence relation. Gluing equivalent things together produces a new set, the quotient, and the one real danger of working with quotients is defining a function on them that secretly depends on a choice. We will learn to check for that.

0.5.1 Relations

Before we can say what a good notion of sameness is, we need a word for any way of comparing two elements of a set, good or bad. Examples are $\leq$ on numbers, “divides” on integers, and “has the same birthday as” on people. What such a comparison really amounts to is the list of pairs that are related, and a list of pairs is a subset of the Cartesian product (Definition 0.3.10).

Definition 0.5.1 — Relation

A **relation** on a set X is a subset $R \subseteq X \times X$. When $(x, y) \in R$ we say x **is related to** y and write $x R y$, or $x \sim y$ when the relation is fixed.

Example. (Relations)

- (a) On $\mathbb{Z}$: $a \leq b$. The pairs $(1, 2)$ and $(2, 2)$ belong to the relation, $(2, 1)$ does not.
- (b) On $\mathbb{Z}$: $a \mid b$ (“ a divides b ”), meaning $b = ka$ for some $k \in \mathbb{Z}$. So $3 \mid 12$ and $3 \mid 0$, but $3 \nmid 5$.
- (c) On $\mathbb{R}$: $x \sim y$ if $x^2 + y^2 = 1$. As a subset of $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$ this relation is the unit circle, so a relation can literally be a picture.
- (d) On any set X : the **empty relation** $R = \emptyset$, in which nothing is related to anything, and the **full relation** $R = X \times X$, in which everything is related to everything. These two extremes are useful test cases for every property below.

Every subset of $X \times X$ is a relation, so the definition has no non-examples. The interesting question is which properties a given relation has.

0.5.2 Equivalence relations

What must any reasonable notion of “ x is the same as y ” satisfy? Think of “has the same birthday as”. Everyone has the same birthday as themselves. If you share a birthday with me, I share one with you. If x shares a birthday with y , and y with z , then x shares one with z . Those three features are exactly what we ask for.

An equivalence relation is a way of declaring elements “the same” that behaves like equality:

everything is the same as itself, sameness goes both ways, and it can be chained.

Definition 0.5.2 — Equivalence relation

A relation $\sim$ on a set X is an **equivalence relation** if it satisfies the following three conditions.

- (E1) **Reflexive:** $x \sim x$ for every $x \in X$.
- (E2) **Symmetric:** for all $x, y \in X$, if $x \sim y$ then $y \sim x$.
- (E3) **Transitive:** for all $x, y, z \in X$, if $x \sim y$ and $y \sim z$ then $x \sim z$.

In words: (E1) asks every single element to be related to itself, with no exceptions. (E2) says the relation cannot be one-way. (E3) says a chain of two links can be shortened to one link, and so, by applying it repeatedly, can any finite chain. When $x \sim y$ we say x and y are **equivalent**.

The standard move for checking an equivalence relation is to write three short paragraphs, one per condition, each starting from its hypothesis (“Let $x \in X$.”, “Suppose $x \sim y$.”, “Suppose $x \sim y$ and $y \sim z$.”).

Example. (Equivalence relations)

Show that each of the following is an equivalence relation.

- (a) Equality on any set X : $x \sim y$ if $x = y$.
- (b) **Congruence modulo n** on $\mathbb{Z}$, for a fixed integer $n \geq 1$: $a \equiv b \pmod{n}$ if $n \mid a - b$.
- (c) “Same length” on the set S of finite strings of the letters a, b : $u \sim v$ if u and v have the same number of letters.
- (d) “Same value”: for a function $f: X \rightarrow Y$, put $x \sim_f x'$ if $f(x) = f(x')$.
- (e) Fractions: on $P = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$, put $(a, b) \sim (c, d)$ if $ad = bc$.

Solution.

- (a) Every x equals itself; $x = y$ gives $y = x$; $x = y$ and $y = z$ give $x = z$. Equality is the degenerate case: it is the equivalence relation that identifies nothing new, and the model the other examples imitate.
- (b) *Reflexive.* Let $a \in \mathbb{Z}$. Then $a - a = 0 = 0 \cdot n$, so $n \mid a - a$. *Symmetric.* Suppose $a - b = kn$ with $k \in \mathbb{Z}$. Then $b - a = (-k)n$, so $n \mid b - a$. *Transitive.* Suppose $a - b = kn$ and $b - c = \ell n$ with $k, \ell \in \mathbb{Z}$. Then $a - c = (a - b) + (b - c) = (k + \ell)n$, so $n \mid a - c$. For $n = 2$ this is “same parity”: $a \equiv b \pmod{2}$ says a and b are both even or both odd.
- (c) This is the special case of (d) in which $f: S \rightarrow \mathbb{N}$ sends a string to its length, so it follows from (d), whose proof does not use (c).
- (d) *Reflexive:* $f(x) = f(x)$. *Symmetric:* $f(x) = f(x')$ gives $f(x') = f(x)$. *Transitive:* $f(x) = f(x')$ and $f(x') = f(x'')$ give $f(x) = f(x'')$. Each condition is inherited from the same property of equality in Y . Many equivalence relations are of this form: “same distance from the origin” on $\mathbb{R}^2$ is $\sim_f$ for $f(x, y) = x^2 + y^2$.
- (e) *Reflexive:* $ab = ba$, so $(a, b) \sim (a, b)$. *Symmetric:* $ad = bc$ gives $cb = da$, which says $(c, d) \sim (a, b)$. *Transitive:* suppose $ad = bc$ and $cf = de$. Multiplying the first

equation by f and the second by b gives $adf = bcf = bde$. Hence $d(af - be) = 0$. Since $d \neq 0$, we get $af = be$, that is, $(a, b) \sim (e, f)$. Transitivity is exactly where second entries must be non-zero: with $d = 0$ allowed, $(1, 1) \sim (0, 0) \sim (2, 1)$ but $(1, 1) \not\sim (2, 1)$.

Minimal changes to these examples show that each condition can fail on its own.

Example. (Relations that are not equivalence relations)

Decide which of (E1)–(E3) each relation satisfies.

- (a) $\leq$ on $\mathbb{Z}$.
- (b) On $\mathbb{R}$: $x \sim y$ if $|x - y| \leq 1$.
- (c) On $\mathbb{R}$: $x \sim y$ if $xy > 0$.

Solution.

- (a) Change “=” to “ $\leq$ ”. Reflexivity survives ($a \leq a$) and so does transitivity ($a \leq b \leq c$ gives $a \leq c$). **Symmetry fails:** $1 \leq 2$ but $2 \not\leq 1$.
- (b) Change “ $|x - y| = 0$ ” (which is equality) to “ $|x - y| \leq 1$ ”. Reflexivity survives, since $|x - x| = 0$, and so does symmetry, since $|y - x| = |x - y|$. **Transitivity fails:** $0 \sim 1$ and $1 \sim 2$, but $|0 - 2| = 2 > 1$. “Close to” cannot be chained; many small steps make a big one.
- (c) Symmetry holds, since $yx = xy$. Transitivity holds: if $xy > 0$ and $yz > 0$, then $xy^2z > 0$, and since $y^2 > 0$ this gives $xz > 0$. **Reflexivity fails, at exactly one point:** $0 \cdot 0 = 0$, so $0 \not\sim 0$. On $\mathbb{R} \setminus \{0\}$ the same rule is an equivalence relation (“same sign”).

Warning. A tempting argument claims that (E1) follows from (E2) and (E3): “if $x \sim y$, then $y \sim x$ by symmetry, so $x \sim x$ by transitivity.” The gap is the word “if”. The argument needs some y with $x \sim y$, and there may be none. In [Example \(Relations that are not equivalence relations\)](#) (c), the element 0 is related to nothing, which is exactly why it escapes. Reflexivity must always be checked on its own.

Quick check. On $\mathbb{Z}$, define $a \sim b$ if $ab \geq 0$. Is $\sim$ an equivalence relation?

Solution. No. It is reflexive ($a^2 \geq 0$) and symmetric, but not transitive: $1 \sim 0$ and $0 \sim -1$ since both products are 0, yet $1 \cdot (-1) = -1 < 0$, so $1 \not\sim -1$. Compare [Example \(Relations that are not equivalence relations\)](#) (c): moving from $>$ to $\geq$ repairs reflexivity at 0 but lets 0 link the positives to the negatives.

Why these three conditions. They are precisely what the next subsection needs. Reflexivity will guarantee that every element lies in its own class, so no class is empty and nothing is lost. Symmetry and transitivity together will guarantee that two classes are either identical or disjoint. Drop any one of them and the picture of X cut cleanly into pieces falls apart, as the non-examples above show.

Remark. Two equivalence relations on matrices are central later: *row equivalence*, which records that one matrix can be reached from another by row operations, and *similarity*, which records that two square matrices describe the same linear map in different

coordinates. Both are introduced once matrices are available, later in this chapter, and studied in Chapters 2 and 3.

0.5.3 Equivalence classes and partitions

Once we have decided which elements count as the same, it is natural to collect everything that is the same as a given element into one set.

Definition 0.5.3 — Equivalence class

Let $\sim$ be an equivalence relation on X , and let $x \in X$. The **equivalence class** of x is

$$[x] := \{y \in X : y \sim x\}.$$

Any element of $[x]$ is called a **representative** of the class.

The word “representative” is chosen carefully. A class is one set, but it usually has many names: $[x] = [y]$ is possible with $x \neq y$.

Example. (Equivalence classes)

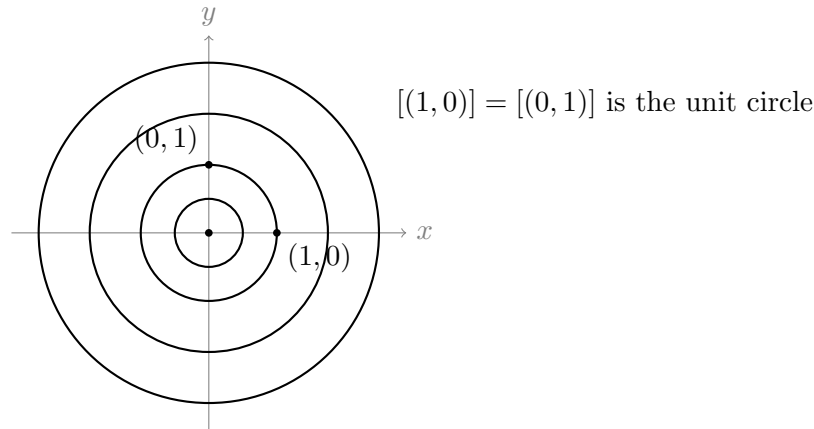
- (a) For equality on X , $[x] = \{x\}$. Every class has one element.
- (b) For congruence modulo 3 on $\mathbb{Z}$:

$$[0] = \{\dots, -6, -3, 0, 3, 6, \dots\}, \quad [1] = \{\dots, -5, -2, 1, 4, 7, \dots\}, \quad [2] = \{\dots, -4, -1, 2, 5, 8, \dots\}.$$

Here $[0] = [3] = [-3]$ and $[2] = [-1]$: one class, many names.

- (c) For “same length” on strings of a, b , the class of ab is $\{aa, ab, ba, bb\}$, and the class of the empty string contains only the empty string.
- (d) For the full relation $X \times X$ on a non-empty set X , there is a single class, $[x] = X$.
- (e) For “same distance from the origin” on $\mathbb{R}^2$, the class of (x, y) is the circle of radius $\sqrt{x^2 + y^2}$ centered at the origin. The class of the origin is the one-point set $\{(0, 0)\}$.

In example (e), the classes look like this. Each circle is one class; the origin on its own is another.



The picture suggests two facts: every point lies on exactly one circle, and two different circles never meet. Both hold for every equivalence relation, and the proof is where the three conditions do their work.

Lemma 0.5.1 — Classes are equal or disjoint

Let $\sim$ be an equivalence relation on X , and let $a, b \in X$.

- (a) $[a] = [b]$ if and only if $a \sim b$.
- (b) Either $[a] = [b]$ or $[a] \cap [b] = \emptyset$.

Proof.

- (a) $(\Rightarrow)$ Suppose $[a] = [b]$. By reflexivity $a \sim a$, so $a \in [a] = [b]$, which means $a \sim b$.
 $(\Leftarrow)$ Suppose $a \sim b$. Let $x \in [a]$, so $x \sim a$. Since $a \sim b$, transitivity gives $x \sim b$, so $x \in [b]$. Hence $[a] \subseteq [b]$. By symmetry $b \sim a$, and swapping the roles of a and b in the argument gives $[b] \subseteq [a]$. By [Theorem 0.3.1](#), $[a] = [b]$.
- (b) Suppose $[a] \cap [b] \neq \emptyset$, and let $c \in [a] \cap [b]$. Then $c \sim a$ and $c \sim b$. By symmetry $a \sim c$, and by transitivity $a \sim b$. By (a), $[a] = [b]$.

■

Part (a) is the working form of the lemma: to show two classes are equal, show one representative is equivalent to the other. The picture of non-overlapping pieces that cover everything has its own name.

Definition 0.5.4 — Partition

A **partition** of a set X is a collection $\mathcal{P}$ of subsets of X , called **blocks**, such that:

- (P1) every block is **non-empty**;
- (P2) **distinct** blocks are disjoint: if $B, B' \in \mathcal{P}$ and $B \neq B'$, then $B \cap B' = \emptyset$;
- (P3) the blocks cover X : every $x \in X$ lies in some block.

By (P2) and (P3), every element of X lies in **exactly one** block. For example, $\{\{1, 3\}, \{2\}\}$ is a partition of $\{1, 2, 3\}$, while $\{\{1, 2\}, \{2, 3\}\}$ is not, because the distinct blocks share 2 and (P2)

fails.

Quick check. Is $\{\{1, 2, 3\}, \emptyset\}$ a partition of $\{1, 2, 3\}$? Is $\{\{1\}, \{2\}\}$?

Solution. Neither. The first contains the empty block, so (P1) fails, even though (P2) and (P3) hold. The second does not cover 3, so (P3) fails.

The main theorem says that equivalence relations and partitions are two descriptions of the same thing: “which elements are the same” and “how X is cut into pieces”.

Theorem 0.5.2 — Equivalence relations and partitions

Let X be a set.

- (a) If $\sim$ is an equivalence relation on X , then the set of its equivalence classes is a partition of X .
- (b) If $\mathcal{P}$ is a partition of X , then the relation “ $x \sim_{\mathcal{P}} y$ if x and y lie in the same block” is an equivalence relation on X , and its equivalence classes are exactly the blocks of $\mathcal{P}$.
- (c) The two constructions undo each other: starting from $\sim$, the relation defined in (b) by the partition in (a) is $\sim$ again.

Idea. Part (a) is the lemma plus reflexivity: (P1) and (P3) come from $x \in [x]$, and (P2) is Lemma 0.5.1 (b). Part (b) runs the other way, and each block condition pays for one equivalence condition: covering gives reflexivity, and disjointness gives transitivity, because a middle element y can lie in only one block. Part (c) is Lemma 0.5.1 (a) read as a sentence about blocks.

Proof.

- (a) Let $\mathcal{P} = \{[x] : x \in X\}$. For every $x \in X$, reflexivity gives $x \in [x]$. Hence every class is non-empty, which is (P1), and every element lies in some class, which is (P3). If $[a] \neq [b]$, then $[a] \cap [b] = \emptyset$ by Lemma 0.5.1 (b), which is (P2). Hence $\mathcal{P}$ is a partition of X .
- (b) *Reflexive.* Let $x \in X$. By (P3), x lies in some block, so $x \sim_{\mathcal{P}} x$. *Symmetric.* “ x and y lie in a common block” does not depend on the order of x and y . *Transitive.* Suppose $x, y \in B$ and $y, z \in B'$ for blocks B, B' . Then $y \in B \cap B'$, so $B = B'$ by (P2). Hence $x, z \in B$, and $x \sim_{\mathcal{P}} z$.
Now let $B \in \mathcal{P}$ and $x \in B$; we show $[x] = B$, where the class is taken for $\sim_{\mathcal{P}}$. ($\subseteq$) If $y \in [x]$, then y and x lie in a common block B' ; since $x \in B \cap B'$, (P2) gives $B' = B$, so $y \in B$. ($\supseteq$) If $y \in B$, then y and x both lie in B , so $y \sim_{\mathcal{P}} x$ and $y \in [x]$. Hence $[x] = B$ by Theorem 0.3.1. Every class $[x]$ is therefore the block containing x , which exists by (P3); and every block B is a class, namely $[x]$ for any $x \in B$, which exists by (P1). So the classes are exactly the blocks.
- (c) Let $\sim$ be an equivalence relation, and let $\approx$ be the relation built by (b) from the classes of $\sim$. If $x \sim y$, then $x, y \in [y]$, so $x \approx y$. Conversely, if $x \approx y$, then $x, y \in [z]$ for some z ; thus $x \sim z$ and $y \sim z$, and symmetry and transitivity give $x \sim y$. Hence $\approx$ and $\sim$ are the same relation. This proves the theorem. ■

So whenever you meet an equivalence relation, you may think of it as a way of cutting the set into non-overlapping pieces, and vice versa. Exercise C2 uses this to count equivalence relations by counting partitions instead.

0.5.4 Quotient sets

Now we take the decisive step: treat each piece as a single new element. The time of day is the classic case. Hours counted from some starting moment are integers, but a clock only records the class of the hour modulo 12.

Definition 0.5.5 — Quotient set

Let $\sim$ be an equivalence relation on X . The **quotient set** of X by $\sim$ is the set of all equivalence classes,

$$X/\sim := \{[x] : x \in X\}.$$

The function $\pi: X \rightarrow X/\sim$, $\pi(x) = [x]$, is the **quotient map**.

The elements of $X/\sim$ are sets, but once the quotient is formed we stop looking inside them and treat each class as a single point. The quotient map is surjective, since every class is $[x] = \pi(x)$ for some x . By Lemma 0.5.1 (a), $\pi(x) = \pi(y)$ exactly when $x \sim y$: the quotient map makes equivalent elements literally equal.

Example. (Quotient sets)

- (a) For an integer $n \geq 1$, the quotient of $\mathbb{Z}$ by congruence modulo n is written $\mathbb{Z}/n\mathbb{Z}$. Show that it has exactly n elements, namely $[0], [1], \dots, [n-1]$.
- (b) Describe the quotient of the set S of strings in a, b by “same length”, and of $\mathbb{R}^2$ by “same distance from the origin”.

Solution.

- (a) Let $a \in \mathbb{Z}$. By division with remainder (Exercise C2: Division with remainder), $a = qn + r$ with $q, r \in \mathbb{Z}$ and $0 \leq r < n$. Then $a - r = qn$, so $a \equiv r \pmod{n}$, and $[a] = [r]$ by Lemma 0.5.1 (a). Hence every class is one of $[0], \dots, [n-1]$. These are distinct: if $0 \leq r < s < n$, then $0 < s - r < n$, so $s - r$ is not a multiple of n , hence $r \not\equiv s$ and $[r] \neq [s]$ by Lemma 0.5.1 (a). So $\mathbb{Z}/n\mathbb{Z} = \{[0], [1], \dots, [n-1]\}$ has exactly n elements.
- (b) Two strings are identified exactly when they have the same length, so there is one class for each length $k \in \mathbb{N}$: the set of all 2^k strings of length k . Similarly, $\mathbb{R}^2$ modulo “same distance” has one class for each radius $r \in [0, \infty)$: the circle of radius r , or the origin when $r = 0$. In both cases the quotient is “the same as” the set of values $\mathbb{N}$ or $[0, \infty)$, and the next subsection makes that precise.

0.5.5 Functions on a quotient

We want to compute with classes. On a clock, “three hours after 11 o’clock” is 2 o’clock, which is really addition on $\mathbb{Z}/12\mathbb{Z}$. The natural way to define such an operation is through representatives: to add two classes, pick an element of each, add them, and take the class of the result. The danger is that a class has many representatives, and different choices might give different answers. When they do, the “definition” does not define a function at all.

Here is that failure in its simplest form.

Example. (The numerator is not well defined)

Let $P = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ with $(a, b) \sim (c, d)$ if $ad = bc$, as in [Example \(Equivalence relations\)](#) (e). Its classes are the rational numbers: the class of (a, b) is the fraction a/b , with all its other names. Is “ $[(a, b)] \mapsto a$ ” a function $P/\sim \rightarrow \mathbb{Z}$?

Solution. No. Since $1 \cdot 4 = 2 \cdot 2$, we have $(1, 2) \sim (2, 4)$, so $[(1, 2)] = [(2, 4)]$ by [Lemma 0.5.1](#) (a). The rule assigns to this one class the value 1 through one name and 2 through the other. So it does not assign **exactly one** value to each element of the domain, and [Definition 0.4.1](#) fails. “The numerator of $\frac{1}{2}$ ” is a statement about how a fraction is written, not about the number.

By contrast, “ $[(a, b)] \mapsto 1$ if $a = 0$, and 0 otherwise” is fine. If $(a, b) \sim (c, d)$ and $a = 0$, then $bc = ad = 0$, so $c = 0$ because $b \neq 0$. By symmetry, $c = 0$ forces $a = 0$. So equivalent pairs always get the same value.

The contrast in that example is the whole story, and the following theorem turns it into a test. A function f on X passes to the quotient exactly when it cannot tell equivalent elements apart.

Theorem 0.5.3 — Functions on a quotient

Let $\sim$ be an equivalence relation on X , let $\pi: X \rightarrow X/\sim$ be the quotient map, and let $f: X \rightarrow Y$ be a function.

- (a) If $f(a) = f(b)$ **whenever** $a \sim b$, then there is **exactly one** function $\bar{f}: X/\sim \rightarrow Y$ with $\bar{f}([x]) = f(x)$ for every $x \in X$, that is, $\bar{f} \circ \pi = f$.
- (b) Conversely, if some function $\bar{f}: X/\sim \rightarrow Y$ satisfies $\bar{f} \circ \pi = f$, then $f(a) = f(b)$ whenever $a \sim b$.

Idea. The formula $\bar{f}([x]) = f(x)$ leaves no freedom: if $\bar{f}$ exists, this is its value. The only question is whether the formula makes sense, since the class $C = [x]$ may also be called $[x']$. So look at **all** the values f takes on C at once, the image $f(C)$. A function needs exactly one value; $f(C)$ is non-empty because C is, and has at most one element because all elements of C are equivalent.

Proof.

- (a) Let $C \in X/\sim$. By [Definition 0.5.5](#), $C = [x]$ for some $x \in X$, so C is non-empty, and hence the image $f(C)$ is non-empty. If $x', x'' \in C$, then $[x'] = C = [x'']$ by [Lemma 0.5.1](#) (a) applied to $x' \sim x$ and $x'' \sim x$; hence $x' \sim x''$ by the same lemma,

and $f(x') = f(x'')$ by hypothesis. Therefore $f(C)$ has exactly one element, and we define $\bar{f}(C)$ to be that element. This assigns exactly one value to each class, so $\bar{f}$ is a function. For $x \in X$, we have $x \in [x]$ by reflexivity, so $\bar{f}([x]) \in f([x]) = \{f(x)\}$, that is, $\bar{f}([x]) = f(x)$.

For uniqueness, suppose $g: X/\sim \rightarrow Y$ also satisfies $g([x]) = f(x)$ for every x . Every class is $[x]$ for some x , and there $g([x]) = f(x) = \bar{f}([x])$. Hence $g = \bar{f}$.

- (b) Suppose $\bar{f} \circ \pi = f$ and $a \sim b$. By [Lemma 0.5.1](#) (a), $[a] = [b]$. Hence $f(a) = \bar{f}([a]) = \bar{f}([b]) = f(b)$, as claimed. ■

In practice we rarely name f and $\bar{f}$ separately. We write down a rule “ $[x] \mapsto (\text{expression in } x)$ ” and check that equivalent representatives give the same value. That check is what people mean when they say a function is **well defined**. For example, “same value” in [Example \(Equivalence relations\)](#) (d) satisfies the hypothesis by its very definition, so every $f: X \rightarrow Y$ induces a function $X/\sim_f \rightarrow Y$. This $\bar{f}$ is injective: $\bar{f}([x]) = \bar{f}([x'])$ means $f(x) = f(x')$, which is $x \sim_f x'$, so $[x] = [x']$. This makes precise the remark that $\mathbb{R}^2$ modulo “same distance” is “the same as” $[0, \infty)$: the induced map $[(x, y)] \mapsto \sqrt{x^2 + y^2}$ is a bijection onto $[0, \infty)$.

The most important application in this chapter is arithmetic modulo n . The operation takes two classes, so we check both representatives at once.

Example. (Addition modulo n is well defined)

Fix an integer $n \geq 1$. Show that the rule $[a] + [b] := [a + b]$ defines a function $\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$. Then compute $[3] + [2]$ in $\mathbb{Z}/4\mathbb{Z}$.

Solution. We must show that the value $[a + b]$ depends only on the classes $[a]$ and $[b]$, not on the representatives a and b . Suppose $[a] = [a']$ and $[b] = [b']$. By [Lemma 0.5.1](#) (a), $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$, so $a - a' = kn$ and $b - b' = \ell n$ for some $k, \ell \in \mathbb{Z}$. Then

$$(a + b) - (a' + b') = (a - a') + (b - b') = (k + \ell)n,$$

so $a + b \equiv a' + b' \pmod{n}$, and $[a + b] = [a' + b']$ by the same lemma. Hence every choice of representatives gives the same class, and the rule defines a function.

In $\mathbb{Z}/4\mathbb{Z}$, $[3] + [2] = [5] = [1]$, since $5 - 1 = 4$. Choosing other names gives the same answer, as it must: $[7] + [-2] = [5] = [1]$.

This is the same principle as [Theorem 0.5.3](#), applied to a function of two variables. The same check ([Exercise \(B3: Multiplication modulo \$n\$ \)](#)) shows multiplication $[a][b] := [ab]$ is well defined. With these operations, $\mathbb{Z}/p\mathbb{Z}$ for a prime p will become our first finite field later in this chapter, and the identical pattern, now with “ $\mathbf{v} \sim \mathbf{w}$ if $\mathbf{v} - \mathbf{w} \in U$ ”, will build quotient vector spaces in Chapter 3.

Warning. A rule written in terms of a representative is not automatically a function on classes. Define it, then check it. For example, “ $[a] \mapsto a$ ” from $\mathbb{Z}/5\mathbb{Z}$ to $\mathbb{Z}$ is not a function, because $[0] = [5]$ would be sent to both 0 and 5. The check is not a formality: forgetting it is the most common error in any argument that uses quotients.

Quick check. Which of these rules defines a function? (i) $\mathbb{Z}/6\mathbb{Z} \rightarrow \mathbb{Z}/3\mathbb{Z}$, $[a]_6 \mapsto [a]_3$. (ii) $\mathbb{Z}/6\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$, $[a]_6 \mapsto [a]_4$. Here the subscript records the modulus.

Solution.

- (i) Yes. If $[a]_6 = [a']_6$, then $a - a' = 6k = 3(2k)$, so $[a]_3 = [a']_3$.
- (ii) No. $[0]_6 = [6]_6$, but $[0]_4 \neq [6]_4 = [2]_4$, since $6 - 0 = 6$ is not a multiple of 4. The first works because $3 \mid 6$; the second fails because $4 \nmid 6$.

0.5.6 Exercises

0.5.6.1 A. Check your understanding

Exercise A1.

- (a) State the three conditions for a relation $\sim$ on X to be an equivalence relation.
- (b) True or false: a symmetric and transitive relation is automatically reflexive. Give a reason.
- (c) True or false: if two equivalence classes $[a]$ and $[b]$ have an element in common, then $a \sim b$.
- (d) You want to define a function on $X/\sim$ by a formula in a representative x . What must you check?
- (e) How many elements does $\mathbb{Z}/5\mathbb{Z}$ have? Is $[12] = [-3]$ in $\mathbb{Z}/5\mathbb{Z}$?

Solution.

- (a) Reflexive: $x \sim x$ for every $x \in X$. Symmetric: $x \sim y$ implies $y \sim x$. Transitive: $x \sim y$ and $y \sim z$ imply $x \sim z$.
- (b) False. On $\mathbb{R}$, $x \sim y$ if $xy > 0$ is symmetric and transitive, but $0 \not\sim 0$; see [Example \(Relations that are not equivalence relations\)](#) (c) and the warning after it.
- (c) True. By [Lemma 0.5.1](#) (b) the classes are equal, and by part (a) of the same lemma $a \sim b$.
- (d) That equivalent representatives give the same value: if $x \sim x'$, the formula gives the same result for x and x' . By [Theorem 0.5.3](#) this is exactly what is needed.
- (e) Five, by [Example \(Quotient sets\)](#) (a). Yes: $12 - (-3) = 15 = 3 \cdot 5$, so $12 \equiv -3 \pmod{5}$, and both classes equal $[2]$.

0.5.6.2 B. Practice

Exercise B1: Which are equivalence relations?.

Determine which of the following are equivalence relations. For those that are, describe the equivalence classes. For those that are not, name a condition that fails with a specific witness.

- (a) On $\mathbb{Z}$: $a \sim b$ if $a + b$ is even.

- (b) On $\mathbb{Z}$: $a \sim b$ if $a - b$ is odd.
- (c) On $\mathbb{R}$: $x \sim y$ if $x - y \in \mathbb{Z}$.
- (d) On $\{1, 2, 3, \dots\}$: $m \sim n$ if m and n have a common divisor greater than 1.
- (e) On the set of all subsets of $\{1, 2, 3\}$: $A \sim B$ if A and B have the same number of elements.

Solution.

- (a) Equivalence relation. *Reflexive*: $a + a = 2a$ is even. *Symmetric*: $b + a = a + b$. *Transitive*: if $a + b$ and $b + c$ are even, then $a + c = (a + b) + (b + c) - 2b$ is a sum of even integers, hence even. There are two classes: the even integers and the odd integers. (In fact $a + b$ is even exactly when $a - b = (a + b) - 2b$ is even, so this is congruence modulo 2.)
- (b) Not an equivalence relation. Reflexivity fails: $0 - 0 = 0$ is even, so $0 \not\sim 0$. (Transitivity also fails: $0 \sim 1$ and $1 \sim 2$, but $0 - 2$ is even.)
- (c) Equivalence relation. *Reflexive*: $x - x = 0 \in \mathbb{Z}$. *Symmetric*: if $x - y = k \in \mathbb{Z}$, then $y - x = -k \in \mathbb{Z}$. *Transitive*: if $x - y = k$ and $y - z = \ell$ with $k, \ell \in \mathbb{Z}$, then $x - z = k + \ell \in \mathbb{Z}$. The class of x is $\{x + k : k \in \mathbb{Z}\}$, the points at integer distance from x in either direction. Each class contains exactly one number in $[0, 1)$, namely $x - \lfloor x \rfloor$.
- (d) Not an equivalence relation. Reflexivity fails at 1: the only positive divisor of 1 is 1. Transitivity also fails: $2 \sim 6$ (common divisor 2) and $6 \sim 3$ (common divisor 3), but 2 and 3 have no common divisor greater than 1.
- (e) Equivalence relation, since it is “same value” (Example (Equivalence relations) (d)) for the function $A \mapsto |A|$. There are four classes: $\{\emptyset\}$; $\{\{1\}, \{2\}, \{3\}\}$; $\{\{1, 2\}, \{1, 3\}, \{2, 3\}\}$; $\{\{1, 2, 3\}\}$.

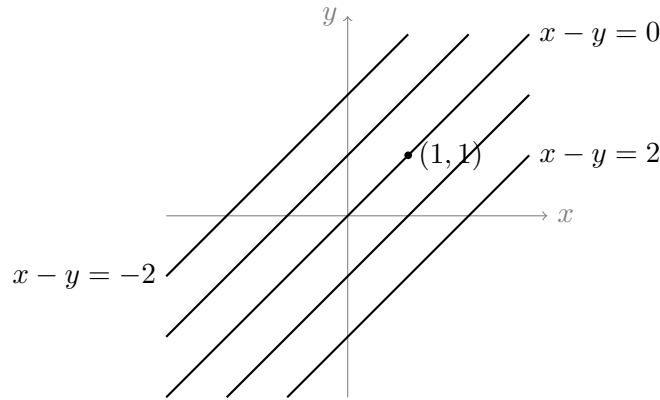
Exercise B2: Parallel lines as classes.

On $\mathbb{R}^2$, define $(x, y) \sim (x', y')$ if $x - y = x' - y'$.

- (a) Prove that $\sim$ is an equivalence relation.
- (b) Describe the class of (x_0, y_0) geometrically, and draw several classes.
- (c) Show that $[(x, y)] \mapsto x - y$ is a well-defined function $\mathbb{R}^2/\sim \rightarrow \mathbb{R}$, and that it is a bijection.

Solution.

- (a) This is “same value” for $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = x - y$, so it is an equivalence relation by Example (Equivalence relations) (d). Directly: $x - y = x - y$; equality of the numbers $x - y$ and $x' - y'$ is symmetric and transitive.
- (b) Put $c = x_0 - y_0$. Then $[(x_0, y_0)] = \{(x, y) \in \mathbb{R}^2 : x - y = c\} = \{(x, y) : y = x - c\}$, the line of slope 1 through (x_0, y_0) , which meets the y -axis at $(0, -c)$. The classes are all the lines parallel to $y = x$: each point of the plane lies on exactly one of them, as Theorem 0.5.2 predicts.



- (c) Let $f(x, y) = x - y$. If $(x, y) \sim (x', y')$, then $f(x, y) = f(x', y')$ by the definition of $\sim$. By [Theorem 0.5.3](#) (a), there is a function $\bar{f}: \mathbb{R}^2/\sim \rightarrow \mathbb{R}$ with $\bar{f}([(x, y)]) = x - y$.

Injective. Suppose $\bar{f}([(x, y)]) = \bar{f}([(x', y')])$. Then $x - y = x' - y'$, so $(x, y) \sim (x', y')$, and $[(x, y)] = [(x', y')]$ by [Lemma 0.5.1](#) (a).

Surjective. Let $c \in \mathbb{R}$. Then $\bar{f}([(c, 0)]) = c - 0 = c$.

Hence $\bar{f}$ is a bijection: the set of parallel lines is labeled, one to one, by the real number c .

Exercise B3: Multiplication modulo n .

Fix an integer $n \geq 1$.

- Prove that $[a][b] := [ab]$ is a well-defined operation on $\mathbb{Z}/n\mathbb{Z}$.
- Write out the multiplication table of $\mathbb{Z}/4\mathbb{Z}$, using the representatives 0, 1, 2, 3.
- Hence find a class $[a] \neq [0]$ in $\mathbb{Z}/4\mathbb{Z}$ with $[a][a] = [0]$.

Solution.

- (a) Suppose $[a] = [a']$ and $[b] = [b']$. By [Lemma 0.5.1](#) (a), $a - a' = kn$ and $b - b' = \ell n$ for some $k, \ell \in \mathbb{Z}$. Then

$$ab - a'b' = ab - a'b + a'b - a'b' = (a - a')b + a'(b - b') = (kb + a'\ell)n,$$

so $ab \equiv a'b' \pmod{n}$ and $[ab] = [a'b']$ by the same lemma. Hence the value does not depend on the representatives, and the operation is well defined.

- (b) Reducing each product modulo 4:

$\cdot$	[0]	[1]	[2]	[3]
[0]	[0]	[0]	[0]	[0]
[1]	[0]	[1]	[2]	[3]
[2]	[0]	[2]	[0]	[2]
[3]	[0]	[3]	[2]	[1]

For instance $[2][3] = [6] = [2]$ and $[3][3] = [9] = [1]$.

- (c) From the table, $[2][2] = [4] = [0]$, while $[2] \neq [0]$ since $4 \nmid 2$. So a product of non-zero classes can be zero in $\mathbb{Z}/4\mathbb{Z}$, something that never happens with ordinary numbers.

0.5.6.3 C. Going deeper

Exercise C1: Squares and powers modulo n .

- (a) Let $n \geq 1$. Prove that $[x] \mapsto [x^2]$ is a well-defined function $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$.
- (b) On $\mathbb{N}$, let $\equiv_3$ be congruence modulo 3. Show that the rule $[x] \mapsto [2^x]_3$ does **not** define a function from $\mathbb{N}/\equiv_3$ to $\mathbb{Z}/3\mathbb{Z}$.
- (c) On $\mathbb{N}$, let $\equiv_2$ be congruence modulo 2. Prove that $[x] \mapsto [2^x]_3$ **does** define a function from $\mathbb{N}/\equiv_2$ to $\mathbb{Z}/3\mathbb{Z}$.

Hint for (c): compute 2^2 modulo 3.

Solution.

- (a) Suppose $[x] = [x']$. By [Exercise \(B3: Multiplication modulo \$n\$ \)](#) (a) with $a = b = x$ and $a' = b' = x'$, we get $[x][x] = [x'][x']$, that is, $[x^2] = [x'^2]$. Hence the value depends only on the class, and the function is well defined.
- (b) In $\mathbb{N}$, $0 \equiv 3 \pmod{3}$, so $[0] = [3]$. The rule sends this class to $[2^0]_3 = [1]_3$ through the representative 0, and to $[2^3]_3 = [8]_3 = [2]_3$ through the representative 3. Since $8 - 1 = 7$ is not a multiple of 3, $[1]_3 \neq [2]_3$. So one class receives two different values, and the rule is not a function.
- (c) By [Theorem 0.5.3](#) (a), it suffices to show that $x \equiv x' \pmod{2}$ implies $2^x \equiv 2^{x'} \pmod{3}$. By symmetry of both relations we may assume $x \leq x'$, so $x' = x + 2k$ with $k \in \mathbb{N}$. First, $4^k \equiv 1 \pmod{3}$ for every $k \in \mathbb{N}$, by induction on k ([Theorem 0.2.7](#)): $4^0 = 1$, and if $4^k - 1 = 3m$, then $4^{k+1} - 1 = 4(4^k - 1) + 3 = 3(4m + 1)$. Hence

$$2^{x'} - 2^x = 2^x(4^k - 1)$$

is a multiple of 3, so $[2^{x'}]_3 = [2^x]_3$, as claimed. The difference between (b) and (c) is that the powers of 2 modulo 3 repeat with period 2, and 3 is not a multiple of 2.

Exercise C2: Counting equivalence relations.

- (a) List all partitions of $\{1, 2, 3\}$.
- (b) Hence deduce that there are exactly 5 equivalence relations on $\{1, 2, 3\}$.
- (c) Write the equivalence relation corresponding to the partition $\{\{1, 3\}, \{2\}\}$ as a subset of $\{1, 2, 3\} \times \{1, 2, 3\}$.

Solution.

- (a) Sort by the number of blocks. One block: $\{\{1, 2, 3\}\}$. Two blocks: one block is a single element and the other the remaining pair, giving $\{\{1\}, \{2, 3\}\}$, $\{\{2\}, \{1, 3\}\}$, $\{\{3\}, \{1, 2\}\}$. Three blocks: $\{\{1\}, \{2\}, \{3\}\}$. Four or more non-empty disjoint blocks are impossible in a three-element set. That is 5 partitions.

- (b) By [Theorem 0.5.2](#) (a), every equivalence relation gives a partition, its set of classes. By [Theorem 0.5.2](#) (c), the relation can be recovered from that partition, so distinct equivalence relations give distinct partitions. By [Theorem 0.5.2](#) (b), every partition arises this way. Hence equivalence relations on $\{1, 2, 3\}$ correspond one to one with partitions, and there are exactly 5 of them.
- (c) Two elements are related exactly when they lie in a common block:

$$\{(1, 1), (2, 2), (3, 3), (1, 3), (3, 1)\}.$$

Exercise C3: Intersections and unions of equivalence relations.

Let R_1 and R_2 be equivalence relations on a set X , viewed as subsets of $X \times X$.

- (a) Prove that $R_1 \cap R_2$ is an equivalence relation on X .
- (b) Show by an example that $R_1 \cup R_2$ need not be an equivalence relation. Which condition fails?
- (c) For congruence modulo 2 and modulo 3 on $\mathbb{Z}$, identify $R_1 \cap R_2$ as a familiar relation.

Hint for (b): try congruence modulo 2 and modulo 3 on $\mathbb{Z}$.

Solution.

- (a) *Reflexive.* Let $x \in X$. Then $(x, x) \in R_1$ and $(x, x) \in R_2$, since both are reflexive, so $(x, x) \in R_1 \cap R_2$. *Symmetric.* Suppose $(x, y) \in R_1 \cap R_2$. Since R_1 and R_2 are symmetric, $(y, x) \in R_1$ and $(y, x) \in R_2$, so $(y, x) \in R_1 \cap R_2$. *Transitive.* Suppose $(x, y), (y, z) \in R_1 \cap R_2$. Both pairs lie in R_1 , which is transitive, so $(x, z) \in R_1$; likewise $(x, z) \in R_2$. Hence $(x, z) \in R_1 \cap R_2$.
- (b) Let R_1 be congruence modulo 2 and R_2 congruence modulo 3 on $\mathbb{Z}$. The union is still reflexive and symmetric, since each of R_1, R_2 is. But transitivity fails: $(0, 2) \in R_1$ since $2 \mid 2$, and $(2, 5) \in R_2$ since $3 \mid 3$, yet $(0, 5)$ lies in neither, because 5 is not a multiple of 2 or of 3.
- (c) $(a, b) \in R_1 \cap R_2$ means $2 \mid a - b$ and $3 \mid a - b$. This holds exactly when $6 \mid a - b$. Indeed, if $a - b = 2k$ and $a - b = 3m$ with $k, m \in \mathbb{Z}$, then $a - b = 3(a - b) - 2(a - b) = 3 \cdot 2k - 2 \cdot 3m = 6(k - m)$. Conversely, if $a - b = 6j$, then $a - b = 2(3j) = 3(2j)$. So $R_1 \cap R_2$ is congruence modulo 6. The same phenomenon, where intersections behave well and unions do not, will reappear for subspaces in Chapter 1.

0.6 Complex Numbers

The equation $x^2 + 1 = 0$ has no real solution, because the square of a real number is never negative. For most of school mathematics this gap is harmless. In linear algebra it is not: later in the book we meet real matrices, such as a rotation of the plane, whose most useful data cannot be found among the real numbers. This section builds the complex numbers, the number system that closes the gap, and collects the facts about them that the rest of the book uses.

0.6.1 The complex numbers as pairs of reals

The quick way to introduce complex numbers is to say “invent a symbol i with $i^2 = -1$ and compute as usual”. That raises a fair worry. How do we know that such a symbol does not lead to a contradiction, the way “invent a number z with $0 \cdot z = 1$ ” does? The answer is to build the new numbers out of objects we already trust: pairs of real numbers.

Here is the recurring expression that suggests the definition. If we pretend $i^2 = -1$ and expand, we get

$$(a + bi)(c + di) = ac + adi + bci + bd i^2 = (ac - bd) + (ad + bc)i.$$

So a “number $a + bi$ ” is really the pair (a, b) , and the product should be the pair $(ac - bd, ad + bc)$. We take this formula as the definition, so nothing is pretended.

A complex number is a pair of real numbers, multiplied as if $i^2 = -1$.

Definition 0.6.1 — Complex Numbers

The set of **complex numbers** is $\mathbb{C} := \mathbb{R}^2 = \{(a, b) : a, b \in \mathbb{R}\}$, with addition and multiplication defined **for every** $(a, b), (c, d) \in \mathbb{C}$ by

$$(a, b) + (c, d) := (a + c, b + d), \quad (a, b)(c, d) := (ac - bd, ad + bc).$$

We write a for $(a, 0)$ and i for $(0, 1)$, so that $(a, b) = a + bi$. For $z = a + bi$ with $a, b \in \mathbb{R}$, the **real part** of z is $\operatorname{Re} z := a$ and the **imaginary part** of z is $\operatorname{Im} z := b$.

In words: addition is coordinate by coordinate, as for points of the plane. Multiplication is the formula found by expanding. The last sentence is a naming convention: the real number a is identified with the point $(a, 0)$ on the horizontal axis, and i is the point one unit up. Note that the imaginary part b is a **real** number; it is not bi .

Two things need checking before this convention is safe.

- **The notation $a + bi$ is honest.** By the definitions, $a + bi = (a, 0) + (b, 0)(0, 1) = (a, 0) + (0, b) = (a, b)$. Two complex numbers are equal exactly when they are equal as pairs, so $a + bi = c + di$ with $a, b, c, d \in \mathbb{R}$ means $a = c$ **and** $b = d$. This is how we “compare real and imaginary parts”.
- **Identifying a with $(a, 0)$ does not change real arithmetic.** We have $(a, 0) + (c, 0) = (a + c, 0)$ and $(a, 0)(c, 0) = (ac - 0, 0 + 0) = (ac, 0)$. So adding or multiplying two real numbers inside $\mathbb{C}$ gives the same answer as in $\mathbb{R}$.

With these in place, the equation that motivated everything is now a computation: $i^2 = (0, 1)(0, 1) = (0 \cdot 0 - 1 \cdot 1, 0 \cdot 1 + 1 \cdot 0) = (-1, 0) = -1$.

Example. (First Computations in $\mathbb{C}$)

Compute $(2 + 3i)(1 - i)$, the powers $i^0, i^1, \dots, i^7$, and the product $(a + bi)(a - bi)$ for real a, b .

Solution. By the multiplication rule with $a = 2, b = 3, c = 1, d = -1$,

$$(2 + 3i)(1 - i) = (2 \cdot 1 - 3 \cdot (-1)) + (2 \cdot (-1) + 3 \cdot 1)i = 5 + i.$$

For the powers, $i^0 = 1$, $i^1 = i$, $i^2 = -1$, $i^3 = i^2 \cdot i = -i$ and $i^4 = (-i)i = -i^2 = 1$. From then on the pattern repeats with period 4: $i^5 = i$, $i^6 = -1$, $i^7 = -i$.

Finally, $(a + bi)(a - bi) = (a^2 - b(-b)) + (a(-b) + ba)i = a^2 + b^2$. This product is a real number, and it is **non-negative**. We will meet it again as $|z|^2$.

The degenerate case deserves a look too. A real number $a = a + 0i$ is a complex number whose imaginary part is zero, and the number $0 = 0 + 0i$ is the only complex number with both parts zero. Every computation with real numbers is also a computation in $\mathbb{C}$, by the second check above.

A non-example by minimal change. Change one sign in the multiplication rule: define $(a, b) * (c, d) := (ac + bd, ad + bc)$, which is what expanding gives if we pretend $j^2 = +1$ instead. Addition is unchanged, and this product is still commutative and has $(1, 0)$ as a unit. But

$$(1, 1) * (1, -1) = (1 - 1, -1 + 1) = (0, 0),$$

so two non-zero elements multiply to zero. Division also breaks down: $(1, 1) * (c, d) = (c + d, c + d)$ can never equal the unit $(1, 0)$, because that would need $c + d = 1$ and $c + d = 0$ at once. So the non-zero element $(1, 1)$ has no inverse. The single sign in $ac - bd$ is what makes the complex numbers a number system in which we can divide, as we are about to see.

Quick check. Using the definition, compute i^{10} and $(1 + i)^2$. Is $\text{Im}(3 - 4i)$ equal to $-4i$?

Solution. Since the powers of i repeat with period 4 and $10 = 8 + 2$, we get $i^{10} = i^2 = -1$. Next, $(1 + i)^2 = (1 - 1) + (1 + 1)i = 2i$. No: the imaginary part is the real number -4 , not $-4i$.

The usual rules of arithmetic hold. Before computing freely, we should confirm that rearranging, regrouping and expanding brackets are legal in $\mathbb{C}$. The definition does not promise this; it has to be checked.

Proposition 0.6.1 — Arithmetic Laws in $\mathbb{C}$

For all $z, w, u \in \mathbb{C}$:

- (a) $z + w = w + z$ and $zw = wz$;
- (b) $(z + w) + u = z + (w + u)$ and $(zw)u = z(wu)$;
- (c) $z(w + u) = zw + zu$;
- (d) $z + 0 = z$, $z \cdot 1 = z$, and $z + (-z) = 0$, where $-(a + bi) := (-a) + (-b)i$.

Idea. Every law reduces to the corresponding law for real numbers once both sides are written as pairs. Addition is coordinate by coordinate, so its laws are immediate. For multiplication the only real work is associativity: expand both sides and see that they produce the same four terms in each coordinate.

Proof. Let $z = a + bi$, $w = c + di$ and $u = e + fi$ with $a, b, c, d, e, f \in \mathbb{R}$. Since addition acts coordinate by coordinate, the laws for addition in (a), (b) and (d) follow from the same laws in $\mathbb{R}$, applied to each coordinate.

- (a) By definition $zw = (ac - bd) + (ad + bc)i$ and $wz = (ca - db) + (cb + da)i$. These

are equal because real multiplication and addition are commutative.

(b) Expanding with the definition twice gives

$$(zw)u = ((ac - bd)e - (ad + bc)f) + ((ac - bd)f + (ad + bc)e)i$$

and

$$z(wu) = (a(ce - df) - b(cf + de)) + (a(cf + de) + b(ce - df))i.$$

Multiplying out, both real parts equal $ace - adf - bcf - bde$ and both imaginary parts equal $acf + ade + bce - bdf$. Hence $(zw)u = z(wu)$.

(c) We have $w + u = (c + e) + (d + f)i$, so

$$z(w+u) = (a(c+e) - b(d+f)) + (a(d+f) + b(c+e))i = ((ac-bd) + (ae-bf)) + ((ad+bc) + (af+be))i,$$

which is $zw + zu$ by the definitions of the product and the sum.

(d) With $1 = 1 + 0i$, we get $z \cdot 1 = (a \cdot 1 - b \cdot 0) + (a \cdot 0 + b \cdot 1)i = a + bi = z$. Also $z + (-z) = (a - a) + (b - b)i = 0$. This proves the proposition. ■

From now on we compute in $\mathbb{C}$ exactly as in $\mathbb{R}$: expand brackets, collect terms, and replace i^2 by -1 . The proposition is what makes that habit legitimate.

0.6.2 Conjugate and modulus

Division is still missing. To divide by $3 + 4i$ we want some number to multiply by that lands in the real numbers, where division is familiar. The computation $(a + bi)(a - bi) = a^2 + b^2$ in [Example \(First Computations in \$\mathbb{C}\$ \)](#) shows exactly which number does the job. Both ingredients get names.

Conjugating reflects a point in the real axis; the modulus is its distance from the origin.

Definition 0.6.2 — Conjugate and Modulus

Let $z = a + bi \in \mathbb{C}$ with $a, b \in \mathbb{R}$. The **complex conjugate** of z is $\bar{z} := a - bi$. The **modulus** (or absolute value) of z is the real number $|z| := \sqrt{a^2 + b^2} \geq 0$, the non-negative square root.

In words: the conjugate keeps the real part and flips the sign of the imaginary part. The modulus is the length of the segment from 0 to the point (a, b) , by Pythagoras. For a real number $a = a + 0i$, we get $\bar{a} = a$ and $|a| = \sqrt{a^2}$, the usual absolute value, so the notation does not clash with its real meaning. For example, $\overline{3 + 4i} = 3 - 4i$, $|3 + 4i| = 5$, $\bar{i} = -i$ and $|i| = 1$.

The following rules are used constantly, in this section and much later for complex inner products.

Theorem 0.6.2 — Properties of Conjugation and Modulus

For all $z, w \in \mathbb{C}$:

- (a) $\overline{z + w} = \bar{z} + \bar{w}$ and $\overline{\bar{z}} = z$;
- (b) $\overline{zw} = \bar{z}\bar{w}$;
- (c) $z\bar{z} = |z|^2$ and $z + \bar{z} = 2\operatorname{Re} z$;
- (d) $|zw| = |z||w|$ and $|\bar{z}| = |z|$.

Idea. Parts (a)–(c) are direct computations with real and imaginary parts. For (d), expanding $|zw|$ directly is legitimate but messy. It is cleaner to square: by (c) a squared modulus is a product with a conjugate, and (b) lets the conjugate split across the product.

Proof. Let $z = a + bi$ and $w = c + di$ with $a, b, c, d \in \mathbb{R}$.

- (a) We have $\overline{z + w} = (a + c) - (b + d)i = (a - bi) + (c - di) = \bar{z} + \bar{w}$, and $\overline{\bar{z}} = \overline{a - bi} = a + bi = z$.
- (b) Since $zw = (ac - bd) + (ad + bc)i$, we get $\overline{zw} = (ac - bd) - (ad + bc)i$. On the other hand,

$$\bar{z}\bar{w} = (a - bi)(c - di) = (ac - (-b)(-d)) + (a(-d) + (-b)c)i = (ac - bd) - (ad + bc)i.$$
- (c) By the computation in [Example \(First Computations in \$\mathbb{C}\$ \)](#), $z\bar{z} = a^2 + b^2 = |z|^2$. Also $z + \bar{z} = 2a = 2\operatorname{Re} z$.
- (d) By (c), then (b), then [Proposition 0.6.1](#) to reorder the four factors,

$$|zw|^2 = (zw)\overline{zw} = zw\bar{z}\bar{w} = (z\bar{z})(w\bar{w}) = |z|^2|w|^2.$$

Both $|zw|$ and $|z||w|$ are non-negative real numbers with the same square, so they are equal. Finally $|\bar{z}| = \sqrt{a^2 + (-b)^2} = |z|$. This proves the theorem. ■

The first payoff is division. If $z \neq 0$, then $|z|^2 = a^2 + b^2$ is a **strictly** positive real number, so we may divide by it.

Proposition 0.6.3 — Inverse of a Complex Number

Let $z \in \mathbb{C}$ with $z \neq 0$. Then $w := \frac{1}{|z|^2} \bar{z}$ satisfies $zw = 1$. We write z^{-1} or $\frac{1}{z}$ for this w .

Proof. Write $z = a + bi$. Since $z \neq 0$, at least one of a, b is non-zero, so $|z|^2 = a^2 + b^2 > 0$. By [Proposition 0.6.1](#) and [Theorem 0.6.2](#) (c),

$$zw = \frac{1}{|z|^2} (z\bar{z}) = \frac{1}{|z|^2} |z|^2 = 1,$$

as claimed. ■

In practice, “multiply top and bottom by the conjugate”. For instance

$$\frac{1}{3+4i} = \frac{3-4i}{(3+4i)(3-4i)} = \frac{3-4i}{25}, \quad \frac{5+i}{1-i} = \frac{(5+i)(1+i)}{(1-i)(1+i)} = \frac{4+6i}{2} = 2+3i,$$

which agrees with $(2+3i)(1-i) = 5+i$ from [Example \(First Computations in \$\mathbb{C}\$ \)](#). Only one inverse exists; this is a special case of a general fact about fields proved in the next section.

The triangle inequality. The modulus measures distance: $|z-w|$ is the distance between the points z and w of the plane. The side of a triangle is never longer than the other two sides together, and this is the algebraic form of that fact.

Theorem 0.6.4 — Triangle Inequality

For all $z, w \in \mathbb{C}$,

$$|z+w| \leq |z| + |w|.$$

Idea. Moduli are square roots, which are awkward to add. Both sides are non-negative, so it suffices to compare their squares. Expand $|z+w|^2$ using $|u|^2 = u\bar{u}$: the cross terms combine into $2\operatorname{Re}(z\bar{w})$. A real part is never bigger than the modulus, and that single estimate finishes the proof.

Proof. First, for any $u = x + yi$ with $x, y \in \mathbb{R}$, we have $\operatorname{Re} u = x \leq |x| = \sqrt{x^2} \leq \sqrt{x^2 + y^2} = |u|$, since the square root is increasing.

Now let $z, w \in \mathbb{C}$. By [Theorem 0.6.2](#) (c), (a) and [Proposition 0.6.1](#),

$$|z+w|^2 = (z+w)(\bar{z}+\bar{w}) = z\bar{z} + z\bar{w} + w\bar{z} + w\bar{w} = |z|^2 + (z\bar{w} + \overline{z\bar{w}}) + |w|^2,$$

where we used $w\bar{z} = \overline{z\bar{w}}$, which follows from [Theorem 0.6.2](#) (b) and (a). By [Theorem 0.6.2](#) (c) the bracket equals $2\operatorname{Re}(z\bar{w})$. Therefore, using $\operatorname{Re} u \leq |u|$ and then [Theorem 0.6.2](#) (d),

$$|z+w|^2 \leq |z|^2 + 2|z\bar{w}| + |w|^2 = |z|^2 + 2|z||w| + |w|^2 = (|z| + |w|)^2.$$

Both $|z+w|$ and $|z|+|w|$ are non-negative, so taking square roots preserves the inequality. This proves the triangle inequality. ■

The route “square, expand, bound the cross term” is worth remembering: it is exactly the route we will take for norms in inner product spaces. The inequality is often strict. For $z = 1$ and $w = -1$, the left side is 0 and the right side is 2.

0.6.3 Two things that do not carry over from the reals

Complex numbers obey the rules of arithmetic, but not every habit from the real line survives.

The first loss is order. On $\mathbb{R}$ we use two facts about positivity: for every x , exactly one of $x > 0$, $x = 0$, $-x > 0$ holds, and sums and products of positive numbers are positive. Suppose

some relation $>$ on $\mathbb{C}$ had both properties. Since $i \neq 0$, either $i > 0$ or $-i > 0$, and in both cases the product rule gives $i^2 = (-i)^2 = -1 > 0$. The same argument applied to $1 \neq 0$ gives $1 = 1^2 = (-1)^2 > 0$. Then both 1 and -1 are positive, so their sum 0 is positive, which contradicts “exactly one”.

Warning. There is no ordering on $\mathbb{C}$ compatible with addition and multiplication, so an inequality such as $z < w$ between complex numbers means nothing. Inequalities in this book compare **real** numbers: moduli $|z|$, real parts, or quantities known to be real. Writing $i > 0$ leads, as shown above, to $0 > 0$.

The second loss is the square-root rule $\sqrt{x}\sqrt{y} = \sqrt{xy}$, which on $\mathbb{R}$ is only claimed for $x, y \geq 0$.

Warning. Every non-zero complex number has **two** square roots, and no choice of “the” square root makes $\sqrt{z}\sqrt{w} = \sqrt{zw}$ true for all z, w . With $\sqrt{-1} = i$: $\sqrt{-1} \cdot \sqrt{-1} = i^2 = -1$, but $\sqrt{(-1)(-1)} = \sqrt{1} = 1$. Avoid the symbol $\sqrt{z}$ for non-real z ; say “a square root of z ” and solve $w^2 = z$ instead.

Solving $w^2 = z$ is a finite computation, done by comparing real and imaginary parts.

Example. (Square Roots of $5 - 12i$)

Find all $w \in \mathbb{C}$ with $w^2 = 5 - 12i$.

Solution. Write $w = x + yi$ with $x, y \in \mathbb{R}$. Then $w^2 = (x^2 - y^2) + 2xyi$, so comparing real and imaginary parts, $w^2 = 5 - 12i$ holds exactly when

$$x^2 - y^2 = 5 \quad \text{and} \quad xy = -6.$$

Taking moduli in $w^2 = 5 - 12i$ and using [Theorem 0.6.2](#) (d) gives $|w|^2 = |5 - 12i| = \sqrt{25 + 144} = 13$, that is $x^2 + y^2 = 13$. Adding and subtracting this and the first equation gives $x^2 = 9$ and $y^2 = 4$, so $x = \pm 3$ and $y = \pm 2$. The condition $xy = -6 < 0$ says x and y have opposite signs. Hence $w = 3 - 2i$ or $w = -3 + 2i$. Both work: $(3 - 2i)^2 = 9 - 12i + 4i^2 = 5 - 12i$, and $(-3 + 2i)^2 = (3 - 2i)^2$.

0.6.4 Polar form: multiplication rotates and scales

Addition in $\mathbb{C}$ is the familiar addition of arrows in the plane. Multiplication has a geometric meaning too, and it only becomes visible when we describe a point by its distance from 0 and its angle, instead of by its coordinates.

Let $z \neq 0$ and put $r = |z| > 0$. The number z/r has modulus 1 by [Theorem 0.6.2](#) (d), so it lies on the unit circle. From trigonometry, every point (x, y) with $x^2 + y^2 = 1$ is $(\cos \theta, \sin \theta)$ for some real θ . We abbreviate

$$e^{i\theta} := \cos \theta + i \sin \theta \quad (\theta \in \mathbb{R}),$$

and then every non-zero complex number can be written as

$$z = re^{i\theta}, \quad r = |z| > 0, \quad \theta \in \mathbb{R}.$$

This is the **polar form** of z , and θ is **an** argument of z . The word “an” matters: θ and $\theta + 2\pi k$ give the same point for every integer k . Conversely, if $re^{i\theta} = se^{i\varphi}$ with $r, s > 0$, then taking

moduli gives $r = s$ (as $|e^{i\theta}| = \sqrt{\cos^2 \theta + \sin^2 \theta} = 1$). Comparing real and imaginary parts then gives $\cos \theta = \cos \varphi$ and $\sin \theta = \sin \varphi$, which from trigonometry means $\theta - \varphi$ is an integer multiple of 2π .

For now $e^{i\theta}$ is only a name for $\cos \theta + i \sin \theta$. The next result is the reason the name is a good one: these numbers multiply the way exponentials do.

Theorem 0.6.5 — Multiplication in Polar Form

For all $\theta, \varphi \in \mathbb{R}$,

$$e^{i\theta} e^{i\varphi} = e^{i(\theta+\varphi)}.$$

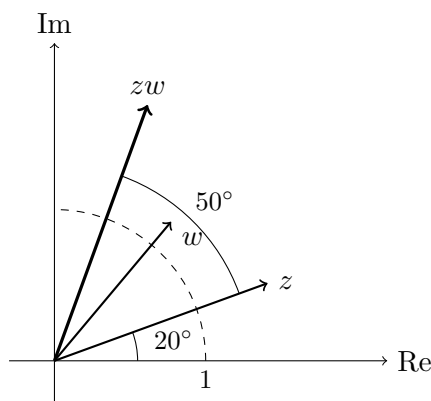
Consequently, if $z = re^{i\theta}$ and $w = se^{i\varphi}$ with $r, s > 0$, then $zw = rs e^{i(\theta+\varphi)}$.

Proof. By the multiplication rule,

$$e^{i\theta} e^{i\varphi} = (\cos \theta \cos \varphi - \sin \theta \sin \varphi) + (\cos \theta \sin \varphi + \sin \theta \cos \varphi)i = \cos(\theta + \varphi) + i \sin(\theta + \varphi),$$

where the last equality uses the addition formulas for cosine and sine. This is $e^{i(\theta+\varphi)}$. For the second statement, $zw = (rs)(e^{i\theta} e^{i\varphi})$ by [Proposition 0.6.1](#), and the first statement finishes the proof. ■

In words: **to multiply two complex numbers, multiply their moduli and add their arguments**. Multiplying by a fixed $w = se^{i\varphi}$ is therefore a rotation of the plane about 0 by the angle φ , combined with a stretch by the factor s . The picture shows z , w and zw for $|z| = 1.5$, $|w| = 1.2$ and arguments 20° and 50° .



The arrow for zw has length $1.5 \times 1.2 = 1.8$ and makes the angle $20^\circ + 50^\circ = 70^\circ$ with the positive real axis: it is the arrow for z , turned by w 's angle and stretched by w 's length. The small arc marks the argument of z , the large arc marks the extra turn by the argument of w , and the dashed arc is the unit circle, where the numbers $e^{i\theta}$ live.

Two special cases are worth isolating. Multiplying by $i = e^{i\pi/2}$ is a quarter turn counterclockwise, which explains $i^2 = -1$: two quarter turns send 1 to -1 . And $e^{i\theta} e^{-i\theta} = e^0 = 1$, so $(e^{i\theta})^{-1} = e^{-i\theta} = \overline{e^{i\theta}}$.

Repeating [Theorem 0.6.5](#) gives powers.

Theorem 0.6.6 — De Moivre's Formula

For every $\theta \in \mathbb{R}$ and every $n \in \mathbb{N}$,

$$(e^{i\theta})^n = e^{in\theta}, \quad \text{that is,} \quad (\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta.$$

Consequently $(re^{i\theta})^n = r^n e^{in\theta}$ for every real $r > 0$.

Proof. We use induction on n (Theorem 0.2.7). For $n = 0$, both sides equal 1, since $e^{i \cdot 0} = \cos 0 + i \sin 0 = 1$. Suppose $(e^{i\theta})^n = e^{in\theta}$ for some $n \in \mathbb{N}$. Then $(e^{i\theta})^{n+1} = (e^{i\theta})^n e^{i\theta} = e^{in\theta} e^{i\theta} = e^{i(n+1)\theta}$ by the induction hypothesis and Theorem 0.6.5. The consequence follows by the same induction, since $(re^{i\theta})^{n+1} = (r^n e^{in\theta})(re^{i\theta}) = r^{n+1} e^{i(n+1)\theta}$ by Theorem 0.6.5. ■

Powers that are painful to expand become one line in polar form.

Example. (A Power via Polar Form)

Compute $(\sqrt{3} + i)^6$.

Solution. The modulus is $|\sqrt{3} + i| = \sqrt{3 + 1} = 2$, and $\sqrt{3} + i = 2\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right) = 2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right) = 2e^{i\pi/6}$. By Theorem 0.6.6,

$$(\sqrt{3} + i)^6 = 2^6 e^{i \cdot 6\pi/6} = 64(\cos \pi + i \sin \pi) = -64.$$

Expanding $(\sqrt{3} + i)^6$ with the binomial theorem gives the same answer after seven terms and some cancellation.

Quick check. Use polar form to compute $(1 + i)^8$.

Solution. We have $|1 + i| = \sqrt{2}$ and $1 + i = \sqrt{2}\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i\right) = \sqrt{2}e^{i\pi/4}$. By Theorem 0.6.6, $(1 + i)^8 = (\sqrt{2})^8 e^{i \cdot 8\pi/4} = 16e^{2\pi i} = 16$. As a cross-check, $(1 + i)^2 = 2i$, so $(1 + i)^8 = (2i)^4 = 16i^4 = 16$.

0.6.5 Roots of unity

Polar form also turns the equation $z^n = c$ into a question about angles. The most important case is $c = 1$.

Example. (Roots of Unity)

Let $n \geq 1$ be an integer. Find all $z \in \mathbb{C}$ with $z^n = 1$, and draw them for $n = 6$.

Solution. Since $0^n = 0 \neq 1$, every solution is non-zero, so we may write $z = re^{i\theta}$ with $r > 0$. By Theorem 0.6.6, $z^n = r^n e^{in\theta}$, and $1 = 1 \cdot e^{i0}$. By uniqueness of the polar form

up to multiples of 2π , $z^n = 1$ holds exactly when $r^n = 1$ and $n\theta = 2\pi k$ for some integer k . Since $r > 0$ is real, $r^n = 1$ forces $r = 1$. Hence the solutions are

$$e^{2\pi i k/n}, \quad k \in \mathbb{Z}.$$

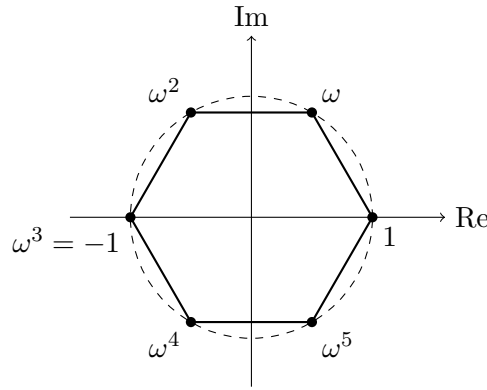
Put $\omega := e^{2\pi i/n}$. By [Theorem 0.6.6](#), $e^{2\pi i k/n} = \omega^k$ for $k \geq 0$. Two integers k, l give the same number exactly when $2\pi(k-l)/n$ is a multiple of 2π , that is, when n divides $k-l$. So there are exactly n distinct solutions,

$$1, \omega, \omega^2, \dots, \omega^{n-1},$$

the **n -th roots of unity**. They sit on the unit circle at equally spaced angles. For $n \geq 3$ they are the corners of a regular polygon with n corners, one of them at 1; for $n = 2$ they are ± 1 .

For $n = 6$, the angles are multiples of 60° , and the roots are

$$1, \quad \frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad -1, \quad -\frac{1}{2} - \frac{\sqrt{3}}{2}i, \quad \frac{1}{2} - \frac{\sqrt{3}}{2}i.$$



The six roots sum to 0: since $\omega^3 = e^{i\pi} = -1$, we have $\omega^{k+3} = -\omega^k$, so the roots cancel in the opposite pairs $\{1, \omega^3\}$, $\{\omega, \omega^4\}$, $\{\omega^2, \omega^5\}$. In the picture, the hexagon is symmetric about its center 0. The sum of the n -th roots of unity is 0 for every $n \geq 2$; see [Exercise \(C2: Roots of Unity Sum to Zero\)](#).

0.6.6 The Fundamental Theorem of Algebra

We built $\mathbb{C}$ so that one polynomial equation, $z^2 + 1 = 0$, has a solution. It turns out that nothing more ever needs to be added: every polynomial equation already has a solution in $\mathbb{C}$. Here a **complex polynomial of degree n** means an expression $a_n z^n + \dots + a_1 z + a_0$ with $a_0, \dots, a_n \in \mathbb{C}$ and $a_n \neq 0$; polynomials are studied properly in a later section of this chapter.

Theorem 0.6.7 — Fundamental Theorem of Algebra, Statement

Let $n \geq 1$ and let $a_0, a_1, \dots, a_n \in \mathbb{C}$ with $a_n \neq 0$. Then there exists $z \in \mathbb{C}$ with

$$a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0 = 0.$$

We do not prove this theorem here. Its proof needs tools we have not built, and it is proved in Chapter 5. Nothing in Chapters 0–4 depends on it.

Remark. Combined with division of polynomials (also in Chapter 5), the theorem gives the form in which it is usually applied: every complex polynomial of degree $n \geq 1$ **splits into linear factors**, $a_n z^n + \cdots + a_0 = a_n(z - z_1)(z - z_2) \cdots (z - z_n)$ for some $z_1, \dots, z_n \in \mathbb{C}$, not necessarily distinct. For linear algebra this is the fact that every complex square matrix has an eigenvalue, the starting point of the theory of canonical forms.

One companion fact needs no heavy machinery. If the coefficients $a_0, \dots, a_n$ are **real** and $p(z) = a_n z^n + \cdots + a_0$ has $p(z_0) = 0$, then applying [Theorem 0.6.2](#) (a) and (b) term by term, and $\overline{a_k} = a_k$, gives $p(\overline{z_0}) = \overline{p(z_0)} = \overline{0} = 0$. So the non-real roots of a real polynomial come in conjugate pairs. For example, $z^2 - 2z + 5$ has the roots $1 + 2i$ and $1 - 2i$: indeed $(1 + 2i)^2 - 2(1 + 2i) + 5 = (-3 + 4i) - (2 + 4i) + 5 = 0$.

0.6.7 Exercises

0.6.7.1 A. Check your understanding

Exercise A1.

- (a) Define the complex conjugate $\bar{z}$ and the modulus $|z|$ of $z = a + bi$, where $a, b \in \mathbb{R}$.
- (b) State the triangle inequality for complex numbers.
- (c) True or false: $|z + w| = |z| + |w|$ for all $z, w \in \mathbb{C}$. Justify your answer.
- (d) True or false: $z\bar{z}$ is a non-negative real number for every $z \in \mathbb{C}$. Justify your answer.
- (e) True or false: if $z, w \in \mathbb{C}$ and $z^2 = w^2$, then $z = w$. Justify your answer.
- (f) State the Fundamental Theorem of Algebra.

Solution.

- (a) See [Definition 0.6.2](#): $\bar{z} = a - bi$ and $|z| = \sqrt{a^2 + b^2} \geq 0$.
- (b) See [Theorem 0.6.4](#): $|z + w| \leq |z| + |w|$ for all $z, w \in \mathbb{C}$.
- (c) False. For $z = 1$ and $w = -1$, $|z + w| = 0$ but $|z| + |w| = 2$.
- (d) True. By [Theorem 0.6.2](#) (c), $z\bar{z} = |z|^2 = a^2 + b^2$, which is real and non-negative.
- (e) False. For $z = 1$ and $w = -1$, $z^2 = w^2 = 1$ but $z \neq w$. (What is true is that $z^2 = w^2$ implies $z = w$ or $z = -w$.)
- (f) See [Theorem 0.6.7](#): every complex polynomial $a_n z^n + \cdots + a_0$ with $n \geq 1$ and $a_n \neq 0$ has a root in $\mathbb{C}$.

0.6.7.2 B. Practice

Exercise B1: Arithmetic.

Write each of the following in the form $a + bi$ with $a, b \in \mathbb{R}$, or as a real number where indicated.

- (a) $(3 - 2i)(4 + i)$.
 (b) $\frac{1 + 2i}{2 - i}$.
 (c) $|(3 + 4i)(5 - 12i)|$, without expanding the product.
 (d) $\operatorname{Re} \frac{1}{1 + i}$ and $\operatorname{Im} \frac{1}{1 + i}$.

Solution.

- (a) Expanding and using $i^2 = -1$: $(3 - 2i)(4 + i) = 12 + 3i - 8i - 2i^2 = 14 - 5i$.
 (b) Multiplying top and bottom by $\overline{2 - i} = 2 + i$ ([Proposition 0.6.3](#)),

$$\frac{1 + 2i}{2 - i} = \frac{(1 + 2i)(2 + i)}{(2 - i)(2 + i)} = \frac{2 + i + 4i + 2i^2}{4 + 1} = \frac{5i}{5} = i.$$

- (c) By [Theorem 0.6.2](#) (d), $|(3 + 4i)(5 - 12i)| = |3 + 4i| |5 - 12i| = \sqrt{25} \cdot \sqrt{169} = 5 \cdot 13 = 65$.
 (d) By [Proposition 0.6.3](#), $\frac{1}{1 + i} = \frac{1 - i}{|1 + i|^2} = \frac{1 - i}{2} = \frac{1}{2} - \frac{1}{2}i$. Hence the real part is $\frac{1}{2}$ and the imaginary part is $-\frac{1}{2}$.

Exercise B2: A Square Root.

- (a) Find all $w \in \mathbb{C}$ with $w^2 = -3 + 4i$.
 (b) Hence solve $z^2 + 2z + 4 - 4i = 0$.

Hint: for (b), complete the square.

Solution.

- (a) Write $w = x + yi$ with $x, y \in \mathbb{R}$. Comparing real and imaginary parts in $(x^2 - y^2) + 2xyi = -3 + 4i$ gives $x^2 - y^2 = -3$ and $xy = 2$. Taking moduli, by [Theorem 0.6.2](#) (d), $x^2 + y^2 = |w|^2 = |-3 + 4i| = 5$. Adding and subtracting, $x^2 = 1$ and $y^2 = 4$. Since $xy = 2 > 0$, x and y have the same sign. Therefore $w = 1 + 2i$ or $w = -1 - 2i$. Check: $(1 + 2i)^2 = 1 + 4i - 4 = -3 + 4i$, and $(-1 - 2i)^2 = (1 + 2i)^2$.
 (b) Completing the square, $z^2 + 2z + 4 - 4i = (z + 1)^2 - 1 + 4 - 4i = (z + 1)^2 - (-3 + 4i)$. So the equation holds exactly when $(z + 1)^2 = -3 + 4i$. By (a), this means $z + 1 = 1 + 2i$ or $z + 1 = -1 - 2i$. Hence the solutions are $z = 2i$ and $z = -2 - 2i$.

Exercise B3.

Let $z \in \mathbb{C}$ with $z \neq 0$. Prove that $\frac{1}{z} = \bar{z}$ if and only if $|z| = 1$.

Solution. By [Proposition 0.6.3](#), $\frac{1}{z} = \frac{1}{|z|^2} \bar{z}$.

($\Leftarrow$) Suppose $|z| = 1$. Then $\frac{1}{z} = \frac{1}{1} \bar{z} = \bar{z}$.

($\Rightarrow$) Suppose $\frac{1}{z} = \bar{z}$. Then $\frac{1}{|z|^2} \bar{z} = \bar{z}$. Since $z \neq 0$, also $\bar{z} \neq 0$, so multiplying both sides by $(\bar{z})^{-1}$ gives $\frac{1}{|z|^2} = 1$, that is, $|z|^2 = 1$. As $|z| \geq 0$, this forces $|z| = 1$. This proves the equivalence.

Exercise B4: Polar Form.

- (a) Compute $(1 - \sqrt{3}i)^6$.
- (b) Find all $z \in \mathbb{C}$ with $z^3 = -8i$, in the form $a + bi$.

Solution.

- (a) We have $|1 - \sqrt{3}i| = \sqrt{1+3} = 2$ and $1 - \sqrt{3}i = 2(\cos(-\frac{\pi}{3}) + i\sin(-\frac{\pi}{3})) = 2e^{-i\pi/3}$. By [Theorem 0.6.6](#), $(1 - \sqrt{3}i)^6 = 64e^{-2\pi i} = 64(\cos 2\pi - i\sin 2\pi) = 64$.
- (b) Since $0^3 \neq -8i$, write $z = re^{i\theta}$ with $r > 0$. Also $-8i = 8e^{-i\pi/2}$. By [Theorem 0.6.6](#), $z^3 = r^3e^{3i\theta}$. By uniqueness of polar form up to multiples of 2π , $z^3 = -8i$ holds exactly when $r^3 = 8$ and $3\theta = -\frac{\pi}{2} + 2\pi k$ for some integer k . Therefore $r = 2$ and $\theta = -\frac{\pi}{6} + \frac{2\pi k}{3}$. The values $k = 0, 1, 2$ give $\theta = -\frac{\pi}{6}, \frac{\pi}{2}, \frac{7\pi}{6}$, and every other k differs from one of these by a multiple of 3, which changes θ by a multiple of 2π . Hence the solutions are

$$2e^{-i\pi/6} = \sqrt{3} - i, \quad 2e^{i\pi/2} = 2i, \quad 2e^{7i\pi/6} = -\sqrt{3} - i.$$

Check: $(2i)^3 = 8i^3 = -8i$.

0.6.7.3 C. Going deeper

Exercise C1: Reverse Triangle Inequality.

Prove that $||z| - |w|| \leq |z - w|$ for all $z, w \in \mathbb{C}$.

Hint: write $z = (z - w) + w$.

Solution. Let $z, w \in \mathbb{C}$. Since $z = (z - w) + w$, the triangle inequality ([Theorem 0.6.4](#)) gives $|z| \leq |z - w| + |w|$, so $|z| - |w| \leq |z - w|$. Swapping the roles of z and w gives $|w| - |z| \leq |w - z|$. Moreover $|w - z| = |(-1)(z - w)| = |-1||z - w| = |z - w|$ by [Theorem 0.6.2](#) (d). The real number $||z| - |w||$ equals one of $|z| - |w|$ and $|w| - |z|$, and both are at most $|z - w|$. This proves the inequality.

Exercise C2: Roots of Unity Sum to Zero.

Let $n \geq 2$ and $\omega = e^{2\pi i/n}$.

- (a) Prove that $1 + \omega + \omega^2 + \cdots + \omega^{n-1} = 0$.
- (b) Deduce that $\sum_{k=0}^{n-1} \cos \frac{2\pi k}{n} = 0$ and $\sum_{k=0}^{n-1} \sin \frac{2\pi k}{n} = 0$.

Hint: compare S with ωS , where S is the sum in (a).

Solution.

- (a) Let $S = 1 + \omega + \cdots + \omega^{n-1}$. By Theorem 0.6.6, $\omega^n = e^{2\pi i} = 1$. Therefore

$$\omega S = \omega + \omega^2 + \cdots + \omega^{n-1} + \omega^n = \omega + \omega^2 + \cdots + \omega^{n-1} + 1 = S,$$

so $(\omega - 1)S = 0$. Since $n \geq 2$, we have $0 < 2\pi/n \leq \pi$, so $\operatorname{Re} \omega = \cos(2\pi/n) < 1$ and $\omega \neq 1$. Hence $\omega - 1 \neq 0$, and multiplying $(\omega - 1)S = 0$ by $(\omega - 1)^{-1}$ (Proposition 0.6.3) gives $S = 0$.

- (b) By Theorem 0.6.6, $\omega^k = \cos \frac{2\pi k}{n} + i \sin \frac{2\pi k}{n}$ for each k . Since addition in $\mathbb{C}$ is coordinate by coordinate, the real part of S is $\sum_{k=0}^{n-1} \cos \frac{2\pi k}{n}$ and the imaginary part is $\sum_{k=0}^{n-1} \sin \frac{2\pi k}{n}$. By (a), $S = 0$, so both sums are 0.

Exercise C3: Symmetries of $\mathbb{C}$ Fixing $\mathbb{R}$.

Let $\sigma: \mathbb{C} \rightarrow \mathbb{C}$ be a function with $\sigma(z + w) = \sigma(z) + \sigma(w)$ and $\sigma(zw) = \sigma(z)\sigma(w)$ for all $z, w \in \mathbb{C}$, and $\sigma(a) = a$ for every $a \in \mathbb{R}$.

- (a) Prove that $\sigma(i)^2 = -1$.
- (b) Prove that the only complex numbers u with $u^2 = -1$ are i and $-i$.
- (c) Deduce that either $\sigma(z) = z$ for every $z \in \mathbb{C}$, or $\sigma(z) = \bar{z}$ for every $z \in \mathbb{C}$.
- (d) Show that $z \mapsto \bar{z}$ does satisfy all three conditions on σ .

Hint: for (b), write $u = x + yi$ and compare real and imaginary parts.

Solution.

- (a) Since σ respects products and fixes the real number -1 , we get $\sigma(i)^2 = \sigma(i \cdot i) = \sigma(-1) = -1$.
- (b) Let $u = x + yi$ with $x, y \in \mathbb{R}$ and $u^2 = -1$. Comparing real and imaginary parts in $(x^2 - y^2) + 2xyi = -1$ gives $x^2 - y^2 = -1$ and $xy = 0$. If $y = 0$, then $x^2 = -1$, which is impossible for real x . Hence $x = 0$, so $y^2 = 1$ and $y = \pm 1$. Therefore $u = i$ or $u = -i$, and both satisfy $u^2 = -1$.
- (c) By (a) and (b), $\sigma(i) = i$ or $\sigma(i) = -i$. Let $z = a + bi$ with $a, b \in \mathbb{R}$. Since σ respects sums and products and fixes a and b ,

$$\sigma(z) = \sigma(a) + \sigma(b)\sigma(i) = a + b\sigma(i).$$

If $\sigma(i) = i$, this gives $\sigma(z) = z$ for every z . If $\sigma(i) = -i$, it gives $\sigma(z) = a - bi = \bar{z}$ for every z .

- (d) By Theorem 0.6.2 (a) and (b), conjugation respects sums and products. For $a \in \mathbb{R}$, $\bar{a} = a + 0i = a - 0i = a$. So conjugation satisfies all three conditions.

0.7 Fields

Linear algebra is about equations like $2x + 3y = 7$, and so far we have met several number systems in which such equations make sense: $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, and the arithmetic of remainders $\mathbb{Z}/n\mathbb{Z}$. Which of them can serve as the numbers of linear algebra? This section answers by isolating the rules that solving linear equations actually uses. A number system obeying those rules is called a field, and fields are the scalars of every later chapter.

0.7.1 What solving linear equations needs

Here is a small system, solved the way you learned at school:

$$\begin{aligned} 2x + 3y &= 7, \\ x - y &= 1. \end{aligned}$$

Subtracting twice the second equation from the first gives $5y = 5$. Dividing by 5 gives $y = 1$, and then $x = 1 + y = 2$. Check: $2 \cdot 2 + 3 \cdot 1 = 7$ and $2 - 1 = 1$.

Look at what we used. We added and subtracted equations, multiplied an equation by a number, and divided by a **non-zero** number. Silently we also rearranged terms, regrouped them and expanded brackets. Every step of Gaussian elimination is of this kind, and every one of them works in $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$.

Now try the integers. The equation $2x = 1$ involves only integers, yet it has no integer solution, because the step “divide by 2” leaves $\mathbb{Z}$. Elimination stalls as soon as a pivot has no inverse. So what separates $\mathbb{Z}$ from $\mathbb{Q}$ is division, and division is exactly what we must demand.

A field is a number system where you can add, subtract, multiply, and divide by anything non-zero, with the usual rules.

0.7.2 The definition

Definition 0.7.1 — Field

A **field** is a set F together with two functions, **addition** $+: F \times F \rightarrow F$, $(a, b) \mapsto a + b$, and **multiplication** $\cdot: F \times F \rightarrow F$, $(a, b) \mapsto a \cdot b = ab$, satisfying the following axioms.

- (F1) $a + b = b + a$ **for all** $a, b \in F$.
- (F2) $(a + b) + c = a + (b + c)$ **for all** $a, b, c \in F$.
- (F3) There exists an element $0 \in F$ such that $a + 0 = a$ **for every** $a \in F$.
- (F4) **For every** $a \in F$ there exists an element $-a \in F$ such that $a + (-a) = 0$.
- (F5) $(ab)c = a(bc)$ **for all** $a, b, c \in F$.
- (F6) $ab = ba$ **for all** $a, b \in F$.
- (F7) There exists an element $1 \in F$ with $1 \neq 0$ such that $a \cdot 1 = a$ **for every** $a \in F$.
- (F8) **For every non-zero** $a \in F$ there exists an element $a^{-1} \in F$ such that $aa^{-1} = 1$.
- (F9) $a(b + c) = ab + ac$ **for all** $a, b, c \in F$.

Read the axioms in groups.

- **(F1)–(F4)** are about addition alone. It is commutative and associative, it has a neutral element 0, and every element has a negative. Subtraction is then defined by $a - b := a + (-b)$.
- **(F5)–(F8)** say the same things about multiplication, with one change: only **non-zero** elements are required to have inverses. Division is defined by $a/b := ab^{-1}$ for $b \neq 0$. We never divide by 0, and the axioms never ask us to.
- **(F7)** also insists that $1 \neq 0$: the two neutral elements are different.
- **(F9)** is the one rule linking the two operations. It is what lets us expand brackets.

Note the order of quantifiers in (F3): **one** element 0 works **for every** a . This is stronger than “for every a there is some element z_a with $a + z_a = a$ ”. In (F4) the order is reversed, as it must be: the negative $-a$ depends on a .

We often say “ F is a field” when the operations are clear, and write $(F, +, \cdot)$ when they need to be named. Since addition and multiplication are functions into F , sums and products of elements of F stay in F ; when we build a field inside a bigger number system, this “closure” is the first thing to check.

0.7.3 First consequences of the axioms

The notation in the definition hides a question. We wrote “0”, “ $-a$ ” and “ a^{-1} ” as if each were a single well-defined element. The axioms only say that such elements **exist**. If a field could contain two different elements behaving like 0, the symbol 0 would be ambiguous, and so would $a - b$ and a/b . The next theorem removes this worry and proves the familiar rules of algebra from the nine axioms.

Theorem 0.7.1 — Basic Properties of Fields

Let F be a field and let $a, b, c \in F$.

- The element 0 in (F3) is unique, and the element 1 in (F7) is unique.
- (Cancellation) If $a + b = a + c$, then $b = c$. If $ab = ac$ and $a \neq 0$, then $b = c$.
- The negative $-a$ in (F4) is unique, and for $a \neq 0$ the inverse a^{-1} in (F8) is unique.
- $0 \cdot a = 0$.
- $(-1)a = -a$.
- If $ab = 0$, then $a = 0$ or $b = 0$.

Idea. Each part is a short chain of axioms, and the chain is found backwards from the goal. For uniqueness we use the standard move: suppose two elements both have the property, and show they are equal. For (d) there is no axiom about $0 \cdot a$ at all, so we must create a situation where one applies. The trick is to rewrite 0 in a fancy way as $0 + 0$, expand with (F9), and cancel. For (f), the hypothesis $a \neq 0$ is there to be spent on (F8).

We prove the uniqueness of 0, additive cancellation, the uniqueness of negatives, (d) and (f). The remaining statements are [Exercise \(B1: The Remaining Basic Properties\)](#).

Proof.

- (a) Suppose 0 and $0'$ both satisfy (F3). Applying (F3) for $0'$ to the element 0, and (F3) for 0 to the element $0'$, and using (F1),

$$0 = 0 + 0' = 0' + 0 = 0'.$$

So the zero element is unique.

- (b) Suppose $a + b = a + c$. Adding $-a$ on the left of both sides and using (F2),

$$((-a) + a) + b = ((-a) + a) + c.$$

By (F1) and (F4), $(-a) + a = a + (-a) = 0$. Hence $0 + b = 0 + c$, and by (F1) and (F3) this says $b = c$.

- (c) Suppose b and c both satisfy (F4) for a , that is, $a + b = 0$ and $a + c = 0$. Then $a + b = a + c$, so $b = c$ by additive cancellation in (b).
 (d) By (F3), $0 + 0 = 0$. Therefore, by (F9),

$$a \cdot 0 = a(0 + 0) = a \cdot 0 + a \cdot 0.$$

By (F3), also $a \cdot 0 = a \cdot 0 + 0$. Comparing, $a \cdot 0 + 0 = a \cdot 0 + a \cdot 0$, and additive cancellation (b) gives $0 = a \cdot 0$. Finally $0 \cdot a = a \cdot 0$ by (F6), so $0 \cdot a = 0$.

- (e) Suppose $ab = 0$. If $a = 0$ there is nothing to prove, so suppose $a \neq 0$. By (F8), a^{-1} exists. Then, by (F7), (F8), (F6) and (F5),

$$b = b \cdot 1 = b(aa^{-1}) = (ba)a^{-1} = (ab)a^{-1} = 0 \cdot a^{-1} = 0,$$

where the last equality is (d). This shows that $a = 0$ or $b = 0$. ■

Two small consequences are used without comment from now on: $0 + a = a$ and $1 \cdot a = a$, by (F1), (F3), (F6) and (F7). More importantly, every identity of school algebra that is proved from these rules, such as $(a + b)^2 = a^2 + 2ab + b^2$ or $(-a)(-b) = ab$, now holds in every field. Part (f) is the property we will use most: a product of non-zero elements is non-zero.

0.7.4 Examples

We check the standard examples first, then an exotic one, then a degenerate case.

Example. (The Fields $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$)

Explain why $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$, with their usual addition and multiplication, are fields.

Solution. For $\mathbb{R}$, the axioms (F1)–(F9) are the rules of arithmetic for real numbers that we take as familiar, and $1 \neq 0$.

For $\mathbb{Q}$, the sum, product and negative of fractions are fractions, and so is the inverse $\frac{q}{p}$ of a non-zero fraction $\frac{p}{q}$. So the operations of $\mathbb{R}$ restrict to functions $\mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$, and $0, 1, -a, a^{-1}$ lie in $\mathbb{Q}$ whenever a does. The identities in (F1), (F2), (F5), (F6) and (F9)

hold for all real numbers, so in particular for rational ones. Hence $\mathbb{Q}$ is a field.

For $\mathbb{C}$, the axioms (F1)–(F6) and (F9), together with the existence of $1 = 1 + 0i$, are [Proposition 0.6.1](#). We have $1 = (1, 0) \neq (0, 0) = 0$. Finally (F8) is [Proposition 0.6.3](#): a non-zero z has the inverse $\bar{z}/|z|^2$. Hence $\mathbb{C}$ is a field.

The next example is a field strictly between $\mathbb{Q}$ and $\mathbb{R}$. It shows that the inverse axiom (F8) can take real work to check.

Example. ($\mathbb{Q}(\sqrt{2})$ Is a Field)

Let $\mathbb{Q}(\sqrt{2}) := \{a + b\sqrt{2} : a, b \in \mathbb{Q}\} \subseteq \mathbb{R}$, with the addition and multiplication of $\mathbb{R}$. Show that $\mathbb{Q}(\sqrt{2})$ is a field, and find the inverse of $1 + 2\sqrt{2}$.

Solution. Let $x = a + b\sqrt{2}$ and $y = c + d\sqrt{2}$ with $a, b, c, d \in \mathbb{Q}$.

Closure. We have $x + y = (a + c) + (b + d)\sqrt{2}$ and $xy = (ac + 2bd) + (ad + bc)\sqrt{2}$, using $\sqrt{2} \cdot \sqrt{2} = 2$. The coefficients are rational, so $x + y, xy \in \mathbb{Q}(\sqrt{2})$.

Axioms that are identities. (F1), (F2), (F5), (F6) and (F9) hold for all real numbers, hence for the elements of $\mathbb{Q}(\sqrt{2})$.

Neutral elements and negatives. $0 = 0 + 0\sqrt{2}$ and $1 = 1 + 0\sqrt{2}$ lie in $\mathbb{Q}(\sqrt{2})$, and $1 \neq 0$. The negative $-x = (-a) + (-b)\sqrt{2}$ lies in $\mathbb{Q}(\sqrt{2})$.

Inverses. Suppose $x \neq 0$. We first claim $a - b\sqrt{2} \neq 0$. If $b \neq 0$ and $a - b\sqrt{2} = 0$, then $\sqrt{2} = a/b$ would be rational, contradicting [Theorem 0.2.3](#). If $b = 0$, then $a = x \neq 0$, so $a - b\sqrt{2} = a \neq 0$. Therefore $a - b\sqrt{2} \neq 0$, and by [Theorem 0.7.1](#) (f) in $\mathbb{R}$,

$$N := (a + b\sqrt{2})(a - b\sqrt{2}) = a^2 - 2b^2 \neq 0.$$

Since N is a non-zero rational number, the element

$$x' := \frac{a}{N} + \frac{-b}{N}\sqrt{2} \in \mathbb{Q}(\sqrt{2})$$

satisfies $xx' = \frac{1}{N}(a + b\sqrt{2})(a - b\sqrt{2}) = 1$. So (F8) holds, and $\mathbb{Q}(\sqrt{2})$ is a field.

For $x = 1 + 2\sqrt{2}$ we get $N = 1 - 8 = -7$, so $x^{-1} = -\frac{1}{7} + \frac{2}{7}\sqrt{2} = \frac{-1+2\sqrt{2}}{7}$. Check: $(1 + 2\sqrt{2})(-1 + 2\sqrt{2}) = -1 + 2\sqrt{2} - 2\sqrt{2} + 8 = 7$.

The method is “rationalize the denominator”, the same move as multiplying by the conjugate in $\mathbb{C}$. The irrationality of $\sqrt{2}$ is exactly what guarantees the new denominator is not zero.

Quick check. Is $\{a + b\sqrt{3} : a, b \in \mathbb{Q}\}$, with the operations of $\mathbb{R}$, a field?

Solution. Yes, by the same argument as [Example \(\$\mathbb{Q}\(\sqrt{2}\)\$ Is a Field\)](#). Products stay in the set because $\sqrt{3} \cdot \sqrt{3} = 3$ is rational. For inverses, the number $a^2 - 3b^2$ is non-zero whenever $a + b\sqrt{3} \neq 0$, because $\sqrt{3}$ is irrational ([Exercise \(B3: \$\sqrt{3}\$ is irrational\)](#)). The inverse is $\frac{a-b\sqrt{3}}{a^2-3b^2}$.

The degenerate case. Let $F = \{0\}$ with $0 + 0 = 0$ and $0 \cdot 0 = 0$. Every axiom except one holds: (F1), (F2), (F5), (F6) and (F9) are identities between elements that all equal 0, (F3) and

(F4) hold with $-0 = 0$, and (F8) holds vacuously because there is no non-zero element. The only failure is (F7): the one element that could serve as 1 is 0, and the axiom demands $1 \neq 0$. So $\{0\}$ is not a field, and every field has at least the two elements 0 and 1. This one-element system is the reason the clause $1 \neq 0$ is there.

0.7.5 Finite fields

Every field so far is infinite. The arithmetic of remainders gives finite ones, and they are genuinely useful: computer science and coding theory do linear algebra over them all the time.

Recall from [Definition 0.5.2](#) and [Definition 0.5.5](#) that for an integer $n \geq 1$, congruence modulo n is an equivalence relation on $\mathbb{Z}$. We write $[a]$ for the class of a and $\mathbb{Z}/n\mathbb{Z}$ for the quotient set. By division with remainder ([Exercise \(C2: Division with remainder\)](#)), $\mathbb{Z}/n\mathbb{Z} = \{[0], [1], \dots, [n-1]\}$, and these n classes are distinct. Also $[a] = [b]$ exactly when n divides $a - b$.

Example. (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$)

Let $n \geq 1$. Define $[a] + [b] := [a + b]$ and $[a][b] := [ab]$. Show that these operations are well defined, that they satisfy every field axiom except possibly (F8) and the clause $1 \neq 0$, and that $1 \neq 0$ holds exactly when $n \geq 2$. Write out the tables for $n = 2$.

Solution. Well-definedness. The formulas use representatives, so we must check that the answer does not depend on which representatives we pick. For addition this is [Example \(Addition modulo \$n\$ is well defined\)](#). For multiplication, suppose $[a] = [a']$ and $[b] = [b']$, so n divides $a - a'$ and $b - b'$. Then

$$ab - a'b' = a(b - b') + (a - a')b'$$

is divisible by n , so $[ab] = [a'b']$.

Axioms. Every identity in (F1), (F2), (F5), (F6), (F9) transfers from $\mathbb{Z}$, because both sides can be computed on representatives. For instance, for (F9),

$$[a]([b] + [c]) = [a][b + c] = [a(b + c)] = [ab + ac] = [ab] + [ac] = [a][b] + [a][c].$$

The same one-line computation works for the other four. The class $[0]$ satisfies (F3) since $[a] + [0] = [a]$, the class $[-a]$ is a negative of $[a]$ as in (F4), and $[1]$ satisfies $[a][1] = [a]$.

The clause $1 \neq 0$. We have $[1] = [0]$ exactly when n divides 1, that is, when $n = 1$. So $[1] \neq [0]$ exactly when $n \geq 2$.

Tables for $n = 2$. Writing 0, 1 for $[0], [1]$, the set $\mathbb{F}_2 := \mathbb{Z}/2\mathbb{Z} = \{0, 1\}$ has the tables

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

The only surprise is $1 + 1 = 0$, since 2 is even. Here (F8) holds as well, since the only non-zero element is 1 and $1 \cdot 1 = 1$. Hence $\mathbb{F}_2$ is a field.

So for $n \geq 2$, whether $\mathbb{Z}/n\mathbb{Z}$ is a field comes down to one question: does every non-zero class have an inverse? To find an inverse of $[a]$ we need an integer x with $[ax] = [1]$, that is, $ax + ny = 1$

for some integer y . The following lemma produces such x and y .

Lemma 0.7.2 — Bézout's Lemma for Integers

Let $a, n \in \mathbb{Z}$ with $n \geq 1$, and suppose that the only positive integer dividing both a and n is 1. Then there exist $x, y \in \mathbb{Z}$ with

$$ax + ny = 1.$$

Idea. Consider all positive integers of the form $ax + ny$. There is a smallest one, d , by well-ordering. The plan: ① show d divides a and n ; ② conclude $d = 1$ from the hypothesis. For ①, divide a by d : the remainder is again of the form $ax + ny$ and is smaller than d , so minimality forces it to be 0.

Proof. Let $S = \{ax + ny : x, y \in \mathbb{Z}\} \cap \{1, 2, 3, \dots\}$. Since $n = a \cdot 0 + n \cdot 1 \geq 1$, we have $n \in S$, so S is a non-empty subset of $\mathbb{N}$. By the well-ordering principle (Theorem 0.2.11), S has a smallest element $d = ax_0 + ny_0$.

By division with remainder (Exercise (C2: Division with remainder)), $a = qd + r$ for some integers q, r with $0 \leq r < d$. Then

$$r = a - q(ax_0 + ny_0) = a(1 - qx_0) + n(-qy_0)$$

has the form $ax + ny$. If $r \geq 1$, then $r \in S$ and $r < d$, contradicting the minimality of d . Hence $r = 0$, and d divides a . The same argument with n in place of a shows that d divides n .

Therefore d is a positive integer dividing both a and n . By hypothesis $d = 1$, so $ax_0 + ny_0 = 1$, as claimed. ■

Now we can decide exactly when $\mathbb{Z}/n\mathbb{Z}$ is a field.

Theorem 0.7.3 — $\mathbb{Z}/n\mathbb{Z}$ Is a Field Exactly for Prime n

Let $n \geq 2$ be an integer. Then $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is prime.

Idea. By Example (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$), only (F8) is in question. If n is prime, a non-zero class $[a]$ has a not divisible by n , so a and n share no factor, and Bézout hands us the inverse. If $n = rs$ factors, then $[r][s] = [n] = [0]$ with both factors non-zero, which no field allows.

Proof. By Example (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$), $\mathbb{Z}/n\mathbb{Z}$ satisfies all field axioms except possibly (F8), including $[1] \neq [0]$ since $n \geq 2$.

($\Leftarrow$) Suppose $n = p$ is prime, and let $[a] \neq [0]$. Then p does not divide a . A positive integer dividing p is 1 or p , and p does not divide a , so the only positive common divisor of a and p is 1. By Lemma 0.7.2 there are $x, y \in \mathbb{Z}$ with $ax + py = 1$. Since $ax - 1 = -py$ is divisible by p ,

$$[a][x] = [ax] = [1].$$

So $[x]$ is an inverse of $[a]$, and (F8) holds. Hence $\mathbb{Z}/p\mathbb{Z}$ is a field.

($\Rightarrow$) We prove the contrapositive. Suppose $n \geq 2$ is not prime. Then $n = rs$ for some integers r, s with $1 < r < n$ and $1 < s < n$. Since $0 < r < n$, n does not divide r , so $[r] \neq [0]$; likewise $[s] \neq [0]$. But $[r][s] = [rs] = [n] = [0]$. By Theorem 0.7.1 (f), this cannot happen in a field. Hence $\mathbb{Z}/n\mathbb{Z}$ is not a field. This proves the theorem. ■

For a prime p we write $\mathbb{F}_p := \mathbb{Z}/p\mathbb{Z}$, and usually drop the brackets, writing its elements as $0, 1, \dots, p-1$ and computing “mod p ”. So $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7, \dots$ are fields, while $\mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/6\mathbb{Z}$ and $\mathbb{Z}/9\mathbb{Z}$ are not.

Quick check. Find the inverse of 3 in $\mathbb{F}_7$.

Solution. We need x with $3x$ leaving remainder 1 on division by 7. Trying $x = 1, 2, \dots$: $3 \cdot 5 = 15 = 2 \cdot 7 + 1$. So $3^{-1} = 5$ in $\mathbb{F}_7$. (By Theorem 0.7.1 (c) this is the only inverse.)

Because $\mathbb{F}_p$ is a field, the elimination from the start of the section runs unchanged in it. Only the arithmetic is new.

Example. (Elimination over $\mathbb{F}_5$)

Solve the system $2x + y = 3$, $3x + 2y = 4$ over $\mathbb{F}_5$.

Solution. In $\mathbb{F}_5$ the inverses are $1^{-1} = 1$, $2^{-1} = 3$, $3^{-1} = 2$, $4^{-1} = 4$, since $2 \cdot 3 = 6 = 5 + 1$ and $4 \cdot 4 = 16 = 15 + 1$. Multiply the first equation by $2^{-1} = 3$: since $3 \cdot 2 = 1$, $3 \cdot 1 = 3$ and $3 \cdot 3 = 9 = 4$, it becomes $x + 3y = 4$. Subtract 3 times this from the second equation: the x -terms cancel, and we get

$$(2 - 9)y = 4 - 12, \quad \text{that is,} \quad 3y = 2,$$

since $-7 = -10 + 3$ and $-8 = -10 + 2$. Multiply by $3^{-1} = 2$: $y = 4$. Then $x = 4 - 3y = 4 - 12 = -8 = 2$. Check: $2 \cdot 2 + 4 = 8 = 3$ and $3 \cdot 2 + 2 \cdot 4 = 14 = 4$ in $\mathbb{F}_5$. Hence $(x, y) = (2, 4)$ is the unique solution.

0.7.6 Non-examples

Each of these fails some part of the slogan, and it pays to name the exact clause that fails.

- $\mathbb{Z}$. Everything but (F8) holds: $\mathbb{Z}$ is closed under $+$ and $\cdot$, contains 0, 1 and negatives, and inherits the identities from $\mathbb{R}$. But $2 \neq 0$ has no inverse in $\mathbb{Z}$, since $2m$ is even and never equals 1. Changing $\mathbb{Z}$ to $\mathbb{Q}$ is the minimal change that repairs it.
- $\mathbb{Z}/6\mathbb{Z}$. This is $\mathbb{F}_5$ with the modulus changed from 5 to 6. All axioms but (F8) still hold (Example (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$)), but $[2][3] = [0]$ with $[2], [3] \neq [0]$. By Theorem 0.7.1 (f), $[2]$ cannot have an inverse; indeed $[2][x]$ is always one of $[0], [2], [4]$.
- 2×2 **real matrices**, with the matrix operations you know from school (they are treated properly later in this chapter). The identity matrix plays the role of 1. Two axioms fail.

(F6) fails:

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

(F8) fails too: $E = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ is not the zero matrix, but every product EX has second row zero, so EX is never the identity.

Warning. “[a] $\neq$ [0]” does not mean [a] can be divided by in $\mathbb{Z}/n\mathbb{Z}$. In $\mathbb{Z}/6\mathbb{Z}$ the equation $[2]x = [1]$ has no solution, and $[2]x = [4]$ has two, $x = [2]$ and $x = [5]$. Before dividing modulo n , check that n is prime, or at least that a and n share no factor.

0.7.7 Why these axioms

Each clause is there because something we rely on breaks without it.

- **Without (F8)**, elimination stalls, as $2x = 1$ over $\mathbb{Z}$ showed. Worse, the theorems of the next chapters fail: over $\mathbb{Z}$, the analogues of vector spaces (called modules) need not have bases, and “a set of n independent vectors in an n -dimensional space spans it” becomes false. This book always works over a field.
- **Without (F6)**, scalars would not commute with each other. The matrices above show such systems exist, and there are even systems with division but without commutativity. Linear algebra constantly swaps scalars, as in $a(b\mathbf{v}) = b(a\mathbf{v})$ for a vector $\mathbf{v}$ (Chapter 1), so we keep commutativity.
- **Without $1 \neq 0$** , the one-element system $\{0\}$ would count, and every statement of the form “a field has a non-zero element” would fail. Keeping it costs nothing.
- **The name** “field” is historical and carries no extra meaning.

Remark. Many identities we are used to are consequences of the axioms and hold in **every** field, including $\mathbb{F}_2$. But properties that involve **order** or **size**, such as “squares are non-negative” or “ $1 + 1 + \cdots + 1 \neq 0$ ”, are not among the axioms. The first already fails in $\mathbb{C}$, where $i^2 = -1$, as the warning on ordering in the previous section explains. The second fails in $\mathbb{F}_p$, which is the subject of the last part of this section.

0.7.8 The characteristic of a field

In $\mathbb{Q}$ we can add 1 to itself as often as we like without reaching 0. In $\mathbb{F}_2$ two copies already give 0, and in $\mathbb{F}_p$ it takes p copies. This number is the most basic invariant of a field, and it decides whether familiar arguments such as “ $2x = 0$ implies $x = 0$ ” are valid.

First some notation. For $a \in F$ and $m \in \mathbb{N}$, define the **integer multiple** $m \cdot a$ by $0 \cdot a := 0$ and $(m+1) \cdot a := m \cdot a + a$. So $m \cdot a = a + a + \cdots + a$ with m terms. This agrees with the field product when $m = 0$, by [Theorem 0.7.1](#) (d).

Lemma 0.7.4 — Integer Multiples

Let F be a field, $a \in F$ and $m, k \in \mathbb{N}$. Then

$$(m \cdot 1) + (k \cdot 1) = (m + k) \cdot 1, \quad (m \cdot 1)(k \cdot 1) = (mk) \cdot 1, \quad (m \cdot 1)a = m \cdot a.$$

Proof. All three follow by induction on one variable ([Theorem 0.2.7](#)). For the first, induct on k : the case $k = 0$ is $m \cdot 1 + 0 = m \cdot 1$ by (F3), and if it holds for k , then by (F2) and the definition,

$$m \cdot 1 + (k + 1) \cdot 1 = (m \cdot 1 + k \cdot 1) + 1 = (m + k) \cdot 1 + 1 = (m + k + 1) \cdot 1.$$

For the third, induct on m : the case $m = 0$ is $0 \cdot a = 0$ by [Theorem 0.7.1](#) (d), and if it holds for m , then by (F6), (F9) and (F7),

$$((m + 1) \cdot 1)a = (m \cdot 1 + 1)a = (m \cdot 1)a + 1 \cdot a = m \cdot a + a = (m + 1) \cdot a.$$

The second is the third with $a = k \cdot 1$, combined with $m \cdot (k \cdot 1) = (mk) \cdot 1$, which follows from the first by induction on m . ■

Definition 0.7.2 — Characteristic

Let F be a field. If there is a **positive** integer n with $n \cdot 1 = 0$, the **characteristic** of F is the **smallest** such n . If there is no such n , the characteristic of F is 0.

In words: count how many copies of 1 must be added to reach 0. If it never happens, the convention is to say “characteristic 0”, not “characteristic ∞ ”: in that case the only integer $m \geq 0$ with $m \cdot 1 = 0$ is $m = 0$.

- $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ have characteristic 0, because $n \cdot 1 = n \neq 0$ for every positive integer n .
- $\mathbb{F}_p$ has characteristic p : $n \cdot [1] = [n]$, which is $[0]$ exactly when p divides n , and the smallest positive such n is p .
- $\mathbb{F}_2$ is the smallest case: characteristic 2, because $1 + 1 = 0$.

Can a field have characteristic 6? In $\mathbb{Z}/6\mathbb{Z}$, six copies of $[1]$ give $[0]$, but $\mathbb{Z}/6\mathbb{Z}$ is not a field. This is no accident.

Theorem 0.7.5 — The Characteristic Is 0 or Prime

The characteristic of a field is either 0 or a prime number.

Idea. Suppose the characteristic n factors as rs with smaller factors. Then the elements $r \cdot 1$ and $s \cdot 1$ are non-zero, by minimality of n , but their product is $n \cdot 1 = 0$. A field has no such pairs. This is the same argument as in [Theorem 0.7.3](#), run inside an arbitrary field.

Proof. Let F be a field of characteristic $n \neq 0$. Since the integer multiple $1 \cdot 1 = 0 + 1 = 1$ is non-zero by (F7), we have $n \neq 1$, so $n \geq 2$. Suppose, for a contradiction, that n is not

prime. Then $n = rs$ with integers $1 < r < n$ and $1 < s < n$. By [Lemma 0.7.4](#),

$$(r \cdot 1)(s \cdot 1) = (rs) \cdot 1 = n \cdot 1 = 0.$$

By [Theorem 0.7.1](#) (f), $r \cdot 1 = 0$ or $s \cdot 1 = 0$. Either way we have a positive integer smaller than n whose multiple of 1 is 0, contradicting the minimality of n . Hence n is prime. ■

So the characteristic is 0, 2, 3, 5, 7, and so on. Together with [Exercise \(C2: Finite Fields Have Prime Characteristic\)](#), this says a finite field always has prime characteristic.

Warning. In a field of characteristic 2, such as $\mathbb{F}_2$, we have $1 + 1 = 0$, so $-1 = 1$ and $x = -x$ for **every** x . The familiar argument “ $x = -x$, so $2x = 0$, so $x = 0$ ” divides by $2 = 1 + 1$, which is 0 there. For example, in $\mathbb{F}_2$ the element 1 satisfies $1 = -1$ but is not 0. This is why later results, notably on symmetric bilinear forms in Chapter 13, carry the hypothesis “characteristic $\neq 2$ ”.

With fields in hand, the stage for linear algebra is set: in Chapter 1, a vector space is defined **over a field** F , and every result proved there holds at once for $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Q}(\sqrt{2})$ and $\mathbb{F}_p$.

0.7.9 Exercises

0.7.9.1 A. Check your understanding

Exercise A1.

- (a) Define what it means for a set F with two operations to be a field.
- (b) True or false: in every field, $a^2 = 0$ implies $a = 0$. Justify your answer.
- (c) True or false: $\mathbb{Z}/8\mathbb{Z}$ is a field. Justify your answer.
- (d) True or false: in every field, $1 + 1 \neq 0$. Justify your answer.
- (e) What are the characteristics of $\mathbb{C}$ and of $\mathbb{F}_7$?
- (f) The set $\mathbb{N}$ with the usual addition and multiplication is not a field. Name two axioms that fail, each with a witness.

Solution.

- (a) See [Definition 0.7.1](#): two operations satisfying (F1)–(F9), including $1 \neq 0$.
- (b) True. If $a \cdot a = 0$, then $a = 0$ or $a = 0$ by [Theorem 0.7.1](#) (f).
- (c) False. Since $8 = 2 \cdot 4$ is not prime, $\mathbb{Z}/8\mathbb{Z}$ is not a field by [Theorem 0.7.3](#); concretely $[2][4] = [0]$ with $[2], [4] \neq [0]$.
- (d) False. In $\mathbb{F}_2$, $1 + 1 = 0$.
- (e) $\mathbb{C}$ has characteristic 0, and $\mathbb{F}_7$ has characteristic 7 (see the examples after [Definition 0.7.2](#)).
- (f) (F4) fails: $1 \in \mathbb{N}$, but no $b \in \mathbb{N}$ has $1 + b = 0$. (F8) fails: $2 \neq 0$, but no $b \in \mathbb{N}$ has $2b = 1$.

0.7.9.2 B. Practice

Exercise B1: The Remaining Basic Properties.

Let F be a field and $a, b, c \in F$. Using only the axioms and the parts of [Theorem 0.7.1](#) proved in the text, prove:

- (a) the element 1 in (F7) is unique;
- (b) if $ab = ac$ and $a \neq 0$, then $b = c$;
- (c) for $a \neq 0$, the inverse a^{-1} is unique;
- (d) $(-1)a = -a$.

Hint: for (d), show that $a + (-1)a = 0$.

Solution.

- (a) Suppose 1 and $1'$ both satisfy (F7). Then, by (F7) for $1'$, (F6), and (F7) for 1, $1 = 1 \cdot 1' = 1' \cdot 1 = 1'$.
- (b) Suppose $ab = ac$ and $a \neq 0$. By (F8), a^{-1} exists, and multiplying both sides by it gives $(ab)a^{-1} = (ac)a^{-1}$. By (F6) and (F5), $(ab)a^{-1} = (ba)a^{-1} = b(aa^{-1}) = b \cdot 1 = b$, using (F8) and (F7) at the end. In the same way $(ac)a^{-1} = c$. Hence $b = c$.
- (c) Suppose b and c both satisfy (F8) for $a \neq 0$: $ab = 1 = ac$. By (b), $b = c$.
- (d) By (F7) and (F6), $a = 1 \cdot a$. Therefore, using (F6) and (F9),

$$a + (-1)a = 1 \cdot a + (-1)a = a \cdot 1 + a(-1) = a(1 + (-1)) = a \cdot 0 = 0,$$

where the last two equalities use (F4) and [Theorem 0.7.1](#) (d). So $(-1)a$ satisfies (F4) for a . By uniqueness of negatives ([Theorem 0.7.1](#) (c)), $(-1)a = -a$.

Exercise B2: Solving Equations in $\mathbb{F}_7$.

- (a) Solve $3x + 5 = 2$ in $\mathbb{F}_7$.
- (b) Solve the system $x + 2y = 3$, $3x + y = 1$ over $\mathbb{F}_7$.

Solution.

- (a) Adding -5 to both sides gives $3x = 2 - 5 = -3 = 4$ in $\mathbb{F}_7$. Since $3 \cdot 5 = 15 = 1$ in $\mathbb{F}_7$, we have $3^{-1} = 5$. Multiplying by 5, $x = 20 = 6$. Check: $3 \cdot 6 + 5 = 23 = 2$ in $\mathbb{F}_7$. Hence $x = 6$.
- (b) Subtract 3 times the first equation from the second: $(1 - 6)y = 1 - 9$, that is, $-5y = -8$, or $2y = 6$ in $\mathbb{F}_7$. Since $2 \cdot 4 = 8 = 1$, $2^{-1} = 4$, so $y = 24 = 3$. Then $x = 3 - 2y = 3 - 6 = -3 = 4$. Check: $4 + 2 \cdot 3 = 10 = 3$ and $3 \cdot 4 + 3 = 15 = 1$ in $\mathbb{F}_7$. Hence $(x, y) = (4, 3)$.

Exercise B3: Which Are Fields?

Determine which of the following are fields, with the stated operations. Justify your answer.

- (a) $\mathbb{Z}/9\mathbb{Z}$, with the operations of [Example \(Arithmetic on \$\mathbb{Z}/n\mathbb{Z}\$ and the Field \$\mathbb{F}_2\$ \)](#).
- (b) $\mathbb{Q}(i) := \{a + bi : a, b \in \mathbb{Q}\}$, with the operations of $\mathbb{C}$.
- (c) $\{a + bi : a, b \in \mathbb{Z}\}$, with the operations of $\mathbb{C}$.
- (d) $\mathbb{F}_5$.

Solution.

- (a) Not a field. Since $9 = 3 \cdot 3$ is not prime, [Theorem 0.7.3](#) applies; concretely $[3][3] = [0]$ with $[3] \neq [0]$, which is impossible in a field by [Theorem 0.7.1 \(f\)](#).
- (b) A field. Let $x = a + bi$ and $y = c + di$ with $a, b, c, d \in \mathbb{Q}$. Then $x + y = (a + c) + (b + d)i$, $xy = (ac - bd) + (ad + bc)i$ and $-x = -a - bi$ have rational parts, and $0, 1 \in \mathbb{Q}(i)$ with $1 \neq 0$. The identities (F1), (F2), (F5), (F6), (F9) hold because they hold in $\mathbb{C}$ ([Example \(The Fields \$\mathbb{Q}\$, \$\mathbb{R}\$ and \$\mathbb{C}\$ \)](#)). If $x \neq 0$, then $a^2 + b^2$ is a non-zero rational number, and by [Proposition 0.6.3](#) $x^{-1} = \frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2}i$, which lies in $\mathbb{Q}(i)$. So (F8) holds.
- (c) Not a field: (F8) fails for 2. Suppose $u = a + bi$ with $a, b \in \mathbb{Z}$ satisfied $2u = 1$. This is also an equation in the field $\mathbb{C}$, where $\frac{1}{2}$ is an inverse of 2; by uniqueness of inverses in $\mathbb{C}$ ([Theorem 0.7.1 \(c\)](#)), $u = \frac{1}{2}$. But $\frac{1}{2}$ has real part $\frac{1}{2} \notin \mathbb{Z}$, a contradiction.
- (d) A field, by [Theorem 0.7.3](#), since 5 is prime.

0.7.9.3 C. Going deeper

Exercise C1: A Field with Four Elements.

Let $F = \{a + b\alpha : a, b \in \mathbb{F}_2\}$, a set of four formal symbols $0, 1, \alpha, \beta$, where $\beta := 1 + \alpha$. Define

$$(a + b\alpha) + (c + d\alpha) := (a + c) + (b + d)\alpha, \quad (a + b\alpha)(c + d\alpha) := (ac + bd) + (ad + bc + bd)\alpha,$$

with all coefficient arithmetic in $\mathbb{F}_2$. (The product is what expanding gives if we replace α^2 by $\alpha + 1$.)

- (a) Write out the addition and multiplication tables of F .
- (b) Prove that F is a field.
- (c) Explain why there is no bijection $\varphi: F \rightarrow \mathbb{Z}/4\mathbb{Z}$ with $\varphi(x + y) = \varphi(x) + \varphi(y)$ and $\varphi(xy) = \varphi(x)\varphi(y)$ for all $x, y \in F$. So this four-element field is not “ $\mathbb{Z}/4\mathbb{Z}$ in disguise”.

Hint: for (c), look for a non-zero element of $\mathbb{Z}/4\mathbb{Z}$ whose square is zero.

Solution.

- (a) Using the formulas, with $1 + 1 = 0$ in $\mathbb{F}_2$: $\begin{array}{c|ccc} & 0 & 1 & \alpha & \beta \\ \hline 0 & 0 & 1 & \alpha & \beta \\ 1 & 1 & 0 & \beta & \alpha \\ \alpha & \alpha & \beta & 0 & 1 \\ \beta & \beta & \alpha & 1 & 0 \end{array}$

• $\alpha^2 = \alpha + 1$

$\alpha^3 = \alpha(\alpha + 1) = \alpha^2 + \alpha = (\alpha + 1) + \alpha = 1$

$\cdot$	0	1	α	β
0	0	0	0	0
1	0	1	α	β
α	0	α	β	1
β	0	β	1	α

] For instance, $\alpha \cdot \alpha = (0 + 1\alpha)(0 + 1\alpha) = (0 + 1) + (0 + 0 + 1)\alpha = 1 + \alpha = \beta$, and $\alpha\beta = (0 + 1\alpha)(1 + 1\alpha) = (0 + 1) + (0 + 1 + 1)\alpha = 1$.

- (b) Let $x = a + b\alpha$, $y = c + d\alpha$, $z = e + f\alpha$ with $a, \dots, f \in \mathbb{F}_2$. Since $\mathbb{F}_2$ is a field (Example (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$)), we may compute with the coefficients using the rules of algebra.

Addition. It acts coordinate by coordinate, so (F1) and (F2) follow from the same laws in $\mathbb{F}_2$, the element $0 = 0 + 0\alpha$ satisfies (F3), and $-x = x$ satisfies (F4) because $a + a = 0$ and $b + b = 0$ in $\mathbb{F}_2$.

(F6). The formula for xy is unchanged when (a, b) and (c, d) are swapped: $ac + bd = ca + db$ and $ad + bc + bd = cb + da + db$.

(F7). With $1 = 1 + 0\alpha$, $x \cdot 1 = (a + 0) + (0 + b + 0)\alpha = x$, and $1 \neq 0$ since the coefficients differ.

(F9). Since $y + z = (c + e) + (d + f)\alpha$,

$$x(y + z) = (a(c + e) + b(d + f)) + (a(d + f) + b(c + e) + b(d + f))\alpha,$$

and expanding the coefficients shows this is $((ac + bd) + (ae + bf)) + ((ad + bc + bd) + (af + be + bf))\alpha = xy + xz$.

(F5). Write $xy = g + h\alpha$ with $g = ac + bd$, $h = ad + bc + bd$. Then $(xy)z = (ge + hf) + (gf + he + hf)\alpha$. Expanding,

$$ge + hf = ace + adf + bcf + bde + bdf, \quad gf + he + hf = acf + ade + adf + bce + bcf + bde,$$

where in the second we used $bdf + bdf = 0$ in $\mathbb{F}_2$. Similarly $yz = k + l\alpha$ with $k = ce + df$, $l = cf + de + df$, and $x(yz) = (ak + bl) + (al + bk + bl)\alpha$. Expanding gives $ak + bl = ace + adf + bcf + bde + bdf$ and $al + bk + bl = acf + ade + adf + bce + bcf + bde$, again using $bdf + bdf = 0$. Hence $(xy)z = x(yz)$.

(F8). From the table, $1 \cdot 1 = 1$, $\alpha\beta = 1$ and $\beta\alpha = 1$, so every non-zero element has an inverse.

This proves that F is a field.

- (c) Suppose such a bijection φ existed. First, $\varphi(0) = \varphi(0 + 0) = \varphi(0) + \varphi(0)$, so additive cancellation in $\mathbb{Z}/4\mathbb{Z}$ (adding $-\varphi(0)$, which only uses (F1)–(F4), valid there by Example (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$)) gives $\varphi(0) = [0]$. Since φ is surjective, there is $x \in F$ with $\varphi(x) = [2]$. As $[2] \neq [0] = \varphi(0)$, we have $x \neq 0$. But

$$\varphi(x \cdot x) = \varphi(x)\varphi(x) = [2][2] = [4] = [0] = \varphi(0),$$

so $x \cdot x = 0$ because φ is injective. This contradicts Theorem 0.7.1 (f) in the field F (from (b)). Hence no such φ exists.

Exercise C2: Finite Fields Have Prime Characteristic.

- (a) Let F be a finite field. Prove that the characteristic of F is a prime p .
 (b) Let F be a field of characteristic $p > 0$. Prove that $p \cdot a = 0$ for every $a \in F$.

Hint: for (a), the elements $1 \cdot 1, 2 \cdot 1, 3 \cdot 1, \dots$ cannot all be different.

Solution.

- (a) The elements $m \cdot 1$ for $m = 1, 2, 3, \dots$ all lie in the finite set F , so they cannot all be distinct. Hence there are positive integers $k < m$ with $m \cdot 1 = k \cdot 1$. By Lemma 0.7.4, $m \cdot 1 = k \cdot 1 + (m - k) \cdot 1$, so $k \cdot 1 + (m - k) \cdot 1 = k \cdot 1 + 0$, and additive cancellation (Theorem 0.7.1 (b)) gives $(m - k) \cdot 1 = 0$ with $m - k \geq 1$. Therefore the characteristic of F is not 0. By Theorem 0.7.5, it is a prime p .
- (b) Let $a \in F$. By Lemma 0.7.4 and the definition of characteristic, $p \cdot a = (p \cdot 1) a = 0 \cdot a = 0$, where the last step is Theorem 0.7.1 (d).

Exercise C3: The Frobenius Identity.

Let p be a prime.

- (a) Let $a, b \in \mathbb{Z}$ with p dividing ab . Prove that p divides a or p divides b .
- (b) Deduce that p divides the binomial coefficient $\binom{p}{k}$ for $0 < k < p$.
- (c) Let F be a field of characteristic p . Prove that $(x + y)^p = x^p + y^p$ for all $x, y \in F$.

You may use the binomial theorem $(x + y)^n = \sum_{k=0}^n \binom{n}{k} \cdot (x^k y^{n-k})$ in any field, with integer multiples as in Lemma 0.7.4; its inductive proof uses only the field axioms.

Hint: for (a), use Lemma 0.7.2. For (b), use $p! = \binom{p}{k} k! (p - k)!$.

Solution.

- (a) Suppose p does not divide a . A positive integer dividing p is 1 or p , so the only positive common divisor of a and p is 1. By Lemma 0.7.2 there are $x, y \in \mathbb{Z}$ with $ax + py = 1$. Multiplying by b , $b = (ab)x + p(by)$. Both terms on the right are divisible by p , the first because p divides ab . Hence p divides b .
- (b) First, by (a) and induction on the number of factors, if p divides a product of integers then it divides one of the factors. The integers $1, 2, \dots, p - 1$ are not divisible by p , so p divides neither $k!$ nor $(p - k)!$ for $0 < k < p$, and hence, by (a) once more, not their product. Now p divides $p! = \binom{p}{k} \cdot (k! (p - k)!)$. By (a), p divides $\binom{p}{k}$.
- (c) Let $x, y \in F$. By the binomial theorem,

$$(x + y)^p = x^p + y^p + \sum_{k=1}^{p-1} \binom{p}{k} \cdot (x^k y^{p-k}),$$

since $\binom{p}{0} = \binom{p}{p} = 1$. Fix $0 < k < p$. By (b), $\binom{p}{k} = pm$ for some $m \in \mathbb{N}$. By Lemma 0.7.4,

$$\binom{p}{k} \cdot (x^k y^{p-k}) = ((pm) \cdot 1)(x^k y^{p-k}) = (p \cdot 1)(m \cdot 1)(x^k y^{p-k}) = 0,$$

because $p \cdot 1 = 0$ in characteristic p and 0 times anything is 0 (Theorem 0.7.1 (d)). So every middle term vanishes, and $(x + y)^p = x^p + y^p$, as claimed.

0.8 Polynomials

We now have fields: number systems in which we can add, subtract, multiply and divide by anything non-zero. The next objects built on top of a field are polynomials. They will be everywhere in this book: as examples of vectors in Chapter 1, as the characteristic and minimal polynomials of a matrix later on, and as the reason some fields behave differently from others. This section says precisely what a polynomial is, introduces its degree, and explains why a polynomial is **not** the same thing as the function it defines.

0.8.1 What a polynomial is

At school, a polynomial is “an expression like $3x^2 - x + 2$ ”. That is a good picture, but it leaves one question open: what is x ? One tempting answer is that x is a number to be plugged in, so a polynomial is really a function. This answer breaks down as soon as we leave the familiar number systems.

Work over the field $\mathbb{F}_2 = \{0, 1\}$ of [Example \(Arithmetic on \$\mathbb{Z}/n\mathbb{Z}\$ and the Field \$\mathbb{F}_2\$ \)](#), where $1 + 1 = 0$. The expressions $x^2 + x$ and 0 look different. But if we plug in the two elements of $\mathbb{F}_2$, we get $0^2 + 0 = 0$ and $1^2 + 1 = 1 + 1 = 0$. As functions on $\mathbb{F}_2$, the two are the same. If a polynomial *were* its function, then $x^2 + x$ would be the zero polynomial, and it would have no sensible degree. We want $x^2 + x$ to be a genuine polynomial of degree 2, so we must define polynomials by their coefficients alone.

A polynomial is its list of coefficients, and nothing else.

Definition 0.8.1 — Polynomial

Let F be a field. A **polynomial over F** in the indeterminate x is a sequence $(a_0, a_1, a_2, \dots)$ of elements of F , indexed by $\mathbb{N}$, such that **only finitely many** a_k are non-zero. That is, there is some $N \in \mathbb{N}$ with $a_k = 0$ **for every** $k > N$.

The element a_k is the **coefficient of x^k** . We write the polynomial as

$$p = a_0 + a_1x + a_2x^2 + \cdots + a_Nx^N = \sum_{k=0}^N a_kx^k.$$

Two polynomials $p = \sum a_kx^k$ and $q = \sum b_kx^k$ are **equal** if and only if $a_k = b_k$ **for every** $k \in \mathbb{N}$.

In words: a polynomial is an infinite list of coefficients that is eventually all zero. The expression $\sum_{k=0}^N a_kx^k$ is only a way of displaying the list: x^k marks the position k , and we stop writing once the remaining coefficients are all zero. Two polynomials are equal exactly when they agree position by position. A sequence indexed by $\mathbb{N}$ is a function $\mathbb{N} \rightarrow F$, so this is just equality of functions in the sense of [Definition 0.4.1](#).

There is one thing to check about the notation. The number N in the display is not unique: if $a_k = 0$ for every $k > N$, the same is true for every $k > N + 1$. Writing $\sum_{k=0}^N a_kx^k$ or $\sum_{k=0}^{N+1} a_kx^k$ only adds a term $0 \cdot x^{N+1}$, which we agree to drop. So the display names the same sequence whichever admissible N we use. We also drop terms with zero coefficient in the middle, write x

for $1 \cdot x^1$, and write a_0 for a_0x^0 .

Example. (Polynomials over various fields)

Read off the coefficient sequence of each polynomial, and check it against [Definition 0.8.1](#).

$$3x^2 - x + 2 \in \mathbb{R}[x], \quad x^5 + \pi x^3 - \sqrt{2} \in \mathbb{R}[x], \quad (1+i)x - i \in \mathbb{C}[x], \quad x^2 + x \in \mathbb{F}_2[x], \quad 7, \quad 0.$$

(The symbol $F[x]$ for the set of all polynomials over F is defined formally in [Definition 0.8.2](#) below.)

Solution. The first is the sequence $(2, -1, 3, 0, 0, \dots)$ of real numbers. It has finitely many non-zero terms, so it is a polynomial over $\mathbb{R}$.

The second is $(-\sqrt{2}, 0, 0, \pi, 0, 1, 0, \dots)$. Its coefficients are real but irrational; the definition only asks that they lie in the field.

The third is $(-i, 1+i, 0, \dots)$, a polynomial over $\mathbb{C}$.

The fourth is $(0, 1, 1, 0, \dots)$ with entries in $\mathbb{F}_2$. It is **not** the zero sequence, so it is not the zero polynomial, even though it vanishes at both points of $\mathbb{F}_2$.

The constant 7 is the sequence $(7, 0, 0, \dots)$. It is a polynomial over $\mathbb{Q}$, $\mathbb{R}$ or $\mathbb{C}$. Over $\mathbb{F}_2$ it would be $(1, 0, 0, \dots)$, because $7 = 1$ there.

The last, $0 = (0, 0, 0, \dots)$, is the **zero polynomial**. It is the degenerate case: every coefficient is zero, and the finiteness condition holds with $N = 0$. It matters because it is the one polynomial whose degree needs a special convention, as we will see.

Now the non-example, by a minimal change. Take the polynomial $1 + x + x^2$, whose sequence is $(1, 1, 1, 0, 0, \dots)$, and change every later 0 to 1. The result $(1, 1, 1, 1, \dots)$, informally “ $1 + x + x^2 + x^3 + \dots$ ”, is still a sequence of elements of $\mathbb{R}$. What fails is the clause **only finitely many** coefficients are non-zero, so this is **not** a polynomial. (It is a *formal power series*, which we will not need.) Expressions such as $x^{-1} + 1$ and $\sqrt{x}$ are not polynomials either: they are not of the form $\sum a_k x^k$ with $k \in \mathbb{N}$ at all.

Why insist on finitely many non-zero coefficients? Because we want to plug elements of the field into polynomials, which means forming the sum $\sum a_k c^k$, and in a field only finite sums make sense. Finiteness is also what gives every non-zero polynomial a highest power, its degree.

0.8.2 Adding and multiplying polynomials

The coefficient picture tells us how to add and multiply. Adding $3x^2 - x + 2$ and $x + 5$ at school means collecting like powers, so we add coefficients position by position. Multiplying means expanding every product $a_i x^i \cdot b_j x^j = a_i b_j x^{i+j}$ and collecting like powers, so the coefficient of x^k collects all pairs (i, j) with $i + j = k$.

Definition 0.8.2 — The polynomial ring $F[x]$

]

Let F be a field. The set of **all** polynomials over F is denoted $F[x]$. For $p = \sum a_k x^k$ and $q = \sum b_k x^k$ in $F[x]$, define the **sum** $p + q$ and the **product** pq by their coefficients:

$$p + q = \sum_k (a_k + b_k) x^k, \quad pq = \sum_k \left(\sum_{i=0}^k a_i b_{k-i} \right) x^k.$$

We must check that $p + q$ and pq are again polynomials, that is, that their coefficient sequences are eventually zero. Suppose $a_i = 0$ for every $i > m$ and $b_j = 0$ for every $j > n$. If $k > \max(m, n)$, then $a_k + b_k = 0 + 0 = 0$. If $k > m + n$, then in each product $a_i b_{k-i}$ either $i > m$ or $k - i > n$, because otherwise $k = i + (k - i) \leq m + n$. So every term is 0, and the coefficient of x^k in pq is 0. Hence both are polynomials, and addition and multiplication are well defined on $F[x]$.

With this multiplication, $x \cdot x$ has coefficient sequence $(0, 0, 1, 0, \dots)$, and by induction on k the k -th power of the polynomial x is the sequence with a 1 in position k and 0 elsewhere. So the notation x^k in [Definition 0.8.1](#) agrees with the actual power, and $\sum a_k x^k$ is an honest sum of products.

Example. (A product, coefficient by coefficient)

In $\mathbb{R}[x]$, compute $(3x^2 - x + 2)(x + 1)$ from the definition.

Solution. Here $(a_0, a_1, a_2) = (2, -1, 3)$ and $(b_0, b_1) = (1, 1)$, with all other coefficients 0. The coefficient of x^k in the product is $\sum_{i=0}^k a_i b_{k-i}$:

$$\begin{aligned} k = 0 : a_0 b_0 &= 2, \\ k = 1 : a_0 b_1 + a_1 b_0 &= 2 - 1 = 1, \\ k = 2 : a_0 b_2 + a_1 b_1 + a_2 b_0 &= 0 - 1 + 3 = 2, \\ k = 3 : a_0 b_3 + a_1 b_2 + a_2 b_1 + a_3 b_0 &= 0 + 0 + 3 + 0 = 3, \end{aligned}$$

and every coefficient with $k \geq 4$ is 0 by the argument above (here $m + n = 3$). Hence $(3x^2 - x + 2)(x + 1) = 3x^3 + 2x^2 + x + 2$, which is what expanding by hand gives.

The operations obey the rules we expect. We record them once, so that later sections can use them freely.

Theorem 0.8.1 — Arithmetic in $F[x]$

]

Let F be a field and $p, q, r \in F[x]$. Then:

1. $p + q = q + p$ and $(p + q) + r = p + (q + r)$;
2. $p + 0 = p$, and $p + (-p) = 0$, where $-p = \sum (-a_k) x^k$;
3. $pq = qp$ and $(pq)r = p(qr)$;
4. $1 \cdot p = p$, where 1 is the constant polynomial $(1, 0, 0, \dots)$;
5. $p(q + r) = pq + pr$.

Idea. Two polynomials are equal when their coefficients agree, so each identity reduces to an identity between coefficients, which the field axioms supply. For products it helps to write the coefficient of x^k in pq symmetrically, as a sum over all pairs (i, j) of natural

numbers with $i + j = k$.

Proof. Let $p = \sum a_k x^k$, $q = \sum b_k x^k$ and $r = \sum c_k x^k$. By Definition 0.8.1, it suffices to compare the coefficient of x^k on both sides, for each $k \in \mathbb{N}$.

Parts 1, 2 and 4 hold because in each position they are the corresponding field axioms of Definition 0.7.1: for example, the coefficient of x^k in $p + q$ is $a_k + b_k = b_k + a_k$, the coefficient in $q + p$. For 4, the coefficient of x^k in $1 \cdot p$ is $\sum_{i+j=k} e_i a_j$ with $e_0 = 1$ and $e_i = 0$ for $i \geq 1$, which equals a_k .

For 3, the coefficient of x^k in pq is $\sum_{i+j=k} a_i b_j$. Since multiplication in F is commutative, this equals $\sum_{i+j=k} b_j a_i$, the coefficient in qp . Next, the coefficient of x^k in $(pq)r$ is

$$\sum_{s+l=k} \left(\sum_{i+j=s} a_i b_j \right) c_l = \sum_{i+j+l=k} a_i b_j c_l,$$

where the equality uses distributivity and associativity in F . The same computation for $p(qr)$ gives $\sum_{i+s=k} a_i \left(\sum_{j+l=s} b_j c_l \right) = \sum_{i+j+l=k} a_i b_j c_l$. Therefore $(pq)r = p(qr)$.

For 5, the coefficient of x^k in $p(q + r)$ is $\sum_{i+j=k} a_i (b_j + c_j) = \sum_{i+j=k} a_i b_j + \sum_{i+j=k} a_i c_j$ by distributivity in F , which is the coefficient in $pq + pr$. This proves the theorem. ■

A set with an addition and a multiplication satisfying these rules is called a **commutative ring**; this is where the name “polynomial ring” comes from. We will not need the word again in this chapter. What matters is that we may now calculate with polynomials exactly as at school.

Quick check. Over $\mathbb{R}$, are $(x + 1)^2$ and $x^2 + 2x + 1$ equal as polynomials? Over $\mathbb{F}_2$, are $(x + 1)^2$ and $x^2 + 1$ equal as polynomials?

Solution. Both answers are yes, but for a reason that depends on the field. Expanding gives $(x + 1)^2 = x^2 + 2x + 1$, with coefficient sequence $(1, 2, 1, 0, \dots)$. Over $\mathbb{R}$ this is exactly the sequence of $x^2 + 2x + 1$. Over $\mathbb{F}_2$ we have $2 = 1 + 1 = 0$, so the sequence is $(1, 0, 1, 0, \dots)$, which is the sequence of $x^2 + 1$.

0.8.3 Degree

The most useful single number attached to a polynomial is the highest power of x that actually occurs. Every non-zero polynomial has one: it has at least one non-zero coefficient and only finitely many, so there is a largest position holding a non-zero coefficient. The zero polynomial has no such position, and we must decide what to do with it.

Definition 0.8.3 — Degree

Let $p = \sum a_k x^k \in F[x]$.

- If $p \neq 0$, the **degree** of p , written $\deg p$, is the **largest** $n \in \mathbb{N}$ with $a_n \neq 0$. The coefficient a_n is the **leading coefficient** of p , and p is **monic** if its leading coefficient is 1.

- The zero polynomial has degree $\deg 0 = -\infty$.

We use the conventions $-\infty < n$ and $-\infty + n = n + (-\infty) = -\infty$ for every $n \in \mathbb{N}$, and $-\infty + (-\infty) = -\infty$.

Examples, simplest first. A non-zero constant c has degree 0; its leading coefficient is c . The polynomial $x + 5 \in \mathbb{R}[x]$ is monic of degree 1. The polynomial $3x^3 + 2x^2 + x + 2$ from [Example \(A product, coefficient by coefficient\)](#) has degree 3 and leading coefficient 3. Over $\mathbb{F}_2$, $x^2 + x$ has degree 2, even though it vanishes at every point of $\mathbb{F}_2$.

Why $-\infty$, and not 0 or -1 ? Look at the formula we expect: the degree of a product is the sum of the degrees. With $\deg 0 = 0$ it fails at once, since $0 \cdot x = 0$ would give $0 = \deg 0 = \deg 0 + \deg x = 1$. With $\deg 0 = -1$ it fails too: $-1 = \deg(0 \cdot x) \neq \deg 0 + \deg x = 0$. The formula needs $\deg 0 + n = \deg 0$ for every n , and $-\infty$ is exactly the symbol with that property. The convention keeps a formula true, and that is its only job.

It turns out that the degree of a sum is controlled by the larger degree, and the degree of a product is the sum of the degrees:

Theorem 0.8.2 — Degree of a sum

Let F be a field and $p, q \in F[x]$. Then

$$\deg(p + q) \leq \max(\deg p, \deg q),$$

with **equality** whenever $\deg p \neq \deg q$.

Proof. Let $p = \sum a_k x^k$, $q = \sum b_k x^k$ and $M = \max(\deg p, \deg q)$. If $M = -\infty$, then $p = q = 0$, so $p + q = 0$ and both sides are $-\infty$. Suppose $M \in \mathbb{N}$. For every $k > M$ we have $k > \deg p$ and $k > \deg q$, so $a_k = b_k = 0$ by [Definition 0.8.3](#). Hence the coefficient $a_k + b_k$ of x^k in $p + q$ is 0 for every $k > M$, and therefore $\deg(p + q) \leq M$.

Now suppose $\deg p \neq \deg q$; by symmetry we may assume $\deg p > \deg q$, so $M = \deg p \in \mathbb{N}$. Then $a_M \neq 0$ and $b_M = 0$, because $M > \deg q$. The coefficient of x^M in $p + q$ is $a_M + 0 = a_M \neq 0$, so $\deg(p + q) \geq M$. Together with the first part, $\deg(p + q) = M$. This proves the theorem. ■

The inequality can be strict: over any field, $\deg(x + (-x)) = \deg 0 = -\infty < 1$. The leading terms cancel. That is the only way the degree of a sum can drop.

Theorem 0.8.3 — Degree of a product

Let F be a field and $p, q \in F[x]$. Then

$$\deg(pq) = \deg p + \deg q.$$

If p and q are non-zero, the leading coefficient of pq is the product of their leading coefficients.

Idea. Expanding $(a_mx^m + \text{lower terms})(b_nx^n + \text{lower terms})$, the only way to reach x^{m+n} is $a_mx^m \cdot b_nx^n$, and nothing reaches beyond. So the top coefficient is a_mb_n . The one real question is whether a_mb_n could be 0. In a field it cannot, and this is exactly where the field axioms enter.

Proof. If $p = 0$ or $q = 0$, then $pq = 0$ by the definition of the product, so both sides equal $-\infty$ by the conventions in Definition 0.8.3. Suppose now $p = \sum a_kx^k$ and $q = \sum b_kx^k$ are non-zero, with $m = \deg p$ and $n = \deg q$. Then $a_m \neq 0$, $b_n \neq 0$, $a_i = 0$ for $i > m$ and $b_j = 0$ for $j > n$.

We showed after Definition 0.8.2 that the coefficient of x^k in pq is 0 for every $k > m+n$. For $k = m+n$, the coefficient is $\sum_{i+j=m+n} a_ib_j$. A term with $i > m$ has $a_i = 0$, and a term with $i < m$ has $j = m+n-i > n$, so $b_j = 0$. Hence only $i = m$, $j = n$ survives, and the coefficient is a_mb_n .

Since $a_m \neq 0$ and $b_n \neq 0$, Theorem 0.7.1 gives $a_mb_n \neq 0$, because a product of non-zero elements of a field is non-zero. Therefore $\deg(pq) = m+n$ and the leading coefficient of pq is a_mb_n , as claimed. ■

Warning. The field axioms are doing real work here. The same definitions make sense with coefficients in $\mathbb{Z}/6\mathbb{Z}$, which is **not** a field. There $(2x+1)(3x^2+x) = 6x^3+5x^2+x = 5x^2+x$, because $2 \cdot 3 = 6 = 0$. The factors have degrees 1 and 2, but the product has degree 2, not 3.

The first consequence is that polynomials over a field behave like integers: a product can only be zero if a factor is, and non-zero factors can be canceled.

Corollary 0.8.4 — No zero divisors in $F[x]$

]

Let F be a field and $p, q, r \in F[x]$.

1. If $pq = 0$, then $p = 0$ or $q = 0$.
2. If $pq = pr$ and $p \neq 0$, then $q = r$.

Proof. For 1, we prove the contrapositive. Suppose $p \neq 0$ and $q \neq 0$. Then $\deg p, \deg q \in \mathbb{N}$, so by Theorem 0.8.3, $\deg(pq) = \deg p + \deg q \in \mathbb{N}$. In particular $\deg(pq) \neq -\infty$, so $pq \neq 0$.

For 2, suppose $pq = pr$ and $p \neq 0$. By Theorem 0.8.1, $p(q-r) = pq - pr = 0$. By part 1 and $p \neq 0$, we get $q-r = 0$, that is, $q = r$. This proves the corollary. ■

Quick check. Let $p, q \in \mathbb{R}[x]$ with $\deg p = 3$ and $\deg q = 2$. What are $\deg(pq)$, $\deg(p+q)$ and $\deg(p^2)$? Can you say what $\deg(p-x^3)$ is?

Solution. By Theorem 0.8.3, $\deg(pq) = 5$ and $\deg(p^2) = 6$. Since the degrees differ, Theorem 0.8.2 gives $\deg(p + q) = 3$. The degree of $p - x^3$ is not determined: if the leading coefficient of p is 1, the x^3 terms cancel and the degree is at most 2 (it could even be $-\infty$ if $p = x^3$); otherwise it is 3.

0.8.4 Polynomials of bounded degree

In linear algebra we constantly need “all polynomials up to a given size”. For instance, the quadratics $ax^2 + bx + c$ are described by three numbers, much like points of $\mathbb{R}^3$. So we give such collections names.

Definition 0.8.4 — Polynomials of degree at most n

Let F be a field and $n \in \mathbb{N}$. The set of polynomials over F of degree **at most** n is

$$F[x]_{\leq n} = \{p \in F[x] : \deg p \leq n\} = \{a_0 + a_1x + \cdots + a_nx^n : a_0, \dots, a_n \in F\}.$$

The two descriptions agree: $\deg p \leq n$ says exactly that every coefficient beyond position n is zero, and the coefficients $a_0, \dots, a_n$ are then free, any of them allowed to be zero. In particular the zero polynomial lies in $F[x]_{\leq n}$, since $-\infty \leq n$.

Example. (Inside $\mathbb{R}[x]_{\leq 2}$)

Which of $3x^2 - 2x + 1$, $5x + 7$, -4 , 0 and $x^3 - x^3 + x$ lie in $\mathbb{R}[x]_{\leq 2}$? Which lie in $\mathbb{R}[x]_{\leq 0}$?

Solution. All five lie in $\mathbb{R}[x]_{\leq 2}$. Their degrees are 2, 1, 0, $-\infty$, and 1 (the last is the polynomial x , since $x^3 - x^3 = 0$ as polynomials). Only -4 and 0 lie in $\mathbb{R}[x]_{\leq 0}$, which is the set of constant polynomials, 0 included.

Sometimes we need the degree to be exactly n , for instance when we speak of “a quadratic” and mean that the x^2 term is really there.

Definition 0.8.5 — Polynomials of degree exactly n

Let F be a field and $n \in \mathbb{N}$. The set of polynomials over F of degree **exactly** n is

$$F[x]_{=n} = \{p \in F[x] : \deg p = n\} = \{a_0 + a_1x + \cdots + a_nx^n : a_0, \dots, a_n \in F, a_n \neq 0\}.$$

Example. (Comparing the two notations)

For each of $x^2 + 1$, $3x - 2$ and 0 , decide whether it lies in $\mathbb{R}[x]_{\leq 2}$ and whether it lies in $\mathbb{R}[x]_{=2}$.

Solution. The polynomial $x^2 + 1$ has degree 2, so it lies in both. The polynomial $3x - 2$ has degree $1 \leq 2$, so it lies in $\mathbb{R}[x]_{\leq 2}$ but **not** in $\mathbb{R}[x]_{=2}$. The zero polynomial has

degree $-\infty \leq 2$, so it lies in $\mathbb{R}[x]_{\leq 2}$, but $-\infty \neq 2$, so it is **not** in $\mathbb{R}[x]_{=2}$. In general $F[x]_{\leq n} = F[x]_{=n} \cup F[x]_{\leq n-1}$ for $n \geq 1$, and the union is disjoint.

The two sets differ in one small clause, $a_n \neq 0$, and that clause changes their behavior completely.

Warning. $F[x]_{\leq n}$ is closed under addition, but $F[x]_{=n}$ is **not**. For example $x^2 + x$ and $-x^2 + 1$ both lie in $\mathbb{R}[x]_{=2}$, but their sum $x + 1$ has degree 1. The leading terms cancel. Also $F[x]_{=n}$ does not contain 0. In Chapter 1 this is why $F[x]_{\leq n}$ is a subspace and $F[x]_{=n}$ is not.

The positive half is [Theorem 0.8.2](#) at work: if $\deg p \leq n$ and $\deg q \leq n$, then $\deg(p + q) \leq \max(\deg p, \deg q) \leq n$. Multiplying by a constant $c \in F$ also cannot raise the degree, by [Theorem 0.8.3](#).

0.8.5 Evaluation and roots

A polynomial is not a function, but every polynomial *defines* a function: plug an element of the field in for x .

Definition 0.8.6 — Evaluation and roots

Let F be a field, $p = \sum_{k=0}^N a_k x^k \in F[x]$ and $c \in F$. The **value** of p at c is the element

$$p(c) = \sum_{k=0}^N a_k c^k = a_0 + a_1 c + \cdots + a_N c^N \in F,$$

where $c^0 = 1$. An element $c \in F$ is a **root** of p if $p(c) = 0$. The function $F \rightarrow F$, $c \mapsto p(c)$, is the **polynomial function** of p .

The value does not depend on the choice of N : enlarging N adds terms $0 \cdot c^k = 0$ (by [Theorem 0.7.1](#)), which do not change the sum. So $p(c)$ is well defined.

For example, $p = x^2 - 3x + 2 \in \mathbb{R}[x]$ has $p(0) = 2$ and $p(1) = 1 - 3 + 2 = 0$, so 1 is a root. Over $\mathbb{F}_3 = \{0, 1, 2\}$, the polynomial $x^3 + 2$ has $1^3 + 2 = 3 = 0$, so 1 is a root; the other two values are $0 + 2 = 2$ and $8 + 2 = 10 = 1$, so 1 is its only root in $\mathbb{F}_3$.

Evaluation turns the operations of $F[x]$ into the operations of F :

Theorem 0.8.5 — Evaluation respects sums and products

Let F be a field, $p, q \in F[x]$ and $c \in F$. Then

$$(p + q)(c) = p(c) + q(c), \quad (pq)(c) = p(c)q(c),$$

and the constant polynomial a has value a at c .

Proof. Let $p = \sum_{i=0}^m a_i x^i$ and $q = \sum_{j=0}^n b_j x^j$, where we choose m and n large enough that all non-zero coefficients are included. The statement about constants is immediate from [Definition 0.8.6](#). For sums, taking $N = \max(m, n)$ and padding with zero coefficients,

$(p + q)(c) = \sum_{k=0}^N (a_k + b_k)c^k = \sum_{k=0}^N a_k c^k + \sum_{k=0}^N b_k c^k$ by distributivity in F .

For products, all non-zero coefficients of pq occur at positions $k \leq m + n$, so

$$(pq)(c) = \sum_{k=0}^{m+n} \left(\sum_{i+j=k} a_i b_j \right) c^k = \sum_{i=0}^m \sum_{j=0}^n a_i b_j c^i c^j = \sum_{i=0}^m \sum_{j=0}^n (a_i c^i)(b_j c^j) = \left(\sum_{i=0}^m a_i c^i \right) \left(\sum_{j=0}^n b_j c^j \right),$$

where the first equality is [Definition 0.8.2](#), the second uses distributivity and $c^{i+j} = c^i c^j$ and regroups the pairs (i, j) , the third uses **commutativity** of multiplication in F to move c^i past b_j , and the last is distributivity again. This shows $(pq)(c) = p(c)q(c)$. ■

Note which step used commutativity. In the next section we will plug a square matrix A into a polynomial. Products of matrices do not commute in general, but the proof only moves **scalars** past powers of A , and multiplies powers of the same A ; both are still allowed for matrices, so the same identity will hold there.

As a first use, a root of a factor is a root of the product: if $q(c) = 0$, then $(pq)(c) = p(c) \cdot 0 = 0$. Conversely, if $(pq)(c) = 0$ then $p(c)q(c) = 0$, so $p(c) = 0$ or $q(c) = 0$ by [Theorem 0.7.1](#). This is how one solves $x^2 - 3x + 2 = (x - 1)(x - 2) = 0$ at school.

Now we can return to the question from the start of the section.

Example. (Different polynomials, same function)

Over $\mathbb{F}_2$, compare the polynomials $x^2 + x$ and 0, and the polynomials x^2 and x , both as polynomials and as functions $\mathbb{F}_2 \rightarrow \mathbb{F}_2$.

Solution. As polynomials, $x^2 + x$ has coefficient sequence $(0, 1, 1, 0, \dots)$ and 0 has $(0, 0, 0, \dots)$. They differ in position 1, so $x^2 + x \neq 0$ by [Definition 0.8.1](#). Its degree is 2.

As functions, $\mathbb{F}_2 = \{0, 1\}$ has two elements, and

$$0^2 + 0 = 0, \quad 1^2 + 1 = 1 + 1 = 0.$$

So the polynomial function of $x^2 + x$ is the zero function, the same as that of 0. Both elements of $\mathbb{F}_2$ are roots of a polynomial of degree 2 that is not the zero polynomial.

Similarly $x^2 \neq x$ as polynomials, but $0^2 = 0$ and $1^2 = 1$, so they define the same function. Indeed, by [Theorem 0.8.5](#), two polynomials define the same function exactly when their difference is a polynomial whose function is zero, and $x^2 - x = x^2 + x$ in $\mathbb{F}_2[x]$.

Warning. A polynomial is not a function. Two different polynomials can define the same polynomial function, as $x^2 + x$ and 0 do over $\mathbb{F}_2$. So “ $p(c) = 0$ for every $c \in F$ ” does **not** imply $p = 0$ in general, and we may never read off coefficients or a degree from the values alone. This happens over every finite field: if $F = \{c_1, \dots, c_q\}$, then $(x - c_1)(x - c_2) \cdots (x - c_q)$ has degree q by [Theorem 0.8.3](#), yet vanishes at every c_i by [Theorem 0.8.5](#). Over $\mathbb{F}_p$, [Exercise \(C2: A non-zero polynomial vanishing everywhere\)](#) below gives the shorter example $x^p - x$.

Over an infinite field such as $\mathbb{Q}$, $\mathbb{R}$ or $\mathbb{C}$, the problem disappears: a polynomial whose function is zero is the zero polynomial, so polynomials and polynomial functions can be identified. We prove this in Chapter 2 (in the section on applications of linear systems), once we can count roots. Until then, whenever we say two polynomials are equal we mean their coefficients agree.

0.8.6 Looking ahead

Two tools will be built on this section later.

Remark. Division with remainder. Over a field, if $f, g \in F[x]$ and $g \neq 0$, there are unique $q, r \in F[x]$ with $f = qg + r$ and $\deg r < \deg g$. This is long division of polynomials, and the convention $\deg 0 = -\infty$ makes “ $\deg r < \deg g$ ” cover the case $r = 0$. We state it here only for orientation; it is proved in Chapter 5 and is not used before then.

Remark. Polynomials in a matrix. In the next section, once matrices and their products are available, we substitute a square matrix A for x , replacing the constant term a_0 by $a_0 I$. This is how the characteristic and minimal polynomials of later chapters act on matrices.

0.8.7 Exercises

0.8.7.1 A. Check your understanding

Exercise A1.

- Define the degree of a polynomial over a field F , including the zero polynomial. State the formula for $\deg(pq)$.
- Is $1 + x + x^2 + x^3 + \cdots$, with every coefficient equal to 1, a polynomial over $\mathbb{R}$? Justify your answer.
- Determine whether the following statement is true: “for all $p, q \in F[x]$, $\deg(p+q) = \max(\deg p, \deg q)$.” Justify your answer.
- Determine whether the following statement is true: “if $p \in \mathbb{F}_2[x]$ and $p(c) = 0$ for every $c \in \mathbb{F}_2$, then $p = 0$.” Justify your answer.
- Explain why $F[x]_{=2}$ is **not** closed under addition, while $F[x]_{\leq 2}$ is.

Solution.

- For $p = \sum a_k x^k \neq 0$, $\deg p$ is the largest n with $a_n \neq 0$; and $\deg 0 = -\infty$. For all $p, q \in F[x]$, $\deg(pq) = \deg p + \deg q$ ([Theorem 0.8.3](#)).
- No. Its coefficient sequence $(1, 1, 1, \dots)$ has infinitely many non-zero terms, so it fails the clause “only finitely many a_k are non-zero” of [Definition 0.8.1](#).
- False. Take $p = x$ and $q = -x$. Then $p + q = 0$ has degree $-\infty$, while $\max(\deg p, \deg q) = 1$. Only the inequality $\leq$ holds in general ([Theorem 0.8.2](#)).
- False. By [Example \(Different polynomials, same function\)](#), $p = x^2 + x$ satisfies $p(0) = p(1) = 0$, but $p \neq 0$.
- The sum of x^2 and $-x^2 + 1$, both of degree exactly 2, is 1, of degree 0. For $F[x]_{\leq 2}$: if $\deg p \leq 2$ and $\deg q \leq 2$, then $\deg(p + q) \leq \max(\deg p, \deg q) \leq 2$ by

Theorem 0.8.2.

0.8.7.2 B. Practice

Exercise B1: Sums, products and degrees.

- (a) In $\mathbb{R}[x]$, let $p = 2x^3 - x + 4$ and $q = -2x^3 + x^2 + 3$. Compute $p + q$ and pq , and give their degrees. Which inequality or formula from this section does each degree illustrate?
- (b) In $\mathbb{F}_3[x]$, compute $(x + 2)(x^2 + x + 1)$. Hence find all roots in $\mathbb{F}_3$ of the product.

Solution.

- (a) Adding coefficients gives $p + q = x^2 - x + 7$, of degree 2. Here $\deg p = \deg q = 3$ and the x^3 terms cancel, so the inequality of Theorem 0.8.2 is strict. Expanding,

$$pq = -4x^6 + 2x^5 + 2x^4 - 3x^3 + 4x^2 - 3x + 12,$$

of degree $6 = 3 + 3$, as Theorem 0.8.3 predicts; its leading coefficient is $2 \cdot (-2) = -4$.

- (b) Expanding, $(x + 2)(x^2 + x + 1) = x^3 + 3x^2 + 3x + 2$. In $\mathbb{F}_3$ we have $3 = 0$, so the product is $x^3 + 2$. By Theorem 0.8.5 and Theorem 0.7.1, c is a root of the product if and only if $c + 2 = 0$ or $c^2 + c + 1 = 0$. The first gives $c = 1$. For the second, the values at 0, 1, 2 are 1, 3 = 0 and 7 = 1. Hence the only root in $\mathbb{F}_3$ is $c = 1$, which occurs in both factors.

Exercise B2: Degree of a composition.

For $p = \sum_{k=0}^m a_k x^k$ and q in $F[x]$, define the **composition** $p \circ q = \sum_{k=0}^m a_k q^k \in F[x]$, where $q^0 = 1$.

- (a) Prove that if p and q are non-constant, then $\deg(p \circ q) = \deg p \cdot \deg q$.
- (b) Give an example showing the formula fails when q is constant.

Hint: first find $\deg(q^k)$.

Solution.

- (a) Let $m = \deg p \geq 1$ and $d = \deg q \geq 1$, so $a_m \neq 0$. By Theorem 0.8.3 and induction on k (Theorem 0.2.7), $\deg(q^k) = kd$ for every $k \in \mathbb{N}$: the case $k = 0$ is $\deg 1 = 0$, and $\deg(q^{k+1}) = \deg(q^k) + \deg q = kd + d$. Hence, again by Theorem 0.8.3, $\deg(a_m q^m) = 0 + md = md$, and for $k < m$, $\deg(a_k q^k) \leq kd < md$ (it is kd or $-\infty$). Let $s = \sum_{k=0}^{m-1} a_k q^k$. Applying Theorem 0.8.2 $m - 1$ times gives $\deg s \leq \max_{k < m} \deg(a_k q^k) < md$. Since $\deg s \neq \deg(a_m q^m)$, the equality case of Theorem 0.8.2 gives $\deg(p \circ q) = \deg(s + a_m q^m) = md$. This proves $\deg(p \circ q) = \deg p \cdot \deg q$.
- (b) Take $p = x - 1$ and $q = 1$. Then $p \circ q = 1 - 1 = 0$ has degree $-\infty$, while $\deg p \cdot \deg q = 1 \cdot 0 = 0$.

Exercise B3: All small polynomials over $\mathbb{F}_2$.

List all polynomials in $\mathbb{F}_2[x]_{\leq 2}$. For each, give its degree and its polynomial function (the values at 0 and 1). Hence determine which of them define the zero function, and explain why every function $\mathbb{F}_2 \rightarrow \mathbb{F}_2$ is the polynomial function of exactly two of them.

Solution. A polynomial $a_0 + a_1x + a_2x^2$ with $a_i \in \mathbb{F}_2$ has value a_0 at 0 and $a_0 + a_1 + a_2$ at 1. With $2^3 = 8$ choices of coefficients:

polynomial	degree	value at 0	value at 1
0	$-\infty$	0	0
1	0	1	1
x	1	0	1
$x + 1$	1	1	0
x^2	2	0	1
$x^2 + 1$	2	1	0
$x^2 + x$	2	0	0
$x^2 + x + 1$	2	1	1

The zero function requires $a_0 = 0$ and $a_0 + a_1 + a_2 = 0$, that is, $a_0 = 0$ and $a_1 = a_2$. Hence exactly 0 and $x^2 + x$ define the zero function.

A function $f: \mathbb{F}_2 \rightarrow \mathbb{F}_2$ is determined by the pair $(f(0), f(1))$. Given such a pair (u, v) , the polynomial $a_0 + a_1x + a_2x^2$ defines f exactly when $a_0 = u$ and $a_1 + a_2 = v - u$. The first fixes a_0 , and for each of the two choices of a_2 the second fixes $a_1 = v - u - a_2$. Therefore exactly two of the eight polynomials define f , and the table confirms this.

0.8.7.3 C. Going deeper**Exercise C1: Roots of a quadratic.**

Let F be a field.

- Let $p = ax^2 + bx + c \in F[x]$ and $r \in F$ with $p(r) = 0$. Prove that $p = (x - r)(ax + (b + ar))$.
- Deduce that a non-zero polynomial in $F[x]_{\leq 2}$ has at most 2 roots in F .
- The same definitions make sense with coefficients in $\mathbb{Z}/8\mathbb{Z}$. Show that $x^2 - 1$ has four roots there, and name the step of (b) that breaks.

Hint for (b): use (a) together with [Theorem 0.8.5](#).

Solution.

- Since $p(r) = 0$, we have $ar^2 + br + c = 0$, so $c = -ar^2 - br$. Expanding with [Theorem 0.8.1](#),

$$(x - r)(ax + (b + ar)) = ax^2 + (b + ar)x - arx - r(b + ar) = ax^2 + bx - (br + ar^2) = ax^2 + bx + c = p.$$

- Let $p \neq 0$ with $\deg p \leq 2$. *Case 1.* $\deg p = 0$. Then $p = c \neq 0$ is constant and $p(r) = c \neq 0$ for every r , so p has no roots. *Case 2.* $\deg p = 1$, say $p = bx + c$

with $b \neq 0$. Then $p(r) = 0$ if and only if $r = -b^{-1}c$, so p has exactly one root. *Case 3.* $\deg p = 2$, say $p = ax^2 + bx + c$ with $a \neq 0$. If p has no root we are done. Otherwise let r be a root. By (a), $p = (x - r)q$ with $q = ax + (b + ar)$, of degree 1 since $a \neq 0$. Let s be any root of p . By [Theorem 0.8.5](#), $0 = p(s) = (s - r)q(s)$, so $s = r$ or $q(s) = 0$ by [Theorem 0.7.1](#). By Case 2, q has exactly one root. Hence every root of p is r or the root of q , and p has at most 2 roots. This proves the claim.

- (c) In $\mathbb{Z}/8\mathbb{Z}$, $1^2 = 1$, $3^2 = 9 = 1$, $5^2 = 25 = 1$ and $7^2 = 49 = 1$, so 1, 3, 5, 7 are four roots of $x^2 - 1$. The argument in (b) breaks at “ $(s - r)q(s) = 0$ implies $s - r = 0$ or $q(s) = 0$ ”: for $r = 1$, $s = 3$ we get $(3 - 1)(3 + 1) = 2 \cdot 4 = 0$ with both factors non-zero. That step needs the field property that a product of non-zero elements is non-zero.

Exercise C2: A non-zero polynomial vanishing everywhere.

Let p be a prime and $F = \mathbb{F}_p$.

- (a) Fix $a \in F \setminus \{0\}$. Prove that the map $F \setminus \{0\} \rightarrow F \setminus \{0\}$, $b \mapsto ab$, is a bijection.
- (b) Deduce that $a^{p-1} = 1$ for every $a \in F \setminus \{0\}$.
- (c) Hence show that the polynomial $x^p - x \in \mathbb{F}_p[x]$ is not the zero polynomial but that its polynomial function is the zero function.

Hint for (b): multiply together all the elements of $F \setminus \{0\}$ in two ways.

Solution.

- (a) The map lands in $F \setminus \{0\}$, since a product of non-zero elements of a field is non-zero ([Theorem 0.7.1](#)). It is injective: if $ab = ab'$, multiplying by a^{-1} gives $b = b'$. The set $F \setminus \{0\}$ is finite with $p - 1$ elements, so an injective map from it to itself is surjective by [Theorem 0.4.9](#), hence bijective.
- (b) Let P be the product of all $p - 1$ elements of $F \setminus \{0\}$; it is non-zero by [Theorem 0.7.1](#). Since $b \mapsto ab$ is a bijection by (a), the elements ab , as b runs over $F \setminus \{0\}$, are the same elements in a different order. Multiplication in F is commutative and associative, so the product does not depend on the order, and

$$P = \prod_{b \neq 0} (ab) = a^{p-1} \prod_{b \neq 0} b = a^{p-1} P.$$

Multiplying by P^{-1} gives $a^{p-1} = 1$.

- (c) The coefficient of x^p in $x^p - x$ is $1 \neq 0$, so the polynomial is non-zero, of degree p . For its function: $0^p - 0 = 0$, and for $a \neq 0$, (b) gives $a^p - a = a \cdot a^{p-1} - a = a - a = 0$. Hence every element of $\mathbb{F}_p$ is a root, and the polynomial function of $x^p - x$ is the zero function. For $p = 2$ this is $x^2 - x = x^2 + x$, the polynomial of [Example \(Different polynomials, same function\)](#).

Exercise C3: Which polynomials have inverses?.

- (a) Let F be a field and $p, q \in F[x]$ with $pq = 1$. Prove that p and q are non-zero constants.

(b) Show that (a) fails with coefficients in $\mathbb{Z}/4\mathbb{Z}$ by computing $(2x + 1)^2$ there.

Solution.

- (a) Since $pq = 1 \neq 0$, neither p nor q is the zero polynomial, so $\deg p, \deg q \in \mathbb{N}$. By [Theorem 0.8.3](#), $\deg p + \deg q = \deg 1 = 0$. Two natural numbers with sum 0 are both 0, so $\deg p = \deg q = 0$. Therefore p and q are non-zero constants.
- (b) In $\mathbb{Z}/4\mathbb{Z}$, $(2x + 1)^2 = 4x^2 + 4x + 1 = 1$, because $4 = 0$. So $2x + 1$ has degree 1 and is its own inverse. The proof of (a) breaks at [Theorem 0.8.3](#), which fails here because the leading coefficient $2 \cdot 2 = 4 = 0$.

0.9 Matrices

You have probably met matrices as tables of numbers that are multiplied by a strange rule. This section rebuilds them over an arbitrary field F and answers the questions a computational course leaves open. Why is the product defined the way it is? Which rules of school algebra survive, and which fail? What does it mean to divide by a matrix? Everything later in the book, from Gaussian elimination in Chapter 2 to the matrix of a linear map in Chapter 3, rests on the answers.

Throughout, F is a field ([Definition 0.7.1](#)). Nothing here uses more about F than the field axioms, so every statement holds over $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ and $\mathbb{F}_p$ alike.

0.9.1 Matrices and their entries

Definition 0.9.1 — Matrix

Let F be a field and $m, n \geq 1$ integers. An $m \times n$ **matrix over** F is a rectangular array

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

with m rows and n columns, whose **entries** a_{ij} lie in F . Formally, it is a function $\{1, \dots, m\} \times \{1, \dots, n\} \rightarrow F$, $(i, j) \mapsto a_{ij}$. The set of all such matrices is $M_{m \times n}(F)$, and we write $M_n(F) = M_{n \times n}(F)$ for the **square** matrices.

The entry a_{ij} sits in row i and column j : **row first, column second**. We write $A = (a_{ij})$, and $(A)_{ij}$ for the (i, j) -entry when the matrix has no letter of its own. Two matrices are **equal** when they have the **same size** and the same entry in every position; this is equality of functions ([Definition 0.4.1](#)).

Two sizes deserve their own names. A matrix in $M_{n \times 1}(F)$ is a **column vector**, and we write

$$F^n = M_{n \times 1}(F), \quad \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

In running text we write $\mathbf{x} = (x_1, \dots, x_n)$ and still mean a column. Columns are printed in bold. A matrix in $M_{1 \times n}(F)$ is a **row vector**. For $j = 1, \dots, n$, the column $\mathbf{e}_j \in F^n$ has 1 in position j and 0 elsewhere.

We will often cut a matrix into its columns. If $\mathbf{a}_j = (a_{1j}, \dots, a_{mj}) \in F^m$ is the j -th column of A , we write

$$A = (\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n).$$

The i -th row of A is the row vector $(a_{i1} \quad \cdots \quad a_{in}) \in M_{1 \times n}(F)$.

Example. (Matrices over different fields)

Identify the size, the field and a few entries of each matrix:

$$A = \begin{pmatrix} 1 & -2 & 0 \\ 3 & 1 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} i & 1+i \\ 0 & 2-i \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad D = (5).$$

Here C is meant over $\mathbb{F}_2$. Also write out the 2×2 real matrix E with $e_{ij} = i + 2j$.

Solution. $A \in M_{2 \times 3}(\mathbb{R})$, with $a_{12} = -2$ and $a_{23} = 4$; its second column is $\mathbf{a}_2 = (-2, 1)$. $B \in M_2(\mathbb{C})$, with $b_{12} = 1+i$ and $b_{21} = 0$. $C \in M_3(\mathbb{F}_2)$; every entry is 0 or 1. $D \in M_1(\mathbb{R})$ is the degenerate case, a 1×1 matrix. It behaves exactly like the number 5, and 1×1 matrices are how numbers will appear inside matrix formulas. Finally $e_{11} = 3$, $e_{12} = 5$, $e_{21} = 4$, $e_{22} = 6$, so $E = \begin{pmatrix} 3 & 5 \\ 4 & 6 \end{pmatrix}$. Swapping the roles of i and j would give a different matrix, which is why the order “row, column” matters.

0.9.2 Special matrices

A few shapes of matrix occur so often that they have names. The **main diagonal** of a square matrix $A \in M_n(F)$ consists of the entries $a_{11}, a_{22}, \dots, a_{nn}$.

Definition 0.9.2 — Zero matrix

The **zero matrix** $0 = 0_{m \times n} \in M_{m \times n}(F)$ is the matrix all of whose entries are 0.

Definition 0.9.3 — Identity matrix

The **identity matrix** $I_n \in M_n(F)$ has entries

$$(I_n)_{ij} = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

That is, 1 on the main diagonal and 0 elsewhere. We write I when the size is clear. The symbol δ_{ij} is the **Kronecker delta**.

The columns of I_n are exactly $\mathbf{e}_1, \dots, \mathbf{e}_n$.

Definition 0.9.4 — Diagonal matrix

A square matrix $A \in M_n(F)$ is **diagonal** if $a_{ij} = 0$ whenever $i \neq j$. We write $\text{diag}(d_1, \dots, d_n)$ for the diagonal matrix with diagonal entries $d_1, \dots, d_n$.

Definition 0.9.5 — Upper triangular matrix

A square matrix $A \in M_n(F)$ is **upper triangular** if $a_{ij} = 0$ whenever $i > j$, that is, every entry **strictly below** the main diagonal is zero. It is **lower triangular** if $a_{ij} = 0$ whenever $i < j$.

Example. (Zero and identity matrices)

Write out $0_{2 \times 3}$, I_2 and I_3 . Which of them are diagonal?

Solution.

$$0_{2 \times 3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

$I_2 = \text{diag}(1, 1)$ and $I_3 = \text{diag}(1, 1, 1)$ are diagonal. The matrix $0_{2 \times 3}$ is **not** diagonal, because the definition only applies to square matrices; the square zero matrix $0_{n \times n} = \text{diag}(0, \dots, 0)$ is diagonal.

Example. (Diagonal and triangular)

Classify each matrix as diagonal, upper triangular, lower triangular, or none of these:

$$P = \begin{pmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}, \quad R = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 7 & 6 \end{pmatrix}.$$

Solution. $P = \text{diag}(3, -1, 0)$ is diagonal. A zero on the diagonal is allowed: the definition only constrains off-diagonal entries. P is also both upper and lower triangular, and in general a matrix is diagonal exactly when it is both. Q is upper triangular but not lower triangular, since $q_{12} = 2 \neq 0$. R differs from Q in one entry, $r_{32} = 7$. All other entries below the diagonal are still zero, but the clause “ $a_{ij} = 0$ whenever $i > j$ ” fails at $(i, j) = (3, 2)$, so R is not upper triangular. It is not lower triangular either.

0.9.3 Addition and scalar multiplication

Matrices of the same size are added, and multiplied by scalars, one entry at a time.

Definition 0.9.6 — Matrix addition

Let $A = (a_{ij})$ and $B = (b_{ij})$ be in $M_{m \times n}(F)$, of the **same size**. Their **sum** is $A + B \in M_{m \times n}(F)$ with $(A + B)_{ij} = a_{ij} + b_{ij}$.

Definition 0.9.7 — Scalar multiplication

Let $A = (a_{ij}) \in M_{m \times n}(F)$ and $c \in F$. The **scalar multiple** $cA \in M_{m \times n}(F)$ has entries $(cA)_{ij} = c a_{ij}$.

Definition 0.9.8 — Additive inverse

The **additive inverse** of $A \in M_{m \times n}(F)$ is $-A = (-1)A$, with entries $-a_{ij}$. We write $A - B = A + (-B)$.

Matrices of different sizes are never added: $\begin{pmatrix} 1 & 2 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is not defined.

Example. (Adding and scaling)

Let $A = \begin{pmatrix} 1 & -2 & 0 \\ 3 & 1 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 2 & -1 \\ 0 & -5 & 1 \end{pmatrix}$ in $M_{2 \times 3}(\mathbb{R})$. Compute $A + B$, $3A$ and $A - 2B$.

Solution. Entry by entry,

$$A+B = \begin{pmatrix} 3 & 0 & -1 \\ 3 & -4 & 5 \end{pmatrix}, \quad 3A = \begin{pmatrix} 3 & -6 & 0 \\ 9 & 3 & 12 \end{pmatrix}, \quad A-2B = \begin{pmatrix} 1-4 & -2-4 & 0+2 \\ 3-0 & 1+10 & 4-2 \end{pmatrix} = \begin{pmatrix} -3 & -6 & 2 \\ 3 & 11 & 2 \end{pmatrix}.$$

Because both operations work entry by entry, they inherit the rules of F :

Theorem 0.9.1 — Properties of addition and scalar multiplication

Let $A, B, C \in M_{m \times n}(F)$ and $c, d \in F$. Then:

1. $A + B = B + A$;
2. $(A + B) + C = A + (B + C)$;
3. $A + 0 = A$;
4. $A + (-A) = 0$;
5. $c(A + B) = cA + cB$ and $(c + d)A = cA + dA$;
6. $(cd)A = c(dA)$ and $1A = A$.

Proof. Two matrices of the same size are equal when their entries agree, so it suffices to compare (i, j) -entries. For 1, $(A + B)_{ij} = a_{ij} + b_{ij} = b_{ij} + a_{ij} = (B + A)_{ij}$, where the middle equality is commutativity of addition in F . For 5, $(c(A + B))_{ij} = c(a_{ij} + b_{ij}) = ca_{ij} + cb_{ij} = (cA + cB)_{ij}$ by distributivity in F . Each remaining identity reduces in the same way to the corresponding field axiom of Definition 0.7.1 in each entry. This proves the theorem. ■

These eight rules will reappear in Chapter 1 as the axioms of a vector space: $M_{m \times n}(F)$ is one of our standard examples.

0.9.4 Matrix multiplication

Addition was entrywise. The obvious guess for multiplication is also entrywise, but that product is almost useless. The product that matters is the one that records **substitution**, and it looks very different.

Suppose new variables y_1, y_2 are given in terms of x_1, x_2 , and z_1, z_2 in terms of y_1, y_2 :

$$\begin{aligned} y_1 &= 2x_1 + x_2 & z_1 &= y_1 + 2y_2 \\ y_2 &= x_1 + 4x_2 & z_2 &= 3y_1 - y_2 \end{aligned}$$

Record each substitution by its table of coefficients: $B = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}$ for $\mathbf{y}$ in terms of $\mathbf{x}$, and $A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$ for $\mathbf{z}$ in terms of $\mathbf{y}$. Substituting,

$$\begin{aligned} z_1 &= (2x_1 + x_2) + 2(x_1 + 4x_2) = (1 \cdot 2 + 2 \cdot 1)x_1 + (1 \cdot 1 + 2 \cdot 4)x_2 = 4x_1 + 9x_2, \\ z_2 &= 3(2x_1 + x_2) - (x_1 + 4x_2) = (3 \cdot 2 + (-1) \cdot 1)x_1 + (3 \cdot 1 + (-1) \cdot 4)x_2 = 5x_1 - x_2. \end{aligned}$$

So $\mathbf{z}$ in terms of $\mathbf{x}$ has coefficient table $\begin{pmatrix} 4 & 9 \\ 5 & -1 \end{pmatrix}$. Look at how each entry arose: the coefficient of x_k in z_i is $a_{i1}b_{1k} + a_{i2}b_{2k}$, row i of A against column k of B . The entrywise product $\begin{pmatrix} 2 & 2 \\ 3 & -4 \end{pmatrix}$ has nothing to do with the substitution. This forces the definition.

To multiply matrices, run each row of the first against each column of the second: the product of two matrices is the table of the combined substitution.

Definition 0.9.9 — Matrix multiplication

Let $A = (a_{ij}) \in M_{m \times n}(F)$ and $B = (b_{jk}) \in M_{n \times p}(F)$, so that **the number of columns of A equals the number of rows of B** . Their **product** is the matrix $AB \in M_{m \times p}(F)$ with entries

$$(AB)_{ik} = \sum_{j=1}^n a_{ij}b_{jk} = a_{i1}b_{1k} + a_{i2}b_{2k} + \cdots + a_{in}b_{nk}$$

for $1 \leq i \leq m$ and $1 \leq k \leq p$. If the sizes do not match, AB is **not defined**.

In words: to get the (i, k) -entry of AB , walk along row i of A and down column k of B at the same time, multiply the entries you meet in pairs, and add. The size condition says that the row and the column have the same length n , so the pairing makes sense. The product has as many rows as A and as many columns as B :

$$\begin{matrix} A & B & = & AB \\ m \times n & n \times p & & m \times p \end{matrix}$$

In particular, for $A \in M_{m \times n}(F)$ and a column $\mathbf{x} \in F^n$, the product $A\mathbf{x} \in F^m$ is defined, with i -th entry $a_{i1}x_1 + \cdots + a_{in}x_n$. A system of m linear equations in n unknowns is therefore a single equation $A\mathbf{x} = \mathbf{b}$.

Example. (A product of square matrices)

Let $A = \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 4 \\ -2 & 1 \end{pmatrix}$ in $M_2(\mathbb{R})$. Compute AB and BA .

Solution. Row i of A against column k of B :

$$AB = \begin{pmatrix} 2 \cdot 1 + (-1)(-2) & 2 \cdot 4 + (-1) \cdot 1 \\ 0 \cdot 1 + 3 \cdot (-2) & 0 \cdot 4 + 3 \cdot 1 \end{pmatrix} = \begin{pmatrix} 4 & 7 \\ -6 & 3 \end{pmatrix}, \quad BA = \begin{pmatrix} 1 \cdot 2 + 4 \cdot 0 & 1 \cdot (-1) + 4 \cdot 3 \\ -2 \cdot 2 + 1 \cdot 0 & (-2)(-1) + 1 \cdot 3 \end{pmatrix} = \begin{pmatrix} 2 & 11 \\ -4 & 5 \end{pmatrix}$$

Both products are defined, and they are different.

Example. (Products of rectangular matrices)

Let $A = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{pmatrix} \in M_{2 \times 3}(\mathbb{R})$ and $B = \begin{pmatrix} 3 & 1 \\ 2 & 1 \\ 1 & 0 \end{pmatrix} \in M_{3 \times 2}(\mathbb{R})$. Determine the sizes of AB and BA and compute them. Then compute $\begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 4 \end{pmatrix} \begin{pmatrix} 1 & 2 \end{pmatrix}$.

Solution. AB is $(2 \times 3)(3 \times 2)$, so it is 2×2 ; BA is $(3 \times 2)(2 \times 3)$, so it is 3×3 . Computing row against column,

$$AB = \begin{pmatrix} 3 + 0 + 2 & 1 + 0 + 0 \\ -3 + 6 + 1 & -1 + 3 + 0 \end{pmatrix} = \begin{pmatrix} 5 & 1 \\ 4 & 2 \end{pmatrix}, \quad BA = \begin{pmatrix} 2 & 3 & 7 \\ 1 & 3 & 5 \\ 1 & 0 & 2 \end{pmatrix}.$$

For instance $(BA)_{13} = 3 \cdot 2 + 1 \cdot 1 = 7$. The last two products show the extreme cases: a row times a column is 1×1 , namely $(1 \cdot 3 + 2 \cdot 4) = (11)$, while a column times a row is 2×2 , namely $\begin{pmatrix} 3 & 6 \\ 4 & 8 \end{pmatrix}$.

Here is a non-example by minimal change. Take the same six entries as B , but arrange them in two rows: $C = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 1 & 0 \end{pmatrix} \in M_{2 \times 3}(\mathbb{R})$. The entries are still real numbers and A is unchanged. But A has 3 columns and C has 2 rows, so forming AC would pair a row of length 3 with a column of length 2. The clause “the number of columns of A equals the number of rows of B ” fails, and AC is **not defined**.

Why this definition, and not the entrywise one? Because it is the only rule under which “the

table of a combined substitution is the product of the tables". In Chapter 3 the same fact becomes: the matrix of a composition of linear maps is the product of their matrices. Every property of matrix multiplication, good or bad, comes from this.

0.9.5 Three views of a product

The entry formula is how one computes a product by hand, but it hides the structure. Two other descriptions, by columns and by rows, are what we will actually think with. It turns out that multiplying by A on the left acts on each column of B separately, and multiplying by B on the right acts on each row of A separately:

Theorem 0.9.2 — Three views of a product

Let $A \in M_{m \times n}(F)$ and $B \in M_{n \times p}(F)$, and write $B = (\mathbf{b}_1 \ \cdots \ \mathbf{b}_p)$ with columns $\mathbf{b}_k \in F^n$. Then:

1. **(Entries)** $(AB)_{ik} = \sum_{j=1}^n a_{ij}b_{jk}$ for all i, k ;
2. **(Columns)** the k -th column of AB is $A\mathbf{b}_k$, that is,

$$AB = (A\mathbf{b}_1 \ A\mathbf{b}_2 \ \cdots \ A\mathbf{b}_p);$$

3. **(Rows)** the i -th row of AB is (row i of A) B .

Proof. Part 1 is [Definition 0.9.9](#). For part 2, fix k . The column $\mathbf{b}_k$ has entries $b_{1k}, \dots, b_{nk}$, so by [Definition 0.9.9](#) the i -th entry of $A\mathbf{b}_k \in F^m$ is $\sum_{j=1}^n a_{ij}b_{jk}$. By part 1 this is $(AB)_{ik}$, the i -th entry of the k -th column of AB . Hence the k -th column of AB is $A\mathbf{b}_k$.

For part 3, fix i , and let $\mathbf{r} = (a_{i1} \ \cdots \ a_{in})$ be row i of A . Then $\mathbf{r}B$ is a $1 \times p$ matrix whose k -th entry is $\sum_{j=1}^n a_{ij}b_{jk} = (AB)_{ik}$, again by [Definition 0.9.9](#). This is the k -th entry of row i of AB . This proves the theorem. ■

The column view reduces every matrix product to products $A\mathbf{x}$ of a matrix with a single column. So it pays to understand $A\mathbf{x}$ well, and here is the fact to understand:

Theorem 0.9.3 — $A\mathbf{x}$ is a combination of the columns of A

Let $A = (\mathbf{a}_1 \ \cdots \ \mathbf{a}_n) \in M_{m \times n}(F)$ with columns $\mathbf{a}_j \in F^m$, and let $\mathbf{x} = (x_1, \dots, x_n) \in F^n$. Then

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

In particular $A\mathbf{e}_j = \mathbf{a}_j$ for each j . Similarly, if $\mathbf{r}_1, \dots, \mathbf{r}_m$ are the rows of A and $\mathbf{y} = (y_1 \ \cdots \ y_m)$ is a row vector, then $\mathbf{y}A = y_1\mathbf{r}_1 + \cdots + y_m\mathbf{r}_m$.

Proof. Both sides are in F^m , so we compare i -th entries. By [Definition 0.9.9](#), the i -th entry of $A\mathbf{x}$ is $\sum_{j=1}^n a_{ij}x_j$. On the right, the i -th entry of $x_j\mathbf{a}_j$ is x_ja_{ij} by [Definition 0.9.7](#), so by [Definition 0.9.6](#) the i -th entry of $\sum_j x_j\mathbf{a}_j$ is $\sum_{j=1}^n x_ja_{ij}$. These agree because multiplication in F is commutative. Taking $\mathbf{x} = \mathbf{e}_j$, whose only non-zero entry is $x_j = 1$,

gives $A\mathbf{e}_j = \mathbf{a}_j$. The statement for rows follows in the same way, comparing the k -th entries $\sum_i y_i a_{ik}$. This proves the theorem. ■

Read the formula as a sentence: **the entries of $\mathbf{x}$ are the weights, and the columns of A are what gets weighted**. So $A\mathbf{x} = \mathbf{b}$ has a solution exactly when $\mathbf{b}$ can be built from the columns of A . This single observation drives much of Chapters 2 and 3. Combined with [Theorem 0.9.2](#), it also says that **every column of AB is a combination of the columns of A** , with weights taken from the corresponding column of B , and **every row of AB is a combination of the rows of B** .

Example. (One product, three ways)

Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \\ 0 & 4 \end{pmatrix} \in M_{3 \times 2}(\mathbb{R}), \quad B = \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix} \in M_2(\mathbb{R}), \quad \mathbf{x} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

Compute $A\mathbf{x}$ as a combination of the columns of A . Then compute AB by columns, check one row by the row view, and check one entry by the entry formula.

Solution. The columns of A are $\mathbf{a}_1 = (1, 3, 0)$ and $\mathbf{a}_2 = (2, -1, 4)$. By [Theorem 0.9.3](#),

$$A\mathbf{x} = 2\mathbf{a}_1 - \mathbf{a}_2 = \begin{pmatrix} 2 \\ 6 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 7 \\ -4 \end{pmatrix}.$$

Columns. The first column of B is $\mathbf{b}_1 = (2, -1) = \mathbf{x}$, so the first column of AB is $A\mathbf{b}_1 = (0, 7, -4)$, just computed. The second column is $\mathbf{b}_2 = (1, 1)$, so $A\mathbf{b}_2 = \mathbf{a}_1 + \mathbf{a}_2 = (3, 2, 4)$. Hence

$$AB = \begin{pmatrix} 0 & 3 \\ 7 & 2 \\ -4 & 4 \end{pmatrix}.$$

Rows. Row 2 of A is $(3 \ -1)$, and by [Theorem 0.9.3](#) (rows version) $(3 \ -1)B = 3(2 \ 1) - (-1 \ 1) = (7 \ 2)$, which is row 2 of AB .

Entries. $(AB)_{32} = a_{31}b_{12} + a_{32}b_{22} = 0 \cdot 1 + 4 \cdot 1 = 4$, as in the answer.

The column view needed nothing beyond scaling and adding columns of A . When the columns of B are simple, it is usually the fastest route.

Quick check. Without multiplying out, what is the second column of $\begin{pmatrix} 5 & -1 & 7 \\ 2 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 4 & 1 \end{pmatrix}$?

Solution. The second column of the right factor is $\mathbf{e}_3 = (0, 0, 1)$. By [Theorem 0.9.2](#) and [Theorem 0.9.3](#), the second column of the product is $A\mathbf{e}_3$, the third column of the left factor: $(7, 3)$.

0.9.6 The rules of multiplication

Matrix multiplication keeps the most important rules of arithmetic: associativity, both distributive laws, and an identity.

Theorem 0.9.4 — Properties of matrix multiplication

Let $c \in F$, and let $A, A' \in M_{m \times n}(F)$, $B, B' \in M_{n \times p}(F)$ and $C \in M_{p \times q}(F)$. Then:

1. **(Associativity)** $(AB)C = A(BC)$;
2. **(Distributivity)** $A(B + B') = AB + AB'$ and $(A + A')B = AB + A'B$;
3. **(Identity)** $I_m A = A = A I_n$;
4. **(Scalars)** $c(AB) = (cA)B = A(cB)$;
5. **(Zero)** $0_{\ell \times m} A = 0_{\ell \times n}$ and $A 0_{n \times r} = 0_{m \times r}$ for all $\ell, r \geq 1$.

Idea. Each identity is checked entry by entry. The only one with content is associativity. Both sides have (i, l) -entry equal to the double sum $\sum_j \sum_k a_{ij} b_{jk} c_{kl}$, added up in two different orders, so the proof is “expand, swap the order of the finite sums, and fold back up”. In the substitution picture it is no surprise: substituting in two stages gives the same result whichever pair of stages we combine first.

Proof. For 1, both sides lie in $M_{m \times q}(F)$. Fix i and l . By Definition 0.9.9 applied twice,

$$((AB)C)_{il} = \sum_{k=1}^p (AB)_{ik} c_{kl} = \sum_{k=1}^p \left(\sum_{j=1}^n a_{ij} b_{jk} \right) c_{kl} = \sum_{k=1}^p \sum_{j=1}^n a_{ij} b_{jk} c_{kl},$$

where the last step uses distributivity in F . The same computation gives

$$(A(BC))_{il} = \sum_{j=1}^n a_{ij} (BC)_{jl} = \sum_{j=1}^n a_{ij} \left(\sum_{k=1}^p b_{jk} c_{kl} \right) = \sum_{j=1}^n \sum_{k=1}^p a_{ij} b_{jk} c_{kl}.$$

The two double sums add the same np terms in different orders, so they are equal, by commutativity and associativity of addition in F . Hence $(AB)C = A(BC)$.

For 2, $(A(B + B'))_{ik} = \sum_j a_{ij} (b_{jk} + b'_{jk}) = \sum_j a_{ij} b_{jk} + \sum_j a_{ij} b'_{jk} = (AB + AB')_{ik}$ by distributivity in F ; the other law is the same with the roles of the factors swapped.

For 3, $(I_m A)_{ik} = \sum_{j=1}^m \delta_{ij} a_{jk} = a_{ik}$, because by Definition 0.9.3 only the term $j = i$ survives. Likewise $(A I_n)_{ik} = \sum_j a_{ij} \delta_{jk} = a_{ik}$.

For 4, $(c(AB))_{ik} = c \sum_j a_{ij} b_{jk} = \sum_j (ca_{ij}) b_{jk} = ((cA)B)_{ik}$, and moving c next to b_{jk} instead, using commutativity in F , gives $(A(cB))_{ik}$. For 5, every term $0 \cdot a_{jk}$ or $a_{ij} \cdot 0$ is 0 by Theorem 0.7.1. This proves the theorem. ■

Associativity means we may write ABC without brackets, and more generally a product of any number of matrices, as long as consecutive sizes match.

Just as important is what is **missing** from the list. There is no commutative law, and no cancellation law. Each of the following failures will cost you a wrong proof if you forget it.

Warning. $AB \neq BA$ in general. Take $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Then $AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ but $BA = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. Before swapping two factors in a proof, you must know that they commute. (And if A is 2×3 and B is 3×2 , then AB and BA do not even have the same size.)

Warning. **A product can be zero with both factors non-zero.** In $M_2(\mathbb{R})$,

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

So $M_n(F)$ is not a field for $n \geq 2$, and “ $AB = 0$ implies $A = 0$ or $B = 0$ ” is false.

Warning. **Cancellation fails: $AC = BC$ does not imply $A = B$.** With

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 \\ 3 & 9 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix},$$

we get $AC = BC = \begin{pmatrix} 1 & 1 \\ 3 & 3 \end{pmatrix}$ although $A \neq B$. By the column view, both columns of C are $\mathbf{e}_1$, so AC and BC only see the first columns of A and B , which agree. Canceling C is legal only when C is invertible, as we will see below.

Quick check. Find a non-zero $A \in M_2(\mathbb{R})$ with $A^2 = AA = 0$.

Solution. Take $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. By the column view, $A\mathbf{e}_1 = \mathbf{0}$ and $A\mathbf{e}_2 = \mathbf{e}_1$, so the columns of A^2 are $A\mathbf{0} = \mathbf{0}$ and $A\mathbf{e}_1 = \mathbf{0}$. Hence $A^2 = 0$ although $A \neq 0$. This small matrix is the source of a remarkable number of counterexamples in this book; it will reappear often.

0.9.7 The transpose

Rows and columns play symmetric roles in a matrix, and the transpose exchanges them.

Definition 0.9.10 — Transpose

Let $A = (a_{ij}) \in M_{m \times n}(F)$. The **transpose** of A is the matrix $A^T \in M_{n \times m}(F)$ with entries

$$(A^T)_{ij} = a_{ji}.$$

In words: row i of A becomes column i of A^T , and the size flips from $m \times n$ to $n \times m$. A square matrix is reflected in its main diagonal.

Definition 0.9.11 — Symmetric and skew-symmetric matrices

A square matrix $A \in M_n(F)$ is **symmetric** if $A^T = A$, that is, $a_{ij} = a_{ji}$ for all i, j . It is **skew-symmetric** if $A^T = -A$, that is, $a_{ij} = -a_{ji}$ for all i, j .

Example. (Transposes, symmetric and skew-symmetric matrices)

Compute the transpose of $A = \begin{pmatrix} 1 & -2 & 0 \\ 3 & 1 & 4 \end{pmatrix}$. Decide whether

$$S = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{pmatrix}, \quad K = \begin{pmatrix} 0 & 2 & -1 \\ -2 & 0 & 3 \\ 1 & -3 & 0 \end{pmatrix}, \quad S' = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 6 & 6 \end{pmatrix}$$

over $\mathbb{R}$ are symmetric or skew-symmetric.

Solution. The rows of A become columns: $A^T = \begin{pmatrix} 1 & 3 \\ -2 & 1 \\ 0 & 4 \end{pmatrix} \in M_{3 \times 2}(\mathbb{R})$.

S is symmetric: $s_{12} = s_{21} = 2$, $s_{13} = s_{31} = 3$, $s_{23} = s_{32} = 5$. K is skew-symmetric: $k_{12} = 2 = -k_{21}$, $k_{13} = -1 = -k_{31}$, $k_{23} = 3 = -k_{32}$, and each diagonal entry satisfies $k_{ii} = -k_{ii}$, so it is 0. The matrix S' differs from S only in the entry $(3, 2)$. Every other symmetric pair still matches, but $s'_{23} = 5 \neq 6 = s'_{32}$, so S' is **not** symmetric. A 1×1 matrix is always symmetric, and the identity I_n and every diagonal matrix are symmetric.

Over $\mathbb{R}$, a skew-symmetric matrix has zero diagonal, since $a_{ii} = -a_{ii}$ forces $2a_{ii} = 0$. Over $\mathbb{F}_2$, where $-1 = 1$, “skew-symmetric” and “symmetric” mean the same thing, and I_2 is skew-symmetric. The characteristic of the field (Definition 0.7.2) matters.

The transpose is compatible with the operations, except that it **reverses** products:

Theorem 0.9.5 — Properties of the transpose

Let $c \in F$. For matrices of sizes for which each expression is defined:

1. $(A^T)^T = A$;
2. $(A + B)^T = A^T + B^T$;
3. $(cA)^T = cA^T$;
4. $(AB)^T = B^T A^T$ for $A \in M_{m \times n}(F)$ and $B \in M_{n \times p}(F)$.

Idea. The order reverses for a size reason before any computation: $(AB)^T$ is $p \times m$, and $B^T A^T$ is $(p \times n)(n \times m) = p \times m$, while $A^T B^T$ is $(n \times m)(p \times n)$, usually not even defined. Then compare entries.

Proof. For 1, $((A^T)^T)_{ij} = (A^T)_{ji} = a_{ij}$ by Definition 0.9.10. For 2 and 3, $((A + B)^T)_{ij} = a_{ji} + b_{ji} = (A^T + B^T)_{ij}$ and $((cA)^T)_{ij} = ca_{ji} = (cA^T)_{ij}$.

For 4, both sides lie in $M_{p \times m}(F)$. Fix $1 \leq i \leq p$ and $1 \leq k \leq m$. By Definition 0.9.10

and Definition 0.9.9,

$$((AB)^\top)_{ik} = (AB)_{ki} = \sum_{j=1}^n a_{kj}b_{ji}, \quad (B^\top A^\top)_{ik} = \sum_{j=1}^n (B^\top)_{ij}(A^\top)_{jk} = \sum_{j=1}^n b_{ji}a_{kj}.$$

The two sums agree term by term because multiplication in F is commutative. Hence $(AB)^\top = B^\top A^\top$. This proves the theorem. ■

For example, for every $A \in M_{m \times n}(F)$, the square matrix $A^\top A$ is symmetric: by parts 4 and 1, $(A^\top A)^\top = A^\top (A^\top)^\top = A^\top A$. Matrices of this form are central to least squares and the singular value decomposition much later on.

0.9.8 The trace

The trace is the simplest number attached to a square matrix, and it has one surprising property.

Definition 0.9.12 — Trace

The **trace** of a square matrix $A = (a_{ij}) \in M_n(F)$ is the sum of its diagonal entries:

$$\operatorname{tr} A = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \cdots + a_{nn} \in F.$$

Example. (Computing traces)

Compute $\operatorname{tr} A$ for $A = \begin{pmatrix} 2 & 7 & -1 \\ 0 & -3 & 5 \\ 4 & 1 & 6 \end{pmatrix} \in M_3(\mathbb{R})$, and $\operatorname{tr} I_n$ over $\mathbb{R}$ and over $\mathbb{F}_2$.

Solution. $\operatorname{tr} A = 2 + (-3) + 6 = 5$; the off-diagonal entries play no role. $\operatorname{tr} I_n = 1 + \cdots + 1 = n \cdot 1$. Over $\mathbb{R}$ this is n . Over $\mathbb{F}_2$ it is 0 when n is even and 1 when n is odd. The trace is **not** defined for a non-square matrix.

Theorem 0.9.6 — Properties of the trace

Let $A, B \in M_n(F)$ and $c \in F$. Then:

1. $\operatorname{tr}(A + B) = \operatorname{tr} A + \operatorname{tr} B$ and $\operatorname{tr}(cA) = c \operatorname{tr} A$;
2. $\operatorname{tr}(A^\top) = \operatorname{tr} A$;
3. $\operatorname{tr}(AB) = \operatorname{tr}(BA)$. More generally, this holds for all $A \in M_{m \times n}(F)$ and $B \in M_{n \times m}(F)$, where AB is $m \times m$ and BA is $n \times n$.

Idea. Part 3 is surprising because $AB \neq BA$ in general. But the trace only sees the diagonal, and $\operatorname{tr}(AB) = \sum_i \sum_j a_{ij}b_{ji}$ is a sum over all pairs (i, j) of the product of a_{ij}

with the “mirror” entry b_{ji} . That description does not care which matrix is written first.

Proof. Parts 1 and 2 follow from the definitions: the diagonal entries of $A + B$, cA and A^T are $a_{ii} + b_{ii}$, ca_{ii} and a_{ii} .

For 3, let $A \in M_{m \times n}(F)$ and $B \in M_{n \times m}(F)$. By [Definition 0.9.12](#) and [Definition 0.9.9](#),

$$\operatorname{tr}(AB) = \sum_{i=1}^m (AB)_{ii} = \sum_{i=1}^m \sum_{j=1}^n a_{ij} b_{ji} = \sum_{j=1}^n \sum_{i=1}^m b_{ji} a_{ij} = \sum_{j=1}^n (BA)_{jj} = \operatorname{tr}(BA),$$

where the middle equality swaps the order of the finite double sum and uses commutativity of multiplication in F . This proves the theorem. ■

A first payoff: for all $A, B \in M_n(F)$, the matrix $AB - BA$ has trace $\operatorname{tr}(AB) - \operatorname{tr}(BA) = 0$. So over $\mathbb{R}$ it can never equal I_n , whose trace is n ([Exercise \(C1: \$AB - BA = I\$ is impossible, or is it?\)](#)). The trace will also turn out to be one of the quantities that does not change when we change coordinates.

It is tempting to conclude that the trace of a product ignores the order of **all** the factors. It does not.

Warning. $\operatorname{tr}(ABC) \neq \operatorname{tr}(BAC)$ **in general**. Take $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $C = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Then $ABC = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ has trace 1, but $BA = 0$, so $BAC = 0$ has trace 0. What **is** true is that a **cyclic** rotation preserves the trace: $\operatorname{tr}(ABC) = \operatorname{tr}((AB)C) = \operatorname{tr}(C(AB)) = \operatorname{tr}(CAB)$ by part 3.

0.9.9 Inverse matrices

In a field we solve $ax = b$ by multiplying by a^{-1} , and this works exactly when $a \neq 0$. We would like to solve a system $A\mathbf{x} = \mathbf{b}$ the same way, by “dividing by A ”. What should that mean, and for which A is it possible?

The naive answer, “every non-zero matrix”, fails. Let $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq 0$. For **any** $B \in M_2(F)$, the first column of BN is B times the first column of N , that is $B\mathbf{0} = \mathbf{0}$, by [Theorem 0.9.2](#). The first column of I_2 is $\mathbf{e}_1 \neq \mathbf{0}$, so $BN \neq I_2$: no matrix undoes N from the left. Dividing needs a genuine two-sided inverse, and not every non-zero matrix has one.

An invertible matrix is a square matrix whose effect can be undone by multiplying, on either side.

Definition 0.9.13 — Invertible matrix

A **square** matrix $A \in M_n(F)$ is **invertible** if there exists $B \in M_n(F)$ such that

$$AB = I_n \quad \text{and} \quad BA = I_n.$$

Such a B is called **an inverse** of A . A square matrix that is not invertible is called **singular**.

In words: A is invertible when some matrix B of the same size cancels it **from the right** ($AB = I_n$) **and** from the left ($BA = I_n$). The definition asks for one B doing both jobs, and it only speaks about square matrices.

Is “an inverse” really “the inverse”? Yes: an inverse, if it exists, is unique. This is part 1 of the next theorem, and it lets us write A^{-1} . The remaining parts are what inverses are for.

Theorem 0.9.7 — Properties of inverses

Let $A, B \in M_n(F)$.

1. **(Uniqueness)** If B and B' are both inverses of A , then $B = B'$. We write A^{-1} for **the** inverse of an invertible A .
2. If A is invertible, then so is A^{-1} , and $(A^{-1})^{-1} = A$.
3. If A and B are invertible, then so is AB , and $(AB)^{-1} = B^{-1}A^{-1}$.
4. If A is invertible, then so is A^T , and $(A^T)^{-1} = (A^{-1})^T$.
5. If A is invertible and $c \in F$ is **non-zero**, then cA is invertible, and $(cA)^{-1} = c^{-1}A^{-1}$.
6. **(Cancellation)** If A is invertible, then $AX = AY$ implies $X = Y$ for all $X, Y \in M_{n \times p}(F)$, and $XA = YA$ implies $X = Y$ for all $X, Y \in M_{p \times n}(F)$.
7. **(Solving)** If A is invertible, then for every $\mathbf{b} \in F^n$ the equation $A\mathbf{x} = \mathbf{b}$ has **exactly one** solution $\mathbf{x} \in F^n$, namely $\mathbf{x} = A^{-1}\mathbf{b}$.

Idea. Uniqueness is the standard trick for “prove unique by assuming two”: sandwich B' between B and A , and use associativity to collapse the product both ways. For the other parts the candidate inverse is written in the statement, so we only multiply it out on **both** sides, as the definition demands. The order in part 3 is the “socks and shoes” rule: to undo “first B , then A ”, undo A first.

Proof. For 1, suppose $AB = BA = I_n$ and $AB' = B'A = I_n$. By Theorem 0.9.4 (identity and associativity),

$$B = BI_n = B(AB') = (BA)B' = I_n B' = B'.$$

For 2, the equations $AA^{-1} = I_n$ and $A^{-1}A = I_n$ say precisely that A is an inverse of A^{-1} . By part 1 it is the inverse.

For 3, by associativity,

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI_n A^{-1} = AA^{-1} = I_n,$$

and similarly $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I_n$. Hence $B^{-1}A^{-1}$ is an inverse of AB , and by part 1 it is $(AB)^{-1}$.

For 4, by Theorem 0.9.5, $A^T(A^{-1})^T = (A^{-1}A)^T = I_n^T = I_n$ and $(A^{-1})^T A^T = (AA^{-1})^T = I_n^T = I_n$, where $I_n^T = I_n$ since I_n is symmetric. Hence $(A^{-1})^T$ is the inverse of A^T .

For 5, since $c \neq 0$, the inverse c^{-1} exists in F . By Theorem 0.9.4 (scalars), $(cA)(c^{-1}A^{-1}) = (cc^{-1})(AA^{-1}) = I_n$, and similarly in the other order.

For 6, suppose $AX = AY$. Multiplying on the left by A^{-1} and using associativity, $X = I_n X = (A^{-1}A)X = A^{-1}(AX) = A^{-1}(AY) = (A^{-1}A)Y = Y$. The statement for $XA = YA$ follows by multiplying on the right by A^{-1} .

For 7, $\mathbf{x} = A^{-1}\mathbf{b}$ is a solution, since $A(A^{-1}\mathbf{b}) = (AA^{-1})\mathbf{b} = \mathbf{b}$. If $\mathbf{x}$ is any solution, then $A\mathbf{x} = \mathbf{b} = A(A^{-1}\mathbf{b})$, and part 6 (with $p = 1$) gives $\mathbf{x} = A^{-1}\mathbf{b}$. This proves the theorem. ■

Part 6 is the repair of the failed cancellation law from the warnings above: **you may cancel an invertible matrix**, on the side where it stands, and nothing else.

Example. (Some invertible matrices)

Show that each of the following is invertible and find its inverse: I_n ; a diagonal matrix $D = \text{diag}(d_1, \dots, d_n)$ with **all** $d_i \neq 0$; $A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix} \in M_2(\mathbb{R})$; and $U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in M_2(\mathbb{F}_2)$.

Solution. $I_n I_n = I_n$, so I_n is its own inverse. This is the degenerate case: I_n plays the role that 1 plays among numbers.

For D , let $E = \text{diag}(d_1^{-1}, \dots, d_n^{-1})$, which exists since every $d_i \neq 0$. Then $(DE)_{ik} = \sum_j d_{ij} e_{jk}$, and only $j = i = k$ can contribute, so $DE = \text{diag}(d_1 d_1^{-1}, \dots, d_n d_n^{-1}) = I_n$; likewise $ED = I_n$. Hence $D^{-1} = E$. For $n = 1$ this says the 1×1 matrix (a) is invertible when $a \neq 0$, with inverse (a^{-1}) , exactly as in the field.

For A , let $B = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$. Then

$$AB = \begin{pmatrix} 6-5 & -2+2 \\ 15-15 & -5+6 \end{pmatrix} = I_2, \quad BA = \begin{pmatrix} 6-5 & 3-3 \\ -10+10 & -5+6 \end{pmatrix} = I_2,$$

so $A^{-1} = B$. (Where B came from is explained by [Theorem 0.9.8](#) below.)

For U over $\mathbb{F}_2$, $U^2 = \begin{pmatrix} 1 & 1+1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, since $1+1=0$. So U is its own inverse.

Over $\mathbb{R}$ the same array would have inverse $\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$: the answer depends on the field.

Now a non-example by minimal change. The matrix $\text{diag}(1, 2)$ is invertible, with inverse $\text{diag}(1, \frac{1}{2})$. Change the entry 2 to 0: $A = \text{diag}(1, 0)$ is still diagonal and still non-zero. But for any $B \in M_2(F)$, the second column of BA is B times the second column of A , which is $B\mathbf{0} = \mathbf{0}$, while the second column of I_2 is $\mathbf{e}_2$. So $BA \neq I_2$ for every B : the clause “**there exists** B with $\dots BA = I_n$ ” fails, and $\text{diag}(1, 0)$ is singular. The same argument shows that any square matrix with a zero column is singular, and in particular so is every zero matrix $0_{n \times n}$.

Why does the definition insist on a **square** matrix and on **both** equations? For non-square matrices one-sided inverses can exist without the other side: with $C = \begin{pmatrix} 1 & 0 \end{pmatrix} \in M_{1 \times 2}(\mathbb{R})$ and $D = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, we get $CD = (1) = I_1$ but $DC = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \neq I_2$. Demanding both equations makes the inverse unique (part 1) and makes both cancellation laws (part 6) hold.

Warning. For square matrices, $AB = I_n$ already implies $BA = I_n$. This is true, but it is not obvious, and we do not know it yet. Nothing in this chapter proves it: the entries of AB and BA are different sums. It is proved in Chapter 2, using row reduction. Until then, to show that B is the inverse of A , check **both** $AB = I_n$ and $BA = I_n$. (The non-square example above shows that some argument is needed.)

For 2×2 matrices, invertibility is decided by one number.

Theorem 0.9.8 — Inverse of a 2×2 matrix

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(F)$. Then A is invertible **if and only if** $ad - bc \neq 0$. In that case,

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Idea. Where does the formula come from? Try to kill the off-diagonal entries of A by multiplying with a matrix built from its own entries. Swapping a and d and negating b and c gives $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = (ad - bc)I_2$. If $ad - bc \neq 0$, divide by it. If $ad - bc = 0$, the same computation shows that A sends a non-zero column to $\mathbf{0}$, and an invertible matrix cannot do that, since its inverse would have to send $\mathbf{0}$ back to a non-zero column.

Proof. Write $\Delta = ad - bc$ and $\tilde{A} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. By Definition 0.9.9 and commutativity of multiplication in F ,

$$A\tilde{A} = \begin{pmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{pmatrix} = \Delta I_2, \quad \tilde{A}A = \begin{pmatrix} da - bc & db - bd \\ -ca + ac & -cb + ad \end{pmatrix} = \Delta I_2.$$

($\Leftarrow$) Suppose $\Delta \neq 0$, so Δ^{-1} exists in F . By Theorem 0.9.4 (scalars), $A(\Delta^{-1}\tilde{A}) = \Delta^{-1}(A\tilde{A}) = \Delta^{-1}\Delta I_2 = I_2$, and likewise $(\Delta^{-1}\tilde{A})A = I_2$. Hence A is invertible with $A^{-1} = \Delta^{-1}\tilde{A}$, which is the stated formula.

($\Rightarrow$) We prove the contrapositive. Suppose $\Delta = 0$. We first find $\mathbf{x} \in F^2$ with $\mathbf{x} \neq \mathbf{0}$ and $A\mathbf{x} = \mathbf{0}$.

Case 1. $A = 0$. Take $\mathbf{x} = \mathbf{e}_1$.

Case 2. $c \neq 0$ or $d \neq 0$. Take $\mathbf{x} = (d, -c) \neq \mathbf{0}$. Then $A\mathbf{x} = (ad - bc, cd - dc) = (\Delta, 0) = \mathbf{0}$.

Case 3. $c = d = 0$ but $a \neq 0$ or $b \neq 0$. Take $\mathbf{x} = (b, -a) \neq \mathbf{0}$. Then $A\mathbf{x} = (ab - ba, cb - da) = (0, 0) = \mathbf{0}$, since $c = d = 0$.

These cases cover every A . Now suppose, for a contradiction, that A is invertible. Then by Theorem 0.9.4,

$$\mathbf{x} = I_2\mathbf{x} = (A^{-1}A)\mathbf{x} = A^{-1}(A\mathbf{x}) = A^{-1}\mathbf{0} = \mathbf{0},$$

contradicting $\mathbf{x} \neq \mathbf{0}$. Hence A is not invertible. This proves the theorem. ■

The number $ad - bc$ will return in Chapter 6 as the determinant of A . Here it arose from a single multiplication, and the proof used nothing beyond the field axioms, so the theorem holds over every field. For instance, $\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$ has $ad - bc = 6 - 5 = 1$, which explains the inverse found in [Example \(Some invertible matrices\)](#); and $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ has $ad - bc = 0$, so it is singular, with $A(2, -1) = \mathbf{0}$.

0.9.10 Powers and polynomials of a matrix

A square matrix can be multiplied by itself, so we can form powers, and then plug the matrix into a polynomial ([Definition 0.8.1](#)). This is how polynomials act on matrices throughout the book.

Definition 0.9.14 — Powers and polynomials of a matrix

Let $A \in M_n(F)$. The **powers** of A are defined recursively by

$$A^0 = I_n, \quad A^{k+1} = A^k A \quad (k \in \mathbb{N}).$$

For a polynomial $p = a_0 + a_1x + \cdots + a_Nx^N \in F[x]$, the matrix $p(A)$ is

$$p(A) = a_0I_n + a_1A + a_2A^2 + \cdots + a_NA^N \in M_n(F).$$

Two points in this definition deserve attention. The **constant term** a_0 becomes a_0I_n , because a_0 alone is a number and cannot be added to matrices; this matches $x^0 = 1$ with $A^0 = I_n$. And $p(A)$ is well defined, because enlarging N only adds terms $0 \cdot A^k = 0$, exactly as for evaluation at a number ([Definition 0.8.6](#)).

The next theorem says that plugging in a matrix respects sums and products, just as plugging in a number does ([Theorem 0.8.5](#)).

Theorem 0.9.9 — Polynomials in one matrix

Let $A \in M_n(F)$ and $p, q \in F[x]$. Then:

1. $A^j A^k = A^{j+k}$ for all $j, k \in \mathbb{N}$;
2. $(p + q)(A) = p(A) + q(A)$ and $(pq)(A) = p(A)q(A)$;
3. $p(A)q(A) = q(A)p(A)$.

Idea. In the proof of [Theorem 0.8.5](#), the only delicate step moved c^i past a coefficient b_j . Here the coefficients are scalars, and scalars can be moved freely through matrix products. The other step, $c^i c^j = c^{i+j}$, becomes part 1, which is associativity plus induction. Part 3 is then free, because $pq = qp$ in $F[x]$.

Proof. For 1, fix j and use induction on k ([Theorem 0.2.7](#)). For $k = 0$, $A^j A^0 = A^j I_n = A^j$. If $A^j A^k = A^{j+k}$, then by [Definition 0.9.14](#) and associativity ([Theorem 0.9.4](#)),

$$A^j A^{k+1} = A^j (A^k A) = (A^j A^k) A = A^{j+k} A = A^{j+k+1}.$$

For 2, let $p = \sum_{i=0}^m a_i x^i$ and $q = \sum_{j=0}^l b_j x^j$. The statement for sums follows from [Theorem 0.9.1](#), since $(a_i + b_i)A^i = a_i A^i + b_i A^i$. For products, by [Definition 0.8.2](#) the coefficient of x^k in pq is $\sum_{i+j=k} a_i b_j$, so

$$(pq)(A) = \sum_k \left(\sum_{i+j=k} a_i b_j \right) A^k = \sum_{i=0}^m \sum_{j=0}^l (a_i b_j) A^i A^j = \left(\sum_{i=0}^m a_i A^i \right) \left(\sum_{j=0}^l b_j A^j \right) = p(A) q(A),$$

where the second equality uses part 1 and [Theorem 0.9.1](#), and the third uses distributivity and the scalar rule $(a_i A^i)(b_j A^j) = (a_i b_j) A^i A^j$ from [Theorem 0.9.4](#).

For 3, $pq = qp$ in $F[x]$ by [Theorem 0.8.1](#). Hence by part 2, $p(A)q(A) = (pq)(A) = (qp)(A) = q(A)p(A)$. This proves the theorem. ■

Example. (Plugging a matrix into a polynomial)

Let $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} \in M_2(\mathbb{R})$ and $p = x^2 - 4x + 3$. Compute $p(A)$ directly, and again using the factorization $p = (x - 1)(x - 3)$.

Solution. First $A^2 = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 8 \\ 0 & 9 \end{pmatrix}$. Hence

$$p(A) = A^2 - 4A + 3I_2 = \begin{pmatrix} 1 & 8 \\ 0 & 9 \end{pmatrix} - \begin{pmatrix} 4 & 8 \\ 0 & 12 \end{pmatrix} + \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

The constant term contributed $3I_2$, not “3”. By [Theorem 0.9.9](#), $p(A) = (A - I_2)(A - 3I_2)$, and indeed

$$(A - I_2)(A - 3I_2) = \begin{pmatrix} 0 & 2 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} -2 & 2 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Note that neither factor is zero; this is the second warning of the multiplication part in action. That the diagonal entries 1, 3 of A are the roots of p is no accident, as we will see with eigenvalues.

[Theorem 0.9.9](#) is about polynomials in **one** matrix. Identities with **two** different matrices are another matter.

Warning. School identities fail for matrices that do not commute. Expanding with distributivity, $(A + B)^2 = A^2 + AB + BA + B^2$, and this equals $A^2 + 2AB + B^2$ only when $AB = BA$. With $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$: $(A + B)^2 = I_2$, but $A^2 + 2AB + B^2 = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$. Before using a polynomial identity valid for numbers, check that all the matrices involved commute; polynomials in a single A always do.

0.9.11 Two equivalence relations on matrices

We close with two relations on matrices that the rest of the book studies in depth. Both are equivalence relations ([Definition 0.5.2](#)), and the proofs are short exercises in using [Theorem 0.9.7](#).

Example. (Similarity)

For $A, B \in M_n(F)$, say A is **similar** to B , written $A \sim B$, if there is an **invertible** $P \in M_n(F)$ with $B = P^{-1}AP$. Prove that similarity is an equivalence relation on $M_n(F)$. Then check that $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ is similar to $\text{diag}(1, 3)$ using $P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, and show that $\text{diag}(1, 2)$ is **not** similar to I_2 over $\mathbb{R}$.

Solution. We check the three conditions of [Definition 0.5.2](#).

Reflexive. Let $A \in M_n(F)$. The identity I_n is invertible with $I_n^{-1} = I_n$ ([Example \(Some invertible matrices\)](#)), and $I_n^{-1}AI_n = A$. Hence $A \sim A$.

Symmetric. Suppose $A \sim B$, so $B = P^{-1}AP$ with P invertible. Multiplying on the left by P and on the right by P^{-1} , and using associativity, $PBP^{-1} = (PP^{-1})A(PP^{-1}) = A$. By [Theorem 0.9.7](#) (part 2), $Q = P^{-1}$ is invertible with $Q^{-1} = P$, so $A = Q^{-1}BQ$. Hence $B \sim A$.

Transitive. Suppose $A \sim B$ and $B \sim C$, so $B = P^{-1}AP$ and $C = Q^{-1}BQ$ with P, Q invertible. Then

$$C = Q^{-1}(P^{-1}AP)Q = (Q^{-1}P^{-1})A(PQ) = (PQ)^{-1}A(PQ),$$

where the last step is [Theorem 0.9.7](#) (part 3), which also says PQ is invertible. Hence $A \sim C$. This shows that similarity is an equivalence relation.

For the numerical part, $P^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$ by [Theorem 0.9.8](#) (here $ad - bc = 1$), and

$$P^{-1}AP = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}.$$

Finally, if $P^{-1}I_2P = \text{diag}(1, 2)$ for some invertible P , then $\text{diag}(1, 2) = P^{-1}P = I_2$, which is false. (Alternatively: similar matrices have equal traces by [Theorem 0.9.6](#), and $\text{tr } \text{diag}(1, 2) = 3 \neq 2 = \text{tr } I_2$.) Hence $\text{diag}(1, 2) \not\sim I_2$.

In Chapter 3 we will see that similar matrices describe the same linear map in two different coordinate systems. That is why a quantity like the trace, which similar matrices share, is really a property of the map and not of the matrix. Deciding when two matrices are similar is one of the main problems of this book.

Example. (Row equivalence)

For $A, B \in M_{m \times n}(F)$, say A is **row equivalent** to B if there is an **invertible** $E \in M_m(F)$ with $B = EA$. Prove that row equivalence is an equivalence relation on $M_{m \times n}(F)$. Then, with $E = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$, compute EA for $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 0 \end{pmatrix}$ and describe

what E did to the rows of A .

Solution. Reflexive. $A = I_m A$ and I_m is invertible, so A is row equivalent to itself.

Symmetric. Suppose $B = EA$ with E invertible. Multiplying on the left by E^{-1} , $E^{-1}B = (E^{-1}E)A = A$, and E^{-1} is invertible by [Theorem 0.9.7](#) (part 2). So B is row equivalent to A .

Transitive. Suppose $B = EA$ and $C = E'B$ with E, E' invertible. Then $C = E'(EA) = (E'E)A$ by associativity, and $E'E$ is invertible by [Theorem 0.9.7](#) (part 3). So A is row equivalent to C . This shows that row equivalence is an equivalence relation.

For the computation, E is invertible, with inverse $\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$ (check both products, or use [Theorem 0.9.8](#) with $ad - bc = 1$). By the row view of [Theorem 0.9.2](#) and [Theorem 0.9.3](#), row 1 of EA is $1 \cdot (\text{row 1 of } A)$ and row 2 is $-2 \cdot (\text{row 1}) + 1 \cdot (\text{row 2})$:

$$EA = \begin{pmatrix} 1 & 2 & 1 \\ 2-2 & 5-4 & 0-2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & -2 \end{pmatrix}.$$

So E subtracted twice row 1 from row 2, which is the first step of Gaussian elimination on A .

In Chapter 2 we will see that every elementary row operation is multiplication on the left by an invertible matrix, and that row equivalence is exactly “reachable by a sequence of row operations”. This is why Gaussian elimination never changes the solutions of a linear system.

0.9.12 Exercises

0.9.12.1 A. Check your understanding

Exercise A1.

- Let $A \in M_{2 \times 3}(F)$ and $B \in M_{3 \times 4}(F)$. Which of AB , BA , $A^T A$ and $B^T A^T$ are defined? Give the size of each one that is.
- Define what it means for $A \in M_n(F)$ to be invertible.
- Determine whether the following statement is true: “if $A, B \in M_n(F)$ and $AB = 0$, then $A = 0$ or $B = 0$.” Justify your answer.
- Determine whether the following statement is true: “ $(A + B)(A - B) = A^2 - B^2$ for all $A, B \in M_2(\mathbb{R})$.” Justify your answer.
- Let $A \in M_3(F)$ have columns $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$. Express $A(2, 0, -1)$ and the second column of A^2 in terms of the columns of A and A itself.

Solution.

- AB is $(2 \times 3)(3 \times 4)$, defined, of size 2×4 . BA would pair rows of length 4 with columns of length 2, so it is not defined. $A^T A$ is $(3 \times 2)(2 \times 3)$, defined, of size 3×3 . $B^T A^T$ is $(4 \times 3)(3 \times 2)$, defined, of size 4×2 ; it equals $(AB)^T$ by [Theorem 0.9.5](#).
- A is invertible if there exists $B \in M_n(F)$ with $AB = I_n$ **and** $BA = I_n$.

- (c) False. $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = 0$ with both factors non-zero.
- (d) False. Expanding gives $(A+B)(A-B) = A^2 - AB + BA - B^2$, which differs from $A^2 - B^2$ unless $AB = BA$. For $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, we have $A^2 = B^2 = 0$, but $(A+B)(A-B) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \neq 0$.
- (e) By [Theorem 0.9.3](#), $A(2, 0, -1) = 2\mathbf{a}_1 - \mathbf{a}_3$. By [Theorem 0.9.2](#), the second column of $A^2 = AA$ is $A\mathbf{a}_2$.

0.9.12.2 B. Practice

Exercise B1: Products and sizes.

Let

$$A = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ 1 & 0 \\ 0 & 4 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

over $\mathbb{R}$.

- Compute AB and BA .
- Determine which of AC and CA is defined, and compute it.
- Compute $A\mathbf{x}$ for $\mathbf{x} = (2, 0, -1)$ as a combination of the columns of A .

Solution.

- (a) AB is 2×2 and BA is 3×3 . Row against column,

$$AB = \begin{pmatrix} 2 - 1 + 0 & 1 + 0 + 8 \\ 0 + 3 + 0 & 0 + 0 + 4 \end{pmatrix} = \begin{pmatrix} 1 & 9 \\ 3 & 4 \end{pmatrix}, \quad BA = \begin{pmatrix} 2 & 1 & 5 \\ 1 & -1 & 2 \\ 0 & 12 & 4 \end{pmatrix}.$$

For instance $(BA)_{12} = 2 \cdot (-1) + 1 \cdot 3 = 1$ and $(BA)_{32} = 0 \cdot (-1) + 4 \cdot 3 = 12$.

- (b) AC would be $(2 \times 3)(2 \times 2)$, and $3 \neq 2$, so it is not defined. CA is $(2 \times 2)(2 \times 3)$, of size 2×3 :

$$CA = \begin{pmatrix} 1 & -1 + 6 & 2 + 2 \\ 3 & -3 + 12 & 6 + 4 \end{pmatrix} = \begin{pmatrix} 1 & 5 & 4 \\ 3 & 9 & 10 \end{pmatrix}.$$

- (c) By [Theorem 0.9.3](#), $A\mathbf{x} = 2\mathbf{a}_1 + 0\mathbf{a}_2 - \mathbf{a}_3 = 2(1, 0) - (2, 1) = (0, -1)$.

Exercise B2: Traces of products.

- With A and B as in [Exercise \(B1: Products and sizes\)](#), verify that $\text{tr}(AB) = \text{tr}(BA)$, although AB and BA have different sizes.
- Let $A = \text{diag}(1, 2)$, $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $C = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Show that $\text{tr}(ABC) \neq \text{tr}(BAC)$, and verify that $\text{tr}(ABC) = \text{tr}(CAB)$.

Solution.

- (a) From [Exercise \(B1: Products and sizes\)](#), $\text{tr}(AB) = 1 + 4 = 5$ and $\text{tr}(BA) = 2 + (-1) + 4 = 5$, as [Theorem 0.9.6](#) predicts.
- (b) By the column view, BC has columns $B\mathbf{e}_2 = \mathbf{e}_1$ and $B\mathbf{0} = \mathbf{0}$, so $BC = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, and $ABC = A(BC) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, with trace 1. Next, $BA = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$, so $BAC = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$, with trace 2. Hence $\text{tr}(ABC) = 1 \neq 2 = \text{tr}(BAC)$. Finally $AB = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $CAB = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$, with trace $1 = \text{tr}(ABC)$, as the cyclic rule predicts.

Exercise B3: Inverses of 2×2 matrices.

- (a) Find the inverse of $\begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix} \in M_2(\mathbb{R})$, and check both products.
- (b) Show that $A = \begin{pmatrix} 2 & 6 \\ 1 & 3 \end{pmatrix}$ is not invertible by finding a non-zero $\mathbf{x}$ with $A\mathbf{x} = \mathbf{0}$.
- (c) Find the inverse of $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \in M_2(\mathbb{F}_5)$.
- (d) For which $t \in \mathbb{R}$ is $\begin{pmatrix} t & 2 \\ 2 & t \end{pmatrix}$ invertible?

Solution.

- (a) Here $ad - bc = 9 - 8 = 1 \neq 0$, so by [Theorem 0.9.8](#) the inverse is $B = \begin{pmatrix} 3 & -4 \\ -2 & 3 \end{pmatrix}$.
Check: $\begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix} B = \begin{pmatrix} 9 - 8 & -12 + 12 \\ 6 - 6 & -8 + 9 \end{pmatrix} = I_2$ and $B \begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 9 - 8 & 12 - 12 \\ -6 + 6 & -8 + 9 \end{pmatrix} = I_2$.
- (b) Here $ad - bc = 6 - 6 = 0$. Following the proof of [Theorem 0.9.8](#), take $\mathbf{x} = (d, -c) = (3, -1) \neq \mathbf{0}$. Then $A\mathbf{x} = (6 - 6, 3 - 3) = \mathbf{0}$. If A were invertible, then $\mathbf{x} = A^{-1}(A\mathbf{x}) = \mathbf{0}$, a contradiction. Hence A is not invertible.
- (c) In $\mathbb{F}_5$, $ad - bc = 4 - 6 = -2 = 3 \neq 0$, and $3^{-1} = 2$ since $3 \cdot 2 = 6 = 1$. By [Theorem 0.9.8](#) the inverse is $2 \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} 8 & -4 \\ -6 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}$. Check:
 $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix} = \begin{pmatrix} 11 & 5 \\ 25 & 11 \end{pmatrix} = I_2$ and $\begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 6 & 10 \\ 10 & 16 \end{pmatrix} = I_2$ in $\mathbb{F}_5$.
- (d) By [Theorem 0.9.8](#), it is invertible if and only if $t^2 - 4 \neq 0$, that is, $t \neq 2$ and $t \neq -2$.

Exercise B4: Transposes of products.

- (a) Prove that $(A_1 A_2 \cdots A_k)^\top = A_k^\top \cdots A_2^\top A_1^\top$ for all $k \geq 1$ and matrices $A_1, \dots, A_k$ whose sizes allow the product.
- (b) Let $A \in M_n(F)$. Prove that $A + A^\top$ is symmetric and $A - A^\top$ is skew-symmetric.

Solution.

- (a) We use induction on k ([Theorem 0.2.7](#)). For $k = 1$ there is nothing to prove. Suppose the formula holds for products of k matrices, and let $A_1, \dots, A_{k+1}$ be given. By associativity, $A_1 \cdots A_{k+1} = (A_1 \cdots A_k)A_{k+1}$. By [Theorem 0.9.5](#) (part 4) and then the induction hypothesis,

$$(A_1 \cdots A_{k+1})^\top = A_{k+1}^\top (A_1 \cdots A_k)^\top = A_{k+1}^\top A_k^\top \cdots A_1^\top.$$

This proves the formula for all $k \geq 1$.

- (b) By [Theorem 0.9.5](#) (parts 1–3), $(A + A^\top)^\top = A^\top + (A^\top)^\top = A^\top + A = A + A^\top$, so $A + A^\top$ is symmetric. Similarly $(A - A^\top)^\top = A^\top - A = -(A - A^\top)$, so $A - A^\top$ is skew-symmetric.

0.9.12.3 C. Going deeper

Exercise C1: $AB - BA = I$ is impossible, or is it?.

- (a) Prove that there are no $A, B \in M_n(\mathbb{R})$, for any $n \geq 1$, with $AB - BA = I_n$.
- (b) Show that the argument of (a) breaks down in $M_2(\mathbb{F}_2)$, and that $A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ satisfy $AB - BA = I_2$ over $\mathbb{F}_2$.

Hint: consider the trace of both sides.

Solution.

- (a) Suppose, for a contradiction, that $AB - BA = I_n$. Taking traces and using [Theorem 0.9.6](#) (parts 1 and 3),

$$n = \text{tr } I_n = \text{tr}(AB - BA) = \text{tr}(AB) - \text{tr}(BA) = 0.$$

This contradicts $n \geq 1$. Hence no such A, B exist.

- (b) Over $\mathbb{F}_2$ the same computation gives $\text{tr } I_2 = 1 + 1 = 0$, so taking traces yields $0 = 0$ and no contradiction. For the given matrices, the column view gives $AB = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ and $BA = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, so $AB - BA = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2$, because $-1 = 1$ in $\mathbb{F}_2$.

Exercise C2: Triangular matrices with non-zero diagonal.

Let F be a field.

- (a) Let $U = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$ with $a, d \neq 0$. Show that U is invertible and that U^{-1} is upper triangular.

- (b) Let $U = \begin{pmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{pmatrix}$ with $a, d, f \neq 0$. Show that

$$V = \begin{pmatrix} a^{-1} & -b(ad)^{-1} & (be - cd)(adf)^{-1} \\ 0 & d^{-1} & -e(df)^{-1} \\ 0 & 0 & f^{-1} \end{pmatrix}$$

is the inverse of U .

- (c) Explain how the entries of V can be found, column by column, by solving $U\mathbf{v}_k = \mathbf{e}_k$.

The general case, for all n , follows from the methods of Chapter 2.

Solution.

- (a) Here $ad - b \cdot 0 = ad \neq 0$, since a product of non-zero elements of a field is non-zero (Theorem 0.7.1). By Theorem 0.9.8, U is invertible with $U^{-1} = (ad)^{-1} \begin{pmatrix} d & -b \\ 0 & a \end{pmatrix} = \begin{pmatrix} a^{-1} & -b(ad)^{-1} \\ 0 & d^{-1} \end{pmatrix}$, which is upper triangular.
- (b) The inverses a^{-1}, d^{-1}, f^{-1} exist, and $ad, df, adf \neq 0$ by Theorem 0.7.1, so V is defined. We check $UV = I_3$ entry by entry. The diagonal entries are $a \cdot a^{-1} = 1$, $d \cdot d^{-1} = 1$, $f \cdot f^{-1} = 1$. Below the diagonal both matrices are zero, so $(UV)_{21} = (UV)_{31} = (UV)_{32} = 0$. Above the diagonal,

$$(UV)_{12} = a \cdot (-b(ad)^{-1}) + bd^{-1} = -bd^{-1} + bd^{-1} = 0,$$

$$(UV)_{23} = d \cdot (-e(df)^{-1}) + ef^{-1} = -ef^{-1} + ef^{-1} = 0,$$

$$(UV)_{13} = a(be - cd)(adf)^{-1} + b(-e(df)^{-1}) + cf^{-1} = (be - cd)(df)^{-1} - be(df)^{-1} + cd(df)^{-1} = 0.$$

Hence $UV = I_3$. For VU , again the diagonal entries are 1 and the entries below the diagonal are 0, and

$$(VU)_{12} = a^{-1}b - b(ad)^{-1}d = a^{-1}b - a^{-1}b = 0,$$

$$(VU)_{23} = d^{-1}e - e(df)^{-1}f = 0,$$

$$(VU)_{13} = a^{-1}c - b(ad)^{-1}e + (be - cd)(adf)^{-1}f = a^{-1}c - be(ad)^{-1} + be(ad)^{-1} - cd(ad)^{-1} = 0,$$

using $cd(ad)^{-1} = ca^{-1}$. Hence $VU = I_3$, and $V = U^{-1}$. Both products had to be checked, since we may not yet use that one implies the other.

- (c) By Theorem 0.9.2, $UV = I_3$ says exactly that the columns $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ of V satisfy $U\mathbf{v}_k = \mathbf{e}_k$. Because U is upper triangular, each system can be solved from the bottom row up. For example, $U\mathbf{v}_3 = \mathbf{e}_3$ reads $fx_3 = 1$, $dx_2 + ex_3 = 0$, $ax_1 + bx_2 + cx_3 = 0$, giving $x_3 = f^{-1}$, then $x_2 = -e(df)^{-1}$, then $x_1 = (be - cd)(adf)^{-1}$, which is the third column of V . (This only finds a candidate satisfying $UV = I_3$; the check $VU = I_3$ in (b) is still needed.)

Exercise C3: Matrices that commute.

Let $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in M_2(\mathbb{R})$.

- (a) Find all $X \in M_2(\mathbb{R})$ with $XN = NX$.
 (b) Deduce that the only matrices $X \in M_2(\mathbb{R})$ with $XY = YX$ **for every** $Y \in M_2(\mathbb{R})$ are the scalar multiples cI_2 , $c \in \mathbb{R}$.

Solution.

- (a) Let $X = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$. By the column view, $XN = (X\mathbf{0} \quad X\mathbf{e}_1) = \begin{pmatrix} 0 & p \\ 0 & r \end{pmatrix}$, and by the row view, NX has rows (row 2 of X) and $\mathbf{0}$: $NX = \begin{pmatrix} r & s \\ 0 & 0 \end{pmatrix}$. Comparing entries, $XN = NX$ if and only if $r = 0$ and $p = s$, that is, $X = \begin{pmatrix} p & q \\ 0 & p \end{pmatrix} = pI_2 + qN$ with $p, q \in \mathbb{R}$ arbitrary.
- (b) Every cI_2 commutes with every Y , by [Theorem 0.9.4](#): $(cI_2)Y = cY = Y(cI_2)$. Conversely, suppose $XY = YX$ for every Y . Taking $Y = N$, part (a) gives $X = \begin{pmatrix} p & q \\ 0 & p \end{pmatrix}$. Taking $Y = N^\top = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, we compute $XN^\top = \begin{pmatrix} q & 0 \\ p & 0 \end{pmatrix}$ and $N^\top X = \begin{pmatrix} 0 & 0 \\ p & q \end{pmatrix}$. Equality forces $q = 0$. Hence $X = pI_2$, as claimed.

0.10 Groups and Permutations

Several structures in this chapter have quietly behaved alike. The invertible $n \times n$ matrices, the non-zero elements of a field, and the bijections of a set can each be combined, have an element that does nothing, and can be undone. This short section gives that common structure a name, proves its basic rules once, and introduces the permutations that the determinant in Chapter 6 is built from. We keep it brief: groups are a language for us here, not a subject.

0.10.1 Groups

Look at four situations side by side. In $\text{GL}_n(F)$, the set of invertible matrices in $M_n(F)$, we multiply matrices, I_n does nothing, and each A is undone by A^{-1} ([Theorem 0.9.7](#)). In a field F , we multiply non-zero elements, 1 does nothing, and each a is undone by a^{-1} . The four rotations of a square about its center (by $0^\circ, 90^\circ, 180^\circ, 270^\circ$) can be performed one after another, the rotation by 0° does nothing, and each rotation is undone by rotating back. The bijections $X \rightarrow X$ of a set are composed, id_X does nothing, and each is undone by its inverse function ([Theorem 0.4.7](#)). The objects differ completely, but the rules we use are the same three. So we give the pattern a name.

A group is a set with one way of combining elements, in which combining is associative, some element does nothing, and every element can be undone.

Definition 0.10.1 — Group

A **group** is a set G together with a **binary operation** $G \times G \rightarrow G$, $(a, b) \mapsto ab$, such that:

- (G1) (**Associativity**) $(ab)c = a(bc)$ for all $a, b, c \in G$;
 (G2) (**Identity**) there is an element $e \in G$ with $ea = ae = a$ for every $a \in G$;
 (G3) (**Inverses**) for every $a \in G$ there is $b \in G$ with $ab = ba = e$, where e is the element of (G2).

The group is **abelian** if moreover $ab = ba$ for all $a, b \in G$.

In words: the operation is a function on pairs, so the combination of two elements of G lies in G again; this “closure” is built into the phrase “binary operation $G \times G \rightarrow G$ ”. (G1) lets us drop brackets. (G2) asks for an element that changes nothing, from **both** sides. (G3) asks that each element can be undone, again from both sides, back to that same e . When the operation is addition we write $a + b$, 0 and $-a$ instead of ab , e and a^{-1} ; this notation is only used for abelian groups.

Example. (Groups)

Check that each of the following is a group, and decide which are abelian: $(\mathbb{Z}, +)$; $(F, +)$ and $(F \setminus \{0\}, \cdot)$ for a field F ; $\text{GL}_n(F)$ under matrix multiplication; the set S_n of bijections $\{1, \dots, n\} \rightarrow \{1, \dots, n\}$ under composition; and a one-element set $\{e\}$ with $ee = e$.

Solution. $(\mathbb{Z}, +)$: the sum of integers is an integer, addition is associative, 0 is an identity, and $-a$ undoes a . It is abelian. $(F, +)$ is a group for the same reasons, by the field axioms (Definition 0.7.1); it is abelian.

$(F \setminus \{0\}, \cdot)$: the product of two non-zero elements is non-zero by Theorem 0.7.1, so the operation lands in $F \setminus \{0\}$. Associativity and the identity $1 \neq 0$ come from the field axioms, and each $a \neq 0$ has an inverse a^{-1} , which is non-zero since $aa^{-1} = 1 \neq 0$. It is abelian.

$\text{GL}_n(F)$: by Theorem 0.9.7 (part 3), a product of invertible matrices is invertible, so the operation lands in $\text{GL}_n(F)$. Associativity holds by Theorem 0.9.4, I_n is invertible and is an identity, and $A^{-1} \in \text{GL}_n(F)$ undoes A by part 2. For $n \geq 2$ it is **not** abelian: for $n = 2$, $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, while the product in the other order is $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$. (Over $\mathbb{F}_2$ read 2 as 0; the two products still differ.) For $n \geq 3$, put these blocks in the top-left corner and 1's further down the diagonal. $\text{GL}_1(F)$ is just $F \setminus \{0\}$, and is abelian.

S_n : a composition of bijections is a bijection (Theorem 0.4.3), composition is associative (Theorem 0.4.2), id is an identity, and each bijection has an inverse function (Theorem 0.4.7), which is again a bijection. We will see below that S_n is not abelian for $n \geq 3$.

$\{e\}$: all three axioms reduce to $ee = e$. This **trivial group** is the degenerate case; it reappears as the smallest subgroup of every group.

Now non-examples, each a minimal change of an example. Replace addition on $\mathbb{Z}$ by multiplication. The operation still lands in $\mathbb{Z}$, it is associative, and 1 is an identity. But (G3) fails: there is no integer b with $2b = 1$. So $(\mathbb{Z}, \cdot)$ is **not** a group. Similarly, replace $\text{GL}_n(F)$ by all of $M_n(F)$ under multiplication: (G1) and (G2) still hold with I_n , but the zero matrix has no inverse, so (G3) fails. Keep $\mathbb{N}$ under $+$: the identity 0 is there, but 1 has no inverse in $\mathbb{N}$.

Warning. **The operation must land in the set.** $\text{GL}_n(F)$ under matrix **addition** is not a group: I_n and $-I_n$ are invertible, but $I_n + (-I_n) = 0$ is not, so addition is not even a binary operation on $\text{GL}_n(F)$. Before checking (G1)–(G3), check that combining two elements of the set gives an element of the set.

The rules we proved separately for fields and for matrices follow from the axioms alone. It is worth proving them once:

Theorem 0.10.1 — Basic properties of groups

Let G be a group and $a, b, c \in G$.

1. The identity element of G is unique.
2. The inverse of a is unique. We denote it by a^{-1} .
3. $(a^{-1})^{-1} = a$ and $(ab)^{-1} = b^{-1}a^{-1}$.
4. **(Cancellation)** If $ab = ac$, then $b = c$. If $ba = ca$, then $b = c$.

Proof. For 1, suppose e and e' both satisfy (G2). Then $e = ee'$, using that e' is an identity, and $ee' = e'$, using that e is one. Hence $e = e'$.

For 2, suppose b and b' both satisfy $ab = ba = e$ and $ab' = b'a = e$. By (G2) and (G1),

$$b = be = b(ab') = (ba)b' = eb' = b'.$$

For 3, the equations $aa^{-1} = a^{-1}a = e$ say that a is an inverse of a^{-1} , so $(a^{-1})^{-1} = a$ by part 2. Next, by (G1),

$$(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = aea^{-1} = aa^{-1} = e,$$

and similarly $(b^{-1}a^{-1})(ab) = b^{-1}(a^{-1}a)b = e$. By part 2, $(ab)^{-1} = b^{-1}a^{-1}$.

For 4, suppose $ab = ac$. Multiplying on the left by a^{-1} and using (G1), $b = (a^{-1}a)b = a^{-1}(ab) = a^{-1}(ac) = (a^{-1}a)c = c$. The second statement follows by multiplying on the right by a^{-1} . This proves the theorem. ■

These are the arguments of Theorem 0.9.7 with matrices erased. That is the point of the definition: a proof that uses only (G1)–(G3) holds in every group at once.

0.10.2 Subgroups

After a structure comes the substructure. The even integers sit inside $(\mathbb{Z}, +)$ and form a group on their own, with the same operation.

Definition 0.10.2 — Subgroup

Let G be a group. A subset $H \subseteq G$ is a **subgroup** of G if $e \in H$, and $ab \in H$ and $a^{-1} \in H$ for all $a, b \in H$.

Then the operation of G restricts to a binary operation on H , and H is a group with it: (G1) holds in H because it holds in G , and (G2), (G3) hold because e and the inverses lie in H . In practice one checks a single condition:

Theorem 0.10.2 — Subgroup test

Let G be a group and $H \subseteq G$. Then H is a subgroup of G if and only if $H \neq \emptyset$ and $ab^{-1} \in H$ for all $a, b \in H$.

Proof. ($\Rightarrow$) If H is a subgroup, then $e \in H$, so $H \neq \emptyset$. For $a, b \in H$ we have $b^{-1} \in H$, and then $ab^{-1} \in H$.

($\Leftarrow$) Suppose $H \neq \emptyset$ and $ab^{-1} \in H$ for all $a, b \in H$. Pick $h \in H$. Taking $a = b = h$ gives $e = hh^{-1} \in H$. For $b \in H$, taking $a = e$ gives $b^{-1} = eb^{-1} \in H$. For $a, b \in H$, we now know $b^{-1} \in H$, so $ab = a(b^{-1})^{-1} \in H$ by Theorem 0.10.1. Hence H is a subgroup. This proves the theorem. ■

For example, $2\mathbb{Z} = \{2k : k \in \mathbb{Z}\}$ is a subgroup of $(\mathbb{Z}, +)$. In additive notation the test reads “ $a - b \in H$ ”, and $2k - 2l = 2(k - l) \in 2\mathbb{Z}$, while $0 \in 2\mathbb{Z}$. By contrast $\mathbb{N} \subseteq \mathbb{Z}$ is non-empty and closed under $+$, but $0 - 1 = -1 \notin \mathbb{N}$: the test fails, because $\mathbb{N}$ lacks inverses. The set of odd integers fails more badly, since it does not contain 0. In $\text{GL}_n(F)$, the invertible diagonal matrices form a subgroup: I_n is one, and by Example (Some invertible matrices) the inverse of $\text{diag}(d_1, \dots, d_n)$ is $\text{diag}(d_1^{-1}, \dots, d_n^{-1})$, so DE^{-1} is again diagonal with non-zero diagonal entries. Every group G has the subgroups $\{e\}$ and G .

0.10.3 Homomorphisms

After objects come the maps between them that respect the structure.

Definition 0.10.3 — Group homomorphism

Let G and H be groups. A function $\varphi: G \rightarrow H$ is a **homomorphism** if $\varphi(ab) = \varphi(a)\varphi(b)$ for all $a, b \in G$. A **bijective** homomorphism is an **isomorphism**. The **kernel** of φ is $\ker \varphi = \{g \in G : \varphi(g) = e_H\}$, where e_H is the identity of H .

The product ab on the left is taken in G , and $\varphi(a)\varphi(b)$ on the right in H . For example, $F \setminus \{0\} \rightarrow \text{GL}_2(F)$, $a \mapsto \text{diag}(a, 1)$, is a homomorphism, since $\text{diag}(ab, 1) = \text{diag}(a, 1) \text{diag}(b, 1)$; its kernel is $\{1\}$. The map $\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$, $a \mapsto [a]$, is a homomorphism of additive groups, because $[a + b] = [a] + [b]$ is exactly how addition on $\mathbb{Z}/n\mathbb{Z}$ was defined (Example (Addition modulo n is well defined)); its kernel is $n\mathbb{Z}$. On the other hand, $x \mapsto x + 1$ from $(\mathbb{R}, +)$ to itself is not a homomorphism: writing $\varphi(x) = x + 1$, we get $\varphi(0 + 0) = 1$ but $\varphi(0) + \varphi(0) = 2$.

Theorem 0.10.3 — Basic properties of homomorphisms

Let $\varphi: G \rightarrow H$ be a homomorphism of groups. Then $\varphi(e_G) = e_H$, $\varphi(a^{-1}) = \varphi(a)^{-1}$ for every $a \in G$, and $\ker \varphi$ is a subgroup of G .

Proof. Since $\varphi(e_G)\varphi(e_G) = \varphi(e_G e_G) = \varphi(e_G) = \varphi(e_G)e_H$, cancellation (Theorem 0.10.1) gives $\varphi(e_G) = e_H$. Next, $\varphi(a)\varphi(a^{-1}) = \varphi(aa^{-1}) = \varphi(e_G) = e_H$, and similarly $\varphi(a^{-1})\varphi(a) = e_H$, so $\varphi(a^{-1}) = \varphi(a)^{-1}$ by uniqueness of inverses. Finally, $e_G \in \ker \varphi$, and for $a, b \in \ker \varphi$,

$$\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = e_H e_H^{-1} = e_H,$$

so $ab^{-1} \in \ker \varphi$. By Theorem 0.10.2, $\ker \varphi$ is a subgroup. This proves the theorem. ■

Two homomorphisms will matter later: the determinant $\text{GL}_n(F) \rightarrow F \setminus \{0\}$ and the sign $S_n \rightarrow \{1, -1\}$, both in Chapter 6. Linear maps in Chapter 3 are, among other things, homomorphisms of additive groups, and their kernels will be one of our main tools.

0.10.4 Permutations

A **permutation** of $\{1, \dots, n\}$ is a bijection $\{1, \dots, n\} \rightarrow \{1, \dots, n\}$, and S_n is the group of all of them, the **symmetric group**. It has $n!$ elements: there are n choices for $\sigma(1)$, then $n-1$ for $\sigma(2)$, and so on.

A permutation σ can be written in **two-line notation**, with $\sigma(i)$ below i . A more compact notation lists what happens along a loop. For **distinct** $i_1, \dots, i_k \in \{1, \dots, n\}$, the **cycle** $(i_1 i_2 \cdots i_k)$ is the permutation sending

$$i_1 \mapsto i_2 \mapsto \cdots \mapsto i_k \mapsto i_1$$

and fixing every other element. A cycle of length 2 is a **transposition**: $(i j)$ swaps i and j . The same cycle can be written starting at any of its entries: $(1 2 3) = (2 3 1) = (3 1 2)$, but $(1 3 2)$ is different.

The product $\sigma\tau$ in S_n means the composition $\sigma \circ \tau$: **first apply τ , then σ** . So products of cycles are read **from right to left**.

Example. (Computing with cycles)

In S_3 , compute $(1 3)(1 2)$ and $(1 2)(1 3)$. In S_5 , write $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 1 & 2 & 4 \end{pmatrix}$ (two-line notation) as a product of cycles.

Solution. For $(1 3)(1 2)$, follow each element through $(1 2)$ first, then $(1 3)$: $1 \mapsto 2 \mapsto 2$, $2 \mapsto 1 \mapsto 3$, $3 \mapsto 3 \mapsto 1$. So $(1 3)(1 2) = (1 2 3)$. For $(1 2)(1 3)$: $1 \mapsto 3 \mapsto 3$, $3 \mapsto 1 \mapsto 2$, $2 \mapsto 2 \mapsto 1$. So $(1 2)(1 3) = (1 3 2)$. The two products differ, so S_3 is not abelian; the same two transpositions show that S_n is not abelian for every $n \geq 3$.

For σ , start at 1 and follow: $1 \mapsto 3 \mapsto 1$, a loop $(1 3)$. The smallest element not yet used

is 2: $2 \mapsto 5 \mapsto 4 \mapsto 2$, a loop $(2\ 5\ 4)$. Every element is now used, so $\sigma = (1\ 3)(2\ 5\ 4)$. These two cycles move disjoint sets of elements, so they can be applied in either order.

Warning. Mind the order. $\sigma\tau$ applies τ first, exactly as for functions, and as for matrices: $(AB)\mathbf{x} = A(B\mathbf{x})$ applies B first. Reading a product of cycles from left to right gives the wrong answer, as the two different products $(1\ 3)(1\ 2) \neq (1\ 2)(1\ 3)$ above show.

Quick check. Compute $(1\ 2\ 3)(1\ 2)$ in S_3 .

Solution. Apply $(1\ 2)$ first: $1 \mapsto 2 \mapsto 3$, $2 \mapsto 1 \mapsto 2$, $3 \mapsto 3 \mapsto 1$. So 1 and 3 are swapped and 2 is fixed: $(1\ 2\ 3)(1\ 2) = (1\ 3)$.

The example showed $(1\ 2\ 3) = (1\ 3)(1\ 2)$: a 3-cycle is a product of transpositions. Every permutation is, and the proof is a model induction on n .

Theorem 0.10.4 — Transpositions generate S_n

Let $n \geq 2$. Every permutation in S_n is a product of transpositions.

Idea. Induction on n : what is the smaller object inside a permutation of $n+1$ elements? If σ fixes $n+1$, it is really a permutation of $\{1, \dots, n\}$. If not, one transposition repairs that: swap $\sigma(n+1)$ back to $n+1$. The result fixes $n+1$, so the induction hypothesis applies, and we undo the swap.

Proof. We use induction on $n \geq 2$ (Theorem 0.2.7). For $n = 2$, $S_2 = \{\text{id}, (1\ 2)\}$, and $\text{id} = (1\ 2)(1\ 2)$.

Suppose every permutation in S_n is a product of transpositions, and let $\sigma \in S_{n+1}$.

Case 1. $\sigma(n+1) = n+1$. Since σ is injective, $\sigma(i) \neq n+1$ for $i \leq n$, so σ restricts to a function $\sigma': \{1, \dots, n\} \rightarrow \{1, \dots, n\}$. It is injective because σ is, and hence bijective by Theorem 0.4.9. By the induction hypothesis, $\sigma' = \tau_1 \cdots \tau_r$ with transpositions $\tau_j \in S_n$. Regard each τ_j as a transposition in S_{n+1} fixing $n+1$. Then $\tau_1 \cdots \tau_r$ agrees with $\sigma' = \sigma$ on $\{1, \dots, n\}$ and fixes $n+1$, as σ does. Hence $\sigma = \tau_1 \cdots \tau_r$.

Case 2. $\sigma(n+1) = k \neq n+1$. Let $\tau = (k\ n+1)$. Then $(\tau\sigma)(n+1) = \tau(k) = n+1$, so by Case 1, $\tau\sigma = \tau_1 \cdots \tau_r$ for some transpositions τ_j . Since $\tau\tau = \text{id}$, multiplying on the left by τ gives $\sigma = \tau\tau_1 \cdots \tau_r$, a product of transpositions.

This completes the induction, and proves the theorem. ■

The product is not unique: $(1\ 2\ 3) = (1\ 3)(1\ 2) = (1\ 2)(2\ 3)$, and one can always insert $(1\ 2)(1\ 2) = \text{id}$. What turns out to be unique is whether the number of transpositions is even or odd. That fact defines the **sign** of a permutation, which we do not define here; it is the subject of Chapter 6, where it builds the determinant.

0.10.5 Exercises

0.10.5.1 A. Check your understanding

Exercise A1.

- (a) State the three axioms of a group.
- (b) Is $\mathbb{Z}$ with the operation $(a, b) \mapsto a - b$ a group? Justify your answer.
- (c) Determine whether the following statement is true: “in every group, $(ab)^{-1} = a^{-1}b^{-1}$.” Justify your answer.
- (d) Compute $(1\ 3)(1\ 2\ 3)$ in S_3 .
- (e) Explain why $\{A \in M_2(\mathbb{R}) : A \text{ is not invertible}\}$ is **not** a subgroup of $GL_2(\mathbb{R})$.

Solution.

- (a) (G1) $(ab)c = a(bc)$ for all a, b, c ; (G2) there is e with $ea = ae = a$ for all a ; (G3) for every a there is b with $ab = ba = e$. The operation must be a binary operation $G \times G \rightarrow G$.
- (b) No. Associativity fails: $(1 - 1) - 1 = -1$, but $1 - (1 - 1) = 1$.
- (c) False. By [Theorem 0.10.1](#), $(ab)^{-1} = b^{-1}a^{-1}$, and this can differ from $a^{-1}b^{-1}$. In S_3 take $a = (1\ 2)$ and $b = (1\ 3)$, which are their own inverses. By [Example \(Computing with cycles\)](#), $ab = (1\ 3\ 2)$, whose inverse is $(1\ 2\ 3)$, while $a^{-1}b^{-1} = ab = (1\ 3\ 2)$. The statement is true in abelian groups.
- (d) Apply $(1\ 2\ 3)$ first: $1 \mapsto 2 \mapsto 2$, $2 \mapsto 3 \mapsto 1$, $3 \mapsto 1 \mapsto 3$. Hence $(1\ 3)(1\ 2\ 3) = (1\ 2)$.
- (e) It is not even a subset of $GL_2(\mathbb{R})$, since its elements are not invertible; for instance it does not contain the identity I_2 .

0.10.5.2 B. Practice

Exercise B1: The table of S_3 .

List the six elements of S_3 in cycle notation and write out the table whose entry in row x and column y is xy . Hence find all elements $x \in S_3$ with $x^2 = \text{id}$, and all x that commute with every element of S_3 .

Solution. The elements are $\text{id}, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)$. Computing each product right to left, as in [Example \(Computing with cycles\)](#), the entry in row x , column y is xy :

xy	id	$(1\ 2)$	$(1\ 3)$	$(2\ 3)$	$(1\ 2\ 3)$	$(1\ 3\ 2)$
id	id	$(1\ 2)$	$(1\ 3)$	$(2\ 3)$	$(1\ 2\ 3)$	$(1\ 3\ 2)$
$(1\ 2)$	$(1\ 2)$	id	$(1\ 3\ 2)$	$(1\ 2\ 3)$	$(2\ 3)$	$(1\ 3)$
$(1\ 3)$	$(1\ 3)$	$(1\ 2\ 3)$	id	$(1\ 3\ 2)$	$(1\ 2)$	$(2\ 3)$
$(2\ 3)$	$(2\ 3)$	$(1\ 3\ 2)$	$(1\ 2\ 3)$	id	$(1\ 3)$	$(1\ 2)$
$(1\ 2\ 3)$	$(1\ 2\ 3)$	$(1\ 3)$	$(2\ 3)$	$(1\ 2)$	$(1\ 3\ 2)$	id
$(1\ 3\ 2)$	$(1\ 3\ 2)$	$(2\ 3)$	$(1\ 2)$	$(1\ 3)$	id	$(1\ 2\ 3)$

For example, the entry in row $(1\ 2)$, column $(2\ 3)$ is $(1\ 2)(2\ 3)$: $1 \mapsto 1 \mapsto 2$, $2 \mapsto 3 \mapsto 3$, $3 \mapsto 2 \mapsto 1$, which is $(1\ 2\ 3)$.

Reading the diagonal, $x^2 = \text{id}$ exactly for $x = \text{id}, (1\ 2), (1\ 3), (2\ 3)$. An element x commutes with every element exactly when row x equals column x . This holds for id . It fails for every other element: $(1\ 2)(1\ 3) = (1\ 3\ 2) \neq (1\ 2\ 3) = (1\ 3)(1\ 2)$ handles $(1\ 2)$ and $(1\ 3)$; $(2\ 3)(1\ 2) = (1\ 3\ 2) \neq (1\ 2\ 3) = (1\ 2)(2\ 3)$ handles $(2\ 3)$; and $(1\ 2\ 3)(1\ 2) = (1\ 3) \neq (2\ 3) = (1\ 2)(1\ 2\ 3)$ handles $(1\ 2\ 3)$ and, since $(1\ 3\ 2)(1\ 2) = (2\ 3) \neq (1\ 3) = (1\ 2)(1\ 3\ 2)$, also $(1\ 3\ 2)$. Hence only id commutes with everything.

Exercise B2: Invertible upper triangular matrices.

Let F be a field and $T = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} : a, b, d \in F, a \neq 0, d \neq 0 \right\}$.

- (a) Prove that T is a subgroup of $\text{GL}_2(F)$.
- (b) For which fields F is T abelian? Justify your answer.

Solution.

- (a) Let $X = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in T$. Since $a, d \neq 0$, $ad - b \cdot 0 = ad \neq 0$ by [Theorem 0.7.1](#), so X is invertible by [Theorem 0.9.8](#). Hence $T \subseteq \text{GL}_2(F)$, and $T \neq \emptyset$ since $I_2 \in T$. Now let $Y = \begin{pmatrix} a' & b' \\ 0 & d' \end{pmatrix} \in T$. By [Theorem 0.9.8](#), $Y^{-1} = (a'd')^{-1} \begin{pmatrix} d' & -b' \\ 0 & a' \end{pmatrix} = \begin{pmatrix} a'^{-1} & -b'(a'd')^{-1} \\ 0 & d'^{-1} \end{pmatrix}$. Therefore

$$XY^{-1} = \begin{pmatrix} aa'^{-1} & -ab'(a'd')^{-1} + bd'^{-1} \\ 0 & dd'^{-1} \end{pmatrix},$$

whose $(2, 1)$ -entry is 0 and whose diagonal entries aa'^{-1} , dd'^{-1} are non-zero by [Theorem 0.7.1](#). So $XY^{-1} \in T$, and T is a subgroup by [Theorem 0.10.2](#).

- (b) It depends on the field. *Case 1:* F has an element c with $c \neq 0$ and $c \neq 1$ (for example $c = 2$ in $\mathbb{R}$ or $\mathbb{F}_3$). Let $X = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $Y = \text{diag}(1, c)$, both in T . Then $XY = \begin{pmatrix} 1 & c \\ 0 & c \end{pmatrix}$ and $YX = \begin{pmatrix} 1 & 1 \\ 0 & c \end{pmatrix}$, which differ in the $(1, 2)$ -entry since $c \neq 1$. So T is **not** abelian. *Case 2:* $F = \{0, 1\}$, that is, $F = \mathbb{F}_2$. Then $a = d = 1$ is forced, so $T = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} : b \in \mathbb{F}_2 \right\}$, and $\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & b' \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & b+b' \\ 0 & 1 \end{pmatrix}$ is symmetric in b, b' . So T is abelian over $\mathbb{F}_2$, and over every other field it is not.

Exercise B3: The exponential is an isomorphism.

Let $\mathbb{R}_{>0} = \{x \in \mathbb{R} : x > 0\}$. Show that $(\mathbb{R}_{>0}, \cdot)$ is a group, and that $\exp: (\mathbb{R}, +) \rightarrow (\mathbb{R}_{>0}, \cdot)$, $x \mapsto e^x$, is an isomorphism. What is its kernel? You may use the familiar facts $e^{x+y} = e^x e^y$, $e^x > 0$, and that $\ln: \mathbb{R}_{>0} \rightarrow \mathbb{R}$ satisfies $\ln(e^x) = x$ and $e^{\ln y} = y$.

Solution. A product of positive reals is positive, so multiplication is a binary operation on $\mathbb{R}_{>0}$. It is associative, $1 \in \mathbb{R}_{>0}$ is an identity, and for $y > 0$ the inverse $1/y$ is positive. Hence $(\mathbb{R}_{>0}, \cdot)$ is a group.

Since $e^x > 0$, $\exp$ maps $\mathbb{R}$ into $\mathbb{R}_{>0}$, and $e^{x+y} = e^x e^y$ says exactly that it is a homomorphism from $(\mathbb{R}, +)$ to $(\mathbb{R}_{>0}, \cdot)$. The function $\ln$ satisfies $\ln \circ \exp = \text{id}_{\mathbb{R}}$ and $\exp \circ \ln = \text{id}_{\mathbb{R}_{>0}}$, so $\exp$ is bijective by [Theorem 0.4.7](#). Hence $\exp$ is an isomorphism. Its kernel is $\{x \in \mathbb{R} : e^x = 1\} = \{0\}$, since $\exp$ is injective and $e^0 = 1$.

0.10.5.3 C. Going deeper

Exercise C1: Every element its own inverse.

- (a) Let G be a group in which $g^2 = gg = e$ for every $g \in G$. Prove that G is abelian.
- (b) Show that $(M_2(\mathbb{F}_2), +)$ is an example of such a group, written additively.

Solution.

- (a) Let $a, b \in G$. The hypothesis $gg = e$ says that every g is its own inverse, by uniqueness of inverses ([Theorem 0.10.1](#)). Applying this to ab , a and b , and using [Theorem 0.10.1](#),

$$ab = (ab)^{-1} = b^{-1}a^{-1} = ba.$$

Hence G is abelian.

- (b) $(M_2(\mathbb{F}_2), +)$ is a group by [Theorem 0.9.1](#): addition is associative, 0 is an identity, and $-A$ is an inverse. In additive notation, “ $g^2 = e$ ” reads $A + A = 0$, and indeed $(A + A)_{ij} = a_{ij} + a_{ij} = (1 + 1)a_{ij} = 0$ in $\mathbb{F}_2$. (This group is abelian anyway, consistent with (a).)

Exercise C2: The order of a product of disjoint cycles.

The **order** of $\sigma \in S_n$ is the smallest integer $m \geq 1$ with $\sigma^m = \text{id}$, where σ^m is the product of m copies of σ . Two cycles are **disjoint** if no element appears in both.

- (a) Let $\sigma = (i_1 \cdots i_k)$ be a k -cycle. Prove that $\sigma^m = \text{id}$ if and only if k divides m . Deduce that σ has order k .
- (b) Prove that disjoint cycles σ and τ commute, and deduce $(\sigma\tau)^m = \sigma^m\tau^m$ for all $m \geq 1$.
- (c) Let σ be a k -cycle and τ an l -cycle, disjoint. Prove that $\sigma\tau$ has order $\text{lcm}(k, l)$.
- (d) Find the order of $(1\ 2)(3\ 4\ 5) \in S_5$.

Hint for (c): σ^m moves only elements that appear in σ .

Solution.

- (a) Index the entries of the cycle modulo k , so $\sigma(i_j) = i_{j+1}$ with $i_{k+1} = i_1$. By induction on m , $\sigma^m(i_j) = i_{j+m}$, indices taken modulo k , and σ^m fixes every element not in the cycle. Hence $\sigma^m = \text{id}$ if and only if $i_{j+m} = i_j$ for all j . Since $i_1, \dots, i_k$ are distinct, this holds if and only if $j + m \equiv j \pmod{k}$, that is, $k \mid m$. The smallest $m \geq 1$ with $k \mid m$ is k , so σ has order k .

- (b) Let A and B be the disjoint sets of elements appearing in σ and τ . Note that σ maps A to A and fixes everything outside A , and τ does the same with B . Let $i \in \{1, \dots, n\}$. If $i \in A$, then $i \notin B$ and $\sigma(i) \in A$, so $\sigma(i) \notin B$; hence $\sigma\tau(i) = \sigma(i) = \tau\sigma(i)$. The case $i \in B$ is the same with the roles swapped. If $i \notin A \cup B$, both sides give i . Hence $\sigma\tau = \tau\sigma$. Then $(\sigma\tau)^m = \sigma^m\tau^m$ follows by induction on m : $(\sigma\tau)^{m+1} = \sigma^m\tau^m\sigma\tau = \sigma^m\sigma\tau^m\tau = \sigma^{m+1}\tau^{m+1}$, moving σ past τ one factor at a time, which is allowed because they commute.
- (c) By (b), $(\sigma\tau)^m = \sigma^m\tau^m$. We claim this is id if and only if $\sigma^m = \text{id}$ and $\tau^m = \text{id}$. One direction is clear. Conversely, suppose $\sigma^m\tau^m = \text{id}$, and let $i \in A$. Then $i \notin B$, so $\tau^m(i) = i$ by the hint, and $\sigma^m(i) = \sigma^m\tau^m(i) = i$. Since σ^m also fixes every $i \notin A$, $\sigma^m = \text{id}$, and then $\tau^m = \sigma^m\tau^m = \text{id}$ as well. By (a), $(\sigma\tau)^m = \text{id}$ if and only if $k \mid m$ and $l \mid m$. The smallest such $m \geq 1$ is $\text{lcm}(k, l)$, which is therefore the order of $\sigma\tau$.
- (d) $(1\ 2)$ and $(3\ 4\ 5)$ are disjoint cycles of lengths 2 and 3, so by (c) the order is $\text{lcm}(2, 3) = 6$.

Chapter 1

Vector Spaces

Arrows in the plane can be added and stretched. So can solutions of a homogeneous system of equations, polynomials, matrices, and real-valued functions. In each case the same handful of rules governs the arithmetic, and none of the arguments we care about uses anything beyond those rules. This chapter names the structure, a **vector space**, so that every theorem proved once holds for all of these examples at the same time.

The second half of the chapter answers a question that sounds innocent: how big is a vector space? The spaces $\mathbb{R}^3$, $\mathbb{R}[x]_{\leq 2}$ and the symmetric 2×2 matrices are all infinite sets, yet each is “three-dimensional” in a precise sense. Making that sense precise, and proving that it does not depend on any choices, takes the Exchange Theorem, the most important proof in the chapter. Everything later in the book measures with the dimension defined here.

What you need. From Chapter 0: fields ([Definition 0.7.1](#)) and their basic arithmetic ([Theorem 0.7.1](#)), sets and double inclusion ([Theorem 0.3.1](#)), functions, and induction ([Theorem 0.2.7](#)). Polynomials ([Definition 0.8.1](#)) and matrices ([Definition 0.9.1](#)) supply many of the examples.

1.0.1 Roadmap

- **Vector spaces.** The eight axioms, a gallery of examples and non-examples, and the first consequences of the axioms.
- **Subspaces.** A subset that is a vector space in its own right, recognized by three checks.
- **Linear combinations and span.** Everything that can be built from a list of vectors, and the smallest subspace containing them.
- **Linear independence.** When a list contains no redundant vector, and the Linear Dependence Lemma that finds the redundant one when it does.
- **Bases and coordinates.** Lists that span without waste; every vector gets unique coordinates, and every finite spanning list can be trimmed to a basis.
- **The Exchange Theorem and dimension.** Independent lists are never longer than spanning lists, so all bases have the same length; extending bases and counting instead of checking.
- **Sums and direct sums.** Combining subspaces, the dimension formula, and splitting a space into independent pieces.
- **Infinite-dimensional spaces.** Every vector space has a basis (via Zorn’s lemma), and which finite-dimensional facts fail without finiteness.
- **Changing the field.** The same set viewed over a smaller or larger field: restriction of

scalars, the tower law, and complexification.

1.0.2 Named moves

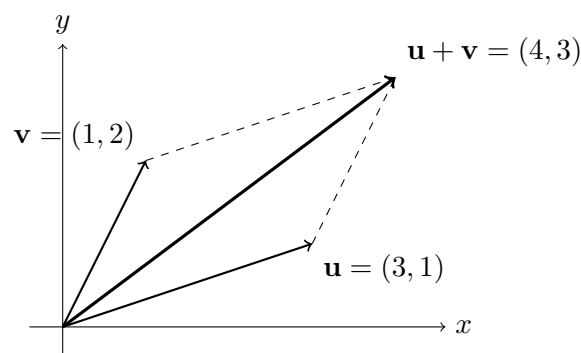
- **The three-check subspace test.** Zero vector, closed under addition, closed under scaling, each check ending with its own sentence.
- **“Let $a_1\mathbf{v}_1 + \cdots + a_k\mathbf{v}_k = \mathbf{0}$.”** The first line of every independence proof; the goal is then to show every a_i is zero.
- **Exchange one vector at a time.** Swap a vector of an independent list into a spanning list without losing the span.
- **Count instead of check.** In a space of dimension n , an independent list of length n is automatically a basis, and so is a spanning list of length n .
- **Basis of the smallest space, then extend.** For any dimension formula, start from a basis of the smallest subspace in sight, extend it, and count.

1.1 Vector Spaces

Chapter 0 met several kinds of objects that can be added and scaled: pairs of real numbers, polynomials, matrices. Each time we checked the same handful of rules by hand. This section collects those rules into one definition, the vector space, and proves the first consequences once, so that they hold in every example at the same time.

1.1.1 One set of rules in three places

Start with the plane $\mathbb{R}^2$. A pair $\mathbf{u} = (3, 1)$ can be pictured as an arrow from the origin to the point $(3, 1)$. Two arrows are added tip to tail, which is the same as taking the diagonal of the parallelogram they span, and an arrow is scaled by stretching it.



The picture shows one rule at once: going along $\mathbf{u}$ and then $\mathbf{v}$ ends at the same corner as going along $\mathbf{v}$ and then $\mathbf{u}$, so $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$. Other rules are just as familiar. Scaling a sum is the same as adding the scaled arrows, $2(\mathbf{u} + \mathbf{v}) = 2\mathbf{u} + 2\mathbf{v}$. Adding the zero arrow changes nothing, and every arrow is undone by its reverse.

Now look at polynomials of degree at most 2. We add them coefficient by coefficient,

$$(1 + 2x - x^2) + (3 - x + 4x^2) = 4 + x + 3x^2,$$

and scale them the same way, $2(1 + 2x - x^2) = 2 + 4x - 2x^2$. There is no picture of arrows here, yet the rules are identical: order does not matter in a sum, the zero polynomial changes nothing, and $-1 - 2x + x^2$ undoes $1 + 2x - x^2$.

The same happens for 2×2 real matrices. [Theorem 0.9.1](#) lists the rules for adding matrices and multiplying them by numbers, and they are again the same rules.

Arrows, polynomials and matrices are very different objects, but many arguments about them only use these rules. If we prove a statement from the rules alone, it holds for arrows, polynomials and matrices at once, and for every other system obeying the rules. So we list the rules once and give the systems that obey them a name.

A vector space is a set whose elements can be added and scaled by numbers from a field, with the usual rules of arithmetic.

1.1.2 The definition

Definition 1.1.1 — Vector Space

Let F be a field. A **vector space over F** is a set V together with two functions

$$+ : V \times V \rightarrow V, (\mathbf{u}, \mathbf{v}) \mapsto \mathbf{u} + \mathbf{v}, \quad \cdot : F \times V \rightarrow V, (a, \mathbf{v}) \mapsto a\mathbf{v},$$

called **addition** and **scalar multiplication**, such that the following axioms hold.

- (VS1) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ for all $\mathbf{u}, \mathbf{v} \in V$.
- (VS2) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$.
- (VS3) There exists an element $\mathbf{0} \in V$ such that $\mathbf{v} + \mathbf{0} = \mathbf{v}$ for every $\mathbf{v} \in V$.
- (VS4) For every $\mathbf{v} \in V$ there exists an element $\mathbf{v}' \in V$ such that $\mathbf{v} + \mathbf{v}' = \mathbf{0}$.
- (VS5) $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$ for all $a \in F$ and $\mathbf{u}, \mathbf{v} \in V$.
- (VS6) $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$ for all $a, b \in F$ and $\mathbf{v} \in V$.
- (VS7) $a(b\mathbf{v}) = (ab)\mathbf{v}$ for all $a, b \in F$ and $\mathbf{v} \in V$.
- (VS8) $1\mathbf{v} = \mathbf{v}$ for every $\mathbf{v} \in V$, where 1 is the identity element of F .

Read the definition clause by clause.

- **The two functions.** Addition takes two elements of V and returns an element of V . Scalar multiplication takes a number from F and an element of V and returns an element of V . So “the sum stays in the set” and “a multiple stays in the set” are not extra axioms: they are built into the word “function into V ”. When we test a candidate, this closure is the first thing to check.
- **(VS1)–(VS4)** concern addition alone. It is commutative and associative, there is a vector $\mathbf{0}$ that changes nothing, and every vector can be undone. These are the axioms (F1)–(F4) of [Definition 0.7.1](#), now for vectors.
- **(VS5) and (VS6)** are the two distributive laws. In (VS6) the symbol $+$ appears twice with two meanings: on the left it adds scalars in F , on the right it adds vectors in V .
- **(VS7)** says scaling by b and then by a is scaling by the product ab , computed in F .
- **(VS8)** says the scalar 1 acts as it should. It looks too obvious to state, but we will see below that it cannot be dropped.

Note the order of quantifiers in (VS3): **one** vector $\mathbf{0}$ works for **every** $\mathbf{v}$. In (VS4) the vector $\mathbf{v}'$ is allowed to depend on $\mathbf{v}$.

Definition 1.1.2 — Vectors and Scalars

Let V be a vector space over F . The elements of V are called **vectors**, and the elements of F are called **scalars**. The element $\mathbf{0}$ in (VS3) is a **zero vector** of V , and an element $\mathbf{v}'$ as in (VS4) is an **additive inverse** of $\mathbf{v}$.

A vector space over $\mathbb{R}$ is also called a **real vector space**, and one over $\mathbb{C}$ a **complex vector space**. When the operations are clear from context, we say “ V is a vector space over F ” instead of naming the triple $(V, +, \cdot)$. The field is part of the data, and we will see shortly that changing it changes the space.

We wrote “**a** zero vector” and “**an** additive inverse” on purpose. The axioms only say these exist. That there is only one of each is a theorem, proved later in this section.

Warning. Two different zeros. The scalar $0 \in F$ and the zero vector $\mathbf{0} \in V$ are different objects. In $M_2(\mathbb{R})$, the scalar is the number 0 and the zero vector is the zero matrix; in $\mathbb{R}^3$, the zero vector is $(0, 0, 0)$. This book writes the zero vector in bold. An equation such as $0\mathbf{v} = \mathbf{0}$ has a scalar on the left and a vector on the right, and mixing them up produces statements that make no sense, such as “ $\mathbf{v} = 0$ ” for a matrix $\mathbf{v}$.

1.1.3 Examples

We start with the families we will use throughout the book. For each, the work is to say what the operations are and why the axioms hold. We check some axioms in full and point out the pattern for the rest.

Example. (The Standard Vector Spaces)

Let F be a field. Explain why each of the following is a vector space over F .

- (a) F^n , with entrywise addition and scalar multiplication.
- (b) $M_{m \times n}(F)$, with the matrix operations of Definition 0.9.6 and Definition 0.9.7.
- (c) $F[x]$, with the addition of Definition 0.8.2 and $c \sum a_k x^k := \sum (ca_k) x^k$.
- (d) For a set X , the set F^X of **all** functions $X \rightarrow F$, with the **pointwise** operations

$$(f + g)(t) := f(t) + g(t), \quad (af)(t) := af(t) \quad \text{for all } t \in X.$$

- (e) The set $F^{\mathbb{N}}$ of all sequences $(s_0, s_1, s_2, \dots)$ of elements of F , with termwise operations.

Solution.

- (b) The sum and scalar multiple of $m \times n$ matrices are $m \times n$ matrices, so both operations are functions into $M_{m \times n}(F)$. The eight axioms are exactly the rules of Theorem 0.9.1, with the zero matrix as $\mathbf{0}$ and $-A$ as an additive inverse of A .
- (c) Chapter 0 defines $F^n = M_{n \times 1}(F)$, the column vectors, and the entrywise opera-

tions are the matrix operations. So (a) is the case of (b) with n rows and 1 column. The zero vector is $(0, \dots, 0)$.

- (d) The formulas define functions $f + g$ and af from X to F , so the operations land in F^X . Two functions $X \rightarrow F$ are equal when they take the same value at every $t \in X$ (Definition 0.4.1), so each axiom is checked one point at a time. For (VS3), let $\mathbf{0}$ be the zero function, $\mathbf{0}(t) = 0$ for all t . Then $(f + \mathbf{0})(t) = f(t) + 0 = f(t)$ for every t , by (F3), so $f + \mathbf{0} = f$. For (VS4), let $f'(t) := -f(t)$. Then $(f + f')(t) = f(t) + (-f(t)) = 0 = \mathbf{0}(t)$ by (F4), so $f + f' = \mathbf{0}$. For (VS6), let $a, b \in F$. For every $t \in X$,

$$((a + b)f)(t) = (a + b)f(t) = af(t) + bf(t) = (af)(t) + (bf)(t) = (af + bf)(t),$$

where the second equality is (F9) together with (F6) in F . Hence $(a + b)f = af + bf$. The other five axioms reduce in the same way to a field axiom at each point: (VS1) to (F1), (VS2) to (F2), (VS5) to (F9), (VS7) to (F5), and (VS8) to (F7) with (F6).

- (e) A sequence $(s_0, s_1, \dots)$ is a function $\mathbb{N} \rightarrow F$, $k \mapsto s_k$, and the termwise operations are the pointwise ones. So (e) is (d) with $X = \mathbb{N}$.
- (f) Let $p = \sum a_k x^k \in F[x]$ and $c \in F$. Only finitely many a_k are non-zero, and $a_k = 0$ forces $ca_k = 0$, so cp is again a polynomial. The sum of polynomials is a polynomial by Definition 0.8.2. By Theorem 0.8.1 (parts 1 and 2), addition satisfies (VS1)–(VS4), with the zero polynomial as $\mathbf{0}$ and $-p$ as an additive inverse. Since polynomials are equal when their coefficients agree (Definition 0.8.1), (VS5)–(VS8) are checked coefficient by coefficient. For instance, the coefficient of x^k in $(c + d)p$ is $(c + d)a_k = ca_k + da_k$, which is the coefficient of x^k in $cp + dp$. The others reduce in the same way: (VS5) to (F9), (VS7) to (F5), and (VS8) to (F7) with (F6).

The pattern is worth naming. Whenever the operations are defined **entry by entry**, **point by point** or **coefficient by coefficient**, each axiom for vectors is a field axiom applied in each slot. In fact (a), (b) and (e) are all instances of (d): a column in F^n is a function $\{1, \dots, n\} \rightarrow F$, and a matrix in $M_{m \times n}(F)$ is a function on the set of positions $\{1, \dots, m\} \times \{1, \dots, n\}$.

A **subset** of a known space often works just as well, provided the operations do not lead out of it.

The set $F[x]_{\leq n}$ of Definition 0.8.4 is a vector space over F , with the operations of $F[x]$. Closure is the only real issue. If $\deg p \leq n$ and $\deg q \leq n$, then $\deg(p + q) \leq \max(\deg p, \deg q) \leq n$ by Theorem 0.8.2. If $a_k = 0$ for all $k > n$, then $ca_k = 0$ for all $k > n$, so $\deg(cp) \leq n$. The zero polynomial has degree $-\infty \leq n$, and $-p$ has the same coefficients as p up to sign, so it stays in $F[x]_{\leq n}$. The six axioms (VS1), (VS2), (VS5)–(VS8) are equations that hold for **all** polynomials, so in particular for those of degree at most n .

The next example applies the same pattern to a set of real numbers, with a smaller field of scalars.

Example. $(\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$)

Let $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$, with the addition of $\mathbb{R}$ and with scalar multiplication $c(a + b\sqrt{2}) := ca + cb\sqrt{2}$ for $c \in \mathbb{Q}$. Show that $\mathbb{Q}(\sqrt{2})$ is a vector space over $\mathbb{Q}$.

Solution. Let $\mathbf{u} = a + b\sqrt{2}$ and $\mathbf{v} = a' + b'\sqrt{2}$ with $a, b, a', b' \in \mathbb{Q}$, and let $c \in \mathbb{Q}$.

Closure. We have $\mathbf{u} + \mathbf{v} = (a + a') + (b + b')\sqrt{2}$ and $c\mathbf{u} = ca + cb\sqrt{2}$. Sums and

products of rationals are rational, so both lie in $\mathbb{Q}(\sqrt{2})$. Hence addition is a function $\mathbb{Q}(\sqrt{2}) \times \mathbb{Q}(\sqrt{2}) \rightarrow \mathbb{Q}(\sqrt{2})$, and scalar multiplication is a function $\mathbb{Q} \times \mathbb{Q}(\sqrt{2}) \rightarrow \mathbb{Q}(\sqrt{2})$.

Zero and additive inverses. The number $0 = 0 + 0\sqrt{2}$ lies in $\mathbb{Q}(\sqrt{2})$ and satisfies $\mathbf{u} + 0 = \mathbf{u}$, so (VS3) holds. The number $-\mathbf{u} = (-a) + (-b)\sqrt{2}$ lies in $\mathbb{Q}(\sqrt{2})$ and satisfies $\mathbf{u} + (-\mathbf{u}) = 0$, so (VS4) holds.

The identities. Scalar multiplication by $c \in \mathbb{Q}$ is the multiplication of real numbers, since $c(a + b\sqrt{2}) = ca + cb\sqrt{2}$ in $\mathbb{R}$. So each of (VS1), (VS2) and (VS5)–(VS8) is an identity between real numbers, and it holds by the field axioms of $\mathbb{R}$: (VS1) is (F1), (VS2) is (F2), (VS5) is (F9), (VS6) is (F9) with (F6), (VS7) is (F5), and (VS8) is (F7) with (F6). Hence $\mathbb{Q}(\sqrt{2})$ is a vector space over $\mathbb{Q}$.

We saw in [Example \(\$\mathbb{Q}\(\sqrt{2}\)\$ Is a Field\)](#) that $\mathbb{Q}(\sqrt{2})$ is even a field. As a vector space over $\mathbb{Q}$, we simply ignore how to multiply two of its elements together and keep only multiplication by rationals.

The same set can carry more than one field of scalars. The next example makes this concrete.

Example. (The Same Set over Different Fields)

Explain why $\mathbb{C}$ is a vector space over $\mathbb{C}$ and also over $\mathbb{R}$, and why $\mathbb{R}$ is a vector space over $\mathbb{R}$ and also over $\mathbb{Q}$. In each case the addition is the usual one and the scalar multiplication is the usual product.

Solution. $\mathbb{C}$ over $\mathbb{C}$ is F^1 for $F = \mathbb{C}$, covered by [Example \(The Standard Vector Spaces\)](#) (a). For $\mathbb{C}$ over $\mathbb{R}$, a real number r is the complex number $r + 0i$, and rz is their complex product, which lies in $\mathbb{C}$. Each axiom is an identity among complex numbers in which some of the numbers happen to be real, so it holds by [Proposition 0.6.1](#): (VS5) and (VS6) by distributivity together with commutativity, (VS7) by associativity of multiplication, and (VS8) since $1 \cdot z = z$. The argument for $\mathbb{R}$ over $\mathbb{Q}$ is the same, with the field axioms of $\mathbb{R}$ in place of the laws of $\mathbb{C}$.

As sets, $\mathbb{C}$ over $\mathbb{C}$ and $\mathbb{C}$ over $\mathbb{R}$ are the same. As vector spaces they differ, because the scalars available differ. In $\mathbb{C}$ over $\mathbb{C}$, the vector i is a scalar multiple of the vector 1 . In $\mathbb{C}$ over $\mathbb{R}$ it is not, since $r \cdot 1 = i$ has no real solution r . Later in this chapter this difference becomes a difference in dimension. That is why we always say **over which field**.

The degenerate case. The set $\{\mathbf{0}\}$ with one element, with $\mathbf{0} + \mathbf{0} = \mathbf{0}$ and $a\mathbf{0} = \mathbf{0}$ for every $a \in F$, is a vector space over any field. Every axiom is an equation between elements that all equal $\mathbf{0}$, and $\mathbf{0}$ serves as the zero vector and as its own additive inverse. It is called the **zero space**. It matters because it turns up constantly as an answer: the only solution of a system, or the smallest vector space inside every other one. Unlike a field, which must contain $1 \neq 0$ ([Definition 0.7.1](#)), a vector space is allowed to have a single element, since no axiom asks for a vector different from $\mathbf{0}$.

A final example looks strange, but it satisfies every axiom.

Example. (Positive Reals with Exotic Operations)

Let $V = \mathbb{R}_{>0} = \{t \in \mathbb{R} : t > 0\}$. For $u, v \in V$ and $c \in \mathbb{R}$, define

$$u \boxplus v := uv, \quad c \boxminus v := v^c.$$

Show that these operations are functions into V , find the zero vector and the additive inverse of v , and check (VS6) and (VS8).

Solution. Closure. For $u, v > 0$ and $c \in \mathbb{R}$, the product uv is positive and the power v^c is positive. So $\boxplus: V \times V \rightarrow V$ and $\boxminus: \mathbb{R} \times V \rightarrow V$.

(VS3). We need $z \in V$ with $v \boxplus z = v$, that is, $vz = v$, for every $v > 0$. The number $z = 1$ works, since $v \cdot 1 = v$. So the zero vector of V is the real number 1.

(VS4). For $v \in V$, we need $v' \in V$ with $v \boxplus v' = 1$, that is, $vv' = 1$. The number $v' = 1/v$ is positive and works. So the additive inverse of v is $1/v$.

(VS6). For $a, b \in \mathbb{R}$ and $v \in V$, by the laws of exponents,

$$(a + b) \boxminus v = v^{a+b} = v^a v^b = (a \boxminus v) \boxplus (b \boxminus v).$$

(VS8). $1 \boxminus v = v^1 = v$.

The remaining axioms (VS1), (VS2), (VS5) and (VS7) are [Exercise \(B3: Finishing the Positive Reals\)](#). So V is a real vector space.

This example earns its place for one reason. The words “zero vector” and “addition” describe **roles** fixed by the axioms, not the familiar number 0 or the familiar $+$. Here the vector playing the role of **0** is the number 1, and “adding” means multiplying.

Quick check. Explain why each of the following, with the usual operations, is a vector space over $\mathbb{R}$.

- (a) The set S of twice-differentiable functions $f: \mathbb{R} \rightarrow \mathbb{R}$ with $f'' + f = 0$.
- (b) The set of symmetric matrices $A \in M_2(\mathbb{R})$, that is, those with $A^\top = A$.

Solution. Both are subsets of known spaces, so we use the pattern after [Example \(The Standard Vector Spaces\)](#): check closure, the zero vector and additive inverses; the other six axioms are identities inherited from the big space.

- (a) $S \subseteq \mathbb{R}^\mathbb{R}$. If $f, g \in S$ and $c \in \mathbb{R}$, then by the sum and constant multiple rules of calculus, $f + g$ and cf are twice differentiable, with $(f + g)'' + (f + g) = (f'' + f) + (g'' + g) = 0$ and $(cf)'' + cf = c(f'' + f) = 0$. So both lie in S . The zero function **0** satisfies $\mathbf{0}'' + \mathbf{0} = \mathbf{0}$, so $\mathbf{0} \in S$. If $f \in S$, its additive inverse in $\mathbb{R}^\mathbb{R}$ is $t \mapsto -f(t)$, and $-f(t) = (-1)f(t)$ by [Theorem 0.7.1](#) (e); so it is the function $(-1)f$, which lies in S by closure under scaling. The identities hold in $\mathbb{R}^\mathbb{R}$ ([Example \(The Standard Vector Spaces\)](#) (d)), hence in S . For instance $\sin$ and $\cos$ lie in S .
- (b) By [Theorem 0.9.5](#), $(A + B)^\top = A^\top + B^\top = A + B$ and $(cA)^\top = cA^\top = cA$ when $A^\top = A$ and $B^\top = B$. The zero matrix is symmetric, and the additive inverse $-A$, which Chapter 0 defines as $(-1)A$, is symmetric by closure under scaling. The identities hold in $M_2(\mathbb{R})$, hence for symmetric matrices.

1.1.4 Non-examples

Each non-example below is a small change to an example. We say what still works and name the exact clause that fails.

- **Change scalar multiplication on $\mathbb{R}^2$ to $c(x, y) := (cx, 0)$** , keeping the usual addition. Addition is untouched, so (VS1)–(VS4) hold. (VS5) holds: $c((x, y) + (x', y')) = (c(x + x'), 0) = (cx, 0) + (cx', 0)$. In the same way (VS6) and (VS7) hold. But $1(0, 1) = (0, 0) \neq (0, 1)$, so **(VS8) fails**.
- **Change the line $x + 2y = 0$ to $x + 2y = 3$** . Let $L = \{(x, y) \in \mathbb{R}^2 : x + 2y = 3\}$ with the operations of $\mathbb{R}^2$. The points $(3, 0)$ and $(1, 1)$ lie on L , but their sum $(4, 1)$ has $4 + 2 \cdot 1 = 6 \neq 3$. So addition is **not a function** $L \times L \rightarrow L$, and L is not a vector space with these operations.
- **Change “degree at most 2” to “degree exactly 2”**. In $\mathbb{R}[x]_{=2}$ (Definition 0.8.5), $x^2 + 2x$ and $-x^2 + 3$ have degree 2, but their sum $2x + 3$ has degree 1. Again addition leaves the set. There is no zero vector either: the zero polynomial has degree $-\infty$, not 2.
- **Change the entries from real to integer**, keeping $\mathbb{R}$ as the field. For $\mathbb{Z}^2 \subseteq \mathbb{R}^2$, addition is fine, but $\frac{1}{2}(1, 0) = (\frac{1}{2}, 0) \notin \mathbb{Z}^2$. So scalar multiplication is **not a function** $\mathbb{R} \times \mathbb{Z}^2 \rightarrow \mathbb{Z}^2$.

1.1.5 Why these axioms

Each axiom is there for a reason, and a few conventions deserve a comment.

- **(VS8) cannot be dropped**. The first non-example satisfies all other axioms. Without (VS8), the “scalar multiplication” $c\mathbf{v} = \mathbf{0}$ for all c and $\mathbf{v}$ would also be allowed, and scaling would carry no information at all.
- **(VS1) is in fact redundant**. Commutativity of addition follows from the other seven axioms, by expanding $(1 + 1)(\mathbf{u} + \mathbf{v})$ in two ways (Exercise (C1: Commutativity Comes for Free)). It is kept because the proof is fiddly and nobody wants to repeat it.
- **Closure is not optional**. Two of the non-examples fail only because an operation leads out of the set. That is why the definition asks for **functions into V** .
- **The scalars form a field**. We use division by non-zero scalars in the zero product law at the end of the next subsection. The warning after it shows what goes wrong without division.
- **The name**. “Vector” originally meant an arrow in physics. The word is kept for elements of any vector space, even when they are polynomials or functions.

1.1.6 First consequences of the axioms

The definition says a zero vector and additive inverses **exist**, not that they are unique. Before writing $\mathbf{0}$ and $-\mathbf{v}$ as if each named one thing, we prove uniqueness. We also prove that familiar rules such as $0\mathbf{v} = \mathbf{0}$ follow from the axioms. None of this is obvious from the axioms: in Example (Positive Reals with Exotic Operations), $0 \boxplus v = v^0 = 1$, which happens to be the zero vector there, and it takes a proof to see that this is no coincidence.

Throughout, we use one fact about functions without further comment: if two inputs are equal, the outputs are equal. So from $\mathbf{v} = \mathbf{w}$ we may conclude $\mathbf{u} + \mathbf{v} = \mathbf{u} + \mathbf{w}$ and $a\mathbf{v} = a\mathbf{w}$.

The first tool lets us cancel a vector from both sides of an equation.

Theorem 1.1.1 — Left Cancellation Law

Let V be a vector space over F and let $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$. If $\mathbf{u} + \mathbf{v} = \mathbf{u} + \mathbf{w}$, then $\mathbf{v} = \mathbf{w}$.

Proof. Suppose $\mathbf{u} + \mathbf{v} = \mathbf{u} + \mathbf{w}$. By (VS4) there is $\mathbf{u}' \in V$ with $\mathbf{u} + \mathbf{u}' = \mathbf{0}$. Adding $\mathbf{u}'$ on the left of both sides gives $\mathbf{u}' + (\mathbf{u} + \mathbf{v}) = \mathbf{u}' + (\mathbf{u} + \mathbf{w})$, and by (VS2),

$$(\mathbf{u}' + \mathbf{u}) + \mathbf{v} = (\mathbf{u}' + \mathbf{u}) + \mathbf{w}.$$

By (VS1) and the choice of $\mathbf{u}'$, $\mathbf{u}' + \mathbf{u} = \mathbf{u} + \mathbf{u}' = \mathbf{0}$. Hence $\mathbf{0} + \mathbf{v} = \mathbf{0} + \mathbf{w}$. By (VS1) this is $\mathbf{v} + \mathbf{0} = \mathbf{w} + \mathbf{0}$, and by (VS3), $\mathbf{v} = \mathbf{w}$. ■

Theorem 1.1.2 — Right Cancellation Law

Let V be a vector space over F and let $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$. If $\mathbf{v} + \mathbf{u} = \mathbf{w} + \mathbf{u}$, then $\mathbf{v} = \mathbf{w}$.

Proof. By (VS1), the hypothesis says $\mathbf{u} + \mathbf{v} = \mathbf{u} + \mathbf{w}$. By [Theorem 1.1.1](#), $\mathbf{v} = \mathbf{w}$. ■

Uniqueness uses the standard move from Chapter 0: suppose there are two, and show they are equal.

Theorem 1.1.3 — Zero Vector Is Unique

Let V be a vector space over F . There is exactly one element $\mathbf{0} \in V$ satisfying (VS3).

Proof. Existence is (VS3). Suppose $\mathbf{0}$ and $\mathbf{0}'$ both satisfy (VS3). Applying (VS3) for $\mathbf{0}'$ to the vector $\mathbf{0}$, then (VS1), then (VS3) for $\mathbf{0}$ to the vector $\mathbf{0}'$,

$$\mathbf{0} = \mathbf{0} + \mathbf{0}' = \mathbf{0}' + \mathbf{0} = \mathbf{0}'.$$

This shows the zero vector is unique. ■

Theorem 1.1.4 — Additive Inverse Is Unique

Let V be a vector space over F and $\mathbf{v} \in V$. There is exactly one $\mathbf{v}' \in V$ with $\mathbf{v} + \mathbf{v}' = \mathbf{0}$.

Proof. Existence is (VS4). Suppose $\mathbf{v} + \mathbf{v}' = \mathbf{0}$ and $\mathbf{v} + \mathbf{v}'' = \mathbf{0}$. Then $\mathbf{v} + \mathbf{v}' = \mathbf{v} + \mathbf{v}''$, and [Theorem 1.1.1](#) gives $\mathbf{v}' = \mathbf{v}''$. ■

Since the inverse is unique, it deserves a name.

Definition 1.1.3 — Notation for the Additive Inverse

Let V be a vector space over F and $\mathbf{v} \in V$. The unique additive inverse of $\mathbf{v}$ is denoted $-\mathbf{v}$. For $\mathbf{u}, \mathbf{v} \in V$ we write $\mathbf{u} - \mathbf{v} := \mathbf{u} + (-\mathbf{v})$.

For example, $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$ and $(-\mathbf{v}) + \mathbf{v} = \mathbf{0}$ by (VS1), so $\mathbf{v}$ is the additive inverse of $-\mathbf{v}$. By Theorem 1.1.4, $-(-\mathbf{v}) = \mathbf{v}$.

Remark. Axioms (VS1)–(VS4) say that V with addition is an abelian group (Definition 0.10.1). So the four results above are also special cases of Theorem 0.10.1. We proved them directly to practice citing the axioms.

Now to scalars. No axiom mentions $0\mathbf{v}$, so we must create a situation in which one applies. The move is to write the scalar 0 in a fancy way, as $0 + 0$, expand with a distributive law, and cancel. We will call it **expand zero and cancel**.

Theorem 1.1.5 — Zero Scalar Gives the Zero Vector

Let V be a vector space over F . Then $0\mathbf{v} = \mathbf{0}$ for every $\mathbf{v} \in V$.

Proof. Let $\mathbf{v} \in V$. Since $0 + 0 = 0$ in F by (F3), (VS6) gives

$$0\mathbf{v} + 0\mathbf{v} = (0 + 0)\mathbf{v} = 0\mathbf{v} = 0\mathbf{v} + \mathbf{0},$$

where the last equality is (VS3). By Theorem 1.1.1, $0\mathbf{v} = \mathbf{0}$. ■

The same move, with the zero vector in place of the zero scalar, handles $a\mathbf{0}$.

Theorem 1.1.6 — Scalar Multiple of the Zero Vector

Let V be a vector space over F . Then $a\mathbf{0} = \mathbf{0}$ for every $a \in F$.

Proof. Let $a \in F$. Since $\mathbf{0} + \mathbf{0} = \mathbf{0}$ by (VS3), (VS5) gives

$$a\mathbf{0} + a\mathbf{0} = a(\mathbf{0} + \mathbf{0}) = a\mathbf{0} = a\mathbf{0} + \mathbf{0},$$

where the last equality is (VS3). By Theorem 1.1.1, $a\mathbf{0} = \mathbf{0}$. ■

To identify a vector as $-\mathbf{v}$, it is enough to show it adds with $\mathbf{v}$ to $\mathbf{0}$, by Theorem 1.1.4. This gives the next two results.

Theorem 1.1.7 — Negation as Scalar Multiplication

Let V be a vector space over F . Then $(-1)\mathbf{v} = -\mathbf{v}$ for every $\mathbf{v} \in V$.

Proof. Let $\mathbf{v} \in V$. By (VS8), (VS6), (F4) and [Theorem 1.1.5](#),

$$\mathbf{v} + (-1)\mathbf{v} = 1\mathbf{v} + (-1)\mathbf{v} = (1 + (-1))\mathbf{v} = 0\mathbf{v} = \mathbf{0}.$$

So $(-1)\mathbf{v}$ is an additive inverse of $\mathbf{v}$, and by [Theorem 1.1.4](#) it equals $-\mathbf{v}$. ■

Theorem 1.1.8 — Negative Scalars

Let V be a vector space over F . Then $(-a)\mathbf{v} = -(a\mathbf{v})$ for every $a \in F$ and $\mathbf{v} \in V$.

Proof. Let $a \in F$ and $\mathbf{v} \in V$. By (VS6), (F4) and [Theorem 1.1.5](#),

$$a\mathbf{v} + (-a)\mathbf{v} = (a + (-a))\mathbf{v} = 0\mathbf{v} = \mathbf{0}.$$

So $(-a)\mathbf{v}$ is an additive inverse of $a\mathbf{v}$, and by [Theorem 1.1.4](#) it equals $-(a\mathbf{v})$. ■

The companion rules $a(-\mathbf{v}) = -(a\mathbf{v})$ and $a(\mathbf{u} - \mathbf{v}) = a\mathbf{u} - a\mathbf{v}$ are [Exercise \(B2: More Rules for Signs\)](#). From now on, we compute with $+$, $-$ and scalars as in school algebra, citing these results when a step is not a single axiom.

The last result is the one we will use most. In a field, a product of non-zero numbers is non-zero ([Theorem 0.7.1](#) (f)). The same holds for a scalar times a vector.

Theorem 1.1.9 — Zero Product Law

Let V be a vector space over F , let $a \in F$ and $\mathbf{v} \in V$. Then $a\mathbf{v} = \mathbf{0}$ if and only if $a = 0$ or $\mathbf{v} = \mathbf{0}$.

Idea. The direction $(\Leftarrow)$ is the two zero results above, [Theorem 1.1.5](#) and [Theorem 1.1.6](#). For $(\Rightarrow)$, the only way to get rid of a is to multiply by a^{-1} , and that is exactly where the hypothesis $a \neq 0$ and the field axiom (F8) are spent.

Proof. $(\Leftarrow)$ If $a = 0$, then $a\mathbf{v} = \mathbf{0}$ by [Theorem 1.1.5](#). If $\mathbf{v} = \mathbf{0}$, then $a\mathbf{v} = \mathbf{0}$ by [Theorem 1.1.6](#).

$(\Rightarrow)$ Suppose $a\mathbf{v} = \mathbf{0}$. If $a = 0$ there is nothing to prove, so suppose $a \neq 0$. By (F8), a has an inverse $a^{-1} \in F$. Then

$$\mathbf{v} = 1\mathbf{v} = (a^{-1}a)\mathbf{v} = a^{-1}(a\mathbf{v}) = a^{-1}\mathbf{0} = \mathbf{0},$$

by (VS8), (F8) with (F6), (VS7), the hypothesis, and [Theorem 1.1.6](#). This shows $a = 0$ or $\mathbf{v} = \mathbf{0}$.

Warning. The zero product law needs division by scalars. Suppose we allowed integer “scalars” and let $V = \mathbb{F}_2 = \{0, 1\}$ with its own addition, where $n \cdot v$ is v for odd n and 0 for even n (for $n \geq 0$ this is $v + \cdots + v$ with n terms). Every rule of the shape (VS1)–(VS8) still holds. But $2 \cdot 1 = 1 + 1 = 0$, although $2 \neq 0$ in $\mathbb{Z}$ and $1 \neq 0$ in V . The proof above breaks exactly where it multiplies by a^{-1} : the integer 2 has no inverse in $\mathbb{Z}$. This is one reason vector spaces are defined **over a field**.

As a first use, here is a fact about sizes. Let $\mathbf{v} \neq \mathbf{0}$ be a vector in a space over $\mathbb{R}$. If $a\mathbf{v} = b\mathbf{v}$, then $(a - b)\mathbf{v} = a\mathbf{v} - b\mathbf{v} = \mathbf{0}$ by (VS6) and Theorem 1.1.8, so $a = b$ by Theorem 1.1.9. So the multiples $a\mathbf{v}$, $a \in \mathbb{R}$, are all different, and a real vector space other than $\{\mathbf{0}\}$ has infinitely many elements (Exercise (C2: How Big Is a Vector Space?) treats a general field). This is our first glimpse of the question at the heart of this chapter: all these spaces are infinite sets, so how should we measure their size? The next section starts on the answer by looking at the vector spaces that sit inside a given one.

1.1.7 Exercises

1.1.7.1 A. Check your understanding

Exercise A1.

- State the eight axioms of a vector space over a field F , and say which of them involve scalars.
- Identify the zero vector in each of: $M_{2 \times 3}(\mathbb{R})$; $\mathbb{R}[x]_{\leq 2}$; $\mathbb{R}^{\mathbb{R}}$; the space $\mathbb{R}_{>0}$ of Example (Positive Reals with Exotic Operations).
- True or false: a vector space can have exactly two elements. Justify your answer.
- True or false: in every vector space, $\mathbf{v} + \mathbf{v} = \mathbf{0}$ implies $\mathbf{v} = \mathbf{0}$. Justify your answer.
- True or false: if $a\mathbf{v} = b\mathbf{v}$ and $\mathbf{v} \neq \mathbf{0}$, then $a = b$. Justify your answer.

Solution.

- See Definition 1.1.1. The axioms (VS5)–(VS8) involve scalars; (VS1)–(VS4) involve only addition.
- The zero matrix of size 2×3 ; the zero polynomial; the zero function $t \mapsto 0$; the number 1 (Example (Positive Reals with Exotic Operations)).
- True. Let $F = \mathbb{F}_2$. By Example (The Standard Vector Spaces) (a), $\mathbb{F}_2^1 = \mathbb{F}_2 = \{0, 1\}$ is a vector space over $\mathbb{F}_2$, and it has exactly two elements.
- False. In the vector space $\mathbb{F}_2$ over $\mathbb{F}_2$ from (c), the vector $1 \neq 0$ satisfies $1 + 1 = 0$. (Over $\mathbb{R}$ the statement is true: $\mathbf{v} + \mathbf{v} = (1 + 1)\mathbf{v} = 2\mathbf{v}$ by (VS8) and (VS6), and $2 \neq 0$, so $\mathbf{v} = \mathbf{0}$ by Theorem 1.1.9.)
- True. By (VS6) and Theorem 1.1.8, $(a - b)\mathbf{v} = a\mathbf{v} + (-b)\mathbf{v} = a\mathbf{v} - b\mathbf{v} = \mathbf{0}$. Since $\mathbf{v} \neq \mathbf{0}$, Theorem 1.1.9 gives $a - b = 0$, so $a = b$.

1.1.7.2 B. Practice

Exercise B1: Which Are Vector Spaces?.

Determine which of the following are vector spaces. Justify your answer.

- (a) $\mathbb{R}^2$ over $\mathbb{R}$, with the usual scalar multiplication but with addition $(x, y) + (x', y') := (x + x', 0)$.
- (b) The set of upper triangular matrices in $M_2(\mathbb{R})$, over $\mathbb{R}$, with the matrix operations.
- (c) $\{p \in \mathbb{R}[x] : p(1) = 0\}$, over $\mathbb{R}$, with the operations of $\mathbb{R}[x]$.
- (d) $\{p \in \mathbb{R}[x] : p(1) = 1\}$, over $\mathbb{R}$, with the operations of $\mathbb{R}[x]$.
- (e) $\mathbb{R}^2$ over $\mathbb{C}$, with $c(x, y) := (cx, cy)$ and the usual addition.

Solution.

- (a) Not a vector space: (VS3) fails. Suppose $\mathbf{0} = (z_1, z_2)$ satisfied (VS3). Then $(0, 1) + (z_1, z_2) = (z_1, 0)$ would equal $(0, 1)$, which is impossible since the second entries differ.
- (b) A vector space. Let $A = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$ and $B = \begin{pmatrix} a' & b' \\ 0 & d' \end{pmatrix}$ be upper triangular and $c \in \mathbb{R}$. Then $A + B = \begin{pmatrix} a + a' & b + b' \\ 0 & d + d' \end{pmatrix}$ and $cA = \begin{pmatrix} ca & cb \\ 0 & cd \end{pmatrix}$ are upper triangular, so the operations land in the set. The zero matrix is upper triangular, and so is $-A = (-1)A$. The identities (VS1), (VS2), (VS5)–(VS8) hold for all matrices in $M_2(\mathbb{R})$ by [Example \(The Standard Vector Spaces\)](#) (b), hence for upper triangular ones.
- (c) A vector space. Let $U = \{p \in \mathbb{R}[x] : p(1) = 0\}$, $p, q \in U$ and $c \in \mathbb{R}$. By [Theorem 0.8.5](#), $(p + q)(1) = p(1) + q(1) = 0$. Also cp is the product of the constant polynomial c with p , since by the product formula of [Definition 0.8.2](#) that product has coefficients ca_k . So $(cp)(1) = cp(1) = 0$. Hence $p + q, cp \in U$. The zero polynomial has value 0 at 1, and $-p = (-1)p \in U$ by closure under scaling. The identities hold in $\mathbb{R}[x]$ by [Example \(The Standard Vector Spaces\)](#) (c), hence in U .
- (d) Not a vector space: addition leaves the set. The constant polynomial 1 lies in the set, but $(1 + 1)(1) = 2 \neq 1$, so $1 + 1$ does not.
- (e) Not a vector space: scalar multiplication leaves the set. We have $i \in \mathbb{C}$ and $(1, 0) \in \mathbb{R}^2$, but $i(1, 0) = (i, 0) \notin \mathbb{R}^2$. So the rule is not a function $\mathbb{C} \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$.

Exercise B2: More Rules for Signs.

Let V be a vector space over F , let $a, b \in F$ and $\mathbf{u}, \mathbf{v} \in V$. Prove that:

- (a) $a(-\mathbf{v}) = -(a\mathbf{v})$;
- (b) $(-a)(-\mathbf{v}) = a\mathbf{v}$;
- (c) $a(\mathbf{u} - \mathbf{v}) = a\mathbf{u} - a\mathbf{v}$ and $(a - b)\mathbf{v} = a\mathbf{v} - b\mathbf{v}$.

Hint: to show a vector equals $-(a\mathbf{v})$, use [Theorem 1.1.4](#).

Solution.

- (a) By (VS5), (VS4) and [Theorem 1.1.6](#),

$$a\mathbf{v} + a(-\mathbf{v}) = a(\mathbf{v} + (-\mathbf{v})) = a\mathbf{0} = \mathbf{0}.$$

So $a(-\mathbf{v})$ is an additive inverse of $a\mathbf{v}$. By [Theorem 1.1.4](#), $a(-\mathbf{v}) = -(a\mathbf{v})$.

- (b) By (a) applied with the scalar $-a$, then [Theorem 1.1.8](#), then the remark after [Definition 1.1.3](#),

$$(-a)(-\mathbf{v}) = -((-a)\mathbf{v}) = -(-(a\mathbf{v})) = a\mathbf{v}.$$

- (c) By [Definition 1.1.3](#), (VS5) and (a),

$$a(\mathbf{u} - \mathbf{v}) = a(\mathbf{u} + (-\mathbf{v})) = a\mathbf{u} + a(-\mathbf{v}) = a\mathbf{u} + (-(a\mathbf{v})) = a\mathbf{u} - a\mathbf{v}.$$

Similarly, since $a - b = a + (-b)$ in F , (VS6) and [Theorem 1.1.8](#) give $(a - b)\mathbf{v} = a\mathbf{v} + (-b)\mathbf{v} = a\mathbf{v} + (-(b\mathbf{v})) = a\mathbf{v} - b\mathbf{v}$.

Exercise B3: Finishing the Positive Reals.

Let $V = \mathbb{R}_{>0}$ with the operations $\boxplus$ and $\boxminus$ of [Example \(Positive Reals with Exotic Operations\)](#).

- (a) Verify (VS1), (VS2), (VS5) and (VS7). Hence V is a vector space over $\mathbb{R}$.
- (b) Compute the vector $(2 \boxminus 3) \boxplus ((-1) \boxminus 6)$.
- (c) Find the vector $-(2 \boxminus 3)$, and check your answer against [Theorem 1.1.8](#).

Solution.

- (a) Let $u, v, w \in V$ and $a, b \in \mathbb{R}$. By the laws of real multiplication and of exponents for positive bases:
 - (VS1): $u \boxplus v = uv = vu = v \boxplus u$.
 - (VS2): $(u \boxplus v) \boxplus w = (uv)w = u(vw) = u \boxplus (v \boxplus w)$.
 - (VS5): $a \boxminus (u \boxplus v) = (uv)^a = u^a v^a = (a \boxminus u) \boxplus (a \boxminus v)$.
 - (VS7): $a \boxminus (b \boxminus v) = (v^b)^a = v^{ab} = (ab) \boxminus v$.

Together with closure, (VS3), (VS4), (VS6) and (VS8) from [Example \(Positive Reals with Exotic Operations\)](#), all eight axioms hold. Hence V is a vector space over $\mathbb{R}$.

- (b) $2 \boxminus 3 = 3^2 = 9$ and $(-1) \boxminus 6 = 6^{-1} = \frac{1}{6}$. So the vector is $9 \boxplus \frac{1}{6} = 9 \cdot \frac{1}{6} = \frac{3}{2}$.
- (c) By [Example \(Positive Reals with Exotic Operations\)](#), the additive inverse of a vector v is $1/v$. Since $2 \boxminus 3 = 9$, we get $-(2 \boxminus 3) = \frac{1}{9}$. By [Theorem 1.1.8](#), this should equal $(-2) \boxminus 3 = 3^{-2} = \frac{1}{9}$, and it does.

1.1.7.3 C. Going deeper

Exercise C1: Commutativity Comes for Free.

Let V be a set with operations $+: V \times V \rightarrow V$ and $\cdot: F \times V \rightarrow V$ satisfying (VS2)–(VS8), but **not necessarily** (VS1). Fix $\mathbf{0}$ as in (VS3).

- (a) Let $\mathbf{v} \in V$ and $\mathbf{v}'$ as in (VS4). Put $\mathbf{w} = \mathbf{v}' + \mathbf{v}$. Show that $\mathbf{w} + \mathbf{w} = \mathbf{w}$, and deduce that $\mathbf{w} = \mathbf{0}$.
- (b) Show that $\mathbf{0} + \mathbf{v} = \mathbf{v}$ for every $\mathbf{v} \in V$.
- (c) Deduce that both cancellation laws hold: $\mathbf{u} + \mathbf{v} = \mathbf{u} + \mathbf{w}$ implies $\mathbf{v} = \mathbf{w}$, and $\mathbf{v} + \mathbf{u} = \mathbf{w} + \mathbf{u}$ implies $\mathbf{v} = \mathbf{w}$.
- (d) By expanding $(1+1)(\mathbf{u} + \mathbf{v})$ in two ways, prove that $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ for all $\mathbf{u}, \mathbf{v} \in V$.

Hint: in (a), add an additive inverse of $\mathbf{w}$ on the right. The proof of Theorem 1.1.1 used (VS1), so it cannot be quoted in (c).

Solution. Throughout we use only (VS2)–(VS8), and we drop brackets in sums whenever (VS2) allows it.

- (a) Since $\mathbf{v} + \mathbf{v}' = \mathbf{0}$ and $\mathbf{v}' + \mathbf{0} = \mathbf{v}'$ by (VS3),

$$\mathbf{w} + \mathbf{w} = \mathbf{v}' + (\mathbf{v} + \mathbf{v}') + \mathbf{v} = \mathbf{v}' + \mathbf{0} + \mathbf{v} = \mathbf{v}' + \mathbf{v} = \mathbf{w}.$$

By (VS4) choose $\mathbf{w}'$ with $\mathbf{w} + \mathbf{w}' = \mathbf{0}$. Adding $\mathbf{w}'$ on the right of $\mathbf{w} + \mathbf{w} = \mathbf{w}$ gives $\mathbf{w} + (\mathbf{w} + \mathbf{w}') = \mathbf{w} + \mathbf{w}'$, that is, $\mathbf{w} + \mathbf{0} = \mathbf{0}$. By (VS3), $\mathbf{w} = \mathbf{0}$. So $\mathbf{v}' + \mathbf{v} = \mathbf{0}$ as well.

- (b) Let $\mathbf{v} \in V$ and $\mathbf{v}'$ as in (VS4). By (a), (VS2) and (VS3),

$$\mathbf{0} + \mathbf{v} = (\mathbf{v} + \mathbf{v}') + \mathbf{v} = \mathbf{v} + (\mathbf{v}' + \mathbf{v}) = \mathbf{v} + \mathbf{0} = \mathbf{v}.$$

- (c) Suppose $\mathbf{u} + \mathbf{v} = \mathbf{u} + \mathbf{w}$, and let $\mathbf{u}'$ be as in (VS4). Adding $\mathbf{u}'$ on the left and using (VS2), $(\mathbf{u}' + \mathbf{u}) + \mathbf{v} = (\mathbf{u}' + \mathbf{u}) + \mathbf{w}$. By (a), $\mathbf{u}' + \mathbf{u} = \mathbf{0}$, so $\mathbf{0} + \mathbf{v} = \mathbf{0} + \mathbf{w}$, and (b) gives $\mathbf{v} = \mathbf{w}$. Now suppose $\mathbf{v} + \mathbf{u} = \mathbf{w} + \mathbf{u}$. Adding $\mathbf{u}'$ on the right and using (VS2), $\mathbf{v} + (\mathbf{u} + \mathbf{u}') = \mathbf{w} + (\mathbf{u} + \mathbf{u}')$, so $\mathbf{v} + \mathbf{0} = \mathbf{w} + \mathbf{0}$, and (VS3) gives $\mathbf{v} = \mathbf{w}$.
- (d) Let $\mathbf{u}, \mathbf{v} \in V$. On the one hand, by (VS6) and (VS8),

$$(1+1)(\mathbf{u} + \mathbf{v}) = 1(\mathbf{u} + \mathbf{v}) + 1(\mathbf{u} + \mathbf{v}) = \mathbf{u} + \mathbf{v} + \mathbf{u} + \mathbf{v}.$$

On the other hand, by (VS5), then (VS6) and (VS8),

$$(1+1)(\mathbf{u} + \mathbf{v}) = (1+1)\mathbf{u} + (1+1)\mathbf{v} = \mathbf{u} + \mathbf{u} + \mathbf{v} + \mathbf{v}.$$

Hence $\mathbf{u} + (\mathbf{v} + \mathbf{u}) + \mathbf{v} = \mathbf{u} + (\mathbf{u} + \mathbf{v}) + \mathbf{v}$, grouping by (VS2). Canceling $\mathbf{u}$ on the left and then $\mathbf{v}$ on the right, by (c), gives $\mathbf{v} + \mathbf{u} = \mathbf{u} + \mathbf{v}$. This proves (VS1).

Exercise C2: How Big Is a Vector Space?.

- (a) Let F be an infinite field and $V \neq \{\mathbf{0}\}$ a vector space over F . Prove that V is an infinite set.
- (b) Let p be a prime and $n \geq 1$. How many elements does the vector space $\mathbb{F}_p^n$ over $\mathbb{F}_p$ have? Explain why this does not contradict (a).

Hint: for (a), fix $\mathbf{v} \neq \mathbf{0}$ and consider the function $F \rightarrow V$, $a \mapsto a\mathbf{v}$.

Solution.

- (a) Since $V \neq \{\mathbf{0}\}$ and $\mathbf{0} \in V$, there is $\mathbf{v} \in V$ with $\mathbf{v} \neq \mathbf{0}$. Consider $\varphi: F \rightarrow V$, $\varphi(a) = a\mathbf{v}$. Suppose $\varphi(a) = \varphi(b)$, so $a\mathbf{v} = b\mathbf{v}$. By [Exercise \(B2: More Rules for Signs\)](#) (c), $(a - b)\mathbf{v} = a\mathbf{v} - b\mathbf{v} = \mathbf{0}$. Since $\mathbf{v} \neq \mathbf{0}$, [Theorem 1.1.9](#) gives $a - b = 0$, so $a = b$. Therefore φ is injective. An injective function from an infinite set has an infinite image, so V , which contains the image of φ , is infinite.
- (b) An element of $\mathbb{F}_p^n$ is a list $(x_1, \dots, x_n)$ with each x_i chosen freely from the p elements of $\mathbb{F}_p$. So $\mathbb{F}_p^n$ has p^n elements. There is no contradiction, because (a) assumes the field is infinite and $\mathbb{F}_p$ is finite. The proof of (a) still shows that $a \mapsto a\mathbf{v}$ is injective, so a non-zero space over $\mathbb{F}_p$ has at least p elements.

Exercise C3: Shrinking the Field.

Let F be a field and let $K \subseteq F$ be a subset with $0, 1 \in K$ that is closed under addition, multiplication, negatives, and inverses of non-zero elements. (So K is itself a field with the operations of F ; for example $\mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$.)

- (a) Let V be a vector space over F . Prove that V , with the same addition and with scalar multiplication restricted to $K \times V$, is a vector space over K .
- (b) Deduce that $\mathbb{C}^n$ is a vector space over $\mathbb{R}$ and over $\mathbb{Q}$.
- (c) Explain why $\mathbb{R}$, with its usual addition and multiplication, is **not** a vector space over $\mathbb{C}$.

Solution.

- (a) The addition is unchanged, so it is a function $V \times V \rightarrow V$ satisfying (VS1)–(VS4). Scalar multiplication $F \times V \rightarrow V$ restricts to a function $K \times V \rightarrow V$, since $K \subseteq F$. The axioms (VS5)–(VS7) are statements about **all** scalars in F , so they hold in particular for all scalars in K ; the operations $a + b$ and ab appearing in them are the same in K as in F . For (VS8), the identity of K is the identity 1 of F , and $1\mathbf{v} = \mathbf{v}$ holds in V over F . Hence V is a vector space over K .
- (b) $\mathbb{C}^n$ is a vector space over $\mathbb{C}$ by [Example \(The Standard Vector Spaces\)](#) (a). The subsets $\mathbb{R} \subseteq \mathbb{C}$ and $\mathbb{Q} \subseteq \mathbb{C}$ contain 0 and 1 and are closed under the four operations, so (a) applies to both.
- (c) The only candidate for scalar multiplication is $(c, t) \mapsto ct$, the product in $\mathbb{C}$. But it does not land in $\mathbb{R}$: $i \in \mathbb{C}$ and $1 \in \mathbb{R}$, while $i \cdot 1 = i \notin \mathbb{R}$. So there is no function $\mathbb{C} \times \mathbb{R} \rightarrow \mathbb{R}$ given by this rule, and $\mathbb{R}$ is not a vector space over $\mathbb{C}$ with these operations.

1.2 Subspaces

After defining a structure, the next step is to look at the smaller copies of it that sit inside. In the previous section we showed three times, for $F[x]_{\leq n}$, $\mathbb{Q}(\sqrt{2})$ and the solutions of $f'' + f = 0$, that a subset of a known vector space is itself a vector space, and each time the same few checks did the work. This section turns that pattern into a theorem, the subspace test, and makes its three checks a routine we will use for the rest of the book.

1.2.1 Spaces inside spaces

Consider the set

$$P = \{(x, y, z) \in \mathbb{R}^3 : x + y - z = 0\},$$

a plane through the origin in $\mathbb{R}^3$. It contains $(1, 0, 1)$ and $(0, 1, 1)$, and also their sum $(1, 1, 2)$, since $1 + 1 - 2 = 0$. Stretching a vector of P keeps it in the plane. So P , with the addition and scalar multiplication of $\mathbb{R}^3$, looks like a vector space in its own right.

To prove it we could check all eight axioms again. That is wasteful, and the previous section already showed why. The axioms (VS1), (VS2) and (VS5)–(VS8) are equations that hold for **all** vectors of $\mathbb{R}^3$, so they hold in particular for the vectors of P . What can go wrong is different: a sum or a multiple might leave P , and the zero vector or an additive inverse might not lie in P . The question of this section is which of these checks are really needed.

A subspace is a subset that is a vector space in its own right, using the same addition and scalar multiplication.

Definition 1.2.1 — Subspace

Let V be a vector space over F . A subset $U \subseteq V$ is a **subspace** of V if

- $\mathbf{u} + \mathbf{w} \in U$ and $a\mathbf{u} \in U$ **for all** $\mathbf{u}, \mathbf{w} \in U$ and $a \in F$, so that the operations of V restrict to functions $U \times U \rightarrow U$ and $F \times U \rightarrow U$; and
- U , with these restricted operations, is a vector space **over** F .

In words: the first clause says that adding or scaling vectors of U never leads out of U , so that U has an addition and a scalar multiplication at all. These are **the same** operations as in V , only applied to fewer vectors. The second clause says that with these operations, all eight axioms hold in U . In particular U must contain **a** zero vector of its own, and every vector of U must have an additive inverse **in** U . The field does not change: a subspace of a space over F is again a space over F .

There is a question hidden in “a zero vector of its own”. Could U have a zero vector different from the zero vector of V ? The proof of the subspace test below shows that it cannot.

The two extreme cases come first.

Example. (Trivial Subspaces)

Let V be a vector space over F . Show that V and $\{\mathbf{0}\}$ are subspaces of V , and that the empty set $\emptyset$ is not.

Solution. For V itself, the restricted operations are the operations of V , and V is a vector space by assumption.

For $\{\mathbf{0}\}$, the only sum is $\mathbf{0} + \mathbf{0} = \mathbf{0}$ by (VS3), and $a\mathbf{0} = \mathbf{0}$ for every $a \in F$ by [Theorem 1.1.6](#). So both operations stay in $\{\mathbf{0}\}$, and $\{\mathbf{0}\}$ with them is the zero space, a vector space as noted in the previous section.

For $\emptyset$, axiom (VS3) demands that some element $\mathbf{0}$ exists in the set. The empty set has no elements, so (VS3) fails and $\emptyset$ is not a subspace.

The subspace $\{0\}$ is the **zero subspace**. A subspace other than V is called **proper**. The zero subspace is small, but it is the answer to many questions later: the solutions of a system with only the trivial solution, or the intersection of two lines through the origin that are not equal.

1.2.2 The subspace test

The following theorem says that three checks suffice. Two of them are the closure conditions of the definition, and the third replaces all of the axioms.

Theorem 1.2.1 — Subspace Test

Let V be a vector space over F and let $U \subseteq V$. Then U is a subspace of V if and only if the following three conditions hold.

- (1) $0 \in U$, where 0 is the zero vector of V .
- (2) U is **closed under addition**: $u + w \in U$ for all $u, w \in U$.
- (3) U is **closed under scalar multiplication**: $au \in U$ for all $a \in F$ and $u \in U$.

Idea. Sort the eight axioms into two kinds. Six of them, (VS1), (VS2) and (VS5)–(VS8), say “an equation holds **for all** vectors”; they are true in V , so they are automatically true for the fewer vectors in U . The other two, (VS3) and (VS4), say “**there exists** a vector”, and a witness must be found **inside** U . Condition (1) supplies the zero. For inverses, the results of the previous section give $-u = (-1)u$, so closure under scaling supplies them for free. For $(\Rightarrow)$, the only non-obvious point is that the zero vector of U is the zero vector of V ; cancellation in V settles it.

Proof. $(\Rightarrow)$ Suppose U is a subspace of V . Conditions (2) and (3) are the first clause of Definition 1.2.1. By (VS3) in U , there is $z \in U$ with $u + z = u$ for every $u \in U$; in particular $z + z = z$. By (VS3) in V , also $z + 0 = z$. Hence $z + z = z + 0$, and Theorem 1.1.1 in V gives $z = 0$. Therefore $0 \in U$, which is (1).

$(\Leftarrow)$ Suppose (1), (2) and (3) hold. By (2) and (3), addition and scalar multiplication of V restrict to functions $U \times U \rightarrow U$ and $F \times U \rightarrow U$. We check the axioms for U with these operations.

- (VS1), (VS2), (VS5), (VS6), (VS7) and (VS8) state that certain equations hold for all vectors and scalars. They hold for all vectors of V , since V is a vector space, so in particular for all vectors of U .
- (VS3): by (1), $0 \in U$, and $u + 0 = u$ for every $u \in U$ by (VS3) in V .
- (VS4): let $u \in U$. By (3), $(-1)u \in U$. By Theorem 1.1.7, $(-1)u = -u$, so $u + (-1)u = 0$. Thus u has an additive inverse in U .

Hence U is a vector space over F with the operations of V , that is, a subspace of V . ■

From now on we never check eight axioms for a subset of a known vector space. We check three things, always in the same order and with the same wording. This is the first named move of the chapter.

The three-check move. To show that a subset U of a vector space V is a subspace:

- make sure U is a subset of V (what are its elements, and do they lie in V ?);
- **(1) Zero vector.** Show $\mathbf{0} \in U$ by checking that $\mathbf{0}$ satisfies the condition defining U . End with “Therefore $\mathbf{0} \in U$.”
- **(2) Closed under addition.** Let $\mathbf{u}, \mathbf{w} \in U$. Write down what membership in U means for each, then regroup $\mathbf{u} + \mathbf{w}$ so that those conditions appear. End with “This shows that U is closed under addition.”
- **(3) Closed under scalar multiplication.** Let $a \in F$ and $\mathbf{u} \in U$, and do the same for $a\mathbf{u}$. End with “This shows that U is closed under scalar multiplication.”
- Conclude: “Hence U is a subspace of V by the subspace test ([Theorem 1.2.1](#)).”

To show that U is **not** a subspace, one failure suffices. Check (1) first, since it is the cheapest, and if it holds, look for a concrete pair of vectors or a concrete scalar that leads out of U .

Remark. Condition (1) may be replaced by “ U is non-empty”. Indeed, if U contains some $\mathbf{u}$, then (3) gives $0\mathbf{u} \in U$, and $0\mathbf{u} = \mathbf{0}$ by [Theorem 1.1.5](#). We still prefer (1): the zero vector is the easiest element to test, and when it is missing we have a refutation in one line.

Warning. **The empty set is never a subspace.** It is tempting to check only (2) and (3). But $\emptyset$ satisfies both, vacuously: there are no vectors in it to add or scale. It fails only (1). So whichever version of the test you use, do not skip the first condition, and remember that “ $\mathbf{0} \in U$ ” and “ U is non-empty” are interchangeable only in the presence of (3).

1.2.3 Examples: the three-check move

Every example below follows the move exactly. The first is the model.

Example. (A Line through the Origin)

Show that $U = \{(x, y) \in \mathbb{R}^2 : x + 2y = 0\}$ is a subspace of $\mathbb{R}^2$.

Solution. By definition, U is a subset of $\mathbb{R}^2$.

(1) Zero vector. The zero vector $(0, 0)$ satisfies $0 + 2 \cdot 0 = 0$. Therefore $\mathbf{0} \in U$.

(2) Closed under addition. Let $\mathbf{u} = (x_1, y_1)$ and $\mathbf{w} = (x_2, y_2)$ be in U , so that $x_1 + 2y_1 = 0$ and $x_2 + 2y_2 = 0$. Then $\mathbf{u} + \mathbf{w} = (x_1 + x_2, y_1 + y_2)$, and

$$(x_1 + x_2) + 2(y_1 + y_2) = (x_1 + 2y_1) + (x_2 + 2y_2) = 0 + 0 = 0.$$

Therefore $\mathbf{u} + \mathbf{w} \in U$. This shows that U is closed under addition.

(3) Closed under scalar multiplication. Let $a \in \mathbb{R}$ and $\mathbf{u} = (x, y) \in U$, so that $x + 2y = 0$. Then $a\mathbf{u} = (ax, ay)$, and

$$ax + 2(ay) = a(x + 2y) = a \cdot 0 = 0.$$

Therefore $a\mathbf{u} \in U$. This shows that U is closed under scalar multiplication.

Hence U is a subspace of $\mathbb{R}^2$ by the subspace test (Theorem 1.2.1).

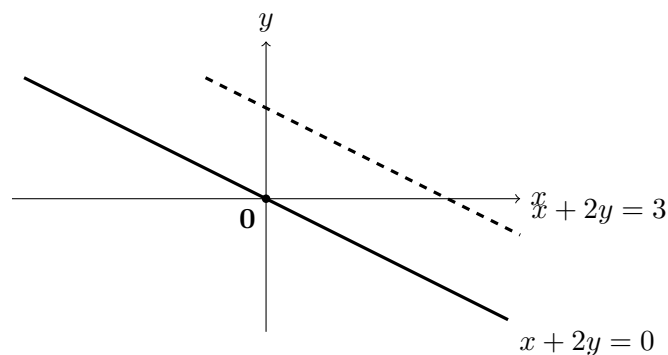
Change the right-hand side from 0 to 3, and everything breaks.

Example. (A Line Missing the Origin)

Show that $L = \{(x, y) \in \mathbb{R}^2 : x + 2y = 3\}$ is **not** a subspace of $\mathbb{R}^2$.

Solution. Condition (1) fails: the zero vector has $0 + 2 \cdot 0 = 0 \neq 3$, so $\mathbf{0} \notin L$. By the subspace test (Theorem 1.2.1), L is not a subspace of $\mathbb{R}^2$.

The other two conditions fail as well, as we saw in the previous section: $(3, 0) + (1, 1) = (4, 1) \notin L$, and $2(3, 0) = (6, 0) \notin L$. One failure is enough, though, and (1) is the cheapest to find.



The solid line U passes through the origin; the dashed line L is parallel to it and misses the origin. The line L is a translate of a subspace, a kind of set we will study in Chapter 3, but it is not a subspace.

The line U is the set of solutions of one homogeneous equation. The same three checks work for any number of homogeneous equations, written as one matrix equation.

Example. (Solutions of a Homogeneous System)

Let $A \in M_{m \times n}(F)$. Show that $N = \{\mathbf{x} \in F^n : A\mathbf{x} = \mathbf{0}\}$ is a subspace of F^n .

Solution. By definition, N is a subset of F^n . Recall that $F^n = M_{n \times 1}(F)$, so $A\mathbf{x} \in F^m$ is a matrix product, and we may use Theorem 0.9.4.

(1) Zero vector. By the zero rule of Theorem 0.9.4, $A\mathbf{0} = \mathbf{0}$, where the first $\mathbf{0}$ is the zero column in F^n and the second is the zero column in F^m . Therefore $\mathbf{0} \in N$.

(2) Closed under addition. Let $\mathbf{x}, \mathbf{y} \in N$, so that $A\mathbf{x} = \mathbf{0}$ and $A\mathbf{y} = \mathbf{0}$. By distributivity (Theorem 0.9.4),

$$A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y} = \mathbf{0} + \mathbf{0} = \mathbf{0}.$$

Therefore $\mathbf{x} + \mathbf{y} \in N$. This shows that N is closed under addition.

(3) Closed under scalar multiplication. Let $a \in F$ and $\mathbf{x} \in N$. By the scalar rule

of [Theorem 0.9.4](#) and [Theorem 1.1.6](#) in F^m ,

$$A(a\mathbf{x}) = a(A\mathbf{x}) = a\mathbf{0} = \mathbf{0}.$$

Therefore $a\mathbf{x} \in N$. This shows that N is closed under scalar multiplication.

Hence N is a subspace of F^n by the subspace test ([Theorem 1.2.1](#)).

With $A = \begin{pmatrix} 1 & 2 \end{pmatrix}$ and $F = \mathbb{R}$, this is [Example \(A Line through the Origin\)](#) again, and with $A = \begin{pmatrix} 1 & 1 & -1 \end{pmatrix}$ it is the plane P from the start of the section. The word **homogeneous** (right-hand side $\mathbf{0}$) is essential: [Example \(A Line Missing the Origin\)](#) is a system with right-hand side 3. This subspace will be called the null space of A in Chapter 2.

Next, polynomials. The pair “degree at most n ” and “degree exactly n ” is another minimal change. The previous section checked the first of them by hand; here it is as a three-check move.

Example. (Polynomials of Bounded Degree)

Let $n \in \mathbb{N}$. Show that $F[x]_{\leq n}$ is a subspace of $F[x]$.

Solution. By [Definition 0.8.4](#), $F[x]_{\leq n}$ is a subset of $F[x]$.

(1) Zero vector. The zero polynomial has degree $-\infty \leq n$ ([Definition 0.8.3](#)). Therefore $\mathbf{0} \in F[x]_{\leq n}$.

(2) Closed under addition. Let $p, q \in F[x]_{\leq n}$, so that $\deg p \leq n$ and $\deg q \leq n$. By [Theorem 0.8.2](#), $\deg(p + q) \leq \max(\deg p, \deg q) \leq n$. Therefore $p + q \in F[x]_{\leq n}$. This shows that $F[x]_{\leq n}$ is closed under addition.

(3) Closed under scalar multiplication. Let $c \in F$ and $p = a_0 + a_1x + \cdots + a_nx^n \in F[x]_{\leq n}$. Then $cp = ca_0 + ca_1x + \cdots + ca_nx^n$, whose coefficients of x^k for $k > n$ are all 0. So $\deg(cp) \leq n$, and therefore $cp \in F[x]_{\leq n}$. This shows that $F[x]_{\leq n}$ is closed under scalar multiplication.

Hence $F[x]_{\leq n}$ is a subspace of $F[x]$ by the subspace test ([Theorem 1.2.1](#)).

Example. (Polynomials of Exact Degree)

Let $n \in \mathbb{N}$. Show that $F[x]_{=n}$ is **not** a subspace of $F[x]$.

Solution. Condition (1) fails: the zero polynomial has degree $-\infty \neq n$, so $\mathbf{0} \notin F[x]_{=n}$. By the subspace test ([Theorem 1.2.1](#)), $F[x]_{=n}$ is not a subspace of $F[x]$.

Even if we added the zero polynomial by hand, condition (2) would still fail for $n \geq 1$: the polynomials $x^n + 1$ and $-x^n$ have degree exactly n , but their sum is the constant 1, of degree 0. The leading terms cancel, and nothing in “degree exactly n ” prevents it.

The same move works in function spaces. Here the facts we need come from calculus.

Example. (Differentiable Functions)

Let D be the set of differentiable functions $\mathbb{R} \rightarrow \mathbb{R}$. Show that D is a subspace of $\mathbb{R}^{\mathbb{R}}$.

Solution. Every element of D is a function $\mathbb{R} \rightarrow \mathbb{R}$, so $D \subseteq \mathbb{R}^{\mathbb{R}}$.

(1) Zero vector. The zero vector of $\mathbb{R}^{\mathbb{R}}$ is the zero function ([Example \(The Standard Vector Spaces\)](#)), which is constant and hence differentiable. Therefore $\mathbf{0} \in D$.

(2) Closed under addition. Let $f, g \in D$. By the sum rule of calculus, $f + g$ is differentiable, with $(f + g)' = f' + g'$. Therefore $f + g \in D$. This shows that D is closed under addition.

(3) Closed under scalar multiplication. Let $a \in \mathbb{R}$ and $f \in D$. By the constant multiple rule, af is differentiable, with $(af)' = af'$. Therefore $af \in D$. This shows that D is closed under scalar multiplication.

Hence D is a subspace of $\mathbb{R}^{\mathbb{R}}$ by the subspace test ([Theorem 1.2.1](#)).

The same three checks show that the continuous functions $\mathbb{R} \rightarrow \mathbb{R}$ form a subspace of $\mathbb{R}^{\mathbb{R}}$, and that D is a subspace of it. A subspace of a subspace is a subspace, since the operations are the same throughout.

Finally, two families of matrices. Each uses one rule from Chapter 0 for all three checks.

Example. (Symmetric Matrices)

Show that the set $S = \{A \in M_n(F) : A^T = A\}$ of symmetric matrices ([Definition 0.9.11](#)) is a subspace of $M_n(F)$.

Solution. By definition, S is a subset of $M_n(F)$.

(1) Zero vector. Every entry of the zero matrix is 0, so $(0)_{ji} = 0 = (0)_{ij}$ for all i, j , and the zero matrix is symmetric. Therefore $\mathbf{0} \in S$.

(2) Closed under addition. Let $A, B \in S$, so that $A^T = A$ and $B^T = B$. By [Theorem 0.9.5](#),

$$(A + B)^T = A^T + B^T = A + B.$$

Therefore $A + B \in S$. This shows that S is closed under addition.

(3) Closed under scalar multiplication. Let $c \in F$ and $A \in S$. By [Theorem 0.9.5](#), $(cA)^T = cA^T = cA$. Therefore $cA \in S$. This shows that S is closed under scalar multiplication.

Hence S is a subspace of $M_n(F)$ by the subspace test ([Theorem 1.2.1](#)).

Example. (Trace-Free Matrices)

Show that the set $T = \{A \in M_n(F) : \text{tr } A = 0\}$ of trace-free matrices is a subspace of $M_n(F)$.

Solution. By definition, T is a subset of $M_n(F)$.

(1) Zero vector. The zero matrix has trace $0 + \cdots + 0 = 0$. Therefore $\mathbf{0} \in T$.

(2) Closed under addition. Let $A, B \in T$, so that $\operatorname{tr} A = \operatorname{tr} B = 0$. By [Theorem 0.9.6](#), $\operatorname{tr}(A + B) = \operatorname{tr} A + \operatorname{tr} B = 0 + 0 = 0$. Therefore $A + B \in T$. This shows that T is closed under addition.

(3) Closed under scalar multiplication. Let $c \in F$ and $A \in T$. By [Theorem 0.9.6](#), $\operatorname{tr}(cA) = c \operatorname{tr} A = c \cdot 0 = 0$. Therefore $cA \in T$. This shows that T is closed under scalar multiplication.

Hence T is a subspace of $M_n(F)$ by the subspace test ([Theorem 1.2.1](#)).

The pattern in the last two examples, and in [Example \(Solutions of a Homogeneous System\)](#), is the same: the condition defining the set is “some rule that respects sums and scalar multiples sends the element to zero”. Chapter 3 gives such rules a name, linear maps, and this pattern becomes a single theorem about their kernels.

1.2.4 Non-examples

Here are two more subsets of $\mathbb{R}^2$ that fail, each in a different way.

- **The unit circle** $\{(x, y) : x^2 + y^2 = 1\}$ fails (1), since $0^2 + 0^2 \neq 1$. It fails (2) and (3) as well: $(1, 0) + (0, 1) = (1, 1)$ and $2(1, 0) = (2, 0)$ are not on the circle.
- **The set** $Q = \{(x, y) : xy \geq 0\}$, the first and third quadrants together with both axes. Here (1) holds, since $0 \cdot 0 = 0 \geq 0$. Condition (3) holds too: if $xy \geq 0$ and $a \in \mathbb{R}$, then $(ax)(ay) = a^2xy \geq 0$. But (2) fails: $(1, 0)$ and $(0, -1)$ lie in Q , while their sum $(1, -1)$ has $1 \cdot (-1) = -1 < 0$. So Q is not a subspace.

The set Q shows that (3) does not imply (2). The next check shows that (2) does not imply (3).

Quick check. Find a subset of $\mathbb{R}^2$ that contains $\mathbf{0}$ and is closed under addition, but is not a subspace. Which condition of the subspace test fails?

Solution. Take the closed first quadrant $U = \{(x, y) : x \geq 0 \text{ and } y \geq 0\}$. It contains $(0, 0)$, and if $x_1, y_1, x_2, y_2 \geq 0$, then $x_1 + x_2 \geq 0$ and $y_1 + y_2 \geq 0$, so it is closed under addition. Condition (3) fails: $(1, 0) \in U$ but $(-1)(1, 0) = (-1, 0) \notin U$. Another answer is $\mathbb{Z}^2$: it contains $\mathbf{0}$ and is closed under addition, but $\frac{1}{2}(1, 0) \notin \mathbb{Z}^2$.

1.2.5 Why this definition

The definition insists on **the same** operations. This is not a formality. The positive reals $\mathbb{R}_{>0}$ form a vector space over $\mathbb{R}$ ([Example \(Positive Reals with Exotic Operations\)](#)), and $\mathbb{R}_{>0}$ is a subset of the vector space $\mathbb{R}$. But $\mathbb{R}_{>0}$ is **not** a subspace of $\mathbb{R}$: with the operations of $\mathbb{R}$, it does not contain the zero vector 0 , and $(-1) \cdot 1 = -1$ leaves it. Its vector space structure uses different operations, multiplication and powers, and those do not count.

The definition also keeps the field. For example, $\mathbb{R}^2$ is a subset of $\mathbb{C}^2$, and it is closed under addition and under multiplication by **real** scalars. But it is not a subspace of $\mathbb{C}^2$ over $\mathbb{C}$, because $i(1, 0) = (i, 0) \notin \mathbb{R}^2$. Condition (3) quantifies over **every** scalar of the field.

Finally, why “(1) $\mathbf{0} \in U$ ” rather than “ U is non-empty”? The two are equivalent given (3), as the remark after the test explains. We choose the version that is quicker to check and quicker

to refute.

1.2.6 Intersections of subspaces

Given several subspaces, can we build new ones? The first way is to take the vectors they have in common. Recall from [Definition 0.3.9](#) that the intersection of a **non-empty** family of sets $(U_i)_{i \in I}$ consists of the elements lying in every U_i .

Theorem 1.2.2 — Intersection of Subspaces

Let V be a vector space over F , and let $(U_i)_{i \in I}$ be a family of subspaces of V with I non-empty. Then $\bigcap_{i \in I} U_i$ is a subspace of V .

Proof. Let $U = \bigcap_{i \in I} U_i$. Each U_i is a subset of V , so $U \subseteq V$. We use the subspace test ([Theorem 1.2.1](#)) for U , and for each U_i its direction ($\Rightarrow$).

- (1) For every $i \in I$, $\mathbf{0} \in U_i$, since U_i is a subspace. Therefore $\mathbf{0} \in U$.
- (2) Let $\mathbf{u}, \mathbf{w} \in U$. Then $\mathbf{u}, \mathbf{w} \in U_i$ for every $i \in I$. Since each U_i is closed under addition, $\mathbf{u} + \mathbf{w} \in U_i$ for every $i \in I$. Therefore $\mathbf{u} + \mathbf{w} \in U$. This shows that U is closed under addition.
- (3) Let $a \in F$ and $\mathbf{u} \in U$. Then $\mathbf{u} \in U_i$ for every $i \in I$, and since each U_i is closed under scalar multiplication, $a\mathbf{u} \in U_i$ for every $i \in I$. Therefore $a\mathbf{u} \in U$. This shows that U is closed under scalar multiplication.

Hence U is a subspace of V by the subspace test ([Theorem 1.2.1](#)). ■

The index set may be infinite, and this matters. In the next section we will build the smallest subspace containing a given set of vectors, and one way to describe it is as the intersection of **all** subspaces containing that set.

For two subspaces, the theorem gives familiar pictures: two different planes through the origin in $\mathbb{R}^3$ meet in a line through the origin. It also explains [Example \(Solutions of a Homogeneous System\)](#) from a new angle. The solution set of $A\mathbf{x} = \mathbf{0}$ is the intersection of the solution sets of its m individual equations, each of which is a subspace of F^n . An algebraic example: in $M_2(F)$, the matrices that are both symmetric and trace-free ([Example \(Symmetric Matrices\)](#), [Example \(Trace-Free Matrices\)](#)) are those of the form $\begin{pmatrix} a & b \\ b & -a \end{pmatrix}$, and they form a subspace.

1.2.7 Unions of subspaces

Intersections behave well. Unions do not, and the reason is visible in the simplest picture.

Example. (The Union of the Two Axes)

In $\mathbb{R}^2$, let $U = \{(x, 0) : x \in \mathbb{R}\}$ be the x -axis and $W = \{(0, y) : y \in \mathbb{R}\}$ the y -axis. Show that U and W are subspaces of $\mathbb{R}^2$, but $U \cup W$ is not.

Solution. The x -axis is the solution set of the homogeneous equation $y = 0$, that is, of $A\mathbf{x} = \mathbf{0}$ with $A = \begin{pmatrix} 0 & 1 \end{pmatrix}$. By [Example \(Solutions of a Homogeneous System\)](#), U is a subspace. In the same way W is the solution set of $x = 0$, so it is a subspace.

For the union, condition (2) fails. We have $(1, 0) \in U \subseteq U \cup W$ and $(0, 1) \in W \subseteq U \cup W$, but

$$(1, 0) + (0, 1) = (1, 1),$$

which is on neither axis, since both of its entries are non-zero. So $(1, 1) \notin U \cup W$, and $U \cup W$ is not a subspace of $\mathbb{R}^2$.

The union passes (1) and (3): it contains $\mathbf{0}$, and scaling a vector on an axis keeps it on that axis. Only addition breaks it, by mixing a vector from each axis. That observation is the whole proof of the general fact.

Proposition 1.2.3 — Union of Two Subspaces

Let V be a vector space over F , and let U and W be subspaces of V . Then $U \cup W$ is a subspace of V if and only if $U \subseteq W$ or $W \subseteq U$.

Idea. The direction $(\Leftarrow)$ is immediate, because the union is then just the bigger of the two. For $(\Rightarrow)$, a statement of the form “one of two inclusions holds” invites contradiction. If neither inclusion holds, there is a vector $\mathbf{u}$ of U outside W and a vector $\mathbf{w}$ of W outside U , exactly like $(1, 0)$ and $(0, 1)$ for the axes. Look at $\mathbf{u} + \mathbf{w}$. It must lie in U or in W . If it lies in U , subtracting $\mathbf{u}$ keeps us in U and leaves $\mathbf{w}$, which is not in U .

Proof. $(\Leftarrow)$ If $U \subseteq W$, then $U \cup W = W$, which is a subspace. If $W \subseteq U$, then $U \cup W = U$, which is a subspace.

$(\Rightarrow)$ Suppose $U \cup W$ is a subspace of V , and suppose, for a contradiction, that $U \not\subseteq W$ and $W \not\subseteq U$. Then there exist $\mathbf{u} \in U$ with $\mathbf{u} \notin W$, and $\mathbf{w} \in W$ with $\mathbf{w} \notin U$. Both lie in $U \cup W$, so $\mathbf{u} + \mathbf{w} \in U \cup W$ by condition (2) of the subspace test ([Theorem 1.2.1](#)).

Case 1: $\mathbf{u} + \mathbf{w} \in U$. Since U is a subspace and $\mathbf{u} \in U$, condition (3) gives $(-1)\mathbf{u} \in U$, and $(-1)\mathbf{u} = -\mathbf{u}$ by [Theorem 1.1.7](#). By condition (2) for U , $(\mathbf{u} + \mathbf{w}) + (-\mathbf{u}) \in U$. By (VS1), (VS2), (VS4) and (VS3),

$$(\mathbf{u} + \mathbf{w}) + (-\mathbf{u}) = \mathbf{w} + (\mathbf{u} + (-\mathbf{u})) = \mathbf{w} + \mathbf{0} = \mathbf{w}.$$

So $\mathbf{w} \in U$, contradicting the choice of $\mathbf{w}$.

Case 2: $\mathbf{u} + \mathbf{w} \in W$. Swapping the roles of U and W , and of $\mathbf{u}$ and $\mathbf{w}$, in Case 1 gives $\mathbf{u} \in W$, contradicting the choice of $\mathbf{u}$.

Both cases lead to a contradiction. Hence $U \subseteq W$ or $W \subseteq U$. ■

Warning. Do not combine subspaces by taking their union. Apart from the case where one subspace already contains the other, the union of two subspaces is never a subspace, by [Proposition 1.2.3](#). The two axes in $\mathbb{R}^2$ are the standard failing case: their union misses $(1, 1)$, and in fact misses every vector with two non-zero entries.

The right way to combine two subspaces U and W is to take all sums $\mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in U$ and $\mathbf{w} \in W$. For the two axes this gives all of $\mathbb{R}^2$. This construction, the sum of subspaces, is the subject of a later section of this chapter; [Exercise \(C2: The Sum of Two Subspaces\)](#) gives a preview.

We now have a supply of vector spaces for free: every subspace of a known space is one, and every theorem of the previous section applies to it. In the next section we ask a sharper question. Given a few vectors, what is the smallest subspace that contains them? Its answer, the span, is the first step toward measuring the size of a vector space.

1.2.8 Exercises

1.2.8.1 A. Check your understanding

Exercise A1.

- Define what it means for a subset U of a vector space V over F to be a subspace.
- State the subspace test.
- True or false: $\emptyset$ is a subspace of every vector space. Justify your answer.
- True or false: if U is a subspace of V and W is a subspace of U , then W is a subspace of V . Justify your answer.
- True or false: the union of two subspaces is never a subspace. Justify your answer.
- Which condition of the subspace test would you check first to show that $\{(x, y, z) \in \mathbb{R}^3 : x + y + z = 1\}$ is not a subspace of $\mathbb{R}^3$? Carry out the check.

Solution.

- See [Definition 1.2.1](#): U is closed under the addition and scalar multiplication of V , and with these operations it is a vector space over F .
- See [Theorem 1.2.1](#): $U \subseteq V$ is a subspace if and only if $\mathbf{0} \in U$, U is closed under addition, and U is closed under scalar multiplication.
- False. The empty set does not contain $\mathbf{0}$, so condition (1) fails ([Example \(Trivial Subspaces\)](#)).
- True. Since W is a subspace of U , it contains the zero vector of U , which is $\mathbf{0}$ (the proof of [Theorem 1.2.1](#) shows the zero vector of a subspace is the zero vector of the big space). The operations of U are those of V , so closure of W under the operations of U is closure under the operations of V . By the subspace test ([Theorem 1.2.1](#)), W is a subspace of V .
- False. If $U \subseteq W$, then $U \cup W = W$ is a subspace ([Proposition 1.2.3](#)). For example, $\{\mathbf{0}\} \cup V = V$.
- Condition (1). The zero vector has $0 + 0 + 0 = 0 \neq 1$, so it is not in the set, and the set is not a subspace.

1.2.8.2 B. Practice

Exercise B1: Which Are Subspaces?.

Determine which of the following are subspaces of the given vector space over $\mathbb{R}$. Justify your answer.

- (a) $\{(x, y, z) \in \mathbb{R}^3 : x - 2y + z = 0\}$ in $\mathbb{R}^3$.
- (b) $\{(x, y, z) \in \mathbb{R}^3 : x = y^2\}$ in $\mathbb{R}^3$.
- (c) $\{A \in M_2(\mathbb{R}) : A \text{ is invertible}\}$ in $M_2(\mathbb{R})$.
- (d) $\{A \in M_2(\mathbb{R}) : a_{12} = 2a_{21}\}$ in $M_2(\mathbb{R})$.
- (e) $\{p \in \mathbb{R}[x] : p(2) = 0 \text{ and } p'(2) = 0\}$ in $\mathbb{R}[x]$, where p' is the derivative of p .
- (f) The even functions $\{f \in \mathbb{R}^{\mathbb{R}} : f(-t) = f(t) \text{ for all } t \in \mathbb{R}\}$ in $\mathbb{R}^{\mathbb{R}}$.
- (g) $\{f \in \mathbb{R}^{\mathbb{R}} : f(0) = 1\}$ in $\mathbb{R}^{\mathbb{R}}$.
- (h) The continuous functions $\mathbb{R} \rightarrow \mathbb{R}$, in $\mathbb{R}^{\mathbb{R}}$.

Solution.

- (a) A subspace. It is $\{\mathbf{x} \in \mathbb{R}^3 : A\mathbf{x} = \mathbf{0}\}$ with $A = \begin{pmatrix} 1 & -2 & 1 \end{pmatrix}$, a subspace by [Example \(Solutions of a Homogeneous System\)](#).
- (b) Not a subspace: (3) fails. The vector $(1, 1, 0)$ lies in the set, since $1 = 1^2$. But $2(1, 1, 0) = (2, 2, 0)$ does not, since $2 \neq 2^2$.
- (c) Not a subspace: (1) fails. The zero matrix is not invertible, because $0 \cdot B = 0 \neq I_2$ for every $B \in M_2(\mathbb{R})$.
- (d) A subspace. Call the set U .
 - (1) The zero matrix has $a_{12} = 0 = 2 \cdot 0 = 2a_{21}$. Therefore $\mathbf{0} \in U$.
 - (2) Let $A, B \in U$, so $a_{12} = 2a_{21}$ and $b_{12} = 2b_{21}$. The $(1, 2)$ -entry of $A + B$ is $a_{12} + b_{12} = 2a_{21} + 2b_{21} = 2(a_{21} + b_{21})$, which is twice its $(2, 1)$ -entry. Therefore $A + B \in U$. This shows that U is closed under addition.
 - (3) Let $c \in \mathbb{R}$ and $A \in U$. The $(1, 2)$ -entry of cA is $ca_{12} = c(2a_{21}) = 2(ca_{21})$, twice its $(2, 1)$ -entry. Therefore $cA \in U$. This shows that U is closed under scalar multiplication.
 Hence U is a subspace of $M_2(\mathbb{R})$ by the subspace test ([Theorem 1.2.1](#)).
- (e) A subspace. Call the set U . We use [Theorem 0.8.5](#) for values, and the sum and constant multiple rules for derivatives.
 - (1) The zero polynomial has value 0 at 2, and its derivative is the zero polynomial, also with value 0 at 2. Therefore $\mathbf{0} \in U$.
 - (2) Let $p, q \in U$. Then $(p+q)(2) = p(2) + q(2) = 0$ and $(p+q)'(2) = p'(2) + q'(2) = 0$. Therefore $p+q \in U$. This shows that U is closed under addition.
 - (3) Let $c \in \mathbb{R}$ and $p \in U$. Then $(cp)(2) = cp(2) = 0$ and $(cp)'(2) = cp'(2) = 0$. Therefore $cp \in U$. This shows that U is closed under scalar multiplication.
 Hence U is a subspace of $\mathbb{R}[x]$ by the subspace test ([Theorem 1.2.1](#)).
- (f) A subspace. Call the set E .
 - (1) The zero function satisfies $\mathbf{0}(-t) = 0 = \mathbf{0}(t)$ for all t . Therefore $\mathbf{0} \in E$.
 - (2) Let $f, g \in E$. For every $t \in \mathbb{R}$, $(f+g)(-t) = f(-t) + g(-t) = f(t) + g(t) = (f+g)(t)$. Therefore $f+g \in E$. This shows that E is closed under addition.
 - (3) Let $a \in \mathbb{R}$ and $f \in E$. For every $t \in \mathbb{R}$, $(af)(-t) = af(-t) = af(t) = (af)(t)$. Therefore $af \in E$. This shows that E is closed under scalar multiplication.
 Hence E is a subspace of $\mathbb{R}^{\mathbb{R}}$ by the subspace test ([Theorem 1.2.1](#)).

- (g) Not a subspace: (1) fails. The zero function has value $0 \neq 1$ at 0.
- (h) A subspace. **(1)** The zero function is constant, hence continuous. **(2)** A sum of continuous functions is continuous. **(3)** A constant multiple of a continuous function is continuous. Hence the continuous functions form a subspace of $\mathbb{R}^{\mathbb{R}}$ by the subspace test ([Theorem 1.2.1](#)).

Exercise B2: Matrices Commuting with a Given Matrix.

Fix $B \in M_n(F)$, and let $C(B) = \{A \in M_n(F) : AB = BA\}$.

- (a) Prove that $C(B)$ is a subspace of $M_n(F)$.
- (b) Let $n = 2$, $F = \mathbb{R}$ and $B = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$. Find all $A \in C(B)$. Hence describe $C(B)$ in words.

Solution.

- (a) By definition, $C(B) \subseteq M_n(F)$. We use [Theorem 0.9.4](#) throughout.
- (1)** By the zero rule, $0B = 0 = B0$. Therefore $\mathbf{0} \in C(B)$.
- (2)** Let $A_1, A_2 \in C(B)$, so $A_1B = BA_1$ and $A_2B = BA_2$. By distributivity,

$$(A_1 + A_2)B = A_1B + A_2B = BA_1 + BA_2 = B(A_1 + A_2).$$

Therefore $A_1 + A_2 \in C(B)$. This shows that $C(B)$ is closed under addition.

(3) Let $c \in F$ and $A \in C(B)$. By the scalar rule, $(cA)B = c(AB) = c(BA) = B(cA)$. Therefore $cA \in C(B)$. This shows that $C(B)$ is closed under scalar multiplication.

Hence $C(B)$ is a subspace of $M_n(F)$ by the subspace test ([Theorem 1.2.1](#)).

- (b) Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then

$$AB = \begin{pmatrix} a & 2b \\ c & 2d \end{pmatrix}, \quad BA = \begin{pmatrix} a & b \\ 2c & 2d \end{pmatrix}.$$

So $AB = BA$ if and only if $2b = b$ and $c = 2c$, that is, $b = 0$ and $c = 0$. Hence $C(B)$ is the set of diagonal matrices $\begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}$ with $a, d \in \mathbb{R}$.

Exercise B3: Intersection and Union of Two Planes.

In $\mathbb{R}^3$, let $U = \{(x, y, z) : x + y + z = 0\}$ and $W = \{(x, y, z) : x - y = 0\}$.

- (a) Explain why U and W are subspaces of $\mathbb{R}^3$, and find $U \cap W$.
- (b) Find $\mathbf{u} \in U$ with $\mathbf{u} \notin W$ and $\mathbf{w} \in W$ with $\mathbf{w} \notin U$. Hence show directly that $U \cup W$ is not a subspace of $\mathbb{R}^3$.

Solution.

- (a) U and W are the solution sets of $A\mathbf{x} = \mathbf{0}$ for $A = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}$ and $A = \begin{pmatrix} 1 & -1 & 0 \end{pmatrix}$, so they are subspaces by [Example \(Solutions of a Homogeneous System\)](#). A vector

(x, y, z) lies in $U \cap W$ exactly when $x + y + z = 0$ and $x = y$. Substituting $y = x$ gives $z = -2x$. Hence

$$U \cap W = \{(t, t, -2t) : t \in \mathbb{R}\},$$

a line through the origin, which is a subspace by [Theorem 1.2.2](#).

- (b) Take $\mathbf{u} = (1, -1, 0)$, which lies in U since $1 - 1 + 0 = 0$, but not in W since $1 - (-1) = 2 \neq 0$. Take $\mathbf{w} = (1, 1, 0)$, which lies in W , but not in U since $1 + 1 + 0 = 2 \neq 0$. Both lie in $U \cup W$. Their sum is $(2, 0, 0)$, which is not in U (as $2 + 0 + 0 \neq 0$) and not in W (as $2 - 0 \neq 0$). Hence $U \cup W$ is not closed under addition, and it is not a subspace of $\mathbb{R}^3$.

1.2.8.3 C. Going deeper

Exercise C1: Covering a Space by Subspaces.

- (a) Let V be a vector space over any field F . Prove that V is not the union of two proper subspaces.
- (b) Show that $\mathbb{F}_2^2$ is the union of three proper subspaces.

Hint: for (a), use [Proposition 1.2.3](#).

Solution.

- (a) Suppose, for a contradiction, that $V = U \cup W$ with U and W proper subspaces of V . Then $U \cup W = V$ is a subspace of V , so by [Proposition 1.2.3](#), $U \subseteq W$ or $W \subseteq U$. In the first case $V = U \cup W = W$, and in the second $V = U$. Either way one of the subspaces equals V , contradicting that both are proper. Hence V is not the union of two proper subspaces.
- (b) The space $\mathbb{F}_2^2$ has the four elements $(0, 0), (1, 0), (0, 1), (1, 1)$. Let $\mathbf{v}$ be one of the three non-zero elements, and let $U_{\mathbf{v}} = \{\mathbf{0}, \mathbf{v}\}$. We check that $U_{\mathbf{v}}$ is a subspace with the three-check move.

(1) $\mathbf{0} \in U_{\mathbf{v}}$ by construction.

(2) The possible sums are $\mathbf{0} + \mathbf{0} = \mathbf{0}$, $\mathbf{0} + \mathbf{v} = \mathbf{v} + \mathbf{0} = \mathbf{v}$, and $\mathbf{v} + \mathbf{v} = (1 + 1)\mathbf{v} = 0\mathbf{v} = \mathbf{0}$, using (VS8), (VS6), $1 + 1 = 0$ in $\mathbb{F}_2$, and [Theorem 1.1.5](#). All lie in $U_{\mathbf{v}}$, so $U_{\mathbf{v}}$ is closed under addition.

(3) The only scalars are 0 and 1. For $\mathbf{u} \in U_{\mathbf{v}}$, $0\mathbf{u} = \mathbf{0}$ by [Theorem 1.1.5](#) and $1\mathbf{u} = \mathbf{u}$ by (VS8), both in $U_{\mathbf{v}}$. So $U_{\mathbf{v}}$ is closed under scalar multiplication.

Hence each $U_{\mathbf{v}}$ is a subspace. Each has two elements, so it is proper. Every element of $\mathbb{F}_2^2$ is $\mathbf{0}$ or one of the three non-zero vectors, so

$$\mathbb{F}_2^2 = U_{(1,0)} \cup U_{(0,1)} \cup U_{(1,1)}.$$

So three proper subspaces can cover a vector space, although (a) shows two never can.

Exercise C2: The Sum of Two Subspaces.

Let U and W be subspaces of a vector space V over F , and let

$$U + W = \{\mathbf{u} + \mathbf{w} : \mathbf{u} \in U, \mathbf{w} \in W\}.$$

- (a) Prove that $U + W$ is a subspace of V containing both U and W .
- (b) Prove that if X is a subspace of V with $U \cup W \subseteq X$, then $U + W \subseteq X$.
- (c) For the two axes U and W of [Example \(The Union of the Two Axes\)](#), show that $U + W = \mathbb{R}^2$.

Solution.

- (a) Every $\mathbf{u} + \mathbf{w}$ is a vector of V , so $U + W \subseteq V$.
 (1) Since $\mathbf{0} \in U$ and $\mathbf{0} \in W$, and $\mathbf{0} + \mathbf{0} = \mathbf{0}$ by (VS3), we get $\mathbf{0} \in U + W$.
 (2) Let $\mathbf{u}_1 + \mathbf{w}_1$ and $\mathbf{u}_2 + \mathbf{w}_2$ be in $U + W$, with $\mathbf{u}_1, \mathbf{u}_2 \in U$ and $\mathbf{w}_1, \mathbf{w}_2 \in W$. By (VS1) and (VS2),

$$(\mathbf{u}_1 + \mathbf{w}_1) + (\mathbf{u}_2 + \mathbf{w}_2) = (\mathbf{u}_1 + \mathbf{u}_2) + (\mathbf{w}_1 + \mathbf{w}_2).$$

Since U and W are closed under addition, $\mathbf{u}_1 + \mathbf{u}_2 \in U$ and $\mathbf{w}_1 + \mathbf{w}_2 \in W$. Therefore the sum lies in $U + W$. This shows that $U + W$ is closed under addition.

(3) Let $a \in F$ and $\mathbf{u} + \mathbf{w} \in U + W$. By (VS5), $a(\mathbf{u} + \mathbf{w}) = a\mathbf{u} + a\mathbf{w}$, with $a\mathbf{u} \in U$ and $a\mathbf{w} \in W$ by closure. Therefore $a(\mathbf{u} + \mathbf{w}) \in U + W$. This shows that $U + W$ is closed under scalar multiplication.

Hence $U + W$ is a subspace of V . For $\mathbf{u} \in U$, we have $\mathbf{u} = \mathbf{u} + \mathbf{0}$ with $\mathbf{0} \in W$, so $\mathbf{u} \in U + W$; thus $U \subseteq U + W$. Swapping the roles of U and W gives $W \subseteq U + W$, using (VS1).

- (b) Let $\mathbf{u} + \mathbf{w} \in U + W$ with $\mathbf{u} \in U$ and $\mathbf{w} \in W$. Then $\mathbf{u}, \mathbf{w} \in U \cup W \subseteq X$. Since X is a subspace, it is closed under addition, so $\mathbf{u} + \mathbf{w} \in X$. Hence $U + W \subseteq X$.
- (c) By (a), $U + W \subseteq \mathbb{R}^2$. Conversely, let $(x, y) \in \mathbb{R}^2$. Then $(x, y) = (x, 0) + (0, y)$ with $(x, 0) \in U$ and $(0, y) \in W$, so $(x, y) \in U + W$. Hence $U + W = \mathbb{R}^2$, whereas $U \cup W$ is only the two axes.

Exercise C3: A One-Line Test, and a Test over $\mathbb{F}_2$.

Let V be a vector space over F and $U \subseteq V$.

- (a) Prove that U is a subspace of V if and only if $U \neq \emptyset$ and $a\mathbf{u} + \mathbf{w} \in U$ for all $a \in F$ and $\mathbf{u}, \mathbf{w} \in U$.
- (b) Suppose $F = \mathbb{F}_2$. Prove that U is a subspace of V if and only if $\mathbf{0} \in U$ and U is closed under addition.
- (c) Show by an example that the statement in (b) is false over $\mathbb{R}$.

Solution.

- (a) ($\Rightarrow$) Suppose U is a subspace. By the subspace test ([Theorem 1.2.1](#)), $\mathbf{0} \in U$, so $U \neq \emptyset$. For $a \in F$ and $\mathbf{u}, \mathbf{w} \in U$, condition (3) gives $a\mathbf{u} \in U$, and then condition (2) gives $a\mathbf{u} + \mathbf{w} \in U$.
 ($\Leftarrow$) Suppose $U \neq \emptyset$ and $a\mathbf{u} + \mathbf{w} \in U$ for all $a \in F$, $\mathbf{u}, \mathbf{w} \in U$. Pick $\mathbf{u}_0 \in U$.

Taking $a = -1$ and $\mathbf{u} = \mathbf{w} = \mathbf{u}_0$, and using [Theorem 1.1.7](#) and (VS4),

$$(-1)\mathbf{u}_0 + \mathbf{u}_0 = -\mathbf{u}_0 + \mathbf{u}_0 = \mathbf{0} \in U,$$

where $-\mathbf{u}_0 + \mathbf{u}_0 = \mathbf{u}_0 + (-\mathbf{u}_0)$ by (VS1). This is condition (1). For $\mathbf{u}, \mathbf{w} \in U$, taking $a = 1$ and using (VS8) gives $\mathbf{u} + \mathbf{w} = 1\mathbf{u} + \mathbf{w} \in U$, which is (2). For $a \in F$ and $\mathbf{u} \in U$, taking $\mathbf{w} = \mathbf{0} \in U$ and using (VS3) gives $a\mathbf{u} = a\mathbf{u} + \mathbf{0} \in U$, which is (3). By the subspace test ([Theorem 1.2.1](#)), U is a subspace.

- (b) ($\Rightarrow$) This is part of the subspace test ([Theorem 1.2.1](#)). ($\Leftarrow$) Suppose $\mathbf{0} \in U$ and U is closed under addition. The only scalars in $\mathbb{F}_2$ are 0 and 1. For $\mathbf{u} \in U$, $0\mathbf{u} = \mathbf{0} \in U$ by [Theorem 1.1.5](#), and $1\mathbf{u} = \mathbf{u} \in U$ by (VS8). So U is closed under scalar multiplication, and by the subspace test ([Theorem 1.2.1](#)) U is a subspace.
- (c) Take $V = \mathbb{R}^2$ and $U = \mathbb{Z}^2$. Then $\mathbf{0} \in U$, and the sum of two integer vectors is an integer vector. But $\frac{1}{2}(1, 0) = (\frac{1}{2}, 0) \notin U$, so U is not a subspace. Over $\mathbb{R}$ there are many more scalars to test than 0 and 1.

1.3 Linear Combinations and Span

We now have vector spaces and, inside them, subspaces. So far every subspace has been handed to us by a condition, such as “ $x + 2y = 0$ ” or “ $A^T = A$ ”. This section goes the other way. We start from a few vectors, build everything that can be built from them, and show that the result is always a subspace: the smallest one containing the vectors we started with. Along the way we meet the first way of measuring a space, by asking whether finitely many vectors are enough to build all of it.

1.3.1 Linear combinations

Take two vectors in $\mathbb{R}^3$, say $\mathbf{v}_1 = (1, 0, 1)$ and $\mathbf{v}_2 = (0, 1, 1)$. Which vectors can we build from them, using only the two operations a vector space gives us? Scaling gives $a\mathbf{v}_1$ and $b\mathbf{v}_2$, and adding gives

$$a\mathbf{v}_1 + b\mathbf{v}_2 = (a, b, a + b).$$

Adding or scaling again produces nothing new: for instance $2(a\mathbf{v}_1 + b\mathbf{v}_2) + (c\mathbf{v}_1 + d\mathbf{v}_2) = (2a + c)\mathbf{v}_1 + (2b + d)\mathbf{v}_2$ has the same shape. So the vectors we can reach are exactly those of the form $(a, b, a + b)$: the plane $z = x + y$ through the origin. The expression $a\mathbf{v}_1 + b\mathbf{v}_2$ will appear in almost every argument from now on, so it deserves a name.

The vectors we combine come as a **list** $(\mathbf{v}_1, \dots, \mathbf{v}_k)$: a finite sequence of vectors, in order, in which repeats are allowed. A list of length $k = 0$ is the **empty list** $()$. Why lists rather than sets becomes important in the next section; for now it costs nothing.

Definition 1.3.1 — Linear Combination

Let V be a vector space over F , and let $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ be a list of vectors in V , where $k \geq 0$. A **linear combination** of $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ is a vector of the form

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_k\mathbf{v}_k, \quad \text{where } a_1, \dots, a_k \in F.$$

The scalars a_i are the **coefficients** of the combination. A linear combination of the empty list is, by convention, the **empty sum** $\mathbf{0}$.

In words: scale each vector of the list by a scalar from F , then add up the results. Two small points hide in the notation. First, a sum of k vectors needs no brackets, because addition in V is associative (VS2); so the expression names a single vector. Second, the coefficients must lie in the field F we are working over. Changing the field can change what counts as a combination, as [Exercise \(C2: The Span Depends on the Field\)](#) shows.

The empty sum convention is the additive twin of $0! = 1$ and $x^0 = 1$, where a product with no factors is the neutral element 1 for multiplication: a sum with no terms is the neutral element $\mathbf{0}$ for addition. It will make several statements true without exceptions.

Example. (Linear Combinations in Four Families)

Show that each vector is a linear combination of the given list.

- (a) $(3, -2, 1) \in \mathbb{R}^3$, of the list $((1, 0, 1), (0, 1, 1))$.
- (b) $x^2 - 1 \in \mathbb{R}[x]$, of the list $((x+1)^2, x+1)$.
- (c) The function $\sin^2$ in $\mathbb{R}^{\mathbb{R}}$, of the list $(1, \cos 2x)$, where 1 is the constant function.
- (d) The zero vector of any vector space V , of any list $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ in V .

Solution.

- (a) We need $a(1, 0, 1) + b(0, 1, 1) = (a, b, a+b)$ to equal $(3, -2, 1)$. The first two entries force $a = 3$ and $b = -2$, and then the third entry is $a + b = 1$, as required. Hence $(3, -2, 1) = 3(1, 0, 1) - 2(0, 1, 1)$.
- (b) Expanding, $(x+1)^2 - 2(x+1) = x^2 + 2x + 1 - 2x - 2 = x^2 - 1$. So the coefficients are 1 and -2 .
- (c) The double-angle formula $\cos 2x = 1 - 2\sin^2 x$ holds for every real x . Hence $\sin^2 x = \frac{1}{2} \cdot 1 + (-\frac{1}{2})\cos 2x$ for every x , which is an equality of functions by [Definition 0.4.1](#). The coefficients $\frac{1}{2}$ and $-\frac{1}{2}$ are real, as they must be.
- (d) Take every coefficient to be 0. By [Theorem 1.1.5](#), $0\mathbf{v}_i = \mathbf{0}$ for each i , so $0\mathbf{v}_1 + \dots + 0\mathbf{v}_k = \mathbf{0}$. For the empty list, $\mathbf{0}$ is the empty sum. This degenerate case matters: it says the zero vector can always be built, from anything, even from nothing.

Notice that the question in (a) was a small system of equations in the unknown coefficients. That is typical: deciding whether a vector is a combination of a list means deciding whether a system of linear equations has a solution. Here we solve such systems by hand; Chapter 2 gives a systematic method.

1.3.2 The span

The span of some vectors is everything you can build from them by scaling and adding.

We have two kinds of input to cover. Usually we start from a list, as above. But sometimes the natural input is a set, possibly infinite, such as all the powers $1, x, x^2, \dots$ in $F[x]$. A linear combination only ever uses finitely many vectors, so for a set we allow every finite list drawn from it.

Definition 1.3.2 — Span

Let V be a vector space over F .

- The **span** of a list $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ of vectors in V is the set of **all** its linear combinations:

$$\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k) = \{a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k : a_1, \dots, a_k \in F\}.$$

- The **span** of a subset $S \subseteq V$, possibly infinite, is the set of all vectors that are linear combinations of **some finite** list of vectors in S :

$$\text{span}(S) = \{a_1\mathbf{s}_1 + \dots + a_k\mathbf{s}_k : k \geq 0, \mathbf{s}_1, \dots, \mathbf{s}_k \in S, a_1, \dots, a_k \in F\}.$$

In particular $\text{span}(\emptyset) = \text{span}(\emptyset) = \{\mathbf{0}\}$. When the field matters we write span_F .

In words: for a list, the span collects every choice of coefficients. For a set, we may first pick any finite number of vectors from S (the same vector more than once is allowed), and then any coefficients. The value $k = 0$ is allowed, and it contributes the empty sum $\mathbf{0}$. So $\mathbf{0} \in \text{span}(S)$ for **every** S , including $S = \emptyset$.

One thing to check. A list $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ now has two spans: the span of the list, and the span of the set $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ of its entries. They agree. A combination of the list is a combination of a finite list from the set, so the first is contained in the second. Conversely, take a combination $b_1\mathbf{w}_1 + \dots + b_m\mathbf{w}_m$ where each $\mathbf{w}_r$ is one of the $\mathbf{v}_i$; for each r , choose one index $i(r)$ with $\mathbf{w}_r = \mathbf{v}_{i(r)}$. Group the terms with the same index together, using (VS1), (VS2) and $b\mathbf{v} + b'\mathbf{v} = (b + b')\mathbf{v}$ (VS6). The result is $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k$, where a_i is the sum of the b_r with $i(r) = i$ (and $a_i = 0$ if there are none, by [Theorem 1.1.5](#)). So the second is contained in the first. In particular the span of a list does not change if we reorder the list or repeat an entry.

Why $\text{span}(\emptyset) = \{\mathbf{0}\}$? The definition forces it through the empty sum, but it is also the only sensible choice. We will prove that the span of S is the smallest subspace containing S . The smallest subspace containing nothing in particular is $\{\mathbf{0}\}$, since every subspace contains $\mathbf{0}$. Any other convention would make that theorem false for $S = \emptyset$, and the empty list will turn out to be a basis of the zero space.

Example. (Computing Spans)

Describe each span.

- $\text{span}((1, 0, 1), (0, 1, 1))$ in $\mathbb{R}^3$. Is $(1, 2, 3)$ in it? Is $(1, 2, 4)$?
- $\text{span}(\mathbf{v})$ for a single vector $\mathbf{v}$ in a vector space V .
- $\text{span}(1, x^2)$ in $\mathbb{R}[x]$.
- $\text{span}(\{1, x, x^2, x^3, \dots\})$ in $F[x]$.

Solution.

- As computed at the start of the section, the combinations are $a(1, 0, 1) + b(0, 1, 1) = (a, b, a + b)$. So the span is $\{(x, y, z) \in \mathbb{R}^3 : z = x + y\}$, a plane through the origin. For membership, a vector (x, y, z) is in the span exactly when the entries fit: the first two entries force $a = x$ and $b = y$, and then we need $z = x + y$. Since $3 = 1 + 2$, the vector $(1, 2, 3) = 1(1, 0, 1) + 2(0, 1, 1)$ is in the span. Since $4 \neq 1 + 2$, the vector

$(1, 2, 4)$ is not.

- (b) $\text{span}(\mathbf{v}) = \{a\mathbf{v} : a \in F\}$. If $\mathbf{v} \neq \mathbf{0}$ and $V = \mathbb{R}^2$ or $\mathbb{R}^3$, this is the line through the origin in the direction of $\mathbf{v}$. If $\mathbf{v} = \mathbf{0}$, then $a\mathbf{0} = \mathbf{0}$ for every a (Theorem 1.1.6), so $\text{span}(\mathbf{0}) = \{\mathbf{0}\}$. This degenerate case shows that a span can be as small as the zero space even when the list is not empty.
- (c) The combinations are $a + bx^2$ with $a, b \in \mathbb{R}$: the polynomials of degree at most 2 with no x term. For example $x^2 + x$ is **not** in the span, because its coefficient of x is $1 \neq 0$ and polynomials are equal only when all coefficients agree (Definition 0.8.1).
- (d) Every polynomial $a_0 + a_1x + \cdots + a_Nx^N$ is a combination of the finite list $(1, x, \dots, x^N)$ of vectors in the set. Conversely every combination of finitely many powers of x is a polynomial. So the span is all of $F[x]$. No single polynomial here is a combination of *infinitely* many powers: each one uses only the powers up to its degree.

Quick check. Is $\text{span}((1, 2))$ a subset of $\text{span}((2, 4))$ in $\mathbb{R}^2$? Are they equal?

Solution. Yes to both. Since $(2, 4) = 2(1, 2)$, every $a(2, 4) = (2a)(1, 2)$ lies in $\text{span}((1, 2))$. Since $(1, 2) = \frac{1}{2}(2, 4)$, every $a(1, 2) = \frac{a}{2}(2, 4)$ lies in $\text{span}((2, 4))$. Both spans are the line $y = 2x$. Different lists can have the same span.

1.3.3 The span is the smallest subspace

The plane in Example (Computing Spans) (a) was a subspace. That was no accident. Adding two combinations gives a combination, and so does scaling one. And any subspace that contains the vectors must contain all their combinations, because a subspace is closed under exactly the operations that build them. So the span is both a subspace and the smallest one available:

Theorem 1.3.1 — Span Is the Smallest Subspace

Let V be a vector space over F and let $S \subseteq V$. Then:

- (1) $\text{span}(S)$ is a subspace of V ;
- (2) $S \subseteq \text{span}(S)$;
- (3) if U is a subspace of V with $S \subseteq U$, then $\text{span}(S) \subseteq U$.

The same holds for the span of a list, which equals the span of the set of its entries.

Idea. Part (1) is the three-check subspace test. Part (2) is the combination 1s. Part (3) is “a subspace is closed under the operations that build combinations”; since a combination has k terms for an arbitrary k , we add one term at a time, by induction on k .

Proof.

- (1) We apply Theorem 1.2.1.

- *Zero.* The empty list is a finite list of vectors in S , and its only combination is $\mathbf{0}$. Hence $\mathbf{0} \in \text{span}(S)$.
- *Addition.* Let $\mathbf{x}, \mathbf{y} \in \text{span}(S)$. Then $\mathbf{x} = a_1\mathbf{s}_1 + \cdots + a_k\mathbf{s}_k$ and $\mathbf{y} = b_1\mathbf{t}_1 + \cdots + b_l\mathbf{t}_l$.

$\cdots + b_m \mathbf{t}_m$ for some $\mathbf{s}_i, \mathbf{t}_j \in S$ and scalars $a_i, b_j \in F$. Therefore

$$\mathbf{x} + \mathbf{y} = a_1 \mathbf{s}_1 + \cdots + a_k \mathbf{s}_k + b_1 \mathbf{t}_1 + \cdots + b_m \mathbf{t}_m,$$

which is a combination of the list $(\mathbf{s}_1, \dots, \mathbf{s}_k, \mathbf{t}_1, \dots, \mathbf{t}_m)$ of vectors in S . (If some $\mathbf{t}_j$ equals some $\mathbf{s}_i$, the list has a repeat, which lists allow.) Hence $\mathbf{x} + \mathbf{y} \in \text{span}(S)$. This shows $\text{span}(S)$ is closed under addition.

- *Scaling.* Let $\mathbf{x} = a_1 \mathbf{s}_1 + \cdots + a_k \mathbf{s}_k \in \text{span}(S)$ and $c \in F$. By (VS5), applied $k - 1$ times, and by (VS7) for each term,

$$c\mathbf{x} = (ca_1)\mathbf{s}_1 + \cdots + (ca_k)\mathbf{s}_k,$$

which is a combination of the same list. Hence $c\mathbf{x} \in \text{span}(S)$. This shows $\text{span}(S)$ is closed under scalar multiplication.

Therefore $\text{span}(S)$ is a subspace of V .

- (2) Let $\mathbf{s} \in S$. By (VS8), $\mathbf{s} = 1\mathbf{s}$, which is a combination of the list $(\mathbf{s})$. Hence $\mathbf{s} \in \text{span}(S)$.
- (3) Let U be a subspace with $S \subseteq U$. By [Theorem 1.2.1](#), U contains $\mathbf{0}$ and is closed under addition and scalar multiplication. We prove by induction on $k \geq 0$ ([Theorem 0.2.7](#)) that every combination $a_1 \mathbf{s}_1 + \cdots + a_k \mathbf{s}_k$ with $\mathbf{s}_i \in S$ lies in U . For $k = 0$ the combination is $\mathbf{0} \in U$. Suppose the claim holds for k , and consider $a_1 \mathbf{s}_1 + \cdots + a_{k+1} \mathbf{s}_{k+1}$. By the induction hypothesis $\mathbf{x} = a_1 \mathbf{s}_1 + \cdots + a_k \mathbf{s}_k \in U$. Since $\mathbf{s}_{k+1} \in S \subseteq U$ and U is closed under scaling, $a_{k+1} \mathbf{s}_{k+1} \in U$. Since U is closed under addition, $\mathbf{x} + a_{k+1} \mathbf{s}_{k+1} \in U$. This completes the induction, and shows $\text{span}(S) \subseteq U$.

■

In one phrase: $\text{span}(S)$ is a subspace containing S , and it sits inside every other one. So “the smallest subspace containing S ” makes sense, and it is $\text{span}(S)$. This gives a second, top-down description of the span, which does not mention combinations at all.

Corollary 1.3.2 — Span as an Intersection

Let V be a vector space and $S \subseteq V$. Let $\mathcal{U}$ be the set of all subspaces U of V with $S \subseteq U$. Then

$$\text{span}(S) = \bigcap_{U \in \mathcal{U}} U.$$

Proof. The index set $\mathcal{U}$ is non-empty, because V is a subspace of itself containing S ; so the intersection is defined ([Definition 0.3.9](#)).

($\subseteq$) By [Theorem 1.3.1](#) (3), $\text{span}(S) \subseteq U$ for every $U \in \mathcal{U}$. Hence $\text{span}(S)$ is contained in the intersection.

($\supseteq$) By [Theorem 1.3.1](#) (1) and (2), $\text{span}(S)$ is itself a member of $\mathcal{U}$. An intersection is contained in each of its members, so the intersection is contained in $\text{span}(S)$.

By [Theorem 0.3.1](#), the two sets are equal.

■

By [Theorem 1.2.2](#), an intersection of subspaces is a subspace, so the right-hand side is a subspace for a reason that never mentions combinations. The two descriptions complement each other. The bottom-up one (combinations) tells us what the elements look like. The top-down one (intersection) is convenient for proving that the span lies inside something.

Warning. The span of S is **not** S , and it is a subspace **even when** S is **not**. For example $S = \{(1, 0), (0, 1)\}$ is a two-element set, and not a subspace of $\mathbb{R}^2$ (it does not contain $(0, 0)$). Its span is the whole plane $\mathbb{R}^2$, since $(x, y) = x(1, 0) + y(0, 1)$. By [Theorem 1.3.1](#), $\text{span}(S) = S$ holds exactly when S is already a subspace. Also keep the two uses of the word apart: “the span of S ” is a noun, a subspace; “ S spans V ” is a verb, defined next.

1.3.4 Spanning lists and finite-dimensional spaces

The most important case is when the span is everything. Then the list is a complete toolkit for the space: every vector can be built from it.

Definition 1.3.3 — Spanning List, Spanning Set

Let V be a vector space over F . A list $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ in V **spans** V , or is a **spanning list** of V , if $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k) = V$. A subset $S \subseteq V$ **spans** V , or is a **spanning set** of V , if $\text{span}(S) = V$.

Since every span is contained in V , to show that a list spans V we must show one thing: **every** $\mathbf{v} \in V$ can be written as $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k$ for **some** scalars. The scalars may depend on $\mathbf{v}$, and they need not be unique.

Example. (Spanning Lists)

Show that each list spans the given space.

- (a) $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ in F^n , where $\mathbf{e}_i$ has 1 in position i and 0 elsewhere.
- (b) $(1, x, x^2)$ in $F[x]_{\leq 2}$.
- (c) $((1, 1), (1, -1))$ in $\mathbb{R}^2$.
- (d) $(E_{11}, E_{12}, E_{21}, E_{22})$ in $M_2(F)$, where E_{ij} has 1 in entry (i, j) and 0 elsewhere.
- (e) The empty list, in the zero space $\{\mathbf{0}\}$.

Solution.

- (a) Let $(x_1, \dots, x_n) \in F^n$. Adding entry by entry, $x_1\mathbf{e}_1 + \dots + x_n\mathbf{e}_n = (x_1, \dots, x_n)$. So every vector is a combination, with its own entries as coefficients.
- (b) Every $p \in F[x]_{\leq 2}$ has the form $a_0 + a_1x + a_2x^2$ ([Definition 0.8.4](#)), which is the combination $a_0 \cdot 1 + a_1 \cdot x + a_2 \cdot x^2$.
- (c) Let $(p, q) \in \mathbb{R}^2$. We look for a, b with $a(1, 1) + b(1, -1) = (p, q)$, that is,

$$a + b = p, \quad a - b = q.$$

Adding the equations gives $2a = p + q$, and subtracting gives $2b = p - q$. So

- $a = \frac{p+q}{2}$ and $b = \frac{p-q}{2}$. Check: $\frac{p+q}{2} + \frac{p-q}{2} = p$ and $\frac{p+q}{2} - \frac{p-q}{2} = q$. Hence $(p, q) = \frac{p+q}{2}(1, 1) + \frac{p-q}{2}(1, -1)$, and the list spans $\mathbb{R}^2$. For instance $(3, 5) = 4(1, 1) - (1, -1)$.
- (d) For any matrix, $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = aE_{11} + bE_{12} + cE_{21} + dE_{22}$, by adding entry by entry.
- (e) $\text{span}() = \{\mathbf{0}\}$ by [Definition 1.3.2](#). So the empty list spans the zero space. (So does the list $(\mathbf{0})$.)

The solution of (3) divided by 2, which is fine in $\mathbb{R}$. Over $\mathbb{F}_2$, where $2 = 0$, the same list fails: there $1 = -1$, so both vectors equal $(1, 1)$, and their span is $\{(0, 0), (1, 1)\} \neq \mathbb{F}_2^2$. Whether a list spans depends on the field.

Now the non-example, by a minimal change of (3). Replace the list by $((1, 2), (2, 4))$. It still has two vectors, and neither is zero. But every combination is

$$a(1, 2) + b(2, 4) = (a + 2b)(1, 2),$$

a vector whose second entry is twice its first. So $(0, 1)$ is **not** reached, and the list does **not** span $\mathbb{R}^2$. The clause that fails is “**every** $\mathbf{v} \in V$ is a combination”. The two vectors point along the same line, so their span is only the line $y = 2x$.

Some spaces are spanned by a short list; others need infinitely many vectors. That distinction is the first coarse measurement of size in this book.

Definition 1.3.4 — Finite-Dimensional

A vector space V over F is **finite-dimensional** if **some** finite list of vectors in V spans V . Otherwise V is **infinite-dimensional**.

By [Example \(Spanning Lists\)](#), F^n , $F[x]_{\leq 2}$, $M_2(F)$ and $\{\mathbf{0}\}$ are finite-dimensional. The same argument as in (2) and (4) shows that $F[x]_{\leq n}$ and $M_{m \times n}(F)$ are finite-dimensional too. The word “dimension” is used before we have defined a number called dimension; that number, and the fact that it is well defined, are the subject of section 6 of this chapter.

For the whole polynomial space no finite list is enough. The reason is degree: finitely many polynomials have a largest degree, and combining them never climbs above it.

Theorem 1.3.3 — $F[x]$ Is Infinite-Dimensional

For every field F , the vector space $F[x]$ is infinite-dimensional.

Proof. Let $(p_1, \dots, p_k)$ be any finite list in $F[x]$. We show it does not span $F[x]$. Each $\deg p_i$ is either a natural number or $-\infty$, and there are finitely many, so we can choose $N \in \mathbb{N}$ with $\deg p_i \leq N$ for every i .

Let $a_1, \dots, a_k \in F$. The coefficients of $a_i p_i$ are a_i times those of p_i ([Example \(The Standard Vector Spaces\)](#) (c)). Every coefficient of x^m with $m > N$ in p_i is 0, by [Definition 0.8.3](#), and $a_i \cdot 0 = 0$ by (F6) and [Theorem 0.7.1](#) (d); hence $\deg(a_i p_i) \leq N$. By [Theorem 0.8.2](#), $\deg(p + q) \leq \max(\deg p, \deg q)$, so a sum of two polynomials of degree at most N has degree at most N . Applying this $k - 1$ times, $\deg(a_1 p_1 + \dots + a_k p_k) \leq N$. (For $k = 0$ the combination is 0, of degree $-\infty \leq N$.)

Hence every element of $\text{span}(p_1, \dots, p_k)$ has degree at most N . But $x^{N+1} \in F[x]$ has

degree $N + 1 > N$, so $x^{N+1} \notin \text{span}(p_1, \dots, p_k)$. Since the list was arbitrary, no finite list spans $F[x]$. This shows $F[x]$ is infinite-dimensional. ■

(In the proof we wrote polynomials in plain italic, p , as in Chapter 0; they are vectors of $F[x]$ all the same.) The proof is the move “find a quantity that combinations cannot increase”. The space $\mathbb{R}^{\mathbb{R}}$ of all functions is infinite-dimensional too, and so are the sequence spaces; we return to such spaces in section 8 of this chapter.

Remark. For an arbitrary subset $S \subseteq V$, finite or infinite, “ S spans V ” means that **every** vector of V is a combination of **some finite** list of vectors in S . Different vectors may use different finite lists, as in [Example \(Computing Spans\)](#) (d). We never form infinite sums: a vector space only lets us add two vectors at a time, so only finite sums have a meaning. The same “every finite list” reading will be used for independence of an arbitrary set in the next section.

1.3.5 Working with spans

Three facts make spans easy to manipulate. Each follows from [Theorem 1.3.1](#) without writing a single combination.

Proposition 1.3.4 — Basic Properties of Span

Let V be a vector space and $S, T \subseteq V$.

- (1) If $T \subseteq \text{span}(S)$, then $\text{span}(T) \subseteq \text{span}(S)$.
- (2) (Monotonicity) If $T \subseteq S$, then $\text{span}(T) \subseteq \text{span}(S)$.
- (3) $\text{span}(\text{span}(S)) = \text{span}(S)$.

Proof.

- (1) By [Theorem 1.3.1](#) (1), $\text{span}(S)$ is a subspace, and by hypothesis it contains T . By [Theorem 1.3.1](#) (3) applied to T , $\text{span}(T) \subseteq \text{span}(S)$.
- (2) By [Theorem 1.3.1](#) (2), $T \subseteq S \subseteq \text{span}(S)$. Now apply (1).
- (3) Taking $T = \text{span}(S)$ in (1) gives $\text{span}(\text{span}(S)) \subseteq \text{span}(S)$. The reverse inclusion is [Theorem 1.3.1](#) (2) applied to the set $\text{span}(S)$. ■

Part (1) is the workhorse. To show $\text{span}(T) \subseteq \text{span}(S)$, it is enough to check that each vector **of** T is a combination of vectors of S ; we do not have to handle every combination of T .

The next fact is the one later sections lean on most. If a vector can already be built from S , then adding it to S builds nothing new.

Theorem 1.3.5 — Absorbing a Vector into a Span

Let V be a vector space, $S \subseteq V$ and $\mathbf{v} \in V$. If $\mathbf{v} \in \text{span}(S)$, then

$$\text{span}(S \cup \{\mathbf{v}\}) = \text{span}(S).$$

In particular, for a list: if $\mathbf{v} \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$, then $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{v}) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$, and the same holds with $\mathbf{v}$ inserted at any position of the list.

Proof. ($\supseteq$) Since $S \subseteq S \cup \{\mathbf{v}\}$, monotonicity (Proposition 1.3.4 (2)) gives $\text{span}(S) \subseteq \text{span}(S \cup \{\mathbf{v}\})$.

($\subseteq$) By Theorem 1.3.1 (2), $S \subseteq \text{span}(S)$, and $\mathbf{v} \in \text{span}(S)$ by hypothesis. Hence $S \cup \{\mathbf{v}\} \subseteq \text{span}(S)$, and Proposition 1.3.4 (1) gives $\text{span}(S \cup \{\mathbf{v}\}) \subseteq \text{span}(S)$.

For the list version, the span of a list equals the span of the set of its entries, whatever the order. The entries of $(\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{v})$ form the set $\{\mathbf{v}_1, \dots, \mathbf{v}_k\} \cup \{\mathbf{v}\}$, so the first part applies. ■

Concretely, the proof substitutes. If $\mathbf{v} = c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k$, then any combination $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k + b\mathbf{v}$ equals $(a_1 + bc_1)\mathbf{v}_1 + \dots + (a_k + bc_k)\mathbf{v}_k$, which no longer uses $\mathbf{v}$. Read backwards, the lemma says that a vector lying in the span of the others can be **deleted** from a list without shrinking its span. The next section turns this into the idea of a redundant vector, and section 6 of this chapter uses it at every step of the Exchange Theorem.

Example. (Two Lists with the Same Span)

Show that $\text{span}(1+x, 1-x) = \text{span}(1, x)$ in $\mathbb{R}[x]$.

Solution. ($\subseteq$) Both $1+x = 1 \cdot 1 + 1 \cdot x$ and $1-x = 1 \cdot 1 + (-1) \cdot x$ lie in $\text{span}(1, x)$. By Proposition 1.3.4 (1), $\text{span}(1+x, 1-x) \subseteq \text{span}(1, x)$.

($\supseteq$) Conversely,

$$1 = \frac{1}{2}(1+x) + \frac{1}{2}(1-x), \quad x = \frac{1}{2}(1+x) - \frac{1}{2}(1-x),$$

so both 1 and x lie in $\text{span}(1+x, 1-x)$. By Proposition 1.3.4 (1) again, $\text{span}(1, x) \subseteq \text{span}(1+x, 1-x)$.

Hence the spans are equal; both are $\mathbb{R}[x]_{\leq 1}$. Notice that we only checked four vectors, not all combinations. We also used $\frac{1}{2}$, so the argument needs $2 \neq 0$ in the field.

Quick check. Is $x^2 + 1 \in \text{span}(x^2 + x, x + 1)$ in $\mathbb{R}[x]$?

Solution. No. Suppose $a(x^2 + x) + b(x + 1) = x^2 + 1$. The left side is $ax^2 + (a+b)x + b$. Comparing coefficients (Definition 0.8.1): $a = 1$ from x^2 , $b = 1$ from the constant term, and $a + b = 0$ from x . But then $a + b = 2 \neq 0$, a contradiction. So no such a, b exist.

1.3.6 Exercises

1.3.6.1 A. Check your understanding

Exercise A1.

- Define the span of a subset S of a vector space V . What is $\text{span}(\emptyset)$, and why?
- Determine whether the following statement is true: “for a subset $S \subseteq V$, $\text{span}(S) = S$ if and only if S is a subspace of V .” Justify your answer.
- Determine whether the following statement is true: “if $\text{span}(S) = \text{span}(T)$, then $S = T$.” Justify your answer.
- Is the zero vector in the span of every list? Is it in the span of the empty list?
- State what it means for V to be finite-dimensional, and name the quantity used to show that $F[x]$ is not.

Solution.

- $\text{span}(S)$ is the set of all linear combinations $a_1\mathbf{s}_1 + \cdots + a_k\mathbf{s}_k$ with $k \geq 0$, $\mathbf{s}_i \in S$ and $a_i \in F$. $\text{span}(\emptyset) = \{\mathbf{0}\}$: the only combination with no vectors is the empty sum $\mathbf{0}$. This is also the smallest subspace containing $\emptyset$, as [Theorem 1.3.1](#) requires.
- True. If $\text{span}(S) = S$, then S is a subspace by [Theorem 1.3.1](#) (1). Conversely, if S is a subspace, then $\text{span}(S) \subseteq S$ by [Theorem 1.3.1](#) (3) with $U = S$, and $S \subseteq \text{span}(S)$ by (2).
- False. In $\mathbb{R}^2$, take $S = \{(1, 0), (0, 1)\}$ and $T = \{(1, 1), (1, -1)\}$. Both span $\mathbb{R}^2$ ([Example \(Spanning Lists\)](#)), but $S \neq T$.
- Yes to both: take all coefficients 0, or, for the empty list, the empty sum. See [Example \(Linear Combinations in Four Families\)](#) (d).
- V is finite-dimensional if some finite list of vectors in V spans V . For $F[x]$ the quantity is the degree: combinations of a finite list have degree at most the largest degree in the list ([Theorem 1.3.3](#)).

1.3.6.2 B. Practice

Exercise B1: Membership in a Span.

Determine whether each vector lies in the given span. If it does, write it as a linear combination. Justify your answer.

- $(2, -1, 5)$ and $(3, 4, 1)$, in $\text{span}((1, 1, 0), (0, 1, 1)) \subseteq \mathbb{R}^3$.
- $2x^2 - 3x + 1$ and $x^2 + x + 1$, in $\text{span}(x^2 - x, x - 1) \subseteq \mathbb{R}[x]$.

Solution.

- A combination is $a(1, 1, 0) + b(0, 1, 1) = (a, a + b, b)$. For $(2, -1, 5)$, the first and third entries force $a = 2$ and $b = 5$; then the middle entry would be $7 \neq -1$. Hence $(2, -1, 5)$ is **not** in the span. For $(3, 4, 1)$, we get $a = 3$, $b = 1$, and the middle entry is 4, as required. Hence $(3, 4, 1) = 3(1, 1, 0) + (0, 1, 1)$ is in the span.
- A combination is $a(x^2 - x) + b(x - 1) = ax^2 + (b - a)x - b$. Compare coefficients

([Definition 0.8.1](#)). For $2x^2 - 3x + 1$: $a = 2$, $-b = 1$ so $b = -1$, and then $b - a = -3$, as required. Hence $2x^2 - 3x + 1 = 2(x^2 - x) - (x - 1)$ is in the span. For $x^2 + x + 1$: $a = 1$, $b = -1$, and then $b - a = -2 \neq 1$. Hence $x^2 + x + 1$ is **not** in the span. (Every element p of this span has $p(1) = a - a + b - b = 0$, and indeed $x^2 + x + 1$ has value 3 at 1.)

Exercise B2: Two Lists with the Same Span.

Prove that $\text{span}((1, 1, 0), (0, 1, 1)) = \text{span}((1, 2, 1), (1, 0, -1))$ in $\mathbb{R}^3$.

Solution. Write $\mathbf{u} = (1, 1, 0)$, $\mathbf{w} = (0, 1, 1)$.

($\subseteq$) We have $\mathbf{u} = \frac{1}{2}(1, 2, 1) + \frac{1}{2}(1, 0, -1)$ and $\mathbf{w} = \frac{1}{2}(1, 2, 1) - \frac{1}{2}(1, 0, -1)$: indeed $\frac{1}{2}(2, 2, 0) = (1, 1, 0)$ and $\frac{1}{2}(0, 2, 2) = (0, 1, 1)$. So $\mathbf{u}, \mathbf{w} \in \text{span}((1, 2, 1), (1, 0, -1))$, and [Proposition 1.3.4](#) (1) gives the inclusion.

($\supseteq$) We have $(1, 2, 1) = \mathbf{u} + \mathbf{w}$ and $(1, 0, -1) = \mathbf{u} - \mathbf{w}$. So both lie in $\text{span}(\mathbf{u}, \mathbf{w})$, and [Proposition 1.3.4](#) (1) gives the reverse inclusion.

Hence the two spans are equal.

Exercise B3: Spans in $\mathbb{R}^3$, Geometrically.

Describe each span as the origin, a line through the origin, a plane through the origin (give an equation), or all of $\mathbb{R}^3$. Justify your answer.

- (a) $\text{span}((1, 2, 3))$.
- (b) $\text{span}((1, 0, 0), (0, 1, 0), (1, 1, 0))$.
- (c) $\text{span}((1, -1, 0), (0, 1, -1))$.
- (d) $\text{span}((0, 0, 0))$.

Solution.

- (a) The span is $\{(t, 2t, 3t) : t \in \mathbb{R}\}$: the line through the origin and $(1, 2, 3)$.
- (b) Since $(1, 1, 0) = (1, 0, 0) + (0, 1, 0)$, [Theorem 1.3.5](#) gives $\text{span}((1, 0, 0), (0, 1, 0), (1, 1, 0)) = \text{span}((1, 0, 0), (0, 1, 0)) = \{(a, b, 0) : a, b \in \mathbb{R}\}$: the plane $z = 0$.
- (c) A combination is $a(1, -1, 0) + b(0, 1, -1) = (a, b - a, -b)$, whose entries sum to 0. Conversely, if $x + y + z = 0$, take $a = x$ and $b = -z$; then $b - a = -z - x = y$, so $(x, y, z) = x(1, -1, 0) - z(0, 1, -1)$. Hence the span is the plane $x + y + z = 0$.
- (d) $\text{span}((0, 0, 0)) = \{\mathbf{0}\}$, by [Example \(Computing Spans\)](#) (b): the origin alone.

1.3.6.3 C. Going deeper

Exercise C1: Spans of Intersections and Unions.

Let S, T be subsets of a vector space V .

- (a) Prove that $\text{span}(S \cap T) \subseteq \text{span}(S) \cap \text{span}(T)$.

- (b) Give an example in $\mathbb{R}^2$ where the inclusion in (a) is strict.
 (c) Prove that $\text{span}(S \cup T) = \{\mathbf{s} + \mathbf{t} : \mathbf{s} \in \text{span}(S), \mathbf{t} \in \text{span}(T)\}$.

Hint for (c): in a combination of vectors of $S \cup T$, sort the terms into two groups.

Solution.

- (a) Since $S \cap T \subseteq S$, monotonicity (Proposition 1.3.4 (2)) gives $\text{span}(S \cap T) \subseteq \text{span}(S)$. By symmetry, $\text{span}(S \cap T) \subseteq \text{span}(T)$. Hence $\text{span}(S \cap T) \subseteq \text{span}(S) \cap \text{span}(T)$.
 (b) Take $S = \{(1, 0), (0, 1)\}$ and $T = \{(1, 1)\}$. Then $S \cap T = \emptyset$, so $\text{span}(S \cap T) = \{\mathbf{0}\}$. But $\text{span}(S) = \mathbb{R}^2$ (Example (Spanning Lists)), so $\text{span}(S) \cap \text{span}(T) = \text{span}(T)$, the line $y = x$, which contains $(1, 1) \neq \mathbf{0}$. The inclusion is strict.
 (c) Write $W = \{\mathbf{s} + \mathbf{t} : \mathbf{s} \in \text{span}(S), \mathbf{t} \in \text{span}(T)\}$.
 ($\subseteq$) Let $\mathbf{v} \in \text{span}(S \cup T)$, say $\mathbf{v} = a_1 \mathbf{u}_1 + \cdots + a_k \mathbf{u}_k$ with each $\mathbf{u}_i \in S \cup T$. Let $\mathbf{s}$ be the sum of the terms $a_i \mathbf{u}_i$ with $\mathbf{u}_i \in S$, and $\mathbf{t}$ the sum of the remaining terms, whose vectors lie in T . Since addition is commutative and associative, $\mathbf{v} = \mathbf{s} + \mathbf{t}$. Now $\mathbf{s} \in \text{span}(S)$ and $\mathbf{t} \in \text{span}(T)$; if a group is empty, its sum is $\mathbf{0}$, which lies in every span. Hence $\mathbf{v} \in W$.
 ($\supseteq$) Let $\mathbf{s} \in \text{span}(S)$ and $\mathbf{t} \in \text{span}(T)$. By monotonicity, both lie in $\text{span}(S \cup T)$, which is closed under addition by Theorem 1.3.1 (1). Hence $\mathbf{s} + \mathbf{t} \in \text{span}(S \cup T)$. By Theorem 0.3.1, $\text{span}(S \cup T) = W$. (So W is the set $\text{span}(S) + \text{span}(T)$ of Exercise (C2: The Sum of Two Subspaces), the **sum** of the two subspaces, studied in section 7 of this chapter.)

Exercise C2: The Span Depends on the Field.

The set $\mathbb{C}^2$ is a vector space over $\mathbb{C}$. With the same addition, and scalar multiplication restricted to real scalars, it is also a vector space over $\mathbb{R}$, by Exercise (C3: Shrinking the Field): every axiom holds for all complex scalars, so in particular for real ones.

- (a) Show that $((1, 0), (0, 1))$ spans $\mathbb{C}^2$ over $\mathbb{C}$.
 (b) Explain why $((1, 0), (0, 1))$ does **not** span $\mathbb{C}^2$ over $\mathbb{R}$.
 (c) Find a finite list that spans $\mathbb{C}^2$ over $\mathbb{R}$. Hence $\mathbb{C}^2$ is finite-dimensional over $\mathbb{R}$ as well.

Solution.

- (a) For $(z, w) \in \mathbb{C}^2$, $(z, w) = z(1, 0) + w(0, 1)$ with $z, w \in \mathbb{C}$.
 (b) Over $\mathbb{R}$ the coefficients must be real, and $a(1, 0) + b(0, 1) = (a, b)$ with $a, b \in \mathbb{R}$ has real entries. So $(i, 0)$ is not in $\text{span}_{\mathbb{R}}((1, 0), (0, 1))$.
 (c) Take $((1, 0), (i, 0), (0, 1), (0, i))$. Write $z = a + bi$ and $w = c + di$ with $a, b, c, d \in \mathbb{R}$ (Definition 0.6.1). Then

$$(z, w) = a(1, 0) + b(i, 0) + c(0, 1) + d(0, i),$$

a combination with real coefficients. So this list spans $\mathbb{C}^2$ over $\mathbb{R}$, and $\mathbb{C}^2$ is finite-dimensional over $\mathbb{R}$ by Definition 1.3.4.

1.4 Linear Independence

A spanning list tells us that every vector of a space can be built. It says nothing about waste. The list $((1, 0), (0, 1), (1, 1))$ spans $\mathbb{R}^2$, but its third vector adds nothing that the first two did not already reach. To measure a space by the length of a list, we need lists with no such waste. This section makes “no vector is wasted” precise, proves the lemma that finds a wasted vector whenever there is one, and shows that lists without waste give every vector in their span a unique description.

1.4.1 Redundant vectors

The previous section gave a way to shorten a list: if a vector lies in the span of the others, we may delete it without changing the span (Theorem 1.3.5). Call such a vector **redundant**. We want a test for “this list has no redundant vector”.

A first guess is that a vector is wasted when it repeats the direction of another one, so the test would be “no vector is a scalar multiple of another”. This guess fails. In $\mathbb{R}^3$, take

$$\mathbf{u} = (1, 0, 1), \quad \mathbf{v} = (0, 1, 1), \quad \mathbf{w} = (1, 1, 2).$$

No one of them is a multiple of another: a multiple of $\mathbf{u}$ has second entry 0, a multiple of $\mathbf{v}$ has first entry 0, a multiple $(c, c, 2c)$ of $\mathbf{w}$ has equal first and second entries, and in each case neither of the other two vectors has that shape. Yet $\mathbf{w} = \mathbf{u} + \mathbf{v}$. All three lie in the plane $z = x + y$, and by Theorem 1.3.5, $\text{span}(\mathbf{u}, \mathbf{v}, \mathbf{w}) = \text{span}(\mathbf{u}, \mathbf{v})$. The vector $\mathbf{w}$ is redundant even though it is not parallel to anything.

A second guess is correct but clumsy: “no vector is a linear combination of the others”. For a list of length k this is k separate conditions, one for each vector. Here is a way to fold them into one. Move everything in $\mathbf{w} = \mathbf{u} + \mathbf{v}$ to one side:

$$1\mathbf{u} + 1\mathbf{v} + (-1)\mathbf{w} = \mathbf{0}.$$

This is a way to build $\mathbf{0}$ from the list with coefficients that are **not all zero**. Conversely, suppose $a\mathbf{u} + b\mathbf{v} + c\mathbf{w} = \mathbf{0}$ with, say, $c \neq 0$. Then $\mathbf{w} = -\frac{a}{c}\mathbf{u} - \frac{b}{c}\mathbf{v}$, so $\mathbf{w}$ is redundant. The same works whichever coefficient is non-zero. So a redundant vector exists exactly when $\mathbf{0}$ can be built in a non-trivial way. Building $\mathbf{0}$ trivially, with every coefficient 0, is always possible, so the condition we want is that this is the only way.

A list is independent when the only way to build the zero vector from it is the trivial way.

1.4.2 Definition and examples

Definition 1.4.1 — Linear Independence

Let V be a vector space over F and let $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ be a list of vectors in V , where $k \geq 0$. The list is **linearly independent** (over F) if, for all $a_1, \dots, a_k \in F$,

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_k\mathbf{v}_k = \mathbf{0} \implies a_1 = a_2 = \dots = a_k = 0.$$

The list is **linearly dependent** if it is not linearly independent: that is, if there exist $a_1, \dots, a_k \in F$, **not all zero**, with $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = \mathbf{0}$. Such an equation is a **linear**

relation among the vectors. The empty list is linearly independent.

In words: the hypothesis is an arbitrary linear combination that happens to equal $\mathbf{0}$. The conclusion says that **every** coefficient in it is 0. Dependence is the negation, obtained by the rules of Definition 0.1.4: “for all a , (combination = $\mathbf{0} \Rightarrow$ all $a_i = 0$)” becomes “there exist a with combination = $\mathbf{0}$ and **not** all $a_i = 0$ ”. This is exactly the negation worked out in Example (Negating “only the zero combination”). The empty list is independent because the condition is vacuous: there are no coefficients, so “all of them are 0” holds automatically.

The definition also dictates how every independence proof starts. It is a “for all” statement about coefficients, so we take arbitrary coefficients and assume the hypothesis:

Let $a_1\mathbf{v}_1 + \cdots + a_k\mathbf{v}_k = \mathbf{0}$. Then ... hence all $a_i = 0$.

Between “Let” and “hence”, the work is to turn one vector equation into scalar equations. How we do that depends on what $\mathbf{0}$ means in the space: in F^n we compare entries, in $F[x]$ we compare coefficients, in a function space we evaluate at points, and in $M_{m \times n}(F)$ we compare entries.

Example. (Independent Lists in Four Families)

Show that each list is linearly independent.

- (a) $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ in F^n .
- (b) $(1, x, x^2)$ in $F[x]$.
- (c) $((1, 1), (1, -1))$ in $\mathbb{R}^2$.
- (d) $(\sin, \cos)$ in the space $\mathbb{R}^{\mathbb{R}}$ of all functions $\mathbb{R} \rightarrow \mathbb{R}$.

Solution.

- (a) Let $a_1\mathbf{e}_1 + \cdots + a_n\mathbf{e}_n = \mathbf{0}$. The left side is the vector $(a_1, \dots, a_n)$, and $\mathbf{0} = (0, \dots, 0)$. Comparing entries, $a_1 = \cdots = a_n = 0$. Hence the list is independent.
- (b) Let $a + bx + cx^2 = 0$, where 0 is the **zero polynomial**, the polynomial with every coefficient 0. Two polynomials are equal exactly when all their coefficients agree (Definition 0.8.1). Comparing the coefficients of $1, x, x^2$ gives $a = b = c = 0$. Hence the list is independent.
- (c) Let $a(1, 1) + b(1, -1) = (0, 0)$. Comparing entries, $a + b = 0$ and $a - b = 0$. Adding the two equations gives $2a = 0$, so $a = 0$ since $2 \neq 0$ in $\mathbb{R}$; then $b = -a = 0$. Hence the list is independent.
- (d) Let $a \sin + b \cos = 0$, where 0 is the **zero function**. Equality of functions means equality at every point (Definition 0.4.1), so $a \sin t + b \cos t = 0$ for **every** $t \in \mathbb{R}$. We may choose convenient points. At $t = 0$: $a \cdot 0 + b \cdot 1 = 0$, so $b = 0$. At $t = \pi/2$: $a \cdot 1 + b \cdot 0 = 0$, so $a = 0$. Hence the list is independent.

In (d) we did not need the equation at every point, only at two well-chosen ones. That is the general move for functions: **choose good points**, where most terms vanish. Note the logic. Evaluating at a point can only prove independence; if the equations at our chosen points had allowed non-zero solutions, we would have learned nothing, and would have to try other points or another method.

The degenerate cases are worth settling once, because they come up inside every later argument.

Example. (Degenerate Lists)

Let V be a vector space over F . Decide whether each list is independent.

- (a) A list $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ with some $\mathbf{v}_j = \mathbf{0}$.
- (b) A list $(\mathbf{v})$ of length one.
- (c) A list $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ with a repeat: $\mathbf{v}_i = \mathbf{v}_j$ for some $i \neq j$.

Solution.

- (a) Dependent. Take $a_j = 1$ and every other coefficient 0. By [Theorem 1.1.5](#) and [Theorem 1.1.6](#), every term is $\mathbf{0}$, so the combination is $\mathbf{0}$, and the coefficient $a_j = 1$ is not zero.
- (b) Independent if and only if $\mathbf{v} \neq \mathbf{0}$. If $\mathbf{v} = \mathbf{0}$, use (a). If $\mathbf{v} \neq \mathbf{0}$, let $a\mathbf{v} = \mathbf{0}$; by [Theorem 1.1.9](#), $a = 0$ or $\mathbf{v} = \mathbf{0}$, and the second is false, so $a = 0$.
- (c) Dependent. Take $a_i = 1$, $a_j = -1$, and every other coefficient 0. The combination is $\mathbf{v}_i + (-1)\mathbf{v}_j = \mathbf{v}_i - \mathbf{v}_j = \mathbf{0}$ by [Theorem 1.1.7](#), since $\mathbf{v}_i = \mathbf{v}_j$, and $a_i = 1 \neq 0$.

Now the non-example, by a minimal change. Add one vector to the independent list $((1, 0), (0, 1))$ of [Example \(Independent Lists in Four Families\)](#) (a), to get $((1, 0), (0, 1), (1, 1))$. Each vector is still non-zero, and any two of the three still form an independent list. But

$$1(1, 0) + 1(0, 1) + (-1)(1, 1) = (0, 0)$$

with coefficients not all zero, so the clause “ $\implies a_1 = \dots = a_k = 0$ ” fails, and the list is **dependent**. The polynomial version behaves the same way: $(1, x)$ is independent, but $(1, x, 1 + x)$ is dependent, because $1 \cdot 1 + 1 \cdot x + (-1)(1 + x) = 0$.

Quick check. Is the list $((1, 2, 0), (0, 1, 1), (1, 0, -2))$ in $\mathbb{R}^3$ linearly independent?

Solution. No. Let $a(1, 2, 0) + b(0, 1, 1) + c(1, 0, -2) = \mathbf{0}$. Comparing entries: $a + c = 0$, $2a + b = 0$, $b - 2c = 0$. The first two give $c = -a$ and $b = -2a$, and then the third reads $-2a + 2a = 0$, which holds for every a . Taking $a = 1$ gives the relation $(1, 2, 0) - 2(0, 1, 1) - (1, 0, -2) = \mathbf{0}$, with coefficients not all zero.

Two more kinds of question come up constantly: lists that depend on a parameter, and lists of matrices. The method does not change.

Example. (A Parameter and a List of Matrices)

- (a) For which $k \in \mathbb{R}$ is the list $((1, 1, 0), (1, 0, 1), (0, 1, k))$ in $\mathbb{R}^3$ linearly independent?
- (b) Is the list $\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right)$ in $M_2(\mathbb{R})$ linearly independent?

Solution.

- (a) Let $a(1, 1, 0) + b(1, 0, 1) + c(0, 1, k) = \mathbf{0}$. Comparing entries gives $a + b = 0$, $a + c = 0$ and $b + kc = 0$. The first two give $b = -a$ and $c = -a$; substituting into the third, $-a - ka = -(1 + k)a = 0$. If $k \neq -1$, then $1 + k \neq 0$, so $a = 0$ by [Theorem 0.7.1](#), and then $b = c = 0$. The

list is independent.

If $k = -1$, the equations hold for every a ; taking $a = 1$ gives $(1, 1, 0) - (1, 0, 1) - (0, 1, -1) = \mathbf{0}$ with coefficients not all zero. The list is dependent.

So the list is independent exactly when $k \neq -1$. Notice the shape of the argument: the parameter ends up as a factor multiplying one unknown, and the only question is when that factor can be zero.

- (b) Let $a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$. The left side is $\begin{pmatrix} a & b+c \\ b & a \end{pmatrix}$. Two matrices are equal when their entries agree (Definition 0.9.1), so $a = 0$, $b + c = 0$ and $b = 0$; hence $c = 0$ too. The list is independent.

For matrices, “compare entries” turns one matrix equation into four scalar equations, one per entry.

1.4.3 Why lists, and why this definition

Remark. Why lists, not sets. In a set, repeats disappear: $\{\mathbf{v}, \mathbf{v}\} = \{\mathbf{v}\}$. If independence were defined for sets, the “list” $(\mathbf{v}, \mathbf{v})$, which has the obvious relation $\mathbf{v} - \mathbf{v} = \mathbf{0}$, would silently become the independent set $\{\mathbf{v}\}$ when $\mathbf{v} \neq \mathbf{0}$. The repeat can also be hidden: $x + 1$ and $1 + x$ look different on the page but are the same vector. A list keeps both entries and so records the dependence. The second reason is order. Coordinates with respect to a basis (next section) are read off in order, so $(\mathbf{e}_1, \mathbf{e}_2)$ and $(\mathbf{e}_2, \mathbf{e}_1)$ must be different objects. The Linear Dependence Lemma below also refers to “the vectors before $\mathbf{v}_j$ ”, which needs an order. Reordering a list does not change whether it is independent, since a relation can be rearranged term by term.

For an arbitrary subset $S \subseteq V$, finite or infinite, we say S is **linearly independent** if **every** finite list of **distinct** vectors of S is linearly independent. The word “distinct” is essential: without it, the list $(\mathbf{s}, \mathbf{s})$ would make every non-empty set dependent. For a finite set $S = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ with the $\mathbf{v}_i$ distinct, this agrees with independence of the list $(\mathbf{v}_1, \dots, \mathbf{v}_k)$: every list of distinct vectors of S is a reordered sub-list, and a sub-list of an independent list is independent (Exercise (B3: Sub-Lists)). Together with “ S spans V ” from the previous section, this is the language section 8 of this chapter uses for spaces with no finite spanning list.

Why “all $a_i = 0$ ”, and why is the empty list independent? Suppose we only asked that **some** a_i be 0. The list $((1, 0), (0, 1), (0, 2))$ has the redundant vector $(0, 2) = 2(0, 1)$. But in any relation $a(1, 0) + b(0, 1) + c(0, 2) = \mathbf{0}$, comparing first entries forces $a = 0$, so every relation has a zero coefficient, and the list would pass the weaker test. Only “all $a_i = 0$ ” rules out every non-trivial relation. The empty list is independent because it wastes nothing, and this convention makes the empty list a basis of the zero space in the next section, just as $\text{span}() = \{\mathbf{0}\}$ makes it a spanning list.

Warning. “Not all zero” is not “all non-zero”. A dependence relation needs **one** non-zero coefficient, not every coefficient non-zero. The list $((1, 0), (0, 1), (0, 2))$ is dependent, by $0(1, 0) + 2(0, 1) - (0, 2) = \mathbf{0}$. But no relation has all three coefficients non-zero: comparing first entries in $a(1, 0) + b(0, 1) + c(0, 2) = \mathbf{0}$ forces $a = 0$. Hunting for a relation with no zero coefficient would wrongly conclude that the list is independent.

Warning. Pairwise independence is not independence. Checking pairs is not enough. In the list $((1, 0, 1), (0, 1, 1), (1, 1, 2))$ from the start of the section, every two of the three vectors form an independent list, but the three together are dependent, since

$(1, 0, 1) + (0, 1, 1) - (1, 1, 2) = \mathbf{0}$. For three or more vectors, independence is a condition on all of them at once.

1.4.4 The Linear Dependence Lemma

The definition says that a dependent list has a relation. It does not say which vector is redundant, and the discussion at the start of the section only found a redundant vector where the coefficient was non-zero. The next lemma makes a definite choice, and chooses it so that the redundant vector depends only on vectors **before** it in the list. This one-directional form is what later proofs need, because they build lists from left to right.

Theorem 1.4.1 — Linear Dependence Lemma

Let V be a vector space over F , and let $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ be a linearly dependent list in V . Then:

- (1) there is an index $j \in \{1, \dots, k\}$ with $\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$. For $j = 1$ this means $\mathbf{v}_1 = \mathbf{0}$, so if $\mathbf{v}_1 \neq \mathbf{0}$, then $j \geq 2$;
- (2) for any such j , removing $\mathbf{v}_j$ does not change the span:

$$\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1}, \mathbf{v}_{j+1}, \dots, \mathbf{v}_k) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k).$$

Idea. Take a relation with coefficients not all zero. We want to solve for one vector in terms of the vectors before it, so we need a vector whose coefficient is non-zero and which comes **after** every other vector with a non-zero coefficient. That is the **largest** index j with $a_j \neq 0$. Everything after it has coefficient zero and drops out, and we divide by a_j . Part (2) is then [Theorem 1.3.5](#) read backwards.

Proof.

- (1) Since the list is dependent, there are $a_1, \dots, a_k \in F$, not all zero, with $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = \mathbf{0}$. The set $\{i : a_i \neq 0\}$ is non-empty and finite, so it has a largest element; call it j . Then $a_j \neq 0$, and $a_i = 0$ for every $i > j$. By [Theorem 1.1.5](#) the terms with $i > j$ are $\mathbf{0}$, so

$$a_1\mathbf{v}_1 + \dots + a_{j-1}\mathbf{v}_{j-1} + a_j\mathbf{v}_j = \mathbf{0}.$$

Adding the negative of $a_1\mathbf{v}_1 + \dots + a_{j-1}\mathbf{v}_{j-1}$ to both sides gives $a_j\mathbf{v}_j = -(a_1\mathbf{v}_1 + \dots + a_{j-1}\mathbf{v}_{j-1})$. Since $a_j \neq 0$ and F is a field, a_j has an inverse $a_j^{-1} \in F$ ([Definition 0.7.1](#)). Multiplying both sides by a_j^{-1} , and using $a_j^{-1}(a_j\mathbf{v}_j) = (a_j^{-1}a_j)\mathbf{v}_j = 1\mathbf{v}_j = \mathbf{v}_j$ on the left, and $-\mathbf{x} = (-1)\mathbf{x}$ ([Theorem 1.1.7](#)) with the distributive laws on the right, we get

$$\mathbf{v}_j = (-a_j^{-1}a_1)\mathbf{v}_1 + \dots + (-a_j^{-1}a_{j-1})\mathbf{v}_{j-1}.$$

Hence $\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$. When $j = 1$ the right side is the empty sum, so $\mathbf{v}_1 = \mathbf{0}$; therefore if $\mathbf{v}_1 \neq \mathbf{0}$, then $j \geq 2$.

- (2) Let j be any index with $\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$, and let $L = (\mathbf{v}_1, \dots, \mathbf{v}_{j-1}, \mathbf{v}_{j+1}, \dots, \mathbf{v}_k)$. Each of $\mathbf{v}_1, \dots, \mathbf{v}_{j-1}$ is an entry of L , so by monotonicity ([Proposition 1.3.4](#) (2))

$\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1}) \subseteq \text{span}(L)$, and hence $\mathbf{v}_j \in \text{span}(L)$. By [Theorem 1.3.5](#), inserting $\mathbf{v}_j$ back into L at position j does not change the span. That is, $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k) = \text{span}(L)$, as claimed. ■

The index j need not be unique, and the lemma does **not** say that every vector depends on earlier ones. A small example shows both.

Example. (The Lemma on a Concrete List)

In $\mathbb{R}^2$, let $\mathbf{v}_1 = (1, 1)$, $\mathbf{v}_2 = (1, -1)$, $\mathbf{v}_3 = (2, 0)$, $\mathbf{v}_4 = (0, 1)$. Find two different linear relations, and the index j that the proof of [Theorem 1.4.1](#) extracts from each. Which indices j satisfy $\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$?

Solution. First relation: $\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3 = (1 + 1 - 2, 1 - 1 - 0) = (0, 0)$. The largest index with a non-zero coefficient is $j = 3$, and solving gives $\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2 \in \text{span}(\mathbf{v}_1, \mathbf{v}_2)$.

Second relation: $\mathbf{v}_1 - \mathbf{v}_2 - 2\mathbf{v}_4 = (1 - 1 - 0, 1 + 1 - 2) = (0, 0)$. Here the largest index is $j = 4$, and $\mathbf{v}_4 = \frac{1}{2}\mathbf{v}_1 - \frac{1}{2}\mathbf{v}_2 \in \text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$.

So $j = 3$ and $j = 4$ both qualify, and by part (2) of the lemma either $\mathbf{v}_3$ or $\mathbf{v}_4$ can be removed without changing the span. On the other hand, $j = 2$ does not qualify: $\mathbf{v}_2 = (1, -1)$ is not a multiple of $\mathbf{v}_1 = (1, 1)$, because a multiple (c, c) has equal entries. And $j = 1$ does not occur since $\mathbf{v}_1 \neq \mathbf{0}$. Removing $\mathbf{v}_3$ and then (by the lemma applied to $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4)$, which is still dependent by the second relation) $\mathbf{v}_4$ leaves $(\mathbf{v}_1, \mathbf{v}_2)$, whose span is $\mathbb{R}^2$ by [Example \(Spanning Lists\)](#).

The lemma only locates vectors that depend on **earlier** ones. Here $\mathbf{v}_1$ and $\mathbf{v}_2$ could also be removed without changing the span, since $\mathbf{v}_1 = \mathbf{v}_3 - \mathbf{v}_2$ and $\mathbf{v}_2 = \mathbf{v}_3 - \mathbf{v}_1$, but the lemma never points at them.

Read the other way round, part (1) is a test for independence that works left to right.

Remark. Independence, one vector at a time. A list $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ is linearly independent **if and only if** no $\mathbf{v}_j$ lies in $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$. The “if” direction is the contrapositive of part (1). For “only if”, suppose $\mathbf{v}_j = c_1\mathbf{v}_1 + \dots + c_{j-1}\mathbf{v}_{j-1}$. Then $c_1\mathbf{v}_1 + \dots + c_{j-1}\mathbf{v}_{j-1} + (-1)\mathbf{v}_j = \mathbf{0}$, with zero coefficients on $\mathbf{v}_{j+1}, \dots, \mathbf{v}_k$, is a relation whose coefficient -1 is non-zero. So the list is dependent. In words: a list is independent exactly when each vector reaches something new beyond the vectors before it. For instance $(1, x, x^2)$ passes: $1 \neq 0$, x is not a constant, and x^2 has degree too high to be in $\text{span}(1, x)$.

Combining the two parts of the lemma gives a clean statement about spans alone: a list is dependent exactly when it contains a vector that can be thrown away for free.

Theorem 1.4.2 — Span Preservation

Let V be a vector space and $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ a list in V with $k \geq 1$. Then the list is linearly

dependent **if and only if** there is an index j such that

$$\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1}, \mathbf{v}_{j+1}, \dots, \mathbf{v}_k) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k).$$

Proof. ($\Rightarrow$) By [Theorem 1.4.1](#) (1) there is an index j with $\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$, and by part (2) removing that $\mathbf{v}_j$ does not change the span.

($\Leftarrow$) Let L be the list with $\mathbf{v}_j$ removed, and suppose $\text{span}(L) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$. By [Theorem 1.3.1](#) (2), $\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k) = \text{span}(L)$. So $\mathbf{v}_j = \sum_{i \neq j} c_i \mathbf{v}_i$ for some scalars c_i , and therefore $\sum_{i \neq j} c_i \mathbf{v}_i + (-1)\mathbf{v}_j = \mathbf{0}$. The coefficient of $\mathbf{v}_j$ is $-1 \neq 0$, so the list is dependent. ■

So a spanning list that is dependent can always be shortened and still span. Repeating this until the list becomes independent is the idea behind the next section's proof that every finite spanning list contains a basis.

1.4.5 Unique representations

Independence was introduced to rule out waste. Its most useful consequence is about uniqueness. If a vector could be built from a list in two different ways, subtracting the two recipes would give a non-trivial way to build $\mathbf{0}$. So for an independent list, every vector in the span has exactly one recipe.

Theorem 1.4.3 — Independence and Uniqueness of Coefficients

Let V be a vector space over F and $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ a list in V . The following are equivalent:

- (a) $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ is linearly independent;
- (b) every $\mathbf{v} \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ has **exactly one** representation as a linear combination: if $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = b_1\mathbf{v}_1 + \dots + b_k\mathbf{v}_k$, then $a_i = b_i$ for every i .

Proof. (a $\Rightarrow$ b) Suppose the list is independent, and let $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = b_1\mathbf{v}_1 + \dots + b_k\mathbf{v}_k$. Subtracting the right side from both sides and using $a\mathbf{v} - b\mathbf{v} = (a - b)\mathbf{v}$ ([Theorem 1.1.8](#) and [Definition 1.1.1](#)) gives

$$(a_1 - b_1)\mathbf{v}_1 + \dots + (a_k - b_k)\mathbf{v}_k = \mathbf{0}.$$

By independence, $a_i - b_i = 0$ for every i , that is, $a_i = b_i$.

(b $\Rightarrow$ a) Suppose (b) holds, and let $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = \mathbf{0}$. By [Theorem 1.1.5](#), also $0\mathbf{v}_1 + \dots + 0\mathbf{v}_k = \mathbf{0}$. These are two representations of $\mathbf{0} \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$, so by (b), $a_i = 0$ for every i . Hence the list is independent. ■

Every vector of a span has **at least** one representation, by definition of the span; independence is what makes it **at most** one. For a dependent list uniqueness fails: with the list

$((1, 0), (0, 1), (1, 1)),$

$$(1, 1) = 1(1, 0) + 1(0, 1) + 0(1, 1) = 0(1, 0) + 0(0, 1) + 1(1, 1).$$

In the next section, a list that is independent **and** spans is called a basis. By this theorem every vector then has a unique list of coefficients, its coordinates.

1.4.6 Polynomials of distinct degrees

For polynomials there is a test that needs no equations at all. The list $(x^2 + x, x - 3, 5)$ is independent, and the reason is visible: the degrees are 2, 1 and 0, and combining lower-degree polynomials can never produce a higher degree.

Theorem 1.4.4 — Polynomials of Distinct Degrees Are Independent

Let F be a field and let $p_1, \dots, p_k \in F[x]$ be **non-zero** polynomials whose degrees are **pairwise distinct**. Then the list $(p_1, \dots, p_k)$ is linearly independent.

Idea. Put the polynomials in order of increasing degree. If the list were dependent, the Linear Dependence Lemma would give a polynomial lying in the span of the ones before it. Those all have smaller degree, so their span consists of polynomials of smaller degree, and cannot contain it.

Proof. A linear relation among the p_i can be rearranged term by term, so reordering the list does not affect independence. Hence we may assume $\deg p_1 < \deg p_2 < \dots < \deg p_k$.

Suppose, for a contradiction, that the list is dependent. Since $p_1 \neq 0$, Theorem 1.4.1 gives an index $j \geq 2$ with $p_j \in \text{span}(p_1, \dots, p_{j-1})$. Let $n = \deg p_{j-1}$, a natural number since $p_{j-1} \neq 0$. For every $i \leq j-1$ we have $\deg p_i \leq n$, so $p_1, \dots, p_{j-1}$ lie in $F[x]_{\leq n}$, which is a subspace of $F[x]$ by Example (Polynomials of Bounded Degree). By Theorem 1.3.1 (3), $\text{span}(p_1, \dots, p_{j-1}) \subseteq F[x]_{\leq n}$. Hence $\deg p_j \leq n = \deg p_{j-1}$, contradicting $\deg p_{j-1} < \deg p_j$. Therefore the list is linearly independent. ■

Both hypotheses matter. The zero polynomial must be excluded, because a list containing 0 is dependent (Example (Degenerate Lists)). And the converse fails: distinct degrees are sufficient, not necessary. The list $(x, x + 1)$ has equal degrees but is independent, since $ax + b(x + 1) = (a + b)x + b = 0$ forces $b = 0$ and then $a = 0$.

1.4.7 Independence depends on the field

The same set can be a vector space over two fields, and the answer to “independent?” can change with the field, because the field decides which coefficients are allowed.

Warning. Always ask: independent over which field? In $\mathbb{C}^2$, let $\mathbf{v}_1 = (1, i)$ and $\mathbf{v}_2 = (i, -1)$. Over $\mathbb{C}$ the list $(\mathbf{v}_1, \mathbf{v}_2)$ is **dependent**: $i\mathbf{v}_1 = (i, i^2) = (i, -1) = \mathbf{v}_2$, so $i\mathbf{v}_1 + (-1)\mathbf{v}_2 = \mathbf{0}$ with coefficient $i \neq 0$. Over $\mathbb{R}$ (the same addition, but only real

scalars, as in [Exercise \(C2: The Span Depends on the Field\)](#) the list is **independent**: if $a\mathbf{v}_1 + b\mathbf{v}_2 = (a + bi, ai - b) = (0, 0)$ with $a, b \in \mathbb{R}$, then the first entry $a + bi = 0$ forces $a = b = 0$, since a complex number is zero only when its real and imaginary parts are. The relation over $\mathbb{C}$ used the coefficient i , which is not available over $\mathbb{R}$.

A second example of the same phenomenon: $((1, 1), (1, -1))$ is independent in $\mathbb{R}^2$ ([Example \(Independent Lists in Four Families\) \(c\)](#)), but in $\mathbb{F}_2^2$, where $-1 = 1$, it is the list $((1, 1), (1, 1))$ with a repeat, hence dependent. The proof over $\mathbb{R}$ divided by 2, which is 0 in $\mathbb{F}_2$.

1.4.8 Exercises

1.4.8.1 A. Check your understanding

Exercise A1.

- Define what it means for a list $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ in a vector space over F to be linearly independent, and what it means for it to be linearly dependent.
- Determine whether the following statement is true: “a list of vectors is linearly dependent if and only if one of its vectors is a scalar multiple of another.” Justify your answer.
- Is the empty list linearly independent? Is the list $(\mathbf{0})$?
- Determine whether the following statement is true: “if $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ is linearly dependent, then $\mathbf{v}_3 \in \text{span}(\mathbf{v}_1, \mathbf{v}_2)$.” Justify your answer.
- To prove that a list of functions $\mathbb{R} \rightarrow \mathbb{R}$ is independent, what does “ $= 0$ ” mean, and what is a standard way to extract scalar equations?

Solution.

- It is independent if for all $a_1, \dots, a_k \in F$, $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = \mathbf{0}$ implies $a_1 = \dots = a_k = 0$. It is dependent if there are $a_1, \dots, a_k \in F$, not all zero, with $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = \mathbf{0}$.
- False. The list $((1, 0, 1), (0, 1, 1), (1, 1, 2))$ in $\mathbb{R}^3$ is dependent, since the third vector is the sum of the first two, but no vector is a multiple of another. (The “if” direction is true: $\mathbf{v}_i = c\mathbf{v}_j$ with $i \neq j$ gives the relation $\mathbf{v}_i - c\mathbf{v}_j = \mathbf{0}$ with coefficient $1 \neq 0$.)
- The empty list is independent: there are no coefficients, so the condition holds vacuously. The list $(\mathbf{0})$ is dependent: $1 \cdot \mathbf{0} = \mathbf{0}$ with coefficient $1 \neq 0$ ([Example \(Degenerate Lists\)](#)).
- False. Take $\mathbf{v}_1 = (1, 0)$, $\mathbf{v}_2 = (2, 0)$, $\mathbf{v}_3 = (0, 1)$ in $\mathbb{R}^2$. The list is dependent, since $2\mathbf{v}_1 - \mathbf{v}_2 = \mathbf{0}$. But every combination of $\mathbf{v}_1, \mathbf{v}_2$ has second entry 0, so $\mathbf{v}_3 \notin \text{span}(\mathbf{v}_1, \mathbf{v}_2)$. [Theorem 1.4.1](#) only promises **some** j ; here $j = 2$.
- It means the zero function: the combination takes the value 0 at **every** point. Evaluating at well-chosen points turns it into scalar equations for the coefficients.

1.4.8.2 B. Practice

Exercise B1: Independent or Not?

Determine which of the following lists are linearly independent. Justify your answer.

- (a) $((1, 0, 2), (0, 1, 1), (1, 1, 3))$ in $\mathbb{R}^3$.
- (b) $((1, 1, 0), (0, 1, 1), (1, 0, 1))$ in $\mathbb{R}^3$.
- (c) $(x^2 + 1, x^2 - 1, x)$ in $\mathbb{R}[x]$.
- (d) $\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\right)$ in $M_2(\mathbb{R})$.
- (e) (f, g) in $\mathbb{R}^{\mathbb{R}}$, where $f(t) = e^t$ and $g(t) = e^{2t}$.

Solution.

- (a) Dependent. Since $(1, 0, 2) + (0, 1, 1) = (1, 1, 3)$, we have $(1, 0, 2) + (0, 1, 1) - (1, 1, 3) = \mathbf{0}$, with coefficients not all zero.
- (b) Independent. Let $a(1, 1, 0) + b(0, 1, 1) + c(1, 0, 1) = \mathbf{0}$. Comparing entries: $a + c = 0$, $a + b = 0$, $b + c = 0$. The first two give $c = -a$ and $b = -a$; then the third gives $-2a = 0$, so $a = 0$, and hence $b = c = 0$.
- (c) Independent. Let $a(x^2 + 1) + b(x^2 - 1) + cx = 0$. The left side is $(a+b)x^2 + cx + (a-b)$. Comparing coefficients ([Definition 0.8.1](#)): $a + b = 0$, $c = 0$, $a - b = 0$. Adding the first and third gives $2a = 0$, so $a = 0$, and then $b = 0$.
- (d) Dependent. The first two matrices add up to the third, so $1 \cdot \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + 1 \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + (-1) \cdot \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ is the zero matrix.
- (e) Independent. Let $af + bg = 0$, the zero function. Then $ae^t + be^{2t} = 0$ for every $t \in \mathbb{R}$. At $t = 0$: $a + b = 0$. At $t = \ln 2$: $2a + 4b = 0$, so $a + 2b = 0$. Subtracting the first equation gives $b = 0$, and then $a = 0$.

Exercise B2: A Parameter.

For which $t \in \mathbb{R}$ is the list $((1, t), (t, 1))$ in $\mathbb{R}^2$ linearly independent? Justify your answer.

Solution. Let $a(1, t) + b(t, 1) = (0, 0)$, that is, $a + tb = 0$ and $ta + b = 0$. The first equation gives $a = -tb$. Substituting into the second, $-t^2b + b = (1 - t^2)b = 0$.

If $t \neq \pm 1$, then $1 - t^2 \neq 0$, so $b = 0$ by [Theorem 0.7.1](#), and then $a = -tb = 0$. The list is independent.

If $t = 1$, the list is $((1, 1), (1, 1))$, which has a repeat and is dependent. If $t = -1$, the list is $((1, -1), (-1, 1))$, and $1 \cdot (1, -1) + 1 \cdot (-1, 1) = (0, 0)$, so it is dependent.

Hence the list is linearly independent exactly when $t \neq 1$ and $t \neq -1$.

Exercise B3: Sub-Lists.

A **sub-list** of $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ is a list $(\mathbf{v}_{i_1}, \dots, \mathbf{v}_{i_m})$ with $1 \leq i_1 < i_2 < \dots < i_m \leq k$.

- (a) Prove that every sub-list of a linearly independent list is linearly independent.

- (b) Deduce that if some sub-list of a list is linearly dependent, then the whole list is linearly dependent.

Solution.

- (a) Let $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ be independent, and let $b_1\mathbf{v}_{i_1} + \dots + b_m\mathbf{v}_{i_m} = \mathbf{0}$. Define $a_i = b_r$ if $i = i_r$ for some r (the indices i_r are distinct, so this is well defined), and $a_i = 0$ otherwise. By [Theorem 1.1.5](#) the extra terms are $\mathbf{0}$, so $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = b_1\mathbf{v}_{i_1} + \dots + b_m\mathbf{v}_{i_m} = \mathbf{0}$. By independence, every $a_i = 0$; in particular every $b_r = a_{i_r} = 0$. Hence the sub-list is independent.
- (b) This is the contrapositive of (a): if the whole list were independent, every sub-list would be independent by (a).

1.4.8.3 C. Going deeper

Exercise C1: Sums of Pairs, and the Field $\mathbb{F}_2$.

Let V be a vector space over F and $(\mathbf{u}, \mathbf{v}, \mathbf{w})$ a linearly independent list in V .

- (a) Suppose $F = \mathbb{R}$. Prove that $(\mathbf{u} + \mathbf{v}, \mathbf{v} + \mathbf{w}, \mathbf{w} + \mathbf{u})$ is linearly independent.
- (b) Suppose $F = \mathbb{F}_2$. Prove that $(\mathbf{u} + \mathbf{v}, \mathbf{v} + \mathbf{w}, \mathbf{w} + \mathbf{u})$ is linearly dependent, and give an example of an independent list $(\mathbf{u}, \mathbf{v}, \mathbf{w})$ over $\mathbb{F}_2$, so that the statement is not vacuous.
- (c) Which property of $\mathbb{R}$ did (a) use that $\mathbb{F}_2$ lacks? Deduce that (a) holds over every field in which $1 + 1 \neq 0$.

Solution.

- (a) Let $a(\mathbf{u} + \mathbf{v}) + b(\mathbf{v} + \mathbf{w}) + c(\mathbf{w} + \mathbf{u}) = \mathbf{0}$. Expanding and regrouping with the axioms of [Definition 1.1.1](#) gives $(a + c)\mathbf{u} + (a + b)\mathbf{v} + (b + c)\mathbf{w} = \mathbf{0}$. By independence of $(\mathbf{u}, \mathbf{v}, \mathbf{w})$, $a + c = 0$, $a + b = 0$ and $b + c = 0$. The first two give $c = -a$ and $b = -a$, and then the third gives $-2a = 0$. Since $2 \neq 0$ in $\mathbb{R}$, $a = 0$, and hence $b = c = 0$. This proves independence.
- (b) Over $\mathbb{F}_2$,

$$1(\mathbf{u} + \mathbf{v}) + 1(\mathbf{v} + \mathbf{w}) + 1(\mathbf{w} + \mathbf{u}) = (1+1)\mathbf{u} + (1+1)\mathbf{v} + (1+1)\mathbf{w} = 0\mathbf{u} + 0\mathbf{v} + 0\mathbf{w} = \mathbf{0},$$

using $1 + 1 = 0$ in $\mathbb{F}_2$ and [Theorem 1.1.5](#). The coefficients are $1 \neq 0$, so the list is dependent. For an example, $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ in $\mathbb{F}_2^3$ is independent by the argument of [Example \(Independent Lists in Four Families\)](#) (a), which works over any field.

- (c) Part (a) divided by $2 = 1 + 1$, that is, used $2 \neq 0$, to pass from $-2a = 0$ to $a = 0$. In $\mathbb{F}_2$, $2 = 0$. In any field with $1 + 1 \neq 0$, the element 2 has an inverse, and the proof of (a) goes through word for word.

Exercise C2: Adding a Vector outside the Span.

Let $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ be a linearly independent list in V , and let $\mathbf{v} \in V$ with $\mathbf{v} \notin \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$. Prove that $(\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{v})$ is linearly independent. Give an example showing that the hypothesis $\mathbf{v} \notin \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ cannot be dropped.

Solution. Let $a_1\mathbf{v}_1 + \cdots + a_k\mathbf{v}_k + a\mathbf{v} = \mathbf{0}$. Suppose $a \neq 0$. Then, as in the proof of [Theorem 1.4.1](#), $\mathbf{v} = (-a^{-1}a_1)\mathbf{v}_1 + \cdots + (-a^{-1}a_k)\mathbf{v}_k \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$, contradicting the hypothesis. Hence $a = 0$, and the relation becomes $a_1\mathbf{v}_1 + \cdots + a_k\mathbf{v}_k = \mathbf{0}$. By independence of $(\mathbf{v}_1, \dots, \mathbf{v}_k)$, $a_1 = \cdots = a_k = 0$. So all coefficients are zero, and the longer list is independent.

For the example, take $(\mathbf{e}_1)$ in $\mathbb{R}^2$ and $\mathbf{v} = 2\mathbf{e}_1 \in \text{span}(\mathbf{e}_1)$. The list $(\mathbf{e}_1)$ is independent, but $(\mathbf{e}_1, 2\mathbf{e}_1)$ is dependent, by $2\mathbf{e}_1 - 2\mathbf{e}_1 = \mathbf{0}$.

Exercise C3: $1, \sqrt{2}, \sqrt{3}$ over $\mathbb{Q}$.

Regard $\mathbb{R}$ as a vector space over $\mathbb{Q}$. Prove that the list $(1, \sqrt{2}, \sqrt{3})$ is linearly independent over $\mathbb{Q}$. Is it linearly independent over $\mathbb{R}$?

Hint: you may use without proof that $\sqrt{3}$ and $\sqrt{6}$ are irrational (the proof of [Theorem 0.2.3](#) adapts). Isolate one square root and square.

Solution. Let $a + b\sqrt{2} + c\sqrt{3} = 0$ with $a, b, c \in \mathbb{Q}$. Then $a + b\sqrt{2} = -c\sqrt{3}$, and squaring both sides gives $a^2 + 2b^2 + 2ab\sqrt{2} = 3c^2$.

If $ab \neq 0$, then $\sqrt{2} = \frac{3c^2 - a^2 - 2b^2}{2ab} \in \mathbb{Q}$, contradicting [Theorem 0.2.3](#). Hence $ab = 0$, so $a = 0$ or $b = 0$.

Case 1: $b = 0$. Then $a + c\sqrt{3} = 0$. If $c \neq 0$, then $\sqrt{3} = -a/c \in \mathbb{Q}$, a contradiction. So $c = 0$, and then $a = 0$.

Case 2: $a = 0$. Then $b\sqrt{2} + c\sqrt{3} = 0$. Multiplying by $\sqrt{2}$ gives $2b + c\sqrt{6} = 0$. If $c \neq 0$, then $\sqrt{6} = -2b/c \in \mathbb{Q}$, a contradiction. So $c = 0$, then $b\sqrt{2} = 0$, and $b = 0$ since $\sqrt{2} \neq 0$.

In both cases $a = b = c = 0$, so the list is linearly independent over $\mathbb{Q}$.

Over $\mathbb{R}$ it is dependent: $\sqrt{2} \cdot 1 + (-1) \cdot \sqrt{2} + 0 \cdot \sqrt{3} = 0$, and the coefficient $\sqrt{2}$ is a non-zero real number. Over $\mathbb{R}$ any two non-zero real numbers form a dependent list.

1.5 Bases and Coordinates

Spanning lists and independent lists each solve half of a problem. A spanning list reaches every vector of the space, but it may reach some vectors in more than one way. An independent list never wastes a vector, but it may not reach everything. This section puts the two halves together. The result, a basis, turns an abstract vector space into columns of numbers, and we prove that every space spanned by a finite list has one.

1.5.1 What makes coordinates work

Start with the plane. Every vector of $\mathbb{R}^2$ can be written as

$$(x, y) = x\mathbf{e}_1 + y\mathbf{e}_2,$$

and the pair (x, y) is its address. Two separate facts make this addressing system work. First, **every** vector has an address: the list $(\mathbf{e}_1, \mathbf{e}_2)$ spans $\mathbb{R}^2$. Second, every vector has **only one** address: if $x\mathbf{e}_1 + y\mathbf{e}_2 = x'\mathbf{e}_1 + y'\mathbf{e}_2$, then $(x, y) = (x', y')$, so $x = x'$ and $y = y'$.

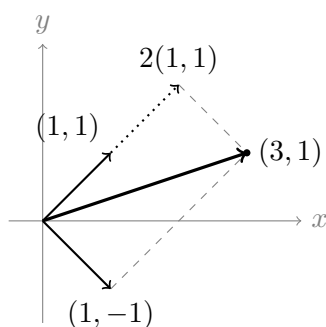
Nothing about this is special to $\mathbf{e}_1$ and $\mathbf{e}_2$. Take instead the list $((1, 1), (1, -1))$. To find an address for (x, y) we solve

$$a(1, 1) + b(1, -1) = (x, y), \quad \text{that is,} \quad a + b = x, \quad a - b = y.$$

Adding and subtracting the two equations gives $2a = x + y$ and $2b = x - y$. So there is a solution, and it is the only one:

$$a = \frac{x + y}{2}, \quad b = \frac{x - y}{2}.$$

For example $(3, 1) = 2(1, 1) + 1(1, -1)$, and no other pair of scalars works. The grid is slanted, but every point still has exactly one address.



Look at which property did which job. **Existence** of an address is spanning. **Uniqueness** of the address is independence: by [Theorem 1.4.3](#), a list is independent exactly when every vector in its span has only one expression as a combination of it. A coordinate system is a list that does both jobs at once.

A basis is a list with enough vectors to build everything and no vector to spare.

1.5.2 Bases

Definition 1.5.1 — Basis

Let V be a vector space over F . A **basis** of V is a **list** $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ of vectors in V such that

- (B1) $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is **linearly independent**, and
- (B2) $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ **spans** V , that is, $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_n) = V$.

We usually name a basis with a calligraphic letter, $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$, and call n its **length**. The plural of basis is **bases**.

In words: (B1) says that the only combination of the $\mathbf{v}_i$ equal to $\mathbf{0}$ is the one with **all** coefficients zero, so no vector in the list is wasted. (B2) says that **every** vector of V is some combination of the $\mathbf{v}_i$, so nothing is out of reach. The basis is a list, so its entries come in a definite order, and a basis is always a basis **of** a particular space: $(\mathbf{e}_1, \mathbf{e}_2)$ is a basis of $\mathbb{R}^2$, but not of $\mathbb{R}^3$.

We start with the bases that come with the running examples.

Example. (Standard Bases)

Show that each of the following lists is a basis. They are called the **standard bases**.

- (a) $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ in F^n , where $\mathbf{e}_i$ has 1 in position i and 0 elsewhere.
- (b) $(1, x, x^2, \dots, x^n)$ in $F[x]_{\leq n}$.
- (c) $(E_{11}, E_{12}, \dots, E_{1n}, E_{21}, \dots, E_{mn})$ in $M_{m \times n}(F)$, where the **matrix unit** E_{ij} has 1 in entry (i, j) and 0 elsewhere, listed row by row.

Solution.

- (a) (B2): every $(x_1, \dots, x_n) \in F^n$ equals $x_1\mathbf{e}_1 + \dots + x_n\mathbf{e}_n$. (B1): let $a_1\mathbf{e}_1 + \dots + a_n\mathbf{e}_n = \mathbf{0}$. The left side is the column $(a_1, \dots, a_n)$, so comparing entries gives $a_1 = \dots = a_n = 0$.
- (b) (B2): by [Definition 0.8.4](#), every element of $F[x]_{\leq n}$ has the form $a_0 + a_1x + \dots + a_nx^n$, which is a combination of $1, x, \dots, x^n$. (B1): let $a_0 \cdot 1 + a_1x + \dots + a_nx^n = 0$. Here 0 is the zero polynomial, all of whose coefficients are 0. Two polynomials are equal exactly when their coefficients agree ([Definition 0.8.1](#)), so $a_0 = a_1 = \dots = a_n = 0$.
- (c) (B2): a matrix $A = (a_{ij})$ equals $\sum_{i,j} a_{ij}E_{ij}$, since both sides have a_{ij} in entry (i, j) . (B1): if $\sum_{i,j} c_{ij}E_{ij}$ is the zero matrix, then its (i, j) entry c_{ij} is 0 for every i, j .

The next examples show that a space has other bases, that the field matters, and what happens in the smallest space.

Example. (Three More Bases)

Show that each list is a basis.

- (a) $((1, 1), (1, -1))$ in $\mathbb{R}^2$.
- (b) $(1, i)$ in $\mathbb{C}$, regarded as a vector space **over** $\mathbb{R}$.
- (c) The empty list $()$ in the zero space $\{\mathbf{0}\}$.

Solution.

- (a) We showed above that for every $(x, y) \in \mathbb{R}^2$ the equation $a(1, 1) + b(1, -1) = (x, y)$ has a solution, so (B2) holds. Taking $(x, y) = (0, 0)$, the only solution is $a = \frac{0+0}{2} = 0$ and $b = 0$, so (B1) holds.
- (b) The scalars are now real. (B2): every complex number is $a + bi = a \cdot 1 + b \cdot i$ with $a, b \in \mathbb{R}$ ([Definition 0.6.1](#)). (B1): let $a \cdot 1 + b \cdot i = 0$ with $a, b \in \mathbb{R}$. A complex number is 0 exactly when its real and imaginary parts are 0, so $a = b = 0$.
- (c) The empty list is linearly independent: there are no coefficients, so the condition “all coefficients are zero” holds vacuously. Its span is $\{\mathbf{0}\}$ by the convention for the empty span. So both (B1) and (B2) hold.

The degenerate case (c) matters more than it looks. Without it, the zero space would be the one space with no basis, and every theorem below would need an exception for it.

Non-examples by minimal change. Add one vector to the standard basis of $\mathbb{R}^2$, making $((1, 0), (0, 1), (1, 1))$. Spanning (B2) still holds, since the first two vectors already span. But (B1) fails:

$$1 \cdot (1, 0) + 1 \cdot (0, 1) + (-1) \cdot (1, 1) = (0, 0),$$

with coefficients **not all zero**. The effect is exactly the loss of unique addresses: $(3, 1) = 3(1, 0) + 1(0, 1) + 0(1, 1) = 2(1, 0) + 0(0, 1) + 1(1, 1)$.

Now remove a vector instead. In $\mathbb{R}^3$, the list $((1, 0, 0), (0, 1, 0))$ is still independent, since $a(1, 0, 0) + b(0, 1, 0) = (a, b, 0) = \mathbf{0}$ forces $a = b = 0$. But (B2) fails: every combination $(a, b, 0)$ has third entry 0, so $(0, 0, 1)$ is out of reach.

Quick check. Is $((1, 2), (2, 1), (0, 0))$ a basis of $\mathbb{R}^2$? If not, which of (B1) and (B2) fails?

Solution. No. (B2) holds: $(1, 2)$ and $(2, 1)$ already span $\mathbb{R}^2$, because solving $a(1, 2) + b(2, 1) = (x, y)$ gives $a = \frac{2y-x}{3}$ and $b = \frac{2x-y}{3}$. But (B1) fails, because the list contains the zero vector: $0(1, 2) + 0(2, 1) + 1(0, 0) = (0, 0)$ with a non-zero coefficient.

Why this definition. Drop (B1), and addresses stop being unique, as the list $((1, 0), (0, 1), (1, 1))$ showed. Drop (B2), and some vectors have no address at all. Keeping both is exactly what an addressing system needs. The name comes from the everyday word: the basis is what everything else is built on.

Remark. Why a list and not a set? Two reasons. Coordinates, defined below, are a column of numbers, and a column needs to know which scalar comes first. And a set would hide repeats: the list $((1, 0), (1, 0), (0, 1))$ is dependent, since $1(1, 0) - 1(1, 0) = \mathbf{0}$, but as a set $\{(1, 0), (1, 0), (0, 1)\} = \{(1, 0), (0, 1)\}$ it would look like the standard basis.

Warning. Say **a** basis, not **the** basis. A space usually has many bases; a non-zero space over an infinite field such as $\mathbb{R}$ has infinitely many. For instance $((1, 0), (c, 1))$ is a basis of $\mathbb{R}^2$ for **every** real number c , since $(x, y) = (x - cy)(1, 0) + y(c, 1)$, and this is the only solution. The standard basis is a convenient choice, not a canonical one.

1.5.3 Unique representation and coordinates

We found the definition by asking for addresses that exist and are unique. The first result confirms that this is exactly what a basis delivers, and nothing more.

Theorem 1.5.1 — Unique Representation

Let V be a vector space over F , and let $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a list in V . Then $\mathcal{B}$ is a basis of V if and only if every $\mathbf{v} \in V$ can be written as

$$\mathbf{v} = a_1 \mathbf{v}_1 + \dots + a_n \mathbf{v}_n \quad (a_1, \dots, a_n \in F)$$

in **exactly one** way.

Proof. ($\Rightarrow$) Let $\mathbf{v} \in V$. Since $\mathcal{B}$ spans V , there are scalars with $\mathbf{v} = a_1 \mathbf{v}_1 + \dots + a_n \mathbf{v}_n$. Suppose also $\mathbf{v} = b_1 \mathbf{v}_1 + \dots + b_n \mathbf{v}_n$. Subtracting the two expressions and regrouping

with the vector space axioms,

$$(a_1 - b_1)\mathbf{v}_1 + \cdots + (a_n - b_n)\mathbf{v}_n = \mathbf{v} - \mathbf{v} = \mathbf{0}.$$

Since $\mathcal{B}$ is linearly independent, $a_i - b_i = 0$ for every i . Hence the two expressions are the same.

($\Leftarrow$) Every $\mathbf{v} \in V$ is a combination of $\mathcal{B}$, so $\mathcal{B}$ spans V . For independence, let $a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n = \mathbf{0}$. We also have $\mathbf{0} = 0\mathbf{v}_1 + \cdots + 0\mathbf{v}_n$. By uniqueness of the expression for the vector $\mathbf{0}$, all $a_i = 0$. Hence $\mathcal{B}$ is linearly independent, and $\mathcal{B}$ is a basis. ■

This is [Theorem 1.4.3](#) in the special case where the span is the whole space. The theorem also gives a second way to check that a list is a basis: solve for the coefficients of a general vector, and see that there is always exactly one solution. We did exactly that for $((1, 1), (1, -1))$.

Whenever something exists and is unique, it deserves a name.

Definition 1.5.2 — Coordinates

Let $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of a vector space V over F , and let $\mathbf{v} \in V$. The unique scalars $a_1, \dots, a_n$ with $\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$ are the **coordinates** of $\mathbf{v}$ with respect to $\mathcal{B}$. The column

$$[\mathbf{v}]_{\mathcal{B}} := \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in F^n$$

is the **coordinate vector** of $\mathbf{v}$ with respect to $\mathcal{B}$.

In the standard basis of F^n , the coordinate vector of $(x_1, \dots, x_n)$ is the column $(x_1, \dots, x_n)$ itself. That is why we rarely notice coordinates in F^n . In other bases and other spaces they carry real information.

Example. (Coordinates of a Polynomial)

Show that $\mathcal{B} = (1, 1+x, 1+x+x^2)$ is a basis of $\mathbb{R}[x]_{\leq 2}$, and find $[p]_{\mathcal{B}}$ for $p = 3 - x + 2x^2$.

Solution. We use [Theorem 1.5.1](#). Let $q = q_0 + q_1x + q_2x^2 \in \mathbb{R}[x]_{\leq 2}$ be arbitrary. For scalars a, b, c ,

$$a \cdot 1 + b(1+x) + c(1+x+x^2) = (a+b+c) + (b+c)x + cx^2.$$

Two polynomials are equal exactly when their coefficients agree, so this equals q if and only if

$$c = q_2, \quad b+c = q_1, \quad a+b+c = q_0.$$

Solving from the top, $c = q_2$, then $b = q_1 - q_2$, then $a = q_0 - q_1$. So every q has exactly one expression, and $\mathcal{B}$ is a basis of $\mathbb{R}[x]_{\leq 2}$.

For $p = 3 - x + 2x^2$ we have $q_0 = 3$, $q_1 = -1$, $q_2 = 2$. Hence $c = 2$, $b = -1 - 2 = -3$,

and $a = 3 - (-1) = 4$:

$$[p]_{\mathcal{B}} = \begin{pmatrix} 4 \\ -3 \\ 2 \end{pmatrix}.$$

Check: $4 - 3(1 + x) + 2(1 + x + x^2) = (4 - 3 + 2) + (-3 + 2)x + 2x^2 = 3 - x + 2x^2$. In the standard basis $(1, x, x^2)$, the same polynomial has coordinate vector $(3, -1, 2)$.

Warning. Coordinates depend on the basis **and on its order**. Take $\mathbf{v} = (3, 1) \in \mathbb{R}^2$. With respect to the standard basis $\mathcal{E} = (\mathbf{e}_1, \mathbf{e}_2)$, $[\mathbf{v}]_{\mathcal{E}} = (3, 1)$. With respect to $\mathcal{B} = ((1, 1), (1, -1))$, $[\mathbf{v}]_{\mathcal{B}} = (2, 1)$. With respect to $\mathcal{B}' = ((1, -1), (1, 1))$, the same two vectors in the other order, $[\mathbf{v}]_{\mathcal{B}'} = (1, 2)$. A column of numbers means nothing until you say which ordered basis it refers to.

The point of coordinates is that they turn computations in V into computations with columns. That works because they respect the two operations.

Theorem 1.5.2 — Coordinates Respect the Operations

Let $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of a vector space V over F . Then for all $\mathbf{u}, \mathbf{v} \in V$ and $c \in F$,

$$[\mathbf{u} + \mathbf{v}]_{\mathcal{B}} = [\mathbf{u}]_{\mathcal{B}} + [\mathbf{v}]_{\mathcal{B}}, \quad [c\mathbf{v}]_{\mathcal{B}} = c[\mathbf{v}]_{\mathcal{B}}.$$

Moreover, every column $(a_1, \dots, a_n) \in F^n$ is the coordinate vector of exactly one vector of V , namely $a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n$.

Proof. Let $[\mathbf{u}]_{\mathcal{B}} = (a_1, \dots, a_n)$ and $[\mathbf{v}]_{\mathcal{B}} = (b_1, \dots, b_n)$. By the vector space axioms,

$$\mathbf{u} + \mathbf{v} = \sum_{i=1}^n a_i \mathbf{v}_i + \sum_{i=1}^n b_i \mathbf{v}_i = \sum_{i=1}^n (a_i + b_i) \mathbf{v}_i, \quad c\mathbf{v} = c \sum_{i=1}^n b_i \mathbf{v}_i = \sum_{i=1}^n (cb_i) \mathbf{v}_i.$$

By Theorem 1.5.1 these are the only expressions of $\mathbf{u} + \mathbf{v}$ and $c\mathbf{v}$ in $\mathcal{B}$, so their coefficients are the coordinates: $[\mathbf{u} + \mathbf{v}]_{\mathcal{B}} = (a_i + b_i)_i$ and $[c\mathbf{v}]_{\mathcal{B}} = (cb_i)_i$, as claimed.

For the last statement, let $(a_1, \dots, a_n) \in F^n$ and $\mathbf{w} = a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n$. By Definition 1.5.2, $[\mathbf{w}]_{\mathcal{B}} = (a_1, \dots, a_n)$. If also $[\mathbf{w}']_{\mathcal{B}} = (a_1, \dots, a_n)$, then by definition $\mathbf{w}' = a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n = \mathbf{w}$. ■

So a basis of length n sets up a one-to-one correspondence $\mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}}$ between V and F^n that respects addition and scaling. Any question about sums and multiples in V can be answered in F^n instead. In Chapter 3 we will call such a correspondence an **isomorphism**.

1.5.4 Every finite spanning list contains a basis

Where do bases come from? A finite-dimensional space is, by Definition 1.3.4, a space spanned by some finite list. That list reaches everything but may contain waste, so the natural plan is to throw the waste away.

Care is needed about **what** to throw away. In $((1, 0), (0, 1), (1, 1))$ each vector is a combination of the other two. Any single one can go, but not two of them. So redundancy is not a property of one vector alone: it depends on which other vectors stay. The fix is to decide one vector at a time, reading from left to right, and to compare each vector only with the vectors already kept.

First, a small lemma that tells us when a kept list stays independent.

Lemma 1.5.3 — Appending a Vector to an Independent List

Let $(\mathbf{u}_1, \dots, \mathbf{u}_k)$ be a linearly independent list in a vector space V , and let $\mathbf{v} \in V$. Then $(\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{v})$ is linearly independent **if and only if** $\mathbf{v} \notin \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$.

Proof. $(\Rightarrow)$ Suppose $\mathbf{v} \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$, say $\mathbf{v} = c_1\mathbf{u}_1 + \dots + c_k\mathbf{u}_k$. Then $c_1\mathbf{u}_1 + \dots + c_k\mathbf{u}_k + (-1)\mathbf{v} = \mathbf{0}$, with the coefficient $-1 \neq 0$. So the longer list is dependent. This proves $(\Rightarrow)$ by contraposition.

$(\Leftarrow)$ Suppose $\mathbf{v} \notin \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$, and let $a_1\mathbf{u}_1 + \dots + a_k\mathbf{u}_k + b\mathbf{v} = \mathbf{0}$. If $b \neq 0$, then b^{-1} exists and

$$\mathbf{v} = -b^{-1}a_1\mathbf{u}_1 - \dots - b^{-1}a_k\mathbf{u}_k \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k),$$

a contradiction. Hence $b = 0$. Then $a_1\mathbf{u}_1 + \dots + a_k\mathbf{u}_k = \mathbf{0}$, and the independence of $(\mathbf{u}_1, \dots, \mathbf{u}_k)$ gives $a_1 = \dots = a_k = 0$. So all coefficients are zero, and the longer list is independent. ■

The direction $(\Leftarrow)$ is [Exercise \(C2: Adding a Vector outside the Span\)](#). When $k = 0$, the lemma says that the one-vector list $(\mathbf{v})$ is independent exactly when $\mathbf{v} \notin \text{span}() = \{\mathbf{0}\}$, that is, when $\mathbf{v} \neq \mathbf{0}$, in agreement with [Example \(Degenerate Lists\)](#).

Now the procedure. Given a list $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ in V , **sifting** it means the following.

Start with the empty list $S_0 = ()$. For $r = 1, 2, \dots, m$ in turn:

- if $\mathbf{v}_r \notin \text{span}(S_{r-1})$, **keep** it: $S_r = (S_{r-1}, \mathbf{v}_r)$;
- if $\mathbf{v}_r \in \text{span}(S_{r-1})$, **discard** it: $S_r = S_{r-1}$.

The final list S_m is the **sifted list** of $(\mathbf{v}_1, \dots, \mathbf{v}_m)$.

Here $(S_{r-1}, \mathbf{v}_r)$ means the list S_{r-1} with $\mathbf{v}_r$ appended at the end. Each S_r is a **sub-list** of $(\mathbf{v}_1, \dots, \mathbf{v}_m)$: it is obtained by deleting some entries and keeping the rest in their original order.

Theorem 1.5.4 — Sifting Theorem

Let $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ be a list in a vector space V , and let S_m be its sifted list. Then S_m is linearly independent and

$$\text{span}(S_m) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_m).$$

In particular, every finite list that spans V contains a sub-list that is a basis of V .

Idea. Sifting protects two things at every step. A vector is kept only when it lies outside the span of what was kept before, so by [Lemma 1.5.3](#) the kept list stays independent. A vector is discarded only when it is already in that span, so by [Theorem 1.3.5](#) discarding it loses nothing. We turn “at every step” into an induction on r , the number of vectors read so far, with the invariant: S_r is independent and has the same span as $\mathbf{v}_1, \dots, \mathbf{v}_r$.

Proof. We prove by induction on $r \in \{0, 1, \dots, m\}$ the statement

$$P(r): \text{ the list } S_r \text{ is linearly independent and } \text{span}(S_r) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_r).$$

Base case. S_0 is the empty list, which is linearly independent, and both sides of the span equation are the span of the empty list. So $P(0)$ holds.

Inductive step. Let $1 \leq r \leq m$ and suppose $P(r-1)$ holds. Put $W = \text{span}(S_{r-1})$, so that $W = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{r-1})$ by $P(r-1)$.

Case 1: $\mathbf{v}_r \in W$. Then $S_r = S_{r-1}$, which is independent. Since $\mathbf{v}_r \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{r-1})$, [Theorem 1.3.5](#) gives $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_r) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{r-1}) = W = \text{span}(S_r)$.

Case 2: $\mathbf{v}_r \notin W$. Then $S_r = (S_{r-1}, \mathbf{v}_r)$. Since S_{r-1} is independent and $\mathbf{v}_r \notin \text{span}(S_{r-1})$, [Lemma 1.5.3](#) shows that S_r is independent. For the spans, recall from [Theorem 1.3.1](#) that the span of a list is a subspace, and is contained in every subspace containing the entries of the list. Every entry of S_r is one of $\mathbf{v}_1, \dots, \mathbf{v}_r$, so $\text{span}(S_r) \subseteq \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_r)$. Conversely, $\text{span}(S_r)$ is a subspace containing the entries of S_{r-1} , so it contains W , and with it $\mathbf{v}_1, \dots, \mathbf{v}_{r-1}$; it also contains $\mathbf{v}_r$. Hence $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_r) \subseteq \text{span}(S_r)$.

In both cases $P(r)$ holds. By induction $P(m)$ holds, which is the first statement.

Finally, suppose $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ spans V . Then S_m is a sub-list of it that is independent and satisfies $\text{span}(S_m) = V$. So S_m is a basis of V . ■

Alternatively, one can delete instead of keep. As long as the list is dependent, the Linear Dependence Lemma ([Theorem 1.4.1](#)) locates an entry lying in the span of the entries before it, and deleting that entry does not change the span. Each deletion shortens the list, so the process stops after at most m steps. Sifting performs the same idea in a single left-to-right pass.

The decision “is $\mathbf{v}_r$ in the span of the kept vectors?” is a small system of equations. For now we solve these systems by hand; Chapter 2 gives a systematic method.

Example. (Sifting a List in $\mathbb{R}^3$)

Sift the list $(\mathbf{v}_1, \dots, \mathbf{v}_6)$ in $\mathbb{R}^3$, where

$$\mathbf{v}_1 = (1, 1, 1), \quad \mathbf{v}_2 = (2, 2, 2), \quad \mathbf{v}_3 = (1, 0, 0), \quad \mathbf{v}_4 = (3, 2, 2), \quad \mathbf{v}_5 = (1, 1, 0), \quad \mathbf{v}_6 = (0, 0, 1).$$

Hence find a basis of $\mathbb{R}^3$ inside the list.

Solution. We go through the list once, comparing each vector with the vectors kept so far.

- $\mathbf{v}_1$: the kept list is empty, with span $\{\mathbf{0}\}$. Since $\mathbf{v}_1 \neq \mathbf{0}$, **keep** $\mathbf{v}_1$.
- $\mathbf{v}_2$: $\mathbf{v}_2 = 2\mathbf{v}_1 \in \text{span}(\mathbf{v}_1)$. **Discard**.
- $\mathbf{v}_3$: if $(1, 0, 0) = a(1, 1, 1) = (a, a, a)$, the first entry gives $a = 1$ and the second gives $a = 0$, a contradiction. So $\mathbf{v}_3 \notin \text{span}(\mathbf{v}_1)$. **Keep**.
- $\mathbf{v}_4$: $2\mathbf{v}_1 + \mathbf{v}_3 = (2, 2, 2) + (1, 0, 0) = (3, 2, 2) = \mathbf{v}_4$, so $\mathbf{v}_4 \in \text{span}(\mathbf{v}_1, \mathbf{v}_3)$. **Discard**.
- $\mathbf{v}_5$: suppose $(1, 1, 0) = a(1, 1, 1) + b(1, 0, 0) = (a + b, a, a)$. The second entry gives $a = 1$ and the third gives $a = 0$, a contradiction. So $\mathbf{v}_5 \notin \text{span}(\mathbf{v}_1, \mathbf{v}_3)$. **Keep**.
- $\mathbf{v}_6$: $\mathbf{v}_1 - \mathbf{v}_5 = (1, 1, 1) - (1, 1, 0) = (0, 0, 1) = \mathbf{v}_6$, so $\mathbf{v}_6 \in \text{span}(\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_5)$. **Discard**.

The sifted list is $(\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_5) = ((1, 1, 1), (1, 0, 0), (1, 1, 0))$. By [Theorem 1.5.4](#) it is independent and has the same span as the original list. That span is all of $\mathbb{R}^3$, since it contains $\mathbf{e}_1 = \mathbf{v}_3$, $\mathbf{e}_2 = \mathbf{v}_5 - \mathbf{v}_3$ and $\mathbf{e}_3 = \mathbf{v}_6$, and hence every combination of them. So $(\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_5)$ is a basis of $\mathbb{R}^3$.

Sifting has no memory of the future, so its output depends on the order of the input.

Example. (The Order Matters)

Sift $((1, 0), (0, 1), (1, 1))$ and $((1, 1), (1, 0), (0, 1))$ in $\mathbb{R}^2$.

Solution. For the first list: $(1, 0) \neq \mathbf{0}$, so keep it. Every multiple of $(1, 0)$ has second entry 0, so $(0, 1) \notin \text{span}((1, 0))$; keep it. Then $(1, 1) = (1, 0) + (0, 1)$; discard it. The result is $((1, 0), (0, 1))$.

For the second list: $(1, 1) \neq \mathbf{0}$, so keep it. If $(1, 0) = a(1, 1) = (a, a)$, then $a = 1$ and $a = 0$, which is impossible; keep $(1, 0)$. Then $(0, 1) = (1, 1) - (1, 0)$; discard it. The result is $((1, 1), (1, 0))$.

Both results are bases of $\mathbb{R}^2$ (the lists span $\mathbb{R}^2$ because they contain $\mathbf{e}_1$ and $\mathbf{e}_2$), but they are different bases. A zero vector, by contrast, is discarded wherever it stands, because $\mathbf{0}$ lies in every span.

Quick check. Sift the list $(x, 2x, 1 + x, 1)$ in $\mathbb{R}[x]_{\leq 1}$. Which vectors are kept?

Solution. Keep x , since $x \neq 0$. Discard $2x = 2 \cdot x$. Keep $1 + x$: every multiple ax has constant term 0, but $1 + x$ has constant term 1. Discard $1 = (1 + x) - x$. The sifted list is $(x, 1 + x)$.

The theorem answers the question we started with.

Corollary 1.5.5 — Existence of Bases

Every finite-dimensional vector space has a basis.

Proof. Let V be finite-dimensional. By [Definition 1.3.4](#), V is spanned by some finite list. By [Theorem 1.5.4](#), that list contains a sub-list that is a basis of V . ■

For $V = \{\mathbf{0}\}$ the proof still works: $\{\mathbf{0}\}$ is spanned by the empty list, and sifting it returns the

empty list, which is the basis found in [Example \(Three More Bases\)](#).

We now know that bases exist, and we can find one inside any finite spanning list. But a space has many bases. The next section proves the fact that makes them measurable: all bases of the same space have the same length.

1.5.5 Exercises

1.5.5.1 A. Check your understanding

Exercise A1.

- (a) Define what it means for a list $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ to be a basis of a vector space V over F .
- (b) True or false: every list that spans $\mathbb{R}^2$ is a basis of $\mathbb{R}^2$. Justify your answer.
- (c) True or false: $((1, 0), (0, 1))$ and $((0, 1), (1, 0))$ are the same basis of $\mathbb{R}^2$. Justify your answer.
- (d) What is a basis of the zero space $\{\mathbf{0}\}$?
- (e) When sifting a list, under what condition is the vector $\mathbf{v}_r$ discarded?
- (f) True or false: if $\mathcal{B}$ is a basis of V and $[\mathbf{u}]_{\mathcal{B}} = [\mathbf{v}]_{\mathcal{B}}$, then $\mathbf{u} = \mathbf{v}$. Justify your answer.

Solution.

- (a) See [Definition 1.5.1](#): the list is linearly independent and spans V .
- (b) False. $((1, 0), (0, 1), (1, 1))$ spans $\mathbb{R}^2$, but it is dependent, since $(1, 0) + (0, 1) - (1, 1) = \mathbf{0}$.
- (c) False. A basis is a list, and the two lists have their entries in different orders. The difference is visible in coordinates: $(3, 1)$ has coordinate vector $(3, 1)$ in the first basis and $(1, 3)$ in the second.
- (d) The empty list $()$, by [Example \(Three More Bases\)](#) (c).
- (e) When $\mathbf{v}_r$ lies in the span of the vectors kept before it, that is, $\mathbf{v}_r \in \text{span}(S_{r-1})$.
- (f) True. If both coordinate vectors equal $(a_1, \dots, a_n)$, then $\mathbf{u} = a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n = \mathbf{v}$ by [Definition 1.5.2](#), where $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$.

1.5.5.2 B. Practice

Exercise B1: A Basis of $\mathbb{R}^3$ and Coordinates.

Let $\mathcal{B} = ((1, 1, 0), (0, 1, 1), (1, 0, 1))$. Show that $\mathcal{B}$ is a basis of $\mathbb{R}^3$. Hence find $[\mathbf{v}]_{\mathcal{B}}$ for $\mathbf{v} = (2, 3, 5)$.

Solution. Let $(p, q, r) \in \mathbb{R}^3$. For scalars a, b, c ,

$$a(1, 1, 0) + b(0, 1, 1) + c(1, 0, 1) = (a + c, a + b, b + c),$$

so this equals (p, q, r) if and only if $a + c = p$, $a + b = q$ and $b + c = r$. Adding the three

equations gives $2(a + b + c) = p + q + r$, so $a + b + c = \frac{p+q+r}{2}$. Subtracting each equation from this one,

$$b = \frac{-p + q + r}{2}, \quad c = \frac{p - q + r}{2}, \quad a = \frac{p + q - r}{2}.$$

Conversely, these values satisfy the three equations: for instance $a + c = \frac{2p}{2} = p$, and the other two are checked the same way. So every vector of $\mathbb{R}^3$ has exactly one expression in $\mathcal{B}$, and $\mathcal{B}$ is a basis by [Theorem 1.5.1](#).

For $(p, q, r) = (2, 3, 5)$: $a = \frac{2+3-5}{2} = 0$, $b = \frac{-2+3+5}{2} = 3$, $c = \frac{2-3+5}{2} = 2$. Check: $0(1, 1, 0) + 3(0, 1, 1) + 2(1, 0, 1) = (2, 3, 5)$. Hence $[\mathbf{v}]_{\mathcal{B}} = (0, 3, 2)$.

Exercise B2: Sifting Polynomials.

Sift the list $(1 + x, 2 + 2x, x^2, 1 + x + x^2, x, 1)$ in $\mathbb{R}[x]_{\leq 2}$, justifying each decision. Hence find a basis of $\mathbb{R}[x]_{\leq 2}$ contained in the list.

Solution. We read the list from left to right, comparing coefficients.

- $1 + x \neq 0$: **keep**.
- $2 + 2x = 2(1 + x)$: **discard**.
- x^2 : every multiple $a(1 + x)$ has x^2 -coefficient 0, while x^2 has x^2 -coefficient 1. So $x^2 \notin \text{span}(1 + x)$: **keep**.
- $1 + x + x^2 = (1 + x) + x^2$: **discard**.
- x : suppose $x = a(1 + x) + bx^2 = a + ax + bx^2$. The constant terms give $a = 0$ and the x -coefficients give $a = 1$, a contradiction. So $x \notin \text{span}(1 + x, x^2)$: **keep**.
- $1 = (1 + x) - x$: **discard**.

The sifted list is $(1 + x, x^2, x)$. The original list contains $1, x$ and x^2 , so its span contains every combination of them, which is all of $\mathbb{R}[x]_{\leq 2}$ by [Example \(Standard Bases\)](#). Hence the list spans $\mathbb{R}[x]_{\leq 2}$, and by [Theorem 1.5.4](#) the sifted list $(1 + x, x^2, x)$ is a basis of $\mathbb{R}[x]_{\leq 2}$.

Exercise B3: Symmetric Matrices.

Let $U = \{A \in M_2(\mathbb{R}) : A^T = A\}$. Show that $(E_{11}, E_{12} + E_{21}, E_{22})$ is a basis of U , and find the coordinate vector of $\begin{pmatrix} 2 & -1 \\ -1 & 5 \end{pmatrix}$ with respect to it.

Solution. Each of the three matrices is symmetric, so the list lies in U .

Spanning. Let $A \in U$. Writing $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the condition $A^T = A$ says $c = b$. Hence

$$A = \begin{pmatrix} a & b \\ b & d \end{pmatrix} = aE_{11} + b(E_{12} + E_{21}) + dE_{22}.$$

Independence. Let $aE_{11} + b(E_{12} + E_{21}) + dE_{22} = 0$. The left side is $\begin{pmatrix} a & b \\ b & d \end{pmatrix}$, so comparing entries with the zero matrix gives $a = b = d = 0$.

Hence the list is a basis of U . Reading off $a = 2$, $b = -1$, $d = 5$, the coordinate vector is $(2, -1, 5)$.

1.5.5.3 C. Going deeper

Exercise C1: Polynomials Vanishing at 1.

Let F be a field, $n \geq 1$, and $U = \{p \in F[x]_{\leq n} : p(1) = 0\}$.

- (a) Prove that $(x - 1, x^2 - 1, \dots, x^n - 1)$ is a basis of U .
- (b) For $n = 3$, find the coordinate vector of $x^3 - 2x + 1$ with respect to this basis.

Hint: for $p = a_0 + a_1x + \dots + a_nx^n$, what does $p(1) = 0$ say about a_0 ?

Solution.

- (a) For $p = a_0 + a_1x + \dots + a_nx^n$, evaluation gives $p(1) = a_0 + a_1 + \dots + a_n$, since every power of 1 is 1. In particular each $x^k - 1$ lies in U , because $1 - 1 = 0$.

Spanning. Let $p \in U$. Then $a_0 = -(a_1 + \dots + a_n)$, so

$$p = a_1x + \dots + a_nx^n - (a_1 + \dots + a_n) = a_1(x - 1) + a_2(x^2 - 1) + \dots + a_n(x^n - 1).$$

Independence. Let $c_1(x - 1) + \dots + c_n(x^n - 1) = 0$. For $1 \leq k \leq n$, the coefficient of x^k on the left is c_k , since x^k appears in no other term. The zero polynomial has all coefficients 0, so $c_k = 0$ for every k .

Hence the list is a basis of U .

- (b) For $p = x^3 - 2x + 1$ we have $p(1) = 0$, and $(a_1, a_2, a_3) = (-2, 0, 1)$. By the formula in (a), $p = -2(x - 1) + 0(x^2 - 1) + 1(x^3 - 1)$. Check: $-2x + 2 + x^3 - 1 = x^3 - 2x + 1$. The coordinate vector is $(-2, 0, 1)$.

Exercise C2: The Same List over a Different Field.

- (a) Explain why $((1, 1), (1, -1))$ is **not** a basis of $\mathbb{F}_2^2$.
- (b) Let F be any field. Prove that $((1, 1), (1, -1))$ is a basis of F^2 if and only if $1 + 1 \neq 0$ in F .

Solution.

- (a) In $\mathbb{F}_2$ we have $-1 = 1$, so the list is $((1, 1), (1, 1))$. It is dependent: $1 \cdot (1, 1) + 1 \cdot (1, 1) = (1 + 1, 1 + 1) = (0, 0)$ with non-zero coefficients. (It also fails to span: every combination is (c, c) , so $(1, 0)$ is out of reach.)

- (b) Write $2 = 1 + 1 \in F$.

($\Leftarrow$) Suppose $2 \neq 0$, so 2^{-1} exists. Let $(x, y) \in F^2$. The equation $a(1, 1) + b(1, -1) = (x, y)$ says $a + b = x$ and $a - b = y$. If it holds, adding and subtracting gives $2a = x + y$ and $2b = x - y$, so $a = 2^{-1}(x + y)$ and $b = 2^{-1}(x - y)$. Conversely these values satisfy $a + b = 2^{-1} \cdot 2x = x$ and $a - b = 2^{-1} \cdot 2y = y$. So every vector has exactly one expression, and the list is a basis by [Theorem 1.5.1](#).

($\Rightarrow$) Suppose $2 = 0$. Then $1 = -1$, and the argument of (a) applies word for word:

$1 \cdot (1, 1) + 1 \cdot (1, -1) = (2, 0) = (0, 0)$, so the list is dependent and is not a basis. This proves $(\Rightarrow)$ by contraposition.

Exercise C3: Rescaling a Basis.

Let $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of V over F , and let $c_1, \dots, c_n \in F$ be **non-zero**.

- (a) Prove that $\mathcal{B}' = (c_1\mathbf{v}_1, \dots, c_n\mathbf{v}_n)$ is a basis of V .
- (b) Express $[\mathbf{v}]_{\mathcal{B}'}$ in terms of $[\mathbf{v}]_{\mathcal{B}}$.
- (c) Explain why the conclusion of (a) can fail if some $c_i = 0$.

Solution.

- (a) Let $\mathbf{v} \in V$ and $[\mathbf{v}]_{\mathcal{B}} = (a_1, \dots, a_n)$. Since each $c_i \neq 0$, c_i^{-1} exists, and

$$\mathbf{v} = \sum_{i=1}^n a_i \mathbf{v}_i = \sum_{i=1}^n (a_i c_i^{-1})(c_i \mathbf{v}_i).$$

So $\mathbf{v}$ has an expression in $\mathcal{B}'$. Suppose $\mathbf{v} = \sum_i b_i (c_i \mathbf{v}_i) = \sum_i (b_i c_i) \mathbf{v}_i$. By uniqueness of the expression in $\mathcal{B}$ (Theorem 1.5.1), $b_i c_i = a_i$ for each i , so $b_i = a_i c_i^{-1}$. Hence the expression in $\mathcal{B}'$ is unique, and $\mathcal{B}'$ is a basis by Theorem 1.5.1.

- (b) By (a), $[\mathbf{v}]_{\mathcal{B}'} = (a_1 c_1^{-1}, \dots, a_n c_n^{-1})$: divide the i -th coordinate by c_i .
- (c) If $c_1 = 0$, then $\mathcal{B}'$ contains $0\mathbf{v}_1 = \mathbf{0}$, and $1 \cdot \mathbf{0} + 0 \cdot c_2 \mathbf{v}_2 + \dots + 0 \cdot c_n \mathbf{v}_n = \mathbf{0}$ is a combination with a non-zero coefficient. So $\mathcal{B}'$ is dependent and is not a basis.

1.6 The Exchange Theorem and Dimension

Every finite-dimensional space has a basis, and usually a great many of them. This section proves that they all have the same length, so that the length is a property of the space and not of the basis we happened to pick. That number is the dimension. The key step is the Steinitz Exchange Theorem, the most important proof of the chapter. With it we build the everyday tools of dimension counting: size bounds, “count instead of check”, extension of independent lists to bases, and dimensions of subspaces.

1.6.1 How big is a vector space?

We would like to say that a plane is smaller than space, and that $M_2(\mathbb{R})$ and $\mathbb{R}[x]_{\leq 3}$ are, in some sense, the same size. The naive measure is the number of elements. It fails at once: $\mathbb{R}^2$, $M_2(\mathbb{R})$ and $\mathbb{R}[x]_{\leq 3}$ are all infinite sets, and counting elements cannot tell them apart.

A better measure counts **how many vectors it takes to build everything, without waste**, which is the length of a basis. The standard bases give

$$\mathbb{R}^2 : 2, \quad M_2(\mathbb{R}) : 4 \quad (E_{11}, E_{12}, E_{21}, E_{22}), \quad \mathbb{R}[x]_{\leq 3} : 4 \quad (1, x, x^2, x^3).$$

This matches intuition: a matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and a polynomial $a + bx + cx^2 + dx^3$ are both described

by four free numbers.

But there is a gap. A space has many bases. We counted the standard basis of $\mathbb{R}^2$; the basis $((1, 1), (1, -1))$ also has length 2, but what guarantees that no basis of $\mathbb{R}^2$ has length 3? If some space had bases of two different lengths, “the length of a basis” would be meaningless. So before we can define anything, we need the following.

Goal. Any two bases of a finite-dimensional vector space have the same length.

A basis is independent **and** spanning, and the goal follows from a single inequality about the two properties separately: *an independent list is never longer than a spanning list*. Granting this, take two bases $\mathcal{B}$ and $\mathcal{C}$. Since $\mathcal{B}$ is independent and $\mathcal{C}$ spans, $\mathcal{B}$ is not longer than $\mathcal{C}$. Swapping the roles, $\mathcal{C}$ is not longer than $\mathcal{B}$. So they have the same length. Everything rests on the inequality.

1.6.2 The Steinitz Exchange Theorem

Why should an independent list be no longer than a spanning list? In $\mathbb{R}^2$ the spanning list $(\mathbf{e}_1, \mathbf{e}_2)$ has length 2, and three vectors in the plane always satisfy a linear relation, but “the plane is two-dimensional” is what we are trying to define, so we cannot use it. The idea is instead to feed the independent vectors into the spanning list one at a time, pushing out one spanning vector each time, while keeping the list spanning. If the independent vectors outnumbered the spanning ones, we would run out of vectors to push out.

Theorem 1.6.1 — Steinitz Exchange Theorem

Let V be a vector space over F . Suppose that $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ is a **linearly independent** list in V and that $(\mathbf{w}_1, \dots, \mathbf{w}_n)$ is a list that **spans** V . Then

- (a) $m \leq n$, and
- (b) after renumbering $\mathbf{w}_1, \dots, \mathbf{w}_n$ suitably, the list $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_{m+1}, \dots, \mathbf{w}_n)$ spans V .

In words: an independent list is at most as long as a spanning list, and the independent vectors can **replace** m of the spanning vectors without losing the spanning property. “Renumbering” means listing the $\mathbf{w}$ ’s in a different order; which $\mathbf{w}$ ’s get replaced depends on the $\mathbf{u}$ ’s.

Idea. Picture a row of n slots holding $\mathbf{w}_1, \dots, \mathbf{w}_n$; the row spans V . We push $\mathbf{u}_1$ into the row and push one $\mathbf{w}$ out, then $\mathbf{u}_2$, and so on, keeping the row spanning after every exchange. Here is the picture for $m = 3$, $n = 4$, with the $\mathbf{w}$ ’s numbered in the order they leave.

after 0	$\mathbf{w}_1$	$\mathbf{w}_2$	$\mathbf{w}_3$	$\mathbf{w}_4$	spans V
after 1	$\mathbf{u}_1$	$\mathbf{w}_2$	$\mathbf{w}_3$	$\mathbf{w}_4$	$\mathbf{u}_1$ in, $\mathbf{w}_1$ out
after 2	$\mathbf{u}_1$	$\mathbf{u}_2$	$\mathbf{w}_3$	$\mathbf{w}_4$	$\mathbf{u}_2$ in, $\mathbf{w}_2$ out
after 3	$\mathbf{u}_1$	$\mathbf{u}_2$	$\mathbf{u}_3$	$\mathbf{w}_4$	$\mathbf{u}_3$ in, $\mathbf{w}_3$ out
every row spans V					

Which $\mathbf{w}$ may leave? Write the incoming $\mathbf{u}_{k+1}$ as a combination of the current row. Any $\mathbf{w}$ with a **non-zero** coefficient can be solved for, so it lies in the span of the other entries together with $\mathbf{u}_{k+1}$, and removing it costs nothing. Such a $\mathbf{w}$ must exist: otherwise $\mathbf{u}_{k+1}$ would be a combination of $\mathbf{u}_1, \dots, \mathbf{u}_k$ alone, which independence forbids. In particular, while $\mathbf{u}$'s remain to be inserted, a $\mathbf{w}$ remains to be removed, and that is why $m \leq n$.

Step roadmap. Induction on the number k of exchanges, with the invariant

$P(k)$: $k \leq n$, and after renumbering the $\mathbf{w}$'s, the list $(\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_{k+1}, \dots, \mathbf{w}_n)$ spans V .

Step 1 is the base case $k = 0$. **Step 2** writes $\mathbf{u}_{k+1}$ in the current list. **Step 3** is the Claim that some remaining $\mathbf{w}$ has a non-zero coefficient. **Step 4** exchanges that $\mathbf{w}$ for $\mathbf{u}_{k+1}$ and checks the span.

Proof. We prove $P(k)$ for $k = 0, 1, \dots, m$ by induction on k . The case $k = m$ is the theorem.

Step 1 (base case). For $k = 0$ the list in $P(0)$ is $(\mathbf{w}_1, \dots, \mathbf{w}_n)$, which spans V by hypothesis, and $0 \leq n$. So $P(0)$ holds.

Step 2. Let $0 \leq k < m$ and suppose $P(k)$ holds. Renumber the $\mathbf{w}$'s as $P(k)$ allows, so that

$$L_k = (\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_{k+1}, \dots, \mathbf{w}_n)$$

spans V . Since $k < m$, the vector $\mathbf{u}_{k+1}$ exists, and $\mathbf{u}_{k+1} \in V = \text{span}(L_k)$. Hence there are scalars $a_1, \dots, a_k, b_{k+1}, \dots, b_n \in F$ with

$$\mathbf{u}_{k+1} = a_1 \mathbf{u}_1 + \dots + a_k \mathbf{u}_k + b_{k+1} \mathbf{w}_{k+1} + \dots + b_n \mathbf{w}_n. \quad (*)$$

(If $k = n$, the list L_k contains no $\mathbf{w}$'s and the second group of terms is empty.)

Step 3.

Claim. We have $k + 1 \leq n$, and $b_j \neq 0$ for some j with $k + 1 \leq j \leq n$.

Proof. Suppose not. Then either $k = n$, so that there are no b 's at all, or $k < n$ and $b_{k+1} = \cdots = b_n = 0$. In both cases $(*)$ reads $\mathbf{u}_{k+1} = a_1\mathbf{u}_1 + \cdots + a_k\mathbf{u}_k$. Therefore

$$a_1\mathbf{u}_1 + \cdots + a_k\mathbf{u}_k + (-1)\mathbf{u}_{k+1} + 0\mathbf{u}_{k+2} + \cdots + 0\mathbf{u}_m = \mathbf{0}$$

is a linear relation among $\mathbf{u}_1, \dots, \mathbf{u}_m$ in which the coefficient of $\mathbf{u}_{k+1}$ is $-1 \neq 0$. This contradicts the linear independence of $(\mathbf{u}_1, \dots, \mathbf{u}_m)$. Hence there is at least one $\mathbf{w}$ in L_k , so $k+1 \leq n$, and some b_j with $j \geq k+1$ is non-zero. ■

Step 4. Renumber $\mathbf{w}_{k+1}, \dots, \mathbf{w}_n$ among themselves so that $b_{k+1} \neq 0$. This only changes the order of entries of L_k , so L_k still spans V (the span of a list does not depend on the order of its entries), and $(*)$ still holds with the coefficients renumbered in the same way. Put

$$L_{k+1} = (\mathbf{u}_1, \dots, \mathbf{u}_{k+1}, \mathbf{w}_{k+2}, \dots, \mathbf{w}_n).$$

Since $b_{k+1} \neq 0$, the scalar b_{k+1}^{-1} exists, and solving $(*)$ for $\mathbf{w}_{k+1}$ gives

$$\mathbf{w}_{k+1} = b_{k+1}^{-1}(\mathbf{u}_{k+1} - a_1\mathbf{u}_1 - \cdots - a_k\mathbf{u}_k - b_{k+2}\mathbf{w}_{k+2} - \cdots - b_n\mathbf{w}_n).$$

The right side is a combination of the entries of L_{k+1} , so $\mathbf{w}_{k+1} \in \text{span}(L_{k+1})$. By [Theorem 1.3.5](#), inserting $\mathbf{w}_{k+1}$ into L_{k+1} does not change the span:

$$\text{span}(L_{k+1}) = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{k+1}, \mathbf{w}_{k+1}, \mathbf{w}_{k+2}, \dots, \mathbf{w}_n).$$

The list on the right contains every entry of L_k , so its span is a subspace containing all entries of L_k , and hence contains $\text{span}(L_k) = V$ by [Theorem 1.3.1](#). Therefore $\text{span}(L_{k+1}) = V$. Together with $k+1 \leq n$ from the Claim, this is $P(k+1)$.

By induction, $P(m)$ holds: $m \leq n$, and after renumbering the $\mathbf{w}$'s, $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_{m+1}, \dots, \mathbf{w}_n)$ spans V . This proves the theorem. ■

Part (a) is the inequality our goal needed. Part (b), the exchange itself, is a bonus that we will spend later in this section, to extend independent lists to bases.

Warning. The $\mathbf{w}$ that leaves must have a **non-zero** coefficient; an arbitrary $\mathbf{w}$ will not do. In $\mathbb{R}^2$, take the spanning list $(\mathbf{w}_1, \mathbf{w}_2) = ((1, 0), (0, 1))$ and the independent vector $\mathbf{u}_1 = (1, 0) = 1\mathbf{w}_1 + 0\mathbf{w}_2$. Exchanging $\mathbf{u}_1$ for $\mathbf{w}_1$ gives $((1, 0), (0, 1))$, which spans. Exchanging $\mathbf{u}_1$ for $\mathbf{w}_2$, whose coefficient is 0, gives $((1, 0), (1, 0))$, which does not reach $(0, 1)$.

The theorem already settles questions that would otherwise need a computation.

Quick check. Can the four polynomials $1 + x$, $x + x^2$, $x^2 + 1$, $1 + x + x^2$ be linearly independent in $\mathbb{R}[x]_{\leq 2}$?

Solution. No. The list $(1, x, x^2)$ spans $\mathbb{R}[x]_{\leq 2}$ and has length 3, so by [Theorem 1.6.1](#)

every independent list there has length at most $3 < 4$. Indeed $(1+x) + (x+x^2) + (x^2+1) - 2(1+x+x^2) = 0$.

1.6.3 Dimension

The goal of the section is now one line away.

Theorem 1.6.2 — Basis Theorem

Let V be a finite-dimensional vector space. Then V has a basis, and any two bases of V have the same length.

Proof. By Corollary 1.5.5, V has a basis. Let $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ and $(\mathbf{w}_1, \dots, \mathbf{w}_m)$ be bases of V . The first is linearly independent and the second spans V , so $n \leq m$ by Theorem 1.6.1. Swapping the roles of the two bases gives $m \leq n$. Hence $m = n$. ■

We started from a question: how big is a vector space? The Basis Theorem says that one number, the length of a basis, answers it without depending on any choice. So we give that number a name.

The dimension of a space is the number of vectors in any basis: how many independent directions the space has.

Definition 1.6.1 — Dimension

Let V be a finite-dimensional vector space over F . The **dimension** of V over F , written $\dim_F V$ or simply $\dim V$, is the length of **any** basis of V .

In words: pick a basis of V , any basis at all, and count its vectors. The dimension is defined only for finite-dimensional spaces (Definition 1.3.4); a space spanned by no finite list is infinite-dimensional, and we do not assign it a number here. The subscript F records the field of scalars, which, as we will see, matters.

Well-definedness. Two things could go wrong with “the length of any basis”: there might be no basis to count, or different bases might give different counts. The first is ruled out by Corollary 1.5.5 and the second by Theorem 1.6.2. That is exactly what the word **any** in the definition needs.

Example. (Dimensions of the Running Examples)

Find the dimension of each space.

- (a) F^n , $M_{m \times n}(F)$ and $F[x]_{\leq n}$.
- (b) The zero space $\{\mathbf{0}\}$.
- (c) $\mathbb{C}$ over $\mathbb{C}$, and $\mathbb{C}$ over $\mathbb{R}$.
- (d) $U = \{A \in M_2(\mathbb{R}) : A^T = A\}$, the symmetric 2×2 real matrices.

(e) $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ over $\mathbb{Q}$ (Example ($\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$)).

Solution. In each case we exhibit one basis and count it.

- (a) The standard bases of Example (Standard Bases) have lengths n , mn and $n + 1$. So $\dim F^n = n$, $\dim M_{m \times n}(F) = mn$, and $\dim F[x]_{\leq n} = n + 1$.
- (b) The empty list is a basis of $\{\mathbf{0}\}$ (Example (Three More Bases)), so $\dim\{\mathbf{0}\} = 0$.
- (c) Over $\mathbb{C}$, the list (1) is a basis of $\mathbb{C}$: every z equals $z \cdot 1$, and $z \cdot 1 = 0$ forces $z = 0$. So $\dim_{\mathbb{C}} \mathbb{C} = 1$. Over $\mathbb{R}$, the list $(1, i)$ is a basis (Example (Three More Bases)), so $\dim_{\mathbb{R}} \mathbb{C} = 2$.
- (d) A symmetric matrix has the form $\begin{pmatrix} a & b \\ b & d \end{pmatrix} = aE_{11} + b(E_{12} + E_{21}) + dE_{22}$, and this combination is the zero matrix only when $a = b = d = 0$. So $(E_{11}, E_{12} + E_{21}, E_{22})$ is a basis of U , as in Exercise (B3: Symmetric Matrices), and $\dim U = 3$.
- (e) Every element is $a \cdot 1 + b\sqrt{2}$ with $a, b \in \mathbb{Q}$, so $(1, \sqrt{2})$ spans. Let $a + b\sqrt{2} = 0$ with $a, b \in \mathbb{Q}$. If $b \neq 0$, then $\sqrt{2} = -a/b \in \mathbb{Q}$, contradicting Theorem 0.2.3. So $b = 0$, and then $a = 0$. Hence $(1, \sqrt{2})$ is a basis, and $\dim_{\mathbb{Q}} \mathbb{Q}(\sqrt{2}) = 2$.

The zero space in (b) is the degenerate case: it has dimension 0 because its basis is empty, not because it has no elements. It has one element, $\mathbf{0}$.

Non-example by minimal change. Remove the degree bound from $F[x]_{\leq n}$. The space $F[x]$ is not spanned by any finite list (Theorem 1.3.3), so it has no finite basis and no dimension in the sense of Definition 1.6.1. The hypothesis “finite-dimensional” in the Basis Theorem is exactly what fails.

Why this definition. Why a basis, and not some other list? Spanning lists can be made as long as we like by repeating vectors, and independent lists can be made as short as we like by deleting vectors, so neither length alone is an invariant. A basis sits exactly where the two meet: it is the longest possible independent list and the shortest possible spanning list, as we prove next. And why not count elements? Over $\mathbb{R}$ every non-zero space is infinite, so the count says nothing. Over a finite field the count does carry information, but it is determined by the dimension; see Exercise (C3: Counting Elements over a Finite Field).

Warning. The dimension of $F[x]_{\leq n}$ is $n + 1$, **not** n . The basis $(1, x, \dots, x^n)$ has one vector for each exponent $0, 1, \dots, n$, and the constant 1 is easy to forget. For instance $\dim \mathbb{R}[x]_{\leq 2} = 3$.

Warning. Dimension depends on the field. The set $\mathbb{C}$ has dimension 1 over $\mathbb{C}$ but dimension 2 over $\mathbb{R}$, because over $\mathbb{R}$ the scalar i is no longer available and i must be counted as a separate basis vector. When more than one field is in sight, write $\dim_F$.

Quick check. What are $\dim M_{2 \times 3}(F)$ and $\dim F[x]_{\leq 5}$? Which of them equals $\dim F^6$?

Solution. $\dim M_{2 \times 3}(F) = 2 \cdot 3 = 6$ and $\dim F[x]_{\leq 5} = 5 + 1 = 6$. Both equal $\dim F^6 = 6$.

1.6.4 Size bounds

The first payoff turns the Exchange Theorem into statements about the number $\dim V$.

Theorem 1.6.3 — Size Bounds

Let V be a vector space over F with $\dim V = n$.

- (a) Every linearly independent list in V has length at most n . Hence every list in V of length **greater than** n is linearly dependent.
- (b) Every list that spans V has length at least n . Hence no list of length **less than** n spans V .

Proof. Let $\mathcal{B}$ be a basis of V ; it has length n .

- (a) Let $\mathcal{L}$ be an independent list of length ℓ . Since $\mathcal{B}$ spans V , [Theorem 1.6.1](#) gives $\ell \leq n$. The second sentence is the contrapositive.
- (b) Let $\mathcal{L}$ be a list of length ℓ that spans V . Since $\mathcal{B}$ is independent, [Theorem 1.6.1](#) gives $n \leq \ell$. The second sentence is the contrapositive.

■

So in a space of dimension n ,

$$\text{length of an independent list} \leq n \leq \text{length of a spanning list}.$$

This settles many questions with no computation. The four vectors $(1, 2, 3), (4, 5, 6), (7, 8, 9), (1, 0, 0)$ in $\mathbb{R}^3$ are dependent, because $4 > 3$. Three matrices never span $M_2(F)$, because $3 < 4$.

1.6.5 Count instead of check

To show a list is a basis we normally check two things, independence and spanning. When the length of the list equals the dimension, one check is enough. This is one of the most used facts in the book.

Theorem 1.6.4 — Count Instead of Check

Let V be a vector space with $\dim V = n$, and let $\mathcal{L} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a list in V of length exactly n .

- (a) If $\mathcal{L}$ is linearly independent, then $\mathcal{L}$ is a basis of V .
- (b) If $\mathcal{L}$ spans V , then $\mathcal{L}$ is a basis of V .

Idea. For (a), the exchange part of [Theorem 1.6.1](#) does the work: we can push all n vectors of $\mathcal{L}$ into a basis of length n , and then no old vectors are left, so $\mathcal{L}$ alone spans. For (b), sift $\mathcal{L}$: the result is a basis, so it has length n , so nothing was discarded.

Proof.

- (a) Let $(\mathbf{w}_1, \dots, \mathbf{w}_n)$ be a basis of V ; in particular it spans V . Since $\mathcal{L}$ is independent of length n , [Theorem 1.6.1](#) (b) with $m = n$ says that, after renumbering the $\mathbf{w}$'s, the list $(\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{w}_{n+1}, \dots, \mathbf{w}_n)$ spans V . This list contains no $\mathbf{w}$'s: it is $\mathcal{L}$. So $\mathcal{L}$ is independent and spans V , and it is a basis.

- (b) By [Theorem 1.5.4](#), $\mathcal{L}$ contains a sub-list that is a basis of V . By [Theorem 1.6.2](#), that sub-list has length $\dim V = n$. The only sub-list of length n of a list of length n is the list itself. Hence $\mathcal{L}$ is a basis. ■

The theorem lets us choose the easier check, and independence is almost always easier: it is a single equation equal to $\mathbf{0}$, while spanning asks us to solve for an arbitrary vector.

Example. (A Basis by Counting)

Show that $(1, 1+x, (1+x)^2)$ is a basis of $\mathbb{R}[x]_{\leq 2}$.

Solution. By [Example \(Dimensions of the Running Examples\)](#), $\dim \mathbb{R}[x]_{\leq 2} = 3$, and the list has length 3. So by [Theorem 1.6.4](#) (a) it suffices to check independence. Let

$$a \cdot 1 + b(1+x) + c(1+x)^2 = 0.$$

Expanding, the left side is $(a+b+c) + (b+2c)x + cx^2$. Comparing coefficients with the zero polynomial: $c = 0$; then $b+2c = 0$ gives $b = 0$; then $a+b+c = 0$ gives $a = 0$. Hence the list is independent, with $3 = \dim \mathbb{R}[x]_{\leq 2}$ elements, and therefore it is a basis.

We never had to show that every polynomial of degree at most 2 is a combination of $1, 1+x, (1+x)^2$. The dimension count did it for us.

1.6.6 Extending to a basis

Sifting finds a basis **inside** a spanning list. The opposite direction starts from an independent list and grows it into a basis. This is the tool behind every dimension formula later in the book: start from a basis of the smallest space, then extend.

Theorem 1.6.5 — Basis Extension Theorem

Let V be a finite-dimensional vector space with $\dim V = n$, and let $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ be a linearly independent list in V . Then $m \leq n$, and there are vectors $\mathbf{v}_{m+1}, \dots, \mathbf{v}_n \in V$ such that

$$(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_{m+1}, \dots, \mathbf{v}_n)$$

is a basis of V . Moreover, the $\mathbf{v}$'s can be chosen from any given basis of V .

Proof. Let $(\mathbf{w}_1, \dots, \mathbf{w}_n)$ be a basis of V . It spans V , so by [Theorem 1.6.1](#), $m \leq n$, and after renumbering the $\mathbf{w}$'s the list $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_{m+1}, \dots, \mathbf{w}_n)$ spans V . This list has length $m + (n - m) = n = \dim V$, so it is a basis by [Theorem 1.6.4](#) (b). Take $\mathbf{v}_j = \mathbf{w}_j$ for $m < j \leq n$. ■

Alternatively, sift the list $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_n)$. It spans V , and sifting keeps every $\mathbf{u}_k$: the sub-list $(\mathbf{u}_1, \dots, \mathbf{u}_k)$ is independent ([Exercise \(B3: Sub-Lists\)](#)), so $\mathbf{u}_k \notin \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{k-1})$ by

Lemma 1.5.3. Sifting has the advantage of telling us **which** $\mathbf{w}$'s to add.

Example. (Extending an Independent List)

Extend the list $((1, 1, 0), (0, 1, 1))$ to a basis of $\mathbb{R}^3$.

Solution. The list is independent: $a(1, 1, 0) + b(0, 1, 1) = (a, a+b, b) = \mathbf{0}$ forces $a = b = 0$. By [Theorem 1.6.5](#), one vector from the standard basis will do, so we try $\mathbf{e}_1$. Let

$$a(1, 1, 0) + b(0, 1, 1) + c(1, 0, 0) = (a + c, a + b, b) = (0, 0, 0).$$

Then $b = 0$, so $a = 0$ from the second entry, so $c = 0$ from the first. Hence $((1, 1, 0), (0, 1, 1), (1, 0, 0))$ is independent, with $3 = \dim \mathbb{R}^3$ elements, and therefore a basis by [Theorem 1.6.4](#) (a).

The extension is far from unique. The same computation with $\mathbf{e}_2$ gives $(a, a + b + c, b) = \mathbf{0}$, and with $\mathbf{e}_3$ gives $(a, a + b, b + c) = \mathbf{0}$; both force $a = b = c = 0$. So any one of $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ completes this list to a basis.

1.6.7 Largest independent lists and smallest spanning lists

We claimed that a basis is where independence and spanning meet. Here is the precise statement. It does not need finite dimension, and it is the form of “basis” that generalizes to infinite-dimensional spaces later in this chapter.

Call an independent list $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ in V **maximal independent** if for **every** $\mathbf{v} \in V$ the longer list $(\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{v})$ is dependent. Call a spanning list **minimal spanning** if deleting **any one** of its entries gives a list that does not span V .

Theorem 1.6.6 — Maximal Independent and Minimal Spanning Lists

Let V be a vector space and $\mathcal{L} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ a list in V .

- (a) $\mathcal{L}$ is a basis of V if and only if $\mathcal{L}$ is maximal independent.
- (b) $\mathcal{L}$ is a basis of V if and only if $\mathcal{L}$ is minimal spanning.

Proof.

- (a) $(\Rightarrow)$ Let $\mathcal{L}$ be a basis and $\mathbf{v} \in V$. Then $\mathbf{v} \in \text{span}(\mathcal{L}) = V$, so $(\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{v})$ is dependent by [Lemma 1.5.3](#). Hence $\mathcal{L}$ is maximal independent.
 $(\Leftarrow)$ Let $\mathcal{L}$ be maximal independent and $\mathbf{v} \in V$. The list $(\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{v})$ is dependent while $\mathcal{L}$ is independent, so $\mathbf{v} \in \text{span}(\mathcal{L})$ by [Lemma 1.5.3](#). Hence $\mathcal{L}$ spans V , and $\mathcal{L}$ is a basis.
- (b) $(\Rightarrow)$ Let $\mathcal{L}$ be a basis, and suppose that deleting $\mathbf{v}_j$ gives a list that still spans V . Then $\mathbf{v}_j = \sum_{i \neq j} c_i \mathbf{v}_i$ for some scalars, so $\sum_{i \neq j} c_i \mathbf{v}_i + (-1) \mathbf{v}_j = \mathbf{0}$, a linear relation with the coefficient $-1 \neq 0$. This contradicts independence. Hence $\mathcal{L}$ is minimal spanning.

($\Leftarrow$) Let $\mathcal{L}$ be minimal spanning. By [Theorem 1.5.4](#), some sub-list S of $\mathcal{L}$ is a basis of V . Suppose $S \neq \mathcal{L}$. Then S omits some entry $\mathbf{v}_j$, so every entry of S is an entry of the list $\mathcal{L}'$ obtained by deleting $\mathbf{v}_j$. By [Theorem 1.3.1](#), $\text{span}(\mathcal{L}')$ contains $\text{span}(S) = V$, so $\mathcal{L}'$ spans V , contradicting minimality. Hence $S = \mathcal{L}$, and $\mathcal{L}$ is a basis. ■

“Maximal” means “cannot be enlarged”, which is different from “longest possible”. In a finite-dimensional space the two agree: by [Theorem 1.6.3](#) a longest independent list has length $\dim V$, and by [Theorem 1.6.4](#) every independent list of that length is a basis.

1.6.8 Dimensions of subspaces

A subspace should be no bigger than the space containing it. Proving this needs care, because of a tempting circular argument.

Theorem 1.6.7 — Dimension of a Subspace

Let V be a finite-dimensional vector space and U a subspace of V . Then U is finite-dimensional, and $\dim U \leq \dim V$.

Idea. The tempting proof is: “take a basis of U ; it is independent in V , so its length is at most $\dim V$ ”. But we do not yet know that U has a basis. [Corollary 1.5.5](#) needs a finite spanning list of U , and that is exactly what we are trying to prove. So we build from below instead. Independent lists inside U are independent in V , so their lengths are bounded by $\dim V$. Take one of the largest possible length. If it did not span U , we could append a vector of U outside its span and get a longer one.

Proof. Let $n = \dim V$. Independence of a list depends only on its vectors and the operations of V , which U shares, so every independent list in U is an independent list in V , and has length at most n by [Theorem 1.6.3](#). The empty list is independent in U . Hence the set of lengths of independent lists in U is a non-empty set of integers in $\{0, 1, \dots, n\}$, and it has a largest element $k \leq n$. Choose an independent list $(\mathbf{u}_1, \dots, \mathbf{u}_k)$ in U of this length.

We claim $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k) = U$. Since U is a subspace containing every $\mathbf{u}_i$, [Theorem 1.3.1](#) gives $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k) \subseteq U$. Conversely, let $\mathbf{u} \in U$, and suppose $\mathbf{u} \notin \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$. By [Lemma 1.5.3](#), $(\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{u})$ is an independent list in U of length $k + 1$, contradicting the maximality of k . Hence $\mathbf{u} \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$.

Therefore $(\mathbf{u}_1, \dots, \mathbf{u}_k)$ spans U , so U is finite-dimensional ([Definition 1.3.4](#)), and the list is a basis of U . Hence $\dim U = k \leq n = \dim V$. ■

When the dimensions are equal, there is no room left.

Theorem 1.6.8 — Equal Dimension Forces Equality

Let V be a finite-dimensional vector space and U a subspace of V . If $\dim U = \dim V$, then $U = V$.

Proof. Let $n = \dim U = \dim V$; U is finite-dimensional by [Theorem 1.6.7](#). Let $(\mathbf{u}_1, \dots, \mathbf{u}_n)$ be a basis of U ([Corollary 1.5.5](#)). It is an independent list in V of length $n = \dim V$, so it is a basis of V by [Theorem 1.6.4](#) (a). Hence $V = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_n) \subseteq U$, where the inclusion is [Theorem 1.3.1](#) (3) applied to the subspace U . Since also $U \subseteq V$, we get $U = V$. ■

This is the dimension version of “count instead of check”: to prove two subspaces are equal, prove **one** inclusion and compare dimensions. It is usually much less work than proving both inclusions.

Quick check. Let U be a subspace of $\mathbb{R}[x]_{\leq 2}$ that contains 1 , $1+x$ and $(1+x)^2$. Must $U = \mathbb{R}[x]_{\leq 2}$?

Solution. Yes. By [Example \(A Basis by Counting\)](#), $(1, 1+x, (1+x)^2)$ is independent, and it lies in U , so $\dim U \geq 3$ by [Theorem 1.6.3](#) applied in U (which is finite-dimensional by [Theorem 1.6.7](#)). Also $\dim U \leq \dim \mathbb{R}[x]_{\leq 2} = 3$. So $\dim U = 3$, and [Theorem 1.6.8](#) gives $U = \mathbb{R}[x]_{\leq 2}$.

Warning. Both hypotheses of [Theorem 1.6.8](#) matter. Containment is needed: the xy -plane and the xz -plane in $\mathbb{R}^3$ both have dimension 2, but neither contains the other, and they are different. Without finite dimension it fails too. In $F[x]$, the subspace U of polynomials with constant coefficient 0 has the basis-like list $x, x^2, x^3, \dots$, which matches $1, x, x^2, \dots$ one for one; yet $1 \notin U$. Later in this chapter we make this precise for infinite-dimensional spaces.

1.6.9 Exercises

1.6.9.1 A. Check your understanding

Exercise A1.

- State the Steinitz Exchange Theorem.
- True or false: $\dim \mathbb{R}[x]_{\leq 3} = 3$. Justify your answer.
- True or false: in a vector space of dimension 3, every spanning list of length 3 is a basis. Justify your answer.
- True or false: $M_2(\mathbb{R})$ contains a linearly independent list of length 5. Justify your answer.
- True or false: if U and W are subspaces of $\mathbb{R}^4$ with $\dim U = \dim W = 2$, then $U = W$. Justify your answer.
- What are $\dim_{\mathbb{C}} \mathbb{C}^2$ and $\dim_{\mathbb{R}} \mathbb{C}^2$?

Solution.

- (a) See [Theorem 1.6.1](#): if $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ is independent and $(\mathbf{w}_1, \dots, \mathbf{w}_n)$ spans V , then $m \leq n$, and after renumbering the $\mathbf{w}$'s, $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_{m+1}, \dots, \mathbf{w}_n)$ spans V .
- (b) False. The basis $(1, x, x^2, x^3)$ has length 4, so $\dim \mathbb{R}[x]_{\leq 3} = 4$.
- (c) True, by [Theorem 1.6.4](#) (b).
- (d) False. $\dim M_2(\mathbb{R}) = 4$, and by [Theorem 1.6.3](#) (a) every list of length $5 > 4$ is dependent.
- (e) False. The subspaces $\text{span}(\mathbf{e}_1, \mathbf{e}_2)$ and $\text{span}(\mathbf{e}_3, \mathbf{e}_4)$ both have dimension 2, but $\mathbf{e}_1$ lies in the first and not in the second. [Theorem 1.6.8](#) needs one subspace to be contained in the other.
- (f) $\dim_{\mathbb{C}} \mathbb{C}^2 = 2$, with basis $(\mathbf{e}_1, \mathbf{e}_2)$. Over $\mathbb{R}$, the list $(\mathbf{e}_1, i\mathbf{e}_1, \mathbf{e}_2, i\mathbf{e}_2)$ is a basis: every $(a + bi, c + di)$ with $a, b, c, d \in \mathbb{R}$ equals $a\mathbf{e}_1 + b(i\mathbf{e}_1) + c\mathbf{e}_2 + d(i\mathbf{e}_2)$, and this is $\mathbf{0}$ only if $a + bi = 0$ and $c + di = 0$, that is, $a = b = c = d = 0$. So $\dim_{\mathbb{R}} \mathbb{C}^2 = 4$.

1.6.9.2 B. Practice

Exercise B1: Dimension and a Basis of a Subspace.

Find a basis and the dimension of each subspace.

- (a) $U_1 = \{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : x_1 + x_2 = 0 \text{ and } x_3 - x_4 = 0\}$.
- (b) $U_2 = \{A \in M_3(\mathbb{R}) : \text{tr } A = 0\}$.
- (c) $U_3 = \{p \in \mathbb{R}[x]_{\leq 3} : p(1) = p(-1) = 0\}$.

Solution.

- (a) A vector lies in U_1 exactly when $x_2 = -x_1$ and $x_3 = x_4$, that is, when it equals

$$(x_1, -x_1, x_4, x_4) = x_1(1, -1, 0, 0) + x_4(0, 0, 1, 1).$$

So $((1, -1, 0, 0), (0, 0, 1, 1))$ spans U_1 . It is independent: $a(1, -1, 0, 0) + b(0, 0, 1, 1) = (a, -a, b, b) = \mathbf{0}$ forces $a = b = 0$. Hence it is a basis, and $\dim U_1 = 2$.

- (b) Write $A = (a_{ij})$. The condition $a_{11} + a_{22} + a_{33} = 0$ says $a_{33} = -a_{11} - a_{22}$, so

$$A = \sum_{i \neq j} a_{ij} E_{ij} + a_{11}(E_{11} - E_{33}) + a_{22}(E_{22} - E_{33}).$$

All eight matrices on the right have trace 0, so the list of the six E_{ij} with $i \neq j$, followed by $E_{11} - E_{33}$ and $E_{22} - E_{33}$, spans U_2 . For independence, let $\sum_{i \neq j} c_{ij} E_{ij} + d_1(E_{11} - E_{33}) + d_2(E_{22} - E_{33}) = 0$. The entry (i, j) with $i \neq j$ of the left side is c_{ij} , the entry $(1, 1)$ is d_1 , and the entry $(2, 2)$ is d_2 ; all must be 0. Hence the list is a basis, and $\dim U_2 = 8$.

- (c) Let $p = a_0 + a_1x + a_2x^2 + a_3x^3$. Then $p(1) = a_0 + a_1 + a_2 + a_3$ and $p(-1) = a_0 - a_1 + a_2 - a_3$. Adding and subtracting, $p \in U_3$ exactly when $a_0 + a_2 = 0$ and $a_1 + a_3 = 0$. Hence $p \in U_3$ exactly when

$$p = -a_2 - a_3x + a_2x^2 + a_3x^3 = a_2(x^2 - 1) + a_3(x^3 - x).$$

So $(x^2 - 1, x^3 - x)$ spans U_3 . It is independent: $a(x^2 - 1) + b(x^3 - x) = 0$ has x^3 -coefficient b and x^2 -coefficient a , so $a = b = 0$. Hence it is a basis, and $\dim U_3 = 2$.

Exercise B2: Extending to a Basis of $\mathbb{R}^4$.

Show that $((1, 1, 0, 0), (0, 1, 1, 0))$ is linearly independent, and extend it to a basis of $\mathbb{R}^4$ using vectors from the standard basis.

Solution. Let $\mathbf{u}_1 = (1, 1, 0, 0)$ and $\mathbf{u}_2 = (0, 1, 1, 0)$. If $a\mathbf{u}_1 + b\mathbf{u}_2 = (a, a + b, b, 0) = \mathbf{0}$, then $a = b = 0$, so the list is independent. By [Theorem 1.6.5](#), two standard basis vectors complete it; we sift $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4)$ to find which.

- $\mathbf{u}_1 \neq \mathbf{0}$: keep. $\mathbf{u}_2 \notin \text{span}(\mathbf{u}_1)$, since a multiple $(a, a, 0, 0)$ of $\mathbf{u}_1$ has third entry 0: keep.
- $\mathbf{e}_1$: if $(a, a + b, b, 0) = (1, 0, 0, 0)$, then $a = 1$ and $b = 0$, but then $a + b = 1 \neq 0$. So $\mathbf{e}_1 \notin \text{span}(\mathbf{u}_1, \mathbf{u}_2)$: keep.
- $\mathbf{e}_2 = \mathbf{u}_1 - \mathbf{e}_1$: discard.
- $\mathbf{e}_3 = -\mathbf{u}_1 + \mathbf{u}_2 + \mathbf{e}_1$, since $-(1, 1, 0, 0) + (0, 1, 1, 0) + (1, 0, 0, 0) = (0, 0, 1, 0)$: discard.
- $\mathbf{e}_4$: every combination of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{e}_1$ has fourth entry 0: keep.

The sifted list $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{e}_1, \mathbf{e}_4)$ is independent by [Theorem 1.5.4](#). It has $4 = \dim \mathbb{R}^4$ entries, so it is a basis of $\mathbb{R}^4$ by [Theorem 1.6.4 \(a\)](#).

Exercise B3: Count Instead of Check.

Show that $(1, x - 1, (x - 1)^2)$ is a basis of $\mathbb{R}[x]_{\leq 2}$. Hence find the coordinate vector of x^2 with respect to this basis.

Solution. Let $a + b(x - 1) + c(x - 1)^2 = 0$. Expanding, $(a - b + c) + (b - 2c)x + cx^2 = 0$. Comparing coefficients, $c = 0$, then $b = 2c = 0$, then $a = b - c = 0$. So the list is independent, with $3 = \dim \mathbb{R}[x]_{\leq 2}$ elements, and it is a basis by [Theorem 1.6.4 \(a\)](#).

Writing $x = (x - 1) + 1$,

$$x^2 = ((x - 1) + 1)^2 = 1 + 2(x - 1) + (x - 1)^2.$$

Hence the coordinate vector of x^2 is $(1, 2, 1)$.

1.6.9.3 C. Going deeper

Exercise C1: Symmetric and Skew-Symmetric Matrices.

Let $n \geq 1$. For a field F , let U_+ and U_- be the subspaces of symmetric and of skew-symmetric matrices in $M_n(F)$ ([Definition 0.9.11](#)): $U_+ = \{A : A^T = A\}$ and $U_- = \{A : A^T = -A\}$.

- (a) For $F = \mathbb{R}$, prove that $\dim U_+ = \frac{n(n+1)}{2}$.

- (b) For $F = \mathbb{R}$, prove that $\dim U_- = \frac{n(n-1)}{2}$.
 (c) For $F = \mathbb{F}_2$, show that $U_- = U_+$, and find its dimension. Which step of (b) fails?

Solution.

- (a) Consider the list $\mathcal{S}$ consisting of E_{ii} for $1 \leq i \leq n$ and $E_{ij} + E_{ji}$ for $1 \leq i < j \leq n$. Each is symmetric. If $A = (a_{ij})$ is symmetric, then $a_{ji} = a_{ij}$, so

$$A = \sum_i a_{ii} E_{ii} + \sum_{i < j} a_{ij} (E_{ij} + E_{ji}),$$

because both sides have a_{ij} in entry (i, j) and $a_{ij} = a_{ji}$ in entry (j, i) . So $\mathcal{S}$ spans U_+ . If $\sum_i c_{ii} E_{ii} + \sum_{i < j} c_{ij} (E_{ij} + E_{ji}) = 0$, then for $i \leq j$ the entry (i, j) of the left side is c_{ij} , so every coefficient is 0. Hence $\mathcal{S}$ is a basis, of length $n + \frac{n(n-1)}{2} = \frac{n(n+1)}{2}$.

- (b) If $A^T = -A$, then comparing diagonal entries gives $a_{ii} = -a_{ii}$, so $2a_{ii} = 0$, and $a_{ii} = 0$ because $2 \neq 0$ in $\mathbb{R}$. Off the diagonal, $a_{ji} = -a_{ij}$. Hence

$$A = \sum_{i < j} a_{ij} (E_{ij} - E_{ji}).$$

Each $E_{ij} - E_{ji}$ is skew-symmetric, so these matrices span U_- , and as in (a) the entry (i, j) with $i < j$ of a combination is its coefficient, so they are independent. There are $\frac{n(n-1)}{2}$ pairs $i < j$, so $\dim U_- = \frac{n(n-1)}{2}$.

- (c) In $\mathbb{F}_2$, $-1 = 1$, so $-A = A$ for every matrix A , and the conditions $A^T = -A$ and $A^T = A$ coincide. Hence $U_- = U_+$ over $\mathbb{F}_2$, which has dimension $\frac{n(n+1)}{2}$ by the argument of (a), which works over any field. The step of (b) that fails is “ $2a_{ii} = 0$ implies $a_{ii} = 0$ ”, which divides by $2 = 0$. For example, $I_n \in U_-$ over $\mathbb{F}_2$, although its diagonal entries are not 0.

Exercise C2: A complex space viewed over the reals.

Let V be a vector space over $\mathbb{C}$ with $\dim_{\mathbb{C}} V = n$. Restricting the scalars to real numbers makes V a vector space over $\mathbb{R}$. Prove that $\dim_{\mathbb{R}} V = 2n$.

Hint: start from a basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ over $\mathbb{C}$, and consider the vectors $i\mathbf{v}_k$.

Solution. Let $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of V over $\mathbb{C}$. We show that $\mathcal{B} = (\mathbf{v}_1, i\mathbf{v}_1, \dots, \mathbf{v}_n, i\mathbf{v}_n)$ is a basis of V over $\mathbb{R}$.

Spanning. Let $\mathbf{v} \in V$. Then $\mathbf{v} = \sum_k z_k \mathbf{v}_k$ with $z_k \in \mathbb{C}$. Write $z_k = a_k + b_k i$ with $a_k, b_k \in \mathbb{R}$. By the distributive and associative axioms of scalar multiplication, $z_k \mathbf{v}_k = a_k \mathbf{v}_k + (b_k i) \mathbf{v}_k = a_k \mathbf{v}_k + b_k (i \mathbf{v}_k)$. Hence $\mathbf{v} = \sum_k (a_k \mathbf{v}_k + b_k (i \mathbf{v}_k))$, a real combination of $\mathcal{B}$.

Independence. Let $\sum_k (a_k \mathbf{v}_k + b_k (i \mathbf{v}_k)) = \mathbf{0}$ with $a_k, b_k \in \mathbb{R}$. Reading the same axioms backwards, $\sum_k (a_k + b_k i) \mathbf{v}_k = \mathbf{0}$. Since $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is independent over $\mathbb{C}$, every $a_k + b_k i = 0$, so $a_k = b_k = 0$.

Hence $\mathcal{B}$ is a basis over $\mathbb{R}$ of length $2n$, and $\dim_{\mathbb{R}} V = 2n$.

Exercise C3: Counting Elements over a Finite Field.

Let p be a prime.

- (a) Let V be a vector space over $\mathbb{F}_p$ with $\dim V = k$. Prove that V has exactly p^k elements.
- (b) Deduce that no vector space over $\mathbb{F}_2$ has exactly 6 elements.
- (c) Let K be a finite field that contains $\mathbb{F}_p$, with the same addition and multiplication. Explain why K is a finite-dimensional vector space over $\mathbb{F}_p$, and deduce that K has p^k elements for some integer $k \geq 1$.

Hint: for (a), use coordinates.

Solution.

- (a) Let $\mathcal{B}$ be a basis of V , of length k . By [Theorem 1.5.2](#), $\mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}}$ is a bijection from V to $\mathbb{F}_p^k$: every column is the coordinate vector of exactly one vector. So V has as many elements as $\mathbb{F}_p^k$. A column in $\mathbb{F}_p^k$ is a choice of k entries, each from the p elements of $\mathbb{F}_p$, so there are p^k columns. Hence V has p^k elements.
- (b) A vector space over $\mathbb{F}_2$ with finitely many elements is spanned by the finite list of all its elements, so it is finite-dimensional, of some dimension k . By (a) it has 2^k elements. Since 6 is not a power of 2, no such space has 6 elements.
- (c) Define scalar multiplication $\mathbb{F}_p \times K \rightarrow K$ by the multiplication of K . The vector space axioms are then instances of the field axioms of K : addition in K is commutative and associative with zero and negatives, multiplication is associative, $1 \cdot a = a$, and multiplication distributes over addition. So K is a vector space over $\mathbb{F}_p$. It is spanned by the finite list of all its elements, so it is finite-dimensional; let $k = \dim_{\mathbb{F}_p} K$. By (a), K has p^k elements. Since K contains $0 \neq 1$, it is not the zero space, so $k \geq 1$.

1.7 Sums and Direct Sums

We can now measure a single subspace: take a basis and count. This section asks how two or more subspaces combine. The intersection of two subspaces is again a subspace, but the union usually is not, so we need a better way to join them. The answer is the sum $U + W$. We then prove the dimension formula for sums and single out the best case, the direct sum, in which every vector splits into pieces in exactly one way.

1.7.1 The sum of subspaces

Recall from [Theorem 1.2.2](#) that $U \cap W$ is a subspace whenever U and W are. It is the largest subspace inside both. The natural companion would be the smallest subspace containing both, and the first guess is the union $U \cup W$.

That guess fails. In $\mathbb{R}^2$, let U be the x -axis and W the y -axis. Then $(1, 0) \in U$ and $(0, 1) \in W$, but their sum $(1, 1)$ lies on neither axis, so $U \cup W$ is not closed under addition ([Example \(The](#)

Union of the Two Axes)). In fact Proposition 1.2.3 shows that $U \cup W$ is a subspace only in the dull case $U \subseteq W$ or $W \subseteq U$.

The failure tells us what is missing. Any subspace containing U and W must contain $\mathbf{u} + \mathbf{w}$ for every $\mathbf{u} \in U$ and $\mathbf{w} \in W$, because subspaces are closed under addition. So we put all these sums in from the start.

The sum of two subspaces is everything you can reach by adding one vector from each.

Definition 1.7.1 — Sum of Subspaces

Let V be a vector space over F , and let U and W be subspaces of V . The **sum** of U and W is the subset

$$U + W := \{\mathbf{u} + \mathbf{w} : \mathbf{u} \in U \text{ and } \mathbf{w} \in W\}.$$

More generally, for subspaces $U_1, \dots, U_k$ of V (with $k \geq 1$), their **sum** is

$$U_1 + \dots + U_k := \{\mathbf{u}_1 + \dots + \mathbf{u}_k : \mathbf{u}_i \in U_i \text{ for each } i = 1, \dots, k\}.$$

In words: a vector $\mathbf{v}$ lies in $U + W$ when **there exist** some $\mathbf{u} \in U$ and some $\mathbf{w} \in W$ with $\mathbf{v} = \mathbf{u} + \mathbf{w}$. The definition asks for one such pair. It does not say the pair is unique, and in general it is not. For k subspaces we pick **one** vector from **each** U_i and add them. We also write $\sum_{i=1}^k U_i$, and for $k = 1$ the sum is just U_1 .

Here are some first examples.

- **The two axes.** In $\mathbb{R}^2$, with U the x -axis and W the y -axis, every (a, b) equals $(a, 0) + (0, b)$, so $U + W = \mathbb{R}^2$. The union is two lines; the sum is the whole plane.
- **Two planes in $\mathbb{R}^3$.** Let $P_{xy} = \{(x, y, 0)\}$ and $P_{yz} = \{(0, y, z)\}$. Every $(a, b, c) = (a, b, 0) + (0, 0, c)$ is a sum of a vector in P_{xy} and one in P_{yz} , so $P_{xy} + P_{yz} = \mathbb{R}^3$.
- **Polynomials.** In $F[x]_{\leq 2}$, let U be the constant polynomials and $W = \{bx + cx^2 : b, c \in F\}$. Then $a + bx + cx^2 = a + (bx + cx^2)$, so $U + W = F[x]_{\leq 2}$.
- **Matrices.** In $M_2(F)$, the upper triangular matrices plus the lower triangular matrices give everything:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ c & 0 \end{pmatrix}.$$

- **Degenerate cases.** For any subspace U , we have $U + \{\mathbf{0}\} = U$ and $U + U = U$. In both, “ $\supseteq$ ” holds because $\mathbf{u} = \mathbf{u} + \mathbf{0}$, and “ $\subseteq$ ” holds because U is closed under addition. So adding a subspace to itself produces nothing new: a sum is not a count.

Non-example by minimal change. The definition makes sense for any two subsets $S, T \subseteq V$, but the result need not be a subspace. Take $S = \{(1, 0)\}$ and $T = \{(0, 1)\}$ in $\mathbb{R}^2$. Then $S + T = \{(1, 1)\}$, which does not contain $\mathbf{0}$. What changed is that S and T are not subspaces: the hypothesis “subspace” is exactly what makes the next theorem work.

Warning. The sum $U + W$ is **not** the union $U \cup W$. A vector of $U + W$ need not lie in U or in W : $(1, 1)$ lies in the sum of the two axes but on neither axis. Conversely, $U + W$ is usually far bigger than $U \cup W$.

The sum does what we built it to do.

Theorem 1.7.1 — Sum of Subspaces Theorem

Let $U_1, \dots, U_k$ be subspaces of a vector space V . Then $U_1 + \dots + U_k$ is a subspace of V that contains each U_i . Moreover, if X is any subspace of V containing every U_i , then $U_1 + \dots + U_k \subseteq X$. In other words, $U_1 + \dots + U_k$ is the **smallest** subspace of V containing $U_1, \dots, U_k$.

Proof. Write $S = U_1 + \dots + U_k$. We apply the Subspace Test (Theorem 1.2.1).

- (1) Each U_i contains $\mathbf{0}$, so $\mathbf{0} = \mathbf{0} + \dots + \mathbf{0} \in S$.
- (2) Let $\mathbf{u}_1 + \dots + \mathbf{u}_k$ and $\mathbf{u}'_1 + \dots + \mathbf{u}'_k$ be in S , with $\mathbf{u}_i, \mathbf{u}'_i \in U_i$. Regrouping with (VS1) and (VS2),

$$(\mathbf{u}_1 + \dots + \mathbf{u}_k) + (\mathbf{u}'_1 + \dots + \mathbf{u}'_k) = (\mathbf{u}_1 + \mathbf{u}'_1) + \dots + (\mathbf{u}_k + \mathbf{u}'_k),$$

and $\mathbf{u}_i + \mathbf{u}'_i \in U_i$ because U_i is closed under addition. Hence S is closed under addition.

- (3) Let $c \in F$. By the distributive axiom, $c(\mathbf{u}_1 + \dots + \mathbf{u}_k) = c\mathbf{u}_1 + \dots + c\mathbf{u}_k$, and $c\mathbf{u}_i \in U_i$ because U_i is closed under scaling. Hence S is closed under scalar multiplication.

Therefore S is a subspace. It contains U_j : for $\mathbf{u} \in U_j$, take $\mathbf{u}_j = \mathbf{u}$ and $\mathbf{u}_i = \mathbf{0} \in U_i$ for $i \neq j$. Finally, let X be a subspace containing every U_i . An element $\mathbf{u}_1 + \dots + \mathbf{u}_k$ of S is a sum of vectors $\mathbf{u}_i \in U_i \subseteq X$, so it lies in X since X is closed under addition. This shows $S \subseteq X$. ■

So the sum plays the role that the union could not. It is the subspace analogue of span, and the two notions fit together exactly. Recall from Theorem 1.3.1 that $\text{span}(S)$ is the smallest subspace containing S .

Proposition 1.7.2 — Sum of Spans

Let $L_1, \dots, L_k$ be finite lists of vectors in V , and let L be the list obtained by writing $L_1, \dots, L_k$ one after another. Then

$$\text{span}(L_1) + \dots + \text{span}(L_k) = \text{span}(L).$$

Proof. ($\subseteq$) The subspace $\text{span}(L)$ contains every vector of each L_i , so by Theorem 1.3.1 it contains each $\text{span}(L_i)$. By Theorem 1.7.1 it therefore contains $\text{span}(L_1) + \dots + \text{span}(L_k)$.

($\supseteq$) The sum $\text{span}(L_1) + \dots + \text{span}(L_k)$ is a subspace by Theorem 1.7.1, and it contains each $\text{span}(L_i)$, hence every vector of L . By Theorem 1.3.1 it contains $\text{span}(L)$. ■

In practice this means: **to find a spanning list of a sum, concatenate spanning lists of the summands.** That list may be dependent, and the dimension formula below tells us by how much.

Quick check. In $\mathbb{R}^3$, what is $\text{span}((1, 0, 0)) + \text{span}((0, 1, 0))$? Is it equal to $\text{span}((1, 0, 0)) \cup \text{span}((0, 1, 0))$?

Solution. By [Proposition 1.7.2](#) the sum is $\text{span}((1, 0, 0), (0, 1, 0)) = \{(a, b, 0) : a, b \in \mathbb{R}\}$, the xy -plane. It is not the union: $(1, 1, 0)$ lies in the plane but on neither axis.

1.7.2 The dimension formula

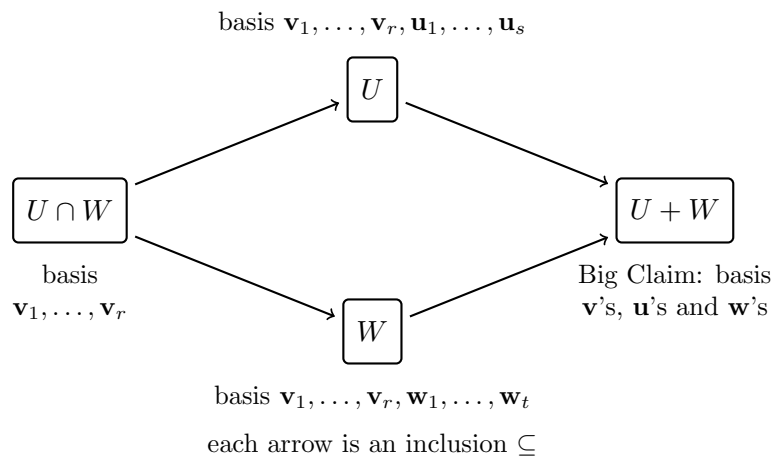
If U and W are finite-dimensional, how big is $U + W$? Concatenating a basis of U and a basis of W gives a spanning list of $U + W$ of length $\dim U + \dim W$ ([Proposition 1.7.2](#)). That count is too large when U and W overlap. For the two planes P_{xy} and P_{yz} in $\mathbb{R}^3$, it would give $2 + 2 = 4$, but $P_{xy} + P_{yz} = \mathbb{R}^3$ has dimension 3. The overlap is the y -axis, of dimension 1, and it has been counted twice. The theorem says this is the whole story.

Theorem 1.7.3 — Dimension Formula for Sums

Let U and W be finite-dimensional subspaces of a vector space V . Then $U + W$ is finite-dimensional, and

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

The picture behind the proof is the inclusion diagram. The smallest space, $U \cap W$, sits inside both U and W , and both sit inside $U + W$. We build bases from the bottom up.



Idea. This is a dimension formula, so we start from the **smallest** space and extend outward.

① Take a basis $(\mathbf{v}_1, \dots, \mathbf{v}_r)$ of $U \cap W$. ② Extend it to a basis of U by adding $\mathbf{u}_1, \dots, \mathbf{u}_s$, and separately to a basis of W by adding $\mathbf{w}_1, \dots, \mathbf{w}_t$. ③ **Big Claim:** the combined list of all $r + s + t$ vectors is a basis of $U + W$. Spanning is easy. For independence, move the $\mathbf{w}$ -part to the other side: the common vector lies in U and in W , hence in $U \cap W$, where the $\mathbf{v}$'s are a basis. ④ Count: $r + s + t = (r + s) + (r + t) - r$.

Proof. Step 1: bases. By [Theorem 1.2.2](#), $U \cap W$ is a subspace of V ; it lies inside U and uses the same operations, so it is also a subspace of U . Since U is finite-dimensional, so is $U \cap W$ ([Theorem 1.6.7](#)), and it has a basis $(\mathbf{v}_1, \dots, \mathbf{v}_r)$, where $r = \dim(U \cap W)$ ([Corollary 1.5.5](#)). This list is independent and lies in U , so by the Basis Extension Theorem ([Theorem 1.6.5](#)) there are vectors $\mathbf{u}_1, \dots, \mathbf{u}_s \in U$ such that $(\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{u}_1, \dots, \mathbf{u}_s)$ is a basis of U . In the same way, since the list also lies in W , there are $\mathbf{w}_1, \dots, \mathbf{w}_t \in W$ such that $(\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{w}_1, \dots, \mathbf{w}_t)$ is a basis of W . Thus $\dim U = r + s$ and $\dim W = r + t$.

Step 2: the Big Claim.

Claim. The list $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{u}_1, \dots, \mathbf{u}_s, \mathbf{w}_1, \dots, \mathbf{w}_t)$ is a basis of $U + W$.

Proof. Spanning. By [Proposition 1.7.2](#), $U + W = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{u}_1, \dots, \mathbf{u}_s) + \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{w}_1, \dots, \mathbf{w}_t)$ is the span of the concatenated list. That list contains each $\mathbf{v}_i$ twice; removing the second copies does not change the span, since every removed vector already lies in the span of what remains ([Theorem 1.3.5](#)). Hence $\mathcal{B}$ spans $U + W$.

Independence. Let $a_i, b_j, c_k \in F$ satisfy

$$a_1\mathbf{v}_1 + \dots + a_r\mathbf{v}_r + b_1\mathbf{u}_1 + \dots + b_s\mathbf{u}_s + c_1\mathbf{w}_1 + \dots + c_t\mathbf{w}_t = \mathbf{0}.$$

Move the $\mathbf{w}$ -part across and name the common vector:

$$\mathbf{x} := a_1\mathbf{v}_1 + \dots + a_r\mathbf{v}_r + b_1\mathbf{u}_1 + \dots + b_s\mathbf{u}_s = -c_1\mathbf{w}_1 - \dots - c_t\mathbf{w}_t.$$

The middle expression lies in U , because all the $\mathbf{v}_i$ and $\mathbf{u}_j$ do. The right-hand expression lies in W , because all the $\mathbf{w}_k$ do. Hence $\mathbf{x} \in U \cap W$. Since $(\mathbf{v}_1, \dots, \mathbf{v}_r)$ spans $U \cap W$, there are $d_1, \dots, d_r \in F$ with $\mathbf{x} = d_1\mathbf{v}_1 + \dots + d_r\mathbf{v}_r$. Comparing with the right-hand expression for $\mathbf{x}$,

$$d_1\mathbf{v}_1 + \dots + d_r\mathbf{v}_r + c_1\mathbf{w}_1 + \dots + c_t\mathbf{w}_t = \mathbf{0}.$$

This is a combination of the basis $(\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{w}_1, \dots, \mathbf{w}_t)$ of W , which is independent, so all $d_i = 0$ and all $c_k = 0$. Putting $c_1 = \dots = c_t = 0$ into the original equation leaves

$$a_1\mathbf{v}_1 + \dots + a_r\mathbf{v}_r + b_1\mathbf{u}_1 + \dots + b_s\mathbf{u}_s = \mathbf{0},$$

a combination of the basis $(\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{u}_1, \dots, \mathbf{u}_s)$ of U . By its independence, all $a_i = 0$ and all $b_j = 0$. Hence $\mathcal{B}$ is linearly independent ([Definition 1.4.1](#)). ■

Step 3: count. By the claim, $U + W$ has a finite basis of length $r + s + t$, so it is finite-dimensional with $\dim(U + W) = r + s + t$ ([Definition 1.6.1](#)). Therefore

$$\dim(U + W) = r + s + t = (r + s) + (r + t) - r = \dim U + \dim W - \dim(U \cap W),$$

as claimed. ■

The proof also works when some of r, s, t are 0: an empty list is independent and spans $\{\mathbf{0}\}$, and every step reads correctly with empty sums. The move “basis of the smallest space, extend, Big Claim, count” is worth learning as a unit. It proves Rank–Nullity in Chapter 3 with the same four blocks.

The formula is often used to find one of the four numbers from the other three. Here are two uses, one by counting and one by computing.

Example. (Two planes in $\mathbb{R}^3$)

Let $U = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$ and $W = \{(x, y, z) \in \mathbb{R}^3 : x = z\}$. Show that $U + W = \mathbb{R}^3$, and find a basis of $\mathbb{R}^3$ of the shape used in the proof.

Solution. Solving $x + y + z = 0$ for y , every vector of U is $(x, -x - z, z) = x(1, -1, 0) + z(0, -1, 1)$. The two vectors are independent (look at the first and third entries), so $\dim U = 2$. Similarly every vector of W is $(x, y, x) = x(1, 0, 1) + y(0, 1, 0)$, and $\dim W = 2$.

A vector lies in $U \cap W$ when $x = z$ and $x + y + z = 0$, that is, $y = -2x$. So $U \cap W = \{(x, -2x, x)\} = \text{span}((1, -2, 1))$, which has dimension 1. By the dimension formula,

$$\dim(U + W) = 2 + 2 - 1 = 3 = \dim \mathbb{R}^3.$$

Since $U + W \subseteq \mathbb{R}^3$ and the dimensions agree, $U + W = \mathbb{R}^3$ by [Theorem 1.6.8](#).

Following the proof: $(1, -2, 1)$ extends to a basis of U by $(1, -1, 0) \in U$. Indeed $(1, -1, 0)$ is not a multiple $(c, -2c, c)$ of $(1, -2, 1)$, since its first and third entries differ, so the list $((1, -2, 1), (1, -1, 0))$ is independent by [Lemma 1.5.3](#); it has length $2 = \dim U$, so it is a basis of U ([Theorem 1.6.4](#)). It extends to a basis of W by $(0, 1, 0) \in W$, for the same reason: a multiple of $(1, -2, 1)$ with first entry 0 is $\mathbf{0}$. The Big Claim says $((1, -2, 1), (1, -1, 0), (0, 1, 0))$ is a basis of $U + W = \mathbb{R}^3$.

Example. (Two planes in $\mathbb{R}^4$ from spanning lists)

Let $U = \text{span}((1, 1, 0, 0), (0, 0, 1, 1))$ and $W = \text{span}((1, 0, 1, 0), (0, 1, 0, 1))$ in $\mathbb{R}^4$. Find $\dim(U \cap W)$ and $\dim(U + W)$, and hence a basis of $U + W$.

Solution. Each spanning list is independent: in $a(1, 1, 0, 0) + b(0, 0, 1, 1) = (a, a, b, b) = \mathbf{0}$, both a and b are 0, and similarly $c(1, 0, 1, 0) + d(0, 1, 0, 1) = (c, d, c, d)$. So $\dim U = \dim W = 2$.

A vector lies in $U \cap W$ exactly when it can be written in both forms, $(a, a, b, b) = (c, d, c, d)$. Comparing entries gives $a = c$, $a = d$, $b = c$, $b = d$, so $a = b = c = d$. Hence $U \cap W = \text{span}((1, 1, 1, 1))$ and $\dim(U \cap W) = 1$. By the dimension formula, $\dim(U + W) = 2 + 2 - 1 = 3$.

For a basis, follow the proof. The vector $(1, 1, 1, 1)$ together with $(1, 1, 0, 0)$ is a list in U , and it is independent by [Lemma 1.5.3](#), because a multiple (c, c, c, c) of $(1, 1, 1, 1)$ never equals $(1, 1, 0, 0)$. It has length $2 = \dim U$, hence it is a basis of U ([Theorem 1.6.4](#)). Likewise $((1, 1, 1, 1), (1, 0, 1, 0))$ is a basis of W . By the Big Claim, $((1, 1, 1, 1), (1, 1, 0, 0), (1, 0, 1, 0))$ is a basis of $U + W$. In particular $U + W \neq \mathbb{R}^4$.

1.7.3 Direct sums

In a basis, every vector has **exactly one** expression as a combination (Theorem 1.5.1). Read that for F^n with the standard basis: $F^n = \text{span}(\mathbf{e}_1) + \cdots + \text{span}(\mathbf{e}_n)$, and each vector splits into pieces from these lines in exactly one way. We want the same property for subspaces that are not lines.

Sums do not always have it. In $\mathbb{R}^3 = P_{xy} + P_{yz}$,

$$(1, 1, 1) = (1, 1, 0) + (0, 0, 1) = (1, 0, 0) + (0, 1, 1),$$

two different splittings, because the planes share the y -axis and the y -component can be handed to either side. If we replace P_{yz} by the z -axis Z , then $(a, b, c) = (a, b, 0) + (0, 0, c)$ is the **only** splitting, since the Z -piece must have first two entries 0. We give this situation a name.

A sum is direct when every vector in it breaks into one piece from each subspace in exactly one way.

Definition 1.7.2 — Direct Sum

Let $U_1, \dots, U_k$ be subspaces of a vector space V over F . The sum $U_1 + \cdots + U_k$ is **direct** if **every** vector $\mathbf{v} \in U_1 + \cdots + U_k$ can be written in **exactly one** way as

$$\mathbf{v} = \mathbf{u}_1 + \cdots + \mathbf{u}_k \quad \text{with } \mathbf{u}_i \in U_i \text{ for each } i.$$

In that case we write the sum as $U_1 \oplus \cdots \oplus U_k$. We write $V = U_1 \oplus \cdots \oplus U_k$ to mean that $V = U_1 + \cdots + U_k$ **and** the sum is direct.

In words: existence of a splitting is automatic, since that is what membership in the sum means. The content of “direct” is **uniqueness**: if $\mathbf{u}_1 + \cdots + \mathbf{u}_k = \mathbf{u}'_1 + \cdots + \mathbf{u}'_k$ with $\mathbf{u}_i, \mathbf{u}'_i \in U_i$, then $\mathbf{u}_i = \mathbf{u}'_i$ for **every** i . The symbol $\oplus$ does not build a new space. It is the same subspace $U_1 + \cdots + U_k$, with a property recorded in the notation. (Chapter 3 builds an “external” direct sum out of spaces that do not sit inside a common V .)

Examples.

- **Standard basis.** $F^n = \text{span}(\mathbf{e}_1) \oplus \cdots \oplus \text{span}(\mathbf{e}_n)$. More generally, if $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is a basis of V , then $V = \text{span}(\mathbf{v}_1) \oplus \cdots \oplus \text{span}(\mathbf{v}_n)$. A splitting $\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$ is a choice of coefficients, and Theorem 1.5.1 says there is exactly one.
- **Two splittings of the plane.** Let X be the x -axis, Y the y -axis and $D = \{(t, t) : t \in \mathbb{R}\}$ the line $y = x$. Then $\mathbb{R}^2 = X \oplus Y$, with $(a, b) = (a, 0) + (0, b)$ the only splitting. Also $\mathbb{R}^2 = X \oplus D$: if $(a, b) = (s, 0) + (t, t)$, the second entry forces $t = b$, and then the first forces $s = a - b$. So $(a, b) = (a - b, 0) + (b, b)$, and there is no other choice.
- **Polynomials.** $F[x]_{\leq 2} = \text{span}(1) \oplus \text{span}(x, x^2)$. If $a + bx + cx^2 = a' + (b'x + c'x^2)$, comparing coefficients gives $a = a'$, $b = b'$, $c = c'$.
- **Degenerate.** $U \oplus \{\mathbf{0}\} = U$ for every subspace U : the only possible $\{\mathbf{0}\}$ -piece is $\mathbf{0}$, so the U -piece of $\mathbf{v}$ must be $\mathbf{v}$.

Non-example by minimal change. Replace the z -axis in $\mathbb{R}^3 = P_{xy} \oplus Z$ by the plane P_{yz} . The sum is still all of $\mathbb{R}^3$, so existence still holds. What fails is uniqueness, as the two splittings of $(1, 1, 1)$ above show. So $\mathbb{R}^3 = P_{xy} + P_{yz}$, but the sum is **not** direct. Likewise upper plus lower triangular 2×2 matrices give $M_2(F)$, but a diagonal matrix can be put in either summand, so that sum is not direct.

Checking uniqueness for **every** vector looks like a lot of work. For two subspaces it reduces to one condition you can check in a line.

Theorem 1.7.4 — Direct Sum Criteria for Two Subspaces

Let U and W be subspaces of a vector space V . The following are equivalent:

- (a) the sum $U + W$ is direct;
- (b) $U \cap W = \{\mathbf{0}\}$.

If moreover U and W are finite-dimensional, they are also equivalent to

- (c) $\dim(U + W) = \dim U + \dim W$.

Idea. Uniqueness questions go through differences. If $\mathbf{u} + \mathbf{w} = \mathbf{u}' + \mathbf{w}'$, then $\mathbf{u} - \mathbf{u}' = \mathbf{w}' - \mathbf{w}$ is a single vector lying in both U and W . So “no overlap” is exactly “no freedom in splitting”. For (c), the dimension formula says the overcount $\dim U + \dim W - \dim(U + W)$ is $\dim(U \cap W)$, which is 0 exactly when $U \cap W = \{\mathbf{0}\}$.

Proof.

- (a) $\Rightarrow$ (b). Suppose the sum is direct and let $\mathbf{v} \in U \cap W$. Then $\mathbf{v} = \mathbf{v} + \mathbf{0}$ with $\mathbf{v} \in U$, $\mathbf{0} \in W$, and $\mathbf{v} = \mathbf{0} + \mathbf{v}$ with $\mathbf{0} \in U$, $\mathbf{v} \in W$. These are two splittings of $\mathbf{v} \in U + W$, so by uniqueness their U -pieces agree: $\mathbf{v} = \mathbf{0}$. Hence $U \cap W = \{\mathbf{0}\}$.
- (b) $\Rightarrow$ (a). Suppose $U \cap W = \{\mathbf{0}\}$, and let $\mathbf{u} + \mathbf{w} = \mathbf{u}' + \mathbf{w}'$ with $\mathbf{u}, \mathbf{u}' \in U$ and $\mathbf{w}, \mathbf{w}' \in W$. Then $\mathbf{u} - \mathbf{u}' = \mathbf{w}' - \mathbf{w}$. The left side lies in U and the right side in W , since both are subspaces. So this common vector lies in $U \cap W = \{\mathbf{0}\}$, which gives $\mathbf{u} = \mathbf{u}'$ and $\mathbf{w} = \mathbf{w}'$. Hence every vector of $U + W$ has only one splitting.
- (c) $\Leftrightarrow$ (c). Now let U and W be finite-dimensional. By the dimension formula (Theorem 1.7.3), (c) holds if and only if $\dim(U \cap W) = 0$. A space has dimension 0 exactly when the empty list is a basis of it, that is, when it equals the span of the empty list, which is $\{\mathbf{0}\}$. So (c) holds if and only if $U \cap W = \{\mathbf{0}\}$. ■

This is the everyday test: to show $V = U \oplus W$, show $V = U + W$ and $U \cap W = \{\mathbf{0}\}$. In finite dimension there is a shortcut that avoids checking $V = U + W$ at all: if $U \cap W = \{\mathbf{0}\}$ and $\dim U + \dim W = \dim V$, then $\dim(U + W) = \dim V$ by (c), so $U + W = V$ by Theorem 1.6.8. Count instead of check.

Example. (A plane and a line in $\mathbb{R}^3$)

Let $U = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$ and $L = \text{span}((1, 1, 1))$. Show that $\mathbb{R}^3 = U \oplus L$, and split (a, b, c) into its two pieces.

Solution. From Example (Two planes in $\mathbb{R}^3$), $\dim U = 2$, and $\dim L = 1$ since $(1, 1, 1) \neq \mathbf{0}$. If $t(1, 1, 1) \in U$, then $t + t + t = 3t = 0$, so $t = 0$. Hence $U \cap L = \{\mathbf{0}\}$. By Theorem 1.7.4, $\dim(U + L) = 2 + 1 = 3$, so $U + L = \mathbb{R}^3$ by Theorem 1.6.8. Therefore $\mathbb{R}^3 = U \oplus L$.

To split (a, b, c) , look for t with $(a, b, c) - t(1, 1, 1) \in U$, that is, $(a-t) + (b-t) + (c-t) = 0$. This gives $t = \frac{a+b+c}{3}$, and

$$(a, b, c) = \left(a - \frac{a+b+c}{3}, b - \frac{a+b+c}{3}, c - \frac{a+b+c}{3}\right) + \frac{a+b+c}{3}(1, 1, 1).$$

The L -piece is the average of the entries times $(1, 1, 1)$; the U -piece is what is left over.

The same splitting trick, “average with a reflected copy”, works for functions and matrices.

Example. (Even and odd functions)

Let $V = \mathbb{R}^{\mathbb{R}}$ be the space of all functions $\mathbb{R} \rightarrow \mathbb{R}$, with $E = \{f : f(-x) = f(x) \text{ for all } x\}$ and $O = \{f : f(-x) = -f(x) \text{ for all } x\}$. Then $V = E \oplus O$.

Solution. Both are subspaces by the Subspace Test ([Theorem 1.2.1](#)): the zero function is even and odd, and the defining equations are preserved by sums and scalar multiples. For instance, if $f, g \in O$, then $(f+g)(-x) = -f(x) - g(x) = -(f+g)(x)$.

Sum. For any $f \in V$, put $f_E(x) = \frac{f(x)+f(-x)}{2}$ and $f_O(x) = \frac{f(x)-f(-x)}{2}$. Then $f = f_E + f_O$, and $f_E(-x) = \frac{f(-x)+f(x)}{2} = f_E(x)$, $f_O(-x) = \frac{f(-x)-f(x)}{2} = -f_O(x)$. So $V = E + O$.

Intersection. If $f \in E \cap O$, then $f(x) = f(-x) = -f(x)$ for every x , so $2f(x) = 0$ and $f(x) = 0$. Hence $E \cap O = \{0\}$, and $V = E \oplus O$ by [Theorem 1.7.4](#).

For example, $e^x = \cosh x + \sinh x$ is the splitting of the exponential function.

Example. (Symmetric and skew-symmetric matrices)

Let F be a field of characteristic not 2 (for instance $\mathbb{R}$ or $\mathbb{C}$), and let $n \geq 1$. In $M_n(F)$, let $U_+ = \{A : A^T = A\}$ be the symmetric matrices and $U_- = \{A : A^T = -A\}$ the skew-symmetric ones ([Definition 0.9.11](#)). Then $M_n(F) = U_+ \oplus U_-$.

Solution. Both are subspaces: $0^T = 0 = -0$, and by [Theorem 0.9.5](#), $(A+B)^T = A^T + B^T$ and $(cA)^T = cA^T$, so the equations $A^T = \pm A$ survive sums and scalar multiples.

Since the characteristic is not 2, we have $2 = 1+1 \neq 0$ in F , so $\frac{1}{2}$ exists ([Definition 0.7.2](#)). For $A \in M_n(F)$,

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T).$$

By [Theorem 0.9.5](#), $(\frac{1}{2}(A + A^T))^T = \frac{1}{2}(A^T + A)$ and $(\frac{1}{2}(A - A^T))^T = \frac{1}{2}(A^T - A) = -\frac{1}{2}(A - A^T)$. So the first piece is in U_+ , the second in U_- , and $M_n(F) = U_+ + U_-$.

If $A \in U_+ \cap U_-$, then $A = A^T = -A$, so $2A = 0$. Multiplying by $\frac{1}{2}$ gives $A = 0$. Hence $U_+ \cap U_- = \{0\}$, and $M_n(F) = U_+ \oplus U_-$ by [Theorem 1.7.4](#). The pattern is the same as even plus odd: transpose plays the role of $x \mapsto -x$.

The hypothesis on the characteristic is used twice, both times to divide by 2. [Exercise \(C1: Symmetric Plus Skew over \$\mathbb{F}_2\$ \)](#) shows what goes wrong over $\mathbb{F}_2$.

Quick check. Can two 2-dimensional subspaces of $\mathbb{R}^3$ form a direct sum?

Solution. No. If the sum $U + W$ were direct, [Theorem 1.7.4](#) (c) would give $\dim(U + W) = 2 + 2 = 4$. But $U + W \subseteq \mathbb{R}^3$, so $\dim(U + W) \leq 3$ by [Theorem 1.6.7](#). So two planes through $\mathbf{0}$ in $\mathbb{R}^3$ always share a non-zero vector.

1.7.4 Direct sums of several subspaces

For $k \geq 3$ subspaces, the tidy condition $U \cap W = \{\mathbf{0}\}$ has no direct replacement by pairwise intersections; we will see why in a moment. What does carry over is the zero vector test, which looks just like linear independence, and a condition that compares each subspace with the sum of **all the others**.

Theorem 1.7.5 — Direct Sum Criteria for Several Subspaces

Let $U_1, \dots, U_k$ be subspaces of a vector space V , and write $S = U_1 + \dots + U_k$. The following are equivalent:

- (a) the sum S is direct;
- (b) the only way to write $\mathbf{0} = \mathbf{u}_1 + \dots + \mathbf{u}_k$ with $\mathbf{u}_i \in U_i$ is $\mathbf{u}_1 = \dots = \mathbf{u}_k = \mathbf{0}$;
- (c) for **every** $j = 1, \dots, k$,

$$U_j \cap \sum_{i \neq j} U_i = \{\mathbf{0}\}.$$

If moreover each U_i is finite-dimensional with a basis $\mathcal{B}_i$, and $\mathcal{B}$ is the list obtained by writing $\mathcal{B}_1, \dots, \mathcal{B}_k$ one after another, then (a)–(c) are also equivalent to

- (d) $\mathcal{B}$ is a basis of S ;
- (e) $\dim S = \dim U_1 + \dots + \dim U_k$.

Here $\sum_{i \neq j} U_i$ is the sum of the $k - 1$ subspaces other than U_j (and $\{\mathbf{0}\}$ if $k = 1$).

Idea. Condition (b) is uniqueness for the single vector $\mathbf{0}$, and just as for independence, uniqueness for $\mathbf{0}$ gives uniqueness everywhere by taking differences. For (b) $\Leftrightarrow$ (c), a non-trivial way of writing $\mathbf{0}$ is the same thing as a non-zero vector $\mathbf{u}_j$ that equals minus the sum of the other pieces. For (d), a combination of $\mathcal{B}$ groups into one piece per U_i ; independence of each $\mathcal{B}_i$ turns “piece is $\mathbf{0}$ ” into “coefficients are 0”. Finally (d) $\Leftrightarrow$ (e) is count instead of check: $\mathcal{B}$ always spans S , so it is a basis exactly when its length is right.

Proof.

- (a) $\Rightarrow$ (b). The vector $\mathbf{0} \in S$ has the splitting $\mathbf{0} = \mathbf{0} + \dots + \mathbf{0}$. If the sum is direct, it has no other, which is (b).
- (b) $\Rightarrow$ (a). Assume (b), and let $\mathbf{u}_1 + \dots + \mathbf{u}_k = \mathbf{u}'_1 + \dots + \mathbf{u}'_k$ with $\mathbf{u}_i, \mathbf{u}'_i \in U_i$. Subtracting and regrouping,

$$(\mathbf{u}_1 - \mathbf{u}'_1) + \dots + (\mathbf{u}_k - \mathbf{u}'_k) = \mathbf{0},$$

where $\mathbf{u}_i - \mathbf{u}'_i \in U_i$ because U_i is a subspace. By (b), $\mathbf{u}_i - \mathbf{u}'_i = \mathbf{0}$, that is, $\mathbf{u}_i = \mathbf{u}'_i$,

for every i . Hence every vector of S has only one splitting.

- (c) $\Rightarrow$ (c). Assume (b), fix j , and let $\mathbf{v} \in U_j \cap \sum_{i \neq j} U_i$. Then $\mathbf{v} = \sum_{i \neq j} \mathbf{u}_i$ for some $\mathbf{u}_i \in U_i$, and so

$$\mathbf{u}_1 + \cdots + \mathbf{u}_{j-1} + (-\mathbf{v}) + \mathbf{u}_{j+1} + \cdots + \mathbf{u}_k = \mathbf{0},$$

with $-\mathbf{v} \in U_j$ in the j -th slot. By (b), every piece is $\mathbf{0}$; in particular $-\mathbf{v} = \mathbf{0}$, so $\mathbf{v} = \mathbf{0}$.

- (d) $\Rightarrow$ (b). Assume (c), and let $\mathbf{u}_1 + \cdots + \mathbf{u}_k = \mathbf{0}$ with $\mathbf{u}_i \in U_i$. Fix j . Then

$$\mathbf{u}_j = -\sum_{i \neq j} \mathbf{u}_i = \sum_{i \neq j} (-\mathbf{u}_i).$$

The left side lies in U_j , and the right side lies in $\sum_{i \neq j} U_i$ since each $-\mathbf{u}_i \in U_i$. By (c), $\mathbf{u}_j = \mathbf{0}$. As j was arbitrary, all pieces are $\mathbf{0}$.

For the rest, assume each U_i is finite-dimensional with basis $\mathcal{B}_i = (\mathbf{b}_{i1}, \dots, \mathbf{b}_{im_i})$, where $m_i = \dim U_i$. By [Proposition 1.7.2](#), $\mathcal{B}$ spans $S = \text{span}(\mathcal{B}_1) + \cdots + \text{span}(\mathcal{B}_k)$. In particular S is finite-dimensional.

- (b) $\Rightarrow$ (d). Since $\mathcal{B}$ spans S , it remains to prove independence. Let $\sum_{i=1}^k \sum_{r=1}^{m_i} a_{ir} \mathbf{b}_{ir} = \mathbf{0}$, and group the terms by subspace: put $\mathbf{u}_i = \sum_{r=1}^{m_i} a_{ir} \mathbf{b}_{ir} \in U_i$. Then $\mathbf{u}_1 + \cdots + \mathbf{u}_k = \mathbf{0}$, so each $\mathbf{u}_i = \mathbf{0}$ by (b). Since $\mathcal{B}_i$ is independent, $a_{i1} = \cdots = a_{im_i} = 0$ for each i . Hence $\mathcal{B}$ is independent, and so a basis of S .
- (c) $\Rightarrow$ (b). Assume (d), and let $\mathbf{u}_1 + \cdots + \mathbf{u}_k = \mathbf{0}$ with $\mathbf{u}_i \in U_i$. Since $\mathcal{B}_i$ spans U_i , write $\mathbf{u}_i = \sum_{r=1}^{m_i} a_{ir} \mathbf{b}_{ir}$. Then $\sum_{i=1}^k \sum_{r=1}^{m_i} a_{ir} \mathbf{b}_{ir} = \mathbf{0}$ is a combination of the independent list $\mathcal{B}$, so every $a_{ir} = 0$, and therefore every $\mathbf{u}_i = \mathbf{0}$.
- (d) $\Rightarrow$ (e). The list $\mathcal{B}$ has length $m_1 + \cdots + m_k$, so if it is a basis of S , then $\dim S = \dim U_1 + \cdots + \dim U_k$ by [Definition 1.6.1](#).
- (e) $\Rightarrow$ (d). The list $\mathcal{B}$ spans S and has length $m_1 + \cdots + m_k$, which equals $\dim S$ by (e). A spanning list whose length is the dimension is a basis ([Theorem 1.6.4](#)). This proves the theorem. ■

For $k = 2$, condition (c) reads $U_1 \cap U_2 = \{\mathbf{0}\}$ and $U_2 \cap U_1 = \{\mathbf{0}\}$, so we recover [Theorem 1.7.4](#). The criteria (d) and (e) say that **bases of the pieces assemble into a basis of a direct sum, and dimensions add**.

For $k \geq 3$, the tempting shortcut is to check only the intersections of the subspaces two at a time. It does not work.

Warning. Pairwise intersections $\{\mathbf{0}\}$ do not make a sum direct. In $\mathbb{R}^2$, take the three lines X (the x -axis), Y (the y -axis) and D (the line $y = x$). Any two of them meet only in $\mathbf{0}$. But

$$(1, 0) + (0, 1) + (-1, -1) = (0, 0)$$

writes $\mathbf{0}$ with non-zero pieces from X , Y , D , so (b) fails and $X + Y + D$ is **not** direct. Condition (c) sees the problem: $D \cap (X + Y) = D \cap \mathbb{R}^2 = D \neq \{\mathbf{0}\}$.

The same example breaks another tempting generalization. For finite sets, $|A \cup B \cup C|$ is given by inclusion–exclusion, and the dimension formula looks like its two-set case. For three subspaces

the analogous formula would be

$$\dim(U_1 + U_2 + U_3) \stackrel{?}{=} \sum_i \dim U_i - \sum_{i < j} \dim(U_i \cap U_j) + \dim(U_1 \cap U_2 \cap U_3).$$

For X, Y, D the left side is $\dim \mathbb{R}^2 = 2$, and the right side is $1 + 1 + 1 - 0 - 0 - 0 + 0 = 3$. So the formula is **false**. The reason is that $U + W$ is not a union: the dimension of $(U_1 + U_2) \cap U_3$ is not determined by the pairwise intersections. [Exercise \(C3: Three Subspaces\)](#) gives the correct replacement and a second failure.

1.7.5 Complements

Given a subspace U of V , can we always find a subspace W with $V = U \oplus W$? Such a W is called a **complement** of U in V . In $\mathbb{R}^3$, the plane U of [Example \(A plane and a line in \$\mathbb{R}^3\$ \)](#) has the complement $\text{span}((1, 1, 1))$. In finite dimension, complements always exist, and the proof is again “basis of the smaller space, then extend”.

Theorem 1.7.6 — Existence of Complements

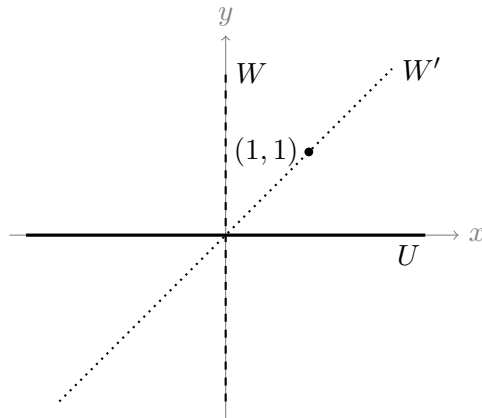
Let V be a finite-dimensional vector space and U a subspace of V . Then there is a subspace W of V with $V = U \oplus W$. Every such W satisfies $\dim W = \dim V - \dim U$.

Proof. By [Theorem 1.6.7](#), U is finite-dimensional; let $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ be a basis of U ([Corollary 1.5.5](#)). This list is independent in V , so by the Basis Extension Theorem ([Theorem 1.6.5](#)) there are $\mathbf{w}_1, \dots, \mathbf{w}_r \in V$ such that $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_r)$ is a basis of V . Put $W = \text{span}(\mathbf{w}_1, \dots, \mathbf{w}_r)$, a subspace by [Theorem 1.3.1](#).

By [Proposition 1.7.2](#), $U + W = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_r) = V$. The list $(\mathbf{w}_1, \dots, \mathbf{w}_r)$ is independent, since a relation $c_1 \mathbf{w}_1 + \dots + c_r \mathbf{w}_r = \mathbf{0}$ is also a relation $0\mathbf{u}_1 + \dots + 0\mathbf{u}_m + c_1 \mathbf{w}_1 + \dots + c_r \mathbf{w}_r = \mathbf{0}$ among the basis vectors of V , forcing all $c_k = 0$. So $(\mathbf{w}_1, \dots, \mathbf{w}_r)$ is a basis of W . The concatenation of the bases of U and W is a basis of $U + W = V$, so the sum is direct by [Theorem 1.7.5](#) ((d) $\Rightarrow$ (a)). Hence $V = U \oplus W$.

Finally, if W is any subspace with $V = U \oplus W$, then W is finite-dimensional ([Theorem 1.6.7](#)), and [Theorem 1.7.4](#) (c) gives $\dim V = \dim U + \dim W$, as claimed. ■

The proof made a choice: the vectors $\mathbf{w}_1, \dots, \mathbf{w}_r$ that extend the basis. Different choices give different complements.



Warning. A complement is not unique. In $\mathbb{R}^2$, the x -axis U (solid) has the complement W , the y -axis (dashed), and also the complement W' , the line $y = x$ (dotted), as shown in the examples after Definition 1.7.2. So “the complement of U ” is meaningless, and $U \oplus W = U \oplus W'$ does **not** allow us to cancel U and conclude $W = W'$. What all complements share is their dimension. Also, a complement is not the set complement $V \setminus U$, which does not even contain $\mathbf{0}$.

Complements matter because a splitting $V = U \oplus W$ lets us study V one piece at a time. In Chapter 3 each such splitting gives a projection of V onto U along W , and the warning above becomes visible there: different complements give different projections. In Chapter 8 the eigenspaces of a diagonalizable operator split V as a direct sum, and that is what makes such operators easy to understand.

1.7.6 Exercises

1.7.6.1 A. Check your understanding

Exercise A1.

- Define the sum $U_1 + \cdots + U_k$ of subspaces of V , and what it means for the sum to be direct.
- State the dimension formula for $\dim(U + W)$, with its hypotheses.
- True or false: if $V = U + W$ and $U \cap W = \{\mathbf{0}\}$, then $V = U \oplus W$. Justify your answer.
- True or false: if U_1, U_2, U_3 are subspaces with $U_i \cap U_j = \{\mathbf{0}\}$ for all $i \neq j$, then the sum $U_1 + U_2 + U_3$ is direct. Justify your answer.
- True or false: if U is a subspace of V with $U \neq V$, then $V \setminus U$ is a complement of U . Justify your answer.
- Name the method used to prove the dimension formula, in four steps.

Solution.

- See Definition 1.7.1: $U_1 + \cdots + U_k = \{\mathbf{u}_1 + \cdots + \mathbf{u}_k : \mathbf{u}_i \in U_i\}$. By Definition 1.7.2, the sum is direct if every vector in it can be written as $\mathbf{u}_1 + \cdots + \mathbf{u}_k$ with $\mathbf{u}_i \in U_i$ in exactly one way.

- (b) If U and W are finite-dimensional subspaces of a vector space V , then $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$ ([Theorem 1.7.3](#)).
- (c) True. By [Theorem 1.7.4](#), $U \cap W = \{\mathbf{0}\}$ means the sum $U + W$ is direct, and it equals V .
- (d) False. The three lines X, Y, D in $\mathbb{R}^2$ (the axes and the line $y = x$) meet pairwise only in $\mathbf{0}$, but $(1, 0) + (0, 1) + (-1, -1) = \mathbf{0}$ is a non-trivial way to write $\mathbf{0}$, so the sum is not direct by [Theorem 1.7.5](#).
- (e) False. $V \setminus U$ does not contain $\mathbf{0}$, because $\mathbf{0} \in U$. So it is not a subspace, and cannot be a complement.
- (f) Take a basis of the smallest space $U \cap W$; extend it to bases of U and of W (Basis Extension); prove the Big Claim that all the vectors together form a basis of $U + W$ (spanning, then independence by moving one block across and noting the common vector lies in $U \cap W$); count.

1.7.6.2 B. Practice

Exercise B1: Two Planes in $\mathbb{R}^4$.

Let $U = \text{span}((1, 2, 0, 1), (0, 1, 1, 0))$ and $W = \text{span}((1, 3, 1, 1), (0, 0, 1, 1))$ in $\mathbb{R}^4$.

- (a) Show that $\dim U = \dim W = 2$.
- (b) Find $U \cap W$ and its dimension.
- (c) Hence find $\dim(U + W)$ and a basis of $U + W$.

Solution.

- (a) If $a(1, 2, 0, 1) + b(0, 1, 1, 0) = (a, 2a + b, b, a) = \mathbf{0}$, then $a = 0$ and $b = 0$ from the first and third entries. So the spanning list of U is independent, hence a basis, and $\dim U = 2$. Similarly $c(1, 3, 1, 1) + d(0, 0, 1, 1) = (c, 3c, c + d, c + d) = \mathbf{0}$ forces $c = 0$, then $d = 0$, so $\dim W = 2$.
- (b) A vector lies in $U \cap W$ exactly when $(a, 2a + b, b, a) = (c, 3c, c + d, c + d)$ for some $a, b, c, d \in \mathbb{R}$. The first entries give $a = c$. The second gives $2a + b = 3a$, so $b = a$. The third gives $a = a + d$, so $d = 0$; the fourth, $a = a + 0$, then holds. So the common vectors are $(a, 3a, a, a) = a(1, 3, 1, 1)$, and $U \cap W = \text{span}((1, 3, 1, 1))$, of dimension 1.
- (c) By the dimension formula ([Theorem 1.7.3](#)), $\dim(U + W) = 2 + 2 - 1 = 3$. Following its proof: $((1, 3, 1, 1), (0, 1, 1, 0))$ is a list in U (by (b), $(1, 3, 1, 1) \in U \cap W$). It is independent by [Lemma 1.5.3](#), since a multiple $c(1, 3, 1, 1)$ with first entry 0 is $\mathbf{0} \neq (0, 1, 1, 0)$. It has length $2 = \dim U$, hence it is a basis of U by [Theorem 1.6.4](#). Likewise $((1, 3, 1, 1), (0, 0, 1, 1))$ is a basis of W . By the Big Claim in the proof, $((1, 3, 1, 1), (0, 1, 1, 0), (0, 0, 1, 1))$ is a basis of $U + W$.

Exercise B2: A Plane and a Line.

Let $U = \{(x, y, z) \in \mathbb{R}^3 : x - 2y + z = 0\}$ and $L = \text{span}((1, 0, 1))$.

- (a) Prove that $\mathbb{R}^3 = U \oplus L$.
 (b) Write $(2, 0, 4)$ as $\mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in U$ and $\mathbf{w} \in L$.

Solution.

- (a) Solving for x , every vector of U is $(2y - z, y, z) = y(2, 1, 0) + z(-1, 0, 1)$. These two vectors are independent (look at the second and third entries), so $\dim U = 2$. Since $(1, 0, 1) \neq \mathbf{0}$, $\dim L = 1$. If $t(1, 0, 1) \in U$, then $t - 0 + t = 2t = 0$, so $t = 0$; hence $U \cap L = \{\mathbf{0}\}$. By [Theorem 1.7.4](#), $\dim(U + L) = 2 + 1 = 3$, so $U + L = \mathbb{R}^3$ by [Theorem 1.6.8](#). Therefore $\mathbb{R}^3 = U \oplus L$.
 (b) We need t with $(2, 0, 4) - t(1, 0, 1) = (2 - t, 0, 4 - t) \in U$, that is, $(2 - t) - 0 + (4 - t) = 0$. This gives $t = 3$. So

$$(2, 0, 4) = (-1, 0, 1) + 3(1, 0, 1),$$

and indeed $-1 - 0 + 1 = 0$, so $(-1, 0, 1) \in U$. By (a) this is the only such splitting.

Exercise B3: Direct or Not?.

Determine which of the following sums are direct. Justify your answer.

- (a) $\text{span}(1, x^2) + \text{span}(x, x^3)$ in $\mathbb{R}[x]_{\leq 3}$.
 (b) $T_{\text{up}} + T_{\text{low}}$ in $M_2(\mathbb{R})$, where T_{up} and T_{low} are the upper and lower triangular matrices.
 (c) $\text{span}((1, 0, 0)) + \text{span}((0, 1, 0)) + \text{span}((1, 1, 1))$ in $\mathbb{R}^3$.
 (d) $\text{span}((1, 1, 0)) + \text{span}((0, 1, 1)) + \text{span}((1, 0, -1))$ in $\mathbb{R}^3$.

Solution.

- (a) Direct. If $p \in \text{span}(1, x^2) \cap \text{span}(x, x^3)$, then $p = a + bx^2 = cx + dx^3$. Comparing coefficients of $1, x, x^2, x^3$ gives $a = c = b = d = 0$, so $p = 0$. By [Theorem 1.7.4](#) the sum is direct.
 (b) Not direct. The identity matrix I_2 is both upper and lower triangular, so $I_2 \in T_{\text{up}} \cap T_{\text{low}} \neq \{\mathbf{0}\}$, and [Theorem 1.7.4](#) applies.
 (c) Direct. Each line has the basis given by its spanning vector, and we check that the concatenated list $((1, 0, 0), (0, 1, 0), (1, 1, 1))$ is independent. If $a(1, 0, 0) + b(0, 1, 0) + c(1, 1, 1) = (a + c, b + c, c) = \mathbf{0}$, then $c = 0$, hence $a = 0$ and $b = 0$. The list spans the sum by [Proposition 1.7.2](#), so it is a basis of the sum, and the sum is direct by [Theorem 1.7.5](#) ((d) $\Rightarrow$ (a)).
 (d) Not direct, although any two of the three lines meet only in $\mathbf{0}$ (no spanning vector is a multiple of another). Indeed

$$(1, 1, 0) + (-1)(0, 1, 1) + (-1)(1, 0, -1) = (0, 0, 0)$$

writes $\mathbf{0}$ as a sum of **non-zero** pieces from the three lines, so condition (b) of [Theorem 1.7.5](#) fails.

1.7.6.3 C. Going deeper

Exercise C1: Symmetric Plus Skew over $\mathbb{F}_2$.

Let U_+ and U_- be the symmetric and skew-symmetric matrices in $M_2(\mathbb{F}_2)$.

- (a) Show that $U_+ = U_-$.
- (b) Deduce that $M_2(\mathbb{F}_2) \neq U_+ \oplus U_-$. Which step of [Example \(Symmetric and skew-symmetric matrices\)](#) fails, and why?

Solution.

- (a) In $\mathbb{F}_2$ we have $1 + 1 = 0$, so $-a = a$ for every $a \in \mathbb{F}_2$, and hence $-A = A$ for every $A \in M_2(\mathbb{F}_2)$. Therefore $A^T = -A$ holds if and only if $A^T = A$, which says $U_- = U_+$.
- (b) By (a), $U_+ \cap U_- = U_+$, which contains $I_2 \neq 0$. So the sum is not direct ([Theorem 1.7.4](#)). It is not even all of $M_2(\mathbb{F}_2)$: $U_+ + U_- = U_+ + U_+ = U_+$, and $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ is not symmetric. In [Example \(Symmetric and skew-symmetric matrices\)](#) both halves of the proof divide by 2: the splitting $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$ and the step from $2A = 0$ to $A = 0$. In $\mathbb{F}_2$, $2 = 1 + 1 = 0$ has no inverse, so neither step is available. This is exactly the case of characteristic 2 excluded by the hypothesis.

Exercise C2: Subspaces That Must Meet.

- (a) Let V be a vector space of dimension n , and let U, W be subspaces with $\dim U + \dim W > n$. Prove that $U \cap W \neq \{\mathbf{0}\}$.
- (b) Deduce that any two 3-dimensional subspaces of $\mathbb{R}[x]_{\leq 4}$ contain a common non-zero polynomial.
- (c) Give an example showing that (a) fails if “ $>$ ” is replaced by “ $=$ ”.

Solution.

- (a) By [Theorem 1.6.7](#), U, W and $U + W$ are finite-dimensional, and $\dim(U + W) \leq \dim V = n$. By the dimension formula ([Theorem 1.7.3](#)),

$$\dim(U \cap W) = \dim U + \dim W - \dim(U + W) \geq \dim U + \dim W - n > 0.$$

A subspace of positive dimension has a basis with at least one vector, which is non-zero because a list containing $\mathbf{0}$ is dependent. Hence $U \cap W \neq \{\mathbf{0}\}$.

- (b) $\dim \mathbb{R}[x]_{\leq 4} = 5$ and $3 + 3 = 6 > 5$. By (a), the two subspaces meet in a non-zero polynomial.
- (c) In $\mathbb{R}^2$ (with $n = 2$), the x -axis and the y -axis have $1 + 1 = 2 = n$, but their intersection is $\{\mathbf{0}\}$.

Exercise C3: Three Subspaces.

- (a) Let U_1, U_2, U_3 be subspaces of a vector space V . Show that $(U_1 + U_2) + U_3 = U_1 + U_2 + U_3$.

(b) Now let U_1, U_2, U_3 be finite-dimensional. Prove that

$$\dim(U_1 + U_2 + U_3) = \dim U_1 + \dim U_2 + \dim U_3 - \dim(U_1 \cap U_2) - \dim((U_1 + U_2) \cap U_3).$$

(c) In $\mathbb{R}^3$, let $U_1 = \{z = 0\}$, $U_2 = \{y = 0\}$ and $U_3 = \{y = z\}$. Compute $\dim(U_1 + U_2 + U_3)$, every $\dim(U_i \cap U_j)$ and $\dim(U_1 \cap U_2 \cap U_3)$. Show that the inclusion–exclusion formula fails, and compare with (b).

Hint: for (b), apply the dimension formula twice.

Solution.

- (a) Both sides are subspaces by [Theorem 1.7.1](#). The right side contains U_1 and U_2 , hence $U_1 + U_2$ ([Theorem 1.7.1](#)), and it contains U_3 ; so it contains $(U_1 + U_2) + U_3$ by [Theorem 1.7.1](#). Conversely, an element $\mathbf{u}_1 + \mathbf{u}_2 + \mathbf{u}_3$ of the right side is the sum of $\mathbf{u}_1 + \mathbf{u}_2 \in U_1 + U_2$ and $\mathbf{u}_3 \in U_3$, so it lies in the left side.
- (b) By [Theorem 1.7.3](#), $U_1 + U_2$ is finite-dimensional with $\dim(U_1 + U_2) = \dim U_1 + \dim U_2 - \dim(U_1 \cap U_2)$. Applying the formula again to $U_1 + U_2$ and U_3 , and using (a),

$$\dim(U_1 + U_2 + U_3) = \dim(U_1 + U_2) + \dim U_3 - \dim((U_1 + U_2) \cap U_3).$$

Substituting the first equation into the second gives the formula.

(c) Each U_i is a plane, so $\dim U_i = 2$: $U_1 = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$, $U_2 = \text{span}(\mathbf{e}_1, \mathbf{e}_3)$, $U_3 = \text{span}(\mathbf{e}_1, \mathbf{e}_2 + \mathbf{e}_3)$, each with an evidently independent spanning list. All three contain the x -axis $\text{span}(\mathbf{e}_1)$.

- $U_1 \cap U_2 = \{y = z = 0\} = \text{span}(\mathbf{e}_1)$.
- $U_1 \cap U_3 = \{z = 0, y = z\} = \{y = z = 0\} = \text{span}(\mathbf{e}_1)$.
- $U_2 \cap U_3 = \{y = 0, y = z\} = \text{span}(\mathbf{e}_1)$.
- $U_1 \cap U_2 \cap U_3 = \text{span}(\mathbf{e}_1)$.

So each of these has dimension 1. Since $U_1 + U_2$ contains $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, it is $\mathbb{R}^3$, and so $U_1 + U_2 + U_3 = \mathbb{R}^3$ has dimension 3. The inclusion–exclusion formula would give

$$2 + 2 + 2 - 1 - 1 - 1 + 1 = 4 \neq 3,$$

so it fails. In (b), the last term is $(U_1 + U_2) \cap U_3 = \mathbb{R}^3 \cap U_3 = U_3$, of dimension 2, and indeed $2 + 2 + 2 - 1 - 2 = 3$. The inclusion–exclusion guess replaces this 2 by $\dim(U_1 \cap U_3) + \dim(U_2 \cap U_3) - \dim(U_1 \cap U_2 \cap U_3) = 1 + 1 - 1 = 1$, which is wrong because $(U_1 + U_2) \cap U_3$ is much bigger than $(U_1 \cap U_3) + (U_2 \cap U_3) = \text{span}(\mathbf{e}_1)$.

1.8 Infinite-Dimensional Spaces

Every theorem about bases so far has assumed a finite spanning list. Yet several of our running examples have none: $F[x]$ is not finite-dimensional ([Theorem 1.3.3](#)), and neither are the sequences or the functions $\mathbb{R} \rightarrow \mathbb{R}$. Do such spaces have a basis at all? This section shows that they always do, using a principle from set theory called Zorn's Lemma, and then looks honestly at the price. The basis exists, but often nobody can write it down, and several finite-dimensional

facts stop being true.

1.8.1 Bases of infinite sets of vectors

An infinite-dimensional space cannot have a finite basis, so a basis now has to be an infinite collection of vectors. We recall the conventions for arbitrary subsets, set up with the span (Definition 1.3.2) and with independence (Definition 1.4.1). A linear combination always involves **finitely many** vectors, because the axioms only let us add two vectors at a time.

Let $S \subseteq V$ be any subset, finite or infinite.

- $\text{span}(S)$ is the set of all linear combinations $a_1\mathbf{v}_1 + \cdots + a_k\mathbf{v}_k$ with $k \in \mathbb{N}$, $a_i \in F$ and $\mathbf{v}_i \in S$.
- S is **linearly independent** if **every finite list of distinct vectors** of S is linearly independent in the sense of Definition 1.4.1.
- S is a **basis** of V if S is linearly independent and $\text{span}(S) = V$.

For a finite set $S = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ of distinct vectors, these agree with the notions for the list $(\mathbf{v}_1, \dots, \mathbf{v}_k)$, since a sub-list of an independent list is independent. The empty set is independent: its only finite list of distinct vectors is the empty list. In the infinite case we lose the ordering of a basis, and we will not need it.

Example. (A Basis of $F[x]$)

Show that $\{x^k : k \in \mathbb{N}\} = \{1, x, x^2, \dots\}$ is a basis of $F[x]$.

Solution. Spanning. By Definition 0.8.1, every polynomial can be written as $a_0 + a_1x + \cdots + a_Nx^N$ for some $N \in \mathbb{N}$. This is a finite linear combination of $1, x, \dots, x^N$.

Independence. Let $x^{k_1}, \dots, x^{k_m}$ be distinct monomials and suppose $a_1x^{k_1} + \cdots + a_mx^{k_m} = 0$. The left side is the polynomial whose coefficient of x^{k_j} is a_j , because the exponents k_j are distinct. The zero polynomial has every coefficient equal to 0, so by the definition of equality of polynomials, $a_j = 0$ for every j .

Hence $\{x^k : k \in \mathbb{N}\}$ is a basis of $F[x]$.

The sequence space behaves differently. Let $F^{\mathbb{N}}$ be the space of all sequences $(a_0, a_1, a_2, \dots)$ with entries in F , with entrywise addition and scaling. For $i \in \mathbb{N}$ let $\mathbf{e}_i$ be the sequence with 1 in position i and 0 everywhere else.

Example. (The Vectors $\mathbf{e}_i$ Do Not Span $F^{\mathbb{N}}$)

Show that $\{\mathbf{e}_i : i \in \mathbb{N}\}$ is linearly independent in $F^{\mathbb{N}}$, but is **not** a basis of $F^{\mathbb{N}}$.

Solution. Independence. Let $i_1, \dots, i_m$ be distinct and suppose $a_1\mathbf{e}_{i_1} + \cdots + a_m\mathbf{e}_{i_m} = \mathbf{0}$. Position i_j of the left side is a_j , since only $\mathbf{e}_{i_j}$ has a non-zero entry there. Position i_j of the zero sequence is 0, so $a_j = 0$ for every j .

Not spanning. A linear combination $a_1\mathbf{e}_{i_1} + \cdots + a_m\mathbf{e}_{i_m}$ has non-zero entries only in

the m positions $i_1, \dots, i_m$. So every vector in $\text{span}\{\mathbf{e}_i : i \in \mathbb{N}\}$ has only finitely many non-zero entries. The sequence $(1, 1, 1, \dots)$ has infinitely many, so it is not in the span.

The same example shows that $F^{\mathbb{N}}$ is infinite-dimensional. For each n , the list $(\mathbf{e}_0, \dots, \mathbf{e}_n)$ is independent of length $n + 1$. If a list of length m spanned $F^{\mathbb{N}}$, then $n + 1 \leq m$ for every n by [Theorem 1.6.1](#), which is impossible.

Warning. It is tempting to write $(1, 1, 1, \dots) = \mathbf{e}_0 + \mathbf{e}_1 + \mathbf{e}_2 + \dots$. In a vector space this sum **means nothing**: the axioms define sums of two vectors, hence of finitely many, and an infinite sum would need a notion of limit. So “the $\mathbf{e}_i$ span” is false, even though every sequence is “made of” its entries. Infinite sums return only in spaces with a notion of distance, much later.

Quick check. Let V be the space of formal power series $a_0 + a_1x + a_2x^2 + \dots$ over F , that is, $F^{\mathbb{N}}$ with x^k standing for $\mathbf{e}_k$. Is $\{1, x, x^2, \dots\}$ a basis of V ?

Solution. No. It is independent, by the argument of [Example \(The Vectors \$\mathbf{e}_i\$ Do Not Span \$F^{\mathbb{N}}\$ \)](#). But its span consists of the series with only finitely many non-zero coefficients, which are the polynomials. The series $1 + x + x^2 + \dots$ is not among them, so the set does not span V .

So does $F^{\mathbb{N}}$ have a basis? In finite dimension we built bases by sifting ([Theorem 1.5.4](#)) or by extending one vector at a time ([Theorem 1.6.5](#)). Both constructions stop after finitely many steps. For $F^{\mathbb{N}}$ no step-by-step process can stop, and we need a different kind of existence argument.

1.8.2 Zorn’s Lemma

Look again at what a basis is in finite dimension: an independent list that cannot be enlarged ([Theorem 1.6.6](#)). “Cannot be enlarged” is a statement about the family of all independent sets, ordered by inclusion. Zorn’s Lemma is a general principle that produces objects which cannot be enlarged. To state it we need words for orderings.

A partial order is a way of comparing elements in which some pairs may be incomparable.

Definition 1.8.1 — Partial Order, Chain, Upper Bound, Maximal Element

Let P be a set. A **partial order** on P is a relation $\leq$ on P ([Definition 0.5.1](#)) such that, for all $p, q, r \in P$:

- (P1) $p \leq p$ (reflexive);
- (P2) if $p \leq q$ and $q \leq p$, then $p = q$ (antisymmetric);
- (P3) if $p \leq q$ and $q \leq r$, then $p \leq r$ (transitive).

Let $(P, \leq)$ be a partially ordered set.

- A **chain** in P is a subset $C \subseteq P$ such that for all $p, q \in C$, $p \leq q$ or $q \leq p$.
- An **upper bound** of a subset $C \subseteq P$ is an element $u \in P$ with $c \leq u$ for every

$c \in C$.

- A **maximal element** of P is an element $m \in P$ such that **no** $p \in P$ satisfies $m \leq p$ and $p \neq m$.

In words: a chain is a subset in which any two elements can be compared. An upper bound of C sits above everything in C , but it only has to lie in P , **not** in C . A maximal element has nothing strictly above it. It is **not** required to lie above everything else.

Example. (Inclusion as a Partial Order)

Let X be a set and P the power set of X (Definition 0.3.12), ordered by $\subseteq$. Check that this is a partial order, and describe chains, upper bounds and maximal elements when $X = \{1, 2\}$. Then do the same for the subsets of $\{1, 2\}$ with **at most one** element.

Solution. Every set satisfies $A \subseteq A$, which is (P1). If $A \subseteq B$ and $B \subseteq A$, then $A = B$ by Theorem 0.3.1, which is (P2). Inclusion is transitive, which is (P3).

For $X = \{1, 2\}$, the set $C = \{\emptyset, \{1\}, \{1, 2\}\}$ is a chain, while $\{1\}$ and $\{2\}$ are incomparable, so $\{\{1\}, \{2\}\}$ is not a chain. The set $\{1, 2\}$ is an upper bound of every subset of P , and it is the only maximal element.

Now let $Q = \{\emptyset, \{1\}, \{2\}\}$, still ordered by $\subseteq$. Both $\{1\}$ and $\{2\}$ are maximal: nothing in Q strictly contains either. Neither is above the other. The subset $\{\{1\}, \{2\}\}$ has no upper bound in Q .

Theorem 1.8.1 — Zorn's Lemma

Let $(P, \leq)$ be a **non-empty** partially ordered set in which **every non-empty chain** has an upper bound in P . Then P has a maximal element.

We do not prove Zorn's Lemma, and no proof is possible from the other everyday rules of set theory. It is equivalent to the **axiom of choice**, the principle already used to build right inverses of surjections in Theorem 0.4.6: from any family of non-empty sets one may choose one element from each. Mathematicians accept it as an axiom, and we do too. In this section we point out each place where it is used.

Warning. “Maximal” is not “largest”. The partially ordered set Q in Example (Inclusion as a Partial Order) has two maximal elements and no largest one. For bases this is the familiar fact that a space has many bases: in $\mathbb{R}^2$, both $\{\mathbf{e}_1, \mathbf{e}_2\}$ and $\{(1, 1), (1, -1)\}$ are maximal independent sets, and neither contains the other.

1.8.3 Every vector space has a basis

We prove the more general extension statement first, since the existence of a basis is the case $S = \emptyset$.

Theorem 1.8.2 — Basis Extension, General Form

Let V be a vector space over F , and let $S \subseteq V$ be a linearly independent subset. Then there is a basis B of V with $S \subseteq B$.

Idea. ① Let P be the family of independent sets containing S , ordered by $\subseteq$. ② A chain has the union as an upper bound, because an independence test uses only finitely many vectors, and finitely many members of a chain all fit inside one of them. ③ Zorn gives a maximal B . ④ If some $\mathbf{v}$ were outside $\text{span}(B)$, then $B \cup \{\mathbf{v}\}$ would be a bigger independent set, so B spans.

Proof. Let P be the set of all linearly independent subsets $T \subseteq V$ with $S \subseteq T$, partially ordered by $\subseteq$ (see [Example \(Inclusion as a Partial Order\)](#)). Since S is independent, $S \in P$, so P is non-empty.

Step 1: every non-empty chain has an upper bound. Let $C \subseteq P$ be a non-empty chain and let $U = \bigcup_{T \in C} T$. Every $T \in C$ contains S and C is non-empty, so $S \subseteq U$, and $T \subseteq U$ for every $T \in C$.

Claim. If $T_1, \dots, T_k \in C$ with $k \geq 1$, then one of them contains all the others.

Proof. We use induction on k . For $k = 1$ there is nothing to prove. Suppose the claim holds for k , and let $T_1, \dots, T_{k+1} \in C$. By the induction hypothesis some T_j contains $T_1, \dots, T_k$. Since C is a chain, $T_j \subseteq T_{k+1}$ or $T_{k+1} \subseteq T_j$. In the first case T_{k+1} contains all of $T_1, \dots, T_{k+1}$, and in the second case T_j does. ■

Now let $(\mathbf{u}_1, \dots, \mathbf{u}_k)$ be a list of distinct vectors of U . If $k = 0$, the empty list is independent. Otherwise, each $\mathbf{u}_i$ lies in some $T_i \in C$, and by the Claim one of $T_1, \dots, T_k$, say T_j , contains all the $\mathbf{u}_i$. Since T_j is independent, the list $(\mathbf{u}_1, \dots, \mathbf{u}_k)$ is independent. Hence U is independent, so $U \in P$, and U is an upper bound of C .

Step 2: a maximal element. By Step 1 and [Theorem 1.8.1](#), P has a maximal element B . Thus B is independent and $S \subseteq B$.

Step 3: B spans V . Suppose not, and let $\mathbf{v} \in V$ with $\mathbf{v} \notin \text{span}(B)$. Then $\mathbf{v} \notin B$, since $B \subseteq \text{span}(B)$. We show $B \cup \{\mathbf{v}\}$ is independent. Take a list of distinct vectors of $B \cup \{\mathbf{v}\}$. If $\mathbf{v}$ is not in the list, the list lies in B and is independent. Otherwise, after reordering (which does not affect independence, as addition is commutative), the list is $(\mathbf{v}, \mathbf{w}_1, \dots, \mathbf{w}_m)$ with distinct $\mathbf{w}_i \in B$. Let

$$a\mathbf{v} + b_1\mathbf{w}_1 + \dots + b_m\mathbf{w}_m = \mathbf{0}.$$

If $a \neq 0$, then $\mathbf{v} = -a^{-1}(b_1\mathbf{w}_1 + \dots + b_m\mathbf{w}_m) \in \text{span}(B)$, a contradiction. So $a = 0$, and then $b_1 = \dots = b_m = 0$ because B is independent. Hence $B \cup \{\mathbf{v}\} \in P$, it contains B , and it is not equal to B . This contradicts the maximality of B . Therefore $\text{span}(B) = V$, and B is a basis of V containing S . ■

Notice where finiteness went. Steps 1 and 3 each look at only finitely many vectors at a time, which is all that linear algebra ever does. The one infinite step, “keep enlarging until you cannot”, is handed to Zorn’s Lemma.

Theorem 1.8.3 — Every Vector Space Has a Basis

Every vector space V over a field F has a basis.

Proof. The empty set is linearly independent. By Theorem 1.8.2 with $S = \emptyset$, V has a basis. For $V = \{\mathbf{0}\}$ that basis is $\emptyset$, since $\text{span}(\emptyset) = \{\mathbf{0}\}$. ■

So $F^{\mathbb{N}}$ has a basis, and so do $\mathbb{R}^{\mathbb{R}}$ and the continuous functions on $[0, 1]$. The theorem does not tell us what these bases are.

1.8.4 Existence without construction: $\mathbb{R}$ over $\mathbb{Q}$

The real numbers form a vector space over $\mathbb{Q}$: add reals as usual, and multiply a real by a rational. By Theorem 1.8.3 this space has a basis, often called a **Hamel basis**. Nobody has ever written one down, and nobody will. It is known that the existence of such a basis cannot be proved without some form of the axiom of choice. What we **can** prove is that this basis is infinite. The tool is counting.

A countable set is one whose elements can be listed as a sequence, possibly with repeats.

Definition 1.8.2 — Countable Set

A set A is **countable** if $A = \emptyset$ or there is a **surjective** function $\mathbb{N} \rightarrow A$. Otherwise A is **uncountable**.

A surjection $\sigma: \mathbb{N} \rightarrow A$ lists A as $\sigma(0), \sigma(1), \sigma(2), \dots$, and every element of A appears somewhere. Repeats are allowed, so every finite non-empty set is countable. The next lemma collects what we need.

Lemma 1.8.4 — Countable Sets (a) If A is countable and $f: A \rightarrow A'$ is surjective, then A' is countable.

(b) $\mathbb{N} \times \mathbb{N}$ is countable.

(c) If A and A' are countable, then $A \times A'$ is countable.

(d) $\mathbb{Q}$ is countable, and $\mathbb{Q}^n$ is countable for every $n \geq 1$.

Proof.

- (a) If $A = \emptyset$, then $A' = f(A) = \emptyset$. Otherwise take a surjection $\sigma: \mathbb{N} \rightarrow A$. Then $f \circ \sigma: \mathbb{N} \rightarrow A'$ is surjective by Theorem 0.4.3 (a composite of surjections is surjective).
- (b) For $n \in \mathbb{N}$, let 2^a be the largest power of 2 dividing $n+1$; it exists because $2^a \leq n+1$ bounds a . Then $(n+1)/2^a$ is odd, so it equals $2b+1$ for a unique $b \in \mathbb{N}$. Put $g(n) = (a, b)$. Given $(a, b) \in \mathbb{N} \times \mathbb{N}$, let $n = 2^a(2b+1) - 1$. The largest power of 2 dividing $2^a(2b+1)$ is 2^a , because $2^{a+1} \mid 2^a(2b+1)$ would force $2 \mid 2b+1$. Hence $g(n) = (a, b)$, and g is surjective.

- (c) If A or A' is empty, so is $A \times A'$. Otherwise take surjections $\sigma: \mathbb{N} \rightarrow A$ and $\tau: \mathbb{N} \rightarrow A'$. The map $\mathbb{N} \times \mathbb{N} \rightarrow A \times A'$, $(m, n) \mapsto (\sigma(m), \tau(n))$, is surjective. By (b) and (a), $A \times A'$ is countable.
- (d) The map $\mathbb{N} \rightarrow \mathbb{Z}$ sending an even m to $m/2$ and an odd m to $-(m+1)/2$ is surjective, so $\mathbb{Z}$ is countable. The map $\mathbb{Z} \times \mathbb{N} \rightarrow \mathbb{Q}$, $(p, q) \mapsto p/(q+1)$, is surjective, so $\mathbb{Q}$ is countable by (c) and (a). Finally $\mathbb{Q}^{n+1}$ is $\mathbb{Q}^n \times \mathbb{Q}$ after identifying $(q_1, \dots, q_{n+1})$ with $((q_1, \dots, q_n), q_{n+1})$, so induction on n with (c) shows that $\mathbb{Q}^n$ is countable for every $n \geq 1$.

■

The real numbers are not countable. This is a fact about $\mathbb{R}$, not about linear algebra, and its proof needs one property of $\mathbb{R}$ that we take from analysis: the **nested interval property**. If $I_0 \supseteq I_1 \supseteq I_2 \supseteq \dots$ are closed bounded intervals $[a_n, b_n]$ with $a_n \leq b_n$, then some real number lies in every I_n .

Lemma 1.8.5 — $\mathbb{R}$ Is Uncountable

Assuming the nested interval property, there is no surjection $\mathbb{N} \rightarrow \mathbb{R}$.

Proof. Let $r: \mathbb{N} \rightarrow \mathbb{R}$ be any function. We build closed intervals $I_0 \supseteq I_1 \supseteq \dots$ of positive length with $r(n) \notin I_n$. Split $[0, 1]$ into the thirds $[0, \frac{1}{3}]$ and $[\frac{2}{3}, 1]$. They are disjoint, so $r(0)$ misses at least one of them; let I_0 be such a third. Given $I_n = [a, b]$ with $a < b$, the intervals $[a, a + \frac{b-a}{3}]$ and $[b - \frac{b-a}{3}, b]$ are disjoint and lie in I_n , so $r(n+1)$ misses one of them; let I_{n+1} be that one. By the nested interval property some x lies in every I_n . For each n , $x \in I_n$ and $r(n) \notin I_n$, so $x \neq r(n)$. Hence x is not a value of r , and r is not surjective.

■

Theorem 1.8.6 — $\mathbb{R}$ Is Infinite-Dimensional over $\mathbb{Q}$

The vector space $\mathbb{R}$ over $\mathbb{Q}$ is not finite-dimensional.

Proof. Suppose a finite list spans $\mathbb{R}$ over $\mathbb{Q}$. Appending 1 keeps it spanning, so we may take a spanning list $(b_1, \dots, b_n)$ with $n \geq 1$. Then the map

$$\mathbb{Q}^n \rightarrow \mathbb{R}, \quad (q_1, \dots, q_n) \mapsto q_1 b_1 + \dots + q_n b_n$$

is surjective, since every real number is a rational linear combination of $b_1, \dots, b_n$. By Lemma 1.8.4 (d) and (a), $\mathbb{R}$ is countable. This contradicts Lemma 1.8.5. Hence $\mathbb{R}$ is not finite-dimensional over $\mathbb{Q}$.

■

The proof really shows more: **every** finite-dimensional vector space over $\mathbb{Q}$ is countable. It is honest about its inputs. Linear algebra supplied only “a spanning list gives a surjection from $\mathbb{Q}^n$ ”; the rest is counting and one property of the real line.

Quick check. Is $\mathbb{C}$ finite-dimensional as a vector space over $\mathbb{Q}$?

Solution. No. If $\mathbb{C}$ were countable, then $\mathbb{R} \subseteq \mathbb{C}$ would be countable too: pick any $y_0 \in \mathbb{R}$ and send $z \in \mathbb{C}$ to z if $z \in \mathbb{R}$ and to y_0 otherwise, a surjection $\mathbb{C} \rightarrow \mathbb{R}$, and apply [Lemma 1.8.4](#) (a). That contradicts [Lemma 1.8.5](#). So $\mathbb{C}$ is uncountable, and the proof of [Theorem 1.8.6](#), with $\mathbb{C}$ in place of $\mathbb{R}$, shows it is not finite-dimensional over $\mathbb{Q}$.

1.8.5 What changes in infinite dimension

Several finite-dimensional theorems use a count, and counting is exactly what fails here.

Warning. **A proper subspace can be as big as the whole space.** In finite dimension, $U \subseteq V$ with $\dim U = \dim V$ forces $U = V$ ([Theorem 1.6.8](#)). Now let $U = \{p \in F[x] : \text{the constant coefficient of } p \text{ is } 0\}$. By the argument of [Example \(A Basis of \$F\[x\]\$ \)](#), U has the basis $\{x, x^2, x^3, \dots\}$, and $x^k \leftrightarrow x^{k+1}$ matches it one-to-one with the basis $\{1, x, x^2, \dots\}$ of $F[x]$. Yet $1 \notin U$, so $U \neq F[x]$. The same happens in $F^{\mathbb{N}}$: the sequences with first entry 0 form a proper subspace, and deleting that first entry matches it one-to-one with all of $F^{\mathbb{N}}$, respecting addition and scaling. In Chapter 3 this becomes the shift map, which is injective but not surjective.

Remark. In an infinite-dimensional space, “ $\dim V = \infty$ ” hides real differences of size: $F[x]$ has a countable basis, while every basis of $\mathbb{R}$ over $\mathbb{Q}$ is uncountable ([Exercise \(C2: Bases of \$\mathbb{R}\$ over \$\mathbb{Q}\$ Are Uncountable\)](#)), and set theory shows that all bases of one space have the same cardinality, a fact we will not develop.

Scope. From here on the book is mostly about finite-dimensional spaces, where bases are finite lists and counting works. Every theorem that needs finite dimension says so in its hypotheses. Infinite-dimensional spaces return with more structure, such as inner products and limits, where infinite sums can finally be given a meaning.

1.8.6 Exercises

1.8.6.1 A. Check your understanding

Exercise A1.

- Define what it means for an arbitrary subset S of a vector space V to be a basis of V .
- State Zorn’s Lemma, including the meaning of “chain” and “upper bound”.
- True or false: $\{\mathbf{e}_i : i \in \mathbb{N}\}$ is a basis of $F^{\mathbb{N}}$. Justify your answer.
- True or false: every linearly independent subset of an infinite-dimensional space is infinite. Justify your answer.
- True or false: in a partially ordered set, a maximal element m satisfies $p \leq m$ for every p . Justify your answer.

Solution.

- (a) S is a basis if every finite list of distinct vectors of S is linearly independent, and every vector of V is a linear combination of finitely many vectors of S .
- (b) See [Theorem 1.8.1](#) and [Definition 1.8.1](#): a non-empty partially ordered set in which every non-empty chain (a subset whose elements are pairwise comparable) has an upper bound (an element of P above every element of the chain) has a maximal element.
- (c) False. The set is independent, but $(1, 1, 1, \dots)$ is not a finite linear combination of the $\mathbf{e}_i$ ([Example \(The Vectors \$\mathbf{e}_i\$ Do Not Span \$F^{\mathbb{N}}\$ \)](#)).
- (d) False. The empty set and $\{x\} \subseteq F[x]$ are finite independent subsets of the infinite-dimensional space $F[x]$.
- (e) False. In $Q = \{\emptyset, \{1\}, \{2\}\}$ ordered by $\subseteq$, the element $\{1\}$ is maximal, but $\{2\} \not\subseteq \{1\}$ ([Example \(Inclusion as a Partial Order\)](#)).

1.8.6.2 B. Practice

Exercise B1: Finitely Supported Sequences.

Let $F^{(\mathbb{N})} \subseteq F^{\mathbb{N}}$ be the set of sequences with only finitely many non-zero entries.

- (a) Prove that $F^{(\mathbb{N})}$ is a subspace of $F^{\mathbb{N}}$.
- (b) Prove that $\{\mathbf{e}_i : i \in \mathbb{N}\}$ is a basis of $F^{(\mathbb{N})}$.

Solution.

- (a) We use [Theorem 1.2.1](#). (1) The zero sequence has no non-zero entries, so $\mathbf{0} \in F^{(\mathbb{N})}$. (2) Let $\mathbf{x}, \mathbf{y} \in F^{(\mathbb{N})}$, with non-zero entries only in finite sets of positions A and A' . If $i \notin A \cup A'$, then $x_i + y_i = 0 + 0 = 0$. So $\mathbf{x} + \mathbf{y}$ has non-zero entries only in the finite set $A \cup A'$, and $\mathbf{x} + \mathbf{y} \in F^{(\mathbb{N})}$. (3) Let $c \in F$. If $i \notin A$, then $cx_i = c \cdot 0 = 0$, so $c\mathbf{x} \in F^{(\mathbb{N})}$. Hence $F^{(\mathbb{N})}$ is a subspace.
- (b) Each $\mathbf{e}_i$ lies in $F^{(\mathbb{N})}$, and the set is independent by [Example \(The Vectors \$\mathbf{e}_i\$ Do Not Span \$F^{\mathbb{N}}\$ \)](#). For spanning, let $\mathbf{x} \in F^{(\mathbb{N})}$ have non-zero entries only in positions $i_1, \dots, i_m$. Then $\mathbf{x}$ and $x_{i_1}\mathbf{e}_{i_1} + \dots + x_{i_m}\mathbf{e}_{i_m}$ agree in positions $i_1, \dots, i_m$ and are both 0 elsewhere, so they are equal. Hence $\{\mathbf{e}_i : i \in \mathbb{N}\}$ is a basis of $F^{(\mathbb{N})}$.

Exercise B2: Exponentials Are Independent.

For $k \in \mathbb{N}$ let $f_k \in \mathbb{R}^{\mathbb{R}}$ be $f_k(x) = e^{kx}$. Prove that $\{f_k : k \in \mathbb{N}\}$ is linearly independent. Hence deduce that $\mathbb{R}^{\mathbb{R}}$ is infinite-dimensional.

Hint: divide by the fastest-growing term.

Solution. Suppose not. Then there are distinct $k_1 < k_2 < \dots < k_m$ and scalars $a_1, \dots, a_m$, not all zero, with $a_1 e^{k_1 x} + \dots + a_m e^{k_m x} = 0$ for every $x \in \mathbb{R}$. Discarding terms with zero coefficient, we may assume $a_m \neq 0$. Dividing by $e^{k_m x} > 0$ gives, for

every $x \in \mathbb{R}$,

$$a_m = - \sum_{j=1}^{m-1} a_j e^{(k_j - k_m)x}.$$

Since $k_j - k_m < 0$ for $j < m$, each $e^{(k_j - k_m)x} \rightarrow 0$ as $x \rightarrow \infty$, so the right side tends to 0. The left side is the constant a_m , so $a_m = 0$, a contradiction. Hence the set is independent.

For each n , the list $(f_0, \dots, f_n)$ is independent of length $n + 1$. If a list of length m spanned $\mathbb{R}^{\mathbb{R}}$, [Theorem 1.6.1](#) would give $n + 1 \leq m$ for every n , which is impossible. Hence $\mathbb{R}^{\mathbb{R}}$ is infinite-dimensional.

Exercise B3: Continuous Functions.

Let $C[0, 1]$ be the space of continuous functions $[0, 1] \rightarrow \mathbb{R}$. Prove that $C[0, 1]$ is infinite-dimensional.

Hint: use “tent” functions supported on disjoint intervals, and choose good points.

Solution. For $k \geq 1$, let $m_k = \frac{1}{2} \left(\frac{1}{k+1} + \frac{1}{k} \right)$ and $r_k = \frac{1}{2} \left(\frac{1}{k} - \frac{1}{k+1} \right) > 0$, the midpoint and half-length of $J_k = [\frac{1}{k+1}, \frac{1}{k}]$. Define

$$t_k(x) = \max(0, r_k - |x - m_k|).$$

This is continuous, as built from continuous functions by $|\cdot|$ and $\max$. Also $t_k(x) > 0$ exactly when $|x - m_k| < r_k$, that is, when x lies in the open interval $(\frac{1}{k+1}, \frac{1}{k})$. These open intervals are pairwise disjoint, and m_j lies in the j -th one. Hence $t_k(m_j) = 0$ for $j \neq k$, while $t_k(m_k) = r_k$.

Let $a_1 t_1 + \dots + a_n t_n = 0$. Evaluating at m_j gives $a_j r_j = 0$, so $a_j = 0$ since $r_j \neq 0$. Hence $(t_1, \dots, t_n)$ is independent for every n . As in [Exercise \(B2: Exponentials Are Independent\)](#), [Theorem 1.6.1](#) shows that no finite list spans $C[0, 1]$.

1.8.6.3 C. Going deeper

Exercise C1: Complements in Any Dimension.

Let U be a subspace of a vector space V , of any dimension. Prove that there is a subspace W of V with $V = U \oplus W$, that is, every $\mathbf{v} \in V$ can be written **uniquely** as $\mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in U$, $\mathbf{w} \in W$.

Hint: [Theorem 1.8.2](#).

Solution. By [Theorem 1.8.3](#), U has a basis B_U . It is independent in V , so by [Theorem 1.8.2](#) there is a basis B of V with $B_U \subseteq B$. Let $W = \text{span}(B \setminus B_U)$, a subspace by [Theorem 1.3.1](#).

Existence. Let $\mathbf{v} \in V$. Since B spans, $\mathbf{v} = \sum_i a_i \mathbf{u}_i + \sum_j b_j \mathbf{w}_j$, a finite sum with $\mathbf{u}_i \in B_U$ and $\mathbf{w}_j \in B \setminus B_U$. The first sum lies in U and the second in W .

$U \cap W = \{\mathbf{0}\}$. Let $\mathbf{v} \in U \cap W$. Since B_U spans U and $B \setminus B_U$ spans W , we have

$\mathbf{v} = \sum_i a_i \mathbf{u}_i = \sum_j b_j \mathbf{w}_j$ with distinct $\mathbf{u}_i \in B_U$ and distinct $\mathbf{w}_j \in B \setminus B_U$. Then $\sum_i a_i \mathbf{u}_i - \sum_j b_j \mathbf{w}_j = \mathbf{0}$ is a combination of a list of distinct vectors of B , because B_U and $B \setminus B_U$ are disjoint. Since B is independent, every a_i and b_j is 0, so $\mathbf{v} = \mathbf{0}$.

Uniqueness. If $\mathbf{u} + \mathbf{w} = \mathbf{u}' + \mathbf{w}'$ with $\mathbf{u}, \mathbf{u}' \in U$ and $\mathbf{w}, \mathbf{w}' \in W$, then $\mathbf{u} - \mathbf{u}' = \mathbf{w}' - \mathbf{w}$ lies in $U \cap W = \{\mathbf{0}\}$. Hence $\mathbf{u} = \mathbf{u}'$ and $\mathbf{w} = \mathbf{w}'$. This proves $V = U \oplus W$.

Exercise C2: Bases of $\mathbb{R}$ over $\mathbb{Q}$ Are Uncountable.

- (a) Let $A_0, A_1, A_2, \dots$ be countable sets. Prove that $\bigcup_{n \in \mathbb{N}} A_n$ is countable. (You may use the axiom of choice.)
- (b) Let V be a vector space over $\mathbb{Q}$ with a countable basis. Prove that V is countable.
- (c) Deduce that every basis of $\mathbb{R}$ over $\mathbb{Q}$ is uncountable.

Hint: for (b), write V as a union of spans of finite lists.

Solution.

- (a) Let $A = \bigcup_n A_n$. If $A = \emptyset$ there is nothing to prove; otherwise fix $a^* \in A$. For each n with $A_n \neq \emptyset$, choose a surjection $\sigma_n: \mathbb{N} \rightarrow A_n$ (this is where choice is used). Define $h: \mathbb{N} \times \mathbb{N} \rightarrow A$ by $h(n, m) = \sigma_n(m)$ if $A_n \neq \emptyset$, and $h(n, m) = a^*$ otherwise. Every $a \in A$ lies in some non-empty A_n and equals $\sigma_n(m)$ for some m , so h is surjective. By Lemma 1.8.4 (b) and (a), A is countable.
- (b) If the basis is empty, $V = \{\mathbf{0}\}$ is countable. Otherwise list the basis as $\sigma(0), \sigma(1), \dots$ for a surjection $\sigma: \mathbb{N} \rightarrow B$. Every $\mathbf{v} \in V$ is a combination of finitely many basis vectors, all among $\sigma(0), \dots, \sigma(n)$ for some n . Hence $V = \bigcup_n V_n$ with $V_n = \text{span}(\sigma(0), \dots, \sigma(n))$. The map $\mathbb{Q}^{n+1} \rightarrow V_n$, $(q_0, \dots, q_n) \mapsto \sum_i q_i \sigma(i)$, is surjective, so each V_n is countable by Lemma 1.8.4 (d) and (a). By (a), V is countable.
- (c) By Theorem 1.8.3, $\mathbb{R}$ has a basis over $\mathbb{Q}$. If some basis were countable, then $\mathbb{R}$ would be countable by (b), contradicting Lemma 1.8.5. Hence every basis is uncountable.

1.9 Changing the Field

Every vector space comes with its field, and we have already seen that the field matters: $\dim_{\mathbb{C}} \mathbb{C} = 1$ but $\dim_{\mathbb{R}} \mathbb{C} = 2$ (Example (Dimensions of the Running Examples)). This section explains the pattern behind that example. Shrinking the field multiplies the dimension by a fixed factor, the **tower law**. Going the other way, every real vector space can be enlarged to a complex one, its **complexification**, without changing the dimension. Both moves are used later, most of all when real matrices need complex eigenvalues.

1.9.1 Restricting the scalars

Start with the standard example. The space $\mathbb{C}^2$ over $\mathbb{C}$ has the basis $(\mathbf{e}_1, \mathbf{e}_2)$, so its dimension is 2. But nothing stops us from multiplying vectors of $\mathbb{C}^2$ only by **real** numbers. The addition is the same, and the axioms still hold, because they hold for all complex scalars. So $\mathbb{C}^2$ is also

a vector space over $\mathbb{R}$. What is its dimension now?

Example. ($\mathbb{C}^2$ as a Real Vector Space)

Show that $(\mathbf{e}_1, i\mathbf{e}_1, \mathbf{e}_2, i\mathbf{e}_2)$ is a basis of $\mathbb{C}^2$ over $\mathbb{R}$, and write $(1 + 2i, 3 - i)$ in it.

Solution. Spanning. Let $(z_1, z_2) \in \mathbb{C}^2$ and write $z_k = a_k + b_k i$ with $a_k, b_k \in \mathbb{R}$. Then

$$(z_1, z_2) = a_1 \mathbf{e}_1 + b_1(i\mathbf{e}_1) + a_2 \mathbf{e}_2 + b_2(i\mathbf{e}_2),$$

a combination with **real** coefficients. For example, $(1 + 2i, 3 - i) = 1\mathbf{e}_1 + 2(i\mathbf{e}_1) + 3\mathbf{e}_2 + (-1)(i\mathbf{e}_2)$.

Independence. Let $a_1, b_1, a_2, b_2 \in \mathbb{R}$ with $a_1 \mathbf{e}_1 + b_1(i\mathbf{e}_1) + a_2 \mathbf{e}_2 + b_2(i\mathbf{e}_2) = \mathbf{0}$. The left side is $(a_1 + b_1 i, a_2 + b_2 i)$. Each entry is 0, and comparing real and imaginary parts (Definition 0.6.1) gives $a_1 = b_1 = a_2 = b_2 = 0$.

Hence $\dim_{\mathbb{R}} \mathbb{C}^2 = 4 = 2 \dim_{\mathbb{C}} \mathbb{C}^2$.

The general setting has a big field L , a smaller field K inside it, and a vector space over L .

Definition 1.9.1 — Subfield, Restriction of Scalars, Degree

Let L be a field. A **subfield** of L is a subset $K \subseteq L$ that contains 0 and 1 and is closed under addition, multiplication, negatives, and inverses of **non-zero** elements.

Let K be a subfield of L , and let V be a vector space over L . Keeping the addition of V and using scalar multiplication only for scalars in K makes V a vector space **over** K . We say V is obtained by **restriction of scalars**.

In particular L itself, a vector space over L , is a vector space over K . If it is finite-dimensional, its dimension is the **degree** $[L : K] := \dim_K L$.

In words: a subfield is a field sitting inside L with the same operations. Restriction of scalars forgets how to multiply by elements of L outside K .

There are two things to check. First, K with the operations of L is a field. Closure makes the operations land in K , the elements 0, 1, negatives and inverses are in K by assumption, and the remaining axioms are identities that hold for all elements of L , so in particular for those of K . Second, V over K is a vector space. Scalar multiplication $K \times V \rightarrow V$ lands in V , and each axiom of Definition 1.1.1 holds **for all** scalars in L , so in particular for all scalars in K .

We always write $\dim_K V$ or $\dim_L V$ when both fields are in play.

Example. (Degrees of Some Fields)

Find $[\mathbb{C} : \mathbb{R}]$, $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}]$ and $[K : K]$ for any field K . Is $[\mathbb{R} : \mathbb{Q}]$ defined?

Solution. $[\mathbb{C} : \mathbb{R}] = 2$. Every $z \in \mathbb{C}$ is $a \cdot 1 + b \cdot i$ with $a, b \in \mathbb{R}$, and $a + bi = 0$ forces $a = b = 0$ by Definition 0.6.1. So $(1, i)$ is a basis of $\mathbb{C}$ over $\mathbb{R}$.

$[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$. The set $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ is a subfield of $\mathbb{R}$ by Example

($\mathbb{Q}(\sqrt{2})$ Is a Field). The list $(1, \sqrt{2})$ spans it over $\mathbb{Q}$ by definition. For independence, let $a + b\sqrt{2} = 0$ with $a, b \in \mathbb{Q}$. If $b \neq 0$, then $\sqrt{2} = -a/b \in \mathbb{Q}$, contradicting [Theorem 0.2.3](#). So $b = 0$, and then $a = 0$.

$[K : K] = 1$. The list (1) is a basis of K over K : every a equals $a \cdot 1$, and $a \cdot 1 = 0$ forces $a = 0$. This degenerate case says that restricting to the same field changes nothing.

$[\mathbb{R} : \mathbb{Q}]$ is not defined, because $\mathbb{R}$ is not finite-dimensional over $\mathbb{Q}$ ([Theorem 1.8.6](#)).

Restriction only goes **down**. The set $\mathbb{R}$ with the operations of $\mathbb{C}$ is **not** a vector space over $\mathbb{C}$: everything about addition works, but scalar multiplication leaves the set, since $i \cdot 1 = i \notin \mathbb{R}$. Enlarging the field needs a construction, which is the second half of this section.

1.9.2 The tower law

In [Example \(\$\mathbb{C}^2\$ as a Real Vector Space\)](#) we took a vector, wrote it in coordinates over $\mathbb{C}$, and then split each complex coordinate into real coordinates. The same two-stage process works for any fields.

Theorem 1.9.1 — Tower Law

Let K be a subfield of L , and let V be a vector space over L . Suppose $(a_1, \dots, a_m)$ is a basis of L over K and $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is a basis of V over L . Then the list of the mn vectors

$$(a_1\mathbf{v}_1, \dots, a_m\mathbf{v}_1, a_1\mathbf{v}_2, \dots, a_m\mathbf{v}_2, \dots, a_1\mathbf{v}_n, \dots, a_m\mathbf{v}_n)$$

is a basis of V over K . In particular,

$$\dim_K V = [L : K] \cdot \dim_L V.$$

Idea. Coordinates in two stages. A vector is an L -combination of the $\mathbf{v}_j$; each L -coefficient is a K -combination of the a_i . Substituting gives a double sum $\sum_j \sum_i b_{ij} a_i \mathbf{v}_j$ with $b_{ij} \in K$. For independence, run the same two stages backwards: regroup the double sum by j , use independence over L , then independence over K .

Proof. Throughout we use two axioms of [Definition 1.1.1](#) for scalars in L : $(c + c')\mathbf{x} = c\mathbf{x} + c'\mathbf{x}$ and $c(c'\mathbf{x}) = (cc')\mathbf{x}$.

Spanning. Let $\mathbf{x} \in V$. Since $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ spans V over L , there are $c_1, \dots, c_n \in L$ with $\mathbf{x} = \sum_{j=1}^n c_j \mathbf{v}_j$. Since $(a_1, \dots, a_m)$ spans L over K , for each j there are $b_{1j}, \dots, b_{mj} \in K$ with $c_j = \sum_{i=1}^m b_{ij} a_i$. Therefore

$$\mathbf{x} = \sum_{j=1}^n \left(\sum_{i=1}^m b_{ij} a_i \right) \mathbf{v}_j = \sum_{j=1}^n \sum_{i=1}^m (b_{ij} a_i) \mathbf{v}_j = \sum_{j=1}^n \sum_{i=1}^m b_{ij} (a_i \mathbf{v}_j),$$

where the second equality uses $(c + c')\mathbf{x} = c\mathbf{x} + c'\mathbf{x}$ and the third uses $c(c'\mathbf{x}) = (cc')\mathbf{x}$. This is a K -linear combination of the vectors $a_i \mathbf{v}_j$.

Independence. Let $b_{ij} \in K$ with $\sum_{j=1}^n \sum_{i=1}^m b_{ij} (a_i \mathbf{v}_j) = \mathbf{0}$. Reading the display above

from right to left,

$$\sum_{j=1}^n \left(\sum_{i=1}^m b_{ij} a_i \right) \mathbf{v}_j = \mathbf{0}.$$

Each coefficient $\sum_i b_{ij} a_i$ lies in L , and $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is independent over L . Hence $\sum_{i=1}^m b_{ij} a_i = 0$ for every j . For each fixed j , this is a K -combination of $(a_1, \dots, a_m)$ equal to 0, and that list is independent over K . Hence $b_{ij} = 0$ for all i and j .

So the mn vectors $a_i \mathbf{v}_j$ form a basis of V over K , and $\dim_K V = mn = [L : K] \dim_L V$. ■

The name comes from the picture of fields stacked in a tower, $K \subseteq L$, with V on top: dimensions multiply as you go down. The most important case is $\mathbb{R} \subseteq \mathbb{C}$.

Theorem 1.9.2 — Complex Spaces as Real Spaces

Let V be a vector space over $\mathbb{C}$ with basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$. Then $(\mathbf{v}_1, i\mathbf{v}_1, \dots, \mathbf{v}_n, i\mathbf{v}_n)$ is a basis of V over $\mathbb{R}$. In particular $\dim_{\mathbb{R}} V = 2 \dim_{\mathbb{C}} V$.

Proof. By [Example \(Degrees of Some Fields\)](#), $(1, i)$ is a basis of $\mathbb{C}$ over $\mathbb{R}$. By [Theorem 1.9.1](#) with $a_1 = 1$ and $a_2 = i$, the list $(1\mathbf{v}_1, i\mathbf{v}_1, \dots, 1\mathbf{v}_n, i\mathbf{v}_n)$ is a basis of V over $\mathbb{R}$, and $1\mathbf{v}_j = \mathbf{v}_j$ by [Definition 1.1.1](#). The dimension is $2n$. ■

This is the result of [Exercise \(C2: A complex space viewed over the reals\)](#), now as a case of the tower law. For example, $\dim_{\mathbb{R}} \mathbb{C}^n = 2n$, and $\dim_{\mathbb{R}} \mathbb{C}[x]_{\leq n} = 2(n+1)$, with basis $(1, i, x, ix, \dots, x^n, ix^n)$.

Warning. A subset of $\mathbb{C}^n$ can be a subspace over $\mathbb{R}$ without being a subspace over $\mathbb{C}$. The real vectors $\mathbb{R}^n \subseteq \mathbb{C}^n$ contain $\mathbf{0}$ and are closed under addition and under multiplication by **real** scalars, so they form an $\mathbb{R}$ -subspace of dimension n . But $\mathbf{e}_1 \in \mathbb{R}^n$ and $i\mathbf{e}_1 \notin \mathbb{R}^n$, so $\mathbb{R}^n$ is **not** a $\mathbb{C}$ -subspace. Always ask: a subspace over which field?

Quick check. What is $\dim_{\mathbb{R}} M_{2 \times 3}(\mathbb{C})$? What is $\dim_{\mathbb{Q}} \mathbb{Q}(\sqrt{2})^2$?

Solution. $M_{2 \times 3}(\mathbb{C})$ has dimension 6 over $\mathbb{C}$ ([Example \(Dimensions of the Running Examples\)](#)), so $\dim_{\mathbb{R}} = 2 \cdot 6 = 12$ by [Theorem 1.9.2](#). The space $\mathbb{Q}(\sqrt{2})^2$ has dimension 2 over the field $\mathbb{Q}(\sqrt{2})$, and $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$ by [Example \(Degrees of Some Fields\)](#), so $\dim_{\mathbb{Q}} = 2 \cdot 2 = 4$ by [Theorem 1.9.1](#).

1.9.3 Complexification

Now go up. A real matrix such as the rotation $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ moves every non-zero vector of $\mathbb{R}^2$ off its own line. In $\mathbb{C}^2$, however, it sends $(1, -i)$ to $(i, 1) = i(1, -i)$. To use such complex vectors for an arbitrary real space V , not just $\mathbb{R}^n$, we need a complex space that contains V the way $\mathbb{C}^n$ contains $\mathbb{R}^n$.

The model is $\mathbb{C}^n$ itself. Every $\mathbf{z} \in \mathbb{C}^n$ splits as $\mathbf{z} = \mathbf{x} + i\mathbf{y}$ with $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, its real and imaginary parts, taken entry by entry. Multiplying by $a + bi$ with $a, b \in \mathbb{R}$ gives

$$(a + bi)(\mathbf{x} + i\mathbf{y}) = (a\mathbf{x} - b\mathbf{y}) + i(a\mathbf{y} + b\mathbf{x}).$$

The right side only uses real scalars acting on real vectors. So we can copy it for any real space, exactly as $\mathbb{C}$ was built from pairs of real numbers in [Definition 0.6.1](#).

The complexification of V is the space of formal expressions $\mathbf{u} + i\mathbf{v}$ with $\mathbf{u}, \mathbf{v} \in V$, multiplied by complex numbers the way the rules of algebra dictate.

Definition 1.9.2 — Complexification

Let V be a vector space over $\mathbb{R}$. The **complexification** of V is the set $V_{\mathbb{C}} := V \times V$ with addition and scalar multiplication by $\mathbb{C}$ defined, for all $(\mathbf{u}, \mathbf{v}), (\mathbf{u}', \mathbf{v}') \in V_{\mathbb{C}}$ and $a, b \in \mathbb{R}$, by

$$(\mathbf{u}, \mathbf{v}) + (\mathbf{u}', \mathbf{v}') := (\mathbf{u} + \mathbf{u}', \mathbf{v} + \mathbf{v}'), \quad (a + bi)(\mathbf{u}, \mathbf{v}) := (a\mathbf{u} - b\mathbf{v}, a\mathbf{v} + b\mathbf{u}).$$

We write $\mathbf{u} + i\mathbf{v}$ for $(\mathbf{u}, \mathbf{v})$, and $\mathbf{u}$ for $(\mathbf{u}, \mathbf{0})$.

In words: addition is entry by entry. Multiplication by $a + bi$ is the formula forced by $i^2 = -1$. The notation is honest: $i(\mathbf{v}, \mathbf{0}) = (0\mathbf{v} - 1\mathbf{0}, 0\mathbf{0} + 1\mathbf{v}) = (\mathbf{0}, \mathbf{v})$, by [Theorem 1.1.5](#) and [Theorem 1.1.6](#), so $(\mathbf{u}, \mathbf{0}) + i(\mathbf{v}, \mathbf{0}) = (\mathbf{u}, \mathbf{v})$. Also a **real** scalar acts entrywise: $a(\mathbf{u}, \mathbf{v}) = (a\mathbf{u}, a\mathbf{v})$. So $V_{\mathbb{C}}$, restricted to real scalars, is just two copies of V .

Why $V_{\mathbb{C}}$ is a vector space over $\mathbb{C}$. The operations land in $V \times V$. Addition is entrywise, so (commutativity, associativity, zero, negatives) follow from the same axioms in each entry of V , with zero $(\mathbf{0}, \mathbf{0})$ and negative $(-\mathbf{u}, -\mathbf{v})$. We check the scalar axioms that involve the multiplication of $\mathbb{C}$; below, $a\mathbf{u} - b\mathbf{v}$ means $a\mathbf{u} + (-b)\mathbf{v}$, which is the same by [Theorem 1.1.8](#). Let $\alpha = a + bi$ and $\beta = c + di$ with $a, b, c, d \in \mathbb{R}$, and $\mathbf{z} = (\mathbf{u}, \mathbf{v})$.

- $1\mathbf{z} = \mathbf{z}$: we get $(1\mathbf{u} - 0\mathbf{v}, 1\mathbf{v} + 0\mathbf{u}) = (\mathbf{u}, \mathbf{v})$, since $0\mathbf{v} = \mathbf{0}$ by [Theorem 1.1.5](#).
- $\alpha(\beta\mathbf{z}) = (\alpha\beta)\mathbf{z}$: since $\beta\mathbf{z} = (c\mathbf{u} - d\mathbf{v}, c\mathbf{v} + d\mathbf{u})$, the axioms of V give

$$\alpha(\beta\mathbf{z}) = (a(c\mathbf{u} - d\mathbf{v}) - b(c\mathbf{v} + d\mathbf{u}), a(c\mathbf{v} + d\mathbf{u}) + b(c\mathbf{u} - d\mathbf{v})) = ((ac - bd)\mathbf{u} - (ad + bc)\mathbf{v}, (ac - bd)\mathbf{v} + (ad + bc)\mathbf{u}).$$

Since $\alpha\beta = (ac - bd) + (ad + bc)i$, this is $(\alpha\beta)\mathbf{z}$.

- $(\alpha + \beta)\mathbf{z} = \alpha\mathbf{z} + \beta\mathbf{z}$: with $\alpha + \beta = (a + c) + (b + d)i$,

$$(\alpha + \beta)\mathbf{z} = ((a + c)\mathbf{u} - (b + d)\mathbf{v}, (a + c)\mathbf{v} + (b + d)\mathbf{u}) = (a\mathbf{u} - b\mathbf{v}, a\mathbf{v} + b\mathbf{u}) + (c\mathbf{u} - d\mathbf{v}, c\mathbf{v} + d\mathbf{u}) = \alpha\mathbf{z} + \beta\mathbf{z}.$$

The remaining axiom, $\alpha(\mathbf{z} + \mathbf{w}) = \alpha\mathbf{z} + \alpha\mathbf{w}$, is [Exercise \(B3: Working in a Complexification\)](#).

Here is the payoff: complexifying keeps the dimension, and even keeps the basis.

Theorem 1.9.3 — A Real Basis Is a Complex Basis of the Complexification

Let V be a vector space over $\mathbb{R}$ with basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$. Then $(\mathbf{v}_1, \dots, \mathbf{v}_n)$, that is, $((\mathbf{v}_1, \mathbf{0}), \dots, (\mathbf{v}_n, \mathbf{0}))$, is a basis of $V_{\mathbb{C}}$ over $\mathbb{C}$. In particular $\dim_{\mathbb{C}} V_{\mathbb{C}} = \dim_{\mathbb{R}} V$.

Idea. A complex coefficient $a_j + b_j i$ on $\mathbf{v}_j$ puts $a_j \mathbf{v}_j$ into the real part and $b_j \mathbf{v}_j$ into the imaginary part. So one complex combination of the $\mathbf{v}_j$ is the same as two real combinations, one for each part, and both questions (spanning, independence) split into two real questions about $(\mathbf{v}_1, \dots, \mathbf{v}_n)$.

Proof. For $a, b \in \mathbb{R}$ and $\mathbf{x} \in V$, [Definition 1.9.2](#) and [Theorem 1.1.6](#) give

$$(a + bi)(\mathbf{x}, \mathbf{0}) = (a\mathbf{x} - b\mathbf{0}, a\mathbf{0} + b\mathbf{x}) = (a\mathbf{x}, b\mathbf{x}).$$

Call this identity $()$.

Spanning. Let $(\mathbf{u}, \mathbf{v}) \in V_{\mathbb{C}}$. Since $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ spans V , there are $a_j, b_j \in \mathbb{R}$ with $\mathbf{u} = \sum_j a_j \mathbf{v}_j$ and $\mathbf{v} = \sum_j b_j \mathbf{v}_j$. By $()$ and entrywise addition,

$$\sum_{j=1}^n (a_j + b_j i)(\mathbf{v}_j, \mathbf{0}) = \sum_{j=1}^n (a_j \mathbf{v}_j, b_j \mathbf{v}_j) = \left(\sum_{j=1}^n a_j \mathbf{v}_j, \sum_{j=1}^n b_j \mathbf{v}_j \right) = (\mathbf{u}, \mathbf{v}).$$

Independence. Let $\alpha_1, \dots, \alpha_n \in \mathbb{C}$ with $\sum_j \alpha_j (\mathbf{v}_j, \mathbf{0}) = (\mathbf{0}, \mathbf{0})$, and write $\alpha_j = a_j + b_j i$ with $a_j, b_j \in \mathbb{R}$. By the same computation, $(\sum_j a_j \mathbf{v}_j, \sum_j b_j \mathbf{v}_j) = (\mathbf{0}, \mathbf{0})$. Since $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is independent over $\mathbb{R}$, all $a_j = 0$ and all $b_j = 0$. Hence every $\alpha_j = 0$.

Therefore $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is a basis of $V_{\mathbb{C}}$, and $\dim_{\mathbb{C}} V_{\mathbb{C}} = n$. ■

As a consistency check, restricting $V_{\mathbb{C}}$ back to $\mathbb{R}$ gives $\dim_{\mathbb{R}} V_{\mathbb{C}} = 2n$ by [Theorem 1.9.2](#), which matches the fact that $V_{\mathbb{C}}$ over $\mathbb{R}$ is two copies of V . The same proof works word for word for an infinite basis set, since every combination involves finitely many vectors.

Example. (The Complexification of $\mathbb{R}^n$ Is $\mathbb{C}^n$)

Show that the map $(\mathbf{x}, \mathbf{y}) \mapsto \mathbf{x} + i\mathbf{y}$, from $(\mathbb{R}^n)_{\mathbb{C}}$ to $\mathbb{C}^n$, is a bijection that respects addition and multiplication by complex scalars.

Solution. Here $\mathbf{x} + i\mathbf{y}$ is computed in $\mathbb{C}^n$: its k -th entry is $x_k + y_k i$. Every $\mathbf{z} \in \mathbb{C}^n$ has unique real and imaginary parts entry by entry ([Definition 0.6.1](#)), so exactly one pair $(\mathbf{x}, \mathbf{y})$ of real vectors maps to $\mathbf{z}$. Hence the map is a bijection.

Addition is entrywise on both sides, so it is respected. For scaling, let $a, b \in \mathbb{R}$. In $(\mathbb{R}^n)_{\mathbb{C}}$, $(a + bi)(\mathbf{x}, \mathbf{y}) = (a\mathbf{x} - b\mathbf{y}, a\mathbf{y} + b\mathbf{x})$, which maps to the vector with entries $(ax_k - by_k) + (ay_k + bx_k)i$. In $\mathbb{C}^n$, the k -th entry of $(a + bi)(\mathbf{x} + i\mathbf{y})$ is

$$(a + bi)(x_k + y_k i) = (ax_k - by_k) + (ay_k + bx_k)i.$$

The two agree, so the map respects scalar multiplication. It sends the basis $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ of $(\mathbb{R}^n)_{\mathbb{C}}$ from [Theorem 1.9.3](#) to the standard basis of $\mathbb{C}^n$.

So $(\mathbb{R}^n)_{\mathbb{C}}$ is $\mathbb{C}^n$ in disguise; in Chapter 3 such a bijection will be called an isomorphism. In the same way $(M_n(\mathbb{R}))_{\mathbb{C}}$ is $M_n(\mathbb{C})$, and $(\mathbb{R}[x]_{\leq n})_{\mathbb{C}}$ is $\mathbb{C}[x]_{\leq n}$. The construction earns its keep in Chapter 8, where real matrices without real eigenvalues, like the rotation above, get complex ones, and in Chapter 11, in the spectral theory of real operators.

1.9.4 Exercises

1.9.4.1 A. Check your understanding

Exercise A1.

- (a) Define the degree $[L : K]$ of a subfield $K \subseteq L$.
- (b) State the Tower Law.
- (c) What is $\dim_{\mathbb{R}} \mathbb{C}^3$? Give a basis.
- (d) True or false: every $\mathbb{C}$ -subspace of $\mathbb{C}^n$ is an $\mathbb{R}$ -subspace. Justify your answer.
- (e) What is $\dim_{\mathbb{C}} (M_2(\mathbb{R}))_{\mathbb{C}}$?

Solution.

- (a) $[L : K] = \dim_K L$, where L is regarded as a vector space over K by restriction of scalars (Definition 1.9.1), when this dimension is finite.
- (b) See Theorem 1.9.1: if (a_i) is a basis of L over K and (v_j) is a basis of V over L , then $(a_i v_j)$ is a basis of V over K , so $\dim_K V = [L : K] \dim_L V$.
- (c) 6, with basis $(e_1, ie_1, e_2, ie_2, e_3, ie_3)$, by Theorem 1.9.2.
- (d) True. A $\mathbb{C}$ -subspace contains $\mathbf{0}$, is closed under addition, and is closed under multiplication by every complex number, in particular by every real number.
- (e) 4: $\dim_{\mathbb{R}} M_2(\mathbb{R}) = 4$ (Example (Dimensions of the Running Examples)), and Theorem 1.9.3 keeps the dimension.

1.9.4.2 B. Practice

Exercise B1: A Tower over $\mathbb{Q}$.

Find $\dim_{\mathbb{Q}} \mathbb{Q}(\sqrt{2})^3$ and a basis of $\mathbb{Q}(\sqrt{2})^3$ over $\mathbb{Q}$.

Solution. By Example ($\mathbb{Q}(\sqrt{2})$ Is a Field), $\mathbb{Q}(\sqrt{2})$ is a field, so $\mathbb{Q}(\sqrt{2})^3$ is a vector space over it with basis (e_1, e_2, e_3) and dimension 3 (Example (Dimensions of the Running Examples)). By Example (Degrees of Some Fields), $(1, \sqrt{2})$ is a basis of $\mathbb{Q}(\sqrt{2})$ over the subfield $\mathbb{Q}$. By Theorem 1.9.1,

$$(e_1, \sqrt{2}e_1, e_2, \sqrt{2}e_2, e_3, \sqrt{2}e_3)$$

is a basis of $\mathbb{Q}(\sqrt{2})^3$ over $\mathbb{Q}$. Hence $\dim_{\mathbb{Q}} \mathbb{Q}(\sqrt{2})^3 = 2 \cdot 3 = 6$.

Exercise B2: Complex Matrices over $\mathbb{R}$.

Let $E_{11}, E_{12}, E_{21}, E_{22}$ be the matrix units of $M_2(\mathbb{C})$. Find a basis of $M_2(\mathbb{C})$ over $\mathbb{R}$, and write $A = \begin{pmatrix} 1+i & 2 \\ -i & 3i \end{pmatrix}$ in it.

Solution. The matrix units form a basis of $M_2(\mathbb{C})$ over $\mathbb{C}$ (Example (Standard Bases)).

By [Theorem 1.9.2](#),

$$(E_{11}, iE_{11}, E_{12}, iE_{12}, E_{21}, iE_{21}, E_{22}, iE_{22})$$

is a basis of $M_2(\mathbb{C})$ over $\mathbb{R}$, so $\dim_{\mathbb{R}} M_2(\mathbb{C}) = 8$. Splitting each entry of A into real and imaginary parts,

$$A = 1E_{11} + 1(iE_{11}) + 2E_{12} + 0(iE_{12}) + 0E_{21} + (-1)(iE_{21}) + 0E_{22} + 3(iE_{22}).$$

Exercise B3: Working in a Complexification.

Let V be a vector space over $\mathbb{R}$.

- (a) Verify that $\alpha(\mathbf{z} + \mathbf{w}) = \alpha\mathbf{z} + \alpha\mathbf{w}$ for all $\alpha \in \mathbb{C}$ and $\mathbf{z}, \mathbf{w} \in V_{\mathbb{C}}$.
- (b) In $(\mathbb{R}[x]_{\leq 1})_{\mathbb{C}}$, compute $(2 - i)((1 + x) + ix)$ from [Definition 1.9.2](#), and compare with the product $(2 - i)(1 + x + ix)$ computed in $\mathbb{C}[x]$.

Solution.

- (a) Let $\alpha = a + bi$ with $a, b \in \mathbb{R}$, $\mathbf{z} = (\mathbf{u}, \mathbf{v})$ and $\mathbf{w} = (\mathbf{u}', \mathbf{v}')$. Then $\mathbf{z} + \mathbf{w} = (\mathbf{u} + \mathbf{u}', \mathbf{v} + \mathbf{v}')$, and by the distributive law in V and [Theorem 1.1.8](#),

$$\alpha(\mathbf{z} + \mathbf{w}) = (a(\mathbf{u} + \mathbf{u}') - b(\mathbf{v} + \mathbf{v}'), a(\mathbf{v} + \mathbf{v}') + b(\mathbf{u} + \mathbf{u}')) = ((a\mathbf{u} - b\mathbf{v}) + (a\mathbf{u}' - b\mathbf{v}'), (a\mathbf{v} + b\mathbf{u}) + (a\mathbf{v}' + b\mathbf{u}')),$$

where we also rearranged the sums using commutativity and associativity of addition in V . The right side is $(a\mathbf{u} - b\mathbf{v}, a\mathbf{v} + b\mathbf{u}) + (a\mathbf{u}' - b\mathbf{v}', a\mathbf{v}' + b\mathbf{u}') = \alpha\mathbf{z} + \alpha\mathbf{w}$.

- (b) Here $a = 2$, $b = -1$, $\mathbf{u} = 1 + x$ and $\mathbf{v} = x$. Then $a\mathbf{u} - b\mathbf{v} = 2 + 2x + x = 2 + 3x$ and $a\mathbf{v} + b\mathbf{u} = 2x - (1 + x) = -1 + x$. So the product is $(2 + 3x) + i(-1 + x)$. In $\mathbb{C}[x]$, $(2 - i)(1 + x + ix) = 2 + 2x + 2ix - i - ix - i^2x = (2 + 3x) + i(-1 + x)$, the same answer.

1.9.4.3 C. Going deeper

Exercise C1: When Is a Real Subspace Complex?.

Let $W \subseteq \mathbb{C}^n$ be a subspace over $\mathbb{R}$.

- (a) Prove that W is a subspace over $\mathbb{C}$ if and only if $i\mathbf{w} \in W$ for every $\mathbf{w} \in W$.
- (b) Deduce that if W is a subspace over $\mathbb{C}$, then $\dim_{\mathbb{R}} W$ is even.
- (c) Show that $W_0 = \{(z, \bar{z}) : z \in \mathbb{C}\} \subseteq \mathbb{C}^2$ is a subspace over $\mathbb{R}$ of even real dimension that is **not** a subspace over $\mathbb{C}$. So the converse of (b) fails.

Solution.

- (a) $(\Rightarrow)$ If W is a $\mathbb{C}$ -subspace, it is closed under multiplication by i .
 $(\Leftarrow)$ Suppose $i\mathbf{w} \in W$ for every $\mathbf{w} \in W$. We use [Theorem 1.2.1](#) over $\mathbb{C}$. Since W is an $\mathbb{R}$ -subspace, $\mathbf{0} \in W$ and W is closed under addition. Let $\mathbf{w} \in W$ and $a + bi \in \mathbb{C}$ with $a, b \in \mathbb{R}$. Then $a\mathbf{w} \in W$ and $b(i\mathbf{w}) \in W$, because W is closed under real scalars, and $(a + bi)\mathbf{w} = a\mathbf{w} + b(i\mathbf{w})$ by the axioms of $\mathbb{C}^n$. This lies in W by closure under addition. Hence W is a $\mathbb{C}$ -subspace.
- (b) W is a subspace of the finite-dimensional $\mathbb{C}$ -space $\mathbb{C}^n$, so by [Theorem 1.6.7](#) it is

finite-dimensional over $\mathbb{C}$, and by [Corollary 1.5.5](#) it has a basis over $\mathbb{C}$, of some length k . Restricting the scalars of W to $\mathbb{R}$ gives the same real space as before, so $\dim_{\mathbb{R}} W = 2k$ by [Theorem 1.9.2](#).

- (c) Write $z = a + bi$ with $a, b \in \mathbb{R}$. Then $(z, \bar{z}) = (a + bi, a - bi) = a(1, 1) + b(i, -i)$, so W_0 is the $\mathbb{R}$ -span of $((1, 1), (i, -i))$ and is an $\mathbb{R}$ -subspace by [Theorem 1.3.1](#). The list is independent over $\mathbb{R}$: $a(1, 1) + b(i, -i) = \mathbf{0}$ gives $a + bi = 0$, so $a = b = 0$. Hence $\dim_{\mathbb{R}} W_0 = 2$. But $(1, 1) \in W_0$ while $i(1, 1) = (i, i) \notin W_0$, because $\bar{i} = -i \neq i$. By (a), W_0 is not a $\mathbb{C}$ -subspace.

Exercise C2: A Tower of Degree Four.

Let $V = \{x + y\sqrt{3} : x, y \in \mathbb{Q}(\sqrt{2})\} \subseteq \mathbb{R}$.

- (a) Prove that $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$.
 (b) Regard $\mathbb{R}$ as a vector space over its subfield $\mathbb{Q}(\sqrt{2})$. Prove that V is a subspace with basis $(1, \sqrt{3})$.
 (c) Deduce that $\dim_{\mathbb{Q}} V = 4$, with basis $(1, \sqrt{2}, \sqrt{3}, \sqrt{6})$.

Hint for (a): square both sides, and argue with remainders on division by 3.

Solution.

- (a) First, there are no integers p, q with $q \neq 0$, not both divisible by 3, and $cp^2 = 3q^2$ for $c \in \{1, 2\}$. Indeed, if $3 \nmid p$, then $p = 3t \pm 1$ and $cp^2 = c(9t^2 \pm 6t + 1)$ leaves remainder $c \in \{1, 2\}$ on division by 3, while $3q^2$ leaves remainder 0. So $3 \mid p$, say $p = 3s$. Then $9cs^2 = 3q^2$, so $q^2 = 3cs^2$, and the same remainder argument (a square not divisible by 3 leaves remainder 1) gives $3 \mid q$, a contradiction.

Now suppose $\sqrt{3} = a + b\sqrt{2}$ with $a, b \in \mathbb{Q}$. Squaring, $3 = a^2 + 2b^2 + 2ab\sqrt{2}$. If $ab \neq 0$, then $\sqrt{2} = (3 - a^2 - 2b^2)/(2ab) \in \mathbb{Q}$, contradicting [Theorem 0.2.3](#). If $b = 0$, then $a^2 = 3$; writing $a = p/q$ in lowest terms gives $p^2 = 3q^2$ with p, q not both divisible by 3, which is impossible. If $a = 0$, then $2b^2 = 3$, and $b = p/q$ in lowest terms gives $2p^2 = 3q^2$, also impossible. Hence $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$.

- (b) $\mathbb{Q}(\sqrt{2})$ is a subfield of $\mathbb{R}$ by [Example \(\$\mathbb{Q}\(\sqrt{2}\)\$ Is a Field\)](#), so $\mathbb{R}$ is a vector space over it by restriction of scalars. We use [Theorem 1.2.1](#). (1) $0 = 0 + 0\sqrt{3} \in V$. (2) $(x + y\sqrt{3}) + (x' + y'\sqrt{3}) = (x + x') + (y + y')\sqrt{3} \in V$, since $\mathbb{Q}(\sqrt{2})$ is closed under addition. (3) For $c \in \mathbb{Q}(\sqrt{2})$, $c(x + y\sqrt{3}) = cx + (cy)\sqrt{3} \in V$, since $\mathbb{Q}(\sqrt{2})$ is closed under multiplication. So V is a subspace.

The list $(1, \sqrt{3})$ spans V by the definition of V . For independence, let $x + y\sqrt{3} = 0$ with $x, y \in \mathbb{Q}(\sqrt{2})$. If $y \neq 0$, then $y^{-1} \in \mathbb{Q}(\sqrt{2})$ and $\sqrt{3} = -xy^{-1} \in \mathbb{Q}(\sqrt{2})$, contradicting (a). So $y = 0$, and then $x = 0$. Hence $(1, \sqrt{3})$ is a basis and $\dim_{\mathbb{Q}(\sqrt{2})} V = 2$.

- (c) By [Example \(Degrees of Some Fields\)](#), $(1, \sqrt{2})$ is a basis of $\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$. By [Theorem 1.9.1](#) applied to $\mathbb{Q} \subseteq \mathbb{Q}(\sqrt{2})$ and the $\mathbb{Q}(\sqrt{2})$ -space V , the list $(1 \cdot 1, \sqrt{2} \cdot 1, 1 \cdot \sqrt{3}, \sqrt{2} \cdot \sqrt{3}) = (1, \sqrt{2}, \sqrt{3}, \sqrt{6})$ is a basis of V over $\mathbb{Q}$. Hence $\dim_{\mathbb{Q}} V = 2 \cdot 2 = 4$.

Chapter 2

Linear Systems and Matrices

Chapter 1 kept asking questions that end in a system of equations. Is this vector in the span of those? Is this list independent? Extend this list to a basis. Each time we solved a small system by hand and promised a systematic method. This chapter keeps that promise. Every question about finitely many vectors in F^n turns into a linear system, and linear systems have a complete algorithm: Gaussian elimination.

We do not treat the algorithm as a recipe. We prove that row operations never change the answer, that the fully reduced form of a matrix does not depend on the choices made along the way, and that the numbers it produces, pivots and free variables, measure exactly the dimensions of Chapter 1. Along the way the invertible matrices get their first full characterization, and we prove a fact Chapter 0 had to postpone: for square matrices, $AB = I$ already forces $BA = I$.

What you need. From Chapter 0, matrix multiplication read column by column ([Theorem 0.9.3](#)) and invertible matrices ([Definition 0.9.13](#)). From Chapter 1, span ([Definition 1.3.2](#)), linear independence ([Definition 1.4.1](#)), bases ([Definition 1.5.1](#)) and dimension ([Definition 1.6.1](#)).

2.0.1 Roadmap

- **Three views of a linear system.** Intersecting lines and planes, combinations of columns, and a single matrix equation; the shape of the solution set as a particular solution plus the homogeneous solutions.
- **Gaussian elimination.** Row operations, echelon and reduced echelon forms, and how to read the solutions off the reduced form.
- **Uniqueness of the reduced form.** Different sequences of row operations always reach the same reduced matrix, so pivot columns are well defined.
- **Elementary matrices and inverses.** Row operations as left multiplication, the invertible matrix theorem, and computing inverses by row reduction.
- **Rank.** Row space, column space and null space; row rank equals column rank; the Rouché–Capelli theorem; and a toolkit for the questions of Chapter 1.
- **LU factorization.** Storing elimination as a product of triangular matrices, with and without row swaps.
- **Applications.** Polynomial interpolation, balancing chemical equations, networks, and a puzzle over $\mathbb{F}_2$.

2.0.2 Named moves

- **Row operations do not change the solution set.** So simplify first, then read the answer.
- **Put it in reduced form, then read off.** Consistency, free variables, bases of the column, row and null spaces all come from the reduced row echelon form.
- **A question about vectors is a question about a matrix.** Put the vectors in as columns; independence, spanning and membership become statements about pivots.
- **Pivots count dimensions.** The number of pivots is the rank, and the free variables count the dimension of the null space.

2.1 Three Views of a Linear System

Chapter 1 kept running into the same kind of question. Is this vector in the span of those? Is this list independent? Every time, the question turned into a handful of equations in unknown coefficients, and we solved them by hand. This chapter gives those equations a name, looks at them from three sides, and turns solving them into an algorithm with a proof behind it. This first section sets up the language and answers a structural question before any computing: what can the set of all solutions look like?

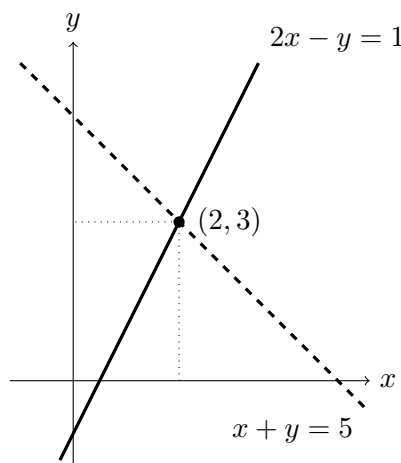
2.1.1 One system, three pictures

Here is a system of two equations in two unknowns, over $\mathbb{R}$:

$$\begin{aligned} 2x - y &= 1, \\ x + y &= 5. \end{aligned}$$

Adding the equations gives $3x = 6$, so $x = 2$, and then $y = 5 - x = 3$. We can read this one small system in three different ways, and each will be useful.

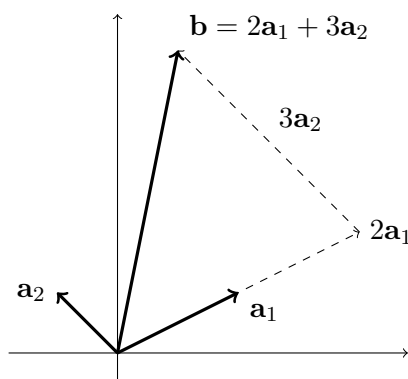
The row picture. Each equation describes a line in the plane: $y = 2x - 1$ and $y = 5 - x$. A solution is a point on **both** lines, so solving the system means intersecting them. The lines meet at $(2, 3)$.



The column picture. Stack the coefficients of x into one column and those of y into another. The system says

$$x \begin{pmatrix} 2 \\ 1 \end{pmatrix} + y \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}.$$

Now the unknowns are **weights**. We are asked how much of the vector $\mathbf{a}_1 = (2, 1)$ and how much of $\mathbf{a}_2 = (-1, 1)$ to combine to reach $\mathbf{b} = (1, 5)$. The answer $x = 2$, $y = 3$ says: walk twice along $\mathbf{a}_1$, then three times along $\mathbf{a}_2$.



The matrix picture. Collect the coefficients into a matrix and the unknowns into a column. By the definition of matrix multiplication ([Definition 0.9.9](#)), the system is the single equation

$$\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}, \quad \text{that is,} \quad A\mathbf{x} = \mathbf{b}.$$

The three pictures say the same thing, and they are tied together by one fact from Chapter 0: $A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$, a combination of the columns of A weighted by the entries of $\mathbf{x}$ ([Theorem 0.9.3](#)). The row picture is the one we draw. The column picture connects to Chapter 1, since it asks whether $\mathbf{b}$ lies in a span. The matrix picture is the one we compute and prove things with.

Remark. With more unknowns the pictures can no longer be drawn, but they do not change. An equation in three unknowns whose coefficients are not all zero describes a plane in $\mathbb{R}^3$, and a system asks where several planes meet. In the column picture, three unknowns mean three columns, and we ask whether a combination of them hits $\mathbf{b}$.

2.1.2 Linear systems

The example had two equations and two unknowns over $\mathbb{R}$. The general notion allows any numbers of each, and any field. Allowing any field is not generality for its own sake. Systems over $\mathbb{F}_2$ describe parity checks and switching puzzles, and systems over $\mathbb{Q}$ are what a computer can solve exactly.

A linear system is a finite list of equations, each saying that a fixed combination of the unknowns equals a fixed scalar.

Definition 2.1.1 — Linear system

Let F be a field and let $m, n \geq 1$. A **system of m linear equations in n unknowns $x_1, \dots, x_n$ over F** is a list of equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2, \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m, \end{aligned}$$

where all a_{ij} and b_i lie in F .

- The **coefficient matrix** is $A = (a_{ij}) \in M_{m \times n}(F)$, the **right-hand side** is $\mathbf{b} = (b_1, \dots, b_m) \in F^m$, and the system is written $A\mathbf{x} = \mathbf{b}$.
- The **augmented matrix** is the $m \times (n+1)$ matrix $[A \mid \mathbf{b}]$ obtained by appending $\mathbf{b}$ to A as a last column.
- A **solution** is a vector $\mathbf{s} = (s_1, \dots, s_n) \in F^n$ such that **every** equation holds when $x_j = s_j$ for all j ; equivalently, $A\mathbf{s} = \mathbf{b}$. The **solution set** is the set of all solutions, a subset of F^n .
- The system is **consistent** if it has **at least one** solution, and **inconsistent** if its solution set is empty.

In words: the coefficient matrix records the left-hand sides, row i for equation i and column j for unknown x_j . The augmented matrix records the whole system, and it is the object we will manipulate in the next section. A solution is a single vector of F^n , and it must satisfy all m equations at once, not just some of them. “Equivalently, $A\mathbf{s} = \mathbf{b}$ ” holds because the i -th entry of $A\mathbf{s}$ is $a_{i1}s_1 + \cdots + a_{in}s_n$ by [Definition 0.9.9](#), and two columns are equal exactly when all their entries agree.

Two small words deserve attention. A solution lives in F^n , with entries in **the field F** : the equation $2x = 1$ has the solution $\frac{1}{2}$ over $\mathbb{Q}$, but over $\mathbb{F}_2$ it reads $0 = 1$ and has none. And “consistent” means **at least one** solution, not exactly one.

Example. (None, one, or a whole line)

Find the solution set of each system over $\mathbb{R}$, and describe it in the row picture.

- (a) $x + 2y = 1, 2x + 4y = 5$.
- (b) $2x - y = 1, x + y = 5$.
- (c) $x + 2y = 1, 2x + 4y = 2$.

Solution.

- (a) Suppose (x, y) is a solution. Doubling the first equation gives $2x + 4y = 2$, while the second says $2x + 4y = 5$. Then $2 = 5$, which is false. So there is no solution, and the system is **inconsistent**. In the row picture, the lines $y = \frac{1}{2} - \frac{1}{2}x$ and $y = \frac{5}{4} - \frac{1}{2}x$ have the same slope and different intercepts: they are parallel and never meet.
- (b) As computed at the start of the section, any solution has $x = 2$ and $y = 3$;

conversely $2 \cdot 2 - 3 = 1$ and $2 + 3 = 5$. The solution set is $\{(2, 3)\}$, the single point where two non-parallel lines cross.

- (c) The second equation is twice the first, so a pair (x, y) satisfies both exactly when it satisfies $x + 2y = 1$, that is, when $x = 1 - 2y$. Writing t for y , the solution set is $\{(1 - 2t, t) : t \in \mathbb{R}\}$. Both equations describe the **same** line, and every point on it is a solution.

The three answers are the three things two lines can do: miss, cross, or coincide. Over $\mathbb{R}$, and over any infinite field, no other sizes are possible: once we know the shape of the solution set (Theorem 2.1.3 below), this is a short argument, which you will give in Exercise (C1: How many solutions?). For instance, no system over $\mathbb{R}$ has exactly two solutions. Over a finite field the picture is different.

Example. (Exactly two solutions over $\mathbb{F}_2$)

Find all solutions of the system $x + y + z = 1$, $y + z = 0$ over $\mathbb{F}_2$. Then compare with the same system over $\mathbb{R}$.

Solution. In $\mathbb{F}_2$ we have $-1 = 1$, so the second equation says $y = -z = z$. Substituting into the first gives $x + 2z = 1$, and $2z = 0$ in $\mathbb{F}_2$, so $x = 1$. Conversely, every $(1, z, z)$ satisfies both equations: $1 + z + z = 1 + 2z = 1$ and $z + z = 0$. Since z can only be 0 or 1, the solution set is

$$\{(1, 0, 0), (1, 1, 1)\},$$

with exactly two elements.

Over $\mathbb{R}$ the second equation gives $y = -z$, and then the first gives $x = 1$. The solution set is $\{(1, -t, t) : t \in \mathbb{R}\}$, a line in $\mathbb{R}^3$ with infinitely many points. The equations are the same, but the field decides how many values the free unknown can take.

Now a non-example, by a minimal change. The equation $x + y^2 = 1$ looks like one of the equations above, and it still has a left side built from the unknowns and a constant on the right. But the unknown y appears squared, so the left side is not of the form $a_1x_1 + a_2x_2$ with **fixed scalars** a_1, a_2 . The clause that fails is “a fixed combination of the unknowns”. Its solution set in $\mathbb{R}^2$ is a parabola, not a line, and nothing in this chapter applies to it.

Warning. “**Consistent**” does not mean “has exactly one solution”. The system $x + 2y = 1$, $2x + 4y = 2$ of Example (None, one, or a whole line) (c) is consistent, and it has infinitely many solutions. Consistency only rules out the empty solution set. Whether the solution is unique is a separate question, answered by Corollary 2.1.4 below.

Why record the augmented matrix at all, when $A\mathbf{x} = \mathbf{b}$ is shorter? Because solving a system never touches the names of the unknowns. Adding two equations adds their coefficients and their right-hand sides, and nothing else. The augmented matrix keeps exactly this data and throws away the symbols $x_1, \dots, x_n$ and the plus signs. In the next section every manipulation is performed on it.

One kind of system is always consistent, because it has an obvious solution.

Definition 2.1.2 — Homogeneous system

A linear system $A\mathbf{x} = \mathbf{b}$ over F is **homogeneous** if $\mathbf{b} = \mathbf{0}$. The vector $\mathbf{0} \in F^n$ is a solution of every homogeneous system $A\mathbf{x} = \mathbf{0}$, called the **trivial solution**. For any system $A\mathbf{x} = \mathbf{b}$, the system $A\mathbf{x} = \mathbf{0}$, with the same coefficient matrix, is its **associated homogeneous system**.

The trivial solution exists by the zero rule of [Theorem 0.9.4](#): $A\mathbf{0} = \mathbf{0}$. So for a homogeneous system the interesting question is never “is there a solution?” but “is there a **non-trivial** one?”. For example, $x + 2y = 0$, $2x + 4y = 0$ has the non-trivial solution $(-2, 1)$, while $2x - y = 0$, $x + y = 0$ has only the trivial one: adding the equations gives $3x = 0$, so $x = 0$, and then $y = -x = 0$.

Quick check. Over $\mathbb{R}$, write the system $x_1 - x_2 + 3x_3 = 4$, $2x_2 + x_3 = 0$ in the matrix picture and the column picture, and give its augmented matrix. Is $(4, 0, 0)$ a solution? Is $(1, 0, 1)$?

Solution. With $A = \begin{pmatrix} 1 & -1 & 3 \\ 0 & 2 & 1 \end{pmatrix}$ and $\mathbf{b} = (4, 0)$, the system is $A\mathbf{x} = \mathbf{b}$. In the column picture it reads $x_1(1, 0) + x_2(-1, 2) + x_3(3, 1) = (4, 0)$. The augmented matrix is

$$[A \mid \mathbf{b}] = \left[\begin{array}{ccc|c} 1 & -1 & 3 & 4 \\ 0 & 2 & 1 & 0 \end{array} \right].$$

For $(4, 0, 0)$: $4 - 0 + 0 = 4$ and $0 + 0 = 0$, so it is a solution. For $(1, 0, 1)$: the first equation gives $1 - 0 + 3 = 4$, but the second gives $0 + 1 = 1 \neq 0$. One failing equation is enough, so it is not a solution.

2.1.3 Consistency is a question about span

The column picture turns “does this system have a solution?” into a question from Chapter 1.

Theorem 2.1.1 — Consistency and the span of the columns

Let $A \in M_{m \times n}(F)$ have columns $\mathbf{a}_1, \dots, \mathbf{a}_n \in F^m$, and let $\mathbf{b} \in F^m$. Then the system $A\mathbf{x} = \mathbf{b}$ is consistent **if and only if** $\mathbf{b} \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$. More precisely, $\mathbf{s} = (s_1, \dots, s_n)$ is a solution exactly when $\mathbf{b} = s_1\mathbf{a}_1 + \dots + s_n\mathbf{a}_n$.

Proof. By [Theorem 0.9.3](#), $A\mathbf{s} = s_1\mathbf{a}_1 + \dots + s_n\mathbf{a}_n$ for every $\mathbf{s} \in F^n$. Hence $\mathbf{s}$ is a solution, that is $A\mathbf{s} = \mathbf{b}$, exactly when $\mathbf{b} = s_1\mathbf{a}_1 + \dots + s_n\mathbf{a}_n$. This proves the second statement.

($\Rightarrow$) If the system is consistent, it has a solution $\mathbf{s}$, and then $\mathbf{b} = s_1\mathbf{a}_1 + \dots + s_n\mathbf{a}_n$ is a linear combination of the columns. By [Definition 1.3.2](#), $\mathbf{b} \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$.

($\Leftarrow$) If $\mathbf{b} \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$, then $\mathbf{b} = s_1\mathbf{a}_1 + \dots + s_n\mathbf{a}_n$ for some $s_1, \dots, s_n \in F$, and $\mathbf{s} = (s_1, \dots, s_n)$ is a solution by the first paragraph. Hence the system is consistent. ■

Read backwards, the theorem is a recipe for Chapter 1 questions: to decide whether $\mathbf{b}$ lies in the span of some vectors of F^m , put the vectors as the columns of a matrix and ask whether the system is consistent.

Example. (Span membership as a system)

In $\mathbb{R}^3$, let $\mathbf{v}_1 = (1, 2, 3)$ and $\mathbf{v}_2 = (0, 1, 2)$. Decide whether $(1, 4, 7)$ and $(1, 4, 8)$ lie in $\text{span}(\mathbf{v}_1, \mathbf{v}_2)$.

Solution. Let A be the 3×2 matrix with columns $\mathbf{v}_1, \mathbf{v}_2$. By [Theorem 2.1.1](#), $(1, 4, 7) \in \text{span}(\mathbf{v}_1, \mathbf{v}_2)$ if and only if the system

$$a = 1, \quad 2a + b = 4, \quad 3a + 2b = 7$$

in the unknowns a, b is consistent. The first equation gives $a = 1$, and then the second gives $b = 2$. The third equation must also hold, and it does: $3 + 4 = 7$. Hence $(1, 4, 7) = \mathbf{v}_1 + 2\mathbf{v}_2$ lies in the span.

For $(1, 4, 8)$, the first two equations are the same, so any solution has $a = 1$ and $b = 2$. The third equation now demands $3 + 4 = 8$, which is false. The system is inconsistent, so $(1, 4, 8) \notin \text{span}(\mathbf{v}_1, \mathbf{v}_2)$.

Geometrically, $\text{span}(\mathbf{v}_1, \mathbf{v}_2)$ is a plane through $\mathbf{0}$ in $\mathbb{R}^3$. The first right-hand side lies on that plane and the second does not.

2.1.4 The shape of the solution set

We now describe the solution set of an arbitrary system, starting with the homogeneous case. Chapter 1 already did the work there: in [Example \(Solutions of a Homogeneous System\)](#) we checked, with the three conditions of the subspace test, that the solutions of $A\mathbf{x} = \mathbf{0}$ form a subspace.

Theorem 2.1.2 — Homogeneous solutions form a subspace

Let $A \in M_{m \times n}(F)$. The solution set $N = \{\mathbf{x} \in F^n : A\mathbf{x} = \mathbf{0}\}$ of the homogeneous system $A\mathbf{x} = \mathbf{0}$ is a subspace of F^n .

Proof. This is [Example \(Solutions of a Homogeneous System\)](#): $A\mathbf{0} = \mathbf{0}$, and for $\mathbf{x}, \mathbf{y} \in N$ and $c \in F$, the rules of [Theorem 0.9.4](#) give $A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y} = \mathbf{0}$ and $A(c\mathbf{x}) = c(A\mathbf{x}) = \mathbf{0}$. By the subspace test ([Theorem 1.2.1](#)), N is a subspace of F^n . ■

The solutions of $A\mathbf{x} = \mathbf{b}$ with $\mathbf{b} \neq \mathbf{0}$ behave differently.

Warning. The solution set of $A\mathbf{x} = \mathbf{b}$ with $\mathbf{b} \neq \mathbf{0}$ is never a subspace. A subspace must contain $\mathbf{0}$, but $A\mathbf{0} = \mathbf{0} \neq \mathbf{b}$, so $\mathbf{0}$ is not a solution. For example, the solutions of $x - y = 1$ in $\mathbb{R}^2$ form a line that misses the origin. The sum of the solutions $(1, 0)$ and $(2, 1)$ is $(3, 1)$, and $3 - 1 = 2 \neq 1$: adding two solutions does not give a solution.

What does survive is this: the **difference** of two solutions of $A\mathbf{x} = \mathbf{b}$ solves the homogeneous system, since $A(\mathbf{s} - \mathbf{s}') = \mathbf{b} - \mathbf{b} = \mathbf{0}$. So once one solution is known, every other one differs from it by a homogeneous solution. That is the whole structure.

Theorem 2.1.3 — Structure of the solution set

Let $A \in M_{m \times n}(F)$ and $\mathbf{b} \in F^m$, let N be the solution set of $A\mathbf{x} = \mathbf{0}$, and suppose $\mathbf{p} \in F^n$ is **one** solution of $A\mathbf{x} = \mathbf{b}$. Then the solution set of $A\mathbf{x} = \mathbf{b}$ is

$$\mathbf{p} + N = \{\mathbf{p} + \mathbf{h} : \mathbf{h} \in N\}.$$

Idea. The theorem is “particular plus homogeneous”. To prove equality of two sets, prove both inclusions. Going from a solution $\mathbf{s}$ to $\mathbf{p} + N$ uses the difference $\mathbf{s} - \mathbf{p}$, as observed above; going back uses the sum $A(\mathbf{p} + \mathbf{h}) = A\mathbf{p} + A\mathbf{h}$.

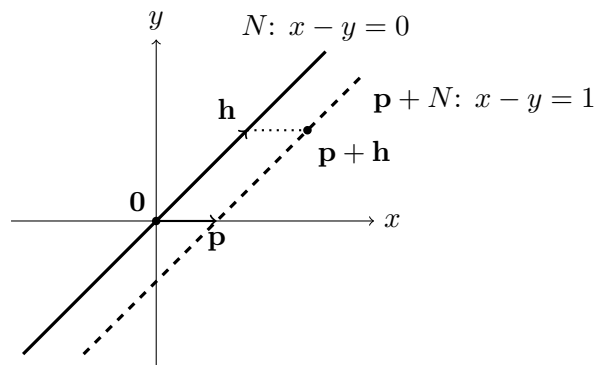
Proof. Let S be the solution set of $A\mathbf{x} = \mathbf{b}$. We use [Theorem 0.3.1](#).

($\subseteq$) Let $\mathbf{s} \in S$, and put $\mathbf{h} = \mathbf{s} - \mathbf{p}$. By distributivity ([Theorem 0.9.4](#)), $A\mathbf{h} = A\mathbf{s} - A\mathbf{p} = \mathbf{b} - \mathbf{b} = \mathbf{0}$, so $\mathbf{h} \in N$. Hence $\mathbf{s} = \mathbf{p} + \mathbf{h} \in \mathbf{p} + N$.

($\supseteq$) Let $\mathbf{h} \in N$. Then $A(\mathbf{p} + \mathbf{h}) = A\mathbf{p} + A\mathbf{h} = \mathbf{b} + \mathbf{0} = \mathbf{b}$, again by distributivity, so $\mathbf{p} + \mathbf{h} \in S$.

Therefore $S = \mathbf{p} + N$, as claimed. ■

The theorem splits solving $A\mathbf{x} = \mathbf{b}$ into two independent jobs: find **one** solution, and find **all** solutions of the homogeneous system. The set $\mathbf{p} + N$ is a translate of the subspace N by the vector $\mathbf{p}$, the kind of set that the picture in [Example \(A Line Missing the Origin\)](#) showed. For $x - y = 1$ in $\mathbb{R}^2$, the homogeneous solutions form the line $N = \text{span}((1, 1))$ through $\mathbf{0}$, one solution is $\mathbf{p} = (1, 0)$, and the solution set is the parallel line through $\mathbf{p}$.



A system solved by substitution fits the same pattern once we collect the free parameter.

Example. (Particular plus homogeneous)

Solve $x_1 + 2x_2 - x_3 = 4$, $x_2 + x_3 = 1$ over $\mathbb{R}$, and write the solution set in the form $\mathbf{p} + \text{span}(\mathbf{h})$.

Solution. The second equation gives $x_2 = 1 - x_3$. Substituting into the first, $x_1 = 4 - 2(1 - x_3) + x_3 = 2 + 3x_3$. The value of x_3 is not constrained, so write $x_3 = t$. Conversely, for every $t \in \mathbb{R}$ the vector $(2 + 3t, 1 - t, t)$ satisfies both equations, since

$(2 + 3t) + 2(1 - t) - t = 4$ and $(1 - t) + t = 1$. Hence the solution set is

$$\{(2 + 3t, 1 - t, t) : t \in \mathbb{R}\} = \{(2, 1, 0) + t(3, -1, 1) : t \in \mathbb{R}\} = \mathbf{p} + \text{span}(\mathbf{h}),$$

with $\mathbf{p} = (2, 1, 0)$ and $\mathbf{h} = (3, -1, 1)$. As [Theorem 2.1.3](#) predicts, $\mathbf{p}$ is a solution (take $t = 0$), and $\mathbf{h}$ solves the homogeneous system: $3 - 2 - 1 = 0$ and $-1 + 1 = 0$. In the row picture, two planes in $\mathbb{R}^3$ meet in a line, and this line does not pass through $\mathbf{0}$.

2.1.5 When is the solution unique?

A consistent system has a unique solution exactly when there is nothing to add to the particular solution. The theorem just proved turns this into a statement about the homogeneous system, and the column picture turns that into independence.

Corollary 2.1.4 — Uniqueness of solutions

Let $A \in M_{m \times n}(F)$ have columns $\mathbf{a}_1, \dots, \mathbf{a}_n$, and let $\mathbf{b} \in F^m$ be such that $A\mathbf{x} = \mathbf{b}$ is **consistent**. The following are equivalent:

- (a) $A\mathbf{x} = \mathbf{b}$ has exactly one solution;
- (b) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution;
- (c) the list $(\mathbf{a}_1, \dots, \mathbf{a}_n)$ is linearly independent.

Proof. Let N be the solution set of $A\mathbf{x} = \mathbf{0}$. Since the system is consistent, it has a solution $\mathbf{p}$, and by [Theorem 2.1.3](#) its solution set is $\mathbf{p} + N$.

(a $\Leftrightarrow$ b) The function $N \rightarrow \mathbf{p} + N$, $\mathbf{h} \mapsto \mathbf{p} + \mathbf{h}$, is surjective by definition of $\mathbf{p} + N$, and injective because $\mathbf{p} + \mathbf{h} = \mathbf{p} + \mathbf{h}'$ implies $\mathbf{h} = \mathbf{h}'$ by adding $-\mathbf{p}$ to both sides. So it is a bijection, and $\mathbf{p} + N$ has exactly one element if and only if N does. Since $\mathbf{0} \in N$ always, N has exactly one element if and only if $N = \{\mathbf{0}\}$, which is (b).

(b $\Leftrightarrow$ c) By [Theorem 0.9.3](#), $A\mathbf{x} = x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n$ for every $\mathbf{x} \in F^n$. So (b) says: for all $x_1, \dots, x_n \in F$, $x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n = \mathbf{0}$ implies $x_1 = \dots = x_n = 0$. This is word for word [Definition 1.4.1](#) for the list $(\mathbf{a}_1, \dots, \mathbf{a}_n)$. ■

Two remarks on the hypotheses. The equivalence of (b) and (c) does not need consistency; only (a) does. And consistency cannot be dropped from (a $\Leftrightarrow$ b): the system $x = 0$, $x = 1$ in one unknown has only the trivial solution in its homogeneous version ($x = 0$, $x = 0$), but it has no solution at all, so it certainly does not have exactly one.

So the three pictures answer the two basic questions about $A\mathbf{x} = \mathbf{b}$ in the language of Chapter 1: a solution **exists** when $\mathbf{b}$ is in the span of the columns, and it is **unique** when the columns are independent. What we still lack is a reliable way to decide either question for a given matrix. That is the job of Gaussian elimination, in the next section.

2.1.6 Exercises

2.1.6.1 A. Check your understanding

Exercise A1.

- Define what it means for a linear system $A\mathbf{x} = \mathbf{b}$ over F to be consistent, and what its augmented matrix is. What is the size of the augmented matrix of a system of m equations in n unknowns?
- Determine whether the following statement is true: “the solution set of $A\mathbf{x} = \mathbf{b}$ is a subspace of F^n .” Justify your answer.
- Determine whether the following statement is true: “if $A\mathbf{x} = \mathbf{b}$ has two different solutions, then $A\mathbf{x} = \mathbf{0}$ has a non-trivial solution.” Justify your answer.
- Let $A \in M_{3 \times 5}(\mathbb{R})$. In which spaces do the unknown vector $\mathbf{x}$ and the right-hand side $\mathbf{b}$ of $A\mathbf{x} = \mathbf{b}$ live?
- Name the theorem that turns “is $\mathbf{b}$ in the span of $\mathbf{a}_1, \dots, \mathbf{a}_n$?” into a question about a linear system, and state which system.

Solution.

- It is consistent if it has at least one solution $\mathbf{s} \in F^n$, that is, some $\mathbf{s}$ with $A\mathbf{s} = \mathbf{b}$. The augmented matrix is $[A \mid \mathbf{b}]$, the matrix A with $\mathbf{b}$ appended as an extra last column. It has size $m \times (n + 1)$.
- False in general. If $\mathbf{b} \neq \mathbf{0}$, then $A\mathbf{0} = \mathbf{0} \neq \mathbf{b}$, so $\mathbf{0}$ is not a solution and the solution set is not a subspace. (If $\mathbf{b} = \mathbf{0}$, it is a subspace, by [Theorem 2.1.2](#).)
- True. If $\mathbf{s} \neq \mathbf{s}'$ are solutions, then $A(\mathbf{s} - \mathbf{s}') = \mathbf{b} - \mathbf{b} = \mathbf{0}$ by distributivity, and $\mathbf{s} - \mathbf{s}' \neq \mathbf{0}$.
- $\mathbf{x} \in \mathbb{R}^5$, one entry per column of A , and $\mathbf{b} \in \mathbb{R}^3$, one entry per row.
- [Theorem 2.1.1](#): $\mathbf{b} \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$ if and only if $A\mathbf{x} = \mathbf{b}$ is consistent, where A is the matrix with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$.

2.1.6.2 B. Practice

Exercise B1: Three views.

- Write the system $x_1 - x_2 + 2x_3 = 3$, $2x_1 + x_3 = 1$ over $\mathbb{R}$ in the matrix picture and in the column picture, and give its augmented matrix.
- Write out, as a list of equations, the system with augmented matrix
$$\left[\begin{array}{cc|c} 1 & 4 & 0 \\ 0 & 1 & 2 \\ 3 & -1 & 5 \end{array} \right]$$
 over $\mathbb{R}$. How many equations and unknowns does it have? Hence decide whether it is consistent.

Solution.

- (a) The matrix picture is

$$\begin{pmatrix} 1 & -1 & 2 \\ 2 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$

The column picture is $x_1(1, 2) + x_2(-1, 0) + x_3(2, 1) = (3, 1)$. The augmented matrix is $\left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 2 & 0 & 1 & 1 \end{array} \right]$.

- (b) The system is $x_1 + 4x_2 = 0$, $x_2 = 2$, $3x_1 - x_2 = 5$: three equations in two unknowns. The second equation gives $x_2 = 2$, and then the first gives $x_1 = -8$. The third equation would require $-24 - 2 = 5$, which is false. Hence no vector satisfies all three equations, and the system is inconsistent.

Exercise B2: In the span or not?

Determine whether the vector $\mathbf{b}$ lies in the span of the given vectors. Justify your answer, and when it does, write $\mathbf{b}$ as a linear combination of them.

- (a) $\mathbf{b} = (2, 1, 4)$ and $(1, 0, 1), (0, 1, 2)$ in $\mathbb{R}^3$.
 (b) $\mathbf{b} = (1, 1, 1)$ and $(1, 2, 0), (2, 4, 1)$ in $\mathbb{R}^3$.
 (c) $\mathbf{b} = (1, 1, 0)$ and $(1, 0, 1), (0, 1, 1)$ in $\mathbb{F}_2^3$. Is the answer the same in $\mathbb{R}^3$?

Solution. By [Theorem 2.1.1](#), in each case we decide whether the system $a\mathbf{v}_1 + b\mathbf{v}_2 = \mathbf{b}$ in the unknowns a, b is consistent.

- (a) The system is $a = 2$, $b = 1$, $a + 2b = 4$. The first two equations force $a = 2$, $b = 1$, and then $a + 2b = 4$ holds. Hence $(2, 1, 4) = 2(1, 0, 1) + (0, 1, 2)$ is in the span.
 (b) The system is $a + 2b = 1$, $2a + 4b = 1$, $b = 1$. Doubling the first equation gives $2a + 4b = 2$, contradicting the second, since $2 \neq 1$. Hence the system is inconsistent and $(1, 1, 1)$ is not in the span.
 (c) The system is $a = 1$, $b = 1$, $a + b = 0$. The first two force $a = b = 1$, and then $a + b = 1 + 1 = 0$ in $\mathbb{F}_2$. Hence $(1, 1, 0) = (1, 0, 1) + (0, 1, 1)$ is in the span over $\mathbb{F}_2$. In $\mathbb{R}^3$ the same first two equations force $a = b = 1$, but $1 + 1 = 2 \neq 0$, so the system is inconsistent and $\mathbf{b}$ is **not** in the span. The answer depends on the field.

Exercise B3: Solve and describe.

Solve each system over $\mathbb{R}$ by substitution, and write its solution set in the form $\mathbf{p} + \text{span}(\dots)$. Check that your $\mathbf{p}$ is a solution and that your spanning vectors solve the associated homogeneous system.

- (a) $x_1 + x_2 + x_3 = 2$, $x_2 - x_3 = 1$.
 (b) $x_1 - 2x_2 + x_3 - x_4 = 1$, $x_3 + 2x_4 = 3$.

Solution.

- (a) The second equation gives $x_2 = 1 + x_3$, and then the first gives $x_1 = 2 - (1 + x_3) - x_3 = 1 - 2x_3$. With $x_3 = t$ free, every solution is $(1 - 2t, 1 + t, t)$, and conversely each such vector satisfies both equations. So the solution set is $(1, 1, 0) + \text{span}((-2, 1, 1))$.

Check: $1 + 1 + 0 = 2$ and $1 - 0 = 1$; and $-2 + 1 + 1 = 0$, $1 - 1 = 0$.

- (b) The second equation gives $x_3 = 3 - 2x_4$. The first then gives $x_1 = 1 + 2x_2 - (3 - 2x_4) + x_4 = -2 + 2x_2 + 3x_4$. The unknowns $x_2 = s$ and $x_4 = t$ are free, and every solution is

$$(-2 + 2s + 3t, s, 3 - 2t, t) = (-2, 0, 3, 0) + s(2, 1, 0, 0) + t(3, 0, -2, 1),$$

and conversely each such vector is a solution. So the solution set is $(-2, 0, 3, 0) + \text{span}((2, 1, 0, 0), (3, 0, -2, 1))$. Check: $-2 - 0 + 3 - 0 = 1$ and $3 + 0 = 3$; for $(2, 1, 0, 0)$: $2 - 2 + 0 - 0 = 0$ and $0 + 0 = 0$; for $(3, 0, -2, 1)$: $3 - 0 - 2 - 1 = 0$ and $-2 + 2 = 0$.

2.1.6.3 C. Going deeper

Exercise C1: How many solutions?

Let $A \in M_{m \times n}(F)$ and $\mathbf{b} \in F^m$, and let N be the solution set of $A\mathbf{x} = \mathbf{0}$.

- Suppose F is **infinite**. Prove that $A\mathbf{x} = \mathbf{b}$ has either no solution, exactly one solution, or infinitely many solutions.
- Suppose F is **finite**. Prove that $A\mathbf{x} = \mathbf{b}$ has either no solution or exactly $|N|$ solutions.
- Give a system over $\mathbb{F}_3$ with exactly three solutions.

Hint: for (a), if N contains a non-zero vector $\mathbf{h}$, look at its multiples.

Solution.

- Suppose the system is consistent, with a solution $\mathbf{p}$. By Theorem 2.1.3 its solution set is $\mathbf{p} + N$, and as in the proof of Corollary 2.1.4 the map $\mathbf{h} \mapsto \mathbf{p} + \mathbf{h}$ is a bijection $N \rightarrow \mathbf{p} + N$. If $N = \{\mathbf{0}\}$, there is exactly one solution. Otherwise, let $\mathbf{h} \in N$ with $\mathbf{h} \neq \mathbf{0}$. Since N is a subspace (Theorem 2.1.2), $c\mathbf{h} \in N$ for every $c \in F$. If $c\mathbf{h} = c'\mathbf{h}$, then $(c - c')\mathbf{h} = \mathbf{0}$, and since $\mathbf{h} \neq \mathbf{0}$, Theorem 1.1.9 gives $c = c'$. So $c \mapsto c\mathbf{h}$ is injective, and N contains infinitely many vectors because F is infinite. Hence $\mathbf{p} + N$ is infinite.
- If the system is consistent with solution $\mathbf{p}$, the bijection $N \rightarrow \mathbf{p} + N$ from (a) shows that the solution set has exactly $|N|$ elements. Otherwise it has none.
- The single equation $x + y = 1$ over $\mathbb{F}_3$ (a system with $m = 1$, $n = 2$). For each of the three values of $y \in \mathbb{F}_3$ there is exactly one $x = 1 - y$, so the solutions are $(1, 0)$, $(0, 1)$, $(2, 2)$. Consistently with (b), the homogeneous equation $x + y = 0$ has the three solutions $(0, 0)$, $(2, 1)$, $(1, 2)$.

Exercise C2: The right-hand sides that work.

Let $A \in M_{m \times n}(F)$ have columns $\mathbf{a}_1, \dots, \mathbf{a}_n$.

- Prove that the set $C = \{\mathbf{b} \in F^m : A\mathbf{x} = \mathbf{b} \text{ is consistent}\}$ is a subspace of F^m .
- Deduce that if $n < m$, then there is some $\mathbf{b} \in F^m$ for which $A\mathbf{x} = \mathbf{b}$ is inconsistent.

- (c) For $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 1 \end{pmatrix}$ over $\mathbb{R}$, find one such $\mathbf{b}$.

Solution.

- (a) By Theorem 2.1.1, $\mathbf{b} \in C$ if and only if $\mathbf{b} \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$. So $C = \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$, which is a subspace of F^m by Theorem 1.3.1.
- (b) By (a), C is spanned by the n vectors $\mathbf{a}_1, \dots, \mathbf{a}_n$. Since $\dim F^m = m$ (Example (Dimensions of the Running Examples)), Theorem 1.6.3 says no list of fewer than m vectors spans F^m . As $n < m$, $C \neq F^m$, so some $\mathbf{b} \in F^m$ is not in C , and for that $\mathbf{b}$ the system is inconsistent.
- (c) Take $\mathbf{b} = (1, 0, 0)$. A solution would satisfy $x + 2y = 1$ and $2x + 4y = 0$. Doubling the first gives $2x + 4y = 2 \neq 0$, a contradiction. So $A\mathbf{x} = (1, 0, 0)$ is inconsistent.

2.2 Gaussian Elimination

Every reader of this book has solved a system of equations by eliminating unknowns: subtract a multiple of one equation from another, and a variable disappears. Done by hand on small systems, this feels like common sense. Done on a system of 50 equations, or over $\mathbb{F}_2$, or in a proof about all systems at once, it needs to be an algorithm, and we need to know that it never changes the answer. This section makes elimination precise, proves that it always reaches a standard shape, and shows how to read every solution from that shape.

2.2.1 Three moves that do not change the answer

Here is a system over $\mathbb{R}$, solved the way one would solve it on paper:

$$\begin{aligned} x + 2y + z &= 4, \\ 2x + 4y + 3z &= 9, \\ x + 3y + z &= 5. \end{aligned}$$

Subtract twice the first equation from the second: $z = 1$. Subtract the first from the third: $y = 1$. Then the first equation gives $x = 4 - 2 - 1 = 1$.

Only three kinds of move were used, or could reasonably be used: **reorder** the equations, **multiply** an equation by a non-zero number, and **add a multiple** of one equation to another. None of them looks at the names x, y, z . Each acts on the coefficients and the right-hand side of whole equations, that is, on the rows of the augmented matrix. The first step above, for instance, replaces row 2, $(2, 4, 3 \mid 9)$, by row 2 minus twice row 1, $(0, 0, 1 \mid 1)$.

Not every move that looks harmless is safe. Multiplying an equation by 0 turns it into $0 = 0$, and a constraint silently disappears. So we list the allowed moves carefully.

An elementary row operation is a reversible move on the rows of a matrix: swap two rows, rescale one row by a non-zero scalar, or add a multiple of one row to a different row.

Definition 2.2.1 — Elementary row operations

Let F be a field and let $M \in M_{m \times k}(F)$ have rows $\mathbf{r}_1, \dots, \mathbf{r}_m$. An **elementary row operation** on M is one of the following.

- (1) **Swap** two rows: for $i \neq j$, exchange $\mathbf{r}_i$ and $\mathbf{r}_j$. Notation: $R_i \leftrightarrow R_j$.
- (2) **Scale** a row: for an index i and a scalar $c \in F$ with $c \neq 0$, replace $\mathbf{r}_i$ by $c\mathbf{r}_i$. Notation: $R_i \rightarrow cR_i$.
- (3) **Replace** a row by adding a multiple of **another** row: for indices $i \neq j$ and any $c \in F$, replace $\mathbf{r}_i$ by $\mathbf{r}_i + c\mathbf{r}_j$. Notation: $R_i \rightarrow R_i + cR_j$.

In each case the rows not mentioned are left unchanged.

In words: (1) reorders, (2) rescales one row by a **non-zero** scalar, and (3) changes **one** row, row i , using a **different** row j , which itself stays as it was. We write $R_i \rightarrow R_i - cR_j$ for $R_i \rightarrow R_i + (-c)R_j$. The notation R_i refers to the current row i at the moment the operation is performed.

Take $M = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 1 & 1 \end{pmatrix}$ over $\mathbb{R}$. Then $R_1 \leftrightarrow R_2$ gives $\begin{pmatrix} 3 & 1 & 1 \\ 1 & 2 & 0 \end{pmatrix}$, $R_1 \rightarrow -2R_1$ gives $\begin{pmatrix} -2 & -4 & 0 \\ 3 & 1 & 1 \end{pmatrix}$, and $R_2 \rightarrow R_2 - 3R_1$ gives $\begin{pmatrix} 1 & 2 & 0 \\ 0 & -5 & 1 \end{pmatrix}$. The degenerate case $m = 1$ is worth a glance: a one-row matrix admits no swap and no replacement, since both need two different rows, so its only elementary row operations are the scalings.

The minimal change that breaks the definition is to drop “ $c \neq 0$ ”. The move $R_2 \rightarrow 0R_2$ still replaces one row by a multiple of itself, but it sends $\begin{pmatrix} 1 & 2 & 0 \\ 3 & 1 & 1 \end{pmatrix}$ to $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, and no move of any kind can recover the lost row $(3, 1, 1)$ from that matrix. The failing clause is $c \neq 0$, and the damage is exactly irreversibility.

Warning. Perform operations one at a time. It is tempting to write “ $R_1 \rightarrow R_1 - R_2$ and $R_2 \rightarrow R_2 - R_1$ ” as one step, using the old rows in both. That is not a sequence of elementary row operations, and it can destroy information. On the system $x = 1$, $x = 1$ it produces $0 = 0$, $0 = 0$, whose solution set is all of F , not $\{1\}$. Done in order, the second operation uses the **new** row 1, $(0 \mid 0)$, and leaves row 2 unchanged, so no information is lost.

Why these three and no others? They are exactly the moves of hand elimination, and each can be undone by a move of the same kind, which is what the theorem below needs. The condition $i \neq j$ in (3) is not fussiness: “add c times row i to row i ” would be scaling by $1 + c$, which is illegal when $c = -1$ (Exercise C1).

An elementary row operation acts on the rows, but it treats each column separately: the new entry in position (i, l) is computed only from entries of column l . We record this, because it will be used again and again.

Lemma 2.2.1 — Row operations act column by column

Let $M = [M_1 \mid M_2] \in M_{m \times k}(F)$ be split into its first k_1 columns M_1 and its remaining columns M_2 . If an elementary row operation turns M into $M' = [M'_1 \mid M'_2]$ (split in the

same place), then the same operation turns M_1 into M'_1 and M_2 into M'_2 . In particular, an elementary row operation turns a zero column into a zero column.

Proof. Let m_{hl} be the entries of M and m'_{hl} those of M' . For a swap $R_i \leftrightarrow R_j$: $m'_{il} = m_{jl}$, $m'_{jl} = m_{il}$, and $m'_{hl} = m_{hl}$ for $h \neq i, j$. For a scaling $R_i \rightarrow cR_i$: $m'_{il} = cm_{il}$, and other rows are unchanged. For a replacement $R_i \rightarrow R_i + cR_j$: $m'_{il} = m_{il} + cm_{jl}$, and other rows are unchanged. In each case, every entry of column l of M' is computed from column l of M alone, by the same rule for every l . Hence applying the operation to M_1 or to M_2 separately gives M'_1 and M'_2 . If column l of M is zero, each formula gives 0, since $c \cdot 0 = 0$ and $0 + c \cdot 0 = 0$ (Theorem 0.7.1). ■

For an augmented matrix, the lemma says: operating on $[A \mid \mathbf{b}]$ is the same as operating on A and on $\mathbf{b}$ with the same moves. Now we name what sequences of moves produce.

Definition 2.2.2 — Row equivalence

Let $A, B \in M_{m \times k}(F)$. We say A is **row equivalent** to B if B can be obtained from A by a **finite** sequence of elementary row operations. The empty sequence is allowed, so every matrix is row equivalent to itself.

In Example (Row equivalence) of Chapter 0 we met a definition that looks different: $B = EA$ for an **invertible** matrix E . In section 4 of this chapter we prove that the two definitions agree (Theorem 2.4.7), by writing each elementary row operation as multiplication by an invertible matrix. Until then, “row equivalent” means the definition above.

2.2.2 Row operations preserve solutions

Here is the promise from the start of the section, now as a theorem.

Theorem 2.2.2 — Row operations preserve solutions

Let F be a field.

- (a) (**Reversibility**) If an elementary row operation turns M into M' , then an elementary row operation **of the same type** turns M' back into M .
- (b) Row equivalence is an equivalence relation on $M_{m \times k}(F)$.
- (c) If $[A \mid \mathbf{b}]$ and $[A' \mid \mathbf{b}']$ in $M_{m \times (n+1)}(F)$ are row equivalent, then the systems $A\mathbf{x} = \mathbf{b}$ and $A'\mathbf{x} = \mathbf{b}'$ have the **same** solution set.

Idea. Reversibility is checked type by type, and the check for scaling is the only place where $c \neq 0$ is used. For (c), every equation of the new system is a combination of equations of the old one, so an old solution satisfies the new system. That gives one inclusion. Instead of repeating the argument backwards, reversibility gives the other inclusion for free.

Proof.

- (a) A swap $R_i \leftrightarrow R_j$ is undone by the same swap. A scaling $R_i \rightarrow cR_i$ is undone by $R_i \rightarrow c^{-1}R_i$, which is a legal scaling because $c \neq 0$ has an inverse $c^{-1} \neq 0$ in F (Definition 0.7.1): the row becomes $c^{-1}(c\mathbf{r}_i) = \mathbf{r}_i$. A replacement $R_i \rightarrow R_i + cR_j$ is undone by $R_i \rightarrow R_i - cR_j$. Here we use $i \neq j$: row j was not changed by the first operation, so the second one gives $(\mathbf{r}_i + c\mathbf{r}_j) - c\mathbf{r}_j = \mathbf{r}_i$.
- (b) Reflexivity holds via the empty sequence. For symmetry, suppose operations $O_1, \dots, O_s$, in this order, turn A into B . By (a), each O_t has an undoing operation O'_t . Then $O'_s, O'_{s-1}, \dots, O'_1$, in this order, turn B back into A . For transitivity, if a sequence turns A into B and another turns B into C , performing the first and then the second turns A into C , and this is still a finite sequence.
- (c) First suppose a **single** operation turns $[A \mid \mathbf{b}]$ into $[A' \mid \mathbf{b}']$. Let S and S' be the two solution sets. Write $\mathbf{r}_h$ for row h of A , so that equation h says $\mathbf{r}_h \mathbf{x} = b_h$, where $\mathbf{r}_h \mathbf{x}$ is a 1×1 product. Let $\mathbf{s} \in S$, so $\mathbf{r}_h \mathbf{s} = b_h$ for every h . Each equation of the new system holds at $\mathbf{s}$:
- after $R_i \leftrightarrow R_j$, the new equations are the old ones in another order;
 - after $R_i \rightarrow cR_i$, the new equation i is $(c\mathbf{r}_i)\mathbf{x} = cb_i$, and $(c\mathbf{r}_i)\mathbf{s} = c(\mathbf{r}_i\mathbf{s}) = cb_i$;
 - after $R_i \rightarrow R_i + cR_j$, the new equation i is $(\mathbf{r}_i + c\mathbf{r}_j)\mathbf{x} = b_i + cb_j$, and $(\mathbf{r}_i + c\mathbf{r}_j)\mathbf{s} = \mathbf{r}_i\mathbf{s} + c(\mathbf{r}_j\mathbf{s}) = b_i + cb_j$, by Theorem 0.9.4.

The other equations are unchanged. Hence $\mathbf{s} \in S'$, and $S \subseteq S'$. By (a), a single operation turns $[A' \mid \mathbf{b}']$ back into $[A \mid \mathbf{b}]$, so the same argument gives $S' \subseteq S$. Therefore $S = S'$.

For a sequence of s operations, the intermediate augmented matrices all have the same solution set, by the single-operation case applied s times (formally, by induction on s). This proves (c). ■

This theorem is the license for everything that follows: we may transform a system as much as we like by row operations, and the solutions of the final system are the solutions of the original one. What it does **not** license is just as important.

Warning. Scaling by 0 is not a row operation, and it changes the solution set. Over $\mathbb{R}$, the system $x + y = 2$, $x - y = 0$ has exactly one solution, $(1, 1)$. Applying the illegal move $R_2 \rightarrow 0R_2$ gives $x + y = 2$, $0 = 0$, whose solutions $(2 - t, t)$ form a whole line. The inclusion $S \subseteq S'$ in the proof above still holds; what fails is reversibility, and with it $S' \subseteq S$.

Warning. Column operations are not allowed. Operations on the columns of $[A \mid \mathbf{b}]$ mix up the unknowns. The system $x + y = 3$, $x - y = 1$ has the unique solution $(2, 1)$. Adding column 1 to column 2 of its augmented matrix gives $\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 1 & 0 & 1 \end{array} \right]$, the system $x + 2y = 3$, $x = 1$, whose only solution is $(1, 1)$. Even swapping two columns of A swaps the corresponding entries of every solution.

2.2.3 Echelon forms

Row operations can change a system a great deal. We now choose a target: a shape of matrix from which the solutions can be read off with no further work. Look at the finished computation from the start of the section. After eliminating, the augmented matrix could be brought to

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right],$$

which says $x = 1$, $y = 1$, $z = 1$ outright. Not every system can be brought to this shape, since not every system has exactly one solution. The general target is a staircase.

For a non-zero row of a matrix, its **leading entry** is its first non-zero entry, reading from the left.

A matrix is in row echelon form when its non-zero rows come first and their leading entries step strictly to the right as we go down.

Definition 2.2.3 — Row echelon form

A matrix $R \in M_{m \times k}(F)$ is in **row echelon form** if:

- (E1) there is an integer r with $0 \leq r \leq m$ such that rows $1, \dots, r$ are non-zero and rows $r+1, \dots, m$ are zero; and
- (E2) if the leading entry of row i lies in column j_i , for $i = 1, \dots, r$, then $j_1 < j_2 < \dots < j_r$, **strictly**.

The leading entries of rows $1, \dots, r$ are the **pivots** of R , and the columns $j_1, \dots, j_r$ are its **pivot columns**.

In words: (E1) says zero rows sit at the bottom, and (E2) says each pivot lies strictly to the right of the pivot above it. Two consequences are used constantly. Each row contains at most one pivot, so there are at most m pivots. And **every entry below a pivot is zero**: if $h > i$, then row h is zero or has its leading entry in column $j_h > j_i$, so its entry in column j_i is 0.

Definition 2.2.4 — Reduced row echelon form

A matrix $R \in M_{m \times k}(F)$ is in **reduced row echelon form (RREF)** if it is in row echelon form, with pivot columns $j_1 < \dots < j_r$, and moreover:

- (E3) every pivot equals 1: $r_{ij_i} = 1$ for $i = 1, \dots, r$; and
- (E4) every pivot is the **only** non-zero entry in its column: $r_{hj_i} = 0$ for all $h \neq i$.

In words: (E3) normalizes the pivots, and (E4) clears the entries **above** each pivot as well as below. Together they say that the pivot column j_i is the standard basis vector $\mathbf{e}_i \in F^m$.

Example. (Echelon or not?)

Decide which of the following real matrices are in row echelon form, and which are in

reduced row echelon form. Give the pivot columns where they exist.

$$P = \begin{pmatrix} 2 & 1 & 3 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 1 & 5 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Solution. P is in row echelon form: rows 1 and 2 are non-zero, row 3 is zero, and the leading entries lie in columns $1 < 3$. The pivot columns are 1 and 3. It is **not** reduced: the pivot 2 is not 1, failing (E3), and column 3 has the non-zero entry 3 above the pivot -1 , failing (E4).

Q is in reduced row echelon form, with pivot columns 1 and 3. Both pivots are 1, and columns 1 and 3 are $\mathbf{e}_1$ and $\mathbf{e}_2$. The entries 5, 2, 3 in the non-pivot columns 2 and 4 are allowed to be anything.

T is **not** in row echelon form. Its leading entries lie in columns 1, 3, 2, and $3 < 2$ fails, so (E2) fails. Swapping rows 2 and 3 would repair it.

Z is in reduced row echelon form, with $r = 0$: there are no pivots, and (E2)–(E4) hold vacuously. This degenerate case matters, because the zero matrix must have an RREF too, and it is itself.

For Q , a minimal change gives a non-example: replacing the 0 in position $(1, 3)$ by 4 keeps (E1)–(E3), but column 3 now has the non-zero entry 4 above its pivot, so (E4) fails.

Why insist on the reduced form, when row echelon form already allows back substitution? Because a system in reduced form needs **no** substitution: each pivot variable appears in exactly one equation. More importantly, the reduced form turns out to be **unique**: a matrix is row equivalent to exactly one matrix in RREF, however the operations are chosen. That is proved in the next section, and it is what makes later definitions, such as the rank, independent of choices. Row echelon form is not unique, as the warning after the worked example shows.

Quick check. Is $\begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ in reduced row echelon form? If it is the coefficient matrix of a homogeneous system in x_1, x_2, x_3, x_4 , which unknowns correspond to pivot columns?

Solution. Yes. Row 3 is the only zero row and it is last; the leading entries lie in columns $1 < 3$; both are 1; and columns 1 and 3 are $\mathbf{e}_1$ and $\mathbf{e}_2$. The pivot columns are 1 and 3, so the unknowns x_1 and x_3 correspond to pivots, and x_2, x_4 do not.

2.2.4 Gauss–Jordan elimination

We now show that every matrix can be brought to reduced row echelon form. The procedure works through the columns from left to right, keeping a count r of the pivots found so far.

Gauss–Jordan elimination. Input: $M \in M_{m \times k}(F)$. Set $r = 0$. For $l = 1, 2, \dots, k$ in turn:

1. Look for a row h with $h > r$ whose entry in column l is non-zero. If there is none, column l gets no pivot; go on to the next l .

2. Otherwise choose such an h . If $h \neq r + 1$, perform $R_{r+1} \leftrightarrow R_h$.
3. Let $a \neq 0$ be the entry now in position $(r + 1, l)$. Perform $R_{r+1} \rightarrow a^{-1}R_{r+1}$, so that this entry becomes 1.
4. For every row $g \neq r + 1$ whose entry c in column l is non-zero, perform $R_g \rightarrow R_g - cR_{r+1}$.
5. Increase r by 1.

Output: the final matrix, which is in reduced row echelon form, with r pivots.

Step 2 involves a choice: any non-zero entry will do. By hand, we choose one that avoids fractions. If we skip the scaling in step 3 and clear only the entries **below** the pivot a , by $R_g \rightarrow R_g - ca^{-1}R_{r+1}$ for the rows $g > r + 1$, the output is a row echelon form instead; this cheaper variant is **Gaussian elimination**, and it is followed by back substitution.

Theorem 2.2.3 — Existence of the reduced row echelon form

Let F be a field. Every matrix $M \in M_{m \times k}(F)$ is row equivalent to a matrix in reduced row echelon form.

Idea. We prove that the algorithm works, by induction on the number l of columns processed. The invariant is: the first l columns of the current matrix form a matrix in RREF. When a new column is processed, all the operations involve only rows below the pivots found so far, or add multiples of the new pivot row, which is zero in the first l columns. So the part already built is never disturbed. After all k columns, the whole matrix is in RREF.

Proof. For $0 \leq l \leq k$, let $P(l)$ be the statement: there is a matrix M_l row equivalent to M and an integer $r \geq 0$ such that the $m \times l$ matrix L formed by the first l columns of M_l is in reduced row echelon form with exactly r non-zero rows. We prove $P(l)$ for all l by induction (Theorem 0.2.7). For $l = 0$ take $M_0 = M$ and $r = 0$; there are no columns, so there is nothing to check.

Suppose $P(l)$ holds for some $l < k$, with M_l , L and r as described, and let $j_1 < \dots < j_r \leq l$ be the pivot columns of L . By (E1) for L , rows $r + 1, \dots, m$ of M_l are zero in columns $1, \dots, l$. Let L' denote the first $l + 1$ columns of the matrix we are about to construct.

Case 1: every entry of column $l + 1$ of M_l in rows $h > r$ is zero (this includes the case $r = m$). Put $M_{l+1} = M_l$. Rows $r + 1, \dots, m$ of L' are zero, and rows $1, \dots, r$ are non-zero with the same leading entries as in L , which lie in columns $\leq l$. So (E1)–(E3) hold for L' with the same r and pivot columns, and so does (E4), since the pivot columns of L' are those of L and are unchanged. Hence L' is in RREF and $P(l + 1)$ holds.

Case 2: some row $h > r$ has a non-zero entry in column $l + 1$. Then $r + 1 \leq h \leq m$. Perform steps 2, 3 and 4 of the algorithm with this h and with column $l + 1$; call the result M_{l+1} . It is row equivalent to M_l , hence to M , by Theorem 2.2.2 (b).

These operations do not change columns $1, \dots, l$. Indeed, the swap in step 2 and the scaling in step 3 only move or rescale rows h and $r + 1$, which are both zero in those columns. Each replacement $R_g \rightarrow R_g - cR_{r+1}$ in step 4 adds a multiple of row $r + 1$, which is still zero in columns $1, \dots, l$. So the first l columns of M_{l+1} form the matrix L . In column $l + 1$, step 3 makes the entry in row $r + 1$ equal to 1, and each replacement in step 4 makes the entry of row g equal to $c - c \cdot 1 = 0$ without changing any other row. So column $l + 1$ of M_{l+1} is $\mathbf{e}_{r+1}$.

We check that L' is in RREF with $r + 1$ non-zero rows. Rows $1, \dots, r$ of L' have the same leading entries as in L , in columns $j_1 < \dots < j_r \leq l$. Row $r + 1$ is zero in columns $1, \dots, l$ and has a 1 in column $l + 1$, so its leading entry is in column $j_{r+1} := l + 1 > j_r$. Rows $r + 2, \dots, m$ are zero in columns $1, \dots, l$ and in column $l + 1$, so they are zero. This gives (E1) and (E2). The pivots are 1, by (E3) for L and the construction, so (E3) holds. For (E4), the columns $j_1, \dots, j_r$ are unchanged from L , and column $l + 1$ is $\mathbf{e}_{r+1}$. Hence $P(l + 1)$ holds.

By induction $P(k)$ holds: M_k is row equivalent to M , and its first k columns, which are all of M_k , form a matrix in reduced row echelon form. This proves the theorem. ■

The proof is also the correctness proof of the algorithm: the matrix M_l is the state of the computation after column l . Note where the field axioms were used. Step 3 divides by $a \neq 0$, which needs (F8), and nothing else in the argument depends on which field we work over.

Example. (Gauss–Jordan on a 3×4 system)

Solve the following system over $\mathbb{R}$ by Gauss–Jordan elimination, naming each operation:

$$\begin{aligned} 2x_2 - 2x_3 + x_4 &= 3, \\ x_1 + x_2 + x_3 &= 3, \\ 2x_1 + x_2 + 3x_3 + x_4 &= 3. \end{aligned}$$

Solution. We work on the augmented matrix. **Column 1.** Row 1 has a 0 there, so we bring up row 2, whose entry is already 1. Then we clear the rest of column 1.

$$\left[\begin{array}{cccc|c} 0 & 2 & -2 & 1 & 3 \\ 1 & 1 & 1 & 0 & 3 \\ 2 & 1 & 3 & 1 & 3 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cccc|c} 1 & 1 & 1 & 0 & 3 \\ 0 & 2 & -2 & 1 & 3 \\ 2 & 1 & 3 & 1 & 3 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 2R_1} \left[\begin{array}{cccc|c} 1 & 1 & 1 & 0 & 3 \\ 0 & 2 & -2 & 1 & 3 \\ 0 & -1 & 1 & 1 & -3 \end{array} \right]$$

Column 2. The candidates below row 1 are 2 and -1 . Choosing -1 avoids fractions, so we swap it up and scale.

$$\xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{cccc|c} 1 & 1 & 1 & 0 & 3 \\ 0 & -1 & 1 & 1 & -3 \\ 0 & 2 & -2 & 1 & 3 \end{array} \right] \xrightarrow{R_2 \rightarrow -R_2} \left[\begin{array}{cccc|c} 1 & 1 & 1 & 0 & 3 \\ 0 & 1 & -1 & -1 & 3 \\ 0 & 2 & -2 & 1 & 3 \end{array} \right]$$

Now clear column 2 above and below the pivot.

$$\xrightarrow{R_1 \rightarrow R_1 - R_2} \left[\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & -1 & -1 & 3 \\ 0 & 2 & -2 & 1 & 3 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 2R_2} \left[\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & -1 & -1 & 3 \\ 0 & 0 & 0 & 3 & -3 \end{array} \right]$$

Column 3. The only entry below row 2 is 0, so column 3 gets no pivot. **Column 4.** The entry 3 in row 3 becomes the pivot.

$$\xrightarrow{R_3 \rightarrow \frac{1}{3}R_3} \left[\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & -1 & -1 & 3 \\ 0 & 0 & 0 & 1 & -1 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 - R_3} \left[\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -1 & -1 & 3 \\ 0 & 0 & 0 & 1 & -1 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 + R_3} \left[\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -1 \end{array} \right]$$

Column 5. No rows remain below row 3, so we stop. The final matrix is in RREF, with pivot columns 1, 2 and 4. It is the system

$$x_1 + 2x_3 = 1, \quad x_2 - x_3 = 2, \quad x_4 = -1,$$

which by [Theorem 2.2.2](#) has the same solutions as the original. The unknown x_3 is unconstrained; writing $x_3 = t$, the solutions are

$$(1 - 2t, 2 + t, t, -1) = (1, 2, 0, -1) + t(-2, 1, 1, 0), \quad t \in \mathbb{R}.$$

Check: $(1, 2, 0, -1)$ gives $4 + 0 - 1 = 3$, $1 + 2 + 0 = 3$ and $2 + 2 + 0 - 1 = 3$; and $(-2, 1, 1, 0)$ gives $2 - 2 + 0 = 0$, $-2 + 1 + 1 = 0$ and $-4 + 1 + 3 + 0 = 0$. This is the “particular plus homogeneous” form of [Theorem 2.1.3](#).

A computer algebra system does the same computation exactly. In the online version the cell below can be edited and run, which is a good way to check a hand computation.

```
from sympy import Matrix

M = Matrix([
    [0, 2, -2, 1, 3],
    [1, 1, 1, 0, 3],
    [2, 1, 3, 1, 3],
])
R, pivots = M.rref() # the RREF and the (0-based) pivot columns
print(R)
print("pivot columns:", [j + 1 for j in pivots])
```

Warning. Echelon forms are not unique; only the reduced form is. Different choices in step 2 give different intermediate matrices. For $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ over $\mathbb{R}$, the operation $R_2 \rightarrow R_2 - 3R_1$ gives the row echelon form $\begin{pmatrix} 1 & 2 \\ 0 & -2 \end{pmatrix}$. Starting instead with $R_1 \leftrightarrow R_2$ and then $R_2 \rightarrow R_2 - \frac{1}{3}R_1$ gives the row echelon form $\begin{pmatrix} 3 & 4 \\ 0 & \frac{2}{3} \end{pmatrix}$. Both continue to the same RREF, I_2 . That this always happens is the theorem of the next section; until it is proved, “the” RREF of a matrix is not yet justified.

2.2.5 Reading off the solutions

Once $[A \mid \mathbf{b}]$ is in reduced row echelon form, the example shows what to do: a pivot in the last column signals a contradiction, and otherwise the unknowns without pivots can be chosen freely. We name the two kinds of unknowns.

Suppose $[A \mid \mathbf{b}]$, with $A \in M_{m \times n}(F)$, is row equivalent to a matrix $[R \mid \mathbf{c}]$ in reduced row echelon form, where R is $m \times n$. For $l \leq n$, the unknown x_l is a **pivot variable** if l is a pivot column of $[R \mid \mathbf{c}]$, and a **free variable** otherwise. (By the uniqueness theorem of the next section, this does not depend on which sequence of operations was used.)

Theorem 2.2.4 — Reading solutions from the RREF

Let $A \in M_{m \times n}(F)$ and $\mathbf{b} \in F^m$, and suppose $[A \mid \mathbf{b}]$ is row equivalent to a matrix $[R \mid \mathbf{c}]$ in reduced row echelon form. Let r_{il} be the entries of R and $\mathbf{c} = (c_1, \dots, c_m)$.

- (a) $A\mathbf{x} = \mathbf{b}$ is inconsistent **if and only if** the last column, column $n + 1$, is a pivot column of $[R \mid \mathbf{c}]$, that is, some row of $[R \mid \mathbf{c}]$ is $(0 \cdots 0 \mid 1)$.
- (b) Suppose there is no pivot in the last column. Let $j_1 < \cdots < j_r$ be the pivot columns. Then for **every** choice of scalars $t_l \in F$, one for each free variable x_l , there is **exactly one** solution with $x_l = t_l$ for all free l , namely the one with

$$x_{j_i} = c_i - \sum_{l \text{ free}} r_{il} t_l \quad (i = 1, \dots, r).$$

In particular, setting every free variable to 0 gives the solution $\mathbf{p}$ with $p_{j_i} = c_i$ and all other entries 0.

- (c) A consistent system $A\mathbf{x} = \mathbf{b}$ has exactly one solution **if and only if** it has no free variables, that is, each of the columns $1, \dots, n$ is a pivot column.

Proof. By Theorem 2.2.2 (c), $A\mathbf{x} = \mathbf{b}$ has the same solution set as $R\mathbf{x} = \mathbf{c}$, so we work with the latter.

(a, $\Leftarrow$) If row i of $[R \mid \mathbf{c}]$ has its pivot in column $n + 1$, then its first n entries are 0 and $c_i = 1$ by (E3). Equation i reads $0 = 1$, which is false in any field (Definition 0.7.1). So there is no solution.

- (b) Suppose no pivot lies in column $n + 1$. Then all pivots of $[R \mid \mathbf{c}]$ lie in columns $j_1, \dots, j_r \leq n$. By (E1), rows $r + 1, \dots, m$ of $[R \mid \mathbf{c}]$ are zero, so the corresponding equations read $0 = 0$ and hold for every $\mathbf{x}$. For $i \leq r$, row i of R has $r_{ij_i} = 1$ by (E3) and $r_{ij_h} = 0$ for $h \neq i$ by (E4). So equation i reads

$$x_{j_i} + \sum_{l \text{ free}} r_{il} x_l = c_i.$$

Hence $\mathbf{x}$ is a solution if and only if $x_{j_i} = c_i - \sum_{l \text{ free}} r_{il} x_l$ for $i = 1, \dots, r$. Each unknown is either a pivot variable or a free variable, not both. So once the free variables are given the values t_l , these equations determine every pivot variable uniquely, and the resulting vector is a solution. This proves (b).

(a, $\Rightarrow$) This is the contrapositive of what (b) just showed: if there is no pivot in the last column, then choosing all $t_l = 0$ produces a solution, so the system is consistent.

- (c) By (a), a consistent system has no pivot in the last column, so (b) applies. If there are no free variables, (b) gives exactly one solution. If x_l is free, the choices $t_l = 0$ and $t_l = 1$, with all other free variables 0, give two solutions whose l -th entries differ, since $0 \neq 1$. This proves (c).

■

In Example (Gauss–Jordan on a 3×4 system) the pivot columns were 1, 2, 4, the free variable was x_3 , and (b) gave $x_1 = 1 - 2t$, $x_2 = 2 + t$, $x_4 = -1$, with $\mathbf{p} = (1, 2, 0, -1)$. The same recipe works over any field, and over a finite field the free variables can take only finitely many values.

Example. (Elimination over $\mathbb{F}_2$)

Solve the following system over $\mathbb{F}_2$:

$$x_1 + x_2 + x_4 = 1, \quad x_2 + x_3 + x_4 = 0, \quad x_1 + x_3 + x_4 = 1.$$

How many solutions are there? What changes if the same equations are read over $\mathbb{R}$?

Solution. In $\mathbb{F}_2$, $-1 = 1$, so $R_i \rightarrow R_i - R_j$ is the same operation as $R_i \rightarrow R_i + R_j$, and every non-zero pivot is already 1. **Column 1** has its pivot in row 1; clear row 3. **Column 2** has its pivot in row 2; clear rows 3 and 1.

$$\begin{aligned} \left[\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \end{array} \right] &\xrightarrow{R_3 \rightarrow R_3 + R_1} \left[\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{array} \right] &\xrightarrow{R_3 \rightarrow R_3 + R_2} \left[\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right] \\ &\xrightarrow{R_1 \rightarrow R_1 + R_2} \left[\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right] \end{aligned}$$

In the row $R_3 + R_1$, the entries $1 + 1$ became 0, and in $R_3 + R_2$ the entry in column 3 is $1 + 1 = 0$. **Column 3** has only a 0 below row 2, so no pivot. **Column 4** has its pivot in row 3; row 1 already has a 0 there, so only row 2 needs clearing.

$$\xrightarrow{R_2 \rightarrow R_2 + R_3} \left[\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right]$$

This is in RREF, with pivot columns 1, 2, 4 and no pivot in the last column, so the system is consistent by [Theorem 2.2.4](#) (a). The free variable is $x_3 = t$, and part (b) gives $x_1 = 1 - t = 1 + t$, $x_2 = -t = t$, $x_4 = 0$. Since $t \in \mathbb{F}_2 = \{0, 1\}$, there are exactly two solutions:

$$(1, 0, 0, 0) \quad \text{and} \quad (0, 1, 1, 0).$$

Check the second: $0 + 1 + 0 = 1$, $1 + 1 + 0 = 0$, $0 + 1 + 0 = 1$ in $\mathbb{F}_2$.

Over $\mathbb{R}$ the first step is $R_3 \rightarrow R_3 - R_1$, giving row 3 = $(0, -1, 1, 0 \mid 0)$, and adding row 2 then gives $(0, 0, 2, 1 \mid 0)$. The entry 2 is non-zero in $\mathbb{R}$, so column 3 now gets a pivot. Completing the reduction gives pivot columns 1, 2, 3, free variable x_4 , and infinitely many real solutions. The same equations lead to different pivot patterns over different fields, because $1 + 1 = 0$ in one and not in the other.

When a coefficient depends on a parameter, whether a column gets a pivot can depend on the parameter too. Then the reduction splits into cases, one for each answer to “is this entry zero?”.

Example. (A system with a parameter)

For which $k \in \mathbb{R}$ does the system $x + ky = 1$, $kx + y = 1$ have no solution, exactly one solution, or infinitely many? Find the solutions when they exist.

Solution. The entry in position $(1, 1)$ is 1 for every k , so it is a pivot with no case

distinction:

$$\left[\begin{array}{cc|c} 1 & k & 1 \\ k & 1 & 1 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - kR_1} \left[\begin{array}{cc|c} 1 & k & 1 \\ 0 & 1 - k^2 & 1 - k \end{array} \right].$$

(For $k = 0$ this operation changes nothing, which is allowed.) Column 2 gets a pivot in row 2 exactly when $1 - k^2 = (1 - k)(1 + k)$ is non-zero, and only then may we scale by its inverse. So we split into cases.

Case $k \neq \pm 1$. Then $1 - k^2 \neq 0$, and $R_2 \rightarrow (1 - k^2)^{-1}R_2$ turns row 2 into $(0, 1 \mid \frac{1}{1+k})$, since $\frac{1-k}{(1-k)(1+k)} = \frac{1}{1+k}$. Then $R_1 \rightarrow R_1 - kR_2$ turns row 1 into $(1, 0 \mid 1 - \frac{k}{1+k}) = (1, 0 \mid \frac{1}{1+k})$. The RREF is $[I_2 \mid \mathbf{c}]$ with $\mathbf{c} = (\frac{1}{1+k}, \frac{1}{1+k})$, and by [Theorem 2.2.4](#) (c) this is the only solution. Check: $\frac{1}{1+k} + \frac{k}{1+k} = 1$ for both equations.

Case $k = 1$. Row 2 is $(0, 0 \mid 0)$, and $\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 0 & 0 \end{array} \right]$ is in RREF with pivot column 1. The free variable is $y = t$, and [Theorem 2.2.4](#) (b) gives the solutions $(1 - t, t)$, $t \in \mathbb{R}$: infinitely many.

Case $k = -1$. Row 2 is $(0, 0 \mid 2)$, and $R_2 \rightarrow \frac{1}{2}R_2$ makes it $(0, 0 \mid 1)$. The last column is a pivot column, so the system is inconsistent by [Theorem 2.2.4](#) (a). Directly: $x - y = 1$ and $-x + y = 1$ add up to $0 = 2$.

Hence there is no solution for $k = -1$, there are infinitely many for $k = 1$, and there is exactly one otherwise. The trap is to scale by $(1 - k^2)^{-1}$ before asking whether $1 - k^2$ is zero: that step silently assumes $k \neq \pm 1$, and loses exactly the two interesting cases.

2.2.6 More unknowns than equations

Counting pivots gives a first general theorem. A matrix with m rows has at most m pivots, one per non-zero row. If there are more unknowns than that, some unknown must be free.

Corollary 2.2.5 — More unknowns than equations

Let $A \in M_{m \times n}(F)$ with $n > m$.

- (a) The homogeneous system $A\mathbf{x} = \mathbf{0}$ has a **non-trivial** solution.
- (b) Consequently, every list of n vectors in F^m with $n > m$ is linearly dependent.

Proof.

- (a) By [Theorem 2.2.3](#), $[A \mid \mathbf{0}]$ is row equivalent to a matrix $[R \mid \mathbf{c}]$ in RREF. By [Lemma 2.2.1](#), each operation keeps the last column zero, so $\mathbf{c} = \mathbf{0}$, and the last column is not a pivot column. Each pivot is the leading entry of a different row, so there are at most m pivots, hence at most $m < n$ pivot columns among the first n columns. So some x_l is a free variable. By [Theorem 2.2.4](#) (b), with $t_l = 1$ and every other free variable 0, there is a solution whose l -th entry is $1 \neq 0$. This solution is non-trivial.
- (b) Let $\mathbf{v}_1, \dots, \mathbf{v}_n \in F^m$ and let $A \in M_{m \times n}(F)$ have these columns. By (a) there is $\mathbf{x} \neq \mathbf{0}$ with $A\mathbf{x} = \mathbf{0}$, and by [Theorem 0.9.3](#) this says $x_1\mathbf{v}_1 + \dots + x_n\mathbf{v}_n = \mathbf{0}$ with $x_1, \dots, x_n$ not all zero. So the list is linearly dependent.

Part (b) is not new: since $\dim F^m = m$, it is also a case of [Theorem 1.6.3](#), which rests on the Steinitz Exchange Theorem ([Theorem 1.6.1](#)). The proof here is independent of that one. It uses nothing from Chapter 1 beyond the definition of independence, only elimination and a count of pivots. Two proofs by different routes give some confidence that the dimension theory of Chapter 1 and the computations of this chapter will fit together, and the rest of the chapter makes that fit exact. ■

2.2.7 Exercises

2.2.7.1 A. Check your understanding

Exercise A1.

- List the three types of elementary row operation, including the conditions on the scalar and on the indices.
- Determine whether the following statement is true: “every matrix in row echelon form is in reduced row echelon form.” Justify your answer.
- Determine whether the following statement is true: “if the RREF of $[A \mid \mathbf{b}]$ has a zero row, then $A\mathbf{x} = \mathbf{b}$ is inconsistent.” Justify your answer.
- Determine whether the following statement is true: “a homogeneous system of 3 equations in 5 unknowns over $\mathbb{R}$ has infinitely many solutions.” Justify your answer.
- Explain why swapping two columns of A in $[A \mid \mathbf{b}]$ is **not** a safe step when solving $A\mathbf{x} = \mathbf{b}$.

Solution.

- $R_i \leftrightarrow R_j$ with $i \neq j$; $R_i \rightarrow cR_i$ with $c \neq 0$; $R_i \rightarrow R_i + cR_j$ with $i \neq j$ and any $c \in F$.
- False. $\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$ is in row echelon form, but its first pivot is $2 \neq 1$, so (E3) fails; its second column also has a non-zero entry above the pivot, so (E4) fails.
- False. A zero row is the equation $0 = 0$, which every vector satisfies. Inconsistency means a row $(0 \cdots 0 \mid 1)$ ([Theorem 2.2.4](#) (a)). For example $\left[\begin{array}{c|c} 1 & 1 \\ 0 & 0 \end{array} \right]$ is the consistent system $x = 1, 0 = 0$.
- True. By [Corollary 2.2.5](#) ($5 > 3$) there is a non-trivial solution $\mathbf{h}$, and since $\mathbb{R}$ is infinite, the multiples $c\mathbf{h}$ ($c \in \mathbb{R}$) are infinitely many distinct solutions, as in [Exercise \(C1: How many solutions?\)](#).
- It swaps the roles of two unknowns: $\mathbf{s}$ solves the new system exactly when the vector obtained from $\mathbf{s}$ by swapping those two entries solves the old one. So the solution set changes, unless the swap happens to fix it. For instance $x = 1, y = 2$ becomes $y = 1, x = 2$.

2.2.7.2 B. Practice

Exercise B1: Reduce and solve.

Reduce the augmented matrix of each system over $\mathbb{R}$ to reduced row echelon form, naming each operation. Hence decide whether the system is consistent, and if so, describe all solutions.

- (a) $x_1 + 2x_2 - x_3 = 1$, $2x_1 + 5x_2 + x_3 = 4$, $x_1 + 3x_2 + 2x_3 = 4$.
 (b) $x_1 + x_2 + 2x_3 = 3$, $2x_1 + 3x_2 + 5x_3 = 7$, $x_1 + 2x_2 + 4x_3 = 5$.
 (c) $x_1 - x_2 + 2x_3 + x_4 = 2$, $2x_1 - 2x_2 + 5x_3 + 4x_4 = 5$, $-x_1 + x_2 - x_3 + x_4 = -1$.

Solution.

(a)

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & 5 & 1 & 4 \\ 1 & 3 & 2 & 4 \end{array} \right] &\xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 1 & 3 & 2 & 4 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_1} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 1 & 3 & 3 \end{array} \right] \\ &\xrightarrow{R_1 \rightarrow R_1 - 2R_2} \left[\begin{array}{ccc|c} 1 & 0 & -7 & -3 \\ 0 & 1 & 3 & 2 \\ 0 & 1 & 3 & 3 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & -7 & -3 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 1 \end{array} \right] \end{aligned}$$

Column 3 gets no pivot. Column 4 has its pivot in row 3:

$$\xrightarrow{R_1 \rightarrow R_1 + 3R_3} \left[\begin{array}{ccc|c} 1 & 0 & -7 & 0 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_3} \left[\begin{array}{ccc|c} 1 & 0 & -7 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right].$$

The last column is a pivot column, so by [Theorem 2.2.4](#) (a) the system is inconsistent. (Row 3 already said $0 = 1$ before the last two operations.)

(b)

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 2 & 3 & 5 & 7 \\ 1 & 2 & 4 & 5 \end{array} \right] &\xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & 1 & 1 & 1 \\ 1 & 2 & 4 & 5 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 \end{array} \right] \\ &\xrightarrow{R_1 \rightarrow R_1 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 - R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array} \right] \end{aligned}$$

Columns 1, 2, 3 are pivot columns and the last column is not, so by [Theorem 2.2.4](#)

(c) there is exactly one solution, $(1, 0, 1)$. Check: $1 + 0 + 2 = 3$, $2 + 0 + 5 = 7$, $1 + 0 + 4 = 5$.

(c)

$$\left[\begin{array}{cccc|c} 1 & -1 & 2 & 1 & 2 \\ 2 & -2 & 5 & 4 & 5 \\ -1 & 1 & -1 & 1 & -1 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{cccc|c} 1 & -1 & 2 & 1 & 2 \\ 0 & 0 & 1 & 2 & 1 \\ -1 & 1 & -1 & 1 & -1 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 + R_1} \left[\begin{array}{cccc|c} 1 & -1 & 2 & 1 & 2 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 & 1 \end{array} \right]$$

Column 2 has only zeros below row 1, so no pivot. Column 3 has its pivot in row 2:

$$\xrightarrow{R_1 \rightarrow R_1 - 2R_2} \left[\begin{array}{cccc|c} 1 & -1 & 0 & -3 & 0 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 & 1 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_2} \left[\begin{array}{cccc|c} 1 & -1 & 0 & -3 & 0 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

Columns 4 and 5 have only a 0 in row 3, so this is the RREF. The pivot columns are 1 and 3; the free variables are $x_2 = s$ and $x_4 = t$. By [Theorem 2.2.4](#) (b), $x_1 = s + 3t$ and $x_3 = 1 - 2t$. Hence the solution set is

$$\{(0, 0, 1, 0) + s(1, 1, 0, 0) + t(3, 0, -2, 1) : s, t \in \mathbb{R}\}.$$

Exercise B2: A parameter.

For which $k \in \mathbb{R}$ is the system

$$x + y - z = 2, \quad x + 2y + z = 3, \quad x + y + (k^2 - 5)z = k$$

inconsistent, which give exactly one solution, and which give infinitely many? Justify your answer, and find the solutions when they exist.

Solution.

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 1 & 2 & 1 & 3 \\ 1 & 1 & k^2 - 5 & k \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & 2 & 1 \\ 1 & 1 & k^2 - 5 & k \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & k^2 - 4 & k - 2 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & -3 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & k^2 - 4 & k - 2 \end{array} \right]$$

Whether column 3 gets a pivot depends on whether $k^2 - 4 = (k - 2)(k + 2)$ is zero.

Case $k \neq \pm 2$. Then $k^2 - 4 \neq 0$, and $R_3 \rightarrow (k^2 - 4)^{-1}R_3$ turns row 3 into $(0, 0, 1 \mid \frac{1}{k+2})$, since $\frac{k-2}{(k-2)(k+2)} = \frac{1}{k+2}$. Then $R_1 \rightarrow R_1 + 3R_3$ and $R_2 \rightarrow R_2 - 2R_3$ give the RREF $[I_3 \mid \mathbf{c}]$ with

$$\mathbf{c} = \left(\frac{k+5}{k+2}, \frac{k}{k+2}, \frac{1}{k+2} \right),$$

using $1 + \frac{3}{k+2} = \frac{k+5}{k+2}$ and $1 - \frac{2}{k+2} = \frac{k}{k+2}$. By [Theorem 2.2.4](#) (c) there is exactly one solution, $\mathbf{c}$.

Case $k = 2$. Row 3 is zero, and the matrix is in RREF with pivot columns 1, 2 and free variable $z = t$. The solutions are $(1 + 3t, 1 - 2t, t)$, $t \in \mathbb{R}$: infinitely many.

Case $k = -2$. Row 3 is $(0, 0, 0 \mid -4)$, and $R_3 \rightarrow -\frac{1}{4}R_3$ makes it $(0, 0, 0 \mid 1)$. The last column is a pivot column, so the system is inconsistent.

Hence: inconsistent for $k = -2$, infinitely many solutions for $k = 2$, and exactly one solution otherwise.

Exercise B3: Over $\mathbb{F}_3$.

Solve each system over $\mathbb{F}_3$ by reducing to RREF, naming each operation, and list all solutions.

- (a) $x + 2y + z = 1$, $2x + y + z = 0$, $x + z = 2$.
 (b) $x + y + 2z = 1$, $2x + z = 2$.

Solution. In $\mathbb{F}_3$ the inverses are $1^{-1} = 1$ and $2^{-1} = 2$, since $2 \cdot 2 = 4 = 1$, and $-1 = 2$, $-2 = 1$.

(a)

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 2 & 1 & 1 & 0 \\ 1 & 0 & 1 & 2 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ 1 & 0 & 1 & 2 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_1} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 \end{array} \right]$$

Here $1 - 4 = -3 = 0$, $1 - 2 = -1 = 2$, $0 - 2 = 1$, and $0 - 2 = 1$, $2 - 1 = 1$. Column 2 has entries 0, 1 below row 1, so we swap:

$$\xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 1 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 - 2R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 1 \end{array} \right] \xrightarrow{R_3 \rightarrow 2R_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 - R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

There are no free variables, so the unique solution is $(0, 1, 2)$. Check: $0 + 2 + 2 = 4 = 1$, $0 + 1 + 2 = 3 = 0$, $0 + 2 = 2$.

(b)

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 2 & 0 & 1 & 2 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & 0 & 0 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 0 \end{array} \right],$$

using $0 - 2 = 1$, $1 - 4 = 0$, $2 - 2 = 0$. The free variable is $z = t$, and $x = 1 - 2t = 1 + t$, $y = 0$. As t runs through 0, 1, 2, the solutions are $(1, 0, 0)$, $(2, 0, 1)$, $(0, 0, 2)$: exactly three.

2.2.7.3 C. Going deeper

Exercise C1: Why $i \neq j$?

Consider the move “ $R_i \rightarrow R_i + cR_i$ ”, which the definition does **not** allow.

- (a) Show that for $c = -1$ this move is not reversible: give a matrix on which its effect cannot be undone by any sequence of elementary row operations and moves of this kind, and a system whose solution set it changes.
 (b) Show that for $c \neq -1$ the move is an elementary row operation of another type. Explain why the condition $i \neq j$ in type (3) therefore loses nothing.

Solution.

- (a) The move replaces $\mathbf{r}_i$ by $\mathbf{r}_i + (-1)\mathbf{r}_i = \mathbf{0}$. Apply it to the 1×2 augmented matrix $[1 \mid 1]$ of the equation $x = 1$ over any field F : the result is $[0 \mid 0]$, the equation $0 = 0$. The solution set changes from $\{1\}$ to all of F . No sequence of moves can return to $[1 \mid 1]$: every elementary row operation, and every move “ $R_i \rightarrow R_i + cR_i$ ”, sends a zero matrix to a zero matrix, by the formulas in the proof of [Lemma 2.2.1](#) and $c \cdot 0 = 0$.
 (b) The move replaces $\mathbf{r}_i$ by $(1 + c)\mathbf{r}_i$. If $c \neq -1$, then $1 + c \neq 0$, and this is the scaling

$R_i \rightarrow (1+c)R_i$, an operation of type (2). So allowing $i = j$ in type (3) would add only moves that are already available (when $c \neq -1$) and one move that destroys information ($c = -1$). Requiring $i \neq j$ excludes exactly the bad case.

Exercise C2: The cost of elimination.

Let $A \in M_n(F)$ and $\mathbf{b} \in F^n$, and suppose that Gauss–Jordan elimination on $[A \mid \mathbf{b}]$ finds, for each $l = 1, \dots, n$, a non-zero entry in position (l, l) and so never swaps. Count as one **operation** each multiplication or division of two scalars, under these conventions: scaling row l costs one division for each entry in columns $l+1, \dots, n+1$ (the pivot is simply set to 1), and each replacement $R_g \rightarrow R_g - cR_l$ costs one multiplication for each entry in columns $l+1, \dots, n+1$ (the entry in column l is simply set to 0).

- Show that processing column l costs at most $n(n+1-l)$ operations.
- Hence show that the whole elimination costs at most $\frac{1}{2}(n^3 + n^2)$ operations.
- Roughly how many times more work is a system of $2n$ equations than one of n equations, for large n ?

Solution.

- Scaling row l costs $n+1-l$ divisions. At most $n-1$ other rows need a replacement, each costing $n+1-l$ multiplications. The total is at most $(n+1-l) + (n-1)(n+1-l) = n(n+1-l)$.
- Summing (a) over $l = 1, \dots, n$, and substituting $q = n+1-l$, which runs through $1, \dots, n$,

$$\sum_{l=1}^n n(n+1-l) = n \sum_{q=1}^n q = n \cdot \frac{n(n+1)}{2} = \frac{n^3 + n^2}{2},$$

using the formula for $1 + \dots + n$ (Proposition 0.2.8).

- The bound is dominated by $\frac{1}{2}n^3$ for large n , since $\frac{1}{2}n^2$ is small compared with it. Replacing n by $2n$ multiplies n^3 by 8, so doubling the size of the system costs roughly eight times as much.

Exercise C3: Solvable for every right-hand side.

Let $A \in M_{m \times n}(F)$, and suppose that $A\mathbf{x} = \mathbf{b}$ is consistent for **every** $\mathbf{b} \in F^m$. Prove that $m \leq n$.

Hint: suppose $m > n$, reduce A to its RREF R , and run the operations backwards on $[R \mid \mathbf{e}_m]$.

Solution. Suppose, for a contradiction, that $m > n$. By Theorem 2.2.3, some sequence of operations $O_1, \dots, O_s$ turns A into a matrix R in RREF. By (E2), R has at most n pivots, one in each pivot column, so at most $n < m$ non-zero rows, and by (E1) its last row is zero. Therefore the last row of $[R \mid \mathbf{e}_m]$ is $(0 \cdots 0 \mid 1)$, so the system $R\mathbf{x} = \mathbf{e}_m$ has no solution.

By Theorem 2.2.2 (a), each O_t is undone by an operation O'_t , and $O'_s, \dots, O'_1$ turn R back into A . Apply them to $[R \mid \mathbf{e}_m]$. By Lemma 2.2.1 they act on the two blocks

separately, so the result is $[A \mid \mathbf{b}]$ for some $\mathbf{b} \in F^m$. By [Theorem 2.2.2](#) (c), $A\mathbf{x} = \mathbf{b}$ has the same solutions as $R\mathbf{x} = \mathbf{e}_m$, namely none. This contradicts the hypothesis, so $m \leq n$.

2.3 Uniqueness of the Reduced Form

Gauss–Jordan elimination involves choices: which non-zero entry to use as a pivot, and in what order to perform the operations. Different choices give different intermediate matrices, and different row echelon forms. The previous section claimed that they always end at the **same** reduced row echelon form. This matters more than it seems. Pivot columns, free variables, and later the rank of a matrix are all read off from the reduced form; if two people could reduce the same matrix to two different answers, none of these would be well defined. This section proves the claim.

2.3.1 What row operations remember

Row operations change the columns of a matrix. What do they keep? Take

$$A = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 4 & 0 & 2 \\ 3 & 6 & 2 & 7 \end{pmatrix} \quad \text{and its RREF} \quad R = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

over $\mathbb{R}$ (the computation is carried out in an example later in this section). The columns of R look nothing like those of A . But look at how the columns depend on each other. In R , column 2 is twice column 1, and column 4 is column 1 plus twice column 3. In A , with columns $\mathbf{a}_1, \dots, \mathbf{a}_4$,

$$\mathbf{a}_2 = (2, 4, 6) = 2\mathbf{a}_1, \quad \mathbf{a}_4 = (3, 2, 7) = (1, 2, 3) + 2(1, 0, 2) = \mathbf{a}_1 + 2\mathbf{a}_3.$$

The same relations, with the same coefficients. This is no accident. A relation $x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n = \mathbf{0}$ among the columns says $A\mathbf{x} = \mathbf{0}$ ([Theorem 0.9.3](#)), and row operations do not change the solutions of a homogeneous system.

Lemma 2.3.1 — Row equivalent matrices have the same homogeneous solutions

Let $A, B \in M_{m \times n}(F)$ be row equivalent. Then for every $\mathbf{x} \in F^n$, $A\mathbf{x} = \mathbf{0}$ if and only if $B\mathbf{x} = \mathbf{0}$.

Proof. Let a sequence of elementary row operations turn A into B . By [Lemma 2.2.1](#), the same sequence turns $[A \mid \mathbf{0}]$ into $[B \mid \mathbf{0}']$, where $\mathbf{0}'$ is obtained from the zero column by row operations and so is again the zero column. Hence $[A \mid \mathbf{0}]$ and $[B \mid \mathbf{0}']$ are row equivalent, and by [Theorem 2.2.2](#) (c) the systems $A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0}$ have the same solution set. ■

Rewriting $A\mathbf{x}$ as a combination of columns turns the lemma into the statement we observed.

Lemma 2.3.2 — Row operations preserve column relations

Let $A, B \in M_{m \times n}(F)$ have columns $\mathbf{a}_1, \dots, \mathbf{a}_n$ and $\mathbf{b}_1, \dots, \mathbf{b}_n$, and suppose that $A\mathbf{x} = \mathbf{0} \Leftrightarrow B\mathbf{x} = \mathbf{0}$ for every $\mathbf{x} \in F^n$. This holds, for instance, if A and B are row equivalent (Lemma 2.3.1). Then:

- (a) for all $x_1, \dots, x_n \in F$: $x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n = \mathbf{0}$ if and only if $x_1\mathbf{b}_1 + \dots + x_n\mathbf{b}_n = \mathbf{0}$;
- (b) for every index l and all scalars $d_1, \dots, d_{l-1}$: $\mathbf{a}_l = d_1\mathbf{a}_1 + \dots + d_{l-1}\mathbf{a}_{l-1}$ if and only if $\mathbf{b}_l = d_1\mathbf{b}_1 + \dots + d_{l-1}\mathbf{b}_{l-1}$;
- (c) for every index l : $\mathbf{a}_l \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_{l-1})$ if and only if $\mathbf{b}_l \in \text{span}(\mathbf{b}_1, \dots, \mathbf{b}_{l-1})$.

Proof.

- (a) By Theorem 0.9.3, $x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n = A\mathbf{x}$ and $x_1\mathbf{b}_1 + \dots + x_n\mathbf{b}_n = B\mathbf{x}$, where $\mathbf{x} = (x_1, \dots, x_n)$. The hypothesis says $A\mathbf{x} = \mathbf{0} \Leftrightarrow B\mathbf{x} = \mathbf{0}$.
- (b) The equation $\mathbf{a}_l = d_1\mathbf{a}_1 + \dots + d_{l-1}\mathbf{a}_{l-1}$ holds if and only if $d_1\mathbf{a}_1 + \dots + d_{l-1}\mathbf{a}_{l-1} + (-1)\mathbf{a}_l + 0\mathbf{a}_{l+1} + \dots + 0\mathbf{a}_n = \mathbf{0}$. By (a), with these coefficients, this holds if and only if the same combination of the $\mathbf{b}_j$ is $\mathbf{0}$, that is, $\mathbf{b}_l = d_1\mathbf{b}_1 + \dots + d_{l-1}\mathbf{b}_{l-1}$.
- (c) By Definition 1.3.2, $\mathbf{a}_l \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_{l-1})$ means that the left side of (b) holds for **some** scalars $d_1, \dots, d_{l-1}$. By (b) this happens if and only if the right side holds for some scalars, which means $\mathbf{b}_l \in \text{span}(\mathbf{b}_1, \dots, \mathbf{b}_{l-1})$.

■

Warning. Row operations preserve column relations, not columns or their span. For $A = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}$ over $\mathbb{R}$, the operation $R_2 \rightarrow R_2 - R_1$ gives $\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$. The relation “column 2 = 2 · column 1” holds in both. But every column of A lies on the line $\text{span}((1, 1))$, and every column of the new matrix lies on the different line $\text{span}((1, 0))$. So the span of the columns changed. To answer a question about the columns of A themselves, find the relations from the reduced form, and then use them on the columns of A .

2.3.2 The columns of a reduced matrix

In a matrix in RREF, the column relations can be read off at a glance: pivot columns are standard basis vectors, and a non-pivot column **displays its own coefficients**. We record this precisely, together with a description of the pivot columns that does not mention pivots at all.

Lemma 2.3.3 — Columns of a matrix in RREF

Let $R \in M_{m \times n}(F)$ be in reduced row echelon form, with entries r_{il} , columns $\mathbf{u}_1, \dots, \mathbf{u}_n$, and pivot columns $j_1 < \dots < j_r$.

- (a) For each $i = 1, \dots, r$, the pivot column $\mathbf{u}_{j_i}$ is $\mathbf{e}_i$.

(b) If l is **not** a pivot column, then

$$\mathbf{u}_l = \sum_{i: j_i < l} r_{il} \mathbf{e}_i = \sum_{i: j_i < l} r_{il} \mathbf{u}_{j_i}.$$

(c) Column l is a pivot column **if and only if** $\mathbf{u}_l \notin \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{l-1})$.

Proof.

- (a) By (E3) the entry of $\mathbf{u}_{j_i}$ in row i is 1, and by (E4) its other entries are 0.
- (b) Let l be a non-pivot column, and look at the entry r_{il} in each row i . If $i > r$, row i is zero by (E1), so $r_{il} = 0$. If $i \leq r$ and $j_i > l$, then r_{il} lies strictly before the leading entry of row i , so $r_{il} = 0$. The case $j_i = l$ does not occur, since l is not a pivot column. So the only possibly non-zero entries of $\mathbf{u}_l$ are the r_{il} with $j_i < l$, which gives the first expression; the second follows from (a).
- (c) If l is not a pivot column, (b) writes $\mathbf{u}_l$ as a combination of the columns $\mathbf{u}_{j_i}$ with $j_i < l$, so $\mathbf{u}_l \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{l-1})$. Conversely, let $l = j_i$ be a pivot column. For each $l' < j_i$, the entry $r_{il'}$ lies before the leading entry of row i , so it is 0. Hence every vector of $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{l-1})$, being a combination of vectors with i -th entry 0, has i -th entry 0. But $\mathbf{u}_l = \mathbf{e}_i$ has i -th entry 1 $\neq 0$, so $\mathbf{u}_l \notin \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{l-1})$. ■

Part (c) says: reading the columns from left to right, a column gets a pivot exactly when it is **new**, not a combination of the columns before it. That condition is about column relations only, and column relations survive row operations. This is the whole idea of the uniqueness proof.

2.3.3 The uniqueness theorem

We can now prove that the reduced form depends only on the homogeneous solutions, and hence not on the operations used to reach it.

Theorem 2.3.4 — Uniqueness of the reduced row echelon form

Let $R, R' \in M_{m \times n}(F)$ be in reduced row echelon form, and suppose that $R\mathbf{x} = \mathbf{0} \Leftrightarrow R'\mathbf{x} = \mathbf{0}$ for every $\mathbf{x} \in F^n$. Then $R = R'$.

Consequently, every matrix $A \in M_{m \times n}(F)$ is row equivalent to **exactly one** matrix in reduced row echelon form, called **the RREF of A** .

Idea. A column dependence $\sum_j x_j \mathbf{u}_j = \mathbf{0}$ is exactly a null vector $\mathbf{x}$, so R and R' have the same column relations. The pivot columns are exactly the columns that are not combinations of earlier columns (Lemma 2.3.3 (c)), so R and R' have the same pivot columns. A pivot column is a standard basis vector in both. A non-pivot column of an RREF matrix is determined by how it is built from the pivot columns before it (Lemma 2.3.3 (b)), and that recipe is a column relation, so it is the same in both. Hence every column agrees.

Proof. Let $\mathbf{u}_1, \dots, \mathbf{u}_n$ and $\mathbf{u}'_1, \dots, \mathbf{u}'_n$ be the columns of R and R' , and r_{il} the entries of R . By hypothesis, [Lemma 2.3.2](#) applies to R and R' .

Same pivot columns. For each l , by [Lemma 2.3.2](#) (c), $\mathbf{u}_l \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{l-1})$ if and only if $\mathbf{u}'_l \in \text{span}(\mathbf{u}'_1, \dots, \mathbf{u}'_{l-1})$. By [Lemma 2.3.3](#) (c) applied to R and to R' , column l is a pivot column of R if and only if it is a pivot column of R' . So both matrices have the same pivot columns $j_1 < \dots < j_r$.

Pivot columns agree. By [Lemma 2.3.3](#) (a), $\mathbf{u}_{j_i} = \mathbf{e}_i = \mathbf{u}'_{j_i}$ for $i = 1, \dots, r$.

Same non-pivot columns. Let l be a non-pivot column. By [Lemma 2.3.3](#) (b) for R ,

$$\mathbf{u}_l = \sum_{i: j_i < l} r_{il} \mathbf{u}_{j_i}.$$

This is an equation of the form in [Lemma 2.3.2](#) (b), with coefficient r_{il} on column j_i and 0 on the other earlier columns. By that lemma, the same equation holds for R' :

$$\mathbf{u}'_l = \sum_{i: j_i < l} r_{il} \mathbf{u}'_{j_i} = \sum_{i: j_i < l} r_{il} \mathbf{e}_i = \mathbf{u}_l,$$

where the middle equality uses [Lemma 2.3.3](#) (a) for R' , and the last uses [Lemma 2.3.3](#) (b) for R .

Hence $\mathbf{u}_l = \mathbf{u}'_l$ for every l , and $R = R'$.

For the consequence, let $A \in M_{m \times n}(F)$. By [Theorem 2.2.3](#), A is row equivalent to at least one matrix in RREF. Suppose A is row equivalent to R and to R' , both in RREF. Row equivalence is an equivalence relation ([Theorem 2.2.2](#) (b)), so R is row equivalent to R' . By [Lemma 2.3.1](#), $R\mathbf{x} = \mathbf{0} \Leftrightarrow R'\mathbf{x} = \mathbf{0}$ for every $\mathbf{x}$, and by the first part $R = R'$. This proves the theorem. ■

Notice what the proof did **not** do: it never compared the two sequences of operations. A direct comparison of two runs of the algorithm would have to track every choice, and quickly becomes unmanageable. The proof instead found a description of the RREF, “pivots where a new column appears, and coefficients of the old columns elsewhere”, that refers only to the null space, which the operations cannot change.

The theorem also gives a practical test: two matrices of the same size are row equivalent exactly when their RREFs are equal (you will write out the two-line proof in [Exercise \(B2: Row equivalence via the RREF\)](#)). For example, over $\mathbb{R}$, $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ and $\begin{pmatrix} 2 & 4 \\ 0 & 0 \end{pmatrix}$ both reduce to $\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$, by $R_2 \rightarrow R_2 - 2R_1$ and by $R_1 \rightarrow \frac{1}{2}R_1$ respectively, so they are row equivalent. But $\begin{pmatrix} 1 & 3 \\ 2 & 6 \end{pmatrix}$ reduces to $\begin{pmatrix} 1 & 3 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$, so it is row equivalent to neither, even though all three matrices have the same pivot columns.

Example. (Two echelon forms, one reduced form)

Let $A = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 4 & 0 & 2 \\ 3 & 6 & 2 & 7 \end{pmatrix}$ over $\mathbb{R}$.

- Reduce A to a row echelon form in two ways: first using row 1 as the first pivot row, then starting with $R_1 \leftrightarrow R_2$.
- Continue both to reduced row echelon form.
- Read the column relations of A off the RREF.

Solution.

- First route:

$$A \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & -2 & -4 \\ 3 & 6 & 2 & 7 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - 3R_1} \begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & -2 & -4 \\ 0 & 0 & -1 & -2 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - \frac{1}{2}R_2} \begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & -2 & -4 \\ 0 & 0 & 0 & 0 \end{pmatrix} =$$

Second route:

$$A \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 2 & 4 & 0 & 2 \\ 1 & 2 & 1 & 3 \\ 3 & 6 & 2 & 7 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - \frac{1}{2}R_1} \begin{pmatrix} 2 & 4 & 0 & 2 \\ 0 & 0 & 1 & 2 \\ 3 & 6 & 2 & 7 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - \frac{3}{2}R_1} \begin{pmatrix} 2 & 4 & 0 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & 4 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - 2R_2} \begin{pmatrix} 2 & 4 & 0 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Both E_1 and E_2 are in row echelon form, and $E_1 \neq E_2$. Both have pivot columns 1 and 3.

- From E_1 :

$$E_1 \xrightarrow{R_2 \rightarrow -\frac{1}{2}R_2} \begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 - R_2} \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

From E_2 :

$$E_2 \xrightarrow{R_1 \rightarrow \frac{1}{2}R_1} \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The two routes end at the same matrix R , as [Theorem 2.3.4](#) guarantees.

- By [Lemma 2.3.3](#) (b), the non-pivot columns of R are $\mathbf{u}_2 = 2\mathbf{u}_1$ and $\mathbf{u}_4 = 1 \cdot \mathbf{u}_1 + 2\mathbf{u}_3$, reading the coefficients from the entries of each column. By [Lemma 2.3.2](#) (b), the same relations hold for the columns of A : $\mathbf{a}_2 = 2\mathbf{a}_1$ and $\mathbf{a}_4 = \mathbf{a}_1 + 2\mathbf{a}_3$. Check: $2(1, 2, 3) = (2, 4, 6)$ and $(1, 2, 3) + 2(1, 0, 2) = (3, 2, 7)$.

Both echelon forms in the example had the same pivot columns, although they were different matrices. That is also true in general, and it is the practical version of the theorem.

2.3.4 Pivot columns are well defined

Corollary 2.3.5 — Pivot columns are well defined

Let $A \in M_{m \times n}(F)$ have columns $\mathbf{a}_1, \dots, \mathbf{a}_n$. Define the **pivot columns** of A to be the

pivot columns of the RREF of A .

- (a) Column l is a pivot column of A **if and only if** $\mathbf{a}_l \notin \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_{l-1})$.
- (b) If E is **any** matrix in row echelon form that is row equivalent to A , then the pivot columns of E are the pivot columns of A . In particular, every row echelon form of A has the same number of non-zero rows.

Idea. Part (a) combines [Lemma 2.3.3](#) (c) with the fact that column relations pass from the RREF back to A . For (b), finish the reduction of E : scale each pivot to 1 and clear the entries above the pivots. These operations never move a leading entry, so they produce a matrix in RREF with the same pivot columns as E , and by uniqueness it is the RREF of A .

Proof. The RREF of A is well defined by [Theorem 2.3.4](#); let $\mathbf{u}_1, \dots, \mathbf{u}_n$ be its columns.

- (a) A and its RREF are row equivalent, so by [Lemma 2.3.2](#) (c), $\mathbf{a}_l \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_{l-1})$ if and only if $\mathbf{u}_l \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{l-1})$. By [Lemma 2.3.3](#) (c), the latter fails exactly when l is a pivot column.
- (b) Let E have entries e_{hl} and pivot columns $j_1 < \dots < j_s$. First perform $R_i \rightarrow e_{ij_i}^{-1} R_i$ for $i = 1, \dots, s$, which is legal since each pivot e_{ij_i} is non-zero. Scaling a row by a non-zero scalar does not change which of its entries are zero ([Theorem 0.7.1](#) (d), (f)), so the result is still in row echelon form with the same pivot columns, and now every pivot is 1. Next, for $i = 1, 2, \dots, s$ in turn, and for each $h < i$, perform $R_h \rightarrow R_h - cR_i$, where c is the current entry of row h in column j_i . Row i is zero in all columns before j_i , so this changes row h only in columns $\geq j_i > j_h$. Hence the leading entry of row h is still a 1 in column j_h , no zero row is touched, and the entries below every pivot stay 0. The operation makes the entry in position (h, j_i) equal to 0, and it does not spoil the entries $(h, j_{i'})$ with $i' < i$ cleared earlier, because row i has a 0 in column $j_{i'} < j_i$. At the end, every pivot column is a standard basis vector, so the matrix is in RREF, with pivot columns $j_1, \dots, j_s$.

This matrix is row equivalent to E , hence to A ([Theorem 2.2.2](#) (b)). By [Theorem 2.3.4](#) it is the RREF of A . Therefore the pivot columns of E are those of A . Since the non-zero rows of E correspond one-to-one to its pivots, their number s is the number of pivot columns of A , whichever E we chose. ■

In practice, (b) means we may stop at any row echelon form to find the pivot columns, and (a) means we can sometimes find them with no elimination at all. Part (a) also says that the pivot columns of A are exactly the columns kept when the list $(\mathbf{a}_1, \dots, \mathbf{a}_n)$ is sifted as in [Theorem 1.5.4](#): a column is kept exactly when it is not in the span of the columns before it. The number of pivot columns is the first candidate for a measure of “how much a matrix contains”; later in this chapter it becomes the rank.

Quick check. Without computing any row operations, find the pivot columns of $A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \end{pmatrix}$ over $\mathbb{R}$.

Solution. Use [Corollary 2.3.5](#) (a) column by column. $\mathbf{a}_1 = (1, 2) \neq \mathbf{0}$, so it is not in $\text{span}() = \{\mathbf{0}\}$: pivot. $\mathbf{a}_2 = (2, 4) = 2\mathbf{a}_1$: not a pivot. $\mathbf{a}_3 = (0, 1)$ is not a multiple of

$(1, 2)$, and $\text{span}(\mathbf{a}_1, \mathbf{a}_2) = \text{span}((1, 2))$: pivot. $\mathbf{a}_4 = (1, 3) = \mathbf{a}_1 + \mathbf{a}_3$: not a pivot. The pivot columns are 1 and 3.

Warning. Uniqueness is a property of the reduced form only. A matrix has many row echelon forms (Example (Two echelon forms, one reduced form) gave two), and it is only their pivot positions that agree. Also, two different matrices in RREF with the **same** pivot columns need not be row equivalent: $\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$ and $\begin{pmatrix} 1 & 3 \\ 0 & 0 \end{pmatrix}$ are both in RREF with pivot column 1, but they are different RREF matrices, so by Theorem 2.3.4 neither is row equivalent to the other.

2.3.5 Exercises

2.3.5.1 A. Check your understanding

Exercise A1.

- State the uniqueness theorem for the reduced row echelon form.
- Determine whether the following statement is true: “if A and B are row equivalent, then the columns of A and the columns of B span the same subspace.” Justify your answer.
- Determine whether the following statement is true: “if A and B are row equivalent and $\mathbf{a}_3 = \mathbf{a}_1 - \mathbf{a}_2$ for the columns of A , then $\mathbf{b}_3 = \mathbf{b}_1 - \mathbf{b}_2$ for the columns of B .” Justify your answer.
- Determine whether the following statement is true: “any two row echelon forms of the same matrix have the same number of zero rows.” Justify your answer.
- In the RREF of a matrix, what do the entries of a non-pivot column tell you about the original matrix?

Solution.

- If $R, R' \in M_{m \times n}(F)$ are in RREF and $R\mathbf{x} = \mathbf{0} \Leftrightarrow R'\mathbf{x} = \mathbf{0}$ for all $\mathbf{x} \in F^n$, then $R = R'$. Hence every matrix is row equivalent to exactly one matrix in RREF.
- False. $\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}$ and $\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$ are row equivalent, but their columns span $\text{span}((1, 1))$ and $\text{span}((1, 0))$, which are different lines in $\mathbb{R}^2$.
- True, by Lemma 2.3.2 (b) with $l = 3$, $d_1 = 1$, $d_2 = -1$.
- True. By Corollary 2.3.5 (b), every row echelon form of an $m \times n$ matrix A has exactly as many non-zero rows as A has pivot columns, so the number of zero rows is m minus that number.
- If column l is not a pivot column, its entries r_{il} (for the pivot rows i with $j_i < l$) are the coefficients expressing column l of the original matrix as a combination of its pivot columns: $\mathbf{a}_l = \sum_{j_i < l} r_{il} \mathbf{a}_{j_i}$.

2.3.5.2 B. Practice

Exercise B1: Column relations.

Let $A = \begin{pmatrix} 1 & -1 & 0 & 2 & 1 \\ 2 & -2 & 1 & 7 & 0 \\ 1 & -1 & 1 & 5 & 1 \end{pmatrix}$ over $\mathbb{R}$. Find the RREF of A , naming each operation.

Hence find the pivot columns of A , and express each non-pivot column of A as a linear combination of the pivot columns before it. Check your answer on the columns of A .

Solution.

$$\begin{aligned} A &\xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & -1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 3 & -2 \\ 1 & -1 & 1 & 5 & 1 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_1} \begin{pmatrix} 1 & -1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 3 & -2 \\ 0 & 0 & 1 & 3 & 0 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{pmatrix} 1 & -1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 & 2 \end{pmatrix} \\ &\xrightarrow{R_3 \rightarrow \frac{1}{2}R_3} \begin{pmatrix} 1 & -1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 - R_3} \begin{pmatrix} 1 & -1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 + 2R_3} \begin{pmatrix} 1 & -1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \end{aligned}$$

The pivot columns are 1, 3 and 5. By [Lemma 2.3.3](#) (b), in the RREF column 2 is -1 times column 1, and column 4 is 2 times column 1 plus 3 times column 3. By [Lemma 2.3.2](#) (b), the columns of A satisfy

$$\mathbf{a}_2 = -\mathbf{a}_1, \quad \mathbf{a}_4 = 2\mathbf{a}_1 + 3\mathbf{a}_3.$$

Check: $-(1, 2, 1) = (-1, -2, -1) = \mathbf{a}_2$, and $2(1, 2, 1) + 3(0, 1, 1) = (2, 7, 5) = \mathbf{a}_4$.

Exercise B2: Row equivalence via the RREF.

Prove that two matrices $A, B \in M_{m \times n}(F)$ are row equivalent if and only if they have the same RREF.

Solution. ($\Rightarrow$) Suppose A and B are row equivalent, and let R be the RREF of B . Then A is row equivalent to B , and B to R , so A is row equivalent to R by transitivity ([Theorem 2.2.2](#) (b)). By the uniqueness in [Theorem 2.3.4](#), R is the RREF of A .

($\Leftarrow$) Suppose both have the RREF R . Then A is row equivalent to R , and by symmetry R is row equivalent to B ([Theorem 2.2.2](#) (b)). By transitivity A is row equivalent to B .

Exercise B3: Row equivalent or not?.

Determine whether the matrices in each pair are row equivalent over $\mathbb{R}$. Justify your answer.

- (a) $P = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \end{pmatrix}$ and $Q = \begin{pmatrix} 1 & 1 & 2 \\ 3 & 4 & 7 \end{pmatrix}$.
- (b) $P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 0 & 1 & 2 \end{pmatrix}$ and $Q = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 1 & 1 & 2 \end{pmatrix}$.

Solution. By [Exercise \(B2: Row equivalence via the RREF\)](#), two matrices of the same size are row equivalent exactly when their RREFs are equal. So we compute both

RREFs.

$$(a) \quad P \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 - 2R_2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \text{ and } Q \xrightarrow{R_2 \rightarrow R_2 - 3R_1} \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}. \text{ The RREFs are equal, so } P \text{ and } Q \text{ are row equivalent.}$$

$$(b) \quad P \xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

$$Q \xrightarrow{R_3 \rightarrow R_3 - R_1} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 1 & 3 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 + R_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - 2R_3} I_3.$$

I_3 .

The RREFs differ (one has a zero row, the other is I_3), so P and Q are **not** row equivalent. Consistently with [Corollary 2.3.5](#), P has pivot columns 1, 2 while Q has 1, 2, 3.

2.3.5.3 C. Going deeper

Exercise C1: Counting reduced forms.

- List, by pivot columns, all 2×3 matrices over $\mathbb{F}_2$ in reduced row echelon form, and show that there are exactly 15 of them.
- Deduce that $M_{2 \times 3}(\mathbb{F}_2)$, which has 64 elements, splits into exactly 15 classes of row equivalent matrices.
- Show that over a field with q elements there are exactly $2q^2 + 2q + 3$ such matrices.

Solution. We count over a field F with q elements, sorting by the list of pivot columns. In a matrix in RREF, a pivot column is forced (it is $\mathbf{e}_i$), an entry before a leading entry or in a zero row is forced to be 0, and every other entry, one lying in a non-pivot column to the right of its row's pivot, is free.

- With $q = 2$, writing $*$ for a free entry:

- no pivots: $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, 1 matrix;
- pivot column 1 only: $\begin{pmatrix} 1 & * & * \\ 0 & 0 & 0 \end{pmatrix}$, 4 matrices;
- pivot column 2 only: $\begin{pmatrix} 0 & 1 & * \\ 0 & 0 & 0 \end{pmatrix}$, 2 matrices;
- pivot column 3 only: $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, 1 matrix;
- pivot columns 1, 2: $\begin{pmatrix} 1 & 0 & * \\ 0 & 1 & * \end{pmatrix}$, 4 matrices;
- pivot columns 1, 3: $\begin{pmatrix} 1 & * & 0 \\ 0 & 0 & 1 \end{pmatrix}$, 2 matrices;
- pivot columns 2, 3: $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, 1 matrix.

A 2×3 matrix has at most two pivots, so these are all the cases. The total is $1 + 4 + 2 + 1 + 4 + 2 + 1 = 15$.

- (b) Row equivalence is an equivalence relation ([Theorem 2.2.2 \(b\)](#)), so $M_{2 \times 3}(\mathbb{F}_2)$ is partitioned into equivalence classes ([Theorem 0.5.2](#)). By [Theorem 2.3.4](#), each class contains exactly one matrix in RREF: at least one by [Theorem 2.2.3](#), and at most one since two RREF matrices in the same class would be row equivalent, hence equal. So the classes correspond one-to-one to the 15 matrices of (a).
- (c) The same list, with each free entry taking q values, gives $1 + q^2 + q + 1 + q^2 + q + 1 = 2q^2 + 2q + 3$. For $q = 2$ this is $8 + 4 + 3 = 15$, as in (a).

Exercise C2: Deleting a column.

Let $A \in M_{m \times n}(F)$ with $n \geq 2$, and let R be its RREF.

- (a) Let A' and R' be obtained from A and R by deleting the **last** column. Prove that R' is the RREF of A' .
- (b) Show by an example that the corresponding statement for deleting the **first** column is false.

Solution.

- (a) Let a sequence of row operations turn A into R . By [Lemma 2.2.1](#), the same sequence turns A' into R' , so A' is row equivalent to R' . By [Theorem 2.3.4](#) it suffices to show that R' is in RREF. Let $j_1 < \dots < j_r$ be the pivot columns of R . If $j_r < n$, then deleting column n does not change any leading entry or any pivot column, and the zero rows of R stay zero, so (E1)–(E4) still hold. If $j_r = n$, then row r of R is zero except for its pivot in column n , so in R' rows $r, \dots, m$ are zero and rows $1, \dots, r-1$ keep their leading entries in columns $j_1 < \dots < j_{r-1}$ and their pivot columns. Again (E1)–(E4) hold, with $r-1$ pivots. Hence R' is in RREF, and it is the RREF of A' .
- (b) Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ over $\mathbb{R}$. Then $A \xrightarrow{R_1 \rightarrow R_1 - R_2} I_2$, so its RREF is I_2 , and deleting the first column of I_2 gives $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$. But deleting the first column of A gives $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$, whose RREF is $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, via $R_2 \rightarrow R_2 - R_1$. These differ. The pivot of a later column depends on the columns before it, so removing an early column can move pivots.

2.4 Elementary Matrices and Inverses

Gaussian elimination is a list of instructions: swap these rows, scale that one, add a multiple of one row to another. So far it lives outside algebra, as something we do to a matrix. This section turns every row operation into a matrix, so that “apply an operation” becomes “multiply on the left”. With that one observation the algorithm becomes a tool for proofs. We use it to prove the main theorem about invertible matrices, to show that a one-sided inverse of a square matrix is automatically two-sided, which Chapter 0 stated after [Definition 0.9.13](#) but could not prove, and to compute inverses.

Throughout, F is a field, and we use the three elementary row operations of [Definition 2.2.1](#)

with their notation: the swap $R_i \leftrightarrow R_j$ ($i \neq j$), the scaling $R_i \rightarrow cR_i$ (c **non-zero**), and the replacement $R_i \rightarrow R_i + cR_j$ ($i \neq j$, any $c \in F$).

2.4.1 Row operations are matrix multiplications

We have already seen one row operation disguised as a product. In [Example \(Row equivalence\)](#) of Chapter 0, multiplying $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 0 \end{pmatrix}$ on the left by $E = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$ replaced row 2 by row 2 minus twice row 1, that is, it performed $R_2 \rightarrow R_2 - 2R_1$. Look at E itself: it is what that same operation does to the identity matrix I_2 . This is no accident, and it deserves a name.

An elementary matrix is the identity matrix after one row operation.

Definition 2.4.1 — Elementary matrix

Let $m \geq 1$. A matrix $E \in M_m(F)$ is an **elementary matrix** if it is obtained from I_m by performing **exactly one** elementary row operation. Writing E_{ij} for the matrix unit (1 in entry (i, j) , 0 elsewhere; see [Example \(Standard Bases\)](#)), the three kinds are:

1. P_{ij} , obtained by $R_i \leftrightarrow R_j$ with $i \neq j$: the identity with rows i and j swapped;
2. $D_i(c) = \text{diag}(1, \dots, 1, c, 1, \dots, 1)$, with c in position i , obtained by $R_i \rightarrow cR_i$ with $c \neq 0$;
3. $I_m + cE_{ij}$, obtained by $R_i \rightarrow R_i + cR_j$ with $i \neq j$ and $c \in F$: the identity with an extra c in entry (i, j) .

In words: start from I_m , do **one** operation, and record the result. The size m is the number of **rows** of the matrices we plan to operate on. Type 3 puts c in row i , column j : the row that changes is i , and the row that gets copied is j .

Examples. In $M_3(F)$,

$$P_{13} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad D_2(5) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad I_3 - 3E_{31} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{pmatrix}$$

come from $R_1 \leftrightarrow R_3$, $R_2 \rightarrow 5R_2$ (over a field where $5 \neq 0$), and $R_3 \rightarrow R_3 - 3R_1$. The degenerate case is worth a look: $D_1(1) = I_m$ is elementary, since “multiply row 1 by 1” is a legal operation, and so is $I_m + 0E_{12}$. The identity is an elementary matrix. For $m = 1$ the elementary matrices are exactly the 1×1 matrices (c) with $c \neq 0$, since there are no two distinct rows to swap or combine.

Non-example by minimal change. Change $c = 5$ in $D_2(5)$ to $c = 0$. The matrix $\text{diag}(1, 0, 1)$ is still diagonal and still differs from I_3 in one entry, but “multiply row 2 by 0” is not an elementary row operation: the clause $c \neq 0$ fails. Likewise $\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ is a rearrangement of the rows of I_3 , but every one of its rows has moved, and a single swap moves only two rows. So it is not elementary. It is, however, the **product** $P_{12}P_{23}$ of two elementary matrices, as Exercise A1 at the end of this section checks.

The whole point of the definition is the next theorem. To perform a row operation on **any** matrix, multiply on the left by what that operation does to the identity.

Theorem 2.4.1 — Row operations are left multiplications

Let ρ be an elementary row operation on matrices with m rows, and let $E = \rho(I_m)$ be the corresponding elementary matrix. Then for **every** $n \geq 1$ and every $A \in M_{m \times n}(F)$,

$$\rho(A) = EA.$$

Idea. Each operation builds every new row as a combination of the old rows, with coefficients that do not depend on the matrix: for $R_i \rightarrow R_i + cR_j$, new row i is $1 \cdot (\text{row } i) + c \cdot (\text{row } j)$, and every other row is copied. Feed in I_m , whose rows are the standard row vectors, and the coefficients appear as the entries of E . Then the row view of a product says EA forms exactly those combinations of the rows of A .

Proof. Let $\mathbf{r}_1, \dots, \mathbf{r}_m$ be the rows of A , and write $\mathbf{e}_l^\top$ for the l -th row of I_m . By Definition 2.2.1, for each k there are scalars $c_{k1}, \dots, c_{km}$, depending on ρ but **not** on A , such that row k of $\rho(A)$ is $c_{k1}\mathbf{r}_1 + \dots + c_{km}\mathbf{r}_m$. Explicitly, every row that the operation does not touch has $c_{kk} = 1$ and all other $c_{kl} = 0$, and the touched rows are:

- for $R_i \leftrightarrow R_j$: row i is $\mathbf{r}_j$ and row j is $\mathbf{r}_i$;
- for $R_i \rightarrow cR_i$: row i is $c\mathbf{r}_i$;
- for $R_i \rightarrow R_i + cR_j$: row i is $\mathbf{r}_i + c\mathbf{r}_j$.

Applying ρ to I_m , row k of E is $c_{k1}\mathbf{e}_1^\top + \dots + c_{km}\mathbf{e}_m^\top = (c_{k1} \ \dots \ c_{km})$. So the (k, l) -entry of E is c_{kl} . By Theorem 0.9.2 (rows), row k of EA is (row k of E) A , and by Theorem 0.9.3 (rows version) this equals $c_{k1}\mathbf{r}_1 + \dots + c_{km}\mathbf{r}_m$, which is row k of $\rho(A)$. Since every row agrees, $\rho(A) = EA$. This proves the theorem. ■

So a sequence of operations $\rho_1, \rho_2, \dots, \rho_k$, performed in that order, turns A into

$$\rho_k(\dots \rho_2(\rho_1(A)) \dots) = E_k \dots E_2 E_1 A, \quad E_i = \rho_i(I_m),$$

by applying the theorem k times and using associativity. The whole of Gaussian elimination is a single matrix $E_k \dots E_1$ standing on the left.

Warning. The **first** operation stands **rightmost**. Doing $R_1 \leftrightarrow R_2$ and then $R_1 \rightarrow R_1 + R_2$ on I_2 gives $(I_2 + E_{12})P_{12} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, while the product in the other order is $P_{12}(I_2 + E_{12}) = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$. Also, multiplying by E on the **right** performs a **column** operation: $A(I_2 + E_{12})$ adds column 1 to column 2, and column operations change solution sets.

Example. (Three operations, three products)

Let $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 3 & 4 \end{pmatrix} \in M_{3 \times 2}(\mathbb{R})$. Compute $P_{13}A$, $D_2(5)A$ and $(I_3 - 3E_{31})A$, and check that each is the result of the corresponding row operation.

Solution. By the row view, row k of EA is the combination of the rows of A whose weights are row k of E .

- Rows of P_{13} are $\mathbf{e}_3^T, \mathbf{e}_2^T, \mathbf{e}_1^T$, so $P_{13}A = \begin{pmatrix} 3 & 4 \\ 0 & 1 \\ 1 & 2 \end{pmatrix}$: rows 1 and 3 are swapped.
- Row 2 of $D_2(5)$ is $5\mathbf{e}_2^T$, so $D_2(5)A = \begin{pmatrix} 1 & 2 \\ 0 & 5 \\ 3 & 4 \end{pmatrix}$: row 2 is multiplied by 5.
- Row 3 of $I_3 - 3E_{31}$ is $(-3 \ 0 \ 1)$, so row 3 of the product is $-3(1, 2) + (3, 4) = (0, -2)$, and $(I_3 - 3E_{31})A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 0 & -2 \end{pmatrix}$: this is $R_3 \rightarrow R_3 - 3R_1$.

In each case the product agrees with the operation, as [Theorem 2.4.1](#) predicts.

Quick check. Which 3×3 elementary matrix performs $R_2 \rightarrow R_2 + 4R_3$? Which operation undoes it, and what is its matrix?

Solution. Apply the operation to I_3 : row 2 becomes $\mathbf{e}_2^T + 4\mathbf{e}_3^T$, so the matrix is $I_3 + 4E_{23}$, with 4 in entry $(2, 3)$. The operation $R_2 \rightarrow R_2 - 4R_3$ undoes it, since row 3 is unchanged by both; its matrix is $I_3 - 4E_{23}$.

Every row operation can be undone by another row operation, and the matching statement for matrices is:

Theorem 2.4.2 — Elementary matrices are invertible

Every elementary matrix is invertible, and its inverse is an elementary matrix of the same kind:

$$P_{ij}^{-1} = P_{ij}, \quad D_i(c)^{-1} = D_i(c^{-1}), \quad (I_m + cE_{ij})^{-1} = I_m - cE_{ij} \quad (i \neq j).$$

Proof. Let $E = \rho(I_m)$, and let ρ' be the operation on the same line of the display: $\rho' = \rho$ for a swap, $R_i \rightarrow c^{-1}R_i$ for $R_i \rightarrow cR_i$ (legal, since $c \neq 0$ gives $c^{-1} \neq 0$), and $R_i \rightarrow R_i - cR_j$ for $R_i \rightarrow R_i + cR_j$. In each case $\rho'(\rho(A)) = A$ and $\rho(\rho'(A)) = A$ for every A , as in [Theorem 2.2.2](#) (a): swapping twice restores the order; scaling by c and c^{-1} multiplies row i by $cc^{-1} = 1$; and adding $c\mathbf{r}_j$ and then $-c\mathbf{r}_j$ to row i restores row i , because $i \neq j$ means row j is not changed in between. Let $E' = \rho'(I_m)$. By [Theorem 2.4.1](#),

$$E'E = \rho'(E) = \rho'(\rho(I_m)) = I_m \quad \text{and} \quad EE' = \rho(E') = \rho(\rho'(I_m)) = I_m.$$

Hence E is invertible with $E^{-1} = E'$, which is the matrix in the display. This proves the theorem. ■

The hypothesis $i \neq j$ did real work: “add c times row i to row i ” is scaling by $1 + c$, which destroys row i when $c = -1$ and cannot be undone. That is why [Definition 2.2.1](#) excludes it.

2.4.2 The invertible matrix theorem

Chapter 0 defined an invertible matrix by the existence of a two-sided inverse, and gave only one practical test, $ad - bc \neq 0$, for 2×2 matrices (Theorem 0.9.8). For larger matrices, searching for B with $AB = BA = I_n$ directly is hopeless. What we want is a list of conditions, each of which we can check by a method we already have, that are all equivalent to invertibility. Then we may test whichever is cheapest.

We need one small fact about square matrices in reduced form.

Lemma 2.4.3 — Square reduced matrices

Let $R \in M_n(F)$ be in reduced row echelon form. Then either $R = I_n$, or the last row of R is zero.

Proof. Suppose the last row of R is not zero. Since zero rows of a matrix in echelon form come after all non-zero rows (Definition 2.2.3), no row of R is zero, so each of the n rows has a pivot. Let the pivot of row k be in column c_k . Pivots move strictly to the right as we go down, so $1 \leq c_1 < c_2 < \cdots < c_n \leq n$. A strictly increasing list of n integers in $\{1, \dots, n\}$ must be $1, 2, \dots, n$, so $c_k = k$. By Definition 2.2.4, each pivot equals 1 and is the only non-zero entry of its column, so column k of R is $\mathbf{e}_k$ for every k . Hence $R = I_n$. ■

Here is the theorem. It collects, for a square matrix, facts about solving systems, about row reduction, and about the columns as vectors in the sense of Chapter 1.

Theorem 2.4.4 — Invertible Matrix Theorem

Let $A \in M_n(F)$, with columns $\mathbf{a}_1, \dots, \mathbf{a}_n \in F^n$. The following are equivalent.

- (a) A is invertible.
- (b) The homogeneous system $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = \mathbf{0}$.
- (c) The reduced row echelon form of A is I_n .
- (d) A is a product of elementary matrices.
- (e) For **every** $\mathbf{b} \in F^n$, the system $A\mathbf{x} = \mathbf{b}$ has a solution.
- (f) The columns $(\mathbf{a}_1, \dots, \mathbf{a}_n)$ are linearly independent.
- (g) The columns $(\mathbf{a}_1, \dots, \mathbf{a}_n)$ span F^n .
- (h) The columns $(\mathbf{a}_1, \dots, \mathbf{a}_n)$ form a basis of F^n .

Idea. Eight conditions would need 56 implications if proved pairwise; a cycle through them needs far fewer. The plan:

- ① the core cycle $(a) \Rightarrow (b) \Rightarrow (c) \Rightarrow (d) \Rightarrow (a)$, where each arrow uses one earlier tool (cancel by A^{-1} ; row reduction; elementary matrices; products of invertible matrices);
- ② $(b) \Leftrightarrow (f)$, which is the column view $A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$;
- ③ $(f) \Leftrightarrow (h) \Leftrightarrow (g)$, which is “count instead of check” in the n -dimensional space F^n ;

④ (g) $\Leftrightarrow$ (e), which is “consistent means $\mathbf{b}$ is in the column span”.

The only arrow that needs a new argument is (b) $\Rightarrow$ (c); the others quote a result we already have. An RREF that is not I_n has a zero row, so it really has fewer equations than unknowns, and such a homogeneous system has a non-trivial solution.

Proof. (a) $\Rightarrow$ (b). Suppose A is invertible and $A\mathbf{x} = \mathbf{0}$. Multiplying on the left by A^{-1} and using [Theorem 0.9.4](#), $\mathbf{x} = (A^{-1}A)\mathbf{x} = A^{-1}(A\mathbf{x}) = A^{-1}\mathbf{0} = \mathbf{0}$.

(b) $\Rightarrow$ (c). Suppose (b) holds. By [Theorem 2.2.3](#), A is row equivalent to a matrix R in reduced row echelon form, and R is the RREF of A by [Theorem 2.3.4](#). By [Lemma 2.3.1](#), $R\mathbf{x} = \mathbf{0}$ has the same solutions as $A\mathbf{x} = \mathbf{0}$, so it too has only the trivial solution. Suppose, for a contradiction, that $R \neq I_n$. By [Lemma 2.4.3](#), the last row of R is zero. If $n = 1$, then $R = (0)$ and $\mathbf{x} = (1)$ is a non-trivial solution. If $n \geq 2$, the last equation of $R\mathbf{x} = \mathbf{0}$ reads $0 = 0$, so $R\mathbf{x} = \mathbf{0}$ has the same solutions as the homogeneous system $R'\mathbf{x} = \mathbf{0}$ formed by the first $n - 1$ rows. That system has $n - 1$ equations in $n > n - 1$ unknowns, so by [Corollary 2.2.5](#) it has a non-trivial solution. Either way we contradict (b). Hence $R = I_n$.

(c) $\Rightarrow$ (d). Suppose the RREF of A is I_n . By [Definition 2.2.2](#) there are row operations $\rho_1, \dots, \rho_k$ taking A to I_n . If $k = 0$, then $A = I_n = D_1(1)$ is elementary. Otherwise, with $E_i = \rho_i(I_n)$, [Theorem 2.4.1](#) gives $E_k \cdots E_1 A = I_n$. Each E_i is invertible by [Theorem 2.4.2](#), so multiplying on the left by E_k^{-1} , then E_{k-1}^{-1} , and so on, gives

$$A = E_1^{-1} E_2^{-1} \cdots E_k^{-1},$$

and each E_i^{-1} is elementary by [Theorem 2.4.2](#).

(d) $\Rightarrow$ (a). Suppose $A = E_1 \cdots E_k$ with each E_i elementary and $k \geq 1$. Each E_i is invertible by [Theorem 2.4.2](#), and a product of two invertible matrices is invertible by [Theorem 0.9.7](#) (part 3). By induction on k , A is invertible.

So far (a), (b), (c), (d) are equivalent.

(b) $\Leftrightarrow$ (f). By [Theorem 0.9.3](#), $A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$. So “ $A\mathbf{x} = \mathbf{0}$ only for $\mathbf{x} = \mathbf{0}$ ” says exactly that $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{0}$ forces $x_1 = \cdots = x_n = 0$, which is [Definition 1.4.1](#) for the columns.

(f) $\Leftrightarrow$ (h) $\Leftrightarrow$ (g). The columns form a list of exactly n vectors in F^n , and $\dim F^n = n$ ([Example \(Dimensions of the Running Examples\)](#)). By [Theorem 1.6.4](#), such a list is a basis if it is linearly independent, and also if it spans. Conversely a basis is independent and spanning by [Definition 1.5.1](#). Hence (f) $\Rightarrow$ (h) $\Rightarrow$ (g) $\Rightarrow$ (h) $\Rightarrow$ (f).

(g) $\Leftrightarrow$ (e). By [Theorem 2.1.1](#), $A\mathbf{x} = \mathbf{b}$ has a solution if and only if $\mathbf{b} \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$. So (e) says every $\mathbf{b} \in F^n$ lies in this span, that is, $F^n \subseteq \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$. The reverse inclusion always holds, so (e) is equivalent to (g).

Combining the four parts, all eight conditions are equivalent. This proves the theorem. ■

In practice the theorem is a menu. To show a matrix **is** invertible, pick the most convenient item, very often (b): take a solution of $A\mathbf{x} = \mathbf{0}$ and show it is $\mathbf{0}$. To show a matrix is **not** invertible, exhibit one non-zero solution of $A\mathbf{x} = \mathbf{0}$, or one linear relation among the columns.

In later chapters the list grows: kernels and images of linear maps in Chapter 3, determinants in Chapter 6, eigenvalues in Chapter 8, and singular values in Chapter 12.

Quick check. Is $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 1 & 3 & 4 \end{pmatrix} \in M_3(\mathbb{R})$ invertible? Use whichever item of [Theorem 2.4.4](#) is quickest.

Solution. No. The third column is the sum of the first two: $(1, 0, 1) + (2, 1, 3) = (3, 1, 4)$. So $\mathbf{a}_1 + \mathbf{a}_2 - \mathbf{a}_3 = \mathbf{0}$ is a linear relation with coefficients not all zero, item (f) fails, and hence (a) fails. Equivalently, $\mathbf{x} = (1, 1, -1)$ is a non-trivial solution of $A\mathbf{x} = \mathbf{0}$.

Warning. The theorem is about **square** matrices. For $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \in M_{2 \times 3}(\mathbb{R})$, the columns span $\mathbb{R}^2$ (the first two are $\mathbf{e}_1, \mathbf{e}_2$), so (g) holds, but $\mathbf{x} = (1, 1, -1)$ solves $A\mathbf{x} = \mathbf{0}$, so (b) fails. The proof of (f) $\Leftrightarrow$ (g) used that the number of columns equals $\dim F^n$, and that is exactly what breaks.

2.4.3 One-sided inverses are two-sided

Chapter 0 left a gap on purpose. The definition of invertibility ([Definition 0.9.13](#)) asks for **both** $AB = I_n$ and $BA = I_n$. Chapter 0 then stated that, for square matrices, the first equation already implies the second, but that nothing there could prove it: the entries of AB and BA are different sums. Now we can prove it, and the proof uses the menu above instead of entries.

Theorem 2.4.5 — One-sided inverses of square matrices

Let $A, B \in M_n(F)$. If $AB = I_n$, then $BA = I_n$. Hence A and B are both invertible, $B = A^{-1}$ and $A = B^{-1}$.

Idea. We cannot touch BA directly, so we aim at invertibility of one factor instead. Which one? From $AB = I_n$, a vector killed by B is killed by $AB = I_n$, so it is $\mathbf{0}$. That is item (b) for B . Once B^{-1} exists, we may cancel it: $A = (AB)B^{-1} = B^{-1}$.

Proof. Suppose $AB = I_n$, and let $\mathbf{x} \in F^n$ with $B\mathbf{x} = \mathbf{0}$. Then, by [Theorem 0.9.4](#),

$$\mathbf{x} = I_n \mathbf{x} = (AB)\mathbf{x} = A(B\mathbf{x}) = A\mathbf{0} = \mathbf{0}.$$

So $B\mathbf{x} = \mathbf{0}$ has only the trivial solution, and B is invertible by [Theorem 2.4.4](#) ((b) $\Rightarrow$ (a)). Multiplying $AB = I_n$ on the right by B^{-1} and using associativity,

$$A = A(BB^{-1}) = (AB)B^{-1} = I_n B^{-1} = B^{-1}.$$

Therefore $BA = BB^{-1} = I_n$. Since $A = B^{-1}$ is invertible with inverse B by [Theorem 0.9.7](#) (part 2), we get $A^{-1} = B$. This proves the theorem. ■

By symmetry, $BA = I_n$ also implies $AB = I_n$: apply the theorem with the roles of A and B

swapped. So from now on, **to verify that B is the inverse of a square matrix A , one product suffices**. We also get a useful consequence of the same kind: if $A, B \in M_n(F)$ and AB is invertible, then A and B are invertible (Exercise B3 at the end of this section).

Warning. For **non-square** matrices a one-sided inverse need not be two-sided. Let

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then $AB = I_2$, but $BA = \text{diag}(1, 1, 0) \neq I_3$. In fact no $C \in M_{3 \times 2}(F)$ has $CA = I_3$: $A\mathbf{e}_3 = \mathbf{0}$, so $CA\mathbf{e}_3 = \mathbf{0} \neq \mathbf{e}_3 = I_3\mathbf{e}_3$. The proof above breaks because $B\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x} = \mathbf{0}$ no longer gives invertibility: B is not square, so [Theorem 2.4.4](#) does not apply.

2.4.4 Computing the inverse

[Theorem 2.4.4](#) (c) says A is invertible exactly when it row reduces to I_n . The proof of (c) $\Rightarrow$ (d) even produced a matrix: if $E_k \cdots E_1 A = I_n$, then $E_k \cdots E_1$ is a left inverse, hence **the** inverse by [Theorem 2.4.5](#). We do not want to multiply out the E_i by hand. The trick is to let the operations act on I_n at the same time as on A : since $E_k \cdots E_1 I_n = E_k \cdots E_1$, the operations that turn A into I_n turn I_n into A^{-1} . We write $[A \mid I_n]$ for the $n \times 2n$ matrix whose first n columns are those of A and whose last n columns are those of I_n .

Input: $A \in M_n(F)$.

1. Form the $n \times 2n$ matrix $[A \mid I_n]$.
2. Row reduce it to reduced row echelon form $[R \mid C]$, with $R, C \in M_n(F)$.
3. If $R = I_n$, output $A^{-1} = C$. Otherwise, output “ A is not invertible”.

In practice, stop as soon as a zero row appears in the left half: A is then not invertible (see the remark after the theorem).

Theorem 2.4.6 — Inverse by row reduction

Let $A \in M_n(F)$, and let $[R \mid C]$ be the reduced row echelon form of $[A \mid I_n]$, with $R, C \in M_n(F)$. Then R is the reduced row echelon form of A . Consequently, A is invertible if and only if $R = I_n$, and in that case $A^{-1} = C$.

Idea. Two things need checking. First, the operations act on the two halves independently, so the left half is transformed exactly as A alone would be. Second, the left half of a reduced matrix is itself reduced, so it really is the RREF of A . The first is the column view of a product; the second is a small check on the staircase shape. After that, $R = EA$ and $C = EI_n = E$, and [Theorem 2.4.5](#) finishes.

Proof. Let $\rho_1, \dots, \rho_k$ be row operations taking $[A \mid I_n]$ to $[R \mid C]$, and put $E = E_k \cdots E_1$ with $E_i = \rho_i(I_n)$ (and $E = I_n$ if $k = 0$). By [Theorem 2.4.1](#), $[R \mid C] = E[A \mid I_n]$. By [Theorem 0.9.2](#) (columns), each column of $E[A \mid I_n]$ is E times the corresponding column, so

$$R = EA \quad \text{and} \quad C = EI_n = E.$$

Claim. R is in reduced row echelon form.

Proof. If row k of R is non-zero, its first non-zero entry is also the first non-zero entry of row k of $[R \mid C]$, so the pivots of the non-zero rows of R are pivots of $[R \mid C]$, in the same positions. Suppose row k of R is zero but some later row $l > k$ of R is non-zero. Row k of $[R \mid C]$ cannot be zero, since zero rows of $[R \mid C]$ come after its non-zero rows and row l is non-zero. So its pivot lies in a column $> n$, while the pivot of row l lies in a column $\leq n$, to the left: this contradicts the staircase condition of Definition 2.2.3. Hence the zero rows of R come last, the pivots of R move strictly right, and each pivot of R is 1 and the only non-zero entry in its column, because this holds in $[R \mid C]$. So R satisfies Definition 2.2.4. ■

Now $R = E_k \cdots E_1 A$ is obtained from A by the operations $\rho_1, \dots, \rho_k$ (Theorem 2.4.1), so A is row equivalent to R , and R is in reduced form. By Theorem 2.3.4, R is the RREF of A . By Theorem 2.4.4 ((a) $\Leftrightarrow$ (c)), A is invertible if and only if $R = I_n$. If $R = I_n$, then $EA = I_n$, and Theorem 2.4.5 gives $A^{-1} = E = C$. This proves the theorem. ■

Remark. The early stop is justified the same way. If, after some operations, $[A \mid I_n]$ has become $[A' \mid E']$ with a zero row in A' , then, exactly as in the proof, E' is the product of the elementary matrices of the operations used so far and $A' = E'A$. So E' is invertible by Theorem 2.4.2 and Theorem 0.9.7 (part 3). A matrix with a zero row k is not invertible, since row k of $A'X$ is zero for every X , so $A'X \neq I_n$. If A were invertible, $A' = E'A$ would be invertible by Theorem 0.9.7 (part 3). Hence A is not invertible.

Example. (A 3×3 inverse)

Find the inverse of $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 3 & 3 \end{pmatrix} \in M_3(\mathbb{R})$, or show that it does not exist.

Solution. Row reduce $[A \mid I_3]$, naming each operation.

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 3 & 3 & 0 & 0 & 1 \end{array} \right] \xrightarrow[\substack{R_3 \rightarrow R_3 - R_1}]{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_2} \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array} \right]$$

The left half now has three pivots, so no zero row will appear. Clear above the pivots, from the bottom up:

$$\xrightarrow[\substack{R_1 \rightarrow R_1 - R_3}]{R_2 \rightarrow R_2 - R_3} \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & -3 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 - 2R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 6 & -3 & 1 \\ 0 & 1 & 0 & -3 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array} \right].$$

The left half is I_3 , so by [Theorem 2.4.6](#) A is invertible and

$$A^{-1} = \begin{pmatrix} 6 & -3 & 1 \\ -3 & 2 & -1 \\ 1 & -1 & 1 \end{pmatrix}.$$

Check one product: row 1 of A times the columns of A^{-1} gives $6-6+1=1$, $-3+4-1=0$, $1-2+1=0$; rows 2 and 3 give $(0, 1, 0)$ and $(0, 0, 1)$ in the same way. So $AA^{-1} = I_3$, and by [Theorem 2.4.5](#) this single product is enough.

Example. (When the algorithm stops)

Decide whether $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 1 & 3 & 4 \end{pmatrix} \in M_3(\mathbb{R})$ is invertible.

Solution. Row reduce $[A \mid I_3]$:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 7 & 0 & 1 & 0 \\ 1 & 3 & 4 & 0 & 0 & 1 \end{array} \right] \xrightarrow[R_3 \rightarrow R_3 - R_1]{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 1 & 1 & -1 & 0 & 1 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_2} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 \end{array} \right].$$

A zero row has appeared in the left half, so by the remark after [Theorem 2.4.6](#), A is not invertible. The right half of that row is not wasted: it records the combination of rows that produced zero. Row 3 of the left half is $1 \cdot (\text{row } 1) - 1 \cdot (\text{row } 2) + 1 \cdot (\text{row } 3)$ of A , and indeed $(1, 2, 3) - (2, 5, 7) + (1, 3, 4) = (0, 0, 0)$.

Finally, the method writes an invertible matrix as a product of elementary matrices, as in [Theorem 2.4.4](#) (d).

Example. (Factoring into elementary matrices)

Write $A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix} \in M_2(\mathbb{R})$ as a product of elementary matrices, and read off A^{-1} .

Solution. Reduce A to I_2 , keeping integers:

$$\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 - R_2} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The elementary matrices of these operations are $E_1 = I_2 - 2E_{21}$, $E_2 = I_2 - E_{12}$, $E_3 = I_2 - E_{21}$, and by [Theorem 2.4.1](#), $E_3E_2E_1A = I_2$. Multiplying on the left by E_3^{-1} , then E_2^{-1} , then E_1^{-1} , and using [Theorem 2.4.2](#),

$$A = E_1^{-1}E_2^{-1}E_3^{-1} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Multiplying out confirms it: the last two factors give $\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, and then the first factor adds twice row 1 to row 2, giving $\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$. Also $E_3E_2E_1$ is a left inverse of A , hence

A^{-1} by [Theorem 2.4.5](#):

$$A^{-1} = E_3 E_2 E_1 = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix},$$

which agrees with [Theorem 0.9.8](#), since $ad - bc = 1$. The factorization is not unique: a different sequence of operations gives different factors.

2.4.5 Row equivalence, revisited

Two descriptions of “row equivalent” are now on the table. [Example \(Row equivalence\)](#) of Chapter 0 said $B = EA$ for some invertible E . [Definition 2.2.2](#) says B is reachable from A by finitely many row operations. Chapter 0 promised that they agree, and now it is one line each way.

Theorem 2.4.7 — Row equivalence is left multiplication by an invertible matrix

Let $A, B \in M_{m \times n}(F)$. Then A is row equivalent to B (that is, B can be reached from A by row operations) if and only if $B = EA$ for some invertible $E \in M_m(F)$.

Proof. ($\Rightarrow$) Suppose row operations $\rho_1, \dots, \rho_k$ take A to B . By [Theorem 2.4.1](#), $B = E_k \cdots E_1 A$ with $E_i = \rho_i(I_m)$, and $E = E_k \cdots E_1$ is invertible by [Theorem 2.4.2](#) and [Theorem 0.9.7](#) (part 3). (If $k = 0$, take $E = I_m$.)

($\Leftarrow$) Suppose $B = EA$ with E invertible. By [Theorem 2.4.4](#) ((a) $\Rightarrow$ (d)), $E = E_k \cdots E_1$ for some elementary matrices E_i . Let ρ_i be the operation with $\rho_i(I_m) = E_i$. By [Theorem 2.4.1](#), performing ρ_1 , then ρ_2 , $\dots$, then ρ_k on A produces $E_k \cdots E_1 A = B$. Hence A is row equivalent to B . ■

So the equivalence relation of [Example \(Row equivalence\)](#) is the one Gaussian elimination moves along, and “two matrices have the same RREF” is the same as “each is an invertible matrix times the other”. In the next section this form, $B = EA$, is what makes row spaces and ranks easy to compare.

2.4.6 Exercises

2.4.6.1 A. Check your understanding

Exercise A1.

- (a) Define an elementary matrix, and write down the 3×3 elementary matrix that performs $R_3 \rightarrow R_3 + 7R_1$.
- (b) State four of the conditions in [Theorem 2.4.4](#).
- (c) True or false: the product of two elementary matrices is always an elementary matrix. Justify your answer.

- (d) True or false: if $A \in M_{2 \times 3}(F)$, $B \in M_{3 \times 2}(F)$ and $AB = I_2$, then $BA = I_3$. Justify your answer.
- (e) True or false: if $A \in M_n(F)$ and $A\mathbf{x} = \mathbf{e}_j$ has a solution for each $j = 1, \dots, n$, then A is invertible. Justify your answer.
- (f) Describe the method for computing the inverse of a square matrix, and how it detects a matrix that is not invertible.

Solution.

- (a) An elementary matrix in $M_m(F)$ is one obtained from I_m by exactly one elementary row operation (Definition 2.4.1). Applying $R_3 \rightarrow R_3 + 7R_1$ to I_3 gives $I_3 + 7E_{31} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 7 & 0 & 1 \end{pmatrix}$.
- (b) For $A \in M_n(F)$, each of the following is equivalent to A being invertible: $A\mathbf{x} = \mathbf{0}$ has only the trivial solution; the RREF of A is I_n ; A is a product of elementary matrices; the columns of A form a basis of F^n .
- (c) False. A single row operation changes at most two rows of I_3 (a swap changes two, a scaling or an addition changes one), so every elementary 3×3 matrix has at least one row equal to the corresponding row of I_3 . But $P_{12}P_{23} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ (by the row view, its rows are rows 2, 1, 3 of P_{23}) differs from I_3 in every row. Hence it is a product of two elementary matrices that is not elementary.
- (d) False. With $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$, $AB = I_2$ but $BA = \text{diag}(1, 1, 0) \neq I_3$. Theorem 2.4.5 needs square matrices.
- (e) True. Let $\mathbf{x}_j$ solve $A\mathbf{x}_j = \mathbf{e}_j$, and let B have columns $\mathbf{x}_1, \dots, \mathbf{x}_n$. By Theorem 0.9.2, AB has columns $\mathbf{e}_1, \dots, \mathbf{e}_n$, so $AB = I_n$, and A is invertible by Theorem 2.4.5.
- (f) Row reduce $[A \mid I_n]$ to reduced form $[R \mid C]$. If $R = I_n$, then $A^{-1} = C$; if $R \neq I_n$, in particular if a zero row appears in the left half at any stage, A is not invertible (Theorem 2.4.6).

2.4.6.2 B. Practice

Exercise B1: Inverses by row reduction.

For each matrix, find the inverse or show that it is not invertible.

- (a) $A = \begin{pmatrix} 1 & 0 & 2 \\ 2 & 1 & 3 \\ 4 & 1 & 8 \end{pmatrix} \in M_3(\mathbb{R})$.
- (b) $B = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 2 & 5 & -1 \end{pmatrix} \in M_3(\mathbb{R})$.

$$(c) \ C = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \in M_3(\mathbb{F}_2).$$

Solution.

(a) Row reduce $[A \mid I_3]$:

$$\begin{aligned} \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 0 & 1 & 0 \\ 4 & 1 & 8 & 0 & 0 & 1 \end{array} \right] &\xrightarrow[R_3 \rightarrow R_3 - 4R_1]{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & -1 & -2 & 1 & 0 \\ 0 & 1 & 0 & -4 & 0 & 1 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & -1 & -2 & 1 & 0 \\ 0 & 0 & 1 & -2 & -1 & 1 \end{array} \right] \\ &\xrightarrow[R_1 \rightarrow R_1 - 2R_3]{R_2 \rightarrow R_2 + R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 5 & 2 & -2 \\ 0 & 1 & 0 & -4 & 0 & 1 \\ 0 & 0 & 1 & -2 & -1 & 1 \end{array} \right]. \end{aligned}$$

The left half is I_3 , so by [Theorem 2.4.6](#), $A^{-1} = \begin{pmatrix} 5 & 2 & -2 \\ -4 & 0 & 1 \\ -2 & -1 & 1 \end{pmatrix}$. Check: row 1 of

A against the columns of A^{-1} gives $5 - 4 = 1$, $2 - 2 = 0$, $-2 + 2 = 0$, and rows 2 and 3 give $(0, 1, 0)$ and $(0, 0, 1)$, so $AA^{-1} = I_3$, which suffices by [Theorem 2.4.5](#).

(b) Row reduce $[B \mid I_3]$:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 2 & 5 & -1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & -2 & 0 & 1 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_2} \left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -2 & -1 & 1 \end{array} \right]$$

A zero row appears in the left half, so B is not invertible (remark after [Theorem 2.4.6](#)). The right half of that row says $-2(\text{row } 1) - (\text{row } 2) + (\text{row } 3) = \mathbf{0}$, that is, row 3 of B is $2(\text{row } 1) + (\text{row } 2)$.

(c) Over $\mathbb{F}_2$ we have $-1 = 1$, so subtracting a row is adding it. Row reduce $[C \mid I_3]$:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 + R_1} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 + R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 + R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array} \right]$$

Hence $C^{-1} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$. Check in $\mathbb{F}_2$: row 1 of C is $(1, 1, 0)$, and against the columns of C^{-1} it gives $0 + 1 = 1$, $1 + 1 = 0$, $1 + 1 = 0$; rows 2 and 3 give $(0, 1, 0)$ and $(0, 0, 1)$ similarly.

Exercise B2: A product of elementary matrices.

Let $A = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix} \in M_2(\mathbb{R})$.

(a) Write A as a product of elementary matrices.

(b) Hence write A^{-1} as a product of elementary matrices, and compute it.

Solution.

(a) Reduce A to I_2 :

$$\begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & 3 \\ 0 & -1 \end{pmatrix} \xrightarrow{R_2 \rightarrow -R_2} \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 - 3R_2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

With $E_1 = I_2 - 2E_{21}$, $E_2 = D_2(-1)$, $E_3 = I_2 - 3E_{12}$, [Theorem 2.4.1](#) gives $E_3E_2E_1A = I_2$. Multiplying on the left by $E_3^{-1}, E_2^{-1}, E_1^{-1}$ in turn and using [Theorem 2.4.2](#),

$$A = E_1^{-1}E_2^{-1}E_3^{-1} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}.$$

Check: the last two factors give $\begin{pmatrix} 1 & 3 \\ 0 & -1 \end{pmatrix}$, and adding twice row 1 to row 2 gives $\begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}$.

(b) Since $E_3E_2E_1A = I_2$, [Theorem 2.4.5](#) gives

$$A^{-1} = E_3E_2E_1 = \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} -5 & 3 \\ 2 & -1 \end{pmatrix}.$$

(The last two factors give $\begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}$, and $R_1 \rightarrow R_1 - 3R_2$ then gives the answer.)

This agrees with [Theorem 0.9.8](#), since $ad - bc = -1$.

Exercise B3: Invertible products.

Let $A, B \in M_n(F)$, and suppose AB is invertible. Prove that A and B are both invertible.

Solution. Let $C = (AB)^{-1}$. Then $A(BC) = (AB)C = I_n$ by associativity, so A has the right inverse BC , and by [Theorem 2.4.5](#) A is invertible. Similarly $(CA)B = C(AB) = I_n$, so B has the left inverse CA . By [Theorem 2.4.5](#) with the roles of the two matrices swapped (CA times B is I_n), B is invertible. This proves the claim.

2.4.6.3 C. Going deeper

Exercise C1: Triangular matrices.

Let $U = (u_{ij}) \in M_n(F)$ be upper triangular ([Definition 0.9.5](#)).

- (a) Prove that U is invertible if and only if $u_{ii} \neq 0$ for every i .
- (b) Prove that if U is invertible, then U^{-1} is upper triangular.

Hint: for (a), use item (b) or item (f) of [Theorem 2.4.4](#); for (b), consider the systems $U\mathbf{x} = \mathbf{e}_k$.

Solution.

- (a) ($\Leftarrow$) Suppose every $u_{ii} \neq 0$, and let $U\mathbf{x} = \mathbf{0}$. Row i of this system reads $u_{ii}x_i + u_{i,i+1}x_{i+1} + \cdots + u_{in}x_n = 0$, since $u_{ij} = 0$ for $j < i$. We show $x_n = x_{n-1} = \cdots = x_1 = 0$ by downward induction. Row n is $u_{nn}x_n = 0$, and $u_{nn} \neq 0$, so $x_n = 0$. If $x_{i+1} = \cdots = x_n = 0$, row i becomes $u_{ii}x_i = 0$, so $x_i = 0$ since $u_{ii} \neq 0$. Hence $\mathbf{x} = \mathbf{0}$, and U is invertible by [Theorem 2.4.4](#) ((b) $\Rightarrow$ (a)).
- ($\Rightarrow$) Suppose $u_{kk} = 0$ for some k . Every column $j \leq k$ of U has zero entries in rows $k, k+1, \dots, n$: below the diagonal by triangularity, and in row k of column k by assumption. So the k columns $\mathbf{u}_1, \dots, \mathbf{u}_k$ all lie in $\text{span}(\mathbf{e}_1, \dots, \mathbf{e}_{k-1})$, a subspace of dimension $k-1$ (its spanning list is a sub-list of the standard basis, hence independent). By [Theorem 1.6.3](#) (a), a list of $k > k-1$ vectors there is linearly dependent. A non-trivial relation among $\mathbf{u}_1, \dots, \mathbf{u}_k$, extended by zero coefficients on $\mathbf{u}_{k+1}, \dots, \mathbf{u}_n$, is a non-trivial relation among all the columns. So the columns of U are dependent, and U is not invertible by [Theorem 2.4.4](#) ((a) $\Rightarrow$ (f)). (For $k = 1$, the span is $\{\mathbf{0}\}$ and $\mathbf{u}_1 = \mathbf{0}$, so the same conclusion holds.)
- (b) Suppose U is invertible, so every $u_{ii} \neq 0$ by (a). Fix k , and let $\mathbf{x}$ be the k -th column of U^{-1} . By [Theorem 0.9.2](#), $U\mathbf{x}$ is the k -th column of $UU^{-1} = I_n$, so $U\mathbf{x} = \mathbf{e}_k$. For $i > k$, row i of this system reads $u_{ii}x_i + \cdots + u_{in}x_n = 0$. By the same downward induction as in (a), running only over $i = n, n-1, \dots, k+1$, we get $x_n = \cdots = x_{k+1} = 0$. So the k -th column of U^{-1} has zeros below position k . As k was arbitrary, U^{-1} is upper triangular.

Exercise C2: All right inverses.

Let $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \in M_{2 \times 3}(\mathbb{R})$.

- (a) Find all $B \in M_{3 \times 2}(\mathbb{R})$ with $AB = I_2$.
 (b) Show that there is no $C \in M_{3 \times 2}(\mathbb{R})$ with $CA = I_3$.

Solution.

- (a) By [Theorem 0.9.2](#), $AB = I_2$ means that the columns $\mathbf{b}_1, \mathbf{b}_2$ of B satisfy $A\mathbf{b}_1 = \mathbf{e}_1$ and $A\mathbf{b}_2 = \mathbf{e}_2$. The matrix A is already in reduced form with pivots in columns 1 and 2 and x_3 free, so $A\mathbf{x} = \mathbf{0}$ means $x_1 = -x_3$, $x_2 = -x_3$, and its solutions are the multiples of $(-1, -1, 1)$. One solution of $A\mathbf{x} = \mathbf{e}_1$ is $(1, 0, 0)$, and one solution of $A\mathbf{x} = \mathbf{e}_2$ is $(0, 1, 0)$. By [Theorem 2.1.3](#),

$$\mathbf{b}_1 = (1, 0, 0) + s(-1, -1, 1), \quad \mathbf{b}_2 = (0, 1, 0) + t(-1, -1, 1) \quad (s, t \in \mathbb{R}).$$

Hence the right inverses of A are exactly

$$B = \begin{pmatrix} 1-s & -t \\ -s & 1-t \\ s & t \end{pmatrix}, \quad s, t \in \mathbb{R},$$

infinitely many of them.

- (b) Suppose $CA = I_3$, and let $\mathbf{x} = (-1, -1, 1)$, so $A\mathbf{x} = \mathbf{0}$. Then $\mathbf{x} = I_3\mathbf{x} = C(A\mathbf{x}) = C\mathbf{0} = \mathbf{0}$, a contradiction since $\mathbf{x} \neq \mathbf{0}$. Hence A has no left inverse.

Exercise C3: Nilpotent perturbations of the identity.

Let $A \in M_n(F)$ and suppose $A^k = 0$ for some integer $k \geq 1$.

- (a) Prove that $I_n + A$ is invertible, by using item (b) of [Theorem 2.4.4](#).
- (b) Verify that $(I_n + A)(I_n - A + A^2 - \cdots + (-1)^{k-1}A^{k-1}) = I_n$, and deduce a formula for $(I_n + A)^{-1}$.

Solution.

- (a) Let $\mathbf{x} \in F^n$ with $(I_n + A)\mathbf{x} = \mathbf{0}$. Then $A\mathbf{x} = -\mathbf{x}$. Multiplying on the left by A and using this again, $A^2\mathbf{x} = A(-\mathbf{x}) = -A\mathbf{x} = (-1)^2\mathbf{x}$, and inductively $A^j\mathbf{x} = (-1)^j\mathbf{x}$ for every $j \geq 1$. Taking $j = k$, $\mathbf{0} = A^k\mathbf{x} = (-1)^k\mathbf{x}$, so $\mathbf{x} = \mathbf{0}$ (multiply by $(-1)^k$). By [Theorem 2.4.4](#) ((b) $\Rightarrow$ (a)), $I_n + A$ is invertible.
- (b) Expanding with [Theorem 0.9.4](#),

$$(I_n + A) \sum_{j=0}^{k-1} (-1)^j A^j = \sum_{j=0}^{k-1} (-1)^j A^j + \sum_{j=0}^{k-1} (-1)^j A^{j+1} = I_n + (-1)^{k-1} A^k = I_n,$$

since the terms $(-1)^j A^j$ for $1 \leq j \leq k-1$ in the first sum cancel against the terms $(-1)^{j-1} A^j$ of the second, and $A^k = 0$. By [Theorem 2.4.5](#), $(I_n + A)^{-1} = I_n - A + A^2 - \cdots + (-1)^{k-1} A^{k-1}$.

2.5 Row Space, Column Space, Null Space and Rank

We can now reduce any matrix and read off the solutions of any linear system. This section asks a coarser question: how much information does a matrix carry? Three subspaces come attached to every matrix A : the vectors $A\mathbf{x}$ it can produce, the combinations of its rows, and the vectors it sends to zero. Their dimensions are governed by a single number, the rank, which is the number of pivots. With it we settle when a system is solvable and how many solutions it has, and we turn the questions of Chapter 1 about vectors in F^m (independent? spanning? a basis of a sum or an intersection?) into a routine computation, which is the “systematic method” that Chapter 1 promised.

Throughout, F is a field and $A \in M_{m \times n}(F)$ has columns $\mathbf{a}_1, \dots, \mathbf{a}_n \in F^m$.

2.5.1 Three subspaces of a matrix

Here is the matrix we will follow through the section:

$$A = \begin{pmatrix} 1 & 2 & 0 & 0 & -1 \\ 2 & 4 & 1 & 0 & 0 \\ 1 & 2 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 3 \end{pmatrix} \in M_{4 \times 5}(\mathbb{R}).$$

Three questions about it have come up in this chapter, and each answer is a set. For which $\mathbf{b} \in \mathbb{R}^4$ can we solve $A\mathbf{x} = \mathbf{b}$? By [Theorem 2.1.1](#), exactly for $\mathbf{b}$ in the span of the columns. Which $\mathbf{x} \in \mathbb{R}^5$ solve $A\mathbf{x} = \mathbf{0}$? By [Theorem 2.1.2](#), a subspace of $\mathbb{R}^5$. And which new equations can we derive from the four by adding multiples of them? Their left-hand sides are exactly the combinations of the rows. These three sets recur so often that they get names.

The column space is everything A can produce, the null space is everything A kills, and the row space is every left-hand side we can build from the equations of $A\mathbf{x} = \mathbf{b}$.

Definition 2.5.1 — Column space

The **column space** of $A \in M_{m \times n}(F)$ is the span of its columns,

$$\text{col}(A) := \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n) = \{A\mathbf{x} : \mathbf{x} \in F^n\} \subseteq F^m.$$

Definition 2.5.2 — Row space

The **row space** of $A \in M_{m \times n}(F)$ is the span of its rows,

$$\text{row}(A) := \text{span}(\text{row } 1, \dots, \text{row } m) \subseteq M_{1 \times n}(F).$$

Definition 2.5.3 — Null space

The **null space** of $A \in M_{m \times n}(F)$ is the solution set of the homogeneous system,

$$\text{nul}(A) := \{\mathbf{x} \in F^n : A\mathbf{x} = \mathbf{0}\} \subseteq F^n.$$

In words: $\text{col}(A)$ lives among the **outputs**, in F^m (one entry per row); $\text{nul}(A)$ lives among the **inputs**, in F^n (one entry per column); and $\text{row}(A)$ consists of row vectors of length n . The two descriptions of $\text{col}(A)$ agree by [Theorem 0.9.3](#): $A\mathbf{x} = x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n$, and as $\mathbf{x}$ runs over F^n these are all the combinations of the columns.

Well-definedness. All three are subspaces. $\text{col}(A)$ and $\text{row}(A)$ are spans, hence subspaces by [Theorem 1.3.1](#), of F^m and of $M_{1 \times n}(F)$ respectively. $\text{nul}(A)$ is a subspace of F^n by [Theorem 2.1.2](#). The row vectors $M_{1 \times n}(F)$ form an n -dimensional space, with the standard basis $\mathbf{e}_1^\top, \dots, \mathbf{e}_n^\top$ ([Example \(Dimensions of the Running Examples\)](#)). Transposing turns row vectors into columns and respects sums and scalar multiples ([Theorem 0.9.5](#)), so it sends a combination of rows to the same combination of their transposes, and only the zero row transposes to $\mathbf{0}$. The transposes of the rows of A are the columns of $A^\top$. Hence transposing sends $\text{row}(A)$ onto $\text{col}(A^\top)$, a list of rows is independent exactly when the list of their transposes is, and transposing a basis of $\text{row}(A)$ gives a basis of $\text{col}(A^\top)$. Therefore

$$\dim \text{row}(A) = \dim \text{col}(A^\top). \quad (*)$$

Examples.

- **The identity.** $\text{col}(I_n) = F^n$, since the columns $\mathbf{e}_1, \dots, \mathbf{e}_n$ span; $\text{row}(I_n) = M_{1 \times n}(F)$; and $\text{nul}(I_n) = \{\mathbf{0}\}$, since $I_n\mathbf{x} = \mathbf{x}$.
- **The zero matrix** is the degenerate case. For $0 = 0_{m \times n}$, $\text{col}(0) = \text{span}(\mathbf{0}, \dots, \mathbf{0}) = \{\mathbf{0}\}$, $\text{row}(0) = \{\mathbf{0}\}$, and $\text{nul}(0) = F^n$, since every $\mathbf{x}$ is killed. It carries no information, and its

column space is as small as possible while its null space is as large as possible. We will see that this trade-off is exact.

- **A single row.** For $A = \begin{pmatrix} 1 & 1 & -1 \end{pmatrix} \in M_{1 \times 3}(\mathbb{R})$, the columns are the scalars $1, 1, -1 \in \mathbb{R}^1$, so $\text{col}(A) = \mathbb{R}^1$, $\text{row}(A) = \text{span}(\begin{pmatrix} 1 & 1 & -1 \end{pmatrix})$ is a line of row vectors, and $\text{nul}(A)$ is the plane $x + y - z = 0$ in $\mathbb{R}^3$.

Non-example by minimal change. Replace $\mathbf{0}$ by a non-zero right-hand side: the solution set $\{\mathbf{x} \in F^n : A\mathbf{x} = \mathbf{b}\}$ with $\mathbf{b} \neq \mathbf{0}$. It is described by the same equations, and it is still of the form $\mathbf{p} + \text{nul}(A)$ when non-empty (Theorem 2.1.3). But it does not contain $\mathbf{0}$, since $A\mathbf{0} = \mathbf{0} \neq \mathbf{b}$, so it fails the first condition of Theorem 1.2.1.

Warning. The three spaces live in **different places**. For $A \in M_{4 \times 5}(\mathbb{R})$ above, $\text{col}(A) \subseteq \mathbb{R}^4$ but $\text{nul}(A) \subseteq \mathbb{R}^5$, so a statement such as “ $\text{nul}(A) \subseteq \text{col}(A)$ ” does not even make sense. Keep track by size: $A\mathbf{x}$ needs $\mathbf{x}$ with n entries and returns a vector with m entries.

2.5.2 Rank and nullity

The column space measures how much A can reach, and the null space how much it collapses. We measure each by its dimension. Both are subspaces of finite-dimensional spaces, so both have a dimension by Theorem 1.6.7.

The rank counts the independent directions that A can produce.

Definition 2.5.4 — Rank of a matrix

The **rank** of $A \in M_{m \times n}(F)$ is $\text{rank } A := \dim \text{col}(A)$.

Definition 2.5.5 — Nullity of a matrix

The **nullity** of $A \in M_{m \times n}(F)$ is $\text{nullity } A := \dim \text{nul}(A)$.

From the examples above: $\text{rank } I_n = n$ and $\text{nullity } I_n = 0$; $\text{rank } 0_{m \times n} = 0$ and $\text{nullity } 0_{m \times n} = n$; and $\text{rank} \begin{pmatrix} 1 & 1 & -1 \end{pmatrix} = 1$, with nullity 2. In each case rank plus nullity equals the number of columns. Two simple bounds hold at once: $\text{rank } A \leq m$, because $\text{col}(A) \subseteq F^m$, and $\text{rank } A \leq n$, because $\text{col}(A)$ is spanned by n vectors (Theorem 1.6.3). Why define rank through columns and not rows? We could have, and one of the main theorems of this section says the answer would be the same.

2.5.3 What row operations preserve

To compute these spaces we will row reduce. So we first ask what a row operation does to each of them. Recall from Theorem 2.4.7 that A is row equivalent to B exactly when $B = EA$ with $E \in M_m(F)$ invertible; since row equivalence is symmetric (Theorem 2.2.2 (b)), we may simply say that A and B are row equivalent.

Theorem 2.5.1 — Row operations and the three spaces

Let $A, B \in M_{m \times n}(F)$ be row equivalent. Then:

- (a) $\text{row}(B) = \text{row}(A)$;
- (b) $\text{nul}(B) = \text{nul}(A)$. Equivalently, writing $\mathbf{b}_1, \dots, \mathbf{b}_n$ for the columns of B : for all $c_1, \dots, c_n \in F$, the columns of A satisfy $c_1 \mathbf{a}_1 + \dots + c_n \mathbf{a}_n = \mathbf{0}$ if and only if the columns of B satisfy $c_1 \mathbf{b}_1 + \dots + c_n \mathbf{b}_n = \mathbf{0}$;
- (c) but $\text{col}(B)$ and $\text{col}(A)$ **need not** be equal.

Proof.

- (a) Write $B = EA$ with E invertible, by Theorem 2.4.7. By Theorem 0.9.2, row i of EA is (row i of E) A , which by Theorem 0.9.3 (rows version) is a combination of the rows of A . So every row of B lies in $\text{row}(A)$, and $\text{row}(B) \subseteq \text{row}(A)$ by Proposition 1.3.4 (1). Since $A = E^{-1}B$ with E^{-1} invertible, the same argument gives $\text{row}(A) \subseteq \text{row}(B)$.
- (b) The equality $\text{nul}(B) = \text{nul}(A)$ is Lemma 2.3.1. By Theorem 0.9.3, $c_1 \mathbf{a}_1 + \dots + c_n \mathbf{a}_n = A\mathbf{c}$ and $c_1 \mathbf{b}_1 + \dots + c_n \mathbf{b}_n = B\mathbf{c}$ with $\mathbf{c} = (c_1, \dots, c_n)$, so the second statement is the first one written out (this is Lemma 2.3.2).
- (c) Let $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \in M_2(\mathbb{R})$ and $B = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$, obtained by $R_2 \rightarrow R_2 - R_1$. Then $\text{col}(A) = \text{span}((1, 1))$ and $\text{col}(B) = \text{span}((1, 0))$. These lines differ, since $(1, 0)$ is not a multiple of $(1, 1)$.

■

So row reduction keeps the row space and the **relations among the columns**, but it moves the columns themselves.

Warning. Do not read $\text{col}(A)$ off the reduced form. In (c) above, the RREF of A is B , whose column space is the x -axis, while $\text{col}(A)$ is the line $y = x$. What survives row reduction is the *pattern* of dependence: in both matrices column 2 equals column 1. So we use the RREF to decide **which** columns to take, and then take those columns **from** A .

2.5.4 Bases from the reduced form

Fix the notation for the rest of the section. Let $R = (r_{kj})$ be the reduced row echelon form of A (Theorem 2.2.3, Theorem 2.3.4), with columns $\mathbf{r}_1, \dots, \mathbf{r}_n \in F^m$. Let R have r non-zero rows, and let the pivot of row k lie in column j_k , so that $j_1 < j_2 < \dots < j_r$. These are the **pivot columns** of A , well defined by Corollary 2.3.5. The remaining columns are the **free columns**. We use three facts from Definition 2.2.4:

- rows $r + 1, \dots, m$ of R are zero;
- column j_k of R is $\mathbf{e}_k$ (the pivot is 1 and the only non-zero entry of its column);
- in row k , every entry to the left of the pivot is 0.

From the last two, if column j of R has exactly t pivot columns to its left, then **every entry of $\mathbf{r}_j$ in rows $t+1, \dots, m$ is zero, except the pivot entry of row $t+1$ when $j = j_{t+1}$.** Indeed, for $k \geq t+1$, row k is either zero or has its pivot in column $j_k \geq j$, and entries to the left of a pivot vanish. In the running example,

$$R = \begin{pmatrix} 1 & 2 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

with $r = 3$, pivot columns 1, 3, 4 and free columns 2, 5 (the reduction is carried out in [Example \(Bases for the running example\)](#) below).

The column space comes first. In R the answer is visible: the pivot columns are $\mathbf{e}_1, \dots, \mathbf{e}_r$, and every column has zeros below row r . Row reduction does not preserve columns, but by [Theorem 2.5.1](#) (b) it preserves every linear relation among them, and “is a basis” and “equals this combination” are statements about relations.

Theorem 2.5.2 — Pivot columns form a basis of the column space

Let $A \in M_{m \times n}(F)$ have pivot columns $j_1 < \dots < j_r$, and let $R = (r_{kj})$ be its RREF. Then:

- (a) the pivot columns of A , $(\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_r})$, form a basis of $\text{col}(A)$;
- (b) if column j is free and t pivot columns lie to its left, then $\mathbf{a}_j = r_{1j}\mathbf{a}_{j_1} + \dots + r_{tj}\mathbf{a}_{j_t}$;
- (c) the pivot columns of A are exactly the vectors kept when $(\mathbf{a}_1, \dots, \mathbf{a}_n)$ is sifted as in [Theorem 1.5.4](#).

Idea. Both claims are statements about linear relations among columns, so by [Theorem 2.5.1](#) (b) it suffices to prove them for R , where they can be seen by inspection. The pivot columns of R are $\mathbf{e}_1, \dots, \mathbf{e}_r$, which are independent. A free column of R is $\sum_k r_{kj}\mathbf{e}_k$, a combination of the pivot columns to its left with its own entries as coefficients. So we transfer two relations back to A : “ $\sum c_k \mathbf{a}_{j_k} = \mathbf{0}$ forces $c_1 = \dots = c_r = 0$ ” and “ $\mathbf{a}_j - \sum_k r_{kj}\mathbf{a}_{j_k} = \mathbf{0}$ ”. Part (c) is a rereading of [Corollary 2.3.5](#).

Proof. Throughout, we use [Theorem 2.5.1](#) (b) in the form: for scalars $c_1, \dots, c_n$,

$$c_1 \mathbf{a}_1 + \dots + c_n \mathbf{a}_n = \mathbf{0} \iff c_1 \mathbf{r}_1 + \dots + c_n \mathbf{r}_n = \mathbf{0}. \quad (\dagger)$$

- (b) Let column j be free, with t pivot columns $j_1, \dots, j_t$ to its left. By the fact in bold before the theorem, the entries of $\mathbf{r}_j$ in rows $t+1, \dots, m$ are zero, so $\mathbf{r}_j = r_{1j}\mathbf{e}_1 + \dots + r_{tj}\mathbf{e}_t = r_{1j}\mathbf{r}_{j_1} + \dots + r_{tj}\mathbf{r}_{j_t}$, since $\mathbf{r}_{j_k} = \mathbf{e}_k$. This is a linear relation among the columns of R with coefficient 1 on $\mathbf{r}_j$, $-r_{kj}$ on $\mathbf{r}_{j_k}$, and 0 elsewhere. By $(\dagger)$, the columns of A satisfy the same relation, which is (b).
- (c) *Spanning.* Each pivot column $\mathbf{a}_{j_k}$ lies in $\text{span}(\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_r})$, and each free column does by (b). Hence $\text{col}(A) = \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n) \subseteq \text{span}(\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_r})$ by [Proposition 1.3.4](#) (1). The reverse inclusion holds by [Proposition 1.3.4](#) (2), since the pivot columns are among the columns.

Independence. Let $c_1 \mathbf{a}_{j_1} + \dots + c_r \mathbf{a}_{j_r} = \mathbf{0}$. This is a relation among all the columns of A with coefficient 0 on the free columns. By $(\dagger)$, $c_1 \mathbf{r}_{j_1} + \dots + c_r \mathbf{r}_{j_r} = c_1 \mathbf{e}_1 + \dots + c_r \mathbf{e}_r = \mathbf{0}$. The k -th entry of the left side is c_k , so $c_1 = \dots = c_r = 0$. Hence the pivot

columns of A are linearly independent, and with spanning they form a basis of $\text{col}(A)$ by [Definition 1.5.1](#).

- (c) By [Corollary 2.3.5](#) (a), column j is a pivot column if and only if $\mathbf{a}_j \notin \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_{j-1})$. Sifting keeps $\mathbf{a}_j$ exactly when $\mathbf{a}_j$ is not in the span of the vectors kept before it, and by [Theorem 1.5.4](#) (applied to $(\mathbf{a}_1, \dots, \mathbf{a}_{j-1})$) that span is $\text{span}(\mathbf{a}_1, \dots, \mathbf{a}_{j-1})$. Hence the kept vectors are exactly the pivot columns. This proves the theorem. ■

This is sifting ([Theorem 1.5.4](#)) done by elimination. In Chapter 1, each step “is $\mathbf{v}_j$ in the span of the vectors kept so far?” was a separate small system. One row reduction answers all of them at once, and part (b) even gives the coefficients.

The row space and the null space are read off R directly.

Theorem 2.5.3 — Basis of the row space

The non-zero rows of the RREF R of A form a basis of $\text{row}(A)$.

Proof. By [Theorem 2.5.1](#) (a), $\text{row}(A) = \text{row}(R)$. The zero rows of R contribute nothing to a combination, so the r non-zero rows span $\text{row}(R)$. For independence, let $c_1(\text{row } 1) + \dots + c_r(\text{row } r) = 0$. Look at the entry in column j_k : since column j_k of R is $\mathbf{e}_k$, only row k has a non-zero entry there, equal to 1. So that entry of the combination is c_k , and $c_k = 0$ for each k . Hence the non-zero rows are a basis of $\text{row}(A)$. ■

For the null space, each free variable gets one basis vector: set that free variable to 1, the other free variables to 0, and solve for the pivot variables.

Theorem 2.5.4 — Basis of the null space

Let $A \in M_{m \times n}(F)$ have RREF $R = (r_{kj})$ with pivot columns $j_1 < \dots < j_r$. For each free column f , define $\mathbf{s}_f \in F^n$ by

$$(\mathbf{s}_f)_f = 1, \quad (\mathbf{s}_f)_{f'} = 0 \text{ for every other free } f', \quad (\mathbf{s}_f)_{j_k} = -r_{kf} \text{ for } k = 1, \dots, r.$$

Then the vectors $\mathbf{s}_f$, one for each free column f , form a basis of $\text{nul}(A)$. In particular nullity $A = n - r$, the number of free columns.

Proof. By [Theorem 2.5.1](#) (b), $\text{nul}(A) = \text{nul}(R)$. For $k \leq r$, row k of R has entry 1 in column j_k , entry 0 in every other pivot column (since column $j_{k'}$ is $\mathbf{e}_{k'}$), and entries r_{kf} in the free columns. Rows $k > r$ are zero. Hence $R\mathbf{x} = \mathbf{0}$ is equivalent to the r equations

$$x_{j_k} = - \sum_{f \text{ free}} r_{kf} x_f \quad (k = 1, \dots, r). \quad (\ddagger)$$

Spanning. Let $\mathbf{x} \in \text{nul}(A)$, and put $\mathbf{y} = \sum_{f \text{ free}} x_f \mathbf{s}_f$. For a free column f , the f -th entry of $\mathbf{y}$ is x_f , since $\mathbf{s}_f$ has a 1 there and the other $\mathbf{s}_{f'}$ have 0. For a pivot column, the

j_k -th entry of $\mathbf{y}$ is $\sum_f x_f(-r_{kf})$, which equals x_{j_k} by $(\dagger)$. So $\mathbf{x} = \mathbf{y}$ lies in the span of the $\mathbf{s}_f$. Each $\mathbf{s}_f$ itself satisfies $(\dagger)$, so it lies in $\text{nul}(A)$.

Independence. If $\sum_f c_f \mathbf{s}_f = \mathbf{0}$, then the f -th entry of the left side is c_f , so every $c_f = 0$.

Hence the $\mathbf{s}_f$ form a basis of $\text{nul}(A)$, and there are $n - r$ of them. ■

Example. (Bases for the running example)

Find bases of $\text{col}(A)$, $\text{row}(A)$ and $\text{nul}(A)$, and the rank and nullity, for

$$A = \begin{pmatrix} 1 & 2 & 0 & 0 & -1 \\ 2 & 4 & 1 & 0 & 0 \\ 1 & 2 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 3 \end{pmatrix} \in M_{4 \times 5}(\mathbb{R}).$$

Solution. Row reduce:

$$A \xrightarrow[\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1}]{\substack{R_3 \rightarrow R_3 - R_2 \\ R_4 \rightarrow R_4 - R_2}} \begin{pmatrix} 1 & 2 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 1 & 3 \end{pmatrix} \xrightarrow[\substack{R_3 \rightarrow R_3 - R_2 \\ R_4 \rightarrow R_4 - R_2}]{\substack{R_3 \rightarrow R_3 - R_2 \\ R_4 \rightarrow R_4 - R_2}} \begin{pmatrix} 1 & 2 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_4 \rightarrow R_4 - R_3} R = \begin{pmatrix} 1 & 2 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The pivot columns are 1, 3, 4, and the free columns are 2, 5.

Column space. By Theorem 2.5.2 (a), the pivot columns of A ,

$$\mathbf{a}_1 = (1, 2, 1, 0), \quad \mathbf{a}_3 = (0, 1, 1, 1), \quad \mathbf{a}_4 = (0, 0, 1, 1),$$

form a basis of $\text{col}(A)$, so $\text{rank } A = 3$. By part (b), the free columns are $\mathbf{a}_2 = 2\mathbf{a}_1$ (from column 2 of R) and $\mathbf{a}_5 = -\mathbf{a}_1 + 2\mathbf{a}_3 + \mathbf{a}_4$ (from column 5). Check: $-(1, 2, 1, 0) + 2(0, 1, 1, 1) + (0, 0, 1, 1) = (-1, 0, 2, 3) = \mathbf{a}_5$. Note that $\text{col}(A) \neq \text{col}(R)$: every column of R has last entry 0, but $\mathbf{a}_3$ does not.

Row space. By Theorem 2.5.3, a basis is

$$(1 \ 2 \ 0 \ 0 \ -1), \quad (0 \ 0 \ 1 \ 0 \ 2), \quad (0 \ 0 \ 0 \ 1 \ 1),$$

so $\dim \text{row}(A) = 3$.

Null space. The equations $(\dagger)$ read $x_1 = -2x_2 + x_5$, $x_3 = -2x_5$, $x_4 = -x_5$. By Theorem 2.5.4, a basis is

$$\mathbf{s}_2 = (-2, 1, 0, 0, 0), \quad \mathbf{s}_5 = (1, 0, -2, -1, 1),$$

so $\text{nullity } A = 2$. Check: $A\mathbf{s}_2 = -2\mathbf{a}_1 + \mathbf{a}_2 = \mathbf{0}$ and $A\mathbf{s}_5 = \mathbf{a}_1 - 2\mathbf{a}_3 - \mathbf{a}_4 + \mathbf{a}_5 = \mathbf{0}$, which are the two relations found above.

So $\text{rank } A = \dim \text{row}(A) = 3$ and $\text{rank } A + \text{nullity } A = 5$, the number of columns.

Quick check. Without computing anything, what is the largest possible rank of a 3×5 matrix, and the smallest possible nullity?

Solution. $\text{rank } A = \dim \text{col}(A) \leq 3$, since $\text{col}(A) \subseteq F^3$. By Theorem 2.5.4, the nullity is the number of free columns, $5 - r$, where $r \leq 3$ is the number of pivots (one per non-zero row). So the nullity is at least 2. For example $(I_3 \ 0_{3 \times 2})$ attains rank 3 and nullity 2.

2.5.5 Row rank equals column rank

The example came out with $\dim \text{row}(A) = \dim \text{col}(A)$. This is not luck, and it is surprising: the row space and the column space live in different spaces, and the rows and columns of a 4×5 matrix have nothing obvious in common. Counting pivots explains it.

Theorem 2.5.5 — Row rank equals column rank

Let $A \in M_{m \times n}(F)$, and let r be the number of pivots of its RREF. Then

$$\dim \text{row}(A) = \dim \text{col}(A) = r.$$

Consequently $\text{rank } A = \text{rank } A^T$.

Proof. By Theorem 2.5.2 (a), $\text{col}(A)$ has a basis consisting of the r pivot columns, so $\dim \text{col}(A) = r$. By Theorem 2.5.3, $\text{row}(A)$ has a basis consisting of the non-zero rows of the RREF. Each non-zero row contains exactly one pivot and each pivot lies in a non-zero row, so there are r of them, and $\dim \text{row}(A) = r$. Finally, by (*) at the start of this section, $\text{rank } A^T = \dim \text{col}(A^T) = \dim \text{row}(A) = r = \text{rank } A$. ■

The theorem makes the rank independent of **which** echelon form we compute. Gaussian elimination without the final clearing-up phase produces an echelon form that is not reduced, and different sequences of operations produce different echelon forms. They all give the same count.

Corollary 2.5.6 — Rank from any echelon form

Let $A \in M_{m \times n}(F)$, and let G be a matrix in row echelon form that is row equivalent to A . Then the number of non-zero rows of G is $\text{rank } A$, and the columns of G containing its pivots are exactly the pivot columns of A .

Proof. By Corollary 2.3.5 (b), the pivot columns of G are the pivot columns of A . Each non-zero row of G has exactly one pivot, in a different column (Definition 2.2.3), so the number of non-zero rows of G equals the number of pivot columns of A , which is $\text{rank } A$ by Theorem 2.5.5. ■

Warning. Rank is a property of A , not of a computation, but it must be counted from a row echelon form. The number of non-zero rows of A itself is not the rank: $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ has two non-zero rows and rank 1. Count non-zero rows only after reaching an echelon form. And the rank depends on the field when the entries are read in different fields:

$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ has rank 2 over $\mathbb{R}$, but over $\mathbb{F}_2$, where $-1 = 1$, both rows are $(1 \ 1)$ and the rank is 1.

2.5.6 Rank–nullity and the Rouché–Capelli theorem

The trade-off between the zero matrix and the identity was exact, and now we can say why.

Theorem 2.5.7 — Rank–Nullity for Matrices

For every $A \in M_{m \times n}(F)$,

$$\text{rank } A + \text{nullity } A = n,$$

the number of **columns** of A .

Proof. Let r be the number of pivot columns of A . By Theorem 2.5.5, $\text{rank } A = r$. By Theorem 2.5.4, $\text{nullity } A = n - r$. Adding gives n . ■

Each column of A is either a pivot column, contributing a direction to $\text{col}(A)$, or a free column, contributing a direction to $\text{nul}(A)$. Nothing is lost and nothing is counted twice. In Chapter 3 the same equation becomes the Rank–Nullity Theorem for linear maps, with a proof that uses no matrices at all.

Now the question this chapter started with. When does $A\mathbf{x} = \mathbf{b}$ have a solution, and how many solutions are there? The first answer was $\mathbf{b} \in \text{col}(A)$ (Theorem 2.1.1); here is a version that is a pure count.

Theorem 2.5.8 — Rouché–Capelli Theorem

Let $A \in M_{m \times n}(F)$, $\mathbf{b} \in F^m$, and let $[A \mid \mathbf{b}] \in M_{m \times (n+1)}(F)$ be the augmented matrix.

- (a) The system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\text{rank } A = \text{rank } [A \mid \mathbf{b}]$.
- (b) Suppose it is consistent, let $\mathbf{p}$ be one solution, let $r = \text{rank } A$, and let $\mathbf{s}_1, \dots, \mathbf{s}_{n-r}$ be a basis of $\text{nul}(A)$. Then every solution can be written as

$$\mathbf{x} = \mathbf{p} + t_1\mathbf{s}_1 + \dots + t_{n-r}\mathbf{s}_{n-r}$$

for **exactly one** choice of $(t_1, \dots, t_{n-r}) \in F^{n-r}$. In words: the solutions form an $(n - r)$ -parameter family.

- (c) If moreover F is a finite field with q elements (for instance $\mathbb{F}_p$, with $q = p$), a consistent system has exactly $q^{n-\text{rank } A}$ solutions.

Idea. For (a), adding the column $\mathbf{b}$ can only enlarge the column space, and the dimension goes up exactly when $\mathbf{b}$ is new, that is, when $\mathbf{b} \notin \text{col}(A)$. For (b), the structure theorem says solutions are $\mathbf{p}$ plus null vectors, and null vectors have unique coordinates in a basis.

For (c), unique coordinates mean the solutions are in bijection with F^{n-r} , which has q^{n-r} elements.

Proof.

- (a) By [Definition 2.5.1](#), $\text{col}([A \mid \mathbf{b}]) = \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n, \mathbf{b})$, which contains $\text{col}(A)$ by [Proposition 1.3.4](#) (2).
 $(\Rightarrow)$ Suppose the system is consistent. By [Theorem 2.1.1](#), $\mathbf{b} \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$, so by [Theorem 1.3.5](#) $\text{col}([A \mid \mathbf{b}]) = \text{col}(A)$, and the ranks are equal.
 $(\Leftarrow)$ Suppose $\text{rank } A = \text{rank } [A \mid \mathbf{b}]$. Then $\text{col}(A)$ is a subspace of $\text{col}([A \mid \mathbf{b}])$ of the same dimension, so $\text{col}(A) = \text{col}([A \mid \mathbf{b}])$ by [Theorem 1.6.8](#). In particular $\mathbf{b} \in \text{col}(A)$, and the system is consistent by [Theorem 2.1.1](#).
- (b) By [Theorem 2.5.7](#), nullity $A = n - r$, so a basis $\mathbf{s}_1, \dots, \mathbf{s}_{n-r}$ of $\text{nul}(A)$ exists ([Theorem 2.5.4](#)). By [Theorem 2.1.3](#), $\mathbf{x}$ is a solution if and only if $\mathbf{x} - \mathbf{p} \in \text{nul}(A)$. By [Theorem 1.5.1](#), applied in the vector space $\text{nul}(A)$ with this basis, each $\mathbf{x} - \mathbf{p} \in \text{nul}(A)$ equals $t_1\mathbf{s}_1 + \dots + t_{n-r}\mathbf{s}_{n-r}$ for exactly one $(t_1, \dots, t_{n-r})$. Conversely every such expression is in $\text{nul}(A)$, so $\mathbf{p}$ plus it is a solution. This proves (b).
- (c) By (b), the rule $(t_1, \dots, t_{n-r}) \mapsto \mathbf{p} + t_1\mathbf{s}_1 + \dots + t_{n-r}\mathbf{s}_{n-r}$ is a bijection from F^{n-r} onto the solution set: every solution is hit, and by exactly one tuple. A tuple in F^{n-r} is a choice of $n-r$ entries, each one of q elements, so F^{n-r} has q^{n-r} elements. (When $r = n$ there is exactly one tuple, the empty one, and the basis of $\text{nul}(A)$ is empty, so $\mathbf{p}$ is the only solution and $q^0 = 1$.) Hence the solution set has q^{n-r} elements. ■

Over an infinite field such as $\mathbb{Q}$, $\mathbb{R}$ or $\mathbb{C}$, part (b) gives the familiar trichotomy: no solution, exactly one solution (when $\text{rank } A = n$), or infinitely many (when $\text{rank } A < n$, since distinct values of t_1 give distinct solutions). Over a finite field “infinitely many” is replaced by an exact power of q . For example, over $\mathbb{F}_2$ a system can never have exactly 3 or 6 solutions.

Example. (Counting solutions over $\mathbb{F}_3$)

Over $\mathbb{F}_3$, consider

$$A = \begin{pmatrix} 1 & 1 & 0 & 2 \\ 1 & 2 & 1 & 0 \\ 2 & 0 & 1 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{b}' = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

Decide whether $A\mathbf{x} = \mathbf{b}$ and $A\mathbf{x} = \mathbf{b}'$ are consistent, and count their solutions.

Solution. We reduce both augmented matrices at once, carrying two right-hand columns; arithmetic is modulo 3, so $-1 = 2$ and $-2 = 1$.

$$\left[\begin{array}{cccc|c|c} 1 & 1 & 0 & 2 & 1 & 1 \\ 1 & 2 & 1 & 0 & 0 & 0 \\ 2 & 0 & 1 & 2 & 1 & 0 \end{array} \right] \xrightarrow[R_3 \rightarrow R_3 - 2R_1]{R_2 \rightarrow R_2 - R_1} \left[\begin{array}{cccc|c|c} 1 & 1 & 0 & 2 & 1 & 1 \\ 0 & 1 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 2 & 1 \end{array} \right] \xrightarrow[R_1 \rightarrow R_1 - R_2]{R_3 \rightarrow R_3 - R_2} \left[\begin{array}{cccc|c|c} 1 & 0 & 2 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 2 \end{array} \right].$$

Row operations act on each column separately, so deleting either right-hand column gives a reduction of the corresponding augmented matrix. For instance, in row 2, $(1, 2, 1, 0) - (1, 1, 0, 2) = (0, 1, 1, -2) = (0, 1, 1, 1)$, and in row 3, $(2, 0, 1, 2) - 2(1, 1, 0, 2) =$

$$(0, -2, 1, -2) = (0, 1, 1, 1).$$

The left block is an echelon form of A with two non-zero rows, so $\text{rank } A = 2$ by [Corollary 2.5.6](#). With $\mathbf{b}$, the augmented matrix $[A \mid \mathbf{b}]$ has the same two non-zero rows (the last is zero), so $\text{rank } [A \mid \mathbf{b}] = 2 = \text{rank } A$, and by [Theorem 2.5.8](#) (a) the system is consistent. By part (c) it has exactly $3^{4-2} = 9$ solutions. Explicitly, the reduced system is $x_1 = 2 - 2x_3 - x_4 = 2 + x_3 + 2x_4$ and $x_2 = 2 - x_3 - x_4 = 2 + 2x_3 + 2x_4$, with $x_3, x_4 \in \mathbb{F}_3$ free: nine choices.

With $\mathbf{b}'$, the last row $(0, 0, 0, 0 \mid 2)$ is a non-zero row with its pivot in the last column, so $[A \mid \mathbf{b}']$ has three non-zero rows in echelon form and $\text{rank } [A \mid \mathbf{b}'] = 3 \neq 2$. By [Theorem 2.5.8](#) (a), $A\mathbf{x} = \mathbf{b}'$ has no solution. Indeed, row 3 of A is the sum of rows 1 and 2, so any solution would need $b'_3 = b'_1 + b'_2 = 1$, but $b'_3 = 0$.

2.5.7 Rank of a product

Multiplying matrices cannot create information: every column of AB is a combination of the columns of A , and every row is a combination of the rows of B . The rank records this.

Theorem 2.5.9 — Rank of a product

Let $A \in M_{m \times n}(F)$ and $B \in M_{n \times p}(F)$. Then

$$\text{rank}(AB) \leq \min(\text{rank } A, \text{rank } B).$$

Moreover, if $P \in M_m(F)$ and $Q \in M_n(F)$ are invertible, then $\text{rank}(PA) = \text{rank } A = \text{rank}(AQ)$.

Proof. Let $\mathbf{b}_1, \dots, \mathbf{b}_p \in F^n$ be the columns of B . By [Theorem 0.9.2](#), the columns of AB are $A\mathbf{b}_1, \dots, A\mathbf{b}_p$, which lie in $\text{col}(A)$ by [Definition 2.5.1](#). By [Proposition 1.3.4](#) (1), $\text{col}(AB) \subseteq \text{col}(A)$, so $\text{rank}(AB) \leq \text{rank } A$ by [Theorem 1.6.7](#). Similarly, row i of AB is (row i of A) B , a combination of the rows of B by [Theorem 0.9.3](#) (rows version), so $\text{row}(AB) \subseteq \text{row}(B)$. By [Theorem 2.5.5](#), $\text{rank}(AB) = \dim \text{row}(AB) \leq \dim \text{row}(B) = \text{rank } B$.

For the second statement, apply the inequality twice: $\text{rank } A = \text{rank}(P^{-1}(PA)) \leq \text{rank}(PA) \leq \text{rank } A$, so equality holds throughout. The same sandwich with $A = (AQ)Q^{-1}$ gives $\text{rank}(AQ) = \text{rank } A$.

■

The equality case recovers something we already knew in a new form: row operations (left multiplication by an invertible P) do not change the rank. The inequality is often strict. For $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, both factors of $A \cdot A$ have rank 1, but $A^2 = 0$ has rank 0.

2.5.8 A toolkit for questions from Chapter 1

In Chapter 1 we asked, again and again, whether some vectors in F^m are independent, whether they span, how to extend them to a basis, and what the sum and intersection of two subspaces are. Each was answered by solving a small system by hand. Everything now reduces to one move: **put the vectors as columns of a matrix, row reduce, and read off pivots.**

Theorem 2.5.10 — Independence and spanning by rank

Let $\mathbf{v}_1, \dots, \mathbf{v}_k \in F^m$, and let $A = (\mathbf{v}_1 \ \cdots \ \mathbf{v}_k) \in M_{m \times k}(F)$. Then:

- (a) $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ is linearly independent if and only if $\text{rank } A = k$, that is, every column is a pivot column;
- (b) $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ spans F^m if and only if $\text{rank } A = m$, that is, every row of the RREF has a pivot;
- (c) $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ is a basis of F^m if and only if $k = m = \text{rank } A$.

Proof.

- (a) By Theorem 0.9.3, $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = A\mathbf{c}$ with $\mathbf{c} = (c_1, \dots, c_k)$, so the list is independent if and only if $\text{nul}(A) = \{\mathbf{0}\}$, that is, nullity $A = 0$. By Theorem 2.5.7 this means $\text{rank } A = k$.
- (b) The list spans F^m if and only if $\text{col}(A) = F^m$. Since $\text{col}(A) \subseteq F^m$ and $\dim F^m = m$, this holds if and only if $\text{rank } A = m$, by Theorem 1.6.8 for one direction and because equal spaces have equal dimension for the other. The number of pivots is $\text{rank } A$ by Theorem 2.5.5, and the RREF has m rows, each with at most one pivot.
- (c) Combine (a) and (b) with Definition 1.5.1.

■

Example. (Testing independence)

Decide whether $\mathbf{v}_1 = (1, 2, 0, 1)$, $\mathbf{v}_2 = (2, 4, 1, 1)$, $\mathbf{v}_3 = (0, 0, 1, -1)$ are linearly independent in $\mathbb{R}^4$. If not, find a linear relation.

Solution. Row reduce the matrix with these columns:

$$\begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & -1 \end{pmatrix} \xrightarrow[\substack{R_4 \rightarrow R_4 - R_1}]{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & -1 & -1 \end{pmatrix} \xrightarrow[\substack{R_1 \rightarrow R_1 - 2R_2}]{R_4 \rightarrow R_4 + R_2} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Only columns 1 and 2 are pivot columns, so the rank is $2 < 3$, and the vectors are **dependent** by Theorem 2.5.10 (a). Column 3 of the RREF gives the relation, by Theorem 2.5.2 (b): $\mathbf{v}_3 = -2\mathbf{v}_1 + \mathbf{v}_2$. Check: $-2(1, 2, 0, 1) + (2, 4, 1, 1) = (0, 0, 1, -1)$. Moreover $(\mathbf{v}_1, \mathbf{v}_2)$ is a basis of $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ by Theorem 2.5.2 (a).

To **extend** an independent list to a basis of F^m , append the standard basis and keep the pivot columns. The appended vectors guarantee that the columns span F^m . The original vectors all

survive, because by [Theorem 2.5.2](#) (c) a column is discarded only if it lies in the span of the columns before it, and an independent list has no such vector ([Lemma 1.5.3](#)).

Example. (Extending to a basis by elimination)

Extend the independent list $(\mathbf{v}_1, \mathbf{v}_2) = ((1, 2, 0, 1), (2, 4, 1, 1))$ from [Example \(Testing independence\)](#) to a basis of $\mathbb{R}^4$.

Solution. Form $M = (\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3 \ \mathbf{e}_4)$ and row reduce:

$$\begin{aligned} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix} &\xrightarrow[R_4 \rightarrow R_4 - R_1]{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 & 0 & 1 \end{pmatrix} \xrightarrow[\text{then } R_4 \rightarrow R_4 + R_2]{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -2 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & 1 \end{pmatrix} \\ &\xrightarrow[\text{then } R_3 \rightarrow -R_3]{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & -2 & 1 & 0 & 0 \end{pmatrix} \xrightarrow{R_4 \rightarrow R_4 + 2R_3} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 0 & 1 & -2 & -2 \end{pmatrix}. \end{aligned}$$

This is an echelon form with pivots in columns 1, 2, 3, 4. By [Corollary 2.5.6](#) these are the pivot columns of M , so by [Theorem 2.5.2](#) (a) the corresponding columns of M form a basis of $\text{col}(M) = \mathbb{R}^4$:

$$(\mathbf{v}_1, \mathbf{v}_2, \mathbf{e}_1, \mathbf{e}_2).$$

(Clearing above the pivots gives the RREF, whose column 5 says $\mathbf{e}_3 = -\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{e}_1 - 2\mathbf{e}_2$; check: $-(1, 2, 0, 1) + (2, 4, 1, 1) - (1, 0, 0, 0) - 2(0, 1, 0, 0) = (0, 0, 1, 0)$.)

For sums and intersections, one reduction does both jobs.

Proposition 2.5.11 — Sum and intersection by elimination

Let $U, W \subseteq F^m$ be subspaces with bases $(\mathbf{u}_1, \dots, \mathbf{u}_k)$ and $(\mathbf{w}_1, \dots, \mathbf{w}_l)$, and let $M = (\mathbf{u}_1 \ \cdots \ \mathbf{u}_k \ \mathbf{w}_1 \ \cdots \ \mathbf{w}_l) \in M_{m \times (k+l)}(F)$. Then:

- (a) the pivot columns of M form a basis of $U + W$;
- (b) for $\mathbf{x} = (x_1, \dots, x_{k+l}) \in F^{k+l}$ write $\mathbf{u}_{\mathbf{x}} := x_1\mathbf{u}_1 + \cdots + x_k\mathbf{u}_k$, the combination of the $\mathbf{u}$'s with the first k entries of $\mathbf{x}$ as coefficients. If $(\mathbf{n}_1, \dots, \mathbf{n}_d)$ is a basis of $\text{nul}(M)$, then $(\mathbf{u}_{\mathbf{n}_1}, \dots, \mathbf{u}_{\mathbf{n}_d})$ is a basis of $U \cap W$.

In particular $\dim(U + W) = \text{rank } M$ and $\dim(U \cap W) = \text{nullity } M$.

Proof.

- (a) By [Proposition 1.7.2](#), $U + W = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_1, \dots, \mathbf{w}_l) = \text{col}(M)$, and [Theorem 2.5.2](#) (a) applies.
- (b) By [Theorem 0.9.3](#), $\mathbf{x} \in \text{nul}(M)$ means $x_1\mathbf{u}_1 + \cdots + x_k\mathbf{u}_k = -(x_{k+1}\mathbf{w}_1 + \cdots + x_{k+l}\mathbf{w}_l)$, so $\mathbf{u}_{\mathbf{x}}$ lies in U and in W . Also $\mathbf{u}_{\mathbf{x}+\mathbf{y}} = \mathbf{u}_{\mathbf{x}} + \mathbf{u}_{\mathbf{y}}$ and $\mathbf{u}_{c\mathbf{x}} = c\mathbf{u}_{\mathbf{x}}$, by the vector space axioms in F^m .

Spanning. Let $\mathbf{v} \in U \cap W$. Since the $\mathbf{u}$'s span U and the $\mathbf{w}$'s span W , $\mathbf{v} = a_1\mathbf{u}_1 + \cdots +$

$a_k \mathbf{u}_k = b_1 \mathbf{w}_1 + \cdots + b_l \mathbf{w}_l$. Then $\mathbf{x} = (a_1, \dots, a_k, -b_1, \dots, -b_l) \in \text{nul}(M)$ and $\mathbf{u}_{\mathbf{x}} = \mathbf{v}$. Writing $\mathbf{x} = \sum_i t_i \mathbf{n}_i$ gives $\mathbf{v} = \sum_i t_i \mathbf{u}_{\mathbf{n}_i}$.

Independence. Let $\sum_i t_i \mathbf{u}_{\mathbf{n}_i} = \mathbf{0}$, and put $\mathbf{x} = \sum_i t_i \mathbf{n}_i \in \text{nul}(M)$. Then $\mathbf{u}_{\mathbf{x}} = x_1 \mathbf{u}_1 + \cdots + x_k \mathbf{u}_k = \mathbf{0}$, so $x_1 = \cdots = x_k = 0$ because the $\mathbf{u}$'s are independent. Since $M\mathbf{x} = \mathbf{0}$, also $x_{k+1} \mathbf{w}_1 + \cdots + x_{k+l} \mathbf{w}_l = \mathbf{0}$, so the remaining entries vanish because the $\mathbf{w}$'s are independent. Hence $\mathbf{x} = \mathbf{0}$, and $t_1 = \cdots = t_d = 0$ because the $\mathbf{n}_i$ are independent.

The dimension statements follow by counting the two bases, using [Theorem 2.5.5](#) for rank M . ■

Adding the two counts and using [Theorem 2.5.7](#) gives $\dim(U + W) + \dim(U \cap W) = \text{rank } M + \text{nullity } M = k + l = \dim U + \dim W$. This is the Dimension Formula for Sums ([Theorem 1.7.3](#)) again, now with a second, computational proof for subspaces of F^m . (Some texts use the matrix with columns $\mathbf{u}_1, \dots, \mathbf{u}_k, -\mathbf{w}_1, \dots, -\mathbf{w}_l$ instead of M , so that a null vector gives equal combinations $\sum a_i \mathbf{u}_i = \sum b_j \mathbf{w}_j$ with no sign change. The two null spaces differ only by the signs of the last l entries.)

Example. (A basis of a sum)

Let $U = \text{span}(\mathbf{u}_1, \mathbf{u}_2)$ and $W = \text{span}(\mathbf{w}_1, \mathbf{w}_2)$ in $\mathbb{R}^4$, where

$$\mathbf{u}_1 = (1, 0, 1, 2), \quad \mathbf{u}_2 = (0, 1, 1, 1), \quad \mathbf{w}_1 = (1, 1, 0, 1), \quad \mathbf{w}_2 = (1, 3, 0, 1).$$

Find a basis and the dimension of $U + W$.

Solution. Each pair is independent, since neither vector is a multiple of the other (compare the first two entries), so $\dim U = \dim W = 2$. Row reduce $M = (\mathbf{u}_1 \quad \mathbf{u}_2 \quad \mathbf{w}_1 \quad \mathbf{w}_2)$:

$$\begin{aligned} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 3 \\ 1 & 1 & 0 & 0 \\ 2 & 1 & 1 & 1 \end{pmatrix} &\xrightarrow[R_4 \rightarrow R_4 - 2R_1]{R_3 \rightarrow R_3 - R_1} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 1 & -1 & -1 \end{pmatrix} \xrightarrow[R_4 \rightarrow R_4 - R_2]{R_3 \rightarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & -2 & -4 \\ 0 & 0 & -2 & -4 \end{pmatrix} \\ &\xrightarrow[\text{then } R_3 \rightarrow -\frac{1}{2}R_3]{R_4 \rightarrow R_4 - R_3} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[R_2 \rightarrow R_2 - R_3]{R_1 \rightarrow R_1 - R_3} R = \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \end{aligned}$$

The pivot columns are 1, 2, 3. By [Proposition 2.5.11](#) (a), $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{w}_1)$ is a basis of $U + W$, and $\dim(U + W) = 3$. Column 4 of R says $\mathbf{w}_2 = -\mathbf{u}_1 + \mathbf{u}_2 + 2\mathbf{w}_1$; check: $-(1, 0, 1, 2) + (0, 1, 1, 1) + 2(1, 1, 0, 1) = (1, 3, 0, 1)$.

Example. (A basis of an intersection)

For U and W as in [Example \(A basis of a sum\)](#), find a basis of $U \cap W$, and check the Dimension Formula for Sums.

Solution. We reuse the RREF R of M . Its only free column is column 4, so by [Theorem 2.5.4](#) $\text{nul}(M)$ has the basis $\mathbf{n} = (1, -1, -2, 1)$ (set $x_4 = 1$; then $x_1 = 1$, $x_2 = -1$,

$x_3 = -2$). The relation $M\mathbf{n} = \mathbf{0}$ reads $\mathbf{u}_1 - \mathbf{u}_2 - 2\mathbf{w}_1 + \mathbf{w}_2 = \mathbf{0}$, that is,

$$\mathbf{u}_1 - \mathbf{u}_2 = 2\mathbf{w}_1 - \mathbf{w}_2.$$

By [Proposition 2.5.11](#) (b), $\mathbf{u}_\mathbf{n} = \mathbf{u}_1 - \mathbf{u}_2 = (1, -1, 0, 1)$ is a basis of $U \cap W$. Check that it lies in W : $2(1, 1, 0, 1) - (1, 3, 0, 1) = (1, -1, 0, 1)$. So $\dim(U \cap W) = 1$, and

$$\dim U + \dim W - \dim(U \cap W) = 2 + 2 - 1 = 3 = \dim(U + W),$$

in agreement with [Theorem 1.7.3](#).

2.5.9 Exercises

2.5.9.1 A. Check your understanding

Exercise A1.

- (a) Define $\text{col}(A)$, $\text{row}(A)$, $\text{nul}(A)$ and $\text{rank } A$ for $A \in M_{m \times n}(F)$, and say which space each subspace lives in.
- (b) State the Rank–Nullity Theorem for matrices and the Rouché–Capelli Theorem.
- (c) True or false: if R is the RREF of A , then $\text{col}(R) = \text{col}(A)$. Justify your answer.
- (d) True or false: there is a 4×6 real matrix with nullity 1. Justify your answer.
- (e) True or false: for every $A \in M_{m \times n}(F)$, $\text{rank } A = \text{rank } A^T$. Justify your answer.
- (f) Describe how to find a basis of $U \cap W$ for subspaces $U, W \subseteq F^m$ given by bases.

Solution.

- (a) $\text{col}(A)$ is the span of the columns, a subspace of F^m ; $\text{row}(A)$ is the span of the rows, a subspace of $M_{1 \times n}(F)$; $\text{nul}(A) = \{\mathbf{x} \in F^n : A\mathbf{x} = \mathbf{0}\}$, a subspace of F^n ; and $\text{rank } A = \dim \text{col}(A)$ ([Definition 2.5.1](#), [Definition 2.5.2](#), [Definition 2.5.3](#), [Definition 2.5.4](#)).
- (b) For $A \in M_{m \times n}(F)$, $\text{rank } A + \text{nullity } A = n$ ([Theorem 2.5.7](#)). For $\mathbf{b} \in F^m$, $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\text{rank } A = \text{rank } [A \mid \mathbf{b}]$; then the solutions are $\mathbf{p} + \text{nul}(A)$, an $(n - \text{rank } A)$ -parameter family, with exactly $q^{n - \text{rank } A}$ solutions over a field with q elements ([Theorem 2.5.8](#)).
- (c) False. For $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $R = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$, and $(1, 1) \in \text{col}(A)$ but $(1, 1) \notin \text{col}(R) = \text{span}((1, 0))$. Row operations preserve the row space and the column relations, not the column space ([Theorem 2.5.1](#)).
- (d) False. Its rank is at most 4, since the column space lies in $\mathbb{R}^4$, so by [Theorem 2.5.7](#) its nullity is $6 - \text{rank } A \geq 2$.
- (e) True. This is [Theorem 2.5.5](#).
- (f) Put the basis vectors of U and then of W as the columns of M , find a basis of $\text{nul}(M)$ from the RREF, and for each basis vector $\mathbf{n}$ take the combination of the $\mathbf{u}$'s with the first k entries of $\mathbf{n}$ as coefficients ([Proposition 2.5.11](#)).

2.5.9.2 B. Practice

Exercise B1: The three subspaces of a 3×5 matrix.

Let

$$A = \begin{pmatrix} 1 & -1 & 2 & 0 & 3 \\ 2 & -2 & 5 & 1 & 4 \\ -1 & 1 & -1 & 1 & -5 \end{pmatrix} \in M_{3 \times 5}(\mathbb{R}).$$

Find bases of $\text{col}(A)$, $\text{row}(A)$ and $\text{nul}(A)$. Hence verify $\text{rank } A + \text{nullity } A = 5$.

Solution. Row reduce:

$$A \xrightarrow[\substack{R_3 \rightarrow R_3 + R_1}]{\substack{R_2 \rightarrow R_2 - 2R_1}} \begin{pmatrix} 1 & -1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 0 & 1 & 1 & -2 \end{pmatrix} \xrightarrow[\substack{R_1 \rightarrow R_1 - 2R_2}]{R_3 \rightarrow R_3 - R_2} R = \begin{pmatrix} 1 & -1 & 0 & -2 & 7 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The pivot columns are 1 and 3; the free columns are 2, 4, 5.

By [Theorem 2.5.2](#), the pivot columns of A , $(1, 2, -1)$ and $(2, 5, -1)$, form a basis of $\text{col}(A)$. By [Theorem 2.5.3](#), $(1 \ -1 \ 0 \ -2 \ 7)$ and $(0 \ 0 \ 1 \ 1 \ -2)$ form a basis of $\text{row}(A)$. The reduced equations are $x_1 = x_2 + 2x_4 - 7x_5$ and $x_3 = -x_4 + 2x_5$, so by [Theorem 2.5.4](#) a basis of $\text{nul}(A)$ is

$$\mathbf{s}_2 = (1, 1, 0, 0, 0), \quad \mathbf{s}_4 = (2, 0, -1, 1, 0), \quad \mathbf{s}_5 = (-7, 0, 2, 0, 1).$$

For example $A\mathbf{s}_4 = 2\mathbf{a}_1 - \mathbf{a}_3 + \mathbf{a}_4 = (2, 4, -2) - (2, 5, -1) + (0, 1, 1) = \mathbf{0}$. Hence $\text{rank } A = 2$ and $\text{nullity } A = 3$, and $2 + 3 = 5$, as [Theorem 2.5.7](#) requires.

Exercise B2: Independence and extension in $\mathbb{R}^4$.

Let $\mathbf{v}_1 = (1, 0, 2, 1)$, $\mathbf{v}_2 = (2, 1, 3, 1)$, $\mathbf{v}_3 = (0, 1, -1, -1)$ in $\mathbb{R}^4$.

- Determine whether $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ is linearly independent. Justify your answer.
- Find a basis of $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$, and extend it to a basis of $\mathbb{R}^4$ using standard basis vectors.

Solution.

- Row reduce the matrix with columns $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$:

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 3 & -1 \\ 1 & 1 & -1 \end{pmatrix} \xrightarrow[\substack{R_4 \rightarrow R_4 - R_1}]{R_3 \rightarrow R_3 - 2R_1} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \\ 0 & -1 & -1 \end{pmatrix} \xrightarrow[\substack{R_4 \rightarrow R_4 + R_2, \ R_1 \rightarrow R_1 - 2R_2}]{R_3 \rightarrow R_3 + R_2} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The rank is $2 < 3$, so the list is dependent by [Theorem 2.5.10](#) (a). Column 3 gives $\mathbf{v}_3 = -2\mathbf{v}_1 + \mathbf{v}_2$ ([Theorem 2.5.2](#) (b)); check: $(-2, 0, -4, -2) + (2, 1, 3, 1) = (0, 1, -1, -1)$.

- By [Theorem 2.5.2](#) (a), $(\mathbf{v}_1, \mathbf{v}_2)$ is a basis of $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$. To extend it, row

reduce $(\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3 \ \mathbf{e}_4)$:

$$\begin{aligned} & \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow[R_4 \rightarrow R_4 - R_1]{R_3 \rightarrow R_3 - 2R_1} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & -1 & -2 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 & 0 & 1 \end{pmatrix} \xrightarrow[R_4 \rightarrow R_4 + R_2]{R_3 \rightarrow R_3 + R_2} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 1 & 1 & 0 \\ 0 & 0 & -1 & 1 & 0 & 1 \end{pmatrix} \\ & \xrightarrow[\text{then } R_3 \rightarrow -R_3]{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & -2 & 1 & 1 & 0 \end{pmatrix} \xrightarrow{R_4 \rightarrow R_4 + 2R_3} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 0 & -1 & 1 & -2 \end{pmatrix}. \end{aligned}$$

This echelon form has pivots in columns 1, 2, 3, 4, which are the pivot columns by [Corollary 2.5.6](#). By [Theorem 2.5.2](#) (a) the corresponding columns form a basis of the column space, which is $\mathbb{R}^4$ since it contains $\mathbf{e}_1, \dots, \mathbf{e}_4$. Hence $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{e}_1, \mathbf{e}_2)$ is a basis of $\mathbb{R}^4$ extending $(\mathbf{v}_1, \mathbf{v}_2)$.

Exercise B3: Sum and intersection in $\mathbb{R}^5$.

Let $U = \text{span}(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3)$ and $W = \text{span}(\mathbf{w}_1, \mathbf{w}_2)$ in $\mathbb{R}^5$, where

$$\mathbf{u}_1 = (1, 1, 0, 0, 1), \quad \mathbf{u}_2 = (0, 1, 1, 0, 0), \quad \mathbf{u}_3 = (1, 0, 0, 1, 0), \quad \mathbf{w}_1 = (1, 2, 2, 1, 2), \quad \mathbf{w}_2 = (0, 0, 1, 1, 1).$$

Find $\dim U$, $\dim W$, a basis of $U + W$ and a basis of $U \cap W$. Hence verify [Theorem 1.7.3](#).

Solution. Row reduce $M = (\mathbf{u}_1 \ \mathbf{u}_2 \ \mathbf{u}_3 \ \mathbf{w}_1 \ \mathbf{w}_2)$:

$$\begin{aligned} & \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 2 & 1 \end{pmatrix} \xrightarrow[R_5 \rightarrow R_5 - R_1]{R_2 \rightarrow R_2 - R_1} \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & -1 & 1 & 1 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & -1 & 1 & 1 \end{pmatrix} \\ & \xrightarrow[R_5 \rightarrow R_5 + R_3]{R_4 \rightarrow R_4 - R_3} \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 2 \end{pmatrix} \xrightarrow[\text{then } R_4 \rightarrow \frac{1}{2}R_4]{R_4 \leftrightarrow R_5} \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \end{aligned}$$

Clearing above the pivots ($R_3 \rightarrow R_3 - R_4$, $R_2 \rightarrow R_2 - R_4$, $R_1 \rightarrow R_1 - R_4$, then $R_2 \rightarrow R_2 + R_3$, $R_1 \rightarrow R_1 - R_3$) gives

$$R = \begin{pmatrix} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The pivot columns are 1, 2, 3, 4, so $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{w}_1)$ is independent by [Theorem 2.5.2](#) (a), and so are its sub-lists $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3)$ and $(\mathbf{w}_1)$; also $\mathbf{w}_2$ is not a multiple of $\mathbf{w}_1$ (compare first entries). Hence $\dim U = 3$ and $\dim W = 2$, with the given spanning lists as bases. By [Proposition 2.5.11](#) (a), $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{w}_1)$ is a basis of $U + W$, so $\dim(U + W) = 4$.

The only free column is column 5. Setting $x_5 = 1$ gives $\mathbf{n} = (1, 1, 0, -1, 1)$, a basis of $\text{nul}(M)$ by [Theorem 2.5.4](#). So $\mathbf{u}_1 + \mathbf{u}_2 - \mathbf{w}_1 + \mathbf{w}_2 = \mathbf{0}$, and by [Proposition 2.5.11](#) (b), $\mathbf{u}_1 + \mathbf{u}_2 = (1, 2, 1, 0, 1)$ is a basis of $U \cap W$. Check: $\mathbf{w}_1 - \mathbf{w}_2 = (1, 2, 1, 0, 1)$. Hence $\dim(U \cap W) = 1$, and

$$\dim U + \dim W - \dim(U \cap W) = 3 + 2 - 1 = 4 = \dim(U + W).$$

2.5.9.3 C. Going deeper

Exercise C1: Rank of a sum.

Let $A, B \in M_{m \times n}(F)$. Prove that $\text{rank}(A + B) \leq \text{rank } A + \text{rank } B$. Give an example where the inequality is strict.

Solution. The j -th column of $A + B$ is $\mathbf{a}_j + \mathbf{b}_j$, which lies in $\text{col}(A) + \text{col}(B)$ ([Definition 1.7.1](#)). This sum is a subspace ([Theorem 1.7.1](#)), so by [Theorem 1.3.1](#), $\text{col}(A + B) \subseteq \text{col}(A) + \text{col}(B)$. By [Theorem 1.6.7](#) and [Theorem 1.7.3](#),

$$\text{rank}(A+B) \leq \dim(\text{col}(A)+\text{col}(B)) = \text{rank } A + \text{rank } B - \dim(\text{col}(A) \cap \text{col}(B)) \leq \text{rank } A + \text{rank } B.$$

For strictness, take $A = I_2$ and $B = -I_2$ over $\mathbb{R}$: $\text{rank}(A + B) = \text{rank } 0 = 0 < 2 + 2$.

Exercise C2: Matrices of rank one.

Let $A \in M_{m \times n}(F)$. Prove that $\text{rank } A = 1$ if and only if $A = \mathbf{xy}^T$ for some **non-zero** $\mathbf{x} \in F^m$ and $\mathbf{y} \in F^n$.

Solution. ($\Leftarrow$) Suppose $A = \mathbf{xy}^T$ with $\mathbf{x}, \mathbf{y} \neq \mathbf{0}$, and write $\mathbf{y} = (y_1, \dots, y_n)$. By [Theorem 0.9.2](#), the j -th column of $\mathbf{xy}^T$ is $\mathbf{x}$ times the j -th column of $\mathbf{y}^T$, the 1×1 matrix (y_j) ; that is, $y_j \mathbf{x}$. So every column lies in $\text{span}(\mathbf{x})$, and $\text{col}(A) \subseteq \text{span}(\mathbf{x})$. Since $\mathbf{y} \neq \mathbf{0}$, some $y_j \neq 0$, and then $y_j \mathbf{x} \neq \mathbf{0}$ because $\mathbf{x} \neq \mathbf{0}$ ([Theorem 1.1.9](#)). The list $(\mathbf{x})$ is independent, since $a\mathbf{x} = \mathbf{0}$ forces $a = 0$ ([Theorem 1.1.9](#)), so $\text{span}(\mathbf{x})$ has dimension 1, and $\text{rank } A \leq 1$ by [Theorem 1.6.7](#). On the other hand $\text{rank } A \neq 0$: a space of dimension 0 has the empty list as a basis, so it equals $\text{span}() = \{\mathbf{0}\}$, but $\text{col}(A)$ contains the non-zero vector $y_j \mathbf{x}$. Hence $\text{rank } A = 1$.

($\Rightarrow$) Suppose $\text{rank } A = 1$, and let $(\mathbf{x})$ be a basis of $\text{col}(A)$; then $\mathbf{x} \neq \mathbf{0}$. Each column $\mathbf{a}_j \in \text{col}(A)$ equals $y_j \mathbf{x}$ for some $y_j \in F$. Put $\mathbf{y} = (y_1, \dots, y_n)$. By the computation in ($\Leftarrow$), the j -th column of $\mathbf{xy}^T$ is $y_j \mathbf{x} = \mathbf{a}_j$, so $A = \mathbf{xy}^T$. If $\mathbf{y} = \mathbf{0}$, then $A = 0$ would have rank 0, so $\mathbf{y} \neq \mathbf{0}$. This proves the claim.

Exercise C3: The matrix $i + j$.

For $n \geq 1$, let $A_n \in M_n(\mathbb{R})$ have entries $a_{ij} = i + j$. Prove that $\text{rank } A_1 = 1$ and $\text{rank } A_n = 2$ for every $n \geq 2$. Hence find nullity A_n .

Hint: write each row as a combination of two fixed row vectors.

Solution. For $n = 1$, $A_1 = (2) \neq 0$, so $\text{rank } A_1 = 1$. Let $n \geq 2$, and put $\mathbf{1} = (1 \ 1 \ \cdots \ 1)$ and $\mathbf{d} = (1 \ 2 \ \cdots \ n)$ in $M_{1 \times n}(\mathbb{R})$. Row i of A_n is $(i + 1 \ \cdots \ i + n) =$

$i\mathbf{1} + \mathbf{d}$. So $\text{row}(A_n) \subseteq \text{span}(\mathbf{1}, \mathbf{d})$, and $\dim \text{row}(A_n) \leq 2$ by [Theorem 1.6.3](#).

Rows 1 and 2 are $(2 \ 3 \ \cdots)$ and $(3 \ 4 \ \cdots)$. If $c_1(\text{row } 1) + c_2(\text{row } 2) = 0$, the first two entries give $2c_1 + 3c_2 = 0$ and $3c_1 + 4c_2 = 0$. Subtracting the first from the second gives $c_1 + c_2 = 0$, so $c_1 = -c_2$, and then $-2c_2 + 3c_2 = c_2 = 0$, so $c_1 = 0$. Hence $\text{row}(A_n)$ contains two independent vectors, and $\dim \text{row}(A_n) \geq 2$ by [Theorem 1.6.3](#).

So $\dim \text{row}(A_n) = 2$, and $\text{rank } A_n = 2$ by [Theorem 2.5.5](#). By [Theorem 2.5.7](#), nullity $A_n = n - 2$ for $n \geq 2$, and nullity $A_1 = 0$.

Exercise C4: Counting right-hand sides over a finite field.

Let F be a finite field with q elements, and let $A \in M_{m \times n}(F)$ have rank r .

- Prove that exactly q^r vectors $\mathbf{b} \in F^m$ make $A\mathbf{x} = \mathbf{b}$ consistent.
- By counting the pairs $(\mathbf{x}, \mathbf{b}) \in F^n \times F^m$ with $A\mathbf{x} = \mathbf{b}$ in two ways, and using [Theorem 2.1.3](#) but **not** [Theorem 2.5.7](#), prove that $\text{nul}(A)$ has exactly q^{n-r} elements. Deduce nullity $A = n - r$: a counting proof of Rank–Nullity over a finite field.
- Over $\mathbb{F}_2$, let $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$. How many $\mathbf{b} \in \mathbb{F}_2^3$ give a consistent system, and how many solutions does each such system have?

Solution.

- By [Theorem 2.1.1](#), $A\mathbf{x} = \mathbf{b}$ is consistent exactly when $\mathbf{b} \in \text{col}(A)$. Let $(\mathbf{c}_1, \dots, \mathbf{c}_r)$ be a basis of $\text{col}(A)$. By [Theorem 1.5.1](#), $(t_1, \dots, t_r) \mapsto t_1\mathbf{c}_1 + \cdots + t_r\mathbf{c}_r$ is a bijection $F^r \rightarrow \text{col}(A)$: every vector of $\text{col}(A)$ is hit by exactly one tuple. Since F^r has q^r elements, so does $\text{col}(A)$.
- Let $N = |\text{nul}(A)|$. Each $\mathbf{x} \in F^n$ determines exactly one $\mathbf{b} = A\mathbf{x}$, so there are q^n pairs. Now group the pairs by $\mathbf{b}$. If $\mathbf{b} \notin \text{col}(A)$, there are none, by [Theorem 2.1.1](#). If $\mathbf{b} \in \text{col}(A)$, pick one solution $\mathbf{p}$; by [Theorem 2.1.3](#) the solutions are $\mathbf{p} + \text{nul}(A)$, and $\mathbf{h} \mapsto \mathbf{p} + \mathbf{h}$ is a bijection from $\text{nul}(A)$ onto this set (it is onto by definition, and $\mathbf{p} + \mathbf{h} = \mathbf{p} + \mathbf{h}'$ gives $\mathbf{h} = \mathbf{h}'$). So each of the q^r vectors $\mathbf{b}$ of (a) accounts for exactly N pairs, and there are $q^r N$ pairs in total. Hence $q^r N = q^n$, and $N = q^{n-r}$.
Let $d = \text{nullity } A$. The argument of (a), applied to a basis of $\text{nul}(A)$ of length d , gives $N = q^d$. So $q^d = q^{n-r}$. Since $q \geq 2$, distinct exponents give distinct powers of q , so $d = n - r$. This is [Theorem 2.5.7](#), proved here by counting alone.
- Over $\mathbb{F}_2$, $R_3 \rightarrow R_3 + R_1$ gives row 3 equal to $(0, 1, 1)$, and then $R_3 \rightarrow R_3 + R_2$ makes it zero. So an echelon form is $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, and $\text{rank } A = 2$ by [Corollary 2.5.6](#). By (a), exactly $2^2 = 4$ of the 8 vectors $\mathbf{b}$ give a consistent system, and by [Theorem 2.5.8](#) (c) each has $2^{3-2} = 2$ solutions. (The consistent $\mathbf{b}$ are those with $b_1 + b_2 + b_3 = 0$. The rows of A add up to zero, so every consistent $\mathbf{b}$ satisfies this equation; exactly 4 vectors of $\mathbb{F}_2^3$ satisfy it, and 4 are consistent, so the two sets coincide.)

2.6 LU and PLU Factorizations

Gaussian elimination solves one system $A\mathbf{x} = \mathbf{b}$. In practice the matrix often stays fixed while the right-hand side changes: the same network under different loads, the same model fed different data. The row operations depend only on A , so repeating them for every $\mathbf{b}$ wastes almost all of the work. This section stores the elimination as a factorization $A = LU$, a lower triangular matrix that remembers the operations times an upper triangular matrix that is their result. We prove when it exists and when it is unique, and we repair its failures with row swaps.

Throughout, F is a field and all matrices are square, in $M_n(F)$.

2.6.1 Triangular systems are cheap

Some systems need no elimination at all. Take

$$U = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & -2 \end{pmatrix}, \quad U\mathbf{x} = \begin{pmatrix} -1 \\ -1 \\ -4 \end{pmatrix}.$$

The last equation mentions only x_3 : $-2x_3 = -4$, so $x_3 = 2$. The second mentions x_2 and the known x_3 : $3x_2 + 2 = -1$, so $x_2 = -1$. The first then gives $2x_1 - 1 - 2 = -1$, so $x_1 = 1$. This is **back substitution**: in an upper triangular system with non-zero diagonal entries, each equation introduces exactly one new unknown, read from the bottom up. A lower triangular system is solved the same way from the top down, by **forward substitution**.

Now suppose a matrix A factors as $A = LU$ with L lower triangular and U upper triangular. Then $A\mathbf{x} = \mathbf{b}$ reads $L(U\mathbf{x}) = \mathbf{b}$. Naming the inner vector $\mathbf{y} = U\mathbf{x}$ splits the problem in two:

first solve $L\mathbf{y} = \mathbf{b}$ for $\mathbf{y}$, then solve $U\mathbf{x} = \mathbf{y}$ for $\mathbf{x}$.

Both are triangular, so both are cheap. The expensive part, finding L and U , is done once for all right-hand sides.

An LU factorization is Gaussian elimination written down once, so that it never has to be done again.

2.6.2 LU factorizations

Definition 2.6.1 — LU factorization

A matrix $L \in M_n(F)$ is **unit lower triangular** if it is lower triangular and **every** diagonal entry equals 1. Let $A \in M_n(F)$. An **LU factorization** of A is a pair (L, U) of matrices in $M_n(F)$ such that

$$A = LU, \quad L \text{ is **unit** lower triangular,} \quad U \text{ is upper triangular.}$$

In words: L has 1's on the diagonal and zeros above it, and the entries below the diagonal are unrestricted. U has zeros strictly below the diagonal, and its diagonal entries are unrestricted; some of them may be 0. The definition says **an** factorization: whether there is only one is a theorem, and it needs a hypothesis.

Example. (An LU factorization from elimination)

Let

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 4 & 5 & -1 \\ -2 & 8 & 2 \end{pmatrix} \in M_3(\mathbb{Q}).$$

Reduce A to upper triangular form using only operations “subtract a multiple of a higher row from a lower row”, and record each multiplier. Compare the result with the matrix U of the previous subsection.

Solution. The first pivot is 2. To clear column 1 below it we subtract $\ell_{21} = 4/2 = 2$ times row 1 from row 2, and $\ell_{31} = -2/2 = -1$ times row 1 from row 3:

$$\begin{pmatrix} 2 & 1 & -1 \\ 4 & 5 & -1 \\ -2 & 8 & 2 \end{pmatrix} \xrightarrow[R_3 \rightarrow R_3 - (-1)R_1]{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 2 & 1 & -1 \\ 0 & 3 & 1 \\ 0 & 9 & 1 \end{pmatrix}.$$

The second pivot is 3. We subtract $\ell_{32} = 9/3 = 3$ times row 2 from row 3:

$$\begin{pmatrix} 2 & 1 & -1 \\ 0 & 3 & 1 \\ 0 & 9 & 1 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - 3R_2} \begin{pmatrix} 2 & 1 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & -2 \end{pmatrix} = U.$$

Put the multipliers below the diagonal of a unit lower triangular matrix, each in the position of the entry it cleared:

$$L = \begin{pmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{pmatrix}.$$

Multiplying out, row by row,

$$LU = \begin{pmatrix} 2 & 1 & -1 \\ 2 \cdot 2 & 2 \cdot 1 + 3 & 2 \cdot (-1) + 1 \\ -2 & -1 + 9 & 1 + 3 - 2 \end{pmatrix} = \begin{pmatrix} 2 & 1 & -1 \\ 4 & 5 & -1 \\ -2 & 8 & 2 \end{pmatrix} = A.$$

So (L, U) is an LU factorization of A , and U is exactly the triangular matrix of the back substitution above.

The example suggests the general statement: elimination produces U , and the multipliers, **with their signs as subtracted**, fill in L . We prove this below. First, the smallest cases.

- **Upper triangular A .** $A = I \cdot A$, and I is unit lower triangular. So every upper triangular matrix, including 0 and I , has the LU factorization (I, A) .
- 1×1 . Every $(a) \in M_1(F)$ is upper triangular, so $(a) = (1)(a)$. Nothing can go wrong here, and that is exactly why the 2×2 failure below is instructive.

Non-example by minimal change. The matrix $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ has the LU factorization

$$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}.$$

Change the top-left entry from 1 to 0, and everything collapses.

Example. (An invertible matrix with no LU factorization)

Show that $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \in M_2(F)$ has **no** LU factorization, over any field F , although A is invertible.

Solution. A is invertible, since $AA = I_2$. Suppose, for a contradiction, that $A = LU$ with

$$L = \begin{pmatrix} 1 & 0 \\ \ell & 1 \end{pmatrix}, \quad U = \begin{pmatrix} u & v \\ 0 & w \end{pmatrix}.$$

Then

$$LU = \begin{pmatrix} u & v \\ \ell u & \ell v + w \end{pmatrix}.$$

Comparing $(1, 1)$ -entries with A gives $u = 0$. Comparing $(2, 1)$ -entries gives $\ell u = 1$. But $\ell u = \ell \cdot 0 = 0$ by [Theorem 0.7.1](#), and $0 \neq 1$ in a field. This contradiction shows that no LU factorization exists.

What failed is visible: the $(1, 1)$ -entry of any product LU is u_{11} alone, so a zero there forces the whole first column of LU to vanish. In elimination terms, the first pivot is 0 while the entry below it is not, and the only cure is to **swap** rows, which “subtract a multiple of a higher row from a lower row” cannot do. We return to this with permutation matrices.

Why “unit”. Without the 1’s on the diagonal of L , uniqueness would fail even for invertible matrices. If $A = LU$ and D is any invertible diagonal matrix, then also $A = (LD)(D^{-1}U)$, where LD is still lower triangular and $D^{-1}U$ still upper triangular. When L is invertible and $D \neq I$, this is a different pair, and over any field with more than two elements such a D exists. For instance $(2) = (1) \ (2) = (2) \ (1) = (4) \ (1/2)$ over $\mathbb{Q}$. Fixing the diagonal of L removes this freedom and matches what elimination does: it never rescales a row.

Warning. **Invertible does not mean “has an LU factorization”.** The matrix of [Example \(An invertible matrix with no LU factorization\)](#) is invertible and has none. Conversely, a singular matrix can have one: $0 = I \cdot 0$. Existence of an LU factorization is a property of the **order of the rows**, not of invertibility.

2.6.3 Unit lower triangular matrices

Elimination builds L out of elementary matrices, and the uniqueness proof below divides one factorization by another. Both need the same facts: products and inverses of unit lower triangular matrices stay unit lower triangular. We also need to know exactly when a triangular matrix is invertible.

The key observation is a statement about where the non-zero entries of a product can sit.

Lemma 2.6.1 — Unit lower triangular matrices are closed

Let $L, L' \in M_n(F)$ be unit lower triangular. Then:

- (a) LL' is unit lower triangular;
- (b) L is invertible, and L^{-1} is unit lower triangular.

Idea. Write $L = I + N$, where N has non-zero entries only strictly below the diagonal. Multiplying two such matrices pushes the non-zero entries **further** below the diagonal, so $N^n = 0$. Then the geometric series $1/(1+t) = 1 - t + t^2 - \cdots$ stops after n terms and gives the inverse exactly.

Proof. Say that $X \in M_n(F)$ is **p -low** if $x_{ij} = 0$ whenever $i < j + p$. So 0-low means lower triangular, and 1-low means zero on and above the diagonal.

Claim. If X is p -low and Y is q -low, then XY is $(p+q)$ -low. A matrix that is n -low is the zero matrix.

Proof. By Definition 0.9.9, $(XY)_{ij} = \sum_{l=1}^n x_{il}y_{lj}$. A term can be non-zero only if $i \geq l + p$ and $l \geq j + q$, which together give $i \geq j + p + q$. So if $i < j + p + q$, every term is 0 and $(XY)_{ij} = 0$. For the second statement, every pair of indices satisfies $i \leq n < 1 + n \leq j + n$, so every entry of an n -low matrix is 0. ■

Write $L = I + N$ and $L' = I + N'$. Since L and L' are unit lower triangular, N and N' are 1-low.

For (a), $LL' = I + N + N' + NN'$ by Theorem 0.9.4. Here $N + N'$ is 1-low, since a sum of entries that are both 0 is 0, and NN' is 2-low, hence 1-low, by the Claim. So $LL' - I$ is 1-low, which says that LL' is unit lower triangular.

For (b), let $K = I - N + N^2 - \cdots + (-1)^{n-1}N^{n-1}$. Multiplying out, the sum telescopes:

$$(I + N)K = I + (-1)^{n-1}N^n = I,$$

where $N^n = 0$ because N^n is n -low by the Claim applied $n - 1$ times. The same computation gives $K(I + N) = I$, since N commutes with its own powers. Hence L is invertible with $L^{-1} = K$. Finally $K - I = \sum_{p=1}^{n-1} (-1)^p N^p$ is a sum of 1-low matrices, by the Claim, so K is unit lower triangular. This proves the lemma. ■

Transposing everything, the same statements hold for **unit upper triangular** matrices (upper triangular with 1's on the diagonal).

The next lemma decides invertibility of a triangular matrix from its diagonal, without any computation beyond back substitution.

Lemma 2.6.2 — Invertibility of a triangular matrix

An upper triangular matrix $U \in M_n(F)$ is invertible if and only if **every** diagonal entry $u_{11}, \dots, u_{nn}$ is non-zero.

Proof. ($\Leftarrow$) Suppose every $u_{kk} \neq 0$, and let $U\mathbf{x} = \mathbf{0}$. We show $x_k = 0$ for $k = n, n-1, \dots, 1$ in turn. Suppose $x_{k+1} = \cdots = x_n = 0$ (nothing is assumed when $k = n$). Row k of $U\mathbf{x} = \mathbf{0}$ reads $u_{kk}x_k + \sum_{j>k} u_{kj}x_j = 0$, because $u_{kj} = 0$ for $j < k$. The sum vanishes, so $u_{kk}x_k = 0$, and $x_k = 0$ since $u_{kk} \neq 0$. Hence $\mathbf{x} = \mathbf{0}$ is the only solution of $U\mathbf{x} = \mathbf{0}$, and U is invertible by Theorem 2.4.4.

($\Rightarrow$) We prove the contrapositive. Suppose $u_{kk} = 0$ for some k . For each $j \leq k$, the entries of column j in rows $i \geq k$ are zero: if $i > j$ this is upper triangularity, and the only remaining case $i = j = k$ is $u_{kk} = 0$. So the first k columns of U lie in $W = \text{span}(\mathbf{e}_1, \dots, \mathbf{e}_{k-1})$. The list $(\mathbf{e}_1, \dots, \mathbf{e}_{k-1})$ is independent and spans W , so $\dim W = k - 1$ (for $k = 1$, $W = \{\mathbf{0}\}$ has dimension 0). By [Theorem 1.6.3](#) (a), these k columns are linearly dependent, so the columns of U are dependent, and U is not invertible by [Theorem 2.4.4](#). This proves the lemma. ■

2.6.4 Elimination without row swaps

To prove that elimination always produces an LU factorization, we first need to say exactly which elimination we mean. It is Gaussian elimination restricted to one kind of operation: subtracting a multiple of a row from a row **below** it.

Elimination without row swaps. Input: $A \in M_n(F)$. Put $A^{(0)} = A$. For $k = 1, 2, \dots, n-1$, write $A^{(k-1)} = (a_{ij})$ and do the following.

- If $a_{kk} \neq 0$, set $\ell_{ik} = a_{ik}/a_{kk}$ for $i = k+1, \dots, n$.
- If $a_{kk} = 0$ and $a_{ik} = 0$ for **every** $i > k$, set $\ell_{ik} = 0$ for $i = k+1, \dots, n$.
- If $a_{kk} = 0$ but $a_{ik} \neq 0$ for **some** $i > k$, **stop**: a row swap is needed.

If we did not stop, obtain $A^{(k)}$ from $A^{(k-1)}$ by replacing row i with $(\text{row } i) - \ell_{ik} \cdot (\text{row } k)$, for each $i = k+1, \dots, n$.

If the loop finishes without stopping, we say **elimination without row swaps succeeds** on A . The numbers ℓ_{ik} ($i > k$) are its **multipliers**, and $U = A^{(n-1)}$ is its **result**.

In both non-stopping cases the new (i, k) -entry is $a_{ik} - \ell_{ik}a_{kk} = 0$: in the first case by the choice of ℓ_{ik} , in the second because both terms are already 0. The multipliers are well defined: row k is not changed during step k , so it does not matter in which order the rows $i > k$ are updated.

The proof will write each step as a matrix. By [Definition 2.4.1](#), the elementary matrix of the operation $R_i \rightarrow R_i + cR_k$ ($i \neq k$) is $I + cE_{ik}$, where E_{ik} is the matrix unit with 1 in entry (i, k) . We will write the matrix unit as a product of a column and a row,

$$E_{ik} = \mathbf{e}_i \mathbf{e}_k^T,$$

which holds because $\mathbf{e}_i \mathbf{e}_k^T$ has (p, q) -entry $(\mathbf{e}_i)_p (\mathbf{e}_k)_q$, equal to 1 if $p = i$ and $q = k$, and 0 otherwise. We will use one more rule constantly: for a vector $\mathbf{v} \in F^n$, the 1×1 matrix $\mathbf{e}_k^T \mathbf{v}$ is the k -th entry v_k , which we treat as a scalar. In particular $\mathbf{e}_k^T \mathbf{e}_j = 0$ for $j \neq k$.

For $1 \leq k \leq n$ and a vector $\mathbf{v} \in F^n$ whose first k entries are zero, put

$$G_k(\mathbf{v}) = I + \mathbf{v} \mathbf{e}_k^T.$$

Its only non-identity entries are v_i in position (i, k) , for $i > k$: it is the identity with the column vector $\mathbf{v}$ written into column k below the diagonal. So $G_k(\mathbf{v})$ is unit lower triangular.

Lemma 2.6.3 — Multiplier matrices

Let $1 \leq k \leq n$, and let $\mathbf{v}, \mathbf{v}_1, \dots, \mathbf{v}_m \in F^n$ be vectors such that the first k entries of $\mathbf{v}$ are zero and, for each j , the first j entries of $\mathbf{v}_j$ are zero ($m \leq n$). Then:

- (a) $G_k(\mathbf{v})$ is the product, in any order, of the elementary matrices $I + v_i E_{ik}$ for $i = k+1, \dots, n$;
- (b) $G_k(\mathbf{v})G_k(-\mathbf{v}) = G_k(-\mathbf{v})G_k(\mathbf{v}) = I$;
- (c) $G_1(\mathbf{v}_1)G_2(\mathbf{v}_2) \cdots G_m(\mathbf{v}_m) = I + \mathbf{v}_1 \mathbf{e}_1^\top + \mathbf{v}_2 \mathbf{e}_2^\top + \cdots + \mathbf{v}_m \mathbf{e}_m^\top$.

Proof. For (a), let $i, j > k$. By [Theorem 0.9.4](#),

$$(I + a \mathbf{e}_i \mathbf{e}_k^\top)(I + b \mathbf{e}_j \mathbf{e}_k^\top) = I + a \mathbf{e}_i \mathbf{e}_k^\top + b \mathbf{e}_j \mathbf{e}_k^\top + ab \mathbf{e}_i (\mathbf{e}_k^\top \mathbf{e}_j) \mathbf{e}_k^\top,$$

and the last term is zero because $\mathbf{e}_k^\top \mathbf{e}_j = 0$ for $j \neq k$. Repeating this, one factor at a time, the product of the $I + v_i E_{ik} = I + v_i \mathbf{e}_i \mathbf{e}_k^\top$ in any order is $I + \sum_{i>k} v_i \mathbf{e}_i \mathbf{e}_k^\top = I + \mathbf{v} \mathbf{e}_k^\top$, since $\mathbf{v} = \sum_{i>k} v_i \mathbf{e}_i$.

For (b), $(I + \mathbf{v} \mathbf{e}_k^\top)(I - \mathbf{v} \mathbf{e}_k^\top) = I - \mathbf{v}(\mathbf{e}_k^\top \mathbf{v}) \mathbf{e}_k^\top = I$, because $\mathbf{e}_k^\top \mathbf{v} = v_k = 0$. Swapping the signs gives the other order.

For (c), we use induction on m ; the case $m = 1$ is the definition. Suppose the formula holds for $m - 1$. Then

$$\left(I + \sum_{j<m} \mathbf{v}_j \mathbf{e}_j^\top\right)(I + \mathbf{v}_m \mathbf{e}_m^\top) = I + \sum_{j<m} \mathbf{v}_j \mathbf{e}_j^\top + \mathbf{v}_m \mathbf{e}_m^\top + \sum_{j<m} \mathbf{v}_j (\mathbf{e}_j^\top \mathbf{v}_m) \mathbf{e}_m^\top.$$

For $j < m$, $\mathbf{e}_j^\top \mathbf{v}_m$ is entry j of $\mathbf{v}_m$, which is zero because the first m entries of $\mathbf{v}_m$ are zero. So the last sum vanishes, and the formula holds for m . This proves the lemma. ■

Part (c) is the heart of the matter: when the factors come in the order $G_1, G_2, \dots$, they **do not interfere**, and each vector $\mathbf{v}_j$ lands untouched in column j .

Theorem 2.6.4 — LU from elimination

Let $A \in M_n(F)$, and suppose elimination without row swaps succeeds on A , with multipliers ℓ_{ik} and result U . Let L be the unit lower triangular matrix whose (i, k) -entry is ℓ_{ik} for every $i > k$. Then U is upper triangular and

$$A = LU.$$

In particular A has an LU factorization.

Idea. Each step of elimination is left multiplication by a product of elementary matrices, which collapses to one matrix $G_k(-\ell_k)$. So $U = G_{n-1}(-\ell_{n-1}) \cdots G_1(-\ell_1)A$. To get A back, undo the steps in reverse order. Their inverses $G_k(\ell_k)$ then appear in the order $G_1, G_2, \dots$, which is exactly the order in which, by [Lemma 2.6.3](#) (c), the multipliers drop into place.

Proof. Let $A^{(0)}, \dots, A^{(n-1)} = U$ be the matrices produced by the algorithm. For each k , let $\ell_k = \sum_{i>k} \ell_{ik} \mathbf{e}_i \in F^n$, a vector whose first k entries are zero.

Step 1: U is upper triangular. We show by induction on k that in $A^{(k)}$ every entry below the diagonal in columns $1, \dots, k$ is zero. For $k = 0$ there is nothing to prove. Suppose it holds for $A^{(k-1)}$. Step k changes only rows $i > k$, replacing row i by row i minus a multiple of row k . In columns $j < k$, the entry of row k lies below the diagonal ($k > j$), so it is 0 by the induction hypothesis, and the entries of row i in these columns stay 0. In column k , the new entries below the diagonal are 0, as noted after the algorithm. This completes the induction, and for $k = n - 1$ it says that U is upper triangular.

Step 2: each step is a matrix. Step k performs the operations “add $-\ell_{ik}$ times row k to row i ” for $i = k + 1, \dots, n$. By [Theorem 2.4.1](#), applying them one after another multiplies $A^{(k-1)}$ on the left by the corresponding elementary matrices $I - \ell_{ik} E_{ik}$. By [Lemma 2.6.3](#) (a), their product is $G_k(-\ell_k)$. Hence $A^{(k)} = G_k(-\ell_k) A^{(k-1)}$, and therefore

$$U = G_{n-1}(-\ell_{n-1}) \cdots G_2(-\ell_2) G_1(-\ell_1) A.$$

Step 3: undo the steps. Multiply both sides on the left by $G_{n-1}(\ell_{n-1})$, then by $G_{n-2}(\ell_{n-2})$, and so on down to $G_1(\ell_1)$. By [Lemma 2.6.3](#) (b), each one cancels the factor next to it, and we get

$$G_1(\ell_1) G_2(\ell_2) \cdots G_{n-1}(\ell_{n-1}) U = A.$$

By [Lemma 2.6.3](#) (c), the product on the left of U is $I + \sum_{k=1}^{n-1} \ell_k \mathbf{e}_k^\top$. Its (i, k) -entry for $i > k$ is ℓ_{ik} ; its diagonal entries are 1; its entries above the diagonal are 0. That is, it is L . Hence $A = LU$ with L unit lower triangular and U upper triangular, as claimed. ■

So the multipliers are not something to be remembered separately: they **are** the lower triangular factor. When a computer performs elimination it stores each ℓ_{ik} in the slot (i, k) that it has just cleared, and at the end the one array holds L below the diagonal and U on and above it.

Warning. The matrix that performs elimination, L^{-1} , is not L with its multipliers negated. In [Example \(An LU factorization from elimination\)](#) the operations multiply A on the left by

$$G_2(-\ell_2) G_1(-\ell_1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 7 & -3 & 1 \end{pmatrix},$$

and the entry $7 = 1 + (-3)(-2)$ mixes two multipliers. This matrix is L^{-1} , by Step 3 of the proof. It is L , whose factors come in the order G_1, G_2 , that has the multipliers cleanly in place.

Quick check. Find the LU factorization of $A = \begin{pmatrix} 2 & 4 \\ 6 & 5 \end{pmatrix}$ over $\mathbb{Q}$ by elimination, and check it by multiplying.

Solution. The pivot is 2 and the multiplier is $\ell_{21} = 6/2 = 3$. Subtracting 3 times row 1 from row 2 gives $U = \begin{pmatrix} 2 & 4 \\ 0 & 5-12 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 0 & -7 \end{pmatrix}$, and $L = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$. Check:
 $LU = \begin{pmatrix} 2 & 4 \\ 6 & 12-7 \end{pmatrix} = A$.

2.6.5 Solving with an LU factorization

Once $A = LU$ is known, each system $A\mathbf{x} = \mathbf{b}$ costs two substitutions.

Example. (Two right-hand sides, one factorization)

Let $A = \begin{pmatrix} 2 & 1 & -1 \\ 4 & 5 & -1 \\ -2 & 8 & 2 \end{pmatrix}$ with the factorization $A = LU$ of [Example \(An LU factorization from elimination\)](#). Solve $A\mathbf{x} = \mathbf{b}$ for

$$\mathbf{b} = \begin{pmatrix} -1 \\ -3 \\ -6 \end{pmatrix} \quad \text{and} \quad \mathbf{b}' = \begin{pmatrix} 3 \\ 7 \\ -2 \end{pmatrix}.$$

Solution. First right-hand side. Forward substitution in $L\mathbf{y} = \mathbf{b}$:

$$y_1 = -1, \quad 2y_1 + y_2 = -3 \Rightarrow y_2 = -1, \quad -y_1 + 3y_2 + y_3 = -6 \Rightarrow y_3 = -6 - 1 + 3 = -4.$$

Back substitution in $U\mathbf{x} = \mathbf{y}$:

$$-2x_3 = -4 \Rightarrow x_3 = 2, \quad 3x_2 + x_3 = -1 \Rightarrow x_2 = -1, \quad 2x_1 + x_2 - x_3 = -1 \Rightarrow x_1 = 1.$$

So $\mathbf{x} = (1, -1, 2)$. Check: $A\mathbf{x} = (2 - 1 - 2, 4 - 5 - 2, -2 - 8 + 4) = (-1, -3, -6) = \mathbf{b}$.

Second right-hand side. $L\mathbf{y} = \mathbf{b}'$: $y_1 = 3$, $y_2 = 7 - 2 \cdot 3 = 1$, $y_3 = -2 + y_1 - 3y_2 = -2 + 3 - 3 = -2$. Then $U\mathbf{x} = \mathbf{y}$: $-2x_3 = -2$, so $x_3 = 1$; $3x_2 + 1 = 1$, so $x_2 = 0$; $2x_1 + 0 - 1 = 3$, so $x_1 = 2$. So $\mathbf{x}' = (2, 0, 1)$, and indeed $A\mathbf{x}' = (4 + 0 - 1, 8 + 0 - 1, -4 + 0 + 2) = (3, 7, -2)$.

Since the diagonal of U is $2, 3, -2$, all non-zero, U is invertible by [Lemma 2.6.2](#), L is invertible by [Lemma 2.6.1](#), and so $A = LU$ is invertible. Each solution found is therefore the only one.

How much is saved? Count the arithmetic operations $(+, -, \times, \div)$ for an $n \times n$ matrix. At step k of elimination there are $n - k$ rows below the pivot. Each needs one division for its multiplier, then $n - k$ multiplications and $n - k$ subtractions in the columns to the right of column k . With $j = n - k$, the total is

$$\sum_{j=1}^{n-1} j(2j+1) = \frac{n(n-1)(4n+1)}{6} \approx \frac{2}{3}n^3.$$

Forward substitution with a unit lower triangular L uses $n(n-1)$ operations, and back substitution n^2 , so a pair of triangular solves costs $2n^2 - n$. For $n = 1000$ that is about 6.7×10^8 operations for the factorization, done once, against about 2×10^6 for each new right-hand side.

2.6.6 Uniqueness

The factorization of [Example \(An LU factorization from elimination\)](#) came from one particular procedure. Could another procedure give a different pair (L, U) ? For invertible matrices, no.

Theorem 2.6.5 — Uniqueness of the LU factorization

Let $A \in M_n(F)$ be **invertible**. If $A = L_1U_1$ and $A = L_2U_2$ are LU factorizations, then $L_1 = L_2$ and $U_1 = U_2$.

Idea. Take the difference in multiplicative form: from $L_1U_1 = L_2U_2$ we get $(L_2^{-1}L_1)U_1 = U_2$. The matrix $M = L_2^{-1}L_1$ is unit lower triangular, and multiplying U_1 by it must leave an upper triangular matrix. Clearing below the diagonal column by column, the non-zero pivots of U_1 force M to have nothing below its diagonal. Invertibility of A is what provides those non-zero pivots.

Proof. By [Lemma 2.6.1](#) (b), L_1 and L_2 are invertible. Hence $U_1 = L_1^{-1}A$ is a product of invertible matrices, so it is invertible by [Theorem 0.9.7](#), and by [Lemma 2.6.2](#) its diagonal entries $u_{11}, \dots, u_{nn}$ are all non-zero.

Let $M = L_2^{-1}L_1$. By [Lemma 2.6.1](#), $M = (m_{ij})$ is unit lower triangular. Multiplying $L_1U_1 = L_2U_2$ on the left by L_2^{-1} gives $MU_1 = U_2$, so MU_1 is upper triangular. We show $m_{ij} = 0$ for all $i > j$, by induction on the column j .

Fix j , and suppose $m_{ik} = 0$ whenever $k < j$ and $i > k$. Let $i > j$. Since MU_1 is upper triangular and $i > j$,

$$0 = (MU_1)_{ij} = \sum_{k=1}^n m_{ik}u_{kj} = \sum_{k \leq j} m_{ik}u_{kj} = m_{ij}u_{jj},$$

where the third equality uses $u_{kj} = 0$ for $k > j$, and the last uses $m_{ik} = 0$ for $k < j$ (here $i > j > k$). Since $u_{jj} \neq 0$, we get $m_{ij} = 0$. This completes the induction, so M is lower triangular with 1's on the diagonal and nothing below it: $M = I$.

Hence $L_2^{-1}L_1 = I$, and multiplying by L_2 gives $L_1 = L_2$. Then $U_1 = L_1^{-1}A = L_2^{-1}A = U_2$. This proves the theorem. ■

Two consequences are worth saying. The LU factorization of an invertible matrix does not depend on how it was found, so it is legitimate to speak of **the** factors L and U . And if elimination without row swaps succeeds on an invertible A , then every pivot is non-zero: row k is not changed after step k , so the pivot of step k is u_{kk} , which is non-zero by the first paragraph of the proof. So the second case of the algorithm (a zero pivot over a zero column) never occurs.

Remark. The hypothesis “invertible” cannot be dropped. For every $c \in F$,

$$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

since the first column of the right factor is zero. Taking $c = 0$ and $c = 1$ gives two

different LU factorizations. The proof breaks exactly at “ $u_{jj} \neq 0$ ”, here for $j = 1$.

2.6.7 Row swaps and the PLU factorization

Example (An invertible matrix with no LU factorization) showed that some matrices, even invertible ones, have no LU factorization, because elimination meets a zero pivot with a non-zero entry below it. The cure in Gaussian elimination is to swap rows. We now show that the swaps can all be collected in front, as a single matrix that rearranges the rows of A **before** elimination starts. After that rearrangement, no swaps are needed.

Matrices that rearrange rows have a name.

Definition 2.6.2 — Permutation matrix

Let $\sigma \in S_n$ be a permutation of $\{1, \dots, n\}$. The **permutation matrix** of σ is

$$P_\sigma = (\mathbf{e}_{\sigma(1)} \quad \mathbf{e}_{\sigma(2)} \quad \cdots \quad \mathbf{e}_{\sigma(n)}) \in M_n(F),$$

the matrix whose j -th column is $\mathbf{e}_{\sigma(j)}$. A matrix is a **permutation matrix** if it equals P_σ for some $\sigma \in S_n$.

Equivalently, a permutation matrix has exactly one 1 in each row and in each column, and zeros elsewhere. The identity is P_{id} . The matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ of **Example (An invertible matrix with no LU factorization)** is P_τ for the transposition $\tau = (1 \ 2)$. By contrast $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is **not** a permutation matrix: each column is still non-zero, but its first row contains two 1's, and its second column $(1, 1)$ is not a standard basis vector.

Lemma 2.6.6 — Permutation matrices

Let $\sigma, \tau \in S_n$.

- (a) $P_\sigma P_\tau = P_{\sigma \circ \tau}$. In particular a product of permutation matrices is a permutation matrix.
- (b) P_σ is invertible and $P_\sigma^{-1} = P_\sigma^\top$.
- (c) If $\tau = (r \ s)$ is a transposition, then P_τ is the elementary matrix P_{rs} that swaps rows r and s , and $P_\tau^2 = I$.
- (d) Let $1 \leq k < r < s \leq n$, let $\tau = (r \ s)$, and let $\mathbf{v} \in F^n$ have its first k entries zero. Then $P_\tau \mathbf{v}$ has its first k entries zero, and

$$P_\tau G_k(\mathbf{v}) P_\tau = G_k(P_\tau \mathbf{v}).$$

Proof.

- (a) By **Theorem 0.9.3**, $P_\tau \mathbf{e}_j = \mathbf{e}_{\tau(j)}$ and so $P_\sigma P_\tau \mathbf{e}_j = P_\sigma \mathbf{e}_{\tau(j)} = \mathbf{e}_{\sigma(\tau(j))}$. So the j -th column of $P_\sigma P_\tau$ is $\mathbf{e}_{(\sigma \circ \tau)(j)}$, for every j , which is the claim.
- (b) By **Theorem 0.9.2**, $(P_\sigma^\top P_\sigma)_{ij} = \mathbf{e}_{\sigma(i)}^\top \mathbf{e}_{\sigma(j)}$, which is 1 if $\sigma(i) = \sigma(j)$ and 0 otherwise.

Since σ is injective, $\sigma(i) = \sigma(j)$ exactly when $i = j$. Hence $P_\sigma^\top P_\sigma = I$, and by Theorem 2.4.5, P_σ is invertible with inverse $P_\sigma^\top$.

- (c) Swapping rows r and s of I produces a matrix whose column r is $\mathbf{e}_s$, whose column s is $\mathbf{e}_r$, and whose other columns j are $\mathbf{e}_j$. These are the columns $\mathbf{e}_{\tau(j)}$, so the swap matrix is P_τ . Since $\tau \circ \tau = \text{id}$, part (a) gives $P_\tau^2 = P_{\text{id}} = I$.
- (d) Write $P = P_\tau$. Its effect on a vector is to exchange entries r and s , which are both beyond position k , so $P\mathbf{v}$ still has its first k entries zero. Since $k \neq r, s$, we have $P\mathbf{e}_k = \mathbf{e}_{\tau(k)} = \mathbf{e}_k$. By (b) and (c), $P^\top = P^{-1} = P$, so $\mathbf{e}_k^\top P = \mathbf{e}_k^\top P^\top = (P\mathbf{e}_k)^\top = \mathbf{e}_k^\top$ by Theorem 0.9.5. Therefore

$$P(I + \mathbf{v}\mathbf{e}_k^\top)P = P^2 + (P\mathbf{v})(\mathbf{e}_k^\top P) = I + (P\mathbf{v})\mathbf{e}_k^\top = G_k(P\mathbf{v}),$$

using $P^2 = I$ from (c). This proves the lemma. ■

By (c) and Theorem 2.4.1, $P_\tau A$ is A with rows r and s swapped, and by (a) every permutation matrix is a product of such swaps applied in turn, since for $n \geq 2$ every permutation is a product of transpositions (Theorem 0.10.4). So PA is always A with its rows rearranged.

Definition 2.6.3 — PLU factorization

Let $A \in M_n(F)$. A **PLU factorization** of A is a triple (P, L, U) of matrices in $M_n(F)$ such that

$$PA = LU, \quad P \text{ is a permutation matrix,} \quad L \text{ is unit lower triangular,} \quad U \text{ is upper triangular.}$$

In words: after rearranging the rows of A by P , the matrix has an LU factorization. By Lemma 2.6.6 (b) this can also be written $A = P^\top LU$, which is why some books write the factorization as $A = PLU$ with their P our $P^\top$. For example, $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ has the PLU factorization (A, I, I) , since $AA = I = I \cdot I$. An LU factorization is the special case $P = I$.

To produce P , run elimination and allow a swap whenever a zero pivot has a non-zero entry below it.

Elimination with row swaps. Input: $A \in M_n(F)$. Put $A^{(0)} = A$. For $k = 1, 2, \dots, n-1$, with $A^{(k-1)} = (a_{ij})$:

- If $a_{ik} = 0$ for every $i \geq k$, set $P_k = I$ and all $\ell_{ik} = 0$, and let $A^{(k)} = A^{(k-1)}$.
- Otherwise choose some $r_k \geq k$ with $a_{r_k k} \neq 0$. Let P_k be the matrix swapping rows k and r_k (so $P_k = I$ if $r_k = k$), and let $B = P_k A^{(k-1)}$. Set $\ell_{ik} = b_{ik}/b_{kk}$ for $i > k$, and obtain $A^{(k)}$ from B by replacing row i with (row i) $-\ell_{ik} \cdot$ (row k) for each $i > k$.

This version never stops. Here is the result it delivers.

Theorem 2.6.7 — Existence of a PLU factorization

Every $A \in M_n(F)$ has a PLU factorization.

Idea. Step roadmap. ① As before, each step is a matrix: $A^{(k)} = G_k(-\ell_k)P_k A^{(k-1)}$, and the final matrix U is upper triangular. ② The swaps are tangled with the multiplier matrices. A swap of two rows below row k can be pushed to the right past G_k at the price of swapping the entries of the multiplier vector: this is Lemma 2.6.6 (d). ③ Once all swaps sit next to A , the rest is the LU proof word for word. In practice, ② says: when you swap two rows, also swap the multipliers already stored in those rows.

Proof. Run elimination with row swaps on A , producing $A^{(0)}, \dots, A^{(n-1)} = U$, swap matrices $P_1, \dots, P_{n-1}$ and multipliers ℓ_{ik} . Let $\ell_k = \sum_{i>k} \ell_{ik} \mathbf{e}_i$, whose first k entries are zero.

Step 1. Exactly as in Step 1 of the proof of Theorem 2.6.4, every entry of $A^{(k)}$ below the diagonal in columns $1, \dots, k$ is zero. The only new point is the swap in step k : it exchanges rows k and $r_k \geq k$, whose entries in columns $j < k$ are below the diagonal and hence zero, so the swap does not disturb these columns. After the swap the pivot b_{kk} is non-zero, and the new entries below it are $b_{ik} - \ell_{ik}b_{kk} = 0$. Hence U is upper triangular. By Lemma 2.6.6 (c), Theorem 2.4.1 and Lemma 2.6.3 (a), as in Step 2 of that proof,

$$A^{(k)} = G_k(-\ell_k) P_k A^{(k-1)} \quad (k = 1, \dots, n-1).$$

This also holds in the first case of the algorithm, where $P_k = I$ and $G_k(\mathbf{0}) = I$.

Step 2. For $1 \leq k \leq m \leq n-1$ put $\ell_k^{(m)} = P_m P_{m-1} \cdots P_{k+1} \ell_k$, with $\ell_m^{(m)} = \ell_m$. We claim that the first k entries of $\ell_k^{(m)}$ are zero and that

$$G_m(-\ell_m) P_m \cdots G_1(-\ell_1) P_1 = G_m(-\ell_m^{(m)}) \cdots G_1(-\ell_1^{(m)}) P_m \cdots P_1.$$

For $m = 1$ both sides are $G_1(-\ell_1) P_1$. Suppose the claim holds for m , and multiply it on the left by $G_{m+1}(-\ell_{m+1}) P_{m+1}$. The matrix P_{m+1} is I or swaps rows $m+1$ and $r_{m+1} > m+1$, both beyond every $k \leq m$. So by Lemma 2.6.6 (c) and (d), for each $k \leq m$ and each vector $\mathbf{v}$ with its first k entries zero,

$$P_{m+1} G_k(\mathbf{v}) = P_{m+1} G_k(\mathbf{v}) P_{m+1} P_{m+1} = G_k(P_{m+1} \mathbf{v}) P_{m+1},$$

and $P_{m+1} \mathbf{v}$ again has its first k entries zero. Applying this for $k = m, m-1, \dots, 1$ moves P_{m+1} to the right past every factor G_k , replacing $\ell_k^{(m)}$ by $P_{m+1} \ell_k^{(m)} = \ell_k^{(m+1)}$ (note $P_{m+1}(-\mathbf{v}) = -P_{m+1} \mathbf{v}$). This is the claim for $m+1$.

Step 3. By Step 1, $U = G_{n-1}(-\ell_{n-1}) P_{n-1} \cdots G_1(-\ell_1) P_1 A$. Write $\ell'_k = \ell_k^{(n-1)}$ and $P = P_{n-1} \cdots P_1$. By Step 2,

$$U = G_{n-1}(-\ell'_{n-1}) \cdots G_1(-\ell'_1) P A.$$

P is a permutation matrix by Lemma 2.6.6 (a) and (c). Multiplying on the left by $G_{n-1}(\ell'_{n-1})$, then $G_{n-2}(\ell'_{n-2})$, and so on, and using Lemma 2.6.3 (b) and (c) as in Step 3 of the proof of Theorem 2.6.4,

$$PA = LU, \quad L = I + \sum_{k=1}^{n-1} \ell'_k \mathbf{e}_k^T,$$

where L is unit lower triangular because the first k entries of each ℓ'_k are zero. So (P, L, U) is a PLU factorization of A . This proves the theorem. ■

Example. (A swap in the middle)

Find a PLU factorization of

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 7 & 2 \end{pmatrix} \in M_3(\mathbb{Q}).$$

Solution. Step 1. The pivot 1 is non-zero, so no swap. The multipliers are $\ell_{21} = 2$ and $\ell_{31} = 3$:

$$A \xrightarrow[R_3 \rightarrow R_3 - 3R_1]{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}, \quad \ell_1 = \begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix}.$$

Step 2. The pivot position $(2, 2)$ holds 0, but the entry below it is 1. Elimination without row swaps would stop here. We swap rows 2 and 3, so $P_2 = P_{23}$:

$$\xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} = U.$$

The entry below the new pivot is already 0, so $\ell_{32} = 0$, and U is upper triangular.

By Step 2 of the proof, the stored multipliers of step 1 are swapped too: $\ell'_1 = P_2 \ell_1 = (0, 3, 2)$. Hence

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}.$$

Check: PA is A with rows 2 and 3 swapped, and

$$LU = \begin{pmatrix} 1 & 2 & 1 \\ 3 & 6+1 & 3-1 \\ 2 & 4 & 2+1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 3 & 7 & 2 \\ 2 & 4 & 3 \end{pmatrix} = PA.$$

Notice where the multipliers ended up: 3 in row 2 and 2 in row 3, traveling with the rows they belong to. To solve $A\mathbf{x} = \mathbf{b}$, multiply by P and solve $LU\mathbf{x} = P\mathbf{b}$ by two substitutions; since P is invertible, this changes no solutions. For $\mathbf{b} = (2, 3, 8)$, $P\mathbf{b} = (2, 8, 3)$. Forward substitution in $L\mathbf{y} = P\mathbf{b}$ gives $y_1 = 2$, $y_2 = 8 - 3 \cdot 2 = 2$, $y_3 = 3 - 2 \cdot 2 = -1$. Back substitution in $U\mathbf{x} = \mathbf{y}$ gives $x_3 = -1$, $x_2 = 2 + x_3 = 1$, $x_1 = 2 - 2x_2 - x_3 = 1$. Check: $A(1, 1, -1) = (1 + 2 - 1, 2 + 4 - 3, 3 + 7 - 2) = (2, 3, 8)$.

Remark. Over $\mathbb{R}$, computers do not choose just any non-zero pivot. They choose the entry of **largest absolute value** in the column, which is called **partial pivoting**. The reason is rounding. For $A = \begin{pmatrix} \varepsilon & 1 \\ 1 & 1 \end{pmatrix}$ with $\varepsilon = 10^{-20}$, elimination without swaps gives the multiplier 10^{20} and $u_{22} = 1 - 10^{20}$, which in standard double-precision arithmetic is stored as -10^{20} . Multiplying back, the computed LU has $(2, 2)$ -entry $10^{20} - 10^{20} = 0$

instead of 1. Swapping the rows first gives the harmless multiplier ε . Chapter 23 makes this precise.

2.6.8 Symmetric matrices: the LDL factorization

When A is symmetric, the two factors of $A = LU$ are secretly related: up to a diagonal matrix, U is the transpose of L . In [Example \(An LU factorization from elimination\)](#) the factors look unrelated, but that matrix is not symmetric. Try a symmetric one:

$$\begin{pmatrix} 1 & 3 & -2 \\ 3 & 7 & -8 \\ -2 & -8 & 5 \end{pmatrix} \xrightarrow[R_3 \rightarrow R_3 + 2R_1]{R_2 \rightarrow R_2 - 3R_1} \begin{pmatrix} 1 & 3 & -2 \\ 0 & -2 & -2 \\ 0 & -2 & 1 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{pmatrix} 1 & 3 & -2 \\ 0 & -2 & -2 \\ 0 & 0 & 3 \end{pmatrix} = U,$$

with multipliers $\ell_{21} = 3$, $\ell_{31} = -2$, $\ell_{32} = 1$. Divide each row of U by its diagonal entry: the rows become $(1, 3, -2)$, $(0, 1, 1)$, $(0, 0, 1)$. These are the columns of $L = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -2 & 1 & 1 \end{pmatrix}$, read as rows. So $U = DL^T$ with $D = \text{diag}(1, -2, 3)$, and $A = LDL^T$.

Theorem 2.6.8 — LDL factorization

Let $A \in M_n(F)$ be **symmetric** and **invertible**, and suppose $A = LU$ is an LU factorization. Then the diagonal entries $u_{11}, \dots, u_{nn}$ of U are non-zero, and with $D = \text{diag}(u_{11}, \dots, u_{nn})$,

$$U = DL^T \quad \text{and} \quad A = LDL^T.$$

Idea. Pull the diagonal out of U , writing $U = DV$ with V unit upper triangular, so $A = LDV$. Transposing, $A = A^T = V^T(DL^T)$, which is a second LU factorization of the same invertible matrix. Uniqueness forces the two to agree, so $L = V^T$.

Proof. As in the proof of [Theorem 2.6.5](#), $U = L^{-1}A$ is invertible, so its diagonal entries are non-zero by [Lemma 2.6.2](#). Hence D is invertible, with $D^{-1} = \text{diag}(u_{11}^{-1}, \dots, u_{nn}^{-1})$. Let $V = D^{-1}U$. Row i of V is u_{ii}^{-1} times row i of U , so V is upper triangular with $v_{ii} = u_{ii}^{-1}u_{ii} = 1$, and $A = LDV$.

Since A is symmetric and $D^T = D$, [Theorem 0.9.5](#) gives

$$A = A^T = (LDV)^T = V^T DL^T = V^T (DL^T).$$

Here V^T is unit lower triangular, because V is unit upper triangular. And DL^T is upper triangular: its (i, j) -entry is $u_{ii}\ell_{ji}$, and $\ell_{ji} = 0$ when $j < i$, since L is lower triangular. So (V^T, DL^T) is an LU factorization of the invertible matrix A . By [Theorem 2.6.5](#), $V^T = L$ and $DL^T = U$. Therefore $A = LU = LDL^T$, as claimed. ■

The factorization stores about half as much: L and the diagonal D determine everything. Over $\mathbb{R}$, when every entry of D is positive, one can split $D = D^{1/2}D^{1/2}$ and write $A = (LD^{1/2})(LD^{1/2})^T$; this is the **Cholesky factorization**, which Chapter 12 connects with positive definite matrices.

Remark. Symmetry alone does not give an LU factorization: $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is symmetric and invertible, and has none by [Example \(An invertible matrix with no LU factorization\)](#). The theorem assumes the factorization exists and then improves it.

2.6.9 Exercises

2.6.9.1 A. Check your understanding

Exercise A1.

- (a) Define an LU factorization of a matrix $A \in M_n(F)$, and a PLU factorization.
- (b) True or false: every invertible matrix has an LU factorization. Justify your answer.
- (c) Suppose $A = LU$. In which order are the two triangular systems solved to find a solution of $A\mathbf{x} = \mathbf{b}$, and what is each one?
- (d) True or false: the product of two unit lower triangular $n \times n$ matrices is unit lower triangular. Justify your answer.
- (e) True or false: the zero matrix in $M_2(F)$ has more than one LU factorization. Justify your answer.
- (f) What is the inverse of a permutation matrix P ?

Solution.

- (a) See [Definition 2.6.1](#): $A = LU$ with L unit lower triangular and U upper triangular. See [Definition 2.6.3](#): $PA = LU$ with P a permutation matrix, L unit lower triangular and U upper triangular.
- (b) False. $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is invertible but has no LU factorization, by [Example \(An invertible matrix with no LU factorization\)](#).
- (c) First $L\mathbf{y} = \mathbf{b}$, by forward substitution; then $U\mathbf{x} = \mathbf{y}$, by back substitution. Then $A\mathbf{x} = L(U\mathbf{x}) = L\mathbf{y} = \mathbf{b}$.
- (d) True, by [Lemma 2.6.1](#) (a).
- (e) True. For every $c \in F$, $\begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix} 0 = 0$, so $\left(\begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix}, 0\right)$ is an LU factorization; $c = 0$ and $c = 1$ give two different ones. This does not contradict [Theorem 2.6.5](#), because 0 is not invertible.
- (f) $P^{-1} = P^T$, by [Lemma 2.6.6](#) (b).

2.6.9.2 B. Practice

Exercise B1: An LU factorization.

Find the LU factorization of

$$A = \begin{pmatrix} 1 & 2 & 0 \\ -3 & -7 & 4 \\ 2 & 6 & -3 \end{pmatrix} \in M_3(\mathbb{Q})$$

by elimination without row swaps. Hence show that A is invertible.

Solution. Step 1. The pivot is 1, with multipliers $\ell_{21} = -3$ and $\ell_{31} = 2$:

$$A \xrightarrow[R_3 \rightarrow R_3 - 2R_1]{R_2 \rightarrow R_2 - (-3)R_1} \begin{pmatrix} 1 & 2 & 0 \\ 0 & -1 & 4 \\ 0 & 2 & -3 \end{pmatrix}.$$

Step 2. The pivot is -1 , with multiplier $\ell_{32} = 2/(-1) = -2$:

$$\xrightarrow{R_3 \rightarrow R_3 - (-2)R_2} \begin{pmatrix} 1 & 2 & 0 \\ 0 & -1 & 4 \\ 0 & 0 & 5 \end{pmatrix} = U.$$

By [Theorem 2.6.4](#),

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 2 & -2 & 1 \end{pmatrix}, \quad A = LU.$$

$$\text{Check: } LU = \begin{pmatrix} 1 & 2 & 0 \\ -3 & -6-1 & 4 \\ 2 & 4+2 & -8+5 \end{pmatrix} = A.$$

The diagonal entries $1, -1, 5$ of U are non-zero, so U is invertible by [Lemma 2.6.2](#), and L is invertible by [Lemma 2.6.1](#). Hence $A = LU$ is invertible by [Theorem 0.9.7](#).

Exercise B2: Two systems, one factorization.

With $A = LU$ as in [Exercise \(B1: An LU factorization\)](#), solve $A\mathbf{x} = \mathbf{b}$ for $\mathbf{b} = (3, -6, 5)$ and for $\mathbf{b} = (0, 1, -2)$, using only forward and back substitution.

Solution. For $\mathbf{b} = (3, -6, 5)$. $L\mathbf{y} = \mathbf{b}$: $y_1 = 3$; $-3y_1 + y_2 = -6$, so $y_2 = 3$; $2y_1 - 2y_2 + y_3 = 5$, so $y_3 = 5 - 6 + 6 = 5$. Then $U\mathbf{x} = \mathbf{y}$: $5x_3 = 5$, so $x_3 = 1$; $-x_2 + 4x_3 = 3$, so $x_2 = 1$; $x_1 + 2x_2 = 3$, so $x_1 = 1$. Hence $\mathbf{x} = (1, 1, 1)$. Check: $A\mathbf{x} = (1+2, -3-7+4, 2+6-3) = (3, -6, 5)$.

For $\mathbf{b} = (0, 1, -2)$. $L\mathbf{y} = \mathbf{b}$: $y_1 = 0$; $y_2 = 1 + 3y_1 = 1$; $y_3 = -2 - 2y_1 + 2y_2 = 0$. Then $U\mathbf{x} = \mathbf{y}$: $5x_3 = 0$, so $x_3 = 0$; $-x_2 + 0 = 1$, so $x_2 = -1$; $x_1 + 2(-1) = 0$, so $x_1 = 2$. Hence $\mathbf{x} = (2, -1, 0)$. Check: $A\mathbf{x} = (2-2, -6+7, 4-6) = (0, 1, -2)$.

Since A is invertible ([Exercise \(B1: An LU factorization\)](#)), each solution is the only one, by [Theorem 0.9.7](#).

Exercise B3: A PLU factorization.

Let $A = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \\ 2 & 3 & 1 \end{pmatrix} \in M_3(\mathbb{Q})$.

- (a) Explain why A has no LU factorization.
- (b) Find a PLU factorization of A .
- (c) Hence solve $A\mathbf{x} = (0, 2, 7)$.

Solution.

- (a) If $A = LU$, then the $(1, 1)$ -entry gives $u_{11} = a_{11} = 0$, and the $(2, 1)$ -entry gives $a_{21} = \ell_{21}u_{11} = 0$. But $a_{21} = 1 \neq 0$, a contradiction, exactly as in [Example \(An invertible matrix with no LU factorization\)](#).
- (b) The first pivot position holds 0 and the entry below is 1, so swap rows 1 and 2, $P_1 = P_{12}$:

$$A \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix} \xrightarrow[R_3 \rightarrow R_3 - 2R_1]{R_2 \rightarrow R_2 - 0R_1} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & -1 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - 1R_2} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & -3 \end{pmatrix} = U.$$

The multipliers are $\ell_{21} = 0$, $\ell_{31} = 2$, $\ell_{32} = 1$, and no swap happens after step 1, so they need no rearranging. Hence

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 1 & 1 \end{pmatrix}.$$

$$\text{Check: } PA = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix}, \text{ and } LU = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 2 & 2+1 & 2+2-3 \end{pmatrix} = PA.$$

- (c) $A\mathbf{x} = \mathbf{b}$ holds if and only if $PA\mathbf{x} = P\mathbf{b}$, since P is invertible ([Lemma 2.6.6 \(b\)](#)). Swapping the first two entries of $\mathbf{b} = (0, 2, 7)$, we solve $LU\mathbf{x} = P\mathbf{b} = (2, 0, 7)$. $L\mathbf{y} = (2, 0, 7)$: $y_1 = 2$, $y_2 = 0$, $y_3 = 7 - 2 \cdot 2 - 0 = 3$. $U\mathbf{x} = \mathbf{y}$: $-3x_3 = 3$, so $x_3 = -1$; $x_2 + 2x_3 = 0$, so $x_2 = 2$; $x_1 + x_2 + x_3 = 2$, so $x_1 = 1$. Hence $\mathbf{x} = (1, 2, -1)$. Check: $A\mathbf{x} = (0 + 2 - 2, 1 + 2 - 1, 2 + 6 - 1) = (0, 2, 7)$.

2.6.9.3 C. Going deeper

Exercise C1: Leading principal submatrices.

For $A \in M_n(F)$ and $1 \leq k \leq n$, the **leading principal submatrix** $A_k \in M_k(F)$ is the top-left $k \times k$ corner of A , with entries a_{ij} for $i, j \leq k$.

- (a) Let L be lower triangular and U upper triangular in $M_n(F)$. Show that $(LU)_k = L_k U_k$ for every k .
- (b) Deduce that if A has an LU factorization $A = LU$ with U invertible, then every leading principal submatrix $A_1, \dots, A_n$ is invertible.
- (c) Prove the converse: if $A_1, \dots, A_n$ are all invertible, then A has an LU factorization with U invertible.

Hint for (c): induction on n . Write A with A_{n-1} in its top-left corner, a column $\mathbf{b}$, a row $\mathbf{c}^\top$ and a corner entry d , and look for L and U of the same shape.

Solution.

- (a) Let $i, j \leq k$. By [Definition 0.9.9](#), $(LU)_{ij} = \sum_{l=1}^n \ell_{il} u_{lj}$. If $l > k$, then $l > i$, so $\ell_{il} = 0$ because L is lower triangular. Hence $(LU)_{ij} = \sum_{l=1}^k \ell_{il} u_{lj} = (L_k U_k)_{ij}$, as claimed.
- (b) Suppose $A = LU$ with U invertible. By (a), $A_k = L_k U_k$. The matrix L_k is unit lower triangular, so it is invertible by [Lemma 2.6.1](#). By [Lemma 2.6.2](#), $u_{11}, \dots, u_{nn}$ are non-zero; U_k is upper triangular with diagonal $u_{11}, \dots, u_{kk}$, so U_k is invertible by the same lemma. Hence A_k is invertible by [Theorem 0.9.7](#).
- (c) We prove by induction on n that every $A \in M_n(F)$ whose leading principal submatrices are all invertible has an LU factorization with U invertible. For $n = 1$, $A = (a_{11})$ with $a_{11} \neq 0$, and $A = (1)(a_{11})$ works. Let $n \geq 2$ and suppose the statement holds for $n - 1$. Write

$$A = \begin{pmatrix} A_{n-1} & \mathbf{b} \\ \mathbf{c}^\top & d \end{pmatrix}, \quad \mathbf{b}, \mathbf{c} \in F^{n-1}, d \in F.$$

The leading principal submatrices of A_{n-1} are $A_1, \dots, A_{n-1}$, all invertible, so by the induction hypothesis $A_{n-1} = L'U'$ with L' unit lower triangular and U' upper triangular and invertible. Put $\mathbf{y} = L'^{-1}\mathbf{b}$, $\mathbf{x}^\top = \mathbf{c}^\top U'^{-1}$ and $z = d - \mathbf{x}^\top \mathbf{y}$, and let

$$L = \begin{pmatrix} L' & \mathbf{0} \\ \mathbf{x}^\top & 1 \end{pmatrix}, \quad U = \begin{pmatrix} U' & \mathbf{y} \\ \mathbf{0}^\top & z \end{pmatrix}.$$

Then L is unit lower triangular and U is upper triangular. Computing LU entry by entry, splitting each sum $\sum_{l=1}^n$ into $l \leq n-1$ and $l = n$: the top-left block is $L'U' + \mathbf{0}\mathbf{0}^\top = A_{n-1}$; the top-right column is $L'\mathbf{y} + \mathbf{0}z = \mathbf{b}$; the bottom-left row is $\mathbf{x}^\top U' + 1 \cdot \mathbf{0}^\top = \mathbf{c}^\top$; the corner is $\mathbf{x}^\top \mathbf{y} + z = d$. Hence $A = LU$.

Finally $A = A_n$ is invertible by hypothesis, and L is invertible by [Lemma 2.6.1](#), so $U = L^{-1}A$ is invertible by [Theorem 0.9.7](#). This completes the induction.

In the language of elimination: z is the n -th pivot, and (c) shows that invertible leading corners are exactly what keeps every pivot non-zero.

Exercise C2: An LDL factorization.

Let $A = \begin{pmatrix} 4 & -4 & 8 \\ -4 & 5 & -11 \\ 8 & -11 & 27 \end{pmatrix} \in M_3(\mathbb{R})$.

- (a) Find the LU factorization of A , show that A is invertible, and write $A = LDL^\top$.
 (b) Deduce that $\mathbf{x}^\top A \mathbf{x} > 0$ for every non-zero $\mathbf{x} \in \mathbb{R}^3$.

Solution.

- (a) Step 1: pivot 4, multipliers $\ell_{21} = -1$, $\ell_{31} = 2$. Row 2 becomes $(-4, 5, -11) + (4, -4, 8) = (0, 1, -3)$, row 3 becomes $(8, -11, 27) - 2(4, -4, 8) = (0, -3, 11)$. Step 2: pivot 1, multiplier $\ell_{32} = -3$; row 3 becomes $(0, -3, 11) + 3(0, 1, -3) = (0, 0, 2)$.

Hence

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -3 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 4 & -4 & 8 \\ 0 & 1 & -3 \\ 0 & 0 & 2 \end{pmatrix},$$

and $A = LU$ by [Theorem 2.6.4](#). The diagonal $4, 1, 2$ of U is non-zero, so U is invertible ([Lemma 2.6.2](#)), L is invertible ([Lemma 2.6.1](#)), and $A = LU$ is invertible. A is symmetric, so by [Theorem 2.6.8](#), with $D = \text{diag}(4, 1, 2)$,

$$A = LDL^T, \quad DL^T = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & -1 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & -4 & 8 \\ 0 & 1 & -3 \\ 0 & 0 & 2 \end{pmatrix} = U,$$

which confirms the theorem directly.

- (b) Let $\mathbf{x} \neq \mathbf{0}$ and put $\mathbf{y} = L^T \mathbf{x}$. By [Lemma 2.6.1](#), L is invertible, so L^T is invertible by [Theorem 0.9.7](#) (4), so $\mathbf{y} \neq \mathbf{0}$ (otherwise $\mathbf{x} = (L^T)^{-1} \mathbf{y} = \mathbf{0}$). By [Theorem 0.9.5](#), $\mathbf{y}^T = \mathbf{x}^T L$, so

$$\mathbf{x}^T A \mathbf{x} = \mathbf{x}^T L D L^T \mathbf{x} = \mathbf{y}^T D \mathbf{y} = 4y_1^2 + y_2^2 + 2y_3^2,$$

reading the 1×1 matrix as a real number. Each term is ≥ 0 , and at least one $y_i \neq 0$, which makes its term strictly positive. Hence $\mathbf{x}^T A \mathbf{x} > 0$.

Exercise C3: Counting matrices without an LU factorization.

- (a) Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(F)$. Prove that A has an LU factorization if and only if $a \neq 0$ or $c = 0$.
- (b) Let q be a prime. How many matrices in $M_2(\mathbb{F}_q)$ have **no** LU factorization? List them for $q = 2$.

Solution.

- (a) ($\Rightarrow$) Suppose $A = LU$ with $L = \begin{pmatrix} 1 & 0 \\ \ell & 1 \end{pmatrix}$ and $U = \begin{pmatrix} u & v \\ 0 & w \end{pmatrix}$. As in [Example \(An invertible matrix with no LU factorization\)](#), $LU = \begin{pmatrix} u & v \\ \ell u & \ell v + w \end{pmatrix}$, so $a = u$ and $c = \ell u = \ell a$. If $a = 0$, then $c = 0$. Hence $a \neq 0$ or $c = 0$.
- ($\Leftarrow$) If $a \neq 0$, elimination without row swaps succeeds with multiplier $\ell = c/a$, and indeed

$$\begin{pmatrix} 1 & 0 \\ c/a & 1 \end{pmatrix} \begin{pmatrix} a & b \\ 0 & d - cb/a \end{pmatrix} = \begin{pmatrix} a & b \\ c & cb/a + d - cb/a \end{pmatrix} = A.$$

If $c = 0$, then A is upper triangular and $A = IA$.

- (b) By (a), A has no LU factorization exactly when $a = 0$ and $c \neq 0$. There are $q - 1$ choices of c , and q choices each of b and d , so there are $(q - 1)q^2$ such matrices. For $q = 2$ this gives 4 of the 16 matrices: $a = 0$, $c = 1$, and $b, d \in \{0, 1\}$, that is,

$$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

2.7 Applications of Linear Systems

We now have a complete engine: elimination decides whether $A\mathbf{x} = \mathbf{b}$ is consistent, rank–nullity counts the free parameters, and the null space describes the freedom. This section runs the engine on four problems from outside linear algebra. In each one the modeling step is where the thinking happens: which quantities are unknown, why the constraints are linear, and over which field. Each model is also honest about what it leaves out.

2.7.1 Polynomial interpolation

Suppose we know the values of an unknown polynomial at a few points, and want the polynomial. Through two points with different x -values there is exactly one line. Through three there is exactly one parabola, possibly degenerate. The general question: given points $(x_0, y_0), \dots, (x_n, y_n)$ in F^2 , is there a polynomial $p \in F[x]_{\leq n}$ with

$$p(x_0) = y_0, \quad p(x_1) = y_1, \quad \dots, \quad p(x_n) = y_n,$$

and is it unique?

The unknowns are the coefficients. Writing $p = c_0 + c_1x + \dots + c_nx^n$, the condition $p(x_i) = y_i$ reads $c_0 + c_1x_i + \dots + c_nx_i^n = y_i$, by [Definition 0.8.6](#). The x_i and y_i are known numbers, so this is a **linear** equation in $c_0, \dots, c_n$, even though p itself is not linear. The $n + 1$ conditions form a square system

$$V\mathbf{c} = \mathbf{y}, \quad V = \begin{pmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^n \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} c_0 \\ \vdots \\ c_n \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} y_0 \\ \vdots \\ y_n \end{pmatrix}.$$

By [Theorem 2.4.4](#), this has exactly one solution for **every** $\mathbf{y}$ as soon as $V\mathbf{c} = \mathbf{0}$ has only the trivial solution. And $V\mathbf{c} = \mathbf{0}$ says that the polynomial with coefficients $\mathbf{c}$ has the $n + 1$ roots $x_0, \dots, x_n$. So everything hinges on a fact about polynomials: a non-zero polynomial of degree at most n cannot have $n + 1$ distinct roots. Chapter 0 proved this only for degree at most 2, in [Exercise \(C1: Roots of a quadratic\)](#), so we prove it now in general. It takes two lemmas.

The first is long division by $x - c$, carried out one leading term at a time.

Lemma 2.7.1 — Factor Theorem for a Linear Divisor

Let $p \in F[x]$ and $c \in F$. Then there is $q \in F[x]$ such that

$$p = (x - c)q + p(c).$$

In particular, if c is a root of p , then $p = (x - c)q$.

Idea. Divide as at school. If p has leading term a_dx^d , subtract $a_dx^{d-1}(x - c)$, which kills the leading term and leaves a polynomial of smaller degree. Repeat until only a constant remains. To turn “repeat” into a proof, use induction on the degree. The constant left over is found by evaluating at c , where the factor $x - c$ vanishes.

Proof. We first show that $p = (x - c)q + r$ for some $q \in F[x]$ and some constant $r \in F$, by strong induction on $\deg p$ ([Theorem 0.2.10](#)), treating $p = 0$ together with the constants. If $p = 0$ or $\deg p = 0$, then p is a constant, and $q = 0$, $r = p$ work.

Let $d \geq 1$, and suppose the statement holds for every polynomial of degree less than d , and for 0. Let $\deg p = d$ with leading coefficient a_d , and put

$$p_1 = p - a_d x^{d-1}(x - c) = p - a_d x^d + c a_d x^{d-1}.$$

The coefficient of x^d in p_1 is $a_d - a_d = 0$, and all higher coefficients are 0, so $p_1 = 0$ or $\deg p_1 < d$. By the induction hypothesis, $p_1 = (x - c)q_1 + r$ with $r \in F$. Hence

$$p = (x - c)(a_d x^{d-1} + q_1) + r,$$

which is the statement for p , with $q = a_d x^{d-1} + q_1$.

Now evaluate at c . By [Theorem 0.8.5](#), $p(c) = (c - c)q(c) + r = 0 \cdot q(c) + r = r$. So $r = p(c)$, and if $p(c) = 0$, then $p = (x - c)q$. This proves the lemma. ■

Lemma 2.7.2 — Root bound

Let $n \in \mathbb{N}$, and let $p \in F[x]$ be **non-zero** with $\deg p \leq n$. Then p has **at most** n distinct roots in F .

Proof. We use induction on n . If $n = 0$, then $p = a$ is a non-zero constant, and $p(c) = a \neq 0$ for every $c \in F$, so p has no roots.

Let $n \geq 1$, suppose the statement holds for $n - 1$, and let $p \neq 0$ with $\deg p \leq n$. If p has no root, we are done. Otherwise let c be a root. By [Lemma 2.7.1](#), $p = (x - c)q$ for some $q \in F[x]$. Then $q \neq 0$, since otherwise $p = 0$, and by [Theorem 0.8.3](#), $\deg p = 1 + \deg q$, so $\deg q \leq n - 1$. By the induction hypothesis, q has at most $n - 1$ distinct roots.

Let $s \in F$ be any root of p . By [Theorem 0.8.5](#), $0 = p(s) = (s - c)q(s)$, so $s - c = 0$ or $q(s) = 0$ by [Theorem 0.7.1](#). Hence every root of p is c or a root of q , and p has at most $1 + (n - 1) = n$ distinct roots. This completes the induction. ■

The field axioms did real work in the last step: a product of two non-zero elements is non-zero. Over $\mathbb{Z}/8\mathbb{Z}$, which is not a field, $x^2 - 1$ has the four roots 1, 3, 5, 7, as [Exercise \(C1: Roots of a quadratic\)](#) showed. And the lemma is sharp: over $\mathbb{F}_2$, the polynomial $x^2 + x$ of [Example \(Different polynomials, same function\)](#) has degree 2 and the two roots 0 and 1.

The lemma also settles a question left open in Chapter 0. If F is **infinite** and the polynomial function of p is zero, then every element of F is a root of p . If p were non-zero, this would be more than $\deg p$ roots, so $p = 0$. Over an infinite field, a polynomial is therefore determined by its values: if $p(c) = q(c)$ for every $c \in F$, then $p - q$ has zero polynomial function by [Theorem 0.8.5](#), so $p - q = 0$.

Theorem 2.7.3 — Polynomial interpolation

Let $x_0, x_1, \dots, x_n \in F$ be **distinct**, and let $y_0, y_1, \dots, y_n \in F$ be arbitrary. Then there is **exactly one** polynomial $p \in F[x]_{\leq n}$ with $p(x_i) = y_i$ for $i = 0, 1, \dots, n$.

Proof. A polynomial $p \in F[x]_{\leq n}$ is the same thing as its coefficient vector $\mathbf{c} = (c_0, \dots, c_n) \in F^{n+1}$, since two polynomials are equal exactly when their coefficients agree (Definition 0.8.1). As computed above, $p(x_i) = y_i$ for all i if and only if $V\mathbf{c} = \mathbf{y}$, where $V \in M_{n+1}(F)$ has (i, j) -entry x_i^j (rows and columns numbered from 0 to n).

Let $V\mathbf{c} = \mathbf{0}$, and let $q = c_0 + c_1x + \dots + c_nx^n$. Then $q(x_i) = 0$ for every i , so q has the $n+1$ distinct roots $x_0, \dots, x_n$, while $\deg q \leq n$. By Lemma 2.7.2, q cannot be non-zero, so $q = 0$ and $\mathbf{c} = \mathbf{0}$. Hence $V\mathbf{c} = \mathbf{0}$ has only the trivial solution, and V is invertible by Theorem 2.4.4. By Theorem 0.9.7 (7), $V\mathbf{c} = \mathbf{y}$ has exactly one solution $\mathbf{c} = V^{-1}\mathbf{y}$. The polynomial with these coefficients is the unique p required. This proves the theorem. ■

The proof is a model of how this chapter turns questions into systems. Existence for every right-hand side and uniqueness are both properties of the square matrix V , and for a square matrix they come together. So we only had to prove uniqueness for the zero data, and that was a statement about roots. Chapter 5 gives an explicit formula for p , the Lagrange interpolation formula. Here is the computation by elimination.

Example. (A cubic through four points)

Find the polynomial $p \in \mathbb{Q}[x]_{\leq 3}$ whose graph passes through $(-1, 2)$, $(0, 1)$, $(1, 2)$ and $(2, 11)$.

Solution. The x -values $-1, 0, 1, 2$ are distinct, so by Theorem 2.7.3 there is exactly one such $p = c_0 + c_1x + c_2x^2 + c_3x^3$. The conditions $p(x_i) = y_i$ give the augmented matrix below. We first swap the rows for $x = -1$ and $x = 0$, so that the simplest row leads:

$$\left(\begin{array}{cccc|c} 1 & -1 & 1 & -1 & 2 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 4 & 8 & 11 \end{array} \right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 1 & -1 & 1 & -1 & 2 \\ 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 4 & 8 & 11 \end{array} \right) \xrightarrow[R_4 \rightarrow R_4 - R_1]{R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 1 & -1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 2 & 4 & 8 & 10 \end{array} \right)$$

Next clear column 2 below the pivot -1 :

$$\xrightarrow[R_4 \rightarrow R_4 + 2R_2]{R_3 \rightarrow R_3 + R_2} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 1 & -1 & 1 \\ 0 & 0 & 2 & 0 & 2 \\ 0 & 0 & 6 & 6 & 12 \end{array} \right) \xrightarrow{R_4 \rightarrow R_4 - 3R_3} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 1 & -1 & 1 \\ 0 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 6 & 6 \end{array} \right).$$

Back substitution: $6c_3 = 6$, so $c_3 = 1$; $2c_2 = 2$, so $c_2 = 1$; $-c_1 + c_2 - c_3 = 1$, so $c_1 = -1$; and $c_0 = 1$. Hence

$$p = 1 - x + x^2 + x^3.$$

Check: $p(-1) = 1+1+1-1 = 2$, $p(0) = 1$, $p(1) = 1-1+1+1 = 2$, $p(2) = 1-2+4+8 = 11$.

Warning. The x -values must be distinct as elements of F . If $x_0 = x_1$, the first two equations of $V\mathbf{c} = \mathbf{y}$ have the same left-hand side, so the system is inconsistent for any data with $y_0 \neq y_1$. This depends on the field: over $\mathbb{F}_3$ the numbers -1 and 2 are the **same** element, so the four points of [Example \(A cubic through four points\)](#) do not have distinct x -values there. Note also that the theorem promises degree **at most** n : the three points $(0, 0), (1, 1), (2, 2)$ are interpolated by $p = x$, of degree 1.

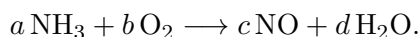
Quick check. Find the polynomial $p \in \mathbb{F}_3[x]_{\leq 2}$ with $p(0) = 1$, $p(1) = 0$ and $p(2) = 0$.

Solution. The x -values $0, 1, 2$ are distinct in $\mathbb{F}_3$, so there is exactly one such p . Since 1 and 2 are roots, try $p = a(x - 1)(x - 2)$, which has degree at most 2 and vanishes at 1 and 2 . Then $p(0) = a \cdot (-1)(-2) = 2a$, and $2a = 1$ gives $a = 2$, since $2 \cdot 2 = 4 = 1$ in $\mathbb{F}_3$. Expanding, $p = 2(x^2 - 3x + 2) = 2(x^2 + 2) = 2x^2 + 1$. Check: $p(0) = 1$, $p(1) = 3 = 0$, $p(2) = 9 = 0$. By uniqueness, this is the answer.

What the model leaves out. Interpolation fits the data **exactly**. When the y_i are measurements with errors, an exact fit of high degree follows the noise and can swing wildly between the data points. The honest tool there is a low-degree polynomial that fits approximately, found by least squares in Chapter 10.

2.7.2 Balancing chemical equations

When ammonia burns in oxygen over a catalyst, it produces nitric oxide and water:



The coefficients a, b, c, d count molecules. The model rests on one physical law: **atoms are neither created nor destroyed**, so each element appears equally often on both sides. For each element this gives one linear equation:

$$\begin{aligned} \text{N: } a &= c, \\ \text{H: } 3a &= 2d, \\ \text{O: } 2b &= c + d. \end{aligned}$$

Moving everything to the left gives a homogeneous system, so the balanced coefficient vectors form the null space of its coefficient matrix. We want positive integers, but we compute over $\mathbb{Q}$. Elimination uses only the four field operations on integer data, so it never leaves $\mathbb{Q}$, and a rational solution can be scaled to an integer one at the end.

Example. (Burning ammonia)

Find all solutions of the system above over $\mathbb{Q}$, and hence the balanced equation with the smallest positive integer coefficients.

Solution. In the order (a, b, c, d) , with rows N, H, O, the coefficient matrix is reduced as follows:

$$\begin{pmatrix} 1 & 0 & -1 & 0 \\ 3 & 0 & 0 & -2 \\ 0 & 2 & -1 & -1 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - 3R_1} \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 3 & -2 \\ 0 & 2 & -1 & -1 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 2 & -1 & -1 \\ 0 & 0 & 3 & -2 \end{pmatrix}.$$

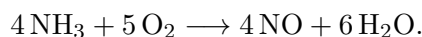
Scale rows 2 and 3 by $\frac{1}{2}$ and $\frac{1}{3}$, then clear column 3 above its pivot:

$$\begin{array}{l} R_2 \rightarrow \frac{1}{2}R_2 \\ R_3 \rightarrow \frac{1}{3}R_3 \end{array} \rightarrow \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{2}{3} \end{pmatrix} \xrightarrow{\begin{array}{l} R_1 \rightarrow R_1 + R_3 \\ R_2 \rightarrow R_2 + \frac{1}{2}R_3 \end{array}} \begin{pmatrix} 1 & 0 & 0 & -\frac{2}{3} \\ 0 & 1 & 0 & -\frac{5}{6} \\ 0 & 0 & 1 & -\frac{2}{3} \end{pmatrix}.$$

This is the RREF. The pivot columns are 1, 2, 3, and d is the only free variable. By [Theorem 2.2.4](#), with $d = t$, the solutions are

$$(a, b, c, d) = t \left(\frac{2}{3}, \frac{5}{6}, \frac{2}{3}, 1 \right), \quad t \in \mathbb{Q}.$$

So the null space is a line: rank 3 and nullity $4 - 3 = 1$, as [Theorem 2.5.7](#) predicts. The coefficients are integers exactly when $\frac{2}{3}t$, $\frac{5}{6}t$ and t are integers, which forces t to be a multiple of 6. The smallest positive choice $t = 6$ gives $(4, 5, 4, 6)$:



Check: N: $4 = 4$. H: $12 = 12$. O: $10 = 4 + 6$.

Nullity 1 is what makes “the balanced equation” meaningful: every balancing is a multiple of one. Positivity was luck of the data here. Nothing in the algebra guarantees that the null space contains a vector with all entries positive, and when it does not, the proposed reaction cannot be balanced at all.

Warning. A null space of dimension 2 or more does not give one balanced equation. For $a\text{H}_2 + b\text{O}_2 \rightarrow c\text{H}_2\text{O} + d\text{H}_2\text{O}_2$, the equations $2a = 2c + 2d$ (hydrogen) and $2b = c + 2d$ (oxygen) have rank 2 and nullity 2. Both $(2, 1, 2, 0)$ and $(1, 1, 0, 1)$ balance, and so does every positive combination of them. The “reaction” is really two reactions, $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ and $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}_2$, and the proportions depend on chemistry, not on algebra.

What the model leaves out. Conservation of atoms is necessary, not sufficient. The equations cannot tell whether a reaction actually happens, and reactions involving ions also conserve electric charge, which adds one more linear equation.

2.7.3 Flows in a network

A small town center has four intersections and five one-way streets. Cars enter and leave the center at some intersections. We measure the external traffic and want to know the traffic on each street.

The model has one assumption: at every intersection, as many cars leave per minute as arrive, since cars do not pile up or vanish. In electrical circuits the same rule for currents is **Kirchhoff’s current law**. To write it down, number the intersections (**vertices**) $1, \dots, m$ and the streets (**edges**) $1, \dots, k$. Each edge e runs from a vertex $\text{tail}(e)$ to a **different** vertex $\text{head}(e)$. The unknown f_e is the flow along edge e . The data s_v is the net external supply at v : cars entering the network there per minute, with a negative sign for cars leaving. Conservation at v says

$$(\text{flow out of } v) - (\text{flow into } v) = s_v.$$

Definition 2.7.1 — Incidence matrix

The **incidence matrix** of a network with vertices $1, \dots, m$ and edges $1, \dots, k$, where $k \geq 1$, is $N \in M_{m \times k}(\mathbb{R})$ with

$$N_{ve} = \begin{cases} 1 & \text{if } v = \text{tail}(e), \\ -1 & \text{if } v = \text{head}(e), \\ 0 & \text{otherwise.} \end{cases}$$

Row v of $N\mathbf{f}$ is $\sum_e N_{ve}f_e$, the flow out of v minus the flow into v . So conservation at every vertex is the single matrix equation $N\mathbf{f} = \mathbf{s}$. Column e of N has exactly one 1 and one -1 , so its entries sum to 0. Adding up all the conservation equations therefore gives $0 = s_1 + \dots + s_m$: **total inflow equals total outflow**. This is necessary for a solution. The next result shows that, for a network in one piece, it is also sufficient.

A network is **connected** if for any two vertices v, w there is a sequence of vertices $v = w_0, w_1, \dots, w_t = w$ in which each consecutive pair is joined by an edge, in either direction.

Proposition 2.7.4 — Rank of an incidence matrix

Let $N \in M_{m \times k}(\mathbb{R})$ be the incidence matrix of a **connected** network with m vertices. Then $\text{rank } N = m - 1$.

Idea. Rank is easier to see through the transpose. A vector $\mathbf{y} \in \mathbb{R}^m$ assigns a number to each vertex, and $N^T\mathbf{y}$ records, for each edge, the difference of the numbers at its two ends. So $N^T\mathbf{y} = \mathbf{0}$ says that the numbers agree across every edge. Along a chain of edges they then agree everywhere, so the null space of N^T is the line of constant vectors. Rank-nullity and “row rank equals column rank” finish.

Proof. Let $\mathbf{y} \in \mathbb{R}^m$. For each edge e , entry e of $N^T\mathbf{y}$ is $\sum_v N_{ve}y_v = y_{\text{tail}(e)} - y_{\text{head}(e)}$, by Definition 0.9.10 and Definition 2.7.1. Hence $N^T\mathbf{y} = \mathbf{0}$ if and only if $y_u = y_w$ whenever u and w are joined by an edge.

Suppose $N^T\mathbf{y} = \mathbf{0}$. Given any vertex w , connectedness gives a sequence $1 = w_0, w_1, \dots, w_t = w$ with consecutive vertices joined by edges, so $y_1 = y_{w_1} = \dots = y_w$. Hence $\mathbf{y} = y_1\mathbf{1}$, where $\mathbf{1} = (1, \dots, 1) \in \mathbb{R}^m$. Conversely $N^T\mathbf{1} = \mathbf{0}$, since all differences vanish. So $\text{nul}(N^T) = \text{span}(\mathbf{1})$, and since $\mathbf{1} \neq \mathbf{0}$, its dimension is 1.

The matrix N^T has m columns, so Theorem 2.5.7 gives $\text{rank } N^T = m - 1$. By Theorem 2.5.5, $\text{rank } N = \text{rank } N^T = m - 1$. This proves the proposition. ■

Corollary 2.7.5 — Solvable flow problems

Let $N \in M_{m \times k}(\mathbb{R})$ be the incidence matrix of a connected network, and let $\mathbf{s} \in \mathbb{R}^m$. Then $N\mathbf{f} = \mathbf{s}$ has a solution if and only if $s_1 + \dots + s_m = 0$. In that case the solutions form $\mathbf{f}_0 + \text{nul}(N)$ for any one solution $\mathbf{f}_0$, and $\dim \text{nul}(N) = k - m + 1$.

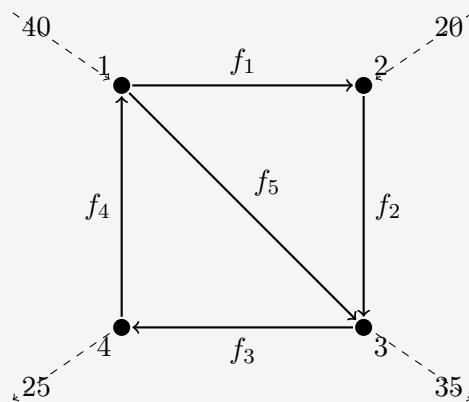
Proof. Let $W = \{\mathbf{z} \in \mathbb{R}^m : z_1 + \cdots + z_m = 0\}$. It is the solution set of the homogeneous system with the $1 \times m$ coefficient matrix $(1 \ \cdots \ 1)$, so it is a subspace by [Theorem 2.1.2](#). That matrix has rank 1, since its column space is spanned by the non-zero column (1) and so equals $\mathbb{R}^1$. By [Theorem 2.5.7](#), $\dim W = m - 1$.

Each column of N has entries summing to 0, so it lies in W , and since W is a subspace, $\text{col}(N) \subseteq W$ by [Theorem 1.3.1](#). Also $\dim \text{col}(N) = \text{rank } N = m - 1 = \dim W$, by [Definition 2.5.4](#) and [Proposition 2.7.4](#). Hence $\text{col}(N) = W$ by [Theorem 1.6.8](#). By [Theorem 2.1.1](#), $N\mathbf{f} = \mathbf{s}$ is consistent if and only if $\mathbf{s} \in \text{col}(N) = W$, which is the first claim. The description of all solutions is [Theorem 2.1.3](#), and $\dim \text{nul}(N) = k - (m - 1)$ by [Theorem 2.5.7](#), since N has k columns. ■

The $k - m + 1$ free parameters have a clear meaning: they are flows around **loops**. Adding the same amount of traffic all the way around a closed circuit of streets changes no intersection's balance.

Example. (Traffic in a town center)

Four intersections are joined by one-way streets $1 \rightarrow 2$, $2 \rightarrow 3$, $3 \rightarrow 4$, $4 \rightarrow 1$ and $1 \rightarrow 3$, carrying unknown flows $f_1, \dots, f_5$ (cars per minute) in that order. From outside, 40 cars per minute enter at intersection 1 and 20 at intersection 2, while 35 leave at intersection 3 and 25 at intersection 4.



Find all flows that satisfy conservation. If street $4 \rightarrow 1$ is closed for repairs and flows cannot be negative, what is the largest flow the diagonal street $1 \rightarrow 3$ may have to carry?

Solution. The network is connected, and $\mathbf{s} = (40, 20, -35, -25)$ sums to 0, so by [Corollary 2.7.5](#) a solution exists and the solutions have $5 - 4 + 1 = 2$ free parameters. We row reduce $[N \mid \mathbf{s}]$, with rows for vertices 1, 2, 3, 4 and columns for $f_1, \dots, f_5$:

$$\left(\begin{array}{ccccc|c} 1 & 0 & 0 & -1 & 1 & 40 \\ -1 & 1 & 0 & 0 & 0 & 20 \\ 0 & -1 & 1 & 0 & -1 & -35 \\ 0 & 0 & -1 & 1 & 0 & -25 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 + R_1} \left(\begin{array}{ccccc|c} 1 & 0 & 0 & -1 & 1 & 40 \\ 0 & 1 & 0 & -1 & 1 & 60 \\ 0 & -1 & 1 & 0 & -1 & -35 \\ 0 & 0 & -1 & 1 & 0 & -25 \end{array} \right)$$

$$\xrightarrow{R_3 \rightarrow R_3 + R_2} \left(\begin{array}{ccccc|c} 1 & 0 & 0 & -1 & 1 & 40 \\ 0 & 1 & 0 & -1 & 1 & 60 \\ 0 & 0 & 1 & -1 & 0 & 25 \\ 0 & 0 & -1 & 1 & 0 & -25 \end{array} \right) \xrightarrow{R_4 \rightarrow R_4 + R_3} \left(\begin{array}{ccccc|c} 1 & 0 & 0 & -1 & 1 & 40 \\ 0 & 1 & 0 & -1 & 1 & 60 \\ 0 & 0 & 1 & -1 & 0 & 25 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right).$$

This is the RREF, with pivots in columns 1, 2, 3, so $\text{rank } N = 3 = m - 1$, as [Proposition 2.7.4](#) predicts. The free variables are $f_4 = \alpha$ and $f_5 = \beta$, and by [Theorem 2.2.4](#),

$$\mathbf{f} = \begin{pmatrix} 40 \\ 60 \\ 25 \\ 0 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} -1 \\ -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad \alpha, \beta \in \mathbb{R}.$$

The two null space vectors are the loops: α sends extra cars around $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$, and β moves cars from the route $1 \rightarrow 2 \rightarrow 3$ onto the diagonal $1 \rightarrow 3$.

With street $4 \rightarrow 1$ closed, $\alpha = f_4 = 0$ and $\mathbf{f} = (40 - \beta, 60 - \beta, 25, 0, \beta)$. Non-negativity requires $\beta \geq 0$, $40 - \beta \geq 0$ and $60 - \beta \geq 0$, that is, $0 \leq \beta \leq 40$. So the diagonal may carry up to 40 cars per minute, which happens when street $1 \rightarrow 2$ is empty. Check, for $\mathbf{f} = (0, 20, 25, 0, 40)$: at vertex 1, out $0 + 40$ minus in 0 is 40; at 2, out 20 minus in 0 is 20; at 3, out 25 minus in $20 + 40$ is -35 ; at 4, out 0 minus in 25 is -25 .

What the model leaves out. Linear algebra finds every flow that conserves cars, including physically meaningless ones with negative entries, and it cannot say which of the many valid flows drivers actually choose. Non-negativity and street capacities turn the problem into one about inequalities, which is the subject of linear programming. The matrix NN^T of this model, the graph Laplacian, returns in Chapter 23.

2.7.4 Lights out over $\mathbb{F}_2$

A puzzle has a 3×3 grid of lights, each on or off. Pressing a light **toggles** it and its horizontal and vertical neighbors (a corner has two neighbors, an edge light three, the center four). Starting from some pattern, the goal is to switch every light off. Can every starting pattern be solved?

The model rests on two observations about the puzzle, and together they say that the right field is $\mathbb{F}_2 = \{0, 1\}$.

1. Encode “on” as 1 and “off” as 0. Toggling a light is adding 1 in $\mathbb{F}_2$, because $0 + 1 = 1$ and $1 + 1 = 0$. The effect of several presses on a light is the sum of their effects, and since addition is commutative, **the order of presses does not matter**.
2. Pressing a button twice toggles each affected light twice, which is $1 + 1 = 0$: nothing. So a strategy is just the **set** of buttons pressed once, a vector $\mathbf{x} \in \mathbb{F}_2^9$.

Number the cells (i, j) , row i , column j , and list them row by row. Pressing button (i, j) adds to the pattern a fixed vector $\mathbf{a}_{ij} \in \mathbb{F}_2^9$, with 1’s at (i, j) and its neighbors. Let $A \in M_9(\mathbb{F}_2)$ be the matrix with these columns. Starting from pattern $\mathbf{b}$ and pressing the buttons in $\mathbf{x}$, the final pattern is $\mathbf{b} + A\mathbf{x}$ by [Theorem 0.9.3](#), and we want it to be $\mathbf{0}$. Since $-\mathbf{b} = \mathbf{b}$ in $\mathbb{F}_2^9$, the puzzle is

the linear system

$$A\mathbf{x} = \mathbf{b} \quad \text{over } \mathbb{F}_2, \quad A = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}.$$

By [Theorem 2.1.1](#), pattern $\mathbf{b}$ is solvable exactly when $\mathbf{b} \in \text{col}(A)$. So “is every pattern solvable?” asks whether $\text{rank } A = 9$.

Example. (Every pattern of the 3×3 puzzle is solvable)

Show that $\text{rank } A = 9$. Deduce that every one of the $2^9 = 512$ starting patterns can be switched off, by exactly one set of presses.

Solution. We show that $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. Write the presses as a grid x_{ij} , and suppose $A\mathbf{x} = \mathbf{0}$: the presses change no light. The equation for light (i, j) says that the presses at (i, j) and its neighbors add up to 0. We solve these equations row by row, remembering that $-u = u$ in $\mathbb{F}_2$. Name the top row $x_{11} = p$, $x_{12} = q$, $x_{13} = r$.

Lights of row 1 determine the presses of row 2. Light $(1, 1)$ gives $x_{11} + x_{12} + x_{21} = 0$, so $x_{21} = p + q$. Likewise lights $(1, 2)$ and $(1, 3)$ give

$$x_{22} = x_{11} + x_{12} + x_{13} = p + q + r, \quad x_{23} = x_{12} + x_{13} = q + r.$$

Lights of row 2 determine the presses of row 3. Light $(2, 1)$ gives $x_{31} = x_{11} + x_{21} + x_{22} = p + (p + q) + (p + q + r) = p + r$, using $3p = p$ and $2q = 0$. Light $(2, 2)$ gives $x_{32} = x_{12} + x_{21} + x_{22} + x_{23} = q + (p + q) + (p + q + r) + (q + r) = 0$. Light $(2, 3)$ gives $x_{33} = x_{13} + x_{22} + x_{23} = r + (p + q + r) + (q + r) = p + r$.

Lights of row 3 give three conditions on p, q, r .

$$\begin{aligned} (3, 1) : \quad & x_{21} + x_{31} + x_{32} = (p + q) + (p + r) + 0 = q + r = 0, \\ (3, 2) : \quad & x_{22} + x_{31} + x_{32} + x_{33} = (p + q + r) + (p + r) + 0 + (p + r) = p + q + r = 0, \\ (3, 3) : \quad & x_{23} + x_{32} + x_{33} = (q + r) + 0 + (p + r) = p + q = 0. \end{aligned}$$

The first and third give $r = q$ and $p = q$, and then the second gives $3q = q = 0$. So $p = q = r = 0$, and the formulas above make every $x_{ij} = 0$.

Hence the null space of A is $\{\mathbf{0}\}$, and $\text{rank } A = 9 - 0 = 9$ by [Theorem 2.5.7](#). So $\text{col}(A)$ is a subspace of $\mathbb{F}_2^9$ of dimension 9, and $\text{col}(A) = \mathbb{F}_2^9$ by [Theorem 1.6.8](#). Every pattern $\mathbf{b}$ lies in $\text{col}(A)$, so every pattern is solvable. The solution is unique by [Corollary 2.1.4](#), since $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. In particular exactly one of the 512 press sets solves each of the 512 patterns.

The same chase finds an actual solution: the constants from $\mathbf{b}$ simply ride along.

Example. (Switching off the center)

Only the center light is on. Which buttons should be pressed?

Solution. Now $\mathbf{b}$ has a 1 at $(2, 2)$ only, so the equation for light $(2, 2)$ has right-hand side 1 and all others 0. Rows 1 and 2 of lights give, as before, $x_{21} = p + q$, $x_{22} = p + q + r$, $x_{23} = q + r$, $x_{31} = p + r$, $x_{33} = p + r$, and now

$$x_{32} = 1 + x_{12} + x_{21} + x_{22} + x_{23} = 1.$$

The lights of row 3 give $(p + q) + (p + r) + 1 = 0$, $(p + q + r) + (p + r) + 1 + (p + r) = 0$ and $(q + r) + 1 + (p + r) = 0$, that is,

$$q + r = 1, \quad p + q + r = 1, \quad p + q = 1.$$

Adding the first two gives $p = 0$; then $q = 1$ and $r = 0$. Substituting, $x_{21} = x_{22} = x_{23} = 1$, $x_{31} = x_{33} = 0$ and $x_{32} = 1$. So we press the center and its four neighbors, a plus sign:

	×	
×	×	×
	×	

Check, counting how often each light is toggled. The center is toggled by all five presses: odd, so it goes from on to off. A corner, say $(1, 1)$, is toggled by presses at $(1, 2)$ and $(2, 1)$: even, so it stays off. An edge light, say $(1, 2)$, is toggled by presses at $(1, 2)$ and $(2, 2)$: even, so it stays off. By the symmetry of the plus sign the other corners and edges behave the same way.

Quick check. Starting from all lights **off**, which set of presses produces the pattern with only the center light **on**?

Solution. The same plus sign. Starting from $\mathbf{0}$, presses $\mathbf{x}$ produce the pattern $A\mathbf{x}$, so we need $A\mathbf{x} = \mathbf{b}$ with $\mathbf{b}$ the center pattern, which is exactly the system just solved. Over $\mathbb{F}_2$, turning a pattern on and turning it off are the same problem.

Remark. Larger boards behave differently. A computation over $\mathbb{F}_2$, by the same elimination, gives rank $A = 12$ for the 4×4 board and rank $A = 23$ for the 5×5 board. On the 5×5 board, $\text{col}(A)$ has dimension 23, so it contains 2^{23} patterns (coordinates in a basis match its elements with $\mathbb{F}_2^{23}$), and only one pattern in four can be switched off.

What the model leaves out. Nothing, which is rare. The puzzle is linear over $\mathbb{F}_2$ by its rules, so the model is exact. It even predicts features that are not obvious when playing, such as the irrelevance of the order of presses.

2.7.5 Exercises

2.7.5.1 A. Check your understanding

Exercise A1.

- (a) State the polynomial interpolation theorem, including its hypotheses.
- (b) True or false: there is a polynomial $p \in \mathbb{R}[x]_{\leq 1}$ with $p(1) = 2$ and $p(1) = 3$. Which hypothesis of the theorem fails?
- (c) In balancing a chemical equation, what does it mean if the null space of the coefficient matrix has dimension 1? Dimension 2?
- (d) Why is lights out modeled over $\mathbb{F}_2$ and not over $\mathbb{R}$?
- (e) A connected network has 6 vertices and 9 edges. What is the rank of its incidence matrix, and how many free parameters does the solution set of a consistent flow problem have?

Solution.

- (a) See [Theorem 2.7.3](#): for **distinct** $x_0, \dots, x_n \in F$ and any $y_0, \dots, y_n \in F$, there is exactly one $p \in F[x]_{\leq n}$ with $p(x_i) = y_i$ for all i .
- (b) False: $p(1)$ is one number, and $2 \neq 3$. The theorem needs distinct x -values, and here $x_0 = x_1 = 1$.
- (c) Dimension 1: all balancings are multiples of one vector, so there is one balanced equation up to scaling (provided a positive multiple exists). Dimension 2 or more: there are independent balancings, and the equation is a mixture of several reactions whose proportions algebra cannot decide.
- (d) A light has two states, and toggling twice restores it, so toggling is adding 1 with $1 + 1 = 0$, which is arithmetic in $\mathbb{F}_2$. Over $\mathbb{R}$, pressing a button twice would count as 2, not 0.
- (e) By [Proposition 2.7.4](#) the rank is $6 - 1 = 5$. By [Corollary 2.7.5](#) a consistent problem has $9 - 6 + 1 = 4$ free parameters.

2.7.5.2 B. Practice**Exercise B1: Interpolating four points.**

Find the polynomial $p \in \mathbb{Q}[x]_{\leq 3}$ with $p(0) = -1$, $p(1) = 1$, $p(2) = 3$ and $p(3) = 11$. Justify that there is only one.

Solution. The x -values 0, 1, 2, 3 are distinct, so there is exactly one such $p = c_0 + c_1x + c_2x^2 + c_3x^3$, by [Theorem 2.7.3](#). Its coefficients solve

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 & 3 \\ 1 & 3 & 9 & 27 & 11 \end{array} \right) \xrightarrow[R_4 \rightarrow R_4 - R_1]{R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 2 & 4 & 8 & 4 \\ 0 & 3 & 9 & 27 & 12 \end{array} \right) \xrightarrow[R_4 \rightarrow R_4 - 3R_2]{R_3 \rightarrow R_3 - 2R_2} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 2 & 6 & 0 \\ 0 & 0 & 6 & 24 & 6 \end{array} \right)$$

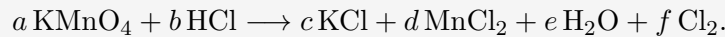
and then $R_4 - 3R_3$ turns the last row into $(0, 0, 0, 6 \mid 6)$. Back substitution: $c_3 = 1$; $2c_2 + 6 = 0$, so $c_2 = -3$; $c_1 - 3 + 1 = 2$, so $c_1 = 4$; $c_0 = -1$. Hence

$$p = -1 + 4x - 3x^2 + x^3.$$

Check: $p(0) = -1$, $p(1) = -1 + 4 - 3 + 1 = 1$, $p(2) = -1 + 8 - 12 + 8 = 3$, $p(3) = -1 + 12 - 27 + 27 = 11$.

Exercise B2: Balancing a redox reaction.

Potassium permanganate reacts with hydrochloric acid:



Write the conservation equations for K, Mn, O, H and Cl, find all rational solutions, and hence the balanced equation with the smallest positive integer coefficients.

Solution. Counting atoms of each element on both sides:

$$\text{K: } a = c, \quad \text{Mn: } a = d, \quad \text{O: } 4a = e, \quad \text{H: } b = 2e, \quad \text{Cl: } b = c + 2d + 2f.$$

In the order (a, b, c, d, e, f) , the coefficient matrix has rows $(1, 0, -1, 0, 0, 0)$, $(1, 0, 0, -1, 0, 0)$, $(4, 0, 0, 0, -1, 0)$, $(0, 1, 0, 0, -2, 0)$ and $(0, 1, -1, -2, 0, -2)$. The equations are nearly solved already, so we eliminate by substitution, which is row reduction written in words. The K, Mn and O equations give $c = a$, $d = a$, $e = 4a$. The H equation gives $b = 2e = 8a$. The Cl equation gives $2f = b - c - 2d = 8a - a - 2a = 5a$, so $a = \frac{2}{5}f$. Taking $f = t$ as the free variable, the RREF of the coefficient matrix is

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & -\frac{2}{5} \\ 0 & 1 & 0 & 0 & 0 & -\frac{16}{5} \\ 0 & 0 & 1 & 0 & 0 & -\frac{1}{5} \\ 0 & 0 & 0 & 1 & 0 & -\frac{1}{5} \\ 0 & 0 & 0 & 0 & 1 & -\frac{1}{5} \end{pmatrix},$$

and the solutions are $(a, b, c, d, e, f) = t(\frac{2}{5}, \frac{16}{5}, \frac{2}{5}, \frac{2}{5}, \frac{8}{5}, 1)$, $t \in \mathbb{Q}$. The rank is 5 and the nullity $6 - 5 = 1$, so the balancing is unique up to scaling. All entries are integers exactly when t is a multiple of 5, and $t = 5$ gives



Check: K $2 = 2$; Mn $2 = 2$; O $8 = 8$; H $16 = 16$; Cl $16 = 2 + 4 + 10$.

Exercise B3: A ring road.

Four intersections lie on a ring of one-way streets $1 \rightarrow 2$, $2 \rightarrow 3$, $3 \rightarrow 4$, $4 \rightarrow 1$, with flows f_1, f_2, f_3, f_4 in that order. From outside, 30 cars per minute enter at intersection 1 and 10 at intersection 3, while 15 leave at intersection 2 and 25 at intersection 4.

- Write down the incidence matrix N and the supply vector $\mathbf{s}$, and explain without computation why the system $N\mathbf{f} = \mathbf{s}$ is consistent.
- Find all solutions.
- If all flows must be non-negative, what is the smallest possible flow on each street?

Solution.

- (a) With rows for the vertices and columns for the edges,

$$N = \begin{pmatrix} 1 & 0 & 0 & -1 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{pmatrix}, \quad \mathbf{s} = \begin{pmatrix} 30 \\ -15 \\ 10 \\ -25 \end{pmatrix}.$$

The ring is connected, and $30 - 15 + 10 - 25 = 0$, so the system is consistent by [Corollary 2.7.5](#).

- (b) Row reducing $[N \mid \mathbf{s}]$: $R_2 + R_1$ gives $(0, 1, 0, -1 \mid 15)$; then $R_3 + R_2$ gives $(0, 0, 1, -1 \mid 25)$; then $R_4 + R_3$ gives the zero row. The pivots are in columns 1, 2, 3, and $f_4 = t$ is free, so

$$\mathbf{f} = (30 + t, 15 + t, 25 + t, t), \quad t \in \mathbb{R}.$$

There is $4 - 4 + 1 = 1$ parameter, the flow around the ring. Check at $t = 0$: vertex 1, $30 - 0 = 30$; vertex 2, $15 - 30 = -15$; vertex 3, $25 - 15 = 10$; vertex 4, $0 - 25 = -25$.

- (c) All flows are non-negative exactly when $t \geq 0$, since then $30 + t, 15 + t, 25 + t \geq 0$ too. Each flow increases with t , so all are smallest at $t = 0$: $f = (30, 15, 25, 0)$. No car needs to use street $4 \rightarrow 1$.

2.7.5.3 C. Going deeper

Exercise C1: The 2×2 puzzle.

Consider lights out on a 2×2 grid, where pressing a light toggles it and its two horizontal and vertical neighbors. Number the cells $1 = (1, 1)$, $2 = (1, 2)$, $3 = (2, 1)$, $4 = (2, 2)$.

- (a) Write down the matrix $A \in M_4(\mathbb{F}_2)$ of the puzzle.
 (b) Show that $A^2 = I$.
 (c) Deduce that every pattern is solvable by exactly one set of presses, and find the presses that switch off the pattern with only light 1 on.

Solution.

- (a) Button j toggles every light except the one diagonally opposite. So

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}.$$

- (b) Let J be the 4×4 all-ones matrix and Q the permutation matrix exchanging $1 \leftrightarrow 4$ and $2 \leftrightarrow 3$, which has 1's on the anti-diagonal. Then $A = J - Q = J + Q$ over $\mathbb{F}_2$. Every entry of J^2 is $1 + 1 + 1 + 1 = 0$, so $J^2 = 0$. Each row and each column of Q has a single 1, so $JQ = J$ and $QJ = J$. And $Q = P_\sigma$ for $\sigma = (1\ 4)(2\ 3)$, with $\sigma \circ \sigma = \text{id}$, so $Q^2 = P_{\text{id}} = I$ by [Lemma 2.6.6](#) (a). By [Theorem 0.9.4](#),

$$A^2 = J^2 + JQ + QJ + Q^2 = 0 + J + J + I = I,$$

because $J + J = 2J = 0$ over $\mathbb{F}_2$.

- (c) By (b), A is invertible with $A^{-1} = A$. By [Theorem 0.9.7](#) (7), $A\mathbf{x} = \mathbf{b}$ has exactly one solution $\mathbf{x} = A\mathbf{b}$ for every $\mathbf{b} \in \mathbb{F}_2^4$. For $\mathbf{b} = \mathbf{e}_1$, $\mathbf{x} = A\mathbf{e}_1 = (1, 1, 1, 0)$: press lights 1, 2, 3. Check: light 1 is toggled by presses 1, 2, 3, three times, so it goes off; light 2 by presses 1 and 2; light 3 by presses 1 and 3; light 4 by presses 2 and 3; each of these is toggled twice and stays off.

Exercise C2: Interpolating values and slopes.

For $p = c_0 + c_1x + c_2x^2 + c_3x^3 \in F[x]_{\leq 3}$, define its (formal) derivative $p' = c_1 + 2c_2x + 3c_3x^2$.

- (a) Prove that for **every** field F and all $a, b, c, d \in F$ there is exactly one $p \in F[x]_{\leq 3}$ with

$$p(0) = a, \quad p'(0) = b, \quad p(1) = c, \quad p'(1) = d.$$

- (b) Find this p over $\mathbb{Q}$ for $(a, b, c, d) = (0, 1, 1, 0)$.

Hint: in (a), check that elimination never divides by an element that could be zero in some field.

Solution.

- (a) The four conditions are linear in $c_0, \dots, c_3$: $p(0) = c_0$, $p'(0) = c_1$, $p(1) = c_0 + c_1 + c_2 + c_3$, $p'(1) = c_1 + 2c_2 + 3c_3$. Row reducing the augmented matrix, using only additions of multiples of rows,

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & a \\ 0 & 1 & 0 & 0 & b \\ 1 & 1 & 1 & 1 & c \\ 0 & 1 & 2 & 3 & d \end{array} \right) \xrightarrow[R_4 \rightarrow R_4 - R_2]{R_3 \rightarrow R_3 - R_1 - R_2} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & a \\ 0 & 1 & 0 & 0 & b \\ 0 & 0 & 1 & 1 & c - a - b \\ 0 & 0 & 2 & 3 & d - b \end{array} \right) \xrightarrow{R_4 \rightarrow R_4 - 2R_3} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & a \\ 0 & 1 & 0 & 0 & b \\ 0 & 0 & 1 & 1 & c - a - b \\ 0 & 0 & 0 & 1 & 2a + b - c \end{array} \right)$$

Here $3 - 2 = 1$ holds in every field. The coefficient matrix is now upper triangular with every diagonal entry equal to 1, so it is invertible by [Lemma 2.6.2](#), and back substitution gives the unique solution without any division:

$$c_3 = 2a + b - 2c + d, \quad c_2 = (c - a - b) - c_3 = -3a - 2b + 3c - d, \quad c_1 = b, \quad c_0 = a.$$

Row operations preserve the solution set ([Theorem 2.2.2](#)), so the original system also has exactly this one solution, for every field F .

- (b) With $(a, b, c, d) = (0, 1, 1, 0)$: $c_3 = 0 + 1 - 2 + 0 = -1$, $c_2 = 0 - 2 + 3 - 0 = 1$, $c_1 = 1$, $c_0 = 0$. So $p = x + x^2 - x^3$. Check: $p(0) = 0$; $p' = 1 + 2x - 3x^2$, so $p'(0) = 1$; $p(1) = 1 + 1 - 1 = 1$; $p'(1) = 1 + 2 - 3 = 0$.

Exercise C3: Networks in several pieces.

Let $N \in M_{m \times k}(\mathbb{R})$ be the incidence matrix of a network with m vertices. Say that vertices v and w are **linked** if they are joined by a sequence of vertices in which consecutive ones share an edge, in either direction (every vertex is linked to itself by the sequence of length zero).

- (a) Show that “linked” is an equivalence relation. Its equivalence classes are called

the **components** of the network; let c be their number.

- (b) Prove that $\text{rank } N = m - c$.

Solution.

- (a) *Reflexive*: by the sequence of length zero. *Symmetric*: reversing a sequence from v to w gives one from w to v , and consecutive vertices still share an edge. *Transitive*: following a sequence from u to v by one from v to w gives a sequence from u to w . So “linked” is an equivalence relation ([Definition 0.5.2](#)).
- (b) Let $C_1, \dots, C_c$ be the components, and let $\mathbf{1}_{C_r} \in \mathbb{R}^m$ have entry 1 at the vertices of C_r and 0 elsewhere. As in the proof of [Proposition 2.7.4](#), $N^T \mathbf{y} = \mathbf{0}$ if and only if $y_u = y_w$ for every edge joining u and w . Following sequences of edges, this holds if and only if $\mathbf{y}$ is constant on each component, that is, $\mathbf{y} = \sum_r \lambda_r \mathbf{1}_{C_r}$ with λ_r the common value on C_r . (Conversely each $\mathbf{1}_{C_r}$ is in the null space, since both ends of an edge lie in the same component.) So $(\mathbf{1}_{C_1}, \dots, \mathbf{1}_{C_c})$ spans $\text{nul}(N^T)$. It is independent: if $\sum_r \lambda_r \mathbf{1}_{C_r} = \mathbf{0}$, then reading the entry at any vertex of C_r , where only the r -th vector is non-zero because the components are disjoint ([Theorem 0.5.2](#)), gives $\lambda_r = 0$. Hence $\dim \text{nul}(N^T) = c$. Since N^T has m columns, [Theorem 2.5.7](#) gives $\text{rank } N^T = m - c$, and [Theorem 2.5.5](#) gives $\text{rank } N = m - c$, as claimed.

Chapter 3

Linear Maps

Chapter 1 studied vector spaces one at a time. This chapter studies the functions between them that respect the structure: **linear maps**, the functions that send sums to sums and scalar multiples to scalar multiples. Differentiation of polynomials, the trace of a matrix, a rotation of the plane and multiplication by a matrix are all linear maps, and one theory covers them all.

Two subspaces measure how far a linear map is from being injective and surjective: its kernel and its image. The Rank–Nullity Theorem ties their dimensions together, and turns many questions about maps into counting. Then we choose bases and write every linear map as a matrix. This explains why matrix multiplication is defined the way it is, and it raises the central question of the rest of the book: how does the matrix change when the bases change, and how simple can we make it?

What you need. From Chapter 1, bases and dimension ([Definition 1.5.1](#), [Definition 1.6.1](#)), basis extension ([Theorem 1.6.5](#)) and counting instead of checking ([Theorem 1.6.4](#)). From Chapter 2, the null space and column space of a matrix ([Definition 2.5.3](#), [Definition 2.5.1](#)), its rank ([Definition 2.5.4](#), [Theorem 2.5.7](#)) and the invertible matrix theorem ([Theorem 2.4.4](#)). From Chapter 0, quotient sets and well-defined functions on them ([Theorem 0.5.3](#)), for the section on quotient spaces.

3.0.1 Roadmap

- **Linear maps.** The definition, a gallery of examples and non-examples, and the fact that a linear map is determined by its values on a basis.
- **Kernel and image.** The two subspaces attached to every linear map, and the kernel test for injectivity.
- **The Rank–Nullity Theorem.** $\dim V = \text{nullity } T + \text{rank } T$, and what counting can decide.
- **The algebra of linear maps.** Adding, scaling and composing maps; polynomials in an operator.
- **Isomorphisms.** When two spaces are “the same”, and why dimension alone decides it.
- **The matrix of a linear map.** Coordinates turn maps into matrices and composition into matrix multiplication.
- **Change of basis and similarity.** How the matrix of a map depends on the chosen bases.
- **Rank, equivalence and factorizations.** With free choice of bases on both sides, every

matrix becomes $\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$.

- **Products and quotient spaces.** Building new spaces from old ones, and the isomorphism theorems.
- **Projections and the trace.** Direct sums seen through operators, and a basis-free trace.

3.0.2 Named moves

- **Injective via the kernel.** To show T is injective, show $T\mathbf{v} = \mathbf{0}$ forces $\mathbf{v} = \mathbf{0}$.
- **Count with Rank–Nullity.** When $\dim V = \dim W$, injective, surjective and bijective are the same; check the easiest one.
- **Basis of the kernel, then extend.** The proof pattern behind Rank–Nullity, the same one that proved the dimension formula for sums.
- **Abstract space $\rightarrow$ write down a matrix.** Choose a basis that makes the map simple, then compute with its matrix.
- **The change-of-basis square.** Go around the square instead of across it: $[T]_{\mathcal{B}'} = P^{-1}[T]_{\mathcal{B}}P$.
- **Check well-definedness on cosets.** Every operation on a quotient space, and every map out of one, starts with this check.

3.1 Linear Maps

Chapter 1 studied vector spaces one at a time, and Chapter 2 studied one kind of question inside F^n : does $A\mathbf{x} = \mathbf{b}$ have a solution? This chapter studies the functions **between** vector spaces that respect addition and scaling. This section defines them, collects examples from geometry, matrices, calculus and algebra, and proves the fact that makes them manageable: a linear map is completely determined by what it does to a basis.

3.1.1 Functions that behave like matrices

Objects come first, then the maps between them that respect their structure. For sets the maps are functions; for groups they are homomorphisms (Definition 0.10.3). For vector spaces the structure is addition and scalar multiplication, so we want functions that respect those two operations.

The standard example is a matrix. For $A \in M_{m \times n}(F)$, the rules of matrix multiplication (Theorem 0.9.4) give

$$A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y}, \quad A(c\mathbf{x}) = c(A\mathbf{x}) \quad \text{for all } \mathbf{x}, \mathbf{y} \in F^n, c \in F.$$

Now look at three functions that are not given by matrices at all.

- **Differentiation** on polynomials: $(p + q)' = p' + q'$ and $(cp)' = cp'$.
- **Trace** on square matrices: $\text{tr}(A + B) = \text{tr} A + \text{tr} B$ and $\text{tr}(cA) = c \text{tr} A$ (Theorem 0.9.6).
- **Rotation** of the plane about the origin: rotating a parallelogram gives a parallelogram, so the rotation of a sum is the sum of the rotations, and rotating a stretched arrow gives the stretched rotated arrow.

The inputs are polynomials, matrices and arrows, not columns. Yet all three obey exactly the two rules that matrices obey. Whatever we can prove from those two rules alone will hold for all of them at once. So we give the two rules a name.

A linear map is a function that preserves linear combinations.

3.1.2 The definition

Definition 3.1.1 — Linear Map

Let V and W be vector spaces **over the same field** F . A function $T: V \rightarrow W$ is a **linear map** (or **linear transformation**) **over** F if

(LT1) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ **for all** $\mathbf{u}, \mathbf{v} \in V$, and

(LT2) $T(c\mathbf{v}) = cT(\mathbf{v})$ **for all** $c \in F$ and **all** $\mathbf{v} \in V$.

A linear map $T: V \rightarrow V$ from a space to itself is also called a **linear operator** on V . We often write $T\mathbf{v}$ for $T(\mathbf{v})$.

In words: (LT1), **additivity**, says that adding first and then applying T gives the same result as applying T first and then adding. The sum on the left is taken in V ; the sum on the right is taken in W . (LT2), **homogeneity**, says the same for scaling: the scalar c can be pulled out of T . Both conditions must hold for **every** choice of vectors and scalars, so one counterexample is enough to show that a function is not linear.

The phrase “over the same field” matters. Scalars from F act on both sides of (LT2), so both spaces must accept them. We will see below that one function can be linear over $\mathbb{R}$ and not linear over $\mathbb{C}$. A general linear map is always named T , S or R in this book, never f ; a specific map may get a letter that recalls what it does, such as D for differentiation.

3.1.3 Examples

We check the examples against (LT1) and (LT2), grouped by where they come from.

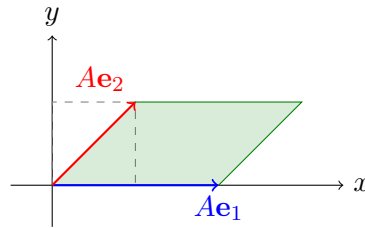
Matrices.

Example. (Multiplication by a Matrix)

Let $A \in M_{m \times n}(F)$. Show that $T_A: F^n \rightarrow F^m$, $T_A(\mathbf{x}) = A\mathbf{x}$, is a linear map.

Solution. Let $\mathbf{x}, \mathbf{y} \in F^n$ and $c \in F$. By the distributive law in [Theorem 0.9.4](#), $T_A(\mathbf{x} + \mathbf{y}) = A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y} = T_A(\mathbf{x}) + T_A(\mathbf{y})$, which is (LT1). By the scalar rule in the same theorem, $T_A(c\mathbf{x}) = A(c\mathbf{x}) = c(A\mathbf{x}) = cT_A(\mathbf{x})$, which is (LT2). Hence T_A is linear.

We will use the name T_A for this map throughout the chapter. For $F = \mathbb{R}$ and $n = m = 2$, the map T_A sends the unit square to the parallelogram spanned by $A\mathbf{e}_1$ and $A\mathbf{e}_2$, the columns of A ([Theorem 0.9.3](#)).



The unit square (dashed) and its image under $A = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$.

Example. (Maps Given by Formulas in Coordinates)

- (a) Show that $T: F^3 \rightarrow F^2$, $T(a_1, a_2, a_3) = (2a_1 + a_3, a_2 - a_1)$, is linear.
- (b) Let $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of a vector space V over F . Show that the **coordinate map** $V \rightarrow F^n$, $\mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}}$, is linear.

Solution.

- (a) The formula is a matrix product:

$$\begin{pmatrix} 2 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} 2a_1 + a_3 \\ -a_1 + a_2 \end{pmatrix}.$$

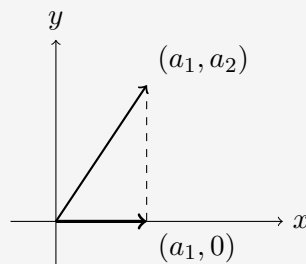
So $T = T_A$ for this 2×3 matrix A , and T is linear by [Example \(Multiplication by a Matrix\)](#). The same argument shows that every map $F^n \rightarrow F^m$ whose output entries are combinations of the input entries, with constant coefficients and no constant terms, is linear.

- (b) By [Theorem 1.5.2](#), $[\mathbf{u} + \mathbf{v}]_{\mathcal{B}} = [\mathbf{u}]_{\mathcal{B}} + [\mathbf{v}]_{\mathcal{B}}$ and $[c\mathbf{v}]_{\mathcal{B}} = c[\mathbf{v}]_{\mathcal{B}}$. These are (LT1) and (LT2).

Geometry.

Example. (Projection onto an Axis)

Show that $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T(a_1, a_2) = (a_1, 0)$, is linear.



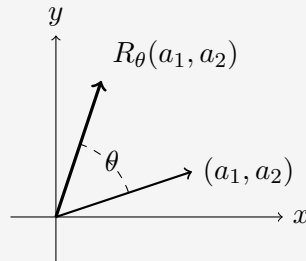
Solution. Let $\mathbf{u} = (a_1, a_2)$, $\mathbf{v} = (b_1, b_2)$ and $c \in \mathbb{R}$. Then

$$T(\mathbf{u} + \mathbf{v}) = (a_1 + b_1, 0) = (a_1, 0) + (b_1, 0) = T(\mathbf{u}) + T(\mathbf{v}), \quad T(c\mathbf{u}) = (ca_1, 0) = c(a_1, 0) = cT(\mathbf{u}).$$

So (LT1) and (LT2) hold. The map drops each point vertically onto the x -axis, like a shadow cast by light from straight above.

Example. (Rotation of the Plane)

Let $\theta \in \mathbb{R}$, and let $R_\theta: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ rotate each vector counterclockwise about the origin by the angle θ . Find a formula for R_θ , and deduce that it is linear.



Solution. Write $(a_1, a_2) = (r \cos \varphi, r \sin \varphi)$ in polar coordinates. Rotating adds θ to the angle and keeps the length r , so by the addition formulas for sine and cosine,

$$R_\theta(a_1, a_2) = (r \cos(\varphi + \theta), r \sin(\varphi + \theta)) = (a_1 \cos \theta - a_2 \sin \theta, a_1 \sin \theta + a_2 \cos \theta).$$

This is T_A for $A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$, so R_θ is linear by [Example \(Multiplication by a Matrix\)](#).

Calculus. For a polynomial $p = a_0 + a_1x + \cdots + a_nx^n \in F[x]$, define its **(formal) derivative** as $p' = a_1 + 2a_2x + \cdots + na_nx^{n-1}$, where ka_k means a_k added to itself k times. Over $\mathbb{R}$ this is the derivative of calculus.

Example. (Differentiation)

Show that $D: F[x] \rightarrow F[x]$, $D(p) = p'$, is linear. Show also that D maps $F[x]_{\leq n}$ into $F[x]_{\leq n-1}$ for $n \geq 1$.

Solution. Let $p = \sum_k a_k x^k$ and $q = \sum_k b_k x^k$, and $c \in F$. The coefficient of x^{k-1} in $D(p+q)$ is $k(a_k + b_k) = ka_k + kb_k$, which is the coefficient of x^{k-1} in $D(p) + D(q)$. The coefficient of x^{k-1} in $D(cp)$ is $k(ca_k) = c(ka_k)$, which is the coefficient in $cD(p)$. Both steps use only the field axioms. Two polynomials with the same coefficients are equal ([Definition 0.8.1](#)), so (LT1) and (LT2) hold. If $\deg p \leq n$, the highest power in p' is at most x^{n-1} , so D restricts to a linear map $F[x]_{\leq n} \rightarrow F[x]_{\leq n-1}$.

Example. (Integration)

Show that $J: \mathbb{R}[x] \rightarrow \mathbb{R}[x]$, $J(p)(x) = \int_0^x p(t) dt$, is linear.

Solution. On coefficients, J sends $a_0 + a_1x + \cdots + a_nx^n$ to $a_0x + \frac{a_1}{2}x^2 + \cdots + \frac{a_n}{n+1}x^{n+1}$,

so it does map polynomials to polynomials. By linearity of the integral from calculus,

$$J(p+q)(x) = \int_0^x (p(t)+q(t)) dt = J(p)(x)+J(q)(x), \quad J(cp)(x) = \int_0^x cp(t) dt = cJ(p)(x)$$

for every $x \in \mathbb{R}$. Hence J is linear. (The coefficient formula divides by $k+1$; over a field such as $\mathbb{F}_2$, where $1+1=0$, it breaks down.)

Algebra and sequences.

Example. (Multiplication by x)

Show that $M: F[x] \rightarrow F[x]$, $M(p) = xp$, is linear.

Solution. By the distributive laws in [Theorem 0.8.1](#), $x(p+q) = xp+xq$, and $x(cp) = c(xp)$ because the product of polynomials is associative and commutative and the constant c is a polynomial. These are (LT1) and (LT2).

Example. (Trace and Transpose)

Show that the trace $\text{tr}: M_n(F) \rightarrow F$ and the transpose $M_{m \times n}(F) \rightarrow M_{n \times m}(F)$, $A \mapsto A^\top$, are linear.

Solution. For the trace, [Theorem 0.9.6](#) (1) says $\text{tr}(A+B) = \text{tr} A + \text{tr} B$ and $\text{tr}(cA) = c \text{tr} A$, which are (LT1) and (LT2); the target is F , regarded as a vector space over itself. For the transpose, the (i, j) entry of $(A+B)^\top$ is $a_{ji} + b_{ji}$, the (i, j) entry of $A^\top + B^\top$; and the (i, j) entry of $(cA)^\top$ is ca_{ji} , the (i, j) entry of $cA^\top$.

Example. (The Shift Maps on Sequences)

On the space $F^\mathbb{N}$ of sequences ([Example \(The Standard Vector Spaces\)](#)), define the **right shift** and the **left shift**

$$R(s_0, s_1, s_2, \dots) = (0, s_0, s_1, \dots), \quad L(s_0, s_1, s_2, \dots) = (s_1, s_2, s_3, \dots).$$

Show that both are linear.

Solution. The operations on $F^\mathbb{N}$ are termwise. The entry in position $k \geq 1$ of $R(\mathbf{s} + \mathbf{t})$ is $s_{k-1} + t_{k-1}$, which is the entry of $R(\mathbf{s}) + R(\mathbf{t})$; in position 0 both have $0 = 0 + 0$. Similarly, $R(c\mathbf{s})$ has entries $0 = c \cdot 0$ and cs_{k-1} , the entries of $cR(\mathbf{s})$. For L , the entry in position k of $L(\mathbf{s} + \mathbf{t})$ is $s_{k+1} + t_{k+1}$, and of $L(c\mathbf{s})$ is cs_{k+1} , as required.

The right shift is the map promised in the warning of Chapter 1, §8, on what changes in infinite dimension: it matches $F^\mathbb{N}$ one-to-one with the sequences whose first entry is 0, respecting addition and scaling. In the next section we measure exactly how R and L fail to be bijective.

Example. (Composition with a Fixed Function)

Let X and Y be sets and $g: X \rightarrow Y$ a function. Show that $C_g: F^Y \rightarrow F^X$, $C_g(h) = h \circ g$,

is linear. Deduce that $\mathbb{R}[x] \rightarrow \mathbb{R}[x]$, $p(x) \mapsto p(x+1)$, is linear.

Solution. The operations on function spaces are pointwise (Example (The Standard Vector Spaces)). For $t \in X$,

$$C_g(h+k)(t) = (h+k)(g(t)) = h(g(t)) + k(g(t)) = (C_g(h) + C_g(k))(t),$$

and $C_g(ch)(t) = ch(g(t)) = (cC_g(h))(t)$. So (LT1) and (LT2) hold. The output is again a function, not a number; linearity is about the function h varying, while g stays fixed.

For the second claim, take $X = Y = \mathbb{R}$ and $g(t) = t+1$, and regard each real polynomial as a function $\mathbb{R} \rightarrow \mathbb{R}$. Then $p(x+1) = C_g(p)$, which is again a polynomial of the same degree. The identities $C_g(p+q) = C_g(p) + C_g(q)$ and $C_g(cp) = cC_g(p)$ hold as functions, hence as polynomials, because two real polynomials with the same values everywhere are equal (a consequence of Lemma 2.7.2). So the map is linear. (It translates the graph of p one unit to the left.)

The degenerate maps. Three maps are always available: the identity on any space, the inclusion of any subspace, and the zero map between any two spaces. They look too simple to matter. They matter because every general statement about linear maps must be true for them, so they are the first test of any conjecture.

Example. (Identity Map)

The identity $\text{id}_V: V \rightarrow V$, $\mathbf{v} \mapsto \mathbf{v}$, is linear: $\text{id}_V(\mathbf{u} + \mathbf{v}) = \mathbf{u} + \mathbf{v} = \text{id}_V(\mathbf{u}) + \text{id}_V(\mathbf{v})$ and $\text{id}_V(c\mathbf{v}) = c\mathbf{v} = c \text{id}_V(\mathbf{v})$.

Example. (Inclusion Map)

If U is a subspace of V , the inclusion $\iota: U \rightarrow V$, $\iota(\mathbf{u}) = \mathbf{u}$, is linear, because the operations of U are those of V . Its formula is the identity's, but its codomain is bigger. For instance $\mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}[x]$ is an inclusion.

Example. (Zero Map)

For any V and W over F , the zero map $0: V \rightarrow W$, $\mathbf{v} \mapsto \mathbf{0}_W$, is linear: $\mathbf{0}_W = \mathbf{0}_W + \mathbf{0}_W$ and $\mathbf{0}_W = c\mathbf{0}_W$ (Theorem 1.1.6).

3.1.4 Three consequences of linearity

The two axioms already force several other rules. The first is the quickest test for non-linearity.

Theorem 3.1.1 — Linear Maps Send Zero to Zero

Let $T: V \rightarrow W$ be linear. Then $T(\mathbf{0}_V) = \mathbf{0}_W$.

Proof. By (LT1), $T(\mathbf{0}_V) = T(\mathbf{0}_V + \mathbf{0}_V) = T(\mathbf{0}_V) + T(\mathbf{0}_V)$. Adding $-T(\mathbf{0}_V)$ to both sides gives $\mathbf{0}_W = T(\mathbf{0}_V)$. ■

The proof used only (LT1). Homogeneity with $c = 0$ would also give it: $T(\mathbf{0}_V) = T(0\mathbf{v}) = 0T(\mathbf{v}) = \mathbf{0}_W$.

Theorem 3.1.2 — Linear Maps Preserve Negatives

Let $T: V \rightarrow W$ be linear. Then $T(-\mathbf{v}) = -T(\mathbf{v})$ and $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v} \in V$.

Proof. By Theorem 1.1.7, $-\mathbf{v} = (-1)\mathbf{v}$. So (LT2) gives $T(-\mathbf{v}) = (-1)T(\mathbf{v}) = -T(\mathbf{v})$, using Theorem 1.1.7 again in W . Then (LT1) gives $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u} + (-\mathbf{v})) = T(\mathbf{u}) + T(-\mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v})$. ■

Checking two conditions separately is often longer than necessary. One combined condition suffices.

Theorem 3.1.3 — One-Line Test for Linearity

Let V and W be vector spaces over F , and let $T: V \rightarrow W$ be a function. Then T is linear **if and only if**

$$T(c\mathbf{u} + \mathbf{v}) = cT(\mathbf{u}) + T(\mathbf{v}) \quad \text{for all } c \in F \text{ and all } \mathbf{u}, \mathbf{v} \in V.$$

Proof. ($\Rightarrow$) Suppose T is linear. By (LT1) and then (LT2), $T(c\mathbf{u} + \mathbf{v}) = T(c\mathbf{u}) + T(\mathbf{v}) = cT(\mathbf{u}) + T(\mathbf{v})$.

($\Leftarrow$) Suppose the condition holds. Taking $c = 1$ gives $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$, which is (LT1). Taking $\mathbf{u} = \mathbf{v} = \mathbf{0}$ and $c = 1$ gives $T(\mathbf{0}) = T(\mathbf{0}) + T(\mathbf{0})$, so $T(\mathbf{0}) = \mathbf{0}$. Now taking $\mathbf{v} = \mathbf{0}$ gives $T(c\mathbf{u}) = cT(\mathbf{u}) + T(\mathbf{0}) = cT(\mathbf{u})$, which is (LT2). ■

Finally, the slogan. Two applications of the axioms handle a combination of two vectors; induction handles any finite combination.

Theorem 3.1.4 — Linear Maps Preserve Linear Combinations

Let $T: V \rightarrow W$ be linear, let $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$ and $a_1, \dots, a_k \in F$, where $k \geq 0$. Then

$$T(a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k) = a_1T(\mathbf{v}_1) + \dots + a_kT(\mathbf{v}_k).$$

Proof. We use induction on k . For $k = 0$ both sides are empty sums, so the claim is $T(\mathbf{0}) = \mathbf{0}$, which is Theorem 3.1.1. Let $k \geq 1$ and suppose the claim holds for $k - 1$

vectors. By (LT1), then the induction hypothesis and (LT2),

$$T\left(\sum_{i=1}^{k-1} a_i \mathbf{v}_i + a_k \mathbf{v}_k\right) = T\left(\sum_{i=1}^{k-1} a_i \mathbf{v}_i\right) + T(a_k \mathbf{v}_k) = \sum_{i=1}^{k-1} a_i T(\mathbf{v}_i) + a_k T(\mathbf{v}_k).$$

This completes the induction. ■

In words: to apply T to a combination, apply it to each vector and keep the same coefficients. This is the property the slogan promised, and it is the property every later theorem about linear maps uses.

3.1.5 Non-examples

A function fails to be linear as soon as **one** instance of (LT1) or (LT2) fails. The cheapest instance to try is $\mathbf{0}$, by [Theorem 3.1.1](#).

Example. (A Constant Function)

Is $T: \mathbb{R} \rightarrow \mathbb{R}$, $T(x) = 5$, linear?

Solution. No. A linear map sends 0 to 0 by [Theorem 3.1.1](#), but $T(0) = 5 \neq 0$.

Example. (Translation)

Let $\mathbf{b} \in \mathbb{R}^2$ be **non-zero**. Is the translation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T(\mathbf{v}) = \mathbf{v} + \mathbf{b}$, linear?

Solution. No: $T(\mathbf{0}) = \mathbf{b} \neq \mathbf{0}$, which is impossible for a linear map by [Theorem 3.1.1](#). This is a minimal change of the identity map ([Example \(Identity Map\)](#)): the formula differs only by the constant $\mathbf{b}$, and that constant is exactly what breaks linearity. For the record, (LT1) fails too: $T(\mathbf{u} + \mathbf{v}) = \mathbf{u} + \mathbf{v} + \mathbf{b}$, while $T(\mathbf{u}) + T(\mathbf{v}) = \mathbf{u} + \mathbf{v} + 2\mathbf{b}$. Maps of the form $\mathbf{v} \mapsto A\mathbf{v} + \mathbf{b}$ are called **affine**; they are linear exactly when $\mathbf{b} = \mathbf{0}$.

Passing the zero test does not make a function linear.

Example. (Squaring)

Is $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T(a_1, a_2) = (a_1^2, 0)$, linear?

Solution. No. Here $T(\mathbf{0}) = \mathbf{0}$, so the quick test says nothing. But (LT2) fails for $\mathbf{v} = (1, 0)$ and $c = 2$: $T(2\mathbf{v}) = T(2, 0) = (4, 0)$, while $2T(\mathbf{v}) = 2(1, 0) = (2, 0)$. Compare with the projection $(a_1, a_2) \mapsto (a_1, 0)$ of [Example \(Projection onto an Axis\)](#): squaring the first entry is the one change, and it destroys homogeneity.

Example. (Length)

Is $T: \mathbb{R}^2 \rightarrow \mathbb{R}$, $T(x, y) = \sqrt{x^2 + y^2}$, linear?

Solution. No. Take $\mathbf{v} = (1, 0)$ and $c = -1$. Then $T(-\mathbf{v}) = 1$, but $-T(\mathbf{v}) = -1$, so (LT2) fails. Length does satisfy $T(c\mathbf{v}) = |c|T(\mathbf{v})$, which agrees with $cT(\mathbf{v})$ only for $c \geq 0$. Additivity fails too: $T((1, 0) + (0, 1)) = \sqrt{2}$, but $T(1, 0) + T(0, 1) = 2$.

The last non-example shows that linearity depends on the field, not only on the function.

Example. (Complex Conjugation: the Field Matters)

Let $T: \mathbb{C} \rightarrow \mathbb{C}$, $T(z) = \bar{z}$. Is T linear when $\mathbb{C}$ is regarded as a vector space **over** $\mathbb{C}$? Over $\mathbb{R}$?

Solution. (LT1) holds in both cases, since $\overline{z + w} = \bar{z} + \bar{w}$ (Theorem 0.6.2).

Over $\mathbb{C}$: not linear. The scalars are all complex numbers. Take $c = i$ and $z = 1$: then $T(i \cdot 1) = \bar{i} = -i$, but $iT(1) = i \cdot 1 = i$. So (LT2) fails.

Over $\mathbb{R}$: linear. Now (LT2) is only required for real c . For $c \in \mathbb{R}$, $T(cz) = \overline{cz} = c\bar{z} = cT(z)$, by Theorem 0.6.2 and $\bar{c} = c$.

Warning. “**Linear**” here is not the “linear function” of school. A function $f(x) = mx + b$ has a straight-line graph, but it is a linear map $\mathbb{R} \rightarrow \mathbb{R}$ only when $b = 0$: $f(0) = b$, and Theorem 3.1.1 demands $f(0) = 0$. In fact every linear map $T: \mathbb{R} \rightarrow \mathbb{R}$ has the form $T(x) = mx$: by (LT2), $T(x) = T(x \cdot 1) = xT(1)$, so $m = T(1)$. And $T(\mathbf{0}) = \mathbf{0}$ is **necessary but not sufficient** for linearity, as Example (Squaring) shows.

Quick check. Which of these maps $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ are linear? (i) $(x, y) \mapsto (y, x)$; (ii) $(x, y) \mapsto (x + y, 1)$; (iii) $(x, y) \mapsto (xy, 0)$.

Solution.

- (i) Linear: it is T_A with $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. (ii) Not linear: $(0, 0) \mapsto (0, 1) \neq \mathbf{0}$. (iii) Not linear, although $\mathbf{0} \mapsto \mathbf{0}$: $(2, 2) \mapsto (4, 0)$, but $2 \cdot T(1, 1) = 2(1, 0) = (2, 0)$, so (LT2) fails.

Why this definition. Could one condition be dropped? Complex conjugation on $\mathbb{C}$ over $\mathbb{C}$ satisfies (LT1) but not (LT2), so additivity alone is not enough. (Over $\mathbb{Q}$, surprisingly, additivity does imply homogeneity; see Exercise (C1: Over $\mathbb{Q}$, Additivity Is Enough).) Conversely, $T: \mathbb{R}^2 \rightarrow \mathbb{R}$, $T(x, y) = \sqrt[3]{x^3 + y^3}$ (the real cube root), satisfies (LT2), since $\sqrt[3]{c^3x^3 + c^3y^3} = c\sqrt[3]{x^3 + y^3}$ for every real c . But $T(1, 0) + T(0, 1) = 2$, while $T(1, 1) = \sqrt[3]{2}$, so (LT1) fails. Neither condition implies the other, and we need both to get Theorem 3.1.4. The name comes from the case $\mathbb{R} \rightarrow \mathbb{R}$ just described: the graph of a linear map is a line through the origin.

3.1.6 A linear map is determined by a basis

A function on an infinite set normally needs infinitely many values to describe it. A linear map needs far fewer. If $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ is a basis of V , every $\mathbf{v} \in V$ is a combination $a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n$, and [Theorem 3.1.4](#) forces

$$T(\mathbf{v}) = a_1T(\mathbf{v}_1) + \dots + a_nT(\mathbf{v}_n).$$

So the n vectors $T(\mathbf{v}_1), \dots, T(\mathbf{v}_n)$ already fix every value of T . The converse question is whether **any** choice of these n vectors comes from a linear map. It does, and that is the theorem.

Theorem 3.1.5 — Linear Maps Are Determined by a Basis

Let V and W be vector spaces over F , let $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of V , and let $\mathbf{w}_1, \dots, \mathbf{w}_n \in W$ be **any** vectors. Then there **exists exactly one** linear map $T: V \rightarrow W$ such that

$$T(\mathbf{v}_i) = \mathbf{w}_i \quad \text{for } i = 1, \dots, n.$$

It is given by $T(a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n) = a_1\mathbf{w}_1 + \dots + a_n\mathbf{w}_n$ for all $a_1, \dots, a_n \in F$.

Idea. **Uniqueness** is the computation above: any linear T with these values must be given by the formula. **Existence** asks us to *use* the formula as a definition. The only danger is that a vector might have two expressions $\sum a_i\mathbf{v}_i$, giving two candidate values; the basis rules that out, because coordinates are unique. So: ① define $T(\mathbf{v})$ from the coordinates of $\mathbf{v}$; ② check (LT1), (LT2), using that coordinates respect the operations; ③ check $T(\mathbf{v}_i) = \mathbf{w}_i$; ④ prove uniqueness by [Theorem 3.1.4](#).

Proof. Existence. Let $\mathbf{v} \in V$. By [Theorem 1.5.1](#), there are **unique** scalars $a_1, \dots, a_n \in F$ with $\mathbf{v} = a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n$; they are the entries of $[\mathbf{v}]_{\mathcal{B}}$ ([Definition 1.5.2](#)). Define

$$T(\mathbf{v}) := a_1\mathbf{w}_1 + \dots + a_n\mathbf{w}_n.$$

Since the scalars are unique, this assigns exactly one vector of W to each $\mathbf{v} \in V$, so $T: V \rightarrow W$ is a well-defined function.

Let $\mathbf{u}, \mathbf{v} \in V$ and $c \in F$, with $[\mathbf{u}]_{\mathcal{B}} = (b_1, \dots, b_n)$ and $[\mathbf{v}]_{\mathcal{B}} = (a_1, \dots, a_n)$. By [Theorem 1.5.2](#), $[c\mathbf{u} + \mathbf{v}]_{\mathcal{B}} = c[\mathbf{u}]_{\mathcal{B}} + [\mathbf{v}]_{\mathcal{B}} = (cb_1 + a_1, \dots, cb_n + a_n)$. Hence, by the definition of T and the vector space axioms in W ,

$$T(c\mathbf{u} + \mathbf{v}) = \sum_{i=1}^n (cb_i + a_i)\mathbf{w}_i = c \sum_{i=1}^n b_i\mathbf{w}_i + \sum_{i=1}^n a_i\mathbf{w}_i = cT(\mathbf{u}) + T(\mathbf{v}).$$

By [Theorem 3.1.3](#), T is linear. Moreover, $\mathbf{v}_i = 0\mathbf{v}_1 + \dots + 1\mathbf{v}_i + \dots + 0\mathbf{v}_n$, so $[\mathbf{v}_i]_{\mathcal{B}} = \mathbf{e}_i$ and $T(\mathbf{v}_i) = \mathbf{w}_i$.

Uniqueness. Let $S: V \rightarrow W$ be any linear map with $S(\mathbf{v}_i) = \mathbf{w}_i$ for all i . For $\mathbf{v} = a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n \in V$, [Theorem 3.1.4](#) gives

$$S(\mathbf{v}) = a_1S(\mathbf{v}_1) + \dots + a_nS(\mathbf{v}_n) = a_1\mathbf{w}_1 + \dots + a_n\mathbf{w}_n = T(\mathbf{v}).$$

Since $S(\mathbf{v}) = T(\mathbf{v})$ for every $\mathbf{v} \in V$, the functions are equal: $S = T$. This proves the theorem.

The theorem has two halves, and they are used differently. Uniqueness says that **to check two linear maps are equal, check them on a basis**. Existence says that **to build a linear map, choose the images of a basis freely**. The same holds for the infinite bases of Chapter 1, §8, because every vector is still a unique **finite** combination of basis vectors; we record this as [Theorem 3.1.6](#) below. Later in this chapter this theorem turns into the matrix of a linear map: once bases are fixed, the n vectors $T(\mathbf{v}_i)$, written in coordinates, are the columns.

Theorem 3.1.6 — Linear Maps on an Arbitrary Basis

Let V and W be vector spaces over F , let B be a basis of V (possibly infinite, in the sense of [Theorem 1.8.2](#)), and for each $\mathbf{b} \in B$ let $\mathbf{w}_{\mathbf{b}} \in W$ be any vector. Then there is exactly one linear map $T: V \rightarrow W$ with $T\mathbf{b} = \mathbf{w}_{\mathbf{b}}$ for every $\mathbf{b} \in B$.

Proof. Let $\mathbf{v} \in V$. Since B spans V and is independent, $\mathbf{v}$ is a finite combination $\mathbf{v} = \sum_{\mathbf{b} \in B} a_{\mathbf{b}} \mathbf{b}$ in which only finitely many $a_{\mathbf{b}}$ are non-zero, and these coefficients are unique: two such expressions differ by a finite combination of distinct elements of B equal to $\mathbf{0}$, whose coefficients all vanish by independence. Define $T\mathbf{v} = \sum_{\mathbf{b} \in B} a_{\mathbf{b}} \mathbf{w}_{\mathbf{b}}$, a finite sum. The coefficients of $\mathbf{v} + \mathbf{u}$ and of $c\mathbf{v}$ are $a_{\mathbf{b}} + a'_{\mathbf{b}}$ and $ca_{\mathbf{b}}$ by uniqueness, so T is linear, and $T\mathbf{b} = \mathbf{w}_{\mathbf{b}}$ because the coefficients of $\mathbf{b}$ are 1 at $\mathbf{b}$ and 0 elsewhere. If S is another linear map with $S\mathbf{b} = \mathbf{w}_{\mathbf{b}}$ for all $\mathbf{b}$, then $S\mathbf{v} = \sum a_{\mathbf{b}} S\mathbf{b} = T\mathbf{v}$ by [Theorem 3.1.4](#). This proves the theorem.

Example. (Finding a Formula from Values on a Basis)

Show that there is exactly one linear map $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ with

$$T(1, 1) = (1, 2, 0), \quad T(1, -1) = (3, 0, 2),$$

and find $T(x, y)$.

Solution. First, $((1, 1), (1, -1))$ is a basis of $\mathbb{R}^2$. The matrix with these columns is $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, and $ad - bc = 1 \cdot (-1) - 1 \cdot 1 = -2 \neq 0$. So the matrix is invertible ([Theorem 0.9.8](#)), and its columns form a basis of $\mathbb{R}^2$ ([Theorem 2.4.4](#)). By [Theorem 3.1.5](#), exactly one such T exists.

To find it, write (x, y) in the basis. Solving $a(1, 1) + b(1, -1) = (x, y)$, that is $a + b = x$ and $a - b = y$, gives $a = \frac{x+y}{2}$ and $b = \frac{x-y}{2}$. By the formula in the theorem,

$$T(x, y) = \frac{x+y}{2}(1, 2, 0) + \frac{x-y}{2}(3, 0, 2) = (2x - y, x + y, x - y).$$

Check: $T(1, 1) = (1, 2, 0)$ and $T(1, -1) = (3, 0, 2)$.

Example. (Too Many Conditions)

Is there a linear map $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ with

$$T(1, 0) = (2, 3, 4), \quad T(1, 1) = (3, 4, 5), \quad T(2, 3) = (4, 5, 6)?$$

Solution. No. The matrix $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ with columns $(1, 0)$ and $(1, 1)$ has $ad - bc = 1 \cdot 1 - 1 \cdot 0 = 1 \neq 0$, so as in [Example \(Finding a Formula from Values on a Basis\)](#) these two vectors form a basis of $\mathbb{R}^2$. By [Theorem 3.1.5](#), the first two conditions already determine T , and the third condition must be checked against them, not imposed. Since $(2, 3) = -1 \cdot (1, 0) + 3 \cdot (1, 1)$, any linear T with the first two values satisfies

$$T(2, 3) = -T(1, 0) + 3T(1, 1) = -(2, 3, 4) + 3(3, 4, 5) = (7, 9, 11)$$

by [Theorem 3.1.4](#). As $(7, 9, 11) \neq (4, 5, 6)$, no such T exists. Had the third value been $(7, 9, 11)$, the answer would be yes, with exactly one such map.

Example. (Recognizing Differentiation)

Show that the differentiation map $D: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$ is the **only** linear map $T: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$ with

$$T(1) = 0, \quad T(x) = 1, \quad T(x^2) = 2x, \quad T(x^3) = 3x^2.$$

Solution. The list $(1, x, x^2, x^3)$ is a basis of $\mathbb{R}[x]_{\leq 3}$ ([Example \(Standard Bases\)](#)). The map D is linear ([Example \(Differentiation\)](#)), and $D(1) = 0$, $D(x) = 1$, $D(x^2) = 2x$, $D(x^3) = 3x^2$, so D has the four prescribed values. By the uniqueness half of [Theorem 3.1.5](#), every linear T with these values equals D . Four values, the derivatives of the monomials, pin down the derivative of every cubic.

Warning. The values must be prescribed on a basis, or at least on an independent list. On a dependent list, the values may contradict each other. There is **no** linear $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with $T(1, 0) = (1, 0)$ and $T(2, 0) = (0, 1)$: since $(2, 0) = 2(1, 0)$, (LT2) forces $T(2, 0) = 2T(1, 0) = (2, 0) \neq (0, 1)$. On the other hand, a list that does not span leaves T undetermined: many linear maps satisfy $T(1, 0) = (1, 0)$ alone.

Quick check. Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}$ be linear with $T(1, 2) = 3$ and $T(2, 5) = 1$. Find $T(x, y)$.

Solution. Here $ad - bc = 1 \cdot 5 - 2 \cdot 2 = 1 \neq 0$, so $((1, 2), (2, 5))$ is a basis. Solving $a(1, 2) + b(2, 5) = (x, y)$, that is $a + 2b = x$ and $2a + 5b = y$, gives $b = y - 2x$ and $a = 5x - 2y$. Hence $T(x, y) = 3(5x - 2y) + 1 \cdot (y - 2x) = 13x - 5y$. Check: $13 - 10 = 3$ and $26 - 25 = 1$.

We now have a supply of linear maps and a way to build new ones from a basis. The next section attaches to every linear map two subspaces that measure how far it is from being injective and from being surjective.

3.1.7 Exercises

3.1.7.1 A. Check your understanding

Exercise A1.

- (a) Define a linear map $T: V \rightarrow W$ over F .
- (b) State the theorem that says a linear map is determined by a basis, including both existence and uniqueness.
- (c) True or false: if $T: V \rightarrow W$ satisfies $T(\mathbf{0}) = \mathbf{0}$, then T is linear. Justify your answer.
- (d) True or false: there is a linear map $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with $T(1, 1) = (1, 0)$ and $T(2, 2) = (3, 0)$. Justify your answer.
- (e) True or false: complex conjugation $\mathbb{C} \rightarrow \mathbb{C}$ is linear over $\mathbb{R}$. Justify your answer.

Solution.

- (a) See [Definition 3.1.1](#): V, W are vector spaces over the same field F , and $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and $T(c\mathbf{v}) = cT(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v} \in V$ and $c \in F$.
- (b) See [Theorem 3.1.5](#): if $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is a basis of V and $\mathbf{w}_1, \dots, \mathbf{w}_n \in W$ are arbitrary, there is one and only one linear $T: V \rightarrow W$ with $T(\mathbf{v}_i) = \mathbf{w}_i$ for all i .
- (c) False. $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T(a_1, a_2) = (a_1^2, 0)$, sends $\mathbf{0}$ to $\mathbf{0}$ but is not linear ([Example \(Squaring\)](#)).
- (d) False. By (LT2), $T(2, 2) = 2T(1, 1) = (2, 0) \neq (3, 0)$.
- (e) True. See [Example \(Complex Conjugation: the Field Matters\)](#): for real c , $\overline{cz} = c\overline{z}$, and $\overline{z + w} = \overline{z} + \overline{w}$.

3.1.7.2 B. Practice

Exercise B1: Which Maps Are Linear?.

Determine which of the following functions are linear maps over $\mathbb{R}$. Justify your answer.

- (a) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $T(x, y, z) = (x - 2z, 3y)$.
- (b) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T(x, y) = (x + 1, y)$.
- (c) $T: M_2(\mathbb{R}) \rightarrow \mathbb{R}$, $T\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$.
- (d) $T: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}[x]_{\leq 3}$, $T(p) = x^2 p'$.
- (e) $T: \mathbb{R}^2 \rightarrow \mathbb{R}$, $T(x, y) = xy$.
- (f) $T: \mathbb{R}^{\mathbb{N}} \rightarrow \mathbb{R}^{\mathbb{N}}$, $T(s_0, s_1, s_2, \dots) = (s_1 - s_0, s_2 - s_1, s_3 - s_2, \dots)$.

Solution.

- (a) Linear. $T = T_A$ with $A = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 3 & 0 \end{pmatrix}$, which is linear by [Example \(Multiplication by a Matrix\)](#).
- (b) Not linear. $T(0, 0) = (1, 0) \neq \mathbf{0}$, contradicting [Theorem 3.1.1](#).

- (c) Not linear. Take I_2 and $c = 2$: $T(2I_2) = 2 \cdot 2 - 0 = 4$, but $2T(I_2) = 2 \cdot 1 = 2$. So (LT2) fails. (Additivity fails too: $T(E_{11}) + T(E_{22}) = 0 + 0 = 0$, but $T(E_{11} + E_{22}) = T(I_2) = 1$.)
- (d) Linear. For $p \in \mathbb{R}[x]_{\leq 2}$, p' has degree at most 1, so $x^2 p' \in \mathbb{R}[x]_{\leq 3}$. For p, q and $c \in \mathbb{R}$, by [Example \(Differentiation\)](#) and the distributive law, $T(cp + q) = x^2(cp' + q') = cx^2 p' + x^2 q' = cT(p) + T(q)$. By [Theorem 3.1.3](#), T is linear.
- (e) Not linear, although $T(0, 0) = 0$. $T(2(1, 1)) = T(2, 2) = 4$, but $2T(1, 1) = 2$.
- (f) Linear. The entry in position k of $T(cs + \mathbf{t})$ is $(cs_{k+1} + t_{k+1}) - (cs_k + t_k) = c(s_{k+1} - s_k) + (t_{k+1} - t_k)$, which is the entry in position k of $cT(\mathbf{s}) + T(\mathbf{t})$. By [Theorem 3.1.3](#), T is linear.

Exercise B2: A Formula from Values on a Basis.

- (a) Show that there is exactly one linear map $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with $T(2, 1) = (1, 4)$ and $T(1, 1) = (0, 3)$. Hence find $T(x, y)$ and a matrix A with $T = T_A$.
- (b) Let $S: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}$ be linear with $S(1) = 1$, $S(1 + x) = 3$ and $S(1 + x + x^2) = 6$. Find $S(a + bx + cx^2)$.

Solution.

- (a) The matrix with columns $(2, 1)$ and $(1, 1)$ has $ad - bc = 2 \cdot 1 - 1 \cdot 1 = 1 \neq 0$, so it is invertible ([Theorem 0.9.8](#)) and its columns form a basis of $\mathbb{R}^2$ ([Theorem 2.4.4](#)). By [Theorem 3.1.5](#), exactly one such T exists. Solving $a(2, 1) + b(1, 1) = (x, y)$, that is $2a + b = x$ and $a + b = y$, gives $a = x - y$ and $b = 2y - x$. Hence

$$T(x, y) = (x - y)(1, 4) + (2y - x)(0, 3) = (x - y, x + 2y).$$

Check: $T(2, 1) = (1, 4)$ and $T(1, 1) = (0, 3)$. Hence $T = T_A$ with $A = \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}$.

- (b) The polynomials $1, 1 + x, 1 + x + x^2$ are non-zero of distinct degrees, so they are independent ([Theorem 1.4.4](#)); there are $3 = \dim \mathbb{R}[x]_{\leq 2}$ of them, so they form a basis ([Theorem 1.6.4](#)). By (LT1) and [Theorem 3.1.2](#), $S(x) = S(1 + x) - S(1) = 2$ and $S(x^2) = S(1 + x + x^2) - S(1 + x) = 3$. By [Theorem 3.1.4](#),

$$S(a + bx + cx^2) = aS(1) + bS(x) + cS(x^2) = a + 2b + 3c.$$

Exercise B3: Evaluation and Integration.

- (a) Prove that $\varphi: \mathbb{R}[x] \rightarrow \mathbb{R}$, $\varphi(p) = p(3)$, is linear.
- (b) Prove that $\psi: \mathbb{R}[x] \rightarrow \mathbb{R}$, $\psi(p) = \int_0^1 p(t) dt$, is linear.
- (c) Hence find $\varphi(a + bx + cx^2)$ and $\psi(a + bx + cx^2)$.

Solution.

- (a) Let $p, q \in \mathbb{R}[x]$ and $c \in \mathbb{R}$. By [Theorem 0.8.5](#), $(cp + q)(3) = cp(3) + q(3)$, where cp is the product of the constant polynomial c with p , whose value at 3 is $cp(3)$. So $\varphi(cp + q) = c\varphi(p) + \varphi(q)$, and φ is linear by [Theorem 3.1.3](#).
- (b) By linearity of the integral, $\int_0^1 (cp(t) + q(t)) dt = c \int_0^1 p(t) dt + \int_0^1 q(t) dt$. So $\psi(cp +$

$q) = c\psi(p) + \psi(q)$, and ψ is linear by [Theorem 3.1.3](#).

- (c) On the basis: $\varphi(1) = 1$, $\varphi(x) = 3$, $\varphi(x^2) = 9$, and $\psi(1) = 1$, $\psi(x) = \frac{1}{2}$, $\psi(x^2) = \frac{1}{3}$. By [Theorem 3.1.4](#),

$$\varphi(a + bx + cx^2) = a + 3b + 9c, \quad \psi(a + bx + cx^2) = a + \frac{1}{2}b + \frac{1}{3}c.$$

3.1.7.3 C. Going deeper

Exercise C1: Over $\mathbb{Q}$, Additivity Is Enough.

Let V and W be vector spaces over $\mathbb{Q}$, and let $T: V \rightarrow W$ be **additive**: $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v} \in V$.

- (a) Prove that $T(n\mathbf{v}) = nT(\mathbf{v})$ for every $n \in \mathbb{N}$ and $\mathbf{v} \in V$.
- (b) Prove that $T(n\mathbf{v}) = nT(\mathbf{v})$ for every $n \in \mathbb{Z}$.
- (c) Prove that $T(q\mathbf{v}) = qT(\mathbf{v})$ for every $q \in \mathbb{Q}$. Deduce that T is linear.

Hint: for (c), apply (b) to the vector $q\mathbf{v}$.

Solution.

- (a) We use induction on n . For $n = 0$: the proof of [Theorem 3.1.1](#) used only additivity, so $T(\mathbf{0}) = \mathbf{0}$, that is, $T(0\mathbf{v}) = 0T(\mathbf{v})$. If $T(n\mathbf{v}) = nT(\mathbf{v})$, then by additivity $T((n+1)\mathbf{v}) = T(n\mathbf{v} + \mathbf{v}) = T(n\mathbf{v}) + T(\mathbf{v}) = nT(\mathbf{v}) + T(\mathbf{v}) = (n+1)T(\mathbf{v})$.
- (b) Let $n \in \mathbb{N}$. By additivity and $T(\mathbf{0}) = \mathbf{0}$, $\mathbf{0} = T(n\mathbf{v} + (-n)\mathbf{v}) = T(n\mathbf{v}) + T((-n)\mathbf{v})$, so $T((-n)\mathbf{v}) = -T(n\mathbf{v}) = -nT(\mathbf{v}) = (-n)T(\mathbf{v})$, using (a). Every integer is n or $-n$ for some $n \in \mathbb{N}$.
- (c) Let $q = m/n$ with $m \in \mathbb{Z}$ and $n \geq 1$. By (b) applied to the vector $q\mathbf{v}$, and then to $\mathbf{v}$,

$$nT(q\mathbf{v}) = T(nq\mathbf{v}) = T(m\mathbf{v}) = mT(\mathbf{v}).$$

Multiplying by $\frac{1}{n} \in \mathbb{Q}$ gives $T(q\mathbf{v}) = \frac{m}{n}T(\mathbf{v}) = qT(\mathbf{v})$. So (LT2) holds for every scalar $q \in \mathbb{Q}$, and with additivity, which is (LT1), T is linear.

Exercise C2: Over $\mathbb{C}$, Additivity Is Not Enough.

- (a) Give an additive map $\mathbb{C} \rightarrow \mathbb{C}$ that is **not** linear over $\mathbb{C}$, and explain which step of [Exercise \(C1: Over \$\mathbb{Q}\$, Additivity Is Enough\)](#) fails for it.
- (b) Let V and W be vector spaces over $\mathbb{C}$. Regarding them also as spaces over $\mathbb{R}$, let $T: V \rightarrow W$ be linear **over** $\mathbb{R}$, and suppose $T(i\mathbf{v}) = iT(\mathbf{v})$ for all $\mathbf{v} \in V$. Prove that T is linear over $\mathbb{C}$.

Solution.

- (a) Complex conjugation $T(z) = \bar{z}$ is additive and not $\mathbb{C}$ -linear ([Example \(Complex Conjugation: the Field Matters\)](#)). The argument of [Exercise \(C1: Over \$\mathbb{Q}\$, Additivity Is Enough\)](#) still shows $T(qz) = qT(z)$ for rational q , and that is true for conjugation. What fails is that the rational numbers are not all the scalars: nothing in that argument reaches a scalar like i , which is not built from 1 by adding,

subtracting, multiplying and dividing.

- (b) (LT1) holds because T is linear over $\mathbb{R}$. For (LT2), let $c = a + bi \in \mathbb{C}$ with $a, b \in \mathbb{R}$, and $\mathbf{v} \in V$. By the vector space axioms, $c\mathbf{v} = a\mathbf{v} + b(i\mathbf{v})$. Using (LT1), then $\mathbb{R}$ -homogeneity, then the hypothesis,

$$T(c\mathbf{v}) = T(a\mathbf{v}) + T(b(i\mathbf{v})) = aT(\mathbf{v}) + bT(i\mathbf{v}) = aT(\mathbf{v}) + b iT(\mathbf{v}) = (a + bi)T(\mathbf{v}) = cT(\mathbf{v}).$$

Hence T is linear over $\mathbb{C}$.

3.2 Kernel and Image

For any function, Chapter 0 asked two questions: is it injective, and is it surjective? For a function between sets, answering them means comparing outputs pair by pair and hunting for preimages point by point. For a linear map both questions reduce to a single subspace each. This section defines the two subspaces, the kernel and the image, shows how to compute them, and proves the test that makes injectivity cheap: a linear map is injective exactly when its kernel is zero.

3.2.1 Two questions, two subspaces

Let $T: V \rightarrow W$ be linear. Start with injectivity. Suppose two inputs collide, $T(\mathbf{u}) = T(\mathbf{v})$. By [Theorem 3.1.2](#) this is the same as

$$T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v}) = \mathbf{0}.$$

So a collision between $\mathbf{u}$ and $\mathbf{v}$ is the same thing as their difference being sent to $\mathbf{0}$. To detect every collision at once, it is enough to know **which vectors are sent to $\mathbf{0}$** . For a general function this trick is unavailable, because there is no subtraction; linearity turns a question about all pairs into a question about one set.

Surjectivity asks whether every $\mathbf{w} \in W$ is an output, so the set to know is **the set of all outputs**. Chapter 2 met both sets for $T_A(\mathbf{x}) = A\mathbf{x}$: the system $A\mathbf{x} = \mathbf{b}$ is consistent exactly when $\mathbf{b}$ is in the column space ([Theorem 2.1.1](#)), and a consistent system has a unique solution exactly when $A\mathbf{x} = \mathbf{0}$ has only the trivial solution ([Corollary 2.1.4](#)). The definitions below give these two sets a name for every linear map. They are defined for what they measure.

The kernel is everything T sends to zero; the image is everything T reaches.

Definition 3.2.1 — Kernel

Let $T: V \rightarrow W$ be a linear map over F . The **kernel** of T is the subset of the **domain**

$$\ker T := \{\mathbf{v} \in V : T(\mathbf{v}) = \mathbf{0}_W\} \subseteq V.$$

Definition 3.2.2 — Image

Let $T: V \rightarrow W$ be a linear map over F . The **image** of T is the subset of the **codomain**

$$\operatorname{im} T := \{T(\mathbf{v}) : \mathbf{v} \in V\} = \{\mathbf{w} \in W : \mathbf{w} = T(\mathbf{v}) \text{ for some } \mathbf{v} \in V\} \subseteq W.$$

In words: a vector $\mathbf{v}$ is in $\ker T$ when $\mathbf{v}$ lives in V **and** T sends it to the zero vector of W . A vector $\mathbf{w}$ is in $\operatorname{im} T$ when $\mathbf{w}$ lives in W **and** there **exists** at least one input with output $\mathbf{w}$; the input need not be unique. In the language of [Definition 0.4.2](#), $\ker T = T^{-1}(\{\mathbf{0}\})$ is the preimage of the zero vector, and $\operatorname{im} T = T(V)$ is the image of the function T . The kernel is also called the **null space** of T , and the image its **range**. For group homomorphisms, the kernel was defined the same way ([Definition 0.10.3](#)), with $\mathbf{0}$ playing the role of the identity element.

There is nothing to check for well-definedness: both sets are described by conditions that either hold or fail. What does need checking is that they are **subspaces**, not just subsets.

Theorem 3.2.1 — The Kernel Is a Subspace

Let $T: V \rightarrow W$ be linear. Then $\ker T$ is a subspace of V .

Proof. We use the Subspace Test ([Theorem 1.2.1](#)).

- (1) By [Theorem 3.1.1](#), $T(\mathbf{0}_V) = \mathbf{0}_W$, so $\mathbf{0}_V \in \ker T$.
- (2) Let $\mathbf{u}, \mathbf{v} \in \ker T$, so $T(\mathbf{u}) = T(\mathbf{v}) = \mathbf{0}_W$. By (LT1), $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) = \mathbf{0}_W + \mathbf{0}_W = \mathbf{0}_W$. Therefore $\mathbf{u} + \mathbf{v} \in \ker T$.
- (3) Let $\mathbf{v} \in \ker T$ and $c \in F$. By (LT2), $T(c\mathbf{v}) = cT(\mathbf{v}) = c\mathbf{0}_W = \mathbf{0}_W$, using [Theorem 1.1.6](#). Therefore $c\mathbf{v} \in \ker T$.

■

Theorem 3.2.2 — The Image Is a Subspace

Let $T: V \rightarrow W$ be linear. Then:

- (a) $\operatorname{im} T$ is a subspace of W ;
- (b) T is surjective if and only if $\operatorname{im} T = W$.

Proof.

- (a) We use the Subspace Test ([Theorem 1.2.1](#)). (1) $\mathbf{0}_W = T(\mathbf{0}_V)$ by [Theorem 3.1.1](#), so $\mathbf{0}_W \in \operatorname{im} T$. (2) Let $\mathbf{w}_1, \mathbf{w}_2 \in \operatorname{im} T$, say $\mathbf{w}_1 = T(\mathbf{v}_1)$ and $\mathbf{w}_2 = T(\mathbf{v}_2)$ with $\mathbf{v}_1, \mathbf{v}_2 \in V$. By (LT1), $\mathbf{w}_1 + \mathbf{w}_2 = T(\mathbf{v}_1 + \mathbf{v}_2)$, and $\mathbf{v}_1 + \mathbf{v}_2 \in V$, so $\mathbf{w}_1 + \mathbf{w}_2 \in \operatorname{im} T$. (3) Let $\mathbf{w} = T(\mathbf{v}) \in \operatorname{im} T$ and $c \in F$. By (LT2), $c\mathbf{w} = T(c\mathbf{v})$ with $c\mathbf{v} \in V$, so $c\mathbf{w} \in \operatorname{im} T$.
- (b) The inclusion $\operatorname{im} T \subseteq W$ always holds. By [Definition 0.4.3](#), T is surjective exactly when every $\mathbf{w} \in W$ equals $T(\mathbf{v})$ for some $\mathbf{v}$, that is, when $W \subseteq \operatorname{im} T$. So T is surjective if and only if $\operatorname{im} T = W$.

■

This pays off a pattern from Chapter 1, §2. The null space of a matrix ([Example \(Solutions of a Homogeneous System\)](#)), the symmetric matrices ([Example \(Symmetric Matrices\)](#)) and the trace-free matrices ([Example \(Trace-Free Matrices\)](#)) were each shown to be subspaces by a separate check, and that section promised a single theorem. Each is the kernel of a linear map (T_A , $A \mapsto A - A^T$, and the trace), so [Theorem 3.2.1](#) covers them all at once.

First examples. Start with the maps that have the most extreme kernels and images.

Example. (The Zero Map and the Identity)

Find the kernel and image of the zero map $0: V \rightarrow W$ and of the identity $\text{id}_V: V \rightarrow V$.

Solution. The zero map sends every vector to $\mathbf{0}_W$, so $\ker 0 = V$ and $\text{im } 0 = \{\mathbf{0}_W\}$. The identity sends $\mathbf{v}$ to $\mathbf{v}$, which is $\mathbf{0}$ only when $\mathbf{v} = \mathbf{0}$, so $\ker \text{id}_V = \{\mathbf{0}\}$. Every $\mathbf{v}$ is its own image, so $\text{im } \text{id}_V = V$.

These two are the ends of the scale. The kernel is always between $\{\mathbf{0}\}$ and V , and the image between $\{\mathbf{0}\}$ and W . The zero map kills everything and reaches nothing but $\mathbf{0}$; the identity kills nothing and reaches everything.

Example. (Projection onto an Axis)

Find the kernel and image of the projection $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T(a_1, a_2) = (a_1, 0)$, of [Example \(Projection onto an Axis\)](#).

Solution. $T(a_1, a_2) = (0, 0)$ exactly when $a_1 = 0$. So $\ker T = \{(0, a_2) : a_2 \in \mathbb{R}\}$, the y -axis. Every output has second entry 0, and every $(a_1, 0)$ is an output, namely $T(a_1, 0)$. So $\text{im } T$ is the x -axis. The kernel is the direction the light shines along; the image is the screen the shadow falls on.

A non-example by minimal change. Replace $\mathbf{0}$ by another vector: for the projection, $\{\mathbf{v} : T(\mathbf{v}) = (1, 0)\} = \{(1, a_2) : a_2 \in \mathbb{R}\}$. This is the vertical line through $(1, 0)$. It does not contain $\mathbf{0}$, and it is not closed under addition, since $(1, 0) + (1, 0) = (2, 0)$ lies off the line. So it is **not** a subspace. It is the kernel shifted by $(1, 0)$. The same thing happened for systems in Chapter 2: the solutions of $A\mathbf{x} = \mathbf{b}$ are $\mathbf{p} + \text{nul}(A)$ ([Theorem 2.1.3](#)).

Why the preimage of $\mathbf{0}$? Because $\mathbf{0}$ is the one output that every linear map is guaranteed to hit, so its preimage is never empty, and because all other preimages are translates of it. If $T(\mathbf{p}) = \mathbf{w}$, then $T(\mathbf{v}) = \mathbf{w}$ exactly when $T(\mathbf{v} - \mathbf{p}) = \mathbf{0}$, that is, when $\mathbf{v} \in \mathbf{p} + \ker T$. The kernel describes the shape of every non-empty preimage at once.

Warning. Keep track of where each subspace lives. The kernel sits inside the **domain** V ; the image sits inside the **codomain** W . For $T_A: F^n \rightarrow F^m$, the kernel consists of columns with n entries and the image of columns with m entries, so “ $\ker T = \text{im } T$ ” does not even make sense unless $V = W$. And the image is **not** the codomain in general: the differentiation map $D: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 3}$ has codomain $\mathbb{R}[x]_{\leq 3}$, but x^3 is not the derivative of any cubic, so $\text{im } D \neq \mathbb{R}[x]_{\leq 3}$.

3.2.2 Injectivity via the kernel

The computation that opened the section now becomes a theorem. It is the reason to prefer the kernel over the definition of injectivity whenever a map is linear.

Theorem 3.2.3 — Kernel Test for Injectivity

Let $T: V \rightarrow W$ be linear. Then T is injective **if and only if** $\ker T = \{0\}$.

Proof. ($\Rightarrow$) Suppose T is injective, and let $\mathbf{v} \in \ker T$. Then $T(\mathbf{v}) = \mathbf{0}_W = T(\mathbf{0}_V)$ by Theorem 3.1.1. Injectivity gives $\mathbf{v} = \mathbf{0}_V$. Hence $\ker T \subseteq \{0\}$, and the reverse inclusion holds because $\ker T$ is a subspace (Theorem 3.2.1).

($\Leftarrow$) Suppose $\ker T = \{0\}$, and let $T(\mathbf{u}) = T(\mathbf{v})$. By Theorem 3.1.2, $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v}) = \mathbf{0}$, so $\mathbf{u} - \mathbf{v} \in \ker T = \{0\}$. Hence $\mathbf{u} = \mathbf{v}$, and T is injective. ■

Compare the work. The definition asks: for **all pairs** $\mathbf{u}, \mathbf{v}$, does $T(\mathbf{u}) = T(\mathbf{v})$ force $\mathbf{u} = \mathbf{v}$? The theorem asks only: for **each** $\mathbf{v}$, does $T(\mathbf{v}) = \mathbf{0}$ force $\mathbf{v} = \mathbf{0}$? The second question is usually a homogeneous linear system or a coefficient comparison. For T_A , the theorem says that $\mathbf{x} \mapsto A\mathbf{x}$ is injective exactly when $A\mathbf{x} = \mathbf{0}$ has only the trivial solution, which is the content of Corollary 2.1.4.

3.2.3 Computing the image

The image is defined by an existence condition, which is hard to check directly. When V has a spanning list, applying T to it gives a spanning list of $\text{im } T$.

Theorem 3.2.4 — The Image Is Spanned by the Images of a Spanning List

Let $T: V \rightarrow W$ be linear, and suppose $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ spans V . Then

$$\text{im } T = \text{span}(T(\mathbf{v}_1), \dots, T(\mathbf{v}_k)).$$

In particular, if V is finite-dimensional, then so is $\text{im } T$, even when W is not.

Proof. ($\supseteq$) Each $T(\mathbf{v}_i)$ lies in $\text{im } T$, which is a subspace by Theorem 3.2.2. By Theorem 1.3.1 (3), the span of these vectors is contained in $\text{im } T$.

($\subseteq$) Let $\mathbf{w} \in \text{im } T$, say $\mathbf{w} = T(\mathbf{v})$ with $\mathbf{v} \in V$. Since $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ spans V , $\mathbf{v} = a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k$ for some $a_i \in F$. By Theorem 3.1.4, $\mathbf{w} = a_1T(\mathbf{v}_1) + \dots + a_kT(\mathbf{v}_k) \in \text{span}(T(\mathbf{v}_1), \dots, T(\mathbf{v}_k))$.

If V is finite-dimensional, some finite list spans V (Definition 1.3.4), and its images form a finite list spanning $\text{im } T$. ■

So the recipe for the image is: apply T to a basis (or any spanning list) of V , then sift the results to a basis if one is wanted ([Theorem 1.5.4](#)). The images of a basis need not be independent; the projection sends the basis $(\mathbf{e}_1, \mathbf{e}_2)$ to $(\mathbf{e}_1, \mathbf{0})$.

3.2.4 Examples

We now compute kernels and images across the families. In each case we then read off injectivity from the kernel and surjectivity from the image.

Example. (A Matrix Map)

Let $A = \begin{pmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 8 \\ -1 & -2 & 1 & -1 \end{pmatrix} \in M_{3 \times 4}(\mathbb{R})$. Find bases of $\ker T_A$ and $\operatorname{im} T_A$ for $T_A: \mathbb{R}^4 \rightarrow \mathbb{R}^3$, and decide whether T_A is injective or surjective.

Solution. By definition, $\ker T_A = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\} = \operatorname{nul}(A)$ ([Definition 2.5.3](#)) and $\operatorname{im} T_A = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^4\} = \operatorname{col}(A)$ ([Definition 2.5.1](#)). So Chapter 2's methods apply. Row reduce:

$$A \xrightarrow[R_3+R_1]{R_2-2R_1} \begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 2 \end{pmatrix} \xrightarrow{R_3-R_2} \begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

This is the RREF, with pivot columns 1 and 3 and free columns 2 and 4. By [Theorem 2.5.4](#), a basis of $\ker T_A$ is

$$((-2, 1, 0, 0), (-3, 0, -2, 1)).$$

By [Theorem 2.5.2](#), the pivot columns of A form a basis of $\operatorname{im} T_A$: $((1, 2, -1), (0, 1, 1))$.

Since $\ker T_A \neq \{\mathbf{0}\}$, T_A is not injective ([Theorem 3.2.3](#)); for instance $T_A(-2, 1, 0, 0) = \mathbf{0} = T_A(\mathbf{0})$. Since $\dim \operatorname{im} T_A = 2 < 3 = \dim \mathbb{R}^3$, $\operatorname{im} T_A \neq \mathbb{R}^3$, so T_A is not surjective ([Theorem 3.2.2](#)).

Example. (Differentiation)

Find the kernel and image of $D: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$, $D(p) = p'$.

Solution. Kernel. Let $p = a_0 + a_1x + a_2x^2 + a_3x^3$. Then $D(p) = a_1 + 2a_2x + 3a_3x^2$, which is the zero polynomial exactly when $a_1 = 2a_2 = 3a_3 = 0$. Since $2 \neq 0$ and $3 \neq 0$ in $\mathbb{R}$, this means $a_1 = a_2 = a_3 = 0$. So $\ker D = \{a_0 : a_0 \in \mathbb{R}\} = \operatorname{span}(1)$, the constants.

Image. By [Theorem 3.2.4](#) applied to the basis $(1, x, x^2, x^3)$,

$$\operatorname{im} D = \operatorname{span}(D(1), D(x), D(x^2), D(x^3)) = \operatorname{span}(0, 1, 2x, 3x^2) = \operatorname{span}(1, x, x^2) = \mathbb{R}[x]_{\leq 2}.$$

So D is surjective but not injective: p and $p + 5$ have the same derivative. Note $\dim \ker D + \dim \operatorname{im} D = 1 + 3 = 4 = \dim \mathbb{R}[x]_{\leq 3}$.

Example. (Trace)

Find the kernel and image of $\text{tr}: M_2(\mathbb{R}) \rightarrow \mathbb{R}$.

Solution. Kernel. A matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ has trace 0 exactly when $d = -a$, that is, when it equals

$$\begin{pmatrix} a & b \\ c & -a \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + bE_{12} + cE_{21}.$$

So $E_{11} - E_{22}, E_{12}, E_{21}$ span $\ker \text{tr}$. They are independent: the combination above is the zero matrix only if $a = b = c = 0$. Hence they form a basis, and $\dim \ker \text{tr} = 3$.

Image. im tr is a subspace of $\mathbb{R}$, and it contains $\text{tr } E_{11} = 1$. The lazy preimage rE_{11} of any $r \in \mathbb{R}$ shows $\text{im tr} = \mathbb{R}$. So the trace is surjective but not injective. Again $3 + 1 = 4 = \dim M_2(\mathbb{R})$.

The next two examples live in infinite-dimensional spaces, and the pattern “one property but not the other” goes the opposite way.

Example. (Integration)

Find the kernel and image of $J: \mathbb{R}[x] \rightarrow \mathbb{R}[x]$, $J(p)(x) = \int_0^x p(t) dt$, of [Example \(Integration\)](#).

Solution. Kernel. For $p = a_0 + a_1x + \cdots + a_nx^n$, $J(p) = a_0x + \frac{a_1}{2}x^2 + \cdots + \frac{a_n}{n+1}x^{n+1}$. This is the zero polynomial only if every $\frac{a_k}{k+1} = 0$, that is, every $a_k = 0$. So $\ker J = \{0\}$, and J is injective by [Theorem 3.2.3](#).

Image. Let $U = \{q \in \mathbb{R}[x] : q(0) = 0\}$, the polynomials with constant coefficient 0. Every $J(p)$ has constant coefficient 0, so $\text{im } J \subseteq U$. Conversely, if $q = b_1x + b_2x^2 + \cdots + b_mx^m \in U$, then $p = b_1 + 2b_2x + \cdots + mb_mx^{m-1}$ satisfies $J(p) = q$. Hence $\text{im } J = U$. Since $1 \notin U$, J is **not** surjective.

Example. (The Shifts: Injective but Not Surjective, and Conversely)

Find the kernels and images of the right and left shifts $R, L: F^{\mathbb{N}} \rightarrow F^{\mathbb{N}}$ of [Example \(The Shift Maps on Sequences\)](#).

Solution. Right shift. $R(s_0, s_1, \dots) = (0, s_0, s_1, \dots)$ is the zero sequence only if every $s_k = 0$. So $\ker R = \{\mathbf{0}\}$, and R is injective. Every output has first entry 0; conversely, $(0, t_1, t_2, \dots) = R(t_1, t_2, \dots)$. So $\text{im } R = \{\mathbf{t} \in F^{\mathbb{N}} : t_0 = 0\} \neq F^{\mathbb{N}}$, and R is not surjective: $\mathbf{e}_0 = (1, 0, 0, \dots)$ is never an output.

Left shift. $L(s_0, s_1, s_2, \dots) = (s_1, s_2, \dots)$ is zero exactly when $s_1 = s_2 = \cdots = 0$, with s_0 arbitrary. So $\ker L = \text{span}(\mathbf{e}_0) \neq \{\mathbf{0}\}$, and L is not injective. Every sequence is an output, since $L(R(\mathbf{t})) = \mathbf{t}$. So $\text{im } L = F^{\mathbb{N}}$, and L is surjective.

This pays off the pointer from Chapter 1, §8: R matches $F^{\mathbb{N}}$ one-to-one with its proper subspace $\text{im } R$. For a map $V \rightarrow V$ on a **finite** set, injective and surjective are equivalent ([Theorem 0.4.9](#)).

The shifts, and the pair J and D on $\mathbb{R}[x]$ (where $D(J(p)) = p$ makes D surjective, while $D(1) = 0$ makes it not injective), show that for linear maps on an infinite-dimensional space this equivalence fails. The next section proves that it holds again when V is finite-dimensional.

Quick check. Find $\ker T$ and $\operatorname{im} T$ for the map $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$, $T(x, y) = (2x - y, x + y, x - y)$, of [Example \(Finding a Formula from Values on a Basis\)](#). Is T injective? Surjective?

Solution. If $T(x, y) = \mathbf{0}$, then $x + y = 0$ and $x - y = 0$, so $x = y = 0$. Hence $\ker T = \{\mathbf{0}\}$, and T is injective. By [Theorem 3.2.4](#), $\operatorname{im} T = \operatorname{span}(T(\mathbf{e}_1), T(\mathbf{e}_2)) = \operatorname{span}((2, 1, 1), (-1, 1, -1))$, a plane in $\mathbb{R}^3$. A span of two vectors has dimension at most $2 < 3$, so $\operatorname{im} T \neq \mathbb{R}^3$ and T is not surjective.

3.2.5 What injective and surjective maps do to lists

Injectivity and surjectivity are statements about vectors. They translate into statements about lists: injective maps keep independent lists independent, and surjective maps keep spanning lists spanning. On a basis, the translation goes both ways.

Theorem 3.2.5 — Injective Maps Preserve Independence, Surjective Maps Preserve Spanning

Let $T: V \rightarrow W$ be linear.

- (a) If T is injective and $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ is linearly independent in V , then $(T(\mathbf{v}_1), \dots, T(\mathbf{v}_k))$ is linearly independent in W .
- (b) If T is surjective and $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ spans V , then $(T(\mathbf{v}_1), \dots, T(\mathbf{v}_k))$ spans W .
- (c) If $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is a basis of V , then T is injective if and only if $(T(\mathbf{v}_1), \dots, T(\mathbf{v}_n))$ is linearly independent, and T is surjective if and only if $(T(\mathbf{v}_1), \dots, T(\mathbf{v}_n))$ spans W .

Proof.

- (a) Let $a_1T(\mathbf{v}_1) + \dots + a_kT(\mathbf{v}_k) = \mathbf{0}$. By [Theorem 3.1.4](#), $T(a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k) = \mathbf{0}$, so $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k \in \ker T$. Since T is injective, $\ker T = \{\mathbf{0}\}$ ([Theorem 3.2.3](#)), so $a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = \mathbf{0}$. The independence of $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ gives $a_1 = \dots = a_k = 0$.
- (b) By [Theorem 3.2.4](#), $\operatorname{span}(T(\mathbf{v}_1), \dots, T(\mathbf{v}_k)) = \operatorname{im} T$, and $\operatorname{im} T = W$ because T is surjective ([Theorem 3.2.2](#)).
- (c) *Injectivity.* $(\Rightarrow)$ is (a). $(\Leftarrow)$ Suppose the images are independent, and let $\mathbf{v} \in \ker T$. Write $\mathbf{v} = a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n$. By [Theorem 3.1.4](#), $\mathbf{0} = T(\mathbf{v}) = a_1T(\mathbf{v}_1) + \dots + a_nT(\mathbf{v}_n)$, so all $a_i = 0$ and $\mathbf{v} = \mathbf{0}$. Hence $\ker T = \{\mathbf{0}\}$, and T is injective by [Theorem 3.2.3](#).

Surjectivity. By [Theorem 3.2.4](#), $\operatorname{im} T = \operatorname{span}(T(\mathbf{v}_1), \dots, T(\mathbf{v}_n))$. By [Theorem 3.2.2](#), T is surjective exactly when this span is W . ■

Warning. Without injectivity, independence can be lost. The projection $T(a_1, a_2) = (a_1, 0)$ sends the independent list $((1, 0), (1, 1))$ to $((1, 0), (1, 0))$, which is dependent. The kernel is to blame: $(1, 1) - (1, 0) = (0, 1)$ lies in $\ker T$. Likewise, a map that is not surjective sends **no** list to a spanning list of W , since all the images lie in $\operatorname{im} T \neq W$: the inclusion $\mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}[x]_{\leq 3}$ sends the basis $(1, x, x^2)$ to a list that misses x^3 .

Part (c) turns a question about a map into a question about n vectors in W , which is a question we can settle by elimination. In the next section, counting dimensions will let us check only **one** of injectivity and surjectivity when $\dim V = \dim W$.

3.2.6 Exercises

3.2.6.1 A. Check your understanding

Exercise A1.

- Define the kernel and the image of a linear map $T: V \rightarrow W$. Of which spaces are they subspaces?
- State the kernel test for injectivity.
- True or false: if $T(\mathbf{u}) = T(\mathbf{v})$, then $\mathbf{u} - \mathbf{v} \in \ker T$. Justify your answer.
- True or false: for $D: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 3}$, $D(p) = p'$, we have $\operatorname{im} D = \mathbb{R}[x]_{\leq 3}$. Justify your answer.
- True or false: for a linear $T: V \rightarrow W$ and **any** $\mathbf{w} \in W$, the set $\{\mathbf{v} \in V : T(\mathbf{v}) = \mathbf{w}\}$ is a subspace of V . Justify your answer.

Solution.

- See [Definition 3.2.1](#) and [Definition 3.2.2](#): $\ker T = \{\mathbf{v} \in V : T(\mathbf{v}) = \mathbf{0}\}$, a subspace of V ([Theorem 3.2.1](#)); $\operatorname{im} T = \{T(\mathbf{v}) : \mathbf{v} \in V\}$, a subspace of W ([Theorem 3.2.2](#)).
- See [Theorem 3.2.3](#): a linear map is injective if and only if its kernel is $\{\mathbf{0}\}$.
- True. By [Theorem 3.1.2](#), $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v}) = \mathbf{0}$.
- False. Every derivative of a polynomial of degree at most 3 has degree at most 2, so $x^3 \notin \operatorname{im} D$.
- False. If $\mathbf{w} \neq \mathbf{0}$, the set does not contain $\mathbf{0}$, since $T(\mathbf{0}) = \mathbf{0} \neq \mathbf{w}$. For example, for the projection of [Example \(Projection onto an Axis\)](#) and $\mathbf{w} = (1, 0)$, the set is the line $\{(1, a_2)\}$. It is a subspace exactly when $\mathbf{w} = \mathbf{0}$, in which case it is $\ker T$.

3.2.6.2 B. Practice

Exercise B1: Kernel and Image of a Matrix Map.

Let $B = \begin{pmatrix} 1 & -1 & 2 & 0 \\ 2 & -2 & 5 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \in M_{3 \times 4}(\mathbb{R})$. Find bases of $\ker T_B$ and $\operatorname{im} T_B$. Hence decide whether $T_B: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ is injective, and whether it is surjective.

Solution. As in [Example \(A Matrix Map\)](#), $\ker T_B = \text{nul}(B)$ and $\text{im } T_B = \text{col}(B)$. Row reduce:

$$B \xrightarrow{R_2-2R_1} \begin{pmatrix} 1 & -1 & 2 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \xrightarrow[R_1-2R_2]{R_3-R_2} \begin{pmatrix} 1 & -1 & 0 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The pivot columns are 1 and 3; the free columns are 2 and 4. By [Theorem 2.5.4](#), $\ker T_B$ has basis $((1, 1, 0, 0), (2, 0, -1, 1))$. Check: $B(2, 0, -1, 1) = (2-2, 4-5+1, -1+1) = \mathbf{0}$. By [Theorem 2.5.2](#), $\text{im } T_B$ has basis $((1, 2, 0), (2, 5, 1))$.

Hence $\ker T_B \neq \{\mathbf{0}\}$, so T_B is not injective ([Theorem 3.2.3](#)). And $\dim \text{im } T_B = 2 < 3$, so $\text{im } T_B \neq \mathbb{R}^3$ and T_B is not surjective ([Theorem 3.2.2](#)).

Exercise B2: Kernels and Images Beyond F^n .

For each map, find a basis of the kernel and a basis of the image, and decide whether the map is injective and whether it is surjective.

- (a) $T: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}[x]_{\leq 2}$, $T(p) = p' + p$.
- (b) $S: M_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})$, $S(A) = A - A^T$.

Solution. Both maps are linear. For $c \in \mathbb{R}$, by linearity of differentiation ([Example \(Differentiation\)](#)), $T(cp + q) = (cp' + q') + (cp + q) = cT(p) + T(q)$. By linearity of the transpose ([Example \(Trace and Transpose\)](#)), $S(cA + B) = cA + B - cA^T - B^T = cS(A) + S(B)$. Both are linear by [Theorem 3.1.3](#).

- (a) *Kernel.* Let $p = a + bx + cx^2$. Then $T(p) = (b + 2cx) + (a + bx + cx^2) = (a + b) + (b + 2c)x + cx^2$. This is zero exactly when $c = 0$, $b + 2c = 0$ and $a + b = 0$, that is, $c = b = a = 0$. So $\ker T = \{0\}$, with the empty list as basis, and T is injective.

Image. We show T is surjective. Given $q \in \mathbb{R}[x]_{\leq 2}$, let $p = q - q' + q''$. Then $p \in \mathbb{R}[x]_{\leq 2}$, and since $q''' = 0$,

$$T(p) = (q' - q'' + q''') + (q - q' + q'') = q.$$

So $\text{im } T = \mathbb{R}[x]_{\leq 2}$, with basis $(1, x, x^2)$, and T is surjective.

- (b) *Kernel.* $S(A) = 0$ exactly when $A^T = A$. A matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is symmetric exactly when $b = c$, that is, when it equals $aE_{11} + dE_{22} + b(E_{12} + E_{21})$. These three matrices are independent (compare entries), so $(E_{11}, E_{22}, E_{12} + E_{21})$ is a basis of $\ker S$. S is not injective.

Image. By [Theorem 3.2.4](#) with the basis $(E_{11}, E_{12}, E_{21}, E_{22})$, $\text{im } S$ is spanned by $S(E_{11}) = 0$, $S(E_{12}) = E_{12} - E_{21}$, $S(E_{21}) = E_{21} - E_{12}$ and $S(E_{22}) = 0$. So $\text{im } S = \text{span}(E_{12} - E_{21})$, with basis $(E_{12} - E_{21})$, the skew-symmetric matrices. Since $I_2 \notin \text{im } S$, S is not surjective.

3.2.6.3 C. Going deeper

In these exercises, for linear maps $T: U \rightarrow V$ and $S: V \rightarrow W$ we write ST for the composite $S \circ T: U \rightarrow W$, which applies T first. It is linear: $ST(c\mathbf{u} + \mathbf{u}') = S(cT(\mathbf{u}) + T(\mathbf{u}')) = cST(\mathbf{u}) + ST(\mathbf{u}')$, by [Theorem 3.1.3](#) for T and then for S .

Exercise C1: Kernels and Images of a Composite.

Let $T: U \rightarrow V$ and $S: V \rightarrow W$ be linear.

- (a) Prove that $\ker T \subseteq \ker(ST)$ and $\operatorname{im}(ST) \subseteq \operatorname{im} S$.
- (b) Give an example, with $U = V = W = \mathbb{R}^2$, where $\ker T \neq \ker(ST)$.
- (c) Deduce that if ST is injective, then T is injective, and if ST is surjective, then S is surjective.

Solution.

- (a) Let $\mathbf{u} \in \ker T$. Then $ST(\mathbf{u}) = S(\mathbf{0}) = \mathbf{0}$ by [Theorem 3.1.1](#), so $\mathbf{u} \in \ker(ST)$. Let $\mathbf{w} \in \operatorname{im}(ST)$, say $\mathbf{w} = S(T(\mathbf{u}))$. Then $\mathbf{w} = S(\mathbf{v})$ with $\mathbf{v} = T(\mathbf{u}) \in V$, so $\mathbf{w} \in \operatorname{im} S$.
- (b) Let $T = \operatorname{id}_{\mathbb{R}^2}$ and let S be the projection $S(a_1, a_2) = (a_1, 0)$. Then $\ker T = \{\mathbf{0}\}$, while $\ker(ST) = \ker S$ is the y -axis ([Example \(Projection onto an Axis\)](#)).
- (c) If ST is injective, then $\ker(ST) = \{\mathbf{0}\}$ ([Theorem 3.2.3](#)), so $\ker T = \{\mathbf{0}\}$ by (a), and T is injective. If ST is surjective, then $W = \operatorname{im}(ST) \subseteq \operatorname{im} S \subseteq W$ by (a), so $\operatorname{im} S = W$ and S is surjective ([Theorem 3.2.2](#)).

Exercise C2: When Is a Composite Injective?.

Let $T: U \rightarrow V$ and $S: V \rightarrow W$ be linear, and suppose T is injective. Prove that ST is injective if and only if $\ker S \cap \operatorname{im} T = \{\mathbf{0}\}$, that is, if and only if S is injective on $\operatorname{im} T$. Give an example where S is not injective but ST is.

Solution. ($\Rightarrow$) Suppose ST is injective, and let $\mathbf{v} \in \ker S \cap \operatorname{im} T$. Then $\mathbf{v} = T(\mathbf{u})$ for some $\mathbf{u} \in U$, and $ST(\mathbf{u}) = S(\mathbf{v}) = \mathbf{0}$. So $\mathbf{u} \in \ker(ST) = \{\mathbf{0}\}$ ([Theorem 3.2.3](#)), and $\mathbf{v} = T(\mathbf{0}) = \mathbf{0}$.

($\Leftarrow$) Suppose $\ker S \cap \operatorname{im} T = \{\mathbf{0}\}$, and let $\mathbf{u} \in \ker(ST)$. Then $T(\mathbf{u}) \in \operatorname{im} T$ and $S(T(\mathbf{u})) = \mathbf{0}$, so $T(\mathbf{u}) \in \ker S \cap \operatorname{im} T = \{\mathbf{0}\}$. Since T is injective, $\mathbf{u} = \mathbf{0}$. Hence $\ker(ST) = \{\mathbf{0}\}$, and ST is injective.

Example: let $T: \mathbb{R} \rightarrow \mathbb{R}^2$, $T(a) = (a, 0)$, which is injective with $\operatorname{im} T$ the x -axis, and let $S: \mathbb{R}^2 \rightarrow \mathbb{R}$, $S(a_1, a_2) = a_1$. Then $\ker S$ is the y -axis, so S is not injective, but $\ker S \cap \operatorname{im} T = \{\mathbf{0}\}$, and indeed $ST(a) = a$ is injective.

Exercise C3: Differentiation over $\mathbb{F}_2$.

Let $D: \mathbb{F}_2[x]_{\leq 3} \rightarrow \mathbb{F}_2[x]_{\leq 3}$ be the formal derivative ([Example \(Differentiation\)](#)). Find bases of $\ker D$ and $\operatorname{im} D$. Which step of [Example \(Differentiation\)](#) fails over $\mathbb{F}_2$?

Solution. For $p = a_0 + a_1x + a_2x^2 + a_3x^3$, $D(p) = a_1 + 2a_2x + 3a_3x^2$. In $\mathbb{F}_2$, $2 = 0$ and $3 = 1$, so $D(p) = a_1 + a_3x^2$.

Kernel. $D(p) = 0$ exactly when $a_1 = a_3 = 0$, so $\ker D = \{a_0 + a_2x^2\}$, with basis $(1, x^2)$. The polynomials 1 and x^2 are independent by [Theorem 1.4.4](#).

Image. By [Theorem 3.2.4](#), $\operatorname{im} D = \operatorname{span}(D(1), D(x), D(x^2), D(x^3)) = \operatorname{span}(0, 1, 0, x^2)$, with basis $(1, x^2)$.

The failing step is “ $2a_2 = 0$ forces $a_2 = 0$ ”, which divides by 2; over $\mathbb{F}_2$, $2 = 0$, and $D(x^2) = 2x = 0$ although x^2 is not constant. So over $\mathbb{F}_2$ the kernel of differentiation is bigger than the constants, and here $\ker D = \operatorname{im} D$.

3.3 The Rank–Nullity Theorem

The examples of the last section kept producing the same coincidence. Differentiation $\mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$ has a kernel of dimension 1 and an image of dimension 3, and $1+3 = 4 = \dim \mathbb{R}[x]_{\leq 3}$. The trace on $M_2(\mathbb{R})$ gives $3+1 = 4$ again. For matrices, Chapter 2 proved that $\operatorname{rank} A + \operatorname{nullity} A$ is the number of columns (Theorem 2.5.7). This section proves the same equation for every linear map on a finite-dimensional space, with no matrices at all, and then turns it into the most useful counting tool of the chapter.

3.3.1 Rank and nullity

The two numbers in the equation are dimensions of the two subspaces of Definition 3.2.1 and Definition 3.2.2, so they are defined whenever those subspaces are finite-dimensional.

Definition 3.3.1 — Nullity

Let $T: V \rightarrow W$ be a linear map with $\ker T$ finite-dimensional. The **nullity** of T is $\operatorname{nullity} T := \dim \ker T$.

Definition 3.3.2 — Rank

Let $T: V \rightarrow W$ be a linear map with $\operatorname{im} T$ finite-dimensional. The **rank** of T is $\operatorname{rank} T := \dim \operatorname{im} T$.

If V is finite-dimensional, both numbers are defined: $\ker T$ is a subspace of V , hence finite-dimensional (Theorem 1.6.7), and $\operatorname{im} T$ is finite-dimensional by Theorem 3.2.4. The nullity counts the independent directions that T destroys; the rank counts the independent directions that survive in the output.

Example. (Rank and Nullity in Examples)

Find the nullity and rank of (a) differentiation $D: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$; (b) the matrix map $T_A: F^n \rightarrow F^m$ for $A \in M_{m \times n}(F)$; (c) the zero map and the identity on a space of dimension n .

Solution.

- (a) By Example (Differentiation), $\ker D = \operatorname{span}(1)$ and $\operatorname{im} D = \mathbb{R}[x]_{\leq 2}$. So $\operatorname{nullity} D = 1$ and $\operatorname{rank} D = 3$.
- (b) By Example (A Matrix Map), $\ker T_A = \operatorname{nul}(A)$ and $\operatorname{im} T_A = \operatorname{col}(A)$. So $\operatorname{nullity} T_A = \dim \operatorname{nul}(A) = \operatorname{nullity} A$ and $\operatorname{rank} T_A = \dim \operatorname{col}(A) = \operatorname{rank} A$ (Definition 2.5.5, Defi-

inition 2.5.4). The new names agree with the old ones.

- (c) By [Example \(The Zero Map and the Identity\)](#), the zero map $V \rightarrow V$ has nullity n and rank 0, and the identity has nullity 0 and rank n .

A non-example. The integration map $J: \mathbb{R}[x] \rightarrow \mathbb{R}[x]$ of [Example \(Integration\)](#) has nullity 0, but its image contains the independent list $(x, x^2, \dots, x^k)$ for every k , so the image is not finite-dimensional ([Theorem 1.6.3](#)) and rank J is not defined. What separates J from the examples above is that its domain is infinite-dimensional.

3.3.2 The theorem

We want to prove that for every linear $T: V \rightarrow W$ with V finite-dimensional,

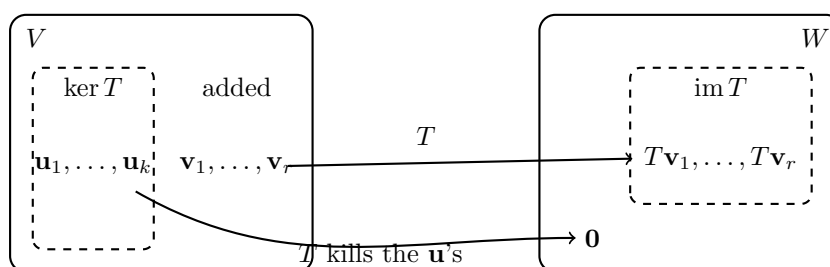
$$\dim \ker T + \dim \operatorname{im} T = \dim V.$$

This is a dimension formula, so we look for the proof pattern that proved the last one, the Dimension Formula for Sums ([Theorem 1.7.3](#)): take a basis of the **smallest** space, extend it, make a Big Claim about the vectors that were added, and count. Which space is the smallest? The kernel lives in V and the image lives in W , different spaces that cannot be compared. The only inclusion available is $\ker T \subseteq V$. So we start with a basis of $\ker T$ and extend it to a basis of V . The image has to be reached by applying T , and the kernel vectors contribute nothing when we do. So the natural candidate for a basis of $\operatorname{im} T$ is the images of the **added** vectors.

Theorem 3.3.1 — Rank–Nullity Theorem

Let V be a **finite-dimensional** vector space, let W be any vector space over the same field, and let $T: V \rightarrow W$ be linear. Then $\ker T$ and $\operatorname{im} T$ are finite-dimensional, and

$$\text{nullity } T + \text{rank } T = \dim V.$$



Idea. The picture is the whole proof. The $\mathbf{u}$'s all go to $\mathbf{0}$; the $\mathbf{v}$'s carry everything else.

① Take a basis $(\mathbf{u}_1, \dots, \mathbf{u}_k)$ of $\ker T$, the smallest space in sight. ② Extend it to a basis $(\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{v}_1, \dots, \mathbf{v}_r)$ of V (Basis Extension). ③ **Big Claim:** $(T\mathbf{v}_1, \dots, T\mathbf{v}_r)$ is a basis of $\operatorname{im} T$. *Spanning:* apply T to a general vector of V ; the $\mathbf{u}$ -terms die because they are in the kernel. *Independence:* the only way to use linearity is backwards, turning $\sum b_j T\mathbf{v}_j = \mathbf{0}$ into $T(\sum b_j \mathbf{v}_j) = \mathbf{0}$. Then $\sum b_j \mathbf{v}_j$ is in the kernel, where the $\mathbf{u}$'s are a basis, and the big basis of V finishes it. ④ Count: $\dim V = k + r = \text{nullity } T + \text{rank } T$.

This is the move of [Theorem 1.7.3](#): basis of the smallest space, extend, Big Claim, count.

The new ingredient is that the Big Claim is about the **images** of the added vectors.

Proof. Step 1: bases. By Theorem 3.2.1, $\ker T$ is a subspace of V . Since V is finite-dimensional, so is $\ker T$ (Theorem 1.6.7), and it has a basis $(\mathbf{u}_1, \dots, \mathbf{u}_k)$ with $k = \text{nullity } T$ (Corollary 1.5.5). This list is linearly independent in V , so by the Basis Extension Theorem (Theorem 1.6.5) there are $\mathbf{v}_1, \dots, \mathbf{v}_r \in V$ such that $(\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{v}_1, \dots, \mathbf{v}_r)$ is a basis of V . Thus $\dim V = k + r$.

Step 2: the Big Claim.

Claim. The list $(T\mathbf{v}_1, \dots, T\mathbf{v}_r)$ is a basis of $\text{im } T$.

Proof. Spanning. By Theorem 3.2.4 applied to the basis of Step 1, $\text{im } T = \text{span}(T\mathbf{u}_1, \dots, T\mathbf{u}_k, T\mathbf{v}_1, \dots, T\mathbf{v}_r)$. Each $T\mathbf{u}_i = \mathbf{0}$, since $\mathbf{u}_i \in \ker T$, and deleting a zero vector from a list does not change its span, because $\mathbf{0}$ lies in every span (Theorem 1.3.5). Hence $\text{im } T = \text{span}(T\mathbf{v}_1, \dots, T\mathbf{v}_r)$.

Independence. Let $b_1, \dots, b_r \in F$ satisfy $b_1 T\mathbf{v}_1 + \dots + b_r T\mathbf{v}_r = \mathbf{0}$. By Theorem 3.1.4, $T(b_1 \mathbf{v}_1 + \dots + b_r \mathbf{v}_r) = \mathbf{0}$, so $b_1 \mathbf{v}_1 + \dots + b_r \mathbf{v}_r \in \ker T$. Since $(\mathbf{u}_1, \dots, \mathbf{u}_k)$ spans $\ker T$, there are $a_1, \dots, a_k \in F$ with

$$b_1 \mathbf{v}_1 + \dots + b_r \mathbf{v}_r = a_1 \mathbf{u}_1 + \dots + a_k \mathbf{u}_k.$$

Rearranging, $(-a_1)\mathbf{u}_1 + \dots + (-a_k)\mathbf{u}_k + b_1 \mathbf{v}_1 + \dots + b_r \mathbf{v}_r = \mathbf{0}$. This is a combination of the basis $(\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{v}_1, \dots, \mathbf{v}_r)$ of V , which is linearly independent, so all its coefficients are 0. In particular $b_1 = \dots = b_r = 0$. Hence $(T\mathbf{v}_1, \dots, T\mathbf{v}_r)$ is linearly independent. ■

Step 3: count. By the claim, $\text{im } T$ has a basis of length r , so it is finite-dimensional with $\text{rank } T = r$ (Definition 1.6.1). Therefore

$$\text{nullity } T + \text{rank } T = k + r = \dim V,$$

as claimed. ■

The proof reads correctly when $k = 0$ (injective T : the $\mathbf{v}$'s are a basis of V) and when $r = 0$ ($\ker T = V$: the claim says $\text{im } T$ has the empty basis). Notice what was **not** used: nothing about W beyond its being a vector space. The theorem counts only in the domain.

Applied to $T_A: F^n \rightarrow F^m$, where $\text{nullity } T_A = \text{nullity } A$ and $\text{rank } T_A = \text{rank } A$ (Example (Rank and Nullity in Examples)), the theorem says $\text{nullity } A + \text{rank } A = n$. This is Theorem 2.5.7 again, now with a second proof. Chapter 2 counted free and pivot columns of a reduced matrix; here we counted basis vectors, and no row operation appeared. In particular, **the solution space of a homogeneous system $A\mathbf{x} = \mathbf{0}$ in n unknowns has dimension $n - \text{rank } A$** , whatever method is used to find the rank.

In the online version you can test the count on any matrix map.

```
from sympy import Matrix
```



```

A = Matrix([
    [1, 2, 0, 3],
    [2, 4, 1, 8],
    [-1, -2, 1, -1],
])
rank = A.rank()                # dim im T_A
nullity = len(A.nullspace())   # dim ker T_A: length of a basis of the kernel
print("rank =", rank, " nullity =", nullity, " dim of domain =", A.cols)
rank + nullity == A.cols

```

3.3.3 Count instead of check

Rank–Nullity links the two questions of the last section. Injectivity is about the kernel, surjectivity about the image, and the theorem says the two dimensions add up to a fixed number. So one controls the other.

Corollary 3.3.2 — Consequences of Rank–Nullity

Let V and W be finite-dimensional vector spaces over F , and let $T: V \rightarrow W$ be linear.

- (a) T is injective if and only if $\text{rank } T = \dim V$.
- (b) T is surjective if and only if $\text{rank } T = \dim W$.
- (c) If $\dim V > \dim W$, then T is **not** injective.
- (d) If $\dim V < \dim W$, then T is **not** surjective.
- (e) If $\dim V = \dim W$, then T is injective $\iff T$ is surjective $\iff T$ is bijective.

Proof.

- (a) By [Theorem 3.2.3](#), T is injective if and only if $\ker T = \{\mathbf{0}\}$. A finite-dimensional space is $\{\mathbf{0}\}$ exactly when its dimension is 0, because the empty list is a basis of $\{\mathbf{0}\}$ and a basis of length 0 spans only $\{\mathbf{0}\}$. So T is injective if and only if $\text{nullity } T = 0$, which by [Theorem 3.3.1](#) means $\text{rank } T = \dim V$.
- (b) If T is surjective, then $\text{im } T = W$ ([Theorem 3.2.2](#)), so $\text{rank } T = \dim W$. Conversely, if $\text{rank } T = \dim W$, then $\text{im } T$ is a subspace of W of the same dimension, so $\text{im } T = W$ by [Theorem 1.6.8](#), and T is surjective.
- (c) Since $\text{im } T$ is a subspace of W , $\text{rank } T \leq \dim W < \dim V$ by [Theorem 1.6.7](#). By (a), T is not injective.
- (d) By [Theorem 3.3.1](#), $\text{rank } T = \dim V - \text{nullity } T \leq \dim V < \dim W$. By (b), T is not surjective.
- (e) If $\dim V = \dim W$, then (a) and (b) have the same right-hand side, so T is injective if and only if it is surjective. Hence either property gives both, which is bijectivity; and bijective maps are injective by definition.

■

Part (e) is the linear version of the pigeonhole principle for finite sets ([Theorem 0.4.9](#)), with dimension in place of the number of elements. It is a **count instead of check**: to show that a map between spaces of the same finite dimension is bijective, check **only** the kernel, usually the

easier half, and let the count supply surjectivity. For T_A with A square, it is the equivalence of (b) and (e) in the Invertible Matrix Theorem (Theorem 2.4.4). Parts (c) and (d) are the linear version of “no injection from a bigger finite set into a smaller one, and no surjection the other way”.

Warning. Both finiteness and equal dimension are needed. Rank–Nullity requires V to be finite-dimensional; W may be anything, as for the inclusion $\mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}[x]$, with nullity 0 and rank 3. Part (e) fails when the spaces are infinite-dimensional, even for an operator $V \rightarrow V$: the right shift R on $F^{\mathbb{N}}$ is injective but not surjective, and the left shift L is surjective but not injective (Example (The Shifts: Injective but Not Surjective, and Conversely)). And the hypothesis $\dim V = \dim W$ cannot be dropped: $T(x, y) = (2x - y, x + y, x - y)$ from $\mathbb{R}^2$ to $\mathbb{R}^3$ is injective but not surjective.

Quick check. Answer each question with a count.

- (a) Can a linear map $\mathbb{R}^5 \rightarrow \mathbb{R}^2$ be injective?
- (b) Is there a linear map $T: \mathbb{R}^6 \rightarrow \mathbb{R}^4$ with nullity $T = 1$?
- (c) A linear operator T on $\mathbb{R}[x]_{\leq 3}$ satisfies $\ker T = \{0\}$. Must T be surjective?

Solution.

- (a) No, by Corollary 3.3.2 (c): $5 > 2$. Concretely, the rank is at most 2, so the nullity is at least 3.
- (b) No. By Theorem 3.3.1 the rank would be $6 - 1 = 5$, but $\operatorname{im} T \subseteq \mathbb{R}^4$ has dimension at most 4.
- (c) Yes. The domain and codomain are both $\mathbb{R}[x]_{\leq 3}$, of dimension 4, and T is injective, so it is surjective by Corollary 3.3.2 (e).

3.3.4 Polynomial interpolation by counting

Chapter 2 proved that through $n + 1$ points with distinct x -values there passes exactly one polynomial of degree at most n (Theorem 2.7.3). Existence for **all** data and uniqueness came together there because a square matrix was involved. Rank–Nullity gives the same result as a count, with no matrix written down.

Example. (Interpolation as a Count)

Let $x_0, \dots, x_n \in F$ be distinct, and define $E: F[x]_{\leq n} \rightarrow F^{n+1}$ by $E(p) = (p(x_0), \dots, p(x_n))$. Show that E is bijective, and deduce Theorem 2.7.3.

Solution. Linear. For $p, q \in F[x]_{\leq n}$ and $c \in F$, Theorem 0.8.5 gives $(cp + q)(x_i) = cp(x_i) + q(x_i)$ in each entry, so $E(cp + q) = cE(p) + E(q)$, and E is linear by Theorem 3.1.3.

Check the kernel. If $E(p) = \mathbf{0}$, then p has the $n + 1$ distinct roots $x_0, \dots, x_n$ while $\deg p \leq n$. A non-zero polynomial of degree at most n has at most n distinct roots (Lemma 2.7.2), so $p = 0$. Hence $\ker E = \{0\}$, and E is injective (Theorem 3.2.3).

Count. $\dim F[x]_{\leq n} = n + 1$, since $(1, x, \dots, x^n)$ is a basis (Example (Standard Bases)), and $\dim F^{n+1} = n + 1$. By Corollary 3.3.2 (e), E is bijective.

Surjectivity of E says that for every $(y_0, \dots, y_n) \in F^{n+1}$ some $p \in F[x]_{\leq n}$ has $p(x_i) = y_i$ for all i ; injectivity says there is only one. This is [Theorem 2.7.3](#).

Compare the two arguments. Both prove uniqueness for the zero data from the root bound. Chapter 2 then invoked the Invertible Matrix Theorem for the Vandermonde matrix; here the count $n+1 = n+1$ does the same job. The proof of existence never constructs a polynomial. That is the price and the power of counting: it tells us a solution exists without saying what it is.

3.3.5 Existence with conditions on derivatives

The same argument works when some conditions are on derivatives, where no Vandermonde matrix is at hand.

Example. (Prescribing Values and Slopes)

Show that for all $a, b, c, d \in \mathbb{R}$ there is **exactly one** $p \in \mathbb{R}[x]_{\leq 3}$ with

$$p(0) = a, \quad p(1) = b, \quad p'(0) = c, \quad p'(1) = d.$$

Hence find the cubic that rises from 0 at $x = 0$ to 1 at $x = 1$ with slope 0 at both ends.

Solution. Define $T: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}^4$ by $T(p) = (p(0), p(1), p'(0), p'(1))$. Each entry is linear in p : evaluation respects sums and scalar multiples ([Theorem 0.8.5](#)), and so does differentiation ([Example \(Differentiation\)](#)). Hence T is linear by [Theorem 3.1.3](#).

Kernel. Let $p = c_0 + c_1x + c_2x^2 + c_3x^3$ with $T(p) = \mathbf{0}$. Then $p(0) = c_0 = 0$ and $p'(0) = c_1 = 0$. So $p = c_2x^2 + c_3x^3$, and the remaining conditions read

$$p(1) = c_2 + c_3 = 0, \quad p'(1) = 2c_2 + 3c_3 = 0.$$

Subtracting twice the first from the second gives $c_3 = 0$, and then $c_2 = 0$. Hence $\ker T = \{0\}$.

Count. $\dim \mathbb{R}[x]_{\leq 3} = 4 = \dim \mathbb{R}^4$, so T is bijective by [Corollary 3.3.2](#) (e). Surjectivity is existence for every (a, b, c, d) , and injectivity is uniqueness.

For the data $(0, 1, 0, 0)$, try $p = \alpha x^2 + \beta x^3$, which already has $p(0) = p'(0) = 0$. Then $p(1) = \alpha + \beta = 1$ and $p'(1) = 2\alpha + 3\beta = 0$ give $\beta = -2$ and $\alpha = 3$. So $p = 3x^2 - 2x^3$; by uniqueness it is the only such cubic.

3.3.6 Dimensions of subspaces cut out by conditions

Many subspaces are defined as “all vectors satisfying some linear conditions”, which is to say as the kernel of a linear map. If the map is **surjective**, Rank–Nullity gives the dimension of the kernel at once, and surjectivity is often easy to see.

Example. (Dimension of a Kernel from a Surjective Map)

Find the dimension of each subspace.

- (a) The trace-free matrices $U = \{A \in M_n(F) : \operatorname{tr} A = 0\}$ (Example (Trace-Free Matrices)).
- (b) $U' = \{p \in \mathbb{R}[x]_{\leq n} : \int_0^1 p(t) dt = 0\}$.

Solution.

- (a) $U = \ker \operatorname{tr}$, where $\operatorname{tr}: M_n(F) \rightarrow F$ is linear (Example (Trace and Transpose)). Its image is a subspace of F containing $\operatorname{tr} E_{11} = 1$, so it contains $c \cdot 1 = c$ for every $c \in F$; it is all of F , and $\operatorname{rank} \operatorname{tr} = 1$. Since $\dim M_n(F) = n^2$ (Example (Standard Bases)), Theorem 3.3.1 gives $\dim U = n^2 - 1$.
- (b) $U' = \ker \psi$, where $\psi: \mathbb{R}[x]_{\leq n} \rightarrow \mathbb{R}$, $\psi(p) = \int_0^1 p(t) dt$, is linear by Exercise (B3: Evaluation and Integration) (b). Since $\psi(1) = 1$, the image is a subspace of $\mathbb{R}$ containing 1, hence all of $\mathbb{R}$, so $\operatorname{rank} \psi = 1$. By Theorem 3.3.1, $\dim U' = (n + 1) - 1 = n$.

Without the theorem, (a) needs an explicit basis of $n^2 - 1$ matrices together with a proof of independence and spanning. With it, one matrix of trace 1 is enough.

3.3.7 Exercises

3.3.7.1 A. Check your understanding

Exercise A1.

- (a) Define the rank and the nullity of a linear map.
- (b) State the Rank–Nullity Theorem, with its hypotheses.
- (c) True or false: there is an injective linear map $\mathbb{R}^4 \rightarrow \mathbb{R}^3$. Justify your answer.
- (d) True or false: if V is finite-dimensional and $T: V \rightarrow V$ is linear with $\operatorname{im} T = V$, then T is injective. Justify your answer.
- (e) True or false: every injective linear map $F^{\mathbb{N}} \rightarrow F^{\mathbb{N}}$ is surjective. Justify your answer.
- (f) Name the four steps of the proof of Rank–Nullity.

Solution.

- (a) See Definition 3.3.1 and Definition 3.3.2: $\operatorname{nullity} T = \dim \ker T$ and $\operatorname{rank} T = \dim \operatorname{im} T$, when these are finite-dimensional.
- (b) See Theorem 3.3.1: if V is finite-dimensional and $T: V \rightarrow W$ is linear, then $\operatorname{nullity} T + \operatorname{rank} T = \dim V$.
- (c) False. $4 > 3$, so no linear map $\mathbb{R}^4 \rightarrow \mathbb{R}^3$ is injective (Corollary 3.3.2 (c)).
- (d) True. T is surjective, and its domain and codomain are the same finite-dimensional space, so T is injective by Corollary 3.3.2 (e).

- (e) False. The right shift R is injective but not surjective ([Example \(The Shifts: Injective but Not Surjective, and Conversely\)](#)). The count fails because $F^{\mathbb{N}}$ is not finite-dimensional.
- (f) Take a basis of $\ker T$; extend it to a basis of V ; prove the Big Claim that the images of the added vectors form a basis of $\operatorname{im} T$ (spanning, then independence); count.

3.3.7.2 B. Practice

Exercise B1: Verifying Rank–Nullity.

For each map, find a basis of the kernel and a basis of the image, and verify that $\operatorname{nullity} T + \operatorname{rank} T = \dim V$.

- (a) $T_C: \mathbb{R}^4 \rightarrow \mathbb{R}^3$, where $C = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 1 & 1 & 2 & 1 \end{pmatrix}$.
- (b) $T: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}^2$, $T(p) = (p(0), p(1))$.
- (c) $T: M_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})$, $T(A) = A + A^T$.

Solution.

- (a) Row reduce: $R_3 - R_1$ gives $(0, 1, 1, -1)$, and then $R_3 - R_2$ gives a zero row, so

$$C \longrightarrow \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

which is the RREF. The free columns are 3 and 4. By [Theorem 2.5.4](#), $\ker T_C$ has basis $((-1, -1, 1, 0), (-2, 1, 0, 1))$. By [Theorem 2.5.2](#), $\operatorname{im} T_C$ has basis $((1, 0, 1), (0, 1, 1))$. So $2 + 2 = 4 = \dim \mathbb{R}^4$.

- (b) T is linear by [Theorem 0.8.5](#). *Kernel:* for $p = a_0 + a_1x + a_2x^2 + a_3x^3$, $T(p) = \mathbf{0}$ means $a_0 = 0$ and $a_1 + a_2 + a_3 = 0$, that is, $p = a_2(x^2 - x) + a_3(x^3 - x)$. The list $(x^2 - x, x^3 - x)$ spans $\ker T$, and it is independent by [Theorem 1.4.4](#), so it is a basis. *Image:* $T(1) = (1, 1)$ and $T(x) = (0, 1)$ are independent, so they span a 2-dimensional subspace of $\mathbb{R}^2$, which is $\mathbb{R}^2$ ([Theorem 1.6.8](#)). So $\operatorname{im} T = \mathbb{R}^2$ with basis $((1, 1), (0, 1))$. Hence $2 + 2 = 4 = \dim \mathbb{R}[x]_{\leq 3}$.
- (c) T is linear because the transpose is ([Example \(Trace and Transpose\)](#)). *Kernel:* $A + A^T = 0$ means $a_{11} = a_{22} = 0$ and $a_{21} = -a_{12}$, so $\ker T = \operatorname{span}(E_{12} - E_{21})$, with basis $(E_{12} - E_{21})$. *Image:* each $A + A^T$ is symmetric, since $(A + A^T)^T = A^T + A$, and each symmetric B equals $T(\frac{1}{2}B)$. So $\operatorname{im} T$ is the space of symmetric matrices, with basis $(E_{11}, E_{22}, E_{12} + E_{21})$ (as in [Exercise \(B2: Kernels and Images Beyond \$F^n\$ \)](#)). Hence $1 + 3 = 4 = \dim M_2(\mathbb{R})$.

Exercise B2: Bijective by the Kernel Alone.

Let $n \geq 0$ and $T: \mathbb{R}[x]_{\leq n} \rightarrow \mathbb{R}[x]_{\leq n}$, $T(p) = p + xp'$.

- (a) Show that T is a linear map into $\mathbb{R}[x]_{\leq n}$ and that $\ker T = \{0\}$. Deduce that T is bijective.

- (b) Hence show that there is exactly one $p \in \mathbb{R}[x]_{\leq 2}$ with $p + xp' = 1 + x + x^2$, and find it.

Solution.

- (a) For $p = \sum_{k=0}^n a_k x^k$, $xp' = \sum_{k=1}^n k a_k x^k$, so

$$T(p) = \sum_{k=0}^n (1+k)a_k x^k \in \mathbb{R}[x]_{\leq n}.$$

Linearity: $T(cp+q) = (cp+q) + x(cp'+q') = c(p+xp') + (q+xq') = cT(p) + T(q)$, using [Example \(Differentiation\)](#). If $T(p) = 0$, then $(1+k)a_k = 0$ for each k ; since $1+k \neq 0$ in $\mathbb{R}$, every $a_k = 0$. So $\ker T = \{0\}$, and T is injective ([Theorem 3.2.3](#)). Domain and codomain have the same finite dimension $n+1$, so T is bijective by [Corollary 3.3.2 \(e\)](#).

- (b) With $n = 2$, surjectivity gives a solution and injectivity makes it unique. By the formula in (a), $T(a_0 + a_1x + a_2x^2) = a_0 + 2a_1x + 3a_2x^2$, so we need $a_0 = 1$, $2a_1 = 1$, $3a_2 = 1$. Hence $p = 1 + \frac{1}{2}x + \frac{1}{3}x^2$. Check: $xp' = \frac{1}{2}x + \frac{2}{3}x^2$, and $p + xp' = 1 + x + x^2$.

Exercise B3: A Dimension via a Surjective Map.

Let $U = \{p \in \mathbb{R}[x]_{\leq 4} : p(1) = p(2) = 0\}$. Use a surjective linear map to show that $\dim U = 3$. Hence find a basis of U .

Solution. Define $T: \mathbb{R}[x]_{\leq 4} \rightarrow \mathbb{R}^2$, $T(p) = (p(1), p(2))$. It is linear by [Theorem 0.8.5](#), and $U = \ker T$. Its image contains $T(1) = (1, 1)$ and $T(x) = (1, 2)$, which are independent, so $\text{im } T$ is a 2-dimensional subspace of $\mathbb{R}^2$ and equals $\mathbb{R}^2$ ([Theorem 1.6.8](#)). Thus $\text{rank } T = 2$, and by [Theorem 3.3.1](#), $\dim U = 5 - 2 = 3$.

The polynomials $(x-1)(x-2)$, $x(x-1)(x-2)$ and $x^2(x-1)(x-2)$ vanish at 1 and 2 and have degrees $2, 3, 4 \leq 4$, so they lie in U . They are non-zero of distinct degrees, hence independent ([Theorem 1.4.4](#)). There are $3 = \dim U$ of them, so they form a basis of U ([Theorem 1.6.4](#)).

3.3.7.3 C. Going deeper

Exercise C1: A Lower Bound for the Rank of a Composite.

Let U and V be finite-dimensional, let W be a vector space, all over F , and let $T: U \rightarrow V$ and $S: V \rightarrow W$ be linear. Write ST for the composite $S \circ T$, which is linear (see the note before [Exercise \(C1: Kernels and Images of a Composite\)](#)). Let $S': \text{im } T \rightarrow W$ be the restriction of S to $\text{im } T$, that is, $S'(\mathbf{v}) = S(\mathbf{v})$ for $\mathbf{v} \in \text{im } T$.

- Show that S' is linear, that $\text{im } S' = \text{im}(ST)$ and that $\ker S' = \ker S \cap \text{im } T$.
- Deduce that $\text{rank}(ST) = \text{rank } T - \dim(\ker S \cap \text{im } T)$.
- Hence prove **Sylvester's inequality**: $\text{rank}(ST) \geq \text{rank } S + \text{rank } T - \dim V$.
- Deduce that $\text{rank}(AB) \geq \text{rank } A + \text{rank } B - n$ for $A \in M_{m \times n}(F)$ and $B \in M_{n \times p}(F)$.

Hint: for (c), compare $\ker S \cap \text{im } T$ with $\ker S$.

Solution.

- (a) S' satisfies (LT1) and (LT2) because S does and $\text{im } T$ is a subspace (Theorem 3.2.2) with the operations of V . A vector $\mathbf{w}$ lies in $\text{im } S'$ exactly when $\mathbf{w} = S(\mathbf{v})$ with $\mathbf{v} = T(\mathbf{u})$ for some $\mathbf{u} \in U$, that is, when $\mathbf{w} = ST(\mathbf{u})$; so $\text{im } S' = \text{im}(ST)$. A vector $\mathbf{v} \in \text{im } T$ lies in $\ker S'$ exactly when $S(\mathbf{v}) = \mathbf{0}$; so $\ker S' = \ker S \cap \text{im } T$.
- (b) $\text{im } T$ is finite-dimensional with $\dim \text{im } T = \text{rank } T$, since U is finite-dimensional (Theorem 3.2.4). By Theorem 3.3.1 applied to $S': \text{im } T \rightarrow W$ and by (a),

$$\text{rank } T = \dim \ker S' + \dim \text{im } S' = \dim(\ker S \cap \text{im } T) + \text{rank}(ST).$$

- (c) $\ker S \cap \text{im } T$ is a subspace of $\ker S$, so $\dim(\ker S \cap \text{im } T) \leq \text{nullity } S = \dim V - \text{rank } S$, by Theorem 1.6.7 and Theorem 3.3.1 for S . Substituting into (b),

$$\text{rank}(ST) \geq \text{rank } T - (\dim V - \text{rank } S) = \text{rank } S + \text{rank } T - \dim V.$$

- (d) Take $T = T_B: F^p \rightarrow F^n$ and $S = T_A: F^n \rightarrow F^m$. By associativity (Theorem 0.9.4), $T_A T_B(\mathbf{x}) = A(B\mathbf{x}) = (AB)\mathbf{x}$, so $ST = T_{AB}$. By Example (Rank and Nullity in Examples) (b), ranks of matrix maps are ranks of matrices, and $\dim F^n = n$. So (c) gives $\text{rank}(AB) \geq \text{rank } A + \text{rank } B - n$.

Exercise C2: When the Kernel Equals the Image.

Let V be finite-dimensional and $T: V \rightarrow V$ linear with $\ker T = \text{im } T$.

- (a) Prove that $\dim V$ is even.
- (b) Give an example of such a T on $\mathbb{R}^2$, and one on $\mathbb{R}^4$.
- (c) Is there such a T on $\mathbb{R}^3$? Justify your answer.

Solution.

- (a) By Theorem 3.3.1, $\dim V = \text{nullity } T + \text{rank } T = \dim \ker T + \dim \text{im } T = 2 \dim \text{im } T$, which is even.
- (b) On $\mathbb{R}^2$, let $T(x, y) = (y, 0)$. Then $T(x, y) = \mathbf{0}$ exactly when $y = 0$, so $\ker T$ is the x -axis; and the outputs $(y, 0)$ fill the x -axis, so $\text{im } T$ is the x -axis too. On $\mathbb{R}^4$, let $T(x_1, x_2, x_3, x_4) = (x_2, 0, x_4, 0)$. Its kernel is $\{x_2 = x_4 = 0\} = \text{span}(\mathbf{e}_1, \mathbf{e}_3)$, and its image is $\{(a, 0, b, 0)\} = \text{span}(\mathbf{e}_1, \mathbf{e}_3)$. Both maps are matrix maps, hence linear.
- (c) No. By (a), $\dim V$ would be even, but $\dim \mathbb{R}^3 = 3$.

3.4 The Algebra of Linear Maps

So far we have studied one linear map at a time: its kernel, its image, and the count that ties them together. But linear maps rarely come alone. The map $p \mapsto p + p'$ is the identity plus differentiation, and rotating then projecting is one map followed by another. This section shows that linear maps can be added, scaled and composed, that these operations obey almost all the rules of arithmetic, and that an operator can be plugged into a polynomial just as a square matrix can.

3.4.1 Adding and scaling linear maps

Look at three maps on $\mathbb{R}[x]_{\leq 2}$ that have already appeared in this chapter or will soon:

$$p \mapsto p + p', \quad p \mapsto 3p', \quad p \mapsto p' - 2p.$$

Each is built from two simpler maps, the identity id and differentiation D , by the recipe “apply both to p , then combine the outputs with the operations of the target space”. The first is “ $\text{id} + D$ ”, the second “ $3D$ ”, the third “ $D - 2\text{id}$ ”. Instead of checking linearity for each such combination separately, we would like to check it once for the recipe. The recipe itself deserves a name.

Linear maps into the same space can be added and scaled output by output, and the results are again linear maps.

Definition 3.4.1 — The space of linear maps

Let V and W be vector spaces over **the same field** F . We write $\mathcal{L}(V, W)$ for the set of **all** linear maps $V \rightarrow W$, and $\mathcal{L}(V) := \mathcal{L}(V, V)$ for the set of linear operators on V (Definition 3.1.1). For $S, T \in \mathcal{L}(V, W)$ and $c \in F$, the **sum** $S + T$ and the **scalar multiple** cT are the functions $V \rightarrow W$ defined by

$$(S + T)(\mathbf{v}) := S\mathbf{v} + T\mathbf{v}, \quad (cT)(\mathbf{v}) := c(T\mathbf{v}) \quad \text{for every } \mathbf{v} \in V.$$

As in Definition 3.1.1, we write $T\mathbf{v}$ for $T(\mathbf{v})$ when no brackets are needed. In words: to evaluate $S + T$ at $\mathbf{v}$, evaluate both maps at $\mathbf{v}$ and add the results **in** W ; to evaluate cT , evaluate T and scale the result in W . The operations of V play no role in the definition. That is why S and T must have the **same codomain**: the sum $S\mathbf{v} + T\mathbf{v}$ has to be computed in one space. Both maps must also be defined on the same V .

We have used these operations before without naming them: the operations on F^X in Example (The Standard Vector Spaces) (d) are defined by the same pointwise recipe, with values added in F instead of W .

It turns out that the recipe never leaves the world of linear maps, and that $\mathcal{L}(V, W)$ is itself a vector space. This is the first sign that maps between vector spaces are as good as vectors.

Theorem 3.4.1 — $\mathcal{L}(V, W)$ Is a Vector Space

Let V and W be vector spaces over F . If $S, T \in \mathcal{L}(V, W)$ and $c \in F$, then $S + T$ and cT are linear. With the operations of Definition 3.4.1, $\mathcal{L}(V, W)$ is a vector space over F . Its zero vector is the zero map $\mathbf{v} \mapsto \mathbf{0}_W$, and the negative of T is $(-1)T: \mathbf{v} \mapsto -T\mathbf{v}$.

Idea. There are two layers. The set W^V of **all** functions $V \rightarrow W$, with the pointwise operations, is a vector space, for the same reason F^X is: each axiom is checked one input at a time, where it becomes an axiom of W . Then $\mathcal{L}(V, W)$ sits inside W^V , so the Subspace Test finishes the job, and its three conditions are exactly “the zero map is linear”, “a sum of linear maps is linear” and “a multiple of a linear map is linear”.

Proof. Let W^V be the set of all functions $V \rightarrow W$, with the operations of [Definition 3.4.1](#) applied to arbitrary functions. The verification in [Example \(The Standard Vector Spaces\)](#) (d) goes through word for word with W in place of F : two functions $V \rightarrow W$ are equal when they agree at every $\mathbf{v} \in V$ ([Definition 0.4.1](#)), and at each $\mathbf{v}$ every axiom for W^V becomes the same axiom in W , with zero function $\mathbf{v} \mapsto \mathbf{0}_W$ and negative $\mathbf{v} \mapsto -f(\mathbf{v})$. Hence W^V is a vector space over F , and $\mathcal{L}(V, W) \subseteq W^V$. We apply [Theorem 1.2.1](#).

- (1) The zero map is linear: $\mathbf{0}_W = \mathbf{0}_W + \mathbf{0}_W$ and $\mathbf{0}_W = c\mathbf{0}_W$ by [Theorem 1.1.6](#). So $\mathbf{0} \in \mathcal{L}(V, W)$.
- (2) Let $S, T \in \mathcal{L}(V, W)$, $\mathbf{u}, \mathbf{v} \in V$ and $a \in F$. By the definition of $S + T$, the linearity of S and T , and the axioms of W ,

$$(S + T)(\mathbf{u} + \mathbf{v}) = S\mathbf{u} + S\mathbf{v} + T\mathbf{u} + T\mathbf{v} = (S + T)\mathbf{u} + (S + T)\mathbf{v},$$

$$(S + T)(a\mathbf{v}) = aS\mathbf{v} + aT\mathbf{v} = a(S\mathbf{v} + T\mathbf{v}) = a(S + T)\mathbf{v}.$$

So $S + T \in \mathcal{L}(V, W)$.

- (3) Let $c \in F$ and $T \in \mathcal{L}(V, W)$. Then $(cT)(\mathbf{u} + \mathbf{v}) = c(T\mathbf{u} + T\mathbf{v}) = (cT)\mathbf{u} + (cT)\mathbf{v}$, and $(cT)(a\mathbf{v}) = c(aT\mathbf{v}) = (ca)T\mathbf{v} = (ac)T\mathbf{v} = a(cT)\mathbf{v}$, using the axioms of W and commutativity of multiplication in F . So $cT \in \mathcal{L}(V, W)$.

By [Theorem 1.2.1](#), $\mathcal{L}(V, W)$ is a subspace of W^V , hence a vector space over F with the same zero and negatives. The negative of T in W^V is $\mathbf{v} \mapsto -T\mathbf{v} = (-1)T\mathbf{v}$, by [Theorem 1.1.7](#). This proves the theorem. ■

The step that used commutativity of F is worth a second look: cT is linear because c can be moved past the scalar a . This is one place where “ F is a field” is used silently.

Example. (Sums of Matrix Maps and Other First Examples)

For $A \in M_{m \times n}(F)$, write $T_A: F^n \rightarrow F^m$, $\mathbf{x} \mapsto A\mathbf{x}$, for the matrix map of [Example \(Multiplication by a Matrix\)](#).

- (a) Show that $T_A + T_B = T_{A+B}$ and $cT_A = T_{cA}$ for $A, B \in M_{m \times n}(F)$ and $c \in F$.
- (b) Describe $\mathcal{L}(V, W)$ when $V = \{\mathbf{0}\}$, and when $W = \{\mathbf{0}\}$.
- (c) On $\mathbb{R}[x]_{\leq 2}$, write $T(p) = p' - 2p$ in terms of id and D , and compute $T(1 + x^2)$.

Solution.

- (a) Let $\mathbf{x} \in F^n$. By [Definition 3.4.1](#) and distributivity ([Theorem 0.9.4](#)),

$$(T_A + T_B)(\mathbf{x}) = A\mathbf{x} + B\mathbf{x} = (A + B)\mathbf{x} = T_{A+B}(\mathbf{x}), \quad (cT_A)(\mathbf{x}) = c(A\mathbf{x}) = (cA)\mathbf{x} = T_{cA}(\mathbf{x}).$$

The maps agree at every $\mathbf{x}$, so they are equal. So adding and scaling matrix maps is adding and scaling matrices.

- (b) If $V = \{\mathbf{0}\}$, a linear map must send $\mathbf{0}$ to $\mathbf{0}_W$, so the zero map is the **only** element: $\mathcal{L}(\{\mathbf{0}\}, W) = \{\mathbf{0}\}$. If $W = \{\mathbf{0}\}$, every function $V \rightarrow \{\mathbf{0}\}$ is the zero map, and it is linear, so again $\mathcal{L}(V, \{\mathbf{0}\}) = \{\mathbf{0}\}$. These degenerate cases matter because formulas

like “ $\dim \mathcal{L}(V, W) = \dim V \cdot \dim W$ ”, which we prove later in this chapter, must give 0 here, and they do.

- (c) $T = D - 2 \text{id}$, which is linear by [Theorem 3.4.1](#). With $p = 1 + x^2$, $p' = 2x$, so $T(p) = 2x - 2 - 2x^2$.

Here is a non-example by minimal change. Keep D on $\mathbb{R}[x]_{\leq 2}$, but replace id by the evaluation map $E(p) = p(0) \in \mathbb{R}$. Both maps are linear and both are defined on $\mathbb{R}[x]_{\leq 2}$. But D lands in $\mathbb{R}[x]_{\leq 2}$ and E lands in $\mathbb{R}$, so $D + E$ would require adding a polynomial to a number. The clause that fails is “ $S, T \in \mathcal{L}(V, W)$ ” with the **same** W : the sum is not defined.

3.4.2 Composition

Addition makes $\mathcal{L}(V, W)$ a vector space, but the operation that really matters for maps is doing one after another. For functions, Chapter 0 wrote this as $S \circ T$ ([Definition 0.4.4](#)). For linear maps we drop the circle.

Remark. Notation. For $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, we write $ST := S \circ T: U \rightarrow W$, so $(ST)\mathbf{u} = S(T\mathbf{u})$. **The map written on the right acts first.** For an operator $T \in \mathcal{L}(V)$, $T^2 = TT$ means “apply T twice”, never “square the output”.

Is ST linear, and how does composition interact with the sum? It turns out that composition behaves like a multiplication: associative, with identities, and distributive over addition on both sides.

Theorem 3.4.2 — Composition of Linear Maps

Let U, V, W, X be vector spaces over F .

- (a) **(Linearity)** If $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, then $ST \in \mathcal{L}(U, W)$.
- (b) **(Associativity)** If $T \in \mathcal{L}(U, V)$, $S \in \mathcal{L}(V, W)$ and $R \in \mathcal{L}(W, X)$, then $R(ST) = (RS)T$.
- (c) **(Identities)** If $T \in \mathcal{L}(U, V)$, then $T \text{id}_U = T = \text{id}_V T$.
- (d) **(Distributivity)** If $T, T_1, T_2 \in \mathcal{L}(U, V)$, $S, S_1, S_2 \in \mathcal{L}(V, W)$ and $c \in F$, then

$$(S_1 + S_2)T = S_1T + S_2T, \quad S(T_1 + T_2) = ST_1 + ST_2, \quad (cS)T = c(ST) = S(cT).$$

Idea. Parts (b) and (c) hold for all functions, so there is nothing new to do. Part (a) is a two-line check. The interesting point is in (d): the three identities are not equally cheap. In $(S_1 + S_2)T$ the map T acts first and the sum is taken afterwards, so the identity is just the definition of the sum. In $S(T_1 + T_2)$ the sum is taken **before** S acts, and pulling it out of S needs S to be linear. So we should watch where the linearity of the **left** factor is spent.

Proof.

- (a) Let $\mathbf{u}, \mathbf{u}' \in U$ and $a \in F$. Since T and then S are linear,

$$ST(\mathbf{u} + \mathbf{u}') = S(T\mathbf{u} + T\mathbf{u}') = ST\mathbf{u} + ST\mathbf{u}', \quad ST(a\mathbf{u}) = S(aT\mathbf{u}) = aST\mathbf{u}.$$

- (b) and (c) hold for arbitrary functions, by [Theorem 0.4.2](#) and the observation after [Definition 0.4.5](#).
 (c) Each side is a function $U \rightarrow W$, so we compare values at $\mathbf{u} \in U$. By [Definition 3.4.1](#),

$$(S_1 + S_2)T\mathbf{u} = S_1(T\mathbf{u}) + S_2(T\mathbf{u}) = (S_1T + S_2T)\mathbf{u},$$

with no linearity used. Next, since S is linear,

$$S(T_1 + T_2)\mathbf{u} = S(T_1\mathbf{u} + T_2\mathbf{u}) = ST_1\mathbf{u} + ST_2\mathbf{u} = (ST_1 + ST_2)\mathbf{u}.$$

Finally $(cS)(T\mathbf{u}) = c(ST\mathbf{u})$ by definition, and $S(cT\mathbf{u}) = c(ST\mathbf{u})$ because S is linear. This proves the theorem. ■

In short, $\mathcal{L}(V)$ carries a vector space structure and a multiplication (composition) that is associative, has the unit id_V , distributes over addition, and lets scalars move freely through products. A vector space with such a multiplication is called an **algebra** over F ; $M_n(F)$ is the other algebra we know ([Theorem 0.9.4](#)). The two will turn out to be the same algebra in disguise when we write operators as matrices later in this chapter.

Example. (Rotate, Then Project)

Let $R: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the rotation by 90° counterclockwise, $R(x, y) = (-y, x)$, and let $P: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $P(x, y) = (x, 0)$, be the projection onto the x -axis. Compute PR and RP .

Solution. In PR the rotation acts first:

$$PR(x, y) = P(-y, x) = (-y, 0), \quad RP(x, y) = R(x, 0) = (0, x).$$

At $(1, 0)$: $PR(1, 0) = (0, 0)$, while $RP(1, 0) = (0, 1)$. So $PR \neq RP$. Geometrically, PR rotates $\mathbf{e}_1$ onto the y -axis and then squashes it to $\mathbf{0}$; RP leaves $\mathbf{e}_1$ alone under the projection and then rotates it to $\mathbf{e}_2$. In matrix form, $P = T_A$ and $R = T_B$ with $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, and indeed $AB = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = BA$.

Warning. Composition is not commutative, and the order is read right to left. In [Example \(Rotate, Then Project\)](#), $PR \neq RP$. It can be worse: for $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^3)$ and $S \in \mathcal{L}(\mathbb{R}^3, \mathbb{R}^2)$, both $ST \in \mathcal{L}(\mathbb{R}^2)$ and $TS \in \mathcal{L}(\mathbb{R}^3)$ are defined, but they live on different spaces, so they cannot even be compared. And for $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^3)$, $S \in \mathcal{L}(\mathbb{R}^3, \mathbb{R}^4)$, ST is defined but TS is not. The thing you do first goes on the **right**.

Quick check. Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ and $S: \mathbb{R}^2 \rightarrow \mathbb{R}^4$ be linear. Which of ST and TS is defined, and between which spaces does it go?

Solution. $ST = S \circ T$ first applies $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$, then $S: \mathbb{R}^2 \rightarrow \mathbb{R}^4$, so $ST \in \mathcal{L}(\mathbb{R}^3, \mathbb{R}^4)$. TS would apply S first, landing in $\mathbb{R}^4$, where T is not defined; so TS is not defined.

Because of the missing commutativity, school identities about products must be rechecked. By

distributivity ([Theorem 3.4.2](#) (d)),

$$(S + T)^2 = (S + T)(S + T) = S^2 + ST + TS + T^2,$$

which equals $S^2 + 2ST + T^2$ only if $ST = TS$. With P and R as above, $PR + RP: (x, y) \mapsto (-y, x)$, while $2PR: (x, y) \mapsto (-2y, 0)$; so $(P + R)^2 \neq P^2 + 2PR + R^2$. This is the operator version of the matrix warning in Chapter 0, §9.

3.4.3 Operators, powers and polynomials

Composition of operators on one space V never leaves $\mathcal{L}(V)$, so we may compose an operator with itself as often as we like. Repeated composition leads to powers, and combinations of powers to polynomials, exactly as for square matrices ([Definition 0.9.14](#)).

Definition 3.4.2 — Powers and Polynomials of an Operator

Let V be a vector space over F and $T \in \mathcal{L}(V)$. The **powers** of T are defined recursively by

$$T^0 := \text{id}_V, \quad T^{k+1} := T^k T \quad (k \in \mathbb{N}).$$

For $p = a_0 + a_1x + \cdots + a_Nx^N \in F[x]$, the operator $p(T) \in \mathcal{L}(V)$ is

$$p(T) := a_0 \text{id}_V + a_1 T + a_2 T^2 + \cdots + a_N T^N.$$

In words: T^k applies T exactly k times, and T^0 does nothing. In $p(T)$, the **constant term becomes** $a_0 \text{id}_V$, because an operator cannot be added to a scalar. Each T^k is linear by [Theorem 3.4.2](#) (a) and induction, and $p(T)$ is linear by [Theorem 3.4.1](#). As for matrices, $p(T)$ does not depend on how many zero coefficients we write, since $0 \cdot T^k$ is the zero map.

Plugging in an operator respects sums and products of polynomials, just as plugging in a matrix does.

Theorem 3.4.3 — Polynomials in One Operator

Let $T \in \mathcal{L}(V)$ and $p, q \in F[x]$. Then:

- (a) $T^j T^k = T^{j+k}$ for all $j, k \in \mathbb{N}$;
- (b) $(p + q)(T) = p(T) + q(T)$ and $(pq)(T) = p(T)q(T)$;
- (c) $p(T)q(T) = q(T)p(T)$.

Idea. The proof of [Theorem 0.9.9](#) used four matrix rules: associativity, identity, distributivity and “scalars move through products”. [Theorem 3.4.2](#) supplies exactly these four rules for operators. So the old proof transfers line by line. Part (c) is the surprise: T need not commute with other operators, but it commutes with every polynomial in itself.

Proof.

- (a) Fix j and use induction on k ([Theorem 0.2.7](#)). For $k = 0$, $T^j T^0 = T^j \text{id}_V = T^j$ by [Theorem 3.4.2](#) (c). If $T^j T^k = T^{j+k}$, then by [Definition 3.4.2](#) and associativity

(Theorem 3.4.2 (b)),

$$T^j T^{k+1} = T^j (T^k T) = (T^j T^k) T = T^{j+k} T = T^{j+k+1}.$$

- (b) Let $p = \sum_{i=0}^m a_i x^i$ and $q = \sum_{j=0}^l b_j x^j$. The statement for sums holds because $(a_i + b_i)T^i = a_i T^i + b_i T^i$ in the vector space $\mathcal{L}(V)$ (Theorem 3.4.1). For products, the coefficient of x^k in pq is $\sum_{i+j=k} a_i b_j$ (Definition 0.8.2), so

$$(pq)(T) = \sum_k \left(\sum_{i+j=k} a_i b_j \right) T^k = \sum_{i=0}^m \sum_{j=0}^l (a_i b_j) T^i T^j = \left(\sum_{i=0}^m a_i T^i \right) \left(\sum_{j=0}^l b_j T^j \right) = p(T) q(T),$$

where the second equality uses (a), and the third uses both distributive laws and $(a_i T^i)(b_j T^j) = (a_i b_j) T^i T^j$, all from Theorem 3.4.2 (d).

- (c) Since $pq = qp$ in $F[x]$ (Theorem 0.8.1), part (b) gives $p(T)q(T) = (pq)(T) = (qp)(T) = q(T)p(T)$. This proves the theorem. ■

In practice, (b) says that **a factorization of p is a factorization of $p(T)$** : for instance $T^2 - 3T + 2\text{id}_V = (T - \text{id}_V)(T - 2\text{id}_V) = (T - 2\text{id}_V)(T - \text{id}_V)$.

Example. (Differentiation Is Nilpotent)

Let $n \geq 0$ and let $D \in \mathcal{L}(F[x]_{\leq n})$ be (formal) differentiation (Example (Differentiation)), $D(a_0 + a_1x + \cdots + a_nx^n) = a_1 + 2a_2x + \cdots + na_nx^{n-1}$, where ka_k means a_k added to itself k times. Show that $D^{n+1} = 0$. Show also that $D^n \neq 0$ when $F = \mathbb{R}$.

Solution. For $0 \leq k \leq n$, $D(x^k) = kx^{k-1}$ if $k \geq 1$ and $D(1) = 0$. So D sends each x^k to a scalar multiple of x^{k-1} , or to 0 when $k = 0$. Applying this $k+1$ times, $D^{k+1}(x^k)$ is a scalar multiple of $D(1) = 0$, so $D^{k+1}(x^k) = 0$. Since $k \leq n$, also $D^{n+1}(x^k) = D^{n-k}(D^{k+1}(x^k)) = D^{n-k}(0) = 0$, by Theorem 3.4.3 (a).

Now let $p = a_0 + a_1x + \cdots + a_nx^n$. Since D^{n+1} is linear, Theorem 3.1.4 gives

$$D^{n+1}(p) = a_0 D^{n+1}(1) + a_1 D^{n+1}(x) + \cdots + a_n D^{n+1}(x^n) = 0.$$

Hence $D^{n+1} = 0$.

Over $\mathbb{R}$, $D^n(x^n) = n(n-1)\cdots 1 = n! \neq 0$, so $D^n \neq 0$. Over $\mathbb{F}_2$ with $n = 2$, however, $D(x^2) = 2x = 0$, and $D^2 = 0$ already; the field matters for the exact power, but never for $D^{n+1} = 0$.

An operator T with $T^k = 0$ for some $k \geq 1$ is called **nilpotent**. The example shows that a non-zero operator can have a power equal to zero, just as the matrix $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ can. Differentiation is the model nilpotent operator, and it returns when we study canonical forms.

Example. (Operators That Satisfy a Polynomial)

With P and R as in Example (Rotate, Then Project), find non-zero polynomials p and q with $p(P) = 0$ and $q(R) = 0$. Also, for an arbitrary $p \in F[x]$, compute $p(\text{id}_V)$ and

$p(Z)$, where $Z \in \mathcal{L}(V)$ is the zero operator.

Solution. For the projection, $P^2(x, y) = P(x, 0) = (x, 0) = P(x, y)$, so $P^2 = P$ and $p = x^2 - x$ satisfies $p(P) = P^2 - P = 0$. By Theorem 3.4.3 (b), this says $P(P - \text{id}) = 0$: whatever $P - \text{id}$ produces is killed by P , and indeed $(P - \text{id})(x, y) = (0, -y)$ lies on the y -axis.

For the rotation, $R^2(x, y) = R(-y, x) = (-x, -y)$, so $R^2 = -\text{id}$, and $q = x^2 + 1$ satisfies $q(R) = R^2 + \text{id} = 0$. Rotating twice by 90° is rotating by 180° . Since q has no real roots, q has no real linear factor, so the trick “factor q , then factor $q(R)$ ” gives nothing over $\mathbb{R}$.

For $p = a_0 + a_1x + \cdots + a_Nx^N$: every power of id_V is id_V , so $p(\text{id}_V) = (a_0 + a_1 + \cdots + a_N)\text{id}_V = p(1)\text{id}_V$. Every power Z^k with $k \geq 1$ is the zero map, but $Z^0 = \text{id}_V$, so $p(Z) = a_0\text{id}_V$, the scalar $p(0)$ times id_V . The second computation is the reason for the convention $T^0 = \text{id}_V$: with it, “the constant term” of $p(T)$ is what survives when T is the zero operator.

Warning. The constant term is $a_0\text{id}_V$, not a_0 . Writing “ $P^2 - P + 3$ ” for an operator is a type error: $P^2 - P$ is a map and 3 is a number. For $p = x^2 - x + 3$, $p(P) = P^2 - P + 3\text{id}$, which is 3id since $P^2 = P$. Forgetting the id leads to “ $p(P) = 3$ ”, which is not an operator at all.

Quick check. With P the projection above, compute $p(P)$ for $p = x^3 - 2x + 1$.

Solution. Since $P^2 = P$, also $P^3 = P^2P = PP = P$. So $p(P) = P^3 - 2P + \text{id} = P - 2P + \text{id} = \text{id} - P$, which is the map $(x, y) \mapsto (0, y)$: the projection onto the y -axis.

Polynomial relations such as $P^2 = P$, $R^2 = -\text{id}$ and $D^{n+1} = 0$ carry real information about an operator. In the next section we use one to write down an inverse without solving any equations, and later in the book the polynomials that kill an operator lead to the minimal polynomial and to canonical forms.

3.4.4 Exercises

3.4.4.1 A. Check your understanding

Exercise A1.

- For $S, T \in \mathcal{L}(V, W)$ and $c \in F$, define $S + T$ and cT . Why must S and T have the same codomain?
- True or false: for all $S, T \in \mathcal{L}(V)$, $ST = TS$. Justify your answer.
- Let $T \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^2)$ and $S \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^5)$. Which of ST , TS are defined, and in which spaces do they lie?
- For $T \in \mathcal{L}(V)$ and $p = 4$ (a constant polynomial), what is $p(T)$?
- True or false: for all $S, T \in \mathcal{L}(V)$, $(S + T)^2 = S^2 + 2ST + T^2$. Justify your answer.

Solution.

- (a) $(S+T)(\mathbf{v}) = S\mathbf{v} + T\mathbf{v}$ and $(cT)(\mathbf{v}) = cT\mathbf{v}$ for every $\mathbf{v} \in V$ (Definition 3.4.1). The sum $S\mathbf{v} + T\mathbf{v}$ must be computed in a single vector space, so both values must lie in the same W .
- (b) False. In Example (Rotate, Then Project), $PR(1,0) = (0,0)$ but $RP(1,0) = (0,1)$.
- (c) Both are defined. ST applies T first: $ST \in \mathcal{L}(\mathbb{R}^5)$. TS applies S first: $TS \in \mathcal{L}(\mathbb{R}^2)$.
- (d) $p(T) = 4\text{id}_V$, by Definition 3.4.2.
- (e) False. By Theorem 3.4.2 (d), $(S+T)^2 = S^2 + ST + TS + T^2$, which differs from $S^2 + 2ST + T^2$ by $TS - ST$. With $S = P$, $T = R$ of Example (Rotate, Then Project), $RP - PR: (x,y) \mapsto (y,x)$ is not the zero map, so the identity fails.

3.4.4.2 B. Practice

Exercise B1: Computing compositions.

Let $S, T \in \mathcal{L}(\mathbb{R}^2)$ be given by $S(x, y) = (x + y, y)$ and $T(x, y) = (y, x)$.

- (a) Compute ST , TS and $2S - T$.
- (b) Compute S^2 and S^k for every $k \in \mathbb{N}$.
- (c) Hence show that $(S - \text{id})^2 = 0$.

Solution.

- (a) Applying the right-hand map first,

$$ST(x, y) = S(y, x) = (x + y, x), \quad TS(x, y) = T(x + y, y) = (y, x + y).$$

So $ST \neq TS$; for instance $ST(1,0) = (1,1)$ and $TS(1,0) = (0,1)$. By Definition 3.4.1, $(2S - T)(x, y) = (2x + 2y, 2y) - (y, x) = (2x + y, 2y - x)$.

- (b) $S^2(x, y) = S(x + y, y) = (x + 2y, y)$. We claim $S^k(x, y) = (x + ky, y)$ for all $k \in \mathbb{N}$, by induction on k . For $k = 0$, $S^0 = \text{id}$ sends (x, y) to $(x + 0y, y)$. If the claim holds for k , then

$$S^{k+1}(x, y) = S^k(S(x, y)) = S^k(x + y, y) = (x + y + ky, y) = (x + (k+1)y, y),$$

where the first equality is Definition 3.4.2, $S^{k+1} = S^k S$.

- (c) $(S - \text{id})(x, y) = (y, 0)$, so $(S - \text{id})^2(x, y) = (S - \text{id})(y, 0) = (0, 0)$. Hence $(S - \text{id})^2 = 0$, that is, $S - \text{id}$ is nilpotent. Expanding with Theorem 3.4.3 (b), $S^2 - 2S + \text{id} = 0$, which matches (b): $(x + 2y, y) - 2(x + y, y) + (x, y) = (0, 0)$.

Exercise B2: Operators that preserve a subspace.

Let U be a subspace of V , and let

$$\mathcal{A}_U := \{T \in \mathcal{L}(V) : T\mathbf{u} \in U \text{ for every } \mathbf{u} \in U\}.$$

Prove that $\mathcal{A}_U$ is a subspace of $\mathcal{L}(V)$, that $\text{id}_V \in \mathcal{A}_U$, and that $ST \in \mathcal{A}_U$ whenever $S, T \in \mathcal{A}_U$.

Solution. We use [Theorem 1.2.1](#) in the vector space $\mathcal{L}(V)$ ([Theorem 3.4.1](#)).

- (1) The zero map sends every $\mathbf{u} \in U$ to $\mathbf{0}$, and $\mathbf{0} \in U$ because U is a subspace. So $0 \in \mathcal{A}_U$.
- (2) Let $S, T \in \mathcal{A}_U$ and $\mathbf{u} \in U$. Then $S\mathbf{u} \in U$ and $T\mathbf{u} \in U$, so $(S+T)\mathbf{u} = S\mathbf{u} + T\mathbf{u} \in U$, since U is closed under addition. Therefore $S + T \in \mathcal{A}_U$.
- (3) Let $c \in F$ and $T \in \mathcal{A}_U$. For $\mathbf{u} \in U$, $(cT)\mathbf{u} = cT\mathbf{u} \in U$, since U is closed under scalar multiplication. Therefore $cT \in \mathcal{A}_U$.

Hence $\mathcal{A}_U$ is a subspace of $\mathcal{L}(V)$. Next, $\text{id}_V \mathbf{u} = \mathbf{u} \in U$ for every $\mathbf{u} \in U$, so $\text{id}_V \in \mathcal{A}_U$. Finally let $S, T \in \mathcal{A}_U$ and $\mathbf{u} \in U$. Then $T\mathbf{u} \in U$, and applying the hypothesis on S to the vector $T\mathbf{u} \in U$ gives $ST\mathbf{u} = S(T\mathbf{u}) \in U$. This shows $ST \in \mathcal{A}_U$.

Exercise B3: A polynomial that kills D .

Let $D \in \mathcal{L}(\mathbb{R}[x]_{\leq 2})$ be differentiation.

- (a) Find a non-zero polynomial p with $p(D) = 0$.
- (b) Show that no non-zero polynomial q of degree at most 2 satisfies $q(D) = 0$.
- (c) Hence, or otherwise, compute $(\text{id} + D)^3$ as a combination of id , D , D^2 , and evaluate $(\text{id} + D)^3(x^2)$.

Solution.

- (a) By [Example \(Differentiation Is Nilpotent\)](#) with $n = 2$, $D^3 = 0$. So $p = x^3$ works.
- (b) Let $q = a + bx + cx^2$ with $q(D) = a\text{id} + bD + cD^2 = 0$. Applying this operator to x^2 , with $D(x^2) = 2x$ and $D^2(x^2) = 2$,

$$0 = ax^2 + 2bx + 2c.$$

A polynomial is zero exactly when all its coefficients are, so $a = 0$, $2b = 0$ and $2c = 0$. Hence $a = b = c = 0$ and $q = 0$.

- (c) The operator $\text{id} + D$ is $q(D)$ for $q = 1 + x$. Applying the product rule of [Theorem 3.4.3](#) (b) twice, $(\text{id} + D)^3 = q(D)q(D)q(D) = (q^3)(D)$, and $q^3 = 1 + 3x + 3x^2 + x^3$. So

$$(\text{id} + D)^3 = \text{id} + 3D + 3D^2 + D^3 = \text{id} + 3D + 3D^2,$$

using $D^3 = 0$ from (a). Therefore $(\text{id} + D)^3(x^2) = x^2 + 3 \cdot 2x + 3 \cdot 2 = x^2 + 6x + 6$.

3.4.4.3 C. Going deeper

Exercise C1: The Heisenberg relation.

Let F be a field, let $D \in \mathcal{L}(F[x])$ be formal differentiation, $D(\sum_k a_k x^k) = \sum_{k \geq 1} k a_k x^{k-1}$ ([Example \(Differentiation\)](#)), and let $S \in \mathcal{L}(F[x])$ be multiplication by x , $S(p) = xp$ (the map M of [Example \(Multiplication by \$x\$ \)](#)).

- (a) Prove that $DS - SD = \text{id}$.
- (b) Let $T \in \mathcal{L}(F^n, F^m)$. Prove that $T = T_A$, where $A \in M_{m \times n}(F)$ is the matrix with

columns $T\mathbf{e}_1, \dots, T\mathbf{e}_n$.

- (c) Now suppose that F has **characteristic** 0 ([Definition 0.7.2](#)), and let $n \geq 1$. Deduce that there are no operators $D', S' \in \mathcal{L}(F^n)$ with $D'S' - S'D' = \text{id}$.
- (d) Show that the assumption on the characteristic in (c) cannot be dropped.

Hint: for (c), use (b) and consider traces.

Solution.

- (a) Both sides are linear ([Theorem 3.4.1](#), [Theorem 3.4.2](#)). We first compare them on x^k , $k \in \mathbb{N}$. We have $DS(x^k) = D(x^{k+1}) = (k+1)x^k$. If $k \geq 1$, $SD(x^k) = x \cdot kx^{k-1} = kx^k$; if $k = 0$, $SD(1) = x \cdot 0 = 0 = 0 \cdot x^0$. In both cases

$$(DS - SD)(x^k) = (k+1)x^k - kx^k = x^k = \text{id}(x^k).$$

Every $p \in F[x]$ is a finite combination $p = \sum_k a_k x^k$, so by [Theorem 3.1.4](#), $(DS - SD)(p) = \sum_k a_k (DS - SD)(x^k) = \sum_k a_k x^k = p$. Hence $DS - SD = \text{id}$.

- (b) Let $\mathbf{x} = (x_1, \dots, x_n) = x_1\mathbf{e}_1 + \dots + x_n\mathbf{e}_n \in F^n$. By [Theorem 3.1.4](#) and [Theorem 0.9.3](#),

$$T\mathbf{x} = x_1T\mathbf{e}_1 + \dots + x_nT\mathbf{e}_n = A\mathbf{x} = T_A(\mathbf{x}).$$

Since $\mathbf{x}$ was arbitrary, $T = T_A$.

- (c) Suppose, for a contradiction, that $D', S' \in \mathcal{L}(F^n)$ satisfy $D'S' - S'D' = \text{id}$. By (b), $D' = T_A$ and $S' = T_B$ for some $A, B \in M_n(F)$, and $\text{id} = T_{I_n}$. By associativity of matrix multiplication, $T_AT_B(\mathbf{x}) = A(B\mathbf{x}) = (AB)\mathbf{x}$, so $T_AT_B = T_{AB}$, and similarly $T_BT_A = T_{BA}$. With [Example \(Sums of Matrix Maps and Other First Examples\)](#) (a),

$$T_{AB-BA} = T_AT_B - T_BT_A = \text{id} = T_{I_n}.$$

Evaluating at $\mathbf{e}_j$ gives equal j -th columns ([Theorem 0.9.3](#)), so $AB - BA = I_n$. Taking traces and using [Theorem 0.9.6](#),

$$n \cdot 1 = \text{tr } I_n = \text{tr}(AB) - \text{tr}(BA) = 0.$$

Since $n \geq 1$ and F has characteristic 0, $n \cdot 1 \neq 0$ by [Definition 0.7.2](#). This is a contradiction, so no such D', S' exist. In words: over $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ or any field of characteristic 0, the relation $DS - SD = \text{id}$ of (a) has no analogue on F^n , $n \geq 1$. Since every n -dimensional space is isomorphic to F^n , as we show in the next section, the same holds on every non-zero finite-dimensional space. (On the zero space $\text{id} = 0$, so the relation holds trivially there; that is why $n \geq 1$.)

- (d) Over $\mathbb{F}_2$ (characteristic 2), take $A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ as in [Exercise](#)

(C1: $AB - BA = I$ is impossible, or is it?). There $AB - BA = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = I_2$,

because $-1 = 1$ in $\mathbb{F}_2$. By the computation in (c), $D' = T_A$ and $S' = T_B$ satisfy $D'S' - S'D' = T_{AB-BA} = \text{id}$ on $\mathbb{F}_2^2$. So the conclusion of (c) fails without the characteristic assumption; the proof breaks exactly at $n \cdot 1 \neq 0$, since $2 \cdot 1 = 0$ in $\mathbb{F}_2$.

Exercise C2: Commuting nilpotent operators.

Let $S, T \in \mathcal{L}(V)$.

- (a) Suppose $ST = TS$. Prove that $(S + T)^n = \sum_{k=0}^n \binom{n}{k} S^k T^{n-k}$ for every $n \in \mathbb{N}$, where $\binom{n}{k} S^k T^{n-k}$ means $S^k T^{n-k}$ added to itself $\binom{n}{k}$ times.
- (b) Deduce that if $ST = TS$, $S^a = 0$ and $T^b = 0$ for some $a, b \geq 1$, then $(S + T)^{a+b-1} = 0$.
- (c) Show that (b) fails without $ST = TS$: give nilpotent $S, T \in \mathcal{L}(\mathbb{R}^2)$ such that $S + T$ is not nilpotent.

Hint: for (c), try maps that send one standard basis vector to the other and the other to 0.

Solution.

- (a) First, $T^j S = ST^j$ for all $j \in \mathbb{N}$, by induction on j : it holds for $j = 0$, and if $T^j S = ST^j$, then $T^{j+1} S = T^j(TS) = T^j(ST) = (T^j S)T = (ST^j)T = ST^{j+1}$, using associativity (Theorem 3.4.2 (b)) and Theorem 3.4.3 (a) in the form $T^{j+1} = T^j T$.

Now induct on n . For $n = 0$ both sides are id_V . Suppose the formula holds for n . By Theorem 3.4.2 (d),

$$(S + T)^{n+1} = (S + T)^n(S + T) = \sum_{k=0}^n \binom{n}{k} S^k T^{n-k} S + \sum_{k=0}^n \binom{n}{k} S^k T^{n-k+1}.$$

By the first paragraph, $S^k T^{n-k} S = S^k (T^{n-k} S) = S^k S T^{n-k} = S^{k+1} T^{n-k}$. Shifting the index in the first sum, the coefficient of $S^k T^{n+1-k}$ becomes $\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}$ (Pascal's rule, with $\binom{n}{-1} = \binom{n}{n+1} = 0$). This is the formula for $n + 1$.

- (b) By (a) with $n = a + b - 1$, $(S + T)^{a+b-1}$ is a sum of multiples of $S^k T^{a+b-1-k}$, $0 \leq k \leq a + b - 1$. If $k \geq a$, then $S^k = S^{k-a} S^a = 0$ by Theorem 3.4.3 (a), so the term is 0 by Theorem 3.4.2 (d). If $k \leq a - 1$, then $a + b - 1 - k \geq b$, so $T^{a+b-1-k} = 0$ likewise. Every term vanishes, so $(S + T)^{a+b-1} = 0$.
- (c) Let $S(x, y) = (y, 0)$ and $T(x, y) = (0, x)$. Then $S^2(x, y) = S(y, 0) = (0, 0)$ and $T^2(x, y) = T(0, x) = (0, 0)$, so both are nilpotent. But $(S + T)(x, y) = (y, x)$, so $(S + T)^2(x, y) = (x, y)$, that is, $(S + T)^2 = \text{id}$. Then $(S + T)^{2k} = \text{id} \neq 0$ and $(S + T)^{2k+1} = S + T \neq 0$ for every k , so $S + T$ is not nilpotent. Here $ST(x, y) = (x, 0) \neq (0, y) = TS(x, y)$, so the hypothesis of (b) indeed fails.

3.5 Invertible Maps and Isomorphisms

In Chapter 1 we noticed that a basis of length n turns every vector of V into a column in F^n , and that sums and multiples survive the translation. Then $\mathbb{R}[x]_{\leq 2}$ and $\mathbb{R}^3$ behave “the same”, although they are different sets. This section makes “the same” precise. We define invertible linear maps and isomorphisms, prove that finite-dimensional spaces are classified by one number, their dimension, and collect everything we know about invertibility of an operator into one theorem.

3.5.1 When are two spaces the same?

The naive notion of sameness is equality of sets, and it fails at once. The polynomial $3 - x + 2x^2$ is not a column vector, so $\mathbb{R}[x]_{\leq 2} \neq \mathbb{R}^3$. Yet every computation in one space has a twin in the other. Match $a_0 + a_1x + a_2x^2$ with (a_0, a_1, a_2) :

$$(1 + 2x) + (x - x^2) = 1 + 3x - x^2 \quad \longleftrightarrow \quad (1, 2, 0) + (0, 1, -1) = (1, 3, -1).$$

A reader who only sees one side cannot tell which space the computation happened in. Three features make this work. The matching is **one-to-one** and **onto**, so nothing is lost or left over. It **respects sums and multiples** in the direction polynomial $\rightarrow$ column. And it respects them in the **reverse** direction too, so a conclusion drawn among columns can be carried back. In the language of this chapter: a linear map with a linear map going back.

An isomorphism is a perfect dictionary between two vector spaces: every word has exactly one translation, and translating respects sums and scalar multiples in both directions.

Definition 3.5.1 — Invertible Linear Map

Let V and W be vector spaces over F , and let $T \in \mathcal{L}(V, W)$. Then T is **invertible** if there **exists a linear** map $S \in \mathcal{L}(W, V)$ such that

$$ST = \text{id}_V \quad \text{and} \quad TS = \text{id}_W.$$

In words: S undoes T on V (first T , then S , and we are back where we started), T undoes S on W , and the undoing map is itself linear. The definition asks for **one** map S doing **both** jobs.

Well-definedness. If S exists, it is an inverse **function** of T in the sense of Definition 0.4.7, and inverse functions are unique by Theorem 0.4.7. So we may speak of **the** inverse and write $T^{-1} := S$. The same theorem says an invertible T is a bijection.

Is the word “linear” in the definition doing any work? One could imagine a linear bijection whose inverse function is not linear, which would then fail the definition. It turns out this never happens:

Theorem 3.5.1 — The Inverse of a Linear Bijection Is Linear

Let $T \in \mathcal{L}(V, W)$ be bijective. Then its inverse function $T^{-1}: W \rightarrow V$ is linear. Consequently, a linear map T is invertible if and only if it is bijective.

Idea. To show $T^{-1}(\mathbf{w}_1 + \mathbf{w}_2) = T^{-1}\mathbf{w}_1 + T^{-1}\mathbf{w}_2$, we cannot compute T^{-1} directly. But we can apply T , which we know is linear, to the right-hand side, and see that it lands on $\mathbf{w}_1 + \mathbf{w}_2$. Since T is injective, only one vector does that, namely $T^{-1}(\mathbf{w}_1 + \mathbf{w}_2)$.

Proof. Let $\mathbf{w}_1, \mathbf{w}_2 \in W$ and $c \in F$, and put $\mathbf{v}_i := T^{-1}\mathbf{w}_i$, so that $T\mathbf{v}_i = \mathbf{w}_i$. Since T is linear,

$$T(\mathbf{v}_1 + \mathbf{v}_2) = \mathbf{w}_1 + \mathbf{w}_2, \quad T(c\mathbf{v}_1) = c\mathbf{w}_1.$$

Applying T^{-1} to both equations gives $T^{-1}(\mathbf{w}_1 + \mathbf{w}_2) = \mathbf{v}_1 + \mathbf{v}_2 = T^{-1}\mathbf{w}_1 + T^{-1}\mathbf{w}_2$ and $T^{-1}(c\mathbf{w}_1) = c\mathbf{v}_1 = cT^{-1}\mathbf{w}_1$. Hence T^{-1} is linear.

For the last statement: if T is invertible, it is bijective by Theorem 0.4.7. Conversely,

if T is bijective, its inverse function $S = T^{-1}$ satisfies $ST = \text{id}_V$ and $TS = \text{id}_W$ by [Definition 0.4.7](#), and S is linear by the first part. So T is invertible. This proves the theorem. ■

So in practice we check invertibility by checking **bijectivity**, and never need to verify linearity of the inverse. Invertible linear maps are important enough to have a second name.

Definition 3.5.2 — Isomorphism

Let V and W be vector spaces over the same field F . An **isomorphism** from V to W is an invertible linear map $T \in \mathcal{L}(V, W)$; equivalently, by [Theorem 3.5.1](#), a **bijective** linear map.

Definition 3.5.3 — Isomorphic Spaces

Vector spaces V and W over F are **isomorphic**, written $V \cong W$, if there **exists an** isomorphism $V \rightarrow W$.

In words: $V \cong W$ says that **some** perfect dictionary exists; it does not name one. The name comes from Greek *isos* (equal) and *morphe* (form): the two spaces have the same form, though not the same elements. An isomorphism $V \rightarrow V$ is also called an **invertible operator**.

Example. (The Standard Isomorphisms)

Show that each map is an isomorphism, and give its inverse.

- (a) $\Phi: \mathbb{R}[x]_{\leq n} \rightarrow \mathbb{R}^{n+1}$, $a_0 + a_1x + \cdots + a_nx^n \mapsto (a_0, a_1, \dots, a_n)$.
- (b) $\Psi: M_{m \times n}(F) \rightarrow F^{mn}$, which lists the entries row by row: $(a_{ij}) \mapsto (a_{11}, \dots, a_{1n}, a_{21}, \dots, a_{mn})$.
- (c) $\Theta: \mathbb{C} \rightarrow \mathbb{R}^2$, $a + bi \mapsto (a, b)$, where $\mathbb{C}$ is a vector space **over** $\mathbb{R}$.
- (d) $T_A: F^n \rightarrow F^n$, $\mathbf{x} \mapsto A\mathbf{x}$, for an invertible $A \in M_n(F)$; and id_V for any V .

Solution.

- (a) *Linear.* Polynomials are added and scaled coefficient by coefficient, and so are columns, so $\Phi(p + q) = \Phi(p) + \Phi(q)$ and $\Phi(cp) = c\Phi(p)$. *Bijective.* The map $(a_0, \dots, a_n) \mapsto a_0 + a_1x + \cdots + a_nx^n$ is a two-sided inverse function, because two polynomials are equal exactly when all their coefficients agree ([Definition 0.8.1](#)). By [Theorem 3.5.1](#), Φ is an isomorphism, and Φ^{-1} is that map. Hence $\mathbb{R}[x]_{\leq n} \cong \mathbb{R}^{n+1}$.
- (b) Matrices are added and scaled entrywise, and so are vectors of F^{mn} , so Ψ is linear. The map that writes the first n entries of a vector into row 1, the next n into row 2, and so on, undoes Ψ on both sides. Hence $M_{m \times n}(F) \cong F^{mn}$.
- (c) For $a, b, a', b', c \in \mathbb{R}$: $(a + bi) + (a' + b'i) = (a + a') + (b + b')i$ and $c(a + bi) = ca + cbi$, so Θ is linear over $\mathbb{R}$. Every complex number has unique real and imaginary parts ([Definition 0.6.1](#)), so $(a, b) \mapsto a + bi$ is a two-sided inverse. Hence $\mathbb{C} \cong \mathbb{R}^2$ **as real vector spaces**. In the same way [Example \(The Complexification of \$\mathbb{R}^n\$ Is \$\mathbb{C}^n\$ \)](#) showed that $(\mathbf{x}, \mathbf{y}) \mapsto \mathbf{x} + i\mathbf{y}$ is a bijection $(\mathbb{R}^n)_{\mathbb{C}} \rightarrow \mathbb{C}^n$ respecting complex scalars;

in our new language, $(\mathbb{R}^n)_{\mathbb{C}} \cong \mathbb{C}^n$ over $\mathbb{C}$.

- (d) T_A is linear ([Example \(Multiplication by a Matrix\)](#)). Since $A(A^{-1}\mathbf{x}) = (AA^{-1})\mathbf{x} = \mathbf{x}$ and $A^{-1}(A\mathbf{x}) = \mathbf{x}$ by [Theorem 0.9.4](#), the linear map $T_{A^{-1}}$ is a two-sided inverse: $T_A^{-1} = T_{A^{-1}}$. The identity id_V is its own inverse. This degenerate example matters: it shows $V \cong V$, and even the zero space is isomorphic to itself, via the zero map $\{\mathbf{0}\} \rightarrow \{\mathbf{0}\}$, which is $\text{id}_{\{\mathbf{0}\}}$.

Here is a non-example by minimal change. Take (a) with $n = 3$, but keep only three coefficients: $\Phi': \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}^3$, $a_0 + a_1x + a_2x^2 + a_3x^3 \mapsto (a_0, a_1, a_2)$. It is still linear, and still onto, since $(a_0, a_1, a_2) = \Phi'(a_0 + a_1x + a_2x^2)$. But $\Phi'(x^3) = \mathbf{0} = \Phi'(0)$, so Φ' is **not injective**. No S can satisfy $S\Phi' = \text{id}$, because $S\Phi'(x^3) = S(\mathbf{0}) = 0 \neq x^3$. The clause $ST = \text{id}_V$ is the one that fails.

Why this definition. Could we drop one of the two equations? The next example says no.

Example. (One Equation Is Not Enough)

On $\mathbb{R}[x]$, let D be differentiation and $J(p) := \int_0^x p(t) dt$, so that $J(x^k) = \frac{x^{k+1}}{k+1}$ ([Example \(Differentiation\)](#), [Example \(Integration\)](#)). On the sequence space $F^{\mathbb{N}}$, let $L(a_0, a_1, a_2, \dots) := (a_1, a_2, a_3, \dots)$ and $R(a_0, a_1, a_2, \dots) := (0, a_0, a_1, \dots)$ be the left and right shifts of [Example \(The Shift Maps on Sequences\)](#). Show that $DJ = \text{id}$ and $LR = \text{id}$, but that none of D, J, L, R is invertible.

Solution. All four maps are linear, by [Example \(Differentiation\)](#), [Example \(Integration\)](#) and [Example \(The Shift Maps on Sequences\)](#). For each $k \in \mathbb{N}$, $DJ(x^k) = D(\frac{x^{k+1}}{k+1}) = x^k$, so DJ and id agree on every x^k . Every polynomial is a finite combination of the x^k , so $DJ = \text{id}$ by [Theorem 3.1.4](#). Directly, $LR(a_0, a_1, \dots) = L(0, a_0, a_1, \dots) = (a_0, a_1, \dots)$, so $LR = \text{id}$.

But $JD(1) = J(0) = 0 \neq 1$, and $RL(1, 0, 0, \dots) = R(0, 0, \dots) = \mathbf{0}$. In fact $D(1) = 0 = D(0)$ and $L(1, 0, 0, \dots) = \mathbf{0} = L(\mathbf{0})$, so D and L are not injective; and $1 \notin \text{im } J$ (every $J(p)$ has constant coefficient 0) and $(1, 0, 0, \dots) \notin \text{im } R$, so J and R are not surjective. None of the four is bijective, so none is invertible ([Theorem 3.5.1](#)).

So a one-sided inverse can exist without invertibility. The spaces here are infinite-dimensional, and that is not an accident: later in this section we prove that in finite dimension one equation does imply the other.

Warning. Isomorphic is not equal, and an isomorphism is a choice. $\mathbb{R}[x]_{\leq 2} \cong \mathbb{R}^3$ does not make a polynomial a column. Worse, there are many isomorphisms, and none is singled out. The coefficient map of [Example \(The Standard Isomorphisms\)](#) sends x to $(0, 1, 0)$. The coordinate map for the basis $(1, 1+x, 1+x+x^2)$ of [Example \(Coordinates of a Polynomial\)](#) is also an isomorphism (we prove this below), and it sends $x = -1 \cdot 1 + 1 \cdot (1+x)$ to $(-1, 1, 0)$. A general n -dimensional space has no “standard” basis, so there is no natural isomorphism $F^n \rightarrow V$; every one depends on a choice. Chapter 4 returns to this point with dual spaces.

The first payoff of the definition is that an isomorphism carries all linear structure across, in both directions. We record the two facts we need.

Theorem 3.5.2 — Isomorphisms Carry Bases to Bases

Let $T: V \rightarrow W$ be an isomorphism, and let $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of V . Then $(T\mathbf{v}_1, \dots, T\mathbf{v}_n)$ is a basis of W .

Proof. Since T is injective, the list $(T\mathbf{v}_1, \dots, T\mathbf{v}_n)$ is linearly independent by [Theorem 3.2.5](#). By [Theorem 3.2.4](#), it spans $\text{im } T$, and $\text{im } T = W$ because T is surjective. Hence it is a basis of W . ■

Theorem 3.5.3 — Isomorphism Is an Equivalence Relation

For vector spaces over F : $V \cong V$; if $V \cong W$ then $W \cong V$; and if $U \cong V$ and $V \cong W$ then $U \cong W$.

Proof. id_V is an isomorphism ([Example \(The Standard Isomorphisms\)](#) (d)). If $T: V \rightarrow W$ is an isomorphism, then T^{-1} is linear ([Theorem 3.5.1](#)) and has inverse T , so it is an isomorphism $W \rightarrow V$. If $T: U \rightarrow V$ and $S: V \rightarrow W$ are isomorphisms, then ST is linear by [Theorem 3.4.2](#) and bijective by [Theorem 0.4.3](#), hence an isomorphism by [Theorem 3.5.1](#), with $(ST)^{-1} = T^{-1}S^{-1}$ by [Theorem 0.4.8](#). This proves the theorem. ■

So $\cong$ sorts vector spaces over F into classes ([Definition 0.5.2](#)). The next question is which spaces land in the same class.

3.5.2 Classification by dimension

By [Theorem 3.5.2](#), isomorphic spaces have bases of the same length. Is that the **only** obstruction? Two spaces of dimension 6, say $M_{2 \times 3}(\mathbb{R})$ and $\mathbb{R}[x]_{\leq 5}$, look nothing alike. The following theorem says that this does not matter: for finite-dimensional spaces, dimension is a **complete invariant**.

Theorem 3.5.4 — Classification of Finite-Dimensional Spaces

Let V and W be finite-dimensional vector spaces over the same field F . Then

$$V \cong W \iff \dim V = \dim W.$$

Idea. $(\Rightarrow)$ is the payoff we just proved: an isomorphism moves a basis to a basis of the same length. For $(\Leftarrow)$ we must **build** an isomorphism, and the only material we have is a basis on each side, $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ and $(\mathbf{w}_1, \dots, \mathbf{w}_n)$. The obvious dictionary sends $\mathbf{v}_i$ to $\mathbf{w}_i$. Linear maps can be prescribed freely on a basis, so this defines T , and the reverse prescription defines S . To see $ST = \text{id}_V$ we do not compute: both sides are linear and agree on the basis, so they are equal. The hypothesis “same dimension” is exactly what makes the two bases the same length, so that “ $\mathbf{v}_i \mapsto \mathbf{w}_i$ ” makes sense.

Proof. ($\Rightarrow$) Suppose $T: V \rightarrow W$ is an isomorphism, and let $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of V , with $n = \dim V$ (Corollary 1.5.5). By Theorem 3.5.2, $(T\mathbf{v}_1, \dots, T\mathbf{v}_n)$ is a basis of W , so $\dim W = n = \dim V$.

($\Leftarrow$) Suppose $\dim V = \dim W = n$, and choose bases $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ of V and $(\mathbf{w}_1, \dots, \mathbf{w}_n)$ of W (Corollary 1.5.5). By Theorem 3.1.5 there are linear maps $T \in \mathcal{L}(V, W)$ with $T\mathbf{v}_i = \mathbf{w}_i$ and $S \in \mathcal{L}(W, V)$ with $S\mathbf{w}_i = \mathbf{v}_i$, for $i = 1, \dots, n$. Then $ST \in \mathcal{L}(V)$ is linear (Theorem 3.4.2) and $ST\mathbf{v}_i = S\mathbf{w}_i = \mathbf{v}_i = \text{id}_V \mathbf{v}_i$ for every i . Two linear maps that agree on a basis are equal, by the uniqueness part of Theorem 3.1.5, so $ST = \text{id}_V$. Swapping the roles of V and W gives $TS = \text{id}_W$. Hence T is invertible, and $V \cong W$. This proves the theorem. ■

The theorem needs both spaces to be over the **same** field; this is part of Definition 3.5.3. $\mathbb{C}$ over $\mathbb{C}$ has dimension 1 and $\mathbb{R}$ over $\mathbb{R}$ has dimension 1, but the question “is $\mathbb{C} \cong \mathbb{R}$?” makes no sense, since a linear map must respect one common set of scalars.

Taking $W = F^n$ with its standard basis, the proof builds a specific isomorphism $V \rightarrow F^n$: the one sending $\mathbf{v}_i$ to $\mathbf{e}_i$. It is an old friend.

Corollary 3.5.5 — The Coordinate Isomorphism

Let $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of a vector space V over F . Then the coordinate map

$$V \rightarrow F^n, \quad \mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}}$$

is an isomorphism, with inverse $(a_1, \dots, a_n) \mapsto a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n$. In particular, every n -dimensional vector space over F is isomorphic to F^n .

Proof. By Theorem 1.5.2, the coordinate map is linear, and every column $(a_1, \dots, a_n) \in F^n$ is the coordinate vector of exactly one vector of V , namely $a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n$. So the coordinate map is bijective, with the stated inverse function, and it is an isomorphism by Theorem 3.5.1. ■

This is the promise of Chapter 1 kept: the correspondence $\mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}}$ from Theorem 1.5.2 is an isomorphism. Its practical meaning is the slogan behind the rest of this chapter: **any computation with sums and multiples in an n -dimensional space can be done with columns in F^n** , at the price of choosing a basis.

Example. (Sorting Spaces by Dimension)

Which of the following real vector spaces are isomorphic to each other?

- (i) $\mathbb{R}[x]_{\leq 5}$;
- (ii) $M_{2 \times 3}(\mathbb{R})$;
- (iii) the upper triangular matrices in $M_3(\mathbb{R})$;
- (iv) $\mathbb{C}^3$, regarded as a vector space over $\mathbb{R}$;

- (v) $U = \{p \in \mathbb{R}[x]_{\leq 6} : p(1) = 0\}$;
- (vi) $\mathbb{R}^5$.

Solution. By [Theorem 3.5.4](#) it suffices to compute dimensions over $\mathbb{R}$.

- (i) The basis $(1, x, \dots, x^5)$ gives dimension 6. (ii) $\dim M_{2 \times 3}(\mathbb{R}) = 2 \cdot 3 = 6$ ([Example \(Dimensions of the Running Examples\)](#)). (iii) An upper triangular matrix is $\sum_{i \leq j} a_{ij} E_{ij}$, and this expression is unique because the coefficient of E_{ij} is the (i, j) -entry. By [Theorem 1.5.1](#) the six matrix units E_{ij} with $1 \leq i \leq j \leq 3$ form a basis, so the dimension is 6. (iv) $\dim_{\mathbb{C}} \mathbb{C}^3 = 3$, so $\dim_{\mathbb{R}} \mathbb{C}^3 = 6$ by [Theorem 1.9.2](#).
- (ii) The evaluation map $E: \mathbb{R}[x]_{\leq 6} \rightarrow \mathbb{R}$, $p \mapsto p(1)$, is linear, and surjective because $E(c) = c$ for each constant c . Its kernel is U . By [Theorem 3.3.1](#), $\dim U = \dim \mathbb{R}[x]_{\leq 6} - \text{rank } E = 7 - 1 = 6$.
- (iii) $\dim \mathbb{R}^5 = 5$.

Hence the spaces (i)–(v) are all isomorphic to each other (and to $\mathbb{R}^6$), while $\mathbb{R}^5$ is isomorphic to none of them. We never wrote down an isomorphism between, say, (iii) and (v); the theorem guarantees one.

Quick check. Is $M_2(\mathbb{R})$ isomorphic to $\mathbb{R}[x]_{\leq 3}$? Is it isomorphic to $\mathbb{R}[x]_{\leq 4}$?

Solution. $\dim M_2(\mathbb{R}) = 4 = \dim \mathbb{R}[x]_{\leq 3}$, so $M_2(\mathbb{R}) \cong \mathbb{R}[x]_{\leq 3}$ by [Theorem 3.5.4](#). But $\dim \mathbb{R}[x]_{\leq 4} = 5 \neq 4$ (not 4: count $1, x, x^2, x^3, x^4$), so $M_2(\mathbb{R}) \not\cong \mathbb{R}[x]_{\leq 4}$.

3.5.3 One-sided inverses

For square matrices, Chapter 2 proved that one equation is enough: if $A, B \in M_n(F)$ and $AB = I_n$, then $BA = I_n$ ([Theorem 2.4.5](#)). [Example \(One Equation Is Not Enough\)](#) shows that this fails for operators on infinite-dimensional spaces. And it fails between spaces of different dimension: for $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$, $(x, y) \mapsto (x, y, 0)$, and $S: \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $(x, y, z) \mapsto (x, y)$, we have $ST = \text{id}_{\mathbb{R}^2}$ but $TS(0, 0, 1) = \mathbf{0}$, so $TS \neq \text{id}_{\mathbb{R}^3}$. What survives is exactly the case that looks like “square”:

Theorem 3.5.6 — One-Sided Inverses of Linear Maps

Let V and W be finite-dimensional vector spaces over F with $\dim V = \dim W$. Let $T \in \mathcal{L}(V, W)$ and $S \in \mathcal{L}(W, V)$. If $ST = \text{id}_V$, then $TS = \text{id}_W$. In that case T and S are both isomorphisms, $S = T^{-1}$ and $T = S^{-1}$.

Idea. Without any dimension hypothesis, $ST = \text{id}_V$ already gives half of what we need: T must be injective, since S recovers $\mathbf{v}$ from $T\mathbf{v}$. The missing half, surjectivity, is exactly what the Rank–Nullity Theorem supplies for free when $\dim V = \dim W$ is finite. Once T is known to be invertible, we cancel it: $S = (ST)T^{-1} = T^{-1}$. So the one hypothesis the infinite-dimensional counterexamples violate, finite equal dimension, is spent on a single line: injective $\Rightarrow$ surjective.

Proof. Suppose $ST = \text{id}_V$, and let $\mathbf{v} \in \ker T$. Then $\mathbf{v} = \text{id}_V \mathbf{v} = S(T\mathbf{v}) = S\mathbf{0} = \mathbf{0}$, since S is linear. So $\ker T = \{\mathbf{0}\}$, and T is injective by [Theorem 3.2.3](#). Since V and W are finite-dimensional with $\dim V = \dim W$, [Corollary 3.3.2](#) shows that T is also surjective. Hence T is bijective, and by [Theorem 3.5.1](#) it has a linear inverse $T^{-1} \in \mathcal{L}(W, V)$. By [Theorem 3.4.2](#) (b) and (c),

$$S = S \text{id}_W = S(TT^{-1}) = (ST)T^{-1} = \text{id}_V T^{-1} = T^{-1}.$$

Therefore $TS = TT^{-1} = \text{id}_W$. Finally $S = T^{-1}$ is an isomorphism with inverse T ([Theorem 3.5.3](#)), so $T = S^{-1}$. This proves the theorem. ■

Remark. Compare the two proofs. For matrices ([Theorem 2.4.5](#)), the step “ $B\mathbf{x} = \mathbf{0}$ has only the trivial solution, hence B is invertible” came from row reduction, through the Invertible Matrix Theorem. Here the same step, “injective, hence bijective”, comes from Rank–Nullity, with no matrices at all. Once we can write maps as matrices (next section), the two theorems become the same theorem seen from two sides.

In practice this means: **to verify that S is the inverse of T between spaces of the same finite dimension, one composition suffices.** By symmetry, $TS = \text{id}_W$ also implies $ST = \text{id}_V$: apply the theorem with the roles of T and S (and of V and W) swapped.

3.5.4 The Invertible Operator Theorem

For an operator on a finite-dimensional space, invertibility can now be detected in many different ways. Chapter 2 collected the matrix versions in the Invertible Matrix Theorem ([Theorem 2.4.4](#)), and said the list would grow. Here is its first growth, stated for operators.

Theorem 3.5.7 — Invertible Operator Theorem

Let V be a **finite-dimensional** vector space over F , and let $T \in \mathcal{L}(V)$. The following are equivalent.

- (a) T is invertible.
- (b) T is injective.
- (c) $\ker T = \{\mathbf{0}\}$, that is, nullity $T = 0$.
- (d) T is surjective.
- (e) $\text{rank } T = \dim V$.
- (f) There is $S \in \mathcal{L}(V)$ with $ST = \text{id}_V$.
- (g) There is $S \in \mathcal{L}(V)$ with $TS = \text{id}_V$.
- (h) T maps **some** basis of V to a basis of V .
- (i) T maps **every** basis of V to a basis of V .

Idea. Nine conditions, but almost every arrow is a citation. The plan:

- ① the core, $(a) \Leftrightarrow (b) \Leftrightarrow (c) \Leftrightarrow (d) \Leftrightarrow (e)$: kernel test for injectivity, Rank–Nullity for “injective $\Leftrightarrow$ surjective”, equal dimension for “image is everything”;

② the one-sided inverses (f), (g): trivially implied by (a), and they imply (a) by [Theorem 3.5.6](#) with $W = V$;

③ the bases, (a) $\Rightarrow$ (i) $\Rightarrow$ (h) $\Rightarrow$ (d): isomorphisms carry bases to bases, some basis exists, and an image containing a basis is everything.

Proof. (a) $\Rightarrow$ (b). An invertible map is bijective ([Theorem 3.5.1](#)).

(b) $\Leftrightarrow$ (c). This is [Theorem 3.2.3](#), and $\ker T = \{\mathbf{0}\}$ is the same as $\dim \ker T = 0$.

(b) $\Leftrightarrow$ (d). Since T maps the finite-dimensional space V to a space of the same dimension, this is [Corollary 3.3.2](#).

(d) $\Leftrightarrow$ (e). If $\operatorname{im} T = V$, then $\operatorname{rank} T = \dim V$. Conversely, if $\operatorname{rank} T = \dim V$, then $\operatorname{im} T$ is a subspace of V of dimension $\dim V$, so $\operatorname{im} T = V$ by [Theorem 1.6.8](#).

(b) and (d) $\Rightarrow$ (a). If T is injective, it is also surjective by (b) $\Rightarrow$ (d), hence bijective, hence invertible by [Theorem 3.5.1](#). Together with the arrows above, (a)–(e) are equivalent.

(a) $\Rightarrow$ (f) and (a) $\Rightarrow$ (g). Take $S = T^{-1}$.

(f) $\Rightarrow$ (a) and (g) $\Rightarrow$ (a). Apply [Theorem 3.5.6](#) with $W = V$: if $ST = \operatorname{id}_V$, then T is invertible; if $TS = \operatorname{id}_V$, the same theorem with the roles of S and T swapped shows that S is invertible with $S^{-1} = T$, so T is invertible.

(a) $\Rightarrow$ (i). This is [Theorem 3.5.2](#).

(i) $\Rightarrow$ (h). V has a basis by [Corollary 1.5.5](#), and by (i) its image is a basis.

(h) $\Rightarrow$ (d). Let $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis whose image $(T\mathbf{v}_1, \dots, T\mathbf{v}_n)$ is a basis of V . Each $T\mathbf{v}_i$ lies in the subspace $\operatorname{im} T$, so $V = \operatorname{span}(T\mathbf{v}_1, \dots, T\mathbf{v}_n) \subseteq \operatorname{im} T$ by [Theorem 1.3.1](#). Hence $\operatorname{im} T = V$.

All nine conditions are equivalent. This proves the theorem. ■

As with the matrix theorem, the list is a menu. To prove that an operator **is** invertible, pick the cheapest item, usually (c): take $\mathbf{v}$ with $T\mathbf{v} = \mathbf{0}$ and show $\mathbf{v} = \mathbf{0}$. To prove that it is **not**, exhibit one non-zero vector in the kernel.

Example. (Invertibility Without Solving Equations)

(a) Let $T \in \mathcal{L}(\mathbb{R}[x]_{\leq 2})$, $T(p) = p + xp'$. Show that T is invertible and find T^{-1} .

(b) Let V be **any** vector space and $T \in \mathcal{L}(V)$ with $T^2 = T + \operatorname{id}_V$. Show that T is invertible, with $T^{-1} = T - \operatorname{id}_V$.

Solution.

(a) Let $p = a + bx + cx^2$. Then $xp' = bx + 2cx^2$, so

$$T(a + bx + cx^2) = a + 2bx + 3cx^2.$$

If $T(p) = 0$, then $a = 2b = 3c = 0$, so $a = b = c = 0$ and $p = 0$. Hence $\ker T = \{0\}$, and T is invertible by [Theorem 3.5.7](#) ((c) $\Rightarrow$ (a)). We did not have to check surjectivity: the count did it for us. For the inverse, let $S(a + bx + cx^2) := a + \frac{b}{2}x + \frac{c}{3}x^2$, which is linear. Then $TS(a + bx + cx^2) = a + 2 \cdot \frac{b}{2}x + 3 \cdot \frac{c}{3}x^2 = a + bx + cx^2$, so $TS = \text{id}$, and by [Theorem 3.5.6](#), $S = T^{-1}$.

- (b) The operators T and $T - \text{id}_V$ are polynomials in T , so they commute ([Theorem 3.4.3](#) (c)). By [Theorem 3.4.2](#) (d) and the hypothesis,

$$T(T - \text{id}_V) = T^2 - T = \text{id}_V = (T - \text{id}_V)T.$$

So $T - \text{id}_V$ is a two-sided inverse, and T is invertible by [Definition 3.5.1](#). Here V may be infinite-dimensional, since we verified **both** equations.

Warning. The Invertible Operator Theorem needs finite dimension, and it is about operators. On $\mathbb{R}[x]$, differentiation D is surjective (every polynomial has an antiderivative $J(p)$) but not injective, and $S(p) = xp$ is injective but not surjective ($1 \notin \text{im } S$). And for a map $T: V \rightarrow W$ between **different** spaces, “injective” does not give “surjective” unless $\dim V = \dim W$: the inclusion $(x, y) \mapsto (x, y, 0)$ of $\mathbb{R}^2$ into $\mathbb{R}^3$ is injective and not surjective.

Quick check. Is $T \in \mathcal{L}(M_2(\mathbb{R}))$, $T(A) = A + A^T$, invertible? Use the cheapest item of [Theorem 3.5.7](#).

Solution. No. Let $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \neq 0$. Then $A^T = -A$, so $T(A) = 0$, and $\ker T \neq \{0\}$. Item (c) fails, so T is not invertible.

Isomorphisms let us move a whole space into F^n . The next section moves **maps** as well: once bases are chosen, every linear map between finite-dimensional spaces becomes a matrix, and composition becomes matrix multiplication.

3.5.5 Exercises

3.5.5.1 A. Check your understanding

Exercise A1.

- Define what it means for $T \in \mathcal{L}(V, W)$ to be an isomorphism, and what $V \cong W$ means.
- True or false: $\mathbb{R}[x]_{\leq 3} \cong M_2(\mathbb{R})$. Justify your answer.
- True or false: if $\dim V = \dim W = 5$ and $T \in \mathcal{L}(V, W)$ is injective, then T is an isomorphism. Justify your answer.
- True or false: for every vector space V and all $S, T \in \mathcal{L}(V)$, $ST = \text{id}_V$ implies $TS = \text{id}_V$. Justify your answer.
- True or false: if $V \cong W$, then every linear map $V \rightarrow W$ is an isomorphism. Justify your answer.

- (f) Which theorem would you use to show that a given operator on a finite-dimensional space is invertible by computing only its kernel?

Solution.

- (a) T is an isomorphism if it is invertible, that is, there is $S \in \mathcal{L}(W, V)$ with $ST = \text{id}_V$ and $TS = \text{id}_W$; equivalently, T is a bijective linear map ([Theorem 3.5.1](#)). $V \cong W$ means that some isomorphism $V \rightarrow W$ exists.
- (b) True. Both have dimension 4, so they are isomorphic by [Theorem 3.5.4](#).
- (c) True. By [Corollary 3.3.2](#) an injective linear map between spaces of the same finite dimension is surjective, so T is bijective, hence an isomorphism.
- (d) False. On $F^\mathbb{N}$, the shifts of [Example \(One Equation Is Not Enough\)](#) satisfy $LR = \text{id}$ but $RL \neq \text{id}$. The statement is true when V is finite-dimensional ([Theorem 3.5.6](#)).
- (e) False. $\mathbb{R}^2 \cong \mathbb{R}^2$, but the zero map $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ is not injective.
- (f) The Invertible Operator Theorem, [Theorem 3.5.7](#), (c) $\Rightarrow$ (a).

3.5.5.2 B. Practice

Exercise B1: An explicit isomorphism.

Let $U = \{A \in M_2(\mathbb{R}) : A^\top = A\}$ be the space of symmetric 2×2 real matrices. Find an explicit isomorphism $U \rightarrow \mathbb{R}[x]_{\leq 2}$, and verify that it is one.

Solution. Every $A \in U$ has the form $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ with $a, b, c \in \mathbb{R}$. Define

$$T \begin{pmatrix} a & b \\ b & c \end{pmatrix} := a + bx + cx^2, \quad S(a + bx + cx^2) := \begin{pmatrix} a & b \\ b & c \end{pmatrix}.$$

Linearity of T . Adding two symmetric matrices adds the entries a, b, c , and adding polynomials adds coefficients, so $T(A + A') = T(A) + T(A')$; similarly $T(\lambda A) = \lambda T(A)$.

Inverse. S takes values in U , and $ST(A) = A$ for every $A \in U$, since A is determined by its entries a, b, c . Also $TS(p) = p$ for every p , since p is determined by its coefficients. So T has a two-sided inverse and is bijective, hence an isomorphism by [Theorem 3.5.1](#). (Alternatively: both spaces have dimension 3 by [Example \(Dimensions of the Running Examples\)](#), so by [Theorem 3.5.6](#) the single equation $ST = \text{id}$ would already suffice.)

Exercise B2: Inverting $\text{id} + D$.

Let $n \geq 0$, and let $T \in \mathcal{L}(\mathbb{R}[x]_{\leq n})$, $T(p) = p + p'$.

- (a) Show that T is an isomorphism by computing its kernel.
- (b) Show that $T^{-1} = \sum_{k=0}^n (-1)^k D^k$, where D is differentiation.
- (c) For $n = 2$, find the polynomial p with $p + p' = x^2$.

Hint: for (b), use the polynomial identity $(1 + x) \sum_{k=0}^n (-x)^k = 1 + (-1)^n x^{n+1}$.

Solution.

- (a) Let $p \in \ker T$, so $p' = -p$. If $p \neq 0$, let $d = \deg p \geq 0$. Differentiation lowers the degree of a non-zero polynomial by at least one (it kills constants), so $\deg p' < d = \deg(-p)$, contradicting $p' = -p$. Hence $p = 0$ and $\ker T = \{0\}$. By [Theorem 3.5.7](#) ((c) $\Rightarrow$ (a)), T is an isomorphism.
- (b) $T = \text{id} + D = q(D)$ with $q = 1 + x$, and $S := \sum_{k=0}^n (-1)^k D^k = r(D)$ with $r = \sum_{k=0}^n (-x)^k$. The identity in the hint is the telescoping sum $\sum_{k=0}^n (-x)^k - \sum_{k=0}^n (-x)^{k+1} = 1 - (-x)^{n+1}$. By [Theorem 3.4.3](#) (b),

$$TS = (qr)(D) = \text{id} + (-1)^n D^{n+1} = \text{id},$$

since $D^{n+1} = 0$ on $\mathbb{R}[x]_{\leq n}$ ([Example \(Differentiation Is Nilpotent\)](#)). By [Theorem 3.5.6](#), $S = T^{-1}$.

- (c) $p = T^{-1}(x^2) = x^2 - D(x^2) + D^2(x^2) = x^2 - 2x + 2$. Check: $p + p' = (x^2 - 2x + 2) + (2x - 2) = x^2$.

Exercise B3: Deciding isomorphism.

Determine which of the following pairs of real vector spaces are isomorphic. Justify your answer.

- (a) $\mathbb{R}^4$ and $M_2(\mathbb{R})$.
 (b) $\mathbb{R}[x]_{\leq 3}$ and $\{A \in M_3(\mathbb{R}) : A^T = -A\}$.
 (c) $\{p \in \mathbb{R}[x]_{\leq 4} : p(2) = 0\}$ and $\mathbb{R}^4$.
 (d) $\mathbb{C}^2$ (over $\mathbb{R}$) and $\mathbb{R}[x]_{\leq 3}$.
 (e) The plane $\{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$ and $\mathbb{C}$ (over $\mathbb{R}$).

Solution. By [Theorem 3.5.4](#), in each case we compare dimensions over $\mathbb{R}$.

- (a) Isomorphic: $\dim \mathbb{R}^4 = 4 = \dim M_2(\mathbb{R})$.
 (b) Not isomorphic. $\dim \mathbb{R}[x]_{\leq 3} = 4$. If $A^T = -A$, then $a_{ii} = -a_{ii}$, so $2a_{ii} = 0$ and $a_{ii} = 0$, and $a_{ji} = -a_{ij}$. Hence $A = a_{12}(E_{12} - E_{21}) + a_{13}(E_{13} - E_{31}) + a_{23}(E_{23} - E_{32})$, and the coefficients are entries of A , so this expression is unique. By [Theorem 1.5.1](#) these three matrices form a basis, and the dimension is $3 \neq 4$.
 (c) Isomorphic. The evaluation $p \mapsto p(2)$, $\mathbb{R}[x]_{\leq 4} \rightarrow \mathbb{R}$, is linear and surjective (constants), with kernel the given space, so by [Theorem 3.3.1](#) its dimension is $5 - 1 = 4$.
 (d) Isomorphic: $\dim_{\mathbb{R}} \mathbb{C}^2 = 2 \dim_{\mathbb{C}} \mathbb{C}^2 = 4$ by [Theorem 1.9.2](#), and $\dim \mathbb{R}[x]_{\leq 3} = 4$.
 (e) Isomorphic. The plane is the kernel of the surjective linear map $(x, y, z) \mapsto x + y + z$ onto $\mathbb{R}$, so it has dimension $3 - 1 = 2$ by [Theorem 3.3.1](#); and $\dim_{\mathbb{R}} \mathbb{C} = 2$ ([Example \(Dimensions of the Running Examples\)](#)).

3.5.5.3 C. Going deeper

Exercise C1: Invertible products.

- (a) Let V be finite-dimensional and $S, T \in \mathcal{L}(V)$. Prove that if ST is invertible, then S and T are both invertible. Deduce that TS is then invertible too.

- (b) Show that (a) fails for $V = F^{\mathbb{N}}$: give $S, T \in \mathcal{L}(F^{\mathbb{N}})$ with ST invertible but neither S nor T invertible, and TS not invertible.

Hint: for (a), regroup $(ST)(ST)^{-1}$ and $(ST)^{-1}(ST)$.

Solution.

- (a) The inverse $(ST)^{-1}$ lies in $\mathcal{L}(V)$ (Theorem 3.5.1), and $(ST)(ST)^{-1} = \text{id}_V = (ST)^{-1}(ST)$. By associativity (Theorem 3.4.2 (b)), $S(T(ST)^{-1}) = \text{id}_V$ and $((ST)^{-1}S)T = \text{id}_V$. By Theorem 3.5.7, (g) $\Rightarrow$ (a) applied to S with right inverse $T(ST)^{-1}$ shows that S is invertible, and (f) $\Rightarrow$ (a) applied to T with left inverse $(ST)^{-1}S$ shows that T is invertible. Then TS is a composition of isomorphisms, hence an isomorphism by Theorem 3.5.3.
- (b) Let $S = L$ and $T = R$ be the left and right shifts of Example (One Equation Is Not Enough). Then $ST = LR = \text{id}$, which is invertible. But L is not injective and R is not surjective, so neither is invertible, and $TS = RL$ sends $(1, 0, 0, \dots)$ to $\mathbf{0}$, so it is not injective and not invertible. The proof of (a) breaks at the appeal to Theorem 3.5.7, which needs finite dimension.

Exercise C2: A space isomorphic to a proper subspace.

- (a) Let V be finite-dimensional and U a subspace of V with $U \cong V$. Prove that $U = V$.
- (b) Let $U = \{p \in F[x] : p(0) = 0\}$, the polynomials with constant coefficient 0. Show that $T: F[x] \rightarrow U$, $T(p) = xp$, is an isomorphism. Hence (a) fails without finite dimension.

Solution.

- (a) U is finite-dimensional by Theorem 1.6.7. Since $U \cong V$, Theorem 3.5.4 gives $\dim U = \dim V$, and then $U = V$ by Theorem 1.6.8.
- (b) For $p = \sum_k a_k x^k$, $xp = \sum_k a_k x^{k+1}$ has constant coefficient 0, so T maps into U , and T is linear because multiplication in $F[x]$ distributes and commutes with scalars (Theorem 0.8.1). *Injective:* if $xp = 0$, then $p = 0$ by Corollary 0.8.4, since $x \neq 0$. *Surjective:* if $q = \sum_{k \geq 1} b_k x^k \in U$, then $q = x \sum_{k \geq 1} b_k x^{k-1} = T(\sum_{k \geq 1} b_k x^{k-1})$. So T is a bijective linear map, an isomorphism by Theorem 3.5.1, and $U \cong F[x]$. But $1 \notin U$, so $U \neq F[x]$. This is the phenomenon noted in Chapter 1, §8, now as an isomorphism.

3.6 The Matrix of a Linear Map

An isomorphism $V \cong F^n$ turns the vectors of an abstract space into columns. This section does the same for **maps**. Once a basis is fixed on each side, every linear map between finite-dimensional spaces becomes a matrix, applying the map becomes multiplying by that matrix, and composing maps becomes multiplying matrices. This last fact finally explains the strange multiplication rule of Chapter 0, and it gives us the working method of the rest of the book: to study an abstract map, write down its matrix, compute there, and translate back.

3.6.1 Writing a map as a matrix

We have seen one family of linear maps that is completely described by a table of numbers: the matrix maps $T_A: F^n \rightarrow F^m$, $\mathbf{x} \mapsto A\mathbf{x}$. In fact **every** linear map $T: F^n \rightarrow F^m$ is one of them. By [Theorem 3.1.4](#) and [Theorem 0.9.3](#), for $\mathbf{x} = x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n$,

$$T\mathbf{x} = x_1T\mathbf{e}_1 + \cdots + x_nT\mathbf{e}_n = A\mathbf{x}, \quad \text{where } A = (T\mathbf{e}_1 \ \cdots \ T\mathbf{e}_n).$$

The j -th column of A is the image of the j -th basis vector. We want the same description for maps between arbitrary finite-dimensional spaces, such as differentiation on polynomials.

Take $D: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$. Then

$$D(a_0 + a_1x + a_2x^2 + a_3x^3) = a_1 + 2a_2x + 3a_3x^2.$$

A polynomial is not a column, but its coordinates are. In the bases $(1, x, x^2, x^3)$ and $(1, x, x^2)$, the input has coordinates (a_0, a_1, a_2, a_3) and the output has coordinates $(a_1, 2a_2, 3a_3)$, and

$$\begin{pmatrix} a_1 \\ 2a_2 \\ 3a_3 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{pmatrix}.$$

Look at the columns of this matrix: $(0, 0, 0)$, $(1, 0, 0)$, $(0, 2, 0)$, $(0, 0, 3)$. They are the coordinates of $D(1) = 0$, $D(x) = 1$, $D(x^2) = 2x$, $D(x^3) = 3x^2$. The standard example generalizes: record the images of the input basis, in coordinates of the output basis, as columns.

The matrix of a linear map lists, column by column, where the input basis vectors go, written in coordinates of the output basis.

Definition 3.6.1 — Matrix of a Linear Map

Let V and W be finite-dimensional vector spaces over F , with **ordered** bases $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ of V and $\mathcal{C} = (\mathbf{w}_1, \dots, \mathbf{w}_m)$ of W , and let $T \in \mathcal{L}(V, W)$. The **matrix of T with respect to $\mathcal{B}$ and $\mathcal{C}$** is the $m \times n$ matrix

$$[T]_{\mathcal{B}}^{\mathcal{C}} := ([T\mathbf{v}_1]_{\mathcal{C}} \ [T\mathbf{v}_2]_{\mathcal{C}} \ \cdots \ [T\mathbf{v}_n]_{\mathcal{C}}) \in M_{m \times n}(F),$$

whose j -th column is the coordinate vector of $T\mathbf{v}_j$ with respect to $\mathcal{C}$. Equivalently, its entries a_{ij} are the **unique** scalars with

$$T\mathbf{v}_j = a_{1j}\mathbf{w}_1 + a_{2j}\mathbf{w}_2 + \cdots + a_{mj}\mathbf{w}_m \quad (j = 1, \dots, n).$$

When $V = W$ and $\mathcal{B} = \mathcal{C}$, we write $[T]_{\mathcal{B}} := [T]_{\mathcal{B}}^{\mathcal{B}}$ and call it the matrix of the operator T with respect to $\mathcal{B}$.

In words, clause by clause:

- **The input basis $\mathcal{B}$ is the subscript, the output basis $\mathcal{C}$ is the superscript.** The subscript sits next to where the input goes, as in $[T]_{\mathcal{B}}^{\mathcal{C}}[\mathbf{v}]_{\mathcal{B}}$ below.
- **Columns correspond to input basis vectors**, so there are $n = \dim V$ of them; **rows correspond to output basis vectors**, so there are $m = \dim W$. The size is “ $\dim(\text{codomain}) \times \dim(\text{domain})$ ”, the same order as $M_{m \times n}$ for $T_A: F^n \rightarrow F^m$.

- Column j answers one question: **where does $\mathbf{v}_j$ go?**

Well-definedness. Each $T\mathbf{v}_j$ lies in W , and $\mathcal{C}$ is a basis, so $[T\mathbf{v}_j]_{\mathcal{C}}$ exists and is unique by [Theorem 1.5.1](#). So the matrix is determined by T , $\mathcal{B}$ and $\mathcal{C}$. It is **not** determined by T alone: change either basis, or even the **order** of the vectors in one of them, and the matrix changes. That is why the bases are ordered.

Example. (Differentiation on Cubic Polynomials)

Let $\mathcal{E}_3 = (1, x, x^2, x^3)$ and $\mathcal{E}_2 = (1, x, x^2)$. Find $[D]_{\mathcal{E}_3}^{\mathcal{E}_2}$ for $D: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$, and $[D]_{\mathcal{E}_3}$ for D regarded as an operator on $\mathbb{R}[x]_{\leq 3}$.

Solution. The images of the input basis are $D(1) = 0$, $D(x) = 1$, $D(x^2) = 2x$, $D(x^3) = 3x^2$. In $\mathcal{E}_2$ their coordinates are $(0, 0, 0)$, $(1, 0, 0)$, $(0, 2, 0)$, $(0, 0, 3)$; in $\mathcal{E}_3$ the same vectors have a fourth coordinate 0. So

$$[D]_{\mathcal{E}_3}^{\mathcal{E}_2} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}, \quad [D]_{\mathcal{E}_3} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The same rule $p \mapsto p'$ gives a 3×4 or a 4×4 matrix, depending on the codomain we declare. The superdiagonal 1, 2, 3 records “ $x^k \mapsto kx^{k-1}$ ”.

Example. (Matrix Maps and the Identity)

- Let $A \in M_{m \times n}(F)$, and let $\mathcal{E}_n, \mathcal{E}_m$ be the standard bases of F^n, F^m . Show that $[T_A]_{\mathcal{E}_n}^{\mathcal{E}_m} = A$.
- Let $\mathcal{B}$ be any basis of an n -dimensional space V . Find $[\text{id}_V]_{\mathcal{B}}$ and the matrix of the zero map $V \rightarrow W$ for any bases.

Solution.

- The j -th column of $[T_A]_{\mathcal{E}_n}^{\mathcal{E}_m}$ is $[A\mathbf{e}_j]_{\mathcal{E}_m} = A\mathbf{e}_j$, since coordinates in the standard basis of F^m are the entries themselves; and $A\mathbf{e}_j$ is the j -th column of A ([Theorem 0.9.3](#)). So the matrix of a matrix map in the standard bases is the matrix we started with.
- $\text{id}_V \mathbf{v}_j = \mathbf{v}_j = 0\mathbf{v}_1 + \cdots + 1\mathbf{v}_j + \cdots + 0\mathbf{v}_n$, so the j -th column of $[\text{id}_V]_{\mathcal{B}}$ is $\mathbf{e}_j$, and $[\text{id}_V]_{\mathcal{B}} = I_n$. The zero map sends every $\mathbf{v}_j$ to $\mathbf{0}$, whose coordinates are all 0, so its matrix is the zero matrix of size $\dim W \times \dim V$, whatever the bases. These degenerate cases fix the expectations “identity map, identity matrix” and “zero map, zero matrix”. The first needs the **same** basis on both sides, as the warning below shows.

Example. (The Transpose Map)

Let $T: M_2(F) \rightarrow M_2(F)$, $T(X) = X^T$, and let $\mathcal{E} = (E_{11}, E_{12}, E_{21}, E_{22})$. Find $[T]_{\mathcal{E}}$.

Solution. T is linear by [Theorem 0.9.5](#). Its values on the basis are $E_{11}^T = E_{11}$, $E_{12}^T = E_{21}$,

$E_{21}^T = E_{12}$, $E_{22}^T = E_{22}$. In $\mathcal{E}$, these have coordinates $\mathbf{e}_1, \mathbf{e}_3, \mathbf{e}_2, \mathbf{e}_4$. So

$$[T]_{\mathcal{E}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

A map on a space of 2×2 matrices is described by a 4×4 matrix, because $\dim M_2(F) = 4$. Do not confuse the matrix X that T acts on with the matrix $[T]_{\mathcal{E}}$ that describes T .

Example. (A Reflection in Two Bases)

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the reflection across the line $y = x$, $T(x, y) = (y, x)$. Find $[T]_{\mathcal{E}}$ for the standard basis $\mathcal{E} = (\mathbf{e}_1, \mathbf{e}_2)$, and $[T]_{\mathcal{B}}$ for $\mathcal{B} = (\mathbf{u}_1, \mathbf{u}_2)$ with $\mathbf{u}_1 = (1, 1)$ and $\mathbf{u}_2 = (1, -1)$.

Solution. $T\mathbf{e}_1 = (0, 1) = \mathbf{e}_2$ and $T\mathbf{e}_2 = \mathbf{e}_1$, so $[T]_{\mathcal{E}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

The vectors $\mathbf{u}_1, \mathbf{u}_2$ are independent (neither is a multiple of the other), hence a basis of $\mathbb{R}^2$ by Theorem 1.6.4. The first lies **on** the mirror line and the second is **perpendicular** to it: $T\mathbf{u}_1 = (1, 1) = 1\mathbf{u}_1 + 0\mathbf{u}_2$ and $T\mathbf{u}_2 = (-1, 1) = 0\mathbf{u}_1 + (-1)\mathbf{u}_2$. So

$$[T]_{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

In the basis adapted to the geometry, the reflection is visibly “keep one direction, flip the other”. The same map has a diagonal matrix in one basis and a non-diagonal one in another. How the two matrices are related is the subject of the next section.

The matrix also works in reverse: from $[T]_{\mathcal{B}}$ alone we can recover T . Take $\mathbf{v} = (3, 1)$. Its coordinates in $\mathcal{B}$ solve $a\mathbf{u}_1 + b\mathbf{u}_2 = (3, 1)$, that is, $a + b = 3$ and $a - b = 1$, so $[\mathbf{v}]_{\mathcal{B}} = (2, 1)$, that is, $\mathbf{v} = 2\mathbf{u}_1 + \mathbf{u}_2$. By linearity, $T\mathbf{v} = 2T\mathbf{u}_1 + T\mathbf{u}_2$. The columns of $[T]_{\mathcal{B}}$ say $T\mathbf{u}_1 = 1\mathbf{u}_1 + 0\mathbf{u}_2$ and $T\mathbf{u}_2 = 0\mathbf{u}_1 + (-1)\mathbf{u}_2$, so $T\mathbf{v} = 2\mathbf{u}_1 - \mathbf{u}_2 = (1, 3)$, which is indeed (y, x) at $(3, 1)$. In coordinates, we turned $(2, 1)$ into $(2, -1) = [T]_{\mathcal{B}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}$; the next subsection proves that this “coordinates in, multiply, combine out” recipe always works.

Here is a non-example by minimal change. In Example (Differentiation on Cubic Polynomials), write the coordinate vectors $[D(x^j)]_{\mathcal{E}_2}$ as **rows** instead of columns. The numbers are the same,

but the result is the 4×3 matrix $\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$, and it cannot even be multiplied by the coordinate

vector (a_0, a_1, a_2, a_3) of an input polynomial. The clause that fails is “the j -th **column** is $[T\mathbf{v}_j]_{\mathcal{C}}$ ”.

Why this definition. Why columns, and why coordinates in $\mathcal{C}$? Because these are the choices under which “apply T ” becomes “multiply by the matrix”, as the D computation above suggested; we prove this next. With rows, we would get the transpose, and the natural formula $[T\mathbf{v}]_{\mathcal{C}} = A[\mathbf{v}]_{\mathcal{B}}$ would fail. The convention that the input basis is the subscript is chosen to make that formula easy to read: the $\mathcal{B}$ of the matrix sits next to the $\mathcal{B}$ of the vector.

Warning. Three ways to get a matrix of a map wrong. (1) **Columns, not rows.** The j -th **column** is $[T\mathbf{v}_j]_{\mathcal{C}}$, as in the non-example above. (2) **Order matters.** In [Example \(A Reflection in Two Bases\)](#), reordering the standard basis as $(\mathbf{e}_2, \mathbf{e}_1)$ leaves $[T] = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ unchanged, but reordering $\mathcal{B}$ as $(\mathbf{u}_2, \mathbf{u}_1)$ gives $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \neq [T]_{\mathcal{B}}$. (3) **Input subscript, output superscript.** $[T]_{\mathcal{B}}^{\mathcal{C}}$ and $[T]_{\mathcal{C}}^{\mathcal{B}}$ are different objects, and for $T: V \rightarrow W$ with $V \neq W$ only one of them makes sense. Also $[\text{id}_V]_{\mathcal{B}}^{\mathcal{C}} \neq I_n$ when $\mathcal{B} \neq \mathcal{C}$: for $\mathcal{B} = (\mathbf{e}_1, \mathbf{e}_2)$ and $\mathcal{C} = (\mathbf{e}_2, \mathbf{e}_1)$ in $\mathbb{R}^2$, $[\text{id}]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Quick check. What is the size of $[T]_{\mathcal{B}}^{\mathcal{C}}$ for a linear map $T: M_{2 \times 3}(\mathbb{R}) \rightarrow \mathbb{R}[x]_{\leq 2}$? And what is the first column of $[T]_{\mathcal{E}'}^{\mathcal{E}}$ for $T: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}^2$, $T(p) = (p(0), p(1))$, with $\mathcal{E} = (1, x, x^2)$ and $\mathcal{E}'$ the standard basis of $\mathbb{R}^2$?

Solution. $\dim \mathbb{R}[x]_{\leq 2} = 3$ and $\dim M_{2 \times 3}(\mathbb{R}) = 6$, so the matrix is 3×6 : rows for the codomain, columns for the domain. For the second map, the first column is the coordinate vector of the image of the first basis vector 1: $T(1) = (1, 1)$, so the column is $(1, 1)$.

3.6.2 The fundamental square

Here is the promise behind the definition: once coordinates are taken on both sides, applying T is multiplying by its matrix.

Theorem 3.6.1 — Coordinates of an Image

Let V and W be finite-dimensional vector spaces over F with ordered bases $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ and $\mathcal{C} = (\mathbf{w}_1, \dots, \mathbf{w}_m)$, and let $T \in \mathcal{L}(V, W)$. Then for every $\mathbf{v} \in V$,

$$[T\mathbf{v}]_{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}} [\mathbf{v}]_{\mathcal{B}}.$$

Moreover, $[T]_{\mathcal{B}}^{\mathcal{C}}$ is the **only** matrix $A \in M_{m \times n}(F)$ with $[T\mathbf{v}]_{\mathcal{C}} = A[\mathbf{v}]_{\mathcal{B}}$ for all $\mathbf{v} \in V$.

The statement says that the following square **commutes**: starting from $\mathbf{v}$ in the top-left corner, the two routes to the bottom-right corner give the same column.

$$\begin{array}{ccc}
 \mathbf{v} & & T\mathbf{v} \\
 V & \xrightarrow{\quad T \quad} & W \\
 \mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}} \downarrow & & \downarrow \mathbf{w} \mapsto [\mathbf{w}]_{\mathcal{C}} \\
 F^n & \xrightarrow{\quad \text{multiply by } [T]_{\mathcal{B}}^{\mathcal{C}} \quad} & F^m \\
 [\mathbf{v}]_{\mathcal{B}} & & [T\mathbf{v}]_{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}} [\mathbf{v}]_{\mathcal{B}}
 \end{array}$$

The top row is the abstract world, where T acts on vectors of V . The bottom row is the matrix world, where a matrix acts on columns. The vertical arrows are the coordinate isomorphisms

of [Corollary 3.5.5](#), so nothing is lost going down or coming back up. **Right then down** (apply T , then take coordinates) equals **down then right** (take coordinates, then multiply by the matrix).

Idea. Both sides are linear in $\mathbf{v}$, so it is enough to check the formula on the basis vectors $\mathbf{v}_j$, and there it is the definition: $[\mathbf{v}_j]_{\mathcal{B}} = \mathbf{e}_j$, and multiplying a matrix by $\mathbf{e}_j$ picks out its j -th column, which was defined to be $[T\mathbf{v}_j]_{\mathcal{C}}$. Written out for a general $\mathbf{v}$, this becomes a one-line chain. Uniqueness uses the same observation backwards: any A that works must have $[T\mathbf{v}_j]_{\mathcal{C}}$ as its j -th column.

Proof. Let $\mathbf{v} \in V$ and $[\mathbf{v}]_{\mathcal{B}} = (x_1, \dots, x_n)$, so that $\mathbf{v} = x_1\mathbf{v}_1 + \dots + x_n\mathbf{v}_n$. Since T is linear, [Theorem 3.1.4](#) gives $T\mathbf{v} = x_1T\mathbf{v}_1 + \dots + x_nT\mathbf{v}_n$. Since taking coordinates respects sums and scalar multiples ([Theorem 1.5.2](#)),

$$[T\mathbf{v}]_{\mathcal{C}} = x_1[T\mathbf{v}_1]_{\mathcal{C}} + \dots + x_n[T\mathbf{v}_n]_{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = [T]_{\mathcal{B}}^{\mathcal{C}}[\mathbf{v}]_{\mathcal{B}},$$

where the middle equality is [Theorem 0.9.3](#) applied to the columns $[T\mathbf{v}_j]_{\mathcal{C}}$ of $[T]_{\mathcal{B}}^{\mathcal{C}}$.

For uniqueness, suppose $A \in M_{m \times n}(F)$ satisfies $[T\mathbf{v}]_{\mathcal{C}} = A[\mathbf{v}]_{\mathcal{B}}$ for all $\mathbf{v}$. Taking $\mathbf{v} = \mathbf{v}_j$, whose coordinate vector is $\mathbf{e}_j$, the j -th column of A is $A\mathbf{e}_j = [T\mathbf{v}_j]_{\mathcal{C}}$ ([Theorem 0.9.3](#)), which is the j -th column of $[T]_{\mathcal{B}}^{\mathcal{C}}$. Hence $A = [T]_{\mathcal{B}}^{\mathcal{C}}$. This proves the theorem. ■

This is the dictionary that lets us “write down a matrix first”. A question about T (its kernel, its image, a preimage) becomes a question about a matrix, which Chapter 2 taught us to answer by row reduction; the coordinate isomorphisms carry the answer back.

Example. (Kernel and Image Through a Matrix)

Let $T: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}^2$, $T(p) = (p(1), p(-1))$. Find $\ker T$ and decide whether T is surjective, by working with the matrix of T in the bases $\mathcal{E} = (1, x, x^2)$ and the standard basis $\mathcal{E}'$ of $\mathbb{R}^2$.

Solution. Write down the matrix. $T(1) = (1, 1)$, $T(x) = (1, -1)$, $T(x^2) = (1, 1)$, so

$$A = [T]_{\mathcal{E}'}^{\mathcal{E}} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix}.$$

Solve in the matrix world. Subtracting the first row from the second gives $\begin{pmatrix} 1 & 1 & 1 \\ 0 & -2 & 0 \end{pmatrix}$, so $A\mathbf{x} = \mathbf{0}$ means $x_2 = 0$ and $x_1 = -x_3$: $\text{nul}(A) = \text{span}((-1, 0, 1))$. Both rows are pivot rows, so $\text{rank } A = 2$.

Translate back. By [Theorem 3.6.1](#), $T(p) = \mathbf{0}$ exactly when $A[p]_{\mathcal{E}} = \mathbf{0}$ (the only vector with zero coordinates is $\mathbf{0}$). The coordinate vector $(-1, 0, 1)$ belongs to $p = -1 + x^2$. So

$$\ker T = \text{span}(x^2 - 1),$$

as a check confirms: $x^2 - 1$ vanishes at 1 and -1 . By [Theorem 3.3.1](#), $\text{rank } T = 3 - 1 =$

$2 = \dim \mathbb{R}^2$, so $\text{im } T = \mathbb{R}^2$ by [Theorem 1.6.8](#), and T is surjective: any two values $p(1)$, $p(-1)$ can be prescribed.

3.6.3 Composition is matrix multiplication

In Chapter 0, §9 we asked why matrices are multiplied row-against-column. The answer given there was **substitution**. If $\mathbf{y} = B\mathbf{x}$ and $\mathbf{z} = A\mathbf{y}$, with

$$B = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix},$$

then substituting gives $\mathbf{z} = \begin{pmatrix} 4 & 9 \\ 5 & -1 \end{pmatrix} \mathbf{x}$, and the table of the combined substitution turned out to be AB . At the time we promised that this fact would become a statement about linear maps. We can now say what it is. Each substitution is a linear map, $T_B: \mathbf{x} \mapsto \mathbf{y}$ and $T_A: \mathbf{y} \mapsto \mathbf{z}$, and substituting one into the other is **composing** them: $\mathbf{z} = T_A(T_B\mathbf{x}) = (T_A T_B)\mathbf{x}$. So the Chapter 0 computation says $T_A T_B = T_{AB}$. Here is the general statement, for arbitrary spaces and bases.

Theorem 3.6.2 — Matrix of a Composition

Let U, V, W be finite-dimensional vector spaces over F with ordered bases $\mathcal{B}$, $\mathcal{C}$, $\mathcal{D}$ respectively. Let $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$. Then

$$[ST]_{\mathcal{B}}^{\mathcal{D}} = [S]_{\mathcal{C}}^{\mathcal{D}} [T]_{\mathcal{B}}^{\mathcal{C}}.$$

Notice how the notation checks itself: the middle basis $\mathcal{C}$ appears as the output of T and the input of S , and it “cancels”, leaving $\mathcal{B}$ below and $\mathcal{D}$ above. The order of the factors is the order of the maps: the map applied first is on the right, in both ST and the product.

Idea. Glue two commutative squares side by side. The left square is T with its matrix, the right square is S with its matrix, and they share the middle column $V \rightarrow F^{\dim V}$. Going along the top from U to W is ST ; going along the bottom is multiplication by $[T]_{\mathcal{B}}^{\mathcal{C}}$ followed by multiplication by $[S]_{\mathcal{C}}^{\mathcal{D}}$, which is multiplication by their product. The big rectangle commutes because both small squares do, and the uniqueness part of [Theorem 3.6.1](#) identifies the product as the matrix of ST .

$$\begin{array}{ccccc} U & \xrightarrow{T} & V & \xrightarrow{S} & W \\ \left[\cdot \right]_{\mathcal{B}} \downarrow & & \left[\cdot \right]_{\mathcal{C}} \downarrow & & \left[\cdot \right]_{\mathcal{D}} \downarrow \\ F^p & \xrightarrow{[T]_{\mathcal{B}}^{\mathcal{C}}} & F^n & \xrightarrow{[S]_{\mathcal{C}}^{\mathcal{D}}} & F^m \end{array}$$

Proof. Let $p = \dim U$ and $m = \dim W$, and let $\mathbf{u} \in U$. Applying [Theorem 3.6.1](#) first to S and the vector $T\mathbf{u}$, then to T and $\mathbf{u}$,

$$[ST\mathbf{u}]_{\mathcal{D}} = [S(T\mathbf{u})]_{\mathcal{D}} = [S]_{\mathcal{C}}^{\mathcal{D}} [T\mathbf{u}]_{\mathcal{C}} = [S]_{\mathcal{C}}^{\mathcal{D}} ([T]_{\mathcal{B}}^{\mathcal{C}} [\mathbf{u}]_{\mathcal{B}}) = ([S]_{\mathcal{C}}^{\mathcal{D}} [T]_{\mathcal{B}}^{\mathcal{C}}) [\mathbf{u}]_{\mathcal{B}},$$

where the last equality is associativity of matrix multiplication ([Theorem 0.9.4](#)). The

map ST is linear ([Theorem 3.4.2](#)), and the $m \times p$ matrix $[S]_{\mathcal{C}}^{\mathcal{D}}[T]_{\mathcal{B}}^{\mathcal{C}}$ satisfies $[ST]\mathbf{u}_{\mathcal{D}} = ([S]_{\mathcal{C}}^{\mathcal{D}}[T]_{\mathcal{B}}^{\mathcal{C}})[\mathbf{u}]_{\mathcal{B}}$ for every $\mathbf{u} \in U$. By the uniqueness part of [Theorem 3.6.1](#), it equals $[ST]_{\mathcal{B}}^{\mathcal{D}}$. This proves the theorem. ■

This is the payoff of the hook in Chapter 0, §9 ([Definition 0.9.9](#)). The row-times-column rule was not an arbitrary convention: it is the **only** rule under which the matrix of a composition is the product of the matrices, because composing linear maps is substituting one coordinate formula into another. Everything Chapter 0 proved about matrix multiplication now has a meaning for maps, and conversely:

- **associativity** of matrix multiplication is associativity of composition ([Theorem 0.4.2](#));
- $AB \neq BA$ in general is $ST \neq TS$ ([Example \(Rotate, Then Project\)](#));
- the size condition “columns of A = rows of B ” is “the codomain of the first map is the domain of the second”;
- in particular $T_A T_B = T_{AB}$: apply the theorem with standard bases and [Example \(Matrix Maps and the Identity\)](#) (a).

Example. (Multiply by x , Then Differentiate)

Let $S: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}[x]_{\leq 3}$, $S(p) = xp$, and $D: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$, $D(p) = p'$, with the monomial bases $\mathcal{E}_2 = (1, x, x^2)$ and $\mathcal{E}_3 = (1, x, x^2, x^3)$. Compute $[DS]_{\mathcal{E}_2}^{\mathcal{E}_2}$ and $[SD]_{\mathcal{E}_3}^{\mathcal{E}_3}$, both directly and as products.

Solution. Matrices of the factors. $S(1) = x$, $S(x) = x^2$, $S(x^2) = x^3$, and $[D]_{\mathcal{E}_3}^{\mathcal{E}_2}$ was found in [Example \(Differentiation on Cubic Polynomials\)](#):

$$[S]_{\mathcal{E}_2}^{\mathcal{E}_3} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad [D]_{\mathcal{E}_3}^{\mathcal{E}_2} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

Products. By [Theorem 3.6.2](#),

$$[DS]_{\mathcal{E}_2}^{\mathcal{E}_2} = [D]_{\mathcal{E}_3}^{\mathcal{E}_2}[S]_{\mathcal{E}_2}^{\mathcal{E}_3} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad [SD]_{\mathcal{E}_3}^{\mathcal{E}_3} = [S]_{\mathcal{E}_2}^{\mathcal{E}_3}[D]_{\mathcal{E}_3}^{\mathcal{E}_2} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

Directly. $DS(x^k) = (x^{k+1})' = (k+1)x^k$ for $k = 0, 1, 2$, giving the diagonal 1, 2, 3. And $SD(x^k) = x \cdot kx^{k-1} = kx^k$ for $k \geq 1$, with $SD(1) = 0$, giving the diagonal 0, 1, 2, 3. Both computations agree. The two composites are not even the same size: DS is an operator on $\mathbb{R}[x]_{\leq 2}$ and SD an operator on $\mathbb{R}[x]_{\leq 3}$.

For operators, the theorem applies with $\mathcal{B} = \mathcal{C} = \mathcal{D}$, and gives powers and polynomials.

Corollary 3.6.3 — Matrices of Powers and Polynomials

Let $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of a finite-dimensional space V , let $T, S \in \mathcal{L}(V)$,

$c \in F$ and $p \in F[x]$. Then $[ST]_{\mathcal{B}} = [S]_{\mathcal{B}}[T]_{\mathcal{B}}$, $[S + T]_{\mathcal{B}} = [S]_{\mathcal{B}} + [T]_{\mathcal{B}}$, $[cT]_{\mathcal{B}} = c[T]_{\mathcal{B}}$, $[T^k]_{\mathcal{B}} = ([T]_{\mathcal{B}})^k$ for $k \in \mathbb{N}$, and

$$[p(T)]_{\mathcal{B}} = p([T]_{\mathcal{B}}).$$

Proof. The first formula is [Theorem 3.6.2](#) with $\mathcal{B} = \mathcal{C} = \mathcal{D}$. For sums and scalar multiples, the j -th column of $[S+T]_{\mathcal{B}}$ is $[S\mathbf{v}_j + T\mathbf{v}_j]_{\mathcal{B}} = [S\mathbf{v}_j]_{\mathcal{B}} + [T\mathbf{v}_j]_{\mathcal{B}}$ by [Theorem 1.5.2](#), and similarly $[cT\mathbf{v}_j]_{\mathcal{B}} = c[T\mathbf{v}_j]_{\mathcal{B}}$. For powers we induct on k : $[T^0]_{\mathcal{B}} = [\text{id}_V]_{\mathcal{B}} = I_n$ by [Example \(Matrix Maps and the Identity\)](#) (b), and if $[T^k]_{\mathcal{B}} = ([T]_{\mathcal{B}})^k$, then $[T^{k+1}]_{\mathcal{B}} = [T^k T]_{\mathcal{B}} = ([T]_{\mathcal{B}})^k [T]_{\mathcal{B}} = ([T]_{\mathcal{B}})^{k+1}$, by the first formula and [Definition 0.9.14](#). Finally, for $p = \sum_k a_k x^k$, combining the three rules gives $[p(T)]_{\mathcal{B}} = \sum_k a_k [T^k]_{\mathcal{B}} = \sum_k a_k ([T]_{\mathcal{B}})^k = p([T]_{\mathcal{B}})$, using [Definition 3.4.2](#) and [Definition 0.9.14](#). ■

For example, with $[D]_{\mathcal{E}_3}$ from [Example \(Differentiation on Cubic Polynomials\)](#), $([D]_{\mathcal{E}_3})^4 = [D^4]_{\mathcal{E}_3} = 0$, recovering [Example \(Differentiation Is Nilpotent\)](#) in matrix form.

3.6.4 Linear maps are matrices

The sums in [Corollary 3.6.3](#) work between different spaces too, and together with the construction of maps from values on a basis they show that, after choosing bases, there is no difference at all between linear maps and matrices.

Theorem 3.6.4 — $\mathcal{L}(V, W) \cong M_{m \times n}(F)$

Let V and W be finite-dimensional vector spaces over F with ordered bases $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ and $\mathcal{C} = (\mathbf{w}_1, \dots, \mathbf{w}_m)$. Then the map

$$\Phi: \mathcal{L}(V, W) \rightarrow M_{m \times n}(F), \quad T \mapsto [T]_{\mathcal{B}}^{\mathcal{C}},$$

is an isomorphism. In particular, $\dim \mathcal{L}(V, W) = mn = (\dim V)(\dim W)$.

Idea. Three things to check. *Linear:* columns of $[S + T]_{\mathcal{B}}^{\mathcal{C}}$ are coordinates of $S\mathbf{v}_j + T\mathbf{v}_j$, and coordinates respect sums. *Injective:* a map whose matrix is zero kills every basis vector, so it is the zero map. *Surjective:* a matrix prescribes the value of each $\mathbf{v}_j$, and a linear map with prescribed values on a basis exists. The last two are the uniqueness and existence halves of the same theorem about bases, which is why a linear map is “nothing but” its values on a basis.

Proof. Linear. Let $S, T \in \mathcal{L}(V, W)$ and $c \in F$. By [Definition 3.4.1](#) and [Theorem 1.5.2](#), the j -th column of $[S + T]_{\mathcal{B}}^{\mathcal{C}}$ is $[S\mathbf{v}_j + T\mathbf{v}_j]_{\mathcal{C}} = [S\mathbf{v}_j]_{\mathcal{C}} + [T\mathbf{v}_j]_{\mathcal{C}}$, and that of $[cT]_{\mathcal{B}}^{\mathcal{C}}$ is $[cT\mathbf{v}_j]_{\mathcal{C}} = c[T\mathbf{v}_j]_{\mathcal{C}}$. Hence $\Phi(S + T) = \Phi(S) + \Phi(T)$ and $\Phi(cT) = c\Phi(T)$.

Injective. Suppose $\Phi(T) = 0$. Then $[T\mathbf{v}_j]_{\mathcal{C}} = \mathbf{0}$, so $T\mathbf{v}_j = \mathbf{0}$ for every j . The zero map also sends every $\mathbf{v}_j$ to $\mathbf{0}$, so $T = 0$ by the uniqueness part of [Theorem 3.1.5](#). Hence $\ker \Phi = \{0\}$, and Φ is injective by [Theorem 3.2.3](#).

Surjective. Let $A = (a_{ij}) \in M_{m \times n}(F)$. By the existence part of [Theorem 3.1.5](#) there is $T \in \mathcal{L}(V, W)$ with $T\mathbf{v}_j = a_{1j}\mathbf{w}_1 + \dots + a_{mj}\mathbf{w}_m$ for each j . By [Definition 3.6.1](#), $[T]_{\mathcal{B}}^{\mathcal{C}} = A$.

So Φ is a bijective linear map, an isomorphism by [Theorem 3.5.1](#). For the dimension, we cannot quote [Theorem 3.5.4](#) directly, since we do not yet know that $\mathcal{L}(V, W)$ is finite-dimensional. Instead, $M_{m \times n}(F)$ has a basis of mn matrix units ([Example \(Dimensions of the Running Examples\)](#)), and the inverse isomorphism Φ^{-1} carries it to a basis of $\mathcal{L}(V, W)$ of length mn ([Theorem 3.5.2](#), applied to Φ^{-1} , which is an isomorphism by [Theorem 3.5.3](#)). Hence $\dim \mathcal{L}(V, W) = mn$. This proves the theorem. ■

With [Theorem 3.6.2](#), the isomorphism Φ also turns composition into multiplication. So, for $V = W$ and $\mathcal{B} = \mathcal{C}$, the algebra $\mathcal{L}(V)$ of the previous sections and the algebra $M_n(F)$ of Chapter 0 are the same algebra, written in two languages. The dictionary depends on the basis, as every isomorphism to F^n does.

3.6.5 Reading rank and invertibility from a matrix

Chapter 2 developed tools for the rank and the invertibility of matrices. The square of [Theorem 3.6.1](#) transfers all of them to linear maps.

Theorem 3.6.5 — Rank and Invertibility from the Matrix

Let V and W be finite-dimensional vector spaces over F with ordered bases $\mathcal{B}$ and $\mathcal{C}$, let $T \in \mathcal{L}(V, W)$, and let $A = [T]_{\mathcal{B}}^{\mathcal{C}}$.

- (a) The coordinate map $\mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}}$ restricts to an isomorphism $\ker T \rightarrow \text{nul}(A)$, and the coordinate map $\mathbf{w} \mapsto [\mathbf{w}]_{\mathcal{C}}$ restricts to an isomorphism $\text{im } T \rightarrow \text{col}(A)$. Hence

$$\text{rank } T = \text{rank } A \quad \text{and} \quad \text{nullity } T = \text{nullity } A.$$

- (b) Suppose $\dim V = \dim W$. Then T is invertible if and only if A is invertible, and in that case $[T^{-1}]_{\mathcal{C}}^{\mathcal{B}} = A^{-1}$.

Idea. For (a), the commutative square says a vector is killed by T exactly when its coordinate column is killed by A , and the outputs of T , in coordinates, are exactly the outputs of A . Isomorphic subspaces have the same dimension. For (b), $T^{-1}T = \text{id}_V$ translates, by the composition theorem, into a matrix equation with I_n on the right; conversely an inverse matrix is the matrix of **some** map, by the isomorphism $\mathcal{L}(W, V) \cong M_n(F)$, and that map is the inverse of T .

Proof. Let $n = \dim V$ and $m = \dim W$, and write $C_{\mathcal{B}}: V \rightarrow F^n$ and $C_{\mathcal{C}}: W \rightarrow F^m$ for the coordinate isomorphisms ([Corollary 3.5.5](#)).

- (a) *Kernel.* For $\mathbf{v} \in V$: $\mathbf{v} \in \ker T$ iff $T\mathbf{v} = \mathbf{0}$ iff $[T\mathbf{v}]_{\mathcal{C}} = \mathbf{0}$ (since $C_{\mathcal{C}}$ is injective) iff $A[\mathbf{v}]_{\mathcal{B}} = \mathbf{0}$ ([Theorem 3.6.1](#)) iff $[\mathbf{v}]_{\mathcal{B}} \in \text{nul}(A)$. So $C_{\mathcal{B}}$ maps $\ker T$ into $\text{nul}(A)$, and every $\mathbf{x} \in \text{nul}(A)$ is $[\mathbf{v}]_{\mathcal{B}}$ for some $\mathbf{v} \in V$ (surjectivity of $C_{\mathcal{B}}$), which then lies in $\ker T$. Thus the restriction of $C_{\mathcal{B}}$ is a linear map $\ker T \rightarrow \text{nul}(A)$, injective because $C_{\mathcal{B}}$ is, and surjective. By [Theorem 3.5.1](#) it is an isomorphism.

Image. By [Theorem 3.6.1](#), $C_{\mathcal{C}}(T\mathbf{v}) = AC_{\mathcal{B}}(\mathbf{v})$ for every $\mathbf{v} \in V$. As $\mathbf{v}$ runs over V , $C_{\mathcal{B}}(\mathbf{v})$ runs over all of F^n , so $\{C_{\mathcal{C}}(\mathbf{w}) : \mathbf{w} \in \text{im } T\} = \{A\mathbf{x} : \mathbf{x} \in F^n\} = \text{col}(A)$ ([Definition 2.5.1](#)). So the restriction of $C_{\mathcal{C}}$ is a linear map $\text{im } T \rightarrow \text{col}(A)$, surjective

by this computation and injective because C_C is; hence an isomorphism.

The subspaces involved are finite-dimensional ([Theorem 1.6.7](#)), so [Theorem 3.5.4](#) gives $\dim \ker T = \dim \text{nul}(A)$ and $\dim \text{im } T = \dim \text{col}(A)$, which are the stated equalities by [Definition 3.3.1](#), [Definition 3.3.2](#), [Definition 2.5.5](#) and [Definition 2.5.4](#).

- (b) ($\Rightarrow$) Suppose T is invertible. By [Theorem 3.6.2](#) and [Example \(Matrix Maps and the Identity\)](#) (b),

$$[T^{-1}]_C^B A = [T^{-1}T]_B^B = [\text{id}_V]_B = I_n, \quad A [T^{-1}]_C^B = [TT^{-1}]_C^C = [\text{id}_W]_C = I_n.$$

So A is invertible with $A^{-1} = [T^{-1}]_C^B$ ([Definition 0.9.13](#)).

($\Leftarrow$) Suppose A is invertible. By [Theorem 3.6.4](#) (for $\mathcal{L}(W, V)$ with bases $\mathcal{C}$ and $\mathcal{B}$), there is $S \in \mathcal{L}(W, V)$ with $[S]_C^B = A^{-1}$. Then by [Theorem 3.6.2](#), $[ST]_B = A^{-1}A = I_n = [\text{id}_V]_B$. Since $T' \mapsto [T']_B$ is injective on $\mathcal{L}(V)$ ([Theorem 3.6.4](#)), $ST = \text{id}_V$. By [Theorem 3.5.6](#), T is invertible with $T^{-1} = S$. This proves the theorem. ■

So we may add one more item to the Invertible Operator Theorem ([Theorem 3.5.7](#)): for $T \in \mathcal{L}(V)$ and **any** basis $\mathcal{B}$ of V , T is invertible if and only if $[T]_B$ is invertible. Every item of the Invertible Matrix Theorem ([Theorem 2.4.4](#)) now becomes a test for operators.

Example. (Rank of $X \mapsto X + X^T$)

Let $T: M_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})$, $T(X) = X + X^T$, and $\mathcal{E} = (E_{11}, E_{12}, E_{21}, E_{22})$. Find $[T]_{\mathcal{E}}$, use it to find $\text{rank } T$, a basis of $\ker T$ and a basis of $\text{im } T$, and decide whether T is invertible. Then do the same for the transpose map of [Example \(The Transpose Map\)](#).

Solution. Matrix. $T(E_{11}) = 2E_{11}$, $T(E_{12}) = E_{12} + E_{21} = T(E_{21})$, $T(E_{22}) = 2E_{22}$. So

$$A = [T]_{\mathcal{E}} = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}.$$

Matrix world. Subtracting row 2 from row 3 turns row 3 into a zero row; moving it to the bottom gives an echelon form with three non-zero rows and pivots in columns 1, 2, 4. So $\text{rank } A = 3$ and $\text{nullity } A = 1$. The free column is column 3, and $A\mathbf{x} = \mathbf{0}$ gives $x_1 = x_4 = 0$, $x_2 = -x_3$, so $\text{nul}(A) = \text{span}((0, -1, 1, 0))$. By [Theorem 2.5.2](#), the pivot columns $(2, 0, 0, 0)$, $(0, 1, 1, 0)$, $(0, 0, 0, 2)$ form a basis of $\text{col}(A)$.

Translate back. By [Theorem 3.6.5](#), $\text{rank } T = 3$. The column $(0, -1, 1, 0)$ is the coordinate vector of $E_{21} - E_{12}$, so $\ker T = \text{span}\left(\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\right)$, the skew-symmetric matrices. The basis of $\text{col}(A)$ translates to $2E_{11}$, $E_{12} + E_{21}$, $2E_{22}$, a basis of $\text{im } T$, which is therefore the space of symmetric matrices. Since $\text{rank } A = 3 < 4$, A is not invertible ([Theorem 2.4.4](#)), and neither is T .

Transpose. The matrix $P = [X \mapsto X^T]_{\mathcal{E}}$ from [Example \(The Transpose Map\)](#) satisfies $P^2 = I_4$ (swapping two coordinates twice does nothing). So P is invertible with $P^{-1} = P$, and by [Theorem 3.6.5](#) (b) the transpose map is invertible and is its own inverse, which is $(X^T)^T = X$ again.

We now have both halves of the working method: a basis turns spaces into F^n and maps into matrices, and every computation can be done with Chapter 2's tools. What we have not yet controlled is the choice. [Example \(A Reflection in Two Bases\)](#) shows that one map can have a complicated matrix in one basis and a diagonal one in another; the next section finds the formula relating the two.

3.6.6 Exercises

3.6.6.1 A. Check your understanding

Exercise A1.

- Define $[T]_{\mathcal{B}}^{\mathcal{C}}$ for $T \in \mathcal{L}(V, W)$ and ordered bases $\mathcal{B}$ of V , $\mathcal{C}$ of W .
- State the formula relating $[T\mathbf{v}]_{\mathcal{C}}$ and $[\mathbf{v}]_{\mathcal{B}}$, and the formula for the matrix of a composition.
- True or false: the j -th **row** of $[T]_{\mathcal{B}}^{\mathcal{C}}$ is $[T\mathbf{v}_j]_{\mathcal{C}}$. Justify your answer.
- True or false: $[\text{id}_V]_{\mathcal{B}}^{\mathcal{C}} = I_n$ for any two bases $\mathcal{B}, \mathcal{C}$ of an n -dimensional space V . Justify your answer.
- What is $\dim \mathcal{L}(\mathbb{R}^3, \mathbb{R}[x]_{\leq 2})$?

Solution.

- If $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$, then $[T]_{\mathcal{B}}^{\mathcal{C}}$ is the $\dim W \times n$ matrix whose j -th column is $[T\mathbf{v}_j]_{\mathcal{C}}$ ([Definition 3.6.1](#)).
- $[T\mathbf{v}]_{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}}[\mathbf{v}]_{\mathcal{B}}$ ([Theorem 3.6.1](#)), and $[ST]_{\mathcal{B}}^{\mathcal{D}} = [S]_{\mathcal{C}}^{\mathcal{D}}[T]_{\mathcal{B}}^{\mathcal{C}}$ ([Theorem 3.6.2](#)).
- False. It is the j -th **column**. For $D: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$ in [Example \(Differentiation on Cubic Polynomials\)](#), $[D(x)]_{\mathcal{E}_2} = (1, 0, 0)$ is the second column, while the second row is $(0, 0, 2, 0)$.
- False. In $\mathbb{R}^2$ with $\mathcal{B} = (\mathbf{e}_1, \mathbf{e}_2)$ and $\mathcal{C} = (\mathbf{e}_2, \mathbf{e}_1)$, $[\mathbf{e}_1]_{\mathcal{C}} = (0, 1)$, so the first column of $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$ is $(0, 1) \neq \mathbf{e}_1$. It is true when $\mathcal{B} = \mathcal{C}$ ([Example \(Matrix Maps and the Identity\)](#) (b)).
- $3 \cdot 3 = 9$, by [Theorem 3.6.4](#).

3.6.6.2 B. Practice

Exercise B1: Matrices in given bases.

Find the matrix of each operator in the given basis. (Both maps are linear; you may take this as given.)

- $T: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}[x]_{\leq 2}$, $T(p) = p(x+1)$, in $\mathcal{E} = (1, x, x^2)$.
- $T: M_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})$, $T(X) = BX$ with $B = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, in $\mathcal{E} = (E_{11}, E_{12}, E_{21}, E_{22})$.

Solution.

- $T(1) = 1$, $T(x) = x+1 = 1+x$, $T(x^2) = (x+1)^2 = 1+2x+x^2$. Their coordinates

are $(1, 0, 0)$, $(1, 1, 0)$, $(1, 2, 1)$, so

$$[T]_{\mathcal{E}} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

(b) Multiplying B by a matrix unit E_{kj} on the right keeps column k of B and moves it to column j :

$$BE_{11} = \begin{pmatrix} 1 & 0 \\ 3 & 0 \end{pmatrix}, \quad BE_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 3 \end{pmatrix}, \quad BE_{21} = \begin{pmatrix} 2 & 0 \\ 4 & 0 \end{pmatrix}, \quad BE_{22} = \begin{pmatrix} 0 & 2 \\ 0 & 4 \end{pmatrix}.$$

Reading entries in the order $(1, 1)$, $(1, 2)$, $(2, 1)$, $(2, 2)$ gives the columns $(1, 0, 3, 0)$, $(0, 1, 0, 3)$, $(2, 0, 4, 0)$, $(0, 2, 0, 4)$, so

$$[T]_{\mathcal{E}} = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 \\ 3 & 0 & 4 & 0 \\ 0 & 3 & 0 & 4 \end{pmatrix}.$$

Exercise B2: Recovering a map from its matrix.

Let $\mathcal{B} = (1, 1+x, x^2)$, a basis of $\mathbb{R}[x]_{\leq 2}$, and let $\mathcal{E}'$ be the standard basis of $\mathbb{R}^2$. A linear map $T: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}^2$ has

$$[T]_{\mathcal{B}}^{\mathcal{E}'} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix}.$$

Find $T(a + bx + cx^2)$. Hence find a basis of $\ker T$.

Solution. First find the coordinates of $p = a + bx + cx^2$ in $\mathcal{B}$: $p = (a-b) \cdot 1 + b(1+x) + cx^2$, so $[p]_{\mathcal{B}} = (a-b, b, c)$, unique by [Theorem 1.5.1](#). By [Theorem 3.6.1](#), and since coordinates in $\mathcal{E}'$ are the entries,

$$T(p) = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} a-b \\ b \\ c \end{pmatrix} = (a-b+2c, b-c).$$

Hence $T(p) = \mathbf{0}$ iff $b = c$ and $a = b - 2c = -c$, that is, $p = c(-1 + x + x^2)$. So $(x^2 + x - 1)$ is a basis of $\ker T$. Check: $[x^2 + x - 1]_{\mathcal{B}} = (-2, 1, 1)$, and $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix} (-2, 1, 1) = (0, 0)$.

Exercise B3: A composition computed two ways.

Let $T: \mathbb{R}[x]_{\leq 1} \rightarrow \mathbb{R}[x]_{\leq 2}$, $T(p) = xp$, and $S: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}^2$, $S(p) = (p(0), p(1))$. Use the monomial bases and the standard basis of $\mathbb{R}^2$. Compute the matrix of ST directly, and again as a product of the matrices of S and T .

Solution. Let $\mathcal{E}_1 = (1, x)$, $\mathcal{E}_2 = (1, x, x^2)$ and $\mathcal{E}'$ the standard basis of $\mathbb{R}^2$.

Factors. $T(1) = x$, $T(x) = x^2$, so $[T]_{\mathcal{E}_1}^{\mathcal{E}_2} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$. $S(1) = (1, 1)$, $S(x) = (0, 1)$,

$S(x^2) = (0, 1)$, so $[S]_{\mathcal{E}_2}^{\mathcal{E}'_1} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$.

Product. By [Theorem 3.6.2](#),

$$[ST]_{\mathcal{E}_1}^{\mathcal{E}'_1} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}.$$

Directly. $ST(1) = S(x) = (0, 1)$ and $ST(x) = S(x^2) = (0, 1)$, so the columns are $(0, 1)$ and $(0, 1)$, in agreement. In words, $ST(a + bx) = (0, a + b)$.

3.6.6.3 C. Going deeper

Exercise C1: Upper triangular matrices and invariant subspaces.

Let V be finite-dimensional and $T \in \mathcal{L}(V)$. A subspace U of V is **T -invariant** if $T\mathbf{u} \in U$ for every $\mathbf{u} \in U$. Let $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of V , and for $1 \leq k \leq n$ let $U_k = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$. Prove that $[T]_{\mathcal{B}}$ is upper triangular if and only if every U_k is T -invariant.

Solution. Let $[T]_{\mathcal{B}} = (a_{ij})$, so $T\mathbf{v}_j = \sum_{i=1}^n a_{ij}\mathbf{v}_i$ with unique coefficients ([Definition 3.6.1](#)). By [Definition 0.9.5](#), $[T]_{\mathcal{B}}$ is upper triangular iff $a_{ij} = 0$ whenever $i > j$, that is, iff

$$T\mathbf{v}_j \in U_j \quad \text{for every } j = 1, \dots, n. \quad (*)$$

Indeed, if $a_{ij} = 0$ for $i > j$, then $T\mathbf{v}_j = \sum_{i \leq j} a_{ij}\mathbf{v}_i \in U_j$; conversely, if $T\mathbf{v}_j = \sum_{i \leq j} b_i\mathbf{v}_i$, then by uniqueness of coordinates $a_{ij} = 0$ for $i > j$.

($\Rightarrow$) Assume (*), and fix k . For $j \leq k$, $T\mathbf{v}_j \in U_j \subseteq U_k$. Let $\mathbf{u} = \sum_{j \leq k} c_j\mathbf{v}_j \in U_k$. Then $T\mathbf{u} = \sum_{j \leq k} c_j T\mathbf{v}_j$ by [Theorem 3.1.4](#), a combination of vectors of the subspace U_k , so $T\mathbf{u} \in U_k$ by [Theorem 1.3.1](#). Hence U_k is T -invariant.

($\Leftarrow$) Assume every U_k is T -invariant. For each j , $\mathbf{v}_j \in U_j$, so $T\mathbf{v}_j \in U_j$ by invariance of U_j . This is (*), so $[T]_{\mathcal{B}}$ is upper triangular.

Exercise C2: Integrating with a matrix.

Let $f_1(x) = e^x$, $f_2(x) = xe^x$, $f_3(x) = x^2e^x$, and let $V = \text{span}(f_1, f_2, f_3)$ inside the real vector space of all functions $\mathbb{R} \rightarrow \mathbb{R}$ ([Example \(The Standard Vector Spaces\)](#) (d)).

- Prove that $\mathcal{B} = (f_1, f_2, f_3)$ is a basis of V .
- Show that differentiation maps V into V , and find $[D]_{\mathcal{B}}$ for $D \in \mathcal{L}(V)$, $D(f) = f'$.
- Hence show that D is invertible on V , and use $([D]_{\mathcal{B}})^{-1}$ to find a function $g \in V$ with $g' = x^2e^x$.

Hint: for (a), divide a vanishing combination by e^x and count roots.

Solution.

- (a) $\mathcal{B}$ spans V by definition. Suppose $af_1 + bf_2 + cf_3 = 0$, the zero function. Then $(a + bx + cx^2)e^x = 0$ for every $x \in \mathbb{R}$, and $e^x \neq 0$, so the polynomial $a + bx + cx^2$ vanishes at every real number. It has infinitely many roots, so it is the zero polynomial by [Lemma 2.7.2](#), and $a = b = c = 0$. Hence $\mathcal{B}$ is independent, and a basis of V .

- (b) By the product rule,

$$f'_1 = e^x = f_1, \quad f'_2 = e^x + xe^x = f_1 + f_2, \quad f'_3 = 2xe^x + x^2e^x = 2f_2 + f_3.$$

Each lies in V , and D is linear, so D maps every combination of f_1, f_2, f_3 into V ([Theorem 3.1.4](#)). Reading off coordinates as columns,

$$[D]_{\mathcal{B}} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

- (c) The matrix is upper triangular with non-zero diagonal entries, and one checks directly that

$$\begin{pmatrix} 1 & -1 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} = I_3,$$

so $[D]_{\mathcal{B}}$ is invertible with this inverse ([Theorem 2.4.5](#)). By [Theorem 3.6.5](#) (b), D is invertible on V and $[D^{-1}]_{\mathcal{B}} = ([D]_{\mathcal{B}})^{-1}$. We want $g = D^{-1}(f_3)$. Since $[f_3]_{\mathcal{B}} = (0, 0, 1)$, [Theorem 3.6.1](#) gives

$$[g]_{\mathcal{B}} = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix},$$

so $g = 2f_1 - 2f_2 + f_3 = (x^2 - 2x + 2)e^x$. Check: $g' = (2x - 2)e^x + (x^2 - 2x + 2)e^x = x^2e^x$. No integration by parts was needed.

3.7 Change of Basis and Similarity

The matrix $[T]_{\mathcal{B}}^{\mathcal{C}}$ of a linear map depends on two choices of basis, and different choices can give very different matrices for the same map. This section answers two questions left open by that fact. How are two matrices of the same map related? And which properties of a matrix belong to the map itself, so that no choice of basis can change them? The answers are the change-of-basis formula, drawn as a commutative square, and the relation of similarity.

3.7.1 Change-of-coordinates matrices

Recall the definition of the matrix of a map ([Definition 3.6.1](#)). If $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ is a basis of V and $\mathcal{C}$ is a basis of W , then $[T]_{\mathcal{B}}^{\mathcal{C}}$ is the matrix whose j -th column is $[T\mathbf{v}_j]_{\mathcal{C}}$, and it turns coordinates into coordinates: $[T\mathbf{v}]_{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}}[\mathbf{v}]_{\mathcal{B}}$ ([Theorem 3.6.1](#)).

Recall the reflection from the previous section ([Example \(A Reflection in Two Bases\)](#)), which motivates this one. Let $R: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the reflection across the line $y = x$, so $R(x, y) = (y, x)$. In the standard basis $\mathcal{E} = (\mathbf{e}_1, \mathbf{e}_2)$, we have $R\mathbf{e}_1 = \mathbf{e}_2$ and $R\mathbf{e}_2 = \mathbf{e}_1$, so

$$[R]_{\mathcal{E}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Now use the basis $\mathcal{B} = ((1, 1), (1, -1))$, one vector on the mirror line and one perpendicular to it. Then $R(1, 1) = (1, 1)$ and $R(1, -1) = (-1, 1) = -(1, -1)$, so

$$[R]_{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The second matrix shows at a glance what R does: it keeps one direction and flips the other. The two matrices describe the same map, so there must be a rule converting one into the other. The first step is to convert the **coordinates** of a single vector.

Take $\mathbf{v} = (3, 1)$. Its standard coordinates are $[\mathbf{v}]_{\mathcal{E}} = (3, 1)$. In the basis $\mathcal{B}$, we solve $a(1, 1) + b(1, -1) = (3, 1)$, which gives $a = 2$, $b = 1$, so $[\mathbf{v}]_{\mathcal{B}} = (2, 1)$. The same vector has two different coordinate columns, and we want a matrix that translates between them. Since the identity map id_V sends every vector to itself, the matrix of id_V , with one basis on the input side and the other on the output side, should do exactly this job.

A change-of-coordinates matrix is the matrix of “do nothing”, read with one basis on the way in and another on the way out.

Definition 3.7.1 — Change-of-coordinates matrix

Let V be a finite-dimensional vector space over F , and let $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ and $\mathcal{C}$ be **ordered** bases of V . The **change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$** is the $n \times n$ matrix

$$[\text{id}]_{\mathcal{B}}^{\mathcal{C}} := [\text{id}_V]_{\mathcal{B}}^{\mathcal{C}},$$

the matrix of the identity map $\text{id}_V: V \rightarrow V$ with respect to the input basis $\mathcal{B}$ and the output basis $\mathcal{C}$. Its j -th column is $[\mathbf{v}_j]_{\mathcal{C}}$.

In words: to build $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$, take the vectors of the **old** basis $\mathcal{B}$ one at a time, write each in the **new** basis $\mathcal{C}$, and put the coordinate columns side by side. The column description is [Definition 3.6.1](#) with $T = \text{id}_V$, since $\text{id}_V(\mathbf{v}_j) = \mathbf{v}_j$. Both bases must be bases of the **same** space V , and both are ordered: reordering either one permutes the columns or the rows.

Examples.

- **Into the standard basis of F^n .** If $\mathcal{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ is a basis of F^n and $\mathcal{E}$ is the standard basis, then $[\mathbf{b}_j]_{\mathcal{E}} = \mathbf{b}_j$. So $[\text{id}]_{\mathcal{B}}^{\mathcal{E}}$ is the matrix whose columns are the vectors of $\mathcal{B}$ themselves. For $\mathcal{B} = ((1, 1), (1, -1))$ in $\mathbb{R}^2$, $[\text{id}]_{\mathcal{B}}^{\mathcal{E}} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, and indeed $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ turns $[\mathbf{v}]_{\mathcal{B}}$ into $[\mathbf{v}]_{\mathcal{E}}$ for $\mathbf{v} = (3, 1)$.
- **Out of the standard basis.** In the other direction, $[\text{id}]_{\mathcal{E}}^{\mathcal{B}}$ has columns $[\mathbf{e}_1]_{\mathcal{B}}$ and $[\mathbf{e}_2]_{\mathcal{B}}$. Solving $a(1, 1) + b(1, -1) = (1, 0)$ gives $a = b = \frac{1}{2}$, and solving for $(0, 1)$ gives $a = \frac{1}{2}$, $b = -\frac{1}{2}$. So $[\text{id}]_{\mathcal{E}}^{\mathcal{B}} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. This is the inverse of the previous matrix, which the next theorem explains.

- **Polynomials.** In $F[x]_{\leq 2}$, let $\mathcal{B} = (1, x, x^2)$ and $\mathcal{C} = (1, x-1, (x-1)^2)$. Expanding $(x-1)^2 = 1 - 2x + x^2$, the vectors of $\mathcal{C}$ have $\mathcal{B}$ -coordinates $(1, 0, 0)$, $(-1, 1, 0)$ and $(1, -2, 1)$, so

$$[\text{id}]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}.$$

- **Degenerate case.** If $\mathcal{C} = \mathcal{B}$, then $[\mathbf{v}_j]_{\mathcal{B}} = \mathbf{e}_j$, so $[\text{id}]_{\mathcal{B}} = I_n$. Changing nothing is recorded by the identity matrix, as it should be.

Non-example by minimal change. Replace the basis $((1, 1), (1, -1))$ by the list $\mathcal{L} = ((1, 1), (2, 2))$. We can still write the matrix with these vectors as columns, but $\mathcal{L}$ is not a basis, so “coordinates with respect to $\mathcal{L}$ ” are not defined: $(2, 2) = 2(1, 1) + 0(2, 2) = 0(1, 1) + 1(2, 2)$ has two coordinate columns, and $(1, 0)$ has none. The clause that fails is “ $\mathcal{B}$ is a basis”, which Definition 1.5.2 needs for coordinates to exist and be unique.

Warning. Watch the direction. The columns of $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$ are the $\mathcal{B}$ -vectors written in $\mathcal{C}$, and the matrix turns $\mathcal{B}$ -coordinates into $\mathcal{C}$ -coordinates. Using it the other way gives nonsense. With $\mathcal{B} = ((1, 1), (1, -1))$, the matrix $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = [\text{id}]_{\mathcal{B}}^{\mathcal{E}}$ applied to $[\mathbf{v}]_{\mathcal{E}} = (3, 1)$ gives $(4, 2)$, which is **not** $[\mathbf{v}]_{\mathcal{B}} = (2, 1)$. A reliable test: the first column must be the first **input** basis vector written in the **output** basis. Here $(1, 1) = 1 \cdot \mathbf{e}_1 + 1 \cdot \mathbf{e}_2$, so the column $(1, 1)$ belongs to $[\text{id}]_{\mathcal{B}}^{\mathcal{E}}$, which accepts $\mathcal{B}$ -coordinates.

The examples suggest three facts: the matrix converts coordinates, converting twice multiplies the matrices, and converting back inverts the matrix. All three follow from results about matrices of maps, applied to the identity.

Theorem 3.7.1 — Change of Coordinates

Let V be a vector space of dimension n , and let $\mathcal{B}, \mathcal{C}, \mathcal{D}$ be bases of V .

- For every $\mathbf{v} \in V$, $[\mathbf{v}]_{\mathcal{C}} = [\text{id}]_{\mathcal{B}}^{\mathcal{C}}[\mathbf{v}]_{\mathcal{B}}$. Moreover, $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$ is the **only** matrix $M \in M_n(F)$ with $[\mathbf{v}]_{\mathcal{C}} = M[\mathbf{v}]_{\mathcal{B}}$ for every $\mathbf{v} \in V$.
- $[\text{id}]_{\mathcal{B}}^{\mathcal{D}} = [\text{id}]_{\mathcal{C}}^{\mathcal{D}}[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$.
- $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$ is invertible, and $([\text{id}]_{\mathcal{B}}^{\mathcal{C}})^{-1} = [\text{id}]_{\mathcal{C}}^{\mathcal{B}}$.

Proof.

- Apply Theorem 3.6.1, including its uniqueness clause, to id_V with input basis $\mathcal{B}$ and output basis $\mathcal{C}$: since $\text{id}_V(\mathbf{v}) = \mathbf{v}$, it says $[\mathbf{v}]_{\mathcal{C}} = [\text{id}]_{\mathcal{B}}^{\mathcal{C}}[\mathbf{v}]_{\mathcal{B}}$ for every $\mathbf{v}$, and that no other matrix has this property.
- Since $\text{id}_V = \text{id}_V \circ \text{id}_V$, Theorem 3.6.2, with $\mathcal{B}$ on the input, $\mathcal{C}$ in the middle and $\mathcal{D}$ on the output, gives $[\text{id}]_{\mathcal{B}}^{\mathcal{D}} = [\text{id}]_{\mathcal{C}}^{\mathcal{D}}[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$.
- Taking $\mathcal{D} = \mathcal{B}$ in (b) gives $[\text{id}]_{\mathcal{B}}^{\mathcal{B}}[\text{id}]_{\mathcal{B}}^{\mathcal{C}} = [\text{id}]_{\mathcal{B}} = I_n$, where the last equality holds because, writing $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$, the j -th column of $[\text{id}]_{\mathcal{B}}$ is $[\mathbf{v}_j]_{\mathcal{B}} = \mathbf{e}_j$. Swapping the roles of $\mathcal{B}$ and $\mathcal{C}$ gives $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}[\text{id}]_{\mathcal{C}}^{\mathcal{B}} = I_n$. By Definition 0.9.13, $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$ is invertible with inverse $[\text{id}]_{\mathcal{C}}^{\mathcal{B}}$. ■

So every change-of-coordinates matrix is invertible. The converse is also true, and it will matter when we compare matrices: every invertible matrix is a change-of-coordinates matrix, for a suitable new basis.

Proposition 3.7.2 — Invertible matrices are changes of basis

Let $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of V , and let $P = (p_{ij}) \in M_n(F)$ be invertible. Put

$$\mathbf{v}'_j := p_{1j}\mathbf{v}_1 + p_{2j}\mathbf{v}_2 + \cdots + p_{nj}\mathbf{v}_n \quad (j = 1, \dots, n).$$

Then $\mathcal{B}' = (\mathbf{v}'_1, \dots, \mathbf{v}'_n)$ is a basis of V , and $[\text{id}]_{\mathcal{B}'}^{\mathcal{B}} = P$.

Proof. By Definition 1.5.2, $[\mathbf{v}'_j]_{\mathcal{B}}$ is the j -th column $\mathbf{p}_j$ of P . Let $c_1, \dots, c_n \in F$ satisfy $c_1\mathbf{v}'_1 + \cdots + c_n\mathbf{v}'_n = \mathbf{0}$. Taking $\mathcal{B}$ -coordinates and using Theorem 1.5.2,

$$c_1\mathbf{p}_1 + \cdots + c_n\mathbf{p}_n = [\mathbf{0}]_{\mathcal{B}} = \mathbf{0},$$

that is, $P\mathbf{c} = \mathbf{0}$ with $\mathbf{c} = (c_1, \dots, c_n)$, by Theorem 0.9.3. Since P is invertible, Theorem 2.4.4 ((a) $\Rightarrow$ (b)) gives $\mathbf{c} = \mathbf{0}$. So $\mathcal{B}'$ is linearly independent. It has length $n = \dim V$, hence it is a basis by Theorem 1.6.4 (a). Finally, the j -th column of $[\text{id}]_{\mathcal{B}'}^{\mathcal{B}}$ is $[\mathbf{v}'_j]_{\mathcal{B}} = \mathbf{p}_j$, so $[\text{id}]_{\mathcal{B}'}^{\mathcal{B}} = P$. ■

Example. (Coordinates around the point 1)

In $\mathbb{R}[x]_{\leq 2}$, let $\mathcal{B} = (1, x, x^2)$ and $\mathcal{C} = (1, x-1, (x-1)^2)$. Find $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$, and use it to write $p = 2 + 3x + x^2$ in powers of $x-1$.

Solution. From the examples above, $[\text{id}]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$. By Theorem 3.7.1 (c), $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$ is its inverse. We can also read it off directly: $1 = 1$, $x = 1 + (x-1)$, and $x^2 = (1 + (x-1))^2 = 1 + 2(x-1) + (x-1)^2$. So

$$[\text{id}]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix},$$

and multiplying the two matrices gives I_3 , as the theorem predicts. By Theorem 3.7.1 (a),

$$[p]_{\mathcal{C}} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 5 \\ 1 \end{pmatrix}.$$

Hence $2 + 3x + x^2 = 6 + 5(x-1) + (x-1)^2$. Check at $x = 1$: both sides equal 6. The $\mathcal{C}$ -coordinates are $p(1)$, $p'(1)$ and $p''(1)/2$, the Taylor coefficients at 1.

3.7.2 Change of basis for linear maps

Now return to a linear map $T: V \rightarrow W$ between finite-dimensional spaces. Suppose we know its matrix $[T]_{\mathcal{B}}^{\mathcal{C}}$ and want the matrix $[T]_{\mathcal{B}'}^{\mathcal{C}'}$ for new bases $\mathcal{B}'$ of V and $\mathcal{C}'$ of W . The trick is to write T in a slightly more elaborate way that does nothing:

$$T = \text{id}_W \circ T \circ \text{id}_V.$$

Each factor can now carry its own pair of bases, and the matrix of the composite is the product of the matrices. The picture keeps track of which basis sits where.

$$\begin{array}{ccc}
 V \text{ with } \mathcal{B} & \xrightarrow{\begin{array}{c} T \\ [T]_{\mathcal{B}}^{\mathcal{C}} \end{array}} & W \text{ with } \mathcal{C} \\
 \begin{array}{c} \uparrow \text{id}_V \\ [\text{id}]_{\mathcal{B}'}^{\mathcal{B}} \end{array} & & \begin{array}{c} \downarrow \text{id}_W \\ [\text{id}]_{\mathcal{C}'}^{\mathcal{C}} \end{array} \\
 V \text{ with } \mathcal{B}' & \xrightarrow{\begin{array}{c} T \\ [T]_{\mathcal{B}'}^{\mathcal{C}'} \text{ (wanted)} \end{array}} & W \text{ with } \mathcal{C}'
 \end{array}$$

Going along the bottom edge applies T . Going up, across the top and down applies $\text{id}_W \circ T \circ \text{id}_V$, which is the same map. So the two routes must have the same matrix. Reading the long route, the step done **first** (up the left side) is written on the **right**.

Theorem 3.7.3 — Change of Basis for Linear Maps

Let V and W be finite-dimensional vector spaces over F , let $\mathcal{B}, \mathcal{B}'$ be bases of V and $\mathcal{C}, \mathcal{C}'$ bases of W , and let $T: V \rightarrow W$ be linear. Then

$$[T]_{\mathcal{B}'}^{\mathcal{C}'} = [\text{id}]_{\mathcal{C}'}^{\mathcal{C}} [T]_{\mathcal{B}}^{\mathcal{C}} [\text{id}]_{\mathcal{B}'}^{\mathcal{B}}.$$

In particular, for an operator $T \in \mathcal{L}(V)$ and bases $\mathcal{B}, \mathcal{B}'$ of V , with $P = [\text{id}]_{\mathcal{B}'}^{\mathcal{B}}$,

$$[T]_{\mathcal{B}'} = P^{-1} [T]_{\mathcal{B}} P.$$

Idea. Nothing is computed: the formula is the change-of-basis square read along its long route. ① Write $T = \text{id}_W \circ T \circ \text{id}_V$. ② Give the middle factor the bases whose matrix we know. ③ Apply the composition rule twice. Do not memorize the formula; redraw the square and read it off, right to left.

Proof. Since $T = \text{id}_W \circ (T \circ \text{id}_V)$, Theorem 3.6.2 with bases $\mathcal{B}', \mathcal{C}, \mathcal{C}'$ gives $[T]_{\mathcal{B}'}^{\mathcal{C}'} = [\text{id}]_{\mathcal{C}'}^{\mathcal{C}} [T \circ \text{id}_V]_{\mathcal{B}'}^{\mathcal{C}}$. Applying Theorem 3.6.2 again, with bases $\mathcal{B}', \mathcal{B}, \mathcal{C}$, gives $[T \circ \text{id}_V]_{\mathcal{B}'}^{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}} [\text{id}]_{\mathcal{B}'}^{\mathcal{B}}$. Combining the two equations proves the first formula.

For an operator, take $W = V$, $\mathcal{C} = \mathcal{B}$ and $\mathcal{C}' = \mathcal{B}'$. The first formula becomes $[T]_{\mathcal{B}'} = [\text{id}]_{\mathcal{B}'}^{\mathcal{B}} [T]_{\mathcal{B}} P$, and $[\text{id}]_{\mathcal{B}'}^{\mathcal{B}} = P^{-1}$ by Theorem 3.7.1 (c). This proves the theorem. ■

We will call this move **the change-of-basis square**: whenever a question involves the same map in two bases, draw the square, label the arrows with bases, and read the formula off the long route. It also explains the word “coordinates”: a matrix is a map seen through a choice of coordinates, and the side arrows re-express those coordinates.

Example. (The reflection in a mirror-adapted basis)

Let $R(x, y) = (y, x)$ on $\mathbb{R}^2$, and $\mathcal{B} = ((1, 1), (1, -1))$. Compute $[R]_{\mathcal{B}}$ from $[R]_{\mathcal{E}}$ using [Theorem 3.7.3](#), and compare with the direct computation at the start of the section.

Solution. Here $P = [\text{id}]_{\mathcal{B}}^{\mathcal{E}} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, with inverse $P^{-1} = [\text{id}]_{\mathcal{E}}^{\mathcal{B}} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, found in the examples after [Definition 3.7.1](#). Then

$$P^{-1}[R]_{\mathcal{E}}P = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

which agrees with the direct computation. The direct route was shorter here, because we chose $\mathcal{B}$ so that R acts simply on it. The formula earns its keep when the new matrix is not visible by inspection, or when we want to go back from the simple matrix to the standard one: $[R]_{\mathcal{E}} = P[R]_{\mathcal{B}}P^{-1}$.

The next example changes basis in a polynomial space, where “standard” does not mean “best”.

Example. (Differentiation with a rescaled basis)

Let $D: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}[x]_{\leq 2}$ be differentiation, $\mathcal{B} = (1, x, x^2)$ and $\mathcal{B}' = (1, x, \frac{1}{2}x^2)$. Find $[D]_{\mathcal{B}'}$.

Solution. In $\mathcal{B}$, $D(1) = 0$, $D(x) = 1$ and $D(x^2) = 2x$, so $A = [D]_{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$.

The vectors of $\mathcal{B}'$ have $\mathcal{B}$ -coordinates $\mathbf{e}_1$, $\mathbf{e}_2$, $\frac{1}{2}\mathbf{e}_3$, so $P = [\text{id}]_{\mathcal{B}'}^{\mathcal{B}} = \text{diag}(1, 1, \frac{1}{2})$ and $P^{-1} = \text{diag}(1, 1, 2)$. Multiplying a matrix on the left by a diagonal matrix scales its rows, and on the right scales its columns, so

$$P^{-1}AP = \begin{pmatrix} 0 & 1 \cdot 1 \cdot 1 & 0 \\ 0 & 0 & 1 \cdot 2 \cdot \frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Directly: $D(1) = 0$, $D(x) = 1$, $D(\frac{1}{2}x^2) = x$, so D shifts each vector of $\mathcal{B}'$ to the previous one, and the matrix has 1's just above the diagonal. The factor 2 has disappeared into the basis.

Quick check. Let $T: V \rightarrow W$, and suppose we change only the basis of W , from $\mathcal{C}$ to $\mathcal{C}'$, keeping $\mathcal{B}$. Which of the two side matrices in [Theorem 3.7.3](#) becomes the identity, and what is the formula?

Solution. With $\mathcal{B}' = \mathcal{B}$, the left side of the square is $[\text{id}]_{\mathcal{B}} = I_n$. The formula becomes $[T]_{\mathcal{B}}^{\mathcal{C}'} = [\text{id}]_{\mathcal{C}'}^{\mathcal{C}} [T]_{\mathcal{B}}^{\mathcal{C}}$: only a multiplication on the left. Changing the output basis acts on the rows; changing the input basis acts on the columns.

3.7.3 Similar matrices

For an operator, the two bases on the two sides of the square are the same, and the formula takes the special shape $P^{-1}AP$. Recall from [Example \(Similarity\)](#) that we already gave this shape a name in Chapter 0 and checked that it defines an equivalence relation. We now record it as a definition, because it is exactly the relation “same operator, different basis”.

Two square matrices are similar when one is obtained from the other by a change of basis applied on both sides at once.

Definition 3.7.2 — Similar matrices

Let $A, B \in M_n(F)$. We say A is **similar** to B , written $A \sim B$, if there **exists** an **invertible** matrix $P \in M_n(F)$ such that

$$B = P^{-1}AP.$$

In words: the **same** P appears on both sides, once inverted. By [Example \(Similarity\)](#), $\sim$ is an equivalence relation, so we may say “ A and B are similar” without worrying about the order. The matrix P is not unique: for instance $A = P^{-1}AP$ holds for every invertible P that commutes with A , such as any non-zero multiple of I_n . Writing $B = QAQ^{-1}$ defines the same relation, with $Q = P^{-1}$.

Examples.

- **The reflection.** $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, by [Example \(The reflection in a mirror-adapted basis\)](#).
- **A triangular matrix.** $\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} \sim \text{diag}(1, 3)$, as computed in [Example \(Similarity\)](#) with $P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.
- **Reordering a basis.** $\text{diag}(1, 2) \sim \text{diag}(2, 1)$: with $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = P^{-1}$, we get $P^{-1} \text{diag}(1, 2)P = \text{diag}(2, 1)$. This is the operator with matrix $\text{diag}(1, 2)$ in $(\mathbf{v}_1, \mathbf{v}_2)$, read in the reordered basis $(\mathbf{v}_2, \mathbf{v}_1)$.
- **Degenerate case: scalar matrices.** For $c \in F$, $P^{-1}(cI_n)P = cP^{-1}P = cI_n$ for every invertible P . So cI_n is similar only to itself. In particular I_n and the zero matrix are alone in their classes: the identity operator looks like I_n in **every** basis.

Non-example by minimal change. Allow two different invertible matrices, $B = QAP$. With $A = I_2$, $Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $P = I_2$, we get $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \neq I_2$. So $B = QAP$ holds, but B is not similar to I_2 , which is similar only to itself. The clause that failed is “the **same** P on both sides, once inverted”. The relaxed relation is also useful; it is the subject of the next section.

The definition was designed to capture one idea, and the next theorem says it captures exactly that idea.

Theorem 3.7.4 — Similar Matrices Represent the Same Operator

Let $A, B \in M_n(F)$.

- (a) If V is a vector space of dimension n , $T \in \mathcal{L}(V)$, and $\mathcal{B}, \mathcal{C}$ are bases of V , then $[T]_{\mathcal{B}} \sim [T]_{\mathcal{C}}$.
- (b) Conversely, suppose $A \sim B$. Let V be any vector space of dimension n with a basis $\mathcal{B}$, and let $T \in \mathcal{L}(V)$ be an operator with $[T]_{\mathcal{B}} = A$. Then there is a basis $\mathcal{C}$ of V with $[T]_{\mathcal{C}} = B$.
- (c) Such an operator T in (b) always exists. In particular, $A \sim B$ if and only if A and B are the matrices of the operator $\mathbf{x} \mapsto A\mathbf{x}$ on F^n in the standard basis and in some basis of F^n , respectively.

Proof.

- (a) By [Theorem 3.7.3](#), $[T]_{\mathcal{C}} = P^{-1}[T]_{\mathcal{B}}P$ with $P = [\text{id}]_{\mathcal{C}}^{\mathcal{B}}$, which is invertible by [Theorem 3.7.1](#) (c).
- (b) Suppose $B = P^{-1}AP$ with P invertible. By [Proposition 3.7.2](#), there is a basis $\mathcal{C}$ of V with $[\text{id}]_{\mathcal{C}}^{\mathcal{B}} = P$. By [Theorem 3.7.3](#), $[T]_{\mathcal{C}} = P^{-1}[T]_{\mathcal{B}}P = P^{-1}AP = B$.
- (c) Write $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$. By [Theorem 3.1.5](#), there is a linear map $T: V \rightarrow V$ with $T\mathbf{v}_j = a_{1j}\mathbf{v}_1 + \dots + a_{nj}\mathbf{v}_n$ for each j . The j -th column of $[T]_{\mathcal{B}}$ is $[T\mathbf{v}_j]_{\mathcal{B}}$, which is the j -th column of A ; so $[T]_{\mathcal{B}} = A$. For $V = F^n$ with the standard basis $\mathcal{E}$, the operator $\mathbf{x} \mapsto A\mathbf{x}$ works, since $A\mathbf{e}_j$ is the j -th column of A ([Theorem 0.9.3](#)). The last statement now follows from (a) and (b). ■

This is the promise of Chapter 0 kept: similar matrices describe the same operator in two coordinate systems. So a question of the form “are A and B similar?” is really the question “is there a basis in which the operator $\mathbf{x} \mapsto A\mathbf{x}$ has matrix B ?”. Deciding it is one of the central problems of the book, and it will take until Chapter 9 to answer it in general.

3.7.4 Similarity invariants

If a quantity computed from a matrix does not change under $A \mapsto P^{-1}AP$, then it is really a property of the operator, and we call it a **similarity invariant**. Invariants are the cheap way to prove that two matrices are **not** similar: find one invariant on which they differ. The trace is the first example, and it is the reason the trace will make sense for operators in this chapter’s last section.

Theorem 3.7.5 — Trace Is a Similarity Invariant

If $A, B \in M_n(F)$ are similar, then $\text{tr } A = \text{tr } B$.

Proof. Let $B = P^{-1}AP$ with P invertible. By [Theorem 0.9.6](#) (3), applied to the matrices P^{-1} and AP ,

$$\text{tr } B = \text{tr } (P^{-1}(AP)) = \text{tr } ((AP)P^{-1}) = \text{tr } (A(PP^{-1})) = \text{tr } A,$$

where the third equality is associativity of matrix multiplication. This proves the theo-

rem. ■

The proof never used that B is “similar” in any geometric sense, only that the trace ignores the order of two factors. Several other quantities survive a change of basis, for simpler reasons.

Proposition 3.7.6 — More similarity invariants

Let $A, B \in M_n(F)$ with $B = P^{-1}AP$ for an invertible P .

- (a) $\text{rank } A = \text{rank } B$.
- (b) A is invertible if and only if B is invertible.
- (c) For every polynomial $p \in F[x]$, $p(B) = P^{-1}p(A)P$. In particular, $p(A) = 0$ if and only if $p(B) = 0$; for example, $A^2 = 0$ if and only if $B^2 = 0$.

Proof.

- (a) By [Theorem 2.5.9](#), multiplying by invertible matrices on either side does not change the rank, so $\text{rank}(P^{-1}AP) = \text{rank}(AP) = \text{rank } A$.
- (b) If A is invertible, then $B = P^{-1}AP$ is a product of invertible matrices, hence invertible by [Theorem 0.9.7](#) (part 3). Since $A = PBP^{-1} = (P^{-1})^{-1}BP^{-1}$, the same argument with the roles swapped gives the converse.
- (c) First, $B^k = P^{-1}A^kP$ for every $k \geq 0$, by induction on k : for $k = 0$ both sides are I_n , and if it holds for k , then $B^{k+1} = B^k B = (P^{-1}A^kP)(P^{-1}AP) = P^{-1}A^k(P P^{-1})AP = P^{-1}A^{k+1}P$. Now let $p = c_0 + c_1x + \cdots + c_dx^d$. By [Definition 0.9.14](#) and the distributive and scalar laws of [Theorem 0.9.4](#),

$$p(B) = \sum_{k=0}^d c_k B^k = \sum_{k=0}^d c_k P^{-1}A^kP = P^{-1} \left(\sum_{k=0}^d c_k A^k \right) P = P^{-1}p(A)P.$$

If $p(A) = 0$, then $p(B) = P^{-1}0P = 0$. Conversely, $A = PBP^{-1}$ is of the same form with P^{-1} in place of P , so $p(B) = 0$ implies $p(A) = 0$. ■

For example, $\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ are **not** similar, because their traces are 4 and 3. And $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ is not similar to the zero matrix, because their ranks are 1 and 0 (or because the zero matrix is similar only to itself).

It is tempting to turn this around and decide similarity by comparing a list of invariants. That fails.

Warning. Shared invariants do not imply similarity. Let $J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Then I_2 and J have the same trace 2, the same rank 2, both are invertible, and both satisfy $(X - I_2)^2 = 0$. But they are **not** similar: for every invertible P , $P^{-1}I_2P = I_2 \neq J$. A sharper invariant does separate them: combining (a) and (c) of [Proposition 3.7.6](#), similar matrices A and B have $\text{rank } p(A) = \text{rank } p(B)$ for every polynomial p , and

$\text{rank}(I_2 - I_2) = 0$ while $\text{rank}(J - I_2) = 1$. Which invariants suffice in general is the question Chapter 9 answers.

Quick check. Are $\text{diag}(2, 5)$ and $\text{diag}(5, 2)$ similar? Are $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ and $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ similar?

Solution. The first pair is similar: with $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, which is its own inverse, $P^{-1} \text{diag}(2, 5)P = \text{diag}(5, 2)$. The second pair is not: $2I_2$ is a scalar matrix, so it is similar only to itself, although both matrices have trace 4 and rank 2.

3.7.5 A basis adapted to the operator

The reflection became diagonal once we chose one basis vector the map keeps and one it flips. The same idea works whenever a map is simple on two complementary pieces of the space. For a map of rank 1 on $\mathbb{R}^2$, the natural pieces are the image and the kernel, provided they are two different lines. (They need not be: for $\mathbf{x} \mapsto N\mathbf{x}$ with $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, both are $\text{span}(\mathbf{e}_1)$, and [Exercise \(C3: Every non-zero \$2 \times 2\$ matrix with square zero\)](#) shows what to do instead.)

Example. (Making a matrix diagonal with a well-chosen basis)

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T(x, y) = (2x - y, 2x - y)$. Find a basis $\mathcal{B}$ of $\mathbb{R}^2$ such that $[T]_{\mathcal{B}}$ is diagonal, and check the answer with [Theorem 3.7.3](#).

Solution. In the standard basis $A = [T]_{\mathcal{E}} = \begin{pmatrix} 2 & -1 \\ 2 & -1 \end{pmatrix}$.

Both entries of $T(x, y)$ are equal, so $\text{im } T \subseteq \text{span}((1, 1))$, and $T(1, 1) = (1, 1)$ shows $\text{im } T = \text{span}((1, 1))$. The kernel is $\{(x, y) : y = 2x\} = \text{span}((1, 2))$. So T fixes $(1, 1)$ and kills $(1, 2)$. These two vectors are not multiples of each other, so $\mathcal{B} = ((1, 1), (1, 2))$ is independent, and it is a basis of $\mathbb{R}^2$ by [Theorem 1.6.4](#). Since $T(1, 1) = 1 \cdot (1, 1) + 0 \cdot (1, 2)$ and $T(1, 2) = \mathbf{0}$,

$$[T]_{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Check: $P = [\text{id}]_{\mathcal{B}}^{\mathcal{E}} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ has $ad - bc = 1$, so $P^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$ by [Theorem 0.9.8](#). Then

$$P^{-1}AP = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

As [Theorem 3.7.5](#) and [Proposition 3.7.6](#) require, $\text{tr } A = 1$ and $\text{rank } A = 1$ match the diagonal matrix. Its form also explains a property of A that is not obvious from its entries: $\text{diag}(1, 0)^2 = \text{diag}(1, 0)$, so $A^2 = A$ by [Proposition 3.7.6](#) (c) applied to $p = x^2 - x$.

The basis in this example consisted of vectors that T sends to multiples of themselves. Chapter 8 turns this into a method: such vectors are called eigenvectors, and an operator has a diagonal matrix exactly when there is a basis of them. Operators with $T^2 = T$, like this one, are the

projections of the last section of this chapter.

3.7.6 Exercises

3.7.6.1 A. Check your understanding

Exercise A1.

- Define the change-of-coordinates matrix $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$, and say what its columns are.
- State the change-of-basis formula for an operator $T \in \mathcal{L}(V)$ and bases $\mathcal{B}, \mathcal{B}'$, saying exactly which matrix P is.
- True or false: $[\text{id}]_{\mathcal{B}}^{\mathcal{C}} = [\text{id}]_{\mathcal{C}}^{\mathcal{B}}$ for all bases $\mathcal{B}, \mathcal{C}$ of V . Justify your answer.
- True or false: if $A \sim B$, then $\text{tr } A = \text{tr } B$ and $\text{rank } A = \text{rank } B$. Justify your answer.
- True or false: if $\text{tr } A = \text{tr } B$ and $\text{rank } A = \text{rank } B$, then $A \sim B$. Justify your answer.
- Describe the method for recovering the change-of-basis formula without memorizing it.

Solution.

- For bases $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ and $\mathcal{C}$ of V , $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$ is the matrix of id_V with input basis $\mathcal{B}$ and output basis $\mathcal{C}$ (Definition 3.7.1). Its j -th column is $[\mathbf{v}_j]_{\mathcal{C}}$.
- $[T]_{\mathcal{B}'} = P^{-1}[T]_{\mathcal{B}}P$ with $P = [\text{id}]_{\mathcal{B}'}^{\mathcal{B}}$, whose columns are the $\mathcal{B}$ -coordinates of the vectors of $\mathcal{B}'$ (Theorem 3.7.3).
- False. By Theorem 3.7.1 (c) the two matrices are inverse to each other, and an invertible matrix usually differs from its inverse: for $\mathcal{B} = ((1, 1), (1, -1))$ and the standard basis $\mathcal{E}$ of $\mathbb{R}^2$, $[\text{id}]_{\mathcal{B}}^{\mathcal{E}} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ but $[\text{id}]_{\mathcal{E}}^{\mathcal{B}} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$.
- True, by Theorem 3.7.5 and Proposition 3.7.6 (a).
- False. I_2 and $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ both have trace 2 and rank 2, but I_2 is similar only to itself.
- Draw the change-of-basis square: T along the top and bottom with the old and new bases, identity maps up the left side and down the right side. Since $T = \text{id}_W \circ T \circ \text{id}_V$, the matrix of the bottom arrow is the product of the matrices along the long route, with the first step on the right.

3.7.6.2 B. Practice

Exercise B1: Change-of-coordinates matrices.

- In $\mathbb{R}^2$, let $\mathcal{B} = ((1, 2), (3, 5))$ and $\mathcal{C} = ((1, 1), (1, 2))$. Find $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$. If $[\mathbf{v}]_{\mathcal{B}} = (1, -1)$, find $[\mathbf{v}]_{\mathcal{C}}$, and check your answer by computing $\mathbf{v}$ both ways.
- In $\mathbb{R}[x]_{\leq 2}$, let $\mathcal{B} = (1, x, x^2)$ and $\mathcal{C} = (1, 1+x, 1+x+x^2)$. Find $[\text{id}]_{\mathcal{C}}^{\mathcal{B}}$ and $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$. Hence find the $\mathcal{C}$ -coordinates of $3 - x + 2x^2$.

Solution.

- (a) The columns are the $\mathcal{C}$ -coordinates of $(1, 2)$ and $(3, 5)$. Solving $a(1, 1) + b(1, 2) = (1, 2)$ gives $a + b = 1$, $a + 2b = 2$, so $b = 1$, $a = 0$. Solving $a(1, 1) + b(1, 2) = (3, 5)$ gives $b = 2$, $a = 1$. Hence

$$[\text{id}]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}, \quad [\mathbf{v}]_{\mathcal{C}} = \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

by [Theorem 3.7.1](#) (a). Check: from $\mathcal{B}$, $\mathbf{v} = (1, 2) - (3, 5) = (-2, -3)$; from $\mathcal{C}$, $\mathbf{v} = -(1, 1) - (1, 2) = (-2, -3)$.

- (b) The vectors of $\mathcal{C}$ have $\mathcal{B}$ -coordinates $(1, 0, 0)$, $(1, 1, 0)$, $(1, 1, 1)$, so $[\text{id}]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.

For the other direction, write the vectors of $\mathcal{B}$ in $\mathcal{C}$: $1 = 1$, $x = (1 + x) - 1$, $x^2 = (1 + x + x^2) - (1 + x)$. So

$$[\text{id}]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix},$$

and multiplying the two matrices gives I_3 , as [Theorem 3.7.1](#) (c) requires. Hence

$$[3 - x + 2x^2]_{\mathcal{C}} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ -3 \\ 2 \end{pmatrix}.$$

Check: $4 - 3(1 + x) + 2(1 + x + x^2) = 3 - x + 2x^2$.

Exercise B2: A matrix in a new basis, two ways.

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T(x, y) = (4x - 2y, x + y)$, and $\mathcal{B} = ((1, 1), (2, 1))$.

- (a) Compute $[T]_{\mathcal{B}}$ directly from [Definition 3.6.1](#).
 (b) Compute it again with [Theorem 3.7.3](#), starting from the standard matrix.

Solution.

- (a) $T(1, 1) = (2, 2) = 2(1, 1) + 0(2, 1)$ and $T(2, 1) = (6, 3) = 0(1, 1) + 3(2, 1)$. Hence $[T]_{\mathcal{B}} = \text{diag}(2, 3)$.

- (b) $A = [T]_{\mathcal{E}} = \begin{pmatrix} 4 & -2 \\ 1 & 1 \end{pmatrix}$ and $P = [\text{id}]_{\mathcal{B}}^{\mathcal{E}} = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$. Here $ad - bc = -1$, so by [Theorem 0.9.8](#) $P^{-1} = \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix}$. Then

$$P^{-1}AP = \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 4 & -2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 4 \\ 3 & -3 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix},$$

in agreement with (a).

Exercise B3: Similar or not?.

Determine which of the following pairs of real matrices are similar. Justify your answer, either with an invariant or with an explicit invertible P .

- (a) $\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$.
- (b) $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.
- (c) $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$.
- (d) $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ and $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$.
- (e) $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$.

Solution.

- (a) Not similar: the traces are 4 and 3, contradicting [Theorem 3.7.5](#).
- (b) Not similar: the ranks are 1 and 0, contradicting [Proposition 3.7.6](#) (a).
- (c) Similar. Let A be the first matrix. We look for a basis in which $\mathbf{x} \mapsto A\mathbf{x}$ is diagonal:

$A\mathbf{e}_1 = \mathbf{e}_1$, and $A(1,1) = (2,2) = 2(1,1)$. With $P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $P^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$ and

$$P^{-1}AP = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

- (d) Not similar: $2I_2$ is a scalar matrix, so $P^{-1}(2I_2)P = 2I_2$ for every invertible P , and the second matrix is not $2I_2$. The traces and ranks agree, so invariants alone would not have decided this.
- (e) Similar. Let A be the first matrix. $A(1,1) = (2,2)$ and $A(1,-1) = (0,0)$. With $P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ and $P^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$,

$$P^{-1}AP = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 2 & 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix},$$

where $AP = \begin{pmatrix} 2 & 0 \\ 2 & 0 \end{pmatrix}$ has columns $A(1,1)$ and $A(1,-1)$.

3.7.6.3 C. Going deeper**Exercise C1: Matrices similar only to themselves.**

Let $n \geq 1$ and $A \in M_n(F)$.

- (a) Prove that A is similar only to itself if and only if $A = cI_n$ for some $c \in F$.
- (b) Show that $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $N^T = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ are similar, and explain the answer in

terms of the operator $\mathbf{x} \mapsto N\mathbf{x}$ and a reordered basis.

Hint: for (a), use the invertible matrices $I_n + E_{ij}$ with $i \neq j$.

Solution.

(a) ($\Leftarrow$) If $A = cI_n$, then $P^{-1}AP = cP^{-1}P = cI_n = A$ for every invertible P , so the only matrix similar to A is A .

($\Rightarrow$) Suppose $P^{-1}AP = A$ for every invertible P , that is, $AP = PA$. If $n = 1$, $A = (a_{11}) = a_{11}I_1$. Let $n \geq 2$, and fix $i \neq j$. The matrix $I_n + E_{ij}$ is invertible, with inverse $I_n - E_{ij}$, since $E_{ij}E_{ij} = 0$ for $i \neq j$. So $A(I_n + E_{ij}) = (I_n + E_{ij})A$, which gives $AE_{ij} = E_{ij}A$. Compare the entries in position (k, l) : by [Definition 0.9.9](#),

$$(AE_{ij})_{kl} = \begin{cases} a_{ki} & l = j \\ 0 & l \neq j \end{cases}, \quad (E_{ij}A)_{kl} = \begin{cases} a_{jl} & k = i \\ 0 & k \neq i. \end{cases}$$

Taking $l = j$ and $k \neq i$ gives $a_{ki} = 0$. Taking $k = i$ and $l = j$ gives $a_{ii} = a_{jj}$. Since $i \neq j$ were arbitrary (and every index i has some $j \neq i$ because $n \geq 2$), all off-diagonal entries of A vanish and all diagonal entries are equal to $c = a_{11}$. Hence $A = cI_n$.

(b) Let $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, so $P^{-1} = P$. Then $NP = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $P^{-1}NP = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = N^T$. In terms of operators: $T\mathbf{x} = N\mathbf{x}$ has $T\mathbf{e}_1 = \mathbf{0}$ and $T\mathbf{e}_2 = \mathbf{e}_1$. In the reordered basis $\mathcal{B} = (\mathbf{e}_2, \mathbf{e}_1)$, the first basis vector goes to the second and the second goes to $\mathbf{0}$, so $[T]_{\mathcal{B}} = N^T$. The same operator has both matrices, as [Theorem 3.7.4](#) predicts.

Exercise C2: AB and BA .

Let $A, B \in M_n(F)$.

- (a) Prove that if A is invertible, then $AB \sim BA$.
- (b) Show that (a) fails without the invertibility hypothesis, even though $\text{tr}(AB) = \text{tr}(BA)$ always holds.

Solution.

(a) Since A is invertible, $A^{-1}(AB)A = (A^{-1}A)(BA) = BA$ by associativity. With $P = A$ this is $BA = P^{-1}(AB)P$, so $AB \sim BA$.

(b) Take $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Then $AB = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, of rank 1, and $BA = 0$, of rank 0. By [Proposition 3.7.6](#) (a), $AB \not\sim BA$. Both traces are 0, in agreement with [Theorem 0.9.6](#); the trace simply cannot detect the difference.

Exercise C3: Every non-zero 2×2 matrix with square zero.

Let $N \in M_2(F)$ with $N \neq 0$ and $N^2 = 0$, and let $T: F^2 \rightarrow F^2$, $T\mathbf{x} = N\mathbf{x}$.

- (a) Explain why there is $\mathbf{v} \in F^2$ with $N\mathbf{v} \neq \mathbf{0}$, and show that $\mathcal{B} = (N\mathbf{v}, \mathbf{v})$ is a basis of F^2 .

- (b) Compute $[T]_{\mathcal{B}}$, and deduce that $N \sim \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$.
- (c) Hence show that $\begin{pmatrix} 2 & -4 \\ 1 & -2 \end{pmatrix} \sim \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ in $M_2(\mathbb{R})$, with an explicit P .

Solution.

- (a) Since $N \neq 0$, some column $N\mathbf{e}_j$ is non-zero ([Theorem 0.9.3](#)); take $\mathbf{v} = \mathbf{e}_j$. Let $aN\mathbf{v} + b\mathbf{v} = \mathbf{0}$. Multiplying by N and using $N^2 = 0$ gives $bN\mathbf{v} = \mathbf{0}$, so $b = 0$ because $N\mathbf{v} \neq \mathbf{0}$ ([Theorem 1.1.9](#)). Then $aN\mathbf{v} = \mathbf{0}$ forces $a = 0$ for the same reason. So $\mathcal{B}$ is linearly independent of length $2 = \dim F^2$, hence a basis by [Theorem 1.6.4](#).
- (b) $T(N\mathbf{v}) = N^2\mathbf{v} = \mathbf{0}$ and $T\mathbf{v} = N\mathbf{v} = 1 \cdot N\mathbf{v} + 0 \cdot \mathbf{v}$. The columns of $[T]_{\mathcal{B}}$ are therefore $(0, 0)$ and $(1, 0)$, so $[T]_{\mathcal{B}} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Also $[T]_{\mathcal{E}} = N$. By [Theorem 3.7.4](#)
- (a), $N \sim \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$.
- (c) Let $N = \begin{pmatrix} 2 & -4 \\ 1 & -2 \end{pmatrix}$. Then $N^2 = \begin{pmatrix} 4-4 & -8+8 \\ 2-2 & -4+4 \end{pmatrix} = 0$ and $N \neq 0$. Take $\mathbf{v} = \mathbf{e}_1$, so $N\mathbf{v} = (2, 1)$, and $\mathcal{B} = ((2, 1), (1, 0))$. By [Theorem 3.7.3](#), $P = [\text{id}]_{\mathcal{B}}^{\mathcal{E}} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$ satisfies $P^{-1}NP = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Check: $P^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix}$ by [Theorem 0.9.8](#) (here $ad - bc = -1$), $NP = \begin{pmatrix} 0 & 2 \\ 0 & 1 \end{pmatrix}$, and $P^{-1}NP = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$.

3.8 Rank, Equivalence and Factorizations

Similarity compares two matrices of one operator, where the same basis is used for input and output, and the last section showed that deciding it is hard. This section asks what happens when the two bases may be chosen independently, as they may for any map $T: V \rightarrow W$. The answer is strikingly simple: the rank is then the only thing that matters. On the way we meet the column–row factorization, which writes every matrix as a product through a space of dimension equal to its rank, and we collect the basic inequalities for the rank of a composition.

3.8.1 The column–row factorization

In Chapter 2 we computed the rank of a matrix by row reduction, and the reduction gave more than a number. It produced a basis of the column space, the pivot columns of A ([Theorem 2.5.2](#)), and it expressed every other column in terms of them, with coefficients read off the reduced row echelon form. Recording both pieces of information as matrices gives a factorization.

Theorem 3.8.1 — Column–Row Factorization

Let $A \in M_{m \times n}(F)$ have rank $r \geq 1$, with pivot columns $j_1 < \cdots < j_r$. Let $C \in M_{m \times r}(F)$ be the matrix whose columns are the pivot columns $\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_r}$ of A , and let

$R \in M_{r \times n}(F)$ consist of the first r rows of the reduced row echelon form of A , that is, its non-zero rows. Then

$$A = CR.$$

Moreover, the columns of C are linearly independent and the rows of R are linearly independent.

Idea. Column l of CR is C times column l of R , that is, a combination of the pivot columns of A with the entries of column l of the RREF as coefficients. For a pivot column the coefficients are a standard basis vector, and for a free column they are exactly the coefficients that Theorem 2.5.2 (b) provides. So the factorization is that theorem written as a matrix identity.

Proof. Let E be the RREF of A , with entries e_{kl} . By Theorem 2.5.5, E has exactly r pivots, so its non-zero rows are rows $1, \dots, r$. Fix a column index l . By Theorem 0.9.3, column l of CR is

$$C \begin{pmatrix} e_{1l} \\ \vdots \\ e_{rl} \end{pmatrix} = e_{1l}\mathbf{a}_{j_1} + \cdots + e_{rl}\mathbf{a}_{j_r}.$$

Case 1: $l = j_i$ is a pivot column. By Lemma 2.3.3 (a), column j_i of E is $\mathbf{e}_i$, so the combination is $\mathbf{a}_{j_i} = \mathbf{a}_l$.

Case 2: l is a free column. Let t be the number of pivot columns to the left of l . For $k > t$, the leading entry of row k of E lies in column $j_k > l$, so $e_{kl} = 0$. The combination is therefore $e_{1l}\mathbf{a}_{j_1} + \cdots + e_{tl}\mathbf{a}_{j_t}$, which equals $\mathbf{a}_l$ by Theorem 2.5.2 (b).

In both cases column l of CR is column l of A ; hence $A = CR$. The columns of C form a basis of $\text{col}(A)$ by Theorem 2.5.2 (a), so they are independent. The rows of R are the non-zero rows of E , which form a basis of $\text{row}(A)$ by Theorem 2.5.3, so they are independent. This proves the theorem. ■

In words: every matrix of rank r is a “tall” matrix with independent columns times a “wide” matrix with independent rows, and the middle size is exactly r . The factorization cannot be squeezed further.

Proposition 3.8.2 — Rank as the size of the thinnest factorization

Let $A \in M_{m \times n}(F)$ be non-zero. Then $\text{rank } A$ is the **smallest** integer $s \geq 1$ such that $A = BD$ for some $B \in M_{m \times s}(F)$ and $D \in M_{s \times n}(F)$.

Proof. Let $r = \text{rank } A$; then $r \geq 1$ since $A \neq 0$. By Theorem 3.8.1, a factorization with $s = r$ exists. Conversely, suppose $A = BD$ with $B \in M_{m \times s}(F)$. By Theorem 2.5.9, $r = \text{rank}(BD) \leq \text{rank } B$, and $\text{rank } B \leq s$ because $\text{col}(B)$ is spanned by s columns (Theorem 1.6.3). Hence $r \leq s$. ■

This gives a second view of “row rank equals column rank”. Let $A \neq 0$, and let B be the

matrix whose $s = \dim \operatorname{col}(A)$ columns form a basis of $\operatorname{col}(A)$. Each column $\mathbf{a}_j$ of A is $B\mathbf{d}_j$ for its coordinate column $\mathbf{d}_j$ (Theorem 0.9.3), so $A = BD$ with $D = (\mathbf{d}_1 \ \cdots \ \mathbf{d}_n)$. Then every row of A is a row of B times D , a combination of the s rows of D (Theorem 0.9.2 (3) and the row form of Theorem 0.9.3), so $\dim \operatorname{row}(A) \leq s = \dim \operatorname{col}(A)$. Applying the same argument to A^T gives the reverse inequality. In the language of maps, the factorization says that $\mathbf{x} \mapsto A\mathbf{x}$ can be done in two steps, $F^n \rightarrow F^r \rightarrow F^m$: first a surjection onto a space of dimension r , then an injection.

Example. (A column–row factorization)

Find the column–row factorization of $A = \begin{pmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 1 & 7 \end{pmatrix} \in M_{3 \times 4}(\mathbb{R})$.

Solution. Row reduce. Subtracting 2 times row 1 from row 2 and 3 times row 1 from row 3 gives rows $(1, 2, 0, 3)$, $(0, 0, 1, -2)$, $(0, 0, 1, -2)$. Subtracting row 2 from row 3 gives the RREF

$$\begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The pivot columns are columns 1 and 3, so $r = 2$, and by Theorem 3.8.1

$$A = CR = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{pmatrix}.$$

Check by rows: row 2 of CR is $2(1, 2, 0, 3) + (0, 0, 1, -2) = (2, 4, 1, 4)$, and row 3 is $3(1, 2, 0, 3) + (0, 0, 1, -2) = (3, 6, 1, 7)$. The last column of R says $\mathbf{a}_4 = 3\mathbf{a}_1 - 2\mathbf{a}_3$, and indeed $3(1, 2, 3) - 2(0, 1, 1) = (3, 4, 7)$.

3.8.2 Equivalent matrices

Go back to the change-of-basis square for a map $T: V \rightarrow W$ (Theorem 3.7.3):

$$[T]_{B'}^{C'} = [\operatorname{id}]_C^{C'} [T]_B^C [\operatorname{id}]_{B'}^B.$$

When $V \neq W$, nothing ties the basis of V to the basis of W . So the two outer matrices are **unrelated** invertible matrices, of sizes $m \times m$ and $n \times n$. Any two matrices of the same map are related in this way, and this recurring shape deserves a name.

Two matrices are equivalent when they describe the same linear map, with the input basis and the output basis each changed freely.

Definition 3.8.1 — Equivalent matrices

Let $A, B \in M_{m \times n}(F)$. We say A is **equivalent** to B if there **exist** invertible matrices $Q \in M_m(F)$ and $P \in M_n(F)$ such that

$$B = QAP.$$

In words: we may multiply on the left by **any** invertible matrix and on the right by **any** invertible matrix, and the two need not be related. Both sizes are forced: Q acts on the m rows and P on the n columns. This is an equivalence relation on $M_{m \times n}(F)$: $A = I_m A I_n$; if $B = QAP$ then $A = Q^{-1}BP^{-1}$; and if also $C = Q'BP'$, then $C = (Q'Q)A(P'P)$, where products of invertible matrices are invertible by [Theorem 0.9.7](#) (part 3).

Examples.

- **Row equivalence.** If B is obtained from A by row operations, then $B = EA$ with E invertible ([Theorem 2.4.7](#)), so A and B are equivalent with $Q = E$ and $P = I_n$. Column operations correspond to multiplying on the right, and they are allowed too.
- **Similarity.** If $B = P^{-1}AP$, then A and B are equivalent, with $Q = P^{-1}$.
- **The identity and a swap.** I_2 and $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ are equivalent, since $S = SI_2I_2$. They are not similar, because I_2 is similar only to itself.
- **Degenerate case.** The zero matrix $0_{m \times n}$ is equivalent only to itself, since $Q0P = 0$. It is the matrix of the zero map in every pair of bases.

Non-example by minimal change. Keep $B = QAP$ but drop the requirement that Q be invertible. Then every matrix could be sent to the zero matrix with $Q = 0$, and the relation would not even be symmetric: $0 = 0 \cdot I_2 \cdot I_2$, but $I_2 \neq Q0P$ for any Q, P . The clause “invertible” is what makes QAP a change of basis rather than an arbitrary composition.

Warning. **Equivalent is much weaker than similar.** The matrices I_2 and $J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ are equivalent, since $J = JI_2I_2$ and J is invertible, but they are not similar (see the warning in the previous section). For square matrices, “equivalent” allows a different change of basis on each side, and that freedom destroys almost all the structure an operator has.

How much structure survives? The rank does, since multiplying by invertible matrices does not change it ([Theorem 2.5.9](#)). The theorem below says that nothing else survives. The target is the simplest matrix of each rank.

For $0 \leq r \leq \min(m, n)$, let $N_r \in M_{m \times n}(F)$ be the block matrix

$$N_r := \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix},$$

with 1 in positions $(1, 1), \dots, (r, r)$ and 0 everywhere else. Its first r columns are $\mathbf{e}_1, \dots, \mathbf{e}_r \in F^m$, and its remaining columns are zero.

Theorem 3.8.3 — Rank Normal Form (a) Let V and W be finite-dimensional with $\dim V = n$ and $\dim W = m$, and let $T: V \rightarrow W$ be linear of rank r . Then there are bases $\mathcal{B}$ of V and $\mathcal{C}$ of W such that $[T]_{\mathcal{B}}^{\mathcal{C}} = N_r$.
 (b) Every $A \in M_{m \times n}(F)$ is equivalent to N_r , where $r = \text{rank } A$.
 (c) Two matrices $A, B \in M_{m \times n}(F)$ are equivalent if and only if $\text{rank } A = \text{rank } B$.

Idea. Work backwards from the goal. $[T]_{\mathcal{B}}^{\mathcal{C}} = N_r$ says: T sends the first r vectors of $\mathcal{B}$ to the first r vectors of $\mathcal{C}$, and sends the last $n - r$ vectors of $\mathcal{B}$ to $\mathbf{0}$. So ① the last $n - r$ vectors of $\mathcal{B}$ should be a basis of $\ker T$, and by Rank–Nullity that is exactly the right number; ② the first r vectors of $\mathcal{B}$ extend it to a basis of V ; ③ their images are

forced to be the first r vectors of $\mathcal{C}$, and we must show they are independent, which follows by counting; ④ extend them to a basis of W . These are the bases of the proof of Rank–Nullity, one on each side.

Proof.

- (a) By the Rank–Nullity Theorem (Theorem 3.3.1), $\dim \ker T = n - r$. Let $(\mathbf{u}_{r+1}, \dots, \mathbf{u}_n)$ be a basis of $\ker T$, a subspace of V by Theorem 3.2.1. By the Basis Extension Theorem (Theorem 1.6.5) there are $\mathbf{u}_1, \dots, \mathbf{u}_r \in V$ such that $\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_r, \mathbf{u}_{r+1}, \dots, \mathbf{u}_n)$ is a basis of V ; the new vectors may be listed first, since reordering a basis gives a basis. Put $\mathbf{w}_i = T\mathbf{u}_i$ for $i = 1, \dots, r$.

By Theorem 3.2.4, $\operatorname{im} T = \operatorname{span}(T\mathbf{u}_1, \dots, T\mathbf{u}_n)$. Since $T\mathbf{u}_j = \mathbf{0}$ for $j > r$, removing these vectors does not change the span, so $(\mathbf{w}_1, \dots, \mathbf{w}_r)$ spans $\operatorname{im} T$. This list has length $r = \dim \operatorname{im} T$, so it is a basis of $\operatorname{im} T$ by Theorem 1.6.4 (b); in particular it is linearly independent in W . By the Basis Extension Theorem there are $\mathbf{w}_{r+1}, \dots, \mathbf{w}_m \in W$ such that $\mathcal{C} = (\mathbf{w}_1, \dots, \mathbf{w}_m)$ is a basis of W .

Now read off the columns of $[T]_{\mathcal{B}}^{\mathcal{C}}$ (Definition 3.6.1). For $i \leq r$, $T\mathbf{u}_i = \mathbf{w}_i$, whose $\mathcal{C}$ -coordinate vector is $\mathbf{e}_i$. For $j > r$, $T\mathbf{u}_j = \mathbf{0}$, whose coordinate vector is $\mathbf{0}$. These are the columns of N_r .

- (b) Let $T: F^n \rightarrow F^m$, $T\mathbf{x} = A\mathbf{x}$, and let $\mathcal{E}_n, \mathcal{E}_m$ be the standard bases. The j -th column of $[T]_{\mathcal{E}_n}^{\mathcal{E}_m}$ is $[A\mathbf{e}_j]_{\mathcal{E}_m} = A\mathbf{e}_j$, the j -th column of A (Theorem 0.9.3), so $[T]_{\mathcal{E}_n}^{\mathcal{E}_m} = A$. By Theorem 3.6.5, $\operatorname{rank} T = \operatorname{rank} A = r$. Take $\mathcal{B}, \mathcal{C}$ as in (a). By Theorem 3.7.3,

$$N_r = [T]_{\mathcal{B}}^{\mathcal{C}} = [\operatorname{id}]_{\mathcal{E}_m}^{\mathcal{C}} A [\operatorname{id}]_{\mathcal{B}}^{\mathcal{E}_n},$$

and both outer matrices are invertible by Theorem 3.7.1 (c). Hence A is equivalent to N_r .

- (c) ($\Rightarrow$) If $B = QAP$ with Q, P invertible, then $\operatorname{rank} B = \operatorname{rank} A$ by Theorem 2.5.9. ($\Leftarrow$) If $\operatorname{rank} A = \operatorname{rank} B = r$, then by (b) both A and B are equivalent to N_r . Since equivalence is symmetric and transitive, A is equivalent to B . This proves the theorem. ■

So the equivalence classes of $M_{m \times n}(F)$ are exactly the sets of matrices of rank r , for $r = 0, 1, \dots, \min(m, n)$: there are $\min(m, n) + 1$ of them, whatever the field. Every linear map between finite-dimensional spaces looks like “keep r coordinates and delete the rest” once the bases on both sides are chosen well. **The right pair of bases makes any single map trivial.**

Remark. Contrast this with operators. For $T \in \mathcal{L}(V)$ we must use **one** basis for input and output, and the relation becomes similarity. There the rank is far from the whole story: I_2 and $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ have the same rank but are not similar. Classifying operators up to a change of one basis is a much harder problem, answered in Chapter 9 by the Jordan and rational canonical forms.

The proof is also an algorithm. To find Q and P for a concrete matrix, compute a basis of $\operatorname{nul}(A)$, extend it, and take images.

Example. (Bringing a 2×3 matrix to normal form)

Let $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \end{pmatrix} \in M_{2 \times 3}(\mathbb{R})$. Find invertible $Q \in M_2(\mathbb{R})$ and $P \in M_3(\mathbb{R})$ with $QAP = N_r$.

Solution. Subtracting 2 times row 1 from row 2 gives $\begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix}$, so $r = \text{rank } A = 2$, and the solutions of $A\mathbf{x} = \mathbf{0}$ satisfy $x_3 = 0$, $x_1 = -2x_2$. Hence $\text{nul}(A) = \text{span}((-2, 1, 0))$, as Rank–Nullity predicts ($3 - 2 = 1$).

Following the proof, let $\mathbf{u}_3 = (-2, 1, 0)$, and extend with $\mathbf{u}_1 = \mathbf{e}_1$, $\mathbf{u}_2 = \mathbf{e}_3$. The list $(\mathbf{e}_1, \mathbf{e}_3, \mathbf{u}_3)$ is independent: in $a\mathbf{e}_1 + b\mathbf{e}_3 + c(-2, 1, 0) = (a - 2c, c, b) = \mathbf{0}$, the second and third entries give $c = b = 0$, and then $a = 0$. So it is a basis $\mathcal{B}$ of $\mathbb{R}^3$. The images are $\mathbf{w}_1 = A\mathbf{e}_1 = (1, 2)$ and $\mathbf{w}_2 = A\mathbf{e}_3 = (1, 3)$, which already form a basis $\mathcal{C}$ of $\mathbb{R}^2$ since $r = m = 2$. Therefore

$$P = [\text{id}]_{\mathcal{B}}^{\mathcal{E}_3} = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad Q = [\text{id}]_{\mathcal{E}_2}^{\mathcal{C}} = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}^{-1} = \begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix},$$

using [Theorem 3.7.1](#) (c) and [Theorem 0.9.8](#). Check: $AP = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 3 & 0 \end{pmatrix}$ (the columns are $\mathbf{w}_1, \mathbf{w}_2, \mathbf{0}$), and

$$QAP = \begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 2 & 3 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = N_2.$$

The pair (Q, P) is far from unique; any choice of extension gives another.

Quick check. How many equivalence classes does $M_{3 \times 2}(F)$ have, and which is the class of $\begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 0 \end{pmatrix}$?

Solution. Ranks in $M_{3 \times 2}(F)$ range over $0, 1, 2$, so there are $\min(3, 2) + 1 = 3$ classes by [Theorem 3.8.3](#) (c). The given matrix is non-zero and its second column is twice the first, so its rank is 1, and it lies in the class of N_1 : all matrices of rank 1.

3.8.3 Rank inequalities for compositions

The rank of a composition cannot exceed the rank of either factor, and it cannot fall too far either. Both facts come from the same observation: ST is S applied to the subspace $\text{im } T$. For a subspace $X \subseteq W$ and a linear map $S: W \rightarrow U$, the **restriction** $S|_X: X \rightarrow U$, $\mathbf{x} \mapsto S\mathbf{x}$, is linear, since its values are those of S . Its kernel is $\ker S \cap X$.

Theorem 3.8.4 — Rank of a Composition

Let $T: V \rightarrow W$ and $S: W \rightarrow U$ be linear, with V and W finite-dimensional. Then

$$\text{rank}(ST) \leq \min(\text{rank } S, \text{rank } T).$$

Proof. Consider the restriction $S' = S|_{\text{im } T}: \text{im } T \rightarrow U$. For $\mathbf{v} \in V$, $(ST)(\mathbf{v}) = S(T\mathbf{v}) = S'(T\mathbf{v})$, so $\text{im}(ST) \subseteq \text{im } S'$; conversely every $S'\mathbf{w}$ with $\mathbf{w} = T\mathbf{v} \in \text{im } T$ equals $(ST)(\mathbf{v})$. Hence $\text{im}(ST) = \text{im } S'$. Since $\text{im } T$ is finite-dimensional (a subspace of W , [Theorem 1.6.7](#)), Rank–Nullity ([Theorem 3.3.1](#)) applied to S' gives

$$\text{rank}(ST) = \dim \text{im } S' = \dim \text{im } T - \dim \ker S' \leq \text{rank } T.$$

Also $\text{im}(ST) = \text{im } S' \subseteq \text{im } S$, and $\text{im } S$ is finite-dimensional by Rank–Nullity for S , so $\text{rank}(ST) \leq \text{rank } S$ by [Theorem 1.6.7](#). This proves the theorem. ■

The lower bound uses the same restriction, but now we control its kernel instead of ignoring it.

Theorem 3.8.5 — Sylvester’s Rank Inequality

Let $T: V \rightarrow W$ and $S: W \rightarrow U$ be linear, with V and W finite-dimensional. Then

$$\text{rank}(ST) \geq \text{rank } S + \text{rank } T - \dim W.$$

In matrix form: for $A \in M_{m \times n}(F)$ and $B \in M_{n \times p}(F)$, $\text{rank}(AB) \geq \text{rank } A + \text{rank } B - n$.

Idea. Of the $\text{rank } T$ dimensions that reach W , the map S can destroy only those lying in $\ker S$, and $\ker S$ has dimension $\dim W - \text{rank } S$. Rank–Nullity for the restriction $S|_{\text{im } T}$ turns this sentence into an inequality.

Proof. Let $S' = S|_{\text{im } T}$, so that $\text{im } S' = \text{im}(ST)$ as in the proof of [Theorem 3.8.4](#), and $\ker S' = \ker S \cap \text{im } T \subseteq \ker S$. By Rank–Nullity for S' and then for S ,

$$\text{rank } T = \text{rank}(ST) + \dim(\ker S \cap \text{im } T) \leq \text{rank}(ST) + \dim \ker S = \text{rank}(ST) + \dim W - \text{rank } S,$$

where the inequality is [Theorem 1.6.7](#). Rearranging gives the claim. For matrices, apply it to $S\mathbf{y} = A\mathbf{y}$ on $W = F^n$ and $T\mathbf{x} = B\mathbf{x}$ on $V = F^p$: the matrix of ST in standard bases is AB by [Theorem 3.6.2](#), and ranks of maps and matrices agree by [Theorem 3.6.5](#). ■

If you worked [Exercise \(C1: A Lower Bound for the Rank of a Composite\)](#), you have seen this proof already; we record the result here as a theorem so that later arguments can cite it. A typical use: if $AB = 0$ with $A \in M_{m \times n}(F)$ and $B \in M_{n \times p}(F)$, then $0 \geq \text{rank } A + \text{rank } B - n$, that is,

$$\text{rank } A + \text{rank } B \leq n.$$

For instance, there are no $A, B \in M_3(\mathbb{R})$ of rank 2 with $AB = 0$.

Warning. The rank of a product depends on the order of the factors. For $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $AB = B$ has rank 1 but $BA = 0$ has rank 0. [Theorem 2.5.9](#)

and Sylvester's inequality bound $\text{rank}(AB)$ from both sides; they do not determine it, and they say nothing that would make $\text{rank}(AB) = \text{rank}(BA)$.

3.8.4 Exercises

3.8.4.1 A. Check your understanding

Exercise A1.

- (a) Define what it means for $A, B \in M_{m \times n}(F)$ to be equivalent.
- (b) State the Rank Normal Form theorem for a matrix $A \in M_{m \times n}(F)$.
- (c) True or false: equivalent square matrices are similar. Justify your answer.
- (d) True or false: $\text{rank}(AB) = \text{rank}(BA)$ for all $A, B \in M_n(F)$. Justify your answer.
- (e) How many equivalence classes does $M_{2 \times 5}(\mathbb{F}_7)$ have?
- (f) In the proof of the normal form for $T: V \rightarrow W$, which vectors form the end of the basis of V , and which vectors form the start of the basis of W ?

Solution.

- (a) There are invertible $Q \in M_m(F)$ and $P \in M_n(F)$ with $B = QAP$ (Definition 3.8.1).
- (b) If $r = \text{rank } A$, there are invertible Q, P with $QAP = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$; and two $m \times n$ matrices are equivalent if and only if they have the same rank (Theorem 3.8.3).
- (c) False. I_2 and $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ both have rank 2, so they are equivalent, but I_2 is similar only to itself.
- (d) False. With $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $\text{rank}(AB) = 1$ and $\text{rank}(BA) = 0$.
- (e) The possible ranks are 0, 1, 2, so there are 3 classes. The field plays no role.
- (f) The last $n - r$ vectors of the basis of V form a basis of $\ker T$. The first r vectors of the basis of W are the images of the first r basis vectors of V , and they form a basis of $\text{im } T$.

3.8.4.2 B. Practice

Exercise B1: A column–row factorization.

Find the column–row factorization of

$$A = \begin{pmatrix} 1 & -1 & 2 & 0 \\ 2 & -2 & 5 & 1 \\ 1 & -1 & 3 & 1 \end{pmatrix} \in M_{3 \times 4}(\mathbb{R}).$$

Hence write the second and fourth columns of A as combinations of its pivot columns.

Solution. Subtract 2 times row 1 from row 2 and row 1 from row 3: the rows become $(1, -1, 2, 0)$, $(0, 0, 1, 1)$, $(0, 0, 1, 1)$. Subtract row 2 from row 3, and then 2 times row 2 from row 1:

$$\text{RREF} = \begin{pmatrix} 1 & -1 & 0 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The pivot columns are 1 and 3, so $\text{rank } A = 2$, and by [Theorem 3.8.1](#)

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 5 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 & -2 \\ 0 & 0 & 1 & 1 \end{pmatrix}.$$

Check by rows: $2(1, -1, 0, -2) + 5(0, 0, 1, 1) = (2, -2, 5, 1)$ and $(1, -1, 0, -2) + 3(0, 0, 1, 1) = (1, -1, 3, 1)$; row 1 is $(1, -1, 0, -2) + 2(0, 0, 1, 1)$. Reading columns 2 and 4 of R : $\mathbf{a}_2 = -\mathbf{a}_1$ and $\mathbf{a}_4 = -2\mathbf{a}_1 + \mathbf{a}_3$. Indeed $-2(1, 2, 1) + (2, 5, 3) = (0, 1, 1) = \mathbf{a}_4$.

Exercise B2: Normal form of a rank-one matrix.

Let $A = \begin{pmatrix} 1 & -1 & 2 \\ -2 & 2 & -4 \end{pmatrix} \in M_{2 \times 3}(\mathbb{R})$. Find invertible $Q \in M_2(\mathbb{R})$ and $P \in M_3(\mathbb{R})$ such that $QAP = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, following the proof of [Theorem 3.8.3](#).

Solution. Row 2 is -2 times row 1, and $A \neq 0$, so $\text{rank } A = 1$. The null space is $\{x_1 - x_2 + 2x_3 = 0\}$; setting the free variables x_2, x_3 to $(1, 0)$ and $(0, 1)$ gives the basis $\mathbf{u}_2 = (1, 1, 0)$, $\mathbf{u}_3 = (-2, 0, 1)$. Extend with $\mathbf{u}_1 = \mathbf{e}_1$: in $a\mathbf{e}_1 + b\mathbf{u}_2 + c\mathbf{u}_3 = (a+b-2c, b, c) = \mathbf{0}$, the last two entries give $b = c = 0$ and then $a = 0$, so $\mathcal{B} = (\mathbf{e}_1, \mathbf{u}_2, \mathbf{u}_3)$ is a basis of $\mathbb{R}^3$. Its first image is $\mathbf{w}_1 = A\mathbf{e}_1 = (1, -2)$, which we extend to the basis $\mathcal{C} = ((1, -2), \mathbf{e}_2)$ of $\mathbb{R}^2$ (independent, since $a(1, -2) + b(0, 1) = (a, -2a + b) = \mathbf{0}$ forces $a = b = 0$). Then

$$P = \begin{pmatrix} 1 & 1 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad Q = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}.$$

Check: AP has columns $A\mathbf{e}_1 = (1, -2)$, $A\mathbf{u}_2 = \mathbf{0}$, $A\mathbf{u}_3 = \mathbf{0}$, and $Q(1, -2) = (1, 0)$. So $QAP = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. Both Q and P are invertible by [Theorem 3.7.1](#) (c).

Exercise B3: Rank-one pieces.

Let $A \in M_{m \times n}(F)$.

- Use [Theorem 3.8.1](#) to show that if $\text{rank } A = 1$, then $A = \mathbf{xy}^T$ for some non-zero $\mathbf{x} \in F^m$ and $\mathbf{y} \in F^n$.
- Show that if $\text{rank } A = r \geq 1$, then A is a sum of r matrices of rank 1, and that it is not a sum of fewer than r matrices of rank 1.
- Write $\begin{pmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 1 & 7 \end{pmatrix}$ as a sum of two rank-one matrices.

Solution.

- (a) With $r = 1$, $C \in M_{m \times 1}(F)$ is a single column $\mathbf{x}$, and $R \in M_{1 \times n}(F)$ is a single row, which we write as $\mathbf{y}^\top$ with $\mathbf{y} \in F^n$. So $A = \mathbf{x}\mathbf{y}^\top$. Both are non-zero: the columns of C and the rows of R are linearly independent by [Theorem 3.8.1](#), and a list containing a zero vector is dependent. (This recovers one direction of [Exercise \(C2: Matrices of rank one\)](#).)
- (b) Write $A = CR$ with columns $\mathbf{c}_1, \dots, \mathbf{c}_r$ of C and rows $\mathbf{r}_1, \dots, \mathbf{r}_r$ of R , and let c_{ik} , r_{kl} be the entries of C , R . The $m \times n$ matrix $\mathbf{c}_k \mathbf{r}_k$ has (i, l) -entry $c_{ik} r_{kl}$, so by [Theorem 0.9.2](#) (1),

$$(CR)_{il} = \sum_{k=1}^r c_{ik} r_{kl} = \sum_{k=1}^r (\mathbf{c}_k \mathbf{r}_k)_{il}, \quad \text{that is,} \quad A = \mathbf{c}_1 \mathbf{r}_1 + \dots + \mathbf{c}_r \mathbf{r}_r.$$

Each $\mathbf{c}_k \mathbf{r}_k$ has rank 1 by the ($\Leftarrow$) direction of [Exercise \(C2: Matrices of rank one\)](#), since $\mathbf{c}_k \neq \mathbf{0}$ (a column of C , whose columns are independent) and $\mathbf{r}_k \neq \mathbf{0}$ (a row of R , whose rows are independent). If $A = A_1 + \dots + A_s$ with each $\text{rank } A_i = 1$, then by [Exercise \(C1: Rank of a sum\)](#) applied repeatedly, $r = \text{rank } A \leq \text{rank } A_1 + \dots + \text{rank } A_s = s$.

- (c) By [Example \(A column–row factorization\)](#), $A = CR$ with $C = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 3 & 1 \end{pmatrix}$ and

$$R = \begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{pmatrix}, \text{ so}$$

$$A = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} (1 \ 2 \ 0 \ 3) + \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} (0 \ 0 \ 1 \ -2) = \begin{pmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 0 & 6 \\ 3 & 6 & 0 & 9 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 1 & -2 \end{pmatrix}.$$

3.8.4.3 C. Going deeper

Exercise C1: Frobenius's rank inequality.

Let $A \in M_{m \times n}(F)$, $B \in M_{n \times p}(F)$ and $C \in M_{p \times q}(F)$. Write $N = \text{nul}(A) \subseteq F^n$, and let $S: F^n \rightarrow F^m$ be $S\mathbf{y} = A\mathbf{y}$.

- (a) Show that $\text{col}(AB) = \{A\mathbf{y} : \mathbf{y} \in \text{col}(B)\}$, and deduce from Rank–Nullity for $S|_{\text{col}(B)}$ that $\text{rank}(AB) = \text{rank } B - \dim(N \cap \text{col}(B))$.
- (b) Show similarly that $\text{rank}(ABC) = \text{rank}(BC) - \dim(N \cap \text{col}(BC))$.
- (c) Prove that $\text{col}(BC) \subseteq \text{col}(B)$, and deduce **Frobenius's inequality**

$$\text{rank}(ABC) + \text{rank } B \geq \text{rank}(AB) + \text{rank}(BC).$$

- (d) Deduce Sylvester's inequality for matrices from (c).

Hint: for (d), look for a special choice of the middle factor.

Solution.

- (a) By [Theorem 0.9.2](#), every vector $AB\mathbf{x}$ equals $A\mathbf{y}$ with $\mathbf{y} = B\mathbf{x} \in \text{col}(B)$; conversely

every $\mathbf{y} \in \text{col}(B)$ is $B\mathbf{x}$ for some $\mathbf{x}$, and then $A\mathbf{y} = (AB)\mathbf{x} \in \text{col}(AB)$. So $\text{col}(AB) = \text{im}(S|_{\text{col}(B)})$. The kernel of $S|_{\text{col}(B)}$ is $N \cap \text{col}(B)$, and $\text{col}(B)$ is finite-dimensional, so by [Theorem 3.3.1](#),

$$\text{rank } B = \dim \text{col}(B) = \text{rank}(AB) + \dim(N \cap \text{col}(B)).$$

- (b) The same argument with BC in place of B gives $\text{col}(ABC) = \text{im}(S|_{\text{col}(BC)})$, with kernel $N \cap \text{col}(BC)$, so $\text{rank}(BC) = \text{rank}(ABC) + \dim(N \cap \text{col}(BC))$.
- (c) Every vector $BC\mathbf{x} = B(C\mathbf{x})$ lies in $\text{col}(B)$, so $\text{col}(BC) \subseteq \text{col}(B)$. Hence $N \cap \text{col}(BC) \subseteq N \cap \text{col}(B)$, and by [Theorem 1.6.7](#), $\dim(N \cap \text{col}(BC)) \leq \dim(N \cap \text{col}(B))$. By (b) and then (a),

$$\text{rank}(ABC) = \text{rank}(BC) - \dim(N \cap \text{col}(BC)) \geq \text{rank}(BC) - \dim(N \cap \text{col}(B)) = \text{rank}(BC) - \text{rank } B + \text{rank } A$$

Rearranging gives Frobenius's inequality.

- (d) Let $A \in M_{m \times n}(F)$ and $C \in M_{n \times q}(F)$, and apply (c) with $p = n$ and $B = I_n$. Then $\text{rank}(AC) + n \geq \text{rank } A + \text{rank } C$, since $\text{rank } I_n = n$. This is [Theorem 3.8.5](#) for matrices.

Exercise C2: Counting over $\mathbb{F}_2$.

- (a) Show that $M_2(\mathbb{F}_2)$ contains exactly 1 matrix of rank 0, 6 of rank 2 and 9 of rank 1.
- (b) Deduce the sizes of the equivalence classes of $M_2(\mathbb{F}_2)$.
- (c) Compare with similarity: how many matrices of $M_2(\mathbb{F}_2)$ are similar to I_2 ? Is the class of I_2 under equivalence a union of several similarity classes?

Solution.

- (a) The only matrix of rank 0 is 0. A matrix has rank 2 exactly when its two columns span $\mathbb{F}_2^2$, that is ([Theorem 1.6.4](#)), when they form a basis of $\mathbb{F}_2^2$. The first column can be any of the 3 non-zero vectors of $\mathbb{F}_2^2$. The second must lie outside the span of the first, which has 2 elements, so there are $4 - 2 = 2$ choices. That gives $3 \cdot 2 = 6$ matrices of rank 2. Since $|M_2(\mathbb{F}_2)| = 2^4 = 16$, the remaining $16 - 1 - 6 = 9$ matrices have rank 1.
- (b) By [Theorem 3.8.3](#) (c), the classes are the sets of matrices of a fixed rank, of sizes 1, 9 and 6.
- (c) I_2 is similar only to itself, so its similarity class has 1 element, while its equivalence class, the invertible matrices, has 6. Since similar matrices are equivalent, each equivalence class is a union of similarity classes, and here the class of the 6 invertible matrices contains the similarity class $\{I_2\}$ together with others. For example $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is invertible but not similar to I_2 , so there are at least two similarity classes inside.

3.9 Products and Quotient Spaces

So far every new vector space in this book has been found **inside** an old one, as a subspace. This section builds new spaces in two other ways. The first glues spaces together side by side, making a product; it is the “external” direct sum promised in Chapter 1. The second, the quotient space V/U , is subtler and more important: it collapses a subspace U to a single point and keeps only what V knows beyond U . Quotients let us state precisely what a linear map remembers and what it forgets, in the three Isomorphism Theorems.

3.9.1 Products of vector spaces

In Chapter 1 we wrote $V = U \oplus W$ for subspaces U and W of one space V (Definition 1.7.2). The two pieces already lived in a common space, and “direct” was a property of how they sat there. But often we want to combine two spaces that have nothing to do with each other, such as $\mathbb{R}^2$ and $\mathbb{R}[x]_{\leq 1}$, into one space in which each keeps its identity. There is no common space to add inside, so we build one.

The model is F^2 . A vector of F^2 is a pair (a, b) of scalars, and we add and scale pairs entry by entry. Nothing in that recipe needs the entries to be scalars: the entries could be vectors, taken from any vector spaces over F .

The product of vector spaces is the space of lists with one entry from each space, added and scaled entry by entry.

Definition 3.9.1 — Product of vector spaces

Let $V_1, \dots, V_k$ be vector spaces over the **same** field F . Their **product** (or **external direct sum**) is the Cartesian product (Definition 0.3.10)

$$V_1 \times \cdots \times V_k = \{(\mathbf{v}_1, \dots, \mathbf{v}_k) : \mathbf{v}_i \in V_i \text{ for each } i\},$$

with addition and scalar multiplication defined **entrywise**: for $a \in F$,

$$(\mathbf{v}_1, \dots, \mathbf{v}_k) + (\mathbf{w}_1, \dots, \mathbf{w}_k) := (\mathbf{v}_1 + \mathbf{w}_1, \dots, \mathbf{v}_k + \mathbf{w}_k), \quad a(\mathbf{v}_1, \dots, \mathbf{v}_k) := (a\mathbf{v}_1, \dots, a\mathbf{v}_k).$$

In words: the i -th entry of a sum is the sum of the i -th entries, computed **in** V_i , and similarly for scalar multiples. Two elements are equal exactly when all their entries are equal. Every operation happens inside one V_i at a time, which is why all the V_i must share the field F : the same scalar a has to act on every entry.

It is a vector space. Both operations land in $V_1 \times \cdots \times V_k$, since $\mathbf{v}_i + \mathbf{w}_i \in V_i$ and $a\mathbf{v}_i \in V_i$. Each axiom of Definition 1.1.1 is an equation between lists, so it holds when it holds in each entry, and in each entry it is the same axiom in V_i . For example, (VS6) reads $(a + b)(\mathbf{v}_1, \dots, \mathbf{v}_k) = ((a + b)\mathbf{v}_1, \dots, (a + b)\mathbf{v}_k) = (a\mathbf{v}_1 + b\mathbf{v}_1, \dots, a\mathbf{v}_k + b\mathbf{v}_k) = a(\mathbf{v}_1, \dots, \mathbf{v}_k) + b(\mathbf{v}_1, \dots, \mathbf{v}_k)$, using (VS6) in each V_i . The zero vector is $(\mathbf{0}, \dots, \mathbf{0})$, with the zero of V_i in the i -th entry, and the additive inverse of $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ is $(-\mathbf{v}_1, \dots, -\mathbf{v}_k)$. The other axioms are checked in exactly the same way.

Examples.

- **The standard example.** $F \times \cdots \times F$ (n factors) is F^n with its usual operations.
- **Unrelated spaces.** An element of $\mathbb{R}^2 \times \mathbb{R}[x]_{\leq 1}$ is a pair $((a, b), c + dx)$, and $((1, 0), x) + ((2, 5), 3 - x) = ((3, 5), 3)$.
- **Augmented data.** A linear system $A\mathbf{x} = \mathbf{b}$ with m equations in n unknowns is a single element $(A, \mathbf{b})$ of $M_{m \times n}(F) \times F^m$.
- **Degenerate cases.** For $k = 1$ the product is V_1 itself, with entries written in parentheses. And $V \times \{\mathbf{0}\}$ consists of the pairs $(\mathbf{v}, \mathbf{0})$; the map $\mathbf{v} \mapsto (\mathbf{v}, \mathbf{0})$ is a linear bijection $V \rightarrow V \times \{\mathbf{0}\}$, so a zero factor adds nothing.

Non-example by minimal change. Take $V_1 = \mathbb{C}$ as a vector space over $\mathbb{C}$ and $V_2 = \mathbb{R}$ over $\mathbb{R}$. The set $\mathbb{C} \times \mathbb{R}$ is fine, but the entrywise rule $i \cdot (z, t) = (iz, it)$ fails: $it \notin \mathbb{R}$ for $t \neq 0$. The clause that fails is “over the **same** field”. Restricting the scalars of $\mathbb{C}$ to $\mathbb{R}$ (Definition 1.9.1) repairs it, and $\mathbb{C} \times \mathbb{R}$ over $\mathbb{R}$ is a vector space of dimension 3.

Bases of the factors combine into a basis of the product. For $\mathbf{v} \in V_i$, write $\iota_i(\mathbf{v})$ for the list with $\mathbf{v}$ in the i -th entry and $\mathbf{0}$ in every other entry. The map $\iota_i: V_i \rightarrow V_1 \times \cdots \times V_k$ is linear, because the operations are entrywise.

Theorem 3.9.1 — Dimension of a Product

Let $V_1, \dots, V_k$ be finite-dimensional vector spaces over F , and let $(\mathbf{b}_{i1}, \dots, \mathbf{b}_{in_i})$ be a basis of V_i for each i . Then the list of all vectors $\iota_i(\mathbf{b}_{ij})$, for $i = 1, \dots, k$ and $j = 1, \dots, n_i$, is a basis of $V_1 \times \cdots \times V_k$. In particular,

$$\dim(V_1 \times \cdots \times V_k) = \dim V_1 + \cdots + \dim V_k.$$

Proof. Spanning. Let $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ be in the product. Since the operations are entrywise, $(\mathbf{v}_1, \dots, \mathbf{v}_k) = \iota_1(\mathbf{v}_1) + \cdots + \iota_k(\mathbf{v}_k)$. Each $\mathbf{v}_i$ is a combination $\sum_j c_{ij} \mathbf{b}_{ij}$, and ι_i is linear, so $\iota_i(\mathbf{v}_i) = \sum_j c_{ij} \iota_i(\mathbf{b}_{ij})$ (Theorem 3.1.4). Hence $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ is a combination of the vectors $\iota_i(\mathbf{b}_{ij})$.

Independence. Let $\sum_{i=1}^k \sum_{j=1}^{n_i} c_{ij} \iota_i(\mathbf{b}_{ij}) = \mathbf{0}$. The i -th entry of the left side is $\sum_j c_{ij} \mathbf{b}_{ij}$, because every $\iota_{i'}(\mathbf{b}_{i'j})$ with $i' \neq i$ has $\mathbf{0}$ there. So $\sum_j c_{ij} \mathbf{b}_{ij} = \mathbf{0}$ in V_i for each i , and the independence of the basis of V_i gives $c_{ij} = 0$ for all j .

The list has $n_1 + \cdots + n_k$ vectors, which gives the dimension (Definition 1.6.1).

■

For instance, $\dim(\mathbb{R}^2 \times \mathbb{R}[x]_{\leq 1}) = 2 + 2 = 4$, so this product is isomorphic to $\mathbb{R}^4$ by Theorem 3.5.4.

How does the external product relate to the internal direct sum of Chapter 1? If $U_1, \dots, U_k$ are subspaces of V , there is a natural linear map from the product to V : add up the entries. The sum is direct exactly when this map loses no information.

Theorem 3.9.2 — Internal and External Direct Sums (a) Let $U_1, \dots, U_k$ be subspaces of a vector space V , and let

$$S: U_1 \times \cdots \times U_k \rightarrow V, \quad S(\mathbf{u}_1, \dots, \mathbf{u}_k) = \mathbf{u}_1 + \cdots + \mathbf{u}_k.$$

Then S is linear with $\text{im } S = U_1 + \cdots + U_k$. It is injective if and only if the sum $U_1 + \cdots + U_k$ is direct. In that case S is an isomorphism from $U_1 \times \cdots \times U_k$ onto

$$U_1 \oplus \cdots \oplus U_k.$$

- (b) Conversely, let $V_1, \dots, V_k$ be vector spaces over F . Then the images $\iota_i(V_i)$ are subspaces of $V_1 \times \cdots \times V_k$, each isomorphic to V_i via ι_i , and

$$V_1 \times \cdots \times V_k = \iota_1(V_1) \oplus \cdots \oplus \iota_k(V_k).$$

Proof.

- (a) For lists $(\mathbf{u}_i)$, $(\mathbf{u}'_i)$ and $a \in F$, regrouping the finite sums gives $S((\mathbf{u}_i) + (\mathbf{u}'_i)) = \sum_i (\mathbf{u}_i + \mathbf{u}'_i) = S((\mathbf{u}_i)) + S((\mathbf{u}'_i))$ and $S(a(\mathbf{u}_i)) = \sum_i a\mathbf{u}_i = aS((\mathbf{u}_i))$, so S is linear. Its image is $\{\mathbf{u}_1 + \cdots + \mathbf{u}_k : \mathbf{u}_i \in U_i\}$, which is $U_1 + \cdots + U_k$ by [Definition 1.7.1](#). Its kernel consists of the lists with $\mathbf{u}_1 + \cdots + \mathbf{u}_k = \mathbf{0}$. By [Theorem 3.2.3](#), S is injective if and only if the only such list is $(\mathbf{0}, \dots, \mathbf{0})$, which is condition (b) of [Theorem 1.7.5](#), equivalent to the sum being direct. In that case S , viewed as a map onto its image, is a linear bijection, hence an isomorphism ([Definition 3.5.2](#)).
- (b) Each ι_i is linear and injective (its kernel is $\{\mathbf{0}\}$, since $\iota_i(\mathbf{v})$ has $\mathbf{v}$ as its i -th entry), so $\iota_i(V_i)$ is a subspace ([Theorem 3.2.2](#)) and ι_i is a bijection onto it. Every list satisfies $(\mathbf{v}_1, \dots, \mathbf{v}_k) = \iota_1(\mathbf{v}_1) + \cdots + \iota_k(\mathbf{v}_k)$, so the product is the sum of the subspaces $\iota_i(V_i)$. If $\iota_1(\mathbf{v}_1) + \cdots + \iota_k(\mathbf{v}_k) = \mathbf{0}$, then $(\mathbf{v}_1, \dots, \mathbf{v}_k) = (\mathbf{0}, \dots, \mathbf{0})$, so every piece $\iota_i(\mathbf{v}_i)$ is $\mathbf{0}$. By [Theorem 1.7.5](#) ((b) $\Rightarrow$ (a)) the sum is direct. ■

So internal and external direct sums are the same thing seen from two sides. An internal direct sum is a product that happens to live inside V ; a product is an internal direct sum of copies of its factors. Many books write $\oplus$ for both. This book keeps $\times$ for the external construction, so that $U \oplus W$ always refers to subspaces of one space.

Quick check. Use [Theorem 3.9.2](#) (a) and Rank–Nullity to explain why, for **finite-dimensional** subspaces U, W of V , the sum $U + W$ is direct if and only if $\dim(U + W) = \dim U + \dim W$.

Solution. The addition map $S: U \times W \rightarrow V$ has image $U + W$, and $\dim(U \times W) = \dim U + \dim W$ by [Theorem 3.9.1](#). By [Theorem 3.3.1](#), $\dim U + \dim W = \dim(U + W) + \dim \ker S$. So $\dim(U + W) = \dim U + \dim W$ exactly when $\ker S = \{\mathbf{0}\}$, that is, when S is injective, which by the theorem means the sum is direct. This recovers [Theorem 1.7.4](#) (a) $\Leftrightarrow$ (c).

3.9.2 Cosets

The second construction starts from a picture you have already seen. In Chapter 1 we compared the line $U = \{x + 2y = 0\}$ through the origin with the parallel line $L = \{x + 2y = 3\}$, which is not a subspace. In Chapter 2 we found that the solutions of a consistent system $A\mathbf{x} = \mathbf{b}$ form a set $\mathbf{p} + \text{nul}(A)$, a subspace shifted by one particular solution ([Theorem 2.1.3](#)). Both are translates of a subspace. They deserve a name.

Definition 3.9.2 — Coset

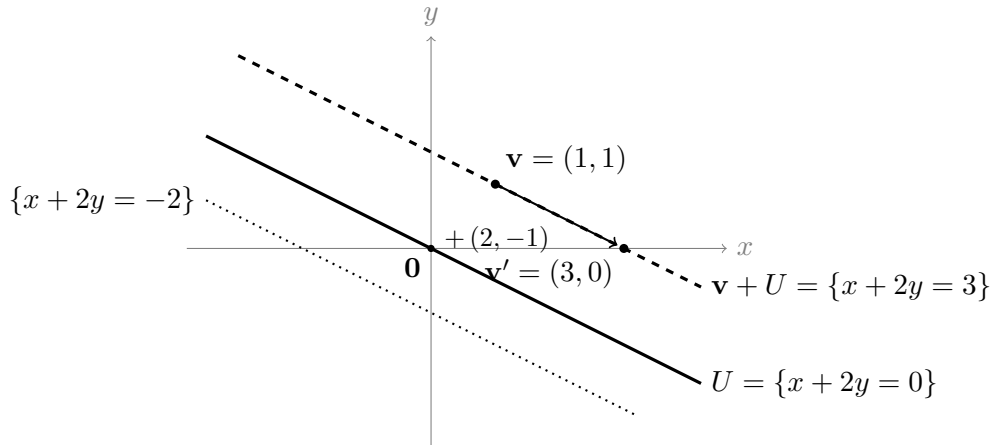
Let U be a subspace of a vector space V , and let $\mathbf{v} \in V$. The **coset** of U through $\mathbf{v}$ is

the subset

$$\mathbf{v} + U := \{\mathbf{v} + \mathbf{u} : \mathbf{u} \in U\} \subseteq V.$$

Any $\mathbf{w} \in \mathbf{v} + U$ is called a **representative** of the coset.

In $\mathbb{R}^2$, with $U = \{x + 2y = 0\} = \text{span}((2, -1))$, the coset through $(3, 0)$ is the line $\{x + 2y = 3\}$: a point $(3, 0) + t(2, -1) = (3 + 2t, -t)$ satisfies $x + 2y = 3$, and every solution of $x + 2y = 3$ is of this form with $t = -y$. In general $(a, b) + U = \{x + 2y = a + 2b\}$, so the cosets of U are exactly the lines parallel to U .



The picture shows three cosets of U : U itself (solid), the line through $\mathbf{v} = (1, 1)$ (dashed), and another parallel line (dotted). The points $\mathbf{v} = (1, 1)$ and $\mathbf{v}' = (3, 0)$ lie on the same dashed line, because $\mathbf{v}' - \mathbf{v} = (2, -1) \in U$; so they are two representatives of **one** coset. That observation is the whole theory of cosets.

Lemma 3.9.3 — When two cosets are equal

Let U be a subspace of V . For $\mathbf{v}, \mathbf{w} \in V$, write $\mathbf{v} \sim \mathbf{w}$ if $\mathbf{v} - \mathbf{w} \in U$.

- (a) $\sim$ is an equivalence relation on V , and the equivalence class of $\mathbf{v}$ is the coset $\mathbf{v} + U$.
- (b) $\mathbf{v} + U = \mathbf{w} + U$ if and only if $\mathbf{v} - \mathbf{w} \in U$.
- (c) Two cosets of U are either equal or disjoint, and every vector of V lies in exactly one coset.

Proof.

- (a) We check [Definition 0.5.2](#). (E1) $\mathbf{v} - \mathbf{v} = \mathbf{0} \in U$. (E2) If $\mathbf{v} - \mathbf{w} \in U$, then $\mathbf{w} - \mathbf{v} = (-1)(\mathbf{v} - \mathbf{w}) \in U$ by [Theorem 1.1.7](#) and closure under scalar multiplication. (E3) If $\mathbf{v} - \mathbf{w} \in U$ and $\mathbf{w} - \mathbf{x} \in U$, then $\mathbf{v} - \mathbf{x} = (\mathbf{v} - \mathbf{w}) + (\mathbf{w} - \mathbf{x}) \in U$ by closure under addition. For the class, $\mathbf{w} \sim \mathbf{v}$ means $\mathbf{w} - \mathbf{v} = \mathbf{u}$ for some $\mathbf{u} \in U$, that is, $\mathbf{w} = \mathbf{v} + \mathbf{u} \in \mathbf{v} + U$; so the class $[\mathbf{v}]$ of [Definition 0.5.3](#) is $\mathbf{v} + U$.
- (b) By (a) and [Lemma 0.5.1](#) (a), $\mathbf{v} + U = \mathbf{w} + U$ if and only if $\mathbf{v} \sim \mathbf{w}$, that is, $\mathbf{v} - \mathbf{w} \in U$.
- (c) This is [Lemma 0.5.1](#) (b), together with $\mathbf{v} \in \mathbf{v} + U$ (take $\mathbf{u} = \mathbf{0}$).

■

More examples.

- **Vertical lines.** In $\mathbb{R}^3$, let $U = \text{span}(\mathbf{e}_3)$, the z -axis. The coset $(a, b, c) + U$ is the vertical line through $(a, b, 0)$, and $(a, b, c) + U = (a', b', c') + U$ exactly when $a = a'$ and $b = b'$. A coset is determined by the shadow of any of its points on the xy -plane.
- **Polynomials with a prescribed value.** In $F[x]_{\leq n}$, let $U = \{p : p(0) = 0\}$, a subspace (the kernel of evaluation at 0). Then $p + U = q + U$ exactly when $(p - q)(0) = 0$, that is, $p(0) = q(0)$. So $p + U$ is the set of all polynomials with the same constant term as p .
- **Degenerate cases.** If $U = \{\mathbf{0}\}$, each coset $\mathbf{v} + U = \{\mathbf{v}\}$ is a single vector. If $U = V$, there is only one coset, $\mathbf{v} + V = V$, because $\mathbf{v} - \mathbf{w} \in V$ always.

Non-example by minimal change. Try to use the line $L = \{x + 2y = 3\}$, which is not a subspace, in place of U . The rule “ $\mathbf{v} \sim \mathbf{w}$ if $\mathbf{v} - \mathbf{w} \in L$ ” is not even reflexive: $\mathbf{v} - \mathbf{v} = \mathbf{0} \notin L$. What fails is the hypothesis that U is a subspace, which (E1), (E2) and (E3) each used.

Warning. A coset has many names, and it is usually not a subspace. In the example above, $(1, 1) + U$ and $(3, 0) + U$ are the **same** set, although $(1, 1) \neq (3, 0)$. Never conclude $\mathbf{v} = \mathbf{w}$ from $\mathbf{v} + U = \mathbf{w} + U$; conclude only $\mathbf{v} - \mathbf{w} \in U$. And $\mathbf{v} + U$ is a subspace only when it contains $\mathbf{0}$, that is, when $\mathbf{v} \in U$ and so $\mathbf{v} + U = U$. The dashed line above is not a subspace.

3.9.3 The quotient space

Now we want to treat each coset as a single point. The motivation is a kind of deliberate forgetting. Suppose we only care about vectors of $\mathbb{R}^2$ “up to” adding something from U , for instance because we only ever measure the quantity $x + 2y$. Then $(1, 1)$ and $(3, 0)$ carry the same information, and the space of all the information we keep is the set of lines parallel to U . We would like this set to be a vector space, so that we can add and scale lines.

The recipe is the one used for $\mathbb{Z}/n\mathbb{Z}$ in Chapter 0 ([Example \(Addition modulo \$n\$ is well defined\)](#)): define the operations through representatives, then check that the choice of representatives does not matter.

The quotient space V/U is V with everything in U declared to be zero.

Definition 3.9.3 — Quotient space

Let U be a subspace of a vector space V over F . The **quotient space** V/U (read “ V modulo U ”) is the set of all cosets of U ,

$$V/U := \{\mathbf{v} + U : \mathbf{v} \in V\},$$

with the operations

$$(\mathbf{v} + U) + (\mathbf{w} + U) := (\mathbf{v} + \mathbf{w}) + U, \quad a(\mathbf{v} + U) := a\mathbf{v} + U \quad (\mathbf{v}, \mathbf{w} \in V, a \in F).$$

The **quotient map** is $\pi: V \rightarrow V/U$, $\pi(\mathbf{v}) = \mathbf{v} + U$.

In words: an element of V/U is a whole coset, a subset of V . By [Lemma 3.9.3 \(a\)](#), V/U is the quotient set of V by the relation $\mathbf{v} \sim \mathbf{w} \Leftrightarrow \mathbf{v} - \mathbf{w} \in U$ ([Definition 0.5.5](#)), and π is its quotient map. To add two cosets, pick a representative of each, add them in V , and take the coset of the

result. To scale a coset, scale a representative. The operations are written with representatives, so before anything else we must check that they are **well defined**.

The danger is real. By [Lemma 3.9.3](#), $(1, 1) + U$ and $(3, 0) + U$ are one element of $\mathbb{R}^2/U$. If the sum $((1, 1) + U) + (\mathbf{w} + U)$ depended on which of the two names we used, the “definition” would assign two values to one pair of cosets, and it would not define a function at all ([Definition 0.4.1](#)). The theorem below rules this out, and then shows that the operations make V/U a vector space.

Theorem 3.9.4 — The Quotient Space Is a Vector Space

Let U be a subspace of a vector space V over F .

- (a) The addition and scalar multiplication in [Definition 3.9.3](#) are well defined: the right-hand sides depend only on the cosets $\mathbf{v} + U$ and $\mathbf{w} + U$, not on the representatives $\mathbf{v}$ and $\mathbf{w}$.
- (b) With these operations V/U is a vector space over F . Its zero vector is $\mathbf{0} + U = U$, and the additive inverse of $\mathbf{v} + U$ is $(-\mathbf{v}) + U$.
- (c) The quotient map $\pi: V \rightarrow V/U$ is linear and surjective, and $\ker \pi = U$.

Idea. Chapter 0’s [Theorem 0.5.3](#) turns a function on V that is constant on classes into a function on V/U . It is stated for functions of one variable, so we use it one variable at a time. ① For scalar multiplication by a fixed a : if $\mathbf{v} - \mathbf{v}' \in U$, then $a\mathbf{v} - a\mathbf{v}' = a(\mathbf{v} - \mathbf{v}') \in U$. ② For addition, first freeze the second summand $\mathbf{w}$ and vary the first; then show that the resulting function on cosets does not change when $\mathbf{w}$ is replaced by an equivalent $\mathbf{w}'$. Each check is one line and uses exactly one closure property of U . Once the operations exist, every axiom for cosets is the same axiom in V , passed through π .

Proof. Throughout, $\mathbf{v} \sim \mathbf{v}'$ means $\mathbf{v} - \mathbf{v}' \in U$, and by [Lemma 3.9.3](#) (b), $\mathbf{v} \sim \mathbf{v}'$ exactly when $\mathbf{v} + U = \mathbf{v}' + U$.

- (a) *Scalar multiplication.* Fix $a \in F$, and let $g_a: V \rightarrow V/U$, $g_a(\mathbf{v}) = a\mathbf{v} + U$. If $\mathbf{v} \sim \mathbf{v}'$, then $a\mathbf{v} - a\mathbf{v}' = a(\mathbf{v} - \mathbf{v}') \in U$ by (VS5) and closure of U under scalar multiplication, so $g_a(\mathbf{v}) = g_a(\mathbf{v}')$. By [Theorem 0.5.3](#) (a), there is exactly one function $\bar{g}_a: V/U \rightarrow V/U$ with $\bar{g}_a(\mathbf{v} + U) = a\mathbf{v} + U$ for every $\mathbf{v} \in V$. We define $a(\mathbf{v} + U) := \bar{g}_a(\mathbf{v} + U)$.

Addition, first variable. Fix $\mathbf{w} \in V$, and let $f_{\mathbf{w}}: V \rightarrow V/U$, $f_{\mathbf{w}}(\mathbf{v}) = (\mathbf{v} + \mathbf{w}) + U$. If $\mathbf{v} \sim \mathbf{v}'$, then $(\mathbf{v} + \mathbf{w}) - (\mathbf{v}' + \mathbf{w}) = \mathbf{v} - \mathbf{v}' \in U$, so $f_{\mathbf{w}}(\mathbf{v}) = f_{\mathbf{w}}(\mathbf{v}')$. By [Theorem 0.5.3](#) (a), there is exactly one function $\bar{f}_{\mathbf{w}}: V/U \rightarrow V/U$ with $\bar{f}_{\mathbf{w}}(\mathbf{v} + U) = (\mathbf{v} + \mathbf{w}) + U$ for every $\mathbf{v}$.

Addition, second variable. Let $\mathcal{F}$ be the set of all functions $V/U \rightarrow V/U$, and let $h: V \rightarrow \mathcal{F}$, $h(\mathbf{w}) = \bar{f}_{\mathbf{w}}$. Suppose $\mathbf{w} \sim \mathbf{w}'$. For every $\mathbf{v} \in V$, $(\mathbf{v} + \mathbf{w}) - (\mathbf{v} + \mathbf{w}') = \mathbf{w} - \mathbf{w}' \in U$, so

$$\bar{f}_{\mathbf{w}}(\mathbf{v} + U) = (\mathbf{v} + \mathbf{w}) + U = (\mathbf{v} + \mathbf{w}') + U = \bar{f}_{\mathbf{w}'}(\mathbf{v} + U).$$

Since every element of V/U has the form $\mathbf{v} + U$, the functions $\bar{f}_{\mathbf{w}}$ and $\bar{f}_{\mathbf{w}'}$ are equal, that is, $h(\mathbf{w}) = h(\mathbf{w}')$. By [Theorem 0.5.3](#) (a) again, there is exactly one function $\bar{h}: V/U \rightarrow \mathcal{F}$ with $\bar{h}(\mathbf{w} + U) = \bar{f}_{\mathbf{w}}$. We define $X + Y := \bar{h}(Y)(X)$ for $X, Y \in V/U$.

Then

$$(\mathbf{v} + U) + (\mathbf{w} + U) = \bar{h}(\mathbf{w} + U)(\mathbf{v} + U) = \bar{f}_{\mathbf{w}}(\mathbf{v} + U) = (\mathbf{v} + \mathbf{w}) + U,$$

which is the formula of [Definition 3.9.3](#), and its value depends only on the two cosets.

- (b) Every element of V/U is $\pi(\mathbf{v}) = \mathbf{v} + U$ for some $\mathbf{v}$, and by (a) we have $\pi(\mathbf{v}) + \pi(\mathbf{w}) = \pi(\mathbf{v} + \mathbf{w})$ and $a\pi(\mathbf{v}) = \pi(a\mathbf{v})$. Let $X = \pi(\mathbf{x})$, $Y = \pi(\mathbf{y})$, $Z = \pi(\mathbf{z})$ be in V/U , and $a, b \in F$. Each axiom follows by moving π outside, using the axiom in V , and moving π back:

- (VS1) $X + Y = \pi(\mathbf{x} + \mathbf{y}) = \pi(\mathbf{y} + \mathbf{x}) = Y + X$.
- (VS2) $(X + Y) + Z = \pi((\mathbf{x} + \mathbf{y}) + \mathbf{z}) = \pi(\mathbf{x} + (\mathbf{y} + \mathbf{z})) = X + (Y + Z)$.
- (VS3) $X + \pi(\mathbf{0}) = \pi(\mathbf{x} + \mathbf{0}) = \pi(\mathbf{x}) = X$, so $\pi(\mathbf{0}) = \mathbf{0} + U = U$ is a zero vector.
- (VS4) $X + \pi(-\mathbf{x}) = \pi(\mathbf{x} + (-\mathbf{x})) = \pi(\mathbf{0})$, so $\pi(-\mathbf{x}) = (-\mathbf{x}) + U$ is an additive inverse of X .
- (VS5) $a(X + Y) = \pi(a(\mathbf{x} + \mathbf{y})) = \pi(a\mathbf{x} + a\mathbf{y}) = aX + aY$.
- (VS6) $(a + b)X = \pi((a + b)\mathbf{x}) = \pi(a\mathbf{x} + b\mathbf{x}) = aX + bX$.
- (VS7) $a(bX) = \pi(a(b\mathbf{x})) = \pi((ab)\mathbf{x}) = (ab)X$.
- (VS8) $1X = \pi(1\mathbf{x}) = \pi(\mathbf{x}) = X$.

Hence V/U is a vector space over F .

- (c) The identities $\pi(\mathbf{v} + \mathbf{w}) = \pi(\mathbf{v}) + \pi(\mathbf{w})$ and $\pi(a\mathbf{v}) = a\pi(\mathbf{v})$ from (b) say that π is linear. It is surjective by the definition of V/U . Finally, $\mathbf{v} \in \ker \pi$ means $\mathbf{v} + U = \mathbf{0} + U$, which by [Lemma 3.9.3](#) (b) means $\mathbf{v} = \mathbf{v} - \mathbf{0} \in U$. So $\ker \pi = U$. ■

Part (c) has a pleasant reading: **every subspace is a kernel**. Given $U \subseteq V$, the quotient map is a linear map whose kernel is exactly U .

Examples.

- **Parallel lines.** In $\mathbb{R}^2/U$ with $U = \{x + 2y = 0\}$, write ℓ_c for the line $\{x + 2y = c\}$. Since $(a, b) \in \ell_{a+2b}$, the definition gives $\ell_c + \ell_d = \ell_{c+d}$ and $t\ell_c = \ell_{tc}$: to add two parallel lines, add their “heights” c and d . The zero vector is $\ell_0 = U$.
- **Constant terms.** In $F[x]_{\leq n}/U$ with $U = \{p : p(0) = 0\}$, the coset $p + U$ is determined by $p(0)$, and $(p + U) + (q + U) = (p + q) + U$ is determined by $p(0) + q(0)$. All that survives of a polynomial is its constant term.
- **Degenerate cases.** $V/\{\mathbf{0}\}$ has the cosets $\{\mathbf{v}\}$, added exactly as in V : nothing is forgotten. V/V has the single element V , so it is the zero space: everything is forgotten.

Non-example by minimal change. Replace the line U by the union of the two axes, $S = \text{span}(\mathbf{e}_1) \cup \text{span}(\mathbf{e}_2) \subseteq \mathbb{R}^2$. It still contains $\mathbf{0}$ and is still closed under scalar multiplication, but it is not closed under addition. Then “ $\mathbf{v} - \mathbf{w} \in S$ ” is not transitive: $(1, 1) - (0, 1) = (1, 0) \in S$ and $(0, 1) - (0, 0) = (0, 1) \in S$, but $(1, 1) - (0, 0) = (1, 1) \notin S$. Concretely, the “cosets” $(1, 1) + S$ and $(0, 0) + S = S$ both contain $(0, 1) = (1, 1) + (-1, 0)$, yet they are different sets, since $\mathbf{0} \in S$ but $(1, 1) + \mathbf{s} = \mathbf{0}$ would need $\mathbf{s} = (-1, -1) \notin S$. The sets $\mathbf{v} + S$ do not split $\mathbb{R}^2$ into disjoint pieces, so there is no set “ $\mathbb{R}^2/S$ ” of the kind the definition needs. The clause that fails is “ U is a subspace”, specifically closure under addition.

Why this definition. Each part of the hypothesis that U is a subspace has a job: $\mathbf{0} \in U$ makes $\sim$ reflexive; closure under scalar multiplication makes $\sim$ symmetric and scaling of cosets

well defined; closure under addition makes $\sim$ transitive and addition of cosets well defined. The non-example shows what goes wrong when one of them is missing. The notation V/U and the phrase “modulo U ” come from $\mathbb{Z}/n\mathbb{Z}$, where integers are identified when their difference lies in $n\mathbb{Z}$.

Warning. V/U is not a subspace of V . Its elements are subsets of V , not vectors of V , and its addition is a new operation. For example, $\mathbb{R}^3/\text{span}(\mathbf{e}_3)$ is not the xy -plane: its elements are vertical lines. The xy -plane is a **complement** of the z -axis, one of many (Chapter 1), whereas V/U involves no choice at all. The two are isomorphic, as we show below, but they are different objects.

In finite dimension the size of a quotient is exactly what one would guess from the picture: we lose $\dim U$ dimensions.

Theorem 3.9.5 — Dimension of a Quotient

Let V be finite-dimensional and U a subspace of V . Then V/U is finite-dimensional and

$$\dim(V/U) = \dim V - \dim U.$$

More precisely, if $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_s)$ is a basis of V such that $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ is a basis of U , then $(\mathbf{w}_1 + U, \dots, \mathbf{w}_s + U)$ is a basis of V/U .

Proof. By Theorem 3.9.4 (c), $\pi: V \rightarrow V/U$ is linear, surjective and has kernel U . By the Rank–Nullity Theorem (Theorem 3.3.1), $\text{im } \pi = V/U$ is finite-dimensional and $\dim V = \dim U + \dim(V/U)$, which is the formula.

For the basis, let $\mathbf{v} + U \in V/U$ and write $\mathbf{v} = \sum_i a_i \mathbf{u}_i + \sum_j b_j \mathbf{w}_j$. Since π is linear and $\pi(\mathbf{u}_i) = U$ is the zero vector, Theorem 3.1.4 gives $\mathbf{v} + U = \sum_j b_j(\mathbf{w}_j + U)$. So the list $(\mathbf{w}_1 + U, \dots, \mathbf{w}_s + U)$ spans V/U . Its length is $s = \dim V - \dim U = \dim(V/U)$, so it is a basis by Theorem 1.6.4 (b). ■

Such a basis always exists: take a basis of U and extend it (Theorem 1.6.5). So to find a basis of a quotient, **extend a basis of U to one of V and keep the cosets of the new vectors.**

Example. (A basis of a quotient of $\mathbb{R}^3$)

Let $U = \text{span}((1, 1, 1)) \subseteq \mathbb{R}^3$. Find a basis of $\mathbb{R}^3/U$, and write $(2, 5, -1) + U$ in it.

Solution. Extend $(1, 1, 1)$ by $\mathbf{e}_1$ and $\mathbf{e}_2$. The list $((1, 1, 1), \mathbf{e}_1, \mathbf{e}_2)$ is independent: in $a(1, 1, 1) + b\mathbf{e}_1 + c\mathbf{e}_2 = (a + b, a + c, a) = \mathbf{0}$, the third entry gives $a = 0$, and then $b = c = 0$. It has length 3, so it is a basis of $\mathbb{R}^3$ (Theorem 1.6.4). By Theorem 3.9.5, $(\mathbf{e}_1 + U, \mathbf{e}_2 + U)$ is a basis of $\mathbb{R}^3/U$, which has dimension $3 - 1 = 2$.

To write $(2, 5, -1) + U$, solve $(2, 5, -1) = a(1, 1, 1) + b\mathbf{e}_1 + c\mathbf{e}_2$: the third entry gives $a = -1$, then $b = 3$ and $c = 6$. Hence

$$(2, 5, -1) + U = 3(\mathbf{e}_1 + U) + 6(\mathbf{e}_2 + U),$$

and indeed $(2, 5, -1) - (3, 6, 0) = (-1, -1, -1) \in U$. The coordinates $(3, 6)$ are $(x -$

$z, y - z$) evaluated at $(2, 5, -1)$: they are exactly the quantities that do not change when a multiple of $(1, 1, 1)$ is added.

The warning above said that a quotient is not a complement. The next result says how they are related: any complement is a concrete model of the quotient.

Proposition 3.9.6 — A complement is a model of the quotient

Let U be a subspace of V , and let W be a subspace with $V = U \oplus W$. Then the restriction of the quotient map, $W \rightarrow V/U$, $\mathbf{w} \mapsto \mathbf{w} + U$, is an isomorphism.

Proof. The restriction of the linear map π to W is linear. *Injective:* if $\mathbf{w} + U = U$ with $\mathbf{w} \in W$, then $\mathbf{w} \in U$ by Lemma 3.9.3 (b), so $\mathbf{w} \in U \cap W = \{\mathbf{0}\}$ (Theorem 1.7.4); by Theorem 3.2.3 the restriction is injective. *Surjective:* given $\mathbf{v} + U$, write $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in U$, $\mathbf{w} \in W$; then $\mathbf{v} - \mathbf{w} = \mathbf{u} \in U$, so $\mathbf{v} + U = \mathbf{w} + U$. A linear bijection is an isomorphism (Definition 3.5.2). ■

So every complement of U is isomorphic to V/U , which gives a second proof that all complements have the same dimension. The quotient is the choice-free object that all the complements imitate.

Quick check. In $\mathbb{R}^2/U$ with $U = \text{span}((1, 1))$, is $(2, 3) + U = (0, 1) + U$? Is $(1, 0) + U = (0, 1) + U$? What is $\dim(\mathbb{R}^2/U)$?

Solution. By Lemma 3.9.3 (b): $(2, 3) - (0, 1) = (2, 2) \in U$, so the first two cosets are equal. $(1, 0) - (0, 1) = (1, -1) \notin U$, so the second pair are different cosets. By Theorem 3.9.5, $\dim(\mathbb{R}^2/U) = 2 - 1 = 1$.

3.9.4 The First Isomorphism Theorem

A linear map $T: V \rightarrow W$ forgets exactly the information in its kernel: $T\mathbf{v} = T\mathbf{v}'$ if and only if $T(\mathbf{v} - \mathbf{v}') = \mathbf{0}$, that is, $\mathbf{v} - \mathbf{v}' \in \ker T$. In the language of this section, T takes the same value at two vectors exactly when they lie in the same coset of $\ker T$. So T is really a function of the coset, and on cosets it forgets nothing.

Theorem 3.9.7 — First Isomorphism Theorem

Let $T: V \rightarrow W$ be a linear map, and let $K = \ker T$. Then the rule

$$\bar{T}: V/K \rightarrow W, \quad \bar{T}(\mathbf{v} + K) = T\mathbf{v}$$

defines a linear map, which is injective with $\text{im } \bar{T} = \text{im } T$, and satisfies $\bar{T} \circ \pi = T$. Hence

$$V/\ker T \cong \text{im } T.$$

Proof. *Well defined.* K is a subspace by Theorem 3.2.1, so V/K is a vector space. If $\mathbf{v} + K = \mathbf{v}' + K$, then $\mathbf{v} - \mathbf{v}' \in K$ by Lemma 3.9.3 (b), so $T\mathbf{v} - T\mathbf{v}' = T(\mathbf{v} - \mathbf{v}') = \mathbf{0}$.

Hence T is constant on each coset of K , and by [Theorem 0.5.3](#) (a) there is exactly one function $\bar{T}: V/K \rightarrow W$ with $\bar{T}(\mathbf{v} + K) = T\mathbf{v}$, that is, $\bar{T} \circ \pi = T$.

Linear. For $\mathbf{v}, \mathbf{w} \in V$ and $a \in F$, by [Definition 3.9.3](#) and the linearity of T ,

$$\bar{T}((\mathbf{v} + K) + (\mathbf{w} + K)) = \bar{T}((\mathbf{v} + \mathbf{w}) + K) = T(\mathbf{v} + \mathbf{w}) = T\mathbf{v} + T\mathbf{w} = \bar{T}(\mathbf{v} + K) + \bar{T}(\mathbf{w} + K),$$

and similarly $\bar{T}(a(\mathbf{v} + K)) = \bar{T}(a\mathbf{v} + K) = T(a\mathbf{v}) = a\bar{T}(\mathbf{v} + K)$.

Injective. If $\bar{T}(\mathbf{v} + K) = \mathbf{0}$, then $T\mathbf{v} = \mathbf{0}$, so $\mathbf{v} \in K$ and $\mathbf{v} + K = K$, the zero of V/K . By [Theorem 3.2.3](#), $\bar{T}$ is injective.

Image. The values of $\bar{T}$ are the vectors $T\mathbf{v}$, $\mathbf{v} \in V$, so $\text{im } \bar{T} = \text{im } T$. Regarded as a map $V/K \rightarrow \text{im } T$, $\bar{T}$ is a linear bijection, hence an isomorphism ([Definition 3.5.2](#)).

■

So every linear map factors as “collapse the kernel, then embed”: $T = \bar{T} \circ \pi$, with π surjective and $\bar{T}$ injective. When V is finite-dimensional, taking dimensions with [Theorem 3.9.5](#) gives $\dim V - \dim \ker T = \dim \text{im } T$, which is Rank–Nullity again. The theorem is stronger than the count, because it names the isomorphism, and it holds in infinite dimension too.

The practical use runs the other way: to identify a quotient V/U , find a surjective linear map out of V whose kernel is exactly U .

Example. (Identifying quotients)

Use the First Isomorphism Theorem to show:

- (a) $\mathbb{R}^3/L \cong \mathbb{R}^2$, where $L = \text{span}((1, 1, 1))$;
- (b) $\mathbb{R}[x]_{\leq n}/U \cong \mathbb{R}$, where $U = \{p : p(0) = 0\}$.

Solution.

- (a) Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $T(x, y, z) = (x - z, y - z)$, which is linear because each entry is a linear expression. It is surjective, since $T(a, b, 0) = (a, b)$. Its kernel is $\{x = z, y = z\} = \{(t, t, t)\} = L$. By [Theorem 3.9.7](#), $(x, y, z) + L \mapsto (x - z, y - z)$ is an isomorphism $\mathbb{R}^3/L \rightarrow \mathbb{R}^2$. This explains the coordinates found in [Example \(A basis of a quotient of \$\mathbb{R}^3\$ \)](#). The check that it is well defined is visible: adding (t, t, t) to (x, y, z) does not change $x - z$ or $y - z$.
- (b) Let $\text{ev}_0: \mathbb{R}[x]_{\leq n} \rightarrow \mathbb{R}$, $p \mapsto p(0)$. It is linear by [Theorem 0.8.5](#), surjective because the constant polynomial c maps to c , and its kernel is U by definition. By [Theorem 3.9.7](#), $p + U \mapsto p(0)$ is an isomorphism $\mathbb{R}[x]_{\leq n}/U \rightarrow \mathbb{R}$. In particular $\dim(\mathbb{R}[x]_{\leq n}/U) = 1$ by [Theorem 3.5.4](#), so $\dim U = (n + 1) - 1 = n$ by [Theorem 3.9.5](#).

3.9.5 The Second and Third Isomorphism Theorems

Two more identifications come from the same method: write down a surjective map, compute its kernel, and apply the First Isomorphism Theorem. The first compares a sum of subspaces with an intersection.

Theorem 3.9.8 — Second Isomorphism Theorem

Let U and W be subspaces of a vector space V . Then $U \cap W$ is a subspace of U , W is a subspace of $U + W$, and

$$U/(U \cap W) \rightarrow (U + W)/W, \quad \mathbf{u} + (U \cap W) \mapsto \mathbf{u} + W$$

is a well-defined isomorphism. Hence $(U + W)/W \cong U/(U \cap W)$.

Idea. Picture two planes U and W in $\mathbb{R}^3$ meeting in a line. Modulo W , a vector of $U + W$ is represented by its U -part alone, since the W -part is forgotten. The only vectors of U that become zero are those already in W . So the map “ $\mathbf{u} \mapsto \mathbf{u} + W$ ” from U is onto, with kernel $U \cap W$.

Proof. $U \cap W$ is a subspace (Theorem 1.2.2) contained in U , and W is a subspace contained in $U + W$ (Theorem 1.7.1), so both quotients exist. Let $\pi_W: U + W \rightarrow (U + W)/W$ be the quotient map, and let $S: U \rightarrow (U + W)/W$, $S\mathbf{u} = \mathbf{u} + W$, be its restriction to U ; S is linear because π_W is (Theorem 3.9.4).

S is surjective. An element of $(U + W)/W$ is $(\mathbf{u} + \mathbf{w}) + W$ with $\mathbf{u} \in U$, $\mathbf{w} \in W$. Since $(\mathbf{u} + \mathbf{w}) - \mathbf{u} = \mathbf{w} \in W$, Lemma 3.9.3 (b) gives $(\mathbf{u} + \mathbf{w}) + W = \mathbf{u} + W = S\mathbf{u}$.

$\ker S = U \cap W$. For $\mathbf{u} \in U$, $S\mathbf{u} = W$ means $\mathbf{u} \in W$ (Lemma 3.9.3 (b)), that is, $\mathbf{u} \in U \cap W$.

By Theorem 3.9.7 applied to S , the rule $\mathbf{u} + (U \cap W) \mapsto S\mathbf{u} = \mathbf{u} + W$ is a well-defined injective linear map with image $\text{im } S = (U + W)/W$, hence an isomorphism. (Directly: if $\mathbf{u} - \mathbf{u}' \in U \cap W$, then $\mathbf{u} - \mathbf{u}' \in W$, so $\mathbf{u} + W = \mathbf{u}' + W$, and the value does not depend on the representative.)

■

If U and W are finite-dimensional, then so is $U + W$ (it is spanned by the union of bases, Proposition 1.7.2), and Theorem 3.9.5 turns the theorem into $\dim(U + W) - \dim W = \dim U - \dim(U \cap W)$. That is the dimension formula for sums (Theorem 1.7.3), now as a consequence of an isomorphism rather than of a basis computation. An exercise below asks for the details.

The third theorem says that quotients can be taken in stages.

Theorem 3.9.9 — Third Isomorphism Theorem

Let $U \subseteq W \subseteq V$ be subspaces. Then $W/U = \{\mathbf{w} + U : \mathbf{w} \in W\}$ is a subspace of V/U ,

and

$$(V/U)/(W/U) \rightarrow V/W, \quad (\mathbf{v} + U) + W/U \mapsto \mathbf{v} + W$$

is a well-defined isomorphism. Hence $(V/U)/(W/U) \cong V/W$.

Proof. Let $S: V/U \rightarrow V/W$, $S(\mathbf{v} + U) = \mathbf{v} + W$.

Well defined. If $\mathbf{v} + U = \mathbf{v}' + U$, then $\mathbf{v} - \mathbf{v}' \in U \subseteq W$, so $\mathbf{v} + W = \mathbf{v}' + W$ by Lemma 3.9.3 (b). By Theorem 0.5.3 (a), applied to $V \rightarrow V/W$, $\mathbf{v} \mapsto \mathbf{v} + W$, the rule defines a function on V/U .

Linear and surjective. $S((\mathbf{v} + U) + (\mathbf{w} + U)) = S((\mathbf{v} + \mathbf{w}) + U) = (\mathbf{v} + \mathbf{w}) + W = S(\mathbf{v} + U) + S(\mathbf{w} + U)$, and similarly for scalars, by Definition 3.9.3 in both quotients. Every $\mathbf{v} + W$ equals $S(\mathbf{v} + U)$.

Kernel. $S(\mathbf{v} + U) = W$ means $\mathbf{v} \in W$. If $\mathbf{v} \in W$, then $\mathbf{v} + U \in W/U$. Conversely, if $\mathbf{v} + U = \mathbf{w} + U$ with $\mathbf{w} \in W$, then $\mathbf{v} - \mathbf{w} \in U \subseteq W$, so $\mathbf{v} \in W$. Hence $\ker S = W/U$. In particular W/U is a subspace of V/U , by Theorem 3.2.1.

By Theorem 3.9.7 applied to S , $(V/U)/(W/U) \cong \text{im } S = V/W$, via $(\mathbf{v} + U) + W/U \mapsto S(\mathbf{v} + U) = \mathbf{v} + W$.

■

The notation W/U is used with two meanings, and they agree: the quotient space of W by U , and the set of cosets $\mathbf{w} + U$ with $\mathbf{w} \in W$, sitting inside V/U . The coset $\mathbf{w} + U$ is the same subset of V in both readings, and the operations agree because both are computed with representatives. For example, with $V = \mathbb{R}^3$, U the z -axis and W the xz -plane, $V/W \cong \mathbb{R}$ through the y -coordinate, and the theorem says that first forgetting z and then forgetting x leaves the same thing as forgetting both at once.

3.9.6 The universal property of the quotient

The proofs of the three theorems repeated one move: a linear map that vanishes on U passes to V/U . We record that move once, in its general form. It is the standard way to **define** linear maps out of a quotient.

Theorem 3.9.10 — Universal Property of the Quotient

Let U be a subspace of V , $\pi: V \rightarrow V/U$ the quotient map, and $T: V \rightarrow W$ a linear map.

- (a) If $U \subseteq \ker T$, then there is **exactly one** linear map $\bar{T}: V/U \rightarrow W$ with $\bar{T} \circ \pi = T$, namely $\bar{T}(\mathbf{v} + U) = T\mathbf{v}$. Its image is $\text{im } T$, and it is injective if and only if $U = \ker T$.
- (b) Conversely, if some function $\bar{T}: V/U \rightarrow W$ satisfies $\bar{T} \circ \pi = T$, then $U \subseteq \ker T$.

Proof.

- (a) If $\mathbf{v} - \mathbf{v}' \in U \subseteq \ker T$, then $T\mathbf{v} = T\mathbf{v}'$. By Theorem 0.5.3 (a) there is exactly one

function $\bar{T}$ with $\bar{T}(\mathbf{v} + U) = T\mathbf{v}$, and it is linear by the computation in the proof of [Theorem 3.9.7](#), which used only this formula. Any linear $\bar{T}$ with $\bar{T} \circ \pi = T$ satisfies this formula, so it is this one. The image is $\{T\mathbf{v}\} = \text{im } T$. The kernel of $\bar{T}$ is $\{\mathbf{v} + U : \mathbf{v} \in \ker T\}$; it is $\{U\}$, the zero subspace, exactly when every $\mathbf{v} \in \ker T$ lies in U , that is, when $\ker T \subseteq U$, which together with $U \subseteq \ker T$ means $U = \ker T$. Apply [Theorem 3.2.3](#).

- (b) Let $\mathbf{u} \in U$. Then $\pi(\mathbf{u}) = U = \pi(\mathbf{0})$, so $T\mathbf{u} = \bar{T}(\pi(\mathbf{u})) = \bar{T}(\pi(\mathbf{0})) = T(\mathbf{0}) = \mathbf{0}$ by [Theorem 3.1.1](#). Hence $U \subseteq \ker T$.

■

In words: **linear maps out of V/U are the same as linear maps out of V that kill U .** The First Isomorphism Theorem is the case $U = \ker T$. For example, the map $T: \mathbb{R}^3 \rightarrow \mathbb{R}$, $T(x, y, z) = x$, kills the z -axis U , so it induces $\bar{T}: \mathbb{R}^3/U \rightarrow \mathbb{R}$, $(x, y, z) + U \mapsto x$. Since $\ker T$ is the yz -plane, strictly larger than U , this $\bar{T}$ is not injective: it also kills the coset $\mathbf{e}_2 + U$.

Quotients will return in Chapter 8, where an operator T with $T(U) \subseteq U$ induces an operator on V/U , and this is the engine of induction arguments on the dimension.

3.9.7 Exercises

3.9.7.1 A. Check your understanding

Exercise A1.

- Define the quotient space V/U and its two operations.
- State the condition, in terms of $\mathbf{v} - \mathbf{w}$, for $\mathbf{v} + U = \mathbf{w} + U$.
- True or false: V/U is a subspace of V . Justify your answer.
- State the First Isomorphism Theorem.
- If U is a 2-dimensional subspace of $\mathbb{R}^5$, what is $\dim(\mathbb{R}^5/U)$? What is $\dim(\mathbb{R}^5 \times U)$?
- What must be checked before the formula $(\mathbf{v} + U) + (\mathbf{w} + U) = (\mathbf{v} + \mathbf{w}) + U$ can be used as a definition, and which property of U makes the check work?

Solution.

- For a subspace U of V , $V/U = \{\mathbf{v} + U : \mathbf{v} \in V\}$, with $(\mathbf{v} + U) + (\mathbf{w} + U) = (\mathbf{v} + \mathbf{w}) + U$ and $a(\mathbf{v} + U) = a\mathbf{v} + U$ ([Definition 3.9.3](#)).
- $\mathbf{v} + U = \mathbf{w} + U$ if and only if $\mathbf{v} - \mathbf{w} \in U$ ([Lemma 3.9.3](#)).
- False. The elements of V/U are cosets, which are subsets of V , not vectors of V .
- If $T: V \rightarrow W$ is linear, then $\mathbf{v} + \ker T \mapsto T\mathbf{v}$ is a well-defined isomorphism $V/\ker T \rightarrow \text{im } T$ ([Theorem 3.9.7](#)).
- $5 - 2 = 3$ by [Theorem 3.9.5](#), and $5 + 2 = 7$ by [Theorem 3.9.1](#).
- That the right-hand side does not depend on the representatives: if $\mathbf{v} - \mathbf{v}' \in U$ and $\mathbf{w} - \mathbf{w}' \in U$, then $(\mathbf{v} + \mathbf{w}) + U = (\mathbf{v}' + \mathbf{w}') + U$. This works because $(\mathbf{v} + \mathbf{w}) - (\mathbf{v}' + \mathbf{w}') = (\mathbf{v} - \mathbf{v}') + (\mathbf{w} - \mathbf{w}')$ and U is closed under addition.

3.9.7.2 B. Practice

Exercise B1: A basis of a quotient of $\mathbb{R}^4$.

Let $U = \text{span}((1, 0, 1, 0), (0, 1, 0, 1)) \subseteq \mathbb{R}^4$.

- (a) Find $\dim(\mathbb{R}^4/U)$ and a basis of $\mathbb{R}^4/U$.
- (b) Write $(3, 1, 4, 1) + U$ in your basis.
- (c) Decide whether $(1, 2, 3, 4) + U = (3, 4, 1, 2) + U$.

Solution.

- (a) The two spanning vectors are independent (look at the first two entries), so $\dim U = 2$ and $\dim(\mathbb{R}^4/U) = 4 - 2 = 2$ by [Theorem 3.9.5](#). Extend by $\mathbf{e}_1, \mathbf{e}_2$. If $a(1, 0, 1, 0) + b(0, 1, 0, 1) + c\mathbf{e}_1 + d\mathbf{e}_2 = (a + c, b + d, a, b) = \mathbf{0}$, then $a = b = 0$ from the last two entries and $c = d = 0$ from the first two. So the four vectors are independent, hence a basis of $\mathbb{R}^4$ ([Theorem 1.6.4](#)), and by [Theorem 3.9.5](#) $(\mathbf{e}_1 + U, \mathbf{e}_2 + U)$ is a basis of $\mathbb{R}^4/U$.
- (b) Solve $(3, 1, 4, 1) = (a + c, b + d, a, b)$: $a = 4, b = 1, c = -1, d = 0$. Since the U -part becomes zero in the quotient, $(3, 1, 4, 1) + U = -1(\mathbf{e}_1 + U) + 0(\mathbf{e}_2 + U)$. Check: $(3, 1, 4, 1) + \mathbf{e}_1 = (4, 1, 4, 1) = 4(1, 0, 1, 0) + (0, 1, 0, 1) \in U$.
- (c) The difference is $(1, 2, 3, 4) - (3, 4, 1, 2) = (-2, -2, 2, 2)$. A vector (a, b, a, b) of U has equal first and third entries, but $-2 \neq 2$. So the difference is not in U , and the cosets are different by [Lemma 3.9.3](#).

Exercise B2: A map that factors through a quotient.

Let $T: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}^2$, $T(p) = (p(1), p'(1))$, and $U = \text{span}((x - 1)^2)$.

- (a) Show that $U \subseteq \ker T$, so that T induces a linear map $\bar{T}: \mathbb{R}[x]_{\leq 2}/U \rightarrow \mathbb{R}^2$ with $\bar{T}(p + U) = T(p)$.
- (b) Prove that $\bar{T}$ is an isomorphism.
- (c) Compute $\bar{T}(x^2 + U)$, and find the coset $\bar{T}^{-1}(0, 1)$.

Solution.

- (a) T is linear, since evaluation and differentiation are. For $q = (x - 1)^2$, $q(1) = 0$ and $q'(1) = 2(1 - 1) = 0$, so $T(q) = \mathbf{0}$, and hence $T(cq) = \mathbf{0}$ for all c . So $U \subseteq \ker T$, and [Theorem 3.9.10](#) (a) gives the linear map $\bar{T}$.
- (b) $T(1) = (1, 0)$ and $T(x - 1) = (0, 1)$, so T is surjective and $\text{rank } T = 2$. By [Theorem 3.3.1](#), $\dim \ker T = 3 - 2 = 1 = \dim U$, so $\ker T = U$ by [Theorem 1.6.8](#). By [Theorem 3.9.10](#) (a), $\bar{T}$ is injective with image $\text{im } T = \mathbb{R}^2$. Hence $\bar{T}$ is a linear bijection, an isomorphism.
- (c) $\bar{T}(x^2 + U) = T(x^2) = (1, 2)$. Since $T(x - 1) = (0, 1)$, $\bar{T}^{-1}(0, 1) = (x - 1) + U$, the set of all $x - 1 + c(x - 1)^2$.

Exercise B3: Matrices modulo scalars.

Let $n \geq 1$ and $S = \{cI_n : c \in F\} \subseteq M_n(F)$.

- (a) Show that S is a subspace and find $\dim(M_n(F)/S)$.
- (b) For $n = 2$, find a basis of $M_2(F)/S$.
- (c) Show that $A + S = B + S$ if and only if $A - B$ is a scalar matrix, and decide whether $\begin{pmatrix} 3 & 1 \\ 0 & 5 \end{pmatrix} + S = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix} + S$.

Solution.

- (a) $S = \text{span}(I_n)$ is a subspace by [Theorem 1.3.1](#), of dimension 1 since $I_n \neq 0$. By [Theorem 3.9.5](#), $\dim(M_n(F)/S) = n^2 - 1$.
- (b) Extend (I_2) by E_{11}, E_{12}, E_{21} . If $aI_2 + bE_{11} + cE_{12} + dE_{21} = \begin{pmatrix} a+b & c \\ d & a \end{pmatrix} = 0$, then $a = c = d = 0$ and then $b = 0$. So the four matrices are independent, hence a basis of $M_2(F)$ ([Theorem 1.6.4](#)), and $(E_{11} + S, E_{12} + S, E_{21} + S)$ is a basis of $M_2(F)/S$ by [Theorem 3.9.5](#).
- (c) This is [Lemma 3.9.3](#) (b) for the subspace S . Here the difference is $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2I_2 \in S$, so the two cosets are equal.

3.9.7.3 C. Going deeper

Exercise C1: The dimension formula from isomorphisms.

Let U and W be finite-dimensional subspaces of a vector space V .

- (a) Using [Theorem 3.9.8](#) and [Theorem 3.9.5](#), prove that $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.
- (b) Give a second proof: apply Rank–Nullity to the addition map $S: U \times W \rightarrow V$, $S(\mathbf{u}, \mathbf{w}) = \mathbf{u} + \mathbf{w}$, after showing that $\mathbf{u} \mapsto (\mathbf{u}, -\mathbf{u})$ is an isomorphism from $U \cap W$ onto $\ker S$.

Solution.

- (a) $U + W$ is finite-dimensional, being spanned by a basis of U followed by a basis of W ([Proposition 1.7.2](#)), and $U \cap W \subseteq U$ is finite-dimensional ([Theorem 1.6.7](#)). By [Theorem 3.9.8](#), $(U + W)/W \cong U/(U \cap W)$, and isomorphic finite-dimensional spaces have equal dimension ([Theorem 3.5.4](#)). Applying [Theorem 3.9.5](#) to both sides,

$$\dim(U + W) - \dim W = \dim U - \dim(U \cap W),$$

which rearranges to the formula.

- (b) The map S is linear with image $U + W$ by [Theorem 3.9.2](#) (a). Let $J: U \cap W \rightarrow U \times W$, $J\mathbf{u} = (\mathbf{u}, -\mathbf{u})$; it is well defined because $\mathbf{u} \in U$ and $-\mathbf{u} \in W$, and linear because the operations are entrywise. It is injective, since $(\mathbf{u}, -\mathbf{u}) = (\mathbf{0}, \mathbf{0})$ forces $\mathbf{u} = \mathbf{0}$. Its image lies in $\ker S$, since $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$. Conversely, if $(\mathbf{u}, \mathbf{w}) \in \ker S$, then $\mathbf{w} = -\mathbf{u}$; so $\mathbf{u} = -\mathbf{w} \in W$, hence $\mathbf{u} \in U \cap W$ and $(\mathbf{u}, \mathbf{w}) = J\mathbf{u}$. Thus J is an isomorphism onto $\ker S$, and $\dim \ker S = \dim(U \cap W)$. By [Theorem 3.9.1](#) and

[Theorem 3.3.1](#),

$$\dim U + \dim W = \dim(U \times W) = \dim(U + W) + \dim(U \cap W).$$

Exercise C2: The operator induced on a quotient.

Let V be finite-dimensional, $T \in \mathcal{L}(V)$, and U a subspace with $T\mathbf{u} \in U$ for every $\mathbf{u} \in U$.

- (a) Prove that $\bar{T}(\mathbf{v} + U) := T\mathbf{v} + U$ is a well-defined linear operator on V/U .
 (b) Let $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ be a basis of U , extended to a basis $\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_s)$ of V . Show that

$$[T]_{\mathcal{B}} = \begin{pmatrix} A & B \\ 0 & D \end{pmatrix},$$

where $A \in M_m(F)$ is the matrix of the restriction $T|_U: U \rightarrow U$ in $(\mathbf{u}_1, \dots, \mathbf{u}_m)$, $D \in M_s(F)$ is the matrix of $\bar{T}$ in $(\mathbf{w}_1 + U, \dots, \mathbf{w}_s + U)$, and 0 is the $s \times m$ zero matrix.

- (c) Check (b) when T is differentiation on $\mathbb{R}[x]_{\leq 2}$, with $U = \mathbb{R}[x]_{\leq 1}$ and $\mathcal{B} = (1, x, x^2)$.

Hint: for (a), apply [Theorem 3.9.10](#) to $\pi \circ T$.

Solution.

- (a) The map $\pi \circ T: V \rightarrow V/U$, $\mathbf{v} \mapsto T\mathbf{v} + U$, is linear as a composition of linear maps ([Theorem 3.4.2](#)). For $\mathbf{u} \in U$, $T\mathbf{u} \in U$, so $(\pi \circ T)(\mathbf{u}) = U$, the zero of V/U ; thus $U \subseteq \ker(\pi \circ T)$. By [Theorem 3.9.10](#) (a) there is exactly one linear map $\bar{T}: V/U \rightarrow V/U$ with $\bar{T}(\mathbf{v} + U) = T\mathbf{v} + U$.
 (b) For $j \leq m$, $T\mathbf{u}_j \in U$, so it is a combination of $\mathbf{u}_1, \dots, \mathbf{u}_m$ only, and its $\mathcal{B}$ -coordinates are 0 in the last s places. These first m columns, restricted to their first m entries, are the columns of the matrix A of $T|_U$ (which maps U to U by hypothesis and is linear). For the remaining columns, write $T\mathbf{w}_k = \sum_i b_{ik}\mathbf{u}_i + \sum_l d_{lk}\mathbf{w}_l$. Passing to V/U , where $\mathbf{u}_i + U = U$ is zero, gives $\bar{T}(\mathbf{w}_k + U) = \sum_l d_{lk}(\mathbf{w}_l + U)$. So the $s \times s$ matrix (d_{lk}) is the matrix D of $\bar{T}$ in the basis $(\mathbf{w}_1 + U, \dots, \mathbf{w}_s + U)$ of [Theorem 3.9.5](#), and the bottom-right block of $[T]_{\mathcal{B}}$ is D .

- (c) $T(1) = 0$, $T(x) = 1$, $T(x^2) = 2x$, so $[T]_{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$. Differentiation maps

$U = \mathbb{R}[x]_{\leq 1}$ into itself. The block $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ is the matrix of T on U in $(1, x)$, the lower-left block is $\begin{pmatrix} 0 & 0 \end{pmatrix}$, and the block D is the 1×1 matrix (0) : indeed $\bar{T}(x^2 + U) = 2x + U = U$, the zero coset, since $2x \in U$.

Exercise C3: Counting cosets over a finite field.

Let $F = \mathbb{F}_q$ be a finite field with q elements, V a vector space over F of dimension n , and U a subspace of dimension m .

- (a) Show that $|V| = q^n$.
 (b) Show that every coset $\mathbf{v} + U$ has exactly $|U|$ elements, and deduce $|V/U| = q^{n-m}$ by counting.

(c) Check that (b) agrees with [Theorem 3.9.5](#).

Solution.

- (a) Choose a basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$. By [Theorem 1.5.1](#), $(a_1, \dots, a_n) \mapsto a_1\mathbf{v}_1 + \dots + a_n\mathbf{v}_n$ is a bijection $F^n \rightarrow V$, and F^n has q^n elements.
- (b) The map $U \rightarrow \mathbf{v} + U$, $\mathbf{u} \mapsto \mathbf{v} + \mathbf{u}$, is surjective by [Definition 3.9.2](#) and injective because $\mathbf{v} + \mathbf{u} = \mathbf{v} + \mathbf{u}'$ implies $\mathbf{u} = \mathbf{u}'$. So $|\mathbf{v} + U| = |U| = q^m$ by (a) applied to U . By [Lemma 3.9.3](#) (c), the cosets partition V into $|V/U|$ disjoint sets of size q^m , so $q^n = |V/U| \cdot q^m$, and $|V/U| = q^{n-m}$.
- (c) By [Theorem 3.9.5](#), V/U is a vector space over $\mathbb{F}_q$ of dimension $n - m$, so by (a) it has q^{n-m} elements, in agreement with (b). For example, $\mathbb{F}_2^3$ modulo a line has $8/2 = 4$ cosets.

3.10 Projections and the Trace of an Operator

Chapter 1 ended its discussion of complements with a promise: a splitting $V = U \oplus W$ should give a projection of V onto U along W , and different complements should give different projections. This section keeps that promise. Projections are the operators that are simplest in a well-chosen basis, and computing their matrices there leads to a second topic: the trace of an operator, a number attached to $T \in \mathcal{L}(V)$ that no choice of basis can change. We end with involutions, which split a space into two pieces in the same way.

3.10.1 Projections

Suppose $V = U \oplus W$. By [Definition 1.7.2](#), every $\mathbf{v} \in V$ can be written as $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in U$ and $\mathbf{w} \in W$ in **exactly one** way. So the rule “ $\mathbf{v} \mapsto \mathbf{u}$ ”, keep the U -piece and throw away the W -piece, is a well-defined function $V \rightarrow V$. In $\mathbb{R}^3 = \{\text{the } xy\text{-plane}\} \oplus \{\text{the } z\text{-axis}\}$, it sends (a, b, c) to $(a, b, 0)$: the shadow of a point on the floor when light falls straight down.

What property does such a map have that singles it out? Apply it twice. The output $\mathbf{u}$ already lies in U , and its splitting is $\mathbf{u} = \mathbf{u} + \mathbf{0}$, so applying the rule again returns $\mathbf{u}$. Doing it twice is the same as doing it once. We take this as the definition, because it can be checked without knowing U and W in advance.

A projection is an operator that leaves alone everything it has already produced.

Definition 3.10.1 — Projection

Let V be a vector space over F . An operator $P \in \mathcal{L}(V)$ is a **projection** if

$$P^2 = P,$$

that is, $P(P\mathbf{v}) = P\mathbf{v}$ for every $\mathbf{v} \in V$.

In words: P^2 is the composite PP (Definition 3.4.2), so the condition says that P fixes each of its own outputs. Equivalently, $P\mathbf{x} = \mathbf{x}$ for every $\mathbf{x} \in \text{im } P$: if $\mathbf{x} = P\mathbf{v}$, then $P\mathbf{x} = P^2(\mathbf{v}) = P\mathbf{v} = \mathbf{x}$, and conversely this for $\mathbf{x} = P\mathbf{v}$ is $P^2(\mathbf{v}) = P\mathbf{v}$. No inner product and no notion of angle is involved. The projections of this section are often called **oblique** projections, to distinguish them from the orthogonal projections met later, in the chapters on inner product spaces.

Examples.

- **Onto an axis.** $P(x, y) = (x, 0)$ on F^2 . Then $P^2(x, y) = P(x, 0) = (x, 0)$, so $P^2 = P$.
- **A rank-one projection.** $T(x, y) = (2x - y, 2x - y)$ on $\mathbb{R}^2$ from Example (Making a matrix diagonal with a well-chosen basis). We saw there that its matrix A satisfies $A^2 = A$; directly, $T(2x - y, 2x - y) = (4x - 2y - 2x + y, \dots) = (2x - y, 2x - y)$.
- **Constant term.** On $F[x]_{\leq n}$, $P(p) = p(0)$, the constant polynomial with value $p(0)$. The constant polynomial c has value c at 0, so $P(P(p)) = P(p)$.
- **Degenerate cases.** The identity id_V and the zero operator are projections, since $\text{id}_V^2 = \text{id}_V$ and $0^2 = 0$. They correspond to keeping everything and keeping nothing.

Non-example by minimal change. Change the first example to $T(x, y) = (2x, 0)$ on $\mathbb{R}^2$. Its image is still the x -axis and its kernel is still the y -axis. But $T^2(x, y) = (4x, 0) \neq T(x, y)$ when $x \neq 0$: T does not fix the vectors of its own image, since $T(1, 0) = (2, 0)$. So having “the right image and kernel” is not enough; the clause $P^2 = P$ is what forces P to be the identity on its image. Likewise the reflection $R(x, y) = (y, x)$ is not a projection: $R^2 = \text{id} \neq R$.

Here is the promised correspondence. It says that projections and direct sum decompositions are two descriptions of the same thing.

Theorem 3.10.1 — Projections and Direct Sums

Let V be a vector space.

- (a) If $P \in \mathcal{L}(V)$ is a projection, then

$$V = \text{im } P \oplus \ker P,$$

P is the identity on $\text{im } P$, and P is zero on $\ker P$.

- (b) Conversely, suppose $V = U \oplus W$. Then there is **exactly one** operator $P \in \mathcal{L}(V)$ with $P\mathbf{u} = \mathbf{u}$ for all $\mathbf{u} \in U$ and $P\mathbf{w} = \mathbf{0}$ for all $\mathbf{w} \in W$, namely $P(\mathbf{u} + \mathbf{w}) = \mathbf{u}$. It is a projection, with $\text{im } P = U$ and $\ker P = W$. It is called the **projection onto U along W** .

Idea. For (a), every vector splits as “what P keeps” plus “what P removes”: $\mathbf{v} = P\mathbf{v} + (\mathbf{v} - P\mathbf{v})$. The first piece is in the image by definition, and $P^2 = P$ is exactly what puts the second piece in the kernel. For (b), uniqueness of the splitting makes the rule a function, and uniqueness again makes it linear.

Proof.

- (a) Let $\mathbf{v} \in V$. Then $\mathbf{v} = P\mathbf{v} + (\mathbf{v} - P\mathbf{v})$, where $P\mathbf{v} \in \text{im } P$ and $P(\mathbf{v} - P\mathbf{v}) = P\mathbf{v} - P^2(\mathbf{v}) = \mathbf{0}$ because $P^2 = P$. Hence $V = \text{im } P + \ker P$. If $\mathbf{x} \in \text{im } P \cap \ker P$, then $\mathbf{x} = P\mathbf{x}$, as shown after Definition 3.10.1, and $P\mathbf{x} = \mathbf{0}$; so $\mathbf{x} = \mathbf{0}$. By Theorem 1.7.4, $V = \text{im } P \oplus \ker P$. We have also seen that P fixes $\text{im } P$, and P is zero on $\ker P$ by definition.

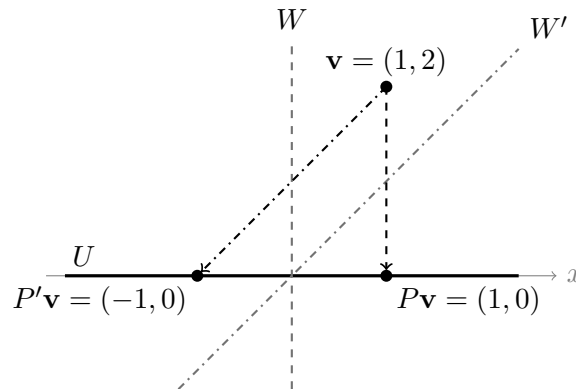
(b) *Existence.* By Definition 1.7.2, each $\mathbf{v} \in V$ has exactly one splitting $\mathbf{v} = \mathbf{u} + \mathbf{w}$, so $P\mathbf{v} := \mathbf{u}$ defines a function $V \rightarrow V$. It is linear: if $\mathbf{v} = \mathbf{u} + \mathbf{w}$ and $\mathbf{v}' = \mathbf{u}' + \mathbf{w}'$, then $\mathbf{v} + \mathbf{v}' = (\mathbf{u} + \mathbf{u}') + (\mathbf{w} + \mathbf{w}')$ and $a\mathbf{v} = a\mathbf{u} + a\mathbf{w}$ are splittings, because U and W are subspaces, so by uniqueness $P(\mathbf{v} + \mathbf{v}') = \mathbf{u} + \mathbf{u}' = P\mathbf{v} + P\mathbf{v}'$ and $P(a\mathbf{v}) = a\mathbf{u} = aP\mathbf{v}$. For $\mathbf{u} \in U$ the splitting is $\mathbf{u} = \mathbf{u} + \mathbf{0}$, so $P\mathbf{u} = \mathbf{u}$; for $\mathbf{w} \in W$ it is $\mathbf{w} = \mathbf{0} + \mathbf{w}$, so $P\mathbf{w} = \mathbf{0}$.

Properties. Every value $P\mathbf{v}$ lies in U , and every $\mathbf{u} \in U$ is $P\mathbf{u}$; so $\text{im } P = U$. Then $P(P\mathbf{v}) = P\mathbf{v}$ because P fixes U ; so $P^2 = P$. Finally $P(\mathbf{u} + \mathbf{w}) = \mathbf{u}$ is $\mathbf{0}$ exactly when $\mathbf{u} = \mathbf{0}$, that is, when $\mathbf{v} = \mathbf{w} \in W$; so $\ker P = W$.

Uniqueness. If $P' \in \mathcal{L}(V)$ is identity on U and zero on W , then $P'(\mathbf{u} + \mathbf{w}) = P'\mathbf{u} + P'\mathbf{w} = \mathbf{u}$ by linearity, so $P' = P$. ■

So a projection is a direct sum decomposition $V = U \oplus W$, recorded as an operator: U is its image and W its kernel. Given P , the decomposition is $\text{im } P \oplus \ker P$; given the decomposition, P is the projection onto U along W . The two constructions undo each other, by the “exactly one” in (b).

Now the promise from Chapter 1. There, the x -axis U in $\mathbb{R}^2$ had many complements, for instance the y -axis W and the line W' spanned by $(1, 1)$. Each gives a projection onto the same line U .



To project $\mathbf{v}$ onto U along a complement, slide $\mathbf{v}$ parallel to that complement until it hits U . Along the y -axis, $(1, 2) = (1, 0) + (0, 2)$ gives $P\mathbf{v} = (1, 0)$ (dashed arrow). Along W' , $(1, 2) = (-1, 0) + (2, 2)$ gives $P'\mathbf{v} = (-1, 0)$ (dash-dotted arrow). In general $P(x, y) = (x, 0)$ and $P'(x, y) = (x - y, 0)$, with standard matrices $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $\begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$.

Warning. “Projection onto U ” is incomplete: the direction “along W ” matters. The two operators above are both projections, both have image U , and they are different: $P(1, 2) = (1, 0)$ but $P'(1, 2) = (-1, 0)$. Since U has infinitely many complements in $\mathbb{R}^2$, it is the image of infinitely many projections. A projection is determined by its image **and** its kernel together.

In finite dimension, the direct sum $V = \text{im } P \oplus \ker P$ hands us a basis in which P is as simple as possible.

Corollary 3.10.2 — The matrix of a projection

Let V be finite-dimensional and $P \in \mathcal{L}(V)$ a projection with $\text{rank } P = r$. Let $(\mathbf{u}_1, \dots, \mathbf{u}_r)$ be a basis of $\text{im } P$ and $(\mathbf{w}_1, \dots, \mathbf{w}_s)$ a basis of $\ker P$. Then $\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_r, \mathbf{w}_1, \dots, \mathbf{w}_s)$ is a basis of V , and

$$[P]_{\mathcal{B}} = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} = \text{diag}(1, \dots, 1, 0, \dots, 0),$$

with r ones.

Proof. By Theorem 3.10.1 (a), $V = \text{im } P \oplus \ker P$, so the concatenated list $\mathcal{B}$ is a basis of V by Theorem 1.7.5 ((a) $\Rightarrow$ (d)). Since $P\mathbf{u}_i = \mathbf{u}_i$ and $P\mathbf{w}_j = \mathbf{0}$, the columns of $[P]_{\mathcal{B}}$ are $\mathbf{e}_1, \dots, \mathbf{e}_r$ followed by s zero columns (Definition 3.6.1). ■

Conversely, an operator whose matrix in some basis is $\text{diag}(1, \dots, 1, 0, \dots, 0)$ is a projection, since that matrix squares to itself. With the change-of-basis square, this gives a way to write down any projection in standard coordinates.

Example. (Projecting onto a plane along a line)

In $\mathbb{R}^3$, let $U = \{(x, y, z) : x + y + z = 0\}$ and $L = \text{span}((1, 1, 1))$, so that $\mathbb{R}^3 = U \oplus L$ by Example (A plane and a line in $\mathbb{R}^3$). Find the standard matrix of the projection P onto U along L , and check that it squares to itself.

Solution. Direct route. By Example (A plane and a line in $\mathbb{R}^3$), (a, b, c) splits as $(a - m, b - m, c - m) + m(1, 1, 1)$ with $m = \frac{a+b+c}{3}$, and the first piece lies in U . So

$$P(a, b, c) = \left(\frac{2a-b-c}{3}, \frac{-a+2b-c}{3}, \frac{-a-b+2c}{3} \right), \quad [P]_{\mathcal{E}} = \frac{1}{3} \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}.$$

Adapted basis. $(1, -1, 0)$ and $(0, -1, 1)$ lie in U and are independent (look at the first and third entries), so they form a basis of the 2-dimensional plane U . With $\mathcal{B} = ((1, -1, 0), (0, -1, 1), (1, 1, 1))$, Corollary 3.10.2 gives $[P]_{\mathcal{B}} = \text{diag}(1, 1, 0)$, and by Theorem 3.7.3 $[P]_{\mathcal{E}} = M \text{diag}(1, 1, 0) M^{-1}$ with $M = [\text{id}]_{\mathcal{B}}^{\mathcal{E}}$, whose columns are the vectors of $\mathcal{B}$. Multiplying out gives the same matrix as the direct route.

Check. Let A be the matrix found. The first column of A^2 is A applied to $\frac{1}{3}(2, -1, -1)$, which is $\frac{1}{9}(4 + 1 + 1, -2 - 2 + 1, -2 + 1 - 2) = \frac{1}{3}(2, -1, -1)$, the first column of A ; the other columns work the same way by symmetry of the entries. Also $A(1, 1, 1) = \mathbf{0}$ and $A(1, -1, 0) = (1, -1, 0)$, as a projection onto U along L must satisfy. The division by 3 needs $3 \neq 0$ in the field, which holds over $\mathbb{R}$.

Quick check. Is $P = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ (acting on $\mathbb{R}^2$) a projection? If so, onto which line, and along which line?

Solution. $P^2 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = P$, so yes. Its image is spanned by its columns, $\text{span}((1, 0))$, the x -axis. Its kernel is $\{x + y = 0\} = \text{span}((1, -1))$. By [Theorem 3.10.1](#), it is the projection onto the x -axis along the line $y = -x$.

3.10.2 The trace of an operator

For a projection of rank r , the diagonal matrix of [Corollary 3.10.2](#) has trace $1 + \cdots + 1 = r \cdot 1$. But traces are defined for matrices, and the same projection has many other matrices, such as the 3×3 matrix of [Example \(Projecting onto a plane along a line\)](#), whose trace is $\frac{2+2+2}{3} = 2$. The agreement is no accident. In [Theorem 3.7.5](#) we proved that similar matrices have the same trace, and by [Theorem 3.7.4](#) all matrices of one operator are similar. So the trace belongs to the operator.

The trace of an operator is the trace of its matrix in any basis; the basis does not matter.

Definition 3.10.2 — Trace of an operator

Let V be a finite-dimensional vector space over F with $\dim V \geq 1$, and let $T \in \mathcal{L}(V)$. The **trace** of T is

$$\text{tr } T := \text{tr}[T]_{\mathcal{B}},$$

where $\mathcal{B}$ is **any** basis of V . (For $V = \{0\}$ we set $\text{tr } T = 0$.)

In words: to find $\text{tr } T$, choose a basis, write down the matrix of T with that **same** basis on input and output, and add the diagonal entries.

Well-definedness. The definition names a basis but claims the answer does not depend on it. Let $\mathcal{B}$ and $\mathcal{C}$ be bases of V . By [Theorem 3.7.3](#), $[T]_{\mathcal{C}} = P^{-1}[T]_{\mathcal{B}}P$ with $P = [\text{id}]_{\mathcal{C}}^{\mathcal{B}}$ invertible, so the two matrices are similar, and by [Theorem 3.7.5](#) they have the same trace. Hence $\text{tr } T$ depends only on T .

Examples.

- **Identity.** In every basis, $[\text{id}_V]_{\mathcal{B}} = I_n$, so $\text{tr id}_V = n \cdot 1$, where $n = \dim V$. Over $\mathbb{R}$ this is n ; over $\mathbb{F}_2$ it is 0 when n is even.
- **Differentiation.** On $F[x]_{\leq n}$ with the basis $(1, x, \dots, x^n)$, $D(x^k) = kx^{k-1}$ has no x^k -component, so every diagonal entry of the matrix is 0 and $\text{tr } D = 0$.
- **A shift of the variable.** On $\mathbb{R}[x]_{\leq 2}$, let $T(p) = p(x+1)$. In $(1, x, x^2)$, $T(1) = 1$, $T(x) = 1 + x$, $T(x^2) = 1 + 2x + x^2$, so the matrix is $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$ and $\text{tr } T = 3$.
- **The reflection.** $R(x, y) = (y, x)$ has trace 0, whether computed from $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ or from $\text{diag}(1, -1)$ ([Example \(The reflection in a mirror-adapted basis\)](#)).

Warning. The trace needs the same basis on both sides. For an operator $T \in \mathcal{L}(V)$, the matrix $[T]_{\mathcal{B}}^{\mathcal{C}}$ with $\mathcal{B} \neq \mathcal{C}$ is square, but its trace depends on the bases. For id on $\mathbb{R}^2$, with $\mathcal{E}$ the standard basis and $\mathcal{C} = ((2, 0), (0, 1))$, we get $[\text{id}]_{\mathcal{E}}^{\mathcal{C}} = \text{diag}(\frac{1}{2}, 1)$, of trace $\frac{3}{2}$, while $\text{tr id} = 2$. Different bases on input and output give equivalent matrices,

not similar ones, and equivalence does not preserve the trace. For the same reason there is no trace of a map $V \rightarrow W$ between different spaces.

The rules for the trace of a matrix carry over at once.

Theorem 3.10.3 — Properties of the Trace of an Operator

Let V be finite-dimensional with $n = \dim V$, let $S, T \in \mathcal{L}(V)$ and $c \in F$. Then

$$\operatorname{tr}(S + T) = \operatorname{tr} S + \operatorname{tr} T, \quad \operatorname{tr}(cT) = c \operatorname{tr} T, \quad \operatorname{tr}(ST) = \operatorname{tr}(TS), \quad \operatorname{tr} \operatorname{id}_V = n \cdot 1.$$

Proof. For $n = 0$ all traces are 0. Otherwise fix a basis $\mathcal{B}$ of V . The map $T \mapsto [T]_{\mathcal{B}}$ is linear by Theorem 3.6.4, so $[S + T]_{\mathcal{B}} = [S]_{\mathcal{B}} + [T]_{\mathcal{B}}$ and $[cT]_{\mathcal{B}} = c[T]_{\mathcal{B}}$; and $[ST]_{\mathcal{B}} = [S]_{\mathcal{B}}[T]_{\mathcal{B}}$ by Theorem 3.6.2, and likewise for TS . The first three identities now follow from Theorem 0.9.6, since $\operatorname{tr} T$ may be computed in $\mathcal{B}$. The last was computed in the examples. ■

A first consequence: on a finite-dimensional space V of dimension $n \geq 1$ over a field in which $n \cdot 1 \neq 0$, such as $\mathbb{R}$, there are no operators with $ST - TS = \operatorname{id}_V$, since the left side has trace 0 and the right side has trace $n \cdot 1$.

Now the trace of a projection. Over $\mathbb{R}$ it is simply the rank, but the precise statement is an equation in the field, and reading it as an equation of integers needs a hypothesis.

Theorem 3.10.4 — Trace of a Projection

Let V be a finite-dimensional vector space over F and $P \in \mathcal{L}(V)$ a projection. Then

$$\operatorname{tr} P = (\operatorname{rank} P) \cdot 1 \quad \text{in } F.$$

If F has characteristic 0, or characteristic $p > \dim V$, then $\operatorname{rank} P$ is determined by $\operatorname{tr} P$: it is the unique integer r with $0 \leq r \leq \dim V$ and $r \cdot 1 = \operatorname{tr} P$. In particular $\operatorname{rank} P = \operatorname{tr} P$ for projections on real or complex spaces.

Proof. Let $r = \operatorname{rank} P$. If $V = \{\mathbf{0}\}$, both sides are 0. Otherwise, by Corollary 3.10.2 there is a basis $\mathcal{B}$ with $[P]_{\mathcal{B}} = \operatorname{diag}(1, \dots, 1, 0, \dots, 0)$ with r ones, so $\operatorname{tr} P = r \cdot 1$ by Definition 3.10.2.

For the second statement, let $n = \dim V$. Since $0 \leq r \leq n$ and $r \cdot 1 = \operatorname{tr} P$, it remains to show uniqueness. Suppose $k \cdot 1 = l \cdot 1$ for integers $0 \leq l \leq k \leq n$. By Lemma 0.7.4, $(k - l) \cdot 1 + l \cdot 1 = k \cdot 1 = l \cdot 1$, and adding $-(l \cdot 1)$ to both sides gives $(k - l) \cdot 1 = 0$. If $k - l > 0$, then by Definition 0.7.2 the characteristic of F is non-zero and at most $k - l \leq n$, which contradicts the hypothesis. Hence $k = l$. In $\mathbb{R}$ and $\mathbb{C}$, $r \cdot 1$ is the number r . ■

The hypothesis cannot be dropped. Over $\mathbb{F}_2$, the identity of $\mathbb{F}_2^2$ is a projection of rank 2, and

its trace is $1 + 1 = 0$, the same as the trace of the zero projection, of rank 0. The equation $\text{tr } P = (\text{rank } P) \cdot 1$ still holds there; what fails is recovering the integer rank P from the field element $\text{tr } P$.

3.10.3 Involutions and reflections

A projection satisfies $P^2 = P$. The other simple polynomial relation is $T^2 = \text{id}_V$: doing T twice changes nothing. Such an operator is called an **involution**. Examples are the reflection $R(x, y) = (y, x)$, the transpose $A \mapsto A^T$ on $M_n(F)$, and $f(x) \mapsto f(-x)$ on functions $\mathbb{R} \rightarrow \mathbb{R}$. In Chapter 1 the last two produced direct sums: symmetric plus skew-symmetric matrices ([Example \(Symmetric and skew-symmetric matrices\)](#)), and even plus odd functions ([Example \(Even and odd functions\)](#)). The next theorem explains both at once. For an involution T , write

$$E_+ := \{\mathbf{v} \in V : T\mathbf{v} = \mathbf{v}\}, \quad E_- := \{\mathbf{v} \in V : T\mathbf{v} = -\mathbf{v}\}.$$

These are subspaces, since $E_+ = \ker(T - \text{id}_V)$ and $E_- = \ker(T + \text{id}_V)$ ([Theorem 3.2.1](#)).

Theorem 3.10.5 — Involutions Split the Space

Let F be a field of characteristic not 2, V a vector space over F , and $T \in \mathcal{L}(V)$ with $T^2 = \text{id}_V$. Then

$$V = E_+ \oplus E_-,$$

and the projection onto E_+ along E_- is $\frac{1}{2}(\text{id}_V + T)$.

Idea. Copy the even/odd split: $\mathbf{v} = \frac{1}{2}(\mathbf{v} + T\mathbf{v}) + \frac{1}{2}(\mathbf{v} - T\mathbf{v})$. Applying T swaps $\mathbf{v}$ and $T\mathbf{v}$, so the first piece is fixed and the second changes sign. Both halves of the argument divide by 2, which is where the characteristic enters.

Proof. Since the characteristic is not 2, $2 = 1 + 1 \neq 0$ in F ([Definition 0.7.2](#)), so $\frac{1}{2}$ exists. Let $\mathbf{v} \in V$, and put $\mathbf{v}_+ = \frac{1}{2}(\mathbf{v} + T\mathbf{v})$ and $\mathbf{v}_- = \frac{1}{2}(\mathbf{v} - T\mathbf{v})$. Then $\mathbf{v} = \mathbf{v}_+ + \mathbf{v}_-$, and by linearity and $T^2 = \text{id}_V$,

$$T\mathbf{v}_+ = \frac{1}{2}(T\mathbf{v} + \mathbf{v}) = \mathbf{v}_+, \quad T\mathbf{v}_- = \frac{1}{2}(T\mathbf{v} - \mathbf{v}) = -\mathbf{v}_-.$$

So $\mathbf{v}_+ \in E_+$, $\mathbf{v}_- \in E_-$, and $V = E_+ + E_-$. If $\mathbf{v} \in E_+ \cap E_-$, then $\mathbf{v} = T\mathbf{v} = -\mathbf{v}$, so $2\mathbf{v} = \mathbf{0}$ and $\mathbf{v} = \frac{1}{2}(2\mathbf{v}) = \mathbf{0}$. By [Theorem 1.7.4](#), $V = E_+ \oplus E_-$.

The operator $\frac{1}{2}(\text{id}_V + T)$ is linear by [Theorem 3.4.1](#), sends $\mathbf{v}$ to $\mathbf{v}_+$, fixes E_+ (where $T\mathbf{v} = \mathbf{v}$) and kills E_- (where $T\mathbf{v} = -\mathbf{v}$). By the uniqueness in [Theorem 3.10.1](#) (b), it is the projection onto E_+ along E_- . ■

For the transpose, E_+ is the symmetric and E_- the skew-symmetric matrices; for $f(x) \mapsto f(-x)$, they are the even and odd functions; for the reflection $R(x, y) = (y, x)$, they are the mirror line $y = x$ and the perpendicular line $y = -x$, and the adapted basis $((1, 1), (1, -1))$ is the one of [Example \(The reflection in a mirror-adapted basis\)](#). In finite dimension, a basis of E_+ followed by a basis of E_- gives $[T]_{\mathcal{B}} = \text{diag}(1, \dots, 1, -1, \dots, -1)$, and so $\text{tr } T = (\dim E_+) \cdot 1 - (\dim E_-) \cdot 1$. In Chapter 8 the non-zero vectors of E_+ and E_- will be called eigenvectors with eigenvalues 1 and -1 .

Warning. In characteristic 2 an involution need not split the space. Over $\mathbb{F}_2$, let $T\mathbf{x} = J\mathbf{x}$ with $J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Then $J^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = I_2$, since $2 = 0$. But $-1 = 1$, so $E_- = E_+ = \ker(J - I_2) = \text{span}(\mathbf{e}_1)$, and $E_+ + E_- = \text{span}(\mathbf{e}_1) \neq \mathbb{F}_2^2$.

3.10.4 Exercises

3.10.4.1 A. Check your understanding

Exercise A1.

- Define a projection $P \in \mathcal{L}(V)$, and state the correspondence between projections and direct sum decompositions.
- True or false: a projection is determined by its image. Justify your answer.
- True or false: an operator T on $\mathbb{R}^2$ with $\text{im } T$ the x -axis and $\ker T$ the y -axis is a projection. Justify your answer.
- Explain why $\text{tr } T$ does not depend on the basis used to compute it.
- True or false: for every field F and every projection P on a finite-dimensional space over F , the integer $\text{rank } P$ can be recovered from $\text{tr } P \in F$. Justify your answer.
- Where does the hypothesis “characteristic not 2” enter the proof of [Theorem 3.10.5](#)?

Solution.

- P is a projection if $P^2 = P$ ([Definition 3.10.1](#)). By [Theorem 3.10.1](#), a projection gives $V = \text{im } P \oplus \ker P$, and each decomposition $V = U \oplus W$ gives exactly one projection with image U and kernel W .
- False. On $\mathbb{R}^2$, $P(x, y) = (x, 0)$ and $P'(x, y) = (x - y, 0)$ are different projections with the same image, the x -axis. A projection is determined by its image and its kernel together.
- False. $T(x, y) = (2x, 0)$ has this image and kernel, but $T^2 \neq T$.
- Matrices of T in two bases are similar by [Theorem 3.7.3](#), and similar matrices have equal traces by [Theorem 3.7.5](#).
- False. Over $\mathbb{F}_2$, id on $\mathbb{F}_2^2$ has rank 2 and the zero operator has rank 0, but both have trace 0. It is true when the characteristic is 0 or larger than $\dim V$ ([Theorem 3.10.4](#)).
- Twice: to define $\mathbf{v}_\pm = \frac{1}{2}(\mathbf{v} \pm T\mathbf{v})$, and to pass from $2\mathbf{v} = \mathbf{0}$ to $\mathbf{v} = \mathbf{0}$. Both need $2 \neq 0$ in F .

3.10.4.2 B. Practice

Exercise B1: A projection in $\mathbb{R}^3$.

Let $U = \{(x, y, z) \in \mathbb{R}^3 : x + y - z = 0\}$ and $L = \text{span}((1, 1, 1))$.

- Show that $\mathbb{R}^3 = U \oplus L$.

- (b) Find the standard matrix A of the projection P onto U along L .
 (c) Verify that $A^2 = A$, and that $\text{tr } A = \text{rank } A$.

Solution.

- (a) U is the kernel of $(x, y, z) \mapsto x + y - z$, which is non-zero; its image is then a non-zero subspace of $\mathbb{R}$, hence all of $\mathbb{R}$ (Theorem 1.6.8), so $\dim U = 3 - 1 = 2$ by Theorem 3.3.1. If $t(1, 1, 1) \in U$, then $t + t - t = t = 0$, so $U \cap L = \{\mathbf{0}\}$. By Theorem 1.7.4, $\dim(U + L) = 2 + 1 = 3$, so $U + L = \mathbb{R}^3$ by Theorem 1.6.8, and $\mathbb{R}^3 = U \oplus L$.
 (b) Split $(a, b, c) = \mathbf{u} + t(1, 1, 1)$ with $\mathbf{u} = (a - t, b - t, c - t) \in U$: the condition $(a - t) + (b - t) - (c - t) = 0$ gives $t = a + b - c$. So

$$P(a, b, c) = (-b + c, -a + c, -a - b + 2c), \quad A = \begin{pmatrix} 0 & -1 & 1 \\ -1 & 0 & 1 \\ -1 & -1 & 2 \end{pmatrix}.$$

Check: $A(1, 1, 1) = \mathbf{0}$, and for the vectors $(1, 0, 1), (0, 1, 1) \in U$, $A(1, 0, 1) = (1, 0, 1)$ and $A(0, 1, 1) = (0, 1, 1)$.

- (c) Row by row,

$$A^2 = \begin{pmatrix} 0+1-1 & 0+0-1 & 0-1+2 \\ 0+0-1 & 1+0-1 & -1+0+2 \\ 0+1-2 & 1+0-2 & -1-1+4 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 1 \\ -1 & 0 & 1 \\ -1 & -1 & 2 \end{pmatrix} = A.$$

$\text{tr } A = 0 + 0 + 2 = 2$, and $\text{rank } A = \text{rank } P = \dim U = 2$ by Theorem 3.10.1 (b), in agreement with Theorem 3.10.4.

Exercise B2: The complementary projection.

Let $P \in \mathcal{L}(V)$ be a projection.

- (a) Prove that $\text{id}_V - P$ is a projection.
 (b) Prove that $\text{im}(\text{id}_V - P) = \ker P$ and $\ker(\text{id}_V - P) = \text{im } P$.
 (c) Deduce that if P is the projection onto U along W , then $\text{id}_V - P$ is the projection onto W along U .

Solution.

- (a) By Theorem 3.4.2 (composition distributes over sums of maps), $(\text{id}_V - P)^2 = \text{id}_V - 2P + P^2 = \text{id}_V - 2P + P = \text{id}_V - P$.
 (b) $(\subseteq)$ $P((\text{id}_V - P)(\mathbf{v})) = P\mathbf{v} - P^2(\mathbf{v}) = \mathbf{0}$, so $\text{im}(\text{id}_V - P) \subseteq \ker P$. $(\supseteq)$ If $P\mathbf{v} = \mathbf{0}$, then $\mathbf{v} = (\text{id}_V - P)(\mathbf{v}) \in \text{im}(\text{id}_V - P)$. For the kernel: $(\text{id}_V - P)(\mathbf{v}) = \mathbf{0}$ means $\mathbf{v} = P\mathbf{v}$. Such $\mathbf{v}$ lies in $\text{im } P$, and conversely every $\mathbf{x} \in \text{im } P$ satisfies $P\mathbf{x} = \mathbf{x}$, as shown after Definition 3.10.1. So $\ker(\text{id}_V - P) = \text{im } P$.
 (c) By Theorem 3.10.1 (b), $\text{im } P = U$ and $\ker P = W$. By (a) and (b), $\text{id}_V - P$ is a projection with image W and kernel U . By the uniqueness in Theorem 3.10.1 (b), applied to $V = W \oplus U$, it is the projection onto W along U .

Exercise B3: Traces of operators on matrices.

- (a) Let $\tau: M_n(F) \rightarrow M_n(F)$, $\tau(A) = A^T$. Show that $\text{tr } \tau = n \cdot 1$.
- (b) Fix $B \in M_2(F)$, and let $L_B: M_2(F) \rightarrow M_2(F)$, $L_B(X) = BX$. Show that $\text{tr } L_B = 2 \text{tr } B$.

Hint: use the basis of matrix units E_{ij} .

Solution.

- (a) Order the basis (E_{ij}) of $M_n(F)$ in any way. Since $\tau(E_{ij}) = E_{ji}$, the coordinate column of $\tau(E_{ij})$ has a 1 in the position of E_{ji} and 0 elsewhere. The diagonal entry of the matrix in the column of E_{ij} is therefore 1 if $E_{ji} = E_{ij}$, that is $i = j$, and 0 otherwise. There are n basis vectors E_{ii} , so $\text{tr } \tau = n \cdot 1$ by [Definition 3.10.2](#). (Consistently with [Theorem 3.10.5](#) over $\mathbb{R}$: symmetric matrices have dimension $\frac{n(n+1)}{2}$, skew-symmetric ones $\frac{n(n-1)}{2}$, and the difference is n .)
- (b) Use the basis $(E_{11}, E_{12}, E_{21}, E_{22})$ and write $B = (b_{ij})$. The product BE_{ij} has the i -th column of B in column j and zeros elsewhere, so $BE_{ij} = b_{1i}E_{1j} + b_{2i}E_{2j}$. The diagonal entry in the column of E_{ij} is the coefficient of E_{ij} itself, which is b_{ii} . Adding over the four basis vectors, $\text{tr } L_B = b_{11} + b_{11} + b_{22} + b_{22} = 2 \text{tr } B$.

3.10.4.3 C. Going deeper**Exercise C1: When is a sum of projections a projection?**

Let F have characteristic not 2, and let $P, Q \in \mathcal{L}(V)$ be projections.

- (a) Prove that $P + Q$ is a projection if and only if $PQ = QP = 0$.
- (b) Show that (a) fails over $\mathbb{F}_2$.
- (c) Suppose V is finite-dimensional over $\mathbb{R}$ and $P + Q$ is a projection. Deduce that $\text{rank}(P + Q) = \text{rank } P + \text{rank } Q$.

Hint: for $(\Rightarrow)$ in (a), use the relation $PQ + QP = 0$ together with $P^2 = P$.

Solution.

- (a) By [Theorem 3.4.2](#), $(P + Q)^2 = P^2 + PQ + QP + Q^2 = P + Q + PQ + QP$.
 $(\Leftarrow)$ If $PQ = QP = 0$, this is $P + Q$.
 $(\Rightarrow)$ If $(P + Q)^2 = P + Q$, then $PQ + QP = 0$. Multiplying on the left by P and using $P^2 = P$ gives $PQ + PQP = 0$; multiplying on the right by P gives $PQP + QP = 0$. Subtracting, $PQ - QP = 0$, so $PQ = QP$. Then $2PQ = PQ + QP = 0$, and since $2 \neq 0$ in F , $PQ = 0$. Hence also $QP = 0$.
- (b) Over $\mathbb{F}_2$, let $P = Q = \text{id}_V$ with $V \neq \{0\}$. Then $P + Q = 2 \text{id}_V = 0$, which is a projection, but $PQ = \text{id}_V \neq 0$. The step from $2PQ = 0$ to $PQ = 0$ is the one that fails.
- (c) By [Theorem 3.10.4](#) over $\mathbb{R}$, and [Theorem 3.10.3](#),

$$\text{rank}(P + Q) = \text{tr}(P + Q) = \text{tr } P + \text{tr } Q = \text{rank } P + \text{rank } Q.$$

Exercise C2: The trace is the only trace-like functional.

Let $n \geq 1$, and let $\varphi: M_n(F) \rightarrow F$ be linear with $\varphi(AB) = \varphi(BA)$ for all $A, B \in M_n(F)$.

- (a) Using $E_{ij}E_{jk} = E_{ik}$ and $E_{jk}E_{ij} = 0$ for $k \neq i$, show that $\varphi(E_{ik}) = 0$ whenever $i \neq k$.
- (b) Show that $\varphi(E_{ii}) = \varphi(E_{jj})$ for all i, j .
- (c) Deduce that $\varphi = c \operatorname{tr}$ for some $c \in F$, and conversely that every $c \operatorname{tr}$ has the property.

Solution. Recall that $E_{ij}E_{kl}$ is E_{il} if $j = k$ and 0 otherwise, by [Definition 0.9.9](#).

- (a) Let $i \neq k$. Then $E_{ik} = E_{ii}E_{ik}$ and $E_{ik}E_{ii} = 0$, because $k \neq i$. By the hypothesis and linearity, $\varphi(E_{ik}) = \varphi(E_{ii}E_{ik}) = \varphi(E_{ik}E_{ii}) = \varphi(0) = 0$.
- (b) $E_{ii} = E_{ij}E_{ji}$ and $E_{jj} = E_{ji}E_{ij}$. Hence $\varphi(E_{ii}) = \varphi(E_{ij}E_{ji}) = \varphi(E_{ji}E_{ij}) = \varphi(E_{jj})$.
- (c) Let $c = \varphi(E_{11})$. For $A = (a_{ij}) = \sum_{i,j} a_{ij}E_{ij}$, linearity with (a) and (b) gives

$$\varphi(A) = \sum_{i,j} a_{ij}\varphi(E_{ij}) = \sum_i a_{ii}\varphi(E_{ii}) = c \sum_i a_{ii} = c \operatorname{tr} A.$$

Conversely, $c \operatorname{tr}$ is linear and $c \operatorname{tr}(AB) = c \operatorname{tr}(BA)$ by [Theorem 0.9.6](#). So the linear functionals with $\varphi(AB) = \varphi(BA)$ are exactly the multiples of the trace.

Chapter 4

Duality

A linear map $V \rightarrow F$ turns each vector into a single number. The i -th coordinate in a basis does this, and so do evaluating a polynomial at a point, integrating a function, and taking the trace of a matrix. These **linear functionals** are the measurements we can make on a vector space, and together they form a vector space of their own, the dual space V^* .

The dual space mirrors V with every inclusion reversed. A subspace corresponds to the functionals that vanish on it. An injective map corresponds to a surjective one. The kernel of a map corresponds to the image of its dual map, whose matrix is the transpose. Passing to the dual turns a hard question into an easy one surprisingly often. It also explains facts we already know from a new angle: why a subspace of F^n is always the solution set of a homogeneous system, and why row rank equals column rank.

What you need. From Chapter 1, bases, dimension and complements ([Definition 1.5.1](#), [Definition 1.6.1](#), [Theorem 1.7.6](#)), and bases of infinite-dimensional spaces ([Theorem 1.8.2](#)) for the remarks on infinite dimension. From Chapter 2, the row space and null space of a matrix ([Definition 2.5.2](#), [Definition 2.5.3](#)) and inverses by row reduction ([Theorem 2.4.6](#)). From Chapter 3, linear maps and the space $\mathcal{L}(V, W)$ ([Definition 3.1.1](#), [Theorem 3.4.1](#)), the Rank–Nullity Theorem ([Theorem 3.3.1](#)), the matrix of a map and change of basis ([Definition 3.6.1](#), [Theorem 3.7.1](#)), and quotient spaces with the First Isomorphism Theorem ([Definition 3.9.3](#), [Theorem 3.9.7](#)).

4.0.1 Roadmap

- **Linear functionals and the dual space.** Measurements as vectors; dual bases; separating vectors by functionals.
- **Annihilators.** The functionals that vanish on a subspace, and the formula $\dim U + \dim U^\circ = \dim V$.
- **The double dual.** A natural isomorphism $V \rightarrow V^{**}$ that needs no basis, and what “natural” means.
- **The dual map and the transpose.** Pulling functionals back along a map; its matrix is the transpose; kernels and images trade places.
- **Duality for subspaces and quotients.** The dual of a subspace is a quotient of the dual, and the dual of a quotient is an annihilator; hyperplanes and codimension.
- **Hyperplanes and linear systems revisited.** Every subspace is cut out by equations; converting between parametric and implicit descriptions.

4.0.2 Named moves

- **Test against functionals.** To show a vector is zero, show that every functional (or every vector of a dual basis) sends it to zero.
- **Pass to the dual and count.** Replace a subspace by its annihilator and use $\dim U + \dim U^0 = \dim V$.
- **Natural or chosen?** Ask whether an isomorphism needs a basis. $V \cong V^*$ does; $V \cong V^{**}$ does not.
- **Dualize a statement.** Injective becomes surjective, kernel becomes image, a matrix becomes its transpose, and inclusions reverse.

4.1 Linear Functionals and the Dual Space

Chapter 3 studied linear maps $V \rightarrow W$ into arbitrary targets. The smallest non-zero target is the field F itself, and maps into it are everywhere: a coordinate, the value of a polynomial at a point, an integral, the trace. This section names these maps and shows that they form a vector space V^* , of the same dimension as V when V is finite-dimensional. It also proves the fact that makes them useful: every non-zero vector is detected by at least one of them.

4.1.1 Measuring a vector by a number

After studying vector spaces and the maps between them, we look at the simplest possible target. Here are four maps we have already met, from four different families.

- **A coordinate.** Fix a basis $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ of V . The map $V \rightarrow F$ sending $\mathbf{v}$ to the first entry of $[\mathbf{v}]_{\mathcal{B}}$ respects sums and multiples, by [Theorem 1.5.2](#).
- **Evaluation.** The map $F[x] \rightarrow F$, $p \mapsto p(3)$, satisfies $(p + q)(3) = p(3) + q(3)$ and $(cp)(3) = cp(3)$, by [Theorem 0.8.5](#) (a constant c is a polynomial with value c).
- **Integration.** The map $\mathbb{R}[x] \rightarrow \mathbb{R}$, $p \mapsto \int_0^1 p(t) dt$, is linear by the linearity of the integral from calculus.
- **Trace.** The map $\text{tr}: M_n(F) \rightarrow F$ is linear ([Example \(Trace and Transpose\)](#)).

The inputs are columns, polynomials and matrices. The outputs are always single scalars. Each map is a way of **measuring** a vector by one number, and each measurement respects sums and scalar multiples. The same expression keeps recurring, “a linear map whose target is F ”, and we will see that these maps form a space with a structure of its own. So we give them a name.

A linear functional is a linear measurement: it assigns a scalar to each vector, and the measurement of a linear combination is the same combination of the measurements.

Definition 4.1.1 — Linear Functional

Let V be a vector space over F , and regard F as a vector space over itself. A **linear functional** on V is a linear map $\varphi: V \rightarrow F$. That is, for all $\mathbf{u}, \mathbf{v} \in V$ and all $c \in F$,

$$\varphi(\mathbf{u} + \mathbf{v}) = \varphi(\mathbf{u}) + \varphi(\mathbf{v}), \quad \varphi(c\mathbf{v}) = c\varphi(\mathbf{v}).$$

Definition 4.1.2 — Dual Space

Let V be a vector space over F . The **dual space** of V is the set of **all** linear functionals on V ,

$$V^* := \mathcal{L}(V, F),$$

with the pointwise operations $(\varphi + \psi)(\mathbf{v}) := \varphi(\mathbf{v}) + \psi(\mathbf{v})$ and $(c\varphi)(\mathbf{v}) := c\varphi(\mathbf{v})$ for $\varphi, \psi \in V^*$, $c \in F$ and $\mathbf{v} \in V$.

In words: a linear functional is a linear map (Definition 3.1.1) whose codomain is the one-dimensional space F . Its values lie in the **same** field F over which V is a vector space. The first condition says that measuring a sum gives the sum of the measurements, and the second says that scaling a vector scales its measurement. The dual space collects **every** such measurement, and it adds them the natural way: the measurement $\varphi + \psi$ reports the sum of the two readings. We write functionals with Greek letters φ, ψ , which keeps them apart from vectors of V .

Well-definedness. Two things need checking: that $\varphi + \psi$ and $c\varphi$ are again linear, and that the vector space axioms hold. Both are the case $W = F$ of Theorem 3.4.1. So V^* is a vector space over F . Its zero vector is the **zero functional** $\mathbf{v} \mapsto 0$, and the negative of φ is $\mathbf{v} \mapsto -\varphi(\mathbf{v})$. Note that V^* is a set of **functions on** V , not a subset of V .

Example. (Four Families of Functionals)

Show that each map is a linear functional.

- (a) $\varepsilon_c: F[x] \rightarrow F$, $p \mapsto p(c)$, for a fixed $c \in F$ (**evaluation at** c).
- (b) $\iota: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}$, $p \mapsto \int_0^1 p(t) dt$.
- (c) $\text{tr}: M_n(F) \rightarrow F$.
- (d) The zero functional on any V . Also find the dual space of the zero space $\{\mathbf{0}\}$.

Solution.

- (a) By Theorem 0.8.5, $(p+q)(c) = p(c) + q(c)$, and, regarding ap as the product of the constant polynomial a with p , $(ap)(c) = a(c)p(c) = ap(c)$, because the constant polynomial a has value a at c . These are the two conditions of Definition 4.1.1.
- (b) By linearity of the integral, $\int_0^1 (p+q) = \int_0^1 p + \int_0^1 q$ and $\int_0^1 ap = a \int_0^1 p$. On coefficients, $\iota(a_0 + a_1x + a_2x^2) = a_0 + \frac{a_1}{2} + \frac{a_2}{3}$.
- (c) This is Theorem 0.9.6 (1): $\text{tr}(A+B) = \text{tr} A + \text{tr} B$ and $\text{tr}(cA) = c \text{tr} A$.
- (d) $0 = 0 + 0$ and $0 = c \cdot 0$ in F , so the zero functional is linear. On $\{\mathbf{0}\}$, a linear functional must send $\mathbf{0}$ to 0 (Theorem 3.1.1), and there is only one vector. So the zero functional is the only element, and $\{\mathbf{0}\}^* = \{0\}$. This degenerate case matters: it is the case $n = 0$ of the count $\dim V^* = \dim V$ proved below.

The family F^n can be described completely. A row vector times a column is a scalar, and every functional on F^n arises this way.

Theorem 4.1.1 — Functionals on F^n Are Row Vectors

For $\mathbf{a} = (a_1, \dots, a_n) \in F^n$, define $\varphi_{\mathbf{a}}: F^n \rightarrow F$ by

$$\varphi_{\mathbf{a}}(\mathbf{x}) := \mathbf{a}^T \mathbf{x} = a_1 x_1 + \dots + a_n x_n.$$

Then $\varphi_{\mathbf{a}} \in (F^n)^*$. Every $\varphi \in (F^n)^*$ equals $\varphi_{\mathbf{a}}$ for **exactly one** $\mathbf{a}$, namely $\mathbf{a} = (\varphi(\mathbf{e}_1), \dots, \varphi(\mathbf{e}_n))$. The map $F^n \rightarrow (F^n)^*$, $\mathbf{a} \mapsto \varphi_{\mathbf{a}}$, is an isomorphism.

Proof. The map $\varphi_{\mathbf{a}}$ is $T_{\mathbf{a}^T}$ for the $1 \times n$ matrix $\mathbf{a}^T$, with a 1×1 matrix read as a scalar, so it is linear by [Example \(Multiplication by a Matrix\)](#). Let $\varphi \in (F^n)^*$ and put $a_i := \varphi(\mathbf{e}_i)$. Then $\varphi_{\mathbf{a}}(\mathbf{e}_i) = a_i = \varphi(\mathbf{e}_i)$ for each i . Two linear maps that agree on the basis $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ are equal by the uniqueness part of [Theorem 3.1.5](#), so $\varphi = \varphi_{\mathbf{a}}$. If $\varphi_{\mathbf{a}} = \varphi_{\mathbf{b}}$, then $a_i = \varphi_{\mathbf{a}}(\mathbf{e}_i) = \varphi_{\mathbf{b}}(\mathbf{e}_i) = b_i$ for every i , so $\mathbf{a} = \mathbf{b}$.

Hence $\mathbf{a} \mapsto \varphi_{\mathbf{a}}$ is a bijection $F^n \rightarrow (F^n)^*$. It is linear, since $\varphi_{\mathbf{a}+c\mathbf{b}}(\mathbf{x}) = \sum_i (a_i + cb_i)x_i = \varphi_{\mathbf{a}}(\mathbf{x}) + c\varphi_{\mathbf{b}}(\mathbf{x})$ for every $\mathbf{x}$. By [Theorem 3.5.1](#) it is an isomorphism. ■

So in F^n a functional **is** a row vector acting by matrix multiplication, and its $1 \times n$ matrix in the standard bases is $(\varphi(\mathbf{e}_1) \ \dots \ \varphi(\mathbf{e}_n))$. This is how we compute with functionals: rows act on columns.

Non-examples by minimal change. Each of the following changes one of the examples slightly. In each case something still works, and we name a condition that fails.

- On $\mathbb{R}^2$, the length $(x, y) \mapsto \sqrt{x^2 + y^2}$ ([Example \(Length\)](#)). It sends $\mathbf{0}$ to 0 and respects **positive** scalars. But with $c = -1$ and $\mathbf{v} = (1, 0)$ it gives 1, not -1 , so the scaling condition fails.
- On $\mathbb{R}[x]$, squaring the evaluation at 0: $p \mapsto p(0)^2$. The outputs are still scalars. But for $p = q = 1$ we get $(1 + 1)^2 = 4 \neq 1 + 1$, so additivity fails.
- On $M_2(F)$, replace the trace $a + d$ of $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ by $ad - bc$. Then $E_{11} \mapsto 0$ and $E_{22} \mapsto 0$, but $E_{11} + E_{22} = I_2 \mapsto 1$, so additivity fails. The expression $ad - bc$ is linear in each **row** separately, which is a different property.

Why this definition. Why single out the target F ? Because nothing is lost. A linear map $T: V \rightarrow F^m$ is the same thing as m functionals: if $\varphi_i(\mathbf{v})$ denotes the i -th entry of $T\mathbf{v}$, then each φ_i is linear and $T\mathbf{v} = (\varphi_1(\mathbf{v}), \dots, \varphi_m(\mathbf{v}))$; conversely any m functionals stacked this way give a linear map. So functionals are the “rows” of linear maps, and understanding them one at a time is a way of understanding all maps into F^m . The name comes from analysis, where the vectors were themselves functions and a map such as $f \mapsto \int_0^1 f$ was called a “functional”, a function of a function. The word “dual” will explain itself in this chapter: every statement about V has a mirror statement about V^* .

Warning. The values must lie in the field of scalars of V . Regard $\mathbb{C}$ as a vector space over $\mathbb{C}$. The real part $z \mapsto \operatorname{Re} z$ is additive, and it respects **real** scalars, but it is **not** a linear functional on $\mathbb{C}$ over $\mathbb{C}$: with $c = i$ and $z = 1$, $\operatorname{Re}(i \cdot 1) = 0 \neq i \cdot \operatorname{Re}(1) = i$. It is a functional on $\mathbb{C}$ regarded as a vector space over $\mathbb{R}$. The dual space depends on the field.

The first payoff is a count that holds for **every** non-zero functional: it hits every scalar, and in finite dimension it kills exactly one dimension.

Proposition 4.1.2 — A Non-Zero Functional Is Surjective

Let V be a vector space over F , and let $\varphi \in V^*$ be **non-zero**. Then φ is surjective. If V is finite-dimensional, then $\dim \ker \varphi = \dim V - 1$.

Proof. Since $\varphi \neq 0$, there is $\mathbf{u} \in V$ with $c := \varphi(\mathbf{u}) \neq 0$. For every $a \in F$, linearity gives $\varphi(ac^{-1}\mathbf{u}) = ac^{-1}c = a$, so φ is surjective. Hence $\text{im } \varphi = F$ and $\text{rank } \varphi = 1$. If V is finite-dimensional, the Rank–Nullity Theorem (Theorem 3.3.1) gives $\dim \ker \varphi = \dim V - 1$. ■

For example, $\ker \text{tr}$ in $M_n(F)$ has dimension $n^2 - 1$, and the polynomials in $F[x]_{\leq n}$ with $p(c) = 0$ form a subspace of dimension n . Every non-zero functional cuts out a subspace one dimension smaller than the whole space. Later in this chapter these kernels, the hyperplanes, become the building blocks of every subspace.

Quick check. Which of the following are linear functionals? (a) $p \mapsto p(0) + 1$ on $\mathbb{R}[x]$. (b) $p \mapsto p(1) - 2p'(0)$ on $\mathbb{R}[x]$. (c) $A \mapsto a_{12}$ on $M_2(F)$.

Solution.

- (a) No. It sends the zero polynomial to 1, but a linear map sends 0 to 0 (Theorem 3.1.1). (b) Yes. Evaluation at 1, differentiation and evaluation at 0 are linear, and V^* is closed under linear combinations and composition with linear maps (Theorem 3.4.1, Theorem 3.4.2). (c) Yes. It is the coordinate of A at E_{12} in the standard basis of Example (Standard Bases), and coordinates are linear (Theorem 1.5.2).

4.1.2 The dual basis

Return to the coordinate functionals. Let $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of V , and let $\varphi_i(\mathbf{v})$ be the i -th entry of $[\mathbf{v}]_{\mathcal{B}}$. The basis vector $\mathbf{v}_j$ has coordinate vector $\mathbf{e}_j$, so

$$\varphi_i(\mathbf{v}_j) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j, \end{cases}$$

with the Kronecker delta of Definition 0.9.3. Each φ_i reads off one coordinate and ignores the others. These n functionals are the natural candidates for a basis of V^* , and the theorem says they are one.

Theorem 4.1.3 — The Dual Basis

Let V be a finite-dimensional vector space over F , and let $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of V .

- (a) There are **unique** $\varphi_1, \dots, \varphi_n \in V^*$ with $\varphi_i(\mathbf{v}_j) = \delta_{ij}$ for all i, j . For every $\mathbf{v} \in V$, $\varphi_i(\mathbf{v})$ is the i -th entry of $[\mathbf{v}]_{\mathcal{B}}$.
 (b) The list $(\varphi_1, \dots, \varphi_n)$ is a basis of V^* .

(c) For every $\mathbf{v} \in V$ and every $\varphi \in V^*$,

$$\mathbf{v} = \sum_{i=1}^n \varphi_i(\mathbf{v}) \mathbf{v}_i, \quad \varphi = \sum_{i=1}^n \varphi(\mathbf{v}_i) \varphi_i.$$

Idea. ① A functional is a linear map into F , and a linear map may be prescribed freely on a basis, so existence and uniqueness in (a) are one citation. ② Independence: apply a vanishing combination $\sum c_i \varphi_i$ to $\mathbf{v}_j$; the delta kills every term but c_j . ③ Spanning: we cannot count, because $\dim V^*$ is exactly what we do not know yet. So we guess the coefficients. If $\varphi = \sum c_i \varphi_i$, then applying both sides to $\mathbf{v}_j$ forces $c_j = \varphi(\mathbf{v}_j)$. With this guess, the two sides are linear maps that agree on a basis, hence equal.

Proof.

- (a) For each i , apply [Theorem 3.1.5](#) with $W = F$ and the prescribed values $\delta_{i1}, \dots, \delta_{in} \in F$. It gives exactly one linear map $\varphi_i: V \rightarrow F$ with $\varphi_i(\mathbf{v}_j) = \delta_{ij}$ for all j . If $\mathbf{v} = a_1 \mathbf{v}_1 + \dots + a_n \mathbf{v}_n$, then by [Theorem 3.1.4](#)

$$\varphi_i(\mathbf{v}) = \sum_{j=1}^n a_j \varphi_i(\mathbf{v}_j) = \sum_{j=1}^n a_j \delta_{ij} = a_i,$$

the i -th entry of $[\mathbf{v}]_{\mathcal{B}}$ ([Definition 1.5.2](#)).

- (b) *Independence.* Let $c_1 \varphi_1 + \dots + c_n \varphi_n = 0$ in V^* . Applying both sides to $\mathbf{v}_j$ and using the pointwise operations of [Definition 4.1.2](#),

$$0 = \sum_{i=1}^n c_i \varphi_i(\mathbf{v}_j) = \sum_{i=1}^n c_i \delta_{ij} = c_j$$

for each j . Hence $(\varphi_1, \dots, \varphi_n)$ is linearly independent.

Spanning. Let $\varphi \in V^*$, and put $\psi := \sum_{i=1}^n \varphi(\mathbf{v}_i) \varphi_i \in V^*$. For each j , $\psi(\mathbf{v}_j) = \sum_i \varphi(\mathbf{v}_i) \delta_{ij} = \varphi(\mathbf{v}_j)$. So ψ and φ are linear maps $V \rightarrow F$ that agree on the basis $\mathcal{B}$, and $\psi = \varphi$ by the uniqueness part of [Theorem 3.1.5](#). Hence $\varphi \in \text{span}(\varphi_1, \dots, \varphi_n)$, and $(\varphi_1, \dots, \varphi_n)$ is a basis of V^* .

- (c) The second formula was proved in the spanning step. The first is the definition of coordinates, since $\varphi_i(\mathbf{v})$ is the i -th coordinate of $\mathbf{v}$ by (a). This proves the theorem. ■

The two formulas in (c) are mirror images. A vector is recovered from its measurements by the dual basis; a functional is recovered from its values on the basis.

Definition 4.1.3 — Dual Basis

Let $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of a finite-dimensional vector space V over F . The basis $(\varphi_1, \dots, \varphi_n)$ of V^* given by [Theorem 4.1.3](#), characterized by $\varphi_i(\mathbf{v}_j) = \delta_{ij}$, is **the dual basis** of $\mathcal{B}$. We write it $\mathcal{B}^*$.

Counting the dual basis gives the dimension of the dual space at once.

Corollary 4.1.4 — Dimension of the Dual Space

Let V be a finite-dimensional vector space over F . Then V^* is finite-dimensional and $\dim V^* = \dim V$. In particular $V^* \cong V$.

Proof. Let $n = \dim V$ and choose a basis of V (Corollary 1.5.5). By Theorem 4.1.3 (b), its dual basis is a basis of V^* of length n . Hence $\dim V^* = n$, and $V^* \cong V$ by Theorem 3.5.4. ■

For instance $\dim M_{m \times n}(F)^* = mn$ and $\dim F[x]_{\leq n}^* = n + 1$.

Remark. Many books write $\mathbf{v}_i^*$ for the dual basis functional φ_i . We avoid that notation for two reasons. It suggests that φ_i is determined by $\mathbf{v}_i$ alone, which is false (see the warning below). And in this book the star on a map is reserved for adjoints, which come much later. We write φ_i , and speak of “the dual basis of $\mathcal{B}$ ”.

How do we find a dual basis in practice? In F^n , functionals are rows, and the conditions $\varphi_i(\mathbf{v}_j) = \delta_{ij}$ turn into a single matrix equation.

Example. (The Dual Basis Is Read Off the Inverse Matrix)

- (a) Find the dual basis of $\mathcal{B} = ((1, 1), (1, -1))$ in $\mathbb{R}^2$.
- (b) More generally, let $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of F^n , and let $P = (\mathbf{v}_1 \ \cdots \ \mathbf{v}_n)$. Show that P is invertible and that the dual basis of $\mathcal{B}$ is $\varphi_i(\mathbf{x}) = \mathbf{r}_i \mathbf{x}$, where $\mathbf{r}_i$ is the i -th **row** of P^{-1} .

Solution.

- (a) By Theorem 4.1.1, $\varphi_1(x, y) = px + qy$ for some $p, q \in \mathbb{R}$, that is, $\varphi_1(\mathbf{x}) = (p \ q) \mathbf{x}$. The conditions $\varphi_1(1, 1) = 1$ and $\varphi_1(1, -1) = 0$ say $(p \ q)P = (1 \ 0)$ with $P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. Likewise the row $\mathbf{r}_2$ of φ_2 satisfies $\mathbf{r}_2 P = (0 \ 1)$. Stacking the two rows into a matrix R , the four conditions become $RP = I_2$, so $R = P^{-1}$. By Theorem 0.9.8, with $ad - bc = -2$,

$$P^{-1} = \frac{1}{-2} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}.$$

Hence

$$\varphi_1(x, y) = \frac{x+y}{2}, \quad \varphi_2(x, y) = \frac{x-y}{2}.$$

Check: $\varphi_1(1, 1) = 1$, $\varphi_1(1, -1) = 0$, $\varphi_2(1, 1) = 0$, $\varphi_2(1, -1) = 1$. And the first formula of Theorem 4.1.3 (c) for $\mathbf{v} = (3, 1)$ reads $(3, 1) = 2(1, 1) + 1(1, -1)$, which is true. The second formula expresses a functional in the dual basis through its values on $\mathcal{B}$: for $\psi(x, y) = x$, the values are $\psi(1, 1) = 1$ and $\psi(1, -1) = 1$, so $\psi = \varphi_1 + \varphi_2$, and indeed $\frac{x+y}{2} + \frac{x-y}{2} = x$.

- (b) The columns of P form a basis of F^n , so P is invertible by Theorem 2.4.4. Let $R \in M_n(F)$ have rows $\mathbf{r}_1, \dots, \mathbf{r}_n$, and let $\psi_i(\mathbf{x}) := \mathbf{r}_i \mathbf{x}$. By the entry formula for a

product (Theorem 0.9.2), $\psi_i(\mathbf{v}_j) = \mathbf{r}_i \mathbf{v}_j = (RP)_{ij}$. Hence $\psi_i(\mathbf{v}_j) = \delta_{ij}$ for all i, j if and only if $RP = I_n$, if and only if $R = P^{-1}$ (multiply by P^{-1} on the right). Now take $R = P^{-1}$: the functionals $\psi_i(\mathbf{x}) = \mathbf{r}_i \mathbf{x}$ are linear (Theorem 4.1.1) and satisfy $\psi_i(\mathbf{v}_j) = \delta_{ij}$, so by the uniqueness in Theorem 4.1.3 (a) they are the dual basis.

This agrees with Chapter 3: $P = [\text{id}]_{\mathcal{B}}^{\mathcal{E}}$ and $P^{-1} = [\text{id}]_{\mathcal{E}}^{\mathcal{B}}$ (Theorem 3.7.1), so $P^{-1}\mathbf{x} = [\mathbf{x}]_{\mathcal{B}}$, and row i of P^{-1} computes the i -th coordinate, as Theorem 4.1.3 (a) predicts. The columns of P are the basis; the rows of P^{-1} are the dual basis.

Warning. A dual basis functional depends on the whole basis, not only on its own vector. In $\mathbb{R}^2$, the basis $((1, 0), (0, 1))$ has first dual functional $\varphi_1(x, y) = x$. The basis $((1, 0), (1, 1))$ has the same first vector, but $P^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$, so its first dual functional is $\varphi_1(x, y) = x - y$. Changing $\mathbf{v}_2$ changed φ_1 , because φ_1 must vanish on $\mathbf{v}_2$.

The same method works away from F^n , once we recognize the coordinates.

Example. (Dual Bases in $\mathbb{R}[x]_{\leq 2}$)

- (a) Show that the dual basis of $(1, x, x^2)$ in $\mathbb{R}[x]_{\leq 2}$ is $(\varphi_0, \varphi_1, \varphi_2)$ with $\varphi_0(p) = p(0)$, $\varphi_1(p) = p'(0)$ and $\varphi_2(p) = \frac{1}{2}p''(0)$. (We index from 0 to match the exponents.)
- (b) Let $\ell_{-1} = \frac{1}{2}x(x-1)$, $\ell_0 = 1 - x^2$ and $\ell_1 = \frac{1}{2}x(x+1)$. Show that $(\ell_{-1}, \ell_0, \ell_1)$ is a basis of $\mathbb{R}[x]_{\leq 2}$ whose dual basis is $(\varepsilon_{-1}, \varepsilon_0, \varepsilon_1)$, the evaluations at $-1, 0, 1$. Deduce that $p = p(-1)\ell_{-1} + p(0)\ell_0 + p(1)\ell_1$ for every $p \in \mathbb{R}[x]_{\leq 2}$.

Solution.

- (a) Each φ_k is linear, as a composition of differentiation (Example (Differentiation)) with evaluation at 0, scaled. For $p = a_0 + a_1x + a_2x^2$, we have $p(0) = a_0$, $p'(0) = a_1$ and $\frac{1}{2}p''(0) = \frac{1}{2} \cdot 2a_2 = a_2$. So $\varphi_k(x^j) = \delta_{kj}$ for $j, k \in \{0, 1, 2\}$, and by the uniqueness in Theorem 4.1.3 (a) these are the dual basis. Over a general field F , the dual basis of $(1, x, x^2)$ is still “the coefficient of x^k ”, but the formula $\frac{1}{2}p''(0)$ needs $2 \neq 0$ in F .
- (b) The values of the ℓ ’s at the three points are

$$\ell_{-1}(-1) = 1, \ell_{-1}(0) = 0, \ell_{-1}(1) = 0; \quad \ell_0(-1) = 0, \ell_0(0) = 1, \ell_0(1) = 0; \quad \ell_1(-1) = 0, \ell_1(0) = 0, \ell_1(1) = 1$$

That is, $\varepsilon_i(\ell_j) = \delta_{ij}$ for $i, j \in \{-1, 0, 1\}$. *Basis.* Let $c_{-1}\ell_{-1} + c_0\ell_0 + c_1\ell_1 = 0$. Evaluating at $j \in \{-1, 0, 1\}$ gives $c_j = 0$. So the list is independent, of length $3 = \dim \mathbb{R}[x]_{\leq 2}$, hence a basis by Theorem 1.6.4. *Dual basis.* The evaluations are linear (Example (Four Families of Functionals)) and satisfy $\varepsilon_i(\ell_j) = \delta_{ij}$, so they are the dual basis by the uniqueness in Theorem 4.1.3 (a). The first formula of Theorem 4.1.3 (c) now reads $p = \sum_i \varepsilon_i(p) \ell_i = p(-1)\ell_{-1} + p(0)\ell_0 + p(1)\ell_1$.

The last formula rebuilds a polynomial from three of its values. It is a preview of the Lagrange interpolation formula of Chapter 5, and it explains where the ℓ ’s come from: they are the basis dual to the evaluations.

Quick check. Let $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ be a basis of V with dual basis $(\varphi_1, \varphi_2, \varphi_3)$. Compute $\varphi_2(3\mathbf{v}_1 - \mathbf{v}_2 + 5\mathbf{v}_3)$ and $(2\varphi_1 + \varphi_3)(\mathbf{v}_3)$.

Solution. $\varphi_2(3\mathbf{v}_1 - \mathbf{v}_2 + 5\mathbf{v}_3) = 3 \cdot 0 - 1 \cdot 1 + 5 \cdot 0 = -1$, the second coordinate. $(2\varphi_1 + \varphi_3)(\mathbf{v}_3) = 2 \cdot 0 + 1 = 1$.

Warning. $V \cong V^*$ is true in finite dimension, but no isomorphism is singled out. Corollary 4.1.4 gives an isomorphism by sending $\mathbf{v}_i \mapsto \varphi_i$, and that depends on the basis. In $\mathbb{R}^2$ with the standard basis, it sends $(1, 0)$ to the functional $(x, y) \mapsto x$. With the basis $((1, 0), (1, 1))$, it sends the **same** vector $(1, 0)$ to $(x, y) \mapsto x - y$, by the warning above. A vector does not come with “its” functional. Later in this chapter we make precise the sense in which V^* needs a choice to be identified with V , while the double dual V^{**} does not.

4.1.3 Testing against functionals

In F^n a vector is zero exactly when all its entries are zero, and the entries are the values of the coordinate functionals. The same holds in any finite-dimensional space, and it gives a flexible way to prove that a vector is zero.

Theorem 4.1.5 — Functionals Separate Vectors

Let V be a finite-dimensional vector space over F .

- (a) If $\mathbf{v} \in V$ and $\mathbf{v} \neq \mathbf{0}$, then there is $\varphi \in V^*$ with $\varphi(\mathbf{v}) = 1$.
- (b) If $\mathbf{v} \in V$ satisfies $\varphi(\mathbf{v}) = 0$ for **every** $\varphi \in V^*$, then $\mathbf{v} = \mathbf{0}$. It suffices that $\varphi_i(\mathbf{v}) = 0$ for the functionals φ_i of **one** dual basis.
- (c) For $\mathbf{u}, \mathbf{w} \in V$: $\mathbf{u} = \mathbf{w}$ if and only if $\varphi(\mathbf{u}) = \varphi(\mathbf{w})$ for every $\varphi \in V^*$.

Idea. We need a functional that does not vanish at $\mathbf{v}$, and the only functionals we can build at will are dual basis functionals. The first one takes the value 1 on the first basis vector. So make $\mathbf{v}$ the first vector of a basis: this is possible exactly because $\mathbf{v} \neq \mathbf{0}$.

Proof.

- (a) Since $\mathbf{v} \neq \mathbf{0}$, the list $(\mathbf{v})$ is linearly independent: if $a\mathbf{v} = \mathbf{0}$, then $a = 0$ by Theorem 1.1.9. By the Basis Extension Theorem (Theorem 1.6.5), there is a basis $(\mathbf{v}, \mathbf{v}_2, \dots, \mathbf{v}_n)$ of V . Let $(\varphi_1, \dots, \varphi_n)$ be its dual basis (Theorem 4.1.3). Then $\varphi_1(\mathbf{v}) = 1$.
- (b) The first statement is the contrapositive of (a). For the second, if $(\varphi_1, \dots, \varphi_n)$ is the dual basis of a basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ and every $\varphi_i(\mathbf{v}) = 0$, then $\mathbf{v} = \sum_i \varphi_i(\mathbf{v})\mathbf{v}_i = \mathbf{0}$ by Theorem 4.1.3 (c).
- (c) If $\mathbf{u} = \mathbf{w}$, the values agree. Conversely, if $\varphi(\mathbf{u}) = \varphi(\mathbf{w})$ for every φ , then $\varphi(\mathbf{u} - \mathbf{w}) = \varphi(\mathbf{u}) - \varphi(\mathbf{w}) = 0$ for every φ , so $\mathbf{u} - \mathbf{w} = \mathbf{0}$ by (b).

■

This is the signature move of the chapter: **to show a vector is zero, test it against every functional, and a dual basis suffices.** It turns one equation in V into scalar equations, which are often easier to reach.

Remark. Parts (a), (b) (first sentence) and (c) hold for **every** vector space V ; only (a) needs a new construction. By [Theorem 1.8.2](#), $\{\mathbf{v}\}$ extends to a basis B of V . Every $\mathbf{w} \in V$ is a finite combination of distinct vectors of B , with coefficients that are unique because B is independent. Let $\varphi(\mathbf{w})$ be the coefficient of $\mathbf{v}$ in this expansion (and 0 if $\mathbf{v}$ does not occur). Adding or scaling expansions gives expansions, so by uniqueness coefficients add and scale, and φ is linear with $\varphi(\mathbf{v}) = 1$. The price is Zorn's Lemma ([Theorem 1.8.1](#)): for $\mathbb{R}$ over $\mathbb{Q}$, no one has an explicit formula for such a functional.

Example. (A Matrix Killed by Every Trace)

Let $A \in M_n(F)$ satisfy $\text{tr}(AX) = 0$ for every $X \in M_n(F)$. Show that $A = 0$. Deduce that every linear functional on $M_n(F)$ is $X \mapsto \text{tr}(AX)$ for exactly one $A \in M_n(F)$.

Solution. The matrix units E_{ij} form a basis of $M_n(F)$ ([Example \(Standard Bases\)](#)), and $A = \sum_{i,j} a_{ij} E_{ij}$. By [Theorem 4.1.3](#) (a), its dual basis consists of the coordinate functionals $\varepsilon_{ij}(A) = a_{ij}$. We compute $\varepsilon_{ij}(A)$ as a trace. By [Definition 0.9.9](#), $(AE_{ji})_{kk} = \sum_l a_{kl} (E_{ji})_{lk}$, and $(E_{ji})_{lk} = 1$ only for $l = j$ and $k = i$. So the only non-zero diagonal entry of AE_{ji} is $(AE_{ji})_{ii} = a_{ij}$, and

$$\text{tr}(AE_{ji}) = a_{ij} = \varepsilon_{ij}(A).$$

Taking $X = E_{ji}$ in the hypothesis gives $\varepsilon_{ij}(A) = 0$ for all i, j . By [Theorem 4.1.5](#) (b), $A = 0$.

For the deduction, let $\Theta: M_n(F) \rightarrow M_n(F)^*$, $\Theta(A)(X) := \text{tr}(AX)$. Each $\Theta(A)$ is linear in X , and Θ is linear in A , both by [Theorem 0.9.6](#) (1) and the distributive laws of [Theorem 0.9.4](#). We just showed $\ker \Theta = \{0\}$, so Θ is injective ([Theorem 3.2.3](#)). Since $\dim M_n(F)^* = n^2 = \dim M_n(F)$ ([Corollary 4.1.4](#)), Θ is bijective by [Corollary 3.3.2](#) (e). This is the matrix analogue of [Theorem 4.1.1](#).

4.1.4 When V is infinite-dimensional

The main results of the last two subsections assumed finite dimension, or used a finite basis. That hypothesis does real work. Consider $F[x]$, with its basis $\{1, x, x^2, \dots\}$ ([Example \(A Basis of \$F\[x\]\$ \)](#)). For $k \in \mathbb{N}$ let

$$\varphi_k(p) := \text{the coefficient of } x^k \text{ in } p.$$

These are the coordinate functionals, and $\varphi_k(x^j) = \delta_{kj}$, exactly as in [Theorem 4.1.3](#). But they are not a basis of the dual space.

Proposition 4.1.6 — The Coordinate Functionals Do Not Span $F[x]^*$

*)

The functionals φ_k ($k \in \mathbb{N}$) lie in $F[x]^*$ and form a linearly independent set. But they

do **not** span $F[x]^*$: the evaluation $\varepsilon_1: p \mapsto p(1)$ is **not** a linear combination of finitely many φ_k .

Proof. Polynomials are added and scaled coefficient by coefficient (Definition 0.8.1), so each φ_k is linear. *Independence.* Let $k_1, \dots, k_m$ be distinct and $c_1\varphi_{k_1} + \dots + c_m\varphi_{k_m} = 0$. Applying both sides to x^{k_j} gives $c_j = 0$, since $\varphi_{k_i}(x^{k_j}) = \delta_{k_i k_j}$.

Not spanning. The evaluation ε_1 is linear by Example (Four Families of Functionals) (a), and $\varepsilon_1(x^N) = 1^N = 1$ for every $N \in \mathbb{N}$. Suppose, for a contradiction, that $\varepsilon_1 = c_1\varphi_{k_1} + \dots + c_m\varphi_{k_m}$ for some $m \in \mathbb{N}$, distinct k_j and $c_j \in F$. Choose $N \in \mathbb{N}$ larger than every k_j . Then $\varphi_{k_j}(x^N) = 0$ for every j , so the right-hand side sends x^N to 0, while $\varepsilon_1(x^N) = 1$. This contradiction shows $\varepsilon_1 \notin \text{span}\{\varphi_k : k \in \mathbb{N}\}$. ■

Where did the proof of Theorem 4.1.3 break? In the spanning step. The candidate $\sum_i \varphi(\mathbf{v}_i)\varphi_i$ for $\varphi = \varepsilon_1$ would be $\varphi_0 + \varphi_1 + \varphi_2 + \dots$, an infinite sum, and infinite sums mean nothing in a vector space (the warning in Chapter 1, §8). A functional on $F[x]$ may take non-zero values on infinitely many basis vectors, while a finite combination of the φ_k cannot.

Remark. More is true, and we state it without proof. For every infinite-dimensional V , a basis of V^* is **strictly larger**, as an infinite set, than a basis of V ; in particular $V^* \not\cong V$. So “ $\dim V^* = \dim V$ ” is a genuinely finite-dimensional fact, and this is why finite dimension appears in the hypotheses of most theorems in this chapter. A first step towards seeing how large $F[x]^*$ is appears in Exercise C4 below.

4.1.5 Exercises

4.1.5.1 A. Check your understanding

Exercise A1.

- Define a linear functional on a vector space V over F , and the dual space V^* .
- True or false: $p \mapsto p(1)^2$ is a linear functional on $\mathbb{R}[x]$. Justify your answer.
- True or false: if $(\mathbf{v}_1, \mathbf{v}_2)$ and $(\mathbf{v}_1, \mathbf{w}_2)$ are bases of V , then the first functionals of their dual bases are equal. Justify your answer.
- What is $\dim M_{2 \times 3}(F)^*$?
- State the formula that expresses $\varphi \in V^*$ in the dual basis of $(\mathbf{v}_1, \dots, \mathbf{v}_n)$.
- Name the move you would use to prove that a vector $\mathbf{v}$ of a finite-dimensional space is $\mathbf{0}$ without computing $\mathbf{v}$ itself.

Solution.

- See Definition 4.1.1 and Definition 4.1.2: a linear functional is a linear map $V \rightarrow F$, and $V^* = \mathcal{L}(V, F)$ with pointwise addition and scaling.
- False. For $p = 1$ and $c = 2$, the polynomial $cp = 2$ is sent to $2^2 = 4$, while c times the image of p is $2 \cdot 1^2 = 2$. So the scaling condition fails.
- False. By the warning after Example (The Dual Basis Is Read Off the Inverse

Matrix), in $\mathbb{R}^2$ the bases $((1, 0), (0, 1))$ and $((1, 0), (1, 1))$ have first dual functionals x and $x - y$.

- (d) 6, by [Corollary 4.1.4](#), since $\dim M_{2 \times 3}(F) = 6$.
- (e) $\varphi = \sum_{i=1}^n \varphi(\mathbf{v}_i) \varphi_i$, by [Theorem 4.1.3](#) (c).
- (f) Test against functionals: show $\varphi(\mathbf{v}) = 0$ for every $\varphi \in V^*$, or only for the functionals of one dual basis, and apply [Theorem 4.1.5](#) (b).

4.1.5.2 B. Practice

Exercise B1: Finding dual bases.

- (a) Find the dual basis of $((1, 0, 1), (0, 1, 1), (1, 1, 1))$ in $\mathbb{R}^3$.
- (b) Show that the dual basis of $(1, x - 1, (x - 1)^2)$ in $\mathbb{R}[x]_{\leq 2}$ is $p \mapsto p(1)$, $p \mapsto p'(1)$, $p \mapsto \frac{1}{2}p''(1)$. Hence write $p = 2 + 3x - x^2$ in this basis.

Solution.

- (a) Let P have the three vectors as columns. Row reduction of $[P \mid I_3]$ ([Theorem 2.4.6](#)) gives

$$P = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} 0 & -1 & 1 \\ -1 & 0 & 1 \\ 1 & 1 & -1 \end{pmatrix},$$

and one checks $P^{-1}P = I_3$. By [Example \(The Dual Basis Is Read Off the Inverse Matrix\)](#) (b), the dual basis is

$$\varphi_1(x, y, z) = -y + z, \quad \varphi_2(x, y, z) = -x + z, \quad \varphi_3(x, y, z) = x + y - z.$$

For instance $\varphi_3(1, 0, 1) = 0$, $\varphi_3(0, 1, 1) = 0$ and $\varphi_3(1, 1, 1) = 1$.

- (b) The three maps are linear, as combinations of differentiation and evaluation at 1. Call them ψ_1, ψ_2, ψ_3 . On the basis:

$$\psi_1(1) = 1, \psi_1(x-1) = 0, \psi_1((x-1)^2) = 0; \quad \psi_2(1) = 0, \psi_2(x-1) = 1, \psi_2((x-1)^2) = 2(1-1) = 0;$$

$$\psi_3(1) = 0, \psi_3(x-1) = 0, \psi_3((x-1)^2) = \frac{1}{2} \cdot 2 = 1.$$

So ψ_i takes the value δ_{ij} on the j -th basis vector, and by the uniqueness in [Theorem 4.1.3](#) (a) they are the dual basis. (The list $(1, x - 1, (x - 1)^2)$ is a basis: it has distinct degrees 0, 1, 2, so it is independent by [Theorem 1.4.4](#), and it has length $3 = \dim \mathbb{R}[x]_{\leq 2}$, so [Theorem 1.6.4](#) applies.) For $p = 2 + 3x - x^2$: $p(1) = 4$, $p'(1) = 3 - 2 = 1$, $\frac{1}{2}p''(1) = -1$. By [Theorem 4.1.3](#) (c),

$$p = 4 + (x - 1) - (x - 1)^2.$$

Check: $4 + x - 1 - (x^2 - 2x + 1) = 2 + 3x - x^2$.

Exercise B2: Which maps are functionals?.

Determine which of the following are linear functionals. Justify your answer.

- (a) $\mathbb{R}^3 \rightarrow \mathbb{R}$, $(x, y, z) \mapsto 2x - z$.

- (b) $\mathbb{R}^3 \rightarrow \mathbb{R}, (x, y, z) \mapsto xy$.
- (c) $M_2(\mathbb{R}) \rightarrow \mathbb{R}, A \mapsto a_{12} - 3a_{21}$.
- (d) $\mathbb{R}[x] \rightarrow \mathbb{R}, p \mapsto \int_0^1 t p(t) dt$.
- (e) $\mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}, p \mapsto p(0) + p'(0) + 1$.
- (f) $\mathbb{C}^2 \rightarrow \mathbb{C}, (z, w) \mapsto z + \bar{w}$, where $\mathbb{C}^2$ is a vector space over $\mathbb{C}$.

Solution.

- (a) Yes: it is $\varphi_{\mathbf{a}}$ with $\mathbf{a} = (2, 0, -1)$, linear by [Theorem 4.1.1](#).
- (b) No. $(1, 1, 0) \mapsto 1$, but $2(1, 1, 0) = (2, 2, 0) \mapsto 4 \neq 2$, so the scaling condition fails.
- (c) Yes: it is $\varepsilon_{12} - 3\varepsilon_{21}$, a combination of coordinate functionals for the standard basis, and V^* is a vector space ([Definition 4.1.2](#)).
- (d) Yes. By linearity of the integral, $\int_0^1 t(p+q)(t) dt = \int_0^1 tp(t) dt + \int_0^1 tq(t) dt$ and $\int_0^1 t(cp)(t) dt = c \int_0^1 tp(t) dt$.
- (e) No. The zero polynomial is sent to $1 \neq 0$, contradicting [Theorem 3.1.1](#).
- (f) No. With $c = i$ and $(z, w) = (0, 1)$: $(0, i) \mapsto \bar{i} = -i$, while $i \cdot (0 + \bar{1}) = i$. The map is additive and respects real scalars, but it is not linear over $\mathbb{C}$.

Exercise B3: Coordinates of a functional.

- (a) Let (φ_1, φ_2) be the dual basis of $((1, 1), (1, -1))$ from [Example \(The Dual Basis Is Read Off the Inverse Matrix\)](#). Write $\psi(x, y) = 3x + 5y$ as a combination of φ_1 and φ_2 .
- (b) Let $(\varphi_0, \varphi_1, \varphi_2)$ be the dual basis of $(1, x, x^2)$ from [Example \(Dual Bases in \$\mathbb{R}\[x\]_{\leq 2}\$ \)](#). Write $\iota(p) = \int_0^1 p(t) dt$ as a combination of them.

Solution.

- (a) By [Theorem 4.1.3](#) (c), $\psi = \psi(1, 1)\varphi_1 + \psi(1, -1)\varphi_2 = 8\varphi_1 - 2\varphi_2$. Check: $8 \cdot \frac{x+y}{2} - 2 \cdot \frac{x-y}{2} = 4x + 4y - x + y = 3x + 5y$.
- (b) By [Theorem 4.1.3](#) (c), $\iota = \iota(1)\varphi_0 + \iota(x)\varphi_1 + \iota(x^2)\varphi_2 = \varphi_0 + \frac{1}{2}\varphi_1 + \frac{1}{3}\varphi_2$. Check: for $p = a_0 + a_1x + a_2x^2$, the right-hand side is $a_0 + \frac{a_1}{2} + \frac{a_2}{3} = \int_0^1 p$.

4.1.5.3 C. Going deeper

Exercise C1: Functionals with the same kernel.

Let V be **any** vector space over F , and let $\varphi, \psi \in V^*$ with $\ker \varphi = \ker \psi$. Prove that $\psi = c\varphi$ for some $c \in F$. Is the converse true?

Hint: if $\varphi \neq 0$, fix $\mathbf{u}$ with $\varphi(\mathbf{u}) = 1$ and consider $\mathbf{v} - \varphi(\mathbf{v})\mathbf{u}$.

Solution. If $\varphi = 0$, then $\ker \psi = \ker \varphi = V$, so $\psi = 0 = 1 \cdot \varphi$. Suppose $\varphi \neq 0$. By [Proposition 4.1.2](#) there is $\mathbf{u} \in V$ with $\varphi(\mathbf{u}) = 1$. Let $\mathbf{v} \in V$. Then

$$\varphi(\mathbf{v} - \varphi(\mathbf{v})\mathbf{u}) = \varphi(\mathbf{v}) - \varphi(\mathbf{v})\varphi(\mathbf{u}) = 0,$$

so $\mathbf{v} - \varphi(\mathbf{v})\mathbf{u} \in \ker \varphi = \ker \psi$. Therefore $0 = \psi(\mathbf{v}) - \varphi(\mathbf{v})\psi(\mathbf{u})$, that is, $\psi(\mathbf{v}) = \psi(\mathbf{u})\varphi(\mathbf{v})$. Since $\mathbf{v}$ was arbitrary, $\psi = c\varphi$ with $c = \psi(\mathbf{u})$.

The converse holds only for $c \neq 0$: if $\psi = c\varphi$ with $c \neq 0$, then $\psi(\mathbf{v}) = 0 \iff \varphi(\mathbf{v}) = 0$. For $c = 0$ and $\varphi \neq 0$, $\ker \psi = V \neq \ker \varphi$.

Exercise C2: Functionals vanishing where others vanish.

Let V be a vector space over F , and let $\varphi_1, \dots, \varphi_k, \psi \in V^*$.

- (a) Prove that if $\ker \varphi_1 \cap \dots \cap \ker \varphi_k \subseteq \ker \psi$, then $\psi \in \text{span}(\varphi_1, \dots, \varphi_k)$.
- (b) Deduce that if V is finite-dimensional, then $(\varphi_1, \dots, \varphi_k)$ spans V^* if and only if $\ker \varphi_1 \cap \dots \cap \ker \varphi_k = \{\mathbf{0}\}$.

Hint: for (a), consider the linear map $T: V \rightarrow F^k$, $\mathbf{v} \mapsto (\varphi_1(\mathbf{v}), \dots, \varphi_k(\mathbf{v}))$.

Solution.

- (a) Let T be as in the hint; it is linear because each φ_i is. Its kernel is $\ker \varphi_1 \cap \dots \cap \ker \varphi_k$. Define $S: \text{im } T \rightarrow F$ by $S(T\mathbf{v}) := \psi(\mathbf{v})$. This is well defined: if $T\mathbf{v} = T\mathbf{v}'$, then $\mathbf{v} - \mathbf{v}' \in \ker T \subseteq \ker \psi$, so $\psi(\mathbf{v}) = \psi(\mathbf{v}')$. It is linear, since $S(cT\mathbf{v} + T\mathbf{v}') = S(T(c\mathbf{v} + \mathbf{v}')) = \psi(c\mathbf{v} + \mathbf{v}') = c\psi(\mathbf{v}) + \psi(\mathbf{v}') = cS(T\mathbf{v}) + S(T\mathbf{v}')$. Now extend S to F^k . The subspace $\text{im } T \subseteq F^k$ has a basis $(\mathbf{y}_1, \dots, \mathbf{y}_r)$ (Theorem 1.6.7, Corollary 1.5.5), which extends to a basis $(\mathbf{y}_1, \dots, \mathbf{y}_k)$ of F^k by Theorem 1.6.5. By Theorem 3.1.5 there is $\tilde{S} \in (F^k)^*$ with $\tilde{S}(\mathbf{y}_j) = S(\mathbf{y}_j)$ for $j \leq r$ and $\tilde{S}(\mathbf{y}_j) = 0$ for $j > r$. Then $\tilde{S}$ and S agree on a basis of $\text{im } T$, so $\tilde{S} = S$ on $\text{im } T$. By Theorem 4.1.1, $\tilde{S}(z_1, \dots, z_k) = a_1 z_1 + \dots + a_k z_k$ for some $a_i \in F$. Hence for every $\mathbf{v} \in V$,

$$\psi(\mathbf{v}) = S(T\mathbf{v}) = \tilde{S}(\varphi_1(\mathbf{v}), \dots, \varphi_k(\mathbf{v})) = a_1 \varphi_1(\mathbf{v}) + \dots + a_k \varphi_k(\mathbf{v}),$$

so $\psi = a_1 \varphi_1 + \dots + a_k \varphi_k$. Note that V was arbitrary; only F^k needed to be finite-dimensional.

- (b) ($\Leftarrow$) If the intersection is $\{\mathbf{0}\}$, it lies in $\ker \psi$ for every $\psi \in V^*$, so every ψ is in the span by (a). ($\Rightarrow$) Suppose the φ_i span V^* , and let $\mathbf{v}$ lie in every $\ker \varphi_i$. Every $\psi \in V^*$ is a combination $\sum c_i \varphi_i$, so $\psi(\mathbf{v}) = \sum c_i \varphi_i(\mathbf{v}) = 0$. By Theorem 4.1.5 (b), $\mathbf{v} = \mathbf{0}$.

Exercise C3: Evaluations as a dual basis.

Let $n \in \mathbb{N}$.

- (a) Let $x_0, \dots, x_n \in F$ be **distinct**. Prove that the evaluations $(\varepsilon_{x_0}, \dots, \varepsilon_{x_n})$ form a basis of $F[x]_{\leq n}^*$, and that they are the dual basis of some basis of $F[x]_{\leq n}$.
- (b) Over $\mathbb{F}_2$ with $n = 2$, show that the evaluations at the elements of $\mathbb{F}_2$ do **not** span $\mathbb{F}_2[x]_{\leq 2}^*$.

Hint: for (a), use Theorem 2.7.3 to build the basis first.

Solution.

- (a) For each j , [Theorem 2.7.3](#) gives $\ell_j \in F[x]_{\leq n}$ with $\ell_j(x_i) = \delta_{ij}$ for $i = 0, \dots, n$, because the x_i are distinct. If $\sum_j c_j \ell_j = 0$, evaluating at x_i gives $c_i = 0$. So $(\ell_0, \dots, \ell_n)$ is independent of length $n+1 = \dim F[x]_{\leq n}$, hence a basis by [Theorem 1.6.4](#). The evaluations are linear ([Example \(Four Families of Functionals\)](#)) and $\varepsilon_{x_i}(\ell_j) = \delta_{ij}$, so by the uniqueness in [Theorem 4.1.3](#) (a) they are the dual basis of $(\ell_0, \dots, \ell_n)$, and by [Theorem 4.1.3](#) (b) a basis of $F[x]_{\leq n}^*$.
- (b) $\mathbb{F}_2 = \{0, 1\}$ has only two elements, so only ε_0 and ε_1 are available, and $\dim \mathbb{F}_2[x]_{\leq 2}^* = 3$ by [Corollary 4.1.4](#). Two functionals cannot span a 3-dimensional space ([Theorem 1.6.3](#)). Concretely, $p = x^2 + x$ is non-zero but $p(0) = 0$ and $p(1) = 1 + 1 = 0$; the coefficient functional of x^2 takes the value 1 on p , so it is not a combination of $\varepsilon_0, \varepsilon_1$. The hypothesis “distinct points” in (a) needs $n+1$ distinct elements of F , which a small field may not have.

Exercise C4: How large is $F[x]^*$?

Let $F^{\mathbb{N}}$ be the space of sequences with entries in F .

- (a) Prove that $\Phi: F[x]^* \rightarrow F^{\mathbb{N}}$, $\varphi \mapsto (\varphi(1), \varphi(x), \varphi(x^2), \dots)$, is an isomorphism.
- (b) Identify $\Phi(\varphi_k)$ for the coordinate functionals φ_k of [Proposition 4.1.6](#), and $\Phi(\varepsilon_1)$. Explain how [Proposition 4.1.6](#) matches [Example \(The Vectors \$\mathbf{e}_i\$ Do Not Span \$F^{\mathbb{N}}\$ \)](#).

Hint: for surjectivity, remember that a polynomial has only finitely many non-zero coefficients.

Solution.

- (a) *Linear.* The k -th entry of $\Phi(c\varphi + \psi)$ is $(c\varphi + \psi)(x^k) = c\varphi(x^k) + \psi(x^k)$, by the pointwise operations of [Definition 4.1.2](#). *Injective.* If $\Phi(\varphi) = \mathbf{0}$, then φ vanishes on the basis $\{x^k\}$ of [Example \(A Basis of \$F\[x\]\$ \)](#). Every polynomial is a finite combination of the x^k , so $\varphi = 0$ by [Theorem 3.1.4](#); hence Φ is injective by [Theorem 3.2.3](#). *Surjective.* Given $(a_0, a_1, \dots) \in F^{\mathbb{N}}$, define $\varphi(b_0 + b_1x + \dots + b_Nx^N) := a_0b_0 + \dots + a_Nb_N$. This is a finite sum, it is well defined because coefficients are unique, and it is linear because coefficients add and scale; it is the linear map with $\varphi(x^k) = a_k$ given by [Theorem 3.1.6](#). Then $\Phi(\varphi) = (a_0, a_1, \dots)$. By [Theorem 3.5.1](#), Φ is an isomorphism.
- (b) $\Phi(\varphi_k) = (\delta_{k0}, \delta_{k1}, \dots) = \mathbf{e}_k$, and $\Phi(\varepsilon_1) = (1, 1, 1, \dots)$. An isomorphism carries spans to spans, so $\varepsilon_1 \in \text{span}\{\varphi_k\}$ would give $(1, 1, 1, \dots) \in \text{span}\{\mathbf{e}_k\}$. [Example \(The Vectors \$\mathbf{e}_i\$ Do Not Span \$F^{\mathbb{N}}\$ \)](#) showed this is false, because finite combinations of the $\mathbf{e}_k$ have only finitely many non-zero entries. So $F[x]$, with its basis $\{x^k\}$ indexed by $\mathbb{N}$, has a dual as large as the whole sequence space $F^{\mathbb{N}}$.

4.2 Annihilators

Section 1 built the dual space V^* , the space of linear measurements on V . Now we connect subspaces of V with subspaces of V^* . A plane in $\mathbb{R}^3$ can be described by a spanning list or by

an equation, and an equation is a functional. This section collects all the equations that a set of vectors satisfies into one subspace of V^* , the annihilator, counts its dimension, and shows that it turns sums into intersections and intersections into sums.

4.2.1 The equations a subspace satisfies

Every object comes with its subobjects. The dual space V^* has subspaces of its own, and the question is which of them are attached to subspaces of V . The standard example shows the way.

Let $U = \text{span}((1, 0, 1), (0, 1, 1))$, a plane in $\mathbb{R}^3$. The functional $\varphi(x, y, z) = x + y - z$ vanishes on both spanning vectors, so by linearity it vanishes on all of U : the plane satisfies the equation $x + y - z = 0$. So do $2x + 2y - 2z = 0$ and the trivial equation $0 = 0$. Which functionals vanish on U ? By [Theorem 4.1.1](#) a functional is $\psi(x, y, z) = ax + by + cz$. It vanishes on the two spanning vectors if and only if $a + c = 0$ and $b + c = 0$, that is, $\psi = -c(x + y - z) = -c\varphi$. So the equations of the plane form the **line** $\text{span}(\varphi)$ in $(\mathbb{R}^3)^*$.

For the z -axis $\text{span}((0, 0, 1))$, the same computation gives $c = 0$: the equations are $ax + by = 0$, a **plane** in $(\mathbb{R}^3)^*$. A bigger subspace satisfies fewer equations, and in both cases the dimensions add up to 3. The set of equations deserves a name, and the pattern deserves a theorem.

The annihilator of a set of vectors is the space of all linear equations that every one of those vectors satisfies.

Definition 4.2.1 — Annihilator

Let V be a vector space over F , and let $S \subseteq V$ be **any subset**. The **annihilator** of S is

$$S^0 := \{\varphi \in V^* : \varphi(\mathbf{s}) = 0 \text{ for all } \mathbf{s} \in S\}.$$

In words: S^0 is a subset of the **dual space** V^* , not of V . A functional belongs to it when it vanishes at **every** vector of S ; a single non-zero value disqualifies it. The set S needs no structure: it may be a subspace, a finite list of vectors, or even empty. The superscript 0 recalls the value that the functionals take on S .

Well-definedness. The definition makes sense for every subset. What needs checking is that the result is a subspace, which is what makes its dimension meaningful.

Proposition 4.2.1 — The Annihilator Is a Subspace

Let V be a vector space over F , and let $S \subseteq V$. Then S^0 is a subspace of V^* .

Proof. We use the subspace test ([Theorem 1.2.1](#)) in V^* . The zero functional sends every $\mathbf{s} \in S$ to 0, so it lies in S^0 . Let $\varphi, \psi \in S^0$ and $c \in F$. For every $\mathbf{s} \in S$, the pointwise operations of [Definition 4.1.2](#) give

$$(\varphi + \psi)(\mathbf{s}) = \varphi(\mathbf{s}) + \psi(\mathbf{s}) = 0 + 0 = 0, \quad (c\varphi)(\mathbf{s}) = c\varphi(\mathbf{s}) = c \cdot 0 = 0.$$

Hence $\varphi + \psi \in S^0$ and $c\varphi \in S^0$, and S^0 is a subspace of V^* . ■

Example. (First Annihilators)

- (a) For any vector space V , find $\{\mathbf{0}\}^0$, $\emptyset^0$ and V^0 .
- (b) In $M_2(F)$, find the annihilator of the single matrix E_{11} .
- (c) In $\mathbb{R}[x]_{\leq 2}$, find the annihilator of $S = \{1, x\}$.

Solution.

- (a) Every functional sends $\mathbf{0}$ to 0 (Theorem 3.1.1), so $\{\mathbf{0}\}^0 = V^*$. The condition “for all $\mathbf{s} \in \emptyset$ ” holds vacuously, so $\emptyset^0 = V^*$. A functional in V^0 takes the value 0 at every vector, so it is the zero functional, and $V^0 = \{0\}$. These degenerate cases fix the direction of everything that follows: the smallest subspace of V has the largest annihilator, and the largest has the smallest.
- (b) Let $\varepsilon_{ij}(A) = a_{ij}$ be the dual basis of the standard basis $(E_{11}, E_{12}, E_{21}, E_{22})$. By Theorem 4.1.3 (c), every $\psi \in M_2(F)^*$ is $\psi = \sum_{i,j} \psi(E_{ij}) \varepsilon_{ij}$. Such a ψ lies in $\{E_{11}\}^0$ exactly when $\psi(E_{11}) = 0$. Hence

$$\{E_{11}\}^0 = \text{span}(\varepsilon_{12}, \varepsilon_{21}, \varepsilon_{22}),$$

the functionals $A \mapsto c_{12}a_{12} + c_{21}a_{21} + c_{22}a_{22}$. It has dimension 3, and $3 + 1 = 4 = \dim M_2(F)$.

- (c) Let $(\varphi_0, \varphi_1, \varphi_2)$ be the dual basis of $(1, x, x^2)$, so φ_k reads off the coefficient of x^k . A functional $\psi = \psi(1)\varphi_0 + \psi(x)\varphi_1 + \psi(x^2)\varphi_2$ lies in S^0 exactly when $\psi(1) = \psi(x) = 0$, that is, when $\psi = \psi(x^2)\varphi_2$. So $S^0 = \text{span}(\varphi_2)$: the only equations satisfied by both 1 and x are the multiples of “the coefficient of x^2 is 0”.

Non-example by minimal change. Shift the plane of the opening example off the origin: $A = \{(x, y, z) \in \mathbb{R}^3 : x + y - z = 1\}$. The set is still cut out by the same left-hand side, but $\varphi(x, y, z) = x + y - z$ takes the value 1, not 0, on A , so the defining clause $\varphi(\mathbf{s}) = 0$ fails and $\varphi \notin A^0$. In fact $A^0 = \{0\}$: the set A contains $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, -1)$, and a functional $ax + by + cz$ vanishing at these three vectors has $a = b = c = 0$. Annihilators record only **homogeneous** equations.

Why this definition. We allow any subset S , not only subspaces, because subspaces are usually handed to us by spanning lists, and we want to test a functional on the list alone. The theorem at the end of this section shows that this loses nothing: $S^0 = (\text{span } S)^0$. The name is literal. To annihilate is to reduce to nothing, and the functionals in S^0 reduce every vector of S to 0.

Warning. U^0 lives in V^* , and it is not a complement of U . It is tempting to turn the functional $ax + by$ into the vector (a, b) and to picture U^0 as a subspace of F^2 “perpendicular” to U . Over $\mathbb{F}_2$ this picture fails completely. Let $U = \text{span}((1, 1)) \subseteq \mathbb{F}_2^2$. A functional $ax + by$ vanishes on $(1, 1)$ exactly when $a + b = 0$, that is, $a = b$, so $U^0 = \{0, x + y\}$. The functional $x + y$ corresponds to the vector $(1, 1)$, which lies in U . So the “picture” of U^0 is U itself, not a complement. Perpendicularity needs an inner product, which is the orthogonal complement of Chapter 10; the annihilator needs nothing.

The payoff of the definition is the pattern we saw in the opening example. The equations of a subspace take up exactly the dimensions that the subspace leaves free.

4.2.2 Counting equations

Theorem 4.2.2 — Dimension of the Annihilator

Let V be a finite-dimensional vector space over F , and let U be a subspace of V . Then

$$\dim U + \dim U^0 = \dim V.$$

More precisely, if $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_{m+1}, \dots, \mathbf{v}_n)$ is a basis of V extending a basis $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ of U , with dual basis $(\varphi_1, \dots, \varphi_n)$, then $(\varphi_{m+1}, \dots, \varphi_n)$ is a basis of U^0 .

Idea. This is a dimension formula, so we start from the smallest subspace, U , take a basis and extend it to V . The dual basis now splits in two. The last $n - m$ functionals vanish on every $\mathbf{u}_i$, by the delta, so they vanish on U . The first m do not. To see that nothing else is in U^0 , expand a functional $\psi \in U^0$ in the dual basis: its coefficients are its values on the basis, and the first m values are $\psi(\mathbf{u}_i) = 0$.

Proof. Since U is a subspace of the finite-dimensional space V , it has a basis $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ (Theorem 1.6.7, Corollary 1.5.5), and by the Basis Extension Theorem (Theorem 1.6.5) it extends to a basis $\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_{m+1}, \dots, \mathbf{v}_n)$ of V , where $n = \dim V$. Let $(\varphi_1, \dots, \varphi_n)$ be the dual basis of $\mathcal{B}$ (Theorem 4.1.3).

The φ_j with $j > m$ lie in U^0 . Let $j > m$ and $\mathbf{u} \in U$, say $\mathbf{u} = a_1\mathbf{u}_1 + \dots + a_m\mathbf{u}_m$. By Theorem 3.1.4, $\varphi_j(\mathbf{u}) = \sum_{i=1}^m a_i\varphi_j(\mathbf{u}_i) = 0$, since $\varphi_j(\mathbf{u}_i) = \delta_{ji} = 0$ for $i \leq m < j$.

They span U^0 . Let $\psi \in U^0$. By Theorem 4.1.3 (c),

$$\psi = \sum_{i=1}^m \psi(\mathbf{u}_i) \varphi_i + \sum_{j=m+1}^n \psi(\mathbf{v}_j) \varphi_j = \sum_{j=m+1}^n \psi(\mathbf{v}_j) \varphi_j,$$

because $\psi(\mathbf{u}_i) = 0$ for $\mathbf{u}_i \in U$. Hence $U^0 \subseteq \text{span}(\varphi_{m+1}, \dots, \varphi_n)$, and the reverse inclusion holds because U^0 is a subspace (Proposition 4.2.1, Theorem 1.3.1).

They are independent, being part of the basis $(\varphi_1, \dots, \varphi_n)$ of V^* . Therefore $(\varphi_{m+1}, \dots, \varphi_n)$ is a basis of U^0 , and $\dim U^0 = n - m = \dim V - \dim U$. This proves the theorem. ■

Alternatively, restrict functionals to U . The map $R: V^* \rightarrow U^*$, $\varphi \mapsto \varphi|_U$, is linear, and its kernel is U^0 by definition. It is surjective: given $\xi \in U^*$, Theorem 3.1.5 gives $\varphi \in V^*$ with $\varphi(\mathbf{u}_i) = \xi(\mathbf{u}_i)$ and $\varphi(\mathbf{v}_j) = 0$, and then $\varphi|_U = \xi$ because both agree on a basis of U . By the Rank-Nullity Theorem (Theorem 3.3.1) and Corollary 4.1.4, $\dim V = \dim V^* = \dim U^0 + \dim U^* = \dim U^0 + \dim U$. This route shows **why** the formula holds: a functional on V is a functional on U plus a choice of values off U , and U^0 measures that choice.

The theorem is the second signature move of the chapter: **pass to the dual and count**. One immediate consequence: if U is a **proper** subspace of a finite-dimensional V , then $\dim U^0 > 0$, so some **non-zero** functional vanishes on U . Equivalently, a list spans a finite-dimensional V if and only if the only functional that kills every vector of the list is 0: apply the previous sentence to the span U of the list, and note that a functional kills every vector of the list exactly when it kills U , by Theorem 3.1.4.

In F^n the whole computation becomes elimination, because functionals are rows.

Example. (An Annihilator as a Null Space)

Find a basis of U^0 for $U = \text{span}((1, 2, 0), (0, 1, 1)) \subseteq \mathbb{R}^3$, and check the dimension formula.

Solution. By [Theorem 4.1.1](#), every functional is $\varphi_{\mathbf{a}}(\mathbf{x}) = \mathbf{a}^T \mathbf{x}$ for a unique $\mathbf{a} = (a_1, a_2, a_3)$. By linearity, $\varphi_{\mathbf{a}}$ vanishes on U if and only if it vanishes on the two spanning vectors (a combination of vectors killed by $\varphi_{\mathbf{a}}$ is killed, by [Theorem 3.1.4](#)). These two conditions are

$$a_1 + 2a_2 = 0, \quad a_2 + a_3 = 0, \quad \text{that is,} \quad A\mathbf{a} = \mathbf{0} \quad \text{with} \quad A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix},$$

the matrix whose **rows** are the spanning vectors. So $\varphi_{\mathbf{a}} \in U^0$ if and only if $\mathbf{a} \in \text{nul}(A)$. Subtracting twice row 2 from row 1 gives the RREF

$$\begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix},$$

with a_3 free. By [Theorem 2.5.4](#), $\text{nul}(A) = \text{span}((2, -1, 1))$. Hence $U^0 = \text{span}(\psi)$ with

$$\psi(x, y, z) = 2x - y + z.$$

Check: $\psi(1, 2, 0) = 2 - 2 + 0 = 0$ and $\psi(0, 1, 1) = -1 + 1 = 0$. The two spanning vectors are independent (neither is a multiple of the other), so $\dim U = 2$, and $\dim U + \dim U^0 = 2 + 1 = 3$, as [Theorem 4.2.2](#) predicts. As a by-product, U is the plane $2x - y + z = 0$: $U \subseteq \ker \psi$, both have dimension 2 ([Proposition 4.1.2](#)), and [Theorem 1.6.8](#) gives equality.

The same recipe works in general. For $U = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_m) \subseteq F^n$, let A have rows $\mathbf{u}_1^T, \dots, \mathbf{u}_m^T$. Then $U^0 = \{\varphi_{\mathbf{a}} : \mathbf{a} \in \text{nul}(A)\}$, and the isomorphism of [Theorem 4.1.1](#) carries a basis of $\text{nul}(A)$ to a basis of U^0 .

Away from F^n , counting often replaces the computation entirely.

Example. (The Equations of a Root Condition)

Let $U = \{p \in \mathbb{R}[x]_{\leq 2} : p(1) = 0\}$. Find U^0 .

Solution. Let $\varepsilon_1(p) = p(1)$, a linear functional ([Example \(Four Families of Functionals\)](#)). By definition $U = \ker \varepsilon_1$, so $\varepsilon_1 \in U^0$. Since $\varepsilon_1(1) = 1$, $\varepsilon_1 \neq 0$, and [Proposition 4.1.2](#) gives $\dim U = 3 - 1 = 2$. By [Theorem 4.2.2](#), $\dim U^0 = 3 - 2 = 1$. Now $\text{span}(\varepsilon_1)$ is a subspace of U^0 ([Proposition 4.2.1](#)) of dimension 1, so $U^0 = \text{span}(\varepsilon_1)$ by [Theorem 1.6.8](#).

We never found a basis of U and never solved an equation. The conclusion is concrete: a functional on $\mathbb{R}[x]_{\leq 2}$ vanishes on every polynomial with a root at 1 **only if** it is a multiple of $p \mapsto p(1)$.

Quick check. Let $U = \text{span}((1, 1, 1)) \subseteq \mathbb{R}^3$. What is $\dim U^0$? Name a basis of U^0 without solving a system.

Solution. $\dim U^0 = 3 - 1 = 2$ by Theorem 4.2.2. The functionals $x - y$ and $y - z$ vanish at $(1, 1, 1)$, hence on U , and they are independent: in $a(x - y) + b(y - z) = 0$, the coefficient of x gives $a = 0$, and then $b = 0$. Two independent vectors in a 2-dimensional space form a basis (Theorem 1.6.4).

4.2.3 Sums, intersections and spans

Taking annihilators reverses inclusions: more vectors satisfy fewer common equations. It also exchanges sums with intersections. In $\mathbb{R}^3$, let U be the xy -plane and W the xz -plane. Then $U^0 = \text{span}(z)$ and $W^0 = \text{span}(y)$, writing x, y, z for the coordinate functionals. The intersection $U \cap W$ is the x -axis, with annihilator $\text{span}(y, z) = U^0 + W^0$. The sum $U + W = \mathbb{R}^3$ has annihilator $\{0\} = U^0 \cap W^0$. The theorem says this always happens.

Theorem 4.2.3 — Properties of Annihilators

Let V be a vector space over F .

- (a) If $S \subseteq T \subseteq V$, then $T^0 \subseteq S^0$.
- (b) For every subset $S \subseteq V$, $S^0 = (\text{span } S)^0$.
- (c) For subspaces U, W of V , $(U + W)^0 = U^0 \cap W^0$.
- (d) If V is **finite-dimensional**, then for subspaces U, W of V , $(U \cap W)^0 = U^0 + W^0$.

Idea. Parts (a)–(c) are unwinding definitions: fix a functional in one side and test it on the vectors of the other. In (d), the inclusion $U^0 + W^0 \subseteq (U \cap W)^0$ is also direct. The reverse inclusion is the hard half. Given φ vanishing on $U \cap W$, we would have to **split** it as $\psi + \chi$ with ψ killing U and χ killing W , and nothing in sight tells us how. So we count instead: one inclusion plus equal dimensions gives equality. The plan is ① compute $\dim(U^0 + W^0)$ by the dimension formula for sums inside V^* ; ② replace $U^0 \cap W^0$ by $(U + W)^0$ using (c); ③ turn every annihilator dimension into a dimension in V with Theorem 4.2.2; ④ recognize the dimension formula in V . Finite dimension is spent in step ③.

Proof.

- (a) Let $\varphi \in T^0$. Every $\mathbf{s} \in S$ lies in T , so $\varphi(\mathbf{s}) = 0$. Hence $\varphi \in S^0$.
- (b) Since $S \subseteq \text{span } S$ (Theorem 1.3.1), part (a) gives $(\text{span } S)^0 \subseteq S^0$. Conversely, let $\varphi \in S^0$ and $\mathbf{v} \in \text{span } S$, say $\mathbf{v} = a_1 \mathbf{s}_1 + \cdots + a_k \mathbf{s}_k$ with $\mathbf{s}_i \in S$ and $k \geq 0$. By Theorem 3.1.4, $\varphi(\mathbf{v}) = \sum_i a_i \varphi(\mathbf{s}_i) = 0$. Hence $\varphi \in (\text{span } S)^0$.
- (c) Since $U \subseteq U + W$ and $W \subseteq U + W$ (Theorem 1.7.1), part (a) gives $(U + W)^0 \subseteq U^0$ and $(U + W)^0 \subseteq W^0$, so $(U + W)^0 \subseteq U^0 \cap W^0$. Conversely, let $\varphi \in U^0 \cap W^0$, and let $\mathbf{u} + \mathbf{w} \in U + W$ with $\mathbf{u} \in U$, $\mathbf{w} \in W$. Then $\varphi(\mathbf{u} + \mathbf{w}) = \varphi(\mathbf{u}) + \varphi(\mathbf{w}) = 0 + 0 = 0$. Hence $\varphi \in (U + W)^0$.
- (d) ($\supseteq$) Let $\varphi = \psi + \chi$ with $\psi \in U^0$ and $\chi \in W^0$, and let $\mathbf{v} \in U \cap W$. Since $\mathbf{v} \in U$ and $\mathbf{v} \in W$, $\varphi(\mathbf{v}) = \psi(\mathbf{v}) + \chi(\mathbf{v}) = 0 + 0 = 0$. Hence $U^0 + W^0 \subseteq (U \cap W)^0$.

($\subseteq$) Let $n = \dim V$. The subspaces U , W , $U \cap W$ (Theorem 1.2.2) and $U + W$ (Theorem 1.7.1) of V are finite-dimensional by Theorem 1.6.7. The space V^* has dimension n

by [Corollary 4.1.4](#), so its subspaces U^0, W^0 are finite-dimensional too. By the Dimension Formula for Sums ([Theorem 1.7.3](#)) in V^* , then (c),

$$\dim(U^0 + W^0) = \dim U^0 + \dim W^0 - \dim(U^0 \cap W^0) = \dim U^0 + \dim W^0 - \dim(U + W)^0.$$

By [Theorem 4.2.2](#) applied to U, W and $U + W$, the right-hand side is

$$(n - \dim U) + (n - \dim W) - (n - \dim(U + W)) = n - (\dim U + \dim W - \dim(U + W)) = n - \dim(U \cap W),$$

where the last equality is [Theorem 1.7.3](#) in V . By [Theorem 4.2.2](#) once more, $n - \dim(U \cap W) = \dim(U \cap W)^0$. So $U^0 + W^0$ is a subspace of $(U \cap W)^0$ (by $(\supseteq)$ and [Theorem 1.7.1](#)) with the same finite dimension, and [Theorem 1.6.8](#) gives $U^0 + W^0 = (U \cap W)^0$. This proves the theorem. ■

Part (b) is what made the computations of this section legitimate: to find the annihilator of a subspace, test functionals on a spanning list only. Parts (c) and (d) form a dictionary. A subspace given as a **sum** has equations obtained by **intersecting** equation spaces; a subspace given as an **intersection** has equations obtained by **adding** them. Later in this chapter this dictionary becomes a practical method for computing sums and intersections of subspaces.

Remark. Parts (a)–(c) and the inclusion $(\supseteq)$ in (d) hold in every vector space; only the counting step used finite dimension. The equality (d) is in fact true for **every** V , but the proof must then construct the splitting $\varphi = \psi + \chi$ by hand; Exercise C1 below does this. In finite dimension, (a) also becomes strict: if $U \subsetneq W$, then $\dim U < \dim W$ ([Theorem 1.6.8](#)), so $\dim W^0 < \dim U^0$ and $W^0 \subsetneq U^0$.

4.2.4 Exercises

4.2.4.1 A. Check your understanding

Exercise A1.

- Define the annihilator S^0 of a subset S of a vector space V .
- True or false: U^0 is a subspace of V . Justify your answer.
- Let U be a 2-dimensional subspace of $\mathbb{R}^5$. What is $\dim U^0$?
- For a finite-dimensional V , what are $\{\mathbf{0}\}^0$ and V^0 ?
- True or false: if $U \subsetneq W$ are subspaces of a finite-dimensional V , then $W^0 \subsetneq U^0$. Justify your answer.
- To compute U^0 for $U = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_m) \subseteq F^n$, which matrix do you row reduce, and why?

Solution.

- $S^0 = \{\varphi \in V^* : \varphi(\mathbf{s}) = 0 \text{ for all } \mathbf{s} \in S\}$ ([Definition 4.2.1](#)).
- False. U^0 is a subspace of the dual space V^* ([Proposition 4.2.1](#)); its elements are functionals, not vectors of V .
- $\dim U^0 = 5 - 2 = 3$, by [Theorem 4.2.2](#).

- (d) $\{\mathbf{0}\}^0 = V^*$ and $V^0 = \{0\}$, by [Example \(First Annihilators\)](#) (a).
- (e) True. By [Theorem 4.2.3](#) (a), $W^0 \subseteq U^0$. Since $U \subsetneq W$, $\dim U < \dim W$ by [Theorem 1.6.8](#), so $\dim W^0 = \dim V - \dim W < \dim V - \dim U = \dim U^0$. Hence the inclusion is strict.
- (f) The matrix A whose **rows** are $\mathbf{u}_1^\top, \dots, \mathbf{u}_m^\top$. By [Theorem 4.1.1](#) a functional is $\mathbf{x} \mapsto \mathbf{a}^\top \mathbf{x}$, and it kills every $\mathbf{u}_i$ exactly when $A\mathbf{a} = \mathbf{0}$; so U^0 corresponds to $\text{nul}(A)$ ([Example \(An Annihilator as a Null Space\)](#)).

4.2.4.2 B. Practice

Exercise B1: Computing annihilators.

- (a) Let $U = \text{span}((1, 2, 1, 0), (2, 4, 3, 1), (1, 2, 2, 1)) \subseteq \mathbb{R}^4$. Find a basis of U^0 , and verify that $\dim U + \dim U^0 = 4$.
- (b) Let $U = \{p \in \mathbb{R}[x]_{\leq 3} : p(1) = p(-1) = 0\}$. Show that $U^0 = \text{span}(\varepsilon_1, \varepsilon_{-1})$, where $\varepsilon_c(p) = p(c)$.

Solution.

- (a) Let A have the three spanning vectors as rows. As in [Example \(An Annihilator as a Null Space\)](#), $\varphi_{\mathbf{a}} \in U^0$ if and only if $A\mathbf{a} = \mathbf{0}$. Subtracting 2 times row 1 from row 2 and row 1 from row 3 gives rows $(1, 2, 1, 0)$, $(0, 0, 1, 1)$, $(0, 0, 1, 1)$; subtracting row 2 from row 3 and then row 2 from row 1 gives the RREF

$$\begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The free variables are a_2 and a_4 , and [Theorem 2.5.4](#) gives $\text{nul}(A) = \text{span}((-2, 1, 0, 0), (1, 0, -1, 1))$. Hence U^0 has the basis

$$\psi_1(x_1, x_2, x_3, x_4) = -2x_1 + x_2, \quad \psi_2(x_1, x_2, x_3, x_4) = x_1 - x_3 + x_4.$$

Check on the first two spanning vectors: $\psi_1(1, 2, 1, 0) = 0$, $\psi_2(1, 2, 1, 0) = 0$, $\psi_1(2, 4, 3, 1) = 0$, $\psi_2(2, 4, 3, 1) = 2 - 3 + 1 = 0$; the third vector is their difference, $(2, 4, 3, 1) - (1, 2, 1, 0)$. The RREF has two pivots, so $\dim U = \dim \text{row}(A) = 2$ ([Theorem 2.5.5](#)), and $2 + 2 = 4$.

- (b) By definition $\varepsilon_1, \varepsilon_{-1} \in U^0$. *Independent:* if $a\varepsilon_1 + b\varepsilon_{-1} = 0$, applying it to $x + 1$ gives $2a = 0$ and applying it to $x - 1$ gives $-2b = 0$. *Dimension of U :* the map $T: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}^2$, $p \mapsto (p(1), p(-1))$, is linear with kernel U , and its image contains $T(x + 1) = (2, 0)$ and $T(x - 1) = (0, -2)$, which span $\mathbb{R}^2$. So $\text{rank } T = 2$ and $\dim U = 4 - 2 = 2$ by [Theorem 3.3.1](#). By [Theorem 4.2.2](#), $\dim U^0 = 2$. Hence the 2-dimensional subspace $\text{span}(\varepsilon_1, \varepsilon_{-1})$ of U^0 is all of U^0 ([Theorem 1.6.8](#)).

Exercise B2: Equations of the symmetric matrices.

Let $U = \{A \in M_2(\mathbb{R}) : A^\top = A\}$. Find U^0 , and verify the dimension formula.

Solution. Every symmetric matrix is $aE_{11} + b(E_{12} + E_{21}) + cE_{22}$, so $U = \text{span}(E_{11}, E_{12} + E_{21}, E_{22})$. Let (ε_{ij}) be the dual basis of the standard basis, so that every $\psi \in M_2(\mathbb{R})^*$ is $\psi = \sum_{i,j} c_{ij}\varepsilon_{ij}$ with $c_{ij} = \psi(E_{ij})$ (Theorem 4.1.3 (c)). By Theorem 4.2.3 (b), $\psi \in U^0$ if and only if ψ kills the three spanning matrices:

$$c_{11} = 0, \quad c_{12} + c_{21} = 0, \quad c_{22} = 0.$$

So $\psi = c_{12}(\varepsilon_{12} - \varepsilon_{21})$, and $U^0 = \text{span}(\varepsilon_{12} - \varepsilon_{21})$: the only equation of the symmetric matrices, up to scaling, is $a_{12} - a_{21} = 0$. The three spanning matrices are independent (they involve disjoint sets of entries), so $\dim U = 3$, and $\dim U + \dim U^0 = 3 + 1 = 4 = \dim M_2(\mathbb{R})$.

Exercise B3: The equations of a hyperplane.

Let V be a finite-dimensional vector space over F , and let $\varphi \in V^*$ be non-zero. Prove that $(\ker \varphi)^0 = \text{span}(\varphi)$.

Solution. By definition of the kernel, φ vanishes on $\ker \varphi$, so $\varphi \in (\ker \varphi)^0$, and $\text{span}(\varphi) \subseteq (\ker \varphi)^0$ by Proposition 4.2.1. Since $\varphi \neq 0$, $\dim \text{span}(\varphi) = 1$, and Proposition 4.1.2 gives $\dim \ker \varphi = \dim V - 1$. By Theorem 4.2.2, $\dim(\ker \varphi)^0 = \dim V - (\dim V - 1) = 1$. A 1-dimensional subspace of a 1-dimensional space is the whole space (Theorem 1.6.8), so $(\ker \varphi)^0 = \text{span}(\varphi)$.

4.2.4.3 C. Going deeper

Exercise C1: The intersection formula without counting.

Let U, W be subspaces of a vector space V over F , and let $\varphi \in (U \cap W)^0$.

- Suppose V is finite-dimensional. Construct $\psi \in U^0$ and $\chi \in W^0$ with $\varphi = \psi + \chi$, without using any dimension formula.
- Explain how to modify the construction when V is infinite-dimensional. Deduce that $(U \cap W)^0 = U^0 + W^0$ holds in every vector space.

Hint: extend a basis of $U \cap W$ to a basis of U and, separately, to a basis of W .

Solution.

- Let $(\mathbf{x}_1, \dots, \mathbf{x}_k)$ be a basis of $U \cap W$ (Theorem 1.2.2, Theorem 1.6.7). Extend it to a basis $(\mathbf{x}_1, \dots, \mathbf{x}_k, \mathbf{u}_1, \dots, \mathbf{u}_p)$ of U and to a basis $(\mathbf{x}_1, \dots, \mathbf{x}_k, \mathbf{w}_1, \dots, \mathbf{w}_q)$ of W (Theorem 1.6.5).

The combined list $(\mathbf{x}_1, \dots, \mathbf{x}_k, \mathbf{u}_1, \dots, \mathbf{u}_p, \mathbf{w}_1, \dots, \mathbf{w}_q)$ is independent. Let $\sum a_i \mathbf{x}_i + \sum b_i \mathbf{u}_i + \sum c_j \mathbf{w}_j = \mathbf{0}$. Then $\sum c_j \mathbf{w}_j = -\sum a_i \mathbf{x}_i - \sum b_i \mathbf{u}_i$ lies in W and in U , hence in $U \cap W$, so $\sum c_j \mathbf{w}_j = \sum d_i \mathbf{x}_i$ for some $d_i \in F$. Since $(\mathbf{x}, \mathbf{w})$ is a basis of W , all c_j and d_i are 0. Then $\sum a_i \mathbf{x}_i + \sum b_i \mathbf{u}_i = \mathbf{0}$, and since $(\mathbf{x}, \mathbf{u})$ is a basis of U , all a_i and b_i are 0.

Extend the combined list to a basis of V by vectors $\mathbf{y}_1, \dots, \mathbf{y}_r$ (Theorem 1.6.5). By Theorem 3.1.5 there is $\psi \in V^*$ with

$$\psi(\mathbf{x}_i) = 0, \quad \psi(\mathbf{u}_i) = 0, \quad \psi(\mathbf{w}_j) = \varphi(\mathbf{w}_j), \quad \psi(\mathbf{y}_l) = 0.$$

Then ψ kills a basis of U , so $\psi \in U^0$ (Theorem 4.2.3 (b)). Put $\chi := \varphi - \psi$. Then $\chi(\mathbf{x}_i) = \varphi(\mathbf{x}_i) - 0 = 0$, because $\mathbf{x}_i \in U \cap W$ and $\varphi \in (U \cap W)^0$; and $\chi(\mathbf{w}_j) = \varphi(\mathbf{w}_j) - \varphi(\mathbf{w}_j) = 0$. So χ kills a basis of W , and $\chi \in W^0$. Hence $\varphi = \psi + \chi \in U^0 + W^0$.

- (b) Replace the finite lists by sets. By Theorem 1.8.3, $U \cap W$ has a basis X , and by Theorem 1.8.2 it extends to a basis $X \cup P$ of U and to a basis $X \cup Q$ of W , with $P \cap X = Q \cap X = \emptyset$. First, $P \cap Q = \emptyset$: a vector of $P \cap Q$ lies in $U \cap W = \text{span } X$, which contradicts the independence of $X \cup P$. So every vector of $X \cup P \cup Q$ lies in exactly one of X , P , Q . A linear relation among finitely many distinct vectors of $X \cup P \cup Q$ splits into an X -part, a P -part and a Q -part, and the independence argument of (a) applies word for word; so $X \cup P \cup Q$ is independent. By Theorem 1.8.2 it extends to a basis B of V . Every $\mathbf{v} \in V$ is a finite combination $\mathbf{v} = \sum_{\mathbf{b}} c_{\mathbf{b}} \mathbf{b}$ of distinct vectors of B , with unique coefficients because B is independent. Define

$$\psi(\mathbf{v}) := \sum_{\mathbf{b} \in Q} c_{\mathbf{b}} \varphi(\mathbf{b}),$$

a finite sum over the vectors of Q that occur. Adding or scaling expansions gives expansions, so by uniqueness the coefficients add and scale, and ψ is linear. It vanishes on $X \cup P$ and agrees with φ on Q , so the rest of (a) is unchanged: $\psi \in U^0$ and $\chi = \varphi - \psi$ kills the basis $X \cup Q$ of W . Together with the inclusion ($\supseteq$) of Theorem 4.2.3 (d), which used no finiteness, this gives $(U \cap W)^0 = U^0 + W^0$ in every vector space.

Exercise C2: Equations of a double root.

Let $U = \{p \in \mathbb{R}[x]_{\leq 3} : p(1) = 0 \text{ and } p'(1) = 0\}$, and let $\varepsilon_1(p) = p(1)$ and $\eta_1(p) = p'(1)$. Prove that $U^0 = \text{span}(\varepsilon_1, \eta_1)$. Deduce that $\psi(p) = p(0) - p(2) + 2p'(1)$ does **not** vanish on U .

Solution. Both ε_1 and η_1 are linear (evaluation, and differentiation followed by evaluation), and they vanish on U by definition. *Independent:* if $a\varepsilon_1 + b\eta_1 = 0$, applying it to 1 gives $a = 0$, and applying it to $x - 1$ gives $b = 0$. *Dimension of U :* the linear map $T: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}^2$, $p \mapsto (p(1), p'(1))$, has kernel U , and $T(1) = (1, 0)$, $T(x - 1) = (0, 1)$, so T is surjective. By Theorem 3.3.1, $\dim U = 4 - 2 = 2$, and by Theorem 4.2.2, $\dim U^0 = 2$. The 2-dimensional subspace $\text{span}(\varepsilon_1, \eta_1)$ of U^0 is therefore all of U^0 (Theorem 1.6.8).

For the deduction, suppose $\psi \in U^0$. Then $\psi = a\varepsilon_1 + b\eta_1$. Applying both sides to 1 gives $1 - 1 + 0 = a$, so $a = 0$; applying them to $x - 1$ gives $-1 - 1 + 2 = b$, so $b = 0$. Then $\psi = 0$. But $\psi(x^2) = 0 - 4 + 4 = 0$ and $\psi(x^3) = 0 - 8 + 6 = -2 \neq 0$, a contradiction. Hence $\psi \notin U^0$. Check with an explicit witness: $p = x(x - 1)^2 \in U$ has $\psi(p) = 0 - 2 + 0 = -2 \neq 0$.

Exercise C3: Dual of a direct sum decomposition.

- (a) Let V be finite-dimensional with $V = U \oplus W$. Prove that $V^* = U^0 \oplus W^0$.
 (b) In $\mathbb{R}^2$, let $U = \text{span}((1, 0))$ and $W = \text{span}((1, 1))$. Write $\psi(x, y) = x$ as $\psi_U + \psi_W$ with $\psi_U \in U^0$ and $\psi_W \in W^0$.

Solution.

- (a) Since the sum is direct, $U \cap W = \{0\}$ (Theorem 1.7.4), and $U + W = V$. By Theorem 4.2.3 (c) and Example (First Annihilators) (a), $U^0 \cap W^0 = (U + W)^0 = V^0 = \{0\}$. By Theorem 4.2.3 (d), $U^0 + W^0 = (U \cap W)^0 = \{0\}^0 = V^*$. So $V^* = U^0 + W^0$, and the sum is direct by Theorem 1.7.4.
- (b) A functional $ax + by$ vanishes on $(1, 0)$ if and only if $a = 0$, so $U^0 = \text{span}(y)$. It vanishes on $(1, 1)$ if and only if $b = -a$, so $W^0 = \text{span}(x - y)$. Solving $x = sy + t(x - y)$ gives $t = 1$ and $s - t = 0$, so $s = 1$:

$$\psi = \underbrace{y}_{\in U^0} + \underbrace{(x - y)}_{\in W^0}.$$

By (a) this decomposition is unique.

4.3 The Double Dual

In a finite-dimensional space, the dual basis gives an isomorphism $V \cong V^*$, sending $\mathbf{v}_i$ to φ_i . But that isomorphism depends on the basis: change the basis and the same vector goes to a different functional. Nothing stops us from taking duals twice. The dual V^* is a vector space, so it has a dual of its own, V^{**} , and two chosen bases give $V \cong V^* \cong V^{**}$. This section shows that the composite isomorphism $V \rightarrow V^{**}$ needs **no** choice at all: it is built from evaluation alone. We use it to make one meaning of the word “natural” precise, and to prove that a subspace can be recovered from its annihilator. At the end we see that in infinite dimension the construction still makes sense, but it is no longer an isomorphism.

4.3.1 Measuring the measurements

We have studied a space V and its space of measurements $V^* = \mathcal{L}(V, F)$ (Definition 4.1.2). The next object is the space of measurements **of** V^* .

Look at the expression $\varphi(\mathbf{v})$, with $\varphi \in V^*$ and $\mathbf{v} \in V$. It has two inputs. If we fix φ and let $\mathbf{v}$ vary, we get the functional φ itself. If we fix $\mathbf{v}$ and let φ vary, we get a rule that takes a functional and returns a number. For example, take $\mathbf{v} = (3, -1) \in \mathbb{R}^2$. A functional on $\mathbb{R}^2$ is $\varphi(x, y) = ax + by$ for unique a, b (Theorem 4.1.1), and the rule returns $\varphi(3, -1) = 3a - b$. So the fixed vector $(3, -1)$ measures each measurement. This rule appears every time we test a vector against functionals, so it deserves a name.

A vector measures a functional by letting the functional act on it.

Definition 4.3.1 — Double Dual and Evaluation Map

Let V be a vector space over F . The **double dual** of V is $V^{**} := (V^*)^*$, the space of linear functionals on V^* .

For $\mathbf{v} \in V$, **evaluation at $\mathbf{v}$** is the function

$$\text{ev}_{\mathbf{v}}: V^* \rightarrow F, \quad \text{ev}_{\mathbf{v}}(\varphi) := \varphi(\mathbf{v}).$$

The **evaluation map** of V is

$$\text{ev}_V: V \rightarrow V^{**}, \quad \mathbf{v} \mapsto \text{ev}_{\mathbf{v}}.$$

When only one space is in play we write ev for ev_V .

In words, clause by clause:

- V^{**} consists of **linear** maps $V^* \rightarrow F$. Its elements take a whole functional as input.
- $\text{ev}_{\mathbf{v}}$ is **defined on** V^* , not on V . The vector $\mathbf{v}$ is fixed once and for all; the variable is φ .
- The evaluation map sends each vector $\mathbf{v}$ to the function $\text{ev}_{\mathbf{v}}$. A bold subscript $\text{ev}_{\mathbf{v}}$ is a single element of V^{**} ; an italic capital ev_V is the whole map.

Well-definedness. The definition claims that $\text{ev}_{\mathbf{v}} \in V^{**}$, so we must check that $\text{ev}_{\mathbf{v}}$ is linear. The operations on $V^* = \mathcal{L}(V, F)$ are pointwise (Definition 3.4.1). So for $\varphi, \psi \in V^*$ and $a, b \in F$,

$$\text{ev}_{\mathbf{v}}(a\varphi + b\psi) = (a\varphi + b\psi)(\mathbf{v}) = a\varphi(\mathbf{v}) + b\psi(\mathbf{v}) = a\text{ev}_{\mathbf{v}}(\varphi) + b\text{ev}_{\mathbf{v}}(\psi).$$

Notice what the definition did **not** use: no basis, no coordinates, no choice of any kind. Only $\mathbf{v}$ and the meaning of “functional” appear.

Example. (Evaluation in Three Spaces)

- Let $V = F^n$. Describe $\text{ev}_{\mathbf{x}}$ for $\mathbf{x} \in F^n$ in terms of row vectors.
- Let $V = \mathbb{R}[x]_{\leq 2}$, and let $(\varphi_0, \varphi_1, \varphi_2)$ be the dual basis of $(1, x, x^2)$, so that $\varphi_k(p)$ is the coefficient of x^k in p (Example (Dual Bases in $\mathbb{R}[x]_{\leq 2}$)). Compute ev_{x^2} on $\varphi_0, \varphi_1, \varphi_2$ and on the functional $\varepsilon_3(p) = p(3)$. Which element of V^{**} is ev_{x^2} ?
- Describe $\text{ev}_{\mathbf{0}}$, and the evaluation map of $V = \{\mathbf{0}\}$.

Solution.

- By Theorem 4.1.1, every $\varphi \in (F^n)^*$ has the form $\varphi(\mathbf{y}) = \mathbf{a}^T \mathbf{y}$ for a unique $\mathbf{a} \in F^n$. So $\text{ev}_{\mathbf{x}}(\varphi) = \mathbf{a}^T \mathbf{x}$: the vector $\mathbf{x}$ measures the row $\mathbf{a}^T$ by standing to its right in the product. The same product $\mathbf{a}^T \mathbf{x}$ is read as “ $\mathbf{a}^T$ measures $\mathbf{x}$ ” or “ $\mathbf{x}$ measures $\mathbf{a}^T$ ”, depending on which factor we hold fixed.
- $\text{ev}_{x^2}(\varphi_k) = \varphi_k(x^2)$, the coefficient of x^k in x^2 . So the values on $\varphi_0, \varphi_1, \varphi_2$ are $0, 0, 1$. Also $\text{ev}_{x^2}(\varepsilon_3) = \varepsilon_3(x^2) = 9$. A linear functional on V^* is determined by its values on the basis $(\varphi_0, \varphi_1, \varphi_2)$ of V^* (Theorem 3.1.5). The values $0, 0, 1$ say that ev_{x^2} is the third vector of the dual basis of $(\varphi_0, \varphi_1, \varphi_2)$ (Theorem 4.1.3). In the same way, ev_1 and ev_x are the first and second vectors of that dual basis. We will see that this is no accident.
- $\text{ev}_{\mathbf{0}}(\varphi) = \varphi(\mathbf{0}) = 0$ for every φ , because a linear map sends $\mathbf{0}$ to 0 (Theorem 3.1.1). So $\text{ev}_{\mathbf{0}}$ is the zero functional on V^* . If $V = \{\mathbf{0}\}$, the only functional on V is the zero functional, so $V^* = \{0\}$ and $V^{**} = \{0\}$, and ev_V is the only map $\{0\} \rightarrow \{0\}$. This degenerate case is a first sign that ev loses nothing: the only vector it sends to zero is $\mathbf{0}$.

Here is a non-example by minimal change. On $V = \mathbb{R}^2$, replace $\text{ev}_{\mathbf{v}}$ by $q_{\mathbf{v}}(\varphi) := \varphi(\mathbf{v})^2$. It is still a function $V^* \rightarrow \mathbb{R}$, and it is still built without any choice. But take $\mathbf{v} = (1, 0)$ and $\varphi(x, y) = x$. Then $q_{\mathbf{v}}(2\varphi) = 2^2 = 4$, while $2q_{\mathbf{v}}(\varphi) = 2$. The clause that fails is **linearity** on V^* , so $q_{\mathbf{v}} \notin V^{**}$.

Why this definition. Given only a vector $\mathbf{v}$ and a functional φ , the one number we can form

without choosing anything is $\varphi(\mathbf{v})$; anything else, such as $\varphi(\mathbf{v})^2$, is built from it and usually breaks linearity. The pointwise operations of [Definition 3.4.1](#) are exactly what make $\varphi \mapsto \varphi(\mathbf{v})$ linear. The name “double dual” is literal: it is the dual of the dual.

Warning. Two different “evaluations”. On $V = \mathbb{R}[x]_{\leq 2}$, the functional $\varepsilon_3: p \mapsto p(3)$ lies in V^* : it evaluates **polynomials at a number**. The element $\text{ev}_p: \varphi \mapsto \varphi(p)$ lies in V^{**} : it evaluates **functionals at a polynomial**. They meet in one number, $\text{ev}_p(\varepsilon_3) = p(3) = \varepsilon_3(p)$, read from opposite sides. Also, $\text{ev}_{\mathbf{v}}$ is not the vector $\mathbf{v}$: it is a function on V^* .

Quick check. Let $\mathbf{v} = (1, 2) \in \mathbb{R}^2$. Compute $\text{ev}_{\mathbf{v}}(\varphi)$ for $\varphi(x, y) = 3x - y$. Is there a **non-zero** $\psi \in (\mathbb{R}^2)^*$ with $\text{ev}_{\mathbf{v}}(\psi) = 0$?

Solution. $\text{ev}_{\mathbf{v}}(\varphi) = \varphi(1, 2) = 3 - 2 = 1$. Yes: $\psi(x, y) = 2x - y$ is non-zero and $\psi(1, 2) = 0$. A single $\text{ev}_{\mathbf{v}}$ kills many functionals; what the next theorem says is that for $\mathbf{v} \neq \mathbf{0}$ it does not kill **all** of them.

4.3.2 The evaluation map never loses information

The first result says that ev is a linear map, and that it is injective in every vector space, of any dimension.

Theorem 4.3.1 — The Evaluation Map Is Linear and Injective

Let V be any vector space over F . Then the evaluation map $\text{ev}_V: V \rightarrow V^{**}$ is linear and injective.

Idea. Linearity is an identity between two elements of V^{**} , that is, between two functions on V^* ; compare them at an arbitrary φ . For injectivity, use the kernel test. If $\text{ev}_{\mathbf{v}} = 0$, then **every** functional vanishes at $\mathbf{v}$. This is the signature move of the chapter: to show a vector is zero, test it against every functional. In finite dimension [Theorem 4.1.5](#) does the job. In general we repeat its proof with the Basis Extension Theorem for arbitrary spaces.

Proof. Linear. Let $\mathbf{u}, \mathbf{v} \in V$ and $a, b \in F$. For every $\varphi \in V^*$, since φ is linear and the operations on V^{**} are pointwise ([Definition 3.4.1](#)),

$$\text{ev}_{a\mathbf{u}+b\mathbf{v}}(\varphi) = \varphi(a\mathbf{u} + b\mathbf{v}) = a\varphi(\mathbf{u}) + b\varphi(\mathbf{v}) = (a\text{ev}_{\mathbf{u}} + b\text{ev}_{\mathbf{v}})(\varphi).$$

Two functions on V^* that agree at every φ are equal, so $\text{ev}_{a\mathbf{u}+b\mathbf{v}} = a\text{ev}_{\mathbf{u}} + b\text{ev}_{\mathbf{v}}$.

Injective. By [Theorem 3.2.3](#) it suffices to show that $\text{ev}_{\mathbf{v}} = 0$ implies $\mathbf{v} = \mathbf{0}$. We prove the contrapositive. Let $\mathbf{v} \neq \mathbf{0}$. If V is finite-dimensional, [Theorem 4.1.5](#) (a) gives $\varphi \in V^*$ with $\varphi(\mathbf{v}) = 1$. In general, $\{\mathbf{v}\}$ is linearly independent, so by [Theorem 1.8.2](#) there is a basis B of V with $\mathbf{v} \in B$. By [Theorem 3.1.6](#), which allows infinite bases, there is a linear map $\varphi: V \rightarrow F$ with $\varphi(\mathbf{v}) = 1$ and $\varphi(\mathbf{b}) = 0$ for every $\mathbf{b} \in B \setminus \{\mathbf{v}\}$. In either case $\varphi \in V^*$ and $\text{ev}_{\mathbf{v}}(\varphi) = \varphi(\mathbf{v}) = 1 \neq 0$, so $\text{ev}_{\mathbf{v}} \neq 0$. This proves that ev_V is injective. ■

In infinite dimension the proof repeats the remark after [Theorem 4.1.5](#), and Zorn’s Lemma enters through [Theorem 1.8.2](#). In finite dimension no choice principle is needed.

So every vector $\mathbf{v}$ is completely remembered by the list of all its measurements $\varphi(\mathbf{v})$. In finite dimension a count turns “injective” into “bijective”.

Theorem 4.3.2 — The Double Dual Isomorphism

Let V be a finite-dimensional vector space over F . Then the evaluation map $\text{ev}_V: V \rightarrow V^{**}$ is an isomorphism. Moreover, if $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ is a basis of V with dual basis $\mathcal{B}^* = (\varphi_1, \dots, \varphi_n)$, then $(\text{ev}_{\mathbf{v}_1}, \dots, \text{ev}_{\mathbf{v}_n})$ is the dual basis of $\mathcal{B}^*$.

Idea. Count instead of check. The dual basis has as many vectors as the basis, so $\dim V^* = \dim V$, and the same holds one level up. An injective linear map between spaces of the same finite dimension is bijective. For the second statement, “dual basis” is the condition $\theta_i(\varphi_j) = \delta_{ij}$, and for $\theta_i = \text{ev}_{\mathbf{v}_i}$ that condition is the definition of φ_j read backwards.

Proof. Let $n = \dim V$. By [Corollary 4.1.4](#), V^* is finite-dimensional with $\dim V^* = n$, and applying it again to V^* gives $\dim V^{**} = n$. By [Theorem 4.3.1](#), ev_V is linear and injective. Since $\dim V = \dim V^{**}$ is finite, [Corollary 3.3.2](#) (e) shows that ev_V is bijective, hence an isomorphism ([Definition 3.5.2](#)).

For the second statement, $\text{ev}_{\mathbf{v}_i}(\varphi_j) = \varphi_j(\mathbf{v}_i) = \delta_{ji} = \delta_{ij}$ for all i, j , by the definition of the dual basis. By the uniqueness in [Theorem 4.1.3](#) (a), applied to the basis $\mathcal{B}^*$ of V^* , the dual basis of $\mathcal{B}^*$ ([Definition 4.1.3](#)) is the only list $(\theta_1, \dots, \theta_n)$ in V^{**} with $\theta_i(\varphi_j) = \delta_{ij}$. Hence it is $(\text{ev}_{\mathbf{v}_1}, \dots, \text{ev}_{\mathbf{v}_n})$, as claimed. ■

The theorem says: **every** linear functional on V^* is evaluation at exactly one vector. To find the vector, read off its coordinates from the values on a dual basis. For instance, on $V = \mathbb{R}[x]_{\leq 2}$ with the coefficient functionals $\varphi_0, \varphi_1, \varphi_2$ of [Example \(Evaluation in Three Spaces\)](#), let $\theta \in V^{**}$ be the functional with $\theta(\varphi_0) = 2$, $\theta(\varphi_1) = 0$, $\theta(\varphi_2) = -1$. By the second statement, $\theta = 2\text{ev}_1 - \text{ev}_{x^2} = \text{ev}_{2-x^2}$, since both sides agree on the basis $(\varphi_0, \varphi_1, \varphi_2)$ and ev is linear. So, for example, $\theta(\varepsilon_2) = \varepsilon_2(2 - x^2) = -2$, where $\varepsilon_2(p) = p(2)$. Part (b) of [Example \(Evaluation in Three Spaces\)](#) was the case $\theta = \text{ev}_{x^2}$.

4.3.3 What “natural” means

Both $V \cong V^*$ and $V \cong V^{**}$ hold in finite dimension, and both follow from counting dimensions. Yet they are very different isomorphisms. The difference is whether a choice is needed.

The isomorphism $V \rightarrow V^*$ depends on the basis. For a basis $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ of V , let $\Theta_{\mathcal{B}}: V \rightarrow V^*$ be the linear map with $\Theta_{\mathcal{B}}(\mathbf{v}_i) = \varphi_i$ ([Theorem 3.1.5](#)). It sends a basis to a basis, so it is an isomorphism, by the argument in the proof of [Theorem 3.5.4](#). Now take $V = \mathbb{R}$, a space of dimension 1. With the basis (1), the dual basis is $\varphi(t) = t$, and $\Theta_{(1)}(1)$ is $t \mapsto t$. With the basis (2), the dual basis is $\varphi(t) = t/2$, since that is the functional sending 2 to 1. So $\Theta_{(2)}(2) = (t \mapsto t/2)$, and by linearity $\Theta_{(2)}(1) = \frac{1}{2}\Theta_{(2)}(2)$ is $t \mapsto t/4$. The same vector 1 goes to two different functionals. Every basis is as good as any other, so none of these isomorphisms is

preferred.

The isomorphism $V \rightarrow V^{}$ does not.** Build it the same way, choosing bases twice. First $\Theta_{\mathcal{B}}: V \rightarrow V^*$ sends $\mathbf{v}_i \mapsto \varphi_i$. Then, using the basis $\mathcal{B}^*$ of V^* , the map $\Theta_{\mathcal{B}^*}: V^* \rightarrow V^{**}$ sends each φ_i to the i -th vector of the dual basis of $\mathcal{B}^*$. By [Theorem 4.3.2](#), that vector is $\text{ev}_{\mathbf{v}_i}$. So the composite $\Theta_{\mathcal{B}^*}\Theta_{\mathcal{B}}$ sends $\mathbf{v}_i \mapsto \text{ev}_{\mathbf{v}_i}$ for every i . The linear map ev_V does the same, so by the uniqueness part of [Theorem 3.1.5](#),

$$\Theta_{\mathcal{B}^*}\Theta_{\mathcal{B}} = \text{ev}_V \quad \text{for every basis } \mathcal{B} \text{ of } V.$$

Each factor depends on $\mathcal{B}$, but the composite does not: the two choices cancel. This is the first meaning of **natural**: an isomorphism is natural if it can be defined without choosing anything, and ev_V is defined by the formula $\text{ev}_{\mathbf{v}}(\varphi) = \varphi(\mathbf{v})$ alone.

There is a second, sharper meaning. A choice-free construction should be compatible with every linear map: if $T: V \rightarrow W$, then first applying T and then identifying W with W^{**} should agree with first identifying V with V^{**} and then applying “the map that T induces on double duals”. To state this we need to know how a linear map acts on functionals. That is the **dual map**, the subject of the next section, where we prove that ev is compatible with every linear map, and see that no family of isomorphisms $V \rightarrow V^*$ can be.

Quick check. In $V = \mathbb{R}$ with the basis $\mathcal{B} = (2)$, compute $\Theta_{\mathcal{B}^*}\Theta_{\mathcal{B}}(1)$ as an element of V^{**} , and evaluate it on $\psi(t) = 5t$. Compare with $\text{ev}_1(\psi)$.

Solution. The dual basis of $\mathcal{B}$ is $\varphi_1(t) = t/2$, so $\Theta_{\mathcal{B}}(1) = \frac{1}{2}\Theta_{\mathcal{B}}(2) = \frac{1}{2}\varphi_1$. The dual basis of $\mathcal{B}^* = (\varphi_1)$ is the $\theta \in V^{**}$ with $\theta(\varphi_1) = 1$, and $\text{ev}_2(\varphi_1) = \varphi_1(2) = 1$, so $\theta = \text{ev}_2$ and $\Theta_{\mathcal{B}^*}(\varphi_1) = \text{ev}_2$. Hence $\Theta_{\mathcal{B}^*}\Theta_{\mathcal{B}}(1) = \frac{1}{2}\text{ev}_2 = \text{ev}_1$, using linearity of ev . On ψ it gives $\text{ev}_1(\psi) = \psi(1) = 5$. The factor $\frac{1}{2}$ from the first step and the factor 2 hidden in ev_2 cancel, exactly as the general computation predicts.

Remark. Because ev_V is natural, it is common to **identify** a finite-dimensional V with V^{**} , and to regard a vector $\mathbf{v}$ as the functional $\varphi \mapsto \varphi(\mathbf{v})$ on V^* . We will do this only with an explicit “under the identification $\mathbf{v} \leftrightarrow \text{ev}_{\mathbf{v}}$ ”, as in the next corollary.

4.3.4 Consequences in finite dimension

Throughout this subsection, V is finite-dimensional. The isomorphism ev_V lets us run arguments about V^* “one level up” and bring the answer back to V .

The dual basis construction goes from a basis of V to a basis of V^* . Does every basis of V^* arise this way? A list of n independent measurements should be the coordinate functions of some basis, and the double dual shows it is.

Corollary 4.3.3 — Every Basis of the Dual Is a Dual Basis

Let V be a finite-dimensional vector space over F , and let $(\psi_1, \dots, \psi_n)$ be a basis of V^* . Then there is exactly one basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ of V whose dual basis is $(\psi_1, \dots, \psi_n)$, that is, with

$$\psi_i(\mathbf{v}_j) = \delta_{ij} \quad \text{for all } i, j.$$

Idea. Go up one level, where the problem is already solved. The basis $(\psi_1, \dots, \psi_n)$ of V^* has a dual basis $(\theta_1, \dots, \theta_n)$ in V^{**} . Every θ_j is evaluation at some vector, because ev is onto. Those vectors are the answer, since $\theta_j(\psi_i) = \delta_{ij}$ says $\psi_i(\mathbf{v}_j) = \delta_{ij}$. Uniqueness is a “test against functionals” argument.

Proof. Existence. Here V^* is finite-dimensional (Corollary 4.1.4). By Theorem 4.1.3 applied to V^* , there is a basis $(\theta_1, \dots, \theta_n)$ of V^{**} with $\theta_j(\psi_i) = \delta_{ij}$. By Theorem 4.3.2, ev_V is an isomorphism, so there are unique $\mathbf{v}_j \in V$ with $\text{ev}_{\mathbf{v}_j} = \theta_j$. Its inverse is an isomorphism (Theorem 3.5.3), so $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is a basis of V by Theorem 3.5.2. For all i, j ,

$$\psi_i(\mathbf{v}_j) = \text{ev}_{\mathbf{v}_j}(\psi_i) = \theta_j(\psi_i) = \delta_{ij}.$$

By the uniqueness in Theorem 4.1.3 (a), the dual basis of $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is $(\psi_1, \dots, \psi_n)$.

Uniqueness. Suppose $(\mathbf{u}_1, \dots, \mathbf{u}_n)$ is also a basis with $\psi_i(\mathbf{u}_j) = \delta_{ij}$. Fix j . Then $\psi_i(\mathbf{v}_j - \mathbf{u}_j) = \delta_{ij} - \delta_{ij} = 0$ for every i . Every $\varphi \in V^*$ is a combination of the ψ_i , so $\varphi(\mathbf{v}_j - \mathbf{u}_j) = 0$ for every $\varphi \in V^*$. By Theorem 4.1.5 (b), $\mathbf{v}_j - \mathbf{u}_j = \mathbf{0}$. Hence $\mathbf{u}_j = \mathbf{v}_j$ for every j . ■

In coordinates this is a matrix inversion, the mirror image of Example (The Dual Basis Is Read Off the Inverse Matrix). There, a basis was given and its dual basis appeared as the rows of an inverse matrix. Here the functionals are given, and the basis appears as the columns of an inverse matrix.

Example. (A Basis With Prescribed Dual Basis)

On $V = \mathbb{R}[x]_{\leq 2}$, consider the functionals

$$\psi_1(p) = p(0), \quad \psi_2(p) = p(1), \quad \psi_3(p) = p'(1).$$

Show that (ψ_1, ψ_2, ψ_3) is a basis of V^* , and find the basis (p_1, p_2, p_3) of V whose dual basis it is.

Solution. Write down a matrix. For $p = a + bx + cx^2$, with $[p]_{\mathcal{E}} = (a, b, c)$ in $\mathcal{E} = (1, x, x^2)$,

$$\psi_1(p) = a, \quad \psi_2(p) = a + b + c, \quad \psi_3(p) = b + 2c.$$

So $\psi_i(p) = \mathbf{r}_i[p]_{\mathcal{E}}$, where $\mathbf{r}_i$ is row i of

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

Translate the condition. Let P be the 3×3 matrix whose j -th column is $[p_j]_{\mathcal{E}}$. By Theorem 0.9.2, the (i, j) entry of AP is $\mathbf{r}_i[p_j]_{\mathcal{E}} = \psi_i(p_j)$. So the condition $\psi_i(p_j) = \delta_{ij}$ says exactly $AP = I_3$.

Solve. Row reduction (Theorem 2.4.6) gives

$$A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{pmatrix},$$

and one checks $AA^{-1} = I_3$, so A is invertible by Theorem 2.4.5. Multiplying $AP = I_3$ on the left by A^{-1} forces $P = A^{-1}$. Since the columns of the invertible matrix P are independent, (p_1, p_2, p_3) is a basis of V (via the coordinate isomorphism, Corollary 3.5.5). By the uniqueness in Theorem 4.1.3 (a), its dual basis is (ψ_1, ψ_2, ψ_3) , which in particular is a basis of V^* . Reading the columns,

$$p_1 = 1 - 2x + x^2 = (1 - x)^2, \quad p_2 = 2x - x^2, \quad p_3 = x^2 - x.$$

Check. $p_1(0) = 1, p_1(1) = 0, p_1'(1) = 0$; $p_2(0) = 0, p_2(1) = 1, p_2'(1) = 2 - 2 = 0$; $p_3(0) = 0, p_3(1) = 0, p_3'(1) = 1$.

So every $p \in \mathbb{R}[x]_{\leq 2}$ is $p = p(0)p_1 + p(1)p_2 + p'(1)p_3$: the coordinates of p in this basis are the three measurements.

The next consequence is the one we will use most. A subspace U determines the set U^0 of functionals vanishing on it. Going back, the functionals in U^0 vanish together on some set of vectors, which contains U . Is it exactly U ?

Corollary 4.3.4 — The Annihilator of the Annihilator

Let V be a finite-dimensional vector space over F , and let U be a subspace of V . Write $U^{00} := (U^0)^0 \subseteq V^{**}$. Then

$$U^{00} = \text{ev}_V(U) = \{\text{ev}_{\mathbf{u}} : \mathbf{u} \in U\}.$$

So, under the identification $\mathbf{v} \leftrightarrow \text{ev}_{\mathbf{v}}$, we have $U^{00} = U$. Equivalently,

$$U = \{\mathbf{v} \in V : \varphi(\mathbf{v}) = 0 \text{ for every } \varphi \in U^0\}.$$

Idea. One inclusion is a tautology: if $\mathbf{u} \in U$ and φ kills U , then $\text{ev}_{\mathbf{u}}(\varphi) = \varphi(\mathbf{u}) = 0$. For the other, count. The dimension formula applied twice, once in V and once in V^* , gives $\dim U^{00} = \dim U$, and ev is injective, so $\dim \text{ev}_V(U) = \dim U$ too. One inclusion plus equal dimension gives equality.

Proof. Let $n = \dim V$, so $\dim V^* = n$ by Corollary 4.1.4.

Inclusion. Let $\mathbf{u} \in U$. For every $\varphi \in U^0$, $\text{ev}_{\mathbf{u}}(\varphi) = \varphi(\mathbf{u}) = 0$ by Definition 4.2.1. Hence $\text{ev}_{\mathbf{u}} \in (U^0)^0$, and $\text{ev}_V(U) \subseteq U^{00}$.

Dimensions. By Theorem 4.2.2 in V , $\dim U^0 = n - \dim U$. By Theorem 4.2.2 in the n -dimensional space V^* , applied to its subspace U^0 ,

$$\dim U^{00} = n - \dim U^0 = n - (n - \dim U) = \dim U.$$

The restriction of ev_V to U is linear and injective (Theorem 4.3.1), so it is an isomorphism onto its image $\text{ev}_V(U)$, and $\dim \text{ev}_V(U) = \dim U$ (Theorem 3.5.4). So $\text{ev}_V(U)$ is a subspace of U^{00} of the same finite dimension, and Theorem 1.6.8 gives $\text{ev}_V(U) = U^{00}$.

The equivalent form. Let $\mathbf{v} \in V$. Then $\varphi(\mathbf{v}) = 0$ for every $\varphi \in U^0$ if and only if $\text{ev}_{\mathbf{v}} \in U^{00} = \text{ev}_V(U)$, if and only if $\text{ev}_{\mathbf{v}} = \text{ev}_{\mathbf{u}}$ for some $\mathbf{u} \in U$. Since ev_V is injective, this happens if and only if $\mathbf{v} \in U$. ■

The last form says: **a subspace is exactly the common zero set of the functionals that vanish on it.** No vector outside U sneaks through all the equations. In section 6 of this chapter this becomes the statement that every subspace of F^n is the solution set of a homogeneous linear system. An immediate consequence is that the annihilator loses no information.

Corollary 4.3.5 — Subspaces Are Determined by Their Annihilators

Let V be a finite-dimensional vector space, and let U and W be subspaces of V . Then $U = W$ if and only if $U^0 = W^0$.

Proof. $(\Rightarrow)$ is immediate. $(\Leftarrow)$ Suppose $U^0 = W^0$. By Corollary 4.3.4, $U = \{\mathbf{v} : \varphi(\mathbf{v}) = 0 \text{ for all } \varphi \in U^0\} = \{\mathbf{v} : \varphi(\mathbf{v}) = 0 \text{ for all } \varphi \in W^0\} = W$. ■

Quick check. Is Corollary 4.3.5 still true if U and W are allowed to be arbitrary subsets of V ?

Solution. No. In $\mathbb{R}^2$, take $U = \{(1, 0)\}$ and $W = \{(2, 0)\}$. A functional $\varphi(x, y) = ax + by$ vanishes at $(1, 0)$ exactly when $a = 0$, and at $(2, 0)$ exactly when $2a = 0$, that is, again when $a = 0$. So $U^0 = W^0$ but $U \neq W$. An annihilator only sees the span, $U^0 = (\text{span } U)^0$ (Theorem 4.2.3 (b)).

4.3.5 Infinite dimension: the evaluation map is not onto

Nothing in Definition 4.3.1 or Theorem 4.3.1 needed finite dimension. The count in Theorem 4.3.2 did.

Warning. $V \cong V^{**}$ **through** ev is a **finite-dimensional fact**. For $V = F[x]$, the evaluation map is injective but **not** surjective: there is a linear functional on $F[x]^*$ that is not evaluation at any polynomial. The next example constructs one.

Example. (A Functional on $F[x]^*$ That Is Not an Evaluation)

Let $\varphi_k \in F[x]^*$ ($k \in \mathbb{N}$) be the coordinate functionals, $\varphi_k(p) =$ the coefficient of x^k in p , and let $\varepsilon_1(p) = p(1)$, as in Proposition 4.1.6. Show that there is $\theta \in F[x]^{**}$ with $\theta \neq \text{ev}_p$ for every $p \in F[x]$.

Solution. Step 1: $S = \{\varphi_k : k \in \mathbb{N}\} \cup \{\varepsilon_1\}$ is linearly independent. By [Proposition 4.1.6](#), the φ_k form an independent set and $\varepsilon_1 \notin \text{span}\{\varphi_k : k \in \mathbb{N}\}$. Let L be a finite list of distinct elements of S . If ε_1 is not in L , then L is independent. Otherwise, list ε_1 last; the others are independent and ε_1 is not in their span, so L is independent by [Lemma 1.5.3](#).

Step 2: build θ . By [Theorem 1.8.2](#) there is a basis B of $F[x]^*$ containing S . By [Theorem 3.1.6](#) for the (infinite) basis B , there is a linear $\theta: F[x]^* \rightarrow F$ with $\theta(\varepsilon_1) = 1$ and $\theta(\beta) = 0$ for every $\beta \in B \setminus \{\varepsilon_1\}$. In particular $\theta(\varphi_k) = 0$ for all k .

Step 3: θ is not an evaluation. Suppose $\theta = \text{ev}_p$ for some $p \in F[x]$. Then for every k , the coefficient of x^k in p is $\varphi_k(p) = \text{ev}_p(\varphi_k) = \theta(\varphi_k) = 0$. So $p = 0$, and $\theta = \text{ev}_0 = 0$. This contradicts $\theta(\varepsilon_1) = 1$. Hence $\theta \notin \text{ev}(F[x])$, and ev is not surjective.

The functional θ exists, but nobody can write it down: it comes from Zorn's Lemma through [Theorem 1.8.2](#). The deeper reason for the failure is size. $F[x]^*$ is so much bigger than $F[x]$ ([Exercise \(C4: How large is \$F\[x\]^*\$?\)](#) identifies it with the whole sequence space $F^{\mathbb{N}}$) that $F[x]^{**}$ is bigger still, and in fact no isomorphism $F[x] \cong F[x]^{**}$ exists at all, a statement about the sizes of bases that we do not prove. In finite dimension, sizes cannot grow, and the evaluation map is the perfect dictionary of [Theorem 4.3.2](#).

4.3.6 Exercises

4.3.6.1 A. Check your understanding

Exercise A1.

- Define $\text{ev}_{\mathbf{v}}$ for $\mathbf{v} \in V$, and the evaluation map ev_V . In which space does $\text{ev}_{\mathbf{v}}$ live?
- State a hypothesis on V under which ev_V is an isomorphism, and a space for which it is not.
- True or false: if V is finite-dimensional and $\mathbf{v} \neq \mathbf{0}$, then $\text{ev}_{\mathbf{v}}(\varphi) \neq 0$ for **every** non-zero $\varphi \in V^*$. Justify your answer.
- True or false: if U and W are subspaces of $\mathbb{R}^5$ with $U^0 = W^0$, then $U = W$. Justify your answer.
- Name the method used to prove $U^{00} = \text{ev}_V(U)$ in [Corollary 4.3.4](#).

Solution.

- $\text{ev}_{\mathbf{v}}: V^* \rightarrow F$, $\text{ev}_{\mathbf{v}}(\varphi) = \varphi(\mathbf{v})$, and $\text{ev}_V: V \rightarrow V^{**}$, $\mathbf{v} \mapsto \text{ev}_{\mathbf{v}}$ ([Definition 4.3.1](#)). $\text{ev}_{\mathbf{v}}$ lives in V^{**} .
- If V is finite-dimensional, ev_V is an isomorphism ([Theorem 4.3.2](#)). For $V = F[x]$ it is not surjective ([Example \(A Functional on \$F\[x\]^*\$ That Is Not an Evaluation\)](#)).
- False. It says only that **some** φ has $\varphi(\mathbf{v}) \neq 0$ ([Theorem 4.3.1](#)). In $\mathbb{R}^2$, $\mathbf{v} = (1, 2)$ and $\varphi(x, y) = 2x - y \neq 0$ give $\text{ev}_{\mathbf{v}}(\varphi) = 0$.
- True, by [Corollary 4.3.5](#), since $\mathbb{R}^5$ is finite-dimensional.
- One inclusion plus equal dimension: $\text{ev}_V(U) \subseteq U^{00}$ directly, then $\dim U^{00} = \dim U = \dim \text{ev}_V(U)$ from the dimension formula for annihilators (twice) and injectivity of ev , and [Theorem 1.6.8](#).

4.3.6.2 B. Practice

Exercise B1: A basis with a given dual basis.

- (a) Let $\psi_1(x, y) = 2x + y$ and $\psi_2(x, y) = x + y$ on $\mathbb{R}^2$. Find the basis $(\mathbf{v}_1, \mathbf{v}_2)$ of $\mathbb{R}^2$ whose dual basis is (ψ_1, ψ_2) .
- (b) Let $\psi_1(p) = p(0)$ and $\psi_2(p) = \int_0^1 p(x) dx$ on $\mathbb{R}[x]_{\leq 1}$. Find the basis (p_1, p_2) of $\mathbb{R}[x]_{\leq 1}$ whose dual basis is (ψ_1, ψ_2) .

Solution.

- (a) As in [Example \(A Basis With Prescribed Dual Basis\)](#), $\psi_i(\mathbf{v}) = \mathbf{r}_i \mathbf{v}$ with $\mathbf{r}_i$ the rows of $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, and the condition $\psi_i(\mathbf{v}_j) = \delta_{ij}$ says $AP = I_2$ for $P = (\mathbf{v}_1 \ \mathbf{v}_2)$. Here $2 \cdot 1 - 1 \cdot 1 = 1 \neq 0$, so by [Theorem 0.9.8](#)

$$A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix},$$

$P = A^{-1}$, so $\mathbf{v}_1 = (1, -1)$ and $\mathbf{v}_2 = (-1, 2)$. These are independent (the columns of an invertible matrix), hence a basis. Check: $\psi_1(\mathbf{v}_1) = 1$, $\psi_2(\mathbf{v}_1) = 0$, $\psi_1(\mathbf{v}_2) = 0$, $\psi_2(\mathbf{v}_2) = 1$. By the uniqueness part of [Theorem 4.1.3](#), the dual basis of $(\mathbf{v}_1, \mathbf{v}_2)$ is (ψ_1, ψ_2) .

- (b) For $p = a + bx$, $\psi_1(p) = a$ and $\psi_2(p) = a + \frac{1}{2}b$. So $A = \begin{pmatrix} 1 & 0 \\ 1 & 1/2 \end{pmatrix}$ and

$$P = A^{-1} = \begin{pmatrix} 1 & 0 \\ -2 & 2 \end{pmatrix},$$

whose columns are the coordinates of $p_1 = 1 - 2x$ and $p_2 = 2x$ in $(1, x)$. Check: $p_1(0) = 1$, $\int_0^1 (1 - 2x) dx = 1 - 1 = 0$, $p_2(0) = 0$, $\int_0^1 2x dx = 1$. As in (a), (p_1, p_2) is a basis with dual basis (ψ_1, ψ_2) .

Exercise B2: $U^{00} = U$ for a line.

Let $U = \text{span}((1, 2, -1)) \subseteq \mathbb{R}^3$.

- (a) Find a basis of U^0 .
- (b) Find all $\mathbf{v} \in \mathbb{R}^3$ with $\varphi(\mathbf{v}) = 0$ for every $\varphi \in U^0$, and check that the answer is U .
- (c) Hence verify that $U^{00} = \text{ev}(U)$, and confirm the dimension count.

Solution.

- (a) A functional $\varphi(x, y, z) = ax + by + cz$ ([Theorem 4.1.1](#)) lies in U^0 iff it vanishes at $(1, 2, -1)$ ([Theorem 4.2.3](#) (b)), that is, $a + 2b - c = 0$. The solutions are $(a, b, c) = s(-2, 1, 0) + t(1, 0, 1)$, so $\varphi_1(x, y, z) = 2x - y$ and $\varphi_2(x, y, z) = x + z$ span U^0 (we replaced $(-2, 1, 0)$ by its negative). They are independent, since neither is a multiple of the other, so (φ_1, φ_2) is a basis of U^0 . This matches $\dim U^0 = 3 - 1 = 2$ ([Theorem 4.2.2](#)).
- (b) Every $\varphi \in U^0$ is a combination of φ_1, φ_2 , so $\varphi(\mathbf{v}) = 0$ for all $\varphi \in U^0$ iff $\varphi_1(\mathbf{v}) =$

$\varphi_2(\mathbf{v}) = 0$, that is, $2x - y = 0$ and $x + z = 0$. With $x = t$: $y = 2t$, $z = -t$, so $\mathbf{v} = t(1, 2, -1)$. The set is exactly U .

- (c) Let $\theta \in U^{00} \subseteq (\mathbb{R}^3)^{**}$. By [Theorem 4.3.2](#), $\theta = \text{ev}_{\mathbf{v}}$ for some $\mathbf{v}$, and $\theta \in (U^0)^0$ means $\varphi(\mathbf{v}) = 0$ for all $\varphi \in U^0$. By (b), $\mathbf{v} \in U$, so $U^{00} \subseteq \text{ev}(U)$; the reverse inclusion is immediate from [Definition 4.2.1](#). Dimensions: $\dim U^{00} = 3 - \dim U^0 = 1 = \dim U$.

Exercise B3: Recognizing an evaluation.

Let $V = \mathbb{R}[x]_{\leq 2}$, with the coefficient functionals $\varphi_0, \varphi_1, \varphi_2$ of [Example \(Evaluation in Three Spaces\)](#). Let $\theta \in V^{**}$ be the linear functional with $\theta(\varphi_0) = 1$, $\theta(\varphi_1) = -3$, $\theta(\varphi_2) = 2$. Find $p \in V$ with $\theta = \text{ev}_p$. Hence compute $\theta(\varepsilon_1)$ and $\theta(\psi)$, where $\varepsilon_1(q) = q(1)$ and $\psi(q) = q'(0)$.

Solution. By [Theorem 4.3.2](#), $(\text{ev}_1, \text{ev}_x, \text{ev}_{x^2})$ is the dual basis of $(\varphi_0, \varphi_1, \varphi_2)$. Hence $\text{ev}_1 - 3\text{ev}_x + 2\text{ev}_{x^2}$ takes the values $1, -3, 2$ on $\varphi_0, \varphi_1, \varphi_2$. So does θ , and a linear functional is determined by its values on a basis ([Theorem 3.1.5](#)). Since ev is linear ([Theorem 4.3.1](#)),

$$\theta = \text{ev}_1 - 3\text{ev}_x + 2\text{ev}_{x^2} = \text{ev}_p, \quad p = 1 - 3x + 2x^2.$$

Hence $\theta(\varepsilon_1) = \varepsilon_1(p) = 1 - 3 + 2 = 0$ and $\theta(\psi) = \psi(p) = p'(0) = -3$.

4.3.6.3 C. Going deeper

Exercise C1: Subspaces of V and of V^* .

Let V be a finite-dimensional vector space over F . Prove that $U \mapsto U^0$ is a bijection from the set of subspaces of V to the set of subspaces of V^* , and that it reverses inclusions in both directions: $U \subseteq W$ if and only if $W^0 \subseteq U^0$.

Hint: for surjectivity, given a subspace $X \subseteq V^*$, consider $U = \{\mathbf{v} \in V : \varphi(\mathbf{v}) = 0 \text{ for all } \varphi \in X\}$.

Solution. Let $n = \dim V = \dim V^*$ ([Corollary 4.1.4](#)). By [Proposition 4.2.1](#), U^0 is a subspace of V^* , so the map is well defined.

Injective. This is [Corollary 4.3.5](#).

Inclusions. If $U \subseteq W$, then $W^0 \subseteq U^0$ by [Theorem 4.2.3](#) (a). Conversely, suppose $W^0 \subseteq U^0$, and let $\mathbf{u} \in U$. Every $\varphi \in W^0$ lies in U^0 , so $\varphi(\mathbf{u}) = 0$. By [Corollary 4.3.4](#) (the equivalent form, for W), $\mathbf{u} \in W$. Hence $U \subseteq W$.

Surjective. Let X be a subspace of V^* , and let $U = \{\mathbf{v} \in V : \varphi(\mathbf{v}) = 0 \text{ for all } \varphi \in X\}$. For $\mathbf{v} \in V$, $\mathbf{v} \in U$ iff $\text{ev}_{\mathbf{v}}(\varphi) = 0$ for all $\varphi \in X$ iff $\text{ev}_{\mathbf{v}} \in X^0$, where $X^0 \subseteq V^{**}$. So $U = \text{ev}_V^{-1}(X^0)$, and since ev_V is an isomorphism ([Theorem 4.3.2](#)), U is a subspace and ev_V restricts to an isomorphism $U \rightarrow X^0$. By [Theorem 4.2.2](#) in V^* , $\dim U = \dim X^0 = n - \dim X$. By definition of U , every $\varphi \in X$ vanishes on U , so $X \subseteq U^0$. By [Theorem 4.2.2](#) in V , $\dim U^0 = n - \dim U = \dim X$. By [Theorem 1.6.8](#), $X = U^0$. This proves the claim.

Exercise C2: Spans cut out by equations.

Let V be finite-dimensional and $S \subseteq V$ any subset.

- (a) Prove that $\{\mathbf{v} \in V : \varphi(\mathbf{v}) = 0 \text{ for every } \varphi \in S^0\} = \text{span}(S)$.
- (b) Deduce: a vector $\mathbf{v}$ lies in $\text{span}(S)$ if and only if every functional that vanishes on S vanishes at $\mathbf{v}$.
- (c) Show by an example that (a) can fail if “ $\text{span}(S)$ ” is replaced by “ S ”.

Solution.

- (a) $\text{span}(S)$ is a subspace (Theorem 1.3.1), and $S^0 = (\text{span } S)^0$ by Theorem 4.2.3 (b). Applying Corollary 4.3.4 to $U = \text{span}(S)$,

$$\{\mathbf{v} : \varphi(\mathbf{v}) = 0 \ \forall \varphi \in S^0\} = \{\mathbf{v} : \varphi(\mathbf{v}) = 0 \ \forall \varphi \in (\text{span } S)^0\} = \text{span}(S).$$

- (b) This is (a) read for a single vector $\mathbf{v}$.
- (c) Take $S = \{(1, 0)\} \subseteq \mathbb{R}^2$. Then S^0 consists of the functionals $(x, y) \mapsto by$, and their common zero set is the whole x -axis, which contains $(2, 0) \notin S$.

Exercise C3: Where finite dimension is needed.

Let $V = F[x]$, let φ_k and ε_1 be as in Example (A Functional on $F[x]^*$ That Is Not an Evaluation), and let $X = \text{span}\{\varphi_k : k \in \mathbb{N}\} \subseteq V^*$.

- (a) Show that $\{p \in F[x] : \varphi(p) = 0 \text{ for all } \varphi \in X\} = \{0\}$.
- (b) Deduce that $X \neq U^0$ for **every** subspace U of $F[x]$. So the map $U \mapsto U^0$ of Exercise (C1: Subspaces of V and of V^*) is not surjective for $F[x]$.
- (c) Which step of the solution of Exercise (C1: Subspaces of V and of V^*) fails here?

Solution.

- (a) If $\varphi(p) = 0$ for all $\varphi \in X$, then in particular $\varphi_k(p) = 0$ for every k , so every coefficient of p is 0 and $p = 0$. Conversely $\varphi(0) = 0$ for every φ .
- (b) Suppose $X = U^0$. Every $\varphi \in X = U^0$ vanishes on U , so $U \subseteq \{p : \varphi(p) = 0 \ \forall \varphi \in X\} = \{0\}$ by (a). Hence $U = \{0\}$ and $U^0 = V^*$ (every functional vanishes at 0). But $\varepsilon_1 \in V^*$ and $\varepsilon_1 \notin X$, by Proposition 4.1.6. So $X \neq V^* = U^0$, a contradiction.
- (c) The dimension count. In Exercise (C1: Subspaces of V and of V^*) we used that ev_V is onto and that $\dim U + \dim U^0 = \dim V$ to force $X = U^0$ from $X \subseteq U^0$. Here $U = \{0\}$ and $X \subsetneq U^0 = V^*$: the inclusion holds, but there is no finite dimension to compare, and a proper subspace of an infinite-dimensional space can be “as large as” the whole space.

4.4 The Dual Map and the Transpose

So far each space has had its own dual, but the duals of different spaces have not talked to each other. Linear algebra is about maps, so the question is: what does a linear map $T: V \rightarrow W$ do to functionals? This section shows that T moves vectors forward but moves measurements **backward**, giving the dual map $T': W^* \rightarrow V^*$. In dual bases its matrix is the transpose of

the matrix of T , which finally explains what the transpose of Chapter 0 means. The kernel and image of T' are annihilators of the image and kernel of T , and counting dimensions gives a third proof that row rank equals column rank. At the end we return to the double dual and prove that the evaluation map is compatible with every linear map.

4.4.1 Pulling functionals back

After spaces come the maps between them. We have a linear map $T: V \rightarrow W$ and the two spaces of measurements V^* and W^* . We want to know how T relates them.

Start with the standard example, a matrix map $T_A: F^n \rightarrow F^m$, $\mathbf{x} \mapsto A\mathbf{x}$, with $A \in M_{m \times n}(F)$. A functional on F^m is $\psi(\mathbf{y}) = \mathbf{b}^\top \mathbf{y}$ for a unique $\mathbf{b} \in F^m$ (Theorem 4.1.1). Given ψ , there is one obvious way to measure a vector $\mathbf{x} \in F^n$: move it to F^m with T_A , then measure. The result is

$$\psi(T_A \mathbf{x}) = \mathbf{b}^\top A \mathbf{x} = (A^\top \mathbf{b})^\top \mathbf{x},$$

using $(A^\top \mathbf{b})^\top = \mathbf{b}^\top A$ (Theorem 0.9.5). So “move, then measure” is again a functional, now on F^n , and it is given by the vector $A^\top \mathbf{b}$. The transpose appears by itself. The recipe “compose with T ” makes sense for any linear map, and it deserves a name.

The dual map pulls a measurement of W back to a measurement of V : first move the vector with T , then measure it in W .

Definition 4.4.1 — Dual Map

Let V and W be vector spaces over F , and let $T \in \mathcal{L}(V, W)$. The **dual map** of T is the map

$$T': W^* \rightarrow V^*, \quad T'(\psi) := \psi T = \psi \circ T.$$

That is, $(T'\psi)(\mathbf{v}) = \psi(T\mathbf{v})$ for every $\psi \in W^*$ and every $\mathbf{v} \in V$.

In words, clause by clause:

- The input of T' is a functional on the **codomain** W , and the output is a functional on the **domain** V . The arrow is reversed.
- The formula $(T'\psi)(\mathbf{v}) = \psi(T\mathbf{v})$ has two steps: to measure $\mathbf{v}$ with $T'\psi$, send $\mathbf{v}$ to W with T , then apply ψ there.
- No basis and no finite dimension are involved.

Well-definedness. Two things must be checked. First, $T'\psi$ must lie in V^* : it is the composite of the linear maps $T: V \rightarrow W$ and $\psi: W \rightarrow F$, so it is linear by Theorem 3.4.2 (a). Second, T' should be linear, so that $T' \in \mathcal{L}(W^*, V^*)$. By the distributive laws in Theorem 3.4.2 (d), for $\psi_1, \psi_2 \in W^*$ and $c \in F$,

$$T'(\psi_1 + \psi_2) = (\psi_1 + \psi_2)T = \psi_1 T + \psi_2 T = T'\psi_1 + T'\psi_2, \quad T'(c\psi) = (c\psi)T = c(\psi T) = cT'\psi.$$

Example. (Dual Maps in Four Settings)

- For $A \in M_{m \times n}(F)$, describe T'_A in terms of the vectors $\mathbf{b}$ and $A^\top \mathbf{b}$ that represent functionals.
- Let $D: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$, $D(p) = p'$, and let $\varepsilon_2 \in (\mathbb{R}[x]_{\leq 2})^*$ be $\varepsilon_2(q) = q(2)$.

Compute $D'(\varepsilon_2)$.

- (c) Let $S: F[x] \rightarrow F[x]$, $S(p) = xp$, and let $\varphi_k \in F[x]^*$ send a polynomial to its coefficient of x^k . Compute $S'(\varphi_k)$ for every $k \in \mathbb{N}$.
- (d) Find $(\text{id}_V)'$, the dual map of the zero map $V \rightarrow W$, and the dual map of the inclusion $\iota: U \rightarrow V$, $\iota(\mathbf{u}) = \mathbf{u}$, of a subspace $U \subseteq V$.

Solution.

- (a) If $\psi(\mathbf{y}) = \mathbf{b}^\top \mathbf{y}$, the computation before the definition shows $(T'_A \psi)(\mathbf{x}) = (A^\top \mathbf{b})^\top \mathbf{x}$. So, identifying a functional with its vector, T'_A is $\mathbf{b} \mapsto A^\top \mathbf{b}$. In row language: T'_A sends the row $\mathbf{b}^\top$ to the row $\mathbf{b}^\top A$.
- (b) For $p \in \mathbb{R}[x]_{\leq 3}$, $(D'\varepsilon_2)(p) = \varepsilon_2(Dp) = p'(2)$. So D' turns “the value at 2” into “the slope at 2”.
- (c) $(S'\varphi_k)(p) = \varphi_k(xp)$. Multiplying by x shifts every coefficient up one place, so the coefficient of x^k in xp is the coefficient of x^{k-1} in p when $k \geq 1$, and 0 when $k = 0$. Hence $S'\varphi_k = \varphi_{k-1}$ for $k \geq 1$, and $S'\varphi_0 = 0$. Here $F[x]$ is infinite-dimensional, and the definition works all the same.
- (d) $(\text{id}_V)'\varphi = \varphi \circ \text{id}_V = \varphi$, so $(\text{id}_V)' = \text{id}_{V^*}$. For the zero map $0: V \rightarrow W$, $(0'\psi)(\mathbf{v}) = \psi(0) = 0$ ([Theorem 3.1.1](#)), so $0'$ is the zero map $W^* \rightarrow V^*$. For the inclusion, $(\iota'\varphi)(\mathbf{u}) = \varphi(\mathbf{u})$ for $\mathbf{u} \in U$, so $\iota'\varphi = \varphi|_U$ is the **restriction** of φ to U . The restriction map $V^* \rightarrow U^*$ is thus a dual map; we study it further in the next section.

Here is a non-example by minimal change. Try to push functionals **forward** instead: for $T: V \rightarrow W$ and $\varphi \in V^*$, look for a $\psi \in W^*$ with $\psi T = \varphi$, and call it “ $T_*\varphi$ ”. Take $T: \mathbb{R} \rightarrow \mathbb{R}^2$, $T(t) = (t, 0)$, and $\varphi(t) = t$. Then $\psi(x, y) = x + cy$ satisfies $\psi(T(t)) = t$ for **every** $c \in \mathbb{R}$, so there is no single answer. And for the zero map $\mathbb{R} \rightarrow \mathbb{R}^2$, no ψ at all satisfies $\psi \circ 0 = \varphi$. What fails is well-definedness: a function must assign **exactly one** value. Composition with T goes only one way, on the right, and that forces the direction $W^* \rightarrow V^*$.

Why this definition. From $T: V \rightarrow W$ and $\psi: W \rightarrow F$, the only composite we can form is ψT ; the non-example shows that the other direction is not a function. Linearity of T' comes for free from the distributive laws. The name “dual map” says that T' is the map T induces between the duals. It is also called the **transpose** of T , for a reason that the matrix theorem below makes exact. We write T' rather than T^* , because T^* is reserved for the adjoint of Chapter 10, which is related to T' but needs an inner product.

Warning. **The dual map goes backwards.** For $T: V \rightarrow W$, the dual map is $T': W^* \rightarrow V^*$, **not** $V^* \rightarrow W^*$. As a result, composition reverses order: if $T: U \rightarrow V$ and $S: V \rightarrow W$, then $(ST)' = T'S'$, just as $(AB)^\top = B^\top A^\top$. The other order $S'T'$ does not even make sense: T' lands in U^* , where S' is not defined.

Quick check. Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $T(x, y, z) = (x - z, 2y)$, and $\psi(u, v) = u + v$. Compute $T'\psi$. If T' is written as a matrix in some bases, what is its size?

Solution. $(T'\psi)(x, y, z) = \psi(x - z, 2y) = x + 2y - z$. The map T' goes from $(\mathbb{R}^2)^*$, of dimension 2, to $(\mathbb{R}^3)^*$, of dimension 3 ([Corollary 4.1.4](#)), so its matrix is 3×2 , the size of the transpose of the 2×3 matrix of T .

The first payoff is that duals behave well under the operations on maps, with one reversal.

Theorem 4.4.1 — Properties of the Dual Map

Let U, V, W be vector spaces over F .

- (a) For $S, T \in \mathcal{L}(V, W)$ and $a, b \in F$, $(aS + bT)' = aS' + bT'$.
- (b) For $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, $(ST)' = T'S'$.
- (c) $(\text{id}_V)' = \text{id}_{V^*}$.
- (d) If $T \in \mathcal{L}(V, W)$ is invertible, then T' is invertible and $(T')^{-1} = (T^{-1})'$.

Idea. Each identity is an equality of maps into a dual space, so we evaluate both sides at a functional, and then evaluate the resulting functionals at a vector. After that, each line is the definition of T' read twice. Part (d) needs no computation: apply (b) and (c) to $T^{-1}T = \text{id}_V$ and $TT^{-1} = \text{id}_W$.

Proof.

- (a) Let $\psi \in W^*$ and $\mathbf{v} \in V$. Using the linearity of ψ and the pointwise operations (Definition 3.4.1),

$$((aS + bT)'\psi)(\mathbf{v}) = \psi(aS\mathbf{v} + bT\mathbf{v}) = a\psi(S\mathbf{v}) + b\psi(T\mathbf{v}) = (aS'\psi + bT'\psi)(\mathbf{v}).$$

Hence $(aS + bT)'\psi = (aS' + bT')\psi$ for every ψ , and $(aS + bT)' = aS' + bT'$.

- (b) Let $\psi \in W^*$. By Definition 4.4.1 and associativity of composition (Theorem 3.4.2 (b)),

$$(ST)'\psi = \psi(ST) = (\psi S)T = T'(\psi S) = T'(S'\psi) = (T'S')\psi.$$

Hence $(ST)' = T'S'$.

- (c) This is Example (Dual Maps in Four Settings) (d).
- (d) Suppose T is invertible with inverse $T^{-1} \in \mathcal{L}(W, V)$. By (b) and (c),

$$T'(T^{-1})' = (T^{-1}T)' = (\text{id}_V)' = \text{id}_{V^*}, \quad (T^{-1})'T' = (TT^{-1})' = (\text{id}_W)' = \text{id}_{W^*}.$$

The map $(T^{-1})'$ is linear, so by Definition 3.5.1 T' is invertible with inverse $(T^{-1})'$. This proves the theorem. ■

Part (a) says that $T \mapsto T'$ is itself a linear map $\mathcal{L}(V, W) \rightarrow \mathcal{L}(W^*, V^*)$. Part (b) is the reversal of the warning. Parts (c) and (d) say that dualizing turns isomorphisms into isomorphisms, so isomorphic spaces have isomorphic duals.

4.4.2 The matrix of the dual map

Now we fix bases and compute. Let $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of V with dual basis $\mathcal{B}^* = (\varphi_1, \dots, \varphi_n)$ (Definition 4.1.3), and $\mathcal{C} = (\mathbf{w}_1, \dots, \mathbf{w}_m)$ a basis of W with dual basis $\mathcal{C}^* = (\psi_1, \dots, \psi_m)$. The map T has the $m \times n$ matrix $[T]_{\mathcal{B}}^{\mathcal{C}}$ (Definition 3.6.1). The dual map $T': W^* \rightarrow V^*$ has input basis $\mathcal{C}^*$ and output basis $\mathcal{B}^*$, so its matrix is $[T']_{\mathcal{C}^*}^{\mathcal{B}^*}$, of size $n \times m$. The size already suggests the answer, and Example (Dual Maps in Four Settings) (a) suggests it for matrix maps.

Theorem 4.4.2 — The Matrix of the Dual Map Is the Transpose

Let V and W be finite-dimensional vector spaces over F , with ordered bases $\mathcal{B}$ of V and $\mathcal{C}$ of W , and let $\mathcal{B}^*$ and $\mathcal{C}^*$ be their dual bases. For every $T \in \mathcal{L}(V, W)$,

$$[T']_{\mathcal{C}^*}^{\mathcal{B}^*} = ([T]_{\mathcal{B}}^{\mathcal{C}})^{\top}.$$

Idea. Both matrices are tables of the same numbers $\psi_i(T\mathbf{v}_j)$. In $[T]_{\mathcal{B}}^{\mathcal{C}}$, the (i, j) entry is the $\mathbf{w}_i$ -coordinate of $T\mathbf{v}_j$, and a dual basis vector **reads off** a coordinate: that coordinate is $\psi_i(T\mathbf{v}_j)$. In $[T']_{\mathcal{C}^*}^{\mathcal{B}^*}$, column i holds the $\mathcal{B}^*$ -coordinates of $T'\psi_i$, and the φ_j -coordinate of a functional is its value at $\mathbf{v}_j$: that is $(T'\psi_i)(\mathbf{v}_j) = \psi_i(T\mathbf{v}_j)$. The same number sits at (i, j) in one matrix and at (j, i) in the other.

Proof. Write $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$, $\mathcal{B}^* = (\varphi_1, \dots, \varphi_n)$, $\mathcal{C} = (\mathbf{w}_1, \dots, \mathbf{w}_m)$, $\mathcal{C}^* = (\psi_1, \dots, \psi_m)$, and $A = (a_{kj}) = [T]_{\mathcal{B}}^{\mathcal{C}}$. By [Definition 3.6.1](#), a_{ij} is the i -th entry of $[T\mathbf{v}_j]_{\mathcal{C}}$, and by [Theorem 4.1.3](#) (a) that entry is read off by ψ_i :

$$\psi_i(T\mathbf{v}_j) = a_{ij}. \quad (*)$$

Next, by [Theorem 4.1.3](#) (c), every $\varphi \in V^*$ satisfies $\varphi = \sum_{j=1}^n \varphi(\mathbf{v}_j)\varphi_j$. Since $\mathcal{B}^*$ is a basis, coordinates in it are unique ([Theorem 1.5.1](#)), so $[\varphi]_{\mathcal{B}^*} = (\varphi(\mathbf{v}_1), \dots, \varphi(\mathbf{v}_n))$.

By [Definition 3.6.1](#), column i of $[T']_{\mathcal{C}^*}^{\mathcal{B}^*}$ is $[T'\psi_i]_{\mathcal{B}^*}$. By the previous paragraph and $(*)$, its j -th entry is

$$(T'\psi_i)(\mathbf{v}_j) = \psi_i(T\mathbf{v}_j) = a_{ij}.$$

So the (j, i) entry of $[T']_{\mathcal{C}^*}^{\mathcal{B}^*}$ is a_{ij} , which is the (j, i) entry of $A^{\top}$ ([Definition 0.9.10](#)). Both matrices are $n \times m$, so $[T']_{\mathcal{C}^*}^{\mathcal{B}^*} = A^{\top}$. This proves the theorem. ■

This is the meaning of the transpose. In Chapter 0, [Definition 0.9.10](#) was a rule for rearranging entries. Now we see it as the matrix shadow of pulling functionals back: **transposing a matrix is dualizing the map**, provided we use dual bases on both sides. Every property of the transpose becomes a property of dual maps, and conversely; for instance, [Theorem 4.4.1](#) (b) and this theorem together give $(AB)^{\top} = B^{\top}A^{\top}$ a second time (Exercise B3 below).

Example. (A 2×3 Example)

Let $T: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}^2$, $T(p) = (p(1), p'(1))$. Use the basis $\mathcal{B} = (1, x, x^2)$, whose dual basis $\mathcal{B}^* = (\varphi_0, \varphi_1, \varphi_2)$ consists of $\varphi_0(p) = p(0)$, $\varphi_1(p) = p'(0)$, $\varphi_2(p) = p''(0)/2$ ([Example \(Dual Bases in \$\mathbb{R}\[x\]_{\leq 2}\$ \)](#)), and the standard basis $\mathcal{C} = (\mathbf{e}_1, \mathbf{e}_2)$ of $\mathbb{R}^2$, with dual basis $\mathcal{C}^* = (\eta_1, \eta_2)$, $\eta_1(u, v) = u$, $\eta_2(u, v) = v$.

- Find $A = [T]_{\mathcal{B}}^{\mathcal{C}}$.
- Compute $T'\eta_1$ and $T'\eta_2$ directly as combinations of $\varphi_0, \varphi_1, \varphi_2$, write down $[T']_{\mathcal{C}^*}^{\mathcal{B}^*}$, and compare with $A^{\top}$.
- Use the matrix to compute $T'\psi$ for $\psi = -2\eta_1 + \eta_2$, and check the answer directly.

Solution.

- (a) $T(1) = (1, 0)$, $T(x) = (1, 1)$, $T(x^2) = (1, 2)$. These are the columns:

$$A = [T]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

- (b) Let $p = a + bx + cx^2$, so $\varphi_0(p) = a$, $\varphi_1(p) = b$, $\varphi_2(p) = c$. Then

$$(T'\eta_1)(p) = \eta_1(Tp) = p(1) = a + b + c, \quad (T'\eta_2)(p) = \eta_2(Tp) = p'(1) = b + 2c.$$

So $T'\eta_1 = \varphi_0 + \varphi_1 + \varphi_2$ and $T'\eta_2 = \varphi_1 + 2\varphi_2$. Their coordinate vectors in $\mathcal{B}^*$ are the columns of

$$[T']_{\mathcal{C}^*}^{\mathcal{B}^*} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix} = A^T,$$

as [Theorem 4.4.2](#) predicts. Read in words, the first column is Taylor's formula $p(1) = p(0) + p'(0) + \frac{1}{2}p''(0)$ for quadratics, and the second is $p'(1) = p'(0) + p''(0)$.

- (c) $[\psi]_{\mathcal{C}^*} = (-2, 1)$. By [Theorem 3.6.1](#),

$$[T'\psi]_{\mathcal{B}^*} = A^T \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \\ 0 \end{pmatrix},$$

so $T'\psi = -2\varphi_0 - \varphi_1$. Directly: $(T'\psi)(p) = -2p(1) + p'(1) = -2(a+b+c) + (b+2c) = -2a - b$, which is $(-2\varphi_0 - \varphi_1)(p)$.

Warning. Dual bases on both sides. The transpose rule needs the **dual** bases $\mathcal{C}^*$ and $\mathcal{B}^*$. With other bases of W^* and V^* the matrix of T' is usually not A^T . In [Example \(A \$2 \times 3\$ Example\)](#), keep $\mathcal{B}^*$ but use the basis $(\eta_1 + \eta_2, \eta_2)$ of $(\mathbb{R}^2)^*$: then $T'(\eta_1 + \eta_2) = \varphi_0 + 2\varphi_1 + 3\varphi_2$, so the first column is $(1, 2, 3)$, not the first column $(1, 1, 1)$ of A^T .

4.4.3 Kernel and image of the dual map

The kernel and image of T' are subspaces of W^* and V^* . The natural candidates for them are annihilators of the subspaces that T defines, and the dictionary swaps kernel and image.

Theorem 4.4.3 — Kernel and Image of the Dual Map

Let V and W be vector spaces over F , and let $T \in \mathcal{L}(V, W)$.

- (a) $\ker T' = (\operatorname{im} T)^0$.
- (b) If V and W are finite-dimensional, then $\operatorname{im} T' = (\ker T)^0$.

Idea.

- (a) is a chain of definitions: $\psi T = 0$ says that ψ vanishes on every $T\mathbf{v}$, that is, on $\operatorname{im} T$. For (b), one inclusion is just as easy: ψT kills every vector that T kills.

The reverse inclusion asks us to **extend** a functional given on V to one of the form ψT , which is a construction. Instead we count. Rank–Nullity for T' , together with (a) and the dimension formula for annihilators, gives $\dim \operatorname{im} T' = \operatorname{rank} T$; the dimension formula in V and Rank–Nullity for T give $\dim(\ker T)^0 = \operatorname{rank} T$. Finite dimension is spent exactly on these counts.

Proof.

- (a) Let $\psi \in W^*$. Then $\psi \in \ker T'$ iff $\psi T = 0$ iff $\psi(T\mathbf{v}) = 0$ for every $\mathbf{v} \in V$ iff $\psi(\mathbf{w}) = 0$ for every $\mathbf{w} \in \operatorname{im} T$ (Definition 3.2.2) iff $\psi \in (\operatorname{im} T)^0$ (Definition 4.2.1).
 (b) Let $n = \dim V$ and $m = \dim W$, so $\dim V^* = n$ and $\dim W^* = m$ (Corollary 4.1.4).

Inclusion. Let $\varphi \in \operatorname{im} T'$, say $\varphi = \psi T$ with $\psi \in W^*$. For $\mathbf{u} \in \ker T$, $\varphi(\mathbf{u}) = \psi(T\mathbf{u}) = \psi(\mathbf{0}) = 0$. Hence $\varphi \in (\ker T)^0$, and $\operatorname{im} T' \subseteq (\ker T)^0$.

Dimensions. By the Rank–Nullity Theorem (Theorem 3.3.1) for $T': W^* \rightarrow V^*$, then (a), then Theorem 4.2.2 for the subspace $\operatorname{im} T$ of W ,

$$\dim \operatorname{im} T' = m - \dim \ker T' = m - \dim(\operatorname{im} T)^0 = m - (m - \dim \operatorname{im} T) = \operatorname{rank} T.$$

By Theorem 4.2.2 for the subspace $\ker T$ of V , and Rank–Nullity for T ,

$$\dim(\ker T)^0 = n - \dim \ker T = \operatorname{rank} T.$$

So $\operatorname{im} T'$ is a subspace of the finite-dimensional space $(\ker T)^0$ with the same dimension, and Theorem 1.6.8 gives $\operatorname{im} T' = (\ker T)^0$. This proves the theorem. ■

For a matrix map, this is a statement about rows. By Example (Dual Maps in Four Settings) (a), T'_A sends the row $\mathbf{b}^\top$ to $\mathbf{b}^\top A$, so $\operatorname{im} T'_A$ is the set of combinations of the rows of A : it is the row space, read as functionals. Part (b) then says that **the functionals vanishing on $\operatorname{nul}(A)$ are exactly the combinations of the rows of A** . Part (a) says that the functionals vanishing on $\operatorname{col}(A)$ are the rows $\mathbf{b}^\top$ with $\mathbf{b}^\top A = 0$, that is, $A^\top \mathbf{b} = \mathbf{0}$.

Example. (Kernel and Image of a Dual Map)

Let $T = T_A: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ with

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{pmatrix}.$$

Find bases of $\ker T'$ and $\operatorname{im} T'$, written as explicit functionals, and check Theorem 4.4.3.

Solution. The matrix world. The third row of A is the sum of the first two, and subtracting them leaves the RREF $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$. So $\operatorname{rank} A = 2$, and $A\mathbf{x} = \mathbf{0}$ gives $x = -2z$, $y = z$: $\ker T = \operatorname{span}((-2, 1, 1))$. The pivot columns give $\operatorname{im} T = \operatorname{span}((1, 0, 1), (0, 1, 1))$ (Theorem 2.5.2).

Kernel of T' . By Theorem 4.4.2 with standard bases and Example (Matrix Maps and

the Identity) (a), a functional $\psi(u, v, w) = b_1u + b_2v + b_3w$ lies in $\ker T'$ iff $A^\top \mathbf{b} = \mathbf{0}$:

$$b_1 + b_3 = 0, \quad b_2 + b_3 = 0, \quad 2b_1 - b_2 + b_3 = 0.$$

The first two give $\mathbf{b} = t(1, 1, -1)$, and these satisfy the third. So $\ker T' = \text{span}(\psi_0)$ with $\psi_0(u, v, w) = u + v - w$. *Check:* $\psi_0(1, 0, 1) = 0$ and $\psi_0(0, 1, 1) = 0$, so ψ_0 vanishes on $\text{im } T$; and $\dim(\text{im } T)^0 = 3 - 2 = 1$ (Theorem 4.2.2). So $\ker T' = (\text{im } T)^0$.

Image of T' . By Example (Dual Maps in Four Settings) (a), $\text{im } T'$ is spanned by the rows of A read as functionals, and the first two rows already span (the third is their sum):

$$\varphi_1(x, y, z) = x + 2z, \quad \varphi_2(x, y, z) = y - z.$$

They are independent (neither is a multiple of the other), so (φ_1, φ_2) is a basis of $\text{im } T'$. *Check:* $\varphi_1(-2, 1, 1) = 0$ and $\varphi_2(-2, 1, 1) = 0$, so both vanish on $\ker T$, and $\dim(\ker T)^0 = 3 - 1 = 2$. So $\text{im } T' = (\ker T)^0$.

Counting dimensions in the proof of Theorem 4.4.3 already gave $\dim \text{im } T' = \text{rank } T$. We record it, together with its most famous consequence.

Corollary 4.4.4 — Rank of the Dual Map (a) Let V and W be finite-dimensional vector spaces over F , and $T \in \mathcal{L}(V, W)$. Then $\text{rank } T' = \text{rank } T$.
 (b) For every $A \in M_{m \times n}(F)$, $\text{rank } A^\top = \text{rank } A$. Equivalently, $\dim \text{row}(A) = \dim \text{col}(A)$: **row rank equals column rank**.

Proof.

- (a) This is the first display in the proof of Theorem 4.4.3 (b).
 (b) Let $\mathcal{E}_n, \mathcal{E}_m$ be the standard bases. By Example (Matrix Maps and the Identity) (a), $[T_A]_{\mathcal{E}_n}^{\mathcal{E}_m} = A$, so by Theorem 4.4.2, $[T'_A]_{\mathcal{E}_m}^{\mathcal{E}_n} = A^\top$. By Theorem 3.6.5 (a), applied to T_A and to T'_A , and by (a),

$$\text{rank } A = \text{rank } T_A = \text{rank } T'_A = \text{rank } A^\top.$$

Finally, the transpose $M_{1 \times n}(F) \rightarrow F^n$ is linear (Theorem 0.9.5) and bijective (it is its own inverse), and it sends row i of A to column i of $A^\top$. By Theorem 3.1.4 it maps $\text{row}(A)$ onto $\text{col}(A^\top)$, so these spaces are isomorphic and $\dim \text{row}(A) = \dim \text{col}(A^\top) = \text{rank } A^\top = \text{rank } A = \dim \text{col}(A)$ (Theorem 3.5.4, Definition 2.5.4). ■

This is the third proof of this fact in the book, and each one explains it differently.

- **Chapter 2** (Theorem 2.5.5) counted pivots. Row reduction preserves the row space and the relations among the columns, and in the RREF both dimensions visibly equal the number of pivots. The proof is an algorithm.
- **Chapter 3, §8** factored $A = BD$ through a space of dimension $\dim \text{col}(A)$, after Proposition 3.8.2, and saw that the rows of A are combinations of the few rows of D . The proof is a factorization.
- **Here** there are no row operations and no factorizations. The equality is the dimension formula $\dim U + \dim U^0 = \dim V$, applied once to $\text{im } T$ in W and once to $\ker T$ in V , glued

together by Rank–Nullity. The rows of A are the functionals that T'_A produces, and the columns span the image of T_A ; the two counts agree because pulling back measurements loses exactly as much as pushing forward vectors.

(The dimension formula for annihilators in section 2 was proved by extending a basis of U , without using ranks of matrices, so this proof does not go in a circle.)

The dictionary also swaps injective and surjective.

Corollary 4.4.5 — Injective and Surjective Swap Under Duality

Let V and W be finite-dimensional vector spaces over F , and $T \in \mathcal{L}(V, W)$. Then:

- (a) T is surjective if and only if T' is injective;
- (b) T is injective if and only if T' is surjective.

Proof. Let $n = \dim V$ and $m = \dim W$. A finite-dimensional space is $\{0\}$ exactly when its dimension is 0.

- (a) By Theorem 3.2.3 and Theorem 4.4.3 (a), T' is injective iff $(\operatorname{im} T)^0 = \{0\}$. By Theorem 4.2.2, $\dim(\operatorname{im} T)^0 = m - \operatorname{rank} T$, so this holds iff $\operatorname{rank} T = m$, that is, iff T is surjective (Corollary 3.3.2 (b)).
- (b) T' is surjective iff $\operatorname{im} T' = V^*$, iff $\dim \operatorname{im} T' = n$ (by Theorem 1.6.8 for one direction, since $\dim V^* = n$). By Corollary 4.4.4 (a), $\dim \operatorname{im} T' = \operatorname{rank} T$, so this holds iff $\operatorname{rank} T = n$, that is, iff T is injective (Corollary 3.3.2 (a)).

■

For T_A , part (a) reads: $A\mathbf{x} = \mathbf{b}$ is solvable for every $\mathbf{b}$ exactly when $A^T\mathbf{y} = \mathbf{0}$ has only the trivial solution. So a question about the columns of A turns into a question about the columns of A^T , the rows of A . Exercise C1 below sharpens this into a test for a single $\mathbf{b}$.

4.4.4 Naturality of the double dual

In the previous section we called $\operatorname{ev}_V: V \rightarrow V^{**}$ natural because it needs no choice, and promised a sharper sense: compatibility with every linear map. With dual maps available, we can state it. A linear map $T: V \rightarrow W$ has a dual $T': W^* \rightarrow V^*$, and that has a dual in turn,

$$T'' := (T')': V^{**} \rightarrow W^{**},$$

which points **forward** again, because we reversed twice.

Theorem 4.4.6 — The Evaluation Map Is Natural

Let V and W be vector spaces over F , and let $T \in \mathcal{L}(V, W)$. Then

$$T'' \operatorname{ev}_V = \operatorname{ev}_W T.$$

That is, $T''(\text{ev}_{\mathbf{v}}) = \text{ev}_{T\mathbf{v}}$ for every $\mathbf{v} \in V$. If V and W are finite-dimensional, then $T'' = \text{ev}_W T \text{ev}_V^{-1}$.

The statement says that the square below commutes, for **every** linear map T : going right then down (apply T , then evaluate) equals going down then right (evaluate, then apply T'').

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ \text{ev}_V \downarrow & & \downarrow \text{ev}_W \\ V^{**} & \xrightarrow{T''} & W^{**} \end{array}$$

Proof. Let $\mathbf{v} \in V$. Both $T''(\text{ev}_{\mathbf{v}})$ and $\text{ev}_{T\mathbf{v}}$ are functionals on W^* , so let $\psi \in W^*$. By [Definition 4.4.1](#) applied to T' , then to T ,

$$(T''(\text{ev}_{\mathbf{v}}))(\psi) = \text{ev}_{\mathbf{v}}(T'\psi) = (T'\psi)(\mathbf{v}) = \psi(T\mathbf{v}) = \text{ev}_{T\mathbf{v}}(\psi).$$

Hence $T''(\text{ev}_{\mathbf{v}}) = \text{ev}_{T\mathbf{v}}$ for every $\mathbf{v}$, which is $T'' \text{ev}_V = \text{ev}_W T$. If V is finite-dimensional, ev_V is an isomorphism ([Theorem 4.3.2](#)); composing on the right with ev_V^{-1} gives $T'' = \text{ev}_W T \text{ev}_V^{-1}$. ■

So, once V is identified with V^{**} and W with W^{**} through evaluation, the double dual map T'' is T . The matrices agree: by [Theorem 4.3.2](#), ev carries $\mathcal{B}$ and $\mathcal{C}$ to the dual bases $\mathcal{B}^{**}$ and $\mathcal{C}^{**}$ of $\mathcal{B}^*$ and $\mathcal{C}^*$, and two applications of [Theorem 4.4.2](#) give $[T'']_{\mathcal{B}^{**}}^{\mathcal{C}^{**}} = (([T]_{\mathcal{B}}^{\mathcal{C}})^{\top})^{\top} = [T]_{\mathcal{B}}^{\mathcal{C}}$ ([Theorem 0.9.5](#)). The rule $(A^{\top})^{\top} = A$ is the matrix form of naturality.

Remark. No such statement holds for $V \cong V^*$. Suppose that for every finite-dimensional real space V we had an isomorphism $\theta_V: V \rightarrow V^*$, compatible with isomorphisms in the only way the directions allow: $\theta_V = T' \theta_W T$ for every isomorphism $T: V \rightarrow W$. Take $V = W = \mathbb{R}$ and $T = 2\text{id}$. Then $T' = 2\text{id}$ by [Theorem 4.4.1](#) (a) and (c), so $\theta_{\mathbb{R}} = 2 \cdot 2 \cdot \theta_{\mathbb{R}} = 4\theta_{\mathbb{R}}$, which forces $\theta_{\mathbb{R}} = 0$, not an isomorphism. The dual space is isomorphic to V , but not naturally.

4.4.5 Exercises

4.4.5.1 A. Check your understanding

Exercise A1.

- Define the dual map T' of $T \in \mathcal{L}(V, W)$, including its domain and codomain.
- State the relation between $[T']_{\mathcal{C}^*}^{\mathcal{B}^*}$ and $[T]_{\mathcal{B}}^{\mathcal{C}}$, with its hypotheses.
- True or false: $(ST)' = S'T'$ whenever ST is defined. Justify your answer.
- True or false: $\ker T' = (\ker T)^0$ for every linear map between finite-dimensional spaces. Justify your answer.

- (e) True or false: for $T \in \mathcal{L}(\mathbb{R}^4, \mathbb{R}^3)$, the dual map T' is never surjective. Justify your answer.
- (f) Name the two tools combined to prove $\text{rank } T' = \text{rank } T$.

Solution.

- (a) For $T: V \rightarrow W$, $T': W^* \rightarrow V^*$, $T'(\psi) = \psi T$ (Definition 4.4.1).
- (b) If V, W are finite-dimensional with bases $\mathcal{B}, \mathcal{C}$ and dual bases $\mathcal{B}^*, \mathcal{C}^*$, then $[T']_{\mathcal{C}^*}^{\mathcal{B}^*} = ([T]_{\mathcal{B}}^{\mathcal{C}})^{\top}$ (Theorem 4.4.2).
- (c) False. The correct rule is $(ST)' = T'S'$ (Theorem 4.4.1 (b)); for $T: U \rightarrow V$, $S: V \rightarrow W$ with $U \neq W$, the composite $S'T'$ is not even defined.
- (d) False. $\ker T' = (\text{im } T)^0$. For the zero map $T: \mathbb{R} \rightarrow \mathbb{R}$, $\ker T = \mathbb{R}$, so $(\ker T)^0 = \{0\}$, while $T' = 0$ has kernel $\mathbb{R}^* \neq \{0\}$.
- (e) True. T' maps $(\mathbb{R}^3)^*$, of dimension 3, to $(\mathbb{R}^4)^*$, of dimension 4, so it cannot be surjective (Corollary 3.3.2 (d)). This matches Corollary 4.4.5: T cannot be injective.
- (f) The Rank–Nullity Theorem and the dimension formula for annihilators, $\dim U + \dim U^0 = \dim V$ (together with $\ker T' = (\text{im } T)^0$).

4.4.5.2 B. Practice

Exercise B1: Differentiation in dual bases.

Let $D: \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}[x]_{\leq 2}$, $D(p) = p'$, and let $\mathcal{B} = (1, x, x^2)$ with dual basis $\mathcal{B}^* = (\varphi_0, \varphi_1, \varphi_2)$, $\varphi_k(p) = \text{coefficient of } x^k \text{ in } p$. Compute $D'\varphi_0$, $D'\varphi_1$, $D'\varphi_2$ directly, write down $[D']_{\mathcal{B}^*}$, and check that it is $([D]_{\mathcal{B}})^{\top}$. Hence describe $\ker D'$ and $\text{im } D'$, and verify Theorem 4.4.3.

Solution. Let $p = a + bx + cx^2$, so $Dp = b + 2cx$. Then $(D'\varphi_0)(p) = \varphi_0(Dp) = b$, $(D'\varphi_1)(p) = 2c$, $(D'\varphi_2)(p) = 0$. Hence

$$D'\varphi_0 = \varphi_1, \quad D'\varphi_1 = 2\varphi_2, \quad D'\varphi_2 = 0.$$

The coordinate vectors $(0, 1, 0)$, $(0, 0, 2)$, $(0, 0, 0)$ are the columns of $[D']_{\mathcal{B}^*}$. On the other side, $D(1) = 0$, $D(x) = 1$, $D(x^2) = 2x$, so

$$[D]_{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}, \quad [D']_{\mathcal{B}^*} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix} = ([D]_{\mathcal{B}})^{\top},$$

as Theorem 4.4.2 predicts.

From the formulas, $D'(c_0\varphi_0 + c_1\varphi_1 + c_2\varphi_2) = c_0\varphi_1 + 2c_1\varphi_2$, which is 0 iff $c_0 = c_1 = 0$. So $\ker D' = \text{span}(\varphi_2)$ and $\text{im } D' = \text{span}(\varphi_1, \varphi_2)$. On the other hand $\text{im } D = \mathbb{R}[x]_{\leq 1} = \text{span}(1, x)$, and φ_2 is exactly the functional (up to scalars) vanishing on it, since $c_0\varphi_0 + c_1\varphi_1 + c_2\varphi_2$ kills 1 and x iff $c_0 = c_1 = 0$. So $\ker D' = (\text{im } D)^0$. Also $\ker D = \text{span}(1)$, and a functional kills 1 iff $c_0 = 0$, so $(\ker D)^0 = \text{span}(\varphi_1, \varphi_2) = \text{im } D'$.

Exercise B2: Kernel and image of a dual map.

Let $A = \begin{pmatrix} 1 & 2 & -1 \\ -2 & -4 & 2 \end{pmatrix}$ and $T = T_A: \mathbb{R}^3 \rightarrow \mathbb{R}^2$. Find bases of $\ker T'$ and $\operatorname{im} T'$ as explicit functionals, and verify $\dim \ker T' + \dim \operatorname{im} T' = 2$ and $\operatorname{rank} T' = \operatorname{rank} T$.

Solution. The second row is -2 times the first, so $\operatorname{rank} A = 1$, $\operatorname{im} T = \operatorname{col}(A) = \operatorname{span}((1, -2))$, and $\ker T = \{(x, y, z) : x + 2y - z = 0\}$, of dimension 2.

Kernel of T' . As in [Example \(Kernel and Image of a Dual Map\)](#), $\psi(u, v) = b_1u + b_2v$ lies in $\ker T'$ iff $A^T \mathbf{b} = \mathbf{0}$, that is, $b_1 - 2b_2 = 0$, $2b_1 - 4b_2 = 0$, $-b_1 + 2b_2 = 0$. So $\mathbf{b} = t(2, 1)$, and $\ker T' = \operatorname{span}(\psi_0)$ with $\psi_0(u, v) = 2u + v$. Check: $\psi_0(1, -2) = 0$, so $\psi_0 \in (\operatorname{im} T)^0$, which has dimension $2 - 1 = 1$.

Image of T' . By [Example \(Dual Maps in Four Settings\)](#) (a), $\operatorname{im} T'$ is spanned by the rows of A read as functionals, $x + 2y - z$ and $-2x - 4y + 2z$. The second is -2 times the first, so $\operatorname{im} T' = \operatorname{span}(\varphi_0)$ with $\varphi_0(x, y, z) = x + 2y - z$. Check: φ_0 vanishes on $\ker T$ by the description of $\ker T$, and $\dim(\ker T)^0 = 3 - 2 = 1$.

Hence $\dim \ker T' + \dim \operatorname{im} T' = 1 + 1 = 2 = \dim(\mathbb{R}^2)^*$, and $\operatorname{rank} T' = 1 = \operatorname{rank} T$.

Exercise B3: The transpose of a product, again.

Let $A \in M_{m \times n}(F)$ and $B \in M_{n \times p}(F)$. Using $T_A T_B = T_{AB}$ (Chapter 3, §6), [Theorem 4.4.1](#) (b) and [Theorem 4.4.2](#), give a new proof that $(AB)^T = B^T A^T$.

Solution. Let $\mathcal{E}_p, \mathcal{E}_n, \mathcal{E}_m$ be the standard bases, with dual bases $\mathcal{E}_p^*, \mathcal{E}_n^*, \mathcal{E}_m^*$. By [Example \(Matrix Maps and the Identity\)](#) (a), $[T_A]_{\mathcal{E}_n}^{\mathcal{E}_m} = A$, $[T_B]_{\mathcal{E}_p}^{\mathcal{E}_n} = B$, and $[T_{AB}]_{\mathcal{E}_p}^{\mathcal{E}_m} = AB$. Since $T_{AB} = T_A T_B$, [Theorem 4.4.1](#) (b) gives $T'_{AB} = T'_B T'_A$. Taking matrices, with [Theorem 3.6.2](#) in the middle and [Theorem 4.4.2](#) on the outside,

$$(AB)^T = [T'_{AB}]_{\mathcal{E}_m^*}^{\mathcal{E}_p^*} = [T'_B T'_A]_{\mathcal{E}_m^*}^{\mathcal{E}_p^*} = [T'_B]_{\mathcal{E}_n^*}^{\mathcal{E}_p^*} [T'_A]_{\mathcal{E}_m^*}^{\mathcal{E}_n^*} = B^T A^T.$$

The reversal of order is now explained: it is the reversal of arrows under duality.

4.4.5.3 C. Going deeper**Exercise C1: The Fredholm alternative.**

- Let V, W be finite-dimensional, $T \in \mathcal{L}(V, W)$ and $\mathbf{w} \in W$. Prove that $T\mathbf{v} = \mathbf{w}$ has a solution $\mathbf{v} \in V$ if and only if $\psi(\mathbf{w}) = 0$ for every $\psi \in \ker T'$.
- Deduce: for $A \in M_{m \times n}(F)$ and $\mathbf{b} \in F^m$, the system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{y}^T \mathbf{b} = 0$ for every $\mathbf{y} \in F^m$ with $A^T \mathbf{y} = \mathbf{0}$.
- Let $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}$. Find a single linear condition on $\mathbf{b} = (b_1, b_2, b_3) \in \mathbb{R}^3$ that is equivalent to consistency of $A\mathbf{x} = \mathbf{b}$.

Hint: for (a), combine [Theorem 4.4.3](#) with the description of a subspace by its annihilator in the previous section.

Solution.

- (a) $T\mathbf{v} = \mathbf{w}$ has a solution iff $\mathbf{w} \in \text{im } T$. By [Theorem 4.4.3](#) (a), $\ker T' = (\text{im } T)^0$. By [Corollary 4.3.4](#) applied to the subspace $\text{im } T$ of the finite-dimensional space W ,

$$\text{im } T = \{\mathbf{w} \in W : \psi(\mathbf{w}) = 0 \text{ for every } \psi \in (\text{im } T)^0\} = \{\mathbf{w} \in W : \psi(\mathbf{w}) = 0 \text{ for every } \psi \in \ker T'\}.$$

This is the claim.

- (b) Apply (a) to $T_A: F^n \rightarrow F^m$. By [Theorem 4.1.1](#), every $\psi \in (F^m)^*$ is $\psi(\mathbf{u}) = \mathbf{y}^T \mathbf{u}$ for a unique $\mathbf{y} \in F^m$. By [Example \(Dual Maps in Four Settings\)](#) (a), $T'_A \psi$ is represented by $A^T \mathbf{y}$, and the zero functional by $\mathbf{0}$; by uniqueness of the representing vector, $\psi \in \ker T'_A$ iff $A^T \mathbf{y} = \mathbf{0}$. For such ψ , $\psi(\mathbf{b}) = \mathbf{y}^T \mathbf{b}$. So (a) says: $A\mathbf{x} = \mathbf{b}$ is consistent iff $\mathbf{y}^T \mathbf{b} = 0$ for every $\mathbf{y}$ with $A^T \mathbf{y} = \mathbf{0}$.
- (c) $A^T \mathbf{y} = \mathbf{0}$ reads $y_1 + y_2 + y_3 = 0$ and $y_1 + 2y_2 + 3y_3 = 0$. Subtracting, $y_2 + 2y_3 = 0$, so $\mathbf{y} = t(1, -2, 1)$. By (b), $A\mathbf{x} = \mathbf{b}$ is consistent iff $\mathbf{y}^T \mathbf{b} = 0$ for all these $\mathbf{y}$, that is, iff

$$b_1 - 2b_2 + b_3 = 0.$$

In words, the system $x_1 + kx_2 = b_k$ ($k = 1, 2, 3$) asks for a line through the points $(1, b_1), (2, b_2), (3, b_3)$, and it exists exactly when the second difference $b_3 - 2b_2 + b_1$ vanishes. No inner product was used: $\mathbf{y}^T \mathbf{b}$ is just a functional applied to $\mathbf{b}$.

Exercise C2: Invariant subspaces and annihilators.

Let V be finite-dimensional, $T \in \mathcal{L}(V)$, and U a subspace of V . Recall that U is *T -invariant* if $T\mathbf{u} \in U$ for every $\mathbf{u} \in U$. Prove that U is T -invariant if and only if U^0 is T' -invariant.

Solution. ($\Rightarrow$) Suppose U is T -invariant, and let $\varphi \in U^0$. For every $\mathbf{u} \in U$, $(T'\varphi)(\mathbf{u}) = \varphi(T\mathbf{u}) = 0$, because $T\mathbf{u} \in U$. Hence $T'\varphi \in U^0$, and U^0 is T' -invariant.

($\Leftarrow$) Suppose U^0 is T' -invariant, and let $\mathbf{u} \in U$. For every $\varphi \in U^0$, $T'\varphi \in U^0$, so $\varphi(T\mathbf{u}) = (T'\varphi)(\mathbf{u}) = 0$. Thus $T\mathbf{u}$ is killed by every functional in U^0 . By [Corollary 4.3.4](#), $T\mathbf{u} \in U$. Hence U is T -invariant.

Exercise C3: Duality without finite dimension.

Let V and W be arbitrary vector spaces over F , and $T \in \mathcal{L}(V, W)$.

- (a) Prove that if T is surjective, then T' is injective.
 (b) Prove that if T is injective, then T' is surjective.
 (c) Check (b) on $S: F[x] \rightarrow F[x]$, $S(p) = xp$, from [Example \(Dual Maps in Four Settings\)](#) (c): given $\varphi \in F[x]^*$, find ψ with $S'\psi = \varphi$. Is S' injective?

Hint: for (b), use [Theorem 1.8.2](#) in W .

Solution.

- (a) By [Theorem 4.4.3](#) (a), which needs no finite dimension, $\ker T' = (\text{im } T)^0 = W^0$. A functional vanishing on all of W is the zero functional, so $\ker T' = \{0\}$ and T' is

injective (Theorem 3.2.3).

- (b) Let $\varphi \in V^*$. By Theorem 1.8.3, V has a basis B . Since T is injective, T is one-to-one on B , and $T(B)$ is linearly independent: a finite combination $\sum c_i T\mathbf{b}_i = T(\sum c_i \mathbf{b}_i) = \mathbf{0}$ with distinct $\mathbf{b}_i$ forces $\sum c_i \mathbf{b}_i = \mathbf{0}$ (Theorem 3.2.3), hence all $c_i = 0$. By Theorem 1.8.2 there is a basis $C \supseteq T(B)$ of W . By Theorem 3.1.6, there is $\psi \in W^*$ with $\psi(T\mathbf{b}) = \varphi(\mathbf{b})$ for $\mathbf{b} \in B$, and $\psi(\mathbf{c}) = 0$ for $\mathbf{c} \in C \setminus T(B)$; this is well defined because each element of $T(B)$ is $T\mathbf{b}$ for exactly one $\mathbf{b}$. Then ψT and φ are linear and agree on the basis B , so $T'\psi = \psi T = \varphi$ by the uniqueness part of Theorem 3.1.6. Hence T' is surjective.
- (c) Here $B = \{1, x, x^2, \dots\}$ and $S(B) = \{x, x^2, \dots\}$, extended to a basis by adding 1. The recipe gives $\psi(x^{k+1}) = \varphi(x^k)$ for $k \geq 0$ and $\psi(1) = 0$. Then $(S'\psi)(x^k) = \psi(x^{k+1}) = \varphi(x^k)$, so $S'\psi = \varphi$. The dual map S' is not injective: $S'\varphi_0 = 0$ by Example (Dual Maps in Four Settings) (c). This matches the contrapositive of (a): S is not surjective, since $1 \notin \text{im } S$, and indeed φ_0 is a non-zero functional vanishing on $\text{im } S$.

4.5 Duality for Subspaces and Quotients

A subspace U of V gives two new vector spaces: U itself, and the quotient V/U of Chapter 3. Each has a dual space, and so far U^* and $(V/U)^*$ have nothing to do with V^* . This section connects them. The answer is a swap: the dual of the quotient is a **subspace** of V^* , namely the annihilator U^0 , and the dual of the subspace is a **quotient** of V^* , namely V^*/U^0 . Both identifications come from the First Isomorphism Theorem, and neither needs a basis. At the end we use them to describe hyperplanes, the subspaces cut out by a single equation, in any dimension.

4.5.1 Restricting functionals to a subspace

Let U be a subspace of V . A functional $\varphi \in V^*$ measures every vector of V , so in particular it measures the vectors of U . Forgetting the rest gives the **restriction** $\varphi|_U: U \rightarrow F$, $\mathbf{u} \mapsto \varphi(\mathbf{u})$, which is a functional on U . This defines the restriction map

$$\rho: V^* \rightarrow U^*, \quad \rho(\varphi) = \varphi|_U.$$

It is not a new kind of map. Let $\iota: U \rightarrow V$, $\iota(\mathbf{u}) = \mathbf{u}$, be the inclusion. Its dual map (Definition 4.4.1) sends φ to $\varphi \circ \iota$, and $(\varphi \circ \iota)(\mathbf{u}) = \varphi(\mathbf{u})$. So $\rho = \iota'$, and in particular ρ is linear.

Two questions decide what ρ does. Which functionals restrict to zero? By Definition 4.2.1, exactly those in U^0 . And does every functional on U come from one on V ? That is an extension problem, and it has a positive answer in every dimension.

Theorem 4.5.1 — Dual of a Subspace

Let V be a vector space over F , of any dimension, and let U be a subspace of V . The restriction map $\rho: V^* \rightarrow U^*$, $\rho(\varphi) = \varphi|_U$, is linear and surjective, and $\ker \rho = U^0$. Hence

$$V^*/U^0 \rightarrow U^*, \quad \varphi + U^0 \mapsto \varphi|_U$$

is a well-defined isomorphism, and $U^* \cong V^*/U^0$.

Idea. The kernel is the definition of U^0 read aloud. The real content is surjectivity: given ψ on U , we must extend it to all of V . A basis tells us how. Extend a basis of U to a basis of V ; a linear map may be prescribed freely on a basis, so declare the extension to agree with ψ on the basis of U and to be 0 on the new basis vectors. Then the First Isomorphism Theorem turns “surjective with kernel U^0 ” into the isomorphism.

Proof. We saw above that $\rho = \iota'$ is linear. For $\varphi \in V^*$, $\rho(\varphi) = 0$ means $\varphi(\mathbf{u}) = 0$ for every $\mathbf{u} \in U$, which is $\varphi \in U^0$ by Definition 4.2.1. So $\ker \rho = U^0$.

Surjective. Let $\psi \in U^*$. By Theorem 1.8.3, U has a basis B_U . It is a linearly independent subset of V , so by Theorem 1.8.2 there is a basis B of V with $B_U \subseteq B$. By Theorem 3.1.6 there is a linear map $\varphi: V \rightarrow F$ with

$$\varphi(\mathbf{b}) = \psi(\mathbf{b}) \quad \text{for } \mathbf{b} \in B_U, \quad \varphi(\mathbf{b}) = 0 \quad \text{for } \mathbf{b} \in B \setminus B_U.$$

Then $\varphi|_U$ and ψ are linear maps $U \rightarrow F$ that agree on the basis B_U of U , so $\varphi|_U = \psi$ by the uniqueness part of Theorem 3.1.6. Thus $\rho(\varphi) = \psi$, and ρ is surjective. (If V is finite-dimensional, the finite versions Corollary 1.5.5, Theorem 1.6.5 and Theorem 3.1.5 suffice, and no choice principle is needed.)

By the First Isomorphism Theorem (Theorem 3.9.7), $\varphi + \ker \rho \mapsto \rho(\varphi)$ is a well-defined isomorphism $V^*/\ker \rho \rightarrow \text{im } \rho$. Since $\ker \rho = U^0$ and $\text{im } \rho = U^*$, this is the claimed isomorphism.

■

In words: **a functional on U is a functional on V , up to adding something that vanishes on U .** Two extensions of the same ψ differ by an element of U^0 , and every element of U^0 can be added.

When V is finite-dimensional, taking dimensions recovers the count of this chapter. By Corollary 4.1.4, $\dim U^* = \dim U$ and $\dim V^* = \dim V$, and by Theorem 3.9.5, $\dim(V^*/U^0) = \dim V - \dim U^0$. So the isomorphism says $\dim U = \dim V - \dim U^0$, which is Theorem 4.2.2 again.

4.5.2 Functionals on a quotient

Now the other space. A functional ψ on V/U measures cosets. Composing with the quotient map $\pi: V \rightarrow V/U$, $\pi(\mathbf{v}) = \mathbf{v} + U$, turns it into a measurement of vectors, $\mathbf{v} \mapsto \psi(\mathbf{v} + U)$. This functional on V cannot tell apart two vectors in the same coset, so in particular it is zero on U . The theorem says that nothing is lost and nothing else appears: the functionals on V/U are **exactly** the functionals on V that vanish on U .

Theorem 4.5.2 — Dual of a Quotient

Let V be a vector space over F , of any dimension, let U be a subspace, and let $\pi: V \rightarrow$

V/U be the quotient map. Then the dual map

$$\pi': (V/U)^* \rightarrow V^*, \quad \pi'(\psi) = \psi \circ \pi,$$

is injective, and its image is U^0 . Hence π' is an isomorphism from $(V/U)^*$ onto U^0 , and $(V/U)^* \cong U^0$.

Idea. Injectivity uses only that π is surjective: if $\psi \circ \pi$ is zero, then ψ is zero on every coset. For the image, one inclusion is “ π kills U ”. The other inclusion asks us to build a functional on V/U from a functional on V that vanishes on U , and that is exactly what the universal property of the quotient does.

Proof. The map π' is linear by [Definition 4.4.1](#).

Injective. Suppose $\pi'(\psi) = \psi \circ \pi = 0$. Every element of V/U is $\pi(\mathbf{v})$ for some $\mathbf{v} \in V$, because π is surjective ([Theorem 3.9.4](#) (c)). Then $\psi(\pi(\mathbf{v})) = 0$ for every $\mathbf{v}$, so $\psi = 0$. By [Theorem 3.2.3](#), π' is injective.

$\text{im } \pi' \subseteq U^0$. For $\mathbf{u} \in U$, $\pi(\mathbf{u}) = U$ is the zero vector of V/U , so $(\psi \circ \pi)(\mathbf{u}) = \psi(\mathbf{0}) = 0$. Hence $\psi \circ \pi \in U^0$.

$U^0 \subseteq \text{im } \pi'$. Let $\varphi \in U^0$. Then $U \subseteq \ker \varphi$, so by [Theorem 3.9.10](#) (a) there is a linear map $\psi: V/U \rightarrow F$ with $\psi \circ \pi = \varphi$. This ψ lies in $(V/U)^*$, and $\pi'(\psi) = \varphi$.

So π' is an injective linear map with image U^0 ; regarded as a map onto U^0 it is a linear bijection, hence an isomorphism. ■

Concretely, the inverse isomorphism $U^0 \rightarrow (V/U)^*$ is $\varphi \mapsto \bar{\varphi}$, where $\bar{\varphi}(\mathbf{v} + U) = \varphi(\mathbf{v})$. The formula is well defined precisely because φ vanishes on U : if $\mathbf{v} - \mathbf{v}' \in U$, then $\varphi(\mathbf{v}) - \varphi(\mathbf{v}') = \varphi(\mathbf{v} - \mathbf{v}') = 0$.

The two theorems fit together as a mirror:

	subspace $U \subseteq V$	quotient V/U
its dual is	a quotient of V^* : $U^* \cong V^*/U^0$	a subspace of V^* : $(V/U)^* \cong U^0$
the map	ι' : restrict	π' : compose with π
the map is	surjective	injective

The last row is no accident: ι is injective and π is surjective, and dualizing swaps the two properties.

Alternatively, in finite dimension both theorems follow from [Theorem 4.4.3](#), which says $\ker T' = (\text{im } T)^0$ and $\text{im } T' = (\ker T)^0$. For $T = \pi$: $\text{im } \pi = V/U$, whose annihilator is $\{0\}$, and $\ker \pi = U$, so π' is injective with image U^0 . For $T = \iota$: $\text{im } \iota = U$, so $\ker \iota' = U^0$, and $\ker \iota = \{0\}$, whose annihilator is all of U^* , so ι' is surjective. The proofs above avoid dimension counts, which is why they work in every dimension.

Quick check. Let V be a vector space of dimension 6 and U a subspace of dimension

2. Find $\dim U^0$, $\dim(V/U)^*$, $\dim U^*$ and $\dim(V^*/U^0)$, and check that the two theorems are consistent with these numbers.

Solution. By Theorem 4.2.2, $\dim U^0 = 6 - 2 = 4$. By Theorem 3.9.5 and Corollary 4.1.4, $\dim(V/U)^* = \dim(V/U) = 4$, matching $(V/U)^* \cong U^0$. Next, $\dim U^* = \dim U = 2$, and $\dim(V^*/U^0) = 6 - 4 = 2$, matching $U^* \cong V^*/U^0$.

4.5.3 A line in $\mathbb{R}^3$, in coordinates

The theorems name their isomorphisms without choosing anything. To see them act, we now choose bases and compute. Recall that a functional on $\mathbb{R}^3$ is determined by its values on $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ (Theorem 4.1.1). So every $\varphi \in (\mathbb{R}^3)^*$ has the form $\varphi(x, y, z) = ax + by + cz$, with $(a, b, c) = (\varphi(\mathbf{e}_1), \varphi(\mathbf{e}_2), \varphi(\mathbf{e}_3))$. We write such a functional simply as $ax + by + cz$.

Example. (Both isomorphisms for a line)

Let $\mathbf{u} = (1, 2, -1)$ and $U = \text{span}(\mathbf{u}) \subseteq \mathbb{R}^3$.

- Find a basis of U^0 .
- Find a basis of $\mathbb{R}^3/U$, its dual basis (ψ_1, ψ_2) of $(\mathbb{R}^3/U)^*$, and the images $\pi'(\psi_1), \pi'(\psi_2)$.
- Describe the restriction map $\rho: (\mathbb{R}^3)^* \rightarrow U^*$ and the isomorphism $(\mathbb{R}^3)^*/U^0 \rightarrow U^*$.

Solution.

- By Theorem 4.2.3 (b), $U^0 = \{\mathbf{u}\}^0$, so $ax + by + cz \in U^0$ exactly when $a + 2b - c = 0$, that is, $c = a + 2b$. Taking $(a, b) = (1, 0)$ and $(0, 1)$ gives

$$\varphi_1 = x + z, \quad \varphi_2 = y + 2z.$$

They are independent (look at the coefficients of x and y), and $\dim U^0 = 3 - 1 = 2$ by Theorem 4.2.2, so (φ_1, φ_2) is a basis of U^0 (Theorem 1.6.4).

- The list $(\mathbf{u}, \mathbf{e}_1, \mathbf{e}_2)$ is independent: in $a\mathbf{u} + b\mathbf{e}_1 + c\mathbf{e}_2 = (a + b, 2a + c, -a) = \mathbf{0}$, the third entry gives $a = 0$, and then $b = c = 0$. It has length 3, so it is a basis of $\mathbb{R}^3$, and by Theorem 3.9.5 $(\mathbf{e}_1 + U, \mathbf{e}_2 + U)$ is a basis of $\mathbb{R}^3/U$. Its dual basis consists of the ψ_i with $\psi_i(\mathbf{e}_j + U) = \delta_{ij}$ (Theorem 4.1.3).

The functional $\pi'(\psi_1) = \psi_1 \circ \pi$ on $\mathbb{R}^3$ takes the values $\psi_1(\mathbf{e}_1 + U) = 1$ at $\mathbf{e}_1$, $\psi_1(\mathbf{e}_2 + U) = 0$ at $\mathbf{e}_2$, and $\psi_1(U) = 0$ at $\mathbf{u}$. Writing it as $ax + by + cz$, this says $a = 1$, $b = 0$ and $a + 2b - c = 0$, so $c = 1$. Similarly $\pi'(\psi_2)$ has $a = 0$, $b = 1$, $c = 2$. Hence

$$\pi'(\psi_1) = x + z = \varphi_1, \quad \pi'(\psi_2) = y + 2z = \varphi_2.$$

So π' carries the dual basis of $(\mathbb{R}^3/U)^*$ onto the basis of U^0 found in (a), as Theorem 4.5.2 predicts. Read backwards: $\psi_1((x, y, z) + U) = x + z$ and $\psi_2((x, y, z) + U) = y + 2z$. These are well defined, since adding $t\mathbf{u} = (t, 2t, -t)$ changes $x + z$ by $t - t = 0$ and $y + 2z$ by $2t - 2t = 0$.

- The space U^* is one-dimensional, with basis the functional ξ determined by $\xi(\mathbf{u}) = 1$ (the dual basis of $(\mathbf{u})$). A functional $\varphi = ax + by + cz$ restricts to $\varphi|_U = \varphi(\mathbf{u})\xi = (a + 2b - c)\xi$, since both sides take the value $a + 2b - c$ at $\mathbf{u}$. So ρ is “evaluate at $\mathbf{u}$ ”. Its kernel is $\{a + 2b - c = 0\} = U^0$, and it is surjective, since $\rho(x) = \xi$. The

isomorphism of [Theorem 4.5.1](#) is

$$(ax + by + cz) + U^0 \mapsto (a + 2b - c)\xi.$$

For instance $x + U^0 = -z + U^0$, because $x - (-z) = \varphi_1 \in U^0$; both restrict to ξ .

4.5.4 Why “natural” matters here

In the example, bases entered only when we wanted numbers. The maps ρ and π' were defined before any basis was chosen: restrict a functional, or compose it with π . That is what makes the two isomorphisms **natural**, in the sense of the discussion of the double dual. They are part of the structure of V and U , not of a coordinate system.

There is a tempting shortcut that is **not** natural. Since U^* is a quotient of V^* , one might try to find a copy of U^* **inside** V^* , as a subspace. This can be done, but only by making a choice. Pick a complement W of U , so $V = U \oplus W$. A functional that vanishes on W is determined by its values on U , so the restriction ρ maps W^0 isomorphically onto U^* ([Exercise \(C2: Complements and a choice of copy of \$U^*\$ \)](#)). But different complements give different subspaces W^0 .

Warning. U^* is **not** a subspace of V^* . Its elements are functionals on U , which cannot be evaluated at vectors outside U . To place U^* inside V^* one must extend each functional, and extensions are not unique. In the example above, the complement $W_1 = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$ of U gives $W_1^0 = \text{span}(z)$, while $W_2 = \text{span}(\mathbf{e}_1, \mathbf{e}_3)$ gives $W_2^0 = \text{span}(y)$. These are different lines in $(\mathbb{R}^3)^*$, and both restrict isomorphically onto U^* : $z|_U = -\xi$ and $y|_U = 2\xi$. The choice-free statement is $U^* \cong V^*/U^0$, a quotient.

The same caution applies on the other side, in reverse. $(V/U)^*$ is naturally a subspace of V^* , namely U^0 , but V/U itself is not a subspace of V . Duality turns the quotient, which has no natural home inside V , into a subspace, which does.

4.5.5 Codimension and hyperplanes

The quotient measures how far a subspace is from filling the whole space. In $\mathbb{R}^3$ a plane through the origin misses “one direction” and a line misses “two directions”. For finite-dimensional V the number of missing directions is $\dim V - \dim U$. But this formula is useless in $F[x]$, where $\dim V$ and $\dim U$ may both be infinite, although $U = \{p : p(0) = 0\}$ misses only one direction: every polynomial is a constant plus an element of U . The quotient gives the right number in every case.

The codimension of U counts the directions in V that U is missing.

Definition 4.5.1 — Codimension

Let U be a subspace of a vector space V over F . If V/U is finite-dimensional, the **codimension** of U in V is

$$\text{codim } U := \dim(V/U).$$

If V/U is infinite-dimensional, we say that U has **infinite codimension**.

In words: we do not compare the sizes of U and V . We collapse U to zero and measure what is left. The codimension depends on the ambient space, so when that is unclear we write “codimension of U in V ”.

Definition 4.5.2 — Hyperplane

A **hyperplane** in V is a subspace H of V of codimension **exactly** 1, that is, with $\dim(V/H) = 1$.

Examples.

- **Finite dimension.** If V is finite-dimensional, then $\operatorname{codim} U = \dim V - \dim U$ by [Theorem 3.9.5](#). The hyperplanes of F^n are its subspaces of dimension $n - 1$: the lines through $\mathbf{0}$ in F^2 and the planes through $\mathbf{0}$ in F^3 .
- **Trace.** In $M_n(F)$ with $n \geq 1$, the trace-zero matrices are the kernel of $\operatorname{tr}: M_n(F) \rightarrow F$. This map is linear and onto, since $\operatorname{tr}(cE_{11}) = c$, so $M_n(F)/\ker \operatorname{tr} \cong F$ by [Theorem 3.9.7](#), and the trace-zero matrices form a hyperplane.
- **Polynomials.** In $F[x]$, let $U = \{p : p(0) = 0\}$. Evaluation at 0 is a linear map $F[x] \rightarrow F$ with kernel U , and it is surjective because the constant c maps to c . By [Theorem 3.9.7](#), $F[x]/U \cong F$, so U is a hyperplane in the infinite-dimensional space $F[x]$. In the same way $\{p : p(0) = p(1) = 0\}$ has codimension 2 ([Exercise \(B3: Dimension computations\)](#)).
- **Degenerate cases.** $\operatorname{codim} V = 0$, since V/V is the zero space, and $\operatorname{codim}\{\mathbf{0}\} = \dim V$ when V is finite-dimensional. In particular V itself is **not** a hyperplane, and the only hyperplane of a one-dimensional space is $\{\mathbf{0}\}$. The zero space $\{\mathbf{0}\}$ has no hyperplanes at all.

Non-example by minimal change. In $\mathbb{R}^3$, the plane $\{x + y + z = 0\}$ is a hyperplane. Replace it by the line $\{x + y + z = 0, x - y = 0\} = \operatorname{span}((1, 1, -2))$. It is still a subspace, but its codimension is $3 - 1 = 2$, so the clause “codimension **exactly** 1” fails. Replace it instead by $\{x + y + z = 1\}$: now it is not a subspace at all, so it has no codimension. It is a coset of a hyperplane, an object we return to in the next section.

Why this definition. Codimension is the dual notion of dimension, and the duality theorems make this precise. If U has finite codimension, then by [Theorem 4.5.2](#), [Corollary 4.1.4](#) and [Theorem 3.5.4](#),

$$\operatorname{codim} U = \dim(V/U) = \dim(V/U)^* = \dim U^0,$$

and this holds even when V is infinite-dimensional. So **dimension counts vectors, and codimension counts independent equations**. For finite-dimensional V this is [Theorem 4.2.2](#) again: $\dim U^0 = \dim V - \dim U$.

Warning. In infinite dimension a hyperplane need not be “smaller” than the whole space. In $F[x]$, the hyperplane $U = \{p : p(0) = 0\} = \{xq : q \in F[x]\}$ is isomorphic to $F[x]$ itself, via $q \mapsto xq$: this map is linear, injective (if $xq = 0$ then $q = 0$) and onto U . So you cannot detect codimension by comparing U with V up to isomorphism, and the formula $\operatorname{codim} U = \dim V - \dim U$ must never be used outside finite dimension. Always compute $\dim(V/U)$, or count equations.

Hyperplanes are exactly the subspaces cut out by **one** non-trivial linear equation. In F^n this is familiar: $a_1x_1 + \cdots + a_nx_n = 0$ with the a_i not all zero defines a subspace of dimension $n - 1$. The quotient makes the statement true in every dimension.

Theorem 4.5.3 — Hyperplanes Are Kernels of Functionals

Let V be a vector space over F , of any dimension, and let H be a subspace of V . Then H is a hyperplane if and only if $H = \ker \varphi$ for some **non-zero** $\varphi \in V^*$.

Idea. A non-zero functional maps onto the one-dimensional space F , so the First Isomorphism Theorem makes $V/\ker \varphi$ one-dimensional. Conversely, if V/H is one-dimensional, it “is” F after choosing an isomorphism, and composing that isomorphism with π gives a functional whose kernel is H .

Proof. ($\Leftarrow$) Suppose $\varphi \in V^*$ is non-zero. Then $\text{im } \varphi$ is a subspace of F (Theorem 3.2.2) containing a non-zero scalar, so $\dim \text{im } \varphi = 1 = \dim F$, and $\text{im } \varphi = F$ by Theorem 1.6.8. By Theorem 3.9.7, $V/\ker \varphi \cong F$, so $\dim(V/\ker \varphi) = 1$, and $\ker \varphi$ is a hyperplane.

($\Rightarrow$) Suppose $\dim(V/H) = 1$. By Theorem 3.5.4 there is an isomorphism $\theta: V/H \rightarrow F$. Let $\varphi = \theta \circ \pi$, where $\pi: V \rightarrow V/H$ is the quotient map. It is linear as a composition of linear maps (Theorem 3.4.2), so $\varphi \in V^*$. For $\mathbf{v} \in V$, $\varphi(\mathbf{v}) = 0$ means $\theta(\mathbf{v} + H) = 0$, which holds exactly when $\mathbf{v} + H$ is the zero coset, since θ is injective; and $\mathbf{v} + H = H$ means $\mathbf{v} \in H$ by Lemma 3.9.3 (b). So $\ker \varphi = H$. Finally $\varphi \neq 0$: since $V/H \neq \{0\}$, we have $H \neq V$, so some $\mathbf{v} \notin H = \ker \varphi$. ■

The functional is not unique, but almost: two functionals with the same kernel are scalar multiples of each other (Exercise (C1: Functionals with the same kernel); Exercise (B2: Functionals with the same kernel) gives a second proof). So hyperplanes of V correspond to non-zero functionals **up to a non-zero scalar**, just as the plane $x + y + z = 0$ is also $2x + 2y + 2z = 0$.

A hyperplane together with any one vector outside it fills the space.

Remark. Let $H = \ker \varphi$ be a hyperplane and $\mathbf{v}_0 \notin H$. Then $V = H \oplus \text{span}(\mathbf{v}_0)$. Indeed $\varphi(\mathbf{v}_0) \neq 0$, and every $\mathbf{v} \in V$ splits as

$$\mathbf{v} = \left(\mathbf{v} - \frac{\varphi(\mathbf{v})}{\varphi(\mathbf{v}_0)} \mathbf{v}_0 \right) + \frac{\varphi(\mathbf{v})}{\varphi(\mathbf{v}_0)} \mathbf{v}_0,$$

where the first bracket lies in $\ker \varphi$ because φ of it is $\varphi(\mathbf{v}) - \varphi(\mathbf{v}) = 0$. And $H \cap \text{span}(\mathbf{v}_0) = \{0\}$, since $\varphi(c\mathbf{v}_0) = c\varphi(\mathbf{v}_0) = 0$ forces $c = 0$.

Quick check. Decide which of the following are hyperplanes, and for each hyperplane give a non-zero functional with that kernel.

- (a) The symmetric matrices in $M_2(F)$.
- (b) The symmetric matrices in $M_3(F)$.
- (c) $\{p \in F[x] : p(1) = 0\}$ in $F[x]$.

Solution.

- (a) Yes. A is symmetric exactly when $a_{12} - a_{21} = 0$, so the symmetric matrices are the kernel of $A \mapsto a_{12} - a_{21}$, which is linear and non-zero (it takes the value 1 at E_{12}). By Theorem 4.5.3 this is a hyperplane.
- (b) No. The symmetric 3×3 matrices form a subspace of dimension 6 (choose the

entries on and above the diagonal freely), so the codimension is $9 - 6 = 3 \neq 1$.

- (c) Yes. It is the kernel of $p \mapsto p(1)$, which is linear ([Theorem 0.8.5](#)) and non-zero (it sends the constant 1 to 1).

4.5.6 Exercises

4.5.6.1 A. Check your understanding

Exercise A1.

- State the theorem identifying $(V/U)^*$ with a subspace of V^* , including the map.
- Define the codimension of a subspace, and define a hyperplane.
- True or false: U^* is a subspace of V^* . Justify your answer.
- Let U be a subspace of dimension 2 in F^5 . Find $\dim(F^5/U)^*$ and $\text{codim } U$.
- Is $\{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : x_1 = x_2 = 0\}$ a hyperplane in $\mathbb{R}^4$? Justify your answer.
- True or false: every non-zero linear functional $\varphi: V \rightarrow F$ is surjective. Justify your answer.

Solution.

- For a subspace U of V with quotient map π , the dual map $\pi': (V/U)^* \rightarrow V^*$, $\psi \mapsto \psi \circ \pi$, is injective with image U^0 ; so $(V/U)^* \cong U^0$ ([Theorem 4.5.2](#)).
- If V/U is finite-dimensional, $\text{codim } U = \dim(V/U)$; otherwise U has infinite codimension. A hyperplane is a subspace of codimension exactly 1.
- False. Elements of U^* are defined only on U , so they are not elements of V^* . What is true is $U^* \cong V^*/U^0$, a quotient ([Theorem 4.5.1](#)).
- $\dim(F^5/U)^* = \dim(F^5/U) = 5 - 2 = 3$, by [Corollary 4.1.4](#) and [Theorem 3.9.5](#). So $\text{codim } U = 3$.
- No. The subspace is $\text{span}(\mathbf{e}_3, \mathbf{e}_4)$, of dimension 2, so its codimension is $4 - 2 = 2$.
- True, in any dimension. This is [Proposition 4.1.2](#): $\text{im } \varphi$ is a subspace of the one-dimensional space F containing a non-zero element, so it is F .

4.5.6.2 B. Practice

Exercise B1: The dual of a quotient of $\mathbb{R}^4$.

Let $U = \text{span}((1, -1, 0, 2), (0, 1, 1, -1)) \subseteq \mathbb{R}^4$, and write functionals on $\mathbb{R}^4$ as $a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4$.

- Find a basis (φ_1, φ_2) of U^0 .
- Let $\psi_1, \psi_2 \in (\mathbb{R}^4/U)^*$ be the functionals with $\pi'(\psi_i) = \varphi_i$. Compute ψ_1 and ψ_2 at the coset $(2, 0, 1, 3) + U$.
- Find a basis of $\mathbb{R}^4/U$ whose dual basis is (ψ_1, ψ_2) .

Solution.

- (a) A functional lies in U^0 exactly when it vanishes at both spanning vectors ([Theorem 4.2.3](#) (b)), that is,

$$a_1 - a_2 + 2a_4 = 0, \quad a_2 + a_3 - a_4 = 0.$$

The second gives $a_3 = a_4 - a_2$, and the first gives $a_1 = a_2 - 2a_4$. Taking $(a_2, a_4) = (1, 0)$ and $(-1, -1)$ gives $(1, 1, -1, 0)$ and $(1, -1, 0, -1)$, that is,

$$\varphi_1 = x_1 + x_2 - x_3, \quad \varphi_2 = x_1 - x_2 - x_4.$$

Check: $\varphi_1(1, -1, 0, 2) = 0$, $\varphi_1(0, 1, 1, -1) = 0$, $\varphi_2(1, -1, 0, 2) = 1 + 1 - 2 = 0$, $\varphi_2(0, 1, 1, -1) = -1 + 1 = 0$. They are independent (compare the coefficients of x_3 and x_4). The spanning vectors of U are independent (compare first entries, then second), so $\dim U^0 = 4 - 2 = 2$ by [Theorem 4.2.2](#), and (φ_1, φ_2) is a basis by [Theorem 1.6.4](#).

- (b) By [Theorem 4.5.2](#), $\psi_i(\mathbf{v} + U) = \varphi_i(\mathbf{v})$. So $\psi_1((2, 0, 1, 3) + U) = 2 + 0 - 1 = 1$ and $\psi_2((2, 0, 1, 3) + U) = 2 - 0 - 3 = -1$.
- (c) We need cosets $\mathbf{c}_1, \mathbf{c}_2$ with $\psi_i(\mathbf{c}_j) = \delta_{ij}$. The vector $-\mathbf{e}_3$ has $\varphi_1(-\mathbf{e}_3) = 1$ and $\varphi_2(-\mathbf{e}_3) = 0$, and $-\mathbf{e}_4$ has $\varphi_1(-\mathbf{e}_4) = 0$, $\varphi_2(-\mathbf{e}_4) = 1$. Put $\mathbf{c}_1 = -\mathbf{e}_3 + U$ and $\mathbf{c}_2 = -\mathbf{e}_4 + U$. If $a\mathbf{c}_1 + b\mathbf{c}_2 = \mathbf{0}$, applying ψ_1 and ψ_2 gives $a = 0$ and $b = 0$; so $(\mathbf{c}_1, \mathbf{c}_2)$ is independent, and since $\dim(\mathbb{R}^4/U) = 2$ it is a basis ([Theorem 1.6.4](#)). By construction $\psi_i(\mathbf{c}_j) = \delta_{ij}$, so (ψ_1, ψ_2) is its dual basis ([Theorem 4.1.3](#)).

Exercise B2: Functionals with the same kernel.

Let V be a vector space over F , of any dimension, and let $\varphi, \psi \in V^*$ be non-zero. Prove that $\ker \varphi = \ker \psi$ if and only if $\psi = c\varphi$ for some non-zero $c \in F$. ([Exercise \(C1: Functionals with the same kernel\)](#) proved $(\Rightarrow)$ by a different route; here use the hyperplane decomposition.)

Hint: the remark after [Theorem 4.5.3](#).

Solution. $(\Leftarrow)$ If $\psi = c\varphi$ with $c \neq 0$, then $\psi(\mathbf{v}) = 0$ if and only if $c\varphi(\mathbf{v}) = 0$, if and only if $\varphi(\mathbf{v}) = 0$, since c is invertible. So $\ker \psi = \ker \varphi$.

$(\Rightarrow)$ Suppose $\ker \varphi = \ker \psi = H$. Since $\varphi \neq 0$, choose $\mathbf{v}_0$ with $\varphi(\mathbf{v}_0) \neq 0$; then $\mathbf{v}_0 \notin H$, so also $\psi(\mathbf{v}_0) \neq 0$. Put $c = \psi(\mathbf{v}_0)/\varphi(\mathbf{v}_0)$, which is non-zero, and $\chi = \psi - c\varphi \in V^*$. Then χ vanishes on H , because both ψ and φ do, and $\chi(\mathbf{v}_0) = \psi(\mathbf{v}_0) - c\varphi(\mathbf{v}_0) = 0$. By [Theorem 4.5.3](#) H is a hyperplane, and by the remark after it $V = H \oplus \text{span}(\mathbf{v}_0)$. A linear map vanishing on H and on $\mathbf{v}_0$ vanishes on every $\mathbf{h} + a\mathbf{v}_0$, hence on V . So $\chi = 0$, that is, $\psi = c\varphi$.

Exercise B3: Dimension computations.

- (a) Let $\dim V = 7$ and $\dim U = 3$. Find $\dim U^*$, $\dim(V^*/U^0)$, $\dim(V/U)^*$ and $\text{codim } U$.
- (b) In $V = F[x]_{\leq 3}$, let $U = \{p : p(0) = p(1) = 0\}$. Using the map $T: V \rightarrow F^2$, $T(p) = (p(0), p(1))$, find $\text{codim } U$ and $\dim U$, and write U as an intersection of two

hyperplanes.

- (c) Hence find a basis of U^0 consisting of evaluation functionals.

Solution.

- (a) $\dim U^* = 3$ by [Corollary 4.1.4](#). $\dim U^0 = 7 - 3 = 4$ by [Theorem 4.2.2](#), so $\dim(V^*/U^0) = 7 - 4 = 3$ ([Theorem 3.9.5](#)), consistent with [Theorem 4.5.1](#). $\dim(V/U)^* = \dim(V/U) = 4$, consistent with [Theorem 4.5.2](#), and $\operatorname{codim} U = 4$.
- (b) T is linear because evaluation is ([Theorem 0.8.5](#)), and $\ker T = U$. It is surjective: $T(1) = (1, 1)$ and $T(x) = (0, 1)$ span F^2 . By [Theorem 3.9.7](#), $V/U \cong F^2$, so $\operatorname{codim} U = 2$, and $\dim U = 4 - 2 = 2$ by [Theorem 3.9.5](#). Writing $\varepsilon_0(p) = p(0)$ and $\varepsilon_1(p) = p(1)$, we have $U = \ker \varepsilon_0 \cap \ker \varepsilon_1$, and both kernels are hyperplanes by [Theorem 4.5.3](#), since both functionals send 1 to 1.
- (c) Both ε_0 and ε_1 lie in U^0 . They are independent: if $a\varepsilon_0 + b\varepsilon_1 = 0$, evaluating at x gives $b = 0$, and then evaluating at 1 gives $a = 0$. Since $\dim U^0 = 4 - 2 = 2$, $(\varepsilon_0, \varepsilon_1)$ is a basis of U^0 ([Theorem 1.6.4](#)).

4.5.6.3 C. Going deeper

Exercise C1: Subspaces of finite codimension are intersections of hyperplanes.

Let V be a vector space over F , of any dimension, and let U be a subspace of finite codimension $k \geq 1$.

- (a) Prove that there are non-zero $\varphi_1, \dots, \varphi_k \in U^0$ with $U = \ker \varphi_1 \cap \dots \cap \ker \varphi_k$. Deduce that U is an intersection of k hyperplanes.
- (b) Prove that U is not an intersection of fewer than k hyperplanes.

Hint: for (a), take a dual basis of $(V/U)^$; for (b), consider the map $V \rightarrow F^m$, $\mathbf{v} \mapsto (\varphi_1(\mathbf{v}), \dots, \varphi_m(\mathbf{v}))$.*

Solution.

- (a) Let $\pi: V \rightarrow V/U$ be the quotient map. Since $\dim(V/U) = k$, choose a basis $(\mathbf{c}_1, \dots, \mathbf{c}_k)$ of V/U and let $(\psi_1, \dots, \psi_k)$ be its dual basis ([Theorem 4.1.3](#)). Put $\varphi_i = \psi_i \circ \pi$. By [Theorem 4.5.2](#), $\varphi_i \in U^0$, and $\varphi_i \neq 0$ because π' is injective and $\psi_i \neq 0$. Since each φ_i lies in U^0 , $U \subseteq \ker \varphi_i$ for each i . Conversely, let $\mathbf{v} \in \ker \varphi_1 \cap \dots \cap \ker \varphi_k$, and write $\mathbf{v} + U = \sum_j a_j \mathbf{c}_j$. Applying ψ_i gives $a_i = \psi_i(\mathbf{v} + U) = \varphi_i(\mathbf{v}) = 0$ for every i . So $\mathbf{v} + U$ is the zero coset, and $\mathbf{v} \in U$ by [Lemma 3.9.3](#) (b). This proves $U = \bigcap_i \ker \varphi_i$. Each $\ker \varphi_i$ is a hyperplane by [Theorem 4.5.3](#).
- (b) Suppose $U = H_1 \cap \dots \cap H_m$ with hyperplanes H_i . By [Theorem 4.5.3](#), $H_i = \ker \varphi_i$ for non-zero $\varphi_i \in V^*$. The map $S: V \rightarrow F^m$, $S(\mathbf{v}) = (\varphi_1(\mathbf{v}), \dots, \varphi_m(\mathbf{v}))$, is linear, since each entry is, and $\ker S = \bigcap_i \ker \varphi_i = U$. By [Theorem 3.9.7](#), $V/U \cong \operatorname{im} S$, a subspace of F^m . Hence $k = \dim(V/U) = \dim \operatorname{im} S \leq m$ by [Theorem 1.6.7](#). So at least k hyperplanes are needed.

Exercise C2: Complements and a choice of copy of U^* .

Let V be finite-dimensional, and let U, W be subspaces with $V = U \oplus W$.

- (a) Prove that $V^* = U^0 \oplus W^0$ without using [Theorem 4.2.3](#) (d). (This is [Exercise \(C3: Dual of a direct sum decomposition\)](#) (a), which used it.)
- (b) Deduce that the restriction map $\rho: V^* \rightarrow U^*$ restricts to an isomorphism $W^0 \rightarrow U^*$.
- (c) In $\mathbb{R}^2$, let $U = \text{span}(\mathbf{e}_1)$, $W_1 = \text{span}(\mathbf{e}_2)$ and $W_2 = \text{span}((1, 1))$. Find W_1^0 and W_2^0 , and find the element of each that restricts to the functional $\xi \in U^*$ with $\xi(\mathbf{e}_1) = 1$. Conclude that the copy of U^* inside V^* depends on the complement.

Solution.

- (a) Let $n = \dim V$. If $\varphi \in U^0 \cap W^0$, then φ vanishes on U and on W , hence on every $\mathbf{u} + \mathbf{w}$, that is, on $U + W = V$; so $\varphi = 0$. By [Theorem 4.2.2](#), $\dim U^0 + \dim W^0 = (n - \dim U) + (n - \dim W) = 2n - n = n$, using $\dim U + \dim W = n$ for a direct sum ([Theorem 1.7.4](#)). By [Theorem 1.7.3](#), $\dim(U^0 + W^0) = n - 0 = n = \dim V^*$, so $U^0 + W^0 = V^*$ ([Theorem 1.6.8](#)), and the sum is direct because the intersection is $\{0\}$.
- (b) The restriction of ρ to W^0 is linear. Its kernel is $W^0 \cap \ker \rho = W^0 \cap U^0 = \{0\}$ by [Theorem 4.5.1](#) and (a), so it is injective. Its domain has dimension $n - \dim W = \dim U = \dim U^*$. An injective linear map between spaces of the same finite dimension is an isomorphism ([Corollary 3.3.2](#) (e)).
- (c) Write functionals on $\mathbb{R}^2$ as $ax + by$. W_1^0 consists of those with $b = 0$, so $W_1^0 = \text{span}(x)$. W_2^0 consists of those with $a + b = 0$, so $W_2^0 = \text{span}(x - y)$. Restricting to U : $x|_U$ and $(x - y)|_U$ both take the value 1 at $\mathbf{e}_1$, so both equal ξ . The elements x and $x - y$ are different, so the subspaces $W_1^0 \neq W_2^0$ of V^* are two different copies of U^* , one for each complement.

4.6 Hyperplanes and Linear Systems Revisited

Chapter 2 met subspaces of F^n in two disguises. A **parametric** description lists spanning vectors, as in $\text{col}(A)$. An **implicit** description lists equations, as in $\text{nul}(A)$. We moved from equations to spanning vectors by elimination, but we never asked whether the reverse is always possible, or how many equations it takes. Duality answers both questions at once: the equations of a subspace U are the elements of U^0 . This section turns that sentence into an algorithm, proves once more that row rank and nullity add up to n , and ends the chapter with a dictionary between a space and its dual.

4.6.1 Row vectors are functionals

Recall from [Theorem 4.1.1](#) how functionals on F^n look. For $\mathbf{a} = (a_1, \dots, a_n) \in F^n$, put

$$\varphi_{\mathbf{a}}: F^n \rightarrow F, \quad \varphi_{\mathbf{a}}(\mathbf{x}) = \mathbf{a}^T \mathbf{x} = a_1 x_1 + \dots + a_n x_n.$$

By that theorem, every $\varphi \in (F^n)^*$ is $\varphi_{\mathbf{a}}$ for exactly one $\mathbf{a}$, namely $a_j = \varphi(\mathbf{e}_j)$, and $\mathbf{a} \mapsto \varphi_{\mathbf{a}}$ is an isomorphism $F^n \rightarrow (F^n)^*$.

The functional $\varphi_{\mathbf{a}}$ is “multiply by the **row** vector $\mathbf{a}^T$ ”. So the identification is really between functionals and row vectors, and that is what makes a system of equations a dual object. Write the rows of $A \in M_{m \times n}(F)$ as $\mathbf{a}_1^T, \dots, \mathbf{a}_m^T$ with $\mathbf{a}_i \in F^n$. The i -th entry of $A\mathbf{x}$ is $\mathbf{a}_i^T \mathbf{x} = \varphi_{\mathbf{a}_i}(\mathbf{x})$, so the homogeneous system $A\mathbf{x} = \mathbf{0}$ says exactly that $\varphi_{\mathbf{a}_1}(\mathbf{x}) = \dots = \varphi_{\mathbf{a}_m}(\mathbf{x}) = 0$. Hence

$$\text{nul}(A) = \ker \varphi_{\mathbf{a}_1} \cap \dots \cap \ker \varphi_{\mathbf{a}_m}.$$

Each non-zero row contributes a hyperplane (Theorem 4.5.3), and the null space is their intersection. **An equation is a functional, and a system is a list of functionals.**

The identification uses the standard basis of F^n , and nothing else. Under it, the coordinate functionals $\mathbf{x} \mapsto x_i$, which form the dual basis of $(\mathbf{e}_1, \dots, \mathbf{e}_n)$, correspond to $\mathbf{e}_1, \dots, \mathbf{e}_n$, that is, to the rows $\mathbf{e}_1^T, \dots, \mathbf{e}_n^T$.

4.6.2 Every subspace is a solution set

To describe U by equations, we need equations that hold on U , that is, functionals in U^0 , and enough of them that they hold **only** on U . The first requirement is the definition of U^0 . The second follows from the separation of points, applied in the quotient.

Lemma 4.6.1 — A subspace is cut out by its annihilator

Let V be a finite-dimensional vector space and U a subspace. For $\mathbf{v} \in V$,

$$\mathbf{v} \in U \iff \varphi(\mathbf{v}) = 0 \text{ for every } \varphi \in U^0.$$

If $(\varphi_1, \dots, \varphi_k)$ is a basis of U^0 , then $U = \ker \varphi_1 \cap \dots \cap \ker \varphi_k$.

Proof. $(\Rightarrow)$ This is Definition 4.2.1.

$(\Leftarrow)$ Suppose $\mathbf{v} \notin U$. Then $\mathbf{v} + U$ is a non-zero vector of V/U by Lemma 3.9.3 (b), and V/U is finite-dimensional (Theorem 3.9.5). By Theorem 4.1.5 there is $\psi \in (V/U)^*$ with $\psi(\mathbf{v} + U) \neq 0$. By Theorem 4.5.2, $\varphi = \psi \circ \pi$ lies in U^0 , and $\varphi(\mathbf{v}) = \psi(\mathbf{v} + U) \neq 0$. This proves the contrapositive.

For the last statement, $U \subseteq \ker \varphi_i$ for each i since $\varphi_i \in U^0$. Conversely, if $\varphi_i(\mathbf{v}) = 0$ for all i , then every $\varphi = \sum_i c_i \varphi_i \in U^0$ has $\varphi(\mathbf{v}) = 0$, so $\mathbf{v} \in U$ by $(\Leftarrow)$. ■

The lemma is Corollary 4.3.4 in disguise. That corollary says $U^{00} = U$ under $V \cong V^{**}$; the lemma unpacks it into a statement about vectors. In F^n , with rows in place of functionals, it becomes the main theorem of this section.

Theorem 4.6.2 — Every Subspace Is a Solution Set

Let U be a subspace of F^n with $\dim U = k$.

- (a) There is a matrix $A \in M_{(n-k) \times n}(F)$ with $U = \text{nul}(A)$. One may take as rows of A the vectors $\mathbf{a}_1^\top, \dots, \mathbf{a}_{n-k}^\top$ for any $\mathbf{a}_i \in F^n$ such that $(\varphi_{\mathbf{a}_1}, \dots, \varphi_{\mathbf{a}_{n-k}})$ is a basis of U^0 .
- (b) If $B \in M_{m \times n}(F)$ and $\text{nul}(B) = U$, then $m \geq n - k$.

So U is the solution set of a homogeneous system of exactly $n - k = \text{codim } U$ equations, and of no system with fewer. (When $k = n$, that is $U = F^n$, this is the empty system, with no equations.)

Proof.

- (a) By [Theorem 4.2.2](#), $\dim U^0 = n - k$. Choose a basis of U^0 ; by the identification above it is $(\varphi_{\mathbf{a}_1}, \dots, \varphi_{\mathbf{a}_{n-k}})$ for some $\mathbf{a}_i \in F^n$. Let A have rows $\mathbf{a}_1^\top, \dots, \mathbf{a}_{n-k}^\top$. Then

$$\text{nul}(A) = \ker \varphi_{\mathbf{a}_1} \cap \dots \cap \ker \varphi_{\mathbf{a}_{n-k}} = U,$$

where the second equality is [Lemma 4.6.1](#).

- (b) By [Theorem 2.5.7](#), $\text{rank } B = n - \text{nullity } B = n - k$. By [Theorem 2.5.5](#), $\text{rank } B = \dim \text{row}(B)$. The row space is spanned by the m rows of B , so $m \geq \dim \text{row}(B)$ by [Theorem 1.6.3](#) (b). Hence $m \geq n - k$.

■

The count is the one promised by codimension: **dimension counts free parameters, codimension counts independent equations**, and in F^n the two add up to n .

Example. (Equations for a plane in $\mathbb{R}^4$)

Let $U = \text{span}(\mathbf{u}_1, \mathbf{u}_2) \subseteq \mathbb{R}^4$, where $\mathbf{u}_1 = (1, 1, 0, 1)$ and $\mathbf{u}_2 = (0, 1, 1, 1)$. Find a system of linear equations whose solution set is U .

Solution. The two vectors are independent (compare first entries, then third), so $\dim U = 2$, and we expect $4 - 2 = 2$ equations. For $\mathbf{a} = (a_1, a_2, a_3, a_4)$, the functional $\varphi_{\mathbf{a}}$ lies in U^0 exactly when it vanishes at both spanning vectors ([Theorem 4.2.3](#) (b)):

$$a_1 + a_2 + a_4 = 0, \quad a_2 + a_3 + a_4 = 0.$$

This is the system $M\mathbf{a} = \mathbf{0}$ for the matrix M whose **rows** are $\mathbf{u}_1^\top$ and $\mathbf{u}_2^\top$. Subtracting the second row from the first gives the RREF

$$M = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix},$$

with free variables a_3, a_4 . By [Theorem 2.5.4](#), a basis of the solutions is $(1, -1, 1, 0)$ (from $a_3 = 1, a_4 = 0$) and $(0, -1, 0, 1)$ (from $a_3 = 0, a_4 = 1$). The corresponding functionals form a basis of U^0 , and by [Theorem 4.6.2](#),

$$U = \{\mathbf{x} \in \mathbb{R}^4 : x_1 - x_2 + x_3 = 0, \quad x_4 - x_2 = 0\}.$$

Check. Both equations hold at $\mathbf{u}_1$: $1 - 1 + 0 = 0$ and $1 - 1 = 0$; and at $\mathbf{u}_2$: $0 - 1 + 1 = 0$ and $1 - 1 = 0$. The coefficient matrix $\begin{pmatrix} 1 & -1 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}$ has rank 2, so its null space has dimension $4 - 2 = 2$, and it contains the independent vectors $\mathbf{u}_1, \mathbf{u}_2$; hence it equals U (Theorem 1.6.8). The implicit description says: U consists of the vectors with $x_4 = x_2$ and $x_1 + x_3 = x_2$.

4.6.3 From spans to equations and back

The example is an instance of a general procedure. Both directions of the conversion are “compute a null space”, applied to transposed data.

Parametric to implicit. Input: vectors $\mathbf{v}_1, \dots, \mathbf{v}_p \in F^n$ with $U = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_p)$.

1. Form $M \in M_{p \times n}(F)$ whose i -th row is $\mathbf{v}_i^T$.
2. Row reduce M and find a basis $\mathbf{n}_1, \dots, \mathbf{n}_d$ of $\text{nul}(M)$ as in Theorem 2.5.4.
3. Output the matrix $A \in M_{d \times n}(F)$ whose j -th row is $\mathbf{n}_j^T$, that is, the equations $\mathbf{n}_j^T \mathbf{x} = 0$ for $j = 1, \dots, d$.

Implicit to parametric. Input: $A \in M_{m \times n}(F)$. Output: a basis of $\text{nul}(A)$, found by row reduction as in Theorem 2.5.4.

The second half was proved correct in Chapter 2. The first half is new.

Theorem 4.6.3 — Correctness of the Conversion

Let $\mathbf{v}_1, \dots, \mathbf{v}_p \in F^n$, $U = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_p)$, and let M and A be as in the algorithm. Then $\text{nul}(A) = U$, and A has exactly $d = n - \dim U$ rows, which are linearly independent.

Idea. The only thing to see is what step 2 computes. The i -th entry of $M\mathbf{n}$ is $\mathbf{v}_i^T \mathbf{n} = \varphi_{\mathbf{n}}(\mathbf{v}_i)$. So $\mathbf{n} \in \text{nul}(M)$ says that the functional $\varphi_{\mathbf{n}}$ vanishes at every $\mathbf{v}_i$, that is, $\varphi_{\mathbf{n}} \in U^0$. Step 2 therefore computes a basis of U^0 , and the theorem of the previous subsection does the rest.

Proof. For $\mathbf{n} \in F^n$, the i -th entry of $M\mathbf{n}$ is $\mathbf{v}_i^T \mathbf{n} = \mathbf{n}^T \mathbf{v}_i = \varphi_{\mathbf{n}}(\mathbf{v}_i)$, where the middle equality holds because both sides equal $\sum_j (\mathbf{v}_i)_j n_j$. Hence

$$\mathbf{n} \in \text{nul}(M) \iff \varphi_{\mathbf{n}}(\mathbf{v}_i) = 0 \text{ for all } i \iff \varphi_{\mathbf{n}} \in \{\mathbf{v}_1, \dots, \mathbf{v}_p\}^0 = U^0,$$

where the last equality is Theorem 4.2.3 (b). So the isomorphism $\mathbf{n} \mapsto \varphi_{\mathbf{n}}$ maps $\text{nul}(M)$ onto U^0 . An isomorphism carries a basis to a basis (Theorem 3.5.2), so $(\varphi_{\mathbf{n}_1}, \dots, \varphi_{\mathbf{n}_d})$ is a basis of U^0 . In particular $d = \dim U^0 = n - \dim U$ by Theorem 4.2.2, and the $\mathbf{n}_j$, hence the rows $\mathbf{n}_j^T$, are independent. By Theorem 4.6.2 (a), the matrix A with rows $\mathbf{n}_1^T, \dots, \mathbf{n}_d^T$ satisfies $\text{nul}(A) = U$. ■

The algorithm does not need the $\mathbf{v}_i$ to be independent. Redundant spanning vectors only produce zero rows in the RREF of M , and the count $d = n - \text{rank } M$ automatically uses $\dim U = \text{rank } M$.

Running both halves in turn is a useful self-check. Starting from A in [Example \(Equations for a plane in \$\mathbb{R}^4\$ \)](#) and computing $\text{nul}(A)$ must return a basis of U ; starting from that basis and running the first half must return equations with the same solution set, though not necessarily the same equations.

Warning. Equations for a subspace are far from unique, and more equations do not mean a smaller solution set. In [Example \(Equations for a plane in \$\mathbb{R}^4\$ \)](#), the system $x_1 - x_2 + x_3 = 0$, $x_4 - x_2 = 0$, $x_1 + x_3 - x_4 = 0$ has the same solution set U : the third equation is the sum of the first and the negative of the second, so it adds a functional already in U^0 . Three equations in $\mathbb{R}^4$ cut out a two-dimensional subspace here, not a one-dimensional one. Each new equation lowers the dimension by one **only if** its functional is not a combination of the previous ones. A theorem later in this section makes this precise.

4.6.4 Affine solution sets

An inhomogeneous equation $\mathbf{a}^T \mathbf{x} = b$ with $\mathbf{a} \neq \mathbf{0}$ defines a set that is not a subspace when $b \neq 0$. It is a translate of the hyperplane $H = \ker \varphi_{\mathbf{a}}$: if $\mathbf{p}$ is one solution, then by [Theorem 2.1.3](#) the solutions are $\mathbf{p} + H$. We call a coset $\mathbf{p} + H$ of a hyperplane H ([Definition 3.9.2](#)) an **affine hyperplane**. In $\mathbb{R}^3$, the plane $x + y + z = 1$ is an affine hyperplane, parallel to the hyperplane $x + y + z = 0$.

A system $A\mathbf{x} = \mathbf{b}$ with rows $\mathbf{a}_i^T$ and right-hand sides b_i asks for the vectors lying on all the sets $\{\mathbf{x} : \mathbf{a}_i^T \mathbf{x} = b_i\}$ at once. Each such set is an affine hyperplane if $\mathbf{a}_i \neq \mathbf{0}$; if $\mathbf{a}_i = \mathbf{0}$, it is all of F^n (when $b_i = 0$) or empty (when $b_i \neq 0$). So **the solution set of a linear system is an intersection of affine hyperplanes**, apart from these trivial rows. The duality theorems give the converse.

Proposition 4.6.4 — Cosets as intersections of affine hyperplanes

Let U be a subspace of F^n of dimension $k < n$, and let $\mathbf{p} \in F^n$. Let $A \in M_{(n-k) \times n}(F)$ be as in [Theorem 4.6.2](#) (a), with rows $\mathbf{a}_1^T, \dots, \mathbf{a}_{n-k}^T$. Then

$$\mathbf{p} + U = \{\mathbf{x} \in F^n : A\mathbf{x} = A\mathbf{p}\} = \bigcap_{i=1}^{n-k} \{\mathbf{x} : \mathbf{a}_i^T \mathbf{x} = \mathbf{a}_i^T \mathbf{p}\},$$

an intersection of $n - k$ affine hyperplanes.

Proof. By [Theorem 4.6.2](#), $\text{nul}(A) = U$. The vector $\mathbf{p}$ is one solution of $A\mathbf{x} = A\mathbf{p}$, so by [Theorem 2.1.3](#) its solution set is $\mathbf{p} + \text{nul}(A) = \mathbf{p} + U$. The i -th equation reads $\mathbf{a}_i^T \mathbf{x} = \mathbf{a}_i^T \mathbf{p}$. Each $\varphi_{\mathbf{a}_i}$ is non-zero, because it belongs to a basis of U^0 ; so $\ker \varphi_{\mathbf{a}_i}$ is a hyperplane ([Theorem 4.5.3](#)), and $\{\mathbf{x} : \mathbf{a}_i^T \mathbf{x} = \mathbf{a}_i^T \mathbf{p}\} = \mathbf{p} + \ker \varphi_{\mathbf{a}_i}$ is an affine hyperplane, again by [Theorem 2.1.3](#) applied to this one equation. ■

Unlike hyperplanes, which all contain $\mathbf{0}$, affine hyperplanes can be disjoint: $x_1 + x_2 = 0$ and $x_1 + x_2 = 1$ are parallel lines in $\mathbb{R}^2$. That is the geometric picture of an inconsistent system.

4.6.5 The row space annihilates the null space

In Chapter 2, [Theorem 2.5.7](#) said $\text{rank } A + \text{nullity } A = n$, and its proof counted pivot and free columns in the RREF. The duality of this chapter explains the same equation without elimination: the rows of A are exactly the equations satisfied by the null space.

Theorem 4.6.5 — The Row Space Is the Annihilator of the Null Space

Let $A \in M_{m \times n}(F)$. Identifying a row vector $\mathbf{a}^\top \in M_{1 \times n}(F)$ with the functional $\varphi_{\mathbf{a}} \in (F^n)^*$,

$$\{\varphi_{\mathbf{a}} : \mathbf{a}^\top \in \text{row}(A)\} = \text{nul}(A)^0.$$

Consequently $\dim \text{row}(A) + \dim \text{nul}(A) = n$.

Idea. Let $T = T_A: F^n \rightarrow F^m$. Its kernel is $\text{nul}(A)$, and [Theorem 4.4.3](#) says $\text{im } T' = (\ker T)^0$. So it is enough to see that $\text{im } T'$ is the row space. Pull back a functional $\varphi_{\mathbf{y}}$ on F^m : the result is $\mathbf{x} \mapsto \mathbf{y}^\top A\mathbf{x}$, which is multiplication by the row $\mathbf{y}^\top A$, the combination of the rows of A with coefficients $y_1, \dots, y_m$. The transpose of Chapter 0 appears exactly here: $\mathbf{y}^\top A = (A^\top \mathbf{y})^\top$.

Proof. Let $T = T_A: F^n \rightarrow F^m$, $T(\mathbf{x}) = A\mathbf{x}$, so that $\ker T = \text{nul}(A)$. Every functional on F^m is $\varphi_{\mathbf{y}}$ for a unique $\mathbf{y} \in F^m$, as recalled at the start of this section. By [Definition 4.4.1](#), associativity of matrix multiplication, and [Theorem 0.9.5](#),

$$T'(\varphi_{\mathbf{y}})(\mathbf{x}) = \varphi_{\mathbf{y}}(A\mathbf{x}) = \mathbf{y}^\top(A\mathbf{x}) = (\mathbf{y}^\top A)\mathbf{x} = (A^\top \mathbf{y})^\top \mathbf{x} = \varphi_{A^\top \mathbf{y}}(\mathbf{x}) \quad (\mathbf{x} \in F^n),$$

so $T'(\varphi_{\mathbf{y}}) = \varphi_{A^\top \mathbf{y}}$. The columns of $A^\top$ are $\mathbf{a}_1, \dots, \mathbf{a}_m$, where $\mathbf{a}_i^\top$ are the rows of A , so $A^\top \mathbf{y} = y_1 \mathbf{a}_1 + \dots + y_m \mathbf{a}_m$ by [Theorem 0.9.3](#). As $\mathbf{y}$ runs over F^m , these are all the combinations of $\mathbf{a}_1, \dots, \mathbf{a}_m$, and since transposing respects combinations ([Theorem 0.9.5](#)), they are exactly the vectors $\mathbf{a}$ with $\mathbf{a}^\top \in \text{row}(A)$ ([Definition 2.5.2](#)). Hence

$$\text{im } T' = \{\varphi_{\mathbf{a}} : \mathbf{a}^\top \in \text{row}(A)\}.$$

By [Theorem 4.4.3](#), $\text{im } T' = (\ker T)^0 = \text{nul}(A)^0$, which is the first claim.

For the count, $\mathbf{a}^\top \mapsto \varphi_{\mathbf{a}}$ is an isomorphism $M_{1 \times n}(F) \rightarrow (F^n)^*$ (transposition followed by $\mathbf{a} \mapsto \varphi_{\mathbf{a}}$), so $\dim \text{row}(A) = \dim \text{nul}(A)^0 = n - \dim \text{nul}(A)$ by [Theorem 4.2.2](#). ■

The count is Rank–Nullity for matrices once more, now as a statement about annihilators: $\text{rank } A = \dim \text{row}(A)$ counts independent equations and $\text{nullity } A$ counts free parameters. It also says that the given equations generate **all** the equations satisfied by $\text{nul}(A)$: any linear equation that holds on the solution set of $A\mathbf{x} = \mathbf{0}$ is a combination of the rows of A .

Corollary 4.6.6 — Same solutions, same row space

Let $A \in M_{m \times n}(F)$ and $B \in M_{p \times n}(F)$. Then $\text{nul}(A) = \text{nul}(B)$ if and only if $\text{row}(A) = \text{row}(B)$.

Proof. ($\Leftarrow$) If $\text{row}(A) = \text{row}(B)$, then $\text{nul}(A)^0 = \text{nul}(B)^0$ by [Theorem 4.6.5](#), so $\text{nul}(A) = \text{nul}(B)$ by [Corollary 4.3.5](#).

($\Rightarrow$) If $\text{nul}(A) = \text{nul}(B)$, then $\{\varphi_{\mathbf{a}} : \mathbf{a}^T \in \text{row}(A)\} = \text{nul}(A)^0 = \{\varphi_{\mathbf{a}} : \mathbf{a}^T \in \text{row}(B)\}$ by [Theorem 4.6.5](#). Since $\mathbf{a}^T \mapsto \varphi_{\mathbf{a}}$ is injective, $\text{row}(A) = \text{row}(B)$. ■

Chapter 2 showed that row operations preserve the null space ([Theorem 2.5.1](#)). The corollary gives a converse in terms of row spaces: two systems in the same unknowns have the same solutions **exactly** when each equation of one is a combination of the equations of the other.

4.6.6 Sums and intersections by duality

Parametric and implicit descriptions are good at different things. A **sum** is easy parametrically: $U + W$ is spanned by a spanning list of U followed by one of W ([Proposition 1.7.2](#)). An **intersection** is easy implicitly: if $U = \text{nul}(A)$ and $W = \text{nul}(B)$, then $\mathbf{x} \in U \cap W$ exactly when $\mathbf{x}$ satisfies both systems, so

$$U \cap W = \text{nul} \begin{pmatrix} A \\ B \end{pmatrix},$$

the null space of the matrix with the rows of A above the rows of B . Duality explains why each is easy where it is. The rows of the stacked matrix span $\text{row}(A) + \text{row}(B)$, which corresponds to $U^0 + W^0 = (U \cap W)^0$ by [Theorem 4.6.5](#) and [Theorem 4.2.3](#) (d); and $(U + W)^0 = U^0 \cap W^0$ says that the equations of a sum are the equations common to both pieces.

Example. (An intersection computed from equations)

In $\mathbb{R}^4$, let $U = \text{span}((1, 0, 1, 2), (0, 1, 1, 1))$ and $W = \text{span}((1, 1, 0, 1), (1, 3, 0, 1))$, as in [Example \(A basis of an intersection\)](#). Find $U \cap W$ by first finding equations for U and for W .

Solution. Equations for U . A row (a_1, a_2, a_3, a_4) annihilates both spanning vectors when $a_1 + a_3 + 2a_4 = 0$ and $a_2 + a_3 + a_4 = 0$. This system is already in RREF, with free a_3, a_4 ; the basic solutions are $(-1, -1, 1, 0)$ and $(-2, -1, 0, 1)$. Changing signs, which does not change the kernels,

$$U = \{x_1 + x_2 - x_3 = 0, \quad 2x_1 + x_2 - x_4 = 0\}.$$

Equations for W . Here $a_1 + a_2 + a_4 = 0$ and $a_1 + 3a_2 + a_4 = 0$. Subtracting gives $a_2 = 0$, and then $a_1 = -a_4$, with a_3 free. The basic solutions are $(0, 0, 1, 0)$ and $(-1, 0, 0, 1)$, so

$$W = \{x_3 = 0, \quad x_4 - x_1 = 0\}.$$

Intersection. Stack the four equations. From W : $x_3 = 0$ and $x_4 = x_1$. Then $x_1 + x_2 - x_3 = 0$ gives $x_2 = -x_1$, and the last equation $2x_1 + x_2 - x_4 = 2x_1 - x_1 - x_1 = 0$ holds automatically. So

$$U \cap W = \{(t, -t, 0, t) : t \in \mathbb{R}\} = \text{span}((1, -1, 0, 1)),$$

in agreement with Chapter 2.

The redundancy of the fourth equation is not a coincidence. The four rows span $U^0 + W^0 = (U \cap W)^0$, which has dimension $4 - 1 = 3$, so exactly one of the four equations must be a consequence of the others.

Contrast with Chapter 2. There we put the four spanning vectors as **columns** of one matrix and read the intersection off a null vector, a relation $\mathbf{u}_1 - \mathbf{u}_2 = 2\mathbf{w}_1 - \mathbf{w}_2$. Here we computed two small null spaces (the equations) and then one more (the intersection). The equation method pays off when the subspaces are given by equations in the first place, or when several intersections with the same U are needed: the equations of U are computed once.

4.6.7 Independent functionals

The warning above said that k equations cut out codimension k only when they are independent. We now make that precise, in any dimension. Given $\varphi_1, \dots, \varphi_k \in V^*$, bundle them into one linear map

$$\Phi: V \rightarrow F^k, \quad \Phi(\mathbf{v}) = (\varphi_1(\mathbf{v}), \dots, \varphi_k(\mathbf{v})),$$

which is linear because each entry is. Its kernel is $\ker \varphi_1 \cap \dots \cap \ker \varphi_k$, the common solution set of the k equations.

Theorem 4.6.7 — Independent Functionals

Let V be a vector space over F , of any dimension, and let $\varphi_1, \dots, \varphi_k \in V^*$. The following are equivalent:

- (a) $(\varphi_1, \dots, \varphi_k)$ is linearly independent in V^* ;
- (b) $\Phi: V \rightarrow F^k$ is surjective;
- (c) $\ker \varphi_1 \cap \dots \cap \ker \varphi_k$ has codimension exactly k .

Idea. A dependence $\sum c_i \varphi_i = 0$ says that $\sum c_i y_i = 0$ for every $\mathbf{y} = \Phi(\mathbf{v})$ in the image: the image lies in the kernel of a non-zero functional on F^k . Conversely, if the image is a proper subspace of F^k , some non-zero functional on F^k kills it, and its coefficients give a dependence. The First Isomorphism Theorem turns “surjective” into “codimension k ”.

Proof.

- (a) $\Rightarrow$ (b). Suppose Φ is not surjective. Then $\text{im } \Phi$ is a proper subspace of F^k , so $\dim \text{im } \Phi < k$ and, by Theorem 4.2.2, $(\text{im } \Phi)^0 \neq \{0\}$. Take a non-zero functional in it; as at the start of this section it is $\varphi_{\mathbf{c}}, \mathbf{y} \mapsto c_1 y_1 + \dots + c_k y_k$, for a non-zero $\mathbf{c} \in F^k$. For every $\mathbf{v} \in V$, $\Phi(\mathbf{v}) \in \text{im } \Phi$, so

$$(c_1 \varphi_1 + \dots + c_k \varphi_k)(\mathbf{v}) = \varphi_{\mathbf{c}}(\Phi(\mathbf{v})) = 0.$$

Thus $c_1 \varphi_1 + \dots + c_k \varphi_k = 0$ with $\mathbf{c} \neq \mathbf{0}$, and the list is dependent. This proves the contrapositive.

- (b) $\Rightarrow$ (a). Suppose Φ is surjective and $c_1 \varphi_1 + \dots + c_k \varphi_k = 0$. For each j , choose $\mathbf{v}_j$ with $\Phi(\mathbf{v}_j) = \mathbf{e}_j$, that is, $\varphi_i(\mathbf{v}_j) = \delta_{ij}$. Evaluating the relation at $\mathbf{v}_j$ gives $c_j = 0$. So the list is independent.

(c) $\Leftrightarrow$ (c). By [Theorem 3.9.7](#), $V/\ker \Phi \cong \operatorname{im} \Phi$, and $\ker \Phi = \bigcap_i \ker \varphi_i$. So the codimension of $\bigcap_i \ker \varphi_i$ is $\dim \operatorname{im} \Phi$, which equals k exactly when $\operatorname{im} \Phi = F^k$ ([Theorem 1.6.8](#)).

■

For $V = F^n$ and $\varphi_i = \varphi_{\mathbf{a}_i}$, the map Φ is T_A , where A has rows $\mathbf{a}_i^\top$. The theorem then says: the rows of A are independent if and only if $A\mathbf{x} = \mathbf{b}$ is solvable for **every** $\mathbf{b} \in F^k$. Independent equations never contradict each other.

Quick check. In $\mathbb{R}^3$, consider $\varphi_1 = x+y$, $\varphi_2 = y+z$, $\varphi_3 = x-z$. What is the codimension of $\ker \varphi_1 \cap \ker \varphi_2 \cap \ker \varphi_3$? Is the system $x+y=1$, $y+z=1$, $x-z=1$ solvable?

Solution. $\varphi_3 = \varphi_1 - \varphi_2$, so the list is dependent, while (φ_1, φ_2) is independent (look at the coefficients of x and z). The three kernels intersect in $\ker \varphi_1 \cap \ker \varphi_2$, of codimension 2 by [Theorem 4.6.7](#), so the intersection is a line, not $\{\mathbf{0}\}$. The system is not solvable: any solution would give $1 = x - z = (x + y) - (y + z) = 1 - 1 = 0$. This is (b) failing: $(1, 1, 1) \notin \operatorname{im} \Phi$.

4.6.8 Summary and transfer

The chapter has built a dictionary. On the left is a finite-dimensional space V of dimension n , or a linear map $T: V \rightarrow W$ between finite-dimensional spaces; on the right is its mirror image in the dual. Every entry has been proved in this chapter, and every arrow reverses inclusions or directions.

In V (vectors)	In V^* (measurements)
a subspace U , $\dim U = k$	its annihilator U^0 , $\dim U^0 = n - k$
$U \subseteq W$	$W^0 \subseteq U^0$
$U + W$	$U^0 \cap W^0$
$U \cap W$	$U^0 + W^0$
U itself, recovered as U^{00}	U^0 determines U
the quotient V/U	the subspace $U^0 \cong (V/U)^*$
the subspace U	the quotient $V^*/U^0 \cong U^*$
a hyperplane H	a non-zero functional, up to scale
a spanning list (parametric form)	a list of equations (implicit form)
dimension (free parameters)	codimension (independent equations)
a basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$	its dual basis $(\varphi_1, \dots, \varphi_n)$
$T: V \rightarrow W$	$T': W^* \rightarrow V^*$, backwards
T injective	T' surjective
T surjective	T' injective
$\ker T$	$\operatorname{im} T' = (\ker T)^0$
$\operatorname{im} T$	$\ker T' = (\operatorname{im} T)^0$
$\operatorname{rank} T$	$\operatorname{rank} T' = \operatorname{rank} T$
the matrix $A = [T]_{\mathcal{B}}^{\mathcal{C}}$	the transpose $A^\top = [T']_{\mathcal{C}^*}^{\mathcal{B}^*}$
$\operatorname{nul}(A)$	$\operatorname{row}(A)$, as $\operatorname{nul}(A)^0$
V	V^{**} , naturally isomorphic via ev_V

Three moves make the dictionary usable, and they transfer to any situation with a pairing

between two spaces.

- **Test against functionals.** To show a vector is zero, or lies in a subspace, check that enough functionals vanish on it ([Theorem 4.1.5](#), [Lemma 4.6.1](#)).
- **Pass to the dual and count.** A question about a subspace can be translated into one about its annihilator, where $\dim U + \dim U^0 = n$ often settles it. The third proof of row rank = column rank ([Corollary 4.4.4](#)) and the count in [Theorem 4.6.5](#) both work this way.
- **Prefer the natural map.** Isomorphisms such as $V \cong V^{**}$, $(V/U)^* \cong U^0$ and $U^* \cong V^*/U^0$ are defined without a basis, so they can be used in any coordinates and survive changes of basis. Isomorphisms such as $V \cong V^*$ depend on a choice, and a proof that uses one must say which.

In Chapter 10, an inner product will give a **chosen** isomorphism $V \cong V^*$, and the annihilator U^0 will reappear inside V itself as the orthogonal complement. The dictionary above is the part of that story that needs no inner product.

4.6.9 Exercises

4.6.9.1 A. Check your understanding

Exercise A1.

- Explain how a row vector $\mathbf{a}^\top$ (with $\mathbf{a} \in F^n$) defines a linear functional on F^n , and why every functional arises this way.
- What is the least number of homogeneous linear equations whose solution set is a given 3-dimensional subspace of F^7 ?
- True or false: every system of three homogeneous equations in $\mathbb{R}^5$ has a solution set of dimension 2. Justify your answer.
- State the relation between $\text{row}(A)$ and $\text{nul}(A)$ in terms of annihilators.
- True or false: if A and B have n columns and $\text{nul}(A) = \text{nul}(B)$, then A and B have the same number of rows. Justify your answer.
- Name the dual counterparts of: a sum of subspaces; an injective linear map; the matrix of a linear map.

Solution.

- $\varphi_{\mathbf{a}}(\mathbf{x}) = \mathbf{a}^\top \mathbf{x}$ is linear by the rules of matrix multiplication ([Theorem 0.9.4](#)). A functional φ equals $\varphi_{\mathbf{a}}$ with $a_j = \varphi(\mathbf{e}_j)$, because both agree on the standard basis ([Theorem 3.1.5](#)); this is [Theorem 4.1.1](#).
- $7 - 3 = 4$, by [Theorem 4.6.2](#).
- False. The equations may be dependent: $x_1 = 0$ written three times has a solution set of dimension 4. The dimension is $5 - \text{rank } A$, which is 2 only when the three rows are independent.
- Identifying rows with functionals, $\text{row}(A) = \text{nul}(A)^0$ ([Theorem 4.6.5](#)).
- False. By [Corollary 4.6.6](#) they have the same row space, but not necessarily the same number of rows: $A = \begin{pmatrix} 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix}$ both have null space $\text{span}(\mathbf{e}_2)$.
- An intersection of annihilators, $(U + W)^0 = U^0 \cap W^0$; a surjective dual map; the

transpose matrix.

4.6.9.2 B. Practice

Exercise B1: Equations for a span.

Let $U = \text{span}((1, 2, 0, -1), (2, 1, 1, 0), (3, 3, 1, -1)) \subseteq \mathbb{R}^4$.

- Find $\dim U$.
- Find a system of homogeneous linear equations with solution set U , using as few equations as possible.
- Hence write the affine plane $(1, 0, 0, 0) + U$ as the solution set of a linear system.

Solution.

- The third vector is the sum of the first two, so $U = \text{span}((1, 2, 0, -1), (2, 1, 1, 0))$ by [Theorem 1.3.5](#). These two vectors are independent, since neither is a multiple of the other, so they form a basis of U and $\dim U = 2$.
- By [Theorem 4.6.3](#), we need a basis of $\text{nul}(M)$, where M has the three vectors as rows. Row reduction gives

$$M = \begin{pmatrix} 1 & 2 & 0 & -1 \\ 2 & 1 & 1 & 0 \\ 3 & 3 & 1 & -1 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 & \frac{2}{3} & \frac{1}{3} \\ 0 & 1 & -\frac{1}{3} & -\frac{2}{3} \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The free variables are the third and fourth. Setting them to $(3, 0)$ and $(0, 3)$ to clear denominators gives the null vectors $(-2, 1, 3, 0)$ and $(-1, 2, 0, 3)$. So

$$U = \{\mathbf{x} \in \mathbb{R}^4 : -2x_1 + x_2 + 3x_3 = 0, \quad -x_1 + 2x_2 + 3x_4 = 0\}.$$

Check on the first two spanning vectors, which suffice since they span U : $-2 + 2 + 0 = 0$, $-1 + 4 - 3 = 0$; $-4 + 1 + 3 = 0$, $-2 + 2 + 0 = 0$. Two equations is the minimum, by [Theorem 4.6.2](#) (b).

- By [Proposition 4.6.4](#), replace each right-hand side by the value of the left-hand side at $(1, 0, 0, 0)$:

$$(1, 0, 0, 0) + U = \{\mathbf{x} : -2x_1 + x_2 + 3x_3 = -2, \quad -x_1 + 2x_2 + 3x_4 = -1\}.$$

Exercise B2: A spanning list for a system.

Let $U = \{\mathbf{x} \in \mathbb{R}^4 : x_1 + x_2 - x_3 + 2x_4 = 0, \quad 2x_1 + 2x_2 + x_3 + x_4 = 0\}$.

- Find a basis of U .
- Run the parametric-to-implicit algorithm on your basis, and check that the resulting equations have the same solution set as the given ones, by comparing row spaces.

Solution.

- Row reduce the coefficient matrix: subtract twice row 1 from row 2 to get $(0, 0, 3, -3)$,

divide by 3, then add row 2 to row 1:

$$\begin{pmatrix} 1 & 1 & -1 & 2 \\ 2 & 2 & 1 & 1 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix}.$$

The pivots are in columns 1 and 3, and x_2, x_4 are free. By [Theorem 2.5.4](#), a basis is $\mathbf{n}_1 = (-1, 1, 0, 0)$ and $\mathbf{n}_2 = (-1, 0, 1, 1)$.

- (b) Put $\mathbf{n}_1^\top, \mathbf{n}_2^\top$ as the rows of M and solve $M\mathbf{a} = \mathbf{0}$: $-a_1 + a_2 = 0$ and $-a_1 + a_3 + a_4 = 0$. So $a_2 = a_1$ and $a_3 = a_1 - a_4$, with a_1, a_4 free; the basic solutions are $(1, 1, 1, 0)$ and $(0, 0, -1, 1)$. The new system is $x_1 + x_2 + x_3 = 0$, $-x_3 + x_4 = 0$. Its rows lie in the row space of the original matrix, which equals the row space of the RREF ([Theorem 2.5.1](#)), since $(1, 1, 1, 0) = (1, 1, 0, 1) + (0, 0, 1, -1)$ and $(0, 0, -1, 1) = -(0, 0, 1, -1)$. Both row spaces are two-dimensional, so they are equal ([Theorem 1.6.8](#)), and the two systems have the same solution set by [Corollary 4.6.6](#).

Exercise B3: Two planes in $\mathbb{R}^4$.

Let $U = \text{span}((1, 0, 0, 1), (0, 1, 1, 0))$ and $W = \text{span}((1, 1, 0, 0), (0, 0, 1, 1))$ in $\mathbb{R}^4$.

- Find two equations for U and two for W .
- Use them to find $U \cap W$.
- Find a single equation for $U + W$, using $(U + W)^0 = U^0 \cap W^0$.

Solution.

- Each pair is independent, so each plane has dimension 2 and needs two equations. The functionals $x_1 - x_4$ and $x_2 - x_3$ vanish at $(1, 0, 0, 1)$ and $(0, 1, 1, 0)$, and they are independent; since $\dim U^0 = 2$ ([Theorem 4.2.2](#)), they form a basis of U^0 . So $U = \{x_1 = x_4, x_2 = x_3\}$ by [Theorem 4.6.2](#). In the same way $x_1 - x_2$ and $x_3 - x_4$ form a basis of W^0 , and $W = \{x_1 = x_2, x_3 = x_4\}$.
- $U \cap W$ is the solution set of all four equations: $x_1 = x_4, x_2 = x_3, x_1 = x_2, x_3 = x_4$. They force $x_1 = x_2 = x_3 = x_4$, so $U \cap W = \text{span}((1, 1, 1, 1))$.
- A functional in $U^0 \cap W^0$ has the form $a(x_1 - x_4) + b(x_2 - x_3) = c(x_1 - x_2) + d(x_3 - x_4)$. Comparing coefficients of x_1, x_2, x_3, x_4 : $a = c, b = -c, -b = d, -a = -d$. So $a = c = d$ and $b = -a$, and $U^0 \cap W^0 = \text{span}(x_1 - x_2 + x_3 - x_4)$. By [Theorem 4.2.3](#) (c) and [Lemma 4.6.1](#),

$$U + W = \{\mathbf{x} : x_1 - x_2 + x_3 - x_4 = 0\},$$

a hyperplane. Consistently, $\dim(U + W) = 2 + 2 - 1 = 3$ by [Theorem 1.7.3](#).

4.6.9.3 C. Going deeper

Exercise C1: Avoiding finitely many subspaces.

Let F be an **infinite** field, V a vector space over F with $1 \leq \dim V = n < \infty$, and $U_1, \dots, U_k$ proper subspaces of V .

- Prove that for each i there is a non-zero $\varphi_i \in V^*$ with $U_i \subseteq \ker \varphi_i$.

- (b) Let $(\mathbf{b}_1, \dots, \mathbf{b}_n)$ be a basis of V , and for $t \in F$ put $\mathbf{v}(t) = \mathbf{b}_1 + t\mathbf{b}_2 + t^2\mathbf{b}_3 + \dots + t^{n-1}\mathbf{b}_n$. Show that $t \mapsto \varphi_i(\mathbf{v}(t))$ is given by a non-zero polynomial of degree at most $n - 1$.
- (c) Deduce that some vector of V lies in none of $U_1, \dots, U_k$. In other words, V is not a union of finitely many proper subspaces.
- (d) Show that the hypothesis that F is infinite cannot be dropped: over $\mathbb{F}_q$, the space $\mathbb{F}_q^2$ is the union of $q + 1$ proper subspaces.

Hint: for (c), use Lemma 2.7.2.

Solution.

- (a) Since $U_i \neq V$, pick $\mathbf{v} \notin U_i$. By Lemma 4.6.1 there is $\varphi_i \in U_i^0$ with $\varphi_i(\mathbf{v}) \neq 0$. Then $\varphi_i \neq 0$ and $U_i \subseteq \ker \varphi_i$.
- (b) By linearity, $\varphi_i(\mathbf{v}(t)) = \sum_{j=1}^n \varphi_i(\mathbf{b}_j) t^{j-1}$, which is the value at t of the polynomial $p_i(x) = \sum_{j=1}^n \varphi_i(\mathbf{b}_j) x^{j-1}$ of degree at most $n - 1$. Its coefficients are $\varphi_i(\mathbf{b}_1), \dots, \varphi_i(\mathbf{b}_n)$, which are not all zero, since otherwise φ_i would vanish on a basis and be zero (Theorem 3.1.5). So $p_i \neq 0$.
- (c) By Lemma 2.7.2, each p_i has at most $n - 1$ roots in F , so together they have at most $k(n - 1)$ roots. Since F is infinite, there is $t_0 \in F$ that is a root of none of them. Then $\varphi_i(\mathbf{v}(t_0)) = p_i(t_0) \neq 0$, so $\mathbf{v}(t_0) \notin \ker \varphi_i \supseteq U_i$, for every i .
- (d) $\mathbb{F}_q^2$ has $q^2 - 1$ non-zero vectors, and each line through $\mathbf{0}$ contains $q - 1$ of them (the non-zero multiples of one vector). Two distinct lines meet only in $\mathbf{0}$, so there are $(q^2 - 1)/(q - 1) = q + 1$ lines, and every non-zero vector lies on one of them. Hence $\mathbb{F}_q^2$ is the union of its $q + 1$ lines, each a proper subspace. For $q = 2$ these are $\text{span}((1, 0))$, $\text{span}((0, 1))$ and $\text{span}((1, 1))$, as in Exercise (C1: Covering a Space by Subspaces). The proof of (c) breaks exactly where it used infinitely many t : here $k(n - 1) = q + 1$ exceeds $|F| = q$.

Exercise C2: The least number of equations for a system.

Let $A \in M_{m \times n}(F)$ and $\mathbf{b} \in F^m$, and suppose the solution set S of $A\mathbf{x} = \mathbf{b}$ is non-empty.

- (a) Prove that if $B \in M_{p \times n}(F)$, $\mathbf{c} \in F^p$ and $B\mathbf{x} = \mathbf{c}$ has the same solution set S , then $\text{nul}(B) = \text{nul}(A)$.
- (b) Deduce that every system with solution set S has at least $\text{rank } A$ equations, and that some system with solution set S has exactly $\text{rank } A$ equations.
- (c) Show by an example that (b) fails without the hypothesis $S \neq \emptyset$.

Solution.

- (a) Fix $\mathbf{p} \in S$. By Theorem 2.1.3, $S = \mathbf{p} + \text{nul}(A)$ and also $S = \mathbf{p} + \text{nul}(B)$. For $\mathbf{h} \in \text{nul}(A)$, $\mathbf{p} + \mathbf{h} \in S = \mathbf{p} + \text{nul}(B)$, so $\mathbf{h} \in \text{nul}(B)$. Swapping the roles of A and B gives the reverse inclusion, so $\text{nul}(A) = \text{nul}(B)$.
- (b) Let $k = \dim \text{nul}(A) = n - \text{rank } A$ (Theorem 2.5.7). If $B\mathbf{x} = \mathbf{c}$ with p equations has solution set S , then $\text{nul}(B) = \text{nul}(A)$ by (a), so $p \geq n - k = \text{rank } A$ by Theorem 4.6.2 (b). Conversely, take $A_0 \in M_{(n-k) \times n}(F)$ with $\text{nul}(A_0) = \text{nul}(A)$, which exists by Theorem 4.6.2 (a) (if $k = n$, take the empty system, whose solution

set is $F^n = S$). By [Proposition 4.6.4](#), or directly by [Theorem 2.1.3](#), the system $A_0\mathbf{x} = A_0\mathbf{p}$ has solution set $\mathbf{p} + \text{nul}(A) = S$, and it has $n - k = \text{rank } A$ equations.

- (c) Over any field, $0x_1 = 1$ (in F^1) has empty solution set and coefficient matrix of rank 0, yet no system with 0 equations has empty solution set, since the empty system is solved by every vector. So at least one equation is needed, which is more than $\text{rank } A = 0$.

Chapter 5

Polynomials

Integers and polynomials share one arithmetic. Both can be divided with a remainder, and from that single tool everything else follows: greatest common divisors, Bézout identities, unique factorization into primes or irreducibles, and the Chinese Remainder Theorem. This chapter develops that arithmetic for $F[x]$ in full, because from Chapter 8 on the structure of an operator is read off from polynomials: which polynomials kill it, how they factor, and which of their factors are coprime.

Along the way we keep two promises made in Chapter 0: we prove division with remainder, stated there only for orientation, and the Fundamental Theorem of Algebra, that every non-constant complex polynomial has a root. We also prove that polynomials factor uniquely into irreducibles, just as integers factor uniquely into primes. The chapter ends by plugging an operator into a polynomial, where the arithmetic of $F[x]$ turns into a splitting of the space into pieces.

What you need. From Chapter 0, induction and well-ordering ([Theorem 0.2.10](#), [Theorem 0.2.11](#)), divisibility of integers and Bézout's lemma ([Definition 0.2.2](#), [Lemma 0.7.2](#)), the fields $\mathbb{F}_p$ ([Theorem 0.7.3](#)), polynomials, degree and evaluation ([Definition 0.8.1](#), [Definition 0.8.3](#), [Theorem 0.8.3](#), [Theorem 0.8.5](#)) and complex numbers ([Theorem 0.6.6](#)). From Chapter 2, the factor theorem and root bound for linear factors ([Lemma 2.7.1](#), [Lemma 2.7.2](#)) and interpolation ([Theorem 2.7.3](#)). From Chapter 3, polynomials of an operator ([Definition 3.4.2](#)), quotient spaces and the First Isomorphism Theorem ([Definition 3.9.3](#), [Theorem 3.9.7](#)). From Chapter 4, dual bases ([Theorem 4.1.3](#)).

5.0.1 Roadmap

- **Division with remainder.** Long division always works over a field, with a unique quotient and remainder; the remainder and factor theorems in general.
- **Ideals and greatest common divisors.** Every ideal of $F[x]$ is generated by one polynomial; gcds, Bézout identities and the Euclidean algorithm.
- **Irreducible polynomials and unique factorization.** The primes of $F[x]$, and why factorization into them is unique.
- **Roots, multiplicity and derivatives.** Repeated roots detected by the formal derivative, and Taylor expansion of a polynomial.
- **The Fundamental Theorem of Algebra.** A complete proof, and what it says about real polynomials.

- **Interpolation and the Chinese Remainder Theorem.** The Lagrange formula, solving simultaneous congruences, and the quotient-space picture behind both.
- **Polynomials of an operator.** Every operator on a finite-dimensional space satisfies a polynomial, and coprime factors split the space.

5.0.2 Named moves

- **Divide and look at the remainder.** Most statements about divisibility reduce to one division.
- **The smallest-degree element generates.** In an ideal, a non-zero element of least degree divides every other element.
- **Evaluate at a root.** Plugging a root of the divisor into $f = qg + r$ kills qg and reads off information about the remainder.
- **Compare degrees.** Uniqueness of quotients, factorizations and interpolants usually comes from a degree count.
- **Bézout gives projections.** When $\gcd(p, q) = 1$, write $1 = ap + bq$; evaluated at an operator, the two terms split the space.

5.1 Division with Remainder

Integers can be divided with a remainder: $17 = 3 \cdot 5 + 2$, and the remainder 2 is smaller than the divisor 5. Polynomials over a field can be divided in the same way, with “smaller” measured by degree. Chapter 0 stated this fact without proof, and Chapter 2 proved the special case of dividing by $x - c$. This section proves division with remainder in general, over any field, and draws the first consequences: the language of divisibility, the Remainder and Factor Theorems for every divisor $x - c$, a sharper root bound, and the fact that over an infinite field a polynomial is determined by its values. Throughout, F is a field unless we say otherwise, and it matters: the warning in the second subsection shows division failing when the coefficients only form $\mathbb{Z}$.

5.1.1 Long division by hand

At school one divides $x^4 + 2x^3 - x + 3$ by $x^2 + x - 1$ with a layout that imitates long division of numbers. Let us do it in words, so that we can see what each step does.

Write $f = x^4 + 2x^3 - x + 3$ and $g = x^2 + x - 1$, in $\mathbb{Q}[x]$.

1. The leading term of f is x^4 , and the leading term of g is x^2 . The quotient $x^4/x^2 = x^2$ is the first term of the answer. Subtract $x^2g = x^4 + x^3 - x^2$:

$$f - x^2g = x^3 + x^2 - x + 3.$$

2. The leading term is now x^3 , and $x^3/x^2 = x$. Subtract $xg = x^3 + x^2 - x$:

$$f - x^2g - xg = 3.$$

3. What is left, 3, has degree 0, smaller than $\deg g = 2$. No multiple $cx^k g$ with $k \geq 0$ can remove it without creating terms of degree at least 2. We stop.

So $f = (x^2 + x)g + 3$: the **quotient** is $x^2 + x$ and the **remainder** is 3. Expanding $(x^2 + x)(x^2 + x - 1) = x^4 + 2x^3 - x$ confirms it.

Two questions are hidden in this routine. **Why does it always stop?** Each step removes the current leading term, so the degree of what is left drops by at least one. Degrees are natural numbers, and a strictly decreasing sequence of natural numbers cannot go on forever. **Why is the answer unique?** Someone else might hope to find a different quotient and a different remainder of degree less than 2. We will see that they cannot, by comparing degrees.

There is also a third, quieter question. In step 1 we divided the leading coefficient of the current polynomial by the leading coefficient of g . Here that coefficient was 1. In general it is some $b \neq 0$, and we need b^{-1} . Over a field this always exists. That is exactly where the field enters.

5.1.2 The Division Theorem

The routine above works for every pair of polynomials with non-zero divisor. Here is the precise statement.

Theorem 5.1.1 — Polynomial Division with Remainder

Let F be a field, and let $f, g \in F[x]$ with $g \neq 0$. Then there exist **unique** polynomials $q, r \in F[x]$ such that

$$f = qg + r \quad \text{and} \quad \deg r < \deg g.$$

The polynomial q is the **quotient** and r is the **remainder** of f on division by g . Because $\deg 0 = -\infty$, the condition $\deg r < \deg g$ allows $r = 0$, which is the case “ g goes into f exactly”.

Idea. Existence is the hand routine turned into an induction on $\deg f$. If $\deg f < \deg g$, there is nothing to do: $q = 0$, $r = f$. Otherwise subtract $(a/b)x^{n-m}g$, where ax^n and bx^m are the leading terms of f and g . This kills the leading term of f , so the induction hypothesis applies to what is left. Uniqueness is “take the difference”: if $qg + r = q'g + r'$, then $(q - q')g = r' - r$. The right side has degree less than $\deg g$, while a non-zero multiple of g has degree at least $\deg g$.

Proof. Let $m = \deg g$, which lies in $\mathbb{N}$ since $g \neq 0$, and let $b \neq 0$ be the leading coefficient of g .

Existence. If $f = 0$, then $q = 0$ and $r = 0$ work. For non-zero f we prove, by strong induction on $n \in \mathbb{N}$ (Theorem 0.2.10 with $n_0 = 0$), the statement

$P(n)$: for every $f \in F[x]$ with $\deg f = n$, there are $q, r \in F[x]$ with $f = qg + r$ and $\deg r < m$.

Let $n \in \mathbb{N}$, assume $P(k)$ for all $k < n$, and let $\deg f = n$ with leading coefficient a .

Case 1: $n < m$. Take $q = 0$ and $r = f$. Then $f = 0 \cdot g + f$ and $\deg r = n < m$.

Case 2: $n \geq m$. Since F is a field and $b \neq 0$, the inverse b^{-1} exists. Put

$$f_1 = f - ab^{-1}x^{n-m}g.$$

By [Theorem 0.8.3](#), the polynomial $ab^{-1}x^{n-m}g$ has degree $(n-m) + m = n$ and leading coefficient $ab^{-1} \cdot b = a$. So the coefficient of x^n in f_1 is $a - a = 0$, and all coefficients of higher powers are 0 as well. Hence $f_1 = 0$ or $\deg f_1 < n$. If $f_1 = 0$, then $f = ab^{-1}x^{n-m}g + 0$, and we take $q = ab^{-1}x^{n-m}$, $r = 0$. Otherwise $k = \deg f_1 < n$, and the induction hypothesis $P(k)$ gives q_1, r_1 with $f_1 = q_1g + r_1$ and $\deg r_1 < m$. Then

$$f = f_1 + ab^{-1}x^{n-m}g = (ab^{-1}x^{n-m} + q_1)g + r_1,$$

so $q = ab^{-1}x^{n-m} + q_1$ and $r = r_1$ work. This proves $P(n)$, and by strong induction $P(n)$ holds for every $n \in \mathbb{N}$.

Uniqueness. Suppose $f = qg + r = q'g + r'$ with $\deg r < m$ and $\deg r' < m$. Then $(q - q')g = r' - r$. By [Theorem 0.8.2](#), $\deg(r' - r) \leq \max(\deg r', \deg r) < m$, since $-r$ has the same degree as r . Suppose $q \neq q'$. Then $\deg(q - q') \geq 0$, and [Theorem 0.8.3](#) gives $\deg((q - q')g) = \deg(q - q') + m \geq m$, a contradiction. Hence $q = q'$. Then $r' - r = 0 \cdot g = 0$, so $r = r'$. This proves the theorem. ■

The field axioms did essential work in two places: the inverse b^{-1} in Case 2, and [Theorem 0.8.3](#) in the uniqueness step, which needs a product of non-zero coefficients to be non-zero. The proof is also an algorithm. Case 2, repeated, is the long division of the previous subsection, and the induction guarantees that it stops after at most $\deg f - \deg g + 1$ subtractions.

The hypothesis “over a field” is not decoration. **The leading coefficient of g must be invertible.**

Warning. **Division with remainder fails over $\mathbb{Z}$.** Write $\mathbb{Z}[x]$ for the polynomials in $\mathbb{Q}[x]$ whose coefficients are all integers. Try to divide $f = x^2$ by $g = 2x$ inside $\mathbb{Z}[x]$: we want $q, r \in \mathbb{Z}[x]$ with $x^2 = 2x \cdot q + r$ and $\deg r < 1$, so r is a constant. Comparing coefficients of x^2 gives $2q_1 = 1$, where q_1 is the coefficient of x in q . No integer q_1 satisfies this, so no such q, r exist. Over $\mathbb{Q}$ there is no problem: $x^2 = 2x \cdot \frac{1}{2}x + 0$. The obstacle is precisely that 2 has no inverse in $\mathbb{Z}$.

Division by a divisor whose leading coefficient **is** invertible still works over $\mathbb{Z}$, for example by a monic polynomial; see Exercise C1.

Quick check. Why must the inequality $\deg r < \deg g$ be strict? Find two different pairs (q, r) with $x^2 = q \cdot (x^2 + 1) + r$ and $\deg r \leq 2$.

Solution. Both $x^2 = 1 \cdot (x^2 + 1) + (-1)$ and $x^2 = 0 \cdot (x^2 + 1) + x^2$ work, with remainders of degree 0 and 2. With $\deg r \leq \deg g$ allowed, uniqueness fails, because we may move one copy of g between qg and r . The strict inequality is exactly what rules this out: in the uniqueness proof, $\deg(r' - r) < \deg g$ is the step that forces $q = q'$.

Now a division in which the leading coefficient of the divisor is not 1, so fractions appear.

Example. (Long Division with a Non-Monic Divisor)

In $\mathbb{Q}[x]$, divide $f = 2x^4 - 3x^3 + x^2 + 4x - 1$ by $g = 2x^2 - x + 3$ with remainder.

Solution. We follow Case 2 of the proof, where $b = 2$ and $b^{-1} = \frac{1}{2}$.

Step 1. The leading term of f is $2x^4$, and $2x^4/2x^2 = x^2$. Subtracting $x^2g = 2x^4 - x^3 + 3x^2$ leaves

$$f - x^2g = -2x^3 - 2x^2 + 4x - 1.$$

Step 2. The leading term is $-2x^3$, and $-2x^3/2x^2 = -x$. Subtracting $-xg = -2x^3 + x^2 - 3x$ leaves

$$-3x^2 + 7x - 1.$$

Step 3. The leading term is $-3x^2$, and $-3x^2/2x^2 = -\frac{3}{2}$. Subtracting $-\frac{3}{2}g = -3x^2 + \frac{3}{2}x - \frac{9}{2}$ leaves

$$(7 - \frac{3}{2})x + (-1 + \frac{9}{2}) = \frac{11}{2}x + \frac{7}{2}.$$

This has degree $1 < 2 = \deg g$, so we stop. Hence

$$2x^4 - 3x^3 + x^2 + 4x - 1 = (x^2 - x - \frac{3}{2})(2x^2 - x + 3) + \frac{11}{2}x + \frac{7}{2},$$

with quotient $q = x^2 - x - \frac{3}{2}$ and remainder $r = \frac{11}{2}x + \frac{7}{2}$. By the uniqueness in [Theorem 5.1.1](#), any other correct method must produce the same q and r . (We checked the identity by expanding the right-hand side with SymPy.)

Notice that the quotient has a non-integer coefficient although f and g have integer coefficients. This is the warning above in action: over $\mathbb{Z}$ this division would be impossible.

5.1.3 Divisibility

With integers, “3 divides 12” means that 12 is an integer multiple of 3 ([Definition 0.2.2](#)), and it is the case “remainder zero” of integer division. Polynomials get the same word. The field is part of the meaning, so we put it in the definition.

g divides f when f is a polynomial multiple of g , with the multiplier taken over the same field.

Definition 5.1.1 — Divisibility, Associates

Let F be a field and $f, g \in F[x]$.

- We say g **divides** f **over** F , and write $g \mid f$, if there **exists** $h \in F[x]$ with $f = gh$. Then g is a **divisor** (or **factor**) of f , and f is a **multiple** of g . We write $g \nmid f$ if g does not divide f .
- We say f and g are **associates** if $f = cg$ for some **non-zero** constant $c \in F$.

In words: $g \mid f$ asks for a multiplier h **in** $F[x]$, so the same pair can behave differently over different fields. Associates differ only by a non-zero scalar factor. Since c^{-1} exists, “associates” is symmetric: $f = cg$ gives $g = c^{-1}f$.

Examples, simplest first.

- **Degenerate cases.** Every g divides 0, since $0 = g \cdot 0$. On the other hand $0 \mid f$ only for $f = 0$, since $0 \cdot h = 0$. Every non-zero constant c divides every f , since $f = c \cdot (c^{-1}f)$. These cases matter: they say that non-zero constants are invisible to divisibility, just as ± 1 are for integers.

- In $\mathbb{Q}[x]$, $x - 1 \mid x^2 - 1$, since $x^2 - 1 = (x - 1)(x + 1)$. Also $2x - 2 \mid x^2 - 1$, since $x^2 - 1 = (2x - 2) \cdot \frac{1}{2}(x + 1)$. The polynomials $x - 1$ and $2x - 2$ are associates.
- In $\mathbb{F}_2[x]$, $x + 1 \mid x^2 + 1$, since $(x + 1)^2 = x^2 + 2x + 1 = x^2 + 1$ there. In $\mathbb{Q}[x]$, however, $x + 1 \nmid x^2 + 1$, as the next paragraph shows.
- In $\mathbb{C}[x]$, $x - i \mid x^2 + 1 = (x - i)(x + i)$.

For a non-example by minimal change, keep $x^2 + 1$ and $x + 1$ but move from $\mathbb{F}_2$ to $\mathbb{Q}$. Both are still polynomials, and division with remainder still applies. What fails is the clause “there **exists** $h \in F[x]$ ”. If $x^2 + 1 = (x + 1)h$ with $h \in \mathbb{Q}[x]$, then evaluating at -1 ([Theorem 0.8.5](#)) gives $2 = 0 \cdot h(-1) = 0$, which is false in $\mathbb{Q}$. Over $\mathbb{F}_2$ the same computation reads $0 = 0$, and there is no contradiction.

Warning. Divisibility depends on where the multiplier may live. In $\mathbb{Q}[x]$, $2x + 2 \mid x + 1$, because $x + 1 = (2x + 2) \cdot \frac{1}{2}$. If we insisted on integer coefficients, $2x + 2$ would **not** divide $x + 1$: any integer multiple $(2x + 2)h$ has all coefficients even. Over a field, a non-zero scalar factor never affects divisibility, so it is pointless to ask whether $x - 1$ or $3x - 3$ is “the” factor of $x^2 - 1$.

Why this definition? Because it makes “divides” the case of remainder zero, which is the link to division we need. The basic rules follow quickly.

Proposition 5.1.2 — Rules for Divisibility

Let F be a field and $f, g, h \in F[x]$.

- If $f \mid g$ and $g \mid h$, then $f \mid h$.
- If $h \mid f$ and $h \mid g$, then $h \mid af + bg$ **for all** $a, b \in F[x]$.
- If $g \mid f$ and $f \neq 0$, then $\deg g \leq \deg f$.
- $f \mid g$ and $g \mid f$ if and only if f and g are associates or both are 0. In particular, if $f \mid g$, $g \mid f$, and f, g are both monic, then $f = g$.
- If $g \neq 0$, then $g \mid f$ if and only if the remainder of f on division by g is 0.

Proof.

- Write $g = fu$ and $h = gv$ with $u, v \in F[x]$. Then $h = f(uv)$ by [Theorem 0.8.1](#), so $f \mid h$.
- Write $f = hu$ and $g = hv$. Then $af + bg = h(au + bv)$ by [Theorem 0.8.1](#), so $h \mid af + bg$.
- Write $f = gu$. Since $f \neq 0$, both g and u are non-zero, so $\deg u \geq 0$. By [Theorem 0.8.3](#), $\deg f = \deg g + \deg u \geq \deg g$.
- ($\Leftarrow$) If $f = cg$ with $c \neq 0$, then $g \mid f$ and $g = c^{-1}f$ gives $f \mid g$. If $f = g = 0$, both divide each other. ($\Rightarrow$) Suppose $g = fu$ and $f = gv$. If $f = 0$, then $g = 0 \cdot u = 0$, so both are 0. If $f \neq 0$, then $f = f(uv)$, so $f(1 - uv) = 0$, and [Corollary 0.8.4](#) gives $uv = 1$. By [Theorem 0.8.3](#), $\deg u + \deg v = \deg 1 = 0$ with both degrees in $\mathbb{N}$, so u and v are non-zero constants. Hence $f = vg$ with $v \neq 0$, and f, g are associates. If both are monic, comparing leading coefficients in $f = vg$ gives $1 = v \cdot 1$, so $f = g$.
- If the remainder is 0, then $f = qg$, so $g \mid f$. Conversely, if $f = hg$, then $f = hg + 0$ with $\deg 0 = -\infty < \deg g$. This is a division with remainder, so by the uniqueness in [Theorem 5.1.1](#), the remainder of f on division by g is 0. This proves the

proposition. ■

Part (d) is the polynomial form of a move we will use again and again: **to show two monic polynomials are equal, show that each divides the other**. It also gives each non-zero polynomial a canonical representative. If $f \neq 0$ has leading coefficient a , then $a^{-1}f$ is monic and an associate of f , and it is the **only** monic associate: two monic associates divide each other, so they are equal by (d). For example, the monic associate of $3x^2 - 6$ in $\mathbb{Q}[x]$ is $x^2 - 2$. The greatest common divisor in the next section will be defined as a monic polynomial for exactly this reason.

5.1.4 The Remainder and Factor Theorems

Division by a polynomial of degree 1 leaves a remainder of degree less than 1, that is, a constant. Which constant? Chapter 2 already answered this. The Factor Theorem for a Linear Divisor (Lemma 2.7.1) showed, by an induction that was long division in disguise, that $p = (x - c)q + p(c)$ for some q . We now read that lemma through Theorem 5.1.1. What is new is that q and the constant are **unique**, so $p(c)$ is **the** remainder in the sense of division.

Theorem 5.1.3 — Remainder Theorem and Factor Theorem

Let F be a field, $f \in F[x]$ and $c \in F$.

- (a) The remainder of f on division by $x - c$ is the constant $f(c)$.
- (b) c is a root of f if and only if $x - c \mid f$.

Proof.

- (a) By Lemma 2.7.1, $f = (x - c)q + f(c)$ for some $q \in F[x]$. The constant $f(c)$ has degree 0 or $-\infty$, which is less than $\deg(x - c) = 1$. So this is a division with remainder, and by the uniqueness in Theorem 5.1.1, $f(c)$ is the remainder.
- (b) By Proposition 5.1.2 (e), $x - c \mid f$ if and only if the remainder is 0, which by (a) means $f(c) = 0$. This proves the theorem. ■

In other words, the Factor Theorem of Chapter 2 is really a statement about division. Division alone would also give it: dividing f by $x - c$ leaves a constant remainder r , and evaluating $f = (x - c)q + r$ at c gives $r = f(c)$. Since Chapter 2 already proved the lemma, we cite it rather than prove it again, and we will use the name “Remainder Theorem” for both parts.

Example. (Remainders Without Dividing)

- (a) Find the remainder of $f = x^5 - 3x^3 + 2x + 7$ on division by $x - 2$ in $\mathbb{Q}[x]$. (b) Show that $x + 1 \nmid x^2 + 1$ in $\mathbb{Q}[x]$, but $x + 1 \mid x^2 + 1$ in $\mathbb{F}_2[x]$. (c) Find the remainder of x^5 on division by $x^2 - x$ in $\mathbb{Q}[x]$, again without long division.

Solution.

- (a) By [Theorem 5.1.3](#) (a), the remainder is $f(2) = 32 - 24 + 4 + 7 = 19$. No long division is needed.
- (b) Here $x+1 = x-(-1)$. In $\mathbb{Q}$, $(-1)^2 + 1 = 2 \neq 0$, so -1 is not a root and $x+1 \nmid x^2+1$ by part (b) of the theorem; the remainder is 2. In $\mathbb{F}_2$, $-1 = 1$ and $1^2 + 1 = 0$, so $x+1 \mid x^2+1$. Indeed $x^2+1 = (x+1)^2$ in $\mathbb{F}_2[x]$. The same polynomial symbols, over two fields, give different answers.
- (c) The divisor has degree 2, so by [Theorem 5.1.1](#), $x^5 = q(x^2 - x) + r$ with $\deg r \leq 1$, say $r = ax + b$. The divisor $x^2 - x = x(x-1)$ vanishes at 0 and at 1, so evaluating both sides there ([Theorem 0.8.5](#)) kills the term $q(x^2 - x)$: at 0, $0 = b$; at 1, $1 = a + b$. Hence $a = 1$, $b = 0$, and the remainder is x . As a check, $x^5 - x = x(x^4 - 1) = (x^2 - x)(x^3 + x^2 + x + 1)$. The move is: **evaluate at the roots of the divisor**, one equation per root.

The Factor Theorem peels off one root at a time. Doing it for several distinct roots at once strengthens the Root bound of Chapter 2 ([Lemma 2.7.2](#)), which said that a non-zero polynomial of degree at most n has at most n distinct roots. The stronger version says **why** there are so few: every root contributes its own linear factor.

Corollary 5.1.4 — Distinct Roots Give a Product of Linear Factors

Let F be a field, $f \in F[x]$, and let $c_1, \dots, c_k \in F$ be **distinct** roots of f , with $k \geq 1$. Then

$$(x - c_1)(x - c_2) \cdots (x - c_k) \mid f.$$

Consequently, if $f \neq 0$, then $k \leq \deg f$: a non-zero polynomial of degree n has at most n distinct roots in F .

Idea. Peel off $x - c_1$ by the Factor Theorem. What remains, h , must still vanish at c_2 , because $f(c_2) = (c_2 - c_1)h(c_2)$ and $c_2 - c_1 \neq 0$. So peel off $x - c_2$ from h , and so on. Distinctness is what makes the factors $c_k - c_i$ non-zero.

Proof. We use induction on k . For $k = 1$, this is [Theorem 5.1.3](#) (b).

Let $k \geq 2$, and assume the statement for $k-1$ distinct roots. By the induction hypothesis, $f = (x - c_1) \cdots (x - c_{k-1}) h$ for some $h \in F[x]$. Evaluating at c_k with [Theorem 0.8.5](#),

$$0 = f(c_k) = (c_k - c_1) \cdots (c_k - c_{k-1}) h(c_k).$$

Since the c_i are distinct, each $c_k - c_i$ with $i < k$ is non-zero, and a product of non-zero elements of a field is non-zero ([Theorem 0.7.1](#)). Hence $h(c_k) = 0$, and [Theorem 5.1.3](#) (b) gives $h = (x - c_k) h_1$ with $h_1 \in F[x]$. Therefore $f = (x - c_1) \cdots (x - c_k) h_1$, which completes the induction.

For the count, suppose $f \neq 0$ and write $f = (x - c_1) \cdots (x - c_k) h_1$ as above. Then $h_1 \neq 0$, and by [Theorem 0.8.3](#), $\deg f = k + \deg h_1 \geq k$. This proves the corollary. ■

The count is [Lemma 2.7.2](#) again, now as a consequence of a divisibility statement. The divisibility

statement is the more useful one. For instance, it tells us at once that a polynomial vanishing at 1, 2 and 3 is a multiple of $(x-1)(x-2)(x-3)$.

5.1.5 Polynomials and polynomial functions

Chapter 0 warned that a polynomial is not the function it defines: over $\mathbb{F}_2$, the non-zero polynomial $x^2 + x$ has the zero function (Example (Different polynomials, same function)). It also promised that over an infinite field the distinction disappears. Chapter 2 kept that promise in a paragraph right after Lemma 2.7.2. We record the fact here as a theorem, in a slightly sharper form that counts how many values are needed.

Theorem 5.1.5 — Values Determine a Polynomial

Let F be a field and $p, q \in F[x]$.

- (a) Let $n \in \mathbb{N}$ with $\deg p \leq n$ and $\deg q \leq n$. If $p(c) = q(c)$ for at least $n+1$ **distinct** elements $c \in F$, then $p = q$.
- (b) If F is **infinite** and $p(c) = q(c)$ for **every** $c \in F$, then $p = q$.

Proof.

- (a) Let $h = p - q$. By Theorem 0.8.2, $\deg h \leq n$. By Theorem 0.8.5, $h(c) = p(c) - q(c) = 0$ for each of the given $n+1$ distinct elements, so h has at least $n+1$ distinct roots. By Lemma 2.7.2, a non-zero polynomial of degree at most n has at most n distinct roots, so h cannot be non-zero. Hence $h = 0$ and $p = q$.
- (b) Let $n = \max(\deg p, \deg q, 0) \in \mathbb{N}$. Since F is infinite, it contains $n+1$ distinct elements, and p and q agree at each of them. Part (a) gives $p = q$. This proves the theorem. ■

So over $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$, we may compare polynomials by comparing their functions, and we may read off coefficients from values. Part (a) holds over **every** field, including finite ones: a quadratic over $\mathbb{F}_5$ is determined by its values at any three distinct points.

Warning. Part (b) is false over every finite field. Over $\mathbb{F}_p$, the polynomial $x^p - x$ is non-zero but vanishes at every element (Exercise (C2: A non-zero polynomial vanishing everywhere)). So x^p and x define the same function $\mathbb{F}_p \rightarrow \mathbb{F}_p$ although they are different polynomials. There is no contradiction with (a): the degrees are at most p , and (a) would need agreement at $p+1$ distinct points, but $\mathbb{F}_p$ has only p elements.

Quick check. Over $\mathbb{F}_3$, do the polynomials $x^3 + 2x$ and 0 define the same function? Does this contradict Theorem 5.1.5?

Solution. Yes. The values of $x^3 + 2x$ at 0, 1, 2 are 0, $1 + 2 = 3 = 0$, and $8 + 4 = 12 = 0$. So the function is zero, although the polynomial is not. There is no contradiction: $\mathbb{F}_3$ is finite, so part (b) does not apply, and part (a) with $n = 3$ would need 4 distinct points. In fact $x^3 + 2x = x^3 - x$, the polynomial of the warning with $p = 3$.

Division with remainder is the one tool this chapter keeps using. In the next section it shows that every set of polynomials closed under the right operations consists of the multiples of a single polynomial, and that gives greatest common divisors.

5.1.6 Exercises

5.1.6.1 A. Check your understanding

Exercise A1.

- State the theorem on division with remainder in $F[x]$, including every hypothesis.
- Determine whether the following statement is true: “for every field F and all $f, g \in F[x]$, if $f \mid g$ and $g \mid f$, then $f = g$.” Justify your answer. Does your answer change if $F = \mathbb{F}_2$?
- Let $c \in F$ be non-zero. What are the quotient and the remainder of f on division by the constant polynomial c ?
- Name the method for finding the remainder of f on division by $x - c$ without dividing.
- Determine whether the following statement is true: “if F is a field and $p \in F[x]$ satisfies $p(c) = 0$ for every $c \in F$, then $p = 0$.” Justify your answer.

Solution.

- Let F be a field and $f, g \in F[x]$ with $g \neq 0$. Then there are unique $q, r \in F[x]$ with $f = qg + r$ and $\deg r < \deg g$ ([Theorem 5.1.1](#)).
- False in general. In $\mathbb{Q}[x]$, take $f = x$ and $g = 2x$. Then $g = 2f$ and $f = \frac{1}{2}g$, so each divides the other, but $f \neq g$. By [Proposition 5.1.2](#) (d), mutual divisibility only gives associates. Over $\mathbb{F}_2$, the statement is **true**: the only non-zero constant in $\mathbb{F}_2$ is 1, so associates are equal, and if $f \mid g$ and $g \mid f$, then either both are 0 or $f = 1 \cdot g$.
- The remainder must have degree less than $\deg c = 0$, so $r = 0$. Then $f = qc$, so $q = c^{-1}f$. In particular every polynomial is divisible by every non-zero constant.
- The Remainder Theorem ([Theorem 5.1.3](#)): the remainder is the constant $f(c)$.
- False. Over $\mathbb{F}_2$, $p = x^2 + x$ is non-zero but $p(0) = p(1) = 0$ ([Example \(Different polynomials, same function\)](#)). The statement is true when F is infinite, by [Theorem 5.1.5](#) (b).

5.1.6.2 B. Practice

Exercise B1: Dividing Over Two Fields.

- In $\mathbb{Q}[x]$, divide $f = x^4 - 3x^3 + 2x - 5$ by $g = 3x^2 - 1$ with remainder.
- In $\mathbb{F}_5[x]$, divide $f = x^4 + 2x^3 + 3x + 1$ by $g = 2x^2 + 3$ with remainder. Hence decide whether $g \mid f$ in $\mathbb{F}_5[x]$.

Hint for (b): first find the inverse of 2 in $\mathbb{F}_5$.

Solution.

- (a) The leading coefficient of g is 3, with inverse $\frac{1}{3}$. First, $x^4/3x^2 = \frac{1}{3}x^2$, and $f - \frac{1}{3}x^2g = -3x^3 + \frac{1}{3}x^2 + 2x - 5$. Next, $-3x^3/3x^2 = -x$, and subtracting $-xg = -3x^3 + x$ leaves $\frac{1}{3}x^2 + x - 5$. Next, $\frac{1}{3}x^2/3x^2 = \frac{1}{9}$, and subtracting $\frac{1}{9}g = \frac{1}{3}x^2 - \frac{1}{9}$ leaves $x - \frac{44}{9}$, of degree $1 < 2$. Hence

$$x^4 - 3x^3 + 2x - 5 = \left(\frac{1}{3}x^2 - x + \frac{1}{9}\right)(3x^2 - 1) + x - \frac{44}{9},$$

with quotient $\frac{1}{3}x^2 - x + \frac{1}{9}$ and remainder $x - \frac{44}{9}$, which are the only ones by [Theorem 5.1.1](#).

- (b) In $\mathbb{F}_5$, $2 \cdot 3 = 6 = 1$, so $2^{-1} = 3$. First, the leading term x^4 divided by $2x^2$ is $3x^2$, and $3x^2g = 6x^4 + 9x^2 = x^4 + 4x^2$. Subtracting leaves $2x^3 - 4x^2 + 3x + 1 = 2x^3 + x^2 + 3x + 1$. Next, $2x^3/2x^2 = x$, and subtracting $xg = 2x^3 + 3x$ leaves $x^2 + 1$. Next, $x^2/2x^2 = 3$, and subtracting $3g = 6x^2 + 9 = x^2 + 4$ leaves $1 - 4 = -3 = 2$. Hence

$$x^4 + 2x^3 + 3x + 1 = (3x^2 + x + 3)(2x^2 + 3) + 2 \quad \text{in } \mathbb{F}_5[x].$$

The remainder 2 is non-zero, so $g \nmid f$ in $\mathbb{F}_5[x]$, by [Proposition 5.1.2](#) (e).

Exercise B2: A Factor of $x^n - c^n$.

Let F be a field, $c \in F$ and $n \geq 1$. Prove that $x - c \mid x^n - c^n$ in $F[x]$, and verify that the quotient is $x^{n-1} + cx^{n-2} + \cdots + c^{n-2}x + c^{n-1}$.

Solution. The value of $f = x^n - c^n$ at c is $c^n - c^n = 0$, so c is a root of f , and [Theorem 5.1.3](#) (b) gives $x - c \mid f$. For the quotient, let $s = \sum_{j=0}^{n-1} c^j x^{n-1-j}$. Then

$$(x - c)s = \sum_{j=0}^{n-1} c^j x^{n-j} - \sum_{j=0}^{n-1} c^{j+1} x^{n-1-j} = \sum_{j=0}^{n-1} c^j x^{n-j} - \sum_{j=1}^n c^j x^{n-j} = x^n - c^n,$$

since the two sums share the terms with $1 \leq j \leq n-1$, leaving $c^0 x^n$ from the first and $c^n x^0$ from the second. This shows $x^n - c^n = (x - c)s$.

Exercise B3: A Remainder Without Long Division.

- (a) In $\mathbb{Q}[x]$, find the remainder of x^{100} on division by $x^2 - 1$ without long division.
 (b) Show that the answer to (a) is the remainder of x^{100} on division by $x^2 - 1$ over **every** field F , including $\mathbb{F}_2$. Explain why the method of (a) alone does not settle the case $F = \mathbb{F}_2$.

Hint for (a): part (c) of [Example \(Remainders Without Dividing\)](#). *Hint for (b):* imitate the identity of B2.

Solution.

- (a) By [Theorem 5.1.1](#), $x^{100} = q(x^2 - 1) + r$ with $\deg r \leq 1$, say $r = ax + b$. Evaluating at 1 and -1 , where $x^2 - 1$ vanishes ([Theorem 0.8.5](#)), gives $1 = a + b$ and $1 = (-1)^{100} = -a + b$. Adding and subtracting, $2b = 2$ and $2a = 0$, so $b = 1$ and $a = 0$. The remainder is 1.

- (b) Over any field, let $s = \sum_{j=0}^{49} x^{2j}$. Then

$$(x^2 - 1)s = \sum_{j=0}^{49} x^{2j+2} - \sum_{j=0}^{49} x^{2j} = x^{100} - 1,$$

because the terms $x^2, x^4, \dots, x^{98}$ occur in both sums. Hence $x^{100} = s \cdot (x^2 - 1) + 1$, and $\deg 1 = 0 < 2$, so by the uniqueness in [Theorem 5.1.1](#) the remainder is 1 over every field. The method of (a) divides by 2, and over $\mathbb{F}_2$ we have $2 = 0$; also $1 = -1$ there, so evaluating at ± 1 gives only the single equation $a + b = 1$, which does not determine a and b .

5.1.6.3 C. Going deeper

Exercise C1: Division by a Monic Polynomial Over $\mathbb{Z}$.

As in the warning after [Theorem 5.1.1](#), $\mathbb{Z}[x]$ denotes the set of polynomials in $\mathbb{Q}[x]$ all of whose coefficients are integers. It is closed under addition and multiplication.

- (a) Let $f, g \in \mathbb{Z}[x]$ with g **monic**. Prove that there exist $q, r \in \mathbb{Z}[x]$ with $f = qg + r$ and $\deg r < \deg g$.
- (b) Prove that q and r in (a) are unique, even among $q, r \in \mathbb{Q}[x]$.
- (c) Does (a) remain true if the leading coefficient of g is -1 instead of 1? Justify your answer.

Hint for (a): find the one step in the proof of [Theorem 5.1.1](#) that leaves $\mathbb{Z}[x]$.

Solution.

- (a) We repeat the existence proof of [Theorem 5.1.1](#), checking that every polynomial stays in $\mathbb{Z}[x]$. Let $m = \deg g$. The only step that could leave $\mathbb{Z}[x]$ is forming $ab^{-1}x^{n-m}g$, where b is the leading coefficient of g . Here $b = 1$, so $ab^{-1} = a \in \mathbb{Z}$ and $f_1 = f - ax^{n-m}g \in \mathbb{Z}[x]$. The strong induction on $\deg f$ ([Theorem 0.2.10](#)) then runs word for word inside $\mathbb{Z}[x]$: in Case 1, $q = 0$ and $r = f$ are in $\mathbb{Z}[x]$; in Case 2, $q = ax^{n-m} + q_1$ and $r = r_1$ are in $\mathbb{Z}[x]$ because q_1, r_1 are, by the induction hypothesis applied to $f_1 \in \mathbb{Z}[x]$. This proves existence in $\mathbb{Z}[x]$.
- (b) Suppose $f = qg + r = q'g + r'$ with $q, q', r, r' \in \mathbb{Q}[x]$ and $\deg r, \deg r' < \deg g$. Since $\mathbb{Q}$ is a field and $g \neq 0$, the uniqueness part of [Theorem 5.1.1](#) in $\mathbb{Q}[x]$ gives $q = q'$ and $r = r'$. In particular the pair found in (a) is the unique one.
- (c) Yes. If the leading coefficient is $b = -1$, then $b^{-1} = -1 \in \mathbb{Z}$, so $ab^{-1} = -a \in \mathbb{Z}$, and the argument of (a) goes through unchanged. What matters is that the leading coefficient has an inverse in $\mathbb{Z}$, which is true exactly for ± 1 .

Exercise C2: Remainders on Division by a Quadratic.

Let F be a field, $f \in F[x]$ and $a, b \in F$.

- (a) Suppose $a \neq b$. Prove that the remainder of f on division by $(x - a)(x - b)$ is

$$r = f(a) \frac{x - b}{a - b} + f(b) \frac{x - a}{b - a},$$

where $\frac{x-b}{a-b}$ means $(a-b)^{-1}(x-b)$.

- (b) Now let $a = b$. Write $f = (x-a)h + f(a)$ by [Theorem 5.1.3](#). Prove that the remainder of f on division by $(x-a)^2$ is $h(a)(x-a) + f(a)$.
- (c) Hence find the remainder of x^{100} on division by $x^2 - 3x + 2$ in $\mathbb{Q}[x]$.

Solution.

- (a) Since $a \neq b$, $a-b$ is invertible, so r is a well-defined polynomial of degree at most 1. By [Theorem 5.1.1](#), $f = q(x-a)(x-b) + s$ with $\deg s \leq 1$. Evaluating at a and at b ([Theorem 0.8.5](#)) gives $s(a) = f(a)$ and $s(b) = f(b)$. Also $r(a) = f(a) \cdot 1 + f(b) \cdot 0 = f(a)$ and $r(b) = f(b)$. So s and r have degree at most 1 and agree at the 2 distinct points a, b . By [Theorem 5.1.5](#) (a) with $n = 1$, $s = r$. This proves the formula.
- (b) By [Theorem 5.1.3](#) (a) applied to h , $h = (x-a)k + h(a)$ for some $k \in F[x]$. Substituting,

$$f = (x-a)((x-a)k + h(a)) + f(a) = k(x-a)^2 + (h(a)(x-a) + f(a)).$$

The last bracket has degree at most $1 < 2 = \deg(x-a)^2$, so by the uniqueness in [Theorem 5.1.1](#) it is the remainder.

- (c) Here $x^2 - 3x + 2 = (x-1)(x-2)$, with $a = 1 \neq b = 2$, $f(1) = 1$ and $f(2) = 2^{100}$. By (a),

$$r = 1 \cdot \frac{x-2}{1-2} + 2^{100} \cdot \frac{x-1}{2-1} = (2^{100} - 1)x + (2 - 2^{100}).$$

As a check, $r(1) = 1$ and $r(2) = 2^{101} - 2 + 2 - 2^{100} = 2^{100}$.

Exercise C3: Equal Functions Over $\mathbb{F}_p$.

Let p be a prime and $f, g \in \mathbb{F}_p[x]$.

- (a) Prove that $f(c) = g(c)$ for every $c \in \mathbb{F}_p$ if and only if $x^p - x \mid f - g$ in $\mathbb{F}_p[x]$.
- (b) Deduce that f and its remainder on division by $x^p - x$ define the same function $\mathbb{F}_p \rightarrow \mathbb{F}_p$. Use this to find a polynomial of degree less than 5 that defines the same function on $\mathbb{F}_5$ as $x^7 + x$.

Hint for (a): [Exercise \(C2: A non-zero polynomial vanishing everywhere\)](#) and [Corollary 5.1.4](#).

Solution.

- (a) ($\Leftarrow$) Suppose $f - g = (x^p - x)h$. By [Exercise \(C2: A non-zero polynomial vanishing everywhere\)](#), $c^p - c = 0$ for every $c \in \mathbb{F}_p$. By [Theorem 0.8.5](#), $f(c) - g(c) = (c^p - c)h(c) = 0$ for every c . ($\Rightarrow$) Suppose $f(c) = g(c)$ for every $c \in \mathbb{F}_p$. Then the p distinct elements $0, 1, \dots, p-1$ of $\mathbb{F}_p$ are roots of $f - g$. Let $P = \prod_{c \in \mathbb{F}_p} (x - c)$. By [Corollary 5.1.4](#), $P \mid f - g$. It remains to show $P = x^p - x$. By [Exercise \(C2: A non-zero polynomial vanishing everywhere\)](#), every $c \in \mathbb{F}_p$ is a root of $x^p - x$, so [Corollary 5.1.4](#) gives $x^p - x = Pu$ for some u . Both P and $x^p - x$ are monic of degree p ([Theorem 0.8.3](#)), so $\deg u = 0$, and comparing leading coefficients gives $u = 1$. Hence $P = x^p - x$, and $x^p - x \mid f - g$.
- (b) Write $f = q(x^p - x) + r$ by [Theorem 5.1.1](#). Then $x^p - x \mid f - r$, so f and r define

the same function by (a). For $f = x^7 + x$ over $\mathbb{F}_5$: $x^7 = x^2 \cdot x^5 = x^2(x^5 - x) + x^3$, so $x^7 + x = x^2(x^5 - x) + (x^3 + x)$ with $\deg(x^3 + x) = 3 < 5$. Hence $x^3 + x$ defines the same function as $x^7 + x$ on $\mathbb{F}_5$.

5.2 Ideals and Greatest Common Divisors

Division with remainder is now available over every field. This section uses it to answer a question about two polynomials f and g : what do all the combinations $af + bg$, with **polynomial** coefficients a and b , look like? The answer is that they are exactly the multiples of one polynomial, and that polynomial is the greatest common divisor of f and g . Along the way we meet ideals, the subsets of $F[x]$ adapted to multiplication, and we prove that each of them is generated by a single polynomial. The Euclidean algorithm then computes the greatest common divisor and the coefficients a, b by repeated division. Throughout, F is a field.

5.2.1 All the combinations of two polynomials

Take $f = x^2 - 1$ and $g = x^2 + x - 2$ in $\mathbb{Q}[x]$, and look at the set

$$\{af + bg : a, b \in \mathbb{Q}[x]\}$$

of all their combinations with polynomial coefficients. It contains f (take $a = 1, b = 0$), g , and $g - f = x - 1$.

Since $f = (x - 1)(x + 1)$ and $g = (x - 1)(x + 2)$, every combination is

$$af + bg = (x - 1)(a(x + 1) + b(x + 2)),$$

a multiple of $x - 1$. Conversely, every multiple $h(x - 1)$ is a combination, namely $(-h)f + hg$. So the set of combinations is **exactly** the set of multiples of $x - 1$. Its non-zero members of smallest degree are the non-zero scalar multiples of $x - 1$. By [Proposition 5.1.2](#) (c), no non-zero constant is a multiple of $x - 1$, so there is nothing of degree 0.

Two features of this set will carry the whole section. It is closed under addition, and it absorbs multiplication by **any** polynomial: if $af + bg$ is in the set, so is $h(af + bg) = (ha)f + (hb)g$. Sets with these two features appear everywhere, so they get a name.

5.2.2 Ideals

In Chapter 1 we studied $F[x]$ as a vector space and its subspaces: subsets closed under addition and under multiplication by **scalars**. But $F[x]$ also has a multiplication of its own. The subsets that respect it are closed under multiplication by **polynomials**. Three recurring examples share this feature:

- the multiples of a fixed polynomial p ;
- the combinations $af + bg$ of two fixed polynomials;
- the polynomials that vanish at a fixed point c : if $f(c) = 0$, then $(hf)(c) = h(c) \cdot 0 = 0$ for every h .

An ideal is a set of polynomials that is closed under addition and swallows multiplication by anything.

Definition 5.2.1 — Ideal of $F[x]$

Let F be a field. A subset $I \subseteq F[x]$ is an **ideal** of $F[x]$ if

- (I1) I is **non-empty**;
- (I2) $f + g \in I$ for all $f, g \in I$;
- (I3) $hf \in I$ for all $f \in I$ and every $h \in F[x]$.

For $f_1, \dots, f_k \in F[x]$, write

$$\langle f_1, \dots, f_k \rangle := \{a_1 f_1 + \dots + a_k f_k : a_1, \dots, a_k \in F[x]\}.$$

In particular $\langle p \rangle = \{hp : h \in F[x]\}$ is the set of all multiples of p .

In words: (I1) rules out the empty set. (I2) says sums stay inside. (I3) is the new clause: a member of I multiplied by **any** polynomial, not just by a constant, is still a member. Taking $h = 0$ in (I3) shows $0 \in I$, and taking $h = -1$ shows $-f \in I$. Taking $h = c$ constant shows I is closed under scalar multiplication, so **every ideal is a subspace** of $F[x]$ by Theorem 1.2.1.

The angle brackets here have nothing to do with the inner products of later chapters: $\langle f, g \rangle$ is a set of polynomials, not a number.

Well-definedness. We have named the set $\langle f_1, \dots, f_k \rangle$, and must check that it is an ideal. It is in fact the smallest ideal containing $f_1, \dots, f_k$.

Proposition 5.2.1 — The Ideal Generated by Finitely Many Polynomials

Let F be a field and $f_1, \dots, f_k \in F[x]$, with $k \geq 1$. Then $\langle f_1, \dots, f_k \rangle$ is an ideal of $F[x]$ containing $f_1, \dots, f_k$. Moreover, every ideal J of $F[x]$ with $f_1, \dots, f_k \in J$ satisfies $\langle f_1, \dots, f_k \rangle \subseteq J$.

Proof. Let $I = \langle f_1, \dots, f_k \rangle$. Taking $a_i = 1$ and $a_j = 0$ for $j \neq i$ shows $f_i \in I$; in particular (I1) holds. For (I2), $\sum a_i f_i + \sum b_i f_i = \sum (a_i + b_i) f_i \in I$. For (I3), $h \sum a_i f_i = \sum (ha_i) f_i \in I$, using Theorem 0.8.1 in both cases. So I is an ideal.

Now let J be an ideal containing every f_i . By (I3) for J , each $a_i f_i \in J$, and by (I2) applied $k - 1$ times, $a_1 f_1 + \dots + a_k f_k \in J$. Hence $I \subseteq J$. This proves the proposition. ■

The ideal $\langle f_1, \dots, f_k \rangle$ is the **ideal generated by** $f_1, \dots, f_k$. An ideal of the form $\langle p \rangle$, generated by one polynomial, is called **principal**.

Example. (First Ideals)

Check each of the following against Definition 5.2.1.

- (a) $\{0\}$ and $F[x]$; and any ideal containing a non-zero constant.
- (b) $I_c = \{f \in F[x] : f(c) = 0\}$ for a fixed $c \in F$.
- (c) For $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in M_2(F)$, the set $\{p \in F[x] : p(A) = 0\}$.

Solution.

- (a) $\{0\} = \langle 0 \rangle$ and $F[x] = \langle 1 \rangle$ are ideals by [Proposition 5.2.1](#). They are the degenerate cases, and the second one is the one to recognize: if an ideal I contains a non-zero constant c , then $1 = c^{-1}c \in I$ by (I3), and then $h = h \cdot 1 \in I$ for every h , so $I = F[x]$. An ideal that contains a non-zero constant is everything.
- (b) $0 \in I_c$. If $f(c) = g(c) = 0$, then $(f+g)(c) = 0$ and $(hf)(c) = h(c)f(c) = 0$ for every $h \in F[x]$, by [Theorem 0.8.5](#). So I_c is an ideal. By [Theorem 5.1.3](#) (b), $f(c) = 0$ exactly when $x - c \mid f$, so $I_c = \langle x - c \rangle$.
- (c) Call the set I . The zero polynomial is in I . If $p(A) = q(A) = 0$, then by [Theorem 0.9.9](#), $(p+q)(A) = p(A) + q(A) = 0$ and $(hp)(A) = h(A)p(A) = h(A) \cdot 0 = 0$ for every h . So I is an ideal. It contains x^2 , since $A^2 = 0$. It contains no non-zero polynomial of degree at most 1: $(a+bx)(A) = \begin{pmatrix} a & b \\ 0 & a \end{pmatrix} = 0$ forces $a = b = 0$. The same argument works for any linear operator T in place of A , with [Theorem 3.4.3](#). In Chapter 8 the ideal of all polynomials killing an operator is where the minimal polynomial comes from.

Now the non-example, by a minimal change. Take $F[x]_{\leq n}$ for some $n \in \mathbb{N}$. It is non-empty, closed under addition, and closed under multiplication by **constants**: it is a subspace of $F[x]$ ([Example \(Polynomials of Bounded Degree\)](#)). But it fails (I3): $x^n \in F[x]_{\leq n}$ and $x \in F[x]$, while $x \cdot x^n = x^{n+1} \notin F[x]_{\leq n}$. The exact clause that fails is “for **every** $h \in F[x]$ ”.

Why insist on every h , and not just constants? Because the main theorem below divides a member f of I by another member d and needs the remainder $f - qd$ to lie in I again. The quotient q is an arbitrary polynomial, so we need $qd \in I$. With only closure under scalars, the argument breaks, and indeed $F[x]_{\leq n}$ is not the set of multiples of any single polynomial. The name “ideal” is historical and says nothing about the definition.

Warning. Do not confuse $\langle f, g \rangle$ with $\text{span}(f, g)$. The span uses **scalar** coefficients, the ideal uses **polynomial** coefficients. For example, $\text{span}(x) = \{cx : c \in F\}$ is a line in $F[x]$, while $\langle x \rangle$ contains $x^2, x^3, \dots$ and is infinite-dimensional. Every ideal is a subspace, but most subspaces, such as $F[x]_{\leq n}$ or $\text{span}(x)$, are not ideals.

Quick check. Is $J = \{f \in \mathbb{R}[x] : f(1) = 0 \text{ or } f(2) = 0\}$ an ideal of $\mathbb{R}[x]$?

Solution. No. J is non-empty and satisfies (I3), since $(hf)(c) = h(c)f(c)$. But it fails (I2): $x - 1 \in J$ and $x - 2 \in J$, while their sum $2x - 3$ takes the values -1 at 1 and 1 at 2 , so $2x - 3 \notin J$. The set with “and” in place of “or” is an ideal.

5.2.3 Every ideal is generated by one polynomial

The payoff of the definition is that ideals of $F[x]$ are very simple: each one is the set of multiples of a single polynomial. The proof is the argument of the opening example, made general. We have met it before for integers, in [Lemma 0.7.2](#), where the smallest positive combination $ax + ny$ turned out to divide a and n .

Theorem 5.2.2 — Every Ideal of $F[x]$ Is Principal

Let F be a field and I an ideal of $F[x]$. If $I \neq \{0\}$, there is **exactly one monic** polynomial d with $I = \langle d \rangle$. Moreover, d is the unique monic polynomial of least degree in I .

Idea. Pick a non-zero $d \in I$ of smallest degree, made monic. Every multiple of d is in I by (I3). For the other inclusion, divide any $f \in I$ by d : the remainder $r = f - qd$ lies in I and has degree smaller than $\deg d$. Minimality leaves only one option, $r = 0$. Uniqueness is mutual divisibility of monic polynomials.

Proof. Suppose $I \neq \{0\}$. The degrees of the non-zero elements of I form a non-empty set of natural numbers, so by [Theorem 0.2.11](#) it has a least element m . Choose $g \in I$ with $\deg g = m$, and let c be its leading coefficient. Put $d = c^{-1}g$. Then $d \in I$ by (I3), and d is monic of degree m .

($\supseteq$) For every $h \in F[x]$, $hd \in I$ by (I3). Hence $\langle d \rangle \subseteq I$.

($\subseteq$) Let $f \in I$. By [Theorem 5.1.1](#), $f = qd + r$ with $\deg r < m$. By (I3), $-qd \in I$, and by (I2), $r = f + (-qd) \in I$. If $r \neq 0$, then r is a non-zero element of I of degree less than m , contradicting the choice of m . Hence $r = 0$, so $f = qd \in \langle d \rangle$.

Therefore $I = \langle d \rangle$. For uniqueness, let d' be monic with $I = \langle d' \rangle$. Then $d \in \langle d' \rangle$ and $d' \in \langle d \rangle$, so $d' \mid d$ and $d \mid d'$. Both are monic, so $d = d'$ by [Proposition 5.1.2](#) (d).

Finally, let $e \in I$ be monic of least degree m . Then $d \mid e$, so $e = ud$ with $\deg u = 0$ by [Theorem 0.8.3](#), and comparing leading coefficients gives $u = 1$. Hence $e = d$. This proves the theorem. ■

For $I = \{0\}$ the only generator is 0, since $\langle p \rangle$ contains p . So every ideal of $F[x]$ is principal, with a canonical generator: 0 for the zero ideal, and the monic polynomial of least degree otherwise. The named move is: **the smallest-degree element of an ideal generates it**. In [Example \(First Ideals\)](#) (c), the ideal contains x^2 and nothing non-zero of smaller degree, so it equals $\langle x^2 \rangle$.

The theorem also turns divisibility into containment. For $p, q \in F[x]$,

$$\langle p \rangle \subseteq \langle q \rangle \iff q \mid p,$$

because $\langle p \rangle \subseteq \langle q \rangle$ holds exactly when $p \in \langle q \rangle$, by [Proposition 5.2.1](#). The bigger the ideal, the smaller the generator.

Quick check. Find the monic generator of $\langle x^2 - 1, x^3 - 1 \rangle \subseteq \mathbb{Q}[x]$ by finding an element of small degree in it.

Solution. The ideal contains $(x^3 - 1) - x(x^2 - 1) = x - 1$. Both generators are multiples of $x - 1$, since $x^2 - 1 = (x - 1)(x + 1)$ and $x^3 - 1 = (x - 1)(x^2 + x + 1)$, so every combination is a multiple of $x - 1$ by Proposition 5.1.2 (b). A non-zero constant is not a multiple of $x - 1$, so $x - 1$ has least degree in the ideal. By Theorem 5.2.2, the ideal is $\langle x - 1 \rangle$.

5.2.4 The greatest common divisor

Apply Theorem 5.2.2 to the ideal $\langle f, g \rangle$ of all combinations. It has a canonical generator, and in the opening example that generator, $x - 1$, was a common divisor of f and g . A naive attempt to define “the greatest common divisor” as “a common divisor of largest degree” runs into a problem at once: $x - 1$, $2x - 2$ and $-\frac{1}{3}x + \frac{1}{3}$ are all common divisors of $x^2 - 1$ and $x^2 + x - 2$ of the largest possible degree. Degree cannot tell associates apart, and it also does not say how the common divisors relate to each other. The generator of the ideal fixes both problems.

The gcd of f and g is the monic polynomial whose multiples are exactly the combinations of f and g .

Definition 5.2.2 — Greatest Common Divisor

Let F be a field and $f, g \in F[x]$. If f and g are **not both zero**, the **greatest common divisor** $\gcd(f, g)$ is the **unique monic** polynomial $d \in F[x]$ with

$$\langle f, g \rangle = \langle d \rangle.$$

We set $\gcd(0, 0) = 0$.

In words: form the ideal of all combinations $af + bg$ with $a, b \in F[x]$; it is generated by one polynomial; take the monic one. The definition refers to $F[x]$ twice, in the coefficients a, b and in the generator d , so at first sight the gcd depends on the field. We show below that it does not change when the field is enlarged.

Well-definedness. By Proposition 5.2.1, $\langle f, g \rangle$ is an ideal. If f and g are not both zero, it contains a non-zero element, so by Theorem 5.2.2 there is exactly one monic d with $\langle f, g \rangle = \langle d \rangle$. The convention $\gcd(0, 0) = 0$ keeps the equation $\langle f, g \rangle = \langle \gcd(f, g) \rangle$ true in the remaining case. The definition is symmetric: $\gcd(f, g) = \gcd(g, f)$.

Example. (First Greatest Common Divisors)

- (a) Let $f \neq 0$ have leading coefficient a . Find $\gcd(f, 0)$ and $\gcd(f, c)$ for a non-zero constant c .
- (b) Find $\gcd(x^2 - 1, x^2 + x - 2)$ in $\mathbb{Q}[x]$.
- (c) Find $\gcd(x^2 + 1, x + 1)$ in $\mathbb{Q}[x]$ and in $\mathbb{F}_2[x]$.

Solution.

- (a) $\langle f, 0 \rangle = \{af : a \in F[x]\} = \langle f \rangle = \langle a^{-1}f \rangle$, and $a^{-1}f$ is monic, so $\gcd(f, 0) = a^{-1}f$, the monic associate of f . This degenerate case shows why the gcd must be made

monic: f itself generates the ideal too, but is not the canonical choice. Next, $\langle f, c \rangle$ contains the non-zero constant c , so it is $F[x] = \langle 1 \rangle$ by [Example \(First Ideals\)](#) (a). Hence $\gcd(f, c) = 1$.

- (b) The opening example showed $\langle x^2 - 1, x^2 + x - 2 \rangle = \langle x - 1 \rangle$, and $x - 1$ is monic. So the gcd is $x - 1$.
- (c) In $\mathbb{Q}[x]$: dividing, $x^2 + 1 = (x - 1)(x + 1) + 2$. So $2 = (x^2 + 1) - (x - 1)(x + 1)$ lies in the ideal, which is therefore $\mathbb{Q}[x]$ by [Example \(First Ideals\)](#) (a). Hence $\gcd = 1$. In $\mathbb{F}_2[x]$: $x^2 + 1 = (x + 1)^2$, so $x^2 + 1 \in \langle x + 1 \rangle$, and $\langle x^2 + 1, x + 1 \rangle = \langle x + 1 \rangle$ by [Proposition 5.2.1](#). Hence $\gcd = x + 1$. The symbols are the same, but $\mathbb{F}_2$ is not a field containing $\mathbb{Q}$, so the two answers need not agree.

For a non-example by minimal change, keep (b) but replace $x - 1$ by the associate $3x - 3$. It is still a common divisor of largest degree and still generates the ideal, but it is **not** $\gcd(x^2 - 1, x^2 + x - 2)$: the clause “monic” fails. Or keep “monic” but lower the degree: 1 is a monic common divisor, and it does not generate the ideal, since $1 \notin \langle x - 1 \rangle$.

Why define the gcd through the ideal, rather than as “the largest common divisor”? Because the ideal definition delivers three things at once: existence and uniqueness (from [Theorem 5.2.2](#)), the fact that the gcd is a combination of f and g , and the fact that it is “greatest” in the strongest sense, namely that every common divisor **divides** it. Here is the last point.

Theorem 5.2.3 — Characterization of the gcd

Let F be a field, $f, g \in F[x]$, and $d = \gcd(f, g)$.

- (a) $d \mid f$ and $d \mid g$.
- (b) If $h \in F[x]$ satisfies $h \mid f$ and $h \mid g$, then $h \mid d$.
- (c) If f, g are not both zero, then d is the **only** monic polynomial with properties (a) and (b).

Proof.

- (a) $f = 1 \cdot f + 0 \cdot g \in \langle f, g \rangle = \langle d \rangle$, so $d \mid f$. Similarly $d \mid g$.
- (b) Since $d \in \langle d \rangle = \langle f, g \rangle$, there are $a, b \in F[x]$ with $d = af + bg$. If $h \mid f$ and $h \mid g$, then $h \mid af + bg = d$ by [Proposition 5.1.2](#) (b).
- (c) Let e be monic with properties (a) and (b). Since d is a common divisor of f and g , property (b) for e gives $d \mid e$. Since e is a common divisor, property (b) for d gives $e \mid d$. Both are monic (e by assumption, and d by [Definition 5.2.2](#), since f, g are not both zero), so $d = e$ by [Proposition 5.1.2](#) (d). This proves the theorem. ■

So $\gcd(f, g)$ is a common divisor that every common divisor divides. If f, g are not both zero, then $d \neq 0$, and by [Proposition 5.1.2](#) (c) every common divisor has degree at most $\deg d$; “greatest” is true for degrees as well. The proof of (b) used one fact that deserves its own statement, because it is the most useful property of the gcd.

Corollary 5.2.4 — Bézout's Identity for Polynomials

Let F be a field and $f, g \in F[x]$. Then there exist $a, b \in F[x]$ with

$$\gcd(f, g) = af + bg.$$

Proof. By [Definition 5.2.2](#), $\gcd(f, g) \in \langle \gcd(f, g) \rangle = \langle f, g \rangle$, and the elements of $\langle f, g \rangle$ are exactly the combinations $af + bg$. ■

The coefficients are not unique: if $d = af + bg$, then also $d = (a + g)f + (b - f)g$. Bézout's identity is the polynomial version of [Lemma 0.7.2](#), and it will be the tool behind the Chinese Remainder Theorem and the splitting of kernels later in this chapter.

5.2.5 The Euclidean algorithm

The definition tells us what the gcd is, but not how to find it. Factoring f and g is usually hard. Division is easy, and one observation turns division into an algorithm: dividing does not change the ideal.

Lemma 5.2.5 — Dividing Does Not Change the Ideal

Let F be a field and $f, g, q, r \in F[x]$ with $f = qg + r$. Then $\langle f, g \rangle = \langle g, r \rangle$. Consequently $\gcd(f, g) = \gcd(g, r)$.

Proof. ($\supseteq$) The ideal $\langle f, g \rangle$ contains g , and $r = f + (-q)g$ by (I2) and (I3). By [Proposition 5.2.1](#), $\langle g, r \rangle \subseteq \langle f, g \rangle$.

($\subseteq$) The ideal $\langle g, r \rangle$ contains g and $f = qg + r$. By [Proposition 5.2.1](#), $\langle f, g \rangle \subseteq \langle g, r \rangle$.

Hence the two ideals are equal. The gcd is the canonical generator of the ideal ([Definition 5.2.2](#)), and f, g are both zero exactly when g, r are, so $\gcd(f, g) = \gcd(g, r)$. ■

Now divide repeatedly. Each division replaces the pair (f, g) by (g, r) , a pair with the same gcd and a smaller second entry.

Theorem 5.2.6 — Euclidean Algorithm

Let F be a field and $f, g \in F[x]$ with $g \neq 0$. Put $r_{-1} = f$ and $r_0 = g$. As long as $r_k \neq 0$, divide r_{k-1} by r_k ([Theorem 5.1.1](#)):

$$r_{k-1} = q_{k+1}r_k + r_{k+1}, \quad \deg r_{k+1} < \deg r_k.$$

Then:

- (a) the process stops: there is an index n with $0 \leq n \leq \deg g$ such that $r_n \neq 0$ and $r_{n+1} = 0$;
- (b) if c is the leading coefficient of this last non-zero remainder r_n , then $\gcd(f, g) = c^{-1}r_n$.

Idea. Two things to see. The degrees $\deg r_0 > \deg r_1 > \deg r_2 > \dots$ strictly decrease, and natural numbers cannot decrease forever, so some remainder is 0. By [Lemma 5.2.5](#), the ideal $\langle r_{k-1}, r_k \rangle$ never changes along the way; at the end it is $\langle r_n, 0 \rangle = \langle r_n \rangle$, and making r_n monic gives the gcd.

Proof.

- (a) As long as $r_1, \dots, r_j$ are all non-zero, their degrees are natural numbers with $\deg g = \deg r_0 > \deg r_1 > \dots > \deg r_j \geq 0$. So $r_1, \dots, r_j$ are j distinct values in $\{0, 1, \dots, \deg g - 1\}$, and $j \leq \deg g$. Hence the remainders cannot all be non-zero for $j = \deg g + 1$: there is a first index $n + 1$ with $r_{n+1} = 0$, and $n \leq \deg g$. Also $n \geq 0$ since $r_0 = g \neq 0$.
- (b) We show by induction on k that $\langle r_{k-1}, r_k \rangle = \langle f, g \rangle$ for $k = 0, 1, \dots, n+1$. For $k = 0$ this is the definition of r_{-1}, r_0 . If it holds for some $k \leq n$, then $r_{k-1} = q_{k+1}r_k + r_{k+1}$, and [Lemma 5.2.5](#) gives $\langle r_k, r_{k+1} \rangle = \langle r_{k-1}, r_k \rangle = \langle f, g \rangle$.

For $k = n + 1$ this reads $\langle f, g \rangle = \langle r_n, 0 \rangle = \langle r_n \rangle$. Since $c \neq 0$, $\langle r_n \rangle = \langle c^{-1}r_n \rangle$, and $c^{-1}r_n$ is monic. Because $g \neq 0$, [Definition 5.2.2](#) gives $\gcd(f, g) = c^{-1}r_n$. This proves the theorem. ■

If $g = 0$, no division is needed: $\gcd(f, 0)$ is the monic associate of f (or 0), as in [Example \(First Greatest Common Divisors\)](#) (a).

The algorithm also produces Bézout coefficients. Each remainder is a combination of the previous two, $r_{k+1} = r_{k-1} - q_{k+1}r_k$, so by induction each remainder is a combination of f and g . Keeping track of the coefficients gives the following procedure.

Extended Euclidean algorithm. Input: $f, g \in F[x]$ with $g \neq 0$.

1. Set $(r_{-1}, a_{-1}, b_{-1}) = (f, 1, 0)$ and $(r_0, a_0, b_0) = (g, 0, 1)$.
2. For $k = 0, 1, 2, \dots$, as long as $r_k \neq 0$: divide $r_{k-1} = q_{k+1}r_k + r_{k+1}$, and set

$$a_{k+1} = a_{k-1} - q_{k+1}a_k, \quad b_{k+1} = b_{k-1} - q_{k+1}b_k.$$

3. When $r_{n+1} = 0$, let c be the leading coefficient of r_n . Output $\gcd(f, g) = c^{-1}r_n$ and the coefficients $a = c^{-1}a_n$, $b = c^{-1}b_n$.

The invariant $r_k = a_k f + b_k g$ holds for $k = -1, 0$ by the choice in step 1. If it holds for $k - 1$ and k , then

$$r_{k+1} = r_{k-1} - q_{k+1}r_k = (a_{k-1} - q_{k+1}a_k)f + (b_{k-1} - q_{k+1}b_k)g = a_{k+1}f + b_{k+1}g.$$

So at the end $\gcd(f, g) = c^{-1}r_n = af + bg$. By hand, it is often quicker to write each division as “remainder = ...” and substitute backwards, which is the same computation.

Example. (The Euclidean Algorithm with Back-Substitution)

In $\mathbb{Q}[x]$, let $f = x^4 - x^3 - 3x^2 + x + 2$ and $g = x^3 - 2x - 1$. Find $\gcd(f, g)$ and polynomials a, b with $\gcd(f, g) = af + bg$.

Solution. Divisions. First divide f by g . The leading terms give x , and $f - xg = -x^3 - x^2 + 2x + 2$; then -1 , and adding g leaves $-x^2 + 1$. So

$$f = (x - 1)g + r_1, \quad r_1 = 1 - x^2.$$

Next divide g by r_1 . The leading terms give $x^3/(-x^2) = -x$, and $g - (-x)r_1 = x^3 - 2x - 1 + x - x^3 = -x - 1$. So

$$g = (-x)r_1 + r_2, \quad r_2 = -x - 1.$$

Next divide r_1 by r_2 . Since $(x - 1)(-x - 1) = -x^2 + 1$,

$$r_1 = (x - 1)r_2 + 0.$$

The last non-zero remainder is $r_2 = -x - 1$, with leading coefficient -1 . By [Theorem 5.2.6](#), $\gcd(f, g) = x + 1$.

Back-substitution. From the first two divisions,

$$r_1 = f - (x - 1)g, \quad r_2 = g + xr_1.$$

Substituting the first into the second,

$$r_2 = g + x(f - (x - 1)g) = xf + (1 + x - x^2)g.$$

Multiplying by $c^{-1} = -1$,

$$x + 1 = (-x)f + (x^2 - x - 1)g.$$

In the notation of the algorithm, $(a_1, b_1) = (1, 1 - x)$ and $(a_2, b_2) = (x, 1 + x - x^2)$.

Checks. Expanding, $-x(x^4 - x^3 - 3x^2 + x + 2) = -x^5 + x^4 + 3x^3 - x^2 - 2x$ and $(x^2 - x - 1)(x^3 - 2x - 1) = x^5 - x^4 - 3x^3 + x^2 + 3x + 1$, which sum to $x + 1$. As an independent check that $x + 1$ divides both, $f(-1) = 1 + 1 - 3 - 1 + 2 = 0$ and $g(-1) = -1 + 2 - 1 = 0$, so [Theorem 5.1.3](#) applies. SymPy's `gcdex` returns the same a , b and $\gcd$.

5.2.6 Coprime polynomials

The most important value a $\gcd$ can take is 1. Then f and g share no non-constant factor, and Bézout's identity writes 1 itself as a combination.

Definition 5.2.3 — Coprime Polynomials

Let F be a field. Polynomials $f, g \in F[x]$ are **coprime** (or **relatively prime**) over F if $\gcd(f, g) = 1$.

Proposition 5.2.7 — Coprime Means 1 Is a Combination

Let F be a field and $f, g \in F[x]$. Then f and g are coprime if and only if there exist $a, b \in F[x]$ with $af + bg = 1$.

Proof. ($\Rightarrow$) This is [Corollary 5.2.4](#) with $\gcd(f, g) = 1$.

($\Leftarrow$) If $af + bg = 1$, then $1 \in \langle f, g \rangle$, so $\langle f, g \rangle = F[x] = \langle 1 \rangle$ by [Example \(First Ideals\)](#) (a). In particular f, g are not both zero, and the monic generator is 1. Hence $\gcd(f, g) = 1$. ■

Examples: for **distinct** $c, c' \in F$, the polynomials $x - c$ and $x - c'$ are coprime, since

$$(c' - c)^{-1}(x - c) - (c' - c)^{-1}(x - c') = (c' - c)^{-1}(c' - c) = 1.$$

A non-zero constant is coprime to every polynomial ([Example \(First Greatest Common Divisors\)](#) (a)). The polynomial 0 is coprime only to the non-zero constants. Non-example: $x^2 - 1$ and $x + 1$ are not coprime, since $x + 1$ divides both, so $x + 1 \mid \gcd$ by [Theorem 5.2.3](#) (b), and the gcd has degree at least 1.

Coprimality is what lets a divisor pass through a product.

Theorem 5.2.8 — A Coprime Factor Can Be Canceled

Let F be a field and $p, q, r \in F[x]$. If $p \mid qr$ and $\gcd(p, q) = 1$, then $p \mid r$.

Proof. By [Proposition 5.2.7](#), there are $a, b \in F[x]$ with $ap + bq = 1$. Multiplying by r , $r = (ar)p + b(qr)$. Here $p \mid p$ and $p \mid qr$ by hypothesis, so $p \mid r$ by [Proposition 5.1.2](#) (b). This proves the theorem. ■

Warning. The coprimality hypothesis cannot be dropped. Take $p = x^2$ and $q = r = x$. Then $p \mid qr = x^2$, but $p \nmid r$, since $\deg x^2 > \deg x$ ([Proposition 5.1.2](#) (c)). Here $\gcd(p, q) = x \neq 1$: part of p is absorbed by q and part by r .

In the next section, [Theorem 5.2.8](#) becomes Euclid's lemma for irreducible polynomials, which is the heart of unique factorization.

Least common multiples, briefly. For $f, g \in F[x]$, the intersection $\langle f \rangle \cap \langle g \rangle$ is again an ideal: it contains 0, so (I1) holds, and (I2) and (I3) hold in each of $\langle f \rangle$ and $\langle g \rangle$, hence in the intersection. Its members are the common multiples of f and g .

Definition 5.2.4 — Least Common Multiple

Let F be a field and $f, g \in F[x]$. The **least common multiple** $\text{lcm}(f, g)$ is the monic generator of the ideal $\langle f \rangle \cap \langle g \rangle$ if this ideal is non-zero, and 0 otherwise.

Exactly as for the gcd, $\text{lcm}(f, g)$ is a common multiple of f and g that divides every common

multiple, by [Theorem 5.2.2](#). If $f = 0$ or $g = 0$, the only common multiple is 0; otherwise fg is a non-zero common multiple. For monic f and g , $\gcd(f, g) \cdot \text{lcm}(f, g) = fg$; this is proved in Exercise C3. For example, in $\mathbb{Q}[x]$, $\gcd(x^2 - 1, x^2 + x - 2) = x - 1$, so $\text{lcm}(x^2 - 1, x^2 + x - 2) = (x - 1)(x + 1)(x + 2)$.

5.2.7 The gcd does not depend on the field

A polynomial with rational coefficients is also a polynomial with real and with complex coefficients. Its gcd with another rational polynomial can be computed in $\mathbb{Q}[x]$, $\mathbb{R}[x]$ or $\mathbb{C}[x]$, and the definition does not obviously give the same answer, because the coefficients a, b in $\langle f, g \rangle$ range over a larger set. The Euclidean algorithm shows the answers agree. Recall that a subfield K of a field L is a subset closed under the field operations ([Definition 1.9.1](#)); then $K[x] \subseteq L[x]$, with the same addition and multiplication.

Proposition 5.2.9 — The gcd Is the Same Over a Larger Field

Let K be a subfield of a field L , and let $f, g \in K[x]$. Then the greatest common divisor of f and g computed in $K[x]$ equals the one computed in $L[x]$. In particular, f and g are coprime over K if and only if they are coprime over L .

Proof. If $g = 0$, both gcds are the monic associate of f (or 0), by [Example \(First Greatest Common Divisors\)](#) (a), and this does not depend on the field. Let $g \neq 0$, and run the Euclidean algorithm in $K[x]$: $r_{k-1} = q_{k+1}r_k + r_{k+1}$ with $\deg r_{k+1} < \deg r_k$, all in $K[x]$. Each of these equations also holds in $L[x]$, with the same degree condition. By the uniqueness in [Theorem 5.1.1](#) over L , they are exactly the divisions that the algorithm performs in $L[x]$. So both runs produce the same last non-zero remainder r_n , and [Theorem 5.2.6](#) gives the same gcd $c^{-1}r_n$ in both. This proves the proposition. ■

Warning. The gcd does not see the field, but **factorization does**. Over $\mathbb{Q}$, the polynomial $x^2 - 2$ has no root, by [Theorem 0.2.3](#), so it has no factor $x - c$ with $c \in \mathbb{Q}$ ([Theorem 5.1.3](#)). Over $\mathbb{R}$ it factors as $(x - \sqrt{2})(x + \sqrt{2})$. Yet $\gcd(x^2 - 2, x^2 - 2x)$, say, is the same over $\mathbb{Q}$ and over $\mathbb{R}$. So “coprime” is a robust notion, while “cannot be factored” depends on the field; the next section makes the second notion precise.

A useful consequence: if $f, g \in \mathbb{Q}[x]$ are coprime over $\mathbb{Q}$, then they have **no common root in $\mathbb{C}$** . Indeed, $af + bg = 1$ with $a, b \in \mathbb{Q}[x] \subseteq \mathbb{C}[x]$, and evaluating at a common root z would give $0 = 1$ by [Theorem 0.8.5](#). So common complex roots can be detected by a computation that never leaves $\mathbb{Q}$.

5.2.8 Exercises

5.2.8.1 A. Check your understanding

Exercise A1.

- (a) Define an ideal of $F[x]$.
- (b) State the theorem describing all ideals of $F[x]$.
- (c) Determine whether the following statement is true: “for all $f, g \in F[x]$, $\text{span}(f, g) = \langle f, g \rangle$.” Justify your answer.
- (d) Is $\text{gcd}(2x - 2, 4x^2 - 4) = 2x - 2$ in $\mathbb{Q}[x]$? Justify your answer.
- (e) Determine whether the following statement is true: “if $af + bg = x - 1$ for some $a, b \in \mathbb{Q}[x]$, then $\text{gcd}(f, g) = x - 1$.” Justify your answer.
- (f) Name an algorithm that computes $\text{gcd}(f, g)$ using only division with remainder (Theorem 5.1.1).

Solution.

- (a) A subset $I \subseteq F[x]$ is an ideal if it is non-empty, $f + g \in I$ for all $f, g \in I$, and $hf \in I$ for all $f \in I$ and every $h \in F[x]$ (Definition 5.2.1).
- (b) If $I \neq \{0\}$ is an ideal of $F[x]$, there is exactly one monic d with $I = \langle d \rangle$, namely the monic polynomial of least degree in I (Theorem 5.2.2).
- (c) False. Take $f = g = x$. Then $\text{span}(x, x) = \{cx : c \in F\}$ does not contain x^2 , while $x^2 = x \cdot x \in \langle x, x \rangle$. The span allows only scalar coefficients.
- (d) No. The gcd is monic by definition. Since $4x^2 - 4 = (2x - 2)(2x + 2)$, we have $\langle 2x - 2, 4x^2 - 4 \rangle = \langle 2x - 2 \rangle = \langle x - 1 \rangle$, so the gcd is $x - 1$.
- (e) False. Take $f = x$ and $g = x - 1$, with $a = 0$, $b = 1$. Then $af + bg = x - 1$, but $\text{gcd}(x, x - 1) = 1$, because $x - (x - 1) = 1$ (Proposition 5.2.7). What is true is that $\text{gcd}(f, g)$ divides every combination $af + bg$, by Theorem 5.2.3 (a) and Proposition 5.1.2 (b).
- (f) The Euclidean algorithm (Theorem 5.2.6).

5.2.8.2 B. Practice

Exercise B1: gcd and Bézout Coefficients of Two Cubics.

In $\mathbb{Q}[x]$, let $f = x^3 + x^2 + x + 1$ and $g = x^3 + x + 2$. Use the Euclidean algorithm to find $d = \text{gcd}(f, g)$ and polynomials $a, b \in \mathbb{Q}[x]$ with $d = af + bg$. Hence find all complex numbers that are roots of both f and g .

Solution. Divide f by g : $f = 1 \cdot g + r_1$ with $r_1 = f - g = x^2 - 1$. Divide g by r_1 : $x \cdot r_1 = x^3 - x$, and $g - xr_1 = 2x + 2$, so $g = xr_1 + r_2$ with $r_2 = 2x + 2$. Divide r_1 by r_2 : $x^2 - 1 = \frac{1}{2}(x - 1)(2x + 2) + 0$. The last non-zero remainder is $r_2 = 2x + 2$, with leading coefficient 2, so $d = x + 1$ by Theorem 5.2.6.

Back-substituting, $r_2 = g - xr_1 = g - x(f - g) = -xf + (1 + x)g$. Dividing by 2,

$$x + 1 = \left(-\frac{x}{2}\right)f + \frac{x+1}{2}g.$$

Check: $-\frac{x}{2}(x^3 + x^2 + x + 1) + \frac{x+1}{2}(x^3 + x + 2) = \frac{1}{2}(-x^4 - x^3 - x^2 - x + x^4 + x^3 + x^2 + 3x + 2) = x + 1$.

Hence: let $z \in \mathbb{C}$. If $f(z) = g(z) = 0$, then evaluating the identity above in $\mathbb{C}$ (Theorem 0.8.5) gives $z + 1 = 0$. Conversely, $x + 1$ divides f and g (Theorem 5.2.3 (a)), so -1

is a root of both. Therefore -1 is the only common complex root.

Exercise B2: Ideal or Not?

Determine which of the following subsets of $\mathbb{R}[x]$ are ideals. For each ideal, find its monic generator. Justify your answers.

- (a) $S_1 = \{f \in \mathbb{R}[x] : f(1) = f(-1)\}$.
- (b) $S_2 = \{f \in \mathbb{R}[x] : f(0) = 0 \text{ and } f(1) = 0\}$.
- (c) $S_3 = \{f \in \mathbb{R}[x] : \deg f \geq 3\} \cup \{0\}$.
- (d) $S_4 = \{f \in \mathbb{R}[x] : f(i) = 0\}$, where $f(i)$ means the value at $i \in \mathbb{C}$ of f regarded as a complex polynomial.

Solution.

- (a) Not an ideal. S_1 is a subspace, and $1 \in S_1$. But $x \cdot 1 = x$ has $x(1) = 1 \neq -1 = x(-1)$, so $x \notin S_1$, and (I3) fails.
- (b) An ideal. The argument of [Example \(First Ideals\)](#) (b), applied at both points, gives (I1)–(I3). If $f \in S_2$, then 0 and 1 are distinct roots of f , so $x(x-1) \mid f$ by [Corollary 5.1.4](#). Conversely every multiple of $x^2 - x$ vanishes at 0 and 1. Hence $S_2 = \langle x^2 - x \rangle$, with monic generator $x^2 - x$.
- (c) Not an ideal. $x^3 \in S_3$ and $-x^3 + x \in S_3$, but their sum x has degree 1 and is non-zero, so (I2) fails.
- (d) An ideal: evaluation at i respects sums and products ([Theorem 0.8.5](#) over $\mathbb{C}$), so the argument of [Example \(First Ideals\)](#) (b) applies. It contains $x^2 + 1$. A non-zero element of degree at most 1 would be $a + bx$ with $a, b \in \mathbb{R}$ not both zero and $a + bi = 0$, which is impossible since the real and imaginary parts give $a = b = 0$. So the least degree of a non-zero element is 2. If d is the monic generator ([Theorem 5.2.2](#)), then $\deg d = 2$ and $d \mid x^2 + 1$, so $x^2 + 1 = ud$ with u constant, and comparing leading coefficients gives $u = 1$. Hence $S_4 = \langle x^2 + 1 \rangle$.

Exercise B3: The gcd of $x^m - 1$ and $x^n - 1$.

Let F be a field. For integers $m, n \geq 0$, not both zero, let $\gcd(m, n)$ denote the largest positive integer dividing both.

- (a) For $m \geq n \geq 1$, verify that $x^m - 1 = x^{m-n}(x^n - 1) + (x^{m-n} - 1)$.
- (b) Prove that $\gcd(x^m - 1, x^n - 1) = x^{\gcd(m, n)} - 1$ in $F[x]$ for all $m, n \geq 0$ not both zero, where $x^0 - 1 = 0$.
- (c) Hence compute $\gcd(x^{12} - 1, x^{18} - 1)$.

Hint for (b): [Theorem 5.2.6](#), run on the exponents.

Solution.

- (a) Expanding, $x^{m-n}(x^n - 1) + x^{m-n} - 1 = x^m - x^{m-n} + x^{m-n} - 1 = x^m - 1$.
- (b) We use strong induction on $s = m + n \geq 1$ ([Theorem 0.2.10](#)). If $n = 0$, then $m \geq 1$, $\gcd(m, 0) = m$, and $\gcd(x^m - 1, 0)$ is the monic associate of $x^m - 1$, which

is $x^m - 1$ itself ([Example \(First Greatest Common Divisors\)](#) (a)). The case $m = 0$ is the same, since both gcds are symmetric. Now let $m, n \geq 1$; by symmetry we may assume $m \geq n$. By (a) and [Lemma 5.2.5](#),

$$\gcd(x^m - 1, x^n - 1) = \gcd(x^n - 1, x^{m-n} - 1).$$

The pair $(n, m - n)$ is not both zero and has sum $m < m + n$, so by the induction hypothesis the right side is $x^{\gcd(n, m-n)} - 1$. Finally, $\gcd(n, m - n) = \gcd(m, n)$: if $k \mid m$ and $k \mid n$, then $k \mid m - n$; and if $k \mid n$ and $k \mid m - n$, then $k \mid (m - n) + n = m$. So the two pairs have the same positive common divisors, and the same largest one. This completes the induction.

(c) By (b), $\gcd(x^{12} - 1, x^{18} - 1) = x^{\gcd(12, 18)} - 1 = x^6 - 1$.

5.2.8.3 C. Going deeper

Exercise C1: Polynomials Vanishing on a Set.

Let F be a field and $S \subseteq F$ non-empty. Put $I(S) = \{f \in F[x] : f(s) = 0 \text{ for all } s \in S\}$.

- (a) Prove that $I(S)$ is an ideal of $F[x]$.
- (b) Suppose $S = \{s_1, \dots, s_k\}$ is finite, with the s_i distinct. Prove that $I(S) = \langle (x - s_1) \cdots (x - s_k) \rangle$.
- (c) Suppose S is infinite. Prove that $I(S) = \{0\}$.
- (d) Deduce from (b) and [Exercise \(C3: Equal Functions Over \$\mathbb{F}_p\$ \)](#) that $I(\mathbb{F}_p) = \langle x^p - x \rangle$ for every prime p .

Solution.

- (a) The zero polynomial vanishes everywhere, so $I(S) \neq \emptyset$. If $f, g \in I(S)$ and $h \in F[x]$, then for every $s \in S$, [Theorem 0.8.5](#) gives $(f + g)(s) = 0 + 0 = 0$ and $(hf)(s) = h(s) \cdot 0 = 0$. So (I2) and (I3) hold.
- (b) Let $P = (x - s_1) \cdots (x - s_k)$. Every multiple hP vanishes at each s_i , since $P(s_i) = 0$; so $\langle P \rangle \subseteq I(S)$. Conversely, if $f \in I(S)$, then $s_1, \dots, s_k$ are distinct roots of f , and [Corollary 5.1.4](#) gives $P \mid f$. Hence $I(S) = \langle P \rangle$, and since P is monic, it is the monic generator.
- (c) Let $f \in I(S)$, and suppose $f \neq 0$. Since S is infinite, it contains $\deg f + 1$ distinct elements, all roots of f . This contradicts [Corollary 5.1.4](#). Hence $f = 0$.
- (d) By (b), $I(\mathbb{F}_p) = \langle P \rangle$ with $P = \prod_{c \in \mathbb{F}_p} (x - c)$. The solution of [Exercise \(C3: Equal Functions Over \$\mathbb{F}_p\$ \)](#) (a) showed $P = x^p - x$. Hence $I(\mathbb{F}_p) = \langle x^p - x \rangle$.

Exercise C2: A Non-Principal Ideal in Two Variables.

A **polynomial in two variables** over F is a finite sum $P = \sum_{i,j} a_{ij} x^i y^j$ with $a_{ij} \in F$, only finitely many non-zero; two are equal when all coefficients agree. They are added coefficient by coefficient and multiplied by expanding and collecting, and the set $F[x, y]$ of all of them satisfies the laws of [Theorem 0.8.1](#), which you may use. Ideals of $F[x, y]$ and $\langle P_1, \dots, P_k \rangle$ are defined exactly as in [Definition 5.2.1](#), and [Proposition 5.2.1](#) holds with the same proof. For $P \neq 0$, let $\deg_y P$ be the largest j with $a_{ij} \neq 0$ for some i , and define $\deg_x P$ similarly.

- (a) Prove that $\deg_y(PQ) = \deg_y P + \deg_y Q$ for non-zero $P, Q \in F[x, y]$. By symmetry the same holds for $\deg_x$.
- (b) Show that $\langle x, y \rangle$ consists of the polynomials whose constant coefficient a_{00} is 0.
- (c) Prove that $\langle x, y \rangle \neq \langle D \rangle$ for every $D \in F[x, y]$.
- (d) Explain precisely which step of the proof of [Theorem 5.2.2](#) has no analogue in $F[x, y]$.

Hint for (a): write $P = \sum_j p_j(x)y^j$ with $p_j \in F[x]$, and use [Corollary 0.8.4](#).

Solution.

- (a) Group terms by the power of y : $P = \sum_{j=0}^s p_j y^j$ and $Q = \sum_{j=0}^t q_j y^j$ with $p_j, q_j \in F[x]$, $s = \deg_y P$, $t = \deg_y Q$, so $p_s \neq 0$ and $q_t \neq 0$. Expanding, the terms of PQ with y -power greater than $s+t$ are all zero, and the terms with y^{s+t} come only from $p_s y^s \cdot q_t y^t$, giving $(p_s q_t) y^{s+t}$. By [Corollary 0.8.4](#), $p_s q_t \neq 0$ in $F[x]$, so some coefficient of $x^i y^{s+t}$ in PQ is non-zero. Hence $\deg_y(PQ) = s+t$.
- (b) Every $Ax + By$ has constant coefficient 0, since each term of Ax and By involves x or y to a positive power. Conversely, if $a_{00} = 0$, then every term $a_{ij}x^i y^j$ with $i \geq 1$ is a multiple of x , and every term with $i = 0, j \geq 1$ is a multiple of y . Grouping them, $P = Ax + By$ for some $A, B \in F[x, y]$.
- (c) Suppose $\langle x, y \rangle = \langle D \rangle$. Since $x \in \langle D \rangle$, $x = HD$ for some H , so $D \neq 0$ and $H \neq 0$. By (a), $0 = \deg_y x = \deg_y H + \deg_y D$, so $\deg_y D = 0$: D involves no y . Similarly $y = KD$ gives $\deg_x D = 0$. So D is a non-zero constant. But $D \in \langle x, y \rangle$, so its constant coefficient is 0 by (b), that is, $D = 0$. This contradiction shows that $\langle x, y \rangle$ is not principal.
- (d) The proof of [Theorem 5.2.2](#) picks a non-zero $d \in I$ of least degree, writes each $f \in I$ as $f = qd + r$ with r smaller than d by [Theorem 5.1.1](#), notes $r = f - qd \in I$, and concludes $r = 0$ from minimality. Measure size in $F[x, y]$ by the **total degree**, the largest $i+j$ with $a_{ij} \neq 0$. Choosing d still works: by (b), every non-zero element of $\langle x, y \rangle$ has total degree at least 1, and those of total degree 1 are the $ax + by$ with $(a, b) \neq (0, 0)$. Let d be any of them. Then $\langle d \rangle \subseteq \langle x, y \rangle$ by (I3), and the inclusion is strict by (c), so some $f \in \langle x, y \rangle$ is not a multiple of d . For this f and **every** $Q \in F[x, y]$, the element $R = f - Qd$ lies in $\langle x, y \rangle$ and is non-zero, so its total degree is at least 1, the total degree of d . So no remainder smaller than d exists: the step with no analogue is **division with remainder**, the appeal to [Theorem 5.1.1](#). Concretely, for $d = x$ and $f = y$, every term of Qx contains x , so the term y survives in $y - Qx$. The underlying reason is that, viewed as a polynomial in y with coefficients in $F[x]$, the leading coefficient x of d is not invertible in $F[x]$, which is exactly the obstacle of the warning after [Theorem 5.1.1](#).

Exercise C3: gcd Times lcm.

Let F be a field and $f, g \in F[x]$ **monic**. Put $d = \gcd(f, g)$ and write $f = df_1$, $g = dg_1$ with $f_1, g_1 \in F[x]$.

- (a) Prove that $\gcd(f_1, g_1) = 1$.
- (b) Prove that $m = df_1 g_1$ is a monic common multiple of f and g that divides every common multiple of f and g .

- (c) Deduce that $\text{lcm}(f, g) = df_1g_1$, and hence $\text{gcd}(f, g) \cdot \text{lcm}(f, g) = fg$.
 (d) Show by an example in $\mathbb{Q}[x]$ that the formula in (c) fails if f and g are not monic.

Hint for (b): use Theorem 5.2.8.

Solution.

- (a) The quotients exist by Theorem 5.2.3 (a), and $d \neq 0$ since $f \neq 0$. By Corollary 5.2.4, $d = af + bg = d(af_1 + bg_1)$. Canceling $d \neq 0$ (Corollary 0.8.4 (2)) gives $af_1 + bg_1 = 1$, so $\text{gcd}(f_1, g_1) = 1$ by Proposition 5.2.7.
 (b) Comparing leading coefficients in $f = df_1$ with f, d monic shows f_1 is monic; likewise g_1 . So m is monic, and $m = fg_1 = gf_1$ is a common multiple. Let M be any common multiple, say $M = fu$. Since $g \mid M$, write $M = gv$. Then $df_1u = dg_1v$, and canceling d gives $f_1u = g_1v$, so $g_1 \mid f_1u$. By (a) and Theorem 5.2.8, $g_1 \mid u$, say $u = g_1w$. Hence $M = fg_1w = mw$, and $m \mid M$.
 (c) By (b), $m \in \langle f \rangle \cap \langle g \rangle$, and every element of this ideal is a multiple of m . So $\langle f \rangle \cap \langle g \rangle = \langle m \rangle$, and m is monic, hence $\text{lcm}(f, g) = m$ by Definition 5.2.4 and Theorem 5.2.2. Then $\text{gcd}(f, g) \cdot \text{lcm}(f, g) = d \cdot df_1g_1 = (df_1)(dg_1) = fg$.
 (d) Take $f = g = 2x$. Then $\text{gcd}(f, g) = x$ and $\text{lcm}(f, g) = x$, both monic, so their product is x^2 , while $fg = 4x^2$. In general the two sides differ by the product of the leading coefficients.

5.3 Irreducible Polynomials and Unique Factorization

Every integer $n \geq 2$ is a product of primes, and Chapter 0 remarked that the convention “1 is not prime” is what makes such factorizations unique. Polynomials over a field behave the same way, with irreducible polynomials in the role of primes and non-zero constants in the role of ± 1 . This section defines irreducible polynomials, proves Euclid’s lemma from the gcd machinery of the previous section, and proves that every non-constant polynomial factors into irreducibles in exactly one way. The catch is that “irreducible” depends on the field: the same polynomial may factor over $\mathbb{R}$ but not over $\mathbb{Q}$, or over $\mathbb{F}_5$ but not over $\mathbb{F}_3$. So we also collect tools for deciding irreducibility over specific fields: roots in low degree, the Rational Root Theorem, and Eisenstein’s criterion. Throughout, F is a field.

5.3.1 Irreducible polynomials

For integers, the building blocks are the primes: integers $p \geq 2$ whose only positive divisors are 1 and p (Definition 0.2.2). To transplant this to $F[x]$, recall from the first section that every non-zero constant divides every polynomial, and that multiplying by a non-zero constant changes nothing for divisibility. So the trivial divisors of p are the non-zero constants and the associates cp . A building block should have no others.

An irreducible polynomial is a non-constant polynomial that cannot be split into two factors of smaller degree.

Definition 5.3.1 — Irreducible Polynomial

Let F be a field. A polynomial $p \in F[x]$ is **irreducible over F** if

- (1) p is **non-constant**, that is, $\deg p \geq 1$; and
- (2) whenever $p = gh$ with $g, h \in F[x]$, **one of** g, h is a constant.

A non-constant polynomial that is not irreducible is **reducible over F** .

In words: condition (1) excludes 0 and the non-zero constants. Condition (2) says every factorization of p is trivial: one factor is a constant, which is non-zero because $p \neq 0$, and the other is then an associate of p . The phrase **over F** is part of the notion, since g and h must lie in $F[x]$.

Since degrees add ([Theorem 0.8.3](#)), a factorization $p = gh$ with $p \neq 0$ has $\deg g + \deg h = \deg p$. So g is constant exactly when $\deg h = \deg p$. This gives a working form of the definition: **a non-constant p is reducible if and only if $p = gh$ with $1 \leq \deg g < \deg p$ and $1 \leq \deg h < \deg p$.**

Example. (First Irreducible and Reducible Polynomials)

- (a) Show that every polynomial of degree 1 is irreducible over every field.
- (b) Show that the constants 0 and 2 are not irreducible over $\mathbb{Q}$.
- (c) Show that $x^2 + x$ is reducible over every field, and that x^3 is reducible over every field.

Solution.

- (a) If $\deg p = 1$ and $p = gh$, then $\deg g + \deg h = 1$ with both degrees in $\mathbb{N}$, so one of them is 0: one factor is a constant. Condition (1) holds since $\deg p = 1$. Hence p is irreducible, whatever the field. For example x , $2x + 4$ and $x - \sqrt{2}$ (over $\mathbb{R}$) are irreducible.
- (b) Both are constants, so condition (1) fails. They are the degenerate cases, and they matter: 2 divides everything in $\mathbb{Q}[x]$, like 1 among the integers, and it is not a building block.
- (c) $x^2 + x = x(x + 1)$ with both factors of degree 1, and $x^3 = x \cdot x^2$ with factors of degrees 1 and 2. In each case both factors are non-constant, so condition (2) fails.

Why insist on “non-constant”? For the same reason 1 is not a prime. If non-zero constants were allowed as building blocks, factorizations could never be unique: $x = 2 \cdot \frac{1}{2}x = 2 \cdot 3 \cdot \frac{1}{6}x = \dots$ over $\mathbb{Q}$, with any number of constant factors. The factorization theorem below pulls out one constant in front and uses only non-constant, monic irreducibles.

For polynomials of degree 2 and 3, irreducibility is decided by roots alone.

Theorem 5.3.1 — Irreducibility in Degree 2 and 3

Let F be a field and $p \in F[x]$.

- (a) p has a factor of degree 1 in $F[x]$ if and only if p has a root in F . In particular, if $\deg p \geq 2$ and p has a root in F , then p is reducible over F .

(b) If $\deg p \in \{2, 3\}$, then p is irreducible over F if and only if p has **no** root in F .

Proof.

- (a) $(\Leftarrow)$ If $c \in F$ is a root, then $p = (x - c)q$ by [Theorem 5.1.3](#) (b). $(\Rightarrow)$ If $p = (ax + b)q$ with $a \neq 0$, then $c = -a^{-1}b$ satisfies $ac + b = 0$, so $p(c) = 0 \cdot q(c) = 0$ by [Theorem 0.8.5](#). For the last sentence, if $\deg p \geq 2$ and $p = (x - c)q$, then $\deg q = \deg p - 1 \geq 1$, so both factors are non-constant of degree less than $\deg p$, and p is reducible.
- (b) We prove both directions by contraposition. $(\Rightarrow)$ If p has a root, it is reducible by (a), since $\deg p \geq 2$. $(\Leftarrow)$ Suppose p is reducible. By the working form of the definition, $p = gh$ with $\deg g, \deg h \geq 1$ and $\deg g + \deg h = \deg p \in \{2, 3\}$. Two integers that are at least 1 and sum to 2 or 3 cannot both be at least 2, so one of g, h has degree 1. By (a), p has a root in F . This proves the theorem. ■

Now the standard examples, each checked with the theorem.

- $x^2 + 1$ **over $\mathbb{R}$ and over $\mathbb{C}$** . For real c , $c^2 + 1 \geq 1 > 0$, so there is no real root, and $x^2 + 1$ is irreducible over $\mathbb{R}$ (and over $\mathbb{Q}$). Over $\mathbb{C}$, $x^2 + 1 = (x - i)(x + i)$ is reducible.
- $x^2 - 2$ **over $\mathbb{Q}$ and over $\mathbb{R}$** . A rational root would be a rational number with square 2, which does not exist ([Theorem 0.2.3](#)). So $x^2 - 2$ is irreducible over $\mathbb{Q}$. Over $\mathbb{R}$, $x^2 - 2 = (x - \sqrt{2})(x + \sqrt{2})$.
- $x^2 + x + 1$ **over $\mathbb{F}_2$** . The values at 0 and 1 are 1 and $1 + 1 + 1 = 1$. No root, so it is irreducible over $\mathbb{F}_2$. It is the only irreducible polynomial of degree 2 over $\mathbb{F}_2$ (Exercise B3).

The non-example by minimal change is $x^2 + x$ over $\mathbb{F}_2$: change the constant term of $x^2 + x + 1$ from 1 to 0, and 0 becomes a root, so $x^2 + x = x(x + 1)$ is reducible. What fails is condition (2).

Warning. “No roots” implies “irreducible” **only in degrees 2 and 3**. The polynomial $(x^2 + 1)^2 = x^4 + 2x^2 + 1$ has no real root, since its values are at least 1, but it is reducible over $\mathbb{R}$: it is a product of two factors of degree 2. In degree 4 and higher, a polynomial can split into non-linear factors without having any root.

Quick check. Over which of the fields $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$ ([Example \(\$\mathbb{Q}\(\sqrt{2}\)\$ Is a Field\)](#)), $\mathbb{R}$ is $x^2 - 2$ irreducible? Is $x^2 + 1$ irreducible over $\mathbb{F}_5$?

Solution. Over $\mathbb{Q}$ only. The field $\mathbb{Q}(\sqrt{2})$ contains $\sqrt{2}$, a root of $x^2 - 2$, so $x^2 - 2 = (x - \sqrt{2})(x + \sqrt{2})$ is reducible there, and likewise over $\mathbb{R}$; over $\mathbb{Q}$ it is irreducible, as shown above. Over $\mathbb{F}_5$, $2^2 + 1 = 5 = 0$, so 2 is a root and $x^2 + 1$ is reducible by [Theorem 5.3.1](#): indeed $x^2 + 1 = (x - 2)(x + 2)$ in $\mathbb{F}_5[x]$.

5.3.2 Euclid’s lemma

What makes primes useful is not their definition but a property the definition implies: if a prime divides a product, it divides one of the factors. For polynomials this follows quickly from the previous section. The first step is that an irreducible polynomial has only two possible gcds with anything.

Lemma 5.3.2 — The gcd with an Irreducible Polynomial

Let F be a field, $p \in F[x]$ irreducible over F , and $f \in F[x]$. Then either $p \mid f$ or $\gcd(p, f) = 1$.

Proof. Let $d = \gcd(p, f)$. Since $p \neq 0$, d is monic, and $d \mid p$ by Theorem 5.2.3 (a), say $p = du$. Since p is irreducible, d or u is a constant. If d is a constant, then $d = 1$, since d is monic. If u is a constant, it is non-zero, and $p = ud$ divides every multiple of d ; since $d \mid f$ by Theorem 5.2.3 (a), Proposition 5.1.2 (a) gives $p \mid f$. This proves the lemma. ■

Theorem 5.3.3 — Euclid's Lemma for Polynomials

Let F be a field and $p \in F[x]$ irreducible over F . If $q, r \in F[x]$ and $p \mid qr$, then $p \mid q$ or $p \mid r$.

Proof. Suppose $p \nmid q$. By Lemma 5.3.2, $\gcd(p, q) = 1$. Since $p \mid qr$, Theorem 5.2.8 gives $p \mid r$. This proves the theorem. ■

The hypothesis “irreducible” is essential. The polynomial x^2 divides $x \cdot x$ but divides neither factor; the lemma does not apply because $x^2 = x \cdot x$ is reducible. Over $\mathbb{F}_2$, $x^2 + 1 = (x + 1)^2$ divides $(x + 1)(x + 1)$ but not $x + 1$, for the same reason.

For unique factorization we need the lemma for products of any length, and we need to know what “divides” means between two monic irreducibles.

Corollary 5.3.4 — Euclid's Lemma for Several Factors

Let F be a field and $p \in F[x]$ irreducible over F .

- (a) If $q_1, \dots, q_k \in F[x]$ with $k \geq 1$ and $p \mid q_1 q_2 \cdots q_k$, then $p \mid q_i$ for some i .
- (b) If q is irreducible over F , $p \mid q$, and p, q are both monic, then $p = q$.

Proof.

- (a) We use induction on k . For $k = 1$ there is nothing to prove. Let $k \geq 2$ and assume the statement for $k - 1$ factors. Since $p \mid (q_1 \cdots q_{k-1}) q_k$, Theorem 5.3.3 gives $p \mid q_1 \cdots q_{k-1}$ or $p \mid q_k$. In the first case the induction hypothesis gives $p \mid q_i$ for some $i \leq k - 1$. Either way the statement holds for k .
- (b) Write $q = pu$. Since q is irreducible, p or u is constant. But p is not constant, being irreducible, so u is a non-zero constant. Comparing leading coefficients of the monic polynomials q and p gives $u = 1$, so $p = q$. ■

5.3.3 Unique factorization

Here is the theorem that Chapter 0's remark about primes was pointing towards, now for polynomials. The constant in front is forced to be the leading coefficient; the rest is a product of monic irreducibles, unique up to order.

Theorem 5.3.5 — Unique Factorization in $F[x]$

Let F be a field and $f \in F[x]$ non-constant.

- (a) **Existence.** There are a non-zero constant $c \in F$, an integer $k \geq 1$, and monic polynomials $p_1, \dots, p_k \in F[x]$, each irreducible over F , with

$$f = c p_1 p_2 \cdots p_k.$$

- (b) **Uniqueness.** If also $f = c' q_1 q_2 \cdots q_l$ with $c' \in F$ non-zero, $l \geq 1$, and $q_1, \dots, q_l$ monic and irreducible over F , then $c = c'$, $k = l$, and after renumbering the q_j we have $p_i = q_i$ for every i .

In both factorizations, the constant is the leading coefficient of f .

Idea. Existence: if f is irreducible, stop; otherwise split it into two factors of smaller degree and factor each of them, which is strong induction on the degree. **Uniqueness:** compare leading coefficients to get $c = c'$. Then p_k divides $q_1 \cdots q_l$, so by Euclid's lemma it equals some q_j . Cancel it from both sides; the two shorter factorizations are equal by induction on the number of factors k . The one thing to watch is that after canceling, the right-hand side must still have at least one factor.

Proof. First a remark used in both parts. A product of monic polynomials is monic, by Theorem 0.8.3 and induction on the number of factors. So if $f = c p_1 \cdots p_k$ with the p_i monic, then the leading coefficient of f is $c \cdot 1 = c$. This proves the final sentence of the theorem.

- (a) We prove by strong induction on $n \geq 1$ (Theorem 0.2.10 with $n_0 = 1$) that every f of degree n has a factorization as in (a). Let $\deg f = n$, assume the statement for all degrees $1 \leq m < n$, and let c be the leading coefficient of f .

Case 1: f is irreducible over F . Then $f = c(c^{-1}f)$. The polynomial $c^{-1}f$ is monic, and it is irreducible: it is non-constant, and if $c^{-1}f = gh$, then $f = (cg)h$, so cg or h is constant, hence g or h is constant. So $k = 1$, $p_1 = c^{-1}f$ works.

Case 2: f is reducible over F . By the working form of the definition, $f = gh$ with $1 \leq \deg g < n$ and $1 \leq \deg h < n$. By the induction hypothesis, $g = a p_1 \cdots p_s$ and $h = b p_{s+1} \cdots p_{s+t}$ with a, b non-zero constants, $s, t \geq 1$, and all p_i monic irreducible. Then $f = (ab) p_1 \cdots p_{s+t}$, with $ab \neq 0$ by Theorem 0.7.1. This is a factorization as in (a), with $k = s + t$.

- (b) By the first remark, c and c' are both the leading coefficient of f , so $c = c'$. Multiplying by c^{-1} , we get $p_1 \cdots p_k = q_1 \cdots q_l$. So it suffices to prove, by induction on $k \geq 1$, the statement

$U(k)$: whenever $p_1 \cdots p_k = q_1 \cdots q_l$ with $l \geq 1$ and all p_i, q_j monic and irreducible over F , we have $k = l$, and after renumbering the q_j , $p_i = q_i$ for

every i .

We first note: if a monic irreducible q equals a product $r_1 \cdots r_m$ of $m \geq 2$ monic irreducibles, then $q = r_1 \cdot (r_2 \cdots r_m)$ with both factors non-constant, since each r_i has degree at least 1; this contradicts the irreducibility of q . So **a monic irreducible is never a product of two or more monic irreducibles.** (*)

Base case $k = 1$. Then $p_1 = q_1 \cdots q_l$. By (*), $l = 1$, and so $p_1 = q_1$.

Inductive step. Let $k \geq 2$ and assume $U(k-1)$. Suppose $p_1 \cdots p_k = q_1 \cdots q_l$. If $l = 1$, then $q_1 = p_1 \cdots p_k$ with $k \geq 2$, which is impossible by (*). So $l \geq 2$. Now $p_k \mid q_1 \cdots q_l$, so by Corollary 5.3.4 (a), $p_k \mid q_j$ for some j , and by Corollary 5.3.4 (b), $p_k = q_j$. Renumber the q 's so that $j = l$. Then

$$(p_1 \cdots p_{k-1}) p_k = (q_1 \cdots q_{l-1}) p_k,$$

and since $p_k \neq 0$, Corollary 0.8.4 (2) lets us cancel it: $p_1 \cdots p_{k-1} = q_1 \cdots q_{l-1}$, with $k-1 \geq 1$ and $l-1 \geq 1$ factors. By $U(k-1)$, $k-1 = l-1$, and after renumbering $q_1, \dots, q_{l-1}$, $p_i = q_i$ for $i \leq k-1$. Together with $p_k = q_k$, this proves $U(k)$.

By induction, $U(k)$ holds for every $k \geq 1$, which proves (b). ■

Grouping equal factors, every non-constant $f \in F[x]$ can be written uniquely as

$$f = c p_1^{e_1} p_2^{e_2} \cdots p_s^{e_s}$$

with c the leading coefficient, $p_1, \dots, p_s$ **distinct** monic irreducibles over F , and exponents $e_i \geq 1$. The monic irreducible factors of the characteristic and minimal polynomials of an operator will organize the structure theory of Chapters 8 and 9.

Warning. Uniqueness needs the normalization “monic”. Without it, constants can move between factors: over $\mathbb{Q}$, $2x^2 - 2 = (2x - 2)(x + 1) = (x - 1)(2x + 2) = (\frac{1}{2}x - \frac{1}{2})(4x + 4)$. All of these are “factorizations into irreducibles”, and they differ only by associates. The theorem removes this freedom by collecting all constants into c . Uniqueness also refers to a **fixed** field: over $\mathbb{Q}$ the factorization of $x^2 - 2$ is $x^2 - 2$ itself, and over $\mathbb{R}$ it is $(x - \sqrt{2})(x + \sqrt{2})$.

The next example shows the field dependence in one table.

Example. (Factoring $x^4 - 1$ Over Six Fields)

Factor $x^4 - 1$ into a constant times monic irreducibles over $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{F}_2$, $\mathbb{F}_3$, and, for contrast, $\mathbb{F}_5$.

Solution. Over every field, $x^4 - 1 = (x^2 - 1)(x^2 + 1) = (x - 1)(x + 1)(x^2 + 1)$, by expanding. The factors $x - 1$ and $x + 1$ are irreducible by Example (First Irreducible and Reducible Polynomials) (a). Everything depends on $x^2 + 1$, which by Theorem 5.3.1 is irreducible exactly when it has no root in the field. The constant in front is always the leading coefficient 1.

- $\mathbb{Q}$ and $\mathbb{R}$. For real c , $c^2 + 1 \geq 1$, so there is no root, and $x^2 + 1$ is irreducible.
- $\mathbb{C}$. $x^2 + 1 = (x - i)(x + i)$.
- $\mathbb{F}_2$. Here $-1 = 1$, so $x - 1 = x + 1$, and $x^2 + 1 = (x + 1)^2$ because $2x = 0$. All four factors coincide.
- $\mathbb{F}_3$. The values of $x^2 + 1$ at $0, 1, 2$ are $1, 2, 5 = 2$. No root, so $x^2 + 1$ is irreducible. Also $x - 1 = x + 2$.
- $\mathbb{F}_5$. $2^2 + 1 = 5 = 0$ and $3^2 + 1 = 10 = 0$, so $x^2 + 1 = (x - 2)(x - 3) = (x + 3)(x + 2)$.

field	factorization of $x^4 - 1$ into monic irreducibles
$\mathbb{Q}$	$(x - 1)(x + 1)(x^2 + 1)$
$\mathbb{R}$	$(x - 1)(x + 1)(x^2 + 1)$
$\mathbb{C}$	$(x - 1)(x + 1)(x - i)(x + i)$
$\mathbb{F}_2$	$(x + 1)^4$
$\mathbb{F}_3$	$(x + 1)(x + 2)(x^2 + 1)$
$\mathbb{F}_5$	$(x + 1)(x + 2)(x + 3)(x + 4)$

By [Theorem 5.3.5](#), each row is **the** factorization over that field, up to order. SymPy's `factor_list`, with `extension=I` over $\mathbb{C}$ and `modulus=p` over $\mathbb{F}_p$, returns the same rows. Notice that the numbers of irreducible factors are $3, 3, 4, 4, 3, 4$, and that over $\mathbb{F}_2$ a repeated factor appears although $x^4 - 1$ has no repeated factor over the other fields.

5.3.4 Tools over $\mathbb{Q}$: rational roots and Eisenstein's criterion

Over $\mathbb{Q}$, [Theorem 5.3.1](#) reduces irreducibility of a quadratic or cubic to the question of rational roots. There are infinitely many rational numbers to try, but for integer coefficients only finitely many candidates can work. We write $\mathbb{Z}[x]$ for the polynomials in $\mathbb{Q}[x]$ whose coefficients are all integers, as in the first section. Every rational number can be written as u/v with $u \in \mathbb{Z}$, $v \geq 1$, and the only positive integer dividing both u and v equal to 1: take $v \geq 1$ as small as possible, since a common divisor $e > 1$ would give the representation $(u/e)/(v/e)$ with a smaller denominator. We say u/v is in **lowest terms**.

Theorem 5.3.6 — Rational Root Theorem

Let $f = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 \in \mathbb{Z}[x]$ with $n \geq 1$ and $a_n \neq 0$. If $u/v \in \mathbb{Q}$ is in lowest terms and $f(u/v) = 0$, then $u \mid a_0$ and $v \mid a_n$.

Idea. Clear denominators: $v^n f(u/v) = a_n u^n + a_{n-1} u^{n-1} v + \cdots + a_0 v^n = 0$. Every term except the last contains u , so u divides $a_0 v^n$; every term except the first contains v , so v divides $a_n u^n$. Since u and v share no factor, the powers of v can be stripped off one at a time, using Bézout for integers.

Proof. Multiplying $f(u/v) = 0$ by v^n gives

$$a_n u^n + a_{n-1} u^{n-1} v + \cdots + a_1 u v^{n-1} + a_0 v^n = 0.$$

Hence $a_0 v^n = -u(a_n u^{n-1} + a_{n-1} u^{n-2} v + \cdots + a_1 v^{n-1})$, so $u \mid a_0 v^n$; and $a_n u^n =$

$-v(a_{n-1}u^{n-1} + \cdots + a_0v^{n-1})$, so $v \mid a_nu^n$.

Since $v \geq 1$ and the only positive common divisor of u and v is 1, [Lemma 0.7.2](#) gives integers s, t with $us + vt = 1$. If $b \in \mathbb{Z}$ and $k \geq 1$ with $u \mid bv^k$, then

$$bv^{k-1} = bv^{k-1}(us + vt) = u(bv^{k-1}s) + t(bv^k)$$

is a sum of two multiples of u , so $u \mid bv^{k-1}$. Applying this n times to $b = a_0$ gives $u \mid a_0$. Similarly, if $v \mid cu^k$ with $k \geq 1$, then $cu^{k-1} = s(cu^k) + v(cu^{k-1}t)$, so $v \mid cu^{k-1}$; applying this n times to $c = a_n$ gives $v \mid a_n$. This proves the theorem. ■

A polynomial in $\mathbb{Q}[x]$ with fractional coefficients can first be multiplied by a common denominator; this does not change its roots or its irreducibility.

Example. (A Cubic That Is Irreducible Over $\mathbb{Q}$)

Decide whether $f = 2x^3 - 3x^2 + 4x - 1$ is irreducible over $\mathbb{Q}$.

Solution. By [Theorem 5.3.6](#), a rational root u/v in lowest terms has $u \mid -1$ and $v \mid 2$ with $v \geq 1$. So $u = \pm 1$ and $v \in \{1, 2\}$, and the candidates are $1, -1, \frac{1}{2}, -\frac{1}{2}$. Evaluating,

$$f(1) = 2, \quad f(-1) = -10, \quad f\left(\frac{1}{2}\right) = \frac{1}{4} - \frac{3}{4} + 2 - 1 = \frac{1}{2}, \quad f\left(-\frac{1}{2}\right) = -\frac{1}{4} - \frac{3}{4} - 2 - 1 = -4.$$

None is 0, so f has no rational root. Since $\deg f = 3$, [Theorem 5.3.1](#) shows that f is irreducible over $\mathbb{Q}$. (SymPy's `factor_list` over $\mathbb{Q}$ agrees.)

Beyond degree 3, roots are not enough, as the warning after [Theorem 5.3.1](#) showed. One classical criterion handles many polynomials of any degree at once. It looks at divisibility of the coefficients by a prime.

Theorem 5.3.7 — Eisenstein's Criterion

Let $f = a_nx^n + a_{n-1}x^{n-1} + \cdots + a_0 \in \mathbb{Z}[x]$ with $n \geq 1$, and let p be a prime such that

- (1) $p \nmid a_n$;
- (2) $p \mid a_i$ for every $i = 0, 1, \dots, n-1$;
- (3) $p^2 \nmid a_0$.

Then f is irreducible over $\mathbb{Q}$.

The proof needs one more idea, **Gauss's lemma**: a polynomial with integer coefficients that factors over $\mathbb{Q}$ already factors over $\mathbb{Z}$, with factors of the same degrees. With it, one reduces the coefficients modulo p and compares. Exercise C2 of this section proves Gauss's lemma and then Eisenstein's criterion, step by step; no result of this chapter depends on the criterion. Here we only use it in examples.

Example. (Irreducible Polynomials of Every Degree Over $\mathbb{Q}$)

- (a) Show that $x^n - 2$ is irreducible over $\mathbb{Q}$ for every $n \geq 1$. (b) Show that $x^4 + 10x + 5$ is irreducible over $\mathbb{Q}$.

Solution.

- (a) Take $p = 2$. The leading coefficient 1 is not divisible by 2; the coefficients $a_1, \dots, a_{n-1}$ are 0 and $a_0 = -2$, all divisible by 2; and $4 \nmid -2$. By [Theorem 5.3.7](#), $x^n - 2$ is irreducible over $\mathbb{Q}$. So over $\mathbb{Q}$ there are irreducible polynomials of every degree. (Over $\mathbb{R}$, by contrast, $x^n - 2$ has the root $\sqrt[n]{2}$ and is reducible for $n \geq 2$.)
- (b) Take $p = 5$: $5 \nmid 1$; 5 divides 0, 0, 10, 5; and $25 \nmid 5$. So $x^4 + 10x + 5$ is irreducible over $\mathbb{Q}$. Roots alone could not have shown this, since a quartic may factor into two quadratics.

Eisenstein's criterion is sufficient, not necessary. The polynomial $x^2 + 1$ is irreducible over $\mathbb{Q}$, but no prime divides its constant term 1, so condition (2) fails for every p .

5.3.5 Exercises

5.3.5.1 A. Check your understanding

Exercise A1.

- (a) Define what it means for $p \in F[x]$ to be irreducible over F .
- (b) State the Unique Factorization Theorem in $F[x]$.
- (c) Determine whether the following statement is true: “a polynomial in $\mathbb{R}[x]$ of degree at least 1 with no real root is irreducible over $\mathbb{R}$.” Justify your answer.
- (d) Is the constant 2 irreducible over $\mathbb{Q}$? Is $2x + 4$?
- (e) Is $x^2 + 1$ irreducible over $\mathbb{F}_5$? Justify your answer.
- (f) Name a tool that reduces the irreducibility of a cubic in $\mathbb{Z}[x]$ over $\mathbb{Q}$ to finitely many evaluations.

Solution.

- (a) p is irreducible over F if $\deg p \geq 1$ and, whenever $p = gh$ with $g, h \in F[x]$, one of g, h is a constant ([Definition 5.3.1](#)).
- (b) Every non-constant $f \in F[x]$ can be written as $f = c p_1 \cdots p_k$ with $c \in F$ non-zero and $p_1, \dots, p_k$ monic and irreducible over F ; the constant c is the leading coefficient of f , and the p_i are unique up to order ([Theorem 5.3.5](#)).
- (c) False. $(x^2 + 1)^2$ has no real root, but it is the product of two non-constant factors, so it is reducible over $\mathbb{R}$. The statement is true in degrees 2 and 3 only ([Theorem 5.3.1](#)).
- (d) No: 2 is constant, and irreducible polynomials are non-constant by definition. Yes: $2x + 4$ has degree 1, so it is irreducible ([Example \(First Irreducible and Reducible Polynomials\)](#) (a)).

- (e) No. In $\mathbb{F}_5$, $2^2 + 1 = 0$, so $x^2 + 1 = (x - 2)(x + 2)$ is reducible by [Theorem 5.3.1](#).
 (f) The Rational Root Theorem ([Theorem 5.3.6](#)), combined with [Theorem 5.3.1](#).

5.3.5.2 B. Practice

Exercise B1: Irreducible or Not?.

Determine whether each polynomial is irreducible over each of the given fields. Justify your answers.

- (a) $x^2 + 2$ over $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{F}_2$ and $\mathbb{F}_3$.
 (b) $x^3 + x + 1$ over $\mathbb{F}_2$ and over $\mathbb{F}_3$.
 (c) $x^3 - 3x - 1$ over $\mathbb{Q}$.
 (d) $x^4 + 4$ over $\mathbb{Q}$.
 (e) $x^5 - 6x + 3$ over $\mathbb{Q}$.

Solution.

- (a) Over $\mathbb{Q}$ and $\mathbb{R}$: for real c , $c^2 + 2 \geq 2 > 0$, so there is no root, and $x^2 + 2$ is irreducible by [Theorem 5.3.1](#). Over $\mathbb{F}_2$: $2 = 0$, so $x^2 + 2 = x \cdot x$ is reducible. Over $\mathbb{F}_3$: $2 = -1$, so $x^2 + 2 = x^2 - 1 = (x - 1)(x + 1)$ is reducible.
 (b) Over $\mathbb{F}_2$: the values at 0 and 1 are 1 and $3 = 1$, so there is no root, and the cubic is irreducible by [Theorem 5.3.1](#). Over $\mathbb{F}_3$: the value at 1 is $3 = 0$, so it is reducible; indeed $x^3 + x + 1 = (x - 1)(x^2 + x + 2)$ in $\mathbb{F}_3[x]$, as expanding shows: $x^3 + x^2 + 2x - x^2 - x - 2 = x^3 + x - 2 = x^3 + x + 1$.
 (c) By [Theorem 5.3.6](#), a rational root u/v in lowest terms has $u \mid -1$ and $v \mid 1$, so it is 1 or -1 . The values are $1 - 3 - 1 = -3$ and $-1 + 3 - 1 = 1$. No rational root, so the cubic is irreducible over $\mathbb{Q}$ by [Theorem 5.3.1](#).
 (d) Reducible, although it has no rational root (its values are at least 4): expanding, $(x^2 + 2x + 2)(x^2 - 2x + 2) = x^4 - 4x^2 + 4 + 4x^2 = x^4 + 4$. This is the warning after [Theorem 5.3.1](#) in action.
 (e) Irreducible by [Theorem 5.3.7](#) with $p = 3$: $3 \nmid 1$; 3 divides the coefficients 0, 0, 0, -6 , 3 of $x^4, x^3, x^2, x, 1$; and $9 \nmid 3$.

Exercise B2: Factoring $x^6 - 1$ Over $\mathbb{Q}$.

Factor $x^6 - 1$ into monic irreducible polynomials over $\mathbb{Q}$, and justify that each factor is irreducible. Hence find its factorization into monic irreducibles over $\mathbb{F}_2$ as well. Is it obtained by reducing the coefficients of each rational factor modulo 2? Explain why, in general, this has to be checked rather than assumed.

Solution. Using $x^6 - 1 = (x^3 - 1)(x^3 + 1)$ and [Exercise \(B2: A Factor of \$x^n - c^n\$ \)](#) (with $c = 1$ and $c = -1$),

$$x^6 - 1 = (x - 1)(x^2 + x + 1)(x + 1)(x^2 - x + 1),$$

which one checks by expanding: $(x - 1)(x^2 + x + 1) = x^3 - 1$ and $(x + 1)(x^2 - x + 1) = x^3 + 1$. The linear factors are irreducible. For the quadratics, $c^2 \pm c + 1 = (c \pm \frac{1}{2})^2 + \frac{3}{4} > 0$ for real

c , so they have no rational root and are irreducible by [Theorem 5.3.1](#). By [Theorem 5.3.5](#) this is the factorization over $\mathbb{Q}$.

Over $\mathbb{F}_2$, reducing coefficients gives $(x+1)(x^2+x+1)(x+1)(x^2+x+1)$, since $-1 = 1$. Here $x+1$ is irreducible and x^2+x+1 is irreducible over $\mathbb{F}_2$ (it has no root), so the factorization over $\mathbb{F}_2$ is $(x+1)^2(x^2+x+1)^2$. So yes, in this case the factorization is obtained by reduction, but only because the reduced factors happen to stay irreducible, and that has to be checked: two distinct rational factors, $x-1$ and $x+1$, became equal, and in general a factor that is irreducible over $\mathbb{Q}$ may become reducible after reduction, as x^2+1 does over $\mathbb{F}_2$ in [Example \(Factoring \$x^4-1\$ Over Six Fields\)](#).

Exercise B3: Small Irreducibles Over $\mathbb{F}_2$.

List all irreducible polynomials over $\mathbb{F}_2$ of degree at most 3. Justify that your list is complete.

Hint: a polynomial over $\mathbb{F}_2$ has the root 0 exactly when its constant term is 0, and the root 1 exactly when its number of non-zero terms is even.

Solution. Every non-zero polynomial over $\mathbb{F}_2$ is monic, since its leading coefficient is 1. For $f = \sum a_k x^k$, $f(0) = a_0$ and $f(1) = \sum a_k$, which is 0 exactly when the number of non-zero coefficients is even.

Degree 1: x and $x+1$, both irreducible ([Example \(First Irreducible and Reducible Polynomials\)](#) (a)).

Degrees 2 and 3: by [Theorem 5.3.1](#), f is irreducible exactly when $f(0) = 1$ and $f(1) = 1$, that is, the constant term is 1 and the number of non-zero terms is odd. In degree 2, the polynomials with constant term 1 are x^2+1 (two terms) and x^2+x+1 (three terms), so only x^2+x+1 is irreducible. In degree 3, they are x^3+1 (two terms), x^3+x+1 and x^3+x^2+1 (three terms), and x^3+x^2+x+1 (four terms). So the irreducible ones are x^3+x+1 and x^3+x^2+1 .

Hence the complete list is x , $x+1$, x^2+x+1 , x^3+x+1 , x^3+x^2+1 .

5.3.5.3 C. Going deeper

Exercise C1: Infinitely Many Irreducibles.

Let F be any field.

- Prove that there are infinitely many monic irreducible polynomials over F .
- Deduce that for every prime p and every $N \in \mathbb{N}$, there is an irreducible polynomial over $\mathbb{F}_p$ of degree greater than N .
- Explain why (a) is immediate when F is infinite, and use [Exercise \(B3: Small Irreducibles Over \$\mathbb{F}_2\$ \)](#) to name an irreducible polynomial of degree 3 over $\mathbb{F}_2$.

Hint for (a): imitate the proof that there are infinitely many primes ([Theorem 0.2.13](#)).

Solution.

- (a) Suppose, for a contradiction, that $p_1, \dots, p_k$ are all the monic irreducible polynomials over F . The list is non-empty, since x is monic and irreducible. Let $f = p_1 p_2 \cdots p_k + 1$. Each p_i has degree at least 1, so $\deg(p_1 \cdots p_k) \geq 1$ by [Theorem 0.8.3](#), and f is non-constant by [Theorem 0.8.2](#). By [Theorem 5.3.5](#) (a), f has a monic irreducible factor, which must be some p_j . Then $p_j \mid f$ and $p_j \mid p_1 \cdots p_k$, so $p_j \mid f - p_1 \cdots p_k = 1$ by [Proposition 5.1.2](#) (b). By [Proposition 5.1.2](#) (c), $\deg p_j \leq \deg 1 = 0$, contradicting $\deg p_j \geq 1$. Hence there are infinitely many monic irreducibles.
- (b) A monic polynomial of degree n over $\mathbb{F}_p$ is determined by its n lower coefficients, so there are exactly p^n of them, and there are $1 + p + \cdots + p^N$ monic polynomials of degree at most N , a finite number. By (a), infinitely many monic irreducibles exist, so some of them have degree greater than N .
- (c) If F is infinite, the polynomials $x - c$ for $c \in F$ are infinitely many distinct monic polynomials of degree 1, all irreducible by [Example \(First Irreducible and Reducible Polynomials\)](#) (a). Over a finite field there are only finitely many polynomials of degree 1, so the content of (a) is that irreducibles of higher and higher degree keep appearing. Over $\mathbb{F}_2$, for example, $x^3 + x + 1$ is irreducible of degree 3 by [Exercise \(B3: Small Irreducibles Over \$\mathbb{F}_2\$ \)](#).

Exercise C2: Gauss's Lemma and Eisenstein's Criterion.

Let $\mathbb{Z}[x]$ be as in [Theorem 5.3.6](#), and let p be a prime. For $g = \sum b_i x^i \in \mathbb{Z}[x]$, define its **reduction modulo p** as $\bar{g} = \sum [b_i] x^i \in \mathbb{F}_p[x]$, where $[b]$ is the class of b in $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ ([Example \(Arithmetic on \$\mathbb{Z}/n\mathbb{Z}\$ and the Field \$\mathbb{F}_2\$ \)](#)).

- (a) Prove that $\overline{g+h} = \bar{g} + \bar{h}$ and $\overline{gh} = \bar{g}\bar{h}$ for all $g, h \in \mathbb{Z}[x]$, and that $\deg \bar{g} \leq \deg g$.
- (b) Let $g, h \in \mathbb{Z}[x]$, and suppose p divides every coefficient of gh . Prove that p divides every coefficient of g or every coefficient of h .
- (c) (**Gauss's lemma.**) Let $f \in \mathbb{Z}[x]$, and suppose $f = gh$ with $g, h \in \mathbb{Q}[x]$. Prove that there is a non-zero $\lambda \in \mathbb{Q}$ with $\lambda g \in \mathbb{Z}[x]$ and $\lambda^{-1}h \in \mathbb{Z}[x]$.
- (d) Let F be a field, $c \in F$ non-zero, $n \in \mathbb{N}$, and $u, v \in F[x]$ with $uv = cx^n$. Prove that $u = \beta x^s$ and $v = \gamma x^t$ for some non-zero $\beta, \gamma \in F$ and $s, t \in \mathbb{N}$.
- (e) Prove [Theorem 5.3.7](#).

Hint for (c): among all ways of writing $kf = (\alpha g)(\beta h)$ with $\alpha, \beta \in \mathbb{Q}$, $\alpha\beta = k$ a positive integer and $\alpha g, \beta h \in \mathbb{Z}[x]$, take k as small as possible. Hint for (e): suppose $f = gh$ with g, h non-constant, use (c), and reduce modulo p .

Solution.

- (a) By [Example \(Arithmetic on \$\mathbb{Z}/n\mathbb{Z}\$ and the Field \$\mathbb{F}_2\$ \)](#), $[a] + [b] = [a + b]$ and $[a][b] = [ab]$. Let $g = \sum b_i x^i$ and $h = \sum c_i x^i$. The coefficient of x^k in $g + h$ is $b_k + c_k$, whose class is $[b_k] + [c_k]$, the coefficient of x^k in $\bar{g} + \bar{h}$. The coefficient of x^k in gh is $\sum_{i+j=k} b_i c_j$, whose class is $\sum_{i+j=k} [b_i][c_j]$, the coefficient of x^k in $\bar{g}\bar{h}$ ([Definition 0.8.2](#)). Finally, if $b_i = 0$ then $[b_i] = 0$, so $\bar{g}$ has no non-zero coefficient beyond position $\deg g$.
- (b) The hypothesis says $\overline{gh} = 0$. By (a), $\bar{g}\bar{h} = 0$ in $\mathbb{F}_p[x]$. Since p is prime, $\mathbb{F}_p$ is a field ([Theorem 0.7.3](#)), so [Corollary 0.8.4](#) gives $\bar{g} = 0$ or $\bar{h} = 0$. That is, p divides every coefficient of g or every coefficient of h .

- (c) Choose positive integers m, n with $mg \in \mathbb{Z}[x]$ and $nh \in \mathbb{Z}[x]$, for instance the products of the denominators of the coefficients. Let K be the set of positive integers k for which there are $\alpha, \beta \in \mathbb{Q}$ with $\alpha\beta = k$, $\alpha g \in \mathbb{Z}[x]$ and $\beta h \in \mathbb{Z}[x]$. Then $mn \in K$, so by [Theorem 0.2.11](#) K has a least element k , with corresponding α, β . Suppose $k > 1$. By [Proposition 0.2.12](#), some prime q divides k . Now $(\alpha g)(\beta h) = \alpha\beta gh = kf$, and every coefficient of kf is divisible by q . By (b) with the prime q , q divides every coefficient of αg or every coefficient of βh ; say the first (the other case is symmetric). Then $(\alpha/q)g \in \mathbb{Z}[x]$, $\beta h \in \mathbb{Z}[x]$, and $(\alpha/q)\beta = k/q$ is a positive integer smaller than k , which lies in K . This contradicts the minimality of k . Hence $k = 1$, so $\beta = \alpha^{-1}$, and $\lambda = \alpha$ works.
- (d) Since $uv \neq 0$, both u, v are non-zero. Write $u = \sum u_i x^i$, $v = \sum v_j x^j$, let i_0 and j_0 be the smallest indices with $u_{i_0} \neq 0$ and $v_{j_0} \neq 0$, and let $s = \deg u$, $t = \deg v$. The coefficient of $x^{i_0+j_0}$ in uv is $\sum_{i+j=i_0+j_0} u_i v_j$. A term with $i < i_0$ has $u_i = 0$, and a term with $i > i_0$ has $j < j_0$, so $v_j = 0$. So this coefficient is $u_{i_0} v_{j_0} \neq 0$. The only non-zero coefficient of cx^n is at x^n , so $i_0 + j_0 = n$. By [Theorem 0.8.3](#), also $s + t = n$. Since $i_0 \leq s$ and $j_0 \leq t$, this forces $i_0 = s$ and $j_0 = t$. So u has exactly one non-zero coefficient, $u = u_s x^s$, and likewise $v = v_t x^t$.
- (e) Suppose f is reducible over $\mathbb{Q}$: $f = gh$ with $g, h \in \mathbb{Q}[x]$, $\deg g = s \geq 1$, $\deg h = t \geq 1$, and $s + t = n$. By (c), replacing g, h by $\lambda g, \lambda^{-1} h$ (which have the same degrees), we may assume $g, h \in \mathbb{Z}[x]$. Write g_0, h_0 for their constant coefficients, so $a_0 = g_0 h_0$. Reduce modulo p . By conditions (1) and (2), $\bar{f} = [a_n]x^n$ with $[a_n] \neq 0$. By (a), $\bar{g}\bar{h} = [a_n]x^n$, so by (d), $\bar{g} = \beta x^{s'}$ and $\bar{h} = \gamma x^{t'}$ with $\beta, \gamma \neq 0$. By [Theorem 0.8.3](#), $s' + t' = n = s + t$, while $s' \leq s$ and $t' \leq t$ by (a). Hence $s' = s \geq 1$ and $t' = t \geq 1$. So the constant coefficients of $\bar{g}$ and $\bar{h}$ are 0, that is, $p \mid g_0$ and $p \mid h_0$. Then $p^2 \mid g_0 h_0 = a_0$, contradicting condition (3). Therefore f is irreducible over $\mathbb{Q}$; it is non-constant since $n \geq 1$.

5.4 Roots, Multiplicity and Derivatives

Chapter 2 bounded the number of roots of a polynomial, and the first section of this chapter restated the bound as [Corollary 5.1.4](#): a non-zero polynomial of degree n has at most n roots. The bound counts each root once, and so it misses something. The polynomial $(x - 1)^2$ has degree 2 but only one root, and the root 1 is somehow “used twice”. This section makes “used twice” precise with the multiplicity of a root, sharpens the root bound, and then gives a test for repeated roots that needs no factoring: the formal derivative. The derivative behaves differently in characteristic p , and we say exactly where. The section ends with Taylor expansion of a polynomial, which turns out to be a dual basis from Chapter 4.

5.4.1 Multiplicity of a root

By the Factor Theorem ([Lemma 2.7.1](#), restated in general as [Theorem 5.1.3](#)), c is a root of f exactly when $x - c$ divides f . A repeated root should mean that a higher power of $x - c$ divides f . So we measure how many factors $x - c$ can be pulled out.

The multiplicity of c counts how many times $x - c$ divides f .

Definition 5.4.1 — Multiplicity of a Root

Let F be a field, $f \in F[x]$ **non-zero** and $c \in F$. The **multiplicity of c as a root of f** , written $\text{mult}_c(f)$, is the **largest** $m \in \mathbb{N}$ such that $(x - c)^m \mid f$.

The root c is **simple** if $\text{mult}_c(f) = 1$, and **multiple** (or **repeated**) if $\text{mult}_c(f) \geq 2$.

In words: keep dividing f by $x - c$ as long as the division leaves no remainder, and count the successful divisions. Divisibility is meant in the sense of [Definition 5.1.1](#), so $(x - c)^m \mid f$ says $f = (x - c)^m g$ for some $g \in F[x]$.

There is something to check: a largest such m must exist. The value $m = 0$ always works, since $(x - c)^0 = 1$ divides every polynomial. And if $f = (x - c)^m g$, then $g \neq 0$ because $f \neq 0$, so [Theorem 0.8.3](#) gives $\deg f = m + \deg g \geq m$. The admissible m form a non-empty set of natural numbers bounded by $\deg f$, so it has a largest element. Moreover $\text{mult}_c(f) \geq 1$ exactly when $x - c \mid f$, that is, exactly when c is a root. So multiplicity 0 means “not a root”.

The hypothesis $f \neq 0$ is not decoration. Every power $(x - c)^m$ divides the zero polynomial, since $0 = (x - c)^m \cdot 0$. There is no largest m , and the zero polynomial has no multiplicities.

A definition by “largest m ” is awkward to use directly, because to confirm m we must rule out all larger powers. The next lemma replaces it with a single test at c .

Lemma 5.4.1 — Multiplicity via the Cofactor

Let $f \in F[x]$ be non-zero, $c \in F$ and $m \in \mathbb{N}$. Then $\text{mult}_c(f) = m$ if and only if

$$f = (x - c)^m g \quad \text{for some } g \in F[x] \text{ with } g(c) \neq 0.$$

Proof. ($\Rightarrow$) Suppose $\text{mult}_c(f) = m$. Then $f = (x - c)^m g$ for some $g \in F[x]$. If $g(c) = 0$, then $g = (x - c)h$ by [Lemma 2.7.1](#), so $f = (x - c)^{m+1}h$, contradicting the maximality of m . Hence $g(c) \neq 0$.

($\Leftarrow$) Suppose $f = (x - c)^m g$ with $g(c) \neq 0$. Then $(x - c)^m \mid f$, so $\text{mult}_c(f) \geq m$. Suppose, for a contradiction, that $(x - c)^{m+1} \mid f$, say $f = (x - c)^{m+1}h$. Then $(x - c)^m g = (x - c)^{m+1}h$. Since $(x - c)^m \neq 0$, cancellation ([Corollary 0.8.4](#)) gives $g = (x - c)h$, and evaluating at c gives $g(c) = 0$ by [Theorem 0.8.5](#), a contradiction. Every higher power $(x - c)^k$ with $k > m + 1$ would also give $(x - c)^{m+1} \mid f$. Hence $\text{mult}_c(f) = m$. ■

The first payoff is that multiplicities add when polynomials multiply, just as exponents do.

Theorem 5.4.2 — Multiplicity of a Product

Let $f, g \in F[x]$ be non-zero and $c \in F$. Then $\text{mult}_c(fg) = \text{mult}_c(f) + \text{mult}_c(g)$.

Proof. Let $m = \text{mult}_c(f)$ and $n = \text{mult}_c(g)$. By [Lemma 5.4.1](#), $f = (x - c)^m u$ and $g = (x - c)^n v$ with $u(c) \neq 0$ and $v(c) \neq 0$. Then $fg = (x - c)^{m+n} uv$, and $(uv)(c) = u(c)v(c) \neq 0$ by [Theorem 0.8.5](#) and [Theorem 0.7.1](#). By [Lemma 5.4.1](#) again, $\text{mult}_c(fg) = m + n$.

Example. (Finding a Multiplicity by Repeated Division)

Let $f = x^4 - 2x^3 + 2x - 1 \in \mathbb{Q}[x]$. Find the multiplicity of 1 and of -1 as roots of f .

Solution. We have $f(1) = 1 - 2 + 2 - 1 = 0$, so we divide by $x - 1$. Long division gives

$$f = (x - 1)(x^3 - x^2 - x + 1).$$

The cofactor $g_1 = x^3 - x^2 - x + 1$ has $g_1(1) = 0$, so we divide again: $g_1 = (x - 1)(x^2 - 1)$. The new cofactor $g_2 = x^2 - 1$ has $g_2(1) = 0$, and $g_2 = (x - 1)(x + 1)$. Now $g_3 = x + 1$ has $g_3(1) = 2 \neq 0$. Hence

$$f = (x - 1)^3(x + 1), \quad g_3(1) \neq 0,$$

and [Lemma 5.4.1](#) gives $\text{mult}_1(f) = 3$. For -1 , write $f = (x + 1) \cdot (x - 1)^3$, whose cofactor $(x - 1)^3$ takes the value $(-2)^3 = -8 \neq 0$ at -1 . So $\text{mult}_{-1}(f) = 1$, a simple root. The multiplicities add up to $3 + 1 = 4 = \deg f$.

Warning. Multiplicity is not visible from the set of roots. The polynomials $(x - 1)^3(x + 1)$ and $(x - 1)(x + 1)^3$ have the same roots $\{1, -1\}$, the same degree and the same leading coefficient, but they are different polynomials. “The roots of f ” as a set loses information; the roots **with their multiplicities** do not, as the next theorem shows for polynomials that split.

Quick check. Over $\mathbb{R}$, what is the multiplicity of 0 as a root of $x^3(x^2 + 1)$? Of 2 as a root of $(x^2 - 4)^2$?

Solution. For $x^3(x^2 + 1)$, the cofactor $x^2 + 1$ is $1 \neq 0$ at 0, so the multiplicity is 3 by [Lemma 5.4.1](#). For $(x^2 - 4)^2 = (x - 2)^2(x + 2)^2$, the cofactor $(x + 2)^2$ is $16 \neq 0$ at 2, so the multiplicity is 2.

5.4.2 Counting roots with multiplicity

In [Example \(Finding a Multiplicity by Repeated Division\)](#) the multiplicities added up to exactly the degree. That cannot always happen: $x^2 + 1 \in \mathbb{R}[x]$ has degree 2 and no real roots at all. The difference is whether f breaks into linear factors, so we give that a name.

Definition 5.4.2 — Splitting into Linear Factors

Let F be a field. A non-zero $f \in F[x]$ of degree n **splits over** F if there are $a \in F$ and $a_1, \dots, a_n \in F$, not necessarily distinct, with

$$f = a(x - a_1)(x - a_2) \cdots (x - a_n).$$

A non-zero constant splits (take $n = 0$ and the empty product 1). The polynomial $x^2 - 1 =$

$(x-1)(x+1)$ splits over $\mathbb{Q}$. The polynomial x^2-2 does not split over $\mathbb{Q}$, since a linear factor $x-a_1$ with $a_1 \in \mathbb{Q}$ would give a rational root, but it splits over $\mathbb{R}$ as $(x-\sqrt{2})(x+\sqrt{2})$. The phrase **over** F carries weight: the a_i must lie in F .

It turns out that the multiplicities always fit inside the degree, and they fill it exactly when f splits:

Theorem 5.4.3 — Roots Counted with Multiplicity

Let F be a field and $f \in F[x]$ non-zero. Let $c_1, \dots, c_k$ be the distinct roots of f in F , with multiplicities $m_1, \dots, m_k$. Then

$$f = (x - c_1)^{m_1} \cdots (x - c_k)^{m_k} g$$

for some $g \in F[x]$ with **no** root in F . Consequently

$$m_1 + \cdots + m_k \leq \deg f,$$

with equality if and only if f splits over F .

Idea. Peel off the roots one at a time. After removing $(x - c_1)^{m_1}$, the other roots c_j are still roots of the cofactor, because $c_j - c_1 \neq 0$, and by Theorem 5.4.2 they keep their multiplicities. Whatever is left at the end has no roots, and its degree is exactly the gap between $\deg f$ and $\sum m_i$. The gap is zero exactly when the leftover is a constant.

Proof. The set of roots is finite by Corollary 5.1.4, so the list $c_1, \dots, c_k$ makes sense. We prove the following claim by induction on $j \in \{0, 1, \dots, k\}$: $f = (x - c_1)^{m_1} \cdots (x - c_j)^{m_j} g_j$ for some non-zero $g_j \in F[x]$ with

$$\text{mult}_{c_i}(g_j) = 0 \text{ for } i \leq j, \quad \text{mult}_{c_i}(g_j) = m_i \text{ for } i > j.$$

For $j = 0$ take $g_0 = f$. Suppose the claim holds for some $j < k$. By Lemma 5.4.1 applied to g_j , we have $g_j = (x - c_{j+1})^{m_{j+1}} g_{j+1}$ with $g_{j+1}(c_{j+1}) \neq 0$, which gives the product formula for $j+1$ and $\text{mult}_{c_{j+1}}(g_{j+1}) = 0$. For $i \neq j+1$, the polynomial $(x - c_{j+1})^{m_{j+1}}$ takes the value $(c_i - c_{j+1})^{m_{j+1}} \neq 0$ at c_i , since the roots are distinct. So its multiplicity at c_i is 0 by Lemma 5.4.1 (with cofactor the polynomial itself), and Theorem 5.4.2 gives

$$\text{mult}_{c_i}(g_j) = 0 + \text{mult}_{c_i}(g_{j+1}).$$

This is 0 for $i \leq j$ and m_i for $i > j+1$, which completes the induction.

Put $g = g_k$. By the claim, $\text{mult}_{c_i}(g) = 0$, that is, $g(c_i) \neq 0$, for every i . If $d \in F$ were a root of g , then $f(d) = (d - c_1)^{m_1} \cdots (d - c_k)^{m_k} g(d) = 0$, so d would be one of the c_i , which is impossible. Hence g has no root in F .

Since $f \neq 0$, also $g \neq 0$, and Theorem 0.8.3 gives $\deg f = m_1 + \cdots + m_k + \deg g \geq m_1 + \cdots + m_k$.

($\Rightarrow$) If equality holds, then $\deg g = 0$, so $g = a$ is a non-zero constant and $f = a(x - c_1)^{m_1} \cdots (x - c_k)^{m_k}$ splits over F .

($\Leftarrow$) Suppose $f = a(x - a_1) \cdots (x - a_n)$ with $n = \deg f$. Group equal factors: $f = a(x - d_1)^{n_1} \cdots (x - d_l)^{n_l}$ with $d_1, \dots, d_l$ distinct and $n_1 + \cdots + n_l = n$. By the computation

above, the cofactor $a \prod_{s \neq r} (x - d_s)^{n_s}$ is non-zero at d_r , so $\text{mult}_{d_r}(f) = n_r$ by Lemma 5.4.1. Every root d of f satisfies $a \prod_s (d - d_s)^{n_s} = 0$, so d is one of the d_r . Hence the roots of f are exactly $d_1, \dots, d_l$, and their multiplicities add up to $n = \deg f$. This proves the theorem. ■

The theorem sharpens Corollary 5.1.4, because the number k of distinct roots is at most $m_1 + \dots + m_k$. It also shows that a split polynomial is determined by its leading coefficient and its roots with multiplicities. Whether a given polynomial splits depends on the field: the fifth-degree polynomial $(x-1)^3(x^2+1)$ has root multiplicities adding up to $3 < 5$ over $\mathbb{R}$, but it splits over $\mathbb{C}$ as $(x-1)^3(x-i)(x+i)$. In the next section we prove that over $\mathbb{C}$ every polynomial splits.

5.4.3 The formal derivative

How do we recognize a repeated root without factoring? Over $\mathbb{R}$ calculus has an answer. At a simple root the graph of f crosses the axis with non-zero slope, while at a double root, as for $(x-1)^2$ at 1, it touches the axis and the slope is 0. We would like to use this over every field, including $\mathbb{F}_p$, where there are no limits and no slopes. So we keep the formula of the derivative and throw away the limit.

The formal derivative is the power rule, applied coefficient by coefficient.

Definition 5.4.3 — Formal Derivative

Let F be a field and $f = a_0 + a_1x + \dots + a_nx^n \in F[x]$. The **formal derivative** of f is

$$f' = a_1 + 2a_2x + 3a_3x^2 + \dots + na_nx^{n-1} = \sum_{k=1}^n ka_kx^{k-1},$$

where ka_k is the integer multiple of a_k (the sum of k copies of a_k). Higher derivatives are defined by $f^{(0)} = f$ and $f^{(k+1)} = (f^{(k)})'$; we also write $f'' = f^{(2)}$.

In words: each monomial a_kx^k becomes ka_kx^{k-1} , and constants disappear. This is the map D of Example (Differentiation), where we already saw that $D: F[x] \rightarrow F[x]$ is linear. Over $\mathbb{R}$ it agrees with the derivative of calculus on polynomial functions, since calculus gives the same power rule. By Lemma 0.7.4, $ka_k = (k \cdot 1)a_k$, so the integer k acts through the element $k \cdot 1$ of F . **That element can be 0**, and this is the source of every difference between characteristic 0 and characteristic p .

Some examples. Over $\mathbb{Q}$, $(x^4 - 2x^3 + 2x - 1)' = 4x^3 - 6x^2 + 2$. Over any field, the derivative of a constant is 0 and $(x - c)' = 1$. Over $\mathbb{F}_3$, $(x^3 + x)' = 3x^2 + 1 = 1$, because $3 \cdot 1 = 0$ there. Over $\mathbb{F}_2$, $(x^2)' = 2x = 0$: a non-constant polynomial with derivative zero.

In characteristic 0 the last example cannot happen. If $\deg f = n \geq 1$, the coefficient of x^{n-1} in f' is $(n \cdot 1)a_n$, a product of two non-zero elements, so $\deg f' = n - 1$ and in particular $f' \neq 0$. In characteristic p the same coefficient vanishes whenever $p \mid n$, because then $n \cdot 1 = (n/p) \cdot (p \cdot 1) = 0$ by Lemma 0.7.4.

The derivative obeys the rules we know from calculus. The proofs cannot use limits, so they go through monomials.

Theorem 5.4.4 — Rules for the Formal Derivative

Let F be a field, $f, g \in F[x]$, $a, b, c \in F$ and $m \geq 1$ an integer. Then:

- (a) $(af + bg)' = af' + bg'$, and $a' = 0$ for the constant polynomial a ;
- (b) $(fg)' = f'g + fg'$ (product rule);
- (c) $(g^m)' = mg^{m-1}g'$; in particular $((x - c)^m)' = m(x - c)^{m-1}$.

Idea. Both sides of the product rule are linear in f for fixed g , and linear in g for fixed f . So it is enough to check the rule when f and g are monomials, where it is the identity $i + j = i + j$ for exponents. Checking on monomials, and extending by linearity, is the standard way to prove any identity about D .

Proof.

- (a) Linearity is [Example \(Differentiation\)](#). A constant has no coefficients in positions $k \geq 1$, so its derivative is 0.
- (b) First let $f = x^i$ and $g = x^j$ with $i, j \in \mathbb{N}$. If $i, j \geq 1$, then

$$(x^{i+j})' = (i+j)x^{i+j-1} = ix^{i-1}x^j + x^i jx^{j-1} = f'g + fg',$$

using $(i+j) \cdot 1 = i \cdot 1 + j \cdot 1$ from [Lemma 0.7.4](#). If $i = 0$, then $f = 1$, $f' = 0$ and both sides equal jx^{j-1} (or 0 when $j = 0$); the case $j = 0$ is symmetric.

Now let $f = \sum_i a_i x^i$ and $g = \sum_j b_j x^j$. By [Theorem 0.8.1](#), $fg = \sum_{i,j} a_i b_j x^i x^j$, a finite sum. Applying (a) and the monomial case,

$$(fg)' = \sum_{i,j} a_i b_j ((x^i)'x^j + x^i(x^j)') = \left(\sum_i a_i (x^i)'\right) \left(\sum_j b_j x^j\right) + \left(\sum_i a_i x^i\right) \left(\sum_j b_j (x^j)'\right) = f'g + fg',$$

where the middle equality regroups with the distributive laws, and the last uses (a) again.

- (c) We use induction on m . For $m = 1$ both sides are g' . If $(g^m)' = mg^{m-1}g'$, then by (b)

$$(g^{m+1})' = (g^m g)' = mg^{m-1}g'g + g^m g' = (m+1)g^m g'.$$

For $g = x - c$ we have $g' = 1$ by (a). This proves the theorem. ■

5.4.4 Repeated roots and the derivative

Now we can turn the calculus picture into a theorem valid over every field.

Theorem 5.4.5 — Repeated Root Test

Let F be a field, $f \in F[x]$ non-zero and $c \in F$.

- (a) c is a multiple root of f if and only if $f(c) = 0$ and $f'(c) = 0$.
- (b) Suppose $\text{mult}_c(f) = m \geq 1$ and $m \cdot 1 \neq 0$ in F (this holds automatically if F has characteristic 0). Then $f' \neq 0$ and $\text{mult}_c(f') = m - 1$.

Idea. Write $f = (x - c)g$ and differentiate: $f' = g + (x - c)g'$, so $f'(c) = g(c)$. The derivative at c reads off exactly the cofactor's value, which is what [Lemma 5.4.1](#) needs. For (b), differentiate $(x - c)^m g$ instead; the factor m appears, and it must not be 0 in F .

Proof.

- (a) ($\Rightarrow$) Suppose $f = (x - c)^2 g$. By [Theorem 5.4.4](#) (b) and (c), $f' = 2(x - c)g + (x - c)^2 g'$, and evaluating at c gives $f(c) = 0$ and $f'(c) = 0$.
 ($\Leftarrow$) Suppose $f(c) = 0$ and $f'(c) = 0$. By [Lemma 2.7.1](#), $f = (x - c)g$ for some $g \in F[x]$. By [Theorem 5.4.4](#), $f' = g + (x - c)g'$, so $0 = f'(c) = g(c)$. By [Lemma 2.7.1](#) again, $g = (x - c)h$, so $f = (x - c)^2 h$ and $\text{mult}_c(f) \geq 2$.
- (b) By [Lemma 5.4.1](#), $f = (x - c)^m g$ with $g(c) \neq 0$. By [Theorem 5.4.4](#),

$$f' = m(x - c)^{m-1}g + (x - c)^m g' = (x - c)^{m-1} h, \quad h := mg + (x - c)g'.$$

Then $h(c) = (m \cdot 1)g(c)$, a product of two non-zero elements of F , so $h(c) \neq 0$. In particular $h \neq 0$, so $f' \neq 0$ by [Corollary 0.8.4](#), and $\text{mult}_c(f') = m - 1$ by [Lemma 5.4.1](#). ■

Part (a) holds over **every** field. Part (b) needs its hypothesis. Over $\mathbb{F}_2$, let $f = x^2(x + 1) = x^3 + x^2$. Then $\text{mult}_0(f) = 2$, and $f' = 3x^2 + 2x = x^2$, so $\text{mult}_0(f') = 2$, not 1. Here $m \cdot 1 = 2 \cdot 1 = 0$, and the factor m killed the term that should have lowered the multiplicity.

Part (a) finds a repeated root at a known point c . To decide whether **any** repeated root exists, without knowing the roots, we use the gcd of f and f' , which the Euclidean algorithm ([Theorem 5.2.6](#)) computes by division alone. Repeated roots are a special case of repeated irreducible factors: c is a multiple root exactly when the irreducible polynomial $x - c$ appears squared.

Corollary 5.4.6 — Repeated Factors and $\gcd(f, f')$

Let F be a field and $f \in F[x]$ non-constant.

- (a) (**Any field.**) If $p \in F[x]$ is irreducible and $p^2 \mid f$, then $p \mid \gcd(f, f')$. Hence, if $\gcd(f, f') = 1$, then no irreducible polynomial divides f twice; in particular f has no multiple root in F .
- (b) (**Characteristic 0.**) If F has characteristic 0 and $\gcd(f, f') \neq 1$, then $p^2 \mid f$ for some irreducible $p \in F[x]$.

Idea. For (a), differentiate $p^2 h$: every term still contains p . For (b), run it backwards. An irreducible p dividing both $f = pg$ and $f' = p'g + pg'$ must divide $p'g$. If p does not

divide p' , then Euclid's lemma pushes p into g , and $p^2 \mid f$. The only question is whether $p \mid p'$ can happen, and that is exactly where the characteristic enters.

Proof.

- (a) Write $f = p^2h$. By [Theorem 5.4.4](#), $f' = 2pp'h + p^2h' = p(2p'h + ph')$. So p is a common divisor of f and f' , and $p \mid \gcd(f, f')$ by [Theorem 5.2.3](#). If $\gcd(f, f') = 1$, this would give $p \mid 1$, which is impossible because p is non-constant ([Definition 5.3.1](#)) and a non-zero multiple of p has degree at least $\deg p \geq 1$ by [Theorem 0.8.3](#). If c were a multiple root, then $(x - c)^2 \mid f$. But $x - c$ is irreducible ([Definition 5.3.1](#)), because a product of two non-constant polynomials has degree at least 2. This is again impossible.
- (b) Let $d = \gcd(f, f') \neq 1$. Since $d \mid f$ and $f \neq 0$, $d \neq 0$; as d is monic and $d \neq 1$, $\deg d \geq 1$. By [Theorem 5.3.5](#), d has a monic irreducible factor p . Then $p \mid f$ and $p \mid f'$ ([Theorem 5.2.3](#)). Write $f = pg$. By [Theorem 5.4.4](#), $f' = p'g + pg'$, so p divides $f' - pg' = p'g$.

Since F has characteristic 0 and $\deg p \geq 1$, we showed after [Definition 5.4.3](#) that $p' \neq 0$ and $\deg p' = \deg p - 1$. A non-zero multiple of p has degree at least $\deg p$, so $p \nmid p'$. By Euclid's lemma ([Theorem 5.3.3](#)), $p \mid g$. Hence $p^2 \mid pg = f$. This proves the corollary. ■

Warning. In characteristic p , a non-constant polynomial can have derivative 0. Over $\mathbb{F}_p$, $(x^p)' = (p \cdot 1)x^{p-1} = 0$. The proof of (b) used that an irreducible factor q of f has $q' \neq 0$ of smaller degree, so that $q \nmid q'$; in characteristic p that step is no longer automatic, because q' may be 0. So the correct statements are: $\gcd(f, f') = 1$ implies no repeated irreducible factor, over **any** field; the converse is proved here **only in characteristic 0**. Do not use “repeated factor if and only if $\gcd(f, f') \neq 1$ ” over a field of characteristic p without checking that no irreducible factor of f has derivative 0.

Remark. The converse really can fail in characteristic p , though not over the fields $\mathbb{F}_p$. Over the field of rational functions in a variable t with coefficients in $\mathbb{F}_p$, which this book does not construct, the polynomial $x^p - t$ is irreducible, so it has no repeated factor, yet its derivative is 0.

Example. (Detecting a Repeated Root with a gcd)

Decide whether $f = x^4 - 2x^3 + 2x - 1 \in \mathbb{Q}[x]$ has a repeated irreducible factor, using only $\gcd(f, f')$. Does f have a multiple root in $\mathbb{C}$?

Solution. Here $f' = 4x^3 - 6x^2 + 2$. Run the Euclidean algorithm ([Theorem 5.2.6](#)). Dividing f by f' ,

$$f = \left(\frac{1}{4}x - \frac{1}{8}\right)f' + \left(-\frac{3}{4}x^2 + \frac{3}{2}x - \frac{3}{4}\right),$$

and the remainder is $-\frac{3}{4}(x^2 - 2x + 1)$. Dividing f' by $x^2 - 2x + 1$,

$$f' = (4x + 2)(x^2 - 2x + 1) + 0.$$

(We divided by the monic multiple of the remainder; scaling a divisor by a non-zero constant changes only the quotient, so the next remainder is still 0.) The last non-zero

remainder, made monic, is the gcd: $\gcd(f, f') = x^2 - 2x + 1 = (x - 1)^2 \neq 1$. By [Corollary 5.4.6](#) (b), since $\mathbb{Q}$ has characteristic 0, f has a repeated irreducible factor. Indeed $x - 1$ divides the gcd, so 1 is a common root of f and f' , and by [Theorem 5.4.5](#) (a) it is a multiple root. This matches [Example \(Finding a Multiplicity by Repeated Division\)](#), where $\text{mult}_1(f) = 3$, and [Theorem 5.4.5](#) (b) gives $\text{mult}_1(f') = 2$, consistent with the factor $(x - 1)^2$ of the gcd.

So yes, f has a multiple root in $\mathbb{C}$, namely $1 \in \mathbb{Q}$. The gcd test is useful over $\mathbb{C}$ even for polynomials with rational coefficients: the Euclidean algorithm uses only divisions in $\mathbb{Q}[x]$, and division with remainder is unique ([Theorem 5.1.1](#)), so running it in $\mathbb{C}[x]$ produces the same remainders and the same gcd. Had the gcd been 1, part (a) over $\mathbb{C}$ would have ruled out every multiple complex root, without our finding a single root.

Quick check. Over $\mathbb{R}$, $f = x^3$ satisfies $f(0) = f'(0) = 0$. Does [Theorem 5.4.5](#) say that 0 has multiplicity 2?

Solution. No. Part (a) says only that the multiplicity is **at least** 2. The actual multiplicity is 3. Part (b) applies, since $\mathbb{R}$ has characteristic 0: $\text{mult}_0(f') = \text{mult}_0(3x^2) = 2 = 3 - 1$. To pin the multiplicity down with derivatives, find the first k with $f^{(k)}(0) \neq 0$: here $f''(0) = 0$ and $f'''(0) = 6 \neq 0$, so $k = 3$.

5.4.5 Taylor expansion of a polynomial

The answer to the check suggests that, in characteristic 0, the derivatives of f at c know everything about f near c . Calculus makes this precise with Taylor's formula. For polynomials the formula is exact, and it has a clean linear-algebra meaning: the numbers $f^{(k)}(c)/k!$ are the coordinates of f in the basis of powers of $x - c$, so the maps $f \mapsto f^{(k)}(c)/k!$ form a dual basis in the sense of [Definition 4.1.3](#). In [Example \(Dual Bases in \$\mathbb{R}\[x\]_{\leq 2}\$ \)](#) we saw this for $c = 0$ and $n = 2$.

We need one derivative computation. For integers $0 \leq j \leq k$, repeated use of [Theorem 5.4.4](#) (c) and (a) gives

$$((x - c)^k)^{(j)} = k(k - 1) \cdots (k - j + 1) (x - c)^{k-j},$$

by induction on j , and $((x - c)^k)^{(j)} = 0$ for $j > k$, since the k -th derivative is the constant $k!$. Evaluating at c , every term with $j < k$ vanishes because of the factor $(x - c)^{k-j}$. Hence

$$((x - c)^k)^{(j)}(c) = \begin{cases} k! \cdot 1 & \text{if } j = k, \\ 0 & \text{if } j \neq k. \end{cases}$$

Theorem 5.4.7 — Taylor Expansion of a Polynomial

Let F be a field, $n \in \mathbb{N}$ and $c \in F$.

- (a) The list $\mathcal{B}_c = (1, x - c, (x - c)^2, \dots, (x - c)^n)$ is a basis of $F[x]_{\leq n}$.
- (b) Suppose F has characteristic 0, or characteristic $p > n$. For $k = 0, \dots, n$ define

$\varphi_k: F[x]_{\leq n} \rightarrow F$ by $\varphi_k(f) = \frac{f^{(k)}(c)}{k!}$. Then $(\varphi_0, \dots, \varphi_n)$ is the dual basis of $\mathcal{B}_c$,

and every $f \in F[x]_{\leq n}$ satisfies

$$f = \sum_{k=0}^n \frac{f^{(k)}(c)}{k!} (x-c)^k.$$

Idea. ① The powers of $x-c$ have degrees $0, 1, \dots, n$, so no combination of them can cancel down to zero; there are $n+1$ of them, so counting finishes (a). ② The computation above says exactly $\varphi_j((x-c)^k) = \delta_{jk}$, once we may divide by $k!$. ③ A list of functionals with the δ property is the dual basis, and the dual basis expansion $\mathbf{v} = \sum \varphi_k(\mathbf{v})\mathbf{v}_k$ of Theorem 4.1.3 is Taylor's formula.

Proof.

- (a) Let $b_0 + b_1(x-c) + \dots + b_n(x-c)^n = 0$, and suppose not all b_k are zero. Let k be the largest index with $b_k \neq 0$. By Theorem 0.8.3, $b_k(x-c)^k$ has degree k , and each earlier term has degree less than k , so by Theorem 0.8.2 the sum has degree $k \geq 0$. This contradicts the sum being 0, of degree $-\infty$. Hence $\mathcal{B}_c$ is linearly independent. It has $n+1$ elements and $\dim F[x]_{\leq n} = n+1$ by Example (Standard Bases), so $\mathcal{B}_c$ is a basis by Theorem 1.6.4 (a).
- (b) First, $k! \cdot 1 \neq 0$ in F for $0 \leq k \leq n$. By Lemma 0.7.4, $k! \cdot 1 = (1 \cdot 1)(2 \cdot 1) \cdots (k \cdot 1)$. In characteristic 0 each factor is non-zero by Definition 0.7.2; in characteristic $p > n$, each factor $j \cdot 1$ with $1 \leq j \leq k < p$ is non-zero by the minimality of p . A product of non-zero elements is non-zero (Theorem 0.7.1). So φ_k is defined.

Each φ_k is linear: $f \mapsto f^{(k)}$ is linear as a composite of copies of D (Example (Differentiation)), evaluation at c is linear by Theorem 0.8.5, and so is multiplication by $(k!)^{-1}$. The derivatives of a polynomial of degree at most n have degree at most n , so $\varphi_k \in F[x]_{\leq n}^*$. By the computation before the theorem,

$$\varphi_j((x-c)^k) = \frac{1}{j!} ((x-c)^k)^{(j)}(c) = \delta_{jk} \quad (0 \leq j, k \leq n).$$

By the uniqueness in Theorem 4.1.3 (a), $(\varphi_0, \dots, \varphi_n)$ is the dual basis of $\mathcal{B}_c$. The first formula of Theorem 4.1.3 (c), $f = \sum_k \varphi_k(f)(x-c)^k$, is the Taylor expansion. This proves the theorem. ■

Example. (Re-centering a Cubic)

Expand $f = x^3 - 2x + 5 \in \mathbb{Q}[x]$ in powers of $x-1$.

Solution. The derivatives are $f' = 3x^2 - 2$, $f'' = 6x$, $f''' = 6$. At $c = 1$: $f(1) = 4$, $f'(1) = 1$, $f''(1) = 6$, $f'''(1) = 6$. By Theorem 5.4.7 (b), since $\mathbb{Q}$ has characteristic 0,

$$f = 4 + 1 \cdot (x-1) + \frac{6}{2}(x-1)^2 + \frac{6}{6}(x-1)^3 = 4 + (x-1) + 3(x-1)^2 + (x-1)^3.$$

Check by expanding: $(x-1)^3 + 3(x-1)^2 + (x-1) + 4 = (x^3 - 3x^2 + 3x - 1) + (3x^2 - 6x + 3) + (x-1) + 4 = x^3 - 2x + 5$.

The expansion also reads off root information at 1: the constant term $f(1) = 4 \neq 0$, so 1 is not a root. In general, $\text{mult}_c(f)$ is the index of the first non-zero coordinate of f in the basis $\mathcal{B}_c$ (over any field, by part (a); in characteristic 0 these coordinates are the Taylor coefficients), because $f = (x - c)^m(b_m + b_{m+1}(x - c) + \cdots)$ and the bracket is b_m at c .

Warning. **Taylor's formula needs $k! \neq 0$.** Over $\mathbb{F}_2$, take $f = x^2$ and $c = 0$. Then $f' = 2x = 0$, so **every** derivative of f , at every point, is 0, even though $f \neq 0$. No formula in the values $f^{(k)}(0)$ can recover the coefficient 1 of x^2 . Part (a) of [Theorem 5.4.7](#) still holds over $\mathbb{F}_2$: x^2 has coordinates $(0, 0, 1)$ in $(1, x, x^2)$. What fails is that these coordinates are derivatives divided by $k!$.

The two halves of this section meet in Chapter 8, where a matrix is diagonalizable exactly when its minimal polynomial splits and has no repeated roots; the gcd test above is how such a condition is checked in practice. Before that, the next section proves that over $\mathbb{C}$ the sum of the multiplicities always equals the degree.

5.4.6 Exercises

5.4.6.1 A. Check your understanding

Exercise A1.

- Define the multiplicity of c as a root of a non-zero $f \in F[x]$. Why is $f \neq 0$ required?
- True or false: if $f(c) = f'(c) = 0$, then c has multiplicity exactly 2. Justify your answer.
- True or false: over every field, $f' = 0$ implies that f is constant. Justify your answer.
- True or false: if $f \in \mathbb{R}[x]$ has degree 4 and the multiplicities of its real roots add up to 4, then f splits over $\mathbb{R}$. Justify your answer.
- True or false: if $f \in \mathbb{Q}[x]$ is non-constant and $\gcd(f, f') \neq 1$, then f has a repeated irreducible factor. Justify your answer.
- Name the dual basis ([Theorem 4.1.3](#)) of $(1, x - c, \dots, (x - c)^n)$ in $\mathbb{R}[x]_{\leq n}$.

Solution.

- It is the largest $m \in \mathbb{N}$ with $(x - c)^m \mid f$ ([Definition 5.4.1](#)). For $f = 0$, every power divides f , so no largest m exists.
- False. For $f = x^3$ and $c = 0$, $f(0) = f'(0) = 0$, but $\text{mult}_0(f) = 3$. [Theorem 5.4.5](#) (a) only gives multiplicity at least 2.
- False. Over $\mathbb{F}_2$, $(x^2)' = 2x = 0$. (In characteristic 0 it is true: $\deg f' = \deg f - 1$ whenever $\deg f \geq 1$.)
- True. By [Theorem 5.4.3](#), the sum of multiplicities equals the degree if and only if f splits over $\mathbb{R}$.

- (e) True. $\mathbb{Q}$ has characteristic 0, so this is [Corollary 5.4.6](#) (b).
 (f) $\varphi_k(f) = f^{(k)}(c)/k!$ for $k = 0, \dots, n$, by [Theorem 5.4.7](#) (b); $\mathbb{R}$ has characteristic 0.

5.4.6.2 B. Practice

Exercise B1: Multiplicities over different fields.

- (a) Let $f = x^5 - 3x^4 + 4x^3 - 4x^2 + 3x - 1$. Find $\text{mult}_1(f)$ in $\mathbb{Q}[x]$. Hence decide whether f splits over $\mathbb{R}$, and whether it splits over $\mathbb{C}$.
 (b) In $\mathbb{F}_2[x]$, find the roots of $g = x^4 + x^2$ and their multiplicities. Does g split over $\mathbb{F}_2$?

Solution.

- (a) Here $f(1) = 1 - 3 + 4 - 4 + 3 - 1 = 0$. Dividing by $x - 1$ repeatedly,

$$f = (x-1)(x^4 - 2x^3 + 2x^2 - 2x + 1), \quad x^4 - 2x^3 + 2x^2 - 2x + 1 = (x-1)(x^3 - x^2 + x - 1),$$

and $x^3 - x^2 + x - 1 = (x-1)(x^2 + 1)$. Since $x^2 + 1$ takes the value $2 \neq 0$ at 1, [Lemma 5.4.1](#) gives $f = (x-1)^3(x^2 + 1)$ and $\text{mult}_1(f) = 3$. Over $\mathbb{R}$, $x^2 + 1$ has no root ($c^2 + 1 \geq 1$), so the only real root of f is 1, and $3 < 5 = \deg f$. By [Theorem 5.4.3](#), f does not split over $\mathbb{R}$. Over $\mathbb{C}$, $x^2 + 1 = (x-i)(x+i)$, so $f = (x-1)^3(x-i)(x+i)$ splits.

- (b) In $\mathbb{F}_2[x]$, $x^4 + x^2 = x^2(x^2 + 1)$, and $x^2 + 1 = (x+1)^2$ because $(x+1)^2 = x^2 + 2x + 1 = x^2 + 1$. Hence $g = x^2(x+1)^2 = (x-0)^2(x-1)^2$, using $-1 = 1$. The cofactor $(x+1)^2$ is 1 at 0, and x^2 is 1 at 1, so $\text{mult}_0(g) = \text{mult}_1(g) = 2$ by [Lemma 5.4.1](#). These are the only elements of $\mathbb{F}_2$, the multiplicities add to $4 = \deg g$, and g splits over $\mathbb{F}_2$.

Exercise B2: A gcd test.

Decide whether $f = x^4 - 2x^2 + 1 \in \mathbb{Q}[x]$ has a repeated root, by computing $\gcd(f, f')$ with the Euclidean algorithm. Hence factor f and give all multiplicities.

Solution. We have $f' = 4x^3 - 4x$. Dividing, $f = \frac{1}{4}x \cdot f' + (-x^2 + 1)$, since $\frac{1}{4}x(4x^3 - 4x) = x^4 - x^2$. Next, $f' = (-4x)(-x^2 + 1) + 0$. So the last non-zero remainder is $-x^2 + 1$, and by [Theorem 5.2.6](#) $\gcd(f, f') = x^2 - 1 \neq 1$. Its roots 1 and -1 are common roots of f and f' , since $x^2 - 1 = (x-1)(x+1)$ divides both. By [Theorem 5.4.5](#) (a), 1 and -1 are multiple roots of f .

To factor, note that $(x^2 - 1)^2 = x^4 - 2x^2 + 1 = f$, so $f = (x-1)^2(x+1)^2$. By [Lemma 5.4.1](#), $\text{mult}_1(f) = 2$ (cofactor $(x+1)^2$, value 4 at 1) and $\text{mult}_{-1}(f) = 2$.

Exercise B3: The multiplicity of 1.

Let $n \geq 2$ and $f = x^n - nx + n - 1 \in \mathbb{Q}[x]$. Prove that 1 is a root of f of multiplicity exactly 2.

Solution. We have $f(1) = 1 - n + n - 1 = 0$, $f' = nx^{n-1} - n$, so $f'(1) = 0$, and $f'' = n(n-1)x^{n-2}$, so $f''(1) = n(n-1) \neq 0$ since $n \geq 2$. In particular $f \neq 0$. By [Theorem 5.4.5](#) (a), $m := \text{mult}_1(f) \geq 2$. Since $\mathbb{Q}$ has characteristic 0, part (b) gives $\text{mult}_1(f') = m - 1 \geq 1$. Apply (a) to $f' \neq 0$: since $f''(1) \neq 0$, 1 is not a multiple root of f' , so $m - 1 = 1$. Hence $m = 2$.

5.4.6.3 C. Going deeper

Exercise C1: Real roots of the derivative.

Let $f \in \mathbb{R}[x]$ have degree $n \geq 2$ and split over $\mathbb{R}$. Prove that f' splits over $\mathbb{R}$. You may use **Rolle's theorem** from calculus: if $a < b$ are real and $f(a) = f(b) = 0$, then $f'(\xi) = 0$ for some ξ with $a < \xi < b$.

Hint: count the roots of f' with multiplicity, and compare with $\deg f'$.

Solution. Let $c_1 < c_2 < \dots < c_k$ be the distinct real roots of f , with multiplicities $m_1, \dots, m_k$. Since f splits over $\mathbb{R}$, [Theorem 5.4.3](#) gives $m_1 + \dots + m_k = n$.

By Rolle's theorem, for each $i = 1, \dots, k-1$ there is ξ_i with $c_i < \xi_i < c_{i+1}$ and $f'(\xi_i) = 0$. These $k-1$ numbers are distinct, since they lie in disjoint open intervals, and none of them is a c_j . Moreover, each c_i with $m_i \geq 2$ is a root of f' of multiplicity $m_i - 1$, by [Theorem 5.4.5](#) (b) (characteristic 0); for $m_i = 1$ the count $m_i - 1 = 0$ is also correct.

Since $\deg f = n \geq 1$ and $\mathbb{R}$ has characteristic 0, $\deg f' = n - 1$, so $f' \neq 0$. Adding the multiplicities of the distinct roots $\xi_1, \dots, \xi_{k-1}, c_1, \dots, c_k$ of f' gives at least

$$(k-1) + \sum_{i=1}^k (m_i - 1) = (k-1) + n - k = n - 1 = \deg f'.$$

By [Theorem 5.4.3](#) the sum of the multiplicities of all roots of f' is at most $\deg f'$, so it equals $\deg f'$, and f' splits over $\mathbb{R}$.

Exercise C2: $x^p - x$ over $\mathbb{F}_p$.

Let p be a prime and $f = x^p - x \in \mathbb{F}_p[x]$.

- Compute f' and deduce that f has no repeated irreducible factor.
- Using [Exercise \(C2: A non-zero polynomial vanishing everywhere\)](#), deduce that $x^p - x = \prod_{a \in \mathbb{F}_p} (x - a)$.
- Explain why the argument of (a) gives nothing for $g = x^p - 1$. Then prove that $g = (x - 1)^p$ in $\mathbb{F}_p[x]$.

Hint for (c): for $0 < k < p$, use the integer identity $k \binom{p}{k} = p \binom{p-1}{k-1}$.

Solution.

- By [Definition 5.4.3](#), $f' = px^{p-1} - 1 = -1$, since $p \cdot 1 = 0$ in $\mathbb{F}_p$. Every common divisor of f and -1 divides 1, so $\gcd(f, f') = 1$, and [Corollary 5.4.6](#) (a), which holds over every field, shows f has no repeated irreducible factor.

(b) By [Exercise \(C2: A non-zero polynomial vanishing everywhere\)](#) (c), every $a \in \mathbb{F}_p$ is a root of f , so each of the p elements has multiplicity at least 1. The multiplicities add up to at least $p = \deg f$, and at most $\deg f$ by [Theorem 5.4.3](#). So they add up to exactly p , and each is 1. By [Theorem 5.4.3](#), $f = \prod_{b \in \mathbb{F}_p} (x - b) g$ with $\deg g = p - p = 0$, so g is a constant; comparing leading coefficients ([Theorem 0.8.3](#)), $g = 1$.

(c) Here $g' = px^{p-1} = 0$, so $\gcd(g, g') = g \neq 1$ (the monic generator of the ideal generated by g and 0). Part (a) of [Corollary 5.4.6](#) needs $\gcd = 1$, so it says nothing, and part (b) is only available in characteristic 0. Still, $g(1) = 0$ and $g'(1) = 0$, so 1 is a multiple root by [Theorem 5.4.5](#) (a).

For the factorization, expand with the binomial theorem (valid in the commutative ring $\mathbb{F}_p[x]$, [Theorem 0.8.1](#)): $(x - 1)^p = \sum_{k=0}^p \binom{p}{k} (-1)^{p-k} x^k$, with the integers $\binom{p}{k}$ acting as integer multiples. For $0 < k < p$, the identity $k \binom{p}{k} = p \binom{p-1}{k-1}$ and [Lemma 0.7.4](#) give $(k \cdot 1) \left(\binom{p}{k} \cdot 1 \right) = (p \cdot 1) \left(\binom{p-1}{k-1} \cdot 1 \right) = 0$ in $\mathbb{F}_p$. Since $0 < k < p$, $k \cdot 1 \neq 0$ by [Definition 0.7.2](#), so $\binom{p}{k} \cdot 1 = 0$ ([Theorem 0.7.1](#)). Hence $(x - 1)^p = x^p + (-1)^p$. For odd p this is $x^p - 1$; for $p = 2$ it is $x^2 + 1 = x^2 - 1$. So $g = (x - 1)^p$, and $\text{mult}_1(g) = p$ by [Lemma 5.4.1](#).

Exercise C3: Removing repeated factors.

Let F be a field and $f = ap_1^{e_1} \cdots p_r^{e_r} \in F[x]$, where $a \in F \setminus \{0\}$, the p_i are **distinct** monic irreducible polynomials and $e_i \geq 1$.

- (a) Suppose F has characteristic 0. Prove that $\gcd(f, f') = p_1^{e_1-1} \cdots p_r^{e_r-1}$, and deduce that $f / \gcd(f, f') = ap_1 \cdots p_r$.
- (b) Over $\mathbb{F}_2$, let $f = x^2(x + 1)$. Compute $f / \gcd(f, f')$ and show that the conclusion of (a) fails.

Hint for (a): write $D = p_1^{e_1-1} \cdots p_r^{e_r-1}$. Show $D \mid \gcd(f, f')$; for the reverse, show $p_i^{e_i} \nmid f'$ using Euclid's lemma.

Solution.

- (a) Let $D = \prod_i p_i^{e_i-1}$ and $d = \gcd(f, f')$. By [Theorem 5.4.4](#) (b) and (c), applied repeatedly,

$$f' = a \sum_{i=1}^r e_i p_i^{e_i-1} p_i' \prod_{j \neq i} p_j^{e_j}.$$

Every term is divisible by D , and so is f . Hence $D \mid d$ by [Theorem 5.2.3](#).

Conversely, $d \mid f$, so $f = dh$. Factoring d and h into monic irreducibles and comparing with the factorization of f by the uniqueness in [Theorem 5.3.5](#), $d = \prod_i p_i^{k_i}$ with $0 \leq k_i \leq e_i$ (as d is monic). Suppose $k_i = e_i$ for some i . Then $p_i^{e_i} \mid f'$. In the formula for f' every term with index $j \neq i$ contains $p_i^{e_i}$, so $p_i^{e_i}$ divides the remaining term $ae_i p_i^{e_i-1} p_i' \prod_{j \neq i} p_j^{e_j}$. Canceling $p_i^{e_i-1}$ ([Corollary 0.8.4](#)), $p_i \mid (ae_i) p_i' \prod_{j \neq i} p_j^{e_j}$. Here $ae_i = a(e_i \cdot 1)$ is a non-zero constant (characteristic 0), $p_i \nmid p_i'$ because $p_i' \neq 0$ has smaller degree, and $p_i \nmid p_j$ for $j \neq i$ because both are monic irreducible and distinct. Applying [Theorem 5.3.3](#) repeatedly gives a contradiction. Hence $k_i \leq e_i - 1$ for all i , so $d \mid D$. Both are monic and divide each other, so they have the same degree and $d = D$. Finally $f = D \cdot ap_1 \cdots p_r$, so $f/d = ap_1 \cdots p_r$.

- (b) In $\mathbb{F}_2[x]$, $f = x^3 + x^2$ and $f' = 3x^2 + 2x = x^2$. Since $x^2 \mid f$, $\gcd(f, f') = x^2$, and $f/\gcd(f, f') = x + 1$. The formula of (a) would predict $\gcd(f, f') = x$ and $f/\gcd(f, f') = x(x + 1)$. The irreducible factor x has been lost. The step that fails is “ $ae_i \neq 0$ ”: here $e_1 = 2$ and $2 \cdot 1 = 0$ in $\mathbb{F}_2$.

5.5 The Fundamental Theorem of Algebra

Chapter 0 stated the Fundamental Theorem of Algebra ([Theorem 0.6.7](#)) and promised a proof. Everything algebraic that the proof needs is now in place: division, the Factor Theorem, multiplicities, and the polar form of complex numbers with De Moivre’s formula. What algebra alone cannot supply is a property of the real numbers that goes beyond the field axioms, and some such property must enter, because $\mathbb{Q}$ is a field too and there $x^2 - 2$ has no root. We use completeness, in the form of the extreme value theorem. This section isolates the analytic input in two clearly labeled facts, proves everything else in full, and then draws the consequences for real polynomials: conjugate pairs, irreducible factors of degree at most 2, and real roots of odd-degree polynomials.

5.5.1 The goal and the plan

Here is the theorem we want, in the language of this chapter. A polynomial $p \in \mathbb{C}[x]$ of degree $n \geq 1$ is exactly the expression $a_n x^n + \cdots + a_0$ with $a_n \neq 0$ of [Theorem 0.6.7](#).

Theorem 5.5.1 — Fundamental Theorem of Algebra

Every non-constant polynomial $p \in \mathbb{C}[x]$ has a root in $\mathbb{C}$.

We prove it after three lemmas. The proof is a minimum argument, and it runs in three steps.

① **Growth.** Far from the origin, $|p(z)|$ is large, because the leading term $a_n z^n$ outweighs all the others. We will prove this with an explicit radius.

② **A minimum exists.** On a large closed disc, the continuous function $z \mapsto |p(z)|$ attains a smallest value. By ①, nothing outside the disc does better, so this is the smallest value of $|p|$ on all of $\mathbb{C}$.

③ **A non-zero value is never the minimum.** If $p(z_0) \neq 0$, we can move from z_0 a little in a well-chosen direction and make $|p|$ strictly smaller. This is d’Alembert’s lemma, and it is pure algebra of $\mathbb{C}$.

Putting ② and ③ together, the minimum value must be 0, and the point where it is attained is a root.

Steps ① and ③ are proved from scratch. Step ② is where analysis enters, and we say exactly how.

Remark. What we assume from analysis. The proof uses exactly two facts from real analysis, and nothing else: no limits, no derivatives, no intermediate value theorem. We identify $\mathbb{C}$ with $\mathbb{R}^2$ as in [Definition 0.6.1](#), so that $|z - w|$ is the usual distance in the plane.

- (A1) **Continuity of polynomial functions.** For every $p \in \mathbb{C}[x]$, the function $\mathbb{C} \rightarrow \mathbb{R}$, $z \mapsto |p(z)|$, is continuous.
- (A2) **Extreme value theorem on a closed disc.** Let $\rho \geq 0$ and $D = \{z \in \mathbb{C} : |z| \leq \rho\}$. If $g: D \rightarrow \mathbb{R}$ is continuous, then there is $w \in D$ with $g(w) \leq g(z)$ for **every** $z \in D$.

Fact (A2) is where the completeness of $\mathbb{R}$ is used. Everything else below is the arithmetic of $\mathbb{C}$ from Chapter 0, including polar form and [Theorem 0.6.6](#).

5.5.2 Step 1: growth

We use three small facts about the modulus repeatedly. For $z, w, z_1, \dots, z_m \in \mathbb{C}$ and $j \in \mathbb{N}$:

$$|z^j| = |z|^j, \quad |z_1 + \dots + z_m| \leq |z_1| + \dots + |z_m|, \quad |z + w| \geq |z| - |w|.$$

The first follows from [Theorem 0.6.2](#) (d) by induction on j , and the second from [Theorem 0.6.4](#) by induction on m . For the third, apply the triangle inequality to $z = (z + w) + (-w)$ and use $|-w| = |w|$.

The leading term wins as soon as $|z|$ exceeds a radius computed from the coefficients.

Lemma 5.5.2 — Growth of a Polynomial

Let $p = a_n x^n + \dots + a_1 x + a_0 \in \mathbb{C}[x]$ with $n \geq 1$ and $a_n \neq 0$. Put

$$A = |a_0| + |a_1| + \dots + |a_{n-1}|, \quad R = \max\left\{1, \frac{2A}{|a_n|}\right\}.$$

Then for every $z \in \mathbb{C}$ with $|z| \geq R$,

$$|p(z)| \geq \frac{1}{2}|a_n||z|^n \geq \frac{1}{2}|a_n||z|.$$

In particular, for every real $M \geq 0$, we have $|p(z)| > M$ whenever $|z| \geq R$ and $|z| > 2M/|a_n|$.

Idea. Compare the leading term with the rest. When $|z| \geq 1$, every lower power $|z|^j$ is at most $|z|^{n-1}$, so the rest is at most $A|z|^{n-1}$, while the leading term is $|a_n||z|^n$. The rest is at most half the leading term once $A \leq \frac{1}{2}|a_n||z|$, which is how R was found.

Proof. Let $|z| \geq R$. Since $|z| \geq 1$, we have $|z|^j \leq |z|^{n-1}$ for $0 \leq j \leq n-1$. By the triangle inequality ([Theorem 0.6.4](#)),

$$\left| \sum_{j=0}^{n-1} a_j z^j \right| \leq \sum_{j=0}^{n-1} |a_j| |z|^j \leq A |z|^{n-1}.$$

Since $|z| \geq 2A/|a_n|$, we have $A \leq \frac{1}{2}|a_n||z|$. By the reverse triangle inequality,

$$|p(z)| \geq |a_n||z|^n - A|z|^{n-1} \geq |a_n||z|^n - \frac{1}{2}|a_n||z|^n = \frac{1}{2}|a_n||z|^n.$$

Since $|z| \geq 1$ and $n \geq 1$, $|z|^n \geq |z|$, which gives the second inequality. Finally, if also $|z| > 2M/|a_n|$, then $|p(z)| \geq \frac{1}{2}|a_n||z| > M$. This proves the lemma.

For example, for $p = x^3 - 5x + 1$ we get $A = 6$ and $R = 12$, so $|p(z)| \geq \frac{1}{2}|z|^3$ whenever $|z| \geq 12$. The radius is far from optimal, and it does not need to be. We only need **some** radius beyond which $|p|$ is large. ■

5.5.3 Step 2: the minimum is attained

Lemma 5.5.3 — The Minimum of $|p|$ Is Attained

Let $p \in \mathbb{C}[x]$. Then there is $z_0 \in \mathbb{C}$ with $|p(z_0)| \leq |p(z)|$ for **every** $z \in \mathbb{C}$.

Idea. Use (A2) on a disc so big that outside it, $|p|$ exceeds $|p(0)|$. The disc contains 0, so its minimum is at most $|p(0)|$, and the points outside cannot compete. Lemma 5.5.2 tells us how big “so big” is.

Proof. If p is constant, every z_0 works. Suppose $p = a_n x^n + \cdots + a_0$ with $n \geq 1$ and $a_n \neq 0$, and let R be the radius of Lemma 5.5.2. Put $M = |p(0)|$ and

$$\rho = R + \frac{2M}{|a_n|}, \quad D = \{z \in \mathbb{C} : |z| \leq \rho\}.$$

By (A1), $z \mapsto |p(z)|$ is continuous on D , so by (A2) there is $z_0 \in D$ with $|p(z_0)| \leq |p(z)|$ for every $z \in D$.

Now let $z \notin D$, so $|z| > \rho$. Then $|z| > R$, and $|z| > 2M/|a_n|$ because $R \geq 1 > 0$. By Lemma 5.5.2, $|p(z)| > M = |p(0)|$. Since $0 \in D$, $|p(0)| \geq |p(z_0)|$. Hence $|p(z)| > |p(z_0)|$ for $z \notin D$, and $|p(z)| \geq |p(z_0)|$ for $z \in D$. This proves the lemma. ■

5.5.4 Step 3: d’Alembert’s lemma

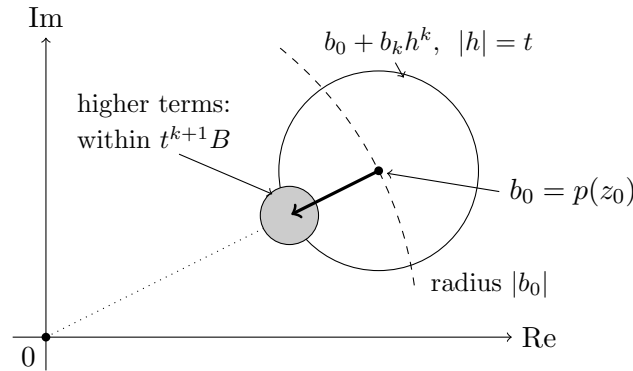
It remains to show that a non-zero value of $|p|$ is never the smallest one. Near z_0 , write $p(z_0 + h) = b_0 + b_k h^k + (\text{terms of higher degree in } h)$, where $b_0 = p(z_0)$ and $b_k h^k$ is the first non-constant term that is actually present. For small h , the term $b_k h^k$ dominates the later ones, so we choose the direction of h to make $b_k h^k$ point from b_0 straight toward 0.

Lemma 5.5.4 — d’Alembert’s Lemma

Let $p \in \mathbb{C}[x]$ be non-constant, and let $z_0 \in \mathbb{C}$ with $p(z_0) \neq 0$. Then there is $z \in \mathbb{C}$ with $|p(z)| < |p(z_0)|$.

Idea. Picture the values $p(z_0 + h)$ in the plane. As $h = tu$ runs over a small circle $|h| = t$, the term $b_0 + b_k h^k$ runs k times around the circle of radius $t^k |b_k|$ centered at $b_0 = p(z_0)$. One point of that circle lies on the segment from b_0 to 0, at distance $|b_0| - t^k |b_k|$ from 0. Reaching it needs a k -th root of a unit complex number, which De Moivre supplies. The

higher terms move the value by at most a multiple of t^{k+1} , a blur much smaller than the gain $t^k|b_k|$ when t is small.



The dashed arc is part of the circle of radius $|b_0|$ about 0, the old value of $|p|$. The solid circle shows where $b_0 + b_k h^k$ goes for $|h| = t$. The arrow is the chosen direction, and the shaded disc is the uncertainty caused by the higher terms. The whole shaded disc lies strictly inside the dashed circle, which is the inequality $|p(z_0 + h)| < |p(z_0)|$.

Proof. Let $n = \deg p \geq 1$.

Step 1: re-center at z_0 . By [Theorem 5.4.7](#) (a), $(1, x - z_0, \dots, (x - z_0)^n)$ is a basis of $\mathbb{C}[x]_{\leq n}$, so $p = \sum_{j=0}^n b_j (x - z_0)^j$ for some $b_0, \dots, b_n \in \mathbb{C}$. Evaluating at z_0 gives $b_0 = p(z_0) \neq 0$. The coefficient of x^n on the right is b_n , since $(x - z_0)^j$ has degree j and $(x - z_0)^n$ is monic; so b_n is the leading coefficient of p , and $b_n \neq 0$. By [Theorem 0.8.5](#), for every $h \in \mathbb{C}$,

$$p(z_0 + h) = \sum_{j=0}^n b_j h^j.$$

Since $b_n \neq 0$ and $n \geq 1$, there is a **smallest** $k \geq 1$ with $b_k \neq 0$. Then $b_1 = \dots = b_{k-1} = 0$, and

$$p(z_0 + h) = b_0 + b_k h^k + \sum_{j=k+1}^n b_j h^j. \quad (*)$$

Step 2: choose the direction. Since $b_0 \neq 0$ and $b_k \neq 0$, the number $w = -b_0/b_k$ is non-zero. Write it in polar form, $w = |w|e^{i\alpha}$ with $\alpha \in \mathbb{R}$, and put $u = e^{i\alpha/k}$. Then $|u| = 1$, and $u^k = e^{i\alpha}$ by [Theorem 0.6.6](#). Using $b_k w = -b_0$ and $|w| = |b_0|/|b_k|$ ([Theorem 0.6.2](#) (d)),

$$b_k u^k = \frac{b_k w}{|w|} = -\frac{|b_k|}{|b_0|} b_0.$$

Hence, for every real $t > 0$,

$$b_0 + b_k (tu)^k = b_0 \left(1 - t^k \frac{|b_k|}{|b_0|} \right).$$

Step 3: choose the size. Put $B = |b_{k+1}| + \dots + |b_n|$ (so $B = 0$ if $k = n$), and

$$t = \min \left\{ 1, \frac{|b_0|}{|b_k|}, \frac{|b_k|}{2B} \right\},$$

where the last entry is omitted if $B = 0$. Then $t > 0$, and t has three properties.

- (i) $0 < t^k |b_k|/|b_0| \leq 1$. Indeed, if $|b_0|/|b_k| \geq 1$, then $t^k \leq 1 \leq |b_0|/|b_k|$; otherwise $t \leq |b_0|/|b_k| < 1$, and $t^k \leq t \leq |b_0|/|b_k|$.
- (ii) $tB \leq \frac{1}{2}|b_k|$. This is trivial if $B = 0$, and follows from $t \leq |b_k|/(2B)$ otherwise.
- (iii) $t^j \leq t^{k+1}$ for every $j \geq k+1$, since $0 < t \leq 1$.

Step 4: estimate. Let $z = z_0 + tu$. By Step 2 and (i), the number $1 - t^k |b_k|/|b_0|$ is real and lies in $[0, 1)$, so

$$|b_0 + b_k(tu)^k| = |b_0| \left(1 - t^k \frac{|b_k|}{|b_0|}\right) = |b_0| - t^k |b_k|.$$

By the triangle inequality (Theorem 0.6.4), $|u| = 1$ and (iii),

$$\left| \sum_{j=k+1}^n b_j(tu)^j \right| \leq \sum_{j=k+1}^n |b_j| t^j \leq t^{k+1} B.$$

Combining these with (*), the triangle inequality and (ii),

$$|p(z)| \leq |b_0| - t^k |b_k| + t^{k+1} B = |b_0| - t^k (|b_k| - tB) \leq |b_0| - \frac{1}{2} t^k |b_k| < |b_0|,$$

where the last step uses $t > 0$ and $b_k \neq 0$. Since $|b_0| = |p(z_0)|$, this proves the lemma. ■

Every hypothesis did a job. Non-constant gave the index k . The condition $p(z_0) \neq 0$ let us divide by $|b_0|$ and fix a direction toward 0. And the field $\mathbb{C}$ supplied the k -th root u , the one step that has no analogue over $\mathbb{R}$.

Example. (One Step Downhill)

Let $p = x^3 + x^2 + 1 \in \mathbb{C}[x]$ and $z_0 = 0$. Follow the proof of Lemma 5.5.4 to find z with $|p(z)| < |p(0)|$.

Solution. Here $p(0) = 1$ and the expansion at $z_0 = 0$ is p itself: $b_0 = 1$, $b_1 = 0$, $b_2 = 1$, $b_3 = 1$. The first non-zero b_j with $j \geq 1$ is b_2 , so $k = 2$ and $B = |b_3| = 1$.

Direction: $w = -b_0/b_2 = -1 = e^{i\pi}$, so $\alpha = \pi$ and $u = e^{i\pi/2} = i$. Indeed $b_2 u^2 = -1$ points from $b_0 = 1$ toward 0.

Size: $t = \min\{1, 1/1, 1/(2 \cdot 1)\} = \frac{1}{2}$. So $z = \frac{i}{2}$, and

$$p\left(\frac{i}{2}\right) = -\frac{i}{8} - \frac{1}{4} + 1 = \frac{3}{4} - \frac{i}{8}, \quad \left|p\left(\frac{i}{2}\right)\right| = \sqrt{\frac{9}{16} + \frac{1}{64}} = \frac{\sqrt{37}}{8} \approx 0.76 < 1.$$

The proof guarantees $|p(z)| \leq 1 - \frac{1}{2} \cdot \frac{1}{4} \cdot 1 = \frac{7}{8}$, and $\sqrt{37}/8 \leq \frac{7}{8}$ since $37 \leq 49$. The direction matters: along the real axis, $p(\frac{1}{2}) = \frac{1}{8} + \frac{1}{4} + 1 = \frac{11}{8} > 1$, because there $b_2 h^2 = \frac{1}{4}$ points away from 0.

5.5.5 The proof of the theorem

The three steps now combine into a proof of [Theorem 5.5.1](#).

Proof. Let $p \in \mathbb{C}[x]$ be non-constant. By [Lemma 5.5.3](#) there is $z_0 \in \mathbb{C}$ with $|p(z_0)| \leq |p(z)|$ for every $z \in \mathbb{C}$. Suppose, for a contradiction, that $p(z_0) \neq 0$. By [Lemma 5.5.4](#) there is z with $|p(z)| < |p(z_0)|$, contradicting the choice of z_0 . Hence $p(z_0) = 0$, and z_0 is a root of p . ■

This proves Chapter 0's [Theorem 0.6.7](#): a polynomial $a_n z^n + \cdots + a_0$ with $n \geq 1$ and $a_n \neq 0$ is a non-constant element of $\mathbb{C}[x]$, and we have just shown that it has a root.

Quick check. The polynomial $p = x^2 + 1 \in \mathbb{R}[x]$ has no real root, yet the function $c \mapsto |p(c)|$ on $\mathbb{R}$ has a minimum, at $c = 0$. Which step of the argument breaks if we try to run it over $\mathbb{R}$?

Solution. Growth and the attained minimum both hold on $\mathbb{R}$: the minimum value is $|p(0)| = 1$. The step that fails is d'Alembert's lemma. At $z_0 = 0$ we have $b_0 = 1$, $k = 2$, $b_2 = 1$, so we need a direction u with $u^2 = w = -1$. There is no such real u , and indeed $p(h) = 1 + h^2 \geq 1$ for every real h . Over $\mathbb{C}$, $u = i$ works and $p(ti) = 1 - t^2 < 1$ for $0 < t \leq 1$.

Warning. **The theorem is an existence theorem.** It says a root exists; it does not say how to find one, and the proof does not produce one. There is no formula for the roots of a general polynomial of degree 5 or more built from the coefficients by $+$, $-$, $\times$, $\div$ and k -th roots (a theorem from Galois theory, which we do not prove). Numerical root-finding algorithms approximate roots; that is a different subject, and the Fundamental Theorem of Algebra is not one of those algorithms.

5.5.6 Complex polynomials split

Combining the theorem with the root count of the previous section gives the form in which it is used most.

Corollary 5.5.5 — Complex Polynomials Split (a) Every non-zero $p \in \mathbb{C}[x]$ splits over $\mathbb{C}$: if $\deg p = n$ and p has leading coefficient a , then $p = a(x - z_1)(x - z_2) \cdots (x - z_n)$ for some $z_1, \dots, z_n \in \mathbb{C}$. In particular the multiplicities of the roots of p add up to n .

(b) The irreducible polynomials in $\mathbb{C}[x]$ are exactly the polynomials of degree 1.

Proof.

- (a) By [Theorem 5.4.3](#), $p = (x - c_1)^{m_1} \cdots (x - c_k)^{m_k} g$, where $c_1, \dots, c_k$ are the distinct roots of p in $\mathbb{C}$ and $g \in \mathbb{C}[x]$ has no root in $\mathbb{C}$. By [Theorem 5.5.1](#), g is constant, and $g \neq 0$ because $p \neq 0$. So p splits over $\mathbb{C}$, and the multiplicities add up to n by the same theorem. Comparing leading coefficients ([Theorem 0.8.3](#)), the constant is a .

- (b) A polynomial of degree 1 is irreducible, since a factorization into two non-constant factors would have degree at least 2. Conversely, let $p \in \mathbb{C}[x]$ be irreducible. It is non-constant (Definition 5.3.1), so by (a) it has a factor $x - z_1$, say $p = (x - z_1)q$. Since p is irreducible and $x - z_1$ is not constant, q is a non-zero constant, and $\deg p = 1$. ■

So in $\mathbb{C}[x]$, unique factorization (Theorem 5.3.5) is completely explicit: the monic irreducible factors are the $x - z$, and the exponent of $x - z$ is the multiplicity of z . In Chapter 8 this is what gives every operator on a non-zero finite-dimensional complex vector space an eigenvalue. (That fact also has a proof using only linear algebra, and one can run the logic in the other direction; we will not need it.)

Example. (Factoring $x^4 + 1$ over $\mathbb{C}$ and over $\mathbb{R}$)

Factor $f = x^4 + 1$ into irreducible polynomials in $\mathbb{C}[x]$ and in $\mathbb{R}[x]$.

Solution. Roots. We solve $z^4 = -1 = e^{i\pi}$ as in Example (Roots of Unity). Write $z = re^{i\theta}$ with $r > 0$; by Theorem 0.6.6, $z^4 = r^4 e^{4i\theta}$. Comparing polar forms, $r^4 = 1$, so $r = 1$, and $4\theta = \pi + 2\pi j$ for an integer j . The values $j = 0, 1, 2, 3$ give four distinct roots

$$z_0 = e^{i\pi/4} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i, \quad z_1 = e^{3i\pi/4} = -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i, \quad z_2 = \overline{z_1}, \quad z_3 = \overline{z_0}.$$

Over $\mathbb{C}$. Four distinct roots, each of multiplicity at least 1, and $\deg f = 4$, so by Theorem 5.4.3 each multiplicity is 1 and

$$x^4 + 1 = (x - z_0)(x - z_1)(x - z_2)(x - z_3),$$

the leading coefficient being 1. The factors are irreducible by Corollary 5.5.5 (b).

Over $\mathbb{R}$. Pair each root with its conjugate. For any $z \in \mathbb{C}$,

$$(x - z)(x - \overline{z}) = x^2 - (z + \overline{z})x + z\overline{z} = x^2 - 2\operatorname{Re}(z)x + |z|^2 \in \mathbb{R}[x],$$

by Theorem 0.6.2 (c). With $\operatorname{Re} z_0 = \frac{\sqrt{2}}{2}$, $\operatorname{Re} z_1 = -\frac{\sqrt{2}}{2}$ and $|z_0| = |z_1| = 1$,

$$x^4 + 1 = (x^2 - \sqrt{2}x + 1)(x^2 + \sqrt{2}x + 1).$$

Check: the product is $(x^2 + 1)^2 - 2x^2 = x^4 + 1$. Each quadratic has non-real roots only, hence no real root, so it is irreducible over $\mathbb{R}$ by Theorem 5.3.1. Note that $x^4 + 1$ has no real root but is **reducible** over $\mathbb{R}$.

5.5.7 Real polynomials

A real polynomial is also a complex polynomial, so the theorem applies to it; the roots it provides may be non-real. They are not scattered arbitrarily. Chapter 0 observed that non-real roots of a real polynomial come in conjugate pairs; we now show that the pairs also share their multiplicity. For $g = \sum_j c_j x^j \in \mathbb{C}[x]$ write $\bar{g} = \sum_j \bar{c}_j x^j$, the polynomial with conjugated coefficients.

Theorem 5.5.6 — Conjugate Roots Have Equal Multiplicity

Let $f \in \mathbb{R}[x]$ be non-zero, regarded as an element of $\mathbb{C}[x]$, and let $z \in \mathbb{C}$. Then $f(\bar{z}) = \overline{f(z)}$. In particular z is a root of f if and only if $\bar{z}$ is, and

$$\text{mult}_{\bar{z}}(f) = \text{mult}_z(f).$$

Idea. Conjugation respects sums and products of complex numbers, so applying it to all coefficients respects sums and products of polynomials. A real polynomial is unchanged by it. So conjugating a factorization $f = (x - z)^m g$ gives a factorization $f = (x - \bar{z})^m \bar{g}$, and the cofactor test transfers.

Proof. Let $g = \sum_j c_j x^j$ and $h = \sum_j d_j x^j$ be in $\mathbb{C}[x]$, and $z \in \mathbb{C}$. By [Theorem 0.6.2](#) (a) and (b), the coefficient of x^k in $\overline{gh}$ is $\overline{\sum_{i+j=k} c_i d_j} = \sum_{i+j=k} \overline{c_i} \overline{d_j}$, which is the coefficient of x^k in $\bar{g} \bar{h}$. So $\overline{gh} = \bar{g} \bar{h}$. By the same parts, $\overline{g(z)} = \sum_j \overline{c_j} \bar{z}^j = \sum_j c_j z^j = g(z)$.

Since f has real coefficients, $\bar{f} = f$. Hence $f(\bar{z}) = \bar{f}(\bar{z}) = \overline{f(z)}$, and $f(z) = 0$ if and only if $f(\bar{z}) = 0$.

Let $m = \text{mult}_z(f)$. By [Lemma 5.4.1](#), $f = (x - z)^m g$ with $g \in \mathbb{C}[x]$ and $g(z) \neq 0$. Conjugating, and using $\overline{x - z} = x - \bar{z}$,

$$f = \bar{f} = (x - \bar{z})^m \bar{g}, \quad \bar{g}(\bar{z}) = \overline{g(z)} \neq 0.$$

By [Lemma 5.4.1](#) again, $\text{mult}_{\bar{z}}(f) = m$. This proves the theorem. ■

For instance, a real polynomial with $2 + i$ as a root of multiplicity 3 also has $2 - i$ as a root of multiplicity 3, so by [Theorem 5.4.3](#) its degree is at least 6. Now the irreducible real polynomials can be listed.

Corollary 5.5.7 — Irreducible Real Polynomials

The irreducible polynomials in $\mathbb{R}[x]$ are exactly the polynomials of degree 1, and the polynomials of degree 2 with no real root. In particular, every irreducible polynomial in $\mathbb{R}[x]$ has degree 1 or 2.

Idea. An irreducible real p has a complex root z by the theorem. If z is real, $x - z$ is a real factor and p is linear. If not, $\bar{z}$ is another root, and $(x - z)(x - \bar{z})$ is a **real** quadratic dividing p . The only subtle point is that it divides p in $\mathbb{R}[x]$ and not merely in $\mathbb{C}[x]$; uniqueness of division settles that.

Proof. ($\Leftarrow$) A polynomial of degree 1 is irreducible, since a product of two non-constant polynomials has degree at least 2 ([Theorem 0.8.3](#)). A polynomial of degree 2 with no real root is irreducible by [Theorem 5.3.1](#).

($\Rightarrow$) Let $p \in \mathbb{R}[x]$ be irreducible. It is non-constant, so by [Theorem 5.5.1](#) it has a root $z \in \mathbb{C}$.

Case 1: $z \in \mathbb{R}$. By [Lemma 2.7.1](#) in $\mathbb{R}[x]$, $p = (x - z)q$ with $q \in \mathbb{R}[x]$. Since p is irreducible

and $x - z$ is not constant, q is a non-zero constant, so $\deg p = 1$.

Case 2: $z \notin \mathbb{R}$. Then $\bar{z} \neq z$, and $\bar{z}$ is also a root by [Theorem 5.5.6](#). Applying [Corollary 5.1.4](#) in $\mathbb{C}[x]$ to the two distinct roots $z, \bar{z}$ gives $p = (x - z)(x - \bar{z})h$ for some $h \in \mathbb{C}[x]$. Put $q = (x - z)(x - \bar{z}) = x^2 - 2\operatorname{Re}(z)x + |z|^2$, which lies in $\mathbb{R}[x]$. Divide in $\mathbb{R}[x]$ ([Theorem 5.1.1](#)): $p = qh_1 + r$ with $h_1, r \in \mathbb{R}[x]$ and $\deg r < 2$. This is also a division with remainder in $\mathbb{C}[x]$, and so is $p = qh + 0$. By the uniqueness in [Theorem 5.1.1](#) over $\mathbb{C}$, $r = 0$ and $h = h_1 \in \mathbb{R}[x]$. So $p = qh_1$ in $\mathbb{R}[x]$. Since p is irreducible and q is not constant, h_1 is a non-zero constant, and $\deg p = 2$. Finally, p has no real root, because a real root c would give a factor $x - c$ and $p = (x - c)q'$ with $\deg q' = 1$, contradicting irreducibility. This proves the corollary. ■

Remark. Completing the square decides which quadratics occur: $x^2 + \beta x + \gamma = (x + \frac{\beta}{2})^2 - \frac{\beta^2 - 4\gamma}{4}$ has a real root if and only if $\beta^2 - 4\gamma \geq 0$. So the monic irreducible real quadratics are those with $\beta^2 < 4\gamma$.

Unique factorization in $\mathbb{R}[x]$ now takes a concrete form.

Corollary 5.5.8 — Factorization over $\mathbb{R}$

Every non-constant $f \in \mathbb{R}[x]$ can be written as

$$f = a(x - c_1) \cdots (x - c_r) q_1 \cdots q_s,$$

where $a \in \mathbb{R}$ is the leading coefficient of f , $c_1, \dots, c_r \in \mathbb{R}$ (not necessarily distinct), and $q_1, \dots, q_s \in \mathbb{R}[x]$ are monic quadratics with no real root. Then $r + 2s = \deg f$, and r is the sum of the multiplicities of the real roots of f . The factorization is unique up to the order of the factors.

Proof. By [Theorem 5.3.5](#), $f = ap_1 \cdots p_k$ with $p_i \in \mathbb{R}[x]$ monic irreducible, unique up to order. By [Corollary 5.5.7](#) each p_i is either $x - c$ with $c \in \mathbb{R}$ or a monic quadratic with no real root; list the first kind as $x - c_1, \dots, x - c_r$ and the second as $q_1, \dots, q_s$. Then $\deg f = r + 2s$ by [Theorem 0.8.3](#). For a real number c , each q_j has multiplicity 0 at c , and $x - c_i$ has multiplicity 1 or 0 according as $c_i = c$ or not; by [Theorem 5.4.2](#), $\operatorname{mult}_c(f)$ is the number of indices i with $c_i = c$. Adding over the real roots of f gives r . ■

The last consequence is one that calculus proves with the intermediate value theorem. Here it falls out of counting degrees.

Corollary 5.5.9 — Odd Degree Forces a Real Root

Every $f \in \mathbb{R}[x]$ of odd degree has a root in $\mathbb{R}$.

Proof. By [Corollary 5.5.8](#), $\deg f = r + 2s$. Since $\deg f$ is odd, r is odd, so $r \geq 1$, and c_1 is a real root of f .

Warning. No real root does not mean irreducible over $\mathbb{R}$. By [Corollary 5.5.7](#), an irreducible real polynomial has degree at most 2; so $x^4 + 1$, which has no real root, is still reducible over $\mathbb{R}$ ([Example \(Factoring \$x^4 + 1\$ over \$\mathbb{C}\$ and over \$\mathbb{R}\$ \)](#)). The test “reducible if and only if it has a root” is only valid in degrees 2 and 3 ([Theorem 5.3.1](#)).

5.5.8 Exercises

5.5.8.1 A. Check your understanding

Exercise A1.

- State the Fundamental Theorem of Algebra, and list the two facts from analysis used in its proof in this section.
- True or false: every polynomial in $\mathbb{R}[x]$ of degree 4 has a real root. Justify your answer.
- True or false: if $f \in \mathbb{R}[x]$ has $2 + i$ as a root of multiplicity 3, then $\deg f \geq 6$. Justify your answer.
- True or false: every polynomial in $\mathbb{R}[x]$ of degree 5 has a real root. Justify your answer.
- True or false: every irreducible polynomial in $\mathbb{F}_2[x]$ has degree at most 2. Justify your answer.
- In the proof of d’Alembert’s lemma ([Lemma 5.5.4](#)), which property of $\mathbb{C}$ is used that $\mathbb{R}$ lacks?

Solution.

- Every non-constant $p \in \mathbb{C}[x]$ has a root in $\mathbb{C}$ ([Theorem 5.5.1](#)). The analytic facts are (A1) continuity of $z \mapsto |p(z)|$ on $\mathbb{C}$, and (A2) the extreme value theorem for continuous real functions on a closed disc.
- False. $x^4 + 1 \geq 1$ at every real number, so it has no real root.
- True. By [Theorem 5.5.6](#), $2 - i$ is also a root of multiplicity 3, and $2 - i \neq 2 + i$. By [Theorem 5.4.3](#), $\deg f \geq 3 + 3 = 6$.
- True, by [Corollary 5.5.9](#).
- False. $x^3 + x + 1$ takes the value 1 at both 0 and 1, so it has no root in $\mathbb{F}_2$, and a cubic without a root is irreducible by [Theorem 5.3.1](#). [Corollary 5.5.7](#) is special to $\mathbb{R}$.
- Every non-zero complex number has a k -th root; the proof takes a k -th root of a complex number of modulus 1 by [Theorem 0.6.6](#). Over $\mathbb{R}$, -1 has no square root.

5.5.8.2 B. Practice

Exercise B1: $x^4 + 4$.

Find all complex roots of $x^4 + 4$, and factor it into irreducible polynomials in $\mathbb{C}[x]$ and

in $\mathbb{R}[x]$.

Solution. Write $z = re^{i\theta}$ with $r > 0$. By [Theorem 0.6.6](#), $z^4 = r^4 e^{4i\theta}$, and $-4 = 4e^{i\pi}$. Comparing polar forms, $r^4 = 4$, so $r = \sqrt{2}$ (the only positive real number with fourth power 4, since $t \mapsto t^4$ is strictly increasing on positive reals), and $4\theta = \pi + 2\pi j$, $j \in \mathbb{Z}$. The values $j = 0, 1, 2, 3$ give the four distinct roots

$$\sqrt{2}e^{i\pi/4} = 1 + i, \quad \sqrt{2}e^{3i\pi/4} = -1 + i, \quad \sqrt{2}e^{5i\pi/4} = -1 - i, \quad \sqrt{2}e^{7i\pi/4} = 1 - i.$$

By [Theorem 5.4.3](#), four distinct roots of a degree-4 monic polynomial each have multiplicity 1, and

$$x^4 + 4 = (x - 1 - i)(x - 1 + i)(x + 1 - i)(x + 1 + i) \quad \text{in } \mathbb{C}[x].$$

Pairing conjugates with $(x - z)(x - \bar{z}) = x^2 - 2\operatorname{Re}(z)x + |z|^2$, and $|1 \pm i|^2 = |-1 \pm i|^2 = 2$,

$$x^4 + 4 = (x^2 - 2x + 2)(x^2 + 2x + 2) \quad \text{in } \mathbb{R}[x].$$

Check: $(x^2 + 2)^2 - (2x)^2 = x^4 + 4x^2 + 4 - 4x^2 = x^4 + 4$. Both quadratics have $\beta^2 - 4\gamma = 4 - 8 < 0$, so they have no real roots and are irreducible by [Corollary 5.5.7](#).

Exercise B2: $x^6 + 1$.

Find the factorization of $x^6 + 1$ into irreducible polynomials in $\mathbb{R}[x]$.

Solution. The roots solve $z^6 = -1 = e^{i\pi}$. As in B1, $z = e^{i\theta}$ with $6\theta = \pi + 2\pi j$, and $j = 0, \dots, 5$ give six distinct roots at the angles $30^\circ, 90^\circ, 150^\circ, 210^\circ, 270^\circ, 330^\circ$:

$$\frac{\sqrt{3}}{2} \pm \frac{1}{2}i, \quad \pm i, \quad -\frac{\sqrt{3}}{2} \pm \frac{1}{2}i.$$

Each has multiplicity 1 by [Theorem 5.4.3](#), and none is real. They form three conjugate pairs, each of modulus 1, with real parts $\frac{\sqrt{3}}{2}, 0, -\frac{\sqrt{3}}{2}$. Pairing them as in [Example \(Factoring \$x^4 + 1\$ over \$\mathbb{C}\$ and over \$\mathbb{R}\$ \)](#),

$$x^6 + 1 = (x^2 + 1)(x^2 - \sqrt{3}x + 1)(x^2 + \sqrt{3}x + 1).$$

Check: $(x^2 - \sqrt{3}x + 1)(x^2 + \sqrt{3}x + 1) = (x^2 + 1)^2 - 3x^2 = x^4 - x^2 + 1$, and $(x^2 + 1)(x^4 - x^2 + 1) = x^6 + 1$. Each quadratic factor has only non-real roots, so it is irreducible over $\mathbb{R}$ by [Corollary 5.5.7](#). By the uniqueness in [Corollary 5.5.8](#), this is the factorization.

Exercise B3: Real cubics.

Let $f \in \mathbb{R}[x]$ have degree 3, and suppose the multiplicities of the real roots of f add up to 1.

- Prove that $f = a(x - c)(x - z)(x - \bar{z})$ for some $a, c \in \mathbb{R}$ and some $z \in \mathbb{C} \setminus \mathbb{R}$, and that z and $\bar{z}$ are simple roots.
- Show that the hypothesis cannot be weakened to “ f has exactly one real root” by considering $(x - 1)^3$.

Solution.

- (a) By [Corollary 5.5.8](#), $f = a(x - c_1) \cdots (x - c_r) q_1 \cdots q_s$ with $r + 2s = 3$ and r equal to the sum of the real multiplicities, which is 1. So $r = 1$, $s = 1$, and $f = a(x - c)q$ with $c \in \mathbb{R}$ and q a monic real quadratic without real roots. By [Theorem 5.5.1](#), q has a root $z \in \mathbb{C}$, which is not real. By [Theorem 5.5.6](#), $\bar{z} \neq z$ is also a root of q . The two multiplicities in q are each at least 1 and add up to at most 2 ([Theorem 5.4.3](#)), so they equal 1 and $q = (x - z)(x - \bar{z})$, the leading coefficient being 1. Finally, by [Theorem 5.4.2](#), $\text{mult}_z(f) = \text{mult}_z(a(x - c)) + \text{mult}_z(q) = 0 + 1 = 1$, since $a(z - c) \neq 0$; similarly for $\bar{z}$.
- (b) $(x - 1)^3$ has exactly one real root, 1, but no non-real roots at all: it already splits over $\mathbb{R}$. Here the real multiplicities add up to 3, not 1.

Exercise B4: Running the proof on an example.

Let $p = x^4 - x^2 + 2 \in \mathbb{C}[x]$.

- (a) Compute the radius R of [Lemma 5.5.2](#) for p .
- (b) Follow the proof of [Lemma 5.5.4](#) at $z_0 = 0$: find k , B , u and t , and verify that $|p(tu)| < |p(0)|$.
- (c) Compute $p(i/2)$, and explain why the direction i does not work here, although it did in [Example \(One Step Downhill\)](#).

Solution.

- (a) Here $a_4 = 1$ and $A = |2| + |0| + |-1| + |0| = 3$, so $R = \max\{1, 6\} = 6$. Hence $|p(z)| \geq \frac{1}{2}|z|^4$ for $|z| \geq 6$.
- (b) At $z_0 = 0$ the coefficients are $b_0 = 2$, $b_1 = 0$, $b_2 = -1$, $b_3 = 0$, $b_4 = 1$. So $k = 2$ and $B = |b_3| + |b_4| = 1$. Then $w = -b_0/b_2 = 2 = 2e^{i \cdot 0}$, so $\alpha = 0$ and $u = 1$. Next $t = \min\{1, 2/1, 1/2\} = \frac{1}{2}$. So $z = \frac{1}{2}$, and $p(\frac{1}{2}) = \frac{1}{16} - \frac{1}{4} + 2 = \frac{29}{16} < 2 = |p(0)|$. The proof guarantees at most $2 - \frac{1}{2} \cdot \frac{1}{4} \cdot 1 = \frac{15}{8} = \frac{30}{16}$, consistent with $\frac{29}{16}$.
- (c) $p(i/2) = \frac{1}{16} + \frac{1}{4} + 2 = \frac{37}{16} > 2$. The right direction depends on the sign of b_k relative to b_0 : we need $b_2 u^2$ to point from b_0 toward 0, that is, $b_2 u^2$ a negative multiple of $b_0 = 2$. With $b_2 = -1$ this means $u^2 > 0$, so $u = \pm 1$; for $u = i$, $b_2 u^2 = 1$ points away from 0. In [Example \(One Step Downhill\)](#), $b_2 = +1$ and $u = i$ was the right choice.

5.5.8.3 C. Going deeper

Exercise C1: Fields where the theorem fails.

Call a field F **algebraically closed** if every non-constant polynomial in $F[x]$ has a root in F .

- (a) Let p be a prime. Show that $x^p - x + 1$ has no root in $\mathbb{F}_p$. More generally, show that no finite field is algebraically closed.
- (b) Give explicit witnesses that $\mathbb{R}$ and $\mathbb{Q}$ are not algebraically closed.
- (c) For your witness over $\mathbb{R}$, explain which step of the proof of the Fundamental Theorem of Algebra ([Theorem 5.5.1](#)) breaks.

- (d) For $f = x^2 - 2$ over $\mathbb{Q}$, prove that $r \mapsto |f(r)|$ has **no** minimum on $\mathbb{Q}$. Which step of the proof breaks now?

Hint for (d): if r_0 were a minimum point with $r_0^2 > \frac{2}{5}$, compare r_0 with $r_1 = \frac{r_0^2 + 2}{2r_0}$.

Solution.

- (a) By [Exercise \(C2: A non-zero polynomial vanishing everywhere\)](#) (c), $a^p - a = 0$ for every $a \in \mathbb{F}_p$, so $a^p - a + 1 = 1 \neq 0$. For a finite field $F = \{c_1, \dots, c_q\}$, let $g = (x - c_1) \cdots (x - c_q) + 1$. It has degree $q \geq 2$ ([Theorem 0.8.3](#)), and $g(c_i) = 0 + 1 = 1 \neq 0$ for every i by [Theorem 0.8.5](#). So g is a non-constant polynomial without a root in F .
- (b) Over $\mathbb{R}$, $x^2 + 1$ has no root, since $c^2 + 1 \geq 1$. Over $\mathbb{Q}$, $x^2 - 2$ has no root, since $\sqrt{2}$ is irrational ([Theorem 0.2.3](#)) and the only real roots are $\pm\sqrt{2}$.
- (c) For $x^2 + 1$ over $\mathbb{R}$, growth holds, and $c \mapsto |c^2 + 1|$ attains its minimum 1 at $c = 0$. The step that breaks is d'Alembert's lemma ([Lemma 5.5.4](#)): at $z_0 = 0$ we have $b_0 = 1$, $k = 2$, $b_2 = 1$, and we need u with $u^2 = -1$, which does not exist in $\mathbb{R}$.
- (d) Suppose $r_0 \in \mathbb{Q}$ satisfies $|r_0^2 - 2| \leq |r^2 - 2|$ for every $r \in \mathbb{Q}$. Put $m = |r_0^2 - 2|$; then $m > 0$ by (b).
Case 1: $r_0^2 \leq \frac{2}{5}$. Then $m = 2 - r_0^2 \geq \frac{8}{5} > 1 = |1^2 - 2|$, contradicting minimality at $r = 1$.
Case 2: $r_0^2 > \frac{2}{5}$. Then $r_0 \neq 0$, and $r_1 = (r_0^2 + 2)/(2r_0) \in \mathbb{Q}$. Expanding,

$$r_1^2 - 2 = \frac{(r_0^2 + 2)^2 - 8r_0^2}{4r_0^2} = \frac{(r_0^2 - 2)^2}{4r_0^2}, \quad \text{so} \quad |r_1^2 - 2| = m \cdot \frac{m}{4r_0^2}.$$

We claim $m < 4r_0^2$. If $r_0^2 \geq 2$, then $m = r_0^2 - 2 < 4r_0^2$. If $\frac{2}{5} < r_0^2 < 2$, then $m = 2 - r_0^2 < \frac{8}{5} < 4r_0^2$. Hence $|r_1^2 - 2| < m$, contradicting minimality.

So there is no minimum. The step of the proof that breaks is [Lemma 5.5.3](#), which rests on the extreme value theorem (A2). That fact is about the completeness of $\mathbb{R}$, and $\mathbb{Q}$ is not complete: the values $|r^2 - 2|$ get arbitrarily close to 0 without reaching it.

Exercise C2: Sums of two squares.

Let $f \in \mathbb{R}[x]$ be monic with **no** real root.

- (a) Prove that $\deg f$ is even and that $f(c) > 0$ for every $c \in \mathbb{R}$.
 (b) Prove that $f = g^2 + h^2$ for some $g, h \in \mathbb{R}[x]$.

Hint for (b): each irreducible factor of f is $(x - z)(x - \bar{z})$. Collect one root from each pair into a complex polynomial G , and split G into real and imaginary parts.

Solution.

- (a) By [Corollary 5.5.8](#), $f = (x - c_1) \cdots (x - c_r) q_1 \cdots q_s$, with leading coefficient 1 since f is monic. A factor $x - c_1$ would give the real root c_1 , so $r = 0$ and $\deg f = 2s$ is even. (If $s = 0$, then $f = 1$ and the claims hold.) Each $q_j = x^2 + \beta x + \gamma$ has no

real root, so by the remark after [Corollary 5.5.7](#), $\beta^2 < 4\gamma$. For real c ,

$$q_j(c) = \left(c + \frac{\beta}{2}\right)^2 + \frac{4\gamma - \beta^2}{4} > 0.$$

By [Theorem 0.8.5](#), $f(c) = q_1(c) \cdots q_s(c)$ is a product of positive reals, so $f(c) > 0$.

- (b) For each j , q_j is non-constant, so by [Theorem 5.5.1](#) it has a root $z_j \in \mathbb{C}$, not real. As in [Exercise \(B3: Real cubics\)](#) (a), $q_j = (x - z_j)(x - \bar{z}_j)$. Let $G = (x - z_1) \cdots (x - z_s) \in \mathbb{C}[x]$. By the product rule for conjugate polynomials proved in [Theorem 5.5.6](#), $\bar{G} = (x - \bar{z}_1) \cdots (x - \bar{z}_s)$, so $f = G\bar{G}$ (commutativity, [Theorem 0.8.1](#)).

Write each coefficient of G as $\alpha_k + i\beta_k$ with $\alpha_k, \beta_k \in \mathbb{R}$, and put $g = \sum_k \alpha_k x^k$ and $h = \sum_k \beta_k x^k$ in $\mathbb{R}[x]$. Then $G = g + ih$ and $\bar{G} = g - ih$, so

$$f = (g + ih)(g - ih) = g^2 - i^2 h^2 + (ihg - igh) = g^2 + h^2.$$

For example, for $x^4 + 1$ take $z_1 = e^{i\pi/4}$ and $z_2 = e^{3i\pi/4}$ from [Example \(Factoring \$x^4 + 1\$ over \$\mathbb{C}\$ and over \$\mathbb{R}\$ \)](#). Then $z_1 + z_2 = i\sqrt{2}$ and $z_1 z_2 = e^{i\pi} = -1$, so $G = x^2 - i\sqrt{2}x - 1$, $g = x^2 - 1$, $h = -\sqrt{2}x$, and $x^4 + 1 = (x^2 - 1)^2 + 2x^2$.

5.6 Interpolation and the Chinese Remainder Theorem

Chapter 2 proved that through $n + 1$ points with distinct x -values there passes exactly one polynomial of degree at most n ([Theorem 2.7.3](#)). The proof went through an invertible matrix, and it never wrote the polynomial down. This section gives the formula. Then it reads interpolation in the arithmetic of this chapter: “ $p(c) = y$ ” says that p leaves remainder y on division by $x - c$. Prescribing remainders modulo several pairwise coprime polynomials at once is the Chinese Remainder Theorem. We prove it twice, once with Bézout’s identity, which gives a formula, and once by counting dimensions in the quotient spaces $F[x]/\langle m \rangle$.

5.6.1 The Lagrange basis

Suppose we want the polynomial of degree at most n taking prescribed values $y_0, \dots, y_n$ at distinct points $c_0, \dots, c_n$. The problem is linear in the data: if p fits the values (y_i) and q fits (y'_i) , then $p + q$ fits $(y_i + y'_i)$, by [Theorem 0.8.5](#). So it is enough to solve the simplest data sets, a single 1 and zeros everywhere else, and add up.

The polynomial that vanishes at every c_j with $j \neq i$ is easy to write: $\prod_{j \neq i} (x - c_j)$. At c_i it takes the value $\prod_{j \neq i} (c_i - c_j)$, which is a product of non-zero scalars, so we can divide by it. In [Example \(Dual Bases in \$\mathbb{R}\[x\]_{\leq 2}\$ \)](#) we met these polynomials for the three points $-1, 0, 1$. Here is the general case.

The i -th Lagrange polynomial is the polynomial of degree n that is 1 at the i -th node and 0 at every other node.

Definition 5.6.1 — Lagrange Basis

Let $n \in \mathbb{N}$, and let $c_0, c_1, \dots, c_n \in F$ be **distinct**. For $i = 0, \dots, n$, the i -th **Lagrange polynomial** for these nodes is

$$\ell_i(x) := \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - c_j}{c_i - c_j} \in F[x].$$

The list $(\ell_0, \dots, \ell_n)$ is the **Lagrange basis** for the nodes $c_0, \dots, c_n$.

In words: the numerator makes ℓ_i vanish at every node except c_i , and the denominator rescales so that the value at c_i is exactly 1. The points c_i are called **nodes**.

Well-definedness. Each denominator $c_i - c_j$ with $j \neq i$ is non-zero because the nodes are **distinct**, so the quotients exist in F . The product has n linear factors, each with leading coefficient $(c_i - c_j)^{-1} \neq 0$, so $\deg \ell_i = n$ by [Theorem 0.8.3](#). The name “basis” is justified by the theorem below.

Examples.

- **One node.** For $n = 0$ the product is empty, so $\ell_0 = 1$. The degenerate case is consistent: a polynomial of degree at most 0 with value y_0 at c_0 is the constant $y_0 = y_0 \ell_0$.
- **Two nodes.** For $c_0 = 0, c_1 = 1$ we get $\ell_0 = \frac{x-1}{0-1} = 1-x$ and $\ell_1 = \frac{x-0}{1-0} = x$. The line through $(0, y_0)$ and $(1, y_1)$ is $y_0(1-x) + y_1x$.
- **Three nodes.** For $-1, 0, 1$, $\ell_{-1} = \frac{x(x-1)}{(-1)(-2)} = \frac{1}{2}x(x-1)$, $\ell_0 = \frac{(x+1)(x-1)}{(1)(-1)} = 1-x^2$ and $\ell_1 = \frac{1}{2}x(x+1)$, the polynomials of [Example \(Dual Bases in \$\mathbb{R}\[x\]_{\leq 2}\$ \)](#). (Here we index the nodes by their values.)

Non-example by minimal change. Take the nodes $0, 1, 1$. The numerator of ℓ_1 is still defined, $(x-0)(x-1)$, but the denominator contains $c_1 - c_2 = 0$, which has no inverse. No rescaling helps: a polynomial that vanishes at $c_2 = 1$ cannot take the value 1 at $c_1 = 1$. The clause that fails is “distinct”.

Everything about the Lagrange polynomials follows from one table of values, the Kronecker delta $\ell_i(c_j) = \delta_{ij}$.

Theorem 5.6.1 — Lagrange Interpolation

Let $n \in \mathbb{N}$, let $c_0, \dots, c_n \in F$ be distinct, and let $(\ell_0, \dots, \ell_n)$ be the Lagrange basis for these nodes.

- (a) $\ell_i(c_j) = \delta_{ij}$ for all $i, j \in \{0, \dots, n\}$.
- (b) $(\ell_0, \dots, \ell_n)$ is a basis of $F[x]_{\leq n}$, and every $p \in F[x]_{\leq n}$ satisfies

$$p = \sum_{i=0}^n p(c_i) \ell_i.$$

- (c) For all $y_0, \dots, y_n \in F$, the unique $p \in F[x]_{\leq n}$ with $p(c_i) = y_i$ for every i is

$$p = \sum_{i=0}^n y_i \ell_i.$$

Idea. ① The delta table is a direct evaluation. ② Evaluating a vanishing combination $\sum a_i \ell_i$ at c_j kills every term but a_j , so the list is independent, and it has the right length, so we count instead of checking spanning. ③ The expansion of p and the interpolation formula are the same computation: both sides of a candidate identity are polynomials of degree at most n with the same values at $n+1$ nodes.

Proof.

- (a) By Theorem 0.8.5, $\ell_i(c_j) = \prod_{k \neq i} \frac{c_j - c_k}{c_i - c_k}$. If $j = i$, every factor is 1, so $\ell_i(c_i) = 1$. If $j \neq i$, the factor with $k = j$ is 0, so $\ell_i(c_j) = 0$.
- (b) Each ℓ_i has degree n , so it lies in $F[x]_{\leq n}$. Let $a_0 \ell_0 + \cdots + a_n \ell_n = 0$. Evaluating at c_j and using (a) gives $0 = \sum_i a_i \delta_{ij} = a_j$ for each j . Hence the list is linearly independent. It has length $n+1 = \dim F[x]_{\leq n}$ (Example (Standard Bases)), so it is a basis by Theorem 1.6.4 (a).
Now let $p \in F[x]_{\leq n}$, and put $q = p - \sum_i p(c_i) \ell_i \in F[x]_{\leq n}$. By (a), $q(c_j) = p(c_j) - p(c_j) = 0$ for every j . So q has the $n+1$ distinct roots $c_0, \dots, c_n$ and degree at most n . By Corollary 5.1.4, $q = 0$, which is the expansion of p .
- (c) By (a), $(\sum_i y_i \ell_i)(c_j) = \sum_i y_i \delta_{ij} = y_j$, so $\sum_i y_i \ell_i$ has the required values, and it lies in $F[x]_{\leq n}$. By Theorem 2.7.3, it is the only such polynomial. This proves the theorem. ■

So Chapter 2's existence statement now comes with a formula, and the uniqueness half of (c) is the expansion in (b) read backwards: a polynomial of degree at most n is determined by its values at the nodes, because those values are its coordinates in the Lagrange basis.

Example. (The Chapter 2 Cubic by Lagrange's Formula)

Use Theorem 5.6.1 to find the $p \in \mathbb{Q}[x]_{\leq 3}$ with $p(-1) = 2$, $p(0) = 1$, $p(1) = 2$ and $p(2) = 11$, and compare with Example (A cubic through four points).

Solution. The nodes $-1, 0, 1, 2$ are distinct. Write $\ell_{-1}, \ell_0, \ell_1, \ell_2$ for the Lagrange polynomials, indexed by the node. The denominators are $\prod_{j \neq i} (c_i - c_j)$:

$$\begin{aligned} \ell_{-1} &= \frac{x(x-1)(x-2)}{(-1)(-2)(-3)} = -\frac{1}{6}x(x-1)(x-2), & \ell_0 &= \frac{(x+1)(x-1)(x-2)}{(1)(-1)(-2)} = \frac{1}{2}(x+1)(x-1)(x-2), \\ \ell_1 &= \frac{(x+1)x(x-2)}{(2)(1)(-1)} = -\frac{1}{2}(x+1)x(x-2), & \ell_2 &= \frac{(x+1)x(x-1)}{(3)(2)(1)} = \frac{1}{6}(x+1)x(x-1). \end{aligned}$$

Expanding,

$$\ell_{-1} = -\frac{1}{6}x^3 + \frac{1}{2}x^2 - \frac{1}{3}x, \quad \ell_0 = \frac{1}{2}x^3 - x^2 - \frac{1}{2}x + 1, \quad \ell_1 = -\frac{1}{2}x^3 + \frac{1}{2}x^2 + x, \quad \ell_2 = \frac{1}{6}x^3 - \frac{1}{6}x.$$

By Theorem 5.6.1 (c), $p = 2\ell_{-1} + \ell_0 + 2\ell_1 + 11\ell_2$. Collecting coefficients:

$$x^3: -\frac{1}{3} + \frac{1}{2} - 1 + \frac{11}{6} = 1, \quad x^2: 1 - 1 + 1 + 0 = 1, \quad x: -\frac{2}{3} - \frac{1}{2} + 2 - \frac{11}{6} = -1, \quad 1: 1.$$

Hence $p = x^3 + x^2 - x + 1$, the polynomial found by elimination in Example (A cubic through four points), as uniqueness demands. The elimination solved a 4×4 system

for these particular values. The Lagrange polynomials were computed once, and would serve for any other values at the same nodes.

Warning. **The interpolant has degree at most n , not exactly n .** Through the three points $(0, 0)$, $(1, 1)$, $(2, 2)$ the formula gives $0 \cdot \ell_0 + 1 \cdot \ell_1 + 2\ell_2$, where $\ell_1 = -x(x-2)$ and $\ell_2 = \frac{1}{2}x(x-1)$; this is $-x^2 + 2x + x^2 - x = x$. The degree-2 terms cancel, because the points lie on a line. Each ℓ_i has degree exactly n , but a combination of them need not.

Quick check. Let $n \geq 1$ and let $(\ell_0, \dots, \ell_n)$ be the Lagrange basis for distinct nodes $c_0, \dots, c_n$. What is $\sum_{i=0}^n c_i \ell_i$?

Solution. It is the polynomial x . Indeed $x \in F[x]_{\leq n}$ since $n \geq 1$, and its values at the nodes are $c_0, \dots, c_n$, so [Theorem 5.6.1](#) (b) gives $x = \sum_i c_i \ell_i$.

5.6.2 Evaluations are the dual basis

The delta table $\ell_i(c_j) = \delta_{ij}$ can be read the other way round. For $c \in F$, evaluation at c is the linear functional $\varepsilon_c: p \mapsto p(c)$ ([Example \(Four Families of Functionals\)](#)), which we restrict to $F[x]_{\leq n}$. Then the table says $\varepsilon_{c_j}(\ell_i) = \delta_{ji}$: the evaluations at the nodes behave on the Lagrange basis exactly as a dual basis must ([Theorem 4.1.3](#)). [Exercise \(C3: Evaluations as a dual basis\)](#) proved from [Theorem 2.7.3](#) that the evaluations form the dual basis of **some** basis. We can now name that basis.

Corollary 5.6.2 — Evaluations Are Dual to the Lagrange Basis

Let $c_0, \dots, c_n \in F$ be distinct, with Lagrange basis $(\ell_0, \dots, \ell_n)$. Then $(\varepsilon_{c_0}, \dots, \varepsilon_{c_n})$ is the dual basis of $(\ell_0, \dots, \ell_n)$ in $F[x]_{\leq n}^*$. In particular, every linear functional φ on $F[x]_{\leq n}$ is a combination of evaluations at the nodes:

$$\varphi = \sum_{i=0}^n \varphi(\ell_i) \varepsilon_{c_i}.$$

Proof. The evaluations are linear ([Example \(Four Families of Functionals\)](#)), and $\varepsilon_{c_i}(\ell_j) = \ell_j(c_i) = \delta_{ij}$ by [Theorem 5.6.1](#) (a). Since $(\ell_0, \dots, \ell_n)$ is a basis ([Theorem 5.6.1](#) (b)), the uniqueness in [Theorem 4.1.3](#) (a) shows that the evaluations are its dual basis. The formula for φ is the second formula of [Theorem 4.1.3](#) (c). ■

The first formula of [Theorem 4.1.3](#) (c), $p = \sum_i \varepsilon_{c_i}(p) \ell_i$, is the expansion in [Theorem 5.6.1](#) (b). So Lagrange's formula is the statement “a vector is recovered from its dual-basis coordinates”, for the space $F[x]_{\leq n}$ and the measurements “value at c_i ”.

The formula for φ has a practical reading: **any** linear measurement of a polynomial of degree at most n can be computed from $n+1$ of its values, with weights $\varphi(\ell_i)$ that do not depend on the polynomial.

Example. (Integrals and Slopes from Values)

- (a) Find $w_0, w_1, w_2 \in \mathbb{R}$ such that $\int_0^1 p(t) dt = w_0 p(0) + w_1 p(\frac{1}{2}) + w_2 p(1)$ for every $p \in \mathbb{R}[x]_{\leq 2}$.
- (b) Express $p \mapsto p'(0)$ on $\mathbb{R}[x]_{\leq 2}$ as a combination of $\varepsilon_{-1}, \varepsilon_0, \varepsilon_1$.

Solution.

- (a) Integration over $[0, 1]$ is a linear functional on $\mathbb{R}[x]_{\leq 2}$ ([Example \(Four Families of Functionals\)](#)). For the nodes $0, \frac{1}{2}, 1$,

$$\ell_0 = \frac{(x - \frac{1}{2})(x - 1)}{(-\frac{1}{2})(-1)} = 2x^2 - 3x + 1, \quad \ell_{1/2} = \frac{x(x - 1)}{(\frac{1}{2})(-\frac{1}{2})} = -4x^2 + 4x, \quad \ell_1 = \frac{x(x - \frac{1}{2})}{(1)(\frac{1}{2})} = 2x^2 - x.$$

Their integrals are $\frac{2}{3} - \frac{3}{2} + 1 = \frac{1}{6}$, $-\frac{4}{3} + 2 = \frac{2}{3}$ and $\frac{2}{3} - \frac{1}{2} = \frac{1}{6}$. By [Corollary 5.6.2](#),

$$\int_0^1 p(t) dt = \frac{1}{6} p(0) + \frac{2}{3} p(\frac{1}{2}) + \frac{1}{6} p(1) \quad \text{for all } p \in \mathbb{R}[x]_{\leq 2}.$$

Check with $p = x^2$: the right side is $\frac{2}{3} \cdot \frac{1}{4} + \frac{1}{6} = \frac{1}{3} = \int_0^1 t^2 dt$. This is Simpson's rule, which here is exact rather than approximate because p has degree at most 2.

- (b) With $\ell_{-1} = \frac{1}{2}x(x - 1)$, $\ell_0 = 1 - x^2$ and $\ell_1 = \frac{1}{2}x(x + 1)$, the derivatives at 0 are $-\frac{1}{2}$, 0 and $\frac{1}{2}$. Hence $p'(0) = \frac{1}{2}(p(1) - p(-1))$ for every $p \in \mathbb{R}[x]_{\leq 2}$, the symmetric difference quotient, exact on quadratics.

5.6.3 Congruences and the Chinese Remainder Theorem

Now we change the point of view. By [Theorem 5.1.3](#), the remainder of p on division by $x - c$ is $p(c)$. So the interpolation conditions $p(c_i) = y_i$ say that p leaves the remainder y_i on division by $x - c_i$, for each i . Nothing in this reformulation needs the divisors to be linear. What happens if we prescribe the remainders modulo arbitrary polynomials $m_1, \dots, m_k$? The integers faced the same question long ago, which explains the name of the answer. We first need the language of congruences, copied from $\mathbb{Z}$.

Definition 5.6.2 — Congruence Modulo a Polynomial

Let $m \in F[x]$. Two polynomials $f, g \in F[x]$ are **congruent modulo m** , written $f \equiv g \pmod{m}$, if $m \mid f - g$.

For example, $x^2 \equiv -1 \pmod{x^2 + 1}$, and $f \equiv f(c) \pmod{x - c}$ for every f , by [Theorem 5.1.3](#). Congruence modulo m is an equivalence relation, and it respects sums and products: if $f \equiv f'$ and $g \equiv g' \pmod{m}$, then $f + g \equiv (f' + g') \pmod{m}$ and $fg \equiv f'g' \pmod{m}$. By [Theorem 5.1.1](#), if $m \neq 0$, every f is congruent modulo m to **exactly one** polynomial of degree less than $\deg m$, its remainder: $f = qm + r$ gives $f \equiv r$, and if $f \equiv r \equiv r'$ with both degrees below $\deg m$, then $m \mid r - r'$ with $\deg(r - r') < \deg m$, which forces $r - r' = 0$.

Interpolation needed distinct nodes. The general condition is that the moduli share no factor, and the proof needs a way to pass from “coprime in pairs” to “coprime to the product”.

Lemma 5.6.3 — Coprime to Each, Coprime to the Product (a) Let

$p, q_1, \dots, q_s \in F[x]$. If p is coprime to each q_j , then there are $a, b \in F[x]$ with $ap + b q_1 \cdots q_s = 1$, and p is coprime to $q_1 \cdots q_s$.

(b) Let $m_1, \dots, m_k \in F[x]$ be non-zero and **pairwise coprime**, that is, $\gcd(m_i, m_j) = 1$ whenever $i \neq j$. If $m_i \mid h$ for every i , then $m_1 \cdots m_k \mid h$.

Proof.

(a) By [Corollary 5.2.4](#), for each j there are a_j, b_j with $a_j p + b_j q_j = 1$. Multiplying these s identities,

$$1 = \prod_{j=1}^s (a_j p + b_j q_j) = ap + (b_1 \cdots b_s) q_1 \cdots q_s$$

for some $a \in F[x]$, because every term of the expanded product except $(b_1 q_1) \cdots (b_s q_s)$ contains a factor $a_j p$. Let $d = \gcd(p, q_1 \cdots q_s)$. By [Theorem 5.2.3](#), d divides p and $q_1 \cdots q_s$, hence divides $ap + b q_1 \cdots q_s = 1$, where $b = b_1 \cdots b_s$. So d is a non-zero constant by [Theorem 0.8.3](#), and being monic, $d = 1$. Hence p and $q_1 \cdots q_s$ are coprime ([Definition 5.2.3](#)).

(b) We use induction on k . For $k = 1$ there is nothing to prove. Let $k \geq 2$. The moduli $m_1, \dots, m_{k-1}$ are pairwise coprime and each divides h , so by the induction hypothesis $M := m_1 \cdots m_{k-1}$ divides h , say $h = Mg$. Since $m_k \mid h = Mg$, and m_k is coprime to M by (a), [Theorem 5.2.8](#) gives $m_k \mid g$, say $g = m_k g'$. Then $h = m_1 \cdots m_k g'$, as claimed. ■

Theorem 5.6.4 — Chinese Remainder Theorem for Polynomials

Let $m_1, \dots, m_k \in F[x]$ be non-zero and pairwise coprime, and put $m = m_1 \cdots m_k$. For all $r_1, \dots, r_k \in F[x]$:

- (a) there is $f \in F[x]$ with $f \equiv r_i \pmod{m_i}$ for every i ;
- (b) if f and g both satisfy (a), then $f \equiv g \pmod{m}$;
- (c) there is **exactly one** such f with $\deg f < \deg m$.

Idea. Copy Lagrange. There, ℓ_i was 1 at c_i and 0 at the other nodes; in the new language, $\ell_i \equiv 1 \pmod{x - c_i}$ and $\ell_i \equiv 0 \pmod{x - c_j}$. So we look for e_i with $e_i \equiv 1 \pmod{m_i}$ and $e_i \equiv 0 \pmod{m_j}$ for $j \neq i$, and then $f = \sum_i r_i e_i$ works. To be 0 modulo every other m_j , e_i should be a multiple of $M_i = \prod_{j \neq i} m_j$; to be 1 modulo m_i , it should be $1 - a m_i$. Bézout’s identity $a m_i + b M_i = 1$ provides exactly such an element: $e_i = b M_i$.

Proof.

(a) Fix i , and let $M_i = \prod_{j \neq i} m_j$. Since m_i is coprime to each m_j with $j \neq i$,

Lemma 5.6.3 (a) gives $a_i, b_i \in F[x]$ with $a_i m_i + b_i M_i = 1$. Put $e_i = b_i M_i$. Then $e_i - 1 = -a_i m_i$, so $e_i \equiv 1 \pmod{m_i}$, and $m_j \mid M_i \mid e_i$ for $j \neq i$, so $e_i \equiv 0 \pmod{m_j}$. Let $f = r_1 e_1 + \cdots + r_k e_k$. Since congruence modulo m_j respects sums and products,

$$f \equiv r_1 \cdot 0 + \cdots + r_j \cdot 1 + \cdots + r_k \cdot 0 = r_j \pmod{m_j}$$

for each j .

- (b) If f and g satisfy (a), then $f - g \equiv r_i - r_i = 0 \pmod{m_i}$, that is, $m_i \mid f - g$, for every i . By **Lemma 5.6.3** (b), $m \mid f - g$.
- (c) Since each $m_i \neq 0$, $m \neq 0$ (**Theorem 0.8.3**). Let f be as in (a), and let f_0 be its remainder on division by m (**Theorem 5.1.1**). Then $m_i \mid m \mid f - f_0$, so $f_0 \equiv f \equiv r_i \pmod{m_i}$ for each i , and $\deg f_0 < \deg m$. If g is another solution with $\deg g < \deg m$, then $m \mid f_0 - g$ by (b), and $\deg(f_0 - g) < \deg m$, so $f_0 = g$. This proves the theorem. ■

Lagrange is the case $m_i = x - c_i$. For distinct nodes, $(x - c_j) - (x - c_i) = c_i - c_j$ is a non-zero constant, so $\frac{1}{c_i - c_j}(x - c_j) - \frac{1}{c_i - c_j}(x - c_i) = 1$, and any common divisor of $x - c_i$ and $x - c_j$ divides 1: the moduli are pairwise coprime. The congruence $f \equiv y_i \pmod{x - c_i}$ means $f(c_i) = y_i$ by **Theorem 5.1.3**, and $\deg m = n + 1$. So part (c) is **Theorem 2.7.3**. The element e_i has degree less than $n + 1$ after reduction modulo m , is 1 at c_i and 0 at the other nodes, so after reduction it is ℓ_i .

Example. (A Linear and a Quadratic Modulus)

Find the polynomial $f \in \mathbb{Q}[x]$ of degree less than 3 with

$$f \equiv 1 \pmod{x - 2}, \quad f \equiv 2x - 1 \pmod{x^2 + 1}.$$

Solution. Bézout. Divide $x^2 + 1$ by $x - 2$: $x^2 + 1 = (x + 2)(x - 2) + 5$. Hence

$$\frac{1}{5}(x^2 + 1) - \frac{1}{5}(x + 2)(x - 2) = 1,$$

and the moduli are coprime, since any common divisor divides 1.

The elements e_i . Put $e_1 = \frac{1}{5}(x^2 + 1)$ and $e_2 = -\frac{1}{5}(x + 2)(x - 2) = 1 - e_1$. Then e_1 is a multiple of $x^2 + 1$ and $e_1 \equiv 1 + \frac{1}{5}(x + 2)(x - 2) \equiv 1 \pmod{x - 2}$; symmetrically, $e_2 \equiv 0 \pmod{x - 2}$ and $e_2 \equiv 1 \pmod{x^2 + 1}$.

Combine and reduce. By the proof of **Theorem 5.6.4**, $f \equiv 1 \cdot e_1 + (2x - 1)e_2 \pmod{(x - 2)(x^2 + 1)}$. Expanding,

$$e_1 + (2x - 1)e_2 = \frac{1}{5}(x^2 + 1) - \frac{1}{5}(2x - 1)(x^2 - 4) = \frac{1}{5}(-2x^3 + 2x^2 + 8x - 3).$$

Dividing by $m = (x - 2)(x^2 + 1) = x^3 - 2x^2 + x - 2$ gives quotient $-\frac{2}{5}$ and remainder

$$f = \frac{1}{5}(-2x^2 + 10x - 7).$$

Check. $f(2) = \frac{1}{5}(-8 + 20 - 7) = 1$, so $f \equiv 1 \pmod{x - 2}$. Modulo $x^2 + 1$, $x^2 \equiv -1$, so $f \equiv \frac{1}{5}(2 + 10x - 7) = 2x - 1$.

Warning. The moduli must be coprime in pairs; coprime as a whole family is not enough. Without coprimality a system can be unsolvable: with $m_1 = x$ and $m_2 = x^2$, the system $f \equiv 0 \pmod{x}$, $f \equiv 1 \pmod{x^2}$ has no solution: the first condition says $f(0) = 0$, the second says $f = 1 + x^2q$, so $f(0) = 1$. For three moduli, x , $x - 1$ and $x(x - 1)$ have no common factor, yet the first and third share x , and $f \equiv 0 \pmod{x}$, $f \equiv 1 \pmod{x(x - 1)}$ is again impossible.

Example. (Values and Slopes by CRT)

- (a) Let $c, a, b \in F$ and $f \in F[x]$. Show that $f \equiv a + b(x - c) \pmod{(x - c)^2}$ if and only if $f(c) = a$ and $f'(c) = b$.
- (b) Find the $p \in \mathbb{Q}[x]_{\leq 3}$ with $p(0) = 2$, $p'(0) = -1$, $p(1) = 1$ and $p'(1) = 3$.

Solution.

- (a) Divide by $(x - c)^2$: $f = q(x - c)^2 + r$ with $\deg r \leq 1$ (Theorem 5.1.1). Writing $r = r_0 + r_1(x - c)$, which is possible because $(1, x - c)$ is a basis of $F[x]_{\leq 1}$ by Theorem 1.4.4 and Theorem 1.6.4, the uniqueness of the remainder says $f \equiv a + b(x - c)$ exactly when $r_0 = a$ and $r_1 = b$. Now $f(c) = r_0$. By Theorem 5.4.4, $f' = q'(x - c)^2 + 2q(x - c) + r_1$, so $f'(c) = r_1$. Hence the congruence holds if and only if $f(c) = a$ and $f'(c) = b$.
- (b) By (a), the conditions are $p \equiv 2 - x \pmod{x^2}$ and $p \equiv 1 + 3(x - 1) \pmod{(x - 1)^2}$. The moduli are coprime, since

$$(2x + 1)(x - 1)^2 - (2x - 3)x^2 = (2x^3 - 3x^2 + 1) - (2x^3 - 3x^2) = 1.$$

So $e_0 = (2x + 1)(x - 1)^2$ is $\equiv 0 \pmod{(x - 1)^2}$ and $\equiv 1 \pmod{x^2}$, while $e_1 = -(2x - 3)x^2 = 3x^2 - 2x^3$ is $\equiv 0 \pmod{x^2}$ and $\equiv 1 \pmod{(x - 1)^2}$. The polynomial e_1 is the smooth step $3x^2 - 2x^3$ of Example (Prescribing Values and Slopes): value 0 and slope 0 at 0, value 1 and slope 0 at 1. By Theorem 5.6.4, p is the remainder of

$$(2 - x)e_0 + (3x - 2)e_1 = -8x^4 + 20x^3 - 12x^2 - x + 2$$

on division by $x^2(x - 1)^2 = x^4 - 2x^3 + x^2$. Adding $8(x^4 - 2x^3 + x^2)$ gives

$$p = 4x^3 - 4x^2 - x + 2.$$

Check. $p(0) = 2$; $p' = 12x^2 - 8x - 1$, so $p'(0) = -1$; $p(1) = 4 - 4 - 1 + 2 = 1$; $p'(1) = 12 - 8 - 1 = 3$.

The same recipe handles any finite set of nodes c_i with multiplicities k_i : the moduli $(x - c_i)^{k_i}$ are pairwise coprime for distinct nodes, by Lemma 5.6.3 (a) applied twice: first $x - c_i$ is coprime to $(x - c_j)^{k_j}$, and then $(x - c_j)^{k_j}$ is coprime to $(x - c_i)^{k_i}$. So for any prescribed remainders modulo $(x - c_i)^{k_i}$, there is exactly one polynomial of degree less than $k_1 + \cdots + k_s$ with those remainders. In characteristic 0, the Taylor expansion of Theorem 5.4.7 translates “the remainder modulo $(x - c)^k$ ” into the values $f(c), f'(c), \dots, f^{(k-1)}(c)$, and this is **Hermite interpolation**.

5.6.4 The quotient space $F[x]/\langle m \rangle$

Congruence modulo m is equality of cosets. The ideal $\langle m \rangle$ is a subspace of $F[x]$: it is non-empty and closed under addition and under multiplication by constant polynomials (Definition 5.2.1). So the quotient space $F[x]/\langle m \rangle$ of Definition 3.9.3 exists, and by Lemma 3.9.3 (b),

$$f + \langle m \rangle = g + \langle m \rangle \iff f - g \in \langle m \rangle \iff f \equiv g \pmod{m}.$$

An element of $F[x]/\langle m \rangle$ is a polynomial “known only modulo m ”. We cannot quote Theorem 3.9.5 for its dimension, because $F[x]$ is infinite-dimensional (Theorem 1.3.3). The remainder does the job instead.

Theorem 5.6.5 — Dimension of $F[x]/\langle m \rangle$

$/\langle m \rangle]$

Let $m \in F[x]$ be non-zero of degree d . Then $F[x]/\langle m \rangle$ is finite-dimensional with

$$\dim F[x]/\langle m \rangle = \deg m = d.$$

If $d \geq 1$, the cosets $(1 + \langle m \rangle, x + \langle m \rangle, \dots, x^{d-1} + \langle m \rangle)$ form a basis.

Proof. If $d = 0$, then m is a non-zero constant, and every $f = (m^{-1}f)m$ lies in $\langle m \rangle$. So $\langle m \rangle = F[x]$ and the quotient is the zero space, of dimension 0.

Let $d \geq 1$, and define $\rho: F[x] \rightarrow F[x]_{\leq d-1}$ by letting $\rho(f)$ be the remainder of f on division by m (Theorem 5.1.1). *Linear.* If $f = q_1m + \rho(f)$ and $g = q_2m + \rho(g)$, then for $c \in F$,

$$cf + g = (cq_1 + q_2)m + (c\rho(f) + \rho(g)),$$

and $\deg(c\rho(f) + \rho(g)) < d$ by Theorem 0.8.2. By the uniqueness in Theorem 5.1.1, $\rho(cf + g) = c\rho(f) + \rho(g)$. *Surjective.* If $\deg r < d$, then $r = 0 \cdot m + r$, so $\rho(r) = r$. *Kernel.* $\rho(f) = 0$ exactly when $f = qm$ for some q , that is, $f \in \langle m \rangle$.

By the First Isomorphism Theorem (Theorem 3.9.7), $\bar{\rho}: F[x]/\langle m \rangle \rightarrow F[x]_{\leq d-1}$, $f + \langle m \rangle \mapsto \rho(f)$, is a well-defined injective linear map with image $\text{im } \bar{\rho} = F[x]_{\leq d-1}$, so it is an isomorphism. Since $\bar{\rho}(x^j + \langle m \rangle) = x^j$ for $j < d$, and $(1, x, \dots, x^{d-1})$ is a basis of $F[x]_{\leq d-1}$ (Example (Standard Bases)), the inverse isomorphism $\bar{\rho}^{-1}$ carries it to the basis $(x^j + \langle m \rangle)_{j < d}$ (Theorem 3.5.2). Hence $\dim F[x]/\langle m \rangle = d$. ■

For example, $\mathbb{R}[x]/\langle x^2 + 1 \rangle$ has basis $(1 + \langle x^2 + 1 \rangle, x + \langle x^2 + 1 \rangle)$, so each element is $a + bx + \langle x^2 + 1 \rangle$ for unique $a, b \in \mathbb{R}$.

Remark. Cosets can also be **multiplied**: $(f + \langle m \rangle)(g + \langle m \rangle) := fg + \langle m \rangle$ is well defined, because congruence modulo m respects products. In $\mathbb{R}[x]/\langle x^2 + 1 \rangle$, $x \cdot x = x^2 \equiv -1$, so the coset of x squares to minus the coset of 1, and $(a + bx)(a' + b'x) \equiv (aa' - bb') + (ab' + a'b)x$: this is multiplication in $\mathbb{C}$, with the coset of x playing the role of i .

The Chinese Remainder Theorem now becomes a statement about linear maps. Reducing a polynomial modulo each m_i records its k remainders; the theorem says that this record, taken

modulo m , is a perfect dictionary. Recall from [Definition 3.9.1](#) that $V_1 \times \cdots \times V_k$ is the space of lists with entrywise operations.

Theorem 5.6.6 — CRT as an Isomorphism

Let $m_1, \dots, m_k \in F[x]$ be non-zero and pairwise coprime, and put $m = m_1 \cdots m_k$. Then

$$\Phi: F[x]/\langle m \rangle \rightarrow F[x]/\langle m_1 \rangle \times \cdots \times F[x]/\langle m_k \rangle, \quad f + \langle m \rangle \mapsto (f + \langle m_1 \rangle, \dots, f + \langle m_k \rangle),$$

is a well-defined isomorphism of vector spaces.

Idea. The map that does not care about well-definedness is $\Psi: f \mapsto (f + \langle m_i \rangle)_i$ on $F[x]$ itself. Its kernel is the set of polynomials divisible by every m_i , which is $\langle m \rangle$ by [Lemma 5.6.3](#). So the First Isomorphism Theorem makes Φ well defined and injective in one stroke. For surjectivity we do **not** construct a preimage. We count: both sides have dimension $\deg m$, because degrees add under products.

Proof. Let $\Psi: F[x] \rightarrow F[x]/\langle m_1 \rangle \times \cdots \times F[x]/\langle m_k \rangle$, $\Psi(f) = (f + \langle m_1 \rangle, \dots, f + \langle m_k \rangle)$. Each entry is a quotient map, which is linear by [Theorem 3.9.4](#) (c), and the operations in the product are entrywise, so Ψ is linear. By [Lemma 3.9.3](#) (b), $\Psi(f) = \mathbf{0}$ if and only if $m_i \mid f$ for every i . If $m \mid f$, then $m_i \mid f$ for all i , since $m_i \mid m$; conversely, if every $m_i \mid f$, then $m \mid f$ by [Lemma 5.6.3](#) (b). Hence $\ker \Psi = \langle m \rangle$.

By the First Isomorphism Theorem ([Theorem 3.9.7](#)), the rule $f + \langle m \rangle \mapsto \Psi(f)$ is a well-defined injective linear map; this is Φ . By [Theorem 5.6.5](#), [Theorem 0.8.3](#) and [Theorem 3.9.1](#),

$$\dim F[x]/\langle m \rangle = \deg m = \sum_{i=1}^k \deg m_i = \sum_{i=1}^k \dim F[x]/\langle m_i \rangle = \dim \left(F[x]/\langle m_1 \rangle \times \cdots \times F[x]/\langle m_k \rangle \right).$$

Since Φ is injective between spaces of the same finite dimension, it is bijective by [Corollary 3.3.2](#) (e), hence an isomorphism ([Definition 3.5.2](#)).

■

Surjectivity of Φ is the existence part of [Theorem 5.6.4](#), and injectivity is the uniqueness part. So the count has given a second proof of the Chinese Remainder Theorem, one that never uses Bézout's identity for existence. It is the same trade as in [Example \(Interpolation as a Count\)](#): counting proves that a solution exists, and the Bézout construction says what it is. By the remark above, Φ also respects products of cosets, since reduction modulo each m_i does. In Chapter 8 this splitting of $F[x]/\langle m \rangle$ into pieces, one for each coprime factor of m , reappears as a splitting of a space on which an operator acts. The next section begins that story.

Quick check. What is $\dim \mathbb{Q}[x]/\langle x^3 - x \rangle$? Into which three spaces does [Theorem 5.6.6](#) split it, and what does Φ do to the coset of x^2 ?

Solution. By [Theorem 5.6.5](#), the dimension is 3. Since $x^3 - x = x(x-1)(x+1)$ with distinct linear factors, which are pairwise coprime, Φ maps it isomorphically onto $\mathbb{Q}[x]/\langle x \rangle \times \mathbb{Q}[x]/\langle x-1 \rangle \times \mathbb{Q}[x]/\langle x+1 \rangle$, a product of three one-dimensional spaces. Modulo $x - c$, a polynomial is congruent to its value at c , so $\Phi(x^2 + \langle x^3 - x \rangle)$ is the list of cosets of the constants 0, 1, 1: the values of x^2 at 0, 1, -1 .

5.6.5 Exercises

5.6.5.1 A. Check your understanding

Exercise A1.

- Define the Lagrange basis for distinct nodes $c_0, \dots, c_n$, and state the value of ℓ_i at c_j .
- True or false: for four distinct nodes, the polynomial of degree at most 3 through the four points always has degree exactly 3. Justify your answer.
- State the Chinese Remainder Theorem for polynomials, with all hypotheses.
- True or false: if $\gcd(m_1, m_2, m_3) = 1$, then every system $f \equiv r_i \pmod{m_i}$, $i = 1, 2, 3$, has a solution. Justify your answer.
- What is $\dim_{\mathbb{R}} \mathbb{R}[x]/\langle x^4 + 1 \rangle$? Give a basis.
- Which functional on $F[x]_{\leq n}$ is the i -th vector of the dual basis (Theorem 4.1.3) of the Lagrange basis?

Solution.

- For distinct $c_0, \dots, c_n \in F$, $\ell_i = \prod_{j \neq i} \frac{x - c_j}{c_i - c_j}$ (Definition 5.6.1), and $\ell_i(c_j) = \delta_{ij}$ (Theorem 5.6.1 (a)).
- False. The points $(0, 0), (1, 0), (2, 0), (3, 0)$ are fitted by the zero polynomial, of degree $-\infty$; by uniqueness (Theorem 2.7.3), that is the interpolant.
- If $m_1, \dots, m_k \in F[x]$ are non-zero and pairwise coprime and $m = m_1 \cdots m_k$, then for all $r_1, \dots, r_k \in F[x]$ there is f with $f \equiv r_i \pmod{m_i}$ for each i , unique modulo m , and exactly one such f has $\deg f < \deg m$ (Theorem 5.6.4).
- False. $x, x - 1, x(x - 1)$ have gcd 1 as a family, but $f \equiv 0 \pmod{x}$, $f \equiv 0 \pmod{x - 1}$, $f \equiv 1 \pmod{x(x - 1)}$ has no solution: the last condition gives $f(0) = 1$, contradicting the first.
- 4, with basis $(1, x, x^2, x^3)$ of cosets modulo $\langle x^4 + 1 \rangle$, by Theorem 5.6.5.
- Evaluation at the i -th node, $\varepsilon_{c_i}: p \mapsto p(c_i)$ (Corollary 5.6.2).

5.6.5.2 B. Practice

Exercise B1: A Lagrange interpolant.

Let $c_0, c_1, c_2, c_3 = 0, 1, 2, 3$. Write down the Lagrange basis for these nodes in factored form, and use it to find the $p \in \mathbb{Q}[x]_{\leq 3}$ with $p(0) = 1$, $p(1) = 0$, $p(2) = 1$ and $p(3) = 10$. Hence compute $p(4)$.

Solution. The denominators $\prod_{j \neq i} (c_i - c_j)$ are $(-1)(-2)(-3) = -6$, $(1)(-1)(-2) = 2$, $(2)(1)(-1) = -2$ and $(3)(2)(1) = 6$. So

$$\ell_0 = -\frac{1}{6}(x-1)(x-2)(x-3), \quad \ell_1 = \frac{1}{2}x(x-2)(x-3), \quad \ell_2 = -\frac{1}{2}x(x-1)(x-3), \quad \ell_3 = \frac{1}{6}x(x-1)(x-2).$$

By Theorem 5.6.1 (c), $p = \ell_0 + 0 \cdot \ell_1 + \ell_2 + 10\ell_3$. Expanding each needed term,

$$\ell_0 = -\frac{1}{6}(x^3 - 6x^2 + 11x - 6), \quad \ell_2 = -\frac{1}{2}(x^3 - 4x^2 + 3x), \quad 10\ell_3 = \frac{5}{3}(x^3 - 3x^2 + 2x).$$

The coefficient of x^3 is $-\frac{1}{6} - \frac{1}{2} + \frac{5}{3} = 1$; of x^2 , $1 + 2 - 5 = -2$; of x , $-\frac{11}{6} - \frac{3}{2} + \frac{10}{3} = 0$; the constant is 1. Hence

$$p = x^3 - 2x^2 + 1.$$

Check: $p(0) = 1$, $p(1) = 0$, $p(2) = 8 - 8 + 1 = 1$, $p(3) = 27 - 18 + 1 = 10$. Hence $p(4) = 64 - 32 + 1 = 33$.

Exercise B2: A CRT system.

Find the polynomial $f \in \mathbb{Q}[x]$ of degree less than 3 with $f \equiv 2 \pmod{x-1}$ and $f \equiv x \pmod{x^2+1}$. Hence describe all $g \in \mathbb{Q}[x]$ satisfying both congruences.

Solution. Bézout. $x^2 + 1 = (x+1)(x-1) + 2$, so $\frac{1}{2}(x^2+1) - \frac{1}{2}(x+1)(x-1) = 1$, and the moduli are coprime.

The elements e_i . $e_1 = \frac{1}{2}(x^2+1)$ is a multiple of x^2+1 and $e_1 = 1 + \frac{1}{2}(x+1)(x-1) \equiv 1 \pmod{x-1}$. And $e_2 = 1 - e_1 = -\frac{1}{2}(x+1)(x-1)$ is a multiple of $x-1$ with $e_2 \equiv 1 \pmod{x^2+1}$.

Combine and reduce. By the proof of [Theorem 5.6.4](#), $f \equiv 2e_1 + xe_2 = (x^2+1) - \frac{1}{2}x(x^2-1) = -\frac{1}{2}x^3 + x^2 + \frac{1}{2}x + 1$. Modulo $m = (x-1)(x^2+1) = x^3 - x^2 + x - 1$, add $\frac{1}{2}m$:

$$f = -\frac{1}{2}x^3 + x^2 + \frac{1}{2}x + 1 + \frac{1}{2}(x^3 - x^2 + x - 1) = \frac{1}{2}x^2 + x + \frac{1}{2} = \frac{1}{2}(x+1)^2.$$

Check. $f(1) = 2$. Modulo x^2+1 , $f \equiv \frac{1}{2}(-1) + x + \frac{1}{2} = x$.

By [Theorem 5.6.4](#) (b), and since every $g \equiv f \pmod{m}$ satisfies both congruences (because $x-1$ and x^2+1 divide m), the solutions are exactly $g = \frac{1}{2}(x+1)^2 + h(x-1)(x^2+1)$ with $h \in \mathbb{Q}[x]$.

Exercise B3: The Lagrange basis sums to one.

Let $c_0, \dots, c_n \in F$ be distinct with Lagrange basis $(\ell_0, \dots, \ell_n)$.

- Prove that $\ell_0 + \ell_1 + \dots + \ell_n = 1$.
- Prove more generally that $\sum_{i=0}^n c_i^k \ell_i = x^k$ for $k = 0, 1, \dots, n$ (with $c^0 = 1$).
- For $n = 2$ and nodes $0, 1, 2$, verify (a) by expanding.

Solution.

- This is (b) with $k = 0$.
- Let $0 \leq k \leq n$. Then $x^k \in F[x]_{\leq n}$, and its value at c_i is c_i^k ([Definition 0.8.6](#)). By [Theorem 5.6.1](#) (b), $x^k = \sum_i c_i^k \ell_i$.
- $\ell_0 = \frac{(x-1)(x-2)}{2} = \frac{1}{2}x^2 - \frac{3}{2}x + 1$, $\ell_1 = \frac{x(x-2)}{-1} = -x^2 + 2x$, $\ell_2 = \frac{x(x-1)}{2} = \frac{1}{2}x^2 - \frac{1}{2}x$. Adding: x^2 has coefficient $\frac{1}{2} - 1 + \frac{1}{2} = 0$, x has $-\frac{3}{2} + 2 - \frac{1}{2} = 0$, and the constant is 1. So the sum is 1.

5.6.5.3 C. Going deeper

Exercise C1: Interpolation over a finite field.

Let p be a prime and $F = \mathbb{F}_p$.

- Explain why [Theorem 5.6.1](#) cannot be applied with more than p nodes.
- Let $E: \mathbb{F}_p[x]_{\leq p} \rightarrow \mathbb{F}_p^p$, $E(f) = (f(0), f(1), \dots, f(p-1))$. Prove that E is linear and surjective but not injective, and find a non-zero polynomial in $\ker E$.
- Deduce that every function $\mathbb{F}_p \rightarrow \mathbb{F}_p$ is the polynomial function of exactly one polynomial of degree at most $p-1$, but of more than one polynomial of degree at most p .

Solution.

- The theorem needs distinct nodes, and $\mathbb{F}_p$ has only p elements, so there are no $p+1$ distinct nodes.
- Linear.* Each entry $f \mapsto f(c)$ is linear by [Theorem 0.8.5](#). *Surjective.* The p elements $0, 1, \dots, p-1$ of $\mathbb{F}_p$ are distinct nodes, so for every $(y_0, \dots, y_{p-1})$, [Theorem 5.6.1](#) (c) with $n = p-1$ gives $f = \sum_i y_i \ell_i \in \mathbb{F}_p[x]_{\leq p-1} \subseteq \mathbb{F}_p[x]_{\leq p}$ with $E(f) = (y_0, \dots, y_{p-1})$. *Not injective.* $\dim \mathbb{F}_p[x]_{\leq p} = p+1 > p = \dim \mathbb{F}_p^p$, so E is not injective by [Corollary 3.3.2](#) (c). Concretely, $g = \prod_{c \in \mathbb{F}_p} (x - c)$ is monic of degree p , hence non-zero, and $g(c) = 0$ for every $c \in \mathbb{F}_p$, because one factor vanishes ([Theorem 0.8.5](#)).
- A function $\mathbb{F}_p \rightarrow \mathbb{F}_p$ is the same as its list of values $(y_0, \dots, y_{p-1})$. By (b), the restriction of E to $\mathbb{F}_p[x]_{\leq p-1}$ is surjective, and it is injective because a polynomial of degree at most $p-1$ vanishing at p distinct points is 0 ([Corollary 5.1.4](#)). So exactly one polynomial of degree at most $p-1$ has the given values. In degree at most p , if f has the given values, so does $f + g \neq f$, with g from (b). Unlike over $\mathbb{R}$, a polynomial over $\mathbb{F}_p$ is not determined by its function.

Exercise C2: Newton's form of the interpolant.

Let $c_0, \dots, c_n \in F$ be distinct. Put $N_0 = 1$ and $N_k = (x - c_0)(x - c_1) \cdots (x - c_{k-1})$ for $1 \leq k \leq n+1$.

- Prove that $(N_0, \dots, N_n)$ is a basis of $F[x]_{\leq n}$.
- Let $p = \sum_{k=0}^n a_k N_k$. Show that $p(c_i) = \sum_{k=0}^i a_k N_k(c_i)$ and $N_i(c_i) \neq 0$. Explain why the coefficients $a_0, a_1, \dots$ of the interpolant can be found one at a time, and show that $a_0 = y_0$ and $a_1 = \frac{y_1 - y_0}{c_1 - c_0}$.
- Let $p_n \in F[x]_{\leq n}$ interpolate $y_0, \dots, y_n$ at $c_0, \dots, c_n$, and let c_{n+1} be a further node, distinct from the others, with a value y_{n+1} . Prove that the interpolant $p_{n+1} \in F[x]_{\leq n+1}$ is

$$p_{n+1} = p_n + \frac{y_{n+1} - p_n(c_{n+1})}{N_{n+1}(c_{n+1})} N_{n+1}.$$

- Find the Newton form of the interpolant of [Exercise \(B1: A Lagrange interpolant\)](#), and then the interpolant after adding the point $(4, 0)$. Why is this more economical than recomputing with Lagrange's formula?

Hint: for (c), check the values of the right-hand side at all $n + 2$ nodes.

Solution.

- (a) N_k is a product of k monic linear factors, so it is monic of degree k (Theorem 0.8.3). The polynomials $N_0, \dots, N_n$ are non-zero with pairwise distinct degrees, hence linearly independent by Theorem 1.4.4. The list has length $n + 1 = \dim F[x]_{\leq n}$, so it is a basis by Theorem 1.6.4 (a).
- (b) For $k > i$, N_k contains the factor $x - c_i$, so $N_k(c_i) = 0$; this gives the formula for $p(c_i)$. And $N_i(c_i) = \prod_{j < i} (c_i - c_j)$ is a product of non-zero scalars, since the nodes are distinct, so it is non-zero. The conditions $p(c_i) = y_i$ therefore form a lower triangular system: the i -th equation involves only $a_0, \dots, a_i$, with coefficient $N_i(c_i) \neq 0$ on a_i . Solving in the order $i = 0, 1, \dots, n$ determines each a_i from the earlier ones. For $i = 0$: $p(c_0) = a_0$, so $a_0 = y_0$. For $i = 1$: $y_1 = a_0 + a_1(c_1 - c_0)$, so $a_1 = \frac{y_1 - y_0}{c_1 - c_0}$, the slope of the chord, the first “divided difference”.
- (c) Let q be the right-hand side; the division is legal because $N_{n+1}(c_{n+1}) \neq 0$ by (b). Then $\deg q \leq n + 1$. For $i \leq n$, $N_{n+1}(c_i) = 0$, so $q(c_i) = p_n(c_i) = y_i$. At c_{n+1} , $q(c_{n+1}) = p_n(c_{n+1}) + y_{n+1} - p_n(c_{n+1}) = y_{n+1}$. By Theorem 2.7.3, $q = p_{n+1}$.
- (d) With nodes $0, 1, 2, 3$: $N_1 = x$, $N_2 = x(x - 1)$, $N_3 = x(x - 1)(x - 2)$. Forward substitution: $a_0 = p(0) = 1$; $0 = p(1) = 1 + a_1$, so $a_1 = -1$; $1 = p(2) = 1 - 2 + 2a_2$, so $a_2 = 1$; $10 = p(3) = 1 - 3 + 6 + 6a_3$, so $a_3 = 1$. Thus

$$p_3 = 1 - x + x(x - 1) + x(x - 1)(x - 2) = x^3 - 2x^2 + 1,$$

agreeing with Exercise (B1: A Lagrange interpolant). Adding $(4, 0)$: $p_3(4) = 33$ and $N_4(4) = 4 \cdot 3 \cdot 2 \cdot 1 = 24$, so by (c)

$$p_4 = x^3 - 2x^2 + 1 - \frac{33}{24} x(x - 1)(x - 2)(x - 3) = x^3 - 2x^2 + 1 - \frac{11}{8} x(x - 1)(x - 2)(x - 3).$$

One checks $p_4(4) = 33 - \frac{11}{8} \cdot 24 = 0$. Adding a node changes **every** Lagrange polynomial, since each ℓ_i acquires a new factor; in Newton’s form the old coefficients $a_0, \dots, a_n$ survive and only one new coefficient is computed.

5.7 Polynomials of an Operator

Chapter 3 taught us to substitute an operator into a polynomial: for $T \in \mathcal{L}(V)$ and $p \in F[x]$, the operator $p(T)$ replaces each x^k by T^k (Definition 3.4.2). That section used the idea for a few examples. With the arithmetic of this chapter in hand, the idea becomes a method. Divisibility, ideals and Bézout’s identity in $F[x]$ turn into statements about T : every operator on a finite-dimensional space satisfies a polynomial equation, the polynomials it satisfies form an ideal, and a factorization of such a polynomial into coprime pieces splits the space. The last statement, the kernel splitting lemma, is the engine that Chapter 8 will run to take operators apart.

5.7.1 Evaluating at an operator

Fix $T \in \mathcal{L}(V)$, where V is any vector space over F . Theorem 3.4.3 already says that substituting T respects sums and products of polynomials, and that any two polynomials in T commute. The

new point of view is to treat “substitute T ” as a single map from $F[x]$ to $\mathcal{L}(V)$, and to ask what it preserves. Both sides are vector spaces, and both have a multiplication. The map respects all of the structure.

Theorem 5.7.1 — Evaluation at an Operator

Let V be a vector space over F and $T \in \mathcal{L}(V)$. Then the map

$$F[x] \rightarrow \mathcal{L}(V), \quad p \mapsto p(T),$$

has the following properties.

- (a) It is **linear**: $(cp + q)(T) = cp(T) + q(T)$ for all $p, q \in F[x]$ and $c \in F$.
- (b) It is **multiplicative** and sends 1 to id_V : $(pq)(T) = p(T)q(T)$ and $1(T) = \text{id}_V$.
- (c) Its image $\{p(T) : p \in F[x]\}$ is a subspace of $\mathcal{L}(V)$, closed under composition, in which any two operators commute.
- (d) If $S \in \mathcal{L}(V)$ commutes with T , then S commutes with $p(T)$ for every $p \in F[x]$.

Proof.

- (a) Let $p = \sum_k a_k x^k$ and $q = \sum_k b_k x^k$. The coefficient of x^k in $cp + q$ is $ca_k + b_k$, so by Definition 3.4.2 and the vector space laws in $\mathcal{L}(V)$ (Theorem 3.4.1),

$$(cp + q)(T) = \sum_k (ca_k + b_k)T^k = c \sum_k a_k T^k + \sum_k b_k T^k = cp(T) + q(T).$$

- (b) The product rule is Theorem 3.4.3 (b). The constant polynomial 1 gives $1 \cdot T^0 = \text{id}_V$.
- (c) The image of a linear map is a subspace (Theorem 3.2.2). For polynomials p, q , the composite $p(T)q(T)$ equals $(pq)(T)$ by (b), so it lies in the image, and $p(T)q(T) = q(T)p(T)$ by Theorem 3.4.3 (c).
- (d) Suppose $ST = TS$. We show $ST^k = T^k S$ by induction on k . For $k = 0$, both sides are S . If $ST^k = T^k S$, then by associativity (Theorem 3.4.2)

$$ST^{k+1} = (ST^k)T = T^k(ST) = T^k(TS) = T^{k+1}S.$$

For $p = \sum_k a_k x^k$, distributivity (Theorem 3.4.2) gives $S p(T) = \sum_k a_k ST^k = \sum_k a_k T^k S = p(T)S$.

■

Parts (a) and (b) together say that $p \mapsto p(T)$ is a **homomorphism of algebras**: it translates every identity built from sums, scalar multiples and products of polynomials into the same identity for operators. (This map should not be confused with the evaluation map $V \rightarrow V^{**}$ of Chapter 4; here we evaluate polynomials, not functionals.) Part (d) is what we will use most. It says that anything commuting with T commutes with the whole family of polynomials in T .

For matrices nothing new is needed. If V is finite-dimensional with basis $\mathcal{B}$, then $[p(T)]_{\mathcal{B}} = p([T]_{\mathcal{B}})$ by Corollary 3.6.3. So every statement in this section holds verbatim for $A \in M_n(F)$, applied to the operator T_A .

Example. (Two Ways to Evaluate)

Let $A = \begin{pmatrix} 3 & 1 \\ -1 & 1 \end{pmatrix} \in M_2(\mathbb{Q})$. Compute $p(A)$ for $p = x^2 - 4x + 4$ and $q = x^2 - 3x$, once by expanding and once by factoring.

Solution. Expanding. $A^2 = \begin{pmatrix} 9-1 & 3+1 \\ -3-1 & -1+1 \end{pmatrix} = \begin{pmatrix} 8 & 4 \\ -4 & 0 \end{pmatrix}$. Hence

$$p(A) = \begin{pmatrix} 8-12+4 & 4-4 \\ -4+4 & 0-4+4 \end{pmatrix} = 0, \quad q(A) = \begin{pmatrix} 8-9 & 4-3 \\ -4+3 & 0-3 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ -1 & -3 \end{pmatrix}.$$

Factoring. $p = (x-2)^2$, so $p(A) = (A-2I)^2$ by [Theorem 5.7.1](#) (b). Here $A-2I = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$, whose square is $\begin{pmatrix} 1-1 & 1-1 \\ -1+1 & -1+1 \end{pmatrix} = 0$. Likewise $q = x(x-3)$, so

$$q(A) = A(A-3I) = \begin{pmatrix} 3 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & -2 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ -1 & -3 \end{pmatrix},$$

in agreement with the expansion. The factored form showed at a glance why $p(A) = 0$: the matrix $A-2I$ squares to zero.

Warning. A product of polynomials in T can vanish without either factor vanishing. In $F[x]$, $pq = 0$ forces $p = 0$ or $q = 0$ ([Corollary 0.8.4](#)). Evaluation at T does not carry this over. For the projection $P(x, y) = (x, 0)$ on F^2 , the polynomials $p = x$ and $q = x-1$ give $p(P)q(P) = P^2 - P = 0$, while $p(P) = P \neq 0$ and $q(P) = P - \text{id} \neq 0$. The map $p \mapsto p(T)$ respects products, but it is not injective, and it is exactly its kernel that makes this possible.

5.7.2 Annihilating polynomials

In Chapter 3 we saw operators satisfying polynomial equations: a projection satisfies $P^2 - P = 0$, the rotation of $\mathbb{R}^2$ by a right angle satisfies $R^2 + \text{id} = 0$ ([Example \(Operators That Satisfy a Polynomial\)](#)), and differentiation on $F[x]_{\leq n}$ satisfies $D^{n+1} = 0$ ([Example \(Differentiation Is Nilpotent\)](#)). Each such equation is a non-zero polynomial in the kernel of $p \mapsto p(T)$. Does every operator have one? We need a word for these polynomials.

Definition 5.7.1 — Annihilating Polynomial

Let $T \in \mathcal{L}(V)$. A polynomial $p \in F[x]$ **annihilates** T if $p(T) = 0$, the zero operator on V .

For example, $x^2 - x$ annihilates every projection, $x^2 + 1$ annihilates the rotation R , and the zero polynomial annihilates every operator. On the zero space $V = \{\mathbf{0}\}$, even the constant 1 annihilates T , because $1(T) = \text{id}_V$ is the zero map there. On $V \neq \{\mathbf{0}\}$, a non-zero constant c never annihilates, since $c \text{id}_V \neq 0$. (The word “annihilate” is also used for the annihilator U^0 of Chapter 4; the two uses share the idea “sends to zero” and nothing more.)

In finite dimension a counting argument settles existence. The space $\mathcal{L}(V)$ is finite-dimensional,

so the powers of T cannot all be independent.

Theorem 5.7.2 — Every Operator Satisfies a Polynomial

Let V be a finite-dimensional vector space over F with $\dim V = n$, and let $T \in \mathcal{L}(V)$. Then there is a **non-zero** $p \in F[x]$ with $\deg p \leq n^2$ and $p(T) = 0$. The same holds for every $A \in M_n(F)$.

Proof. If $n = 0$, the constant 1 works, as noted above. Let $n \geq 1$. By Theorem 3.6.4, $\dim \mathcal{L}(V) = n \cdot n = n^2$. The list $(\text{id}_V, T, T^2, \dots, T^{n^2})$ in $\mathcal{L}(V)$ has length $n^2 + 1 > n^2$, so it is linearly dependent by Theorem 1.6.3 (a). Hence there are $a_0, \dots, a_{n^2} \in F$, **not all zero**, with $a_0 \text{id}_V + a_1 T + \dots + a_{n^2} T^{n^2} = 0$. The polynomial $p = a_0 + a_1 x + \dots + a_{n^2} x^{n^2}$ is non-zero, has degree at most n^2 , and satisfies $p(T) = 0$ by Definition 3.4.2. For $A \in M_n(F)$, apply the same argument in $M_n(F)$, which has dimension n^2 (Example (Dimensions of the Running Examples)).

■

The bound n^2 is crude. In Chapter 8, the Cayley–Hamilton Theorem will produce an annihilating polynomial of degree exactly n . The finite dimension is essential: on $F[x]$, let S be multiplication by x , $S(f) = xf$. Then $S^k(1) = x^k$, so $p(S)(1) = p$ for every $p \in F[x]$, and $p(S) \neq 0$ whenever $p \neq 0$.

The proof is also an algorithm: search for the first power of A that is a combination of the earlier ones. This is a question about linear dependence, which elimination answers (Theorem 2.5.10).

Example. (Finding an Annihilating Polynomial)

Find a non-zero polynomial of least possible degree that annihilates $B = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in M_3(\mathbb{Q})$.

Solution. *Degree 1.* B is not a scalar multiple of I_3 (its $(1, 2)$ -entry is non-zero), so (I_3, B) is independent, and no non-zero polynomial of degree at most 1 annihilates B .

Degree 2. Compute

$$B^2 = \begin{pmatrix} 4 & 3 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

We look for a, b, c with $aI_3 + bB + cB^2 = 0$. Reading entries: position $(1, 2)$ gives $b + 3c = 0$; position $(1, 1)$ gives $a + 2b + 4c = 0$; position $(2, 2)$ gives $a + b + c = 0$. The positions $(1, 3)$ and $(3, 3)$ repeat the $(1, 2)$ and $(2, 2)$ equations, and all other entries are 0 = 0. With $c = 1$: $b = -3$ and $a = 2$, and indeed $2 - 6 + 4 = 0$. So

$$B^2 - 3B + 2I_3 = 0,$$

and $p = x^2 - 3x + 2 = (x - 1)(x - 2)$ annihilates B . Check with the factored form: $B - I_3 =$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \text{ and } B - 2I_3 = \begin{pmatrix} 0 & 1 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \text{ whose product is } \begin{pmatrix} 0 & 1-1 & 1-1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 0.$$

The bound of [Theorem 5.7.2](#) allowed degree 9; the search stopped at degree 2.

Annihilating polynomials are far from unique: if p annihilates T , so does every multiple qp , since $(qp)(T) = q(T)p(T) = q(T) \cdot 0 = 0$. In [Example \(Finding an Annihilating Polynomial\)](#), $x^3 - 3x^2 + 2x$ annihilates B too. This closure under multiplication is the defining property of an ideal, and ideals of $F[x]$ are completely understood ([Theorem 5.2.2](#)).

Theorem 5.7.3 — The Annihilating Polynomials Form an Ideal

Let V be a vector space over F and $T \in \mathcal{L}(V)$, and let

$$I_T := \{p \in F[x] : p(T) = 0\}.$$

- (a) I_T is an ideal of $F[x]$.
- (b) Either $I_T = \{0\}$, or there is a **unique monic** $m \in F[x]$ with $I_T = \langle m \rangle$. In the second case, for every $p \in F[x]$,

$$p(T) = 0 \iff m \mid p,$$

and m has the least degree among the non-zero polynomials annihilating T .

- (c) If V is finite-dimensional, then $I_T \neq \{0\}$, and $\deg m \leq (\dim V)^2$.

Proof.

- (a) By [Theorem 5.7.1](#) (a), I_T is the kernel of a linear map, so it is a subspace ([Theorem 3.2.1](#)); in particular it contains 0 and is closed under addition. If $p \in I_T$ and $q \in F[x]$, then $(qp)(T) = q(T)p(T) = 0$ by [Theorem 5.7.1](#) (b), so $qp \in I_T$. By [Definition 5.2.1](#), I_T is an ideal.
- (b) By [Theorem 5.2.2](#), either $I_T = \{0\}$, or $I_T = \langle m \rangle$ for a unique monic m , which has least degree among the non-zero elements of I_T . In the second case, $p(T) = 0$ means $p \in \langle m \rangle$, that is, $p = qm$ for some q , that is, $m \mid p$.
- (c) By [Theorem 5.7.2](#), I_T contains a non-zero polynomial of degree at most $(\dim V)^2$. So $I_T \neq \{0\}$, and $\deg m \leq (\dim V)^2$ because m has the least degree. ■

This monic generator is one of the most important invariants of an operator. **Chapter 8 names it the minimal polynomial of T** and relates its roots to eigenvalues; here it is simply “the monic generator of I_T ”. Part (b) turns every polynomial identity satisfied by T into a divisibility statement about one fixed polynomial.

Examples.

- **Projections.** Let P be a projection with $P \neq 0$ and $P \neq \text{id}_V$. Then $x^2 - x = x(x-1) \in I_P$, so the monic generator divides $x(x-1)$. By unique factorization ([Theorem 5.3.5](#)), its monic divisors are 1, x , $x-1$ and $x(x-1)$. The first three do not annihilate P : $1(P) = \text{id}_V \neq 0$, $x(P) = P \neq 0$ and $(x-1)(P) = P - \text{id}_V \neq 0$. So the generator is $x^2 - x$.

- **Degenerate cases.** For the identity on $V \neq \{0\}$, the generator is $x - 1$; for the zero operator on $V \neq \{0\}$, it is x ; on $V = \{0\}$, it is 1.
- **The matrix B .** In [Example \(Finding an Annihilating Polynomial\)](#), no non-zero polynomial of degree at most 1 annihilates B , and $x^2 - 3x + 2$ does, so the generator is $x^2 - 3x + 2$.
- **Infinite dimension.** For multiplication by x on $F[x]$, we saw that $I_S = \{0\}$.

Quick check. Let $R(x, y) = (-y, x)$ be the rotation of $\mathbb{R}^2$ by a right angle. What is the monic generator of $I_R \subseteq \mathbb{R}[x]$?

Solution. $R^2 = -\text{id}$, so $x^2 + 1 \in I_R$, and the generator divides $x^2 + 1$. Over $\mathbb{R}$, $x^2 + 1$ has no root, so its only monic divisors are 1 and $x^2 + 1$ (a monic divisor of degree 1 would be $x - c$ with c a root). Since $1(R) = \text{id} \neq 0$, the generator is $x^2 + 1$.

5.7.3 Kernels of polynomials in T

The subspaces we can build from T and a polynomial p are $\ker p(T)$ and $\text{im } p(T)$. For $p = x - \lambda$, the kernel $\ker(T - \lambda \text{id}_V)$ consists of the vectors that T merely scales by λ ; for a projection, $\ker P$ and $\ker(P - \text{id}_V)$ are the two pieces of the direct sum. These subspaces have a property that general subspaces lack: T does not move vectors out of them.

Theorem 5.7.4 — T Preserves Kernels and Images of Polynomials in T (a)

Let $R, S \in \mathcal{L}(V)$ with $SR = RS$. Then $S\mathbf{v} \in \ker R$ for every $\mathbf{v} \in \ker R$, and $S\mathbf{w} \in \text{im } R$ for every $\mathbf{w} \in \text{im } R$.

- (b) In particular, for $T \in \mathcal{L}(V)$ and $p, q \in F[x]$, the operator $q(T)$ maps $\ker p(T)$ into $\ker p(T)$ and $\text{im } p(T)$ into $\text{im } p(T)$. Taking $q = x$: T maps $\ker p(T)$ and $\text{im } p(T)$ into themselves.

Proof.

- (a) Let $\mathbf{v} \in \ker R$. Then $R(S\mathbf{v}) = S(R\mathbf{v}) = S\mathbf{0} = \mathbf{0}$, so $S\mathbf{v} \in \ker R$. Let $\mathbf{w} = R\mathbf{u} \in \text{im } R$. Then $S\mathbf{w} = S(R\mathbf{u}) = R(S\mathbf{u}) \in \text{im } R$.
- (b) By [Theorem 5.7.1](#) (c), $q(T)$ commutes with $p(T)$, so (a) applies with $R = p(T)$ and $S = q(T)$. For $q = x$, $q(T) = T$.

■

In Chapter 8, a subspace that an operator maps into itself will be called an invariant subspace, and the notion will be developed properly there. For now the theorem is what matters: whenever $U = \ker p(T)$, the rule $\mathbf{u} \mapsto T\mathbf{u}$ is an operator on U , and so is $\mathbf{u} \mapsto q(T)\mathbf{u}$.

The inclusion can fail for subspaces not built from T . For the rotation R of the check above, R moves the x -axis to the y -axis, so the x -axis is not of the form $\ker p(R)$ or $\text{im } p(R)$ for any p .

5.7.4 The kernel splitting lemma

Chapter 3 contains two theorems with the same shape. A projection satisfies $P(P - \text{id}_V) = 0$, and [Theorem 3.10.1](#) splits V as $\text{im } P \oplus \ker P$. An involution in characteristic not 2 satisfies

$(T - \text{id}_V)(T + \text{id}_V) = 0$, and [Theorem 3.10.5](#) splits V as $\ker(T - \text{id}_V) \oplus \ker(T + \text{id}_V)$. In both, a polynomial annihilating T factors into two pieces with no common factor, and V is the direct sum of the kernels of the pieces. In both, the projections turned out to be polynomials in T : P itself, and $\frac{1}{2}(\text{id}_V + T)$. The general statement needs only the arithmetic of $F[x]$.

Theorem 5.7.5 — Kernel Splitting Lemma

Let V be a vector space over F , let $T \in \mathcal{L}(V)$, and let $p, q \in F[x]$ be **coprime**. Choose $a, b \in F[x]$ with $ap + bq = 1$ ([Corollary 5.2.4](#)), and put

$$W = \ker(pq)(T), \quad E_1 = b(T)q(T), \quad E_2 = a(T)p(T).$$

Then:

- (a) $W = \ker p(T) \oplus \ker q(T)$;
- (b) $E_1 + E_2 = \text{id}_V$, and for every $\mathbf{w} \in W$, the splitting of $\mathbf{w}$ in (a) is

$$\mathbf{w} = E_1\mathbf{w} + E_2\mathbf{w}, \quad E_1\mathbf{w} \in \ker p(T), \quad E_2\mathbf{w} \in \ker q(T);$$

so E_1 maps W into W and acts on W as the projection onto $\ker p(T)$ along $\ker q(T)$, and E_2 acts on W as the projection onto $\ker q(T)$ along $\ker p(T)$;

- (c) in particular, if $(pq)(T) = 0$, then

$$V = \ker p(T) \oplus \ker q(T),$$

and the projections of V onto the two summands along each other are the polynomials $(bq)(T)$ and $(ap)(T)$ in T .

Idea. Substitute T into Bézout's identity: $\text{id}_V = a(T)p(T) + b(T)q(T)$. Applied to a vector $\mathbf{w}$, this writes $\mathbf{w} = E_1\mathbf{w} + E_2\mathbf{w}$, and it does so for **every** vector of V . Now look at the pieces. ① $p(T)$ kills $E_1\mathbf{w} = b(T)q(T)\mathbf{w}$, because $p(T)b(T)q(T) = b(T)(pq)(T)$ and $(pq)(T)\mathbf{w} = \mathbf{0}$ for $\mathbf{w} \in W$: the factor q in E_1 combines with p to the whole product. Symmetrically $q(T)$ kills $E_2\mathbf{w}$. ② If $\mathbf{w}$ is killed by both $p(T)$ and $q(T)$, then both pieces are $\mathbf{0}$, so $\mathbf{w} = \mathbf{0}$. ③ Direct sum by the intersection criterion; the projections are read off from uniqueness. Coprimality enters only through the identity $ap + bq = 1$, and without it step ② has nothing to stand on.

Proof. Since $ap + bq = 1$, [Theorem 5.7.1](#) gives

$$\text{id}_V = a(T)p(T) + b(T)q(T) = E_2 + E_1.$$

Throughout, polynomials in T commute with each other ([Theorem 5.7.1](#) (c)).

The kernels lie in W . If $p(T)\mathbf{v} = \mathbf{0}$, then $(pq)(T)\mathbf{v} = q(T)p(T)\mathbf{v} = \mathbf{0}$; so $\ker p(T) \subseteq W$, and likewise $\ker q(T) \subseteq W$.

The pieces. Let $\mathbf{w} \in W$. Then $\mathbf{w} = \text{id}_V \mathbf{w} = E_1\mathbf{w} + E_2\mathbf{w}$. Moreover

$$p(T)(E_1\mathbf{w}) = p(T)b(T)q(T)\mathbf{w} = b(T)(pq)(T)\mathbf{w} = b(T)\mathbf{0} = \mathbf{0},$$

so $E_1\mathbf{w} \in \ker p(T)$, and similarly $q(T)(E_2\mathbf{w}) = a(T)(pq)(T)\mathbf{w} = \mathbf{0}$, so $E_2\mathbf{w} \in \ker q(T)$. Hence $W = \ker p(T) + \ker q(T)$.

The intersection. Let $\mathbf{v} \in \ker p(T) \cap \ker q(T)$. Then $E_1\mathbf{v} = b(T)q(T)\mathbf{v} = \mathbf{0}$ and $E_2\mathbf{v} = a(T)p(T)\mathbf{v} = \mathbf{0}$, so $\mathbf{v} = E_1\mathbf{v} + E_2\mathbf{v} = \mathbf{0}$. By [Theorem 1.7.4](#), the sum is direct, which proves (a).

The projections. The splitting of $\mathbf{w} \in W$ in (a) is unique ([Definition 1.7.2](#)), and we have exhibited it as $\mathbf{w} = E_1\mathbf{w} + E_2\mathbf{w}$; this proves the formula in (b). In particular $E_1\mathbf{w} \in \ker p(T) \subseteq W$, so E_1 maps W into W . As an operator on W , E_1 is linear, fixes every $\mathbf{u} \in \ker p(T)$ (whose splitting is $\mathbf{u} = \mathbf{u} + \mathbf{0}$) and kills every vector of $\ker q(T)$ (whose splitting is $\mathbf{0} + \mathbf{v}$). By the uniqueness in [Theorem 3.10.1](#) (b), applied to $W = \ker p(T) \oplus \ker q(T)$, it is the projection onto $\ker p(T)$ along $\ker q(T)$. Swapping the roles of p and q gives the statement for E_2 .

- (c) If $(pq)(T) = 0$, then $W = V$, and (a) and (b) become the statements in (c), with $E_1 = (bq)(T)$ and $E_2 = (ap)(T)$ by [Theorem 5.7.1](#) (b). This proves the theorem. ■

Bézout's coefficients a, b are not unique, but the projections on W are, by [Theorem 3.10.1](#) (b). So different choices of a, b give polynomials bq that may differ, while $(bq)(T)$ always does the same thing on W . When $(pq)(T) = 0$, the lemma turns a factorization of an annihilating polynomial into a direct sum decomposition of the whole space, with explicit projections, and each summand is carried into itself by T ([Theorem 5.7.4](#)).

Example. (Recovering Projections and Direct Sums)

Let $P \in \mathcal{L}(V)$ be a projection. Apply [Theorem 5.7.5](#) with $p = x$ and $q = x - 1$, and compare with [Theorem 3.10.1](#) (a).

Solution. Coprime. $1 \cdot x + (-1)(x - 1) = 1$, so we may take $a = 1$, $b = -1$, and any common divisor of x and $x - 1$ divides 1. *Annihilating.* $(pq)(P) = P^2 - P = 0$ because P is a projection.

By [Theorem 5.7.5](#) (c),

$$V = \ker P \oplus \ker(P - \text{id}_V),$$

with projection $E_1 = b(P)q(P) = -(P - \text{id}_V) = \text{id}_V - P$ onto $\ker P$ along $\ker(P - \text{id}_V)$, and $E_2 = a(P)p(P) = P$ onto $\ker(P - \text{id}_V)$ along $\ker P$.

To compare, note that $\ker(P - \text{id}_V) = \text{im } P$. ($\subseteq$) If $P\mathbf{v} = \mathbf{v}$, then $\mathbf{v} \in \text{im } P$. ($\supseteq$) If $\mathbf{v} = P\mathbf{u}$, then $P\mathbf{v} = P^2\mathbf{u} = P\mathbf{u} = \mathbf{v}$. So the lemma gives $V = \text{im } P \oplus \ker P$, with P the projection onto $\text{im } P$ along $\ker P$: this is [Theorem 3.10.1](#) (a). It also reproves [Exercise \(B2: The complementary projection\)](#), that $\text{id}_V - P$ is the complementary projection.

Example. (Recovering the Involution Decomposition)

Let F have characteristic not 2, and let $T \in \mathcal{L}(V)$ with $T^2 = \text{id}_V$. Apply [Theorem 5.7.5](#) with $p = x - 1$ and $q = x + 1$, and compare with [Theorem 3.10.5](#). What goes wrong in characteristic 2?

Solution. Coprime. Since $2 \neq 0$ in F ([Definition 0.7.2](#)),

$$-\frac{1}{2}(x - 1) + \frac{1}{2}(x + 1) = 1,$$

so $a = -\frac{1}{2}$ and $b = \frac{1}{2}$ work, and a common divisor of p and q divides 1. *Annihilating.* $(pq)(T) = T^2 - \text{id}_V = 0$.

By [Theorem 5.7.5](#) (c),

$$V = \ker(T - \text{id}_V) \oplus \ker(T + \text{id}_V) = E_+ \oplus E_-,$$

and the projection onto E_+ along E_- is $b(T)q(T) = \frac{1}{2}(T + \text{id}_V)$. These are exactly the conclusions of [Theorem 3.10.5](#). The other projection is $a(T)p(T) = \frac{1}{2}(\text{id}_V - T)$.

In characteristic 2, $x + 1 = x - 1$, so $p = q$ and $\gcd(p, q) = x - 1 \neq 1$. The lemma does not apply, and the warning after [Theorem 3.10.5](#) shows that the conclusion really fails there.

The lemma applies just as well when neither factor is linear, and then it finds subspaces that are not visible from single vectors.

Example. (A Splitting with a Repeated Factor)

Let $A = \begin{pmatrix} 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \in M_3(\mathbb{Q})$. Verify that $A^2(A - I_3) = 0$, and use [Theorem 5.7.5](#) with $p = x^2$, $q = x - 1$ to split $\mathbb{Q}^3$, giving both projections as polynomials in A .

Solution. The relation. $A^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$, and $A^3 = A \cdot A^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} = A^2$. So $A^2(A - I_3) = A^3 - A^2 = 0$.

Bézout. $x^2 = (x+1)(x-1) + 1$, so $1 \cdot x^2 + (-(x+1))(x-1) = 1$: take $a = 1$, $b = -(x+1)$. Then

$$E_1 = b(A)q(A) = -(A+I_3)(A-I_3) = I_3 - A^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}, \quad E_2 = a(A)p(A) = A^2.$$

The summands. $\ker A^2 = \{(x, y, z) : z = 0\} = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$, and $\ker(A - I_3)$ is cut out by $-x + y - z = 0$, $-y + z = 0$, $0 = 0$, so it is $\text{span}((0, 1, 1))$. By [Theorem 5.7.5](#) (c),

$$\mathbb{Q}^3 = \text{span}(\mathbf{e}_1, \mathbf{e}_2) \oplus \text{span}((0, 1, 1)),$$

and for $\mathbf{v} = (x, y, z)$, the splitting is $E_1\mathbf{v} + E_2\mathbf{v} = (x, y - z, 0) + (0, z, z)$. One checks directly that $(x, y - z, 0)$ has third entry 0 and $(0, z, z)$ is a multiple of $(0, 1, 1)$.

Note that $\ker A = \text{span}(\mathbf{e}_1)$ is strictly smaller than $\ker A^2$. The factor $p = x^2$ could not be replaced by x : the polynomial $x(x - 1)$ does not annihilate A , since $A^2 - A \neq 0$.

Warning. Without coprimality the kernels need not fill $\ker(pq)(T)$, or even form a direct sum. Let $T = T_N$ with $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ on F^2 , and $p = q = x$. Then $(pq)(T) = N^2 = 0$, so $\ker(pq)(T) = F^2$. But $\ker p(T) = \ker q(T) = \ker N = \text{span}(\mathbf{e}_1)$. The two kernels are equal, so their sum is not direct, and $\text{span}(\mathbf{e}_1) + \text{span}(\mathbf{e}_1) = \text{span}(\mathbf{e}_1) \neq F^2$. Here $\gcd(x, x) = x$, and no identity $ax + bx = 1$ exists, since its left side vanishes at 0.

Quick check. Let $T \in \mathcal{L}(V)$ satisfy $T^2 = 3T$, over a field with $3 \neq 0$. Split V with the lemma, and write the projection onto $\ker T$ as a polynomial in T .

Solution. $x(x-3)$ annihilates T . Since $\frac{1}{3}x - \frac{1}{3}(x-3) = 1$, $p = x$ and $q = x-3$ are coprime with $a = \frac{1}{3}$, $b = -\frac{1}{3}$. By [Theorem 5.7.5](#) (c), $V = \ker T \oplus \ker(T-3\text{id}_V)$, and the projection onto $\ker T$ along $\ker(T-3\text{id}_V)$ is $b(T)q(T) = -\frac{1}{3}(T-3\text{id}_V) = \text{id}_V - \frac{1}{3}T$.

In Chapter 8 the lemma is used with an annihilating polynomial factored into powers of distinct irreducibles, $m = p_1^{e_1} \cdots p_k^{e_k}$. Distinct monic irreducibles are coprime, and so are their powers, so the lemma with k factors ([Exercise \(C2: Splitting with \$k\$ coprime factors\)](#)) splits V into the pieces $\ker p_i(T)^{e_i}$. That is the primary decomposition, and the canonical forms of Chapter 9 are built on it.

5.7.5 Exercises

5.7.5.1 A. Check your understanding

Exercise A1.

- State what it means that $p \mapsto p(T)$ is linear and multiplicative.
- True or false: if $p(T)q(T) = 0$, then $p(T) = 0$ or $q(T) = 0$. Justify your answer.
- Why does every operator on a finite-dimensional space have a non-zero annihilating polynomial? Is the same true on $F[x]$?
- True or false: if p annihilates T and $p \mid q$, then q annihilates T . Justify your answer.
- State the kernel splitting lemma for $(pq)(T) = 0$, and say where coprimality is used in its proof.
- True or false: for every subspace U of V and every $T \in \mathcal{L}(V)$, T maps U into U . Justify your answer.

Solution.

- $(cp+q)(T) = cp(T) + q(T)$ and $(pq)(T) = p(T)q(T)$ for all $p, q \in F[x]$ and $c \in F$ ([Theorem 5.7.1](#)).
- False. For the projection $P(x, y) = (x, 0)$, $P(P - \text{id}) = 0$ while $P \neq 0$ and $P - \text{id} \neq 0$.
- $\mathcal{L}(V)$ has dimension n^2 , so the $n^2 + 1$ operators $\text{id}_V, T, \dots, T^{n^2}$ are linearly dependent, and a dependence is a non-zero annihilating polynomial ([Theorem 5.7.2](#)). On $F[x]$, multiplication by x has no non-zero annihilating polynomial, since $p(S)(1) = p$.
- True. If $q = hp$, then $q(T) = h(T)p(T) = 0$. This is the ideal property ([Theorem 5.7.3](#)).
- If p, q are coprime and $(pq)(T) = 0$, then $V = \ker p(T) \oplus \ker q(T)$, and the projections are polynomials in T ([Theorem 5.7.5](#)). Coprimality gives $ap + bq = 1$, hence $\text{id}_V = a(T)p(T) + b(T)q(T)$, which both produces the splitting of each vector and shows the intersection is $\{0\}$.

- (f) False. The rotation $R(x, y) = (-y, x)$ maps the x -axis to the y -axis. The statement is true for $U = \ker p(T)$ and $U = \operatorname{im} p(T)$ (Theorem 5.7.4).

5.7.5.2 B. Practice

Exercise B1: A polynomial of a 2×2 matrix.

Let $A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix} \in M_2(\mathbb{Q})$.

- (a) Compute $p(A)$ for $p = x^2 - 7$ and for $q = x^2 + x - 5$.
 (b) Hence show that A is invertible and find A^{-1} as a polynomial in A .

Solution.

- (a) $A^2 = \begin{pmatrix} 1+6 & 2-2 \\ 3-3 & 6+1 \end{pmatrix} = 7I_2$. Hence $p(A) = A^2 - 7I_2 = 0$. For q , divide: $q = (x^2 - 7) + (x + 2)$, so by Theorem 5.7.1 $q(A) = p(A) + A + 2I_2 = \begin{pmatrix} 3 & 2 \\ 3 & 1 \end{pmatrix}$. (Directly: $7I_2 + A - 5I_2 = A + 2I_2$.)
 (b) From $A^2 = 7I_2$, $A(\frac{1}{7}A) = (\frac{1}{7}A)A = I_2$. So A is invertible with $A^{-1} = \frac{1}{7}A = \begin{pmatrix} \frac{1}{7} & \frac{2}{7} \\ \frac{3}{7} & -\frac{1}{7} \end{pmatrix}$.

Exercise B2: An annihilating polynomial by elimination.

Let $C = \begin{pmatrix} 0 & 0 & 2 \\ 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix} \in M_3(\mathbb{Q})$.

- (a) Compute C^2 and C^3 , and show that (I_3, C, C^2) is linearly independent.
 (b) Find a linear dependence among I_3, C, C^2, C^3 , and hence a monic polynomial p of degree 3 with $p(C) = 0$.
 (c) Hence show that C is invertible and write C^{-1} as a polynomial in C .

Solution.

- (a) Multiplying out,

$$C^2 = \begin{pmatrix} 0 & 2 & 4 \\ 0 & -1 & 0 \\ 1 & 2 & 3 \end{pmatrix}, \quad C^3 = C \cdot C^2 = \begin{pmatrix} 2 & 4 & 6 \\ -1 & 0 & 1 \\ 2 & 3 & 6 \end{pmatrix}.$$

Let $aI_3 + bC + cC^2 = 0$. The $(2, 1)$ -entry gives $b = 0$; the $(3, 1)$ -entry gives $c = 0$; then the $(1, 1)$ -entry gives $a = 0$. So the list is independent.

- (b) Let $aI_3 + bC + cC^2 + dC^3 = 0$. Reading entries: $(2, 1)$: $b - d = 0$; $(3, 1)$: $c + 2d = 0$; $(1, 1)$: $a + 2d = 0$. With $d = 1$: $b = 1$, $c = -2$, $a = -2$. Check all entries of $-2I_3 + C - 2C^2 + C^3$: row 1 is $(-2 + 0 - 0 + 2, 0 - 4 + 4, 2 - 8 + 6) = (0, 0, 0)$; row 2 is $(1 - 0 - 1, -2 + 0 + 2 + 0, -1 - 0 + 1) = (0, 0, 0)$; row 3 is

$(0 - 2 + 2, 1 - 4 + 3, -2 + 2 - 6 + 6) = (0, 0, 0)$. Hence

$$p = x^3 - 2x^2 + x - 2 = (x - 2)(x^2 + 1)$$

satisfies $p(C) = 0$. By (a) no non-zero polynomial of degree at most 2 annihilates C , so p is the monic generator of I_C (Theorem 5.7.3).

- (c) $C^3 - 2C^2 + C = 2I_3$, that is, $C(C^2 - 2C + I_3) = (C^2 - 2C + I_3)C = 2I_3$, using Theorem 5.7.1 (c) for the second equality. Since $2 \neq 0$ in $\mathbb{Q}$, C is invertible with

$$C^{-1} = \frac{1}{2}(C^2 - 2C + I_3) = \begin{pmatrix} \frac{1}{2} & 1 & 0 \\ -1 & 0 & 1 \\ \frac{1}{2} & 0 & 0 \end{pmatrix}.$$

Exercise B3: Operators with $T^3 = T$.

Let F have characteristic not 2, and let $T \in \mathcal{L}(V)$ with $T^3 = T$.

- (a) Use Theorem 5.7.5 with $p = x$ and $q = x^2 - 1$ to show $V = \ker T \oplus \ker(T^2 - \text{id}_V)$, with projection $\text{id}_V - T^2$ onto $\ker T$.
- (b) Show that $V = \ker T \oplus \ker(T - \text{id}_V) \oplus \ker(T + \text{id}_V)$.
- (c) Verify that the projections onto the three summands in (b) are $\text{id}_V - T^2$, $\frac{1}{2}(T^2 + T)$ and $\frac{1}{2}(T^2 - T)$, in the sense that these three operators sum to id_V and each maps V into the corresponding kernel.

Solution.

- (a) $x \cdot x + (-1)(x^2 - 1) = 1$: take $a = x$, $b = -1$. Then p and q are coprime, and $(pq)(T) = T^3 - T = 0$. By Theorem 5.7.5 (c), $V = \ker T \oplus \ker(T^2 - \text{id}_V)$, and the projection onto $\ker T$ is $b(T)q(T) = -(T^2 - \text{id}_V) = \text{id}_V - T^2$.
- (b) Let $W = \ker(T^2 - \text{id}_V)$. Apply Theorem 5.7.5 (a) with $p_1 = x - 1$, $q_1 = x + 1$, which are coprime as in Example (Recovering the Involution Decomposition): $W = \ker(p_1 q_1)(T) = \ker(T - \text{id}_V) \oplus \ker(T + \text{id}_V)$. Write $K_0 = \ker T$, $K_+ = \ker(T - \text{id}_V)$, $K_- = \ker(T + \text{id}_V)$. Then $V = K_0 + K_+ + K_-$ by (a). If $\mathbf{u}_0 + \mathbf{u}_+ + \mathbf{u}_- = \mathbf{0}$ with $\mathbf{u}_0 \in K_0$, $\mathbf{u}_\pm \in K_\pm$, then $\mathbf{u}_+ + \mathbf{u}_- \in W$, so directness in (a) gives $\mathbf{u}_0 = \mathbf{0}$ and $\mathbf{u}_+ + \mathbf{u}_- = \mathbf{0}$, and directness of $W = K_+ \oplus K_-$ gives $\mathbf{u}_+ = \mathbf{u}_- = \mathbf{0}$. By Theorem 1.7.5 ((b) $\Rightarrow$ (a)), the sum is direct.
- (c) The sum is $\text{id}_V - T^2 + \frac{1}{2}T^2 + \frac{1}{2}T + \frac{1}{2}T^2 - \frac{1}{2}T = \text{id}_V$. Using $T^3 = T$: $T(\text{id}_V - T^2) = T - T^3 = 0$; $(T - \text{id}_V)\frac{1}{2}(T^2 + T) = \frac{1}{2}(T^3 - T) = 0$; $(T + \text{id}_V)\frac{1}{2}(T^2 - T) = \frac{1}{2}(T^3 - T) = 0$. So the three operators map V into K_0 , K_+ , K_- respectively. For $\mathbf{v} \in V$, $\mathbf{v}$ is the sum of its three images, and by the uniqueness of the splitting in (b) these are the components of $\mathbf{v}$. Hence each operator is the projection onto its summand along the sum of the other two.

5.7.5.3 C. Going deeper

Exercise C1: Inverses are polynomials.

Let V be a vector space over F and $T \in \mathcal{L}(V)$.

- (a) Suppose $p(T) = 0$ for some $p \in F[x]$ with $p(0) \neq 0$. Prove that T is invertible and that $T^{-1} = g(T)$ for some $g \in F[x]$.
- (b) Deduce that if V is finite-dimensional and T is invertible, then T^{-1} is a polynomial in T .
- (c) Show that the monic generator m of I_T , for V finite-dimensional and non-zero, satisfies $m(0) \neq 0$ if and only if T is invertible.

Hint: for (b), write a non-zero annihilating polynomial as $x^k p_1$ with $p_1(0) \neq 0$.

Solution.

- (a) Write $p = c + xg_0$, where $c = p(0) \neq 0$ and $g_0 \in F[x]$. Then $0 = p(T) = c \operatorname{id}_V + Tg_0(T)$, so $Tg_0(T) = -c \operatorname{id}_V$. Put $g = -c^{-1}g_0$. Then $Tg(T) = \operatorname{id}_V$, and $g(T)T = Tg(T) = \operatorname{id}_V$ by [Theorem 5.7.1](#) (c). Hence T is invertible with $T^{-1} = g(T)$.
- (b) If $V = \{0\}$, then $T^{-1} = \operatorname{id}_V = 1(T)$. Otherwise, by [Theorem 5.7.2](#) there is a non-zero p with $p(T) = 0$. Let k be the least index with a non-zero coefficient in p ; then $p = x^k p_1$ with $p_1(0) \neq 0$. Now $0 = p(T) = T^k p_1(T)$. Since T is invertible, so is T^k , with inverse $(T^{-1})^k$, and composing on the left with it gives $p_1(T) = 0$. By (a), T^{-1} is a polynomial in T .
- (c) $(\Rightarrow)$ If $m(0) \neq 0$, apply (a) to m . $(\Leftarrow)$ Suppose T is invertible and $m(0) = 0$. Then $m = xm_1$ with m_1 monic of degree $\deg m - 1$, and $0 = m(T) = Tm_1(T)$. Composing with T^{-1} gives $m_1(T) = 0$, so $m \mid m_1$ by [Theorem 5.7.3](#) (b). Since $V \neq \{0\}$, m is not constant, so $m_1 \neq 0$ and $\deg m \leq \deg m_1 = \deg m - 1$ by [Theorem 0.8.3](#), a contradiction. Hence $m(0) \neq 0$.

Exercise C2: Splitting with k coprime factors.

Let $T \in \mathcal{L}(V)$, and let $p_1, \dots, p_k \in F[x]$ be non-zero and pairwise coprime, with $k \geq 2$. Put $p = p_1 \cdots p_k$ and $W = \ker p(T)$.

- (a) Use [Theorem 5.6.4](#) to find $e_1, \dots, e_k \in F[x]$ with $e_i \equiv 1 \pmod{p_i}$ and $e_i \equiv 0 \pmod{p_j}$ for $j \neq i$. Show that $p \mid e_1 + \cdots + e_k - 1$ and that $p \mid p_i e_i$ for each i .
- (b) Prove that for every $\mathbf{w} \in W$, $\mathbf{w} = e_1(T)\mathbf{w} + \cdots + e_k(T)\mathbf{w}$ with $e_i(T)\mathbf{w} \in \ker p_i(T)$.
- (c) Prove that $W = \ker p_1(T) \oplus \cdots \oplus \ker p_k(T)$, and that $e_i(T)$ is the identity on $\ker p_i(T)$ and zero on $\ker p_j(T)$ for $j \neq i$.
- (d) Deduce: if $p(T) = 0$, then $V = \ker p_1(T) \oplus \cdots \oplus \ker p_k(T)$, and the projection onto the i -th summand along the others is the polynomial $e_i(T)$.

Hint: for the second divisibility in (a), use [Lemma 5.6.3](#) (b).

Solution.

- (a) By [Theorem 5.6.4](#) (a), applied to the pairwise coprime moduli $p_1, \dots, p_k$ with $r_i = 1$ and $r_j = 0$ for $j \neq i$, such an e_i exists. The sum $s = e_1 + \cdots + e_k$ satisfies $s \equiv 1 \pmod{p_j}$ for each j , since exactly one summand is $\equiv 1$ and the others are $\equiv 0$. The constant 1 satisfies the same congruences, so $s \equiv 1 \pmod{p}$ by [Theorem 5.6.4](#) (b). For the second claim, $p_j \mid e_i$ for every $j \neq i$, and these p_j are pairwise coprime, so $\prod_{j \neq i} p_j \mid e_i$ by [Lemma 5.6.3](#) (b). Multiplying by p_i , $p \mid p_i e_i$.

- (b) Write $e_1 + \cdots + e_k = 1 + hp$ and $p_i e_i = h_i p$. By [Theorem 5.7.1](#), for $\mathbf{w} \in W$,

$$e_1(T)\mathbf{w} + \cdots + e_k(T)\mathbf{w} = \mathbf{w} + h(T)p(T)\mathbf{w} = \mathbf{w}, \quad p_i(T)(e_i(T)\mathbf{w}) = h_i(T)p(T)\mathbf{w} = \mathbf{0}.$$

- (c) Each $\ker p_i(T) \subseteq W$, since $p(T) = (\prod_{j \neq i} p_j)(T)p_i(T)$. With (b), $W = \ker p_1(T) + \cdots + \ker p_k(T)$. Let $\mathbf{u}_j \in \ker p_j(T)$. For $j \neq i$, $p_j \mid e_i$, say $e_i = g p_j$, so $e_i(T)\mathbf{u}_j = g(T)p_j(T)\mathbf{u}_j = \mathbf{0}$. And $e_i = 1 + g'p_i$, so $e_i(T)\mathbf{u}_i = \mathbf{u}_i + g'(T)p_i(T)\mathbf{u}_i = \mathbf{u}_i$. This proves the second statement. Now suppose $\mathbf{u}_1 + \cdots + \mathbf{u}_k = \mathbf{0}$ with $\mathbf{u}_j \in \ker p_j(T)$. Applying $e_i(T)$ gives $\mathbf{u}_i = \mathbf{0}$, for each i . By [Theorem 1.7.5](#) ((b) $\Rightarrow$ (a)), the sum is direct.
- (d) If $p(T) = 0$, then $W = V$. By (c), V is the direct sum, and $e_i(T) \in \mathcal{L}(V)$ is the identity on the i -th summand and zero on the others. For $\mathbf{v} = \mathbf{u}_1 + \cdots + \mathbf{u}_k$, $e_i(T)\mathbf{v} = \mathbf{u}_i$, which is the projection onto $\ker p_i(T)$ along the sum of the other summands. For $k = 2$ this is [Theorem 5.7.5](#) (c).

Exercise C3: The same rotation over $\mathbb{R}$ and over $\mathbb{C}$.

Let $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, so that $A^2 + I_2 = 0$.

- (a) Over $\mathbb{C}$, apply [Theorem 5.7.5](#) to T_A on $\mathbb{C}^2$ with $p = x - i$ and $q = x + i$. Find both kernels and write the projection onto $\ker(A - iI_2)$ as a polynomial in A .
- (b) Over $\mathbb{R}$, show that the only subspaces of $\mathbb{R}^2$ that T_A maps into themselves are $\{\mathbf{0}\}$ and $\mathbb{R}^2$. Explain why this matches the fact that $x^2 + 1$ admits no factorization into coprime non-constant factors in $\mathbb{R}[x]$.

Solution.

- (a) $(x + i) - (x - i) = 2i$, so $\frac{i}{2}(x - i) - \frac{i}{2}(x + i) = \frac{i}{2}(-2i) = 1$: take $a = \frac{i}{2}$ and $b = -\frac{i}{2}$. So p, q are coprime, and $(pq)(A) = A^2 + I_2 = 0$. By [Theorem 5.7.5](#) (c), $\mathbb{C}^2 = \ker(A - iI_2) \oplus \ker(A + iI_2)$. Solving, $(A - iI_2)(z, w) = (-iz - w, z - iw) = \mathbf{0}$ gives $w = -iz$, so $\ker(A - iI_2) = \text{span}((1, -i))$; similarly $\ker(A + iI_2) = \text{span}((1, i))$. The projection onto $\ker(A - iI_2)$ is

$$b(A)q(A) = -\frac{i}{2}(A + iI_2) = \frac{1}{2}(I_2 - iA) = \begin{pmatrix} \frac{1}{2} & \frac{i}{2} \\ -\frac{i}{2} & \frac{1}{2} \end{pmatrix}.$$

Check: it sends $(1, -i)$ to $(\frac{1}{2} + \frac{1}{2}, -\frac{i}{2} - \frac{i}{2}) = (1, -i)$ and $(1, i)$ to $(\frac{1}{2} - \frac{1}{2}, -\frac{i}{2} + \frac{i}{2}) = \mathbf{0}$.

- (b) A subspace of $\mathbb{R}^2$ other than $\{\mathbf{0}\}$ and $\mathbb{R}^2$ has dimension 1 ([Theorem 1.6.7](#)), so it is $\text{span}(\mathbf{v})$ with $\mathbf{v} = (s, t) \neq \mathbf{0}$. If $A\mathbf{v} \in \text{span}(\mathbf{v})$, then $(-t, s) = c(s, t)$ for some $c \in \mathbb{R}$, so $-t = cs$ and $s = ct$, giving $s = -c^2s$ and $t = -c^2t$. Since $1 + c^2 > 0$, $s = t = 0$, a contradiction. So only $\{\mathbf{0}\}$ and $\mathbb{R}^2$ are mapped into themselves.

Over $\mathbb{R}$, $x^2 + 1$ has no root, so its only monic divisors are 1 and $x^2 + 1$, and the only factorizations $x^2 + 1 = pq$ into monic factors have $p = 1$ or $q = 1$. The lemma then gives only $\ker p(T) \oplus \ker q(T) = \{\mathbf{0}\} \oplus \mathbb{R}^2$, and by (b) there is no finer splitting into subspaces preserved by T_A at all. Enlarging the field to $\mathbb{C}$ creates the coprime factors $x \mp i$, and with them the splitting of (a).

Chapter 6

Determinants

A square matrix is invertible or it is not, and so far deciding which has meant row reducing it. The determinant packages that decision into a single number. The same number measures how the linear map scales areas and volumes, and its sign records whether the map preserves or reverses orientation. It also produces the characteristic polynomial, the gateway to eigenvalues in Chapter 8.

This chapter does not start from a formula. It starts from what a signed volume must do. Scaling one edge scales the volume. Sliding one edge along another does not change it. The unit cube has volume 1. We prove that exactly one function has these properties, write it down, and then derive every property of the determinant from that uniqueness: multiplicativity, the transpose rule, cofactor expansion, the inverse formula and Cramer's rule. Nothing is assumed, and no formula is used before it is proved.

What you need. From Chapter 0, permutations and transpositions ([Theorem 0.10.4](#)) and the characteristic of a field ([Definition 0.7.2](#)). From Chapter 1, linear independence and the Linear Dependence Lemma ([Definition 1.4.1](#), [Theorem 1.4.1](#)). From Chapter 2, elementary matrices and the invertible matrix theorem ([Definition 2.4.1](#), [Theorem 2.4.4](#)). From Chapter 3, the matrix of a linear map and similarity ([Definition 3.6.1](#), [Theorem 3.7.4](#)). From Chapter 4, linear functionals ([Definition 4.1.1](#)). From Chapter 5, the root bound for polynomials ([Corollary 5.1.4](#)).

6.0.1 Roadmap

- **Signed area and volume.** Three rules that any reasonable notion of signed volume obeys, and the formula $ad - bc$ forced by them.
- **Multilinear alternating forms.** The rules made precise for functions of n vectors, and their first consequences.
- **Permutations and sign.** Counting inversions, and why the parity of a permutation is well defined.
- **Existence and uniqueness of the determinant.** Exactly one alternating form takes the value 1 on the identity; the Leibniz formula.
- **Row operations and multiplicativity.** Computing determinants by elimination, $\det(AB) = \det A \det B$, and the determinant of an operator.
- **Cofactor expansion, the adjugate and Cramer's rule.** Expanding along a row or column, and a formula for the inverse.
- **Minors, rank and Cauchy–Binet.** Rank read off from minors, and the determinant of

a product of rectangular matrices.

- **Special determinants.** Vandermonde, tridiagonal and block triangular matrices, and rank-one updates.
- **Orientation and volume.** The determinant as a volume scaling factor, and the two orientations of $\mathbb{R}^n$.
- **The characteristic polynomial.** $\det(xI - A)$, its coefficients, and why similar matrices share it.

6.0.2 Named moves

- **Derive from the rules, not the formula.** Prove determinant facts from multilinearity, alternation and normalization, so no property is assumed before it is proved.
- **Uniqueness does the work.** To prove two expressions agree, show both are alternating multilinear in the columns and agree on the identity.
- **Build from elementary matrices.** Check a claim for elementary matrices, then extend it to products.
- **Look-alike matrices.** A matrix with two equal rows has determinant zero; expanding such a matrix proves the adjugate identity.
- **Induct on size.** Expand along a row or column to reduce an $n \times n$ determinant to smaller ones.

6.1 Signed Area and Volume

In Chapter 0 a single number, $ad - bc$, decided whether a 2×2 matrix is invertible ([Theorem 0.9.8](#)). For an $n \times n$ matrix, Chapter 2 answered the same question by row reduction, which is a procedure, not a number. Is there one number for every square matrix that does the job of $ad - bc$? This section does not define it yet. Instead it asks what such a number should **mean**, finds the answer in area and volume, and extracts a short list of rules. From the rules alone, $ad - bc$ comes out, with no other choice possible.

6.1.1 Area as a function of two vectors

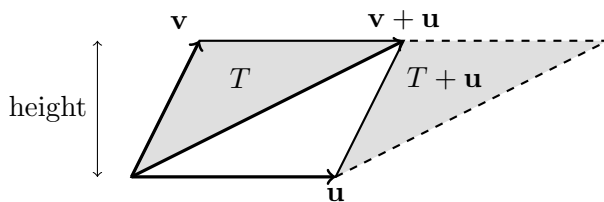
Take two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ and the parallelogram they span,

$$P(\mathbf{u}, \mathbf{v}) = \{s\mathbf{u} + t\mathbf{v} : 0 \leq s \leq 1, 0 \leq t \leq 1\}.$$

Its corners are $\mathbf{0}$, $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{u} + \mathbf{v}$. We want to understand its area as a **function of the two vectors**, without drawing a coordinate grid and counting squares. Four features are visible without any formula.

Scaling a side scales the area. Replacing $\mathbf{v}$ by $2\mathbf{v}$ stacks two copies of $P(\mathbf{u}, \mathbf{v})$ on top of each other, so the area doubles. Replacing $\mathbf{v}$ by $3\mathbf{v}$ triples it, and replacing $\mathbf{v}$ by $\frac{1}{2}\mathbf{v}$ halves it.

Shearing does not change the area. Replace $\mathbf{v}$ by $\mathbf{v} + c\mathbf{u}$. The top edge slides along its own line, parallel to the base $\mathbf{u}$. The base stays the same, and so does the height, so the area stays the same. The picture shows $c = 1$: the triangle cut off on the left of $P(\mathbf{u}, \mathbf{v})$ reappears, moved by $\mathbf{u}$, on the right of $P(\mathbf{u}, \mathbf{v} + \mathbf{u})$.



The solid outline is $P(\mathbf{u}, \mathbf{v})$ and the dashed one completes $P(\mathbf{u}, \mathbf{v} + \mathbf{u})$. Both consist of the unshaded middle triangle plus a shaded triangle, and the two shaded triangles T and $T + \mathbf{u}$ are translates of each other.

Equal sides give area zero. $P(\mathbf{u}, \mathbf{u})$ is a segment, which has no area. More generally, if $\mathbf{v}$ is a multiple of $\mathbf{u}$, the parallelogram is flat.

The unit square has area one. $P(\mathbf{e}_1, \mathbf{e}_2)$ is the unit square. This fixes the unit of measurement.

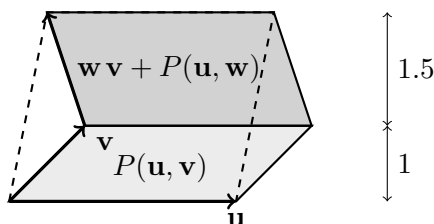
6.1.2 Why the area needs a sign

The first feature is not quite right as stated. Replacing $\mathbf{v}$ by $-\mathbf{v}$ moves the parallelogram to the other side of the line of $\mathbf{u}$ (the point reflection $\mathbf{x} \mapsto \mathbf{u} - \mathbf{x}$ carries $P(\mathbf{u}, \mathbf{v})$ onto $P(\mathbf{u}, -\mathbf{v})$) and does **not** change its area, so ordinary area satisfies $\text{area}(\mathbf{u}, c\mathbf{v}) = |c| \text{area}(\mathbf{u}, \mathbf{v})$, with an absolute value. That absolute value spoils a second, very useful property.

Fix the base $\mathbf{u}$ and let the second vector vary. The area is base times height, and heights along a fixed direction add up: if $\mathbf{v}$ reaches height 1 above the line of $\mathbf{u}$ and $\mathbf{w}$ reaches height 1.5, then $\mathbf{v} + \mathbf{w}$ reaches height 2.5. So we expect

$$\text{area}(\mathbf{u}, \mathbf{v} + \mathbf{w}) = \text{area}(\mathbf{u}, \mathbf{v}) + \text{area}(\mathbf{u}, \mathbf{w}).$$

The picture shows it: $P(\mathbf{u}, \mathbf{w})$, moved up by $\mathbf{v}$, sits exactly on top of $P(\mathbf{u}, \mathbf{v})$, and the two together have the same base and total height as $P(\mathbf{u}, \mathbf{v} + \mathbf{w})$ (dashed).



Now take $\mathbf{w} = -\mathbf{v}$. Then $\mathbf{v} + \mathbf{w} = \mathbf{0}$ and the left side is $\text{area}(\mathbf{u}, \mathbf{0}) = 0$, while the right side is $2 \text{area}(\mathbf{u}, \mathbf{v})$, which is positive. Additivity fails, because $-\mathbf{v}$ points **below** the line of $\mathbf{u}$ and its height should count as negative. The repair is to give area a sign.

Signed area is ordinary area, counted positive when the second vector lies on the counterclockwise side of the first, and negative when it lies on the clockwise side.

So $P(\mathbf{e}_1, \mathbf{e}_2)$ has signed area $+1$, because $\mathbf{e}_2$ is a quarter turn counterclockwise from $\mathbf{e}_1$, and $P(\mathbf{e}_2, \mathbf{e}_1)$ has signed area -1 . With the sign in place, scaling holds with c instead of $|c|$, and heights add with their signs, so additivity holds for all $\mathbf{v}, \mathbf{w}$.

Write $D(\mathbf{u}, \mathbf{v})$ for the signed area. The pictures suggest that $D: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ obeys the following rules, for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^2$ and all $c \in \mathbb{R}$:

- (R1) (**Scaling**) $D(c\mathbf{u}, \mathbf{v}) = cD(\mathbf{u}, \mathbf{v})$ and $D(\mathbf{u}, c\mathbf{v}) = cD(\mathbf{u}, \mathbf{v})$.
- (R2) (**Shear**) $D(\mathbf{u} + c\mathbf{v}, \mathbf{v}) = D(\mathbf{u}, \mathbf{v})$ and $D(\mathbf{u}, \mathbf{v} + c\mathbf{u}) = D(\mathbf{u}, \mathbf{v})$.
- (R3) (**Equal sides**) $D(\mathbf{u}, \mathbf{u}) = 0$.
- (R4) (**Normalization**) $D(\mathbf{e}_1, \mathbf{e}_2) = 1$.
- (R5) (**Additivity**) $D(\mathbf{u} + \mathbf{w}, \mathbf{v}) = D(\mathbf{u}, \mathbf{v}) + D(\mathbf{w}, \mathbf{v})$ and $D(\mathbf{u}, \mathbf{v} + \mathbf{w}) = D(\mathbf{u}, \mathbf{v}) + D(\mathbf{u}, \mathbf{w})$.

These are rules read off from pictures, not yet a definition. The question is what they force.

6.1.3 The rules force $ad - bc$

A 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ has columns $\mathbf{u} = (a, c)$ and $\mathbf{v} = (b, d)$, and $P(\mathbf{u}, \mathbf{v})$ is the image of the unit square under $\mathbf{x} \mapsto A\mathbf{x}$. We now show that the first four rules leave no freedom at all. The method is the column version of elimination: use shears to clear entries until the parallelogram becomes a rectangle.

Example. ($ad - bc$ from the Rules)

Let $D: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ be **any** function satisfying (R1)–(R4). Show that $D((a, c), (b, d)) = ad - bc$ for all $a, b, c, d \in \mathbb{R}$.

Solution. Write $\mathbf{u} = (a, c)$ and $\mathbf{v} = (b, d)$. Two consequences of (R1) come first. Taking the scalar 0 in (R1) gives $D(\mathbf{u}, \mathbf{0}) = D(\mathbf{u}, 0\mathbf{v}) = 0 \cdot D(\mathbf{u}, \mathbf{v}) = 0$, and likewise $D(\mathbf{0}, \mathbf{v}) = 0$. Using (R1) twice and then (R4),

$$D((p, 0), (0, q)) = D(p\mathbf{e}_1, q\mathbf{e}_2) = pq D(\mathbf{e}_1, \mathbf{e}_2) = pq \quad \text{for all } p, q \in \mathbb{R}. \quad (*)$$

Case 1: $a \neq 0$. By the second shear rule with scalar $-b/a$,

$$D(\mathbf{u}, \mathbf{v}) = D(\mathbf{u}, \mathbf{v} - \frac{b}{a}\mathbf{u}) = D(\mathbf{u}, (0, d')), \quad \text{where } d' = d - \frac{bc}{a}.$$

If $d' = 0$, this is $D(\mathbf{u}, \mathbf{0}) = 0 = ad'$. If $d' \neq 0$, the first shear rule with scalar $-c/d'$ clears the second entry of $\mathbf{u}$:

$$D(\mathbf{u}, (0, d')) = D(\mathbf{u} - \frac{c}{d'}(0, d'), (0, d')) = D((a, 0), (0, d')) = ad'$$

by (*). In both sub-cases $D(\mathbf{u}, \mathbf{v}) = ad' = ad - bc$.

Case 2: $a = 0$ and $b \neq 0$. By the first shear rule with scalar 1, $D(\mathbf{u}, \mathbf{v}) = D(\mathbf{u} + \mathbf{v}, \mathbf{v})$, and $\mathbf{u} + \mathbf{v} = (b, c + d)$ has first entry $b \neq 0$. Case 1, applied to the columns $(b, c + d)$ and (b, d) , gives

$$D(\mathbf{u}, \mathbf{v}) = bd - b(c + d) = -bc = ad - bc,$$

since $a = 0$.

Case 3: $a = b = 0$. If $c = 0$, then $\mathbf{u} = \mathbf{0}$ and $D(\mathbf{u}, \mathbf{v}) = 0$. If $c \neq 0$, the second shear rule with scalar $-d/c$ gives $D(\mathbf{u}, \mathbf{v}) = D(\mathbf{u}, \mathbf{v} - \frac{d}{c}\mathbf{u}) = D(\mathbf{u}, \mathbf{0}) = 0$. In both sub-cases $D(\mathbf{u}, \mathbf{v}) = 0 = ad - bc$.

The three cases cover all a, b, c, d , so $D((a, c), (b, d)) = ad - bc$, as claimed.

So **at most one** function obeys (R1)–(R4). Conversely, $(a, c), (b, d) \mapsto ad - bc$ obeys all five rules; for instance, for the second shear rule, replacing (b, d) by $(b + ta, d + tc)$ gives $a(d + tc) - (b + ta)c = ad - bc$. The other checks are just as short. Hence **exactly one** function satisfies (R1)–(R4), and it satisfies (R5) as well.

Two things in this computation are worth noticing. First, (R3) was never used. It follows from the others: $D(\mathbf{u}, \mathbf{u}) = D(\mathbf{u}, \mathbf{u} - \mathbf{u}) = D(\mathbf{u}, \mathbf{0}) = 0$, by a shear and then scaling by 0. Second, the argument used only addition, subtraction, multiplication and division by non-zero numbers. So over **any** field F , exactly one function $F^2 \times F^2 \rightarrow F$ satisfies (R1)–(R4), namely $ad - bc$. The picture needs $\mathbb{R}$; the algebra does not.

There is also a shorter route, if we allow additivity. Expanding $\mathbf{u} = a\mathbf{e}_1 + c\mathbf{e}_2$ and $\mathbf{v} = b\mathbf{e}_1 + d\mathbf{e}_2$ with (R5) and (R1) gives four terms:

$$D(\mathbf{u}, \mathbf{v}) = ab D(\mathbf{e}_1, \mathbf{e}_1) + ad D(\mathbf{e}_1, \mathbf{e}_2) + cb D(\mathbf{e}_2, \mathbf{e}_1) + cd D(\mathbf{e}_2, \mathbf{e}_2).$$

The first and last vanish by (R3), $D(\mathbf{e}_1, \mathbf{e}_2) = 1$ by (R4), and $D(\mathbf{e}_2, \mathbf{e}_1) = -1$ (see the Quick check below). This expansion is the route that generalizes to n vectors later in the chapter, while the elimination route of [Example \(\$ad - bc\$ from the Rules\)](#) becomes the practical way to compute.

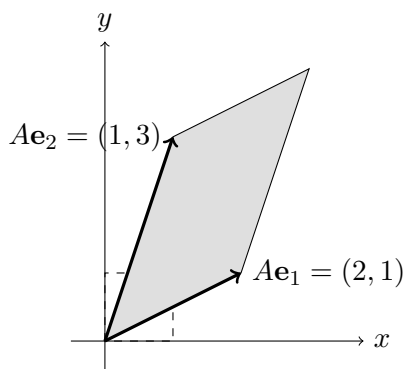
Quick check. Using only (R1) and (R2), show that $D(\mathbf{u}, \mathbf{v}) = -D(\mathbf{v}, \mathbf{u})$ for all $\mathbf{u}, \mathbf{v}$. Deduce $D(\mathbf{e}_2, \mathbf{e}_1)$.

Solution. Apply shears and one scaling:

$$D(\mathbf{u}, \mathbf{v}) \stackrel{(R2)}{=} D(\mathbf{u} + \mathbf{v}, \mathbf{v}) \stackrel{(R2)}{=} D(\mathbf{u} + \mathbf{v}, \mathbf{v} - (\mathbf{u} + \mathbf{v})) = D(\mathbf{u} + \mathbf{v}, -\mathbf{u}) \stackrel{(R2)}{=} D((\mathbf{u} + \mathbf{v}) + (-\mathbf{u}), -\mathbf{u}) = D(\mathbf{v}, -\mathbf{u}) \stackrel{(R1)}{=} -D(\mathbf{v}, \mathbf{u})$$

With $\mathbf{u} = \mathbf{e}_1$, $\mathbf{v} = \mathbf{e}_2$ and (R4), $D(\mathbf{e}_2, \mathbf{e}_1) = -D(\mathbf{e}_1, \mathbf{e}_2) = -1$. This agrees with $ad - bc = 0 \cdot 0 - 1 \cdot 1$ for the columns $(0, 1), (1, 0)$.

Now compare with Chapter 0. By [Theorem 0.9.8](#), A is invertible **if and only if** $ad - bc \neq 0$. In the language of this section: A is invertible exactly when the parallelogram spanned by its columns is **not flat**, that is, when the columns do not lie on a common line through the origin. The number that decides invertibility and the signed area of the image of the unit square are the same number. Drag the images of $\mathbf{e}_1$ and $\mathbf{e}_2$ below and watch the signed area pass through 0 exactly when the two columns line up.



The unit square (dashed) and its image under $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$, a parallelogram of signed area

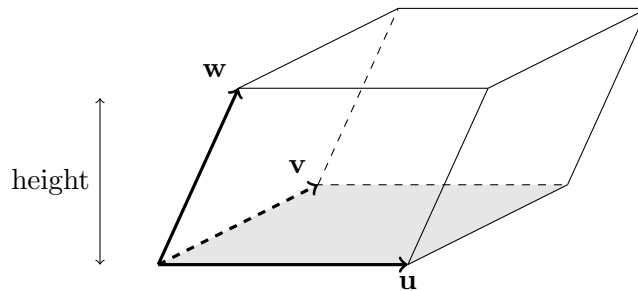
$2 \cdot 3 - 1 \cdot 1 = 5$. The second column lies counterclockwise from the first, so the sign is positive.

6.1.4 Volume in three dimensions

The same story plays out one dimension up. Three vectors $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ span a **parallelepiped**

$$P(\mathbf{u}, \mathbf{v}, \mathbf{w}) = \{r\mathbf{u} + s\mathbf{v} + t\mathbf{w} : 0 \leq r, s, t \leq 1\},$$

a slanted box whose faces are parallelograms.



Its volume is the area of the base $P(\mathbf{u}, \mathbf{v})$ (shaded) times the height of $\mathbf{w}$ above the plane of the base. Each feature from the plane has a counterpart:

- scaling one edge scales the volume;
- replacing $\mathbf{w}$ by $\mathbf{w} + a\mathbf{u} + b\mathbf{v}$ slides the top face within its own plane, so the base and height, hence the volume, are unchanged, and similarly for shears of $\mathbf{u}$ or $\mathbf{v}$;
- if two edges are equal, the box is flat and the volume is 0;
- the unit cube $P(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ has volume 1;
- with a sign attached, heights add, so the signed volume is additive in each edge.

The sign in three dimensions records whether $(\mathbf{u}, \mathbf{v}, \mathbf{w})$ is arranged like $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$, the right-hand rule. We make this precise as orientation later in the chapter. For now the point is that the rules are **the same rules**, now for a function $D(\mathbf{u}, \mathbf{v}, \mathbf{w})$ of three vectors. The Quick check above also carries over word for word: shearing and scaling two of the arguments while the third stays fixed shows that swapping two arguments changes the sign.

6.1.5 What we will prove

Here is the goal for the next three sections. Let F be a field and $n \geq 1$.

There is **exactly one** function D of n vectors in F^n that is linear in each argument separately, is 0 whenever two arguments are equal, and takes the value 1 at $(\mathbf{e}_1, \dots, \mathbf{e}_n)$.

Its value on the columns of a matrix A will be the determinant $\det A$. The plan follows the shorter route for $n = 2$. Expanding every argument in the standard basis gives n^n terms. The rule for equal arguments kills every term with a repeated basis vector, which leaves one term for each rearrangement $(\mathbf{e}_{\sigma(1)}, \dots, \mathbf{e}_{\sigma(n)})$ of the standard basis. Each surviving value is ± 1 , reached from $D(\mathbf{e}_1, \dots, \mathbf{e}_n) = 1$ by swaps, with one sign change per swap. That creates two tasks:

1. Name and study functions of several vectors with these rules. This is the next section, on multilinear alternating forms.
2. Make sure the sign of a rearrangement is well defined, even though it can be reached by different sequences of swaps. This is the section on permutations and sign.

The existence and uniqueness theorem itself comes after both.

Warning. Sarrus's rule is a 3×3 coincidence. For a 3×3 matrix, the “diagonals” shortcut (add the three products along the down-right diagonals of the array with its first two columns copied to the right, subtract the three along the up-right diagonals) gives the correct six terms, because there are exactly $3! = 6$ rearrangements. The same picture for 4×4 gives $4 + 4 = 8$ products, while the expansion above has $4! = 24$ terms. It fails already for the matrix with columns $\mathbf{e}_2, \mathbf{e}_1, \mathbf{e}_3, \mathbf{e}_4$: every one of the eight wrapped diagonals passes through a zero entry, so the shortcut returns 0, but the swap rule gives $D(\mathbf{e}_2, \mathbf{e}_1, \mathbf{e}_3, \mathbf{e}_4) = -D(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4) = -1$. Never use Sarrus's rule beyond 3×3 , and never use it as a proof tool.

6.1.6 Exercises

6.1.6.1 A. Check your understanding

Exercise A1.

- (a) State the four rules (R1)–(R4) for signed area.
- (b) True or false: ordinary (unsigned) area satisfies $\text{area}(\mathbf{u}, \mathbf{v} + \mathbf{w}) = \text{area}(\mathbf{u}, \mathbf{v}) + \text{area}(\mathbf{u}, \mathbf{w})$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^2$. Justify your answer.
- (c) Find the signed area of $P((1, 3), (2, 4))$. Is $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ invertible?
- (d) Explain why any D satisfying (R1) and (R2) has $D(\mathbf{u}, 5\mathbf{u}) = 0$.
- (e) How many terms does the full expansion for 4×4 matrices have, and how many products does the 4×4 analogue of Sarrus's rule use?

Solution.

- (a) For all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ and $c \in \mathbb{R}$: $D(c\mathbf{u}, \mathbf{v}) = cD(\mathbf{u}, \mathbf{v}) = D(\mathbf{u}, c\mathbf{v})$; $D(\mathbf{u} + c\mathbf{v}, \mathbf{v}) = D(\mathbf{u}, \mathbf{v}) = D(\mathbf{u}, \mathbf{v} + c\mathbf{u})$; $D(\mathbf{u}, \mathbf{u}) = 0$; $D(\mathbf{e}_1, \mathbf{e}_2) = 1$.
- (b) False. Take $\mathbf{u} = \mathbf{e}_1$, $\mathbf{v} = \mathbf{e}_2$, $\mathbf{w} = -\mathbf{e}_2$. The left side is the area of $P(\mathbf{e}_1, \mathbf{0})$, which is 0; the right side is $1 + 1 = 2$.
- (c) With $a = 1$, $b = 2$, $c = 3$, $d = 4$, the signed area is $ad - bc = 4 - 6 = -2$. Since $-2 \neq 0$, the matrix is invertible by [Theorem 0.9.8](#).
- (d) By the second shear rule with scalar -5 , $D(\mathbf{u}, 5\mathbf{u}) = D(\mathbf{u}, 5\mathbf{u} - 5\mathbf{u}) = D(\mathbf{u}, \mathbf{0})$, and $D(\mathbf{u}, \mathbf{0}) = D(\mathbf{u}, 0 \cdot \mathbf{u}) = 0 \cdot D(\mathbf{u}, \mathbf{u}) = 0$ by (R1).
- (e) The expansion has $4! = 24$ terms, one for each rearrangement of $\mathbf{e}_1, \dots, \mathbf{e}_4$. The Sarrus-type diagram has only 8 products, so it must miss 16 of them.

6.1.6.2 B. Practice

Exercise B1: Signed Areas from the Rules.

Using only (R1)–(R4), in the manner of [Example \(\$ad - bc\$ from the Rules\)](#), compute the following. Then check each answer against $ad - bc$.

- (a) $D((2, 1), (4, 5))$.
- (b) $D((1, 2), (3, 6))$.
- (c) $D((0, 3), (2, 0))$.

Solution.

- (a) By the second shear rule with scalar -2 , $D((2, 1), (4, 5)) = D((2, 1), (4, 5) - 2(2, 1)) = D((2, 1), (0, 3))$. By the first shear rule with scalar $-\frac{1}{3}$, this equals $D((2, 1) - \frac{1}{3}(0, 3), (0, 3)) = D((2, 0), (0, 3))$. By (R1) twice and (R4), $D(2\mathbf{e}_1, 3\mathbf{e}_2) = 6D(\mathbf{e}_1, \mathbf{e}_2) = 6$. Check: $2 \cdot 5 - 4 \cdot 1 = 6$.
- (b) By the second shear rule with scalar -3 , $D((1, 2), (3, 6)) = D((1, 2), (0, 0))$, and $D(\mathbf{u}, \mathbf{0}) = D(\mathbf{u}, 0 \cdot \mathbf{u}) = 0$ by (R1). So the value is 0. Check: $1 \cdot 6 - 3 \cdot 2 = 0$. The columns lie on one line, and the parallelogram is flat.
- (c) By (R1) twice, $D(3\mathbf{e}_2, 2\mathbf{e}_1) = 6D(\mathbf{e}_2, \mathbf{e}_1)$. By the Quick check (which uses only (R1) and (R2)), $D(\mathbf{e}_2, \mathbf{e}_1) = -D(\mathbf{e}_1, \mathbf{e}_2) = -1$ by (R4). So the value is -6 . Check: with columns $(0, 3)$ and $(2, 0)$, $a = 0$, $b = 2$, $c = 3$, $d = 0$, and $ad - bc = -6$. The sign is negative because $(2, 0)$ lies clockwise from $(0, 3)$.

Exercise B2: Equal Sides Force the Swap Rule.

Let $D: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ satisfy (R1), (R3) and (R5), but not necessarily (R2).

- (a) Prove that $D(\mathbf{u}, \mathbf{v}) = -D(\mathbf{v}, \mathbf{u})$ for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$.
- (b) Suppose moreover that (R4) holds. Deduce that $D((a, c), (b, d)) = ad - bc$.

Hint: expand $D(\mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v})$.

Solution.

- (a) Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$. By (R3), $D(\mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v}) = 0$. Expanding with (R5) in the first argument and then in the second,

$$0 = D(\mathbf{u}, \mathbf{u}) + D(\mathbf{u}, \mathbf{v}) + D(\mathbf{v}, \mathbf{u}) + D(\mathbf{v}, \mathbf{v}) = D(\mathbf{u}, \mathbf{v}) + D(\mathbf{v}, \mathbf{u}),$$

where the second equality uses (R3) twice. Hence $D(\mathbf{u}, \mathbf{v}) = -D(\mathbf{v}, \mathbf{u})$.

- (b) Write $(a, c) = a\mathbf{e}_1 + c\mathbf{e}_2$ and $(b, d) = b\mathbf{e}_1 + d\mathbf{e}_2$. By (R5) and (R1) in each argument,

$$D((a, c), (b, d)) = abD(\mathbf{e}_1, \mathbf{e}_1) + adD(\mathbf{e}_1, \mathbf{e}_2) + cbD(\mathbf{e}_2, \mathbf{e}_1) + cdD(\mathbf{e}_2, \mathbf{e}_2).$$

By (R3) the first and last terms are 0. By (R4), $D(\mathbf{e}_1, \mathbf{e}_2) = 1$, and by (a), $D(\mathbf{e}_2, \mathbf{e}_1) = -1$. Hence $D((a, c), (b, d)) = ad - bc$.

6.1.6.3 C. Going deeper

Exercise C1: Area of a Triangle.

Let $\mathbf{p}, \mathbf{q}, \mathbf{r} \in \mathbb{R}^2$.

- (a) Explain why the triangle with vertices $\mathbf{p}, \mathbf{q}, \mathbf{r}$ has area $\frac{1}{2}|D(\mathbf{q} - \mathbf{p}, \mathbf{r} - \mathbf{p})|$, where $D(\mathbf{u}, \mathbf{v})$ is the signed area.
- (b) Find the area of the triangle with vertices $(1, 2), (4, 3), (2, 6)$.
- (c) Hence decide whether the points $(0, 1), (2, 4), (6, 10)$ lie on a line. Justify your answer.

Solution.

- (a) Translating by $-\mathbf{p}$ does not change areas, so the triangle has the same area as the triangle with vertices $\mathbf{0}, \mathbf{u} = \mathbf{q} - \mathbf{p}, \mathbf{v} = \mathbf{r} - \mathbf{p}$. That triangle is half of the parallelogram $P(\mathbf{u}, \mathbf{v})$: the diagonal from $\mathbf{u}$ to $\mathbf{v}$ cuts $P(\mathbf{u}, \mathbf{v})$ into the triangle with vertices $\mathbf{0}, \mathbf{u}, \mathbf{v}$ and the triangle with vertices $\mathbf{u} + \mathbf{v}, \mathbf{v}, \mathbf{u}$, and the point reflection $\mathbf{x} \mapsto \mathbf{u} + \mathbf{v} - \mathbf{x}$ maps the first onto the second, so they have equal area. The area of $P(\mathbf{u}, \mathbf{v})$ is $|D(\mathbf{u}, \mathbf{v})|$, since the signed area differs from the area only by its sign. Hence the triangle has area $\frac{1}{2}|D(\mathbf{q} - \mathbf{p}, \mathbf{r} - \mathbf{p})|$.
- (b) Here $\mathbf{q} - \mathbf{p} = (3, 1)$ and $\mathbf{r} - \mathbf{p} = (1, 4)$, so $D = 3 \cdot 4 - 1 \cdot 1 = 11$ and the area is $\frac{11}{2}$.
- (c) With $\mathbf{p} = (0, 1), \mathbf{q} - \mathbf{p} = (2, 3)$ and $\mathbf{r} - \mathbf{p} = (6, 9)$, so $D = 2 \cdot 9 - 6 \cdot 3 = 0$ and the “triangle” has area 0. Indeed $(6, 9) = 3(2, 3)$, so $\mathbf{r} - \mathbf{p}$ is a multiple of $\mathbf{q} - \mathbf{p}$, and all three points lie on the line through $(0, 1)$ with direction $(2, 3)$. Yes, they are collinear.

Exercise C2: The 1×1 Case.

For $n = 1$, the rules ask for a function $f: F \rightarrow F$ of one vector in $F^1 = F$.

- (a) Explain why the rule “zero whenever two arguments are equal” imposes no condition when $n = 1$.
- (b) Prove that the only function $f: F \rightarrow F$ that is linear with $f(1) = 1$ is $f(a) = a$. Hence the determinant of a 1×1 matrix (a) must be a .

Solution.

- (a) The rule speaks about two **different** positions $i \neq j$ holding equal vectors. With one argument there is no pair of positions, so the condition is vacuously true ([Example \(A vacuously true implication\)](#)).
- (b) Let f be linear with $f(1) = 1$, and let $a \in F$. Since $a = a \cdot 1$, homogeneity gives $f(a) = f(a \cdot 1) = af(1) = a$. Conversely, $a \mapsto a$ is linear and sends 1 to 1. Hence $f(a) = a$ is the only such function.

6.2 Multilinear Alternating Forms

The previous section found that signed area obeys a few rules: it is linear in each side separately, it vanishes when two sides are equal, and swapping the sides changes its sign. This section gives

these rules names for functions of k vectors in an arbitrary vector space, and derives everything that follows from them before any formula exists. The main result is that “zero on equal arguments” already implies the swap rule, the shear rule and vanishing on dependent lists. The swap rule, in turn, gives back “zero on equal arguments” only when $1 + 1 \neq 0$ in the field.

Throughout, V is a vector space over a field F , $k \geq 1$ is an integer, and $V^k = V \times \cdots \times V$ is the set of k -tuples of vectors (Definition 0.3.11).

6.2.1 Multilinear forms

Signed area is not a linear function of the pair $(\mathbf{u}, \mathbf{v})$: doubling both sides multiplies the area by four, not by two. What it does have is linearity in $\mathbf{u}$ when $\mathbf{v}$ is held fixed, and linearity in $\mathbf{v}$ when $\mathbf{u}$ is held fixed. This “one slot at a time” linearity recurs throughout the book: in the dot product, later in inner products and bilinear forms, and in the determinant. So we give it a name.

A multilinear form is a function of several vectors that is linear in each vector while the others are held fixed.

Definition 6.2.1 — Multilinear Form

A function $f: V^k \rightarrow F$ is k -linear, or a **multilinear form** on V , if **for every** position $i \in \{1, \dots, k\}$ and **every** choice of $\mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{v}_{i+1}, \dots, \mathbf{v}_k \in V$, the function

$$V \rightarrow F, \quad \mathbf{x} \mapsto f(\mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{x}, \mathbf{v}_{i+1}, \dots, \mathbf{v}_k)$$

is linear. A 2-linear form is called **bilinear**.

In words: fix all arguments except the one in position i . What remains is a function of one vector, and it must be a linear functional (Definition 4.1.1). Written out, for all $\mathbf{x}, \mathbf{y} \in V$ and $c \in F$,

$$f(\dots, \mathbf{x} + \mathbf{y}, \dots) = f(\dots, \mathbf{x}, \dots) + f(\dots, \mathbf{y}, \dots), \quad f(\dots, c\mathbf{x}, \dots) = c f(\dots, \mathbf{x}, \dots),$$

where the dots stand for the same fixed vectors on both sides. The condition is imposed for **every** position i , one at a time.

Some examples:

- **Signed area.** On F^2 , $f(\mathbf{u}, \mathbf{v}) = u_1v_2 - u_2v_1$ is bilinear. For fixed $\mathbf{v}$, the map $\mathbf{u} \mapsto u_2u_1 - v_1u_2$ is a linear functional on F^2 , and for fixed $\mathbf{u}$, so is $\mathbf{v} \mapsto u_1v_2 - u_2v_1$.
- **The dot product.** On F^n , $(\mathbf{u}, \mathbf{v}) \mapsto u_1v_1 + \cdots + u_nv_n$ is bilinear, by the same one-slot check.
- **Products of functionals.** If $\varphi_1, \dots, \varphi_k \in V^*$, then $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = \varphi_1(\mathbf{v}_1)\varphi_2(\mathbf{v}_2)\cdots\varphi_k(\mathbf{v}_k)$ is k -linear: in position i it is the linear functional φ_i multiplied by the fixed scalar $\prod_{j \neq i} \varphi_j(\mathbf{v}_j)$. For instance, on $F[x]$, $(p, q) \mapsto p(0)q(1)$ is bilinear, since evaluation at a point is linear (Theorem 0.8.5).
- **The degenerate case $k = 1$.** A 1-linear form is exactly a linear functional on V . So multilinear forms generalize the dual space of Chapter 4.

Warning. Multilinear is not linear. A bilinear form is **not** a linear function of the pair. For the signed area $f(\mathbf{u}, \mathbf{v}) = u_1v_2 - u_2v_1$ on $\mathbb{R}^2$, scaling both arguments by 2 gives $f(2\mathbf{e}_1, 2\mathbf{e}_2) = 4$, not $2f(\mathbf{e}_1, \mathbf{e}_2) = 2$. Adding in both slots at once is not allowed either: $f(\mathbf{e}_1 + \mathbf{0}, \mathbf{e}_1 + \mathbf{e}_2) = f(\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2) = 1$, while $f(\mathbf{e}_1, \mathbf{e}_1) + f(\mathbf{0}, \mathbf{e}_2) = 0$. Pull out scalars and split sums **one slot at a time**.

6.2.2 Alternating and skew-symmetric forms

The second rule of signed area was geometric: if two edges coincide, the parallelogram or box is flat. There are two natural ways to put this into algebra. One says the value is **zero** when two arguments are equal. The other says the value **changes sign** when two arguments are swapped. They look like two phrasings of one idea, but they are not the same, and we must choose.

An alternating form vanishes as soon as two of its arguments coincide.

Definition 6.2.2 — Alternating Form

A k -linear form $f: V^k \rightarrow F$ is **alternating** if $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$ **whenever** $\mathbf{v}_i = \mathbf{v}_j$ for **some** pair of positions $i \neq j$.

Definition 6.2.3 — Skew-Symmetric Form

A k -linear form $f: V^k \rightarrow F$ is **skew-symmetric** if, **for all** $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$ and **all** positions $i < j$, swapping the arguments in positions i and j multiplies the value by -1 :

$$f(\mathbf{v}_1, \dots, \mathbf{v}_j, \dots, \mathbf{v}_i, \dots, \mathbf{v}_k) = -f(\mathbf{v}_1, \dots, \mathbf{v}_i, \dots, \mathbf{v}_j, \dots, \mathbf{v}_k),$$

where on the left $\mathbf{v}_j$ sits in position i , $\mathbf{v}_i$ sits in position j , and all other positions are unchanged.

In words: “alternating” asks for the value 0 as soon as **any two** positions, adjacent or not, hold the same vector, whatever the remaining arguments are. It is a condition on the inputs where two entries coincide, and says nothing directly about other inputs. “Skew-symmetric” is a condition on **every** input: it compares the value at a list with the value at the list with two entries exchanged.

Example. (Alternating Forms)

Check that each of the following is an alternating form.

- (a) On F^2 , for a fixed $c \in F$: $f_c(\mathbf{u}, \mathbf{v}) = c(u_1v_2 - u_2v_1)$.
- (b) On F^3 , for fixed coordinate indices $1 \leq r < s \leq 3$: $g_{rs}(\mathbf{u}, \mathbf{v}) = u_rv_s - u_sv_r$.
- (c) On $F[x]$: $h(p, q) = p(0)q(1) - p(1)q(0)$.
- (d) On any V , with $k = 1$: any linear functional $\varphi \in V^*$.

Solution.

- (a) Bilinearity is the one-slot check above, multiplied by the constant c . With $k = 2$ the only pair of positions is $(1, 2)$, and $f_c(\mathbf{u}, \mathbf{u}) = c(u_1u_2 - u_2u_1) = 0$ by commutativity in F . So f_c is alternating. For $c = 0$ it is the zero form, which is alternating for the silly reason that it is zero everywhere; for $c = 1$ it is the signed area of the previous section.
- (b) In each slot g_{rs} is a linear functional, as in (a), and $g_{rs}(\mathbf{u}, \mathbf{u}) = u_ru_s - u_su_r = 0$. Over $\mathbb{R}$ these three forms are, up to sign and order, the components of the cross product: $\mathbf{u} \times \mathbf{v} = (g_{23}(\mathbf{u}, \mathbf{v}), -g_{13}(\mathbf{u}, \mathbf{v}), g_{12}(\mathbf{u}, \mathbf{v}))$. This example shows that the number of arguments need not equal the dimension: here $k = 2$ and $\dim F^3 = 3$.
- (c) Evaluation at 0 and at 1 are linear in p (Theorem 0.8.5), so for fixed q the map $p \mapsto q(1)p(0) - q(0)p(1)$ is linear, and similarly in q . Also $h(p, p) = p(0)p(1) - p(1)p(0) = 0$. It is not the zero form: $h(1, x) = 1 \cdot 1 - 1 \cdot 0 = 1$.
- (d) A linear functional is 1-linear. With a single position there is **no** pair $i \neq j$, so the alternating condition holds vacuously (Example (A vacuously true implication)). This degenerate case matters because it shows that “alternating” is purely a condition about **pairs** of positions; the interesting content starts at $k = 2$.

Now a non-example, by a minimal change of (a). Over $\mathbb{R}$, replace the minus sign by a plus: $s(\mathbf{u}, \mathbf{v}) = u_1v_2 + u_2v_1$. Bilinearity survives, by the same one-slot check. But $s(\mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2) = 1 + 1 = 2 \neq 0$, so the alternating clause fails. The dot product fails in the same way, $\mathbf{e}_1 \cdot \mathbf{e}_1 = 1$. A different minimal change keeps the alternating clause and breaks the other one. The unsigned area $|u_1v_2 - u_2v_1|$ on $\mathbb{R}^2$ is still 0 whenever $\mathbf{u} = \mathbf{v}$, but it is not linear in $\mathbf{u}$: at $\mathbf{v} = \mathbf{e}_2$ it sends $\mathbf{e}_1 \mapsto 1$ and $-\mathbf{e}_1 \mapsto 1 \neq -1$. This is the failure that forced a sign onto area in the previous section.

Quick check. Is $f(\mathbf{u}, \mathbf{v}) = u_1v_2$ on F^2 alternating? Is it skew-symmetric?

Solution. It is bilinear, but neither. With $\mathbf{u} = \mathbf{v} = \mathbf{e}_1 + \mathbf{e}_2$, $f(\mathbf{u}, \mathbf{u}) = 1 \cdot 1 = 1 \neq 0$, so it is not alternating. And $f(\mathbf{e}_1, \mathbf{e}_2) = 1$ while $f(\mathbf{e}_2, \mathbf{e}_1) = 0$; skew-symmetry would need $0 = -1$, which fails in every field.

Why this definition. We chose “zero when two arguments are equal” as the basic notion, and in the form “**any** two positions”. There are two reasons. First, it is what the picture says directly: two equal edges make a flat box. Second, as we prove next, it implies the swap rule over **every** field, while the swap rule implies it only when $1 + 1 \neq 0$. Allowing any pair of positions, not just adjacent ones, also spares us a separate argument moving equal entries next to each other.

6.2.3 Consequences of the alternating condition

Everything that elimination does to a determinant will come from the following theorem. Its four parts are the column operations of Chapter 2 seen through a form: swap two arguments, add a multiple of one to another, scale one, and the extreme case of a dependent list.

Theorem 6.2.1 — Properties of Alternating Forms

Let $f: V^k \rightarrow F$ be an alternating k -linear form, let $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$, and let $i \neq j$ be positions in $\{1, \dots, k\}$.

- (a) **(Swap)** Exchanging the arguments in positions i and j multiplies $f(\mathbf{v}_1, \dots, \mathbf{v}_k)$ by -1 . In particular, f is skew-symmetric.
- (b) **(Shear)** For every $c \in F$, replacing $\mathbf{v}_i$ by $\mathbf{v}_i + c\mathbf{v}_j$ does not change $f(\mathbf{v}_1, \dots, \mathbf{v}_k)$.
- (c) **(Scaling)** For every $c \in F$, replacing $\mathbf{v}_i$ by $c\mathbf{v}_i$ multiplies $f(\mathbf{v}_1, \dots, \mathbf{v}_k)$ by c .
- (d) **(Dependence)** If the list $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ is linearly dependent, then $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$. In particular, $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$ if some $\mathbf{v}_i = \mathbf{0}$.

Idea. Parts (a)–(c) involve at most the two slots i and j , so freeze the other $k - 2$ arguments. What is left is a bilinear form $g(\mathbf{x}, \mathbf{y})$ with $g(\mathbf{x}, \mathbf{x}) = 0$, and the k -argument statements become two-argument statements about g . For the swap, the move is to feed g a sum in both slots, $g(\mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y}) = 0$, and expand: the two “square” terms die and the two cross terms must cancel. For (d), write one vector as a combination of the others and expand in its slot: every resulting term has a repeated argument.

Proof. Suppose first $i < j$; the case $i > j$ is the same with the roles of i and j exchanged. For $\mathbf{x}, \mathbf{y} \in V$, let

$$g(\mathbf{x}, \mathbf{y}) = f(\mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{x}, \mathbf{v}_{i+1}, \dots, \mathbf{v}_{j-1}, \mathbf{y}, \mathbf{v}_{j+1}, \dots, \mathbf{v}_k),$$

the value of f with $\mathbf{x}$ in position i , $\mathbf{y}$ in position j , and $\mathbf{v}_m$ in every other position m . Since f is k -linear, g is bilinear, and since f is alternating, $g(\mathbf{x}, \mathbf{x}) = 0$ for every $\mathbf{x} \in V$. In this notation $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = g(\mathbf{v}_i, \mathbf{v}_j)$.

- (c) This is linearity of f in position i : $g(c\mathbf{v}_i, \mathbf{v}_j) = c g(\mathbf{v}_i, \mathbf{v}_j)$.
- (d) By linearity in the first slot of g and $g(\mathbf{v}_j, \mathbf{v}_j) = 0$,

$$g(\mathbf{v}_i + c\mathbf{v}_j, \mathbf{v}_j) = g(\mathbf{v}_i, \mathbf{v}_j) + c g(\mathbf{v}_j, \mathbf{v}_j) = g(\mathbf{v}_i, \mathbf{v}_j).$$

- (e) By bilinearity of g ,

$$0 = g(\mathbf{v}_i + \mathbf{v}_j, \mathbf{v}_i + \mathbf{v}_j) = g(\mathbf{v}_i, \mathbf{v}_i) + g(\mathbf{v}_i, \mathbf{v}_j) + g(\mathbf{v}_j, \mathbf{v}_i) + g(\mathbf{v}_j, \mathbf{v}_j) = g(\mathbf{v}_i, \mathbf{v}_j) + g(\mathbf{v}_j, \mathbf{v}_i),$$

where the first and last equalities use $g(\mathbf{x}, \mathbf{x}) = 0$. Hence $g(\mathbf{v}_j, \mathbf{v}_i) = -g(\mathbf{v}_i, \mathbf{v}_j)$, which is the swap statement. Since $\mathbf{v}_1, \dots, \mathbf{v}_k$ and $i < j$ were arbitrary, f is skew-symmetric (Definition 6.2.3).

- (f) Suppose $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ is linearly dependent. By the Linear Dependence Lemma (Theorem 1.4.1), there is an index j with $\mathbf{v}_j \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$. If $j = 1$, then $\mathbf{v}_1 = \mathbf{0} = 0 \cdot \mathbf{v}_1$, and by (c) with $c = 0$, $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0 \cdot f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$. If $j \geq 2$, write $\mathbf{v}_j = a_1\mathbf{v}_1 + \dots + a_{j-1}\mathbf{v}_{j-1}$ with $a_m \in F$. By linearity of f in position j ,

$$f(\mathbf{v}_1, \dots, \mathbf{v}_k) = \sum_{m=1}^{j-1} a_m f(\mathbf{v}_1, \dots, \mathbf{v}_{j-1}, \mathbf{v}_m, \mathbf{v}_{j+1}, \dots, \mathbf{v}_k).$$

In the m -th term, the vector $\mathbf{v}_m$ sits both in position m and in position $j \neq m$, so the term is 0 because f is alternating. Hence $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$. The last sentence of (d) holds because a list containing $\mathbf{0}$ is dependent. This proves the theorem. ■

For the determinant, (a)–(c) will be the rules for column operations, and (d) explains at once why a matrix with dependent columns must have determinant 0. Part (d) also bounds the number

of arguments: on a space of dimension n , any $k > n$ vectors are dependent, so a non-zero alternating form needs $k \leq n$ (Exercise C1).

6.2.4 Alternating versus skew-symmetric

Part (a) says alternating forms are skew-symmetric. The converse depends on the field.

Theorem 6.2.2 — Skew-Symmetric and Alternating Forms

Let $f: V^k \rightarrow F$ be a k -linear form.

- (a) If f is alternating, then f is skew-symmetric.
- (b) If f is skew-symmetric and the characteristic of F is **not** 2, then f is alternating.

Proof.

- (a) is Theorem 6.2.1 (a).
- (b) Suppose $\mathbf{v}_i = \mathbf{v}_j$ for some $i < j$. Exchanging the arguments in positions i and j does not change the list, so skew-symmetry gives $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = -f(\mathbf{v}_1, \dots, \mathbf{v}_k)$, that is, $(1+1)f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$. If $1+1 = 0$ in F , then $2 \cdot 1 = 0$ while $1 \cdot 1 = 1 \neq 0$, so the smallest positive n with $n \cdot 1 = 0$ would be 2, and the characteristic of F would be 2 (Definition 0.7.2). So $1+1 \neq 0$, and multiplying by $(1+1)^{-1}$ gives $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$. Hence f is alternating. ■

Warning. **Skew-symmetric does not imply alternating over $\mathbb{F}_2$.** In $\mathbb{F}_2$ we have $-1 = 1$, so “skew-symmetric” just means “symmetric”. Take $f(\mathbf{u}, \mathbf{v}) = u_1 v_1$ on $\mathbb{F}_2^2$. It is bilinear, and $f(\mathbf{v}, \mathbf{u}) = v_1 u_1 = u_1 v_1 = -f(\mathbf{u}, \mathbf{v})$, so it is skew-symmetric. But $f(\mathbf{e}_1, \mathbf{e}_1) = 1 \neq 0$, so it is **not** alternating. Checking all 16 pairs $(\mathbf{u}, \mathbf{v})$ confirms both claims: $f(\mathbf{u}, \mathbf{v}) = f(\mathbf{v}, \mathbf{u})$ always, and $f(\mathbf{u}, \mathbf{u}) = u_1$ is 1 for $\mathbf{u} = (1, 0)$ and $\mathbf{u} = (1, 1)$. This is why the determinant is built on alternating forms, not skew-symmetric ones: the construction must work over $\mathbb{F}_2$ too.

6.2.5 The space of alternating forms

Alternating forms can be added and scaled, and the result is again alternating. This is what makes the uniqueness question later in the chapter a question about dimension.

Theorem 6.2.3 — Alternating Forms Form a Vector Space

The k -linear forms on V form a subspace of the vector space F^{V^k} of all functions $V^k \rightarrow F$, with the pointwise operations of Example (The Standard Vector Spaces) (d). The alternating k -linear forms on V form a subspace of it. In particular, both are vector spaces over F .

Proof. We use the Subspace Test (Theorem 1.2.1) twice.

The zero function is k -linear, since in each position it is the zero functional, and it is alternating, since all its values are 0.

Let f and g be k -linear and $a \in F$. Fix a position i and vectors in the other positions, and write $f(\dots, \mathbf{x}, \dots)$ for the value with $\mathbf{x}$ in position i . For $\mathbf{x}, \mathbf{y} \in V$ and $c \in F$, the pointwise definition and linearity of f and g in position i give

$$(f+g)(\dots, \mathbf{x}+\mathbf{y}, \dots) = f(\dots, \mathbf{x}, \dots) + f(\dots, \mathbf{y}, \dots) + g(\dots, \mathbf{x}, \dots) + g(\dots, \mathbf{y}, \dots) = (f+g)(\dots, \mathbf{x}, \dots) + (f+g)(\dots, \mathbf{y}, \dots)$$

and

$$(f+g)(\dots, c\mathbf{x}, \dots) = cf(\dots, \mathbf{x}, \dots) + cg(\dots, \mathbf{x}, \dots) = c(f+g)(\dots, \mathbf{x}, \dots).$$

So $f+g$ is k -linear. In the same way $(af)(\dots, \mathbf{x}+\mathbf{y}, \dots) = af(\dots, \mathbf{x}, \dots) + af(\dots, \mathbf{y}, \dots)$ and $(af)(\dots, c\mathbf{x}, \dots) = c(af)(\dots, \mathbf{x}, \dots)$, using commutativity in F , so af is k -linear.

If moreover f and g are alternating and $\mathbf{v}_i = \mathbf{v}_j$ for some $i \neq j$, then $(f+g)(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0 + 0 = 0$ and $(af)(\mathbf{v}_1, \dots, \mathbf{v}_k) = a \cdot 0 = 0$. So the alternating forms are closed under both operations. This proves the theorem. ■

For $V = F^2$ and $k = 2$, Exercise B2 shows that every alternating bilinear form is one of the forms f_c of Example (Alternating Forms) (a), with $c = f(\mathbf{e}_1, \mathbf{e}_2)$. So that space is spanned by the single non-zero form f_1 , and has dimension 1. Later in this chapter we prove the same for all n : the alternating n -linear forms on F^n form a space of dimension exactly 1, spanned by the determinant. The missing ingredient is how the value of an alternating form changes when its arguments are rearranged by more than one swap, which is the subject of the next section.

6.2.6 Exercises

6.2.6.1 A. Check your understanding

Exercise A1.

- Define an alternating k -linear form on a vector space V over F .
- True or false: a bilinear form f on $\mathbb{R}^2$ with $f(\mathbf{e}_1, \mathbf{e}_1) = f(\mathbf{e}_2, \mathbf{e}_2) = 0$ is alternating. Justify your answer.
- True or false: every skew-symmetric bilinear form on $\mathbb{R}^5$ is alternating. Justify your answer.
- Let f be an alternating 3-linear form on $\mathbb{R}^3$ and $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$. What is $f(\mathbf{u}, \mathbf{v}, 2\mathbf{u} - \mathbf{v})$?
- Name the move that proves an alternating form changes sign under a swap.

Solution.

- A function $f: V^k \rightarrow F$ that is linear in each argument when the others are fixed (Definition 6.2.1), and satisfies $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$ whenever $\mathbf{v}_i = \mathbf{v}_j$ for some $i \neq j$ (Definition 6.2.2).

- (b) False. The form $s(\mathbf{u}, \mathbf{v}) = u_1v_2 + u_2v_1$ is bilinear with $s(\mathbf{e}_1, \mathbf{e}_1) = s(\mathbf{e}_2, \mathbf{e}_2) = 0$, but $s(\mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2) = 2 \neq 0$. Checking the basis vectors is not enough, because the alternating condition is about **all** vectors.
- (c) True. $\mathbb{R}$ has characteristic 0, so [Theorem 6.2.2](#) (b) applies.
- (d) 0. The list $(\mathbf{u}, \mathbf{v}, 2\mathbf{u} - \mathbf{v})$ is linearly dependent, since $2\mathbf{u} - \mathbf{v} - (2\mathbf{u} - \mathbf{v}) = \mathbf{0}$ with coefficients 2, -1, -1 not all zero. By [Theorem 6.2.1](#) (d) the value is 0.
- (e) Put the sum of the two arguments into both positions, use $f(\dots, \mathbf{x} + \mathbf{y}, \dots, \mathbf{x} + \mathbf{y}, \dots) = 0$, and expand by linearity in both positions; the terms with equal arguments vanish and the two cross terms must cancel.

6.2.6.2 B. Practice

Exercise B1: Multilinear or Alternating?.

Determine which of the following functions are multilinear, and which of those are alternating. Justify your answer.

- (a) $\mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$, $(\mathbf{u}, \mathbf{v}) \mapsto 3u_1v_2 - 3u_2v_1$.
- (b) $\mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$, $(\mathbf{u}, \mathbf{v}) \mapsto u_1v_2 - u_2v_1 + 1$.
- (c) $\mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}$, $(\mathbf{u}, \mathbf{v}, \mathbf{w}) \mapsto u_1v_2w_3$.
- (d) $M_2(\mathbb{R}) \times M_2(\mathbb{R}) \rightarrow \mathbb{R}$, $(A, B) \mapsto \text{tr}(AB) - \text{tr}(BA)$.
- (e) $\mathbb{R}[x] \times \mathbb{R}[x] \rightarrow \mathbb{R}$, $(p, q) \mapsto p(1)q'(1) - p'(1)q(1)$.

Solution.

- (a) Multilinear and alternating: it is f_3 from [Example \(Alternating Forms\)](#) (a) with $c = 3$.
- (b) Not multilinear. A linear functional sends $\mathbf{0}$ to 0, but at $\mathbf{u} = \mathbf{0}$ the map $\mathbf{v} \mapsto 0 - 0 + 1 = 1$ does not. So it is not an alternating form either; in any case its value at $(\mathbf{u}, \mathbf{u})$ is $1 \neq 0$.
- (c) Multilinear: in each position it is a coordinate functional times a fixed scalar, as for products of functionals. Not alternating: with $\mathbf{u} = \mathbf{v} = \mathbf{e}_1 + \mathbf{e}_2$ and $\mathbf{w} = \mathbf{e}_3$, the value is $1 \cdot 1 \cdot 1 = 1 \neq 0$.
- (d) By [Theorem 0.9.6](#), $\text{tr}(AB) = \text{tr}(BA)$, so this is the zero function. The zero function is multilinear and alternating ([Theorem 6.2.3](#)). This is the degenerate case.
- (e) Multilinear: $p \mapsto p(1)$ and $p \mapsto p'(1)$ are linear, the second as a composition of differentiation ([Example \(Differentiation\)](#)) with evaluation ([Theorem 0.8.5](#)). So for fixed q , $p \mapsto q'(1)p(1) - q(1)p'(1)$ is linear, and similarly in q . Alternating: $p(1)p'(1) - p'(1)p(1) = 0$.

Exercise B2: Alternating Bilinear Forms on F^2 .

Let f be an alternating bilinear form on F^2 . Prove that

$$f(\mathbf{u}, \mathbf{v}) = f(\mathbf{e}_1, \mathbf{e}_2) (u_1v_2 - u_2v_1) \quad \text{for all } \mathbf{u}, \mathbf{v} \in F^2.$$

Hence show that the alternating bilinear forms on F^2 form a space of dimension 1.

Solution. Write $\mathbf{u} = u_1\mathbf{e}_1 + u_2\mathbf{e}_2$ and $\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2$. By linearity in the first and then in the second argument,

$$f(\mathbf{u}, \mathbf{v}) = u_1v_1f(\mathbf{e}_1, \mathbf{e}_1) + u_1v_2f(\mathbf{e}_1, \mathbf{e}_2) + u_2v_1f(\mathbf{e}_2, \mathbf{e}_1) + u_2v_2f(\mathbf{e}_2, \mathbf{e}_2).$$

Since f is alternating, $f(\mathbf{e}_1, \mathbf{e}_1) = f(\mathbf{e}_2, \mathbf{e}_2) = 0$, and by [Theorem 6.2.1](#) (a), $f(\mathbf{e}_2, \mathbf{e}_1) = -f(\mathbf{e}_1, \mathbf{e}_2)$. Hence $f(\mathbf{u}, \mathbf{v}) = f(\mathbf{e}_1, \mathbf{e}_2)(u_1v_2 - u_2v_1)$.

By [Theorem 6.2.3](#) these forms form a vector space, and we have just shown $f = f(\mathbf{e}_1, \mathbf{e}_2)f_1$, where $f_1(\mathbf{u}, \mathbf{v}) = u_1v_2 - u_2v_1$ is alternating by [Example \(Alternating Forms\)](#) (a). So (f_1) spans the space. It is non-zero, since $f_1(\mathbf{e}_1, \mathbf{e}_2) = 1$, so the list (f_1) is linearly independent and hence a basis. The dimension is 1.

Exercise B3: The Triple Product.

For $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ let $\mathbf{v} \times \mathbf{w} = (v_2w_3 - v_3w_2, v_3w_1 - v_1w_3, v_1w_2 - v_2w_1)$, and define $f(\mathbf{u}, \mathbf{v}, \mathbf{w}) = u_1(\mathbf{v} \times \mathbf{w})_1 + u_2(\mathbf{v} \times \mathbf{w})_2 + u_3(\mathbf{v} \times \mathbf{w})_3$. Show that f is an alternating 3-linear form on $\mathbb{R}^3$ with $f(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3) = 1$.

Solution. Multilinear. Each component of $\mathbf{v} \times \mathbf{w}$ is $g_{rs}(\mathbf{v}, \mathbf{w})$ or $-g_{rs}(\mathbf{v}, \mathbf{w})$ for a form of [Example \(Alternating Forms\)](#) (b), hence bilinear in $(\mathbf{v}, \mathbf{w})$. For fixed $\mathbf{v}, \mathbf{w}$, f is a linear functional of $\mathbf{u}$ with coefficients $(\mathbf{v} \times \mathbf{w})_m$. For fixed $\mathbf{u}$ and $\mathbf{w}$, $f = \sum_m u_m(\mathbf{v} \times \mathbf{w})_m$ is a combination of linear functionals of $\mathbf{v}$, hence linear in $\mathbf{v}$; likewise in $\mathbf{w}$.

Alternating. There are three pairs of positions. If $\mathbf{v} = \mathbf{w}$, every component $v_rv_s - v_sv_r$ is 0, so $\mathbf{v} \times \mathbf{v} = \mathbf{0}$ and $f = 0$. If $\mathbf{u} = \mathbf{v}$,

$$f(\mathbf{u}, \mathbf{u}, \mathbf{w}) = u_1(u_2w_3 - u_3w_2) + u_2(u_3w_1 - u_1w_3) + u_3(u_1w_2 - u_2w_1) = 0,$$

since the six terms cancel in pairs: $u_1u_2w_3$ with $-u_2u_1w_3$, $-u_1u_3w_2$ with $u_3u_1w_2$, and $u_2u_3w_1$ with $-u_3u_2w_1$. If $\mathbf{u} = \mathbf{w}$, the same expansion $u_1(v_2u_3 - v_3u_2) + u_2(v_3u_1 - v_1u_3) + u_3(v_1u_2 - v_2u_1)$ cancels in pairs in the same way.

Value. $\mathbf{e}_2 \times \mathbf{e}_3 = (1 \cdot 1 - 0 \cdot 0, 0 \cdot 0 - 0 \cdot 1, 0 \cdot 0 - 1 \cdot 0) = (1, 0, 0)$, so $f(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3) = 1$.

6.2.6.3 C. Going deeper

Exercise C1: Too Many Arguments.

Let V be finite-dimensional with $\dim V = n$.

- Prove that every alternating k -linear form on V with $k > n$ is the zero form.
- Deduce that the only alternating 3-linear form on F^2 is zero, and give a non-zero alternating 2-linear form on F^2 to show that the bound in (a) cannot be lowered to $k \geq n$.

Solution.

- Let f be an alternating k -linear form with $k > n$, and let $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$. By [Theorem 1.6.3](#) (a), a list of length $k > \dim V$ is linearly dependent. By [Theorem 6.2.1](#) (d), $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$. Since the vectors were arbitrary, f is the zero form.

- (b) Here $\dim F^2 = 2 < 3$, so (a) applies. For $k = n = 2$, the form $f_1(\mathbf{u}, \mathbf{v}) = u_1v_2 - u_2v_1$ of [Example \(Alternating Forms\)](#) (a) is alternating with $f_1(\mathbf{e}_1, \mathbf{e}_2) = 1 \neq 0$.

Exercise C2: Bilinear Forms over $\mathbb{F}_2$.

- (a) Let f be a bilinear form on F^2 and write $b_{rs} = f(\mathbf{e}_r, \mathbf{e}_s)$. Show that $f(\mathbf{u}, \mathbf{v}) = \sum_{r,s=1}^2 b_{rs}u_rv_s$, and that conversely every choice of four scalars b_{rs} defines a bilinear form this way. Hence there are exactly 16 bilinear forms on $\mathbb{F}_2^2$.
- (b) Prove that a bilinear form on $\mathbb{F}_2^2$ is skew-symmetric if and only if $b_{12} = b_{21}$. How many are there?
- (c) Prove that a bilinear form on $\mathbb{F}_2^2$ is alternating if and only if $b_{11} = b_{22} = 0$ and $b_{12} = b_{21}$. How many are there?
- (d) Over $\mathbb{R}$, how do the analogous two sets of bilinear forms on $\mathbb{R}^2$ compare? Justify your answer.

Solution.

- (a) Expanding $\mathbf{u} = u_1\mathbf{e}_1 + u_2\mathbf{e}_2$ and $\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2$ by linearity in each argument gives $f(\mathbf{u}, \mathbf{v}) = \sum_{r,s} u_rv_sf(\mathbf{e}_r, \mathbf{e}_s) = \sum_{r,s} b_{rs}u_rv_s$. Conversely, for any scalars b_{rs} , the function $\sum_{r,s} b_{rs}u_rv_s$ is linear in $\mathbf{u}$ for fixed $\mathbf{v}$ (a combination of the coordinates u_r with coefficients $\sum_s b_{rs}v_s$) and similarly in $\mathbf{v}$, and it takes the value b_{rs} at $(\mathbf{e}_r, \mathbf{e}_s)$. So bilinear forms on F^2 correspond exactly to quadruples $(b_{11}, b_{12}, b_{21}, b_{22})$, and over $\mathbb{F}_2$ there are $2^4 = 16$.
- (b) In $\mathbb{F}_2$, $-a = a$ for every a , so f is skew-symmetric exactly when $f(\mathbf{v}, \mathbf{u}) = f(\mathbf{u}, \mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$. ($\Rightarrow$) Taking $\mathbf{u} = \mathbf{e}_1$, $\mathbf{v} = \mathbf{e}_2$ gives $b_{21} = b_{12}$. ($\Leftarrow$) If $b_{12} = b_{21}$, then $b_{rs} = b_{sr}$ for all r, s , and $f(\mathbf{v}, \mathbf{u}) = \sum_{r,s} b_{rs}v_ru_s = \sum_{r,s} b_{sr}u_sv_r = f(\mathbf{u}, \mathbf{v})$. The entries b_{11}, b_{22} and the common value $b_{12} = b_{21}$ are free, so there are $2^3 = 8$ skew-symmetric forms.
- (c) ($\Rightarrow$) If f is alternating, then $b_{11} = f(\mathbf{e}_1, \mathbf{e}_1) = 0$ and $b_{22} = 0$, and $0 = f(\mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2) = b_{11} + b_{12} + b_{21} + b_{22} = b_{12} + b_{21}$, so $b_{21} = -b_{12} = b_{12}$. ($\Leftarrow$) With $b_{11} = b_{22} = 0$ and $b_{12} = b_{21} = b$, $f(\mathbf{u}, \mathbf{u}) = b(u_1u_2 + u_2u_1) = (1+1)b u_1u_2 = 0$. Only $b \in \{0, 1\}$ is free, so there are 2 alternating forms. Hence $8 - 2 = 6$ skew-symmetric forms over $\mathbb{F}_2$ are not alternating; u_1v_1 (with $b_{11} = 1$, all others 0) is one of them, as in the warning. A brute-force check of all 16 forms against all pairs in $\mathbb{F}_2^2$ gives the same counts, 8 and 2.
- (d) Over $\mathbb{R}$ the two sets are **equal**. Alternating forms are skew-symmetric by [Theorem 6.2.2](#) (a), and since $\mathbb{R}$ has characteristic 0, skew-symmetric forms are alternating by part (b) of that theorem. Concretely, both conditions say $b_{11} = b_{22} = 0$ and $b_{21} = -b_{12}$.

6.3 Permutations and Sign

An alternating form changes sign when two of its arguments are swapped ([Theorem 6.2.1](#)). To compare $f(\mathbf{v}_3, \mathbf{v}_1, \mathbf{v}_2)$ with $f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$, we reach one list from the other by swaps and collect a factor -1 per swap. But the same rearrangement can be reached by different sequences of swaps, and Chapter 0 showed that even their number is not unique. If one route used an even number

of swaps and another an odd number, we would get $f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = -f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ for every alternating form, which is absurd for signed volume. This section rules that out. It attaches to each permutation a sign defined without any choices, proves that the sign is multiplicative, and deduces that the parity of the number of transpositions is well defined, as Chapter 0 promised.

6.3.1 Permutations, recalled

Recall from Chapter 0 that S_n is the group of bijections $\{1, \dots, n\} \rightarrow \{1, \dots, n\}$, with $n!$ elements. A permutation can be written in two-line notation, with $\sigma(i)$ below i , or as a product of cycles. The product $\sigma\tau$ is the composition $\sigma \circ \tau$: **first** τ , **then** σ , so products of cycles are read from right to left (Example (Computing with cycles)). A transposition $(i\ j)$ swaps i and j and fixes everything else, and for $n \geq 2$ every permutation is a product of transpositions (Theorem 0.10.4). The product is far from unique, since

$$(1\ 2\ 3) = (1\ 3)(1\ 2) = (1\ 2)(2\ 3) = (1\ 3)(1\ 2)(1\ 2)(1\ 2).$$

Here the numbers of transpositions are 2, 2, 4, all even. The goal is to show this is no accident.

Why can we not simply **define** the sign of σ as $(-1)^r$, where σ is a product of r transpositions? Because that definition depends on a choice of product, and we would have to prove that every choice gives the same answer. That is the statement we are trying to prove. We need a definition that involves no choice at all.

6.3.2 Inversions and the sign

To measure how much a permutation scrambles $1, 2, \dots, n$, look at the bottom row $\sigma(1), \sigma(2), \dots, \sigma(n)$ of its two-line notation and count the pairs of entries that are in the wrong order. A sorted row has none, and each swap of two neighboring entries changes the count by exactly one. So the count is a natural candidate for tracking the parity of swaps.

The sign of a permutation records whether it puts an even or an odd number of pairs out of order.

Definition 6.3.1 — Inversion

Let $\sigma \in S_n$. An **inversion** of σ is a pair (i, j) of positions with $1 \leq i < j \leq n$ and $\sigma(i) > \sigma(j)$. We write $\text{inv}(\sigma)$ for the number of inversions of σ .

Definition 6.3.2 — Sign of a Permutation

The **sign** of $\sigma \in S_n$ is

$$\text{sgn } \sigma = (-1)^{\text{inv}(\sigma)} \in \{1, -1\}.$$

The permutation σ is **even** if $\text{sgn } \sigma = 1$ and **odd** if $\text{sgn } \sigma = -1$.

In words: an inversion is a pair of positions whose order σ **reverses**. The condition $i < j$ makes each pair of positions count once, and $\sigma(i) > \sigma(j)$ says the values come out in the opposite order. Since $\text{inv}(\sigma)$ is the size of a subset of the finite set of pairs $i < j$, it is a well-defined

integer between 0 and $n(n-1)/2$, and there is nothing to choose. That is the whole reason for this definition.

The identity. Its bottom row is sorted, so $\text{inv}(\text{id}) = 0$ and id is even. In $S_1 = \{\text{id}\}$ this is the only permutation, with sign 1. The degenerate case matters: the identity is the empty product of transpositions, and its sign $1 = (-1)^0$ is what a product of 0 swaps should give.

All of S_3 . Here is each permutation with its bottom row and inversions.

σ	id	(1 2)	(2 3)	(1 3)	(1 2 3)	(1 3 2)
bottom row	1 2 3	2 1 3	1 3 2	3 2 1	2 3 1	3 1 2
inversions	none	(1, 2)	(2, 3)	(1, 2), (1, 3), (2, 3)	(2, 3)	(1, 2), (1, 3)
sgn	1	-1	-1	-1	1	1

For (1 2 3), the bottom row is $\sigma(1), \sigma(2), \sigma(3) = 2, 3, 1$, and the out-of-order pairs of positions are (1, 3), since $2 > 1$, and (2, 3), since $3 > 1$.

Two things stand out. The three transpositions are odd, although (1 3) has three inversions, not one. And the two 3-cycles are even, consistent with $(1\ 2\ 3) = (1\ 3)(1\ 2)$ being a product of two transpositions.

Why all pairs, not just neighbors. A tempting shortcut counts only **adjacent** positions $(i, i+1)$ with $\sigma(i) > \sigma(i+1)$. For (1 3), with bottom row 3, 2, 1, this count is 2, which would make (1 3) “even”. But $(1\ 3) = (1\ 2)(2\ 3)(1\ 2)$, as one checks by following $1 \mapsto 2 \mapsto 3 \mapsto 3$, $2 \mapsto 1 \mapsto 1 \mapsto 2$, $3 \mapsto 3 \mapsto 2 \mapsto 1$, and each factor has adjacent count 1. A count that is to be multiplicative must give $(-1)^3 = -1$ here, so the adjacent count is the wrong one.

Quick check. Find $\text{sgn}(1\ 4)$ in S_4 by counting inversions.

Solution. The bottom row is 4, 2, 3, 1. The pairs of positions in the wrong order are (1, 2), (1, 3), (1, 4), since 4 is larger than everything after it, and (2, 4), (3, 4), since $2 > 1$ and $3 > 1$. The pair (2, 3) is in order. So $\text{inv}(1\ 4) = 5$ and $\text{sgn}(1\ 4) = -1$.

6.3.3 The sign is multiplicative

The key property is that signs multiply when permutations are composed.

Theorem 6.3.1 — The Sign Is Multiplicative

For all $\sigma, \tau \in S_n$,

$$\text{sgn}(\sigma\tau) = \text{sgn}\sigma \cdot \text{sgn}\tau.$$

Idea. Count inversions one pair at a time, and record each pair as +1 (order kept) or -1 (order reversed). The sign is the product of these ± 1 's. Applying τ and then σ moves a pair of positions $\{i, j\}$ to $\{\tau(i), \tau(j)\}$ and then to $\{\sigma\tau(i), \sigma\tau(j)\}$. The order is reversed overall exactly when it is reversed at one of the two stages but not both, which

is multiplication of ± 1 's. Finally, as $\{i, j\}$ runs over all pairs, so does $\{\tau(i), \tau(j)\}$, so the contributions of σ multiply to $\text{sgn } \sigma$, just in a different order.

Proof. For a non-zero integer m let $s(m) = 1$ if $m > 0$ and $s(m) = -1$ if $m < 0$. Then $s(mm') = s(m)s(m')$ for non-zero integers m, m' , by the sign rules for products of integers.

Let $\mathcal{P}$ be the set of 2-element subsets $\{i, j\}$ of $\{1, \dots, n\}$. For $\rho \in S_n$ and $\{i, j\} \in \mathcal{P}$, define

$$\varepsilon_\rho(\{i, j\}) = s((\rho(i) - \rho(j))(i - j)).$$

This is defined: $i \neq j$ and $\rho(i) \neq \rho(j)$ since ρ is injective, so the product is non-zero; and exchanging the names i and j negates both factors, so the product does not depend on the order in which the subset is written. If $i < j$, then $i - j < 0$, so $\varepsilon_\rho(\{i, j\}) = -1$ exactly when $\rho(i) > \rho(j)$, that is, when (i, j) is an inversion of ρ . Hence, by [Definition 6.3.2](#),

$$\text{sgn } \rho = \prod_{P \in \mathcal{P}} \varepsilon_\rho(P). \quad (1)$$

Now let $\{i, j\} \in \mathcal{P}$. Since $\tau(i) \neq \tau(j)$, we have $(\tau(i) - \tau(j))^2 > 0$. Multiplying inside s by this positive integer does not change the value, so

$$\varepsilon_{\sigma\tau}(\{i, j\}) = s\left((\sigma(\tau(i)) - \sigma(\tau(j))) (\tau(i) - \tau(j)) \cdot (\tau(i) - \tau(j)) (i - j)\right) = \varepsilon_\sigma(\{\tau(i), \tau(j)\}) \varepsilon_\tau(\{i, j\}).$$

Let $\hat{\tau}: \mathcal{P} \rightarrow \mathcal{P}$, $\{i, j\} \mapsto \{\tau(i), \tau(j)\}$; the image has two elements because τ is injective. The same construction with τ^{-1} gives an inverse function, so $\hat{\tau}$ is a bijection ([Theorem 0.4.7](#)). Taking the product over all $P \in \mathcal{P}$ and using (1) three times,

$$\text{sgn}(\sigma\tau) = \prod_{P \in \mathcal{P}} \varepsilon_\sigma(\hat{\tau}(P)) \cdot \prod_{P \in \mathcal{P}} \varepsilon_\tau(P) = \prod_{Q \in \mathcal{P}} \varepsilon_\sigma(Q) \cdot \text{sgn } \tau = \text{sgn } \sigma \cdot \text{sgn } \tau,$$

where the second equality substitutes $Q = \hat{\tau}(P)$, which runs over $\mathcal{P}$ exactly once because $\hat{\tau}$ is a bijection, and reorders the finite product of integers. This proves the theorem. ■

By induction on the number of factors, $\text{sgn}(\sigma_1\sigma_2 \cdots \sigma_r) = \text{sgn } \sigma_1 \cdots \text{sgn } \sigma_r$. In the language of Chapter 0, the set $\{1, -1\}$ is a group under multiplication (it contains 1, and it is closed under products and inverses, since $(-1)^{-1} = -1$), and the theorem says exactly that

$$\text{sgn}: S_n \rightarrow \{1, -1\}$$

is a group homomorphism ([Definition 0.10.3](#)).

6.3.4 Transpositions and parity

Now we can compute the sign of the building blocks.

Corollary 6.3.2 — Sign of a Transposition

Every transposition is odd: $\text{sgn}(i\ j) = -1$ for all $i \neq j$ in $\{1, \dots, n\}$.

Proof. Since $(i\ j) = (j\ i)$, we may assume $i < j$. Let $\tau = (i\ j)$ and let $p < q$ be positions. We decide when $\tau(p) > \tau(q)$.

- If neither p nor q lies in $\{i, j\}$, then $\tau(p) = p < q = \tau(q)$: no inversion.
- If $(p, q) = (i, j)$, then $\tau(i) = j > i = \tau(j)$: one inversion.
- If $p = i$ and $q \neq j$, then $\tau(p) = j$ and $\tau(q) = q$, an inversion exactly when $i < q < j$. This gives $j - i - 1$ inversions.
- If $q = j$ and $p \neq i$, then $\tau(p) = p$ and $\tau(q) = i$, an inversion exactly when $i < p < j$. This gives $j - i - 1$ inversions.
- If $p = j$ (so $q > j$), then $\tau(p) = i < q = \tau(q)$; if $q = i$ (so $p < i$), then $\tau(p) = p < j = \tau(q)$: no inversion.

These cases cover every pair, so $\text{inv}(\tau) = 1 + 2(j - i - 1)$, an odd number. Hence $\text{sgn } \tau = -1$. ■

The Quick check above is the case $(i, j) = (1, 4)$: $1 + 2 \cdot 2 = 5$ inversions. Combining the corollary with multiplicativity gives the fact that Chapter 0 left open.

Corollary 6.3.3 — Parity Is Well Defined

Let $n \geq 2$ and $\sigma \in S_n$. If $\sigma = \tau_1 \cdots \tau_r$ is a product of r transpositions, then $\text{sgn } \sigma = (-1)^r$. Consequently, if also $\sigma = \tau'_1 \cdots \tau'_s$ with transpositions τ'_m , then r and s are both even or both odd.

Proof. By Theorem 6.3.1, extended to r factors, and Corollary 6.3.2, $\text{sgn } \sigma = \text{sgn } \tau_1 \cdots \text{sgn } \tau_r = (-1)^r$. In the same way $\text{sgn } \sigma = (-1)^s$. Hence $(-1)^r = (-1)^s$, so $r - s$ is even. ■

So a permutation is even exactly when it is a product of an even number of transpositions, and the “number of swaps” has a well-defined parity even though the number itself is not defined.

Proposition 6.3.4 — Sign of a Cycle

A k -cycle has sign $(-1)^{k-1}$. In particular, cycles of even length are odd and cycles of odd length are even.

Proof. For $k = 1$ the cycle is id, with sign $1 = (-1)^0$. Let $k \geq 2$ and let $\gamma = (i_1\ i_2\ \cdots\ i_k)$. We claim

$$\gamma = (i_1\ i_k)(i_1\ i_{k-1}) \cdots (i_1\ i_3)(i_1\ i_2).$$

Apply the factors from right to left. The element i_1 is sent to i_2 by $(i_1\ i_2)$, and no later factor involves i_2 . For $2 \leq m \leq k - 1$, the element i_m is fixed by the factors before $(i_1\ i_m)$, sent to i_1 by it, then to i_{m+1} by $(i_1\ i_{m+1})$, and no later factor involves i_{m+1} . The element i_k is fixed until the last factor $(i_1\ i_k)$ sends it to i_1 . Elements outside the cycle are fixed by every factor. This agrees with γ everywhere, proving the claim. The

product has $k - 1$ transpositions, so $\text{sgn } \gamma = (-1)^{k-1}$ by [Corollary 6.3.3](#). ■

Warning. Only the parity of the number of transpositions is determined, and it is not the parity of the cycle length. The identity in S_3 equals $(1\ 2)(1\ 2)$ and also $(1\ 2)(1\ 3)(1\ 2)(1\ 3)(1\ 2)(1\ 3)$, since $((1\ 2)(1\ 3))^3 = (1\ 3\ 2)^3 = \text{id}$; the numbers 2 and 6 differ, only their parity agrees. And a 3-cycle, “of length 3”, is **even**: $(1\ 2\ 3) = (1\ 3)(1\ 2)$. One more trap concerns order: $(1\ 3)(1\ 2) = (1\ 2\ 3)$ but $(1\ 2)(1\ 3) = (1\ 3\ 2)$. Multiplying in the wrong order gives the wrong permutation and the wrong bottom row, even though, by [Theorem 6.3.1](#), it happens to give the right sign.

6.3.5 The alternating group

Since sgn is a homomorphism, its kernel is a subgroup of S_n ([Theorem 0.10.3](#)).

Definition 6.3.3 — Alternating Group

The **alternating group** A_n is the kernel of $\text{sgn}: S_n \rightarrow \{1, -1\}$, that is, the subgroup of **even** permutations in S_n .

For example, $A_3 = \{\text{id}, (1\ 2\ 3), (1\ 3\ 2)\}$ by the table above, and $A_2 = \{\text{id}\}$. For $n \geq 2$, exactly half of the permutations are even (Exercise C1). The odd permutations do **not** form a subgroup: they do not contain id . The name is no coincidence: as we see below and in the next section, rearranging the arguments of an alternating form by an even permutation leaves its value unchanged.

6.3.6 Back to alternating forms

Here is why this section exists. Let f be an alternating 3-linear form and compare $f(\mathbf{v}_3, \mathbf{v}_1, \mathbf{v}_2)$ with $f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$. Swapping the first two arguments of $(\mathbf{v}_3, \mathbf{v}_1, \mathbf{v}_2)$ gives $(\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_2)$, and swapping the last two then gives $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$. By [Theorem 6.2.1](#) (a) each swap costs a factor -1 , so $f(\mathbf{v}_3, \mathbf{v}_1, \mathbf{v}_2) = (-1)^2 f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$. The list $(\mathbf{v}_3, \mathbf{v}_1, \mathbf{v}_2)$ is $(\mathbf{v}_{\sigma(1)}, \mathbf{v}_{\sigma(2)}, \mathbf{v}_{\sigma(3)})$ for $\sigma = (1\ 3\ 2)$, with bottom row 3, 1, 2, and the factor 1 we found is $\text{sgn } \sigma$.

A different sequence of swaps could have been used, possibly a longer one. By [Corollary 6.3.3](#), every such sequence has an even number of swaps, so every route gives the same factor. In general, the next section proves that $f(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(k)}) = \text{sgn } \sigma \cdot f(\mathbf{v}_1, \dots, \mathbf{v}_k)$ for every alternating k -linear form, and uses it to turn each of the $n!$ surviving terms of the expansion sketched in the first section of this chapter into $\pm D(\mathbf{e}_1, \dots, \mathbf{e}_n)$. Without the corollary, the factor computed swap by swap could depend on the route, and could not be identified with $\text{sgn } \sigma$.

6.3.7 Computing signs in practice

There are now two independent ways to find a sign: count inversions, or factor into cycles and use [Proposition 6.3.4](#) with multiplicativity. Doing both is a good check.

Example. (A Sign Computed Two Ways)

Find the sign of the permutation $\sigma \in S_6$ with two-line notation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 5 & 3 & 6 & 2 & 1 \end{pmatrix}$, first by counting inversions and then from its cycles.

Solution. Inversions. Go through the bottom row 4, 5, 3, 6, 2, 1 from left to right, and for each entry count the **later** entries that are smaller:

position i	1	2	3	4	5	6
$\sigma(i)$	4	5	3	6	2	1
later and smaller	3, 2, 1	3, 2, 1	2, 1	2, 1	1	none
count	3	3	2	2	1	0

So $\text{inv}(\sigma) = 3 + 3 + 2 + 2 + 1 + 0 = 11$, which is odd, and $\text{sgn } \sigma = -1$.

Cycles. Following elements as in [Example \(Computing with cycles\)](#): $1 \mapsto 4 \mapsto 6 \mapsto 1$, then $2 \mapsto 5 \mapsto 2$, and $3 \mapsto 3$. So $\sigma = (1\ 4\ 6)(2\ 5)$. By [Proposition 6.3.4](#), the 3-cycle has sign 1 and the transposition has sign -1 , so by [Theorem 6.3.1](#), $\text{sgn } \sigma = 1 \cdot (-1) = -1$. The two methods agree.

The following cell checks the example, and also tests [Theorem 6.3.1](#) and [Proposition 6.3.4](#) on all of S_5 by brute force. Permutations are stored as tuples of values $(\sigma(1), \dots, \sigma(n))$, shifted to start at 0.

```
from itertools import permutations, combinations

def inv(s):
    return sum(1 for i, j in combinations(range(len(s)), 2) if s[i] > s[j])

def sgn(s):
    return (-1) ** inv(s)

def compose(s, t):
    # (s t)(i) = s(t(i)): first t, then s
    return tuple(s[t[i]] for i in range(len(t)))

def cycle(entries, n):
    # the cycle (entries[0] entries[1] ...), 0-based
    s = list(range(n))
    for a, b in zip(entries, entries[1:] + entries[:1]):
        s[a] = b
    return tuple(s)

sigma = (4, 5, 3, 6, 2, 1)
sigma0 = tuple(x - 1 for x in sigma)
print("inversions:", inv(sigma0), " sgn:", sgn(sigma0))
print("(1 4 6)(2 5) equals sigma:", compose(cycle([0, 3, 5], 6), cycle([1, 4], 6))
      == sigma0)

S5 = list(permutations(range(5)))
print("multiplicative on S_5:", all(sgn(compose(s, t)) == sgn(s) * sgn(t) for s in
    S5 for t in S5))
print("k-cycles have sign (-1)^(k-1):", all(sgn(cycle(list(range(k)), 5)) == (-1) **
    (k - 1) for k in range(1, 6)))
```

6.3.8 Exercises

6.3.8.1 A. Check your understanding

Exercise A1.

- (a) Define an inversion of $\sigma \in S_n$ and the sign $\operatorname{sgn} \sigma$.
- (b) True or false: every 3-cycle is odd. Justify your answer.
- (c) True or false: some permutation is a product of 5 transpositions and also a product of 8 transpositions. Justify your answer.
- (d) What is the kernel of $\operatorname{sgn}: S_n \rightarrow \{1, -1\}$? Is the set of odd permutations a subgroup?
- (e) True or false: $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\tau\sigma)$ for all $\sigma, \tau \in S_n$, even when $\sigma\tau \neq \tau\sigma$. Justify your answer.
- (f) Let f be an alternating 3-linear form. Express $f(\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_1)$ in terms of $f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$.

Solution.

- (a) An inversion of σ is a pair (i, j) with $1 \leq i < j \leq n$ and $\sigma(i) > \sigma(j)$ ([Definition 6.3.1](#)); $\operatorname{sgn} \sigma = (-1)^{\operatorname{inv}(\sigma)}$ ([Definition 6.3.2](#)).
- (b) False. By [Proposition 6.3.4](#) a 3-cycle has sign $(-1)^2 = 1$, so it is even; for instance $(1\ 2\ 3) = (1\ 3)(1\ 2)$.
- (c) False. By [Corollary 6.3.3](#) the numbers of transpositions in two such products have the same parity, and 5 and 8 do not.
- (d) The kernel is the alternating group A_n of even permutations ([Definition 6.3.3](#)). The odd permutations do not form a subgroup: id is even, so it is not among them.
- (e) True. By [Theorem 6.3.1](#) both sides equal $\operatorname{sgn} \sigma \cdot \operatorname{sgn} \tau$, and multiplication of integers is commutative. For example $(1\ 3)(1\ 2) \neq (1\ 2)(1\ 3)$, but both are 3-cycles, with sign 1.
- (f) Swapping the first two arguments of $(\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_1)$ gives $(\mathbf{v}_3, \mathbf{v}_2, \mathbf{v}_1)$, and swapping the first and third then gives $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$. Each swap costs a factor -1 by [Theorem 6.2.1 \(a\)](#), so $f(\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_1) = f(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$. This matches the sign of the rearrangement σ with bottom row 2, 3, 1, namely $\sigma = (1\ 2\ 3)$, which is even.

6.3.8.2 B. Practice

Exercise B1: Computing Signs.

Find the sign of each permutation, both by counting inversions and by using cycles.

- (a) $\sigma \in S_4$ with two-line notation $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix}$.
- (b) $(1\ 2)(3\ 4\ 5) \in S_5$.
- (c) $(1\ 2\ 3)(2\ 3\ 4) \in S_4$.

Solution.

- (a) *Inversions.* In the bottom row 2, 4, 1, 3, the entry 2 is followed by the smaller entry 1, the entry 4 by 1 and 3, and the entry 1 by nothing smaller. So $\text{inv}(\sigma) = 1 + 2 + 0 = 3$ and $\text{sgn } \sigma = -1$. *Cycles.* $1 \mapsto 2 \mapsto 4 \mapsto 3 \mapsto 1$, so $\sigma = (1\ 2\ 4\ 3)$, a 4-cycle, with $\text{sgn } (-1)^3 = -1$ by Proposition 6.3.4.
- (b) *Cycles.* By Proposition 6.3.4 and Theorem 6.3.1, the sign is $(-1) \cdot (-1)^2 = -1$. *Inversions.* The permutation sends $1 \mapsto 2$, $2 \mapsto 1$, $3 \mapsto 4$, $4 \mapsto 5$, $5 \mapsto 3$, so the bottom row is 2, 1, 4, 5, 3. The inversions are (1, 2), (3, 5) and (4, 5), so there are 3, and the sign is -1 .
- (c) *Cycles.* Both factors are 3-cycles, so the sign is $1 \cdot 1 = 1$. *Inversions.* Apply (2 3 4) first: $1 \mapsto 1 \mapsto 2$, $2 \mapsto 3 \mapsto 1$, $3 \mapsto 4 \mapsto 4$, $4 \mapsto 2 \mapsto 3$. The bottom row is 2, 1, 4, 3, with inversions (1, 2) and (3, 4). So the sign is $(-1)^2 = 1$, and in fact the product is (1 2)(3 4).

Exercise B2: Sign of the Inverse.

Prove that $\text{sgn}(\sigma^{-1}) = \text{sgn } \sigma$ for every $\sigma \in S_n$.

Solution. Let $\sigma \in S_n$. By Theorem 6.3.1, $\text{sgn } \sigma \cdot \text{sgn}(\sigma^{-1}) = \text{sgn}(\sigma\sigma^{-1}) = \text{sgn}(\text{id}) = 1$, since id has no inversions. Since $\text{sgn } \sigma \in \{1, -1\}$ and $(\pm 1)^{-1} = \pm 1$, multiplying both sides by $\text{sgn } \sigma$ gives $\text{sgn}(\sigma^{-1}) = \text{sgn } \sigma$.

Exercise B3: The Reversal Permutation.

Let $\rho \in S_n$ be the reversal, $\rho(i) = n + 1 - i$. Show that $\text{inv}(\rho) = n(n-1)/2$. Hence find the n for which ρ is even, and decide the sign of ρ for $n = 6$ and $n = 9$.

Solution. For $1 \leq i < j \leq n$, $\rho(i) = n + 1 - i > n + 1 - j = \rho(j)$. So **every** pair $i < j$ is an inversion, and $\text{inv}(\rho)$ is the number of 2-element subsets of $\{1, \dots, n\}$, which is $n(n-1)/2$. Hence $\text{sgn } \rho = (-1)^{n(n-1)/2}$.

Write $n = 4q + t$ with $t \in \{0, 1, 2, 3\}$. Then $n(n-1)/2$ is $2q(4q-1)$ for $t = 0$, $2q(4q+1)$ for $t = 1$, $(4q+1)(2q+1)$ for $t = 2$, and $(2q+1)(4q+3)$ for $t = 3$. The first two are even and the last two are odd. So ρ is even exactly when n leaves remainder 0 or 1 on division by 4. For $n = 6$, $\text{inv}(\rho) = 15$ and ρ is odd; for $n = 9$, $\text{inv}(\rho) = 36$ and ρ is even.

6.3.8.3 C. Going deeper

Exercise C1: Half the Permutations Are Even.

Let $n \geq 2$. Prove that exactly $n!/2$ permutations in S_n are even.

Hint: consider $\sigma \mapsto (1\ 2)\sigma$.

Solution. Let E and O be the sets of even and odd permutations; they are disjoint and $E \cup O = S_n$, so $|E| + |O| = n!$. Let $\tau = (1\ 2)$, which exists since $n \geq 2$, and define $L: S_n \rightarrow S_n$, $L(\sigma) = \tau\sigma$. By Theorem 6.3.1 and Corollary 6.3.2, $\text{sgn } L(\sigma) = -\text{sgn } \sigma$, so L maps E into O and O into E . Since $\tau\tau = \text{id}$, $L(L(\sigma)) = \sigma$, so L restricted to E is a

bijection $E \rightarrow O$ with inverse L restricted to O (Theorem 0.4.7). Hence $|E| = |O| = n!/2$.

Exercise C2: Sign from the Number of Cycles.

Suppose $\sigma \in S_n$ is written as a product $\sigma = \gamma_1 \gamma_2 \cdots \gamma_c$ of **disjoint** cycles (no element appears in two of them) whose lengths $k_1, \dots, k_c$ add up to n , where fixed points are included as cycles of length 1. (Such a product always exists: follow each element around its loop, as in Example (Computing with cycles).)

- (a) Prove that $\text{sgn } \sigma = (-1)^{n-c}$.
- (b) Use (a) to find the sign of $\sigma = (1\ 4\ 6)(2\ 5) \in S_6$, and compare with Example (A Sign Computed Two Ways).
- (c) Deduce that an n -cycle in S_n and the identity in S_n have the same sign exactly when n is odd.

Solution.

- (a) By Theorem 6.3.1 and Proposition 6.3.4,

$$\text{sgn } \sigma = \prod_{m=1}^c \text{sgn } \gamma_m = \prod_{m=1}^c (-1)^{k_m-1} = (-1)^{(k_1+\cdots+k_c)-c} = (-1)^{n-c}.$$

- (b) Including the fixed point 3 as the cycle (3), $\sigma = (1\ 4\ 6)(2\ 5)(3)$, with lengths $3 + 2 + 1 = 6$ and $c = 3$. So $\text{sgn } \sigma = (-1)^{6-3} = -1$, as in Example (A Sign Computed Two Ways).
- (c) An n -cycle has $c = 1$ and sign $(-1)^{n-1}$; the identity consists of n cycles of length 1, so $c = n$ and its sign is $(-1)^0 = 1$. These agree exactly when $n - 1$ is even, that is, when n is odd.

6.4 Existence and Uniqueness of the Determinant

In the first section of this chapter, signed area obeyed three rules: it is linear in each side, it vanishes when two sides are equal, and the unit square has area 1. In the second section we named functions obeying the first two rules, alternating multilinear forms, and in the third we built the sign of a permutation. This section puts the pieces together. We prove that on F^n there is **exactly one** alternating n -linear form that takes the value 1 on the standard basis, and we find a formula for it. That form is the determinant.

6.4.1 The goal and the plan

Here is the statement we are after.

There is exactly one function $D: F^n \times \cdots \times F^n \rightarrow F$ (with n factors) that is n -linear (Definition 6.2.1), alternating (Definition 6.2.2), and satisfies $D(\mathbf{e}_1, \dots, \mathbf{e}_n) = 1$.

It has two halves of a very different nature. **Uniqueness** says: if such a D exists, its values are forced. **Existence** says: some function really has all three properties. We prove uniqueness first, because its proof is how we *find* the formula. Suppose D exists, expand everything using the rules, and see what is left. The formula that comes out is then the only candidate, and existence reduces to checking that the candidate obeys the rules.

Throughout, a list of n columns $\mathbf{a}_1, \dots, \mathbf{a}_n \in F^n$ and the matrix $A = (\mathbf{a}_1 \ \cdots \ \mathbf{a}_n) \in M_n(F)$ carry the same information, and we pass between them freely. We write a_{ij} for the i -th entry of $\mathbf{a}_j$, so that

$$\mathbf{a}_j = a_{1j}\mathbf{e}_1 + a_{2j}\mathbf{e}_2 + \cdots + a_{nj}\mathbf{e}_n = \sum_{i=1}^n a_{ij}\mathbf{e}_i.$$

6.4.2 Rearranging the arguments

Expanding a form on basis vectors will produce values like $f(\mathbf{e}_3, \mathbf{e}_1, \mathbf{e}_2)$: the standard basis in a scrambled order. Alternating forms handle scrambling by a sign, and the sign is exactly the one built from inversions.

Lemma 6.4.1 — Permuting the arguments of an alternating form

Let f be an alternating n -linear form on a vector space V over F , let $\mathbf{v}_1, \dots, \mathbf{v}_n \in V$, and let $\sigma \in S_n$. Then

$$f(\mathbf{v}_{\sigma(1)}, \mathbf{v}_{\sigma(2)}, \dots, \mathbf{v}_{\sigma(n)}) = \text{sgn}(\sigma) f(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n).$$

Idea. A transposition swaps two arguments, and each swap costs a factor -1 by [Theorem 6.2.1](#). Every permutation is a product of transpositions, so the value picks up $(-1)^k$ for a product of k of them. Multiplicativity of the sign says that $(-1)^k$ is $\text{sgn}(\sigma)$, whichever product we chose.

Proof. If $n = 1$, then $S_1 = \{\text{id}\}$ and $\text{sgn}(\text{id}) = 1$, since the identity has no inversions ([Definition 6.3.2](#)); there is nothing to prove. Let $n \geq 2$. For $\pi \in S_n$, write $f_\pi = f(\mathbf{v}_{\pi(1)}, \dots, \mathbf{v}_{\pi(n)})$.

First let $\pi \in S_n$ and let $\tau = (p \ q)$ be a transposition. The j -th argument of $f_{\pi\tau}$ is $\mathbf{v}_{\pi(\tau(j))}$. For $j \neq p, q$ this is $\mathbf{v}_{\pi(j)}$; for $j = p$ it is $\mathbf{v}_{\pi(q)}$; for $j = q$ it is $\mathbf{v}_{\pi(p)}$. So the arguments of $f_{\pi\tau}$ are those of f_π with positions p and q exchanged. By [Theorem 6.2.1](#), swapping two arguments changes the sign, so

$$f_{\pi\tau} = -f_\pi. \quad (*)$$

Now let $\sigma \in S_n$. By [Theorem 0.10.4](#), $\sigma = \tau_1\tau_2 \cdots \tau_k$ for some transpositions $\tau_1, \dots, \tau_k$ with $k \geq 1$. Put $\pi_0 = \text{id}$ and $\pi_m = \tau_1 \cdots \tau_m$, so that $\pi_m = \pi_{m-1}\tau_m$. Applying $(*)$ k times,

$$f_\sigma = f_{\pi_k} = -f_{\pi_{k-1}} = \cdots = (-1)^k f_{\pi_0} = (-1)^k f(\mathbf{v}_1, \dots, \mathbf{v}_n).$$

On the other hand, by [Theorem 6.3.1](#) and [Corollary 6.3.2](#), $\text{sgn}(\sigma) = \text{sgn}(\tau_1) \cdots \text{sgn}(\tau_k) = (-1)^k$. This proves the lemma. ■

The proof chose a factorization of σ into transpositions, and different choices can have different lengths. The answer does not depend on the choice, because only the parity of k enters, and [Corollary 6.3.3](#) says the parity is determined by σ . In the proof this is absorbed into the equation $(-1)^k = \text{sgn}(\sigma)$.

6.4.3 Uniqueness

We now expand an alternating form completely. It helps to see the case $n = 2$ first.

Theorem 6.4.2 — Uniqueness of Alternating Forms

Let f be an alternating n -linear form on F^n , and let $\mathbf{a}_1, \dots, \mathbf{a}_n \in F^n$ have entries a_{ij} as above. Then

$$f(\mathbf{a}_1, \dots, \mathbf{a}_n) = f(\mathbf{e}_1, \dots, \mathbf{e}_n) \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma(1)1} a_{\sigma(2)2} \cdots a_{\sigma(n)n}.$$

In particular, an alternating n -linear form on F^n is determined by the single value $f(\mathbf{e}_1, \dots, \mathbf{e}_n)$.

Idea. For $n = 2$, write $\mathbf{a}_1 = a_{11}\mathbf{e}_1 + a_{21}\mathbf{e}_2$ and $\mathbf{a}_2 = a_{12}\mathbf{e}_1 + a_{22}\mathbf{e}_2$. Linearity in the first argument, then in the second, gives four terms:

$$\begin{aligned} f(\mathbf{a}_1, \mathbf{a}_2) &= a_{11} f(\mathbf{e}_1, \mathbf{a}_2) + a_{21} f(\mathbf{e}_2, \mathbf{a}_2) \\ &= a_{11}a_{12} f(\mathbf{e}_1, \mathbf{e}_1) + a_{11}a_{22} f(\mathbf{e}_1, \mathbf{e}_2) + a_{21}a_{12} f(\mathbf{e}_2, \mathbf{e}_1) + a_{21}a_{22} f(\mathbf{e}_2, \mathbf{e}_2). \end{aligned}$$

The first and last terms vanish because two arguments are equal, and $f(\mathbf{e}_2, \mathbf{e}_1) = -f(\mathbf{e}_1, \mathbf{e}_2)$. What is left is $(a_{11}a_{22} - a_{21}a_{12}) f(\mathbf{e}_1, \mathbf{e}_2)$, the formula for $n = 2$.

In general the four terms become one term for each way of choosing, for every argument j , which basis vector $\mathbf{e}_{\varphi(j)}$ to take from $\mathbf{a}_j$. A choice is a function $\varphi: \{1, \dots, n\} \rightarrow \{1, \dots, n\}$, and there are n^n of them. The ones that pick some basis vector twice die. The survivors pick every basis vector once, so they are permutations, and [Lemma 6.4.1](#) converts each into a sign.

Proof. Write $[k] = \{1, \dots, k\}$ for $k \geq 1$. We first expand the arguments one at a time.

Claim. For each $k \in \{0, 1, \dots, n\}$,

$$f(\mathbf{a}_1, \dots, \mathbf{a}_n) = \sum_{\varphi: [k] \rightarrow [n]} a_{\varphi(1)1} \cdots a_{\varphi(k)k} f(\mathbf{e}_{\varphi(1)}, \dots, \mathbf{e}_{\varphi(k)}, \mathbf{a}_{k+1}, \dots, \mathbf{a}_n),$$

where the sum runs over **all** functions $\varphi: [k] \rightarrow [n]$. For $k = 0$ we read the right side as the single term $f(\mathbf{a}_1, \dots, \mathbf{a}_n)$.

Proof. We use induction on k . For $k = 0$ both sides are $f(\mathbf{a}_1, \dots, \mathbf{a}_n)$. Suppose the claim holds for some $k < n$. In each term, the argument in position $k + 1$ is $\mathbf{a}_{k+1} = \sum_{i=1}^n a_{i,k+1} \mathbf{e}_i$. Since f is linear in argument $k + 1$,

$$f(\mathbf{e}_{\varphi(1)}, \dots, \mathbf{e}_{\varphi(k)}, \mathbf{a}_{k+1}, \dots, \mathbf{a}_n) = \sum_{i=1}^n a_{i,k+1} f(\mathbf{e}_{\varphi(1)}, \dots, \mathbf{e}_{\varphi(k)}, \mathbf{e}_i, \mathbf{a}_{k+2}, \dots, \mathbf{a}_n).$$

Substituting this into the induction hypothesis gives a sum over all pairs (φ, i) with $\varphi: [k] \rightarrow [n]$ and $i \in [n]$. Such a pair is the same thing as a function $\psi: [k+1] \rightarrow [n]$, namely $\psi(j) = \varphi(j)$ for $j \leq k$ and $\psi(k+1) = i$; and each ψ arises from exactly one pair. The term for the pair (φ, i) is $a_{\psi(1)1} \cdots a_{\psi(k+1),k+1} f(\mathbf{e}_{\psi(1)}, \dots, \mathbf{e}_{\psi(k+1)}, \mathbf{a}_{k+2}, \dots, \mathbf{a}_n)$. This is the claim for $k + 1$. ■

Taking $k = n$ in the claim,

$$f(\mathbf{a}_1, \dots, \mathbf{a}_n) = \sum_{\varphi: [n] \rightarrow [n]} a_{\varphi(1)1} \cdots a_{\varphi(n)n} f(\mathbf{e}_{\varphi(1)}, \dots, \mathbf{e}_{\varphi(n)}). \quad (**)$$

We sort the functions φ into two kinds.

Case 1: φ is not injective. Then $\varphi(p) = \varphi(q)$ for some $p \neq q$, so the arguments in positions p and q of $f(\mathbf{e}_{\varphi(1)}, \dots, \mathbf{e}_{\varphi(n)})$ are equal. Since f is alternating, this value is 0, and the whole term is 0.

Case 2: φ is injective. Since $[n]$ is finite, φ is bijective by [Theorem 0.4.9](#), so $\varphi = \sigma$ for some $\sigma \in S_n$. By [Lemma 6.4.1](#) with $\mathbf{v}_i = \mathbf{e}_i$,

$$f(\mathbf{e}_{\sigma(1)}, \dots, \mathbf{e}_{\sigma(n)}) = \text{sgn}(\sigma) f(\mathbf{e}_1, \dots, \mathbf{e}_n).$$

Hence only the terms of Case 2 survive in (**), there is exactly one for each $\sigma \in S_n$, and

$$f(\mathbf{a}_1, \dots, \mathbf{a}_n) = \sum_{\sigma \in S_n} a_{\sigma(1)1} \cdots a_{\sigma(n)n} \text{sgn}(\sigma) f(\mathbf{e}_1, \dots, \mathbf{e}_n),$$

which is the stated formula after moving the scalar $f(\mathbf{e}_1, \dots, \mathbf{e}_n)$ to the front. This proves the theorem. ■

So the three rules leave no freedom at all. If a normalized alternating form exists, it **must** be

$$\mathbf{a}_1, \dots, \mathbf{a}_n \longmapsto \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1)1} \cdots a_{\sigma(n)n}.$$

Out of n^n terms, the rules killed all but $n!$. For $n = 3$ that is 6 out of 27.

6.4.4 Existence

The candidate has been found. Now we check it against the rules, forgetting where it came from.

Theorem 6.4.3 — Existence: the Leibniz Formula

Define $D: F^n \times \cdots \times F^n \rightarrow F$ by

$$D(\mathbf{a}_1, \dots, \mathbf{a}_n) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1)1} a_{\sigma(2)2} \cdots a_{\sigma(n)n},$$

where a_{ij} is the i -th entry of $\mathbf{a}_j$. Then D is an alternating n -linear form on F^n , and $D(\mathbf{e}_1, \dots, \mathbf{e}_n) = 1$.

Idea. Linearity is visible: each product takes exactly one entry from each column. For alternation, suppose columns p and q are equal. Pair each σ with $\sigma(p \ q)$. The two permutations have opposite signs. Their products use the same entries, except that the entries taken from columns p and q trade places, and those two columns are equal. So each pair contributes $x - x = 0$.

Why pair terms instead of arguing “swapping the equal columns changes D to $-D$, so $D = -D$ and $D = 0$ ”? Because $D = -D$ gives $2D = 0$, and over a field with $1 + 1 = 0$, such as $\mathbb{F}_2$, that says nothing. The pairing argument never divides by 2.

Proof. Multilinearity. Fix $k \in [n]$ and fix all columns except the k -th. For a fixed σ , the product $a_{\sigma(1)1} \cdots a_{\sigma(n)n}$ contains exactly one entry of $\mathbf{a}_k$, namely $a_{\sigma(k)k}$, the $\sigma(k)$ -th coordinate of $\mathbf{a}_k$. So the term for σ equals $c_\sigma a_{\sigma(k)k}$, where $c_\sigma = \operatorname{sgn}(\sigma) \prod_{j \neq k} a_{\sigma(j)j}$ does not depend on $\mathbf{a}_k$. The map $\mathbf{a}_k \mapsto a_{\sigma(k)k}$ is linear, since coordinates of a sum are sums of coordinates and coordinates of $c\mathbf{a}_k$ are c times the coordinates. So each term is linear in $\mathbf{a}_k$, and hence so is their sum D .

Alternation. Suppose $\mathbf{a}_p = \mathbf{a}_q$ with $p < q$, that is, $a_{ip} = a_{iq}$ for all i . Let $\tau = (p \ q)$, let $A_n = \{\sigma \in S_n : \operatorname{sgn}(\sigma) = 1\}$ be the even permutations (Definition 6.3.3), so that $S_n \setminus A_n = \{\sigma \in S_n : \operatorname{sgn}(\sigma) = -1\}$ consists of the odd ones. Since $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma) \operatorname{sgn}(\tau) = -\operatorname{sgn}(\sigma)$ by Theorem 6.3.1 and Corollary 6.3.2, the map $\sigma \mapsto \sigma\tau$ sends A_n to $S_n \setminus A_n$. It is a bijection $A_n \rightarrow S_n \setminus A_n$, because $\rho \mapsto \rho\tau$ sends $S_n \setminus A_n$ back to A_n and $\sigma\tau\tau = \sigma$. Every $\sigma \in S_n$ has sign 1 or -1 , so

$$D(\mathbf{a}_1, \dots, \mathbf{a}_n) = \sum_{\sigma \in A_n} \left(\prod_{j=1}^n a_{\sigma(j)j} - \prod_{j=1}^n a_{\sigma\tau(j)j} \right).$$

Fix $\sigma \in A_n$ and compare the two products factor by factor. For $j \neq p, q$, $\sigma\tau(j) = \sigma(j)$, so the factors agree. For $j = p$, the second product has $a_{\sigma(q)p} = a_{\sigma(q)q}$, which is the factor of the first product at $j = q$. For $j = q$, the second product has $a_{\sigma(p)q} = a_{\sigma(p)p}$, the

factor of the first product at $j = p$. So the two products consist of the same n factors in a different order, and they are equal because multiplication in F is commutative. Every bracket is 0, and $D(\mathbf{a}_1, \dots, \mathbf{a}_n) = 0$.

Normalization. For $\mathbf{a}_j = \mathbf{e}_j$, a_{ij} is 1 if $i = j$ and 0 otherwise. The product $a_{\sigma(1)1} \cdots a_{\sigma(n)n}$ is therefore 1 if $\sigma(j) = j$ for all j , and 0 otherwise. Only $\sigma = \text{id}$ contributes, and $\text{sgn}(\text{id}) = 1$. Hence $D(\mathbf{e}_1, \dots, \mathbf{e}_n) = 1$. This proves the theorem. ■

Uniqueness and existence together answer the question of the first section: the rules of signed area single out exactly one function.

6.4.5 The determinant

We now have a function of square matrices that is forced on us by three reasonable rules, and it deserves a name. It is the central notion of this chapter.

The determinant is the only way to attach a number to n vectors in F^n that is linear in each vector, vanishes when two vectors coincide, and gives 1 on the standard basis.

Definition 6.4.1 — Determinant

Let $n \geq 1$ and $A = (a_{ij}) \in M_n(F)$. The **determinant** of A is

$$\det A := \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma(1)1} a_{\sigma(2)2} \cdots a_{\sigma(n)n} \in F.$$

For columns $\mathbf{a}_1, \dots, \mathbf{a}_n \in F^n$ we also write $\det(\mathbf{a}_1, \dots, \mathbf{a}_n)$ for the determinant of the matrix $(\mathbf{a}_1 \cdots \mathbf{a}_n)$.

In words: run over **all** $n!$ permutations σ . For each, pick from column j the entry in row $\sigma(j)$; this picks one entry in every column and, because σ is a bijection, one entry in every row. Multiply the n picked entries, attach the sign of σ , and add up. The determinant is defined **only for square matrices**, and its value lies in the field F of the entries.

Well-definedness. The formula is a finite sum, so it always gives a number. What the name “**the** determinant” really claims is that nothing else satisfies the three rules, and that is [Theorem 6.4.2](#); [Theorem 6.4.3](#) shows $\det$ does satisfy them. The sign $\text{sgn}(\sigma)$ is fixed by counting inversions ([Definition 6.3.2](#)), and by [Corollary 6.3.3](#) any factorization of σ into transpositions gives the same sign, so either method may be used to evaluate the formula.

Examples.

- $n = 1$. $S_1 = \{\text{id}\}$, so $\det(a) = a$. A degenerate case, but it is the base of every induction on size.
- $n = 2$. $S_2 = \{\text{id}, (1\ 2)\}$, with signs 1 and -1 . The identity picks $a_{11}a_{22}$, and $(1\ 2)$ picks $a_{21}a_{12}$. So

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - cb,$$

the value derived from the rules in [Example \(\$ad - bc\$ from the Rules\)](#), and the number

whose non-vanishing decides invertibility in [Theorem 0.9.8](#).

- $n = 3$. The six permutations of S_3 and their signs (a 3-cycle has sign 1, a transposition -1) give

$$\det A = a_{11}a_{22}a_{33} - a_{21}a_{12}a_{33} - a_{31}a_{22}a_{13} - a_{11}a_{32}a_{23} \\ + a_{21}a_{32}a_{13} + a_{31}a_{12}a_{23},$$

the terms coming from $\text{id}, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)$ in this order. For instance $\sigma = (1\ 2\ 3)$ has $\sigma(1) = 2, \sigma(2) = 3, \sigma(3) = 1$, and picks $a_{21}a_{32}a_{13}$.

- **A numerical 3×3 .** For $A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & 4 & 1 \\ 5 & 2 & 1 \end{pmatrix}$, the six terms in the same order are 8, -0 , -60 , -4 , 0 and 5, so $\det A = 8 - 60 - 4 + 5 = -51$.
- **Identity and zero.** $\det I_n = 1$ by the normalization in [Theorem 6.4.3](#), and $\det 0 = 0$, since every product contains a zero factor.

Non-example by minimal change. Delete the sign. The **permanent** $\text{per } A = \sum_{\sigma} a_{\sigma(1)1} \cdots a_{\sigma(n)n}$ is still n -linear in the columns, by the same argument as in [Theorem 6.4.3](#), and still gives 1 at I_n . But it is not alternating: $\text{per} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = 1 + 1 = 2$, which is non-zero over $\mathbb{R}$ although the two columns are equal. The sign is exactly what makes the pairing argument cancel.

Why this definition. Each rule has a job. Without normalization, the uniqueness theorem only pins the form down up to the scalar $f(\mathbf{e}_1, \dots, \mathbf{e}_n)$. Without alternation, the permanent and many other functions qualify. Multilinearity is what let us expand at all. The choice of **columns** rather than rows is a convention: we show below that the transpose has the same determinant, so rows would give the same function. We have avoided “Sarrus’s rule” on purpose. It is a picture of the six-term formula for $n = 3$ and nothing more, and for $n = 4$ its diagonal pattern would produce 8 products while the Leibniz formula has 24.

Warning. **The determinant is not linear.** It is linear in **each column separately**, not in the matrix. For $n = 2$, $\det I_2 = 1$ but $\det(I_2 + I_2) = \det(2I_2) = 4 \neq 1 + 1$. In general each product in the Leibniz formula has n factors, and scaling A scales every factor, so

$$\det(cA) = c^n \det A, \quad \text{in particular} \quad \det(-A) = (-1)^n \det A.$$

So $\det(-A) = \det A$ for 2×2 matrices, not $-\det A$.

Quick check. Which permutations give a non-zero term in the determinant of $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 0 \end{pmatrix}$, and what is the determinant?

Solution. Column 1 has its only non-zero entry in row 3, column 2 in row 2, and column 3 in row 1. A non-zero term needs $a_{\sigma(j)j} \neq 0$ for every j , so $\sigma(1) = 3, \sigma(2) = 2, \sigma(3) = 1$, that is, $\sigma = (1\ 3)$, of sign -1 . The determinant is $-3 \cdot 2 \cdot 1 = -6$.

The payoff for the forms of the second section is immediate. There we saw that the alternating n -linear forms on F^n form a vector space ([Theorem 6.2.3](#)) and promised its dimension.

Corollary 6.4.4 — Alternating n -Forms on F^n Form a Line

The vector space of alternating n -linear forms on F^n has dimension 1, with basis $(\det)$. Explicitly, every alternating n -linear form f on F^n satisfies

$$f(\mathbf{a}_1, \dots, \mathbf{a}_n) = f(\mathbf{e}_1, \dots, \mathbf{e}_n) \det(\mathbf{a}_1, \dots, \mathbf{a}_n).$$

Proof. By [Theorem 6.4.3](#), $\det$ is an element of this space, and it is not the zero form since $\det(\mathbf{e}_1, \dots, \mathbf{e}_n) = 1$. So the list $(\det)$ is linearly independent. By [Theorem 6.4.2](#) and [Definition 6.4.1](#), every f in the space satisfies the displayed formula, so $f = f(\mathbf{e}_1, \dots, \mathbf{e}_n) \det$ lies in the span of $\det$. Hence $(\det)$ is a basis, and the dimension is 1. ■

This corollary is the most useful form of uniqueness. To prove that some function g of n columns is a multiple of the determinant, check that g is multilinear and alternating, and compute $g(\mathbf{e}_1, \dots, \mathbf{e}_n)$. The multiplicativity of the determinant in the next section is proved exactly this way.

6.4.6 Transposes and triangular matrices

The definition read the matrix column by column. Reading it row by row gives nothing new.

Theorem 6.4.5 — Determinant of the Transpose

For every $A \in M_n(F)$, $\det A^\top = \det A$. Consequently, $\det A$ is also an alternating n -linear function of the **rows** of A , with value 1 at I_n .

Idea. A term of $\det A^\top$ picks the entries $a_{j\sigma(j)}$, one in each row. The same n positions are also “one in each column”: position $(j, \sigma(j))$ is $(\sigma^{-1}(i), i)$ with $i = \sigma(j)$. So the term is the term of $\det A$ for σ^{-1} , and σ^{-1} has the same sign as σ .

Proof. The (i, j) -entry of $A^\top$ is a_{ji} ([Definition 0.9.10](#)), so by [Definition 6.4.1](#)

$$\det A^\top = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)}.$$

Fix σ . As j runs through $[n]$, $i = \sigma(j)$ runs through $[n]$ exactly once, and $j = \sigma^{-1}(i)$. Since multiplication in F is commutative, we may reorder the factors by i :

$$\prod_{j=1}^n a_{j\sigma(j)} = \prod_{i=1}^n a_{\sigma^{-1}(i) i}.$$

Next, $\operatorname{sgn}(\sigma) \operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\sigma\sigma^{-1}) = \operatorname{sgn}(\operatorname{id}) = 1$ by [Theorem 6.3.1](#), and both signs lie in $\{1, -1\}$, so $\operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\sigma)$. Therefore

$$\det A^\top = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma^{-1}) \prod_{i=1}^n a_{\sigma^{-1}(i) i} = \sum_{\rho \in S_n} \operatorname{sgn}(\rho) \prod_{i=1}^n a_{\rho(i) i} = \det A,$$

where the second equality substitutes $\rho = \sigma^{-1}$; this is allowed because $\sigma \mapsto \sigma^{-1}$ is a bijection of S_n onto itself, being its own inverse.

For the second statement, let $\mathbf{r}_1, \dots, \mathbf{r}_n$ be the rows of A . The columns of $A^\top$ are $\mathbf{r}_1^\top, \dots, \mathbf{r}_n^\top$, so $\det A = \det A^\top = \det(\mathbf{r}_1^\top, \dots, \mathbf{r}_n^\top)$. The map $\mathbf{r} \mapsto \mathbf{r}^\top$ is linear ([Theorem 0.9.5](#)), so by [Theorem 6.4.3](#) the function $(\mathbf{r}_1, \dots, \mathbf{r}_n) \mapsto \det(\mathbf{r}_1^\top, \dots, \mathbf{r}_n^\top)$ is linear in each $\mathbf{r}_i$. If $\mathbf{r}_p = \mathbf{r}_q$ for some $p \neq q$, then $\mathbf{r}_p^\top = \mathbf{r}_q^\top$, and the value is 0. Finally $I_n^\top = I_n$. This proves the theorem. ■

So every statement about columns has a twin about rows. In particular [Theorem 6.2.1](#) applies to rows: swapping two rows changes the sign of $\det$, and adding a multiple of one row to another does not change it. The next section turns this into a method of computation.

Before that, one class of matrices whose determinant can be read off directly. Recall that A is upper triangular if $a_{ij} = 0$ whenever $i > j$ ([Definition 0.9.5](#)).

Theorem 6.4.6 — Determinant of a Triangular Matrix

If $A \in M_n(F)$ is upper triangular or lower triangular, then

$$\det A = a_{11} a_{22} \cdots a_{nn}.$$

Idea. Which permutations can pick only entries on or above the diagonal? Picking $a_{\sigma(j)j}$ with $\sigma(j) \leq j$ for every j forces $\sigma = \text{id}$: the numbers $\sigma(1), \dots, \sigma(n)$ are $1, \dots, n$ in some order, so they have the same sum, and none of them is allowed to exceed its partner.

Proof. Suppose A is upper triangular, and let $\sigma \in S_n$ with $\sigma \neq \text{id}$. We claim $\sigma(j) > j$ for some j . Otherwise $\sigma(j) \leq j$ for all j . Since σ is a bijection of $[n]$, $\sum_{j=1}^n \sigma(j) = \sum_{j=1}^n j$, so $\sum_{j=1}^n (j - \sigma(j)) = 0$ is a sum of non-negative integers equal to 0. Hence $\sigma(j) = j$ for every j , contradicting $\sigma \neq \text{id}$. For such j , $a_{\sigma(j)j} = 0$ by upper triangularity, so the term of σ is 0. Only the identity remains, and $\det A = \text{sgn}(\text{id}) a_{11} \cdots a_{nn} = a_{11} \cdots a_{nn}$.

If A is lower triangular, then $A^\top$ is upper triangular with the same diagonal, and $\det A = \det A^\top = a_{11} \cdots a_{nn}$ by [Theorem 6.4.5](#). This proves the theorem. ■

For example, $\det \begin{pmatrix} 2 & 7 & -1 \\ 0 & 3 & 5 \\ 0 & 0 & -4 \end{pmatrix} = 2 \cdot 3 \cdot (-4) = -24$: six terms, five of which vanish. Diagonal

matrices are a special case, $\det \text{diag}(d_1, \dots, d_n) = d_1 \cdots d_n$. Together with Chapter 2's result that an upper triangular matrix is invertible exactly when its diagonal entries are non-zero ([Lemma 2.6.2](#)), this already shows that for upper triangular matrices, A is invertible if and only if $\det A \neq 0$. The next section proves this for all square matrices.

6.4.7 Determinants over a commutative ring

The characteristic polynomial at the end of this chapter is the determinant of a matrix whose entries are **polynomials**, such as $\begin{pmatrix} x-1 & -2 \\ -3 & x-4 \end{pmatrix}$. Polynomials form no field, since x has no inverse in $F[x]$. So we record exactly what this section used.

Remark. Commutative rings. A **commutative ring** is a set R with an addition and a multiplication satisfying all the axioms of a field (Definition 0.7.1) except, possibly, the existence of multiplicative inverses (F8). The integers $\mathbb{Z}$ and the polynomials $F[x]$ are examples (Theorem 0.8.1). For $A \in M_n(R)$, the Leibniz formula of Definition 6.4.1 still defines $\det A \in R$, reading $\operatorname{sgn}(\sigma)r$ as r or $-r$.

Every proof in this section uses only addition, subtraction and multiplication, together with commutativity, and never a multiplicative inverse. So the following hold verbatim over any commutative ring R , with “linear in each argument” meaning additive and compatible with multiplication by elements of R : Lemma 6.4.1, Theorem 6.4.2, Theorem 6.4.3, Theorem 6.4.5 and Theorem 6.4.6. The only outside input is the swap rule of Theorem 6.2.1, and it too is division-free: expanding $0 = f(\dots, \mathbf{u} + \mathbf{v}, \dots, \mathbf{u} + \mathbf{v}, \dots)$ by additivity leaves $f(\dots, \mathbf{u}, \dots, \mathbf{v}, \dots) + f(\dots, \mathbf{v}, \dots, \mathbf{u}, \dots) = 0$. The parts of Theorem 6.2.1 about dependent arguments are not used; they involve solving for one vector in terms of others, which needs division.

In the characteristic-polynomial section we will apply these results with $R = F[x]$. The next two sections flag, in the same way, which of their proofs survive the passage to a ring.

6.4.8 Exercises

6.4.8.1 A. Check your understanding

Exercise A1.

- State the three properties that characterize $\det: M_n(F) \rightarrow F$, and write down the Leibniz formula.
- How many terms does the Leibniz formula have for $n = 4$? How many of the corresponding permutations have sign 1?
- True or false: $\det(A + B) = \det A + \det B$ for all $A, B \in M_2(\mathbb{R})$. Justify your answer.
- True or false: $\det(-A) = -\det A$ for all $A \in M_3(\mathbb{R})$. Justify your answer.
- In the proof of Theorem 6.4.2, why does a term with a non-injective φ vanish?
- Explain why the existence proof pairs σ with $\sigma(p\ q)$ instead of arguing “ $D = -D$, so $D = 0$ ”.

Solution.

- $\det$ is n -linear in the columns, alternating (zero whenever two columns are equal), and $\det I_n = 1$. By Theorem 6.4.2 and Theorem 6.4.3 it is the only such function, and $\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1)1} \cdots a_{\sigma(n)n}$.
- One term per permutation: $4! = 24$. In the proof of Theorem 6.4.3, $\sigma \mapsto \sigma(1\ 2)$ is a bijection from the permutations of sign 1 to those of sign -1 , so each set has 12 elements.
- False. For $A = B = I_2$, $\det(A + B) = \det(2I_2) = 4$ but $\det A + \det B = 2$.
- True. Each Leibniz product has 3 factors, each multiplied by -1 , so $\det(-A) = (-1)^3 \det A = -\det A$. (For $n = 2$ it is false.)

- (e) If $\varphi(p) = \varphi(q)$ with $p \neq q$, the form is evaluated with the equal arguments $\mathbf{e}_{\varphi(p)}$ in positions p and q , and an alternating form vanishes there.
- (f) $D = -D$ only gives $2D = 0$. Over a field in which $2 = 0$, such as $\mathbb{F}_2$, this does not imply $D = 0$. The pairing shows that the terms cancel in pairs, which works over every field.

6.4.8.2 B. Practice

Exercise B1: Six-term determinants.

Using the Leibniz formula, listing all six terms, compute the determinants of

$$A = \begin{pmatrix} 3 & 1 & 2 \\ 1 & 0 & 4 \\ 2 & 5 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \\ 4 & 2 & -2 \end{pmatrix} \in M_3(\mathbb{R}).$$

Then compute $\det A^T$ in the same way, and hence verify [Theorem 6.4.5](#) for this A .

Solution. We use the order $\text{id}, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)$ and the six-term formula from the examples after [Definition 6.4.1](#):

$$a_{11}a_{22}a_{33}, -a_{21}a_{12}a_{33}, -a_{31}a_{22}a_{13}, -a_{11}a_{32}a_{23}, a_{21}a_{32}a_{13}, a_{31}a_{12}a_{23}.$$

For A : $3 \cdot 0 \cdot 1 = 0$; $-1 \cdot 1 \cdot 1 = -1$; $-2 \cdot 0 \cdot 2 = 0$; $-3 \cdot 5 \cdot 4 = -60$; $1 \cdot 5 \cdot 2 = 10$; $2 \cdot 1 \cdot 4 = 8$. Hence $\det A = 0 - 1 + 0 - 60 + 10 + 8 = -43$.

For B : $1 \cdot 3 \cdot (-2) = -6$; $-0 \cdot (-1) \cdot (-2) = 0$; $-4 \cdot 3 \cdot 2 = -24$; $-1 \cdot 2 \cdot 1 = -2$; $0 \cdot 2 \cdot 2 = 0$; $4 \cdot (-1) \cdot 1 = -4$. Hence $\det B = -6 + 0 - 24 - 2 + 0 - 4 = -36$.

For $A^T = \begin{pmatrix} 3 & 1 & 2 \\ 1 & 0 & 5 \\ 2 & 4 & 1 \end{pmatrix}$: $3 \cdot 0 \cdot 1 = 0$; $-1 \cdot 1 \cdot 1 = -1$; $-2 \cdot 0 \cdot 2 = 0$; $-3 \cdot 4 \cdot 5 = -60$; $1 \cdot 4 \cdot 2 = 8$; $2 \cdot 1 \cdot 5 = 10$. The sum is $-43 = \det A$, as [Theorem 6.4.5](#) predicts; the six terms are the same numbers, with the two 3-cycles exchanged, because $(1\ 2\ 3)^{-1} = (1\ 3\ 2)$.

Exercise B2: The determinant of P_σ .

Let $\sigma \in S_n$ and let $P_\sigma = (\mathbf{e}_{\sigma(1)} \ \cdots \ \mathbf{e}_{\sigma(n)})$ be its permutation matrix ([Definition 2.6.2](#)).

- (a) Prove that $\det P_\sigma = \text{sgn}(\sigma)$.

(b) Hence compute $\det \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ and $\det \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$.

Solution.

- (a) By [Theorem 6.4.3](#), $\det$ is an alternating n -linear form on F^n . By [Lemma 6.4.1](#) with $\mathbf{v}_i = \mathbf{e}_i$,

$$\det P_\sigma = \det(\mathbf{e}_{\sigma(1)}, \dots, \mathbf{e}_{\sigma(n)}) = \text{sgn}(\sigma) \det(\mathbf{e}_1, \dots, \mathbf{e}_n) = \text{sgn}(\sigma) \det I_n = \text{sgn}(\sigma).$$

- (b) The first matrix has columns $\mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_1$, so it is P_σ with $\sigma(1) = 2, \sigma(2) = 3, \sigma(3) = 1$, that is, $\sigma = (1\ 2\ 3)$. A 3-cycle is $(1\ 3)(1\ 2)$ (Example (Computing with cycles)), a product of two transpositions, so its sign is 1 and the determinant is 1. The second matrix has columns $\mathbf{e}_2, \mathbf{e}_1, \mathbf{e}_4, \mathbf{e}_3$, so it is P_σ with $\sigma = (1\ 2)(3\ 4)$, of sign $(-1)^2 = 1$ by Theorem 6.3.1 and Corollary 6.3.2. Its determinant is 1.

Exercise B3: A zero row.

Let $A \in M_n(F)$ have a zero row. Prove directly from Definition 6.4.1, without using Theorem 6.4.5, that $\det A = 0$. Then prove the same for a zero column.

Solution. Suppose row i of A is zero, and let $\sigma \in S_n$. Since σ is surjective, there is j with $\sigma(j) = i$. The factor $a_{\sigma(j)j} = a_{ij}$ of the term for σ is 0, so the term is 0. This holds for every σ , so $\det A = 0$.

Suppose column j is zero. For every σ , the term for σ contains the factor $a_{\sigma(j)j} = 0$, so again every term vanishes and $\det A = 0$.

6.4.8.3 C. Going deeper

Exercise C1: Too many zeros.

Let $A \in M_n(F)$.

- Prove that if more than $n^2 - n$ entries of A are zero, then $\det A = 0$.
- Show that the bound is sharp: give, for each n , a matrix with exactly $n^2 - n$ zero entries and non-zero determinant.

Solution.

- If more than $n^2 - n$ entries are zero, then fewer than n entries are non-zero. For $\sigma \in S_n$, the term for σ is a product of the n entries in positions $(\sigma(1), 1), \dots, (\sigma(n), n)$. These positions are distinct, since their column indices are distinct. They cannot all hold non-zero entries, because there are fewer than n such entries. So at least one factor is 0, and the term is 0. Hence $\det A = 0$.
- I_n has exactly $n^2 - n$ zero entries, and $\det I_n = 1 \neq 0$.

Exercise C2: Counting the cost.

A computer performs about 10^9 multiplications per second.

- Show that evaluating the Leibniz formula term by term for an $n \times n$ matrix takes $n!(n-1)$ multiplications. Estimate the time this takes for $n = 20$.
- Gaussian elimination (Chapter 2) brings A to upper triangular form. Suppose no row swaps are needed. At step k (for $k = 1, \dots, n-1$), each of the $n-k$ rows below the pivot requires one division to find its multiplier and $n-k$ multiplications to update its entries in columns $k+1, \dots, n$. Show that the total number of multiplications and divisions is $\frac{(n-1)n(n+1)}{3}$, and evaluate it for $n = 20$.

In the next section we will see that the determinant can be read off from this triangular form with $n - 1$ further multiplications.

Solution.

- (a) There is one term for each $\sigma \in S_n$, that is, $n!$ terms, and each is a product of n entries, which takes $n-1$ multiplications. For $n = 20$, $20! \cdot 19 = 46\,225\,138\,155\,356\,160\,000 \approx 4.6 \times 10^{19}$. At 10^9 per second this takes about 4.6×10^{10} seconds, roughly 1 460 years.
- (b) Step k costs $(n - k)(1 + (n - k))$ operations. With $m = n - k$ running from 1 to $n - 1$, the total is

$$\sum_{m=1}^{n-1} m(m+1) = \sum_{m=1}^{n-1} m^2 + \sum_{m=1}^{n-1} m = \frac{(n-1)n(2n-1)}{6} + \frac{(n-1)n}{2} = \frac{(n-1)n((2n-1)+3)}{6} = \frac{(n-1)n}{3}$$

For $n = 20$ this is $\frac{19 \cdot 20 \cdot 21}{3} = 2660$, a few microseconds. The Leibniz formula is a definition and a proof tool, not an algorithm.

Exercise C3: The permanent, and a change of field.

For $A \in M_n(F)$ define $\text{per } A = \sum_{\sigma \in S_n} a_{\sigma(1)1} \cdots a_{\sigma(n)n}$.

- (a) Prove that per is n -linear in the columns and $\text{per } I_n = 1$.
- (b) Let $n \geq 2$. Prove that over $\mathbb{R}$, per is **not** alternating.
- (c) Prove that over $\mathbb{F}_2$, $\text{per } A = \det A$ for every $A \in M_n(\mathbb{F}_2)$. Explain why this does not contradict (b) and [Theorem 6.4.2](#).

Solution.

- (a) The proof of multilinearity in [Theorem 6.4.3](#) never used the signs: for fixed k , the term for σ is a constant times $a_{\sigma(k)k}$, which is linear in the k -th column. The normalization argument there also applies: only $\sigma = \text{id}$ contributes at I_n , with product 1.
- (b) Let $J \in M_n(\mathbb{R})$ have every entry equal to 1. Its first two columns are equal, but every term of $\text{per } J$ is 1, so $\text{per } J = n! \neq 0$ in $\mathbb{R}$.
- (c) In $\mathbb{F}_2$, $-1 = 1$, so $\text{sgn}(\sigma) \cdot r = r$ for every r , and the two formulas agree term by term. There is no contradiction: (b) is a statement over $\mathbb{R}$, where $n! \neq 0$. Over $\mathbb{F}_2$, $\text{per } J = n! \cdot 1 = 0$ for $n \geq 2$, and $\text{per} = \det$ is alternating, as [Theorem 6.4.3](#) guarantees. Uniqueness is about a fixed field, and over each field there is exactly one normalized alternating form.

6.5 Row Operations and Multiplicativity

The previous section defined the determinant by the Leibniz formula and proved it is the only normalized alternating form. The formula settles every theoretical question in principle, but with $n!$ terms it is useless for computing, and it hides the two facts that make determinants

worth having. The first is that $\det A \neq 0$ exactly when A is invertible. The second is that $\det(AB) = \det A \det B$. This section proves both. Along the way we get a fast algorithm, and we extend the determinant from matrices to operators.

6.5.1 Row operations

Row reduction is the workhorse of Chapter 2, and the three rules of a determinant say precisely how it reacts to each elementary row operation. Everything follows from [Theorem 6.4.5](#), which lets us treat $\det$ as a function of the rows.

Theorem 6.5.1 — Row Operations and the Determinant

Let $A \in M_n(F)$ and let B be obtained from A by one operation.

- (a) **(Swap)** If B is obtained by $R_i \leftrightarrow R_j$ with $i \neq j$, then $\det B = -\det A$.
- (b) **(Scale)** If B is obtained by $R_i \rightarrow cR_i$ with $c \in F$, then $\det B = c \det A$.
- (c) **(Replace)** If B is obtained by $R_i \rightarrow R_i + cR_j$ with $i \neq j$ and $c \in F$, then $\det B = \det A$.
- (d) The same three statements hold for the corresponding operations on **columns**.
- (e) For the elementary matrices of [Definition 2.4.1](#),

$$\det P_{ij} = -1, \quad \det D_i(c) = c, \quad \det(I_n + cE_{ij}) = 1 \quad (i \neq j),$$

and $\det(EA) = \det E \det A$ for every elementary matrix $E \in M_n(F)$.

Proof. By [Theorem 6.4.5](#), $\det$ is an alternating n -linear function of the rows $\mathbf{r}_1, \dots, \mathbf{r}_n$ of A . We write $\det(\mathbf{r}_1, \dots, \mathbf{r}_n)$ for it in this proof.

- (b) This is linearity in row i .
- (c) By linearity in row i ,

$$\det B = \det(\dots, \mathbf{r}_i + c\mathbf{r}_j, \dots, \mathbf{r}_j, \dots) = \det(\dots, \mathbf{r}_i, \dots, \mathbf{r}_j, \dots) + c \det(\dots, \mathbf{r}_j, \dots, \mathbf{r}_j, \dots),$$

where the displayed arguments sit in positions i and j . The last determinant has two equal rows, so it is 0. Hence $\det B = \det A$.

- (d) This is the swap rule of [Theorem 6.2.1](#), applied to the rows.
- (e) By [Theorem 6.4.3](#), $\det$ is an alternating n -linear function of the columns, and the arguments for (a), (b), (c) apply word for word.
- (f) The matrices P_{ij} , $D_i(c)$ and $I_n + cE_{ij}$ are obtained from I_n by a swap, a scaling and a replacement respectively. By (a), (b), (c) and $\det I_n = 1$, their determinants are -1 , c and 1 . Now let $E = \rho(I_n)$ be elementary. By [Theorem 2.4.1](#), $EA = \rho(A)$. By (a), (b) or (c), $\det \rho(A)$ is $\det A$ times -1 , c or 1 respectively, and this factor is $\det E$. Hence $\det(EA) = \det E \det A$. This proves the theorem. ■

Read part (e) as a bookkeeping rule. A swap flips the sign, a scaling by c multiplies by c , and a replacement changes nothing. Replacement, the operation used most in elimination, is free.

Warning. A combined move is not a replacement. The move $R_2 \rightarrow 3R_2 - R_1$ looks harmless but multiplies the determinant by 3: it is the scaling $R_2 \rightarrow 3R_2$ followed by the replacement $R_2 \rightarrow R_2 - R_1$. For $A = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$, with $\det A = 1$, it produces $\begin{pmatrix} 1 & 2 \\ 2 & 7 \end{pmatrix}$, whose determinant is 3. Keep replacements in the form “row i plus a multiple of another row”, with coefficient 1 on row i .

Since elimination brings any square matrix to an upper triangular one, and triangular determinants are products of diagonal entries ([Theorem 6.4.6](#)), we have an algorithm.

Example. (A 4×4 Determinant by Elimination)

Compute $\det A$ for

$$A = \begin{pmatrix} 0 & 2 & 4 & 2 \\ 1 & 1 & 0 & 3 \\ 2 & 3 & 1 & 5 \\ -1 & 2 & 3 & 1 \end{pmatrix} \in M_4(\mathbb{R}).$$

Solution. We record each operation and its effect, using [Theorem 6.5.1](#).

Step 1. The $(1, 1)$ -entry is 0, so swap: $R_1 \leftrightarrow R_2$. This multiplies the determinant by -1 :

$$\det A = -\det \begin{pmatrix} 1 & 1 & 0 & 3 \\ 0 & 2 & 4 & 2 \\ 2 & 3 & 1 & 5 \\ -1 & 2 & 3 & 1 \end{pmatrix}.$$

Step 2. Clear column 1 with the replacements $R_3 \rightarrow R_3 - 2R_1$ and $R_4 \rightarrow R_4 + R_1$, which change nothing. Row 2 has a common factor 2; pulling it out is linearity in row 2:

$$\det A = -\det \begin{pmatrix} 1 & 1 & 0 & 3 \\ 0 & 2 & 4 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 3 & 3 & 4 \end{pmatrix} = -2 \det \begin{pmatrix} 1 & 1 & 0 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 3 & 3 & 4 \end{pmatrix}.$$

Step 3. Clear column 2: $R_3 \rightarrow R_3 - R_2$ and $R_4 \rightarrow R_4 - 3R_2$. Then clear column 3: $R_4 \rightarrow R_4 - 3R_3$. None of these changes the determinant:

$$\det A = -2 \det \begin{pmatrix} 1 & 1 & 0 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & -3 & 1 \end{pmatrix} = -2 \det \begin{pmatrix} 1 & 1 & 0 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & 7 \end{pmatrix}.$$

Step 4. The last matrix is upper triangular, so by [Theorem 6.4.6](#) its determinant is $1 \cdot 1 \cdot (-1) \cdot 7 = -7$. Hence $\det A = -2 \cdot (-7) = 14$.

Elimination costs roughly $n^3/3$ arithmetic operations, and reading off the triangular determinant takes only $n - 1$ more multiplications, against $n!(n - 1)$ multiplications for the Leibniz formula ([Exercise \(C2: Counting the cost\)](#)). For $n = 20$ that is a few thousand operations against about 4.6×10^{19} . Every practical determinant computation is some form of elimination.

6.5.2 Determinant and invertibility

The same bookkeeping answers the invertibility question. Row reduction multiplies the determinant by non-zero factors only, so it cannot turn a non-zero determinant into zero or back. And the end of the reduction is either I_n or a matrix with a zero row.

Theorem 6.5.2 — Invertible if and only if Non-Zero Determinant

A matrix $A \in M_n(F)$ is invertible if and only if $\det A \neq 0$.

Proof. Let R be the reduced row echelon form of A (Theorem 2.2.3, Theorem 2.3.4). By Definition 2.2.2 and Theorem 2.4.1, there are elementary matrices $E_1, \dots, E_k$ (possibly $k = 0$) with $R = E_k \cdots E_1 A$. Applying Theorem 6.5.1 (e) k times,

$$\det R = \det E_k \cdots \det E_1 \det A.$$

Each $\det E_m$ is -1 , 1 , or some $c \neq 0$, since a scaling operation requires $c \neq 0$. A product of non-zero elements of a field is non-zero (Theorem 0.7.1), so $\det R = 0$ if and only if $\det A = 0$.

By Lemma 2.4.3, either $R = I_n$, and then $\det R = 1 \neq 0$; or the last row of R is zero. In the second case, scaling that row by 0 leaves R unchanged, so $\det R = 0 \cdot \det R = 0$ by Theorem 6.5.1 (b). Hence $\det A \neq 0$ if and only if $R = I_n$, which by Theorem 2.4.4 holds if and only if A is invertible. ■

This is the promised addition to the list of equivalent conditions. We collect it with the conditions from Chapter 3 about the map $T_A: \mathbf{x} \mapsto A\mathbf{x}$, and with two conditions about rows that the determinant makes easy.

Theorem 6.5.3 — Invertible Matrix Theorem, with Determinants

Let $A \in M_n(F)$ and $T_A: F^n \rightarrow F^n$, $\mathbf{x} \mapsto A\mathbf{x}$. The following are equivalent, and each is equivalent to each of the eight conditions of Theorem 2.4.4.

- (a) A is invertible.
- (b) T_A is injective.
- (c) T_A is surjective.
- (d) $\text{rank } A = n$.
- (e) $A^\top$ is invertible.
- (f) The rows of A are linearly independent.
- (g) $\det A \neq 0$.

Proof. Item (a) is item (a) of Theorem 2.4.4, so it suffices to connect each new item with that theorem.

(a) $\Leftrightarrow$ (g). This is Theorem 6.5.2.

(a) $\Leftrightarrow$ (b). By Theorem 3.5.7 ((b) $\Leftrightarrow$ (c)) with $V = F^n$, T_A is injective if and only if $\ker T_A = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$ is $\{\mathbf{0}\}$. That is item (b) of Theorem 2.4.4.

(b) $\Leftrightarrow$ (c). This is Theorem 3.5.7 ((b) $\Leftrightarrow$ (d)), since F^n is finite-dimensional.

(c) $\Leftrightarrow$ (d). By Theorem 0.9.3, $\text{im } T_A$ is the set of combinations of the columns of A , that is, $\text{col}(A)$. So T_A is surjective if and only if $\text{col}(A) = F^n$. Since $\text{col}(A)$ is a subspace of F^n , this holds if and only if $\text{rank } A = \dim \text{col}(A) = n$ (Definition 2.5.4, Theorem 1.6.8).

(a) $\Leftrightarrow$ (e). By Theorem 6.4.5, $\det A^\top = \det A$. Applying (a) $\Leftrightarrow$ (g) to A and to $A^\top$, both “ A invertible” and “ $A^\top$ invertible” are equivalent to $\det A \neq 0$.

(e) $\Leftrightarrow$ (f). Let $\mathbf{r}_1, \dots, \mathbf{r}_n$ be the rows of A . The columns of $A^\top$ are $\mathbf{r}_1^\top, \dots, \mathbf{r}_n^\top$. By Theorem 0.9.5, $c_1\mathbf{r}_1 + \dots + c_n\mathbf{r}_n = \mathbf{0}$ if and only if $c_1\mathbf{r}_1^\top + \dots + c_n\mathbf{r}_n^\top = \mathbf{0}$, so the rows of A are independent exactly when the columns of $A^\top$ are. By Theorem 2.4.4 ((a) $\Leftrightarrow$ (f)) applied to $A^\top$, this holds exactly when $A^\top$ is invertible. This proves the theorem. ■

The determinant is now one more item on the menu, and for the right problems it is the best one. For a concrete matrix with a parameter, one polynomial in the parameter decides invertibility at once. For an abstract matrix, such as $A + I$ when $A^k = 0$, the determinant is usually hopeless, and the kernel is the item to try first.

Quick check. For which $t \in \mathbb{R}$ is $\begin{pmatrix} 1 & t \\ t & 4 \end{pmatrix}$ invertible? Over $\mathbb{F}_3$, for which t ?

Solution. The determinant is $4 - t^2 = (2 - t)(2 + t)$. Over $\mathbb{R}$ the matrix is invertible exactly for $t \neq \pm 2$. Over $\mathbb{F}_3$, $4 = 1$, so the determinant is $1 - t^2$, which is 1 for $t = 0$ and 0 for $t = 1, 2$; the matrix is invertible only for $t = 0$.

6.5.3 Multiplicativity

Now the second fact. A natural first attempt is to multiply out the Leibniz formula for AB , whose entries are sums, and regroup. That is legitimate, but the bookkeeping is horrible. The uniqueness theorem does the work instead: we exhibit $B \mapsto \det(AB)$ as an alternating form in the columns of B , and then it has no choice.

Theorem 6.5.4 — Multiplicativity of the Determinant

For all $A, B \in M_n(F)$,

$$\det(AB) = \det A \det B.$$

Idea. Fix A . The j -th column of AB is $A\mathbf{b}_j$, so $g(\mathbf{b}_1, \dots, \mathbf{b}_n) = \det(A\mathbf{b}_1, \dots, A\mathbf{b}_n)$ is $\det(AB)$ viewed as a function of the columns of B . It inherits linearity in each $\mathbf{b}_j$ from $\det$ and from matrix multiplication, and equal columns $\mathbf{b}_p = \mathbf{b}_q$ give equal columns $A\mathbf{b}_p = A\mathbf{b}_q$. So g is an alternating form, hence a multiple of $\det$, and the multiple is $g(\mathbf{e}_1, \dots, \mathbf{e}_n) = \det A$.

Proof. Fix $A \in M_n(F)$, and define $g: F^n \times \cdots \times F^n \rightarrow F$ by

$$g(\mathbf{b}_1, \dots, \mathbf{b}_n) = \det(A\mathbf{b}_1, \dots, A\mathbf{b}_n).$$

By [Theorem 0.9.2](#) (columns), if B has columns $\mathbf{b}_1, \dots, \mathbf{b}_n$, then AB has columns $A\mathbf{b}_1, \dots, A\mathbf{b}_n$, so $g(\mathbf{b}_1, \dots, \mathbf{b}_n) = \det(AB)$.

g is n -linear. Fix k and all $\mathbf{b}_j$ with $j \neq k$. The map $\mathbf{b}_k \mapsto A\mathbf{b}_k$ is linear by [Theorem 0.9.4](#), and $\det$ is linear in its k -th argument by [Theorem 6.4.3](#). A composite of linear maps is linear, so g is linear in $\mathbf{b}_k$.

g is alternating. If $\mathbf{b}_p = \mathbf{b}_q$ with $p \neq q$, then $A\mathbf{b}_p = A\mathbf{b}_q$, so $g(\mathbf{b}_1, \dots, \mathbf{b}_n) = 0$ since $\det$ is alternating.

By [Corollary 6.4.4](#), $g(\mathbf{b}_1, \dots, \mathbf{b}_n) = g(\mathbf{e}_1, \dots, \mathbf{e}_n) \det(\mathbf{b}_1, \dots, \mathbf{b}_n)$. Finally $A\mathbf{e}_j$ is the j -th column of A ([Theorem 0.9.3](#)), so $g(\mathbf{e}_1, \dots, \mathbf{e}_n) = \det A$. Hence $\det(AB) = \det A \det B$. ■

Remark. Over a commutative ring. This proof uses only [Corollary 6.4.4](#) (in the explicit form of [Theorem 6.4.2](#)), linearity of matrix multiplication, and the Leibniz formula. None of these divides. So $\det(AB) = \det A \det B$ holds for $A, B \in M_n(R)$ over any commutative ring R , such as $F[x]$.

Alternatively, the building blocks of Chapter 2 give a proof. If A is invertible, then $A = E_1 \cdots E_k$ with elementary E_m ([Theorem 2.4.4](#) (d)), and peeling off one factor at a time with [Theorem 6.5.1](#) (e) gives $\det(AB) = \det E_1 \cdots \det E_k \det B$; the case $B = I_n$ of the same computation shows $\det A = \det E_1 \cdots \det E_k$. If A is not invertible, neither is AB : otherwise $(AB)C = I_n$ for some C , so $A(BC) = I_n$ and A would be invertible by [Theorem 2.4.5](#). Then both sides are 0 by [Theorem 6.5.2](#). This route leans on row reduction, which divides, so unlike the first proof it needs a field.

For reference, here are the two identities used most often, now both proved.

Theorem 6.5.5 — Properties of Determinants

Let $A, B \in M_n(F)$. Then:

1. $\det(AB) = \det A \det B$;
2. $\det A^T = \det A$.

Proof. Part 1 is [Theorem 6.5.4](#) and part 2 is [Theorem 6.4.5](#). ■

The first consequences are immediate.

Corollary 6.5.6 — Determinant of the Inverse

If $A \in M_n(F)$ is invertible, then $\det A \neq 0$ and $\det(A^{-1}) = (\det A)^{-1}$.

Proof. By [Theorem 6.5.4](#), $\det A \det(A^{-1}) = \det(AA^{-1}) = \det I_n = 1$. Hence $\det A \neq 0$, and $\det(A^{-1})$ is the inverse of $\det A$ in F . ■

Warning. Multiplicativity does not make det additive, and it does not make products commute. For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 5 \end{pmatrix}$, $AB = \begin{pmatrix} 2 & 11 \\ 4 & 23 \end{pmatrix} \neq \begin{pmatrix} 3 & 4 \\ 16 & 22 \end{pmatrix} = BA$, yet $\det(AB) = \det(BA) = 2$, since both equal $\det A \det B = (-2)(-1)$. On the other hand, $\det(A + B)$ is not determined by $\det A$ and $\det B$ at all: I_2 and I_2 have determinants 1, 1 and $\det(I_2 + I_2) = 4$, while I_2 and $C = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ also have determinants 1, 1 but $\det(I_2 + C) = \det \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} = 2$.

Quick check. Let $A = \begin{pmatrix} 2 & 1 \\ 7 & 4 \end{pmatrix}$. Without multiplying any matrices, find $\det(A^5)$, $\det(A^{-1})$ and $\det(A^T A)$.

Solution. $\det A = 8 - 7 = 1$. By [Theorem 6.5.4](#), $\det(A^5) = (\det A)^5 = 1$. By [Corollary 6.5.6](#), $\det(A^{-1}) = 1^{-1} = 1$. By [Theorem 6.5.5](#), $\det(A^T A) = \det A^T \det A = (\det A)^2 = 1$.

6.5.4 The determinant of an operator

In Chapter 3 the trace passed from matrices to operators because similar matrices have the same trace ([Definition 3.10.2](#)). Multiplicativity gives the same for determinants.

Corollary 6.5.7 — Determinant Is a Similarity Invariant

If $A, B \in M_n(F)$ are similar, then $\det A = \det B$.

Proof. Let $B = P^{-1}AP$ with P invertible. By [Theorem 6.5.4](#) and [Corollary 6.5.6](#),

$$\det B = \det(P^{-1}) \det A \det P = (\det P)^{-1} \det P \det A = \det A,$$

where the middle step uses commutativity of multiplication in F . ■

The converse fails: I_2 and $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ both have determinant 1, but the only matrix similar to I_2 is $P^{-1}I_2P = I_2$ itself.

By [Theorem 3.7.4](#), all matrices of one operator in the various bases are similar to each other. So the corollary lets us define:

Definition 6.5.1 — Determinant of an operator

Let V be a finite-dimensional vector space over F with $\dim V \geq 1$, and let $T \in \mathcal{L}(V)$. The **determinant** of T is

$$\det T := \det[T]_{\mathcal{B}},$$

where $\mathcal{B}$ is **any** basis of V . For $V = \{\mathbf{0}\}$ we set $\det T = 1$.

Well-definedness. If $\mathcal{B}$ and $\mathcal{C}$ are bases of V , then $[T]_{\mathcal{C}} = P^{-1}[T]_{\mathcal{B}}P$ with $P = [\text{id}]_{\mathcal{C}}^{\mathcal{B}}$ invertible, by [Theorem 3.7.3](#) and [Theorem 3.7.4](#) (a). By [Corollary 6.5.7](#) the two determinants agree. The convention for $V = \{\mathbf{0}\}$ is the empty product, and it keeps the next theorem true in that case.

Examples.

- **Scalars.** For $c \in F$ and $\dim V = n \geq 1$, $[c \text{id}_V]_{\mathcal{B}} = cI_n$ in every basis, so $\det(c \text{id}_V) = c^n$ by [Theorem 6.4.6](#).
- **A reflection.** $R(x, y) = (y, x)$ on $\mathbb{R}^2$ has standard matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, of determinant -1 . In the basis $((1, 1), (1, -1))$ its matrix is $\text{diag}(1, -1)$ ([Example \(The reflection in a mirror-adapted basis\)](#)), again of determinant -1 , as it must be.
- **Differentiation.** On $\mathbb{R}[x]_{\leq 3}$ with basis $(1, x, x^2, x^3)$, the matrix of D is upper triangular with zero diagonal ([Example \(Differentiation on Cubic Polynomials\)](#)), so $\det D = 0$.

The determinant of a map $T: V \rightarrow W$ between **different** spaces is not defined, even when $\dim V = \dim W$. A matrix $[T]_{\mathcal{B}}^{\mathcal{C}}$ exists, but changing $\mathcal{C}$ alone multiplies it on the left by an arbitrary invertible matrix, which can change its determinant to any non-zero value. For $\text{id}: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with input basis $\mathcal{E}$ and output basis $((2, 0), (0, 1))$, the matrix is $\text{diag}(\frac{1}{2}, 1)$, of determinant $\frac{1}{2}$, not 1.

Theorem 6.5.8 — Properties of the Determinant of an Operator

Let V be a finite-dimensional vector space over F and $S, T \in \mathcal{L}(V)$. Then

$$\det(ST) = \det S \det T, \quad \det \text{id}_V = 1,$$

and T is invertible if and only if $\det T \neq 0$.

Proof. If $V = \{\mathbf{0}\}$, all determinants are 1 and the only operator is id_V , which is invertible. Otherwise fix a basis $\mathcal{B}$ of V . By [Theorem 3.6.2](#), $[ST]_{\mathcal{B}} = [S]_{\mathcal{B}}[T]_{\mathcal{B}}$, so $\det(ST) = \det S \det T$ by [Theorem 6.5.4](#). Next $[\text{id}_V]_{\mathcal{B}} = I_n$, of determinant 1. Finally, by [Theorem 3.6.5](#) (b), T is invertible if and only if $[T]_{\mathcal{B}}$ is, which by [Theorem 6.5.2](#) holds if and only if $\det T = \det[T]_{\mathcal{B}} \neq 0$. ■

So the determinant is a property of the operator, not of the coordinates. Differentiation on $\mathbb{R}[x]_{\leq 3}$ is not invertible because its determinant is 0, whichever basis we compute in. In the section on orientation and volume we will see what the number $\det T$ measures geometrically.

6.5.5 The determinant as a homomorphism

In Chapter 0 we met groups and homomorphisms, and promised that the determinant would be one. The general linear group $\mathrm{GL}_n(F)$ is the group of invertible matrices in $M_n(F)$ under multiplication, and $F \setminus \{0\}$ is a group under multiplication ([Example \(Groups\)](#)).

Theorem 6.5.9 — The Determinant Is a Group Homomorphism

Let $n \geq 1$. The determinant restricts to a **surjective** group homomorphism

$$\det: \mathrm{GL}_n(F) \rightarrow F \setminus \{0\}.$$

Its kernel is the **special linear group** $\mathrm{SL}_n(F) := \{A \in M_n(F) : \det A = 1\}$, which is therefore a subgroup of $\mathrm{GL}_n(F)$.

Proof. By [Theorem 6.5.2](#), $\det A \neq 0$ for $A \in \mathrm{GL}_n(F)$, so $\det$ maps $\mathrm{GL}_n(F)$ into $F \setminus \{0\}$, and [Theorem 6.5.4](#) says it is a homomorphism ([Definition 0.10.3](#)). For $c \neq 0$, the matrix $\mathrm{diag}(c, 1, \dots, 1)$ has determinant c by [Theorem 6.4.6](#), which is non-zero, so it lies in $\mathrm{GL}_n(F)$; hence $\det$ is surjective. A matrix with determinant 1 is invertible by [Theorem 6.5.2](#), so the kernel $\{A \in \mathrm{GL}_n(F) : \det A = 1\}$ is $\mathrm{SL}_n(F)$. It is a subgroup by [Theorem 0.10.3](#). ■

For example, $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and the elementary matrices $I_n + cE_{ij}$ lie in $\mathrm{SL}_n(F)$, while P_{12} , of determinant -1 , does not. Over $\mathbb{R}$, the rotation matrices $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ have determinant $\cos^2 \theta + \sin^2 \theta = 1$ and lie in $\mathrm{SL}_2(\mathbb{R})$. The sign $\mathrm{sgn}: S_n \rightarrow \{1, -1\}$ of the third section is the other homomorphism promised in Chapter 0, and the two are linked: $\det P_\sigma = \mathrm{sgn}(\sigma)$ ([Exercise \(B2: The determinant of \$P_\sigma\$ \)](#)), and $P_\sigma P_\tau = P_{\sigma\tau}$ ([Lemma 2.6.6](#)).

6.5.6 Which proofs survive over a ring

Remark. Commutative rings. Over a commutative ring R (as in the remark at the end of the previous section), the following proofs of this section use no division and remain valid for matrices in $M_n(R)$: [Theorem 6.5.1](#), with $c \in R$ arbitrary in (b) and (c); [Theorem 6.5.4](#), by its first proof; and [Theorem 6.5.5](#). The proofs of [Corollary 6.5.6](#) and [Corollary 6.5.7](#) also survive for matrices that have an inverse in $M_n(R)$ (then $\det A \det(A^{-1}) = 1$, so $\det A$ has an inverse in R), for instance an invertible $P \in M_n(F)$ regarded inside $M_n(F[x])$; this is the case needed for characteristic polynomials. What fails over a ring is the converse direction: a matrix whose determinant is non-zero need not be invertible, since $\det$ may be a non-zero element with no inverse, such as $2 \in \mathbb{Z}$ or $x \in F[x]$. By contrast, [Theorem 6.5.2](#), [Theorem 6.5.3](#) and the “Alternatively” proof of multiplicativity run through row reduction, which divides by pivots, and they are proved here only over a field.

6.5.7 Exercises

6.5.7.1 A. Check your understanding

Exercise A1.

- State how each of the three elementary row operations changes the determinant.
- True or false: the move $R_2 \rightarrow 3R_2 - R_1$ does not change the determinant. Justify your answer.
- True or false: if $A, B \in M_n(F)$ have $\det A = \det B$, then A and B are similar. Justify your answer.
- True or false: $\det(AB) = \det(BA)$ for all $A, B \in M_n(F)$. Justify your answer.
- Explain why $\det T$ does not depend on the basis used to compute it.
- Is $\det: M_n(F) \rightarrow F$ a group homomorphism? Explain.

Solution.

- By [Theorem 6.5.1](#): a swap multiplies it by -1 , scaling a row by c multiplies it by c , and adding a multiple of one row to a different row leaves it unchanged.
- False. The move is $R_2 \rightarrow 3R_2$ followed by $R_2 \rightarrow R_2 - R_1$, so it multiplies the determinant by 3. For I_2 it gives $\begin{pmatrix} 1 & 0 \\ -1 & 3 \end{pmatrix}$, of determinant $3 \neq 1$.
- False. I_2 and $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ both have determinant 1, but $P^{-1}I_2P = I_2$ for every invertible P , so I_2 is similar only to itself.
- True. Both equal $\det A \det B$ by [Theorem 6.5.4](#), since multiplication in F is commutative.
- Matrices of T in two bases are similar ([Theorem 3.7.3](#)), and similar matrices have equal determinants ([Corollary 6.5.7](#)).
- No: $M_n(F)$ is not a group under multiplication, since the zero matrix has no inverse. The restriction $\det: \mathrm{GL}_n(F) \rightarrow F \setminus \{0\}$ is a homomorphism ([Theorem 6.5.9](#)).

6.5.7.2 B. Practice

Exercise B1: Determinants by elimination.

Compute the determinants of the following real matrices by row reduction, recording the effect of each operation.

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 3 & 1 & 4 \\ 0 & 1 & 3 & 2 \\ -1 & 1 & 2 & 5 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & -1 & 3 & 1 \\ 4 & -2 & 7 & 3 \\ -2 & 2 & -1 & 4 \\ 6 & -3 & 9 & 5 \end{pmatrix}.$$

Solution. For A : the replacements $R_2 \rightarrow R_2 - 2R_1$ and $R_4 \rightarrow R_4 + R_1$ give rows $(1, 2, 0, 1)$, $(0, -1, 1, 2)$, $(0, 1, 3, 2)$, $(0, 3, 2, 6)$. Then $R_3 \rightarrow R_3 + R_2$ and $R_4 \rightarrow R_4 + 3R_2$ give $(0, 0, 4, 4)$ and $(0, 0, 5, 12)$. Pulling the factor 4 out of row 3 leaves $(0, 0, 1, 1)$, and $R_4 \rightarrow R_4 - 5R_3$ gives $(0, 0, 0, 7)$. None of the replacements changes the determinant

(Theorem 6.5.1), so

$$\det A = 4 \det \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 7 \end{pmatrix} = 4 \cdot (1 \cdot (-1) \cdot 1 \cdot 7) = -28,$$

by Theorem 6.4.6.

For B : $R_2 \rightarrow R_2 - 2R_1$, $R_3 \rightarrow R_3 + R_1$, $R_4 \rightarrow R_4 - 3R_1$ give rows $(2, -1, 3, 1)$, $(0, 0, 1, 1)$, $(0, 1, 2, 5)$, $(0, 0, 0, 2)$, without changing the determinant. The $(2, 2)$ -entry is 0, so swap $R_2 \leftrightarrow R_3$, which multiplies the determinant by -1 . The result is upper triangular with diagonal $2, 1, 1, 2$, so

$$\det B = -(2 \cdot 1 \cdot 1 \cdot 2) = -4.$$

Exercise B2: A parameter.

Let $t \in \mathbb{R}$ and $A_t = \begin{pmatrix} t & 1 & 1 \\ 1 & t & 1 \\ 1 & 1 & t \end{pmatrix}$. Find $\det A_t$ as a factored polynomial in t . Hence determine for which t the matrix A_t is invertible.

Solution. The replacements $R_1 \rightarrow R_1 + R_2$ and $R_1 \rightarrow R_1 + R_3$ do not change the determinant and make row 1 equal to $(t+2, t+2, t+2)$. By linearity in row 1 (this holds for every value of $t+2$, including 0),

$$\det A_t = (t+2) \det \begin{pmatrix} 1 & 1 & 1 \\ 1 & t & 1 \\ 1 & 1 & t \end{pmatrix} = (t+2) \det \begin{pmatrix} 1 & 1 & 1 \\ 0 & t-1 & 0 \\ 0 & 0 & t-1 \end{pmatrix} = (t+2)(t-1)^2,$$

where the second step uses $R_2 \rightarrow R_2 - R_1$, $R_3 \rightarrow R_3 - R_1$, and the last uses Theorem 6.4.6. By Theorem 6.5.2, A_t is invertible if and only if $t \neq 1$ and $t \neq -2$.

Exercise B3: The shift operator.

Let $T: \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 3}$, $T(p) = p(x+1)$. Find $\det T$. Hence show that T is invertible.

Solution. In the basis $\mathcal{B} = (1, x, x^2, x^3)$, $T(1) = 1$, $T(x) = 1 + x$, $T(x^2) = 1 + 2x + x^2$ and $T(x^3) = 1 + 3x + 3x^2 + x^3$. So

$$[T]_{\mathcal{B}} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

which is upper triangular with ones on the diagonal. By Definition 6.5.1 and Theorem 6.4.6, $\det T = 1$. Since $\det T \neq 0$, T is invertible by Theorem 6.5.8. (Its inverse is $p \mapsto p(x-1)$.)

6.5.7.3 C. Going deeper

Exercise C1: Orthogonal-type matrices.

- (a) Let $A \in M_n(\mathbb{R})$ satisfy $A^\top A = I_n$. Prove that $\det A = 1$ or $\det A = -1$.
- (b) Show that both values occur, for every $n \geq 1$.
- (c) Over $\mathbb{F}_2$, what can $\det A$ be if $A^\top A = I_n$?

Solution.

- (a) By [Theorem 6.5.5](#), $1 = \det I_n = \det(A^\top A) = \det A^\top \det A = (\det A)^2$. A real number with square 1 is 1 or -1 .
- (b) $I_n^\top I_n = I_n$ and $\det I_n = 1$. For $D = \text{diag}(-1, 1, \dots, 1)$, $D^\top D = D^2 = I_n$ and $\det D = -1$ by [Theorem 6.4.6](#).
- (c) The same computation gives $(\det A)^2 = 1$. In $\mathbb{F}_2$, $\det A \in \{0, 1\}$ and $0^2 = 0$, so $\det A = 1$. This agrees with (a), since $-1 = 1$ in $\mathbb{F}_2$.

Exercise C2: Left multiplication on matrices.

Let $A \in M_n(F)$, and let $L_A: M_n(F) \rightarrow M_n(F)$, $L_A(X) = AX$. This is a linear operator on a space of dimension n^2 . Prove that

$$\det L_A = (\det A)^n.$$

- (a) Order the matrix units column by column, $\mathcal{B} = (E_{11}, E_{21}, \dots, E_{n1}, E_{12}, \dots, E_{n2}, \dots, E_{1n}, \dots, E_{nn})$. Show that $[L_A]_{\mathcal{B}}$ consists of n copies of A placed along the diagonal, with zeros elsewhere.
- (b) Show that $L_{AB} = L_A L_B$, and prove the formula when A is an elementary matrix.
- (c) Prove the formula for invertible A , and then for non-invertible A .

Hint: for a swap matrix, identify $[L_{P_{ij}}]_{\mathcal{B}}$ as a permutation matrix and use [Exercise \(B2: The determinant of \$P_\sigma\$ \)](#).

Solution.

- (a) By [Theorem 0.9.2](#), the l -th column of AE_{kl} is Ae_k , the k -th column of A , and the other columns are zero. So $AE_{kl} = \sum_i a_{ik} E_{il}$. In $\mathcal{B}$, the coordinates of this vector are zero outside the l -th block of n positions, and inside that block they are the k -th column of A . Hence the columns of $[L_A]_{\mathcal{B}}$ belonging to the l -th block are those of A , placed in the rows of the l -th block: $[L_A]_{\mathcal{B}}$ has n copies of A along the diagonal.
- (b) $L_{AB}(X) = ABX = L_A(L_B(X))$ by associativity. Now let E be elementary. If $E = I_n + cE_{ij}$ with $i \neq j$, then by (a) $[L_E]_{\mathcal{B}}$ is triangular (upper if $i < j$, lower if $i > j$) with all diagonal entries 1, so $\det L_E = 1 = (\det E)^n$ by [Theorem 6.4.6](#) and [Theorem 6.5.1](#) (e). If $E = D_i(c)$, then $[L_E]_{\mathcal{B}}$ is diagonal with c appearing n times and 1 elsewhere, so $\det L_E = c^n = (\det E)^n$. If $E = P_{ij}$, then $[L_E]_{\mathcal{B}}$ has exactly one 1 in each row and column, so it is a permutation matrix P_σ ([Definition 2.6.2](#)), and σ exchanges the positions of E_{il} and E_{jl} for each l : σ is a product of n disjoint transpositions. By [Exercise \(B2: The determinant of \$P_\sigma\$ \)](#), [Theorem 6.3.1](#) and [Corollary 6.3.2](#), $\det L_E = \text{sgn}(\sigma) = (-1)^n = (\det E)^n$.

- (c) If A is invertible, then $A = E_1 \cdots E_k$ with elementary E_m (Theorem 2.4.4 (d)). By (b) and Theorem 6.5.8, $\det L_A = \det L_{E_1} \cdots \det L_{E_k} = (\det E_1)^n \cdots (\det E_k)^n = (\det E_1 \cdots \det E_k)^n = (\det A)^n$, where the last step is Theorem 6.5.4. If A is not invertible, there is $\mathbf{v} \neq \mathbf{0}$ with $A\mathbf{v} = \mathbf{0}$ (Theorem 2.4.4 (b)). The matrix $X = (\mathbf{v} \ \mathbf{0} \ \cdots \ \mathbf{0})$ is non-zero and $AX = \mathbf{0}$, so L_A is not injective, hence not invertible, and $\det L_A = 0$ by Theorem 6.5.8. Also $\det A = 0$ by Theorem 6.5.2, so $(\det A)^n = 0$ as $n \geq 1$.

Exercise C3: Non-square factors.

- (a) Give $A \in M_{2 \times 1}(\mathbb{R})$ and $B \in M_{1 \times 2}(\mathbb{R})$ with $\det(BA) \neq \det(AB)$.
 (b) Let $m > n$, $A \in M_{m \times n}(F)$ and $B \in M_{n \times m}(F)$. Prove that $\det(AB) = 0$.
 (c) Explain why (a) does not contradict Theorem 6.5.4.

Solution.

- (a) Let $A = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \end{pmatrix}$. Then $BA = (1)$, of determinant 1, and $AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, of determinant 0.
 (b) By Theorem 2.5.9, $\text{rank}(AB) \leq \text{rank } A \leq n$, since $\text{col}(A)$ is spanned by n columns. So $\text{rank}(AB) < m$, and the $m \times m$ matrix AB is not invertible by Theorem 6.5.3 (d). Hence $\det(AB) = 0$ by Theorem 6.5.2.
 (c) Theorem 6.5.4 concerns two **square** matrices of the same size, and then $\det(AB) = \det(BA)$. Here A and B are not square, $\det A$ and $\det B$ are not defined, and AB , BA have different sizes.

6.6 Cofactor Expansion, the Adjugate and Cramer's Rule

We can now compute determinants by elimination and we know what they decide. This section adds a third view: a determinant of size n is a combination of determinants of size $n - 1$. This recursive formula, the Laplace expansion, is rarely the fastest way to compute, but it has two important consequences. It gives an explicit formula for the inverse of a matrix, through the adjugate, and an explicit formula for the solution of a square linear system, Cramer's rule. Both rest on one identity built from the entries alone, with no division, and that identity holds for matrices of polynomials too, which is what later chapters need.

6.6.1 Minors and cofactors

Group the six terms of a 3×3 determinant (from the examples after Definition 6.4.1) by the entry they take from column 1:

$$\det A = a_{11}(a_{22}a_{33} - a_{32}a_{23}) - a_{21}(a_{12}a_{33} - a_{32}a_{13}) + a_{31}(a_{12}a_{23} - a_{22}a_{13}).$$

Each bracket is a 2×2 determinant. The bracket beside a_{11} is the determinant of what remains of A after deleting row 1 and column 1; the bracket beside a_{21} is what remains after deleting row 2 and column 1; and similarly for a_{31} . The signs alternate. These smaller determinants recur constantly, so they get names.

A minor is the determinant of what is left after crossing out one row and one column; a cofactor is that minor with a checkerboard sign.

Definition 6.6.1 — Minor and cofactor

Let $n \geq 2$, $A \in M_n(F)$ and $i, j \in \{1, \dots, n\}$. The (i, j) -**minor** of A is

$M_{ij} := \det(\text{the } (n-1) \times (n-1) \text{ matrix obtained from } A \text{ by deleting row } i \text{ and column } j).$

The (i, j) -**cofactor** of A is $C_{ij} := (-1)^{i+j} M_{ij}$.

The rows and columns that remain keep their original order. The sign $(-1)^{i+j}$ depends only on the position, and follows a checkerboard starting with $+$ in the top-left corner:

$$\begin{pmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

When the matrix matters we write $M_{ij}(A)$ and $C_{ij}(A)$.

Examples. For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the minors are 1×1 determinants: $M_{11} = d$, $M_{12} = c$, $M_{21} = b$, $M_{22} = a$, and the cofactors are $C_{11} = d$, $C_{12} = -c$, $C_{21} = -b$, $C_{22} = a$. For the matrix $A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & 4 & 1 \\ 5 & 2 & 1 \end{pmatrix}$ of the previous sections, the cofactors along column 1 are

$$C_{11} = +\det \begin{pmatrix} 4 & 1 \\ 2 & 1 \end{pmatrix} = 2, \quad C_{21} = -\det \begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix} = 5, \quad C_{31} = +\det \begin{pmatrix} 1 & 3 \\ 4 & 1 \end{pmatrix} = -11,$$

and $a_{11}C_{11} + a_{21}C_{21} + a_{31}C_{31} = 4 + 0 - 55 = -51$, which is $\det A$ as computed before. For the identity I_n , $C_{ii} = \det I_{n-1} = 1$, and $C_{ij} = 0$ for $i \neq j$, because deleting row i and a different column j leaves a matrix with a zero column.

Non-example by minimal change. A minor and a cofactor differ only by a sign, and forgetting the sign is the commonest slip. For the 2×2 matrix above, $M_{12} = c$ but $C_{12} = -c$; using minors throughout would turn $a_{11}C_{11} + a_{21}C_{21} = ad - cb$ into $ad + cb$, which is the permanent, not the determinant.

6.6.2 The Laplace expansion

The 3×3 grouping above is a special case of a general identity, valid along **every** row and **every** column.

Theorem 6.6.1 — Laplace Expansion

Let $n \geq 2$ and $A \in M_n(F)$, with cofactors C_{ij} .

(a) **(Along column j)** For each $j \in \{1, \dots, n\}$,

$$\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj} = \sum_{i=1}^n a_{ij}C_{ij}.$$

(b) **(Along row i)** For each $i \in \{1, \dots, n\}$,

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in} = \sum_{j=1}^n a_{ij}C_{ij}.$$

Idea. The plan for (a):

- ① Write column j as $\sum_i a_{ij}\mathbf{e}_i$ and use linearity in column j . This leaves n determinants, the i -th with column j replaced by $\mathbf{e}_i$.
- ② In such a determinant, move the lone 1 to the top-left corner. To keep the other rows and columns in their original order, use **adjacent** swaps: $j - 1$ column swaps and $i - 1$ row swaps, costing $(-1)^{(i-1)+(j-1)} = (-1)^{i+j}$.
- ③ A matrix whose first column is $\mathbf{e}_1$ has the same determinant as its lower-right $(n - 1) \times (n - 1)$ block: only permutations fixing 1 contribute to the Leibniz formula. After ②, that block is exactly the matrix defining M_{ij} .

Part (b) is part (a) for the transpose.

Proof.

- (a) Fix j . For $i \in \{1, \dots, n\}$, let $B^{(i)}$ be the matrix obtained from A by replacing column j with $\mathbf{e}_i$.

Claim. $\det B^{(i)} = C_{ij}$.

Proof. Write $B = B^{(i)}$, with columns $\mathbf{b}_1, \dots, \mathbf{b}_n$, so $\mathbf{b}_j = \mathbf{e}_i$ and $\mathbf{b}_k$ is the k -th column of A for $k \neq j$.

Moving the column. Swap columns $j-1$ and j , then columns $j-2$ and $j-1$, and so on, ending with columns 1 and 2. After these $j-1$ adjacent swaps the columns are $\mathbf{b}_j, \mathbf{b}_1, \dots, \mathbf{b}_{j-1}, \mathbf{b}_{j+1}, \dots, \mathbf{b}_n$: the column $\mathbf{b}_j$ is first and the others are in their original order. Call this matrix B_1 . By [Theorem 6.5.1](#) (d), each swap multiplies the determinant by -1 , so $\det B_1 = (-1)^{j-1} \det B$.

Moving the row. In the same way, move row i of B_1 to the top by $i-1$ adjacent row swaps, keeping the other rows in their original order. Call the result B_2 . By [Theorem 6.5.1](#) (a), $\det B_2 = (-1)^{i-1} \det B_1$.

The shape of B_2 . The first column of B_1 is $\mathbf{e}_i$, and moving row i to the top turns it into $\mathbf{e}_1$. For $p, q \geq 2$, the (p, q) -entry of B_2 comes from the rows of A other than i and the columns of A other than j , both in their original order. So the lower-right $(n-1) \times (n-1)$ block of B_2 is the matrix A' obtained from A by deleting row i and column j . Writing b'_{pq} for the entries of B_2 , we have $b'_{11} = 1$, $b'_{p1} = 0$ for $p \geq 2$, and $b'_{pq} = a'_{p-1, q-1}$ for $p, q \geq 2$.

The block determinant. In the Leibniz formula for $\det B_2$, the factor $b'_{\sigma(1)1}$ is 0 unless $\sigma(1) = 1$. Let $G = \{\sigma \in S_n : \sigma(1) = 1\}$. For $\sigma \in G$, define $\rho \in S_{n-1}$ by $\rho(k) = \sigma(k+1) - 1$; this is a bijection of $\{1, \dots, n-1\}$, because σ maps $\{2, \dots, n\}$ bijectively onto itself. The map $\sigma \mapsto \rho$ is a bijection $G \rightarrow S_{n-1}$, with inverse $\rho \mapsto \sigma$, $\sigma(1) = 1$, $\sigma(k+1) = \rho(k) + 1$. An inversion of σ ([Definition 6.3.1](#)) is a pair $p < q$ with $\sigma(p) > \sigma(q)$; it never has $p = 1$, since $\sigma(1) = 1$ is smallest, and for $p, q \geq 2$ it is exactly an inversion $(p-1, q-1)$ of ρ . So $\text{sgn}(\sigma) = \text{sgn}(\rho)$ by [Definition 6.3.2](#). Therefore

$$\det B_2 = \sum_{\sigma \in G} \text{sgn}(\sigma) \cdot 1 \cdot \prod_{q=2}^n b'_{\sigma(q)q} = \sum_{\rho \in S_{n-1}} \text{sgn}(\rho) \prod_{k=1}^{n-1} a'_{\rho(k)k} = \det A' = M_{ij}.$$

Combining, $\det B = (-1)^{j-1} (-1)^{i-1} \det B_2 = (-1)^{i+j} M_{ij} = C_{ij}$. ■

Now the j -th column of A is $\sum_{i=1}^n a_{ij} \mathbf{e}_i$, and the other columns of A and of each $B^{(i)}$ agree. By linearity of $\det$ in column j ([Theorem 6.4.3](#)) and the claim,

$$\det A = \sum_{i=1}^n a_{ij} \det B^{(i)} = \sum_{i=1}^n a_{ij} C_{ij}.$$

- (b) Fix i . The (k, i) -entry of $A^\top$ is a_{ik} . Deleting row k and column i from $A^\top$ gives the transpose of the matrix obtained from A by deleting row i and column k , so $M_{ki}(A^\top) = M_{ik}(A)$ by [Theorem 6.4.5](#), and hence $C_{ki}(A^\top) = C_{ik}(A)$. Applying (a) to $A^\top$ along column i , and [Theorem 6.4.5](#) again,

$$\det A = \det A^\top = \sum_{k=1}^n a_{ik} C_{ki}(A^\top) = \sum_{k=1}^n a_{ik} C_{ik}(A).$$

This proves the theorem. ■

Quick check. In step ② of the Idea, why not move column j to the front with the single swap of columns 1 and j ?

Solution. A single swap costs only one factor -1 , but it puts the old first column in position j , so the lower-right block is no longer the matrix defining M_{ij} : its columns are out of order. For $j \geq 3$, putting that column back in place within the block takes $j - 2$ further adjacent swaps, and the total sign is again $(-1)^{1+(j-2)} = (-1)^{j-1}$. Adjacent swaps keep the count honest from the start.

Remark. Over a commutative ring. The proof uses multilinearity and the swap rule for columns and rows, the Leibniz formula, and [Theorem 6.4.5](#). All of these are division-free (see the remarks at the ends of the previous two sections). So the Laplace expansion holds for matrices with entries in any commutative ring R , for instance $F[x]$.

The expansion is most useful along a row or column with many zeros, since each zero entry kills its term without computing the minor.

Example. (A 4×4 Expansion)

Compute $\det A$ for

$$A = \begin{pmatrix} 3 & 1 & 0 & 2 \\ 1 & 0 & 2 & 0 \\ 2 & 4 & 1 & 3 \\ 0 & 1 & 5 & 2 \end{pmatrix} \in M_4(\mathbb{R})$$

by expanding along a suitable row.

Solution. Row 2 has two zeros. Its signs in the checkerboard are $-$, $+$, $-$, $+$. By [Theorem 6.6.1](#) (b) along row 2,

$$\det A = 1 \cdot C_{21} + 0 \cdot C_{22} + 2 \cdot C_{23} + 0 \cdot C_{24} = -M_{21} - 2M_{23}.$$

Deleting row 2 and column 1, and then row 2 and column 3,

$$M_{21} = \det \begin{pmatrix} 1 & 0 & 2 \\ 4 & 1 & 3 \\ 1 & 5 & 2 \end{pmatrix}, \quad M_{23} = \det \begin{pmatrix} 3 & 1 & 2 \\ 2 & 4 & 3 \\ 0 & 1 & 2 \end{pmatrix}.$$

Expand M_{21} along its first row, which has a zero:

$$M_{21} = 1 \cdot \det \begin{pmatrix} 1 & 3 \\ 5 & 2 \end{pmatrix} - 0 + 2 \cdot \det \begin{pmatrix} 4 & 1 \\ 1 & 5 \end{pmatrix} = (2 - 15) + 2(20 - 1) = 25.$$

Expand M_{23} along its third row $(0, 1, 2)$, with signs $+, -, +$:

$$M_{23} = -1 \cdot \det \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix} + 2 \cdot \det \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix} = -(9 - 4) + 2(12 - 2) = 15.$$

Hence $\det A = -25 - 2 \cdot 15 = -55$.

Warning. **The sign belongs to the position, not to the place in the row.** Along row 2 the signs start with $-$. Using the first-row pattern $+, -, +, -$ in the example above would give $1 \cdot 25 + 2 \cdot 15 = 55$, the wrong sign. Always read the sign as $(-1)^{i+j}$.

Expansion alone is slow: expanding fully down to 1×1 determinants reproduces all $n!$ Leibniz terms. In practice one first creates zeros with replacements, which do not change the determinant, and then expands along the line of zeros.

6.6.3 The adjugate

In Chapter 0 we inverted a 2×2 matrix by multiplying it by $\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Theorem 0.9.8). Compare with the cofactors listed after Definition 6.6.1: $C_{11} = d$, $C_{12} = -c$, $C_{21} = -b$, $C_{22} = a$. The matrix of Chapter 0 is the matrix of cofactors, **transposed**. That recurring matrix deserves a name in every size.

The adjugate is the transposed matrix of cofactors.

Definition 6.6.2 — Adjugate

Let $n \geq 2$ and $A \in M_n(F)$. The **adjugate** of A is the matrix $\operatorname{adj} A \in M_n(F)$ with entries

$$(\operatorname{adj} A)_{ij} := C_{ji},$$

that is, the **transpose** of the matrix (C_{ij}) of cofactors. For $n = 1$ we set $\operatorname{adj} A := (1)$.

In words: the entry in row i , column j of $\operatorname{adj} A$ is the cofactor from row j , column i of A . The adjugate is defined for **every** square matrix, invertible or not.

Examples. For $n = 2$, $\operatorname{adj} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. For I_n , the cofactors computed above give $\operatorname{adj} I_n = I_n$. For

$$B = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 3 \end{pmatrix},$$

the nine cofactors, row by row, are $C_{11} = 3$, $C_{12} = 1$, $C_{13} = -1$; $C_{21} = -6$, $C_{22} = 3$, $C_{23} = 2$; $C_{31} = 2$, $C_{32} = -1$, $C_{33} = 1$. Transposing,

$$\operatorname{adj} B = \begin{pmatrix} 3 & -6 & 2 \\ 1 & 3 & -1 \\ -1 & 2 & 1 \end{pmatrix}, \quad B \operatorname{adj} B = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix},$$

and $\det B = 1 \cdot 3 + 2 \cdot 1 + 0 \cdot (-1) = 5$, by expansion along row 1.

Non-example by minimal change. Skip the transpose. The cofactor matrix $\begin{pmatrix} 3 & 1 & -1 \\ -6 & 3 & 2 \\ 2 & -1 & 1 \end{pmatrix}$ has the same entries in other places, and B times it is $\begin{pmatrix} -9 & 7 & 3 \\ -4 & 2 & 3 \\ 9 & -2 & 2 \end{pmatrix}$, which is not a multiple of I_3 . The transpose is what lines up row i of B with the cofactors **of row** i .

The product $B \operatorname{adj} B = (\det B)I_3$ is no accident. Its diagonal entries are row expansions of $\det B$. Its off-diagonal entries are expansions too, of a matrix that looks like B but has a repeated row.

Theorem 6.6.2 — The Adjugate Identity

For every $A \in M_n(F)$,

$$A \operatorname{adj} A = \operatorname{adj} A A = (\det A) I_n.$$

Idea. The (i, k) -entry of $A \operatorname{adj} A$ is $\sum_j a_{ij} C_{kj}$: the entries of row i against the cofactors of row k . If $i = k$, this is the expansion of $\det A$ along row i . If $i \neq k$, build a look-alike matrix $\tilde{A}$: copy A , but overwrite row k with row i . The cofactors along row k never see row k , because computing them deletes it, so $\tilde{A}$ has the same cofactors along row k as A . Expanding $\det \tilde{A}$ along row k gives exactly $\sum_j a_{ij} C_{kj}$, and $\det \tilde{A} = 0$ because two rows are equal.

Proof. For $n = 1$, $A = (a)$, $\operatorname{adj} A = (1)$, and both products are $(a) = (\det A)I_1$. Let $n \geq 2$.

The product $A \operatorname{adj} A$. By Theorem 0.9.2 and Definition 6.6.2, its (i, k) -entry is

$$\sum_{j=1}^n a_{ij} (\operatorname{adj} A)_{jk} = \sum_{j=1}^n a_{ij} C_{kj}.$$

If $i = k$, this is $\det A$ by Theorem 6.6.1 (b) along row i . Suppose $i \neq k$, and let $\tilde{A}$ be the matrix obtained from A by replacing row k with a copy of row i . Rows i and k of $\tilde{A}$ are equal, so $\det \tilde{A} = 0$ by Theorem 6.4.5. Expanding along row k by Theorem 6.6.1 (b),

$$0 = \det \tilde{A} = \sum_{j=1}^n \tilde{a}_{kj} C_{kj}(\tilde{A}).$$

Here $\tilde{a}_{kj} = a_{ij}$. The minor $M_{kj}(\tilde{A})$ is computed from $\tilde{A}$ with row k deleted, and outside row k the matrices $\tilde{A}$ and A agree; so $C_{kj}(\tilde{A}) = C_{kj}(A)$. Hence $\sum_j a_{ij} C_{kj} = 0$. Therefore $A \operatorname{adj} A = (\det A)I_n$.

The product $\operatorname{adj} A A$. Its (i, k) -entry is $\sum_j C_{ji} a_{jk}$. If $i = k$, this is $\det A$ by Theorem 6.6.1 (a) along column i . If $i \neq k$, let $\hat{A}$ be A with column i replaced by a copy of column k . It has two equal columns, so $\det \hat{A} = 0$ by Theorem 6.4.3. Expanding along column i , $0 = \sum_j \hat{a}_{ji} C_{ji}(\hat{A}) = \sum_j a_{jk} C_{ji}(A)$, since deleting column i removes the only column where $\hat{A}$ and A differ. Therefore $\operatorname{adj} A A = (\det A)I_n$. This proves the theorem. ■

Remark. Over a commutative ring. This proof uses only [Theorem 6.6.1](#) and the alternating property, so the adjugate identity holds for $A \in M_n(R)$ over any commutative ring R . For $R = F[x]$, the matrix $\text{adj}(xI - A)$ has polynomial entries and $(xI - A) \text{adj}(xI - A) = \det(xI - A)I$. This identity reappears in Chapter 8. Over R the identity also shows that A has an inverse in $M_n(R)$ as soon as $\det A$ has an inverse in R , namely $(\det A)^{-1} \text{adj } A$.

Dividing by the determinant, when we may, gives the inverse.

Corollary 6.6.3 — Inverse via the Adjugate

Let $A \in M_n(F)$ with $\det A \neq 0$. Then A is invertible and

$$A^{-1} = \frac{1}{\det A} \text{adj } A.$$

Proof. By [Theorem 6.6.2](#), $A((\det A)^{-1} \text{adj } A) = (\det A)^{-1}(\det A)I_n = I_n$, using that scalars commute with matrices ([Theorem 0.9.4](#)). By [Theorem 2.4.5](#), A is invertible with inverse $(\det A)^{-1} \text{adj } A$. ■

For $n = 2$ this is [Theorem 0.9.8](#) again. For the matrix B above, $B^{-1} = \frac{1}{5} \begin{pmatrix} 3 & -6 & 2 \\ 1 & 3 & -1 \\ -1 & 2 & 1 \end{pmatrix}$.

Warning. The adjugate exists for singular matrices too. Nothing in [Definition 6.6.2](#) needs $\det A \neq 0$. For $S = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 0 & 1 \end{pmatrix}$, with $\det S = 0$, one computes

$\text{adj } S = \begin{pmatrix} 4 & -2 & 0 \\ 4 & -2 & 0 \\ -4 & 2 & 0 \end{pmatrix} \neq 0$, and $S \text{adj } S = 0$, as [Theorem 6.6.2](#) predicts. So “adj A exists” says nothing about invertibility, and the formula $A^{-1} = \text{adj } A / \det A$ must never be used before checking $\det A \neq 0$.

Quick check. What are $\text{adj}(cI_n)$ for $c \in F$ and $n \geq 2$, and $\text{adj } 0$ for the zero matrix in $M_n(F)$, $n \geq 2$? What is $\text{adj } 0$ for $n = 1$?

Solution. Deleting row i and column i from cI_n leaves cI_{n-1} , of determinant c^{n-1} ; deleting row i and a different column leaves a zero column, of determinant 0. So $\text{adj}(cI_n) = c^{n-1}I_n$; in particular $\text{adj } 0 = 0$ for $n \geq 2$, since $0^{n-1} = 0$. For $n = 1$, $\text{adj}(0) = (1)$ by definition, and indeed $(0)(1) = (0) = (\det 0)I_1$.

6.6.4 Cramer's rule

For an invertible A , the system $A\mathbf{x} = \mathbf{b}$ has exactly one solution. The adjugate expresses it in closed form, and the closed form has a direct proof by linearity that shows where it comes from.

For $k \in \{1, \dots, n\}$, write $A_k(\mathbf{b})$ for the matrix obtained from A by replacing column k with $\mathbf{b}$.

Theorem 6.6.4 — Cramer's Rule

Let $A \in M_n(F)$ be invertible and $\mathbf{b} \in F^n$. The unique solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{b}$ is given by

$$x_k = \frac{\det A_k(\mathbf{b})}{\det A}, \quad k = 1, \dots, n.$$

Idea. If $A\mathbf{x} = \mathbf{b}$, then $\mathbf{b} = x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n$. Put this combination into column k and expand by linearity. Every term except the k -th has the column $\mathbf{a}_j$ twice, so it vanishes, and what remains is $x_k \det A$.

Proof. By Theorem 0.9.7 (7), $A\mathbf{x} = \mathbf{b}$ has exactly one solution $\mathbf{x}$. By Theorem 0.9.3, $\mathbf{b} = \sum_{j=1}^n x_j \mathbf{a}_j$, where $\mathbf{a}_j$ is the j -th column of A . Since $\det$ is linear in column k (Theorem 6.4.3),

$$\det A_k(\mathbf{b}) = \sum_{j=1}^n x_j \det A_k(\mathbf{a}_j).$$

For $j \neq k$, the matrix $A_k(\mathbf{a}_j)$ has the column $\mathbf{a}_j$ in positions j and k , so its determinant is 0. For $j = k$, $A_k(\mathbf{a}_k) = A$. Hence $\det A_k(\mathbf{b}) = x_k \det A$. Since A is invertible, $\det A \neq 0$ by Theorem 6.5.2, and dividing gives the formula. ■

Alternatively, $\mathbf{x} = A^{-1}\mathbf{b} = (\det A)^{-1} \operatorname{adj} A \mathbf{b}$ by Corollary 6.6.3, and the k -th entry of $\operatorname{adj} A \mathbf{b}$ is $\sum_j C_{jk} b_j$, which is the expansion of $\det A_k(\mathbf{b})$ along column k (Theorem 6.6.1 (a)); the cofactors along column k of $A_k(\mathbf{b})$ and of A agree, because computing them deletes column k .

For example, the system $2x + y = 5$, $7x + 4y = 18$ has coefficient matrix of determinant $8 - 7 = 1$, so

$$x = \det \begin{pmatrix} 5 & 1 \\ 18 & 4 \end{pmatrix} = 2, \quad y = \det \begin{pmatrix} 2 & 5 \\ 7 & 18 \end{pmatrix} = 1,$$

and indeed $2 \cdot 2 + 1 = 5$ and $7 \cdot 2 + 4 = 18$.

Warning. Cramer's rule is a theoretical tool, not an algorithm. It needs $n + 1$ determinants of size n . Evaluated by the Leibniz formula this is about $(n + 1)!$ products; even with each determinant computed by elimination it costs about $n^4/3$ operations, while Gaussian elimination solves the system once in about $n^3/3$. For $n = 20$, term-by-term Leibniz evaluation at 10^9 multiplications per second would take tens of thousands of years. Use Cramer's rule to understand **how** the solution depends on A and $\mathbf{b}$, and elimination to find it.

6.6.5 Integer matrices

The adjugate formula shows that each entry of A^{-1} is a polynomial in the entries of A divided by the single polynomial $\det A$. In particular, the entries of A^{-1} change continuously with those of A over $\mathbb{R}$, as long as $\det A \neq 0$. Here is an arithmetic consequence. A matrix with integer entries lies in $M_n(\mathbb{Q})$, and we may ask when its inverse again has integer entries.

Corollary 6.6.5 — Integer Matrices with Integer Inverses

Let $A \in M_n(\mathbb{Q})$ have all entries in $\mathbb{Z}$.

- (a) $\det A \in \mathbb{Z}$, and $\operatorname{adj} A$ has all entries in $\mathbb{Z}$.
- (b) A is invertible and A^{-1} has all entries in $\mathbb{Z}$ if and only if $\det A = 1$ or $\det A = -1$.

Proof.

- (a) The Leibniz formula (Definition 6.4.1) is a sum of products of entries with signs ± 1 , so the determinant of a matrix with integer entries is an integer. Each cofactor is $\pm$ such a determinant, so $\operatorname{adj} A$ has integer entries (for $n = 1$, $\operatorname{adj} A = (1)$).
- (b) $(\Leftarrow)$ If $\det A = \pm 1$, then A is invertible and $A^{-1} = \pm \operatorname{adj} A$ by Corollary 6.6.3, which has integer entries by (a). $(\Rightarrow)$ Suppose A^{-1} has integer entries. By Theorem 6.5.4, $\det A \det(A^{-1}) = \det I_n = 1$, and both factors are integers by (a). The only integers whose product is 1 are $1 \cdot 1$ and $(-1)(-1)$, so $\det A = \pm 1$.

■

For example, $\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$ has determinant 1 and inverse $\begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$, while $\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$ has determinant 2 and inverse $\begin{pmatrix} 1/2 & -1/2 \\ 0 & 1 \end{pmatrix}$. In the language of the remark after Theorem 6.6.2, part (b) says that a matrix over the ring $\mathbb{Z}$ is invertible over $\mathbb{Z}$ exactly when its determinant is invertible in $\mathbb{Z}$.

6.6.6 Exercises**6.6.6.1 A. Check your understanding****Exercise A1.**

- (a) Define the (i, j) -minor, the (i, j) -cofactor and the adjugate of $A \in M_n(F)$, $n \geq 2$.
- (b) What are the signs $(-1)^{i+j}$ of the cofactors in positions $(2, 3)$ and $(4, 4)$?
- (c) True or false: $\operatorname{adj} A$ is defined only when A is invertible. Justify your answer.
- (d) True or false: Cramer's rule (Theorem 6.6.4) gives the solutions of every square linear system $A\mathbf{x} = \mathbf{b}$. Justify your answer.
- (e) In the proof of Theorem 6.6.2, which matrix is expanded to show that $\sum_j a_{ij}C_{kj} = 0$ for $i \neq k$, and why is its determinant 0?
- (f) True or false: if A has integer entries and $\det A = 2$, then A^{-1} has integer entries. Justify your answer.

Solution.

- (a) M_{ij} is the determinant of the $(n-1) \times (n-1)$ matrix obtained by deleting row i and column j ; $C_{ij} = (-1)^{i+j}M_{ij}$; $\operatorname{adj} A$ is the matrix with (i, j) -entry C_{ji} (Definition 6.6.1, Definition 6.6.2).

- (b) $(-1)^5 = -1$ and $(-1)^8 = 1$.
- (c) False. It is defined for every square matrix; for instance $\text{adj} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$, although the matrix is singular.
- (d) False. It requires $\det A \neq 0$. For $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $\mathbf{b} = (1, 1)$, the system has infinitely many solutions, and the formula would divide by 0.
- (e) The matrix $\tilde{A}$ obtained from A by replacing row k with row i , expanded along row k . It has two equal rows, i and k , so its determinant is 0.
- (f) False. By [Corollary 6.6.5](#), an integer inverse forces $\det A = \pm 1$. For instance $\text{diag}(2, 1)$ has inverse $\text{diag}(\frac{1}{2}, 1)$.

6.6.6.2 B. Practice

Exercise B1: Adjugate and inverse.

Let $A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \\ 2 & 1 & 1 \end{pmatrix} \in M_3(\mathbb{R})$. Compute all nine cofactors, $\text{adj } A$ and $\det A$. Verify that $A \text{adj } A = (\det A)I_3$, and hence write down A^{-1} .

Solution. The cofactors are

$$\begin{aligned} C_{11} &= +(1 \cdot 1 - 3 \cdot 1) = -2, & C_{12} &= -(0 \cdot 1 - 3 \cdot 2) = 6, & C_{13} &= +(0 \cdot 1 - 1 \cdot 2) = -2, \\ C_{21} &= -(2 \cdot 1 - (-1) \cdot 1) = -3, & C_{22} &= +(1 \cdot 1 - (-1) \cdot 2) = 3, & C_{23} &= -(1 \cdot 1 - 2 \cdot 2) = 3, \\ C_{31} &= +(2 \cdot 3 - (-1) \cdot 1) = 7, & C_{32} &= -(1 \cdot 3 - (-1) \cdot 0) = -3, & C_{33} &= +(1 \cdot 1 - 2 \cdot 0) = 1. \end{aligned}$$

Transposing the cofactor matrix ([Definition 6.6.2](#)),

$$\text{adj } A = \begin{pmatrix} -2 & -3 & 7 \\ 6 & 3 & -3 \\ -2 & 3 & 1 \end{pmatrix}.$$

Expanding along row 1 ([Theorem 6.6.1](#)), $\det A = 1 \cdot (-2) + 2 \cdot 6 + (-1)(-2) = 12$. Row 1 of A against the columns of $\text{adj } A$ gives $-2 + 12 + 2 = 12$, $-3 + 6 - 3 = 0$, $7 - 6 - 1 = 0$; row 2 gives $0 + 6 - 6 = 0$, $0 + 3 + 9 = 12$, $0 - 3 + 3 = 0$; row 3 gives $-4 + 6 - 2 = 0$, $-6 + 3 + 3 = 0$, $14 - 3 + 1 = 12$. So $A \text{adj } A = 12I_3$. By [Corollary 6.6.3](#),

$$A^{-1} = \frac{1}{12} \begin{pmatrix} -2 & -3 & 7 \\ 6 & 3 & -3 \\ -2 & 3 & 1 \end{pmatrix}.$$

Exercise B2: Cramer's rule.

Use Cramer's rule to solve

$$x_1 + 2x_2 - x_3 = -3, \quad x_2 + 3x_3 = 5, \quad 2x_1 + x_2 + x_3 = 3$$

over $\mathbb{R}$. Check your answer by substitution.

Solution. The coefficient matrix is the matrix A of [Exercise \(B1: Adjugate and inverse\)](#), with $\det A = 12 \neq 0$, and $\mathbf{b} = (-3, 5, 3)$. Expanding each $A_k(\mathbf{b})$ along its first row,

$$\det A_1(\mathbf{b}) = \det \begin{pmatrix} -3 & 2 & -1 \\ 5 & 1 & 3 \\ 3 & 1 & 1 \end{pmatrix} = -3(1 - 3) - 2(5 - 9) - 1(5 - 3) = 6 + 8 - 2 = 12,$$

$$\det A_2(\mathbf{b}) = \det \begin{pmatrix} 1 & -3 & -1 \\ 0 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix} = 1(5 - 9) + 3(0 - 6) - 1(0 - 10) = -4 - 18 + 10 = -12,$$

$$\det A_3(\mathbf{b}) = \det \begin{pmatrix} 1 & 2 & -3 \\ 0 & 1 & 5 \\ 2 & 1 & 3 \end{pmatrix} = 1(3 - 5) - 2(0 - 10) - 3(0 - 2) = -2 + 20 + 6 = 24.$$

By [Theorem 6.6.4](#), $x_1 = 12/12 = 1$, $x_2 = -12/12 = -1$, $x_3 = 24/12 = 2$. Check: $1 - 2 - 2 = -3$, $-1 + 6 = 5$, $2 - 1 + 2 = 3$.

Exercise B3: Choosing the best line.

Compute $\det N$ for

$$N = \begin{pmatrix} 2 & 0 & 0 & 3 \\ 1 & 4 & 0 & -1 \\ 0 & 5 & 0 & 2 \\ 3 & 1 & 2 & 0 \end{pmatrix} \in M_4(\mathbb{R}),$$

expanding along the row or column that needs the least work.

Solution. Column 3 has a single non-zero entry, $a_{43} = 2$, whose sign is $(-1)^{4+3} = -1$. By [Theorem 6.6.1](#) (a) along column 3, $\det N = -2M_{43}$, where

$$M_{43} = \det \begin{pmatrix} 2 & 0 & 3 \\ 1 & 4 & -1 \\ 0 & 5 & 2 \end{pmatrix} = 2(4 \cdot 2 - (-1) \cdot 5) - 0 + 3(1 \cdot 5 - 4 \cdot 0) = 26 + 15 = 41,$$

expanding along the first row. Hence $\det N = -82$.

6.6.6.3 C. Going deeper

Exercise C1: The determinant of the adjugate.

Let $n \geq 2$ and $A \in M_n(F)$. Prove that $\det(\operatorname{adj} A) = (\det A)^{n-1}$.

Hint: treat $\det A \neq 0$ and $\det A = 0$ separately; in the second case, ask whether $\operatorname{adj} A$ can be invertible.

Solution. *Case 1:* $\det A \neq 0$. By [Theorem 6.6.2](#) and [Theorem 6.5.4](#), $\det A \det(\operatorname{adj} A) = \det((\det A)I_n) = (\det A)^n$, where the last step is [Theorem 6.4.6](#). Dividing by $\det A \neq 0$ gives $\det(\operatorname{adj} A) = (\det A)^{n-1}$.

Case 2: $\det A = 0$. Then $A \operatorname{adj} A = 0$ by [Theorem 6.6.2](#). Suppose, for a contradiction, that $\operatorname{adj} A$ is invertible. Then $A = (A \operatorname{adj} A)(\operatorname{adj} A)^{-1} = 0$. But every minor of the zero matrix is the determinant of a zero matrix of size $n-1 \geq 1$, which is 0, so $\operatorname{adj} 0 = 0$, which

is not invertible. This contradiction shows that $\text{adj } A$ is not invertible, so $\det(\text{adj } A) = 0 = (\det A)^{n-1}$ by [Theorem 6.5.2](#), since $n - 1 \geq 1$.

Exercise C2: The rank of the adjugate.

Let $n \geq 2$ and $A \in M_n(F)$. Prove that

$$\text{rank}(\text{adj } A) = \begin{cases} n & \text{if } \text{rank } A = n, \\ 1 & \text{if } \text{rank } A = n - 1, \\ 0 & \text{if } \text{rank } A \leq n - 2. \end{cases}$$

- (a) Prove the case $\text{rank } A = n$.
- (b) Suppose $\text{rank } A \leq n - 2$. Prove that every minor M_{ij} is 0.
- (c) Suppose $\text{rank } A = n - 1$. Prove that every column of $\text{adj } A$ lies in $\text{nul}(A)$, and deduce $\text{rank}(\text{adj } A) \leq 1$.
- (d) Suppose $\text{rank } A = n - 1$. Prove that some minor M_{ij} is non-zero, and conclude.

Hint for (d): choose $n - 1$ independent columns of A , then $n - 1$ independent rows of the resulting $n \times (n - 1)$ matrix.

Solution.

- (a) If $\text{rank } A = n$, then A is invertible by [Theorem 6.5.3](#), so $\det A \neq 0$ and $\text{adj } A = (\det A)A^{-1}$ by [Corollary 6.6.3](#). This is invertible, since A^{-1} is and $\det A \neq 0$ ([Theorem 0.9.7](#)), so $\text{rank}(\text{adj } A) = n$.
- (b) Fix i, j , and let A' be A with row i and column j deleted. The $n - 1$ columns of A other than column j lie in $\text{col}(A)$, of dimension $\text{rank } A \leq n - 2$, so they are linearly dependent by [Theorem 1.6.3](#): $\sum_{k \neq j} c_k \mathbf{a}_k = \mathbf{0}$ with the c_k not all zero. Deleting the i -th entry of every vector preserves this relation, so the columns of A' are dependent. By [Theorem 2.4.4](#), A' is not invertible, and $M_{ij} = \det A' = 0$ by [Theorem 6.5.2](#). Hence $\text{adj } A = 0$ and $\text{rank}(\text{adj } A) = 0$.
- (c) Since $\text{rank } A < n$, $\det A = 0$ by [Theorem 6.5.3](#), so $A \text{adj } A = 0$ by [Theorem 6.6.2](#). By [Theorem 0.9.2](#), A kills each column of $\text{adj } A$, so $\text{col}(\text{adj } A) \subseteq \text{nul}(A)$. By [Theorem 2.5.7](#), $\dim \text{nul}(A) = n - (n - 1) = 1$. Hence $\text{rank}(\text{adj } A) \leq 1$.
- (d) By [Theorem 2.5.2](#), the pivot columns of A form a basis of $\text{col}(A)$; there are $n - 1$ of them, so exactly one column index j is not a pivot column. Let $A_0 \in M_{n \times (n-1)}(F)$ consist of the other $n - 1$ columns. They are independent, so $\text{rank } A_0 = n - 1$, and $\dim \text{row}(A_0) = n - 1$ by [Theorem 2.5.5](#). The n rows of A_0 span $\text{row}(A_0)$, so by [Theorem 1.5.4](#) some $n - 1$ of them are independent; since $n - 1$ independent vectors in a space of dimension $n - 1$ form a basis, sifting leaves out exactly one row, say row i . Deleting row i from A_0 gives an $(n - 1) \times (n - 1)$ matrix A' with independent rows, so $\det A' \neq 0$ by [Theorem 6.5.3](#). But A' is A with row i and column j deleted, so $M_{ij} \neq 0$, hence $C_{ij} \neq 0$ and $\text{adj } A \neq 0$. With (c), $\text{rank}(\text{adj } A) = 1$.

Exercise C3: An adjugate over $F[x]$

Work in $M_2(\mathbb{R}[x])$, using the remarks after [Theorem 6.6.1](#) and [Theorem 6.6.2](#).

- (a) For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, compute $\det(xI - A)$ and $\text{adj}(xI - A)$, and verify $(xI - A)\text{adj}(xI - A) = \det(xI - A)I$ by multiplying out.
- (b) For a general $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{R})$, show that $\text{adj}(xI - A) = xI - \text{adj } A$ and $A + \text{adj } A = (\text{tr } A)I$.
- (c) Deduce from (b), without using [Theorem 6.6.2](#) over $\mathbb{R}[x]$, that $(xI - A)\text{adj}(xI - A) = (x^2 - (\text{tr } A)x + \det A)I$, and check that $x^2 - (\text{tr } A)x + \det A = \det(xI - A)$.

Solution.

- (a) $xI - A = \begin{pmatrix} x-1 & -2 \\ -3 & x-4 \end{pmatrix}$, so $\det(xI - A) = (x-1)(x-4) - 6 = x^2 - 5x - 2$, and by [Definition 6.6.2](#), $\text{adj}(xI - A) = \begin{pmatrix} x-4 & 2 \\ 3 & x-1 \end{pmatrix}$. The product has entries $(x-1)(x-4) - 6 = x^2 - 5x - 2$, $2(x-1) - 2(x-1) = 0$, $-3(x-4) + 3(x-4) = 0$ and $-6 + (x-4)(x-1) = x^2 - 5x - 2$, so it is $(x^2 - 5x - 2)I$.
- (b) $xI - A = \begin{pmatrix} x-a & -b \\ -c & x-d \end{pmatrix}$, so $\text{adj}(xI - A) = \begin{pmatrix} x-d & b \\ c & x-a \end{pmatrix} = xI - \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = xI - \text{adj } A$. Also $A + \text{adj } A = \begin{pmatrix} a+d & 0 \\ 0 & a+d \end{pmatrix} = (\text{tr } A)I$.
- (c) Since xI commutes with every matrix, expanding with (b) gives

$$(xI - A)(xI - \text{adj } A) = x^2I - x(A + \text{adj } A) + A \text{adj } A = x^2I - (\text{tr } A)xI + (ad - bc)I,$$

where $A \text{adj } A = (ad - bc)I$ is the computation of [Theorem 0.9.8](#) over $\mathbb{R}$. Finally $\det(xI - A) = (x-a)(x-d) - bc = x^2 - (a+d)x + (ad - bc)$, which is $x^2 - (\text{tr } A)x + \det A$.

6.7 Minors, Rank and Cauchy–Binet

The determinant of a square matrix answers a yes-or-no question: is the matrix invertible? A rectangular matrix has no determinant, and a singular square matrix has determinant 0, which says nothing about *how* singular it is. Yet every matrix is full of square pieces, and each piece has a determinant. This section shows that these small determinants measure the rank exactly. It then answers a second question left open by multiplicativity: if A is $m \times n$ and B is $n \times m$, the product AB is square, so what is $\det(AB)$? The answer, the Cauchy–Binet formula, is proved by the same multilinear expansion that gave uniqueness of the determinant in [Theorem 6.4.2](#).

6.7.1 Submatrices and minors

In [Definition 6.6.1](#) we deleted one row and one column of a square matrix and took the determinant of what was left. That is a special case of a more flexible operation: keep **any** k rows and **any** k columns. The recurring object is the square block sitting at the crossings of the chosen rows and columns, and it deserves a name.

A *minor* is the determinant of a square block cut out of a matrix by choosing some rows and the same number of columns.

For a positive integer m , write $[m] = \{1, 2, \dots, m\}$.

Definition 6.7.1 — Submatrix and Minor

Let $A = (a_{ij}) \in M_{m \times n}(F)$, and let $1 \leq k \leq \min(m, n)$. Let $I = \{i_1 < i_2 < \dots < i_k\} \subseteq [m]$ and $J = \{j_1 < j_2 < \dots < j_k\} \subseteq [n]$ be sets with **exactly** k elements each, listed in **increasing** order. The **submatrix** of A on rows I and columns J is

$$A_{I,J} := (a_{i_p j_q})_{p,q=1}^k \in M_k(F),$$

the $k \times k$ matrix whose (p, q) -entry is $a_{i_p j_q}$. Its determinant $\det A_{I,J}$ is a $k \times k$ **minor** of A .

In words: cross out every row not in I and every column not in J , and push the surviving entries together **without changing their order**. The chosen rows need not be adjacent, and neither need the columns. The sets I and J have the **same** size, so the block is square and has a determinant.

A 1×1 minor is a single entry, $\det A_{\{i\},\{j\}} = a_{ij}$. For a square $A \in M_n(F)$ with $n \geq 2$, the minor M_{ij} of Definition 6.6.1 is the $(n-1) \times (n-1)$ minor with $I = [n] \setminus \{i\}$ and $J = [n] \setminus \{j\}$. The only $n \times n$ minor of a square matrix is $\det A$ itself. An $m \times n$ matrix has $\binom{m}{k} \binom{n}{k}$ minors of size k , one for each choice of I and J .

Example. Let

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{pmatrix} \in M_{2 \times 3}(\mathbb{Q}).$$

Its 2×2 minors all use $I = \{1, 2\}$, and there are three choices of J :

$$\det A_{I,\{1,2\}} = \det \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = 1, \quad \det A_{I,\{1,3\}} = \det \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} = 3, \quad \det A_{I,\{2,3\}} = \det \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix} = 6.$$

Its six 1×1 minors are its entries. It has no 3×3 minors, since it has only two rows.

A degenerate example. Every minor of the zero matrix $0_{m \times n}$ is 0: each block is a zero matrix, whose determinant is 0 by Theorem 6.4.6. This case matters because it is exactly the matrix of rank 0, and the theorem below has to treat it separately.

Non-example by minimal change. Take the same rows and columns but list the columns of A above in the order 3, 1: the block $\begin{pmatrix} 0 & 1 \\ 3 & 0 \end{pmatrix}$ has determinant -3 , not 3. Every entry is still an entry of A from rows $\{1, 2\}$ and columns $\{1, 3\}$; what fails is the clause **listed in increasing order**. Reordering columns changes the determinant by a sign (Theorem 6.2.1), so the order convention is what makes “the” minor on (I, J) a single well-defined number.

Why this definition. Insisting on increasing order costs nothing and removes all sign ambiguity. Insisting on $|I| = |J|$ is forced: only square blocks have determinants.

Warning. **Rank is not read off principal minors.** A **principal** minor uses the same index set for rows and columns ($I = J$). It is tempting to test rank using only these, but they can all vanish for a non-zero matrix. For $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, the principal minors

are 0, 0 and $\det N = 0$, yet $\text{rank } N = 1$. The non-zero minor is the off-diagonal entry $\det N_{\{1\},\{2\}} = 1$. The theorem below uses **all** minors.

Quick check. How many 2×2 minors does a 3×4 matrix have? How many 3×3 minors?

Solution. A 2×2 minor needs 2 of the 3 rows and 2 of the 4 columns: $\binom{3}{2} \binom{4}{2} = 3 \cdot 6 = 18$. A 3×3 minor uses all three rows and 3 of the 4 columns: $\binom{3}{3} \binom{4}{3} = 4$.

6.7.2 Rank via minors

A non-zero determinant of a square block means the block is invertible, so its columns are independent. Independent columns inside a block come from independent columns of the whole matrix. So a large non-zero minor forces large rank. The converse, that large rank produces a large non-zero minor, needs us to find the right rows as well as the right columns. Both directions together give a description of rank using only determinants:

Theorem 6.7.1 — Rank via Minors

Let $A \in M_{m \times n}(F)$ be **non-zero**.

- (a) rank A is the **largest** integer k such that A has a **non-zero** $k \times k$ minor.
- (b) Equivalently, for each $1 \leq k \leq \min(m, n)$: rank $A \geq k$ if and only if some $k \times k$ minor of A is non-zero; and rank $A < k$ if and only if **every** $k \times k$ minor of A is zero.

Idea. Two inequalities. ① A non-zero $k \times k$ minor on rows I and columns J makes $A_{I,J}$ invertible; a dependence among the columns of A indexed by J would restrict to a dependence among the columns of $A_{I,J}$. So rank $A \geq k$. ② If rank $A = r$, choose r independent columns (pivot columns). The $m \times r$ matrix they form has rank r , so by row rank = column rank it has r independent rows. Those rows cut out an $r \times r$ block with independent rows, hence a non-zero determinant.

Proof. Step 1: a non-zero $k \times k$ minor forces rank $A \geq k$. Let $I = \{i_1 < \dots < i_k\}$ and $J = \{j_1 < \dots < j_k\}$ with $\det A_{I,J} \neq 0$. Write $\mathbf{a}_1, \dots, \mathbf{a}_n$ for the columns of A . Suppose $c_1 \mathbf{a}_{j_1} + \dots + c_k \mathbf{a}_{j_k} = \mathbf{0}$ with $c_1, \dots, c_k \in F$. Reading only the entries in rows $i_1, \dots, i_k$ of this vector equation gives $\sum_q c_q a_{i_p j_q} = 0$ for $p = 1, \dots, k$, that is, $A_{I,J} \mathbf{c} = \mathbf{0}$ with $\mathbf{c} = (c_1, \dots, c_k)$. By Theorem 6.5.2, $A_{I,J}$ is invertible, so $\mathbf{c} = \mathbf{0}$ by Theorem 2.4.4. Hence $(\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_k})$ is a linearly independent list in $\text{col}(A)$, and by Theorem 1.6.3 (a), $k \leq \dim \text{col}(A) = \text{rank } A$.

Step 2: A has a non-zero $r \times r$ minor, where $r = \text{rank } A$. Since $A \neq 0$, some column is non-zero, so $r \geq 1$. By Theorem 2.5.2 (a), the pivot columns $\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_r}$ ($j_1 < \dots < j_r$) are linearly independent. Let $J = \{j_1, \dots, j_r\}$ and let $A' = A_{[m],J} \in M_{m \times r}(F)$ be the matrix with these columns. By Theorem 2.5.10 (a), rank $A' = r$, so by Theorem 2.5.5 the row space of A' has dimension r . The m rows of A' span its row space, so by Theorem 1.5.4 some r of them, in rows $i_1 < \dots < i_r$, form a basis of it; in particular they are linearly independent. These are exactly the rows of $B = A_{I,J} \in M_r(F)$, where $I = \{i_1, \dots, i_r\}$. Transposition is a linear bijection from $1 \times r$ rows to columns in F^r ,

so the columns of $B^\top$ are linearly independent. By [Theorem 2.4.4](#), $B^\top$ is invertible, so $\det B^\top \neq 0$ by [Theorem 6.5.2](#), and $\det B = \det B^\top \neq 0$ by [Theorem 6.4.5](#).

Conclusion. By Step 2 there is a non-zero minor of size r , and by Step 1 there is none of size larger than r . This proves (a). For (b): if $\text{rank } A \geq k$, then Step 2 gives a non-zero $r \times r$ minor with $r \geq k$, and we still need one of size exactly k . Expanding that $r \times r$ determinant along any row ([Theorem 6.6.1](#)) writes it as a combination of $(r-1) \times (r-1)$ minors of the block, each of which is also an $(r-1) \times (r-1)$ minor of A ; since the determinant is non-zero, one of these minors is non-zero. Repeating $r-k$ times gives a non-zero $k \times k$ minor. Conversely, a non-zero $k \times k$ minor gives $\text{rank } A \geq k$ by Step 1. The second equivalence in (b) is the negation of the first. ■

The theorem turns “rank at most $k-1$ ” into a finite list of polynomial equations in the entries: all $k \times k$ minors vanish. That is why rank can only **drop** when the entries satisfy special equations, as the next example shows.

Example. (Rank of a Family of Matrices)

For $t \in \mathbb{Q}$, let

$$A(t) = \begin{pmatrix} 1 & 2 & 1 \\ 2 & t & 3 \\ 3 & 6 & t \end{pmatrix}.$$

Find $\text{rank } A(t)$ for every t .

Solution. The only 3×3 minor is $\det A(t)$. By cofactor expansion along the first row ([Theorem 6.6.1](#)),

$$\det A(t) = 1 \cdot (t^2 - 18) - 2 \cdot (2t - 9) + 1 \cdot (12 - 3t) = t^2 - 7t + 12 = (t-3)(t-4).$$

So for $t \neq 3, 4$, $A(t)$ has a non-zero 3×3 minor and $\text{rank } A(t) = 3$ by [Theorem 6.7.1](#).

For $t \in \{3, 4\}$ every 3×3 minor is zero, so $\text{rank } A(t) \leq 2$. Look for a 2×2 minor that does not involve t : rows $\{1, 2\}$ and columns $\{1, 3\}$ give

$$\det \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} = 1 \neq 0.$$

Hence $\text{rank } A(3) = \text{rank } A(4) = 2$. The same minor shows $\text{rank } A(t) \geq 2$ for every t , so rank 1 never occurs.

Remark. To **compute** the rank of one given matrix, elimination is far cheaper: a 10×10 matrix has $\binom{10}{5}^2 = 63504$ minors of size 5. Minors earn their keep in theory. They show that rank is unchanged by transposing (the minors of $A^\top$ are the minors of A , by [Theorem 6.4.5](#)) and by enlarging the field ([Exercise \(C2: Rank does not depend on the field\)](#)), and they describe the matrices of rank below k by polynomial equations.

6.7.3 The Cauchy–Binet formula

Multiplicativity, [Theorem 6.5.4](#), handles $\det(AB)$ when both factors are square. Now let $A \in M_{m \times n}(F)$ and $B \in M_{n \times m}(F)$. The product AB is $m \times m$, so it has a determinant, but $\det A$ and $\det B$ do not exist. The natural candidates to replace them are the largest minors: the $m \times m$ minors of A , which keep all rows and choose m columns S , and the $m \times m$ minors of B , which keep all columns and choose the **same** m rows S .

Theorem 6.7.2 — Cauchy–Binet Formula

Let $A \in M_{m \times n}(F)$ and $B \in M_{n \times m}(F)$. Then

$$\det(AB) = \sum_{\substack{S \subseteq [n] \\ |S|=m}} \det A_{[m],S} \det B_{S,[m]},$$

the sum running over all m -element subsets S of $[n]$. In particular:

- (a) if $m > n$, the sum is empty and $\det(AB) = 0$;
- (b) if $m = n$, the only subset is $S = [n]$ and the formula is $\det(AB) = \det A \det B$.

Idea. Treat $\det$ as a function of the m columns of AB . Column k of AB is $A\mathbf{b}_k = \sum_j b_{jk}\mathbf{a}_j$, a combination of the columns of A . Expand by multilinearity in all m columns at once, exactly as in the uniqueness proof: one term for each way of picking a column $\mathbf{a}_{\varphi(k)}$ of A in slot k . ① Terms that pick the same column twice vanish, because the determinant is alternating. ② The surviving picks use m **distinct** columns, forming a set S , in some order; putting them in increasing order costs the sign of the reordering ([Lemma 6.4.1](#)). ③ Collecting all orders of the same S produces a Leibniz sum, which is $\det B_{S,[m]}$.

Proof. Write $\mathbf{a}_1, \dots, \mathbf{a}_n \in F^m$ for the columns of A and b_{jk} for the entries of B . For $\mathbf{x}_1, \dots, \mathbf{x}_m \in F^m$, the function $D(\mathbf{x}_1, \dots, \mathbf{x}_m) = \det(\mathbf{x}_1, \dots, \mathbf{x}_m)$ is an alternating m -linear form by [Theorem 6.4.3](#).

By [Theorem 0.9.2](#), column k of AB is $A\mathbf{b}_k = \sum_{j=1}^n b_{jk}\mathbf{a}_j$. Expanding D by linearity in each of its m arguments in turn,

$$\det(AB) = D\left(\sum_{j=1}^n b_{j1}\mathbf{a}_j, \dots, \sum_{j=1}^n b_{jm}\mathbf{a}_j\right) = \sum_{\varphi} \left(\prod_{k=1}^m b_{\varphi(k)k}\right) D(\mathbf{a}_{\varphi(1)}, \dots, \mathbf{a}_{\varphi(m)}),$$

where φ runs over **all** functions $[m] \rightarrow [n]$. If φ is not injective, two arguments of D are equal and the term is 0 because D is alternating ([Definition 6.2.2](#)). So only injective φ contribute. In particular, if $m > n$ there is no injective φ , and $\det(AB) = 0$, which proves (a).

An injective φ has an image $S = \{s_1 < \dots < s_m\} \subseteq [n]$ with m elements, and then $\varphi(k) = s_{\pi(k)}$ for exactly one $\pi \in S_m$. Conversely each pair (S, π) gives an injective φ . By [Lemma 6.4.1](#) with $\mathbf{v}_p = \mathbf{a}_{s_p}$,

$$D(\mathbf{a}_{\varphi(1)}, \dots, \mathbf{a}_{\varphi(m)}) = \operatorname{sgn}(\pi) D(\mathbf{a}_{s_1}, \dots, \mathbf{a}_{s_m}) = \operatorname{sgn}(\pi) \det A_{[m],S},$$

since $A_{[m],S}$ has columns $\mathbf{a}_{s_1}, \dots, \mathbf{a}_{s_m}$. Hence

$$\det(AB) = \sum_{|S|=m} \det A_{[m],S} \sum_{\pi \in S_m} \operatorname{sgn}(\pi) \prod_{k=1}^m b_{s_{\pi(k)}k}.$$

The matrix $B_{S,[m]}$ has (p, k) -entry b_{s_pk} , so its $(\pi(k), k)$ -entry is $b_{s_{\pi(k)}k}$, and the inner sum is $\det B_{S,[m]}$ by [Definition 6.4.1](#). This proves the formula. For $m = n$, the only S is $[n]$, with $A_{[n],[n]} = A$ and $B_{[n],[n]} = B$, which gives (b). ■

Part (b) is a second proof of multiplicativity, and part (a) matches what rank predicts: if $m > n$, then $\operatorname{rank}(AB) \leq \operatorname{rank} A \leq n < m$ by [Theorem 2.5.9](#), so AB is not invertible. The formula also passes a sanity check at $m = 1$: a $1 \times n$ row times an $n \times 1$ column is the 1×1 matrix $\sum_j a_{1j}b_{j1}$, and the 1×1 minors are single entries.

Remark. The proof uses only sums, differences and products of entries, together with [Lemma 6.4.1](#), which is division-free by the remark in §4. So, like the results flagged in §4–§6, the Cauchy–Binet formula holds for matrices with entries in any commutative ring.

Example. (Cauchy–Binet for a 2×3 times 3×2 Product)

Let

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 2 & -1 \end{pmatrix}$$

over $\mathbb{Q}$. Compute $\det(AB)$ directly and by the Cauchy–Binet formula. Then compute $\det(BA)$.

Solution. Directly. $AB = \begin{pmatrix} 1+2+0 & 0+2+0 \\ 0+1+6 & 0+1-3 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 7 & -2 \end{pmatrix}$, so $\det(AB) = -6 - 14 = -20$.

By Cauchy–Binet. Here $m = 2$, $n = 3$, and there are three subsets S . The minors of A were computed after [Definition 6.7.1](#). The minors of B keep both columns and rows S :

S	$\det A_{[2],S}$	$\det B_{S,[2]}$	product
$\{1, 2\}$	1	$\det \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = 1$	1
$\{1, 3\}$	3	$\det \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix} = -1$	-3
$\{2, 3\}$	6	$\det \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} = -3$	-18

The sum is $1 - 3 - 18 = -20$, in agreement.

The other product. BA is 3×3 , and here the roles are $m = 3 > n = 2$, so $\det(BA) = 0$ by [Theorem 6.7.2](#) (a). Indeed $BA = \begin{pmatrix} 1 & 2 & 0 \\ 1 & 3 & 3 \\ 2 & 3 & -3 \end{pmatrix}$, whose third row is 3 times the first minus the second, so $\det(BA) = 0$.

Warning. For non-square factors, $\det(AB)$ and $\det(BA)$ differ. For square matrices $\det(AB) = \det A \det B = \det(BA)$. In the example above, $\det(AB) = -20$ while $\det(BA) = 0$. An even smaller case: $A = \begin{pmatrix} 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ give $AB = (2)$ and $BA = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, with determinants 2 and 0. Also, do not write “ $\det A \det B$ ” for non-square A, B : those symbols are undefined, and the sum over S is what replaces them.

Quick check. Let $\mathbf{u}, \mathbf{v} \in F^n$ with $n \geq 2$. Without computing, what is $\det(\mathbf{u}\mathbf{v}^\top)$?

Solution. $\mathbf{u}\mathbf{v}^\top$ is the product of the $n \times 1$ matrix $\mathbf{u}$ and the $1 \times n$ matrix $\mathbf{v}^\top$. Here the product is $n \times n$ and the middle size is $1 < n$, so $\det(\mathbf{u}\mathbf{v}^\top) = 0$ by [Theorem 6.7.2](#) (a).

6.7.4 Sums of squares over the reals

Over $\mathbb{R}$, take $B = A^\top$. The minors of $A^\top$ are the minors of A with rows and columns exchanged, so Cauchy–Binet writes $\det(AA^\top)$ as a sum of squares.

Corollary 6.7.3 — Gram Determinants Are Non-negative

Let $A \in M_{m \times n}(\mathbb{R})$. Then

$$\det(AA^\top) = \sum_{\substack{S \subseteq [n] \\ |S|=m}} (\det A_{[m],S})^2 \geq 0,$$

with equality if and only if $\text{rank } A < m$.

Proof. By [Definition 0.9.10](#), the (p, k) -entry of $(A^\top)_{S,[m]}$ is the (k, s_p) -entry of A , so $(A^\top)_{S,[m]} = (A_{[m],S})^\top$. By [Theorem 6.4.5](#), $\det(A^\top)_{S,[m]} = \det A_{[m],S}$. Substituting into [Theorem 6.7.2](#) with $B = A^\top$ gives the formula. A sum of squares of real numbers is ≥ 0 , and it is 0 if and only if every term is 0. Every $m \times m$ minor of A has the form $\det A_{[m],S}$, since an $m \times m$ minor must use all m rows. If $m > n$, the sum is empty and $\text{rank } A \leq n < m$. Otherwise, by [Theorem 6.7.1](#) (b) (or directly if $A = 0$), all these minors vanish if and only if $\text{rank } A < m$. ■

The matrix $AA^\top$ collects the dot products of the rows of A with each other. Such matrices of pairwise inner products are **Gram matrices**; Chapters 10 and 12 study them, and there this

corollary becomes the statement that a Gram matrix is positive semidefinite. For $m = 2$ the corollary is a classical identity about two vectors, [Exercise \(B3: The Lagrange identity\)](#). Over $\mathbb{C}$ or $\mathbb{F}_p$ the conclusion “ ≥ 0 ” has no meaning, and the equality case can fail: over $\mathbb{C}$, $A = \begin{pmatrix} 1 & i \end{pmatrix}$ has rank 1 but $AA^\top = (1 + i^2) = (0)$.

6.7.5 Exercises

6.7.5.1 A. Check your understanding

Exercise A1.

- (a) Define the $k \times k$ minor of $A \in M_{m \times n}(F)$ on rows I and columns J .
- (b) True or false: if every principal minor of a square matrix is zero, the matrix is zero. Justify your answer.
- (c) True or false: if $A \in M_{4 \times 5}(F)$ has a non-zero 3×3 minor, then $\text{rank } A \geq 3$. Justify your answer.
- (d) True or false: if $A \in M_3(F)$ has a zero 2×2 minor, then $\text{rank } A \leq 1$. Justify your answer.
- (e) State the Cauchy–Binet formula, and say what it gives when $m > n$.
- (f) Let $A \in M_{2 \times 3}(\mathbb{R})$. Why is $\det(A^\top A) = 0$?

Solution.

- (a) For $I = \{i_1 < \dots < i_k\} \subseteq [m]$ and $J = \{j_1 < \dots < j_k\} \subseteq [n]$, it is $\det A_{I,J}$, where $A_{I,J} \in M_k(F)$ has (p, q) -entry $a_{i_p j_q}$ ([Definition 6.7.1](#)).
- (b) False. $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has principal minors $0, 0, 0$ but $N \neq 0$.
- (c) True, by Step 1 of the proof of [Theorem 6.7.1](#), or by part (b) of that theorem.
- (d) False. The rank is decided by whether **some** minor is non-zero, not by one zero minor. I_3 has the zero 2×2 minor on rows $\{1, 2\}$ and columns $\{2, 3\}$, namely $\det \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = 0$, but $\text{rank } I_3 = 3$.
- (e) For $A \in M_{m \times n}(F)$ and $B \in M_{n \times m}(F)$, $\det(AB) = \sum_{|S|=m} \det A_{[m],S} \det B_{S,[m]}$, summed over m -subsets $S \subseteq [n]$. If $m > n$ there are no such subsets and $\det(AB) = 0$.
- (f) $A^\top A$ is 3×3 , the product of a 3×2 and a 2×3 matrix. By [Theorem 6.7.2](#) (a) with $m = 3 > n = 2$, $\det(A^\top A) = 0$.

6.7.5.2 B. Practice

Exercise B1: Rank with a parameter.

For $s \in \mathbb{Q}$, let $A(s) = \begin{pmatrix} 1 & s & 1 \\ s & 1 & 1 \\ 1 & 1 & s \end{pmatrix}$. Use minors to find $\text{rank } A(s)$ for every s .

Solution. By cofactor expansion along the first row (Theorem 6.6.1),

$$\det A(s) = 1 \cdot (s-1) - s(s^2-1) + 1 \cdot (s-1) = -s^3 + 3s - 2 = -(s-1)^2(s+2).$$

If $s \neq 1, -2$, this 3×3 minor is non-zero, so $\text{rank } A(s) = 3$ by Theorem 6.7.1.

If $s = -2$, then $\det A(-2) = 0$, so $\text{rank } A(-2) \leq 2$. The minor on rows $\{1, 2\}$ and columns $\{1, 2\}$ is $\det \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix} = 1 - 4 = -3 \neq 0$. Hence $\text{rank } A(-2) = 2$.

If $s = 1$, every entry is 1. Every 2×2 minor is $\det \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = 0$, and the 1×1 minors are non-zero. Hence $\text{rank } A(1) = 1$.

Exercise B2: Checking Cauchy–Binet.

Let $A = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 1 \\ 0 & 1 \\ 1 & -1 \end{pmatrix}$ over $\mathbb{Q}$.

- (a) Compute $\det(AB)$ directly.
- (b) Compute $\det(AB)$ by the Cauchy–Binet formula.
- (c) Hence, without multiplying, find $\det(BA)$.

Solution.

(a) $AB = \begin{pmatrix} 2+0+2 & 1+0-2 \\ -2+0+1 & -1+1-1 \end{pmatrix} = \begin{pmatrix} 4 & -1 \\ -1 & -1 \end{pmatrix}$, so $\det(AB) = -4 - 1 = -5$.

(b) The three subsets of $[3]$ of size 2 give

$$S = \{1, 2\} : \det \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \det \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} = 1 \cdot 2 = 2,$$

$$S = \{1, 3\} : \det \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix} \det \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix} = 3 \cdot (-3) = -9,$$

$$S = \{2, 3\} : \det \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix} \det \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix} = (-2)(-1) = 2.$$

The sum is $2 - 9 + 2 = -5$, agreeing with (a).

(c) $BA \in M_3(\mathbb{Q})$ is a product through F^2 , so $\det(BA) = 0$ by Theorem 6.7.2 (a).

Exercise B3: The Lagrange identity.

Let $\mathbf{a} = (a_1, \dots, a_n)$ and $\mathbf{b} = (b_1, \dots, b_n)$ be vectors in $\mathbb{R}^n$, $n \geq 2$, and write $\mathbf{a} \cdot \mathbf{b} = \sum_i a_i b_i$.

- (a) Let $M \in M_{2 \times n}(\mathbb{R})$ have rows $\mathbf{a}^\top$ and $\mathbf{b}^\top$. Compute $MM^\top$.
- (b) Hence prove the **Lagrange identity**

$$(\mathbf{a} \cdot \mathbf{a})(\mathbf{b} \cdot \mathbf{b}) - (\mathbf{a} \cdot \mathbf{b})^2 = \sum_{1 \leq i < j \leq n} (a_i b_j - a_j b_i)^2.$$

- (c) Check it for $\mathbf{a} = (1, 2, 3)$ and $\mathbf{b} = (4, 5, 6)$, and deduce $(\mathbf{a} \cdot \mathbf{b})^2 \leq (\mathbf{a} \cdot \mathbf{a})(\mathbf{b} \cdot \mathbf{b})$ for all $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$.

Solution.

- (a) By [Theorem 0.9.2](#), the (p, q) -entry of MM^T is row p of M times column q of M^T , which is row q of M transposed. Hence

$$MM^T = \begin{pmatrix} \mathbf{a} \cdot \mathbf{a} & \mathbf{a} \cdot \mathbf{b} \\ \mathbf{b} \cdot \mathbf{a} & \mathbf{b} \cdot \mathbf{b} \end{pmatrix}, \quad \mathbf{b} \cdot \mathbf{a} = \mathbf{a} \cdot \mathbf{b}.$$

- (b) The left side of the identity is $\det(MM^T)$. By [Corollary 6.7.3](#) with $m = 2$, it equals the sum over $S = \{i < j\}$ of $(\det M_{[2],S})^2$, and $M_{[2],\{i,j\}} = \begin{pmatrix} a_i & a_j \\ b_i & b_j \end{pmatrix}$ has determinant $a_i b_j - a_j b_i$. This proves the identity.
- (c) Left side: $\mathbf{a} \cdot \mathbf{a} = 14$, $\mathbf{b} \cdot \mathbf{b} = 77$, $\mathbf{a} \cdot \mathbf{b} = 32$, so $14 \cdot 77 - 32^2 = 1078 - 1024 = 54$. Right side: the pairs $(1, 2), (1, 3), (2, 3)$ give $5 - 8 = -3$, $6 - 12 = -6$, $12 - 15 = -3$, and $9 + 36 + 9 = 54$. In general the right side is a sum of squares of reals, hence ≥ 0 , so $(\mathbf{a} \cdot \mathbf{b})^2 \leq (\mathbf{a} \cdot \mathbf{a})(\mathbf{b} \cdot \mathbf{b})$.

6.7.5.3 C. Going deeper

Exercise C1: Counting spanning trees.

Consider the network with vertices $1, 2, 3, 4$ and five edges, oriented as $e_1 = 1 \rightarrow 2$, $e_2 = 2 \rightarrow 3$, $e_3 = 3 \rightarrow 4$, $e_4 = 4 \rightarrow 1$, $e_5 = 1 \rightarrow 3$. A **spanning tree** is a set of 3 edges that connects all four vertices (ignoring orientation). Let $N \in M_{4 \times 5}(\mathbb{R})$ be its incidence matrix ([Definition 2.7.1](#)), let $L = NN^T$, and let N_0 and L_0 be N and L with the row (and, for L , the column) of vertex 4 deleted.

- Compute L and $\det L_0$.
- List the spanning trees and count them.
- Show that $L_0 = N_0 N_0^T$, and use [Corollary 6.7.3](#) to write $\det L_0$ as a sum over 3-element sets of edges.
- Compute the minor of N_0 for the tree $\{e_1, e_2, e_3\}$ and for the non-tree $\{e_1, e_2, e_5\}$. Explain, for these two sets, why the answers are ± 1 and 0.

The **matrix-tree theorem** says that for every connected network, $\det L_0$ equals the number of spanning trees; parts (c) and (d) are the heart of its proof, which Chapter 23 completes.

Hint: for (d), a set of edges containing a cycle gives dependent columns.

Solution.

- (a) Column e of N has 1 at the tail and -1 at the head:

$$N = \begin{pmatrix} 1 & 0 & 0 & -1 & 1 \\ -1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & -1 \\ 0 & 0 & -1 & 1 & 0 \end{pmatrix}, \quad L = NN^T = \begin{pmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}.$$

(Diagonal entries count the edges at a vertex; off-diagonal entries are -1 for each edge joining two vertices.) Expanding along the first row,

$$\det L_0 = \det \begin{pmatrix} 3 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 3 \end{pmatrix} = 3(6-1) + 1 \cdot (-3-1) - 1 \cdot (1+2) = 15-4-3 = 8.$$

- (b) There are $\binom{5}{3} = 10$ sets of three edges. Three edges on four vertices connect everything exactly when they contain no cycle, and the only 3-edge cycles are the triangles $\{e_1, e_2, e_5\}$ (vertices 1, 2, 3) and $\{e_3, e_4, e_5\}$ (vertices 1, 3, 4). So there are $10 - 2 = 8$ spanning trees, matching $\det L_0$.
- (c) Deleting row 4 of N deletes row 4 of NN^T (row view) and column 4 of NN^T (column view of N^T), by [Theorem 0.9.2](#). So $L_0 = N_0 N_0^T$, with $N_0 \in M_{3 \times 5}(\mathbb{R})$, and [Corollary 6.7.3](#) gives $\det L_0 = \sum_{|S|=3} (\det(N_0)_{[3],S})^2$, one term for each set S of three edges.
- (d) For $S = \{e_1, e_2, e_3\}$, $(N_0)_{[3],S} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$, which is lower triangular with determinant 1 ([Theorem 6.4.6](#)). For $S = \{e_1, e_2, e_5\}$, the columns are $(1, -1, 0)$, $(0, 1, -1)$, $(1, 0, -1)$, and the third is the sum of the first two: going around the cycle $1 \rightarrow 2 \rightarrow 3$ equals going directly $1 \rightarrow 3$. So the columns are dependent and the minor is 0 by [Theorem 6.2.1](#). Thus the non-tree contributes 0^2 and the tree contributes 1^2 . (SymPy confirms that the ten minors are ± 1 for the eight trees and 0 for the two triangles, so the sum is 8.)

Exercise C2: Rank does not depend on the field.

Let K be a subfield of a field L ([Definition 1.9.1](#)), and let $A \in M_{m \times n}(K)$. We may also regard A as a matrix in $M_{m \times n}(L)$.

- Explain why each minor $\det A_{I,J}$ is the same element whether computed in K or in L .
- Prove that the rank of A over K equals its rank over L .
- Deduce: if a system $A\mathbf{x} = \mathbf{0}$ with rational coefficients has a non-zero complex solution, it has a non-zero rational solution.

Solution.

- By [Definition 6.4.1](#), $\det A_{I,J}$ is a finite sum of $\pm$ products of entries of A . The entries lie in K , and K has the same addition and multiplication as L , so the sum is the same element of $K \subseteq L$.
- If $A = 0$, both ranks are 0. Otherwise, by [Theorem 6.7.1](#) (a) applied over K and over L , each rank is the largest k for which some $k \times k$ minor is non-zero. By (a) the list of minors, and which of them are non-zero, is the same in both cases. Hence the two ranks agree.
- Take $K = \mathbb{Q}$, $L = \mathbb{C}$. A non-zero complex solution means nullity $A \geq 1$ over $\mathbb{C}$, so $\text{rank}_{\mathbb{C}} A \leq n-1$ by [Theorem 2.5.7](#). By (b), $\text{rank}_{\mathbb{Q}} A \leq n-1$, so nullity $A \geq 1$ over $\mathbb{Q}$, and a non-zero rational solution exists.

6.8 Special Determinants

A determinant of size n has $n!$ terms, so a general formula for a large matrix is hopeless. Some matrices, though, have so much structure that their determinants collapse to a short product or a recurrence. This section computes the most useful of them: the Vandermonde determinant, which governs polynomial interpolation; tridiagonal determinants, which satisfy a three-term recurrence; block triangular determinants, which split into smaller ones; and the determinant of a rank-one update of the identity. The proofs share one idea: find the structure first (a polynomial with known roots, a row with two non-zero entries, a pattern of zeros) and let the defining properties of the determinant do the rest.

6.8.1 The Vandermonde determinant

Chapter 2 proved that there is exactly one polynomial of degree at most n through $n+1$ points with distinct first coordinates ([Theorem 2.7.3](#)). Behind it sat a square matrix of powers of the nodes. Its determinant turns out to be a product of all the differences of the nodes, and that product explains at a glance why distinct nodes are exactly what is needed.

For $x_1, \dots, x_n \in F$, the **Vandermonde matrix** is

$$V(x_1, \dots, x_n) := \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix} \in M_n(F),$$

whose (i, j) -entry is x_i^{j-1} . Row i lists the powers $1, x_i, \dots, x_i^{n-1}$ of the i -th node. For $n = 1$ it is the 1×1 matrix (1) , and for $n = 2$ its determinant is $x_2 - x_1$.

Theorem 6.8.1 — Vandermonde Determinant

Let $n \geq 1$ and $x_1, \dots, x_n \in F$. Then

$$\det V(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_j - x_i),$$

where the empty product (for $n = 1$) is 1.

Idea. Induction on n . Replace the last node x_n by a variable t . Expanding along the last row shows the determinant is a polynomial $f(t)$ of degree at most $n-1$, with leading coefficient the smaller Vandermonde determinant. Setting $t = x_i$ for $i < n$ makes two rows equal, so $x_1, \dots, x_{n-1}$ are roots of f . A polynomial of degree at most $n-1$ with these $n-1$ roots is its leading coefficient times $\prod_{i < n} (t - x_i)$. Now put $t = x_n$ back. The roots argument needs the x_i distinct; if they are not, both sides are 0 anyway.

Proof. We use induction on n ([Theorem 0.2.7](#)). For $n = 1$, $\det(1) = 1$, the empty product.

Let $n \geq 2$, assume the formula for $n-1$ nodes, and write $V = V(x_1, \dots, x_n)$. Let $C_{n1}, \dots, C_{nn}$ be the cofactors of V along row n ([Definition 6.6.1](#)). Each C_{nj} is computed

after deleting row n , so it does not involve x_n . Define

$$f(t) := \sum_{j=1}^n C_{nj} t^{j-1} \in F[t].$$

For $c \in F$, let V_c be V with row n replaced by $(1, c, \dots, c^{n-1})$. Then V_c has the same cofactors along row n as V , so by [Theorem 6.6.1](#) along row n ,

$$\det V_c = \sum_{j=1}^n c^{j-1} C_{nj} = f(c). \quad (*)$$

In particular $\det V = f(x_n)$. The coefficient of t^{n-1} in f is $C_{nn} = (-1)^{2n} \det V(x_1, \dots, x_{n-1})$, since deleting row n and column n of V leaves the Vandermonde matrix of the first $n-1$ nodes.

Case 1: two of $x_1, \dots, x_{n-1}$ are equal, say $x_p = x_q$ with $p < q < n$. Then rows p and q of V are equal. By [Theorem 6.4.5](#), $\det V = \det V^\top$, and $V^\top$ has two equal columns, so $\det V = 0$ because the determinant is alternating ([Definition 6.2.2](#)). The product also vanishes, since it contains the factor $x_q - x_p = 0$.

Case 2: $x_1, \dots, x_{n-1}$ are distinct. For $i < n$, the matrix V_{x_i} has rows i and n equal, so $f(x_i) = \det V_{x_i} = 0$ by $(*)$ and the argument of Case 1. Hence $x_1, \dots, x_{n-1}$ are $n-1$ distinct roots of f , and by [Corollary 5.1.4](#),

$$g(t) := (t - x_1)(t - x_2) \cdots (t - x_{n-1}) \text{ divides } f.$$

Write $f = hg$ with $h \in F[t]$. If $h \neq 0$, then $\deg h = \deg f - (n-1) \leq 0$ by [Theorem 0.8.3](#), since $\deg f \leq n-1$; so in every case h is a constant $h \in F$. As g is monic of degree $n-1$, the coefficient of t^{n-1} in hg is h , so $h = C_{nn} = \det V(x_1, \dots, x_{n-1})$. Evaluating at x_n ([Theorem 0.8.5](#)) and using the induction hypothesis,

$$\det V = f(x_n) = \det V(x_1, \dots, x_{n-1}) \prod_{i=1}^{n-1} (x_n - x_i) = \prod_{1 \leq i < j \leq n-1} (x_j - x_i) \prod_{i=1}^{n-1} (x_n - x_i).$$

The two products together run over all pairs $i < j \leq n$, the second one collecting exactly the pairs with $j = n$. This proves the formula for n , and the theorem follows by induction. ■

Example. (A 3×3 Vandermonde Determinant)

Compute $\det V(1, 2, 4)$ from the formula and by cofactor expansion.

Solution. The formula gives $(2-1)(4-1)(4-2) = 1 \cdot 3 \cdot 2 = 6$. Directly, expanding

$$V(1, 2, 4) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 16 \end{pmatrix}$$

along the first row gives $1 \cdot (32 - 16) - 1 \cdot (16 - 4) + 1 \cdot (4 - 2) = 16 - 12 + 2 = 6$.

The product vanishes exactly when a factor does, because a field has no zero divisors. So:

Corollary 6.8.2 — Distinct Nodes Give an Invertible Vandermonde Matrix

$V(x_1, \dots, x_n)$ is invertible if and only if $x_1, \dots, x_n$ are **distinct**.

Proof. In a field, a product is non-zero if and only if every factor is non-zero (Theorem 0.7.1 (f), applied repeatedly). So by Theorem 6.8.1, $\det V(x_1, \dots, x_n) \neq 0$ if and only if $x_j \neq x_i$ for all $i < j$. By Theorem 6.5.2, this is equivalent to invertibility. ■

Now the promised two-line proof of Theorem 2.7.3. Given distinct $x_0, \dots, x_n$ and values $y_0, \dots, y_n$, a polynomial $p = c_0 + c_1x + \dots + c_nx^n$ satisfies $p(x_i) = y_i$ for all i if and only if $V(x_0, \dots, x_n) \mathbf{c} = \mathbf{y}$, since row i of this system reads $c_0 + c_1x_i + \dots + c_nx_i^n = y_i$. By Corollary 6.8.2 the matrix is invertible, so the system has exactly one solution $\mathbf{c} = V(x_0, \dots, x_n)^{-1} \mathbf{y}$ (Theorem 2.4.4), and the interpolating polynomial exists and is unique.

Remark. This is not a proof from nothing. The root bound of Chapter 5 did the real work inside Theorem 6.8.1, exactly as the root bound did in Chapter 2. What the determinant adds is an explicit number: it says how close to singular the interpolation problem is, since nodes that nearly coincide make the product small.

Warning. Watch the shape and the order. The formula is for rows $(1, x_i, \dots, x_i^{n-1})$ starting at the power 0, and for the factors $x_j - x_i$ with $i < j$, “later node minus earlier node”. Dropping the column of ones changes the answer: $\det \begin{pmatrix} x_1 & x_1^2 \\ x_2 & x_2^2 \end{pmatrix} = x_1x_2(x_2 - x_1)$, not $x_2 - x_1$. Writing the factors as $x_i - x_j$ instead flips the sign of each of the $\binom{n}{2}$ factors, so it multiplies the answer by $(-1)^{\binom{n}{2}}$: already for $n = 2$ and $n = 3$ this is -1 , a wrong sign.

Quick check. Without expanding, find $\det V(1, 2, 3, 5)$.

Solution. The pairs $i < j$ give $(2-1)(3-1)(5-1)(3-2)(5-2)(5-3) = 1 \cdot 2 \cdot 4 \cdot 1 \cdot 3 \cdot 2 = 48$.

6.8.2 Tridiagonal determinants

Many matrices in applications have non-zero entries only on the diagonal and next to it: discretized differential equations, chains of springs, the recurrences of orthogonal polynomials. Such a matrix is **tridiagonal**. Its determinant is not a closed formula in general, but it obeys a recurrence in the size, because the last row has only two non-zero entries.

Let $T_n \in M_n(F)$ have diagonal entries $a_1, \dots, a_n$, entries $b_1, \dots, b_{n-1}$ just above the diagonal

$((i, i+1)$ -entry b_i), entries $c_1, \dots, c_{n-1}$ just below it $((i+1, i)$ -entry c_i), and zeros elsewhere:

$$T_n = \begin{pmatrix} a_1 & b_1 & & & \\ c_1 & a_2 & b_2 & & \\ & c_2 & \ddots & \ddots & \\ & & \ddots & a_{n-1} & b_{n-1} \\ & & & c_{n-1} & a_n \end{pmatrix}.$$

For $1 \leq k \leq n$, the top-left $k \times k$ block of T_n is the tridiagonal matrix T_k built from $a_1, \dots, a_k, b_1, \dots, b_{k-1}, c_1, \dots, c_{k-1}$. Write $D_k = \det T_k$, and set $D_0 = 1$.

Theorem 6.8.3 — Tridiagonal Recurrence

With this notation, $D_1 = a_1$, and for $2 \leq k \leq n$,

$$D_k = a_k D_{k-1} - b_{k-1} c_{k-1} D_{k-2}.$$

Proof. $D_1 = \det(a_1) = a_1$. For $k = 2$, $D_2 = a_1 a_2 - b_1 c_1 = a_2 D_1 - b_1 c_1 D_0$. Let $k \geq 3$. Row k of T_k has only two possibly non-zero entries, c_{k-1} in column $k-1$ and a_k in column k . By Theorem 6.6.1 along row k ,

$$D_k = c_{k-1}(-1)^{2k-1} M_{k,k-1} + a_k(-1)^{2k} M_{kk} = a_k M_{kk} - c_{k-1} M_{k,k-1},$$

where $M_{kk} = \det T_{k-1} = D_{k-1}$. The minor $M_{k,k-1}$ is the determinant of the $(k-1) \times (k-1)$ matrix obtained by deleting row k and column $k-1$. Its last column consists of rows $1, \dots, k-1$ of column k of T_k , which is b_{k-1} in the last position and 0 elsewhere. Expanding along this last column (Theorem 6.6.1), with sign $(-1)^{(k-1)+(k-1)} = 1$, gives $M_{k,k-1} = b_{k-1} \det T_{k-2} = b_{k-1} D_{k-2}$, because deleting the last row and column leaves rows and columns $1, \dots, k-2$ of T_k . Substituting proves the recurrence. ■

Example. (Fibonacci Numbers as Determinants)

Let T_n have 1 on the diagonal, 1 above it and -1 below it. Find $D_n = \det T_n$.

Solution. Here $a_k = 1$ and $b_{k-1} c_{k-1} = 1 \cdot (-1) = -1$, so Theorem 6.8.3 gives $D_k = D_{k-1} + D_{k-2}$ for $k \geq 2$, with $D_0 = 1$ and $D_1 = 1$. Hence

$$D_1, D_2, D_3, \dots = 1, 2, 3, 5, 8, 13, 21, 34, \dots,$$

the Fibonacci numbers: $D_n = F_{n+1}$, where $F_1 = F_2 = 1$ and $F_k = F_{k-1} + F_{k-2}$. For instance

$$D_3 = \det \begin{pmatrix} 1 & 1 & 0 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{pmatrix} = 1 \cdot (1+1) - 1 \cdot (-1-0) + 0 = 3.$$

(SymPy confirms $D_1, \dots, D_8$.)

Only the products $b_{k-1} c_{k-1}$ enter the recurrence, never b_{k-1} and c_{k-1} separately. For example, for a real number t , putting $2t$ on the diagonal and 1 on both neighboring lines gives $D_k =$

$2t D_{k-1} - D_{k-2}$, so $D_1 = 2t$, $D_2 = 4t^2 - 1$, $D_3 = 8t^3 - 4t$. These are the Chebyshev polynomials of the second kind.

6.8.3 Block triangular matrices

A matrix can be cut into blocks by a horizontal and a vertical line. When the block in the lower left is zero, the determinant should see only the two square blocks on the diagonal, just as a triangular determinant sees only diagonal entries. This result is used constantly, for instance in Chapter 7 for Schur complements.

Theorem 6.8.4 — Block Triangular Determinant

Let $A \in M_k(F)$, $D \in M_l(F)$ and $B \in M_{k \times l}(F)$, with $k, l \geq 1$. Then

$$\det \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = \det A \cdot \det D,$$

where 0 is the $l \times k$ zero matrix. The same holds for $\begin{pmatrix} A & 0 \\ C & D \end{pmatrix}$ with $C \in M_{l \times k}(F)$.

Idea. In the Leibniz formula for the whole matrix, a term picks one entry from each column. The first k columns have zeros below row k , so a non-zero term must pick rows $1, \dots, k$ for the first k columns, and is then forced to pick rows $k+1, \dots, n$ for the last l columns. Such a permutation is a permutation of each block, its inversions are the inversions inside the blocks, and the Leibniz sum factors into two Leibniz sums.

Proof. Let $n = k + l$ and $M = \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = (m_{ij})$. By Definition 6.4.1, $\det M = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{j=1}^n m_{\sigma(j)j}$. Since $m_{ij} = 0$ for $i > k \geq j$, the term of σ is 0 unless $\sigma(j) \leq k$ for every $j \leq k$. For such σ , the injective map σ sends $\{1, \dots, k\}$ into itself, hence onto itself (a finite set), and so it sends $\{k+1, \dots, n\}$ onto itself. Therefore σ determines $\alpha \in S_k$ and $\beta \in S_l$ by

$$\alpha(j) = \sigma(j) \quad (j \leq k), \quad \beta(q) = \sigma(k+q) - k \quad (q \leq l),$$

and every pair (α, β) arises from exactly one such σ .

We compare signs through inversions (Definition 6.3.1). A pair $i < j$ with $i, j \leq k$ is an inversion of σ exactly when it is one of α ; a pair $k+p < k+q$ is an inversion of σ exactly when $p < q$ is one of β ; and a pair $i \leq k < j$ is never an inversion, since $\sigma(i) \leq k < \sigma(j)$. By Definition 6.3.2, $\text{sgn}(\sigma) = \text{sgn}(\alpha) \text{sgn}(\beta)$. Also $m_{\sigma(j)j} = a_{\alpha(j)j}$ for $j \leq k$ and $m_{\sigma(k+q), k+q} = d_{\beta(q)q}$. Hence

$$\det M = \sum_{\alpha \in S_k} \sum_{\beta \in S_l} \text{sgn}(\alpha) \text{sgn}(\beta) \prod_{j=1}^k a_{\alpha(j)j} \prod_{q=1}^l d_{\beta(q)q} = \det A \cdot \det D,$$

using Definition 6.4.1 twice. For the lower block triangular case, the transpose of $\begin{pmatrix} A & 0 \\ C & D \end{pmatrix}$ is $\begin{pmatrix} A^\top & C^\top \\ 0 & D^\top \end{pmatrix}$, so by Theorem 6.4.5 and the first case its determinant is $\det A^\top \det D^\top = \det A \det D$.

The proof uses only sums and products of entries, so, like the results flagged in §4–§6, it holds for matrices over any commutative ring; §10 uses this over $F[x]$. By induction on the number of blocks, a matrix with square blocks $A_1, \dots, A_r$ on the diagonal and zeros below them has determinant $\det A_1 \cdots \det A_r$; [Theorem 6.4.6](#) is the case where every block is 1×1 .

For example,

$$\det \begin{pmatrix} 2 & 1 & 5 & -3 \\ 1 & 1 & 0 & 7 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 4 & 2 \end{pmatrix} = \det \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \det \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix} = 1 \cdot 2 = 2,$$

and the entries 5, -3 , 0, 7 of the upper right block play no role.

Warning. There is no “block $ad-bc$ ” rule. For a general block matrix, $\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} \neq \det A \det D - \det B \det C$. Take $A = D = 0$ and $B = C = I_2$: the 4×4 matrix $\begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}$ is the permutation matrix of $(1\ 3)(2\ 4)$, a product of two transpositions, so its determinant is $+1$; the rule would give $0 \cdot 0 - 1 \cdot 1 = -1$. The zero block in the corner is what the theorem needs. The correct formulas for general blocks use Schur complements, in Chapter 7.

Quick check. Is $\det \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = \det A \det D$ meaningful when A is 2×3 and D is 3×2 (so the whole matrix is 5×5)?

Solution. No. The theorem requires A and D square; here $\det A$ and $\det D$ are undefined. In fact the whole determinant is 0 whatever the entries: the first 3 columns have their non-zero entries only in the first 2 rows, so they are linearly dependent in a 2-dimensional space, and a determinant with dependent columns is 0 ([Theorem 6.2.1](#)).

6.8.4 Rank-one updates

Changing a matrix by a rank-one matrix $\mathbf{u}\mathbf{v}^\top$ is the smallest possible change in the sense of rank, and it happens often: updating one observation in statistics, modifying one row of a system, or adding the same number to every entry. The determinant of the updated identity turns out to be a single dot product.

Theorem 6.8.5 — Matrix Determinant Lemma

Let $\mathbf{u}, \mathbf{v} \in F^n$. Then

$$\det(I_n + \mathbf{u}\mathbf{v}^\top) = 1 + \mathbf{v}^\top \mathbf{u}.$$

Idea. Border $I_n + \mathbf{u}\mathbf{v}^\top$ by an extra column $\mathbf{u}$ and an extra row $(\mathbf{0}^\top, 1)$, giving a block triangular matrix M with the same determinant. Then clear the update by elimination with blocks. Rows: add $\mathbf{v}^\top$ times the top n rows to the new bottom row (multiply by L on the left). Columns: subtract the last column times $\mathbf{v}^\top$ from the first n columns

(multiply by L' on the right). The factors L, L' have determinant 1, and the result is block triangular with I_n and the 1×1 block $1 + \mathbf{v}^\top \mathbf{u}$ on the diagonal.

Proof. Consider the $(n+1) \times (n+1)$ block matrices

$$L = \begin{pmatrix} I_n & \mathbf{0} \\ \mathbf{v}^\top & 1 \end{pmatrix}, \quad M = \begin{pmatrix} I_n + \mathbf{u}\mathbf{v}^\top & \mathbf{u} \\ \mathbf{0}^\top & 1 \end{pmatrix}, \quad L' = \begin{pmatrix} I_n & \mathbf{0} \\ -\mathbf{v}^\top & 1 \end{pmatrix}.$$

Splitting the sum $\sum_j$ in [Theorem 0.9.2](#) (1) into the indices $j \leq n$ and $j = n+1$ shows that matrices partitioned in this way multiply block by block, like 2×2 matrices whose entries are blocks. Hence

$$LM = \begin{pmatrix} I_n + \mathbf{u}\mathbf{v}^\top & \mathbf{u} \\ \mathbf{v}^\top + (\mathbf{v}^\top \mathbf{u})\mathbf{v}^\top & \mathbf{v}^\top \mathbf{u} + 1 \end{pmatrix},$$

where we used $\mathbf{v}^\top(\mathbf{u}\mathbf{v}^\top) = (\mathbf{v}^\top \mathbf{u})\mathbf{v}^\top$ by associativity, $\mathbf{v}^\top \mathbf{u}$ being a 1×1 matrix, that is, a scalar. Multiplying on the right by L' ,

$$LML' = \begin{pmatrix} I_n + \mathbf{u}\mathbf{v}^\top - \mathbf{u}\mathbf{v}^\top & \mathbf{u} \\ \mathbf{v}^\top + (\mathbf{v}^\top \mathbf{u})\mathbf{v}^\top - (1 + \mathbf{v}^\top \mathbf{u})\mathbf{v}^\top & 1 + \mathbf{v}^\top \mathbf{u} \end{pmatrix} = \begin{pmatrix} I_n & \mathbf{u} \\ \mathbf{0}^\top & 1 + \mathbf{v}^\top \mathbf{u} \end{pmatrix}.$$

By [Theorem 6.8.4](#), $\det L = \det I_n \cdot 1 = 1$, $\det L' = 1$, $\det M = \det(I_n + \mathbf{u}\mathbf{v}^\top) \cdot 1$, and $\det(LML') = \det I_n (1 + \mathbf{v}^\top \mathbf{u}) = 1 + \mathbf{v}^\top \mathbf{u}$. By [Theorem 6.5.4](#), $\det(LML') = \det L \det M \det L' = \det(I_n + \mathbf{u}\mathbf{v}^\top)$. This proves the lemma. ■

Example. (a on the Diagonal, 1 Elsewhere)

Let $a \in \mathbb{R}$ and $n \geq 1$, and let $A_n \in M_n(\mathbb{R})$ have every diagonal entry a and every other entry 1. Find $\det A_n$.

Solution. Let $\mathbf{1} = (1, \dots, 1) \in \mathbb{R}^n$. Every entry of $\mathbf{1}\mathbf{1}^\top$ is 1, so $A_n = (a-1)I_n + \mathbf{1}\mathbf{1}^\top$.

Case $a \neq 1$. Then $A_n = (a-1)(I_n + \mathbf{u}\mathbf{1}^\top)$ with $\mathbf{u} = \frac{1}{a-1}\mathbf{1}$. Scaling an $n \times n$ matrix by c multiplies its determinant by c^n (multilinearity in each of the n columns), so by [Theorem 6.8.5](#)

$$\det A_n = (a-1)^n \left(1 + \frac{n}{a-1}\right) = (a-1)^{n-1}(a-1+n),$$

since $\mathbf{1}^\top \mathbf{u} = \frac{n}{a-1}$.

Case $a = 1$. For $n \geq 2$, all columns are equal, so $\det A_n = 0$, which the formula also gives. For $n = 1$, $\det A_1 = 1$, and the formula gives $(a-1)^0 \cdot 1 = 1$.

So $\det A_n = (a-1)^{n-1}(a+n-1)$ for every a . For $n = 3$ this is $(a-1)^2(a+2)$; with $a = 3$ it gives $4 \cdot 5 = 20$, and sympy confirms the formula for $n \leq 5$.

Warning. The lemma is about $I_n + \mathbf{u}\mathbf{v}^\top$, not $\mathbf{u}\mathbf{v}^\top$ or $\mathbf{u}^\top \mathbf{v}$. The rank-one matrix $\mathbf{u}\mathbf{v}^\top$ is $n \times n$ and, for $n \geq 2$, has determinant 0; the scalar $\mathbf{v}^\top \mathbf{u}$ is its trace. Do not confuse $\det(I_n + \mathbf{u}\mathbf{v}^\top) = 1 + \mathbf{v}^\top \mathbf{u}$ with $\det(I_n) + \det(\mathbf{u}\mathbf{v}^\top) = 1$ (for $n \geq 2$), which is false whenever $\mathbf{v}^\top \mathbf{u} \neq 0$: for $\mathbf{u} = \mathbf{v} = \mathbf{e}_1$, $I_n + \mathbf{e}_1\mathbf{e}_1^\top = \text{diag}(2, 1, \dots, 1)$ has determinant 2.

6.8.5 Exercises

6.8.5.1 A. Check your understanding

Exercise A1.

- (a) State the Vandermonde determinant formula, including the order of the factors.
- (b) True or false: $V(x_1, \dots, x_n)$ is invertible whenever $x_1, \dots, x_n$ are non-zero. Justify your answer.
- (c) Write down the recurrence for the determinant of a tridiagonal matrix, and say which row the proof expands along.
- (d) True or false: $\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det A \det D$ whenever A and D are square and $B = 0$. Justify your answer.
- (e) True or false: $\det(I_n + \mathbf{u}\mathbf{v}^\top) = \det(I_n + \mathbf{v}\mathbf{u}^\top)$ for all $\mathbf{u}, \mathbf{v} \in F^n$. Justify your answer.

Solution.

- (a) $\det V(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_j - x_i)$, where V has (i, j) -entry x_i^{j-1} ([Theorem 6.8.1](#)).
- (b) False. $V(1, 1) = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ has non-zero nodes and determinant 0. The condition is that the nodes are **distinct** ([Corollary 6.8.2](#)); 0 is allowed as a node.
- (c) $D_k = a_k D_{k-1} - b_{k-1} c_{k-1} D_{k-2}$, with $D_0 = 1$, $D_1 = a_1$ ([Theorem 6.8.3](#)). The proof expands along the last row, which has only two non-zero entries.
- (d) True. With $B = 0$ the matrix is lower block triangular, and [Theorem 6.8.4](#) applies.
- (e) True. Both equal $1 + \mathbf{v}^\top \mathbf{u} = 1 + \mathbf{u}^\top \mathbf{v}$ by [Theorem 6.8.5](#), since $\mathbf{v}^\top \mathbf{u} = \sum_i v_i u_i = \mathbf{u}^\top \mathbf{v}$.

6.8.5.2 B. Practice

Exercise B1: A 4×4 Vandermonde determinant.

Compute $\det \begin{pmatrix} 1 & -1 & 1 & -1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \end{pmatrix}$. Hence decide whether there is exactly one polynomial $p \in \mathbb{R}[x]_{\leq 3}$ with $p(-1) = 3$, $p(0) = 1$, $p(1) = 0$, $p(2) = 5$.

Solution. Row i is $(1, x_i, x_i^2, x_i^3)$ with nodes $-1, 0, 1, 2$, so this is $V(-1, 0, 1, 2)$. By [Theorem 6.8.1](#) its determinant is

$$(0 - (-1))(1 - (-1))(2 - (-1))(1 - 0)(2 - 0)(2 - 1) = 1 \cdot 2 \cdot 3 \cdot 1 \cdot 2 \cdot 1 = 12.$$

(Check by expanding along row 2, which is $(1, 0, 0, 0)$: the determinant is $(-1)^{2+1} \det \begin{pmatrix} -1 & 1 & -1 \\ 1 & 1 & 1 \\ 2 & 4 & 8 \end{pmatrix} = -(-1(8 - 4) - 1(8 - 2) - 1(4 - 2)) = -(-4 - 6 - 2) = 12$.) Hence the coefficient matrix of the interpolation system is invertible, and there is exactly one $p \in \mathbb{R}[x]_{\leq 3}$ with the

given values.

Exercise B2: The second-difference matrix.

Let $K_n \in M_n(\mathbb{R})$ have 2 on the diagonal, -1 directly above and below it, and 0 elsewhere. Prove that $\det K_n = n + 1$. Hence show that K_n is invertible for every n .

Solution. Here $a_k = 2$ and $b_{k-1}c_{k-1} = (-1)(-1) = 1$. By [Theorem 6.8.3](#), $D_k = 2D_{k-1} - D_{k-2}$ for $k \geq 2$, with $D_0 = 1$, $D_1 = 2$. We show $D_k = k + 1$ by strong induction on k ([Theorem 0.2.10](#)). It holds for $k = 0, 1$. If it holds for $k - 1$ and $k - 2$, then $D_k = 2k - (k - 1) = k + 1$. Hence $\det K_n = D_n = n + 1 \neq 0$, and K_n is invertible by [Theorem 6.5.2](#).

Exercise B3: A rank-one update.

Let $\mathbf{u} = (1, 2, 3)$ and $\mathbf{v} = (1, -1, 2)$ in $\mathbb{Q}^3$.

- (a) Use [Theorem 6.8.5](#) to find $\det(I_3 + \mathbf{u}\mathbf{v}^\top)$.
- (b) Write out $I_3 + \mathbf{u}\mathbf{v}^\top$ and check your answer by elimination.
- (c) Find all $t \in \mathbb{Q}$ for which $I_3 + t\mathbf{u}\mathbf{v}^\top$ is **not** invertible.

Solution.

- (a) $\mathbf{v}^\top \mathbf{u} = 1 - 2 + 6 = 5$, so $\det(I_3 + \mathbf{u}\mathbf{v}^\top) = 6$.

- (b) $\mathbf{u}\mathbf{v}^\top = \begin{pmatrix} 1 & -1 & 2 \\ 2 & -2 & 4 \\ 3 & -3 & 6 \end{pmatrix}$, so $I_3 + \mathbf{u}\mathbf{v}^\top = \begin{pmatrix} 2 & -1 & 2 \\ 2 & -1 & 4 \\ 3 & -3 & 7 \end{pmatrix}$. Subtract row 1 from row 2, which does not change the determinant ([Theorem 6.5.1](#)), to get row 2 equal to $(0, 0, 2)$. Expanding along that row,

$$\det = 2 \cdot (-1)^{2+3} \det \begin{pmatrix} 2 & -1 \\ 3 & -3 \end{pmatrix} = -2(-6 + 3) = 6.$$

- (c) $I_3 + t\mathbf{u}\mathbf{v}^\top = I_3 + (t\mathbf{u})\mathbf{v}^\top$, whose determinant is $1 + t\mathbf{v}^\top \mathbf{u} = 1 + 5t$. It is zero exactly for $t = -\frac{1}{5}$, and by [Theorem 6.5.2](#) that is the only t for which the matrix is not invertible.

6.8.5.3 C. Going deeper

Exercise C1: The Cauchy determinant.

Let $x_1, \dots, x_n, y_1, \dots, y_n \in F$ with $x_i + y_j \neq 0$ for all i, j , and let $Q_n \in M_n(F)$ have (i, j) -entry $\frac{1}{x_i + y_j}$. The goal is

$$\det Q_n = \frac{\prod_{1 \leq i < j \leq n} (x_j - x_i)(y_j - y_i)}{\prod_{i=1}^n \prod_{j=1}^n (x_i + y_j)}.$$

- (a) For $i < n$, subtract row n from row i . Show that the new (i, j) -entry is $\frac{x_n - x_i}{(x_i + y_j)(x_n + y_j)}$, and deduce that $\det Q_n = \frac{\prod_{i < n} (x_n - x_i)}{\prod_j (x_n + y_j)} \det Q'$, where Q' agrees with Q_n in rows $1, \dots, n-1$ and has last row $(1, \dots, 1)$.
- (b) In Q' , for $j < n$, subtract column n from column j . Deduce that $\det Q' = \frac{\prod_{j < n} (y_n - y_j)}{\prod_{i < n} (x_i + y_n)} \det Q_{n-1}$, where Q_{n-1} is built from $x_1, \dots, x_{n-1}$ and $y_1, \dots, y_{n-1}$.
- (c) Prove the formula by induction on n .
- (d) The **Hilbert matrix** H_n has (i, j) -entry $\frac{1}{i+j-1}$. Show that it is a Cauchy matrix, and use (c) to compute $\det H_3$.

Hint: in (a) and (b), factor scalars out of whole rows and columns using multilinearity.

Solution.

- (a) For $i < n$, $\frac{1}{x_i + y_j} - \frac{1}{x_n + y_j} = \frac{(x_n + y_j) - (x_i + y_j)}{(x_i + y_j)(x_n + y_j)} = \frac{x_n - x_i}{(x_i + y_j)(x_n + y_j)}$. These row operations do not change the determinant ([Theorem 6.5.1](#)). In the new matrix, row $i < n$ has the factor $x_n - x_i$ in every entry, and column j has the factor $\frac{1}{x_n + y_j}$ in every entry (in row n , $\frac{1}{x_n + y_j} = \frac{1}{x_n + y_j} \cdot 1$). Pulling these scalars out of each row and column by multilinearity ([Theorem 6.5.1](#), in its row and column forms) leaves the matrix Q' with entries $\frac{1}{x_i + y_j}$ in rows $i < n$ and 1 in row n . This gives the stated formula.
- (b) For $j < n$, subtracting column n from column j leaves the determinant unchanged, makes the last row $(0, \dots, 0, 1)$, and for $i < n$ gives $\frac{1}{x_i + y_j} - \frac{1}{x_i + y_n} = \frac{y_n - y_j}{(x_i + y_j)(x_i + y_n)}$. Pull $y_n - y_j$ out of column j ($j < n$) and $\frac{1}{x_i + y_n}$ out of row i ($i < n$); the last row $(0, \dots, 0, 1)$ is unaffected by the column factors because its first $n-1$ entries are 0, and row factors do not touch row n . What remains has $\frac{1}{x_i + y_j}$ in positions $i, j < n$, entries $(x_i + y_n) \cdot \frac{1}{x_i + y_n} = 1$ in column n above the corner, and last row $(0, \dots, 0, 1)$. Expanding along the last row ([Theorem 6.6.1](#)) gives $\det Q_{n-1}$, with sign $(-1)^{2n} = 1$.
- (c) For $n = 1$, $\det Q_1 = \frac{1}{x_1 + y_1}$, which is the formula with empty products upstairs. For $n \geq 2$, combining (a) and (b),

$$\det Q_n = \frac{\prod_{i < n} (x_n - x_i)(y_n - y_i)}{\prod_{j \leq n} (x_n + y_j) \prod_{i < n} (x_i + y_n)} \det Q_{n-1}.$$

The numerator supplies exactly the factors of $\prod_{i < j} (x_j - x_i)(y_j - y_i)$ with $j = n$, and the denominator supplies exactly the factors $x_i + y_j$ with $\max(i, j) = n$, each once. By the induction hypothesis, $\det Q_{n-1}$ supplies all the remaining factors. This proves the formula.

- (d) With $x_i = i$ and $y_j = j-1$, $x_i + y_j = i+j-1 \geq 1$, so H_n is a Cauchy matrix over $\mathbb{Q}$. For $n = 3$, the numerator is $((2-1)(3-1)(3-2))^2 = 4$, since $y_j - y_i = j - i = x_j - x_i$. The denominator is the product of all $i+j-1$: the rows give $1 \cdot 2 \cdot 3 = 6$, $2 \cdot 3 \cdot 4 = 24$, $3 \cdot 4 \cdot 5 = 60$, in total 8640. Hence $\det H_3 = \frac{4}{8640} = \frac{1}{2160}$, which sympy confirms. (In general $\det H_n = c_n^4 / c_{2n}$ with $c_n = \prod_{k=1}^{n-1} k!$; we state this without proof. These determinants shrink extremely fast, which is why Hilbert matrices are a standard test of numerical methods.)

Exercise C2: A circulant determinant over $\mathbb{C}$.

Let $a, b, c \in \mathbb{C}$, let $\omega = e^{2\pi i/3}$, and let

$$Z = \begin{pmatrix} a & b & c \\ c & a & b \\ b & c & a \end{pmatrix}.$$

- (a) For $k = 0, 1, 2$, let $\mathbf{w}_k = (1, \omega^k, \omega^{2k})$ and $\lambda_k = a + b\omega^k + c\omega^{2k}$. Show that $Z\mathbf{w}_k = \lambda_k\mathbf{w}_k$.
- (b) Let W be the matrix with columns $\mathbf{w}_0, \mathbf{w}_1, \mathbf{w}_2$. Show that $ZW = W \operatorname{diag}(\lambda_0, \lambda_1, \lambda_2)$ and that $\det W \neq 0$. Deduce $\det Z = \lambda_0\lambda_1\lambda_2$.
- (c) Using $\omega^3 = 1$ and $1 + \omega + \omega^2 = 0$, deduce $\det Z = a^3 + b^3 + c^3 - 3abc$, and check this against the Leibniz formula.

Circulant matrices of every size are treated in Chapter 11.

Solution.

- (a) By [Example \(Roots of Unity\)](#), $\omega^3 = 1$, so $\omega^{-k} = \omega^{2k}$ and $\omega^{-2k} = \omega^k$. Row 1 of $Z\mathbf{w}_k$ is $a + b\omega^k + c\omega^{2k} = \lambda_k$. Row 2 is $c + a\omega^k + b\omega^{2k} = \omega^k(c\omega^{-k} + a + b\omega^k) = \omega^k(a + b\omega^k + c\omega^{2k}) = \omega^k\lambda_k$. Row 3 is $b + c\omega^k + a\omega^{2k} = \omega^{2k}(b\omega^{-2k} + c\omega^{-k} + a) = \omega^{2k}\lambda_k$. Hence $Z\mathbf{w}_k = \lambda_k\mathbf{w}_k$.
- (b) By [Theorem 0.9.2](#), column k of ZW is $Z\mathbf{w}_k = \lambda_k\mathbf{w}_k$, which is column k of $W \operatorname{diag}(\lambda_0, \lambda_1, \lambda_2)$. The (i, k) -entry of W is $(\omega^k)^{i-1}$, so $W = V(1, \omega, \omega^2)^\top$. The nodes $1, \omega, \omega^2$ are distinct (they are $e^{2\pi i k/3}$ for $k = 0, 1, 2$), so $\det W = \det V(1, \omega, \omega^2)^\top \neq 0$ by [Theorem 6.4.5](#) and [Corollary 6.8.2](#). By [Theorem 6.5.4](#) and [Theorem 6.4.6](#), $\det Z \det W = \det W \lambda_0\lambda_1\lambda_2$, and canceling $\det W \neq 0$ gives $\det Z = \lambda_0\lambda_1\lambda_2$.
- (c) Multiply out $\lambda_1\lambda_2 = (a + b\omega + c\omega^2)(a + b\omega^2 + c\omega)$. The nine terms are $a^2, ab\omega^2 + ab\omega, ac\omega + ac\omega^2, b^2\omega^3, bc\omega^2, bc\omega^4 = bc\omega, c^2\omega^3$. Using $\omega^3 = 1$ and $\omega + \omega^2 = -1$,

$$\lambda_1\lambda_2 = a^2 + b^2 + c^2 - ab - ac - bc.$$

Then $\det Z = (a + b + c)(a^2 + b^2 + c^2 - ab - ac - bc) = a^3 + b^3 + c^3 - 3abc$, expanding and canceling. The Leibniz formula for Z gives the three even terms $a \cdot a \cdot a, c \cdot c \cdot c$ (from $\sigma = (1 \ 2 \ 3)$) and $b \cdot b \cdot b$, and the three odd terms, each equal to abc with sign $-$: the same answer.

Exercise C3: Rank-one updates of an invertible matrix.

Let $A \in M_n(F)$ be invertible and $\mathbf{u}, \mathbf{v} \in F^n$.

- (a) Prove that $\det(A + \mathbf{u}\mathbf{v}^\top) = \det A (1 + \mathbf{v}^\top A^{-1}\mathbf{u})$.
- (b) Deduce that $A + \mathbf{u}\mathbf{v}^\top$ is invertible if and only if $\mathbf{v}^\top A^{-1}\mathbf{u} \neq -1$.
- (c) Give an example over $\mathbb{Q}$ with $n = 2$ where A is invertible, $\mathbf{u}\mathbf{v}^\top \neq 0$, and $A + \mathbf{u}\mathbf{v}^\top$ is **not** invertible.

Solution.

- (a) Since A is invertible, $A + \mathbf{u}\mathbf{v}^\top = A(I_n + A^{-1}\mathbf{u}\mathbf{v}^\top) = A(I_n + (A^{-1}\mathbf{u})\mathbf{v}^\top)$ by [Theo-](#)

rem 0.9.4. By [Theorem 6.5.4](#) and [Theorem 6.8.5](#) applied to the vectors $A^{-1}\mathbf{u}$ and $\mathbf{v}$, $\det(A + \mathbf{u}\mathbf{v}^T) = \det A (1 + \mathbf{v}^T A^{-1}\mathbf{u})$.

(b) $\det A \neq 0$ by [Theorem 6.5.2](#), so $\det(A + \mathbf{u}\mathbf{v}^T) \neq 0$ exactly when $1 + \mathbf{v}^T A^{-1}\mathbf{u} \neq 0$. Apply [Theorem 6.5.2](#) again.

(c) Take $A = I_2$, $\mathbf{u} = (-1, 0)$, $\mathbf{v} = (1, 0)$. Then $\mathbf{v}^T \mathbf{u} = -1$, and indeed $A + \mathbf{u}\mathbf{v}^T = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ is not invertible.

6.9 Orientation and Volume

The chapter began in §1 with a picture: the area of a parallelogram, with a sign attached, obeys a few simple rules, and those rules pinned down $ad - bc$. We have since built the determinant in every size from the same rules. It is time to return to geometry and say precisely what the determinant measures. There are two halves to the answer. Its absolute value measures volume, and its sign measures orientation, the difference between a left hand and a right hand. Both halves need care. “Volume” in $\mathbb{R}^7$ has no meaning until someone defines it, and “orientation” has no meaning at all over $\mathbb{C}$.

Throughout this section the field is $\mathbb{R}$.

6.9.1 What should the volume of a parallelepiped be?

For vectors $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^n$, the **parallelepiped** they span is the set

$$P(\mathbf{a}_1, \dots, \mathbf{a}_n) := \{t_1\mathbf{a}_1 + \dots + t_n\mathbf{a}_n : 0 \leq t_i \leq 1 \text{ for every } i\}.$$

For $n = 2$ it is the parallelogram with sides $\mathbf{a}_1, \mathbf{a}_2$ at the origin, and for $n = 3$ a slanted box. For $n = 1$ it is the segment from 0 to $\mathbf{a}_1$.

We would like to say “its volume is $|\det(\mathbf{a}_1 \ \dots \ \mathbf{a}_n)|$ ”. But a theorem needs two sides, and one side is missing: we have no definition of the volume of a subset of $\mathbb{R}^n$. School geometry supplies one for $n = 2$ and $n = 3$ through “base times height”, and the proper general notion, Lebesgue measure, belongs to a course in measure theory. So we proceed honestly in three steps. We **define** the volume of a parallelepiped by the determinant. We **prove** that this is the only function obeying the rules that any reasonable notion of volume must obey. And we **state**, without proof, that it agrees with Lebesgue measure.

The volume of a parallelepiped is the size of the determinant of its edge vectors.

Definition 6.9.1 — Volume of a Parallelepiped

Let $n \geq 1$ and $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^n$. The (n -dimensional) **volume** of the parallelepiped spanned by $\mathbf{a}_1, \dots, \mathbf{a}_n$ is

$$\text{vol}(\mathbf{a}_1, \dots, \mathbf{a}_n) := |\det(\mathbf{a}_1 \ \mathbf{a}_2 \ \dots \ \mathbf{a}_n)| \geq 0.$$

In words: stack the n edge vectors as the columns of a square matrix, take its determinant, and forget the sign. There must be **exactly** n vectors in $\mathbb{R}^n$, so that the matrix is square.

Examples.

- The unit cube $P(\mathbf{e}_1, \dots, \mathbf{e}_n)$ has volume $|\det I_n| = 1$.
- A box with edges $d_1\mathbf{e}_1, \dots, d_n\mathbf{e}_n$ has volume $|\det \text{diag}(d_1, \dots, d_n)| = |d_1 \cdots d_n|$ by [Theorem 6.4.6](#): the product of its side lengths, as it should be.
- The parallelepiped spanned by $(1, 2, 0)$, $(0, 1, 1)$, $(1, 0, 3)$ in $\mathbb{R}^3$ has volume

$$\left| \det \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 3 \end{pmatrix} \right| = |1 \cdot (3 - 0) - 0 + 1 \cdot (2 - 0)| = 5,$$

expanding along the first row.

- **Degenerate case.** If $\mathbf{a}_1, \dots, \mathbf{a}_n$ are linearly dependent, the parallelepiped is flattened into a proper subspace, and its volume is 0 by [Theorem 6.2.1](#). For $n = 2$, $P((1, 2), (2, 4))$ is the segment from $(0, 0)$ to $(3, 6)$, with area $|4 - 4| = 0$. This case matters: “volume zero” is the geometric face of “not invertible”.

Non-example by minimal change. Drop one vector: $(1, 2, 0)$ and $(0, 1, 1)$ span a genuine parallelogram in $\mathbb{R}^3$, but the matrix $\begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{pmatrix}$ is 3×2 and has no determinant. The clause that fails is **exactly n vectors in $\mathbb{R}^n$** . The 3-dimensional volume of that flat parallelogram would be 0, and its area needs a different formula ([Exercise \(C1: Area of a triangle in space\)](#)).

Why this definition. Reordering the vectors multiplies the determinant by a sign ([Theorem 6.2.1](#)), which the absolute value removes, so the volume does not depend on the order in which the edges are listed. The absolute value is also what makes volume non-negative. The sign we threw away is not garbage: it is the orientation, the subject of the second half of this section.

Warning. **Volume is not linear.** Scaling **one** edge by 2 doubles the volume, but scaling the whole parallelepiped by 2 multiplies it by 2^n : $\text{vol}(2\mathbf{a}_1, \dots, 2\mathbf{a}_n) = |\det(2A)| = 2^n \text{vol}(\mathbf{a}_1, \dots, \mathbf{a}_n)$, where A has columns $\mathbf{a}_1, \dots, \mathbf{a}_n$. A cube of side 2 in $\mathbb{R}^3$ has volume 8, not 2; this failure persists even with the sign kept, since $\det$ is linear in each edge separately, not in all edges at once. The absolute value also destroys additivity in a single edge: $\text{vol}(\mathbf{a}_1 + \mathbf{b}_1, \mathbf{a}_2, \dots)$ is in general not $\text{vol}(\mathbf{a}_1, \mathbf{a}_2, \dots) + \text{vol}(\mathbf{b}_1, \mathbf{a}_2, \dots)$: with $\mathbf{a}_1 = \mathbf{e}_1$, $\mathbf{b}_1 = -\mathbf{e}_1$ in $\mathbb{R}^2$ and $\mathbf{a}_2 = \mathbf{e}_2$, the left side is 0 and the right side is 2.

6.9.2 Volume is forced by its rules

Why should anyone accept [Definition 6.9.1](#)? Because of what any notion of volume of parallelepipeds must satisfy. The pictures in §1 showed three facts about area, and they have obvious n -dimensional versions: stretching one edge by a factor c multiplies the volume by $|c|$; sliding one edge parallel to another (a shear) does not change the volume; and the unit cube has volume 1. The next theorem says that these three rules alone determine the volume completely. So any definition that obeys them, whether from school geometry or from measure theory, must agree with ours.

Theorem 6.9.1 — Volume Is Determined by Its Rules

Let $n \geq 1$, and let g be a function assigning a real number $g(\mathbf{a}_1, \dots, \mathbf{a}_n)$ to every list of n vectors in $\mathbb{R}^n$. Suppose that for every list, every $i \neq j$ and every $c \in \mathbb{R}$:

- (V1) $g(\mathbf{a}_1, \dots, c\mathbf{a}_i, \dots, \mathbf{a}_n) = |c| g(\mathbf{a}_1, \dots, \mathbf{a}_i, \dots, \mathbf{a}_n)$ (only slot i changes);
 (V2) $g(\mathbf{a}_1, \dots, \mathbf{a}_i + c\mathbf{a}_j, \dots, \mathbf{a}_n) = g(\mathbf{a}_1, \dots, \mathbf{a}_i, \dots, \mathbf{a}_n)$ (only slot i changes);
 (V3) $g(\mathbf{e}_1, \dots, \mathbf{e}_n) = 1$.

Then $g(\mathbf{a}_1, \dots, \mathbf{a}_n) = \text{vol}(\mathbf{a}_1, \dots, \mathbf{a}_n)$ for every list. Moreover, vol itself satisfies (V1)–(V3).

Idea. Think of g as a function of the matrix M with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$. ① If the columns are dependent, shear one column to $\mathbf{0}$ and scale by 0: $g = 0$. ② Each elementary column operation changes g by the factor $|\det E|$. Scaling and shearing are (V1) and (V2); a swap is not among the rules, but three shears and a sign change perform it. ③ An invertible M is a product of elementary matrices, so peel them off one at a time until only I is left.

Proof. For $M \in M_n(\mathbb{R})$ with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$, write $h(M) = g(\mathbf{a}_1, \dots, \mathbf{a}_n)$.

Step 1: dependent columns give $h(M) = 0$. Suppose the columns are linearly dependent. By Theorem 1.4.1, some $\mathbf{a}_i = c_1\mathbf{a}_1 + \dots + c_{i-1}\mathbf{a}_{i-1}$ (for $i = 1$, $\mathbf{a}_1 = \mathbf{0}$). Applying (V2) to slot i with the scalars $-c_1, \dots, -c_{i-1}$, one at a time, does not change g , and it replaces $\mathbf{a}_i$ by $\mathbf{0}$ while the other slots stay fixed. Then (V1) with $c = 0$ gives $g(\dots, \mathbf{0}, \dots) = g(\dots, 0 \cdot \mathbf{0}, \dots) = 0 \cdot g(\dots, \mathbf{0}, \dots) = 0$. Hence $h(M) = 0$.

Step 2: $h(ME) = |\det E| h(M)$ for every elementary matrix E . By Theorem 2.4.1 and Theorem 0.9.5, $ME = (E^\top M^\top)^\top$, and $E^\top$ is the elementary matrix P_{ij} , $D_i(c)$ or $I + cE_{ji}$ when E is P_{ij} , $D_i(c)$ or $I + cE_{ij}$. So right multiplication by E performs a column operation: swap columns i and j ; multiply column i by $c \neq 0$; or add c times column i to column j . The determinants are $\det P_{ij} = -1$, $\det D_i(c) = c$ and $\det(I + cE_{ij}) = 1$ by Theorem 6.5.1.

- For $E = I + cE_{ij}$, (V2) gives $h(ME) = h(M) = |1| h(M)$.
- For $E = D_i(c)$, (V1) gives $h(ME) = |c| h(M)$.
- For $E = P_{ij}$, watch only slots i and j , holding $\mathbf{x}$ and $\mathbf{y}$. By (V2) three times, $(\mathbf{x}, \mathbf{y}) \rightarrow (\mathbf{x} + \mathbf{y}, \mathbf{y}) \rightarrow (\mathbf{x} + \mathbf{y}, -\mathbf{x}) \rightarrow (\mathbf{y}, -\mathbf{x})$ leaves g unchanged, where the second step subtracts slot i from slot j and the third adds slot j to slot i . By (V1) with $c = -1$ on slot j , g at $(\mathbf{y}, \mathbf{x})$ equals $|-1|$ times g at $(\mathbf{y}, -\mathbf{x})$. Hence $h(MP_{ij}) = h(M) = |-1| h(M)$.

Step 3: conclusion. If the columns of M are dependent, then $h(M) = 0$ by Step 1 and $\det M = 0$ by Theorem 6.2.1, so $h(M) = |\det M|$. Otherwise M is invertible by Theorem 2.4.4, hence $M = E_1 E_2 \dots E_k$ is a product of elementary matrices by the same theorem. Applying Step 2 k times, starting from I , and then (V3),

$$h(M) = h(IE_1 \dots E_k) = |\det E_1| \dots |\det E_k| h(I) = |\det(E_1 \dots E_k)| = |\det M|,$$

where the third equality uses Theorem 6.5.4 and $|xy| = |x||y|$. This proves $g = \text{vol}$.

Finally, vol satisfies (V1) because the determinant is linear in each column (Theo-

rem 6.4.3) and $|cd| = |c||d|$; (V2) by [Theorem 6.2.1](#); and (V3) because $\det I_n = 1$.

■

Agreement with area and volume from school. For $n = 2$, let $g(\mathbf{a}_1, \mathbf{a}_2)$ be the area of the parallelogram $P(\mathbf{a}_1, \mathbf{a}_2)$ in the sense of school geometry, base times height. The pictures of §1 are exactly (V1)–(V3) for this g : stretching a side by c stretches the base or the height by $|c|$; a shear $\mathbf{a}_1 \mapsto \mathbf{a}_1 + c\mathbf{a}_2$ slides the opposite side along a line parallel to the base $\mathbf{a}_2$, keeping base and height; and the unit square has area 1. Hence, by [Theorem 6.9.1](#), school area equals $|\det(\mathbf{a}_1 \ \mathbf{a}_2)|$. For $n = 3$ the same argument works with “area of base times height”. What the theorem cannot supply is the geometric input itself: it takes the three rules as given.

Agreement with Lebesgue measure. In measure theory, every reasonable subset $S \subseteq \mathbb{R}^n$ has an n -dimensional volume, its Lebesgue measure. One can prove that the Lebesgue measure of $P(\mathbf{a}_1, \dots, \mathbf{a}_n)$ satisfies (V1)–(V3), and hence equals $\text{vol}(\mathbf{a}_1, \dots, \mathbf{a}_n)$. We state this without proof; nothing in this book depends on it.

Quick check. Which of the rules (V1)–(V3) does the function $g(\mathbf{a}_1, \mathbf{a}_2) = |\det(\mathbf{a}_1 \ \mathbf{a}_2)|^2$ on $\mathbb{R}^2$ break?

Solution. It satisfies (V2), since shears do not change the determinant, and (V3), since $1^2 = 1$. It breaks (V1): $g(2\mathbf{e}_1, \mathbf{e}_2) = 4$, while $|2|g(\mathbf{e}_1, \mathbf{e}_2) = 2$. So the theorem does not apply, and indeed $g \neq \text{vol}$.

6.9.3 Linear maps scale volume

Applying a linear map to a parallelepiped gives another parallelepiped: by linearity, $T(\sum t_i \mathbf{a}_i) = \sum t_i T\mathbf{a}_i$, so $T(P(\mathbf{a}_1, \dots, \mathbf{a}_n)) = P(T\mathbf{a}_1, \dots, T\mathbf{a}_n)$. How does the volume change? Multiplicativity answers this in one line, and the answer does not depend on the parallelepiped.

Theorem 6.9.2 — Linear Maps Scale Volume by $|\det|$

Let $T \in \mathcal{L}(\mathbb{R}^n)$ and $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^n$. Then

$$\text{vol}(T\mathbf{a}_1, \dots, T\mathbf{a}_n) = |\det T| \cdot \text{vol}(\mathbf{a}_1, \dots, \mathbf{a}_n).$$

In particular, the image of the unit cube under T has volume $|\det T|$.

Proof. Let B be the matrix of T in the standard basis, so $T\mathbf{x} = B\mathbf{x}$ for all $\mathbf{x}$ ([Theorem 3.6.1](#)) and $\det T = \det B$ ([Definition 6.5.1](#)). Let $A = (\mathbf{a}_1 \ \dots \ \mathbf{a}_n)$. By [Theorem 0.9.2](#), $BA = (B\mathbf{a}_1 \ \dots \ B\mathbf{a}_n)$. By [Theorem 6.5.4](#),

$$\text{vol}(T\mathbf{a}_1, \dots, T\mathbf{a}_n) = |\det(BA)| = |\det B| |\det A| = |\det T| \text{vol}(\mathbf{a}_1, \dots, \mathbf{a}_n).$$

For the unit cube take $\mathbf{a}_i = \mathbf{e}_i$, with $\text{vol}(\mathbf{e}_1, \dots, \mathbf{e}_n) = 1$.

■

This is the meaning of the determinant of a real operator: **the factor by which it scales every volume**. An operator with $\det T = 0$ squashes $\mathbb{R}^n$ into a lower-dimensional subspace,

where n -dimensional volume is 0. In multivariable calculus the same fact, applied to the linear approximation of a smooth map, becomes the Jacobian factor $|\det J|$ in the change-of-variables formula for integrals.

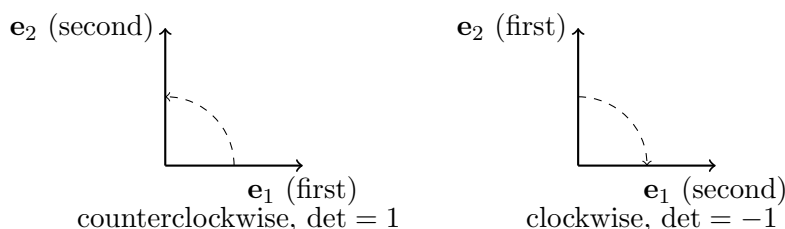
Example. (Area of an Image)

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be $T(x, y) = (3x + y, x + 2y)$. Find the area of the image of the unit square, and of the image of the parallelogram $P((1, 1), (-1, 2))$.

Solution. The standard matrix is $B = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$, with $\det T = 6 - 1 = 5$. By [Theorem 6.9.2](#) the image of the unit square, the parallelogram spanned by $(3, 1)$ and $(1, 2)$, has area 5. The parallelogram $P((1, 1), (-1, 2))$ has area $|2 + 1| = 3$, so its image has area $5 \cdot 3 = 15$. Check directly: $T(1, 1) = (4, 3)$ and $T(-1, 2) = (-1, 3)$, and $|\det \begin{pmatrix} 4 & -1 \\ 3 & 3 \end{pmatrix}| = |12 + 3| = 15$.

6.9.4 Orientation

Now the sign. In $\mathbb{R}^2$, the ordered basis $(\mathbf{e}_1, \mathbf{e}_2)$ turns **counterclockwise** from its first vector to its second, while $(\mathbf{e}_2, \mathbf{e}_1)$ turns clockwise. No rotation carries one arrangement to the other: a rotation of the plane keeps counterclockwise turns counterclockwise. The determinant sees the difference: $\det(\mathbf{e}_1 \ \mathbf{e}_2) = 1 > 0$ and $\det(\mathbf{e}_2 \ \mathbf{e}_1) = -1 < 0$.



To compare two bases of an arbitrary real space, we compare them with each other rather than with a fixed standard basis, using the change-of-coordinates matrix of Chapter 3.

Two bases have the same orientation when the matrix converting one into the other has positive determinant.

Definition 6.9.2 — Orientation

Let V be a vector space **over** $\mathbb{R}$ with $\dim V = n \geq 1$, and let $\mathcal{B}$ and $\mathcal{C}$ be **ordered** bases of V . We say $\mathcal{B}$ and $\mathcal{C}$ have the **same orientation** if

$$\det[\text{id}]_{\mathcal{B}}^{\mathcal{C}} > 0,$$

and **opposite orientations** if $\det[\text{id}]_{\mathcal{B}}^{\mathcal{C}} < 0$.

In words: write each vector of $\mathcal{B}$ in coordinates with respect to $\mathcal{C}$, put the coordinate columns side by side ([Definition 3.7.1](#)), and look at the **sign** of the determinant. Exactly one of the

two cases occurs, because $[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$ is invertible (Theorem 3.7.1 (c)), so its determinant is non-zero (Theorem 6.5.2), and a non-zero real number is either positive or negative. The bases must be **ordered**: reordering a basis can change the answer.

For $V = \mathbb{R}^n$ there is a shortcut. If B and C are the matrices whose columns are the vectors of $\mathcal{B}$ and $\mathcal{C}$, then $[\text{id}]_{\mathcal{B}}^{\mathcal{E}} = B$ and $[\text{id}]_{\mathcal{C}}^{\mathcal{E}} = C$ for the standard basis $\mathcal{E}$. By Theorem 3.7.1 (b) and (c), $[\text{id}]_{\mathcal{B}}^{\mathcal{C}} = C^{-1}B$. Since $\det C \det C^{-1} = \det I = 1$ (Theorem 6.5.4), $\det[\text{id}]_{\mathcal{B}}^{\mathcal{C}} = \det B / \det C$. So $\mathcal{B}$ and $\mathcal{C}$ have the same orientation if and only if $\det B$ and $\det C$ have the same sign.

Examples.

- In $\mathbb{R}^2$, $(\mathbf{e}_1, \mathbf{e}_2)$ and $(\mathbf{e}_2, \mathbf{e}_1)$ have opposite orientations: $\det I_2 = 1$ and $\det \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = -1$.
- In $\mathbb{R}^3$, $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ and $(\mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_1)$ have the **same** orientation. The second matrix is the permutation matrix of a 3-cycle, whose sign is $(-1)^{3-1} = 1$. A cyclic shift of three vectors is two swaps.
- In $V = \mathbb{R}[x]_{\leq 1}$, the bases $\mathcal{B} = (1, x)$ and $\mathcal{C} = (x, 1+x)$ have $1 = -x + (1+x)$ and $x = x + 0 \cdot (1+x)$, so $[\text{id}]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix}$, with determinant -1 . They have opposite orientations. Orientation makes sense in any real space, with no picture at all.
- **Degenerate case, $n = 1$.** Bases of $\mathbb{R}$ are single non-zero numbers (a) , and $[\text{id}]_{(a)}^{(b)} = (a/b)$. So (a) and (b) have the same orientation exactly when a and b have the same sign. The two orientations of a line are its two directions.

Non-example by minimal change. In $\mathbb{R}^2$, $(-\mathbf{e}_1, -\mathbf{e}_2)$ has the same orientation as $(\mathbf{e}_1, \mathbf{e}_2)$: $\det(-I_2) = 1$, and indeed it is the rotation of $(\mathbf{e}_1, \mathbf{e}_2)$ by a half-turn. Change only **one** sign, to $(\mathbf{e}_1, -\mathbf{e}_2)$: the determinant becomes -1 , and the orientation flips. What changed is the sign of the determinant, which is the whole content of the definition. In $\mathbb{R}^3$, by contrast, $-I_3$ has determinant -1 : negating all three vectors reverses orientation.

Why this definition. Using the change-of-coordinates matrix, rather than the matrices B and C above, makes the definition work in every real space, including spaces like $\mathbb{R}[x]_{\leq 1}$ that have no standard picture. Requiring $\det > 0$ rather than $\det = 1$ is what gives a coarse, two-valued answer: rescaling a basis vector by a positive number should not change its handedness.

The payoff is that “same orientation” really sorts all bases into two families, as the pictures suggest.

Theorem 6.9.3 — There Are Exactly Two Orientations

Let V be a vector space over $\mathbb{R}$ with $\dim V = n \geq 1$.

- “Having the same orientation” is an equivalence relation on the set of ordered bases of V .
- It has **exactly two** equivalence classes. If $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$, then $\mathcal{B}^- = (-\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ lies in the class not containing $\mathcal{B}$.

Idea. The three axioms are three facts about change-of-coordinates matrices: $[\text{id}]_{\mathcal{B}}^{\mathcal{B}} = I$; the matrix back is the inverse; and matrices compose along a chain $\mathcal{B} \rightarrow \mathcal{C} \rightarrow \mathcal{D}$. Determinants turn these into $1 > 0$, $1/\text{positive} > 0$ and $\text{positive} \cdot \text{positive} > 0$. For

“exactly two”, flipping one vector gives a second class, and “negative times negative is positive” shows there is no third.

Proof. Write $d(\mathcal{B}, \mathcal{C}) = \det[\text{id}]_{\mathcal{B}}^{\mathcal{C}}$. By [Theorem 3.7.1](#) (b) and [Theorem 6.5.4](#), for bases $\mathcal{B}, \mathcal{C}, \mathcal{D}$,

$$d(\mathcal{B}, \mathcal{D}) = d(\mathcal{C}, \mathcal{D}) d(\mathcal{B}, \mathcal{C}). \quad (*)$$

- (a) *Reflexive:* $[\text{id}]_{\mathcal{B}}^{\mathcal{B}} = I_n$, so $d(\mathcal{B}, \mathcal{B}) = 1 > 0$. *Symmetric:* by $(*)$ with $\mathcal{D} = \mathcal{B}$, $d(\mathcal{C}, \mathcal{B}) d(\mathcal{B}, \mathcal{C}) = d(\mathcal{B}, \mathcal{B}) = 1$, so $d(\mathcal{C}, \mathcal{B}) = 1/d(\mathcal{B}, \mathcal{C})$, which is positive when $d(\mathcal{B}, \mathcal{C})$ is. *Transitive:* if $d(\mathcal{B}, \mathcal{C}) > 0$ and $d(\mathcal{C}, \mathcal{D}) > 0$, then $d(\mathcal{B}, \mathcal{D}) > 0$ by $(*)$, since a product of positive reals is positive. By [Definition 0.5.2](#), this is an equivalence relation.
- (b) *At least two classes.* $\mathcal{B}^-$ is a basis, and $[\text{id}]_{\mathcal{B}^-}^{\mathcal{B}^-} = \text{diag}(-1, 1, \dots, 1)$, because $-\mathbf{v}_1 = (-1)\mathbf{v}_1$ and $\mathbf{v}_j = 1 \cdot \mathbf{v}_j$. Its determinant is -1 by [Theorem 6.4.6](#). So $d(\mathcal{B}^-, \mathcal{B}) < 0$, and $\mathcal{B}^-$ and $\mathcal{B}$ lie in different classes.

At most two classes. Let $\mathcal{C}$ and $\mathcal{D}$ be bases, neither of which has the same orientation as $\mathcal{B}$. Then $d(\mathcal{C}, \mathcal{B}) < 0$ and $d(\mathcal{D}, \mathcal{B}) < 0$, and by symmetry $d(\mathcal{B}, \mathcal{D}) < 0$ as well. By $(*)$, $d(\mathcal{C}, \mathcal{D}) = d(\mathcal{B}, \mathcal{D}) d(\mathcal{C}, \mathcal{B})$ is a product of two negative reals, hence positive. So all bases outside the class of $\mathcal{B}$ lie in one class, and there are exactly two classes. ■

Choosing one of the two classes is called **orienting** V ; its bases are then called **positively oriented**, and the others **negatively oriented**. On $\mathbb{R}^n$ the standard choice is the class of $(\mathbf{e}_1, \dots, \mathbf{e}_n)$. By the shortcut above, $(\mathbf{a}_1, \dots, \mathbf{a}_n)$ is positively oriented if and only if $\det(\mathbf{a}_1 \ \cdots \ \mathbf{a}_n) > 0$. So the determinant of a real square matrix carries two independent pieces of information: its absolute value is a volume, and its sign is an orientation.

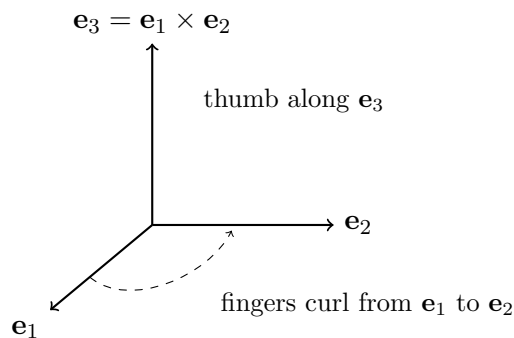
Warning. Orientation needs $\mathbb{R}$. The proof used two facts about the reals: every non-zero number is positive or negative, and products and inverses of positive numbers are positive. Over $\mathbb{C}$ there is no such splitting. Suppose we tried “ $\mathcal{B}$ and $\mathcal{C}$ have the same orientation if $d(\mathcal{B}, \mathcal{C})$ has positive real part”. In $\mathbb{C}^1$, with $z = e^{i\pi/3}$, the bases $(\mathbf{e}_1)$, $(z\mathbf{e}_1)$, $(z^2\mathbf{e}_1)$ give $d((\mathbf{e}_1), (z\mathbf{e}_1)) = z^{-1}$ and $d((z\mathbf{e}_1), (z^2\mathbf{e}_1)) = z^{-1}$, both with real part $\frac{1}{2} > 0$, but $d((\mathbf{e}_1), (z^2\mathbf{e}_1)) = z^{-2}$ has real part $-\frac{1}{2}$: transitivity fails. Over $\mathbb{F}_2$, $-1 = 1$, so the second basis $\mathcal{B}^-$ of the proof is $\mathcal{B}$ itself. The theorem holds verbatim over any field with a compatible order, such as $\mathbb{Q}$, but not over $\mathbb{C}$ or a finite field.

Quick check. Do $((1, 2), (3, 4))$ and $((3, 4), (1, 2))$ have the same orientation? Does $((1, 2), (3, 4))$ have the standard orientation of $\mathbb{R}^2$?

Solution. $\det \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} = 4 - 6 = -2$ and $\det \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix} = 6 - 4 = 2$. The signs differ, so the two bases have opposite orientations. The first has negative determinant, so it is negatively oriented: it is **not** in the class of $(\mathbf{e}_1, \mathbf{e}_2)$.

6.9.5 The right-hand rule

In $\mathbb{R}^3$, the standard orientation is the one described by the right hand: curl the fingers of the right hand from the first basis vector toward the second, and the thumb points along the third. For $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ with the usual picture of the axes, this is the familiar arrangement below.



The **cross product** of $\mathbf{a} = (a_1, a_2, a_3)$ and $\mathbf{b} = (b_1, b_2, b_3)$ in $\mathbb{R}^3$ is

$$\mathbf{a} \times \mathbf{b} := (a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1).$$

Physics defines it by the right-hand rule. Here is the linear-algebra reason the two descriptions match: $(\mathbf{a}, \mathbf{b}, \mathbf{a} \times \mathbf{b})$ is always positively oriented.

Expand $\det(\mathbf{a} \ \mathbf{b} \ \mathbf{a} \times \mathbf{b})$ along its third column (Theorem 6.6.1). The cofactors of that column are $C_{13} = a_2b_3 - a_3b_2$, $C_{23} = -(a_1b_3 - a_3b_1)$ and $C_{33} = a_1b_2 - a_2b_1$, which are exactly the three components of $\mathbf{a} \times \mathbf{b}$. Hence

$$\det(\mathbf{a} \ \mathbf{b} \ \mathbf{a} \times \mathbf{b}) = (a_2b_3 - a_3b_2)^2 + (a_3b_1 - a_1b_3)^2 + (a_1b_2 - a_2b_1)^2 \geq 0.$$

If $\mathbf{a}$ and $\mathbf{b}$ are linearly independent, the 3×2 matrix $(\mathbf{a} \ \mathbf{b})$ has rank 2, so one of its 2×2 minors, which are $\pm$ the components of $\mathbf{a} \times \mathbf{b}$, is non-zero (Theorem 6.7.1). Then the determinant is strictly positive, so $(\mathbf{a}, \mathbf{b}, \mathbf{a} \times \mathbf{b})$ is a basis (Theorem 6.5.2) with the standard orientation. For example, $\mathbf{e}_1 \times \mathbf{e}_2 = (0, 0, 1) = \mathbf{e}_3$, matching the picture.

6.9.6 Orientation-preserving maps

An invertible operator carries each basis to a basis. Does it keep the orientation? The answer is the same for every basis and depends only on the sign of $\det T$.

Proposition 6.9.4 — Determinant Sign and Orientation

Let V be a vector space over $\mathbb{R}$ with $\dim V = n \geq 1$, let $T \in \mathcal{L}(V)$ be invertible, and let $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be any basis of V . Then $T\mathcal{B} = (T\mathbf{v}_1, \dots, T\mathbf{v}_n)$ is a basis of V , and

$$[\text{id}]_{T\mathcal{B}}^{\mathcal{B}} = [T]_{\mathcal{B}}.$$

Hence $\mathcal{B}$ and $T\mathcal{B}$ have the same orientation if and only if $\det T > 0$, and this does not depend on $\mathcal{B}$.

Proof. Since T is invertible, it is an isomorphism, so $T\mathcal{B}$ is a basis by [Theorem 3.5.2](#). By [Definition 3.7.1](#), column j of $[\text{id}]_{T\mathcal{B}}^{\mathcal{B}}$ is $[T\mathbf{v}_j]_{\mathcal{B}}$, which is column j of $[T]_{\mathcal{B}}$ by [Definition 3.6.1](#). Hence the two matrices are equal, and by [Definition 6.5.1](#), $\det[\text{id}]_{T\mathcal{B}}^{\mathcal{B}} = \det T$. By [Definition 6.9.2](#) and the symmetry in [Theorem 6.9.3 \(a\)](#), $\mathcal{B}$ and $T\mathcal{B}$ have the same orientation exactly when $\det T > 0$. ■

An invertible operator with $\det T > 0$ is **orientation-preserving**, and one with $\det T < 0$ is **orientation-reversing**. Rotations of the plane preserve orientation and reflections reverse it ([Exercise \(C2: Rotations and reflections\)](#)). In $\mathbb{R}^3$, the map $\mathbf{x} \mapsto -\mathbf{x}$ reverses orientation, since $\det(-I_3) = -1$: a right hand, pushed through the origin, becomes a left hand.

6.9.7 Exercises

6.9.7.1 A. Check your understanding

Exercise A1.

- State the definition of the volume of the parallelepiped spanned by $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^n$, and say what it is when the vectors are dependent.
- True or false: $\text{vol}(\mathbf{a}_1 + \mathbf{b}, \mathbf{a}_2, \dots, \mathbf{a}_n) = \text{vol}(\mathbf{a}_1, \dots, \mathbf{a}_n) + \text{vol}(\mathbf{b}, \mathbf{a}_2, \dots, \mathbf{a}_n)$. Justify your answer.
- Which three rules characterize volume in [Theorem 6.9.1](#)?
- True or false: if $\det T = -2$, then T doubles every volume. Justify your answer.
- Define “same orientation” for two ordered bases of a real vector space.
- True or false: for every $n \geq 1$, the map $\mathbf{x} \mapsto -\mathbf{x}$ on $\mathbb{R}^n$ reverses orientation. Justify your answer.

Solution.

- $\text{vol}(\mathbf{a}_1, \dots, \mathbf{a}_n) = |\det(\mathbf{a}_1 \ \cdots \ \mathbf{a}_n)|$ ([Definition 6.9.1](#)). For dependent vectors it is 0, by [Theorem 6.2.1](#).
- False. With $n = 2$, $\mathbf{a}_1 = \mathbf{e}_1$, $\mathbf{b} = -\mathbf{e}_1$, $\mathbf{a}_2 = \mathbf{e}_2$: the left side is $\text{vol}(\mathbf{0}, \mathbf{e}_2) = 0$, the right side is $1 + 1 = 2$. The absolute value destroys additivity.
- (V1) scaling one vector by c scales the volume by $|c|$; (V2) adding a multiple of one vector to another does not change it; (V3) the unit cube has volume 1.
- True. By [Theorem 6.9.2](#), $\text{vol}(T\mathbf{a}_1, \dots, T\mathbf{a}_n) = |-2| \text{vol}(\mathbf{a}_1, \dots, \mathbf{a}_n)$. The sign only affects orientation.
- $\mathcal{B}$ and $\mathcal{C}$ have the same orientation if $\det[\text{id}]_{\mathcal{B}}^{\mathcal{C}} > 0$ ([Definition 6.9.2](#)).
- False. $\det(-I_n) = (-1)^n$, so by [Proposition 6.9.4](#) the map reverses orientation for odd n and preserves it for even n . In $\mathbb{R}^2$ it is the rotation by a half-turn.

6.9.7.2 B. Practice

Exercise B1: Volume of a parallelepiped.

Find the volume of the parallelepiped in $\mathbb{R}^3$ spanned by $(2, 1, 0)$, $(1, 3, 1)$ and $(0, 1, 1)$. Hence find the volume of its image under the map $T(x, y, z) = (x + y, y + z, 2z)$.

Solution. Expanding along the first row,

$$\det \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 1 \end{pmatrix} = 2(3 - 1) - 1 \cdot (1 - 0) + 0 = 3,$$

so the volume is 3. The standard matrix of T is $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$, upper triangular with determinant 2 (Theorem 6.4.6). By Theorem 6.9.2 the image has volume $2 \cdot 3 = 6$.

Exercise B2: Comparing orientations.

In $\mathbb{R}^3$, let $\mathcal{B} = ((1, 1, 0), (0, 1, 1), (1, 0, 1))$, $\mathcal{C} = ((0, 1, 0), (1, 0, 0), (1, 1, 1))$ and $\mathcal{D} = ((1, 0, 1), (1, 1, 0), (0, 1, 1))$. Determine which pairs have the same orientation, and which are positively oriented. Justify your answer.

Solution. By the shortcut after Definition 6.9.2, compare signs of the determinants of the matrices with these columns. Expanding along the first row:

$$\det \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} = 1 \cdot (1 - 0) - 0 + 1 \cdot (1 - 0) = 2, \quad \det \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} = -1,$$

the second by expanding along the last row: $1 \cdot \det \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = -1$. The list $\mathcal{D}$ is $\mathcal{B}$ shifted cyclically, $(\mathbf{b}_3, \mathbf{b}_1, \mathbf{b}_2)$; moving the last column to the front takes two column swaps, so by Theorem 6.2.1 its determinant is $(-1)^2 \cdot 2 = 2$. Hence $\mathcal{B}$ and $\mathcal{D}$ are positively oriented and have the same orientation; $\mathcal{C}$ is negatively oriented, opposite to both.

Exercise B3: Areas of images.

Let $T(x, y) = (2x + y, -x + 3y)$ on $\mathbb{R}^2$.

- Find the area of the image of the unit square under T .
- Find the area of the image of the parallelogram $P((1, 0), (1, 2))$, first by Theorem 6.9.2 and then directly.
- Is T orientation-preserving?

Solution.

- The standard matrix $\begin{pmatrix} 2 & 1 \\ -1 & 3 \end{pmatrix}$ has determinant $6 + 1 = 7$, so the image of the unit square has area 7.

- (b) $\text{vol}((1, 0), (1, 2)) = |\det \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}| = 2$, so the image has area $7 \cdot 2 = 14$. Directly,
 $T(1, 0) = (2, -1)$ and $T(1, 2) = (4, 5)$, and $|\det \begin{pmatrix} 2 & 4 \\ -1 & 5 \end{pmatrix}| = |10 + 4| = 14$.
- (c) Yes: $\det T = 7 > 0$, so by [Proposition 6.9.4](#) T preserves orientation.

6.9.7.3 C. Going deeper

Exercise C1: Area of a triangle in space.

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$, let $A = (\mathbf{u} \ \mathbf{v}) \in M_{3 \times 2}(\mathbb{R})$. We **define** the area of the parallelogram $P(\mathbf{u}, \mathbf{v}) \subseteq \mathbb{R}^3$ to be $\sqrt{\det(A^T A)}$; Chapter 10 justifies this with inner products.

- (a) Show that $\det(A^T A) \geq 0$, so the square root exists, and that $\det(A^T A) = (u_2 v_3 - u_3 v_2)^2 + (u_3 v_1 - u_1 v_3)^2 + (u_1 v_2 - u_2 v_1)^2$.
- (b) Show that if $\mathbf{u}$ and $\mathbf{v}$ lie in the plane $z = 0$, this area equals the area $\text{vol}((u_1, u_2), (v_1, v_2))$ of [Definition 6.9.1](#) in $\mathbb{R}^2$.
- (c) Find the area of the triangle with vertices $(1, 0, 0)$, $(0, 2, 0)$, $(0, 0, 3)$.

Hint: apply [Corollary 6.7.3](#) to A^T .

Solution.

- (a) $A^T \in M_{2 \times 3}(\mathbb{R})$ and $(A^T)^T = A$, so by [Corollary 6.7.3](#) applied to A^T , $\det(A^T A) = \sum_{|S|=2} (\det(A^T)_{[2], S})^2 \geq 0$. The 2×2 minors of $A^T = \begin{pmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{pmatrix}$ are $u_1 v_2 - u_2 v_1$, $u_1 v_3 - u_3 v_1$, $u_2 v_3 - u_3 v_2$ for $S = \{1, 2\}, \{1, 3\}, \{2, 3\}$; squaring removes the signs, giving the stated sum.
- (b) If $u_3 = v_3 = 0$, two of the three squares vanish, and $\det(A^T A) = (u_1 v_2 - u_2 v_1)^2$. Hence $\sqrt{\det(A^T A)} = |u_1 v_2 - u_2 v_1| = |\det \begin{pmatrix} u_1 & v_1 \\ u_2 & v_2 \end{pmatrix}|$, which is $\text{vol}((u_1, u_2), (v_1, v_2))$.
- (c) The triangle is half of the parallelogram spanned by $\mathbf{u} = (0, 2, 0) - (1, 0, 0) = (-1, 2, 0)$ and $\mathbf{v} = (0, 0, 3) - (1, 0, 0) = (-1, 0, 3)$. By (a), $\det(A^T A) = (6 - 0)^2 + (0 + 3)^2 + (0 + 2)^2 = 36 + 9 + 4 = 49$. As a check, $A^T A = \begin{pmatrix} 5 & 1 \\ 1 & 10 \end{pmatrix}$ has determinant $50 - 1 = 49$. So the parallelogram has area 7, and the triangle has area $\frac{7}{2}$.

Exercise C2: Rotations and reflections.

For $\theta \in \mathbb{R}$, let $R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ (rotation by θ) and $S_\theta = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$ (reflection in the line through 0 at angle θ).

- (a) Show that every rotation preserves orientation and area, and every reflection reverses orientation and preserves area.
- (b) Show that $S_\theta S_\varphi$ is orientation-preserving for all θ, φ . (It is in fact a rotation, but you need not show this.)

- (c) Prove that no $M \in M_2(\mathbb{R})$ satisfies $M^2 = S_0 = \text{diag}(1, -1)$. Does the same argument rule out $M^2 = -I_2$?

Solution.

- (a) $\det R_\theta = \cos^2 \theta + \sin^2 \theta = 1 > 0$ and $\det S_\theta = -\cos^2 2\theta - \sin^2 2\theta = -1 < 0$. By [Proposition 6.9.4](#), rotations preserve and reflections reverse orientation. Both have $|\det| = 1$, so both preserve area by [Theorem 6.9.2](#).
- (b) By [Theorem 6.5.4](#), $\det(S_\theta S_\varphi) = (-1)(-1) = 1 > 0$.
- (c) If $M^2 = S_0$, then $(\det M)^2 = \det S_0 = -1$ by [Theorem 6.5.4](#), impossible for a real number $\det M$. The argument says nothing about $-I_2$, since $\det(-I_2) = 1$; and indeed $R_{\pi/2}^2 = R_{\pi/2} R_{\pi/2} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^2 = -I_2$. An orientation-reversing map can never be the square of a real map, since the square of anything preserves orientation.

6.10 The Characteristic Polynomial

For a square matrix A over F and a scalar $c \in F$, when is $cI - A$ invertible? The question matters: $cI - A$ fails to be invertible exactly when $A\mathbf{v} = c\mathbf{v}$ for some non-zero vector $\mathbf{v}$, the situation Chapter 8 is built around. The determinant answers the question for each single c . This section lets c vary. Replacing the number c by a variable x turns $\det(cI - A)$ into a polynomial, the characteristic polynomial, and the scalars we are looking for become its roots. The price is that $xI - A$ is a matrix of **polynomials**, not of field elements, so we must check which determinant facts survive there. The remarks at the ends of §4, §5 and §6 did exactly that bookkeeping, and here we cash them in.

6.10.1 From a number to a variable

Take $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \in M_2(\mathbb{Q})$. For each $c \in \mathbb{Q}$,

$$\det(cI - A) = \det \begin{pmatrix} c-1 & -2 \\ -3 & c-4 \end{pmatrix} = (c-1)(c-4) - 6 = c^2 - 5c - 2.$$

The same expression $c^2 - 5c - 2$ appears whatever c is. It is the value at c of one polynomial, and that polynomial deserves a name. To define it directly, without first choosing c , we put the variable x into the matrix and take the determinant of

$$xI - A = \begin{pmatrix} x-1 & -2 \\ -3 & x-4 \end{pmatrix},$$

a matrix whose entries lie in $F[x]$.

Is that allowed? $F[x]$ is not a field, but it is a **commutative ring** in the sense of the remark in §4: all the field axioms hold except the existence of inverses ([Theorem 0.8.1](#)). The Leibniz formula of [Definition 6.4.1](#) uses only sums, differences and products of entries, so it defines $\det M \in F[x]$ for every $M \in M_n(F[x])$.

The characteristic polynomial of A is the determinant of $xI - A$, computed with polynomial entries.

Definition 6.10.1 — Characteristic Polynomial

Let $n \geq 1$ and $A = (a_{ij}) \in M_n(F)$. The **characteristic polynomial** of A is

$$p_A(x) := \det(xI_n - A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^n (x \delta_{\sigma(j)j} - a_{\sigma(j)j}) \in F[x],$$

where $\delta_{ij} = 1$ if $i = j$ and 0 otherwise, so that $x\delta_{ij} - a_{ij}$ is the (i, j) -entry of $xI_n - A$. The determinant is the Leibniz sum of Definition 6.4.1, computed in the commutative ring $F[x]$.

In words: subtract A from x times the identity, so that x appears only on the diagonal; then take the determinant exactly as for a matrix of numbers, multiplying and adding polynomials instead of scalars. The result is **one** polynomial, not a number.

Well-definedness. Each term of the Leibniz sum is a product of n polynomials, and the sum is finite, so p_A is a polynomial. It depends only on A .

How to compute it. By the remarks in §4, §5 and §6, the following results were proved without dividing, and so hold for matrices over $F[x]$: Theorem 6.4.5 and Theorem 6.4.6 (§4), Theorem 6.5.4 by its first proof (§5), Theorem 6.6.1 (§6), and Theorem 6.8.4 (§8). So we may expand $\det(xI - A)$ along any row or column, and read it off from triangular or block triangular shape. What we may **not** do is row reduce $xI - A$ by dividing by a polynomial entry, as the warning below explains.

Examples.

- 1×1 . For $A = (a)$, $p_A(x) = x - a$.
- 2×2 . For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$p_A(x) = (x - a)(x - d) - bc = x^2 - (a + d)x + (ad - bc) = x^2 - (\operatorname{tr} A)x + \det A.$$

For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ this is $x^2 - 5x - 2$, as above.

- **Triangular.** If A is upper or lower triangular, so is $xI - A$, with diagonal entries $x - a_{ii}$. By Theorem 6.4.6 over $F[x]$, $p_A(x) = (x - a_{11})(x - a_{22}) \cdots (x - a_{nn})$.
- **Degenerate cases.** The zero matrix has $p_0(x) = x^n$, and the identity has $p_{I_n}(x) = (x - 1)^n$; both are triangular. These two polynomials are different, although both matrices are as simple as possible, which already shows that p_A records something about A .

Example. (A 3×3 Characteristic Polynomial)

Find p_A for $A = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 3 & 1 \\ 1 & 0 & 2 \end{pmatrix} \in M_3(\mathbb{Q})$.

Solution. We have

$$xI - A = \begin{pmatrix} x-2 & 0 & -1 \\ -1 & x-3 & -1 \\ -1 & 0 & x-2 \end{pmatrix}.$$

Column 2 has a single non-zero entry, $x-3$ in position $(2,2)$. Expanding along it ([Theorem 6.6.1](#) over $F[x]$),

$$p_A(x) = (x-3)(-1)^{2+2} \det \begin{pmatrix} x-2 & -1 \\ -1 & x-2 \end{pmatrix} = (x-3)((x-2)^2 - 1) = (x-1)(x-3)^2.$$

Multiplied out, $p_A(x) = x^3 - 7x^2 + 15x - 9$. Compare with $\text{tr } A = 2 + 3 + 2 = 7$ and $\det A = 9$ (expand A along column 2: $3(4-1) = 9$). Sympy confirms the result.

A companion-type example. Let $A = \begin{pmatrix} 0 & 0 & 6 \\ 1 & 0 & -11 \\ 0 & 1 & 6 \end{pmatrix}$, with ones just below the diagonal, a

last column $(6, -11, 6)$, and zeros elsewhere. Expanding $xI - A = \begin{pmatrix} x & 0 & -6 \\ -1 & x & 11 \\ 0 & -1 & x-6 \end{pmatrix}$ along the first row ([Theorem 6.6.1](#) over $F[x]$),

$$p_A(x) = x \det \begin{pmatrix} x & 11 \\ -1 & x-6 \end{pmatrix} + (-6) \det \begin{pmatrix} -1 & x \\ 0 & -1 \end{pmatrix} = x(x^2 - 6x + 11) - 6 = x^3 - 6x^2 + 11x - 6,$$

which factors as $(x-1)(x-2)(x-3)$. The last column of A lists the coefficients $-6, 11, -6$ of $1, x, x^2$ in p_A , with their signs changed. This is no accident: [Exercise \(C1: Companion matrices\)](#) shows that the same pattern works in every size.

Non-example by minimal change. Swap the order of subtraction: $\det(A - xI)$. Each of the n columns of $A - xI$ is -1 times the corresponding column of $xI - A$, so by multilinearity $\det(A - xI) = (-1)^n p_A(x)$. For $A = (a)$ this is $a - x$, whose leading coefficient is -1 . What still works: the roots are the same. What fails: the result is **not monic** when n is odd.

Why this definition. Some books use $\det(A - xI)$. We use $xI - A$ so that the characteristic polynomial is always **monic** ([Theorem 6.10.2](#) below), which makes statements like “ $p_A = (x-1)(x-3)^2$ ” true without sign bookkeeping. The name comes from the fact that the roots of this polynomial are the “characteristic values” of A , which Chapter 8 calls eigenvalues.

Warning. **Do not row reduce $xI - A$ by dividing by polynomials.** Over a field, a matrix with non-zero determinant is invertible ([Theorem 6.5.2](#)), but that proof runs through elimination, which divides by pivots, and it fails over $F[x]$. For $A = (0) \in M_1(F)$, the matrix $xI - A = (x)$ has determinant $x \neq 0$, yet it has no inverse in $M_1(F[x])$: if $xq(x) = 1$ for a polynomial q , then $q \neq 0$, and $1 + \deg q = \deg 1 = 0$ by [Theorem 0.8.3](#), which is impossible since $\deg q \geq 0$. Likewise, “divide row 1 of $xI - A$ by $x-1$ ” produces entries that are not polynomials and that make no sense at $x=1$. Compute p_A by expansion, by triangular or block shape, or by the Leibniz formula.

Before looking at the coefficients, we record the fact that makes p_A useful: evaluating it at a scalar c gives the same result as putting c in place of x before taking the determinant.

Lemma 6.10.1 — Evaluating the Characteristic Polynomial

Let $A \in M_n(F)$ and $c \in F$. Then $p_A(c) = \det(cI_n - A)$.

Proof. By [Theorem 0.8.5](#), evaluation at c turns sums of polynomials into sums and products into products, and sends a constant a to a . Applying it to the Leibniz sum in [Definition 6.10.1](#) term by term,

$$p_A(c) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^n (c\delta_{\sigma(j)j} - a_{\sigma(j)j}),$$

and $c\delta_{ij} - a_{ij}$ is the (i, j) -entry of $cI_n - A$. By [Definition 6.4.1](#), the right side is $\det(cI_n - A)$. ■

6.10.2 The shape of the characteristic polynomial

The 2×2 example suggests a pattern: the leading coefficient is 1, the next coefficient is minus the trace, and the constant term is the determinant up to sign. The same holds in every size.

Theorem 6.10.2 — Coefficients of the Characteristic Polynomial

Let $n \geq 1$ and $A \in M_n(F)$. Then p_A is **monic of degree n** , and

$$p_A(x) = x^n - (\operatorname{tr} A)x^{n-1} + \cdots + (-1)^n \det A,$$

that is, the coefficient of x^{n-1} is $-\operatorname{tr} A$ and the constant term is $(-1)^n \det A$. (For $n = 1$ these two statements concern the same coefficient, and both say it is $-a_{11}$.)

Idea. Look at the Leibniz sum term by term and count powers of x . A factor $x\delta_{\sigma(j)j} - a_{\sigma(j)j}$ contains x only if $\sigma(j) = j$. A permutation other than the identity moves at least **two** indices, so its term has degree at most $n - 2$ and cannot affect the top two coefficients. The identity contributes $\prod_j (x - a_{jj})$, whose top two coefficients are 1 and $-\sum_j a_{jj}$. For the constant term, set $x = 0$.

Proof. Write $m_{ij} = x\delta_{ij} - a_{ij}$ for the entries of $xI - A$. By [Definition 6.10.1](#), $p_A = \sum_{\sigma} t_{\sigma}$ with $t_{\sigma} = \operatorname{sgn}(\sigma) \prod_j m_{\sigma(j)j}$.

Terms with $\sigma \neq \operatorname{id}$. If $\sigma(j) \neq j$ for some j , then $\sigma(\sigma(j)) \neq \sigma(j)$, because σ is injective. So σ moves at least two indices, and at most $n - 2$ indices are fixed. For a moved index j , $m_{\sigma(j)j} = -a_{\sigma(j)j}$ has degree at most 0; for a fixed index it has degree at most 1. By [Theorem 0.8.3](#), $\deg t_{\sigma} \leq n - 2$.

The term $\sigma = \operatorname{id}$. We show by induction on $k \geq 1$ that $\prod_{j=1}^k (x - a_{jj}) = x^k - s_k x^{k-1} + r_k$ with $s_k = a_{11} + \cdots + a_{kk}$ and $\deg r_k \leq k - 2$. For $k = 1$, take $r_1 = 0$. If it holds for k ,

then with $a = a_{k+1,k+1}$,

$$(x^k - s_k x^{k-1} + r_k)(x - a) = x^{k+1} - (s_k + a)x^k + (s_k a x^{k-1} + r_k x - a r_k),$$

and the bracket has degree at most $k - 1$ by [Theorem 0.8.2](#) and [Theorem 0.8.3](#). This completes the induction. Since $\text{sgn}(\text{id}) = 1$, $t_{\text{id}} = x^n - (\text{tr } A)x^{n-1} + r_n$ with $\deg r_n \leq n - 2$.

Adding, $p_A = x^n - (\text{tr } A)x^{n-1} + (r_n + \sum_{\sigma \neq \text{id}} t_\sigma)$, and the bracket has degree at most $n - 2$ by [Theorem 0.8.2](#). Hence p_A is monic of degree n , with coefficient $-\text{tr } A$ at x^{n-1} ([Definition 0.9.12](#)).

Constant term. The constant term of a polynomial is its value at 0 ([Definition 0.8.6](#)). By [Lemma 6.10.1](#) with $c = 0$, $p_A(0) = \det(-A)$. The matrix $-A$ is obtained from A by multiplying each of its n columns by -1 , so $\det(-A) = (-1)^n \det A$ by [Theorem 6.5.1](#) (d). This proves the theorem. ■

So the characteristic polynomial packages two invariants we already know, the trace and the determinant, and adds $n - 2$ more coefficients in between. For $n = 2$ nothing is in between, which is why $p_A(x) = x^2 - (\text{tr } A)x + \det A$ for every 2×2 matrix.

Quick check. Without expanding any determinant, find the coefficient of x^2 and the constant term of p_A for $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$, given that $\det A = 2$.

Solution. By [Theorem 6.10.2](#), the coefficient of x^2 is $-\text{tr } A = 0$, and the constant term is $(-1)^3 \det A = -2$. (In full, $p_A(x) = x^3 - 3x - 2 = (x - 2)(x + 1)^2$.)

6.10.3 Similar matrices, same polynomial

Similar matrices describe the same operator in different bases ([Theorem 3.7.4](#)), so everything intrinsic to the operator should agree for them. The trace and determinant do ([Theorem 3.7.5](#), [Corollary 6.5.7](#)). The characteristic polynomial, which contains both, does too, and the proof is the proof for the determinant, run over $F[x]$.

Theorem 6.10.3 — Similar Matrices Have the Same Characteristic Polynomial

If $A, B \in M_n(F)$ are similar, then $p_A = p_B$.

Proof. Let $B = P^{-1}AP$ with $P \in M_n(F)$ invertible. Regard P , P^{-1} , A and I as matrices over $F[x]$ whose entries happen to be constants. The laws of matrix arithmetic in [Theorem 0.9.4](#) are proved from the ring axioms alone, so they hold in $M_n(F[x])$. Hence

$$P^{-1}(xI - A)P = xP^{-1}P - P^{-1}AP = xI - B.$$

By [Theorem 6.5.4](#), valid over $F[x]$ by the remark in §5,

$$p_B = \det(P^{-1}(xI - A)P) = \det(P^{-1}) \det(xI - A) \det P = \det(P^{-1}) \det(P) p_A,$$

using commutativity of $F[x]$ in the last step. Finally $\det(P^{-1}) \det P = \det(P^{-1}P) = \det I = 1$ by [Theorem 6.5.4](#) over F . Hence $p_B = p_A$. ■

This is exactly the case singled out in the remark in §5: the proof of [Corollary 6.5.7](#) survives over $F[x]$ because P is invertible already over F . The theorem lets us attach a characteristic polynomial to an operator.

Definition 6.10.2 — Characteristic Polynomial of an Operator

Let V be a vector space over F with $\dim V = n \geq 1$, and let $T \in \mathcal{L}(V)$. The **characteristic polynomial** of T is

$$p_T(x) := p_{[T]_{\mathcal{B}}}(x) = \det(xI_n - [T]_{\mathcal{B}}),$$

where $\mathcal{B}$ is **any** basis of V .

This does not depend on $\mathcal{B}$: for bases $\mathcal{B}$ and $\mathcal{C}$, the matrices $[T]_{\mathcal{B}}$ and $[T]_{\mathcal{C}}$ are similar by [Theorem 3.7.4](#) (a), so they have the same characteristic polynomial by [Theorem 6.10.3](#). By [Theorem 6.10.2](#), p_T is monic of degree $\dim V$, with coefficient $-\operatorname{tr} T$ at x^{n-1} and constant term $(-1)^n \det T$ ([Definition 3.10.2](#), [Definition 6.5.1](#)).

Example. (Characteristic Polynomials of Two Operators)

On $V = \mathbb{R}[x]_{\leq 2}$, find the characteristic polynomials of differentiation $D(q) = q'$ and of the shift $S(q)(x) = q(x+1)$.

Solution. Use the basis $\mathcal{B} = (1, x, x^2)$. Since $D(1) = 0$, $D(x) = 1$, $D(x^2) = 2x$, and $S(1) = 1$, $S(x) = 1 + x$, $S(x^2) = 1 + 2x + x^2$,

$$[D]_{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}, \quad [S]_{\mathcal{B}} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Both are upper triangular, so $p_D(t) = t^3$ and $p_S(t) = (t-1)^3$. (We name the variable t here, since x is already the variable of the polynomials in V .) As a check, $\operatorname{tr} S = 3$ and $\det S = 1$, matching the coefficient -3 of t^2 and the constant term $(-1)^3 \cdot 1$.

Warning. Equal characteristic polynomials do not make matrices similar. The converse of [Theorem 6.10.3](#) is false. Recall the pair from the warning in Chapter 3, §7: I_2 and $J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ both have characteristic polynomial $(x-1)^2$, since both are triangular with diagonal 1, 1. But I_2 is similar only to itself, so $J \not\sim I_2$. In particular the characteristic polynomial does not determine the matrix, not even up to similarity. Deciding similarity needs finer invariants; Chapter 9 finds a complete set.

6.10.4 Roots and singular matrices

We can now answer the opening question. The determinant tests invertibility of $cI - A$ for one c at a time; the characteristic polynomial tests all c at once, and the answer is a finite set.

Theorem 6.10.4 — Roots of the Characteristic Polynomial

Let $A \in M_n(F)$ with $n \geq 1$, and let $c \in F$.

- (a) The following are equivalent: (i) $p_A(c) = 0$; (ii) $cI - A$ is **not** invertible; (iii) there is a **non-zero** $\mathbf{v} \in F^n$ with $A\mathbf{v} = c\mathbf{v}$.
- (b) There are **at most** n scalars $c \in F$ with these properties.

Proof.

- (a) By Lemma 6.10.1, $p_A(c) = \det(cI - A)$, which is a determinant of a matrix over the field F . By Theorem 6.5.2, it is zero if and only if $cI - A$ is not invertible; this is (i) $\Leftrightarrow$ (ii). By Theorem 2.4.4, $cI - A$ is not invertible if and only if $(cI - A)\mathbf{v} = \mathbf{0}$ for some $\mathbf{v} \neq \mathbf{0}$, and $(cI - A)\mathbf{v} = c\mathbf{v} - A\mathbf{v}$ by Theorem 0.9.4; this is (ii) $\Leftrightarrow$ (iii).
- (b) By Theorem 6.10.2, p_A is monic of degree n , in particular non-zero. By Corollary 5.1.4, it has at most n distinct roots in F , and by (a) these are exactly the scalars in question.

■

Notice where the evaluation happens: **first** substitute c , getting a matrix over the field F , **then** use the invertibility test, which is valid over fields. That order is what the warning about $F[x]$ above requires.

The same holds for an operator T on V with $\dim V = n \geq 1$: $p_T(c) = 0$ if and only if $c \operatorname{id}_V - T$ is not invertible, by (a) applied to $[T]_{\mathcal{B}}$ together with Theorem 3.6.5 (b), since $[c \operatorname{id}_V - T]_{\mathcal{B}} = cI - [T]_{\mathcal{B}}$.

The scalars c of this theorem are the **eigenvalues** of A , and the vectors $\mathbf{v}$ in (iii) its eigenvectors. They are the subject of Chapter 8, which starts from this theorem.

Examples.

- For A in Example (A 3×3 Characteristic Polynomial), $p_A = (x - 1)(x - 3)^2$, so $cI - A$ is singular exactly for $c = 1$ and $c = 3$. For instance $3I - A = \begin{pmatrix} 1 & 0 & -1 \\ -1 & 0 & -1 \\ -1 & 0 & 1 \end{pmatrix}$ has a zero column, and $\mathbf{v} = \mathbf{e}_2$ satisfies $A\mathbf{e}_2 = 3\mathbf{e}_2$.
- The rotation $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has $p_R = x^2 + 1$. Over $\mathbb{R}$ this has no root, so $cI - R$ is invertible for **every** real c : no non-zero real vector is merely stretched by a quarter-turn. Over $\mathbb{C}$, the roots $\pm i$ appear, and for example $R(1, -i) = (i, 1) = i(1, -i)$. The field matters.

Quick check. For which $c \in \mathbb{Q}$ is $cI - A$ invertible, where $A = \begin{pmatrix} 5 & -2 \\ 3 & 0 \end{pmatrix}$?

Solution. $p_A(x) = x^2 - (\operatorname{tr} A)x + \det A = x^2 - 5x + 6 = (x-2)(x-3)$. By Theorem 6.10.4, $cI - A$ is invertible exactly for $c \neq 2, 3$.

6.10.5 A tempting false proof

Here is a statement that is true, and important: **every square matrix satisfies its own characteristic polynomial**, $p_A(A) = 0$. This is the Cayley–Hamilton Theorem, proved in Chapter 8. Here $p_A(A)$ means substituting the matrix A into the polynomial p_A , replacing each x^k by A^k and the constant term a_0 by a_0I (Definition 0.9.14). For a 2×2 matrix it says $A^2 - (\operatorname{tr} A)A + (\det A)I = 0$, which you can check by multiplying out.

A one-line “proof” suggests itself, and it is wrong.

Warning. The fake proof of Cayley–Hamilton. “Since $p_A(x) = \det(xI - A)$, put $x = A$: $p_A(A) = \det(AI - A) = \det(0) = 0$.” This is invalid for two reasons. First, the two sides are different kinds of object: $p_A(A)$ is an $n \times n$ **matrix**, while $\det(\cdots)$ is a **scalar**, so “ $= 0$ ” compares a matrix with a number. Second, Lemma 6.10.1 lets us substitute only a **scalar** $c \in F$ into the entries of $xI - A$. Substituting a matrix for x inside the entries would produce a matrix whose entries are matrices, and “ $AI - A$ ” silently treats the x in the entry $x - a_{11}$ as if the entry became $A - a_{11}$, which is not what substitution into a polynomial means.

The same “argument” proves false statements, which shows it is not a proof. Replace $\det$ by tr : let $q(x) = \operatorname{tr}(xI - A) = nx - \operatorname{tr} A$. The fake reasoning gives “ $q(A) = \operatorname{tr}(AI - A) = \operatorname{tr} 0 = 0$ ”. But for $A = \operatorname{diag}(1, 0)$, $q(x) = 2x - 1$ and $q(A) = 2A - I = \operatorname{diag}(1, -1) \neq 0$.

6.10.6 Exercises

6.10.6.1 A. Check your understanding

Exercise A1.

- Define the characteristic polynomial of $A \in M_n(F)$, and say in which ring its determinant is computed.
- State the degree, leading coefficient, coefficient of x^{n-1} and constant term of p_A .
- True or false: $\det(A - xI) = p_A(x)$ for every $A \in M_3(F)$. Justify your answer.
- True or false: if $p_A = p_B$, then $A \sim B$. Justify your answer.
- What does $p_A(c) = 0$ say about the matrix $cI - A$? About vectors?
- Explain in one sentence why “ $p_A(A) = \det(AI - A) = 0$ ” is not a proof.

Solution.

- $p_A(x) = \det(xI_n - A)$, the Leibniz sum computed in the commutative ring $F[x]$ (Definition 6.10.1).
- Degree n , leading coefficient 1, coefficient $-\operatorname{tr} A$ at x^{n-1} , constant term $(-1)^n \det A$ (Theorem 6.10.2).

- (c) False. $\det(A - xI) = (-1)^3 p_A(x) = -p_A(x)$, since each of the three columns changes sign. For $A = 0$, $\det(-xI_3) = -x^3 \neq x^3 = p_A(x)$.
- (d) False. I_2 and $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ both have $p = (x - 1)^2$ but are not similar.
- (e) $cI - A$ is not invertible; equivalently there is a non-zero $\mathbf{v}$ with $A\mathbf{v} = c\mathbf{v}$ ([Theorem 6.10.4](#)).
- (f) $p_A(A)$ is a matrix obtained by substituting A into a polynomial, while $\det(AI - A)$ is a scalar, and substitution into the entries of $xI - A$ is only justified for scalars ([Lemma 6.10.1](#)).

6.10.6.2 B. Practice

Exercise B1: Computing characteristic polynomials.

Find the characteristic polynomial of each matrix over $\mathbb{Q}$, factor it as far as possible over $\mathbb{Q}$, and check the trace and determinant against [Theorem 6.10.2](#).

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 1 & -1 \end{pmatrix}.$$

Hence list the $c \in \mathbb{Q}$ for which $cI - A$, respectively $cI - B$, is not invertible.

Solution. For A . Expanding $xI - A = \begin{pmatrix} x-1 & -1 & 0 \\ 0 & x-1 & -1 \\ -1 & 0 & x-1 \end{pmatrix}$ along the first row ([Theorem 6.6.1](#) over $F[x]$),

$$p_A(x) = (x-1)((x-1)^2 - 0) - (-1)(0-1) + 0 = (x-1)^3 - 1 = x^3 - 3x^2 + 3x - 2.$$

Check: $-\operatorname{tr} A = -3$; $\det A = 1 \cdot (1-0) - 1 \cdot (0-1) + 0 = 2$, and $(-1)^3 \cdot 2 = -2$. Since $p_A(2) = 1 - 1 = 0$, divide by $x - 2$: $p_A = (x-2)(x^2 - x + 1)$. The quadratic has discriminant $1 - 4 = -3 < 0$, so no rational (indeed no real) root. By [Theorem 6.10.4](#), $cI - A$ is singular only for $c = 2$.

For B . $xI - B$ is block upper triangular with diagonal blocks $xI - \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ and $xI - \begin{pmatrix} 1 & 3 \\ 1 & -1 \end{pmatrix}$. By [Theorem 6.8.4](#) over $F[x]$ and the 2×2 formula,

$$p_B(x) = (x-2)^2(x^2 - 0 \cdot x + (-1-3)) = (x-2)^2(x^2 - 4) = (x-2)^3(x+2).$$

Multiplied out, $p_B = x^4 - 4x^3 + 16x - 16$. Check: $-\operatorname{tr} B = -(2+2+1-1) = -4$; $\det B = 4 \cdot (-4) = -16$ by [Theorem 6.8.4](#), and $(-1)^4(-16) = -16$. So $cI - B$ is singular exactly for $c = 2$ and $c = -2$.

Exercise B2: Transpose.

Prove that $p_{A^\top} = p_A$ for every $A \in M_n(F)$. Hence show that $cI - A$ is invertible if and only if $cI - A^\top$ is.

Solution. By [Theorem 0.9.5](#), $(xI - A)^\top = xI^\top - A^\top = xI - A^\top$; these rules are proved entrywise from the ring axioms, so they hold over $F[x]$. By [Theorem 6.4.5](#), which holds over $F[x]$ by the remark in §4,

$$p_{A^\top} = \det(xI - A^\top) = \det((xI - A)^\top) = \det(xI - A) = p_A.$$

By [Theorem 6.10.4](#), $cI - A$ is not invertible exactly when $p_A(c) = 0$, which is exactly when $p_{A^\top}(c) = 0$, that is, when $cI - A^\top$ is not invertible.

Exercise B3: AB and BA .

Let $A, B \in M_n(F)$ with A invertible.

- Show that $BA = A^{-1}(AB)A$, and deduce $p_{AB} = p_{BA}$.
- Deduce that $\text{tr}(AB) = \text{tr}(BA)$ and $\det(AB) = \det(BA)$ in this case.
- Check (a) for $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, and note that $AB \neq BA$.

(Chapter 8 removes the hypothesis that A is invertible.)

Solution.

- By associativity, $A^{-1}(AB)A = (A^{-1}A)(BA) = BA$. So $BA \sim AB$ ([Definition 3.7.2](#), with $P = A$), and $p_{AB} = p_{BA}$ by [Theorem 6.10.3](#).
- By [Theorem 6.10.2](#), the coefficient of x^{n-1} is $-\text{tr}(AB)$ in p_{AB} and $-\text{tr}(BA)$ in p_{BA} ; the constant terms are $(-1)^n \det(AB)$ and $(-1)^n \det(BA)$. Equal polynomials have equal coefficients.
- $AB = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$ and $BA = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$, so $AB \neq BA$. Both have trace 1 and determinant 0, so both characteristic polynomials are $x^2 - x = x(x - 1)$.

6.10.6.3 C. Going deeper

Exercise C1: Companion matrices.

Let $n \geq 2$ and $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 \in F[x]$. The **companion matrix** of p is the matrix $C(p) \in M_n(F)$ with 1 in each position $(i+1, i)$ just below the diagonal, last column $(-a_0, -a_1, \dots, -a_{n-1})$, and 0 elsewhere:

$$C(p) = \begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{pmatrix}.$$

(The matrix C of [Exercise \(B2: An annihilating polynomial by elimination\)](#) is $C(x^3 -$

$2x^2 + x - 2$.)

- (a) Let $M = xI - C(p)$. For each i , show that deleting row i and column n of M leaves a block diagonal matrix whose blocks are an $(i-1) \times (i-1)$ lower triangular matrix with diagonal $x, \dots, x$ and an $(n-i) \times (n-i)$ upper triangular matrix with diagonal $-1, \dots, -1$ (either block may be absent when $i = 1$ or $i = n$). Deduce that the cofactor C_{in} of M equals x^{i-1} .
- (b) Deduce, by expanding along the last column, that $p_{C(p)} = p$.
- (c) Verify (b) for $p = x^4 + x^3 - 2x + 3$ by writing $C(p)$ and checking the coefficients predicted by [Theorem 6.10.2](#).
- (d) Deduce that every monic polynomial of degree $n \geq 1$ over F is the characteristic polynomial of some $n \times n$ matrix over F .

Solution.

- (a) The entries of M are: x at (j, j) for $j < n$; -1 at $(j+1, j)$; a_{j-1} at (j, n) for $j < n$; $x + a_{n-1}$ at (n, n) ; 0 elsewhere. Delete row i and column n . In the remaining matrix, rows $1, \dots, i-1$ have non-zero entries only in columns $j-1$ and j for row j , all at most $i-1$; and rows $i+1, \dots, n$ have non-zero entries only in columns $j-1$ and $j \leq n-1$, all at least i . So the matrix is block diagonal with blocks on rows and columns $\{1, \dots, i-1\}$ and on rows $\{i+1, \dots, n\}$, columns $\{i, \dots, n-1\}$. The first block has x on its diagonal and -1 just below it, so it is lower triangular. In the second block, row j of M (for $j > i$) has -1 in column $j-1$, which is the diagonal position of the block, and x in column j , one step to the right; so it is upper triangular with diagonal $-1, \dots, -1$. By [Theorem 6.8.4](#) and [Theorem 6.4.6](#) over $F[x]$ (both division-free), the minor is $M_{in} = x^{i-1}(-1)^{n-i}$, where an absent block contributes the factor 1 (for $i = 1$ or $i = n$ the matrix is a single triangular block). Hence $C_{in} = (-1)^{i+n}(-1)^{n-i}x^{i-1} = (-1)^{2n}x^{i-1} = x^{i-1}$.
- (b) By [Theorem 6.6.1](#) along column n , valid over $F[x]$,

$$p_{C(p)} = \det M = \sum_{i=1}^{n-1} a_{i-1} x^{i-1} + (x + a_{n-1}) x^{n-1} = a_0 + a_1 x + \dots + a_{n-1} x^{n-1} + x^n = p.$$

- (c) Here $a_3 = 1$, $a_2 = 0$, $a_1 = -2$, $a_0 = 3$, so

$$C(p) = \begin{pmatrix} 0 & 0 & 0 & -3 \\ 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}.$$

Its trace is -1 , so [Theorem 6.10.2](#) predicts the coefficient $1 = a_3$ at x^3 , as in p . Its determinant, expanding along the first row, whose only non-zero entry is -3 in position $(1, 4)$, is $(-3)(-1)^{1+4} \det I_3 = 3$, since deleting row 1 and column 4 leaves I_3 . So the predicted constant term is $(-1)^4 \cdot 3 = 3 = a_0$. Sympy computes $\det(xI - C(p)) = x^4 + x^3 - 2x + 3$ in full.

- (d) For $n \geq 2$, take $C(p)$ and use (b). For $n = 1$, $p = x + a_0$ is $p_{(-a_0)}$.

Exercise C2: Nilpotent 2×2 matrices.

Let $A \in M_2(F)$ satisfy $A^k = 0$ for some $k \geq 1$.

- (a) Show that the constant term of p_A is 0.
- (b) By multiplying out, verify that $A^2 - (\operatorname{tr} A)A + (\det A)I = 0$ for every $A \in M_2(F)$.
- (c) Deduce that $\operatorname{tr} A = 0$, and hence $p_A = x^2$ and $A^2 = 0$.
- (d) Give an example showing that $A^2 = 0$ does not force $A = 0$.

Solution.

- (a) By [Theorem 6.5.4](#), $(\det A)^k = \det(A^k) = \det 0 = 0$, so $\det A = 0$ because a field has no zero divisors ([Theorem 0.7.1](#)). The constant term is $(-1)^2 \det A = 0$ by [Theorem 6.10.2](#).
- (b) Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then $A^2 = \begin{pmatrix} a^2 + bc & ab + bd \\ ca + dc & bc + d^2 \end{pmatrix}$ and $(a+d)A = \begin{pmatrix} a^2 + ad & ab + bd \\ ac + dc & ad + d^2 \end{pmatrix}$. The difference is $\begin{pmatrix} bc - ad & 0 \\ 0 & bc - ad \end{pmatrix} = -(\det A)I$, as claimed.
- (c) By (a) and (b), $A^2 = (\operatorname{tr} A)A$. By induction, $A^m = (\operatorname{tr} A)^{m-1}A$ for all $m \geq 1$: if it holds for m , then $A^{m+1} = (\operatorname{tr} A)^{m-1}A^2 = (\operatorname{tr} A)^m A$. If $A = 0$, then $\operatorname{tr} A = 0$. If $A \neq 0$, then $0 = A^k = (\operatorname{tr} A)^{k-1}A$ forces $(\operatorname{tr} A)^{k-1} = 0$, since a non-zero matrix times a scalar is zero only if the scalar is zero; here $k \geq 2$, because $k = 1$ would give $A = 0$. Hence $\operatorname{tr} A = 0$ by [Theorem 0.7.1](#) (f), applied repeatedly. So $p_A = x^2 - 0 \cdot x + 0 = x^2$, and $A^2 = (\operatorname{tr} A)A = 0$.
- (d) $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has $A^2 = 0$ and $A \neq 0$.

Exercise C3: Odd size over the reals.

Let n be odd and $A \in M_n(\mathbb{R})$.

- (a) Prove that there is $c \in \mathbb{R}$ and a non-zero $\mathbf{v} \in \mathbb{R}^n$ with $A\mathbf{v} = c\mathbf{v}$.
- (b) Show by an example that (a) fails for $n = 2$.
- (c) Show that (a) holds for every $A \in M_n(\mathbb{C})$ and every $n \geq 1$.

Solution.

- (a) By [Theorem 6.10.2](#), $p_A \in \mathbb{R}[x]$ has odd degree n . By [Corollary 5.5.9](#), it has a real root c . By [Theorem 6.10.4](#), there is a non-zero $\mathbf{v} \in \mathbb{R}^n$ with $A\mathbf{v} = c\mathbf{v}$.
- (b) The rotation $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has $p_R = x^2 + 1$, with no real root; by [Theorem 6.10.4](#) no real c and non-zero real $\mathbf{v}$ satisfy $R\mathbf{v} = c\mathbf{v}$.
- (c) $p_A \in \mathbb{C}[x]$ has degree $n \geq 1$, so it has a root $c \in \mathbb{C}$ by [Theorem 5.5.1](#). Apply [Theorem 6.10.4](#) over $F = \mathbb{C}$.